



Cisco Catalyst SD-WAN Systems and Interfaces Configuration Guide, Cisco IOS XE Catalyst SD-WAN Release 17.x

First Published: 2019-04-15

Last Modified: 2025-09-21

Americas Headquarters

Cisco Systems, Inc.
170 West Tasman Drive
San Jose, CA 95134-1706
USA
<http://www.cisco.com>
Tel: 408 526-4000
800 553-NETS (6387)
Fax: 408 527-0883



CHAPTER 1

Read Me First



Note

To achieve simplification and consistency, the Cisco SD-WAN solution has been rebranded as Cisco Catalyst SD-WAN. In addition, from Cisco IOS XE SD-WAN Release 17.12.1a and Cisco Catalyst SD-WAN Release 20.12.1, the following component changes are applicable: **Cisco vManage** to **Cisco Catalyst SD-WAN Manager**, **Cisco vAnalytics** to **Cisco Catalyst SD-WAN Analytics**, **Cisco vBond** to **Cisco Catalyst SD-WAN Validator**, **Cisco vSmart** to **Cisco Catalyst SD-WAN Controller**, and **Cisco Controllers** to **Cisco Catalyst SD-WAN Control Components**. See the latest Release Notes for a comprehensive list of all the component brand name changes. While we transition to the new names, some inconsistencies might be present in the documentation set because of a phased approach to the user interface updates of the software product.

Related References

- [Cisco Catalyst SD-WAN Control Components Compatibility Matrix and Server Recommendations](#)
- [Cisco Catalyst SD-WAN Device Compatibility](#)

User Documentation

- [User Documentation for Cisco IOS XE Catalyst SD-WAN Release 17](#)

Communications, Services, and Additional Information

- Sign up for Cisco email newsletters and other communications at: [Cisco Profile Manager](#).
- For information on the latest technical, advanced, and remote services to increase the operational reliability of your network visit [Cisco Services](#).
- To browse and discover secure, validated enterprise-class apps, products, solutions, and services, visit [Cisco Devnet](#).
- To obtain general networking, training, and certification titles from Cisco Press Publishers, visit [Cisco Press](#).
- To find warranty information for a specific product or product family, visit [Cisco Warranty Finder](#).
- To view open and resolved bugs for a release, access the [Cisco Bug Search Tool](#).
- To submit a service request, visit [Cisco Support](#).

Documentation Feedback

To provide feedback about Cisco technical documentation use the feedback form available in the right pane of every online document.



CHAPTER 2

What's New in Cisco IOS XE (Catalyst SD-WAN)

[What's New in Cisco IOS XE Catalyst SD-WAN Release 17.x](#)

[What's New in Cisco SD-WAN \(vEdge\) Release 20.x](#)



CHAPTER 3

Cisco Catalyst SD-WAN Basic System Overview

- [Cisco Catalyst SD-WAN Basic System Overview, on page 5](#)
- [System and Interfaces Overview, on page 6](#)
- [Basic Settings for Cisco SD-WAN Manager, on page 10](#)
- [Configure Basic System Parameters Using a Configuration Group, on page 18](#)
- [Configure Basic System Parameters, on page 21](#)
- [Configure Global Using a Configuration Group, on page 27](#)
- [Configure Global Parameters, on page 29](#)
- [Configure NTP Servers Using a Configuration Group, on page 33](#)
- [Configure NTP Servers Using Cisco SD-WAN Manager, on page 35](#)
- [Configure a Router as an NTP Primary, on page 38](#)
- [Configure NTP Servers Using CLI Commands, on page 39](#)
- [Configure Time using CLI, on page 41](#)
- [Configure GPS Using a Configuration Group, on page 41](#)
- [Configure GPS Using Cisco SD-WAN Manager, on page 42](#)
- [Configure Automatic Bandwidth Detection, on page 44](#)
- [Configure System Logging Using CLI, on page 46](#)
- [SSH Terminal, on page 46](#)
- [HTTP/HTTPS Proxy Server for Cisco SD-WAN Manager Communication with External Servers, on page 47](#)
- [Bulk API Rate Limit for a Cisco SD-WAN Manager Cluster, on page 49](#)

Cisco Catalyst SD-WAN Basic System Overview

Table 1: Feature History

Feature Name	Release Information	Description
CMAC-AES-128 Authentication for NTP Servers	Cisco Catalyst SD-WAN Control Components Release 20.14.1	Support for cipher-based message authentication code (CMAC) advanced encryption standard (AES) 128-bit (cmac-aes-128) authentication for network time protocol (NTP) server configuration for Cisco SD-WAN Control Components.

Feature Name	Release Information	Description
Statistics Database Upgrade	Cisco Catalyst SD-WAN Manager Release 20.16.1	Cisco SD-WAN Manager utilizes an upgraded database to improve performance

System and Interfaces Overview

Setting up the basic system-wide functionality of network devices is a simple and straightforward process. Basic parameters include defining host properties, such as name and IP address; setting time properties, including NTP; setting up user access to the devices; and defining system log (syslog) parameters.

In addition, the Cisco Catalyst SD-WAN software provides a number of management interfaces for accessing the Cisco Catalyst SD-WAN devices in the overlay network.

Host Properties

All devices have basic system-wide properties that specify information that the Cisco Catalyst SD-WAN software uses to construct a view of the network topology. Each device has a system IP address that provides a fixed location of the device in the overlay network. This address, which functions the same way as a router ID on a router, is independent of any of the interfaces and interface IP addresses on the device. The system IP address is one of the four components of the Transport Location (TLOC) property of each device.

A second host property that must be set on all devices is the IP address of the Cisco SD-WAN Validator for the network domain, or a Domain Name System (DNS) name that resolves to one or more IP addresses for Cisco SD-WAN Validators. A Cisco SD-WAN Validator automatically orchestrates the process of bringing up the overlay network, admitting a new device into the overlay, and providing the introductions that allow the device and Cisco SD-WAN Controllers to locate each other.

Two other system-wide host properties are required on all devices, except for the Cisco SD-WAN Validators, to allow the Cisco Catalyst SD-WAN software to construct a view of the topology—the domain identifier and the site identifier.

To configure the host properties, see [Cisco Catalyst SD-WAN Overlay Network Bring-Up Process](#).

Time and NTP

The Cisco Catalyst SD-WAN software implements the Network Time Protocol (NTP) to synchronize and coordinate time distribution across the Cisco Catalyst SD-WAN overlay network. NTP uses a intersection algorithm to select the applicable time servers and avoid issues caused due to network latency. The servers can also redistribute reference time using local routing algorithms and time daemons. NTP is defined in [RFC 5905, Network Time Protocol Version 4: Protocol and Algorithms Specification](#).

User Authentication and Access with AAA, RADIUS, and TACACS+

The Cisco Catalyst SD-WAN software uses Authentication, Authorization, and Accounting (AAA) to provide security for the devices on a network. AAA, in combination with RADIUS and Terminal Access Controller Access-Control System (TACACS+) user authentication, controls which users are allowed access to devices, and what operations they are authorized to perform after they are logged in or connected to the devices.

Authentication refers to the process by which users trying to access the devices are authenticated. To access devices, users log in with a username and a password. The local device can authenticate users. Alternatively,

authentication can be performed by a remote device, either a RADIUS server or a TACACS+ server, or both in a sequence.

Authorization determines whether a user is authorized to perform a given activity on a device. In the Cisco Catalyst SD-WAN software, authorization is implemented using role-based access. Access is based on groups that are configured on the devices. A user can be a member of one or more groups. User-defined groups are considered when performing authorization, that is, the Cisco Catalyst SD-WAN software uses group names received from RADIUS or TACACS+ servers to check the authorization level of a user. Each group is assigned privileges that authorize the group members to perform specific functions on the corresponding device. These privileges correspond to specific hierarchies of the configuration commands and the corresponding hierarchies of operational commands that members of the group are allowed to view or modify.

Beginning in Cisco IOS XE Catalyst SD-WAN Release 17.5.1a, accounting generates a record of commands that a user executes on a device. Accounting is performed by a TACACS+ server.

For more information, see [Role-Based Access with AAA](#).

Authentication for WANs and WLANs

For wired networks (WANs), Cisco Catalyst SD-WAN devices can run IEEE 802.1X software to prevent unauthorized network devices from gaining access to the WAN. IEEE 802.1X is a port-based network access control (PNAC) protocol that uses a client–server mechanism to provide authentication for devices wishing to connect to the network.

IEEE 802.1X authentication requires three components:

- **Requester:** Client device, such as a laptop, that requests access to the Wide-Area Network (WAN). In the Cisco Catalyst SD-WAN overlay network, a supplicant is any service-side device that is running 802.1X-compliant software. These devices send network access requests to the router.
- **Authenticator:** A network device that provides a barrier to the WAN. In the overlay network, you can configure an interface device to act as an 802.1X authenticator. The device supports both controlled and uncontrolled ports. For controlled ports, the Cisco Catalyst SD-WAN device acts as an 802.1X port access entity (PAE), allowing authorized network traffic and preventing unauthorized network traffic ingressing to and egressing from the controlled port. For uncontrolled ports, Cisco Catalyst SD-WAN, acting as an 802.1X PAE, transmits and receives Extensible Authentication Protocol over IEEE 802 (EAP over LAN, or EAPOL) frames.
- **Authentication server:** Host that is running authentication software that validates and authenticates requesters that want to connect to the WAN. In the overlay network, this host is an external RADIUS server. This RADIUS server authenticates each client connected to the 802.1X port interface Cisco Catalyst SD-WAN device and assigns the interface to a virtual LAN (VLAN) before the client is allowed to access any of the services offered by the router or by the LAN.

For wireless LANs (WLANs), routers can run IEEE 802.11i to prevent unauthorized network devices from gaining access to the WLANs. IEEE 802.11i implements Wi-Fi Protected Access (WPA) and Wi-Fi Protected Access II (WPA2) to provide authentication and encryption for devices that want to connect to a WLAN. WPA authenticates individual users on the WLAN using a username and a password. WPA uses the Temporal Key Integrity Protocol (TKIP), which is based on the RC4 cipher. WPA2 implements the NIST FIPS 140-2–compliant AES encryption algorithm along with IEEE 802.1X-based authentication, to enhance user access security over WPA. WPA2 uses the Counter Mode Cipher Block Chaining Message Authentication Code Protocol (CCMP), which is based on the AES cipher. Authentication is done by either using preshared keys or through RADIUS authentication.

Network Segmentation

The Layer 3 network segmentation in Cisco Catalyst SD-WAN is achieved through VRFs on Cisco IOS XE Catalyst SD-WAN devices. When you configure the network segmentation on a Cisco IOS XE Catalyst SD-WAN device using Cisco SD-WAN Manager, the system automatically maps the VPN configurations to VRF configurations.

Network Interfaces

In the Cisco Catalyst SD-WAN overlay network design, interfaces are associated with VPNs that translate to VRFs. The interfaces that participate in a VPN are configured and enabled in that VPN. Each interface can be present only in a single VPN.



Note Cisco IOS XE Catalyst SD-WAN devices use VRFs in place of VPNs. When you complete the configuration on Cisco SD-WAN Manager, the system automatically maps the VPN configurations to VRF configurations.

The overlay network has the following types of VPNs/VRFs:

- **VPN 0: Transport VPN**, that carries control traffic using the configured WAN transport interfaces. Initially, VPN 0 contains all the interfaces on a device except for the management interface, and all the interfaces are disabled. This is the global VRF on Cisco IOS XE Catalyst SD-WAN software.
- **VPN 512: Management VPN**, that carries out-of-band network management traffic among the Cisco Catalyst SD-WAN devices in the overlay network. The interface used for management traffic resides in VPN 512. By default, VPN 512 is configured and enabled on all Cisco Catalyst SD-WAN devices. For controller devices, by default, VPN 512 is not configured. On Cisco IOS XE Catalyst SD-WAN devices, the management VPN is converted to VRF Mgmt-Intf.

For each network interface, you can configure a number of interface-specific properties, such as DHCP clients and servers, VRRP, interface MTU and speed, and Point-to-Point Protocol over Ethernet (PPPoE). At a high level, for an interface to be operational, you must configure an IP address for the interface and mark it as operational (no shutdown). In practice, you always configure additional parameters for each interface.

Management and Monitoring Options

There are various ways in which you can manage and monitor a router. Management interfaces provide access to devices in the Cisco Catalyst SD-WAN overlay network, allowing you to collect information from the devices in an out-of-band fashion and to perform operations on the devices, such as configuring and rebooting them.

The following management interfaces are available:

- CLI
- IP Flow Information Export (IPFIX)
- RESTful API
- SNMP
- System logging (syslog) messages
- Cisco SD-WAN Manager

CLI

You can access a CLI on each device, and from the CLI, you configure overlay network features on the local device and gather operational status and information regarding that device. Using an available CLI, we strongly recommend that you configure and monitor all the Cisco Catalyst SD-WAN network devices from Cisco SD-WAN Manager, which provides views of network-wide operations and device status, including detailed operational and status data. In addition, Cisco SD-WAN Manager provides straightforward tools for bringing up and configuring overlay network devices, including bulk operations for setting up multiple devices simultaneously.

You can access the CLI by establishing an SSH session to a Cisco Catalyst SD-WAN device.

For a Cisco Catalyst SD-WAN device that is being managed by Cisco SD-WAN Manager, if you create or modify the configuration from the CLI, the changes are overwritten by the configuration that is stored in the Cisco SD-WAN Manager configuration database.

IPFIX

The IP Flow Information Export (IPFIX) protocol, also called cflowd, is a tool for monitoring the traffic flowing through Cisco Catalyst SD-WAN devices in the overlay network and exporting information about the traffic to a flow collector. The exported information is sent in template reports, that contain both information about the flow and the data extracted from the IP headers of the packets in the flow.

Cisco Catalyst SD-WAN cflowd performs 1:1 traffic sampling. Information about all the flows is aggregated in the cflowd records; flows are not sampled.



Note Cisco Catalyst SD-WAN devices do not cache any of the records that are exported to a collector.

The Cisco Catalyst SD-WAN cflowd software implements cflowd Version 10, as specified in RFC 7011 and RFC 7012.

For a list of elements exported by IPFIX, see [Traffic Flow Monitoring with Cflowd](#).

To enable the collection of traffic flow information, you must create data policies that identify the traffic of interest, and then direct that traffic to a cflowd collector. For more information, see [Traffic Flow Monitoring with Cflowd](#).

You can also enable cflowd visibility directly on Cisco Catalyst SD-WAN devices without configuring a data policy, so that you can perform traffic flow monitoring on the traffic coming to the device from all the VPNs in the LAN. You can then monitor the traffic from Cisco SD-WAN Manager or from the device's CLI.

RESTful API

The Cisco Catalyst SD-WAN software provides a RESTful API, which is a programmatic interface for controlling, configuring, and monitoring the Cisco Catalyst SD-WAN devices in an overlay network. You can access the RESTful API through Cisco SD-WAN Manager.

The Cisco Catalyst SD-WAN RESTful API calls expose the functionality of the Cisco Catalyst SD-WAN software and hardware to an application program. Such functionality includes the normal operations you perform to maintain the devices and the overlay network itself.

SNMP

The Simple Network Management Protocol (SNMP) allows you to manage all the Cisco Catalyst SD-WAN devices in the overlay network. The Cisco Catalyst SD-WAN software supports SNMP v2c.

You can configure basic SNMP properties—device name, location, contact, and community—that allow the device to be monitored by an SNMP Network Management System (NMS).

You can configure trap groups and SNMP servers to receive traps.

The object identifier (OID) for the internet port of the SNMP MIB is 1.3.6.1.

SNMP traps are asynchronous notifications that a Cisco Catalyst SD-WAN device sends to an SNMP management server. Traps notify the management server of events, whether normal or significant, that occur on the Cisco Catalyst SD-WAN device. By default, SNMP traps are not sent to an SNMP server. Note that for SNMPv3, the PDU type for notifications, is either SNMPv2c inform (InformRequest-PDU) or trap (Trapv2-PDU).

Syslog Messages

System logging operations use a mechanism that is similar to the UNIX **syslog** command to record system-wide, high-level operations that occur on the Cisco Catalyst SD-WAN devices in the overlay network. The log levels (priorities) of the messages are the same as those in standard UNIX commands, and you can configure the priority of the syslog messages that should be logged. Messages can be logged to files on the Cisco Catalyst SD-WAN device or to a remote host.

Cisco SD-WAN Manager

Cisco SD-WAN Manager is a centralized network management system that allows configuration and management of all the Cisco Catalyst SD-WAN devices in the overlay network, and provides a dashboard displaying the operations of the entire network and of individual devices in the network. Three or more Cisco SD-WAN Manager servers are consolidated into a Cisco SD-WAN Manager cluster to provide scalability and management support for up to 6,000 Cisco Catalyst SD-WAN devices, to distribute Cisco SD-WAN Manager functions across multiple devices, and to provide redundancy of network management operations.

Basic Settings for Cisco SD-WAN Manager

The System template is used to configure system-level Cisco SD-WAN Manager workflows.

Use the Settings screen to view the current settings and configure the setting for Cisco SD-WAN Manager parameters, including the organization name, Cisco SD-WAN Validators DNS name or IP address, certificate settings, and statistics collection.

The current setting for each item is displayed in the bar for each item, immediately following the name.

Configure Organization Name

Before you can generate a Certificate Signing Request (CSR), you must configure the name of your organization. The organization name is included in the CSR.

In public key infrastructure (PKI) systems, a CSR is sent to a certificate authority to apply for a digital identity certificate.

To configure the organization name:

1. From the Cisco SD-WAN Manager menu, choose **Administration > Settings**.
2. Click **Organization Name**. (If you are using Cisco Catalyst SD-WAN Manager Release 20.12.x or earlier, click **Edit**.)
3. In **Organization Name**, enter the name of your organization. The organization name must be identical to the name that is configured on the Cisco SD-WAN Validator.
4. In **Confirm Organization Name**, re-enter and confirm your organization name.
5. Click **Save**.



Note technology-guides

After the control connections are up and running, the organization name bar is no longer editable.

Configure Cisco SD-WAN Validator DNS Name or IP Address

1. From **Validator**, click **Edit**.
2. In **Validator DNS/IP Address: Port**, enter the DNS name that points to the Cisco SD-WAN Validator or the IP address of the Cisco SD-WAN Validator and the port number to use to connect to it.
3. Click **Save**.



Note The DNS cache timeout should be proportional to the number of Cisco Catalyst SD-WAN Validator IP addresses that DNS has to resolve, otherwise the control connection for Cisco SD-WAN Manager might not come up during a link failure. This is because, when there are more than six IP addresses (this is the recommended number since the default DNS cache timeout is currently two minutes) to check, the DNS cache timer expires even as the highest preferred interface tries all Cisco SD-WAN Validator IP addresses, before failing over to a different color. For instance, it takes about 20 seconds to attempt to connect to one IP address. So, if there are eight IP addresses to be resolved, the DNS cache timeout should be $20 \times 8 = 160$ seconds or three minutes.

Configure Controller Certificate Authorization Settings

Signed certificates are used to authenticate devices in the overlay network. Once authenticated, devices can establish secure sessions between each other. It is from the Cisco SD-WAN Manager that you generate these certificates and install them on the controller devices—Cisco SD-WAN Validator, Cisco SD-WAN Manager, and Cisco SD-WAN Controllers. You can use certificates signed by Symantec, or you can use enterprise root certificates.

The controller certification authorization settings establish how the certification generation for all controller devices will be done. They do not generate the certificates.

You need to select the certificate-generation method only once. The method you select is automatically used each time you add a device to the overlay network.

To have the Symantec signing server automatically generate, sign, and install certificates on each controller device:

1. From **Controller Certificate Authorization**, click **Edit**.
2. Click **Symantec Automated** (Recommended). This is the recommended method for handling controller signed certificates.
3. In the **Confirm Certificate Authorization Change** dialog box, click **Proceed** to confirm that you wish to have the Symantec signing server automatically generate, sign, and install certificates on each controller device.
4. Enter the first and last name of the requester of the certificate.
5. Enter the email address of the requester of the certificate. This address is required because the signed certificate and a confirmation email are sent to the requester via email; they are also made available through the customer portal.
6. Specify the validity period for the certificate. It can be 1, 2, or 3 years.
7. Enter a challenge phrase. The challenge phrase is your certificate password and is required when you renew or revoke a certificate.
8. Confirm your challenge phrase.
9. In **Certificate Retrieve Interval**, specify how often the Cisco SD-WAN Manager server checks if the Symantec signing server has sent the certificate.
10. Click **Save**.

To manually install certificates that the Symantec signing server has generated and signed:

1. From **Controller Certificate Authorization**, click **Edit**.
2. Click **Symantec Manual**.
3. In the **Confirm Certificate Authorization Change** dialog box, click **Proceed** to manually install certificates that the Symantec signing server has generated and signed.
4. Click **Save**.

To use enterprise root certificates:

1. From **Controller Certificate Authorization**, click **Edit**.
2. Click **Enterprise Root Certificate**.
3. In the **Confirm Certificate Authorization Change** dialog box, click **Proceed** to confirm that you wish to use enterprise root certificates.
4. In the **Certificate** box, either paste the certificate, or click **Select a file** and upload a file that contains the enterprise root certificate.
5. By default, the enterprise root certificate has the following properties: To view this information, issue the **show certificate signing-request decoded** command on a controller device, and check the output in the Subject line. For example:
 - Country: United States
 - State: California

- City: San Jose
- Organizational unit: ENB
- Organization: CISCO
- Domain Name: cisco.com
- Email: cisco-cloudops-sdwan@cisco.com

```
vSmart# show certificate signing-request decoded
...
Subject: C=US, ST=California, L=San Jose, OU=ENB, O=CISCO, CN=vsmart-uuid
.cisco.com/emailAddress=cisco-cloudops-sdwan@cisco.com
...
```

To change one or more of the default CSR properties:

- Click **Set CSR Properties**.
 - Enter the domain name to include in the CSR. This domain name is appended to the certificate number (CN).
 - Enter the organizational unit (OU) to include in the CSR.
 - Enter the organization (O) to include in the CSR.
 - Enter the city (L), state (ST), and two-letter country code (C) to include in the CSR.
 - Enter the email address (emailAddress) of the certificate requester.
 - Specify the validity period for the certificate. It can be 1, 2, or 3 years.
- Click **Import & Save**.

Enforce Software Version on Devices

If you are using the Cisco Catalyst SD-WAN hosted service, you can enforce a version of the Cisco Catalyst SD-WAN software to run on a router when it first joins the overlay network.

To ensure that templates are in sync after an upgrade that enforces a software version, make sure of the following before you perform the upgrade:

- The bootflash and flash on the router must have enough free space to support the upgrade
- The version of the SD-WAN image that is on the device before the upgrade must be a lower version than the enforced SD-WAN version you specify in the following procedure
- For ZTP enforcement feature to start, initial onboarding has to be done through PNP/ZTP workflow.

To enforce a version of the Cisco Catalyst SD-WAN software to run on a router when it first joins the overlay network, follow these steps:

- Ensure that the software image for the desired device software version is present in the Cisco SD-WAN Manager software image repository:
 - From the Cisco SD-WAN Manager menu, choose **Maintenance > Software Repository**.

The Software Repository screen opens and displays a table of software images. If the desired software image is present in the repository, continue with Step 2.

- b. If you need to add a software image, click **Add New Software**.
 - c. Select the location from which to download the software images, either Cisco SD-WAN Manager, Remote Server, or Remote Server - Cisco SD-WAN Manager.
 - d. Select an x86-based or a MIPS-based software image.
 - e. To place the image in the repository, click **Add**.
2. From the Cisco SD-WAN Manager menu, choose **Administration** > **Settings**.
3. Click **Enforce Software Version (ZTP)**.
(In Cisco Catalyst SD-WAN Manager Release 20.12.x and earlier, locate **Enforce Software Version (ZTP)** and click **Edit**.)
4. For a specific platform, enable enforcing the software version.
5. Do one of the following:
 - Use an image on a local server:
 - a. In the **Image Location** field, choose **Local Server**.
 - b. In the **Version/Image Name** field, choose an image.
 - Use an image on a remote server:
 - a. In the **Image Location** field, choose **Remote Server**.
 - b. In the **Remote Server Name** field, choose a server.
 - c. In the **Image Filename** field, choose an image.
6. Click **Save**.

Configure a login page banner using a configuration group

You can configure the banner text for login pages.

Before you begin

On the **Configuration** > **Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

-
- Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration** > **Configuration Groups**.
 - Step 2** Create and configure a **Banner** feature in a System profile.

Table 2: Banner Basic Settings

Field	Description
Type	Choose a feature from the drop-down list.
Feature Name*	Enter a name for the feature.
Description	Enter a description of the feature. The description can contain any characters and spaces.
Login	Enter the text to display before the login prompt. The string can be up to 2048 characters long. To insert a line break, type \n.
Message of the Day	On a Cisco IOS XE Catalyst SD-WAN device, enter the message-of-the-day text to display before the login banner. The string can be up to 2048 characters long. To insert a line break, type \n.

What to do next

Also see [Deploy a configuration group](#).

Configure a login page banner, using templates

Use the Banner template for Cisco Catalyst SD-WAN Validators, Cisco SD-WAN Managers, Cisco Catalyst SD-WAN Controllers, s, and Cisco IOS XE Catalyst SD-WAN devices.

- To configure the banner text for login screens using Cisco SD-WAN Manager templates, create a Banner feature template to configure PIM parameters, as described in this topic.
- To configure a login banner for the Cisco SD-WAN Manager system, from the Cisco SD-WAN Manager menu, choose **Administration > Settings**.

Configure a Banner

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and click **Create Template**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is called **Device**.

3. From the **Create Template** drop-down list, select **From Feature Template**.
4. From the **Device Model** drop-down list, select the type of device for which you are creating the template.
5. Click **Additional Templates** or scroll to the **Additional Templates** section.
6. From the **Banner** drop-down list, click **Create Template**. The **Banner** template form is displayed. This form contains fields for naming the template, and the fields for defining Banner parameters.

7. In **Template Name**, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.
8. In **Template Description**, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default or to enter a value, click the **Scope** drop-down list.

9. To set a banner, configure the following parameters:

Table 3: Parameters to be configured while setting a banner:

Parameter Name	Description
MOTD Banner	On a Cisco IOS XE Catalyst SD-WAN device enter message-of-the-day text to display prior to the login banner. The string can be up to 2048 characters long. To insert a line break, type \n.
Login Banner	Enter text to display before the login prompt. The string can be up to 2048 characters long. To insert a line break, type \n.

10. To save the feature template, click **Save**.

CLI equivalent:

```
banner{login login-string | motd motd-string}
```

Create a Custom Banner

To create a custom banner that is displayed after you log in to the Cisco SD-WAN Manager:

1. From **Banner**, click **Edit**.
2. In **Enable Banner**, click **Enabled**.
3. In **Banner Info**, enter the text string for the login banner or click **Select a File** to download a file that contains the text string.
4. Click **Save**.

Collect Device Statistics

Enable or disable the collection of statistics for devices in the overlay network. By default, the collection of statistics is enabled for all the devices in the overlay network.

1. From the Cisco SD-WAN Manager menu, choose **Administration > Settings**.
2. To modify the settings for collecting device statistics, click **Statistics Database Configuration**. (If you are using Cisco Catalyst SD-WAN Manager Release 20.12.x or earlier, click **Statistics Setting** and **Edit**.)



Note From Cisco Catalyst SD-WAN Manager Release 20.16.1, upgrading the statistics database improves the performance of statistics processing. This upgrade enhances the Cisco SD-WAN Manager response time and scalability by reducing the software image size.

To delete the previous data from the statistics database, use the REST API provided in the [Cisco Developer Documentation](#).

By default, for every group of statistics (such as **Aggregated SAIE** and **AppHosting**), collection of statistics is enabled for all devices.

3. To enable the collection of a group of statistics for all devices, click **Enable All** for the particular group.
4. To disable the collection of a group of statistics for all devices, click **Disable All** for the particular group.
5. To collect statistics for all devices exclusively for Cisco SD-WAN Analytics, click **vAnalytics** for the specific group.
6. To enable or disable the collection of a group of statistics for specific devices in the overlay network, click **Custom** for the particular group.

In the **Select Devices** dialog box, depending on whether statistics collection is enabled or disabled for a device, the device is listed between **Enabled Devices** or **Disabled Devices** respectively.

- a. To enable statistics collection for one or more devices, choose the devices from **Disabled Devices** and move them to **Enabled Devices**.



Tip To choose all **Disabled Devices**, click **Select All**.

- b. To disable statistics collection for one or more devices, choose the devices from **Enabled Devices** and move them to **Disabled Devices**.



Tip To choose all **Enabled Devices**, click **Select All**.

- c. To save your selections, click **Done**.
To discard your selections, click **Cancel**.

7. To apply the modified settings, click **Save**.
To discard your changes, click **Cancel**.
To revert to the default settings, click **Restore Factory Default**.

Configure the Time Interval To Collect Device Statistics

1. From the Cisco SD-WAN Manager menu, choose **Administration > Settings**.
2. To modify the time interval at which device statistics are collected, click **Statistics Configuration**.
3. Enter the required **Collection Interval** in minutes.

- Default value: 30 minutes
- Minimum value: 5 minutes
- Maximum value: 180 minutes

4. To apply the modified settings, click **Save**.

To discard your changes, click **Cancel**.

To revert to the default settings, click **Restore Factory Default**.



Note Starting from Cisco Catalyst SD-WAN Manager Release 20.9.6, Cisco SD-WAN Manager collects device statistics files at increased frequency, independent of any configured collection intervals.

Configure or Cancel an Cisco SD-WAN Manager Server Maintenance Window

You can set or cancel the start and end times and the duration of the maintenance window for the Cisco SD-WAN Manager server.

1. From the Cisco SD-WAN Manager menu, choose **Administration > Settings**.
2. Click **Maintenance Window**. (If you are using Cisco IOS XE Catalyst SD-WAN Release 17.12.x or earlier, click **Maintenance Window** and then click **Edit**.)
To cancel the maintenance window, click **Cancel**.
3. Click the **Start Date** and **Start Time** drop-down list. Select the date and time when the **Maintenance Window** will start.
4. Click the **End Date** and **EndTime** drop-down list. Select the date and time when the **Maintenance Window** will end.
5. Click **Save**. The start and end times and the duration of the maintenance window are displayed in the **Maintenance Window** bar.

Two days before the start of the window, the Cisco SD-WAN Manager Dashboard displays a maintenance window alert notification.

Configure Basic System Parameters Using a Configuration Group

Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

- Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

Step 2 Create and configure a Basic feature in a System profile.

- a. Configure basic settings.

Table 4: Basic Settings

Field	Description
Time Zone	Choose the time zone to use on the device.
Device Groups	Enter the names of one or more groups to which the device belongs, separated by commas.
Location	Enter a description of the location of the device. It can be up to 128 characters.
Description	Enter any additional descriptive information about the device.
Transport Gateway	(Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.13.1) Enable transport gateway functionality for the device. A transport gateway connects routers that may or may not have direct connectivity. One common use case for transport gateways is to provide connectivity between routers in disjoint networks, such as between public and private WANs. Another use case for transport gateway functionality is to use a transport gateway as the hub in a hub-and-spoke topology.

- b. Configure controller settings.
- c. Configure GPS.

Table 5: GPS

Field	Description
GPS Latitude	Enter the latitude of the device, in the format decimal-degrees.
GPS Longitude	Enter the longitude of the device, in the format decimal-degrees.

- d. Configure track settings.

Table 6: Track Settings

Field	Description
Track Transport	Enable this option to regularly check whether the DTLS connection between the device and a Cisco SD-WAN Validator is up. Default: Enabled
Track Default Gateway	Enable or disable tracking of default gateway. Gateway tracking determines, for static routes, whether the next hop is reachable before adding that route to the route table of the device. Default: Enabled

Field	Description
Track Interface Tag	Set the tag string to include in routes associated with a network that is connected to a non-operational interface. Range: 1 through 4294967295
Tracker DIA Stabilize Status	Enable this option to stabilize interface flaps by using the multiplier to update HTTP or ICMP tracker status from DOWN to UP.

- e. Configure advanced settings.

Table 7: Advanced

Field	Description
Port Hopping	Enable or disable port hopping. When a Cisco Catalyst SD-WAN device is behind a NAT, port hopping rotates through a pool of preselected OMP port numbers (called base ports) to establish DTLS connections with other Cisco Catalyst SD-WAN devices when a connection attempt is unsuccessful. The default base ports are 12346, 12366, 12386, 12406, and 12426. To modify the base ports, set a port offset value. Default: Enabled
Port Offset	Enter a number by which to offset the base port number. Configure this option when multiple Cisco Catalyst SD-WAN devices are behind a single NAT device, to ensure that each device uses a unique base port for DTLS connections. Values: 0 through 19
On Demand Tunnel	Enable dynamic on-demand tunnels between any two Cisco Catalyst SD-WAN spoke devices.
On Demand Tunnel Idle Timeout (In Minute)	Enter the on-demand tunnel idle timeout time. After the configured time, the tunnel between the spoke devices is removed. Range: 1 to 65535 minutes Default: 10 minutes
Control Session PPS	Enter a maximum rate of DTLS control session traffic to police the flow of control traffic. Range: 1 through 65535 pps Default: 300 pps
Multi Tenant	Enable this option to specify the device as multitenant.
Admin Tech On Failure	Enable this option to collect admin-tech information when the device reboots. Default: Enabled

What to do next

Also see [Deploy a configuration group](#).

Configure Basic System Parameters

Use the System template for all Cisco Catalyst SD-WAN devices.

To configure system-wide parameters using Cisco SD-WAN Manager templates:

1. Create a **System** feature template to configure system parameters.
2. Create an **NTP** feature template to configure NTP servers and authentication.
3. Configure the organization name and Cisco Catalyst SD-WAN Validator IP address on the Cisco SD-WAN Manager. These settings are appended to the device templates when the templates are pushed to devices.

Create System Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and click **Create Template**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is called **Device**.

3. From the **Create Template** drop-down list, select **From Feature Template**.
4. From the **Device Model** drop-down list, select the type of device for which you are creating the template.
5. To create a custom template for System, select the **Factory_Default_System_Template** and click **Create Template**.

The System template form is displayed. This form contains fields for naming the template, and fields for defining the System parameters.

6. In **Template Name**, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.
7. In **Template Description**, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default or to enter a value, click the scope drop-down to the left of the parameter field and select one of the following:

Table 8:

Parameter Scope	Scope Description
Device Specific (indicated by a host icon)	Use a device-specific value for the parameter. For device-specific parameters, you cannot enter a value in the feature template. You enter the value when you attach a Cisco Catalyst SD-WAN device to a device template. When you click Device Specific, the Enter Key box opens. This box displays a key, which is a unique string that identifies the parameter in a CSV file that you create. This file is an Excel spreadsheet that contains one column for each key. The header row contains the key names (one key per column), and each row after that corresponds to a device and defines the values of the keys for that device. You upload the CSV file when you attach a Cisco Catalyst SD-WAN device to a device template. To change the default key, type a new string and move the cursor out of the Enter Key box. Examples of device-specific parameters are system IP address, host name, GPS location, and site ID.
Global (indicated by a globe icon)	Enter a value for the parameter, and apply that value to all devices. Examples of parameters that you might apply globally to a group of devices are DNS server, syslog server, and interface MTUs.

Basic System-Wide Configuration

To set up system-wide functionality on a Cisco Catalyst SD-WAN device, select the **Basic Configuration** tab and then configure the following parameters. Parameters marked with an asterisk are required.

Table 9:

Parameter Field	Description
Site ID* (on routers, Cisco SD-WAN Manager instances, and Cisco SD-WAN Controller)	Enter the identifier of the site in the Cisco Catalyst SD-WAN overlay network domain in which the device resides, such as a branch, campus, or data center. The site ID must be the same for all Cisco Catalyst SD-WAN devices that reside in the same site. <i>Range:</i> 1 through 4294967295 ($2^{32} - 1$)
System IP*	Enter the system IP address for the Cisco Catalyst SD-WAN device, in decimal four-part dotted notation. The system IP address provides a fixed location of the device in the overlay network and is a component of the device's TLOC address. It is used as the device's loopback address in the transport VPN (VPN 0). You cannot use this same address for another interface in VPN 0.
Timezone*	Select the timezone to use on the device.
Hostname	Enter a name for the Cisco Catalyst SD-WAN device. It can be up to 32 characters.
Location	Enter a description of the location of the device. It can be up to 128 characters.

Parameter Field	Description
Device Groups	Enter the names of one or more groups to which the device belongs, separated by commas.
Controller Groups	List the Cisco Catalyst SD-WAN Controller groups to which the router belongs.
Description	Enter any additional descriptive information about the device.
Console Baud Rate	Select the baud rate of the console connection on the router. <i>Values:</i> 1200, 2400, 4800, 9600, 19200, 38400, 57600, 115200 baud or bits per second (bps). Starting from Cisco vManage Release 20.3.1, the default value is 9600 on Cisco IOS XE Catalyst SD-WAN devices.
Maximum OMP Sessions	Set the maximum number of OMP sessions that a router can establish to a Cisco Catalyst SD-WAN Controller. <i>Range:</i> 0 through 100. <i>Default:</i> 2

To save the feature template, click **Save**.

To configure the DNS name or IP address of the Cisco Catalyst SD-WAN Validator in your overlay network, go to **Administration > Settings** screen and click **Validator**. (If you are using Cisco Catalyst SD-WAN Manager Release 20.12.x or earlier, click **vBond**.)

Configure the GPS Location

To configure a device location, select the **GPS** tab and configure the following parameters. This location is used to place the device on the Cisco SD-WAN Manager network map. Setting the location also allows Cisco SD-WAN Manager to send a notification if the device is moved to another location.

Table 10:

Parameter Field	Description
Latitude	Enter the latitude of the device, in the format <i>decimal-degrees</i> .
Longitude	Enter the longitude of the device, in the format <i>decimal-degrees</i> .

To save the feature template, click **Save**.

Configure Interface Trackers for NAT Direct Internet Access

The DIA tracker helps determine if the internet or external network becomes unavailable. This feature is useful when NAT is enabled on a transport interface in VPN 0 to allow data traffic from the router to exit directly to the internet.

If the internet or external network becomes unavailable, the router continues to forward traffic based on the NAT route in the service VPN. Traffic that is forwarded to the internet gets dropped. To prevent the internet-bound traffic from being dropped, configure the DIA tracker on the edge router to track the status of the transport interface. The tracker periodically probes the interface IP address of the end point of the tunnel interface to determine the status of the transport interface. The tracker determines the status of the internet and returns the data to the attach points that are associated with the tracker.

When the tracker is configured on the transport interface, the interface IP address is used as a source IP address for probe packets.

IP SLA monitors the status of probes and measures the round trip time of these probe packets and compares the values with the configured latency in the probe. When the latency exceeds the configured threshold value, the tracker considers the network as unavailable.

If the tracker determines that the local internet is unavailable, the router withdraws the NAT route and reroutes the traffic based on the local routing configuration to overlay.

The local router continues to periodically check the status of the path to the interface. When it detects that the path is functioning again, the router reinstalls the NAT route to the internet.

For more information on NAT DIA tracker for Cisco IOS XE Catalyst SD-WAN devices, see the [NAT DIA Tracker](#) section of the *Cisco Catalyst SD-WAN NAT Configuration Guide, Cisco IOS XE Catalyst SD-WAN Release 17.x*.

Configure NAT DIA Tracker

To track the status of transport interfaces that connect to the internet (Network Address Translation Direct Internet Access (NAT DIA)), click **Tracker > Add New Tracker** and configure the following parameters:

Table 11:

Parameter Field	Description
Name	Name of the tracker. The name can be up to 128 alphanumeric characters. You can configure up to eight trackers.
Tracker Type	Choose an interface, static route.
Threshold	How long to wait for the probe to return a response before declaring that the transport interface is down. <i>Range:</i> 100 through 1000 milliseconds. <i>Default:</i> 300 milliseconds.
Interval	How often probes are sent to determine the status of the transport interface. <i>Range:</i> 10 through 600 seconds. <i>Default:</i> 60 seconds (1 minute)
Multiplier	Number of times to resend probes before declaring that the transport interface is down. <i>Range:</i> 1 through 10. <i>Default:</i> 3
End Point Type: IP Address	IP address of the end point of the tunnel interface. This is the destination in the internet to which the router sends probes to determine the status of the transport interface. Note In Cisco SD-WAN Release 20.5.1 and later releases, if the tracker receives an HTTP response status code, which is less than 400, the endpoint is reachable. Prior to Cisco SD-WAN Release 20.5.1, the endpoint is reachable if the tracker receives an HTTP response status code of 200.
End Point Type: DNS Name	DNS name of the end point of the tunnel interface. This is the destination in the internet to which the router sends probes to determine the status of the transport interface.

To save a tracker, click **Add**.

To save the feature template, click **Save**.

Configure NAT DIA Tracker Using the CLI

Configure NAT DIA tracker

```
Device(config)# endpoint-tracker tracker1
  Device(config-endpoint-tracker)# endpoint-ip 10.1.1.1
  Device(config-endpoint-tracker)# threshold 100
  Device(config-endpoint-tracker)# multiplier 5
  Device(config-endpoint-tracker)# interval 10

Device(config)# endpoint-tracker tracker1
  Device(config-endpoint-tracker)# endpoint-api-url https://ip-address:8443/apidocs
  Device(config-endpoint-tracker)# threshold 100
  Device(config-endpoint-tracker)# multiplier 5
  Device(config-endpoint-tracker)# interval 10
```

Apply Tracker to an Interface

To apply a tracker to an interface, configure it in the **VPN Interface Cellular**, **VPN Interface Ethernet**, **VPN Interface NAT Pool**, or **VPN Interface PPP** configuration templates. You can apply only one tracker to an interface.

Monitor NAT DIA Endpoint Tracker Configuration

1. From the Cisco SD-WAN Manager menu, choose **Monitor > Devices**.

Cisco vManage Release 20.6.x and earlier: From the Cisco SD-WAN Manager menu, choose **Monitor > Network**.

2. Choose a device from the list of devices.
3. Click **Real Time**.
4. From the **Device Options** drop-down list, choose **Endpoint Tracker Info**.

Configure Advanced Options

To configure additional system parameters, click **Advanced**:

Table 12:

Parameter Name	Description
Control Session Policer Rate	Specify a maximum rate of DTLS control session traffic, to police the flow of control traffic. <i>Range:</i> 1 through 65535 pps. <i>Default:</i> 300 pps

Parameter Name	Description
Port Hopping	Click On to enable port hopping, or click Off to disable it. When a Cisco Catalyst SD-WAN device is behind a NAT, port hopping rotates through a pool of preselected OMP port numbers (called base ports) to establish DTLS connections with other Cisco Catalyst SD-WAN devices when a connection attempt is unsuccessful. The default base ports are 12346, 12366, 12386, 12406, and 12426. To modify the base ports, set a port offset value. To disable port hopping on an individual TLOC (tunnel interface), use the VPN Interface Ethernet configuration template. <i>Default:</i> Enabled (on routers); disabled (on Cisco SD-WAN Manager devices and Cisco Catalyst SD-WAN Controllers).
Port Offset	Enter a number by which to offset the base port number. Configure this option when multiple Cisco Catalyst SD-WAN devices are behind a single NAT device, to ensure that each device uses a unique base port for DTLS connections. <i>Values:</i> 0 through 19
Track Transport	Click On to regularly check whether the DTLS connection between the device and a Cisco Catalyst SD-WAN Validator is up. Click Off to disable checking. By default, transport checking is enabled.
Track Interface	Set the tag string to include in routes associated with a network that is connected to a non-operational interface. <i>Range:</i> 1 through 4294967295
Gateway Tracking	Click On to enable or click Off to Disable tracking of default gateway. Gateway tracking determines, for static routes, whether the next hop is reachable before adding that route to the device's route table. <i>Default:</i> Enabled
Collect Admin Tech on Reboot	Click On to collect admin-tech information when the device reboots.
Idle CLI Timeout in minutes	Set how long the CLI is inactive on a device before the user is logged out. If a user is connected to the device via an SSH connection, the SSH connection is closed after this time expires. <i>Default:</i> CLI session does not time out. The Idle CLI Timeout in minutes is represented as "exec-timeout" in the CLI equivalent shown next.

To save the feature template, click **Save**.

CLI equivalent:

```

system
  admin-tech-on-failure allow-same-site-tunnels
  control-session-pps rate eco-friendly-mode
  host-policer-pps rate

  icmp-error-pps rate

  exec-timeout seconds multicast-buffer-percent percentage

  port-hop port-offset number
  system-tunnel-mtu bytes timer
  dns-cache-timeout minutes track-default-gateway
  track-interface-tag number

  track-transport upgrade-confirm minutes

```




Note **Exec-Timeout** governs the CLI logouts for devices in controller mode.

Configure Global Using a Configuration Group

Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

Step 1 From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

Step 2 Create and configure a Global feature in a System profile.

a) Configure services.

Table 13: Services

Field	Description
HTTP Server	Enable or disable HTTP server.
HTTPS Server	Enable or disable secure HTTPS server.
FTP Passive	Enable or disable passive FTP.
Domain Lookup	Enable or disable Domain Name System (DNS) lookup.
ARP Proxy	Enable or disable proxy ARP.
RSH/RCP	Enable or disable remote shell (RSH) and remote copy (rcp) on the device.
Line Virtual Teletype (Configure Outbound Telnet)	Enable or disable outbound telnet.
Cisco Discovery Protocol (CDP)	Enable or disable Cisco Discovery Protocol (CDP).
Link Layer Discovery Protocol (LLDP)	Enable or disable Link Layer Discovery Protocol (LLDP).
Specify interface for source address	Enter the address of the source interface in all HTTPS client connections.

b) Configure NAT 64 timeout.

Table 14: NAT 64

Field	Description
UDP Timeout	Specify the NAT64 translation timeout for UDP. Range: 1 to 536870 (seconds) Default: 300 seconds (5 minutes)
TCP Timeout	Specify the NAT64 translation timeout for TCP. Range: 1 to 536870 (seconds) Default: 3600 seconds (1 hour)

- c) Configure Authentication type.

Table 15: Authentication

Field	Description
HTTP Authentication	Choose the HTTP authentication mode. Accepted values: Local, AAA Default: Local

- d) Configure SSH Version.

Table 16: SSH Version

Field	Description
SSH Version	Choose the SSH version. Default: Disabled

- e) Configure Other Settings.

Table 17: Other Settings

Field	Description
TCP Keepalives (In)	Enable or disable generation of keepalive timers when incoming network connections are idle.
TCP Keepalives (Out)	Enable or disable generation of keepalive timers when outgoing network connections are idle.
TCP Small Servers	Enable or disable small TCP servers (for example, ECHO).
UDP Small Servers	Enable or disable small UDP servers (for example, ECHO).
Console Logging	Enable or disable console logging. By default, the router sends all log messages to its console port.

Field	Description
IP Source Routing	Enable or disable IP source routing. IP source routing is a feature that enables the originator of a packet to specify the path for the packet to use to get to the destination.
VTY Line Logging	Enable or disable the device to display log messages to a vty session in real time.
SNMP IFINDEX Persist	Enable or disable SNMP IFINDEX persistence, which provides an interface index (ifIndex) value that is retained and used when the device reboots.
Ignore BOOTP	Enable or disable BOOTP server. When enabled, the device listens for the BOOTP packet that comes in sourced from 0.0.0.0. When disabled, the device ignores these packets.

Configure Global Parameters

Table 18: Feature History

Feature Name	Release Information	Description
Configure Global Parameters	Cisco IOS XE Catalyst SD-WAN Release 17.2.1r	This feature lets you configure HTTP and Telnet server settings, and several other device settings, from Cisco SD-WAN Manager.

Use the Global Settings template to configure a variety of global parameters for all Cisco IOS XE Catalyst SD-WAN devices, including:

- Various services, such as HTTP and Telnet
- NAT64 timeouts
- HTTP authentication mode
- TCP keepalive
- TCP and UDP small servers
- Console logging
- IP source routing
- VTY line logging
- SNMP IFINDEX persistence
- BOOTP server

Before applying the global parameters to a device, you can view the current configuration of the device and view the differences between the parameter values that you have set in the Global Settings template and the current values on a device.

To configure global settings using Cisco SD-WAN Manager:

1. Create a feature template to configure global settings.
2. Create a device template and include the Global Settings feature template.
3. (Recommended) Before applying the device template to a device, use the [Preview Device Configuration and View Configuration Differences, on page 236](#) feature to review the differences between the configuration currently on the device and the configuration to be sent to the device. This step is recommended because applying the device template overwrites the existing configuration on a device.

Limitations

Cisco Catalyst SD-WAN can apply the global settings feature template only to devices running Cisco IOS XE Catalyst SD-WAN Release Amsterdam 17.2.x or later.

Create Global Settings Feature Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

3. Click **Add Template**.
4. In the left pane, select a device type.
5. Select the **Global Settings** template.
6. Provide a name and description for the template.
7. For each of the parameters, use the default or set custom values as desired.

Parameter	Description
Services	
HTTP Server	Enable or disable HTTP server.
HTTPS Server	Enable or disable secure HTTPS server.
Passive FTP	Enable or disable passive FTP.
IP Domain-Lookup	Enable or disable domain name server (DNS) lookup.
Arp Proxy	Enable or disable proxy ARP.
RSH/RCP	Enable or disable remote shell (RSH) and remote copy (RCP) on the device.
Telnet (Outbound)	Enable or disable outbound telnet.
CDP	Enable or disable Cisco Discovery Protocol. Starting from Cisco SD-WAN 17.3 release, CDP on interfaces is enabled when the cdp run command is executed globally on Cisco ASR 1000 series devices.

Parameter	Description
Other Settings	
TCP Keepalives (In)	Enable or disable generation of keepalive timers when incoming network connections are idle.
TCP Keepalives (Out)	Enable or disable generation of keepalive timers when outgoing network connections are idle.
TCP Small Servers	Enable or disable small TCP servers (for example, ECHO).
UDP Small Servers	Enable or disable small UDP servers (for example, ECHO).
Console Logging	Enable or disable console logging. By default, the router sends all log messages to its console port.
IP Source Routing	Enable or disable IP source routing. IP source routing is a feature that enables the originator of a packet to specify the path for the packet to use to get to the destination.
VTY Line Logging	Enable or disable the device to display log messages to a VTY session in real time.
SNMP IFINDEX Persist	Enable or disable SNMP IFINDEX persistence, which provides an interface index (ifIndex) value that is retained and used when the device reboots.
Ignore BOOTP	Enable or disable BOOTP server. When enabled, the device listens for the bootp packet that comes in sourced from 0.0.0.0. When disabled, the device ignores these packets.
NAT64	
UDP Timeout	NAT64 translation timeout for UDP Range: 1 to 65536 (seconds) Default: 300 seconds (5 minutes) Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.6.1a and Cisco vManage Release 20.6.1, the default UDP Timeout value for NAT64 has been changed to 300 seconds (5 minutes).
TCP Timeout	NAT64 translation timeout for TCP Range: 1 to 65536 (seconds) Default: 3600 seconds (1 hour) Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.6.1a and Cisco vManage Release 20.6.1, the default TCP Timeout value for NAT64 has been changed to 3600 seconds (1 hour).
HTTP Authentication	

Parameter	Description
HTTP Authentication	HTTP authentication mode Accepted values: Local, AAA Default: Local
SSH Version	
SSH version	Specify an SSH version. Default value: Version 2

- Enter a name for the template and click **Save**.

CLI Equivalent

Services (enable):

```
system
 ip http server
 ip http secure-server
 ip ftp passive
 ip domain lookup
 ip arp proxy disable
 ip rcmd rsh-enable
 ip rcmd rcp-enable
 cdp run enable
```



Note Starting from Cisco SD-WAN 17.3 release, CDP on interfaces is enabled when the **cdp run** command is executed globally on Cisco ASR 1000 series devices.

Telnet outbound enable:

```
system
 line vty 0 4
 transport input telnet ssh
```

Services (disable):

```
system
 no ip http server
 no ip http secure-server
 no ip ftp passive
 no ip domain lookup
 no ip arp proxy disable
 no ip rcmd rsh-enable
 no ip rcmd rcp-enable
 no cdp run enable
```

Telnet outbound disable:

```
system
 line vty 0 4
 transport input ssh
```

Other settings (enable):

```
system
  service tcp-keepalives-in
  service tcp-keepalives-out
  service tcp-small-servers
  service udp-small-server
  logging console
  ip source-route
  logging monitor
  snmp-server ifindex persist
  ip bootp server
```

Other settings (disable):

```
system
  no service tcp-keepalives-in
  no service tcp-keepalives-out
  no service tcp-small-servers
  no service udp-small-server
  no logging console
  no ip source-route
  no logging monitor
  no snmp-server ifindex persist
  no ip bootp server
```

NAT 64:

```
system
  nat64 translation timeout udp timeout
  nat64 translation timeout tcp timeout
```

HTTP Authentication:

```
system
  ip http authentication {local | aaa}
```

Configure NTP Servers Using a Configuration Group

Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

Step 1 From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

Step 2 Create and configure an NTP feature in a System profile.

a. Configure a server.

Table 19: Server

Field	Description
Add Server	
Hostname/IP address*	Enter the IP address of an NTP server, or a DNS server that knows how to reach the NTP server.

Field	Description
VPN to reach NTP Server*	Enter the number of the VPN that should be used to reach the NTP server, or the VPN in which the NTP server is located. If you have configured multiple NTP servers, they must all be located or be reachable in the same VPN. Range: 0 to 65530
Set authentication key for the server	Specify the MD5 key associated with the NTP server, to enable MD5 authentication. For the key to work, you must mark it as trusted in the Trusted Key field under Authentication .
Set NTP version*	Enter the version number of the NTP protocol software. Range: 1 to 4 Default: 4
Set interface to use to reach NTP server	Enter the name of a specific interface to use for outgoing NTP packets. The interface must be located in the same VPN as the NTP server. If it is not, the configuration is ignored.
Prefer this NTP server*	Enable this option if multiple NTP servers are at the same stratum level and you want one to be preferred. For servers at different stratum levels, Cisco Catalyst SD-WAN chooses the one at the highest stratum level.

b. Configure Authentication.

Table 20: Authentication

Field	Description
Add Authentication Keys	
Key Id*	Enter an MD5 authentication key ID. Range: 1 to 65535
MD5 Value*	Enter an MD5 authentication key. Enter either a cleartext key or an AES-encrypted key.
Trusted Key	Enter the MD5 authentication key to designate the key as trustworthy. To associate this key with a server, enter the same value that you entered for the Set authentication key for the server field under Server .

c. Configure advanced parameters.

Table 21: Advanced

Field	Description
Authoritative NTP Server	Choose Global from the drop-down list, and enable this option if you want to configure one or more supported routers as a primary NTP router. When you enable this option, the following field appears:
Stratum	Enter the stratum value for the primary NTP router. The stratum value defines the hierarchical distance of the router from its reference clock. Valid values: Integers 1 to 15. If you do not enter a value, the system uses the router internal clock default stratum value, which is 8.
Source Interface	Enter the name of the exit interface for NTP communication. If configured, the system sends NTP traffic to this interface. For example, enter GigabitEthernet1 or Loopback0 .

What to do next

Also see [Deploy a configuration group](#).

Configure NTP Servers Using Cisco SD-WAN Manager

Configure NTP servers on your devices in order to synchronize time across all the devices in the Cisco overlay network. You can configure up to four NTP servers, and they must all be located or reachable in the same VPN.

Other devices are allowed to ask a Cisco Catalyst SD-WAN device for the time, but no devices are allowed to use a Cisco Catalyst SD-WAN device as an NTP server.



Note For the NTP to properly function when using Global VRF on the Cisco IOS XE Catalyst SD-WAN devices, you must configure **allow-service ntp** for the tunnel interface on the Cisco VPN Interface Ethernet template.

To configure an NTP server using Cisco SD-WAN Manager templates:

1. Create an NTP feature template to configure NTP parameters, as described in this section.
2. Configure the timezone in the System template.

Name the Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. From the **Create Template** drop-down list, choose **From Feature Template**.
4. From the **Device Model** drop-down list, choose the type of device for which you wish to create the template.
5. Click **Basic Information**.
6. From **Additional Cisco System Templates**, click **NTP**.
7. From the **NTP** drop-down list, choose **Create Template**.

The **Cisco NTP** template form is displayed. This form contains fields for naming the template, and fields for defining NTP parameters.

8. In **Template Name**, enter a name for the template.

The name can be up to 128 characters and can contain only alphanumeric characters.

9. In **Template Description**, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default value or to enter a value, click the scope drop-down list to the left of the parameter field and select one of the following:

Table 22: Setting Parameter Scope

Parameter Scope	Scope Description
Device Specific (indicated by a host icon)	Use a device-specific value for the parameter. For device-specific parameters, you cannot enter a value in the feature template. You enter the value when you attach a device to a device template. When you click Device Specific, the Enter Key box opens. This box displays a key, which is a unique string that identifies the parameter in a CSV file that you create. This file is an Excel spreadsheet that contains one column for each key. The header row contains the key names (one key per column), and each row after that corresponds to a device and defines the values of the keys for that device. You upload the CSV file when you attach a device to a device template. For more information, see Create a Template Variables Spreadsheet. To change the default key, type a new string and move the cursor out of the Enter Key box. Examples of device-specific parameters are system IP address, hostname, GPS location, and site ID.
Global (indicated by a globe icon)	Enter a value for the parameter, and apply that value to all devices. Examples of parameters that you might apply globally to a group of devices are DNS server, syslog server, and interface MTUs.

Configure an NTP Server

To configure an NTP server, click **Server**, and click **Add New Server**, and configure the following parameters. Parameters marked with an asterisk are required to configure an NTP server.

Table 23: Parameters for Configuring an NTP Server

Parameter Name	Description
Hostname/IP Address*	Enter the IP address of an NTP server, or a DNS server that knows how to reach the NTP server.
Authentication Key ID*	Specify the MD5 authentication key associated with the NTP server, to enable authentication. For the key to work, you must mark it as trusted in the Trusted Keys field, under Authentication . Note From Cisco Catalyst SD-WAN Control Components Release 20.14.1, you can use CMAC-AES authentication when configuring NTP servers for Cisco SD-WAN Control Components. This requires configuration using a CLI template.
VPN ID*	Enter the number of the VPN that should be used to reach the NTP server, or the VPN in which the NTP server is located. If you have configured multiple NTP servers, they must all be located or be reachable in the same VPN. The valid range is from 0 through 65530.
Version*	Enter the version number of the NTP protocol software. The range is from 1 through 4. The default is 4.
Source Interface	Enter the name of a specific interface to use for outgoing NTP packets. The interface must be located in the same VPN as the NTP server. If it is not, the configuration is ignored.
Prefer	Click On if multiple NTP servers are at the same stratum level and you want one to be preferred. For servers at different stratum levels, the software chooses the one at the highest stratum level.

To add an NTP server, click **Add**.

To add another NTP server, click **Add New Server**. You can configure up to four NTP servers. The Cisco Catalyst SD-WAN software uses the server at the highest stratum level.

To edit an NTP server, click the pencil icon to the right of the entry.

To delete an NTP server, click the trash icon to the right of the entry.

To save the feature template, click **Save**.

Configure NTP Authentication Keys

To configure the authentication keys used to authenticate NTP servers, click **Authentication**, and then the **Authentication Key**. Then click **New Authentication Key**, and configure the following parameters. Parameters marked with an asterisk are required to configure the authentication keys.

Table 24: Parameters for Configuring NTP Authentication Keys

Parameter Name	Description
Authentication Key ID*	Enter the following values: • Authentication Key: Enter an authentication key ID. Valid range is from 1 to 65535. • Authentication Value: Enter either a cleartext key or an AES-encrypted key.
Authentication Value*	Enter an authentication key. For this key to be used, you must designate it as trusted. To associate a key with a server, enter the same value that you entered in the Authentication Key ID field under Server .

To configure the trusted keys used to authenticate NTP servers, under **Authentication**, click **Trusted Key**, and configure the following parameters.

Table 25: Parameters for Configuring Trusted Keys

Parameter Name	Description
Trusted Keys*	Enter the authentication key to designate the key as trustworthy. To associate this key with a server, enter the same value that you entered for the Authentication Key ID field under Server .

Configure a Router as an NTP Primary

Table 26: Feature History

Feature Name	Release Information	Description
Configuring a Router as an NTP Primary	Cisco IOS XE Catalyst SD-WAN Release 17.3.1a Cisco vManage Release 20.3.1	This feature lets you configure a supported router as an NTP primary router. Other nodes in a Cisco Catalyst SD-WAN deployment synchronize their clocks to the NTP primary router. This configuration is useful if you do not have an NTP server in your deployment.

You can configure one or more supported routers as an NTP primary router in a Cisco Catalyst SD-WAN deployment. A router that is configured in this way acts as the NTP server to which other nodes in the deployment synchronize their clocks.

Configuring a router as an NTP primary router is useful if you do not have an NTP server in your deployment.

To configure a router as an NTP primary router, you create a template that includes configured parameters for the NTP primary router. To do so, follow these steps:

1. From the Cisco SD-WAN Manager menu, choose **Configuration** > **Templates**.

2. Perform either of these actions:

- To create a new template, under **Feature Templates**, click **Add Template**, choose the type of device to be the NTP primary router, and then choose the **NTP** template in the group of **Basic Information** templates.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

- To update an existing template, click ..., and click **Edit**.

3. Configure options for the template as desired, and in the Master tab, perform these actions:

- a. For the Master option, choose **Global** from the drop-down list, and then choose **On**.

- b. (Optional) In the **Stratum** field, enter the stratum value for the NTP primary router.

The stratum value defines the hierarchical distance of the router from its reference clock.

Valid values: Integers 1 through 15. If you do not enter a value, the system uses the router internal clock default stratum value, which is 8.

- c. (Optional) In the **Source** field, enter the name of the exit interface for NTP communication.

If configured, the system sends NTP traffic to this interface.

For example, enter **GigabitEthernet1** or **Loopback0**.

4. Click **Save** (for a new template) or **Update** (for an existing template).

CLI equivalent:

```
ntp master [stratum-number]
ntp source source-interface
```

Configure NTP Servers Using CLI Commands

Before You Begin

For information about using a CLI template, see [CLI Templates](#).

By default, CLI templates execute commands in global configuration mode.

Configure NTP Servers for Cisco SD-WAN Control Components

1. Enter system configuration mode.

```
system
```

2. Enter NTP configuration mode.

```
ntp
```

3. Enter keys configuration mode.

keys

4. Configure an authentication type to use for an NTP server. Assign a key for the authentication type, and assign one of the following authentication methods: MD5, CMAC-AES-128. Using multiple instances of the **authentication** command, you can configure authentication for multiple NTP servers.

```
authentication authentication-key-id {md5 md5-authentication-key | cmac-aes-128
cmac-authentication-key}
```



Note The CMAC-AES option is available from Cisco Catalyst SD-WAN Control Components Release 20.14.1.

5. Designate an authentication type as trusted. Optionally, you can include multiple authentication key IDs.

```
trusted authentication-key-id {authentication-key-id} [authentication-key-id]
```

6. Exit keys configuration mode.

exit

7. Configure an NTP server, including the VPN and version, and optionally an authentication key. You can configure multiple NTP servers.

```
server {server-ip | fully-qualified-domain-name}
key authentication-key
vpn vpn-id
version version-id
exit
```

Example

Here is an example for configuring two authentication types and three NTP servers. Two servers are trusted and use an authentication key, and one server is generic. Authentication key 1001 uses MD5 and key 1002 uses CMAC-AES-128.

```
system ntp
  keys
    authentication 1001 md5 password1
    authentication 1002 cmac-aes-128 password2
    trusted 1001 1002
  !
  server 192.168.10.1
    key 1001
    vpn 512
    version 4
  exit
  server 192.168.10.2
    key 1002
    vpn 512
    version 4
  server us.pool.ntp.org
    vpn 512
    version 4
  exit
  !
  !
```



Note The passwords above are in plain text. When using a CLI template, you can encrypt passwords.

Configure Time using CLI

You can set the time locally on a device without using NTP if you do not need to ensure that time is synchronized across an entire network of devices. You can also set the time locally on any device as it is joining the network, in addition to configuring an NTP server. The local time gets overwritten by the official NTP time once the device contacts the NTP server.

```
clock set 12:00:00 31 May 2019
```

Configure GPS Using a Configuration Group

Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

Step 1 From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

Step 2 Create and configure a GPS feature in a Transport and Management profile.

a. Configure GPS.

Table 27: GPS

Field	Description
Type	Choose a feature from the drop-down list.
Feature Name*	Enter a name for the feature. The name can be up to 128 characters and can contain only alphanumeric characters.
Description	Enter a description of the feature. The description can be up to 2,048 characters and can contain only alphanumeric characters.
GPS	Click On to enable the GPS feature on the router.
GPS Mode	Select the GPS mode: • MS-based: Use mobile station–based assistance, also called assisted GPS mode, when determining position. In this mode, cell tower data is used to enhance the quality and precision in determining location, which is useful when satellite signals are poor. • Standalone: Use satellite information when determining position.

Field	Description
NMEA	Click On to enable the use of NMEA streams to help with determining position. NMEA streams data from the router's cellular module to any marine device, such as a Windows-based PC, that is running a commercially available GPS-based application.
Source Address*	Enter the IP address of the router's interface that connects to the external device reading the NMEA.
Destination Address*	Enter the IP address of the external device's interface that's connected to router.
Destination Port*	Enter the number of the port to use to send NMEA data to the external device's interface.

What to do next

Also see [Deploy a configuration group](#).

Configure GPS Using Cisco SD-WAN Manager

Use the GPS template for all Cisco cellular routers running Cisco Catalyst SD-WAN software.

For Cisco devices running Cisco Catalyst SD-WAN software, you can configure the GPS and National Marine Electronics Association (NMEA) streaming. You enable both these features to allow 4G LTE routers to obtain GPS coordinates.



Note You can configure GPS using Cisco SD-WAN Manager starting from the Cisco vManage Release 20.6.1 and onwards.

Device configuration using the CLI or a CLI template is available starting from the Cisco IOS XE Catalyst SD-WAN Release 17.6.1a only and onwards.

You can configure GPS using a Cisco SD-WAN Manager feature template. For geofencing to work, you need to configure GPS. To configure a GPS feature template, navigate to **Configuration > Templates > Feature Templates > GPS**.

In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

For more information on geofencing, see [Configure Geofencing](#).

Navigate to the Template Screen and Name the Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. Click **Create Template**.
4. From the **Create Template** drop-down list, choose **From Feature Template**.
5. From the **Device Model** drop-down list, choose the type of device for which you wish to create the template.
6. Click **Cellular**.
7. In **Additional Cellular Controller Templates**, click **GPS**.
8. To create a custom template for GPS, click the **GPS** drop-down list and then click **Create Template**. The GPS template form is displayed. This form contains fields for naming the template, and fields for defining the GPS parameters.
9. In the **Template Name** field, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.
10. In the **Template Description** field, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

When you first open a feature template, for each parameter that has a default value, the scope is set to **Default** (indicated by a check mark), and the default setting or value is shown. To change the default or to enter a value, click the scope drop-down list to the left of the parameter field and select either **Device Specific** or **Global**.

Configure GPS

To configure GPS parameters for the cellular router, configure the following parameters. Parameters marked with an asterisk are required to configure the GPS feature.

Table 28:

Parameter Name	Description
GPS	Click On to enable the GPS feature on the router.
GPS Mode	Select the GPS mode: • MS-based—Use mobile station–based assistance, also called assisted GPS mode, when determining position. In this mode, a network data session is used to obtain the GPS satellite locations, resulting in a faster fix of location coordinates. • Standalone—Use satellite information when determining position. Note Standalone mode is currently not supported for geofencing.

Parameter Name	Description
NMEA	Click On to enable the use of NMEA streams to help in determining position. NMEA streams data from the router's 4G LTE Pluggable Interface Module (PIM) to any device, such as a Windows-based PC, that is running a commercially available GPS-based application.
Source Address	(Optional) Enter the IP address of the interface that connects to the router's PIM. Note This option is not used for configuring geofencing.
Destination Address	(Optional) Enter the IP address of the NMEA server. The NMEA server can be local or remote. Note This option is not used for configuring geofencing.
Destination Port	(Optional) Enter the number of the port to use to send NMEA data to the server. Note This option is not used for configuring geofencing.

To save the feature template, click **Save**.

Configure Automatic Bandwidth Detection

Table 29: Feature History

Feature Name	Release Information	Description
Day 0 WAN Interface Automatic Bandwidth Detection	Cisco IOS XE Catalyst SD-WAN Release 17.5.1a Cisco vManage Release 20.5.1	This feature enables a device to automatically determine the bandwidth for WAN interfaces in VPN0 during day 0 onboarding by performing a speed test using an iPerf3 server.

You can configure the Cisco VPN Interface Ethernet template to cause a device to automatically detect the bandwidth for WAN interfaces in VPN0 during its day 0 onboarding. If you configure a template in this way, a Cisco IOS XE Catalyst SD-WAN device attempts to determine the bandwidth for WAN interfaces in VPN0 after completing the PnP process.

Automated bandwidth detection can provide more accurate day 0 bandwidth configuration than manual configuration because there is limited user traffic that can affect results.

A device determines the bandwidth by performing a speed test using an iPerf3 server. iPerf3 is a third-party tool that provides active measurements of bandwidth on IP networks. For more information, see the iperf.fr website.

If a device has a connection to the internet, the device uses a public iPerf3 server for automatic bandwidth detection, unless you specify a private iPerf3 server. If a device has a connection to a private circuit and no internet connection, you must specify a private iPerf3 server for automatic bandwidth detection.

We recommend that you specify a private iPerf3 server. If a private iPerf3 server is not specified, the device pings a system defined set of public iPerf3 servers and selects for the speed test the public server with the minimum hops value or, if all servers have the same minimum hops value, the server with the minimum latency value. If the speed test fails, the device selects another public server from the list. The device continues to select other public iPerf3 servers until the speed test is successful or until it has tried all servers. Therefore, a speed test on a public iPerf3 server can use a server that is far away, resulting in a larger latency than the minimum.

The set of system defined public iPerf3 servers includes the following:

- iperf.scottlinux.com
- iperf.he.net
- bouygues.iperf.fr
- ping.online.net
- iperf.biznetnetworks.com

The following settings on the Cisco SD-WAN Manager VPN Interface Ethernet template control bandwidth detection. These settings are supported for WAN interfaces in VPN0 only.

- **Auto Detect Bandwidth**—When enabled, the device detects the bandwidth.
- **iPerf Server**—To use a private iPerf3 server for automatic bandwidth detection, enter the IPv4 address of the private server. To use a public iPerf3 server for automatic bandwidth detection, leave this field blank.

The private iPerf3 server should run on port 5201, which is the default iPerf3 port.

In addition, automatic bandwidth detection requires that the allow-service all command be configured for the tunnel interface. See “VPN, Interface, and Tunnel Configuration for WAN and LAN interfaces.”

The device writes the results of a speed test to the auto_speedtest.json file in its bootflash directory. It also displays the results in the **Auto Upstream Bandwidth (bps)** and **Auto Downstream Bandwidth (Mbps)** areas on the **Monitor > Devices > Interface** page of Cisco SD-WAN Manager.

If a device does not receive a response from an iPerf3 server, an error is recorded in the auto_speedtest.json file and displays on the **Monitor > Devices > Interface** page of Cisco SD-WAN Manager.



Note In Cisco vManage Release 20.6.x and earlier releases, the speed test results are displayed on the **Monitor > Network > Interface** page.

CLI Equivalent

auto-bandwidth-detect

iperf-server *ipv4-address*

There also is a no auto-bandwidth-detect form of this command.

Example

```
Device# show sdwan running-config sdwan
sdwan
interface GigabitEthernet0/0/0
```

```

tunnel-interface
 encapsulation gre
 allow-service all
 no allow-service bgp
 allow-service dhcp
 allow-service dns
 allow-service icmp
 allow-service sshd
 allow-service netconf
 no allow-service ntp
 no allow-service ospf
 no allow-service stun
 allow-service https
 no allow-service snmp
 no allow-service bfd
 exit
 auto-bandwidth-detect
 iperf-server 192.0.2.255
 exit
 appqoe
 no tcpopt enable
 no dreopt enable

```

Configure System Logging Using CLI

Use the following command to configure system logging on Cisco SDWAN.

```

config-transaction [IP address | description | alarm | buffered | buginf | console |
discriminator
 esm | event | facility | file | history | host | origin-id | persistent | rate-limit |
snmp-authfail | snmp-trap | source-interface
 trap | userinfo]

```

SSH Terminal

Use the SSH Terminal screen to establish an SSH session to a router. From an SSH session, you can issue CLI commands on a router.

Establish an SSH Session to a Device

To establish an SSH session to a device:

1. From the Cisco SD-WAN Manager menu, choose **Tools > SSH Terminal**.
2. Select the device on which you wish to collect statistics:
 - a. Select the device group to which the device belongs.
 - b. If needed, sort the device list by its status, hostname, system IP, site ID, or device type.
 - c. Click the device to select it.
3. Enter the username and password to log in to the device.

You can now issue CLI commands to monitor or configure the device.

HTTP/HTTPS Proxy Server for Cisco SD-WAN Manager Communication with External Servers

Table 30: Feature History

Feature Name	Release Information	Description
HTTP/HTTPS Proxy Server for Cisco SD-WAN Manager Communication with External Servers	Cisco IOS XE Catalyst SD-WAN Release 17.5.1a Cisco vManage Release 20.5.1	Cisco SD-WAN Manager uses HTTP/HTTPS to access some web services and for some REST API calls. With this feature, you can channel the HTTP/HTTPS communication through an HTTP/HTTPS proxy server.
Cisco SD-WAN Manager HTTP/HTTPS Proxy Server Support Over IPv6	Cisco IOS XE Catalyst SD-WAN Release 17.13.1a Cisco Catalyst SD-WAN Manager Release 20.13.1	You can now configure an IPv6 address when configuring an HTTP/HTTPS proxy server.

The following are some instances in which Cisco SD-WAN Manager uses an HTTP/HTTPS connection to an external server:

- Certificate request or renewal
- Cisco Plug and Play integration
- Smart Licensing Using Policy
- Cloud OnRamp
- Software image download
- Data upload to Cisco SD-WAN Analytics

In Cisco vManage Release 20.4.1 and earlier releases, you must permit this HTTP/HTTPS communication in the firewall configured on your on-premises Cisco SD-WAN Manager instance. Beginning Cisco vManage Release 20.5.1, you can channel the HTTP/HTTPS communication via an HTTP/HTTPS proxy server. With the HTTP/HTTPS proxy server configured, you can restrict HTTP/HTTPS communication with external servers while configuring the firewall and secure the system further.

Traffic is directed through the HTTP/HTTPS proxy server in the following cases:

- HTTPS connection for Symantec or Cisco automated certificate request or renewal
- REST API calls to URLs of the following domains:
 - cisco.com
 - amazonaws.com
 - microsoft.com
 - office.com

- microsoftonline.com

Once every 24 hours, Cisco SD-WAN Manager checks whether the configured HTTP/HTTPS proxy server is reachable. If the proxy server is unreachable, Cisco SD-WAN Manager raises the alarm `HTTPS proxy server {IP} not reachable`.

Restrictions

- When configured to communicate with external servers via an HTTP/HTTPS proxy server, Cisco SD-WAN Manager resolves FQDNs locally or through configured DNS servers, bypassing the proxy server. Cisco SD-WAN Manager then sends the HTTP/HTTPS connections resulting from the resolution to the proxy server. DNS queries for the resolution of external server FQDNs must be successful before Cisco SD-WAN Manager can send resulting HTTP/HTTPS connections to the HTTP/HTTPS proxy server.
- Use of the HTTP/HTTPS proxy server is not supported for communication between the SD-AVC container in Cisco SD-WAN Manager and external services.

Configure HTTP/HTTPS Proxy Server

Prerequisites

Enable out of band interface on single node using **Administration > Cluster Management** before configuring proxy server.

1. From the Cisco SD-WAN Manager menu, choose **Administration > Settings**.
2. Open **HTTP/HTTPS Proxy**.
3. For the **Enable HTTP/HTTPS Proxy** setting, click **Enabled**.
4. Enter the **HTTP/HTTPS Proxy IP Address** and **Port** number.

For releases before Cisco Catalyst SD-WAN Manager Release 20.13.1, enter an IPv4 address. For releases from Cisco Catalyst SD-WAN Manager Release 20.13.1, enter an IPv4 or IPv6 address.

5. Enter a **Non Proxy Host/IP List**.

This list is a pipe (|) separated list of IP addresses or hostnames that are not to be proxied.

6. Click **Save**.



Note Cisco SD-WAN Manager uses TCP port 7 echo request to validate reachability of the proxy server. Ensure that you configure your firewall and proxy server to allow the echo requests to make the destination host ports accessible.

Cisco SD-WAN Manager verifies that the HTTP/HTTPS proxy server is reachable and saves the server details in the configuration database. HTTP/HTTPS connections and REST API calls to external servers are directed through the proxy server.

If the HTTP/HTTPS proxy server is not reachable, Cisco SD-WAN Manager displays an error message on the GUI indicating the reason for failure.

Bulk API Rate Limit for a Cisco SD-WAN Manager Cluster

Table 31: Feature History

Feature Name	Release Information	Description
Bulk API Rate Limit for a Cisco SD-WAN Manager Cluster	Cisco vManage Release 20.10.1	For a Cisco SD-WAN Manager cluster, the rate limit for bulk APIs equals (rate-limit per node) * (number of nodes in the cluster). Cisco SD-WAN Manager distributes bulk API requests among the nodes in the cluster. With these changes, you can retrieve data faster from a Cisco SD-WAN Manager cluster through bulk APIs.

In Cisco vManage Release 20.9.x and earlier releases, you send bulk API requests to a node in the Cisco SD-WAN Manager cluster. The bulk API throughput is constrained by the rate-limit per node. To increase the throughput, you must send separate bulk API requests to each node in the cluster and the collate the API responses.

From Cisco vManage Release 20.10.1, send bulk API requests to the Cisco SD-WAN Manager cluster. Cisco SD-WAN Manager distributes the API requests among the clusters in the node. This distribution increases the rate limit to (rate-limit per node) * (number of nodes in the cluster), allowing you to retrieve more data in a shorter duration compared to a bulk API request addressed to a single node. With the distribution, you need not send separate bulk API requests to two or more nodes in the cluster or collate the API responses.

Configure Bulk API Rate Limit

1. Log in to one of the Cisco SD-WAN Manager nodes in the Cisco SD-WAN Manager cluster and configure the following command:

```
vManage# request nms server-proxy set ratelimit
```

2. The command-line displays the following prompt about the rate limit for non-bulk APIs:

```
Do you want to reconfigure rate limit for URL non bulk api [y/n] :
```

Enter **n**.

3. The command-line displays the following prompt about the rate limit for bulk APIs:

```
Do you want to reconfigure rate limit for URL bulk api /dataservice/data/device/statistics [y/n] :
```

Enter **y**.

4. Enter the per-node rate limit in response to a prompt similar to the following:

```
Enter the PER NODE rate limit for URL bulk api /dataservice/data/device/statistics [144 load balanced across all nodes at present] :
```

This prompt is from a three-node Cisco SD-WAN Manager cluster, with the bulk API rate limit configured to the default value of 48 requests per node. Across all the three nodes, the bulk API rate limit is (rate-limit/node) * 3, which is 144 requests.

Before you enter the rate limit, consider its effect on Cisco SD-WAN Manager resources.

5. Enter the unit time for which the rate limit applies in response to a prompt similar to the following.

You can apply a rate limit per second, minute, hour, or day. The default unit is minute.

```
Enter the rate limit unit (second, minute, hour, day) for URL bulk api
/dataservice/data/device/statistics [minute] :
```

Cisco vManage applies the rate limit on all the Cisco SD-WAN Manager instances in the cluster. The command line displays the following message:

```
Propagating rate limit update across all nodes. Please wait.
```

After the rate limit is applied, Cisco SD-WAN Manager prompts you to restart the server-proxy on all nodes and the command line returns to the privileged EXEC mode:

```
Done. Please restart server-proxy on all nodes using "request nms server-proxy restart"
command.
vManage#
```

6. Restart the server-proxy using the following command:
vManage# **request nms server-proxy restart**
7. Log in to the other Cisco SD-WAN Manager nodes in the cluster and restart the server-proxy using the **request nms server-proxy restart** command.

In the following example, the bulk API rate limit per node is set to 50 requests per minute.

```
vManage# request nms server-proxy set ratelimit
Do you want to reconfigure rate limit for URL non bulk api [y/n] : n
Do you want to reconfigure rate limit for URL bulk api /dataservice/data/device/statistics
[y/n] : y
Enter the PER NODE rate limit for URL bulk api /dataservice/data/device/statistics [144
load balanced across all nodes at present] : 50
Enter the rate limit unit (second, minute, hour, day) for URL bulk api
/dataservice/data/device/statistics [minute] : minute
Propagating rate limit update across all nodes. Please wait.
Done. Please restart server-proxy on all nodes using "request nms server-proxy restart"
command.
vManage# request nms server-proxy restart
```

View Bulk API Rate Limit

To view the bulk API rate limit, log in to any node in the Cisco SD-WAN Manager cluster and use the **show nms server-proxy ratelimit** command.

The following is a sample command output:

```
vManage# show nms server-proxy ratelimit
Non Bulk API: 100/second (per node)
Bulk API: 150/minute (across cluster)
```


This sample output is from three-node Cisco SD-WAN Manager cluster with the bulk API rate limit per node configured to 50 requests per minute. Therefore, the bulk API rate limit for the cluster is $50 \times 3 = 150$ requests per minute.



CHAPTER 4

Configure System Logging

Table 32: Feature History

Feature Name	Release Information	Description
Ability to Send Syslog Messages over TLS	Cisco IOS XE Catalyst SD-WAN Release 17.2.1r	This feature allows you to transport syslog messages to external configured hosts by establishing a Transport Layer Security (TLS) connection. Using the TLS protocol enables the content of syslog messages to remain confidential, secure, and untampered or unaltered during each hop.
Remote Logging Over TCP and TLS in Cisco Catalyst SD-WAN Control Components	Cisco Catalyst SD-WAN Control Components Release 20.13.1	The feature allows remote logging of syslog messages through TCP and TLS. This feature is now available on Cisco Catalyst SD-WAN Control Components (Cisco Catalyst SD-WAN Controller, Cisco Catalyst SD-WAN Validator, and Cisco Catalyst SD-WAN Manager) in addition to Cisco IOS XE Catalyst SD-WAN devices.

- [System Logging, on page 54](#)
- [Syslog Message Format, Syslog Message Levels, and System Log Files, on page 54](#)
- [Benefits of Using TLS for Sending Syslog Messages, on page 58](#)
- [Configure Logging in Server Authentication for TLS, on page 58](#)
- [Configure Logging in Mutual Authentication for TLS, on page 59](#)
- [Install Root Certificate Authority on Cisco IOS XE Catalyst SD-WAN Device for Server Authentication , on page 59](#)
- [Install Root Certificate Authority on Syslog Server for Server Authentication, on page 61](#)
- [Install Syslog Root Certificate on Cisco IOS XE Catalyst SD-WAN Device for Mutual Authentication, on page 62](#)
- [Configure Logging Using Configuration Groups, on page 63](#)
- [Configure Logging Feature Template Using Cisco SD-WAN Manager, on page 66](#)
- [Generate Feature Certificate Signing Request and Install Feature Certificates, on page 73](#)
- [Verify Trustpoint Configuration on Cisco IOS XE Catalyst SD-WAN Device, on page 74](#)
- [Export Cisco SD-WAN Manager NMS Audit Log to Syslog Server, on page 75](#)
- [Remote Logging Over TCP and TLS in Cisco Catalyst SD- WAN Control Components, on page 77](#)

System Logging

System logging operations use a mechanism similar to the UNIX syslog command to record system-wide, high-level operations that occur on Cisco Catalyst SD-WAN devices in the overlay network. The log levels (priorities) of the messages are the same as standard UNIX commands, and you can configure the priority of syslog messages. Cisco Catalyst SD-WAN devices can send log messages to a UNIX-style syslog service.

Cisco IOS XE Catalyst SD-WAN devices send syslog messages to syslog servers on configured external hosts using TCP and UDP. When these devices are sending the syslog messages, the messages might transit several hops to reach the output destination. The intermediate networks during the hops might not be trustworthy, be in a different domain, or have a different security level. Therefore, Cisco IOS XE Catalyst SD-WAN devices now support sending secure syslog messages over the Transport Layer Security (TLS) as per RFC5425. To secure the syslog message content from potential tampering, the TLS protocol is used for certificate exchange, mutual authentication, and ciphers negotiation.

Cisco IOS XE Catalyst SD-WAN devices supports both mutual and server authentication for sending syslog messages over TLS.



Note Disabling system logging to disk (`no system logging disk enable`) does not disable `vsyslog`.

Syslog Message Format, Syslog Message Levels, and System Log Files

Syslog Message Format

Syslog messages begin with a percent sign (%) and following are the syslog message formats:

- Syslog message format
seq no:timestamp: %facility-severity-MENEMONIC:description (hostname-n)
- Syslog message format based on RFC5424
<pri>ver timestamp hostname appname procid msgid structured data description/msg



Note In the syslog message format based on RFC5424, the optional fields such as, hostname, appname, procId, msgId, structured data are specified with a -.

The field descriptions of syslog messages are:

Table 33: Field Descriptions of Syslog Message Format

Field	Description
facility	Sets the logging facility to a value other than 20, which UNIX systems expect.

Field	Description
severity	The importance or severity of the message is categorized by the numerical code from 0 through 7. A lower number in this range indicates greater severity of the system condition.
msg or description	A text string that describes the condition of syslog server. This portion of the syslog message sometimes includes IP addresses, interface names, port numbers, or usernames. In syslog message formats based on RFC5424, the description represents: <code>%facility-severity-MENEMONIC:description</code>

Usually, the syslog messages are preceded by extra text.

- The following is an example of a system logging message preceded by a priority value, sequence number, and time stamp:

```
<45>10: polaris-user1: *Jun 21 10:76:84.100: %LINK-5-CHANGED: Interface GigabitEthernet0/0,
changed state to administratively down
```

- Based on RFC5424, the following is an example of a system logging message preceded by a priority value, version of syslog protocol specification, and time stamp:

```
<45>1 2003-10-11T22:14:15.003Z 10.64.48.125 polaris-user1 - - - %LINK-5-CHANGED: Interface
GigabitEthernet0/0, changed state to administratively down
```



Note The time stamp formats are not the same in both the syslog message formats. In the message format based on RFC5424, T, and Z are mandatory where T represents a separator and Z represents zero timezone.

Syslog Message Levels

All syslog messages are associated with priority levels that indicate the severity of syslog messages to save. The default priority value is "informational", so by default, all syslog messages are recorded. The priority level can be one of the following in order of decreasing severity:

- Emergency—System is unusable (corresponds to syslog severity 0).
- Alert—Ensure that you act immediately (corresponds to syslog severity 1).
- Critical—A serious condition (corresponds to syslog severity 2).
- Error—An error condition that does not fully impair system usability (corresponds to syslog severity 3).
- Warning—A minor error condition (corresponds to syslog severity 4).
- Notice—A normal, but significant condition (corresponds to syslog severity 5).
- Informational—Routine condition (the default) (corresponds to syslog severity 6).
- Debug—Issues debug messages that correspond to syslog severity 7.

System Log Files

All syslog messages that are at or above the default or configured priority value are recorded in a number of files in the `/var/log` directory on the local device of the syslog server. The following are the contents of the log files:

- `auth.log`—Login, logout, and superuser access events, and usage of authorization systems
- `kern.log`—Kernel messages
- `messages.log`—Consolidated log file that contains syslog messages from all sources.
- `vconfd.log`—All configuration-related syslog messages
- `vdebug.log`—All debug messages for modules whose debugging is turned on and all syslog messages that are above the default priority value. The debug log messages support various levels of logging based on the module. The different modules implement the logging levels differently. For example, the system manager (sysmgr) has two logging levels (on and off), while the chassis manager (chmgr) has four different logging levels (off, low, normal, and high). You cannot send debug messages to a remote host. Therefore, to enable debugging, use the **debug** operational command.
- `vsyslog.log`—All syslog messages from Cisco Catalyst SD-WAN processes (daemons) that are above the configured priority value. The default priority value is "informational", so by default, all "notice", "warning", "error", "critical", "alert", and "emergency" syslog messages are saved.
- `vmanage-syslog.log`—Cisco SD-WAN Manager NMS Audit log messages

The following are the standard LINUX files that Cisco Catalyst SD-WAN does not use and are available in the `/var/log` directory.

- `cron.log`
- `debug.log`
- `lpr.log`
- `mail.log`
- `syslog`

The messages sent to syslog files are not rate-limited and consequently:

- A storage limit of 10 log files with a capacity of up to 16 MB size is set for each syslog file.
 - When the storage capacity exceeds the 16 MB size limit, the log file is saved as a .GZ file along with the date appended to it.
 - When the storage limit exceeds 10 log files, the oldest log file is dropped.
- If many syslog messages are generated in a short span of time, the overflowing messages are buffered and queued to be stored in the syslog file.

For repeating syslog messages or identical messages that occur multiple times in succession, only one copy of the message is placed in the syslog file. The message is annotated to indicate the number of times the message occurred.

The maximum length of a log message is 1024 bytes. The longer messages are truncated.

The maximum length of a log message for Cisco SD-WAN Manager NMS audit logs is 1024 bytes. The longer messages are truncated into smaller fragments and each of these fragments are indicated by an identifier. The identifiers are, fragment 1/2, fragment 2/2, and so on. For example, a long audit log message when truncated into smaller fragments appears as:

```
local6.info: 18-Oct-2020 17:42:07 vm10 maintenance-fragment-1/2: {"logid":
"d9ed576a-43ae-49ce-921b-a51cled40698", "entry_time":
1576605512190, "statcycletime" 34542398334245, "logmodule": "maintenance", "logfeature":
"upgrade", "loguser": "admin", "logusersrcip":
"10.0.1.1", "logmessage": "Device validation Upgrade to version - Validation success",
"logdeviceid": "Validation", "auditdetails" :
["[18-Oct-2020 17:42:08 UTC] Published messages to vmanage(s)", "auditdetails": "[18-Oct-2020
17:42:07 UTC] Software image: vmanage-99.99.999-
x86_64.tar.gz", "Software image download may take up to 60}

local6.info: 18-Oct-2020 17:42:07 vm10 maintenance-fragment-2/2: { "minutes", "logprocessid":
"software_install-7de0ec44-d290-4429-b24532435324", "tenant":, "default"}
```

The syslog messages related to AAA authentication and Netconf CLI access and usage are placed in the auth.log and messages.log files. Each time a Cisco SD-WAN Manager NMS logs into a router to retrieve statistics and status information and to push files to the router, the router generates AAA and Netconf log messages. So, over time, these messages can fill the log files. To prevent these messages from filling the log files, you can disable the logging of AAA and Netconf syslog messages by using the following commands from Cisco SD-WAN Manager NMS:



Note For information about available syslog messages on Cisco SD-WAN Manager, see [Syslog Messages](#).

How System Log Files Are Deleted

System log file cleanup in /opt/data and /rootfs.rw follows two strategies:

1. Daily Cron Job:

- Automatically deletes logs older than 15 days.
- Targets specific log patterns and directories.

2. Monitor and Truncate:

- Triggered when disk usage exceeds 90%.
- If the above step fails to recover disk space, the system deletes larger files over 1 MB that are less than 2 days old.



Note Critical system files and essential logs are excluded from cleanup, although active log files may be truncated if disk space becomes critically low.

Disable logging of AAA and Netconf Syslog Messages

1. vManage# **config**

Enters the configuration mode terminal

2. `vManage(config)# system aaa logs`
Configures the logging of AAA and Netconf system logging (syslog) messages
3. `vManage(config-logs)# audit-disable`
Disable logging of AAA events
4. `vManage(config-logs)# netconf-disable`
Disable logging of Netconf events
5. `vManage(config-logs)# commit`
Commit complete.

Benefits of Using TLS for Sending Syslog Messages

The benefits of using TLS for sending syslog messages are:

- Confidentiality of message content where each TLS session begins with a handshake between the Cisco IOS XE Catalyst SD-WAN device and the syslog server. The Cisco IOS XE Catalyst SD-WAN device and syslog server agree on the specific security key and the encryption algorithms to be used for that session. The TLS session opposes any disclosure of the contents of the syslog message.
- Integrity-checking of the content of each message to disable modifications to a message during transit on a hop-by-hop basis.
- Mutual authentication between the Cisco IOS XE Catalyst SD-WAN device and syslog server ensures that the syslog server accepts log messages only from authorized clients through certificate exchange.

Configure Logging in Server Authentication for TLS

In server authentication, Cisco IOS XE Catalyst SD-WAN devices verify the identity of the syslog server. If the syslog server and the certificate are legitimate entities, the device establishes a TLS connection with the server. For implementing server authentication, the syslog server shares the public certificate with the Cisco IOS XE Catalyst SD-WAN devices.

Prerequisite

Ensure that Cisco IOS XE Catalyst SD-WAN devices have preinstalled Root Certificate Authority (CA), which you configure using cryptographic module CLIs. See [Install root CA on Cisco IOS XE Catalyst SD-WAN device](#).

To configure TLS profile for syslog server, perform the following steps:

1. [Configure logging feature template using Cisco vManage](#).
 - a. [Configure logging attributes to local disk](#).
 - b. [Configure syslog server for TLS](#).
2. [Create a device template from logging feature template](#).

Configure Logging in Mutual Authentication for TLS

In mutual authentication, both the syslog server and Cisco IOS XE Catalyst SD-WAN device authenticate each other at the same time. Cisco IOS XE Catalyst SD-WAN devices must have root or identity certificates for mutual authentication of the TLS session. To configure TLS profile for syslog server, perform the following steps:

1. [Install Syslog Root Certificate on Cisco IOS XE Catalyst SD-WAN device.](#)
2. [Configure logging feature template using Cisco vManage.](#)
 - a. [Configure logging attributes to local disk.](#)
 - b. [Generate Feature Certificate Signing Request and Install Feature Certificates, on page 73](#)
 - c. [Configure syslog server for TLS.](#)
3. [Create a device template from logging feature template.](#)
4. [Generate Feature Certificate Signing Request and Install Feature Certificates, on page 73.](#)
5. [Verify trustpoint configuration on Cisco IOS XE Catalyst SD-WAN device.](#)

Install Root Certificate Authority on Cisco IOS XE Catalyst SD-WAN Device for Server Authentication

Before you begin

Ensure that you generate the encoded CA certificate on the syslog server. See [Install Root Certificate Authority on Syslog Server for Server Authentication, on page 61.](#)

Procedure

- Step 1** To configure PKI trustpoint for Certificate Authority, use these commands for authorizing and revocation of certificates in PKI.
- a) **enable**
Enables privileged EXEC mode.
Example:

```
Cisco XE SD-WAN> enable
```
 - b) **config-transaction**
Enters the configuration mode.
Example:

```
Cisco XE SD-WAN# config-transaction
```
 - c) **crypto pki trustpoint name**

Declares the trustpoint and a given name and enters CA-trustpoint configuration mode. Also, specifies the enrollment parameters and fingerprint for the CA. Obtain the fingerprint from the fingerprint.txt.

Example:

```
Cisco XE SD-WAN (config)# crypto pki authenticate PROXY-SIGNING-CA
  enrollment url bootflash:
  revocation-check none
  rsakeypair PROXY-SIGNING-CA 2048
  subject-name cn=proxy-signing-cert
  fqdn none
  fingerprint 54F371C8EE2BFB06E2C2D0944245C288FBB07163
```

The Authentication should pass. In case of a failure, reach out to pki-team.

For more information, see [Cisco SD-WAN IOS XE TLS Syslog Configuration on syslog-ng Server](#)

d) **chain-validation [{stop | continue}[parent-trustpoint]]**

Configures the level to which a certificate chain is processed on all certificates.

Example:

```
Cisco XE SD-WAN (ca-trustpoint)# chain-validation stop
```

e) **revocation-check method**

(Optional) Checks the revocation status of a certificate.

Example:

```
Cisco XE SD-WAN (ca-trustpoint)# revocation-check none
```

f) **exit**

Returns to global configuration mode.

Example:

```
Cisco XE SD-WAN (ca-trustpoint)# exit
```

Step 2 Retrieve and authenticate the Root CA before the Cisco IOS XE Catalyst SD-WAN device can be issued a certificate and certificate enrollment occurs.

To authenticate the CA, use the **crypto pki authenticate** command.

Example:

```
Cisco XE SD-WAN (config)# crypto pki authenticate root
```

Step 3 Copy the block of text containing the base 64 encoded CA certificate and paste it at the prompt.

To generate and copy the text containing the encoded CA certificate, see [Install Root Certificate Authority on Syslog Server for Server Authentication, on page 61](#).

Example:

A sample base 64 encoded CA certificate:

```
-----BEGIN CERTIFICATE-----
MIID9jCCAt6gAwIBAgIJAM5b3nyjDAKIMA0GCSqGSIb3DQEBCwUAMIGPMQswCQYD
VQQGEwJUTjESMBAGA1UECAwJS2FybmF0YXVhMRItEAYDVQQHDA1CYW5nYWxvcmlUe
DjAMBgnVBAoMBUNpc2NvMQwwCgYDVQQQLDANDU0cxGzAZBgNVBAMMEmlwQmV5b20w
HhcnMTkwOTIwMTQ1NjAxWjCBjzELMAkGA1UEBhMCU4xEjAQBGNV
BAGMCUthcm5hdGFrYTESMBAGA1UEBwwJQmFuZ2Fsb3JlMQ4wDAYDVQQKDAVDaXNj
bzEMMAoGA1UECwwDQ1NHMRswGQYDVQQDDBJlbWJkLWxueC5jaXNjb20wHTAB
BgkqhkiG9w0BCQEWDMFuYnZAY21zY28uY29tMIIBIjANBgkqhkiG9w0BAQEFAAOC
```

```
AQ8AMIIBCgKCAQEAuof+Dh8EdAQ7bHJPdnXhy9ibTLAQ+OpQrMBoOqeAsU/Jru8y
3ht2Eqci35aNjlDcsTUlZyUHBNAmtL69t1HxTRVCOghOZmipzOS+q8rFykHa+bcA
FqmHyqxNwdQcW3cQZF6rvWTFD9O46ONX3xewpdCR+s+0KFOHdd+RxpAb2NyDWIvn
/1/xwq2a4Zlwgm2d0G8sit0i0D/+6FbZuJjAf+PRTypo4IJyQjcOHpZuslLzPztM
HxLI7pOmR+8+WcInt0l0dyGdpKKHXi6lEbeiYubIym0z0Des5OckDYFejXgXpJDx
9jCVkz+r0bijqbT5PMpSAYYcjdQ0kdH43sykwIDAQABo1MwUTAdBgNVHQ4EFgQU
OcOmN72TyBqD/Ud2qBLUwIdlYv0wHwYDVR0jBBgwFoAUOcOmN72TyBqD/Ud2qBLU
wIdlYv0wHwYDVR0TAQH/BAUwAwEB/zANBgkqhkiG9w0BAQsFAAOCAQEAAUVVWJHwo
rKxfFV2w7jR7mLZSlVtEvZueMXWPvyYP+Qt09MrRqWNUJEvggTxU7lvLwtNITPM
l/dOmpoer8GhDtnxUnjsVeVWgIR74SJCS0GU/03bEJ2sto/eAJEOzI7wDg7Fubgy
Pc3RHbk4JWtWs4JF8+E64p2UzJMuu0eLDPQWx17p2wd3sr4DBHB3qlfbg31T3VHr
PCcuzJmOEdeZYGL1/LFvPx7NZS8lwFAohe6h8ptm3ENG7dzIeyZFZVfcq11Q1rer
+3RcM0VqjScIOzhp97dqfBlHEdqeU/QfKlBt12KU+0sj8yJJC+cuKlHQj5JGmGLI
Y6r7bMcn99Y6Rw==
-----END CERTIFICATE-----
```

Step 4 Type yes to confirm the acceptance of the certificate.

The Root CA certificate is successfully imported.

Example

```
crypto pki trustpoint PROXY-SIGNING-CA
  enrollment url bootflash:
  revocation-check none
  rsa-keypair PROXY-SIGNING-CA 2048
  subject-name cn=proxy-signing-cert
  fqdn none
  fingerprint 54F371C8EE2BFB06E2C2D0944245C288FBB07163 >> The fingerprint configured
was obtained from the fingerprint.txt file above
commit

crypto pki authenticate PROXY-SIGNING-CA
Reading file from bootflash:PROXY-SIGNING-CA.ca
Certificate has the following attributes:
Fingerprint MD5: 7A97B30B 2AE458FF D9E7D91F 66488DCF
Fingerprint SHA1: 21E0F09B B67B2E9D 706DBE69 856E5AA3 D39A268A
Trustpoint Fingerprint: 21E0F09B B67B2E9D 706DBE69 856E5AA3 D39A268A
Certificate validated - fingerprints matched.
Trustpoint CA certificate accepted.
```

What to do next

[Configure Logging Feature Template Using Cisco SD-WAN Manager, on page 66](#)

Install Root Certificate Authority on Syslog Server for Server Authentication

In this document, the following steps describe the procedure to set up syslog-ng server that supports TLS.

Procedure

Step 1 To install syslog-ng on the server, use the following command:

Example:

```
# apt-get install syslog-ng openssl
```

Step 2 To change the directory to syslog-ng folder and create folders to store the root certificates, use the following commands:

Example:

```
# cd /etc/syslog-ng
# mkdir cert.d
# mkdir key.d
# mkdir ca.d
# cd cert.d
# openssl req -new -x509 -out cacert.pem -days 1095 -nodes
# mv privkey.pem ../key.d
```

After using the **openssl** command, an encoded root certificate is available in `cacert.pem` file. The file is located in the `cd/etc/syslog-ng/cert.d` directory.

Step 3 Copy the content from the `cacert.pem` file when installing root certificate on Cisco IOS XE Catalyst SD-WAN Device. See Step 3 of [Install Root Certificate Authority on Cisco IOS XE Catalyst SD-WAN Device for Server Authentication](#), on page 59.

What to do next

[Install Root Certificate Authority on Cisco IOS XE Catalyst SD-WAN Device for Server Authentication](#), on page 59

Install Syslog Root Certificate on Cisco IOS XE Catalyst SD-WAN Device for Mutual Authentication

To configure Cisco IOS XE Catalyst SD-WAN devices with Transport Layer Security (TLS) syslog protocol, the devices must have root or identity certificates for mutual authentication of TLS session. You can either use a third-party Certificate Authority (CA) to get public key infrastructure (PKI) services, or Microsoft Active Directory Certificate Services (AD CS). AD CS allows you to build a PKI and provide public key cryptography, digital certificates, and digital signature capabilities for your requirement.

Procedure

Step 1 Generate the enterprise root certificate using a third party CA or Microsoft Active Directory Certificate Services.

Step 2 Download the root CA in base 64 format, select and copy the content of root CA.

Step 3 From the Cisco SD-WAN Manager menu, choose **Administration > Settings**.

Step 4 Click **Enterprise Feature Certificate Authorization**.

- Step 5** Paste the root CA content in the **Enterprise Root Certificate** box.
- Step 6** (Optional) if you want to generate a Certificate Signing Request (CSR), check the **Set CSR Properties** check box.
- Step 7** Click **Close**.

The root CA is uploaded to Cisco SD-WAN Manager, and Cisco SD-WAN Manager saves the root certificate to the Cisco IOS XE Catalyst SD-WAN device.

What to do next

[Configure Logging Feature Template Using Cisco SD-WAN Manager, on page 66](#)

Configure Logging Using Configuration Groups

Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

-
- Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.
- Step 2** Create and configure Logging in a system profile.
- a) Enter disk information

Table 34: Disk

Field	Description
Enable Disc	Enable this option to allow syslog messages to be saved in a file on the local hard disk, or disable this option to disallow it. By default, logging to a local disk file is enabled on all Cisco IOS XE Catalyst SD-WAN devices.
Max File Size(In Megabytes)	Enter the maximum size of syslog files. The syslog files are rotated on an hourly basis based on the file size. When the file size exceeds the configured value, the file is rotated and the syslog process is notified. Range: 1 to 20 MB Default: 10 MB
Rotations	Enter the number of syslog files to create before discarding the oldest files. Range: 1 to 10 Default: 10

- b) Enter TLS Profile information

Table 35: TLS Profile

Field	Description
Add TLS Profile	
TLS Profile Name*	Enter the name of the TLS profile.
TLS Version	Choose a TLS version: • TLSv1.1 • TLSv1.2
Authentication Type*	Choose Server .
Cipher Suite List	Choose groups of cipher suites (encryption algorithm) based on the TLS version. The following is the list of cipher suites. • aes-128-cbc-sha: Encryption type tls_rsa_with_aes_cbc_128_sha • aes-256-cbc-sha: Encryption type tls_rsa_with_aes_cbc_256_sha • dhe-aes-cbc-sha2: Encryption type tls_dhe_rsa_with_aes_cbc_sha2 (TLS1.2 and above) • dhe-aes-gcm-sha2: Encryption type tls_dhe_rsa_with_aes_gcm_sha2 (TLS1.2 and above) • ecdhe-ecdsa-aes-gcm-sha2: Encryption type tls_ecdhe_ecdsa_aes_gcm_sha2 (TLS1.2 and above) SuiteB • ecdhe-rsa-aes-cbc-sha2: Encryption type tls_ecdhe_rsa_aes_cbc_sha2 (TLS1.2 and above) • ecdhe-rsa-aes-gcm-sha2: Encryption type tls_ecdhe_rsa_aes_gcm_sha2 (TLS1.2 and above) • rsa-aes-cbc-sha2: Encryption type tls_rsa_with_aes_cbc_sha2 (TLS1.2 and above) • rsa-aes-gcm-sha2: Encryption type tls_rsa_with_aes_gcm_sha2 (TLS1.2 and above)

c) Enter server information.

Table 36: Server

Field	Description
Add Server	
Hostname/IPv4 Address*	Enter the DNS name, hostname, or IP address of the system on which to store syslog messages. To add another syslog server, click the plus sign (+). To delete a syslog server, click the trash icon to the right of the entry.

Field	Description
VPN*	Enter the identifier of the VPN in which the syslog server is located or through which the syslog server can be reached. Range: 0 through 65530
Source Interface	Enter the specific interface to use for outgoing system log messages. The interface must be located in the same VPN as the syslog server. Otherwise, the configuration is ignored. If you configure multiple syslog servers, the source interface must be the same for all of them.
Priority	Select the severity of the syslog message to save. The severity indicates the seriousness of the event that generated the message. Priority can be one of the following: • informational: Routine condition (the default) (corresponds to syslog severity 6) • debugging: Prints additional logs to help debugging the issue. • notice: A normal, but significant condition (corresponds to syslog severity 5) • warn: A minor error condition (corresponds to syslog severity 4) • error: An error condition that does not fully impair system usability (corresponds to syslog severity 3) • critical: A serious condition (corresponds to syslog severity 2) • alert: Action must be taken immediately (corresponds to syslog severity 1) • emergency: System is unusable (corresponds to syslog severity 0)
TLS Enable*	Enable this option to allow syslog over TLS. When you enable this option, the following field appears: TLS Properties Custom Profile : Enable this option to choose a TLS profile. When you enable this option, the following field appears: TLS Properties Profile : Choose a TLS profile that you have created for server or mutual authentication in the IPv4 server configuration.
Add IPv6 Server	
Hostname/IPv6 Address*	Enter the DNS name, hostname, or IP address of the system on which to store syslog messages. To add another syslog server, click the plus sign (+). To delete a syslog server, click the trash icon to the right of the entry.
VPN*	Enter the identifier of the VPN in which the syslog server is located or through which the syslog server can be reached. Range: 0 through 65530
Source Interface	Enter the specific interface to use for outgoing system log messages. The interface must be located in the same VPN as the syslog server. Otherwise, the configuration is ignored. If you configure multiple syslog servers, the source interface must be the same for all of them.

Field	Description
Priority	Select the severity of the syslog message to save. The severity indicates the seriousness of the event that generated the message. Priority can be one of the following: • informational: Routine condition (the default) (corresponds to syslog severity 6) • debugging: Prints additional logs to help debugging the issue. • notice: A normal, but significant condition (corresponds to syslog severity 5) • warn: A minor error condition (corresponds to syslog severity 4) • error: An error condition that does not fully impair system usability (corresponds to syslog severity 3) • critical: A serious condition (corresponds to syslog severity 2) • alert: Action must be taken immediately (corresponds to syslog severity 1) • emergency: System is unusable (corresponds to syslog severity 0)
TLS Enable*	Enable this option to allow syslog over TLS.
TLS Properties Custom Profile*	Enable this option to choose a TLS profile.
TLS Properties Profile	Choose a TLS profile that you have created for server or mutual authentication in the IPv6 server configuration.

What to do next

Also see [Deploy a Configuration Group](#).

Configure Logging Feature Template Using Cisco SD-WAN Manager

On Cisco IOS XE Catalyst SD-WAN device, you can log event notification system log (syslog) messages to files on the local device, or you can log them to files on a remote host using Cisco SD-WAN Manager.

Procedure

Step 1 From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.

Step 2 Click **Feature Templates**, and click **Add Template**.

Note

In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

Step 3 From **Select Devices**, choose the device for which you wish to create a template.

Step 4 To create a template for logging, select **Cisco Logging**.

The Cisco Logging template form appears. This form contains fields for naming the template, and fields for defining the Logging parameters. Click a tab or the plus sign (+) to display other fields.

When you first open a feature template, the scope is set to **Default** for those parameters that have a default value. The default setting or value appears next to a parameter. To change the default or to enter a value, click the **Scope** drop-down list to the left of the parameter field.

Step 5 In **Template Name**, enter a name for the template.

The name may contain up to 128 alphanumeric characters.

Step 6 In **Template Description**, enter a description of the template.

The description may contain up to 2048 alphanumeric characters.

What to do next

[Configure Logging Attributes to Local Disk, on page 67](#)

Configure Logging Attributes to Local Disk

1. Click **Disk** and configure the following parameters:

Table 37: Parameter Information

Parameter	Description
Enable Disk	To save syslog messages in a file on the local hard disk, click On or Off to disallow saving. By default, logging to a local disk file is enabled on all devices.
Maximum File Size	Enter the maximum size of syslog files. The syslog files are rotated on an hourly basis based on the file size. When the file size exceeds configured value, the file is rotated and the <i>syslogd</i> process is notified. Range: 1-20 MB Default: 10 MB
Rotations	Enter the number of syslog files to create before discarding the earliest created files. Range: 1-10 MB Default: 10 MB

2. To save the feature template, click **Save**.
3. To associate the feature template with a device template, see [Create a Device Template from Feature Templates](#).

What to Do Next

[Configure TLS Profile for Server Authentication, on page 68](#) or [Configure TLS Profile for Mutual Authentication, on page 70](#)

Configure TLS Profile for Server Authentication

1. Click **TLS Profile**.
2. Click **New Profile**, and configure the following parameters:

Table 38: Parameter Information

Parameter Name	Description
Profile Name	Enter the TLS profile name
TLS Version	Choose TLS versions v1.1 or v1.2
Authentication Type	Choose authentication types as Server .

Parameter Name	Description
Ciphersuites	Choose groups of cipher suites (encryption algorithm) based on the TLS version. The following are the list of cipher suites. • aes-128-cbc-sha Encryption type tls_rsa_with_aes_cbc_128_sha • aes-256-cbc-sha Encryption type tls_rsa_with_aes_cbc_256_sha • dhe-aes-128-cbc-sha Encryption type tls_dhe_rsa_with_aes_128_cbc_sha • dhe-aes-cbc-sha2 Encryption type tls_dhe_rsa_with_aes_cbc_sha2(TLS1.2 & above) • dhe-aes-gcm-sha2 Encryption type tls_dhe_rsa_with_aes_gcm_sha2(TLS1.2 & above) • ecdhe-ecdsa-aes-gcm-sha2 Encryption type tls_ecdhe_ecdsa_aes_gcm_sha2(TLS1.2 & above) SuiteB • ecdhe-rsa-aes-128-cbc-sha Encryption type tls_ecdhe_rsa_with_aes_128_cbc_sha • ecdhe-rsa-aes-cbc-sha2 Encryption type tls_ecdhe_rsa_aes_cbc_sha2(TLS1.2 & above) • ecdhe-rsa-aes-gcm-sha2 Encryption type tls_ecdhe_rsa_aes_gcm_sha2(TLS1.2 & above) • rsa-aes-cbc-sha2 Encryption type tls_rsa_with_aes_cbc_sha2(TLS1.2 & above) • rsa-aes-gcm-sha2 Encryption type tls_rsa_with_aes_gcm_sha2(TLS1.2 & above)

You can use the following cipher suites for each TLS version:

TLS v1.1

```
aes-128-cbc-sha Encryption type tls_rsa_with_aes_cbc_128_sha
aes-256-cbc-sha Encryption type tls_rsa_with_aes_cbc_256_sha
```

TLS v1.2 and later

```
dhe-aes-cbc-sha2 Encryption type tls_dhe_rsa_with_aes_cbc_sha2(TLS1.2 & above)
dhe-aes-gcm-sha2 Encryption type tls_dhe_rsa_with_aes_gcm_sha2(TLS1.2 & above)

ecdhe-ecdsa-aes-gcm-sha2 Encryption type tls_ecdhe_ecdsa_aes_gcm_sha2 (TLS1.2 & above)
ecdhe-rsa-aes-cbc-sha2 Encryption type tls_ecdhe_rsa_aes_cbc_sha2(TLS1.2 & above)
```

```

ecdh-rsa-aes-gcm-sha2 Encryption type tls_ecdhe_rsa_aes_gcm_sha2 (TLS1.2 & above)

rsa-aes-cbc-sha2 Encryption type tls_rsa_with_aes_cbc_sha2 (TLS1.2 & above)
rsa-aes-gcm-sha2 Encryption type tls_rsa_with_aes_gcm_sha2 (TLS1.2 & above)

```

The TLS profiles appear in a table.

3. To create another profile, click **Add**.
4. To edit or delete a TLS profile information, click  or  under **Action**.
5. To save the feature template, click **Save**.
6. To associate the feature template with a device template, see [Create a Device Template from Feature Templates](#).

When you choose the authentication type as **Server**, all information about TLS profiles, except the trustpoint information, is saved.

What to Do Next

[Configure Logging to Remote Servers, on page 72](#)

Configure TLS Profile for Mutual Authentication

1. Click **TLS Profile**.
2. Click **New Profile**, and configure the following parameters:

Table 39: Parameter Information

Parameter Name	Description
Profile Name	Enter the TLS profile name
TLS Version	Choose TLS versions v1.1 or v1.2
Authentication Type	Choose authentication types as Mutual .

Parameter Name	Description
Ciphersuites	Choose groups of cipher suites (encryption algorithm) based on the TLS version that must be used for encryption. The following are the list of cipher suites. • aes-128-cbc-sha Encryption type tls_rsa_with_aes_cbc_128_sha • aes-256-cbc-sha Encryption type tls_rsa_with_aes_cbc_256_sha • dhe-aes-128-cbc-sha Encryption type tls_dhe_rsa_with_aes_128_cbc_sha • dhe-aes-cbc-sha2 Encryption type tls_dhe_rsa_with_aes_cbc_sha2(TLS1.2 & above) • dhe-aes-gcm-sha2 Encryption type tls_dhe_rsa_with_aes_gcm_sha2(TLS1.2 & above) • ecdhe-ecdsa-aes-gcm-sha2 Encryption type tls_ecdhe_ecdsa_aes_gcm_sha2(TLS1.2 & above) SuiteB • ecdhe-rsa-aes-128-cbc-sha Encryption type tls_ecdhe_rsa_with_aes_128_cbc_sha • ecdhe-rsa-aes-cbc-sha2 Encryption type tls_ecdhe_rsa_aes_cbc_sha2(TLS1.2 & above) • ecdhe-rsa-aes-gcm-sha2 Encryption type tls_ecdhe_rsa_aes_gcm_sha2(TLS1.2 & above) • rsa-aes-cbc-sha2 Encryption type tls_rsa_with_aes_cbc_sha2(TLS1.2 & above) • rsa-aes-gcm-sha2 Encryption type tls_rsa_with_aes_gcm_sha2(TLS1.2 & above)

You can use the following cipher suites for each TLS version:

TLS v1.1

```
aes-128-cbc-sha Encryption type tls_rsa_with_aes_cbc_128_sha
aes-256-cbc-sha Encryption type tls_rsa_with_aes_cbc_256_sha
```

TLS v1.2 and later

```
dhe-aes-cbc-sha2 Encryption type tls_dhe_rsa_with_aes_cbc_sha2(TLS1.2 & above)
dhe-aes-gcm-sha2 Encryption type tls_dhe_rsa_with_aes_gcm_sha2(TLS1.2 & above)
```

```

ecdhe-ecdsa-aes-gcm-sha2 Encryption type tls_ecdhe_ecdsa_aes_gcm_sha2 (TLS1.2 & above)
ecdhe-rsa-aes-cbc-sha2 Encryption type tls_ecdhe_rsa_aes_cbc_sha2 (TLS1.2 & above)
ecdhe-rsa-aes-gcm-sha2 Encryption type tls_ecdhe_rsa_aes_gcm_sha2 (TLS1.2 & above)

rsa-aes-cbc-sha2 Encryption type tls_rsa_with_aes_cbc_sha2 (TLS1.2 & above)
rsa-aes-gcm-sha2 Encryption type tls_rsa_with_aes_gcm_sha2 (TLS1.2 & above)

```

The TLS profiles appear in a table.

3. To create another profile, click **Add**.
4. To edit or delete a TLS profile information, click  or  under **Action**.
5. To save the feature template, click **Save**.
6. Associate the feature template with a device template. See [Create a Device Template from Feature Templates](#).

The mutually authenticated feature template is saved on the Cisco IOS XE Catalyst SD-WAN devices, and trustpoint such as, SYSLOG-SIGNING-CA certificate is saved on the device. Cisco SD-WAN Manager can now install the certificate from Cisco IOS XE Catalyst SD-WAN devices.

What to Do Next

[Configure Logging to Remote Servers, on page 72](#)

Configure Logging to Remote Servers

To include the TLS profile in IPV6 or IPV4 server configuration and configure logging of event notification system log messages to a remote server,

1. Click **Server**.
2. Click **Add New Server**, and configure the following parameters for IPv4 or IPv6:

Table 40: Parameter Information

Parameter Name	Description
Hostname/IP Address	Enter the DNS name, hostname, or IPv4, IPv6 address of the system on which to store syslog messages. To add another syslog server, click +. To delete a syslog server, click .
VPN ID	Enter the identifier of the VPN in which the syslog server is located or through which the syslog server can be reached. VPN ID Range: 0-65530

Parameter Name	Description
Source Interface	Enter the specific interface to use for outgoing system log messages. The interface must be located in the same VPN as the syslog server. Otherwise, the configuration of syslog servers is ignored. If you configure multiple syslog servers, the source interface must be same for all of them.
Priority	Choose a severity of the syslog message to be saved. The severity indicates the seriousness of the event that generated the syslog message. See priority levels .
TLS	For Cisco IOS XE Catalyst SD-WAN devices, click On to enable syslog over TLS.
Custom Profile	For Cisco IOS XE Catalyst SD-WAN devices, click On to enable choosing a TLS profile, or click Off to disable choosing a TLS profile.
TLS Profile	For Cisco IOS XE Catalyst SD-WAN devices, choose a TLS profile that you have created for server or mutual authentication in IPv4 or IPv6 server configuration.

The server entries appear in a table.

- To create another entry for a server, click **Add**.
- To edit a logging server, click .
- To remove a logging server, click .
- To save the feature template, click **Save**.
- To associate the feature template with a device template, see [Create a Device Template from Feature Templates](#).

Generate Feature Certificate Signing Request and Install Feature Certificates

To validate and authenticate Cisco IOS XE Catalyst SD-WAN devices and syslog server, perform the following operation on the Cisco SD-WAN Manager Certificates screen. See [Cisco Catalyst SD-WAN Getting Started Guide](#) for information about enterprise certificates.

Procedure

-
- Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration > Certificates**.
- Step 2** From **Certificates**, choose a Cisco IOS XE Catalyst SD-WAN device.
- a) [Generate Feature Certificate Signing Request \(CSR\)](#).
After you generate the Feature CSR, the **View Feature CSR** and **Install Feature certificate** options are available.
 - b) [View Feature CSR](#).
 - c) To download the feature CSR, click **Download**.
- Step 3** To sign the certificate, send the certificate to a third-party signing authority.
- Step 4** To import the certificate into Cisco IOS XE Catalyst SD-WAN devices, [Install feature certificate](#).
- The Install Feature Certificate screen uses the signed certificate and installs it on Cisco IOS XE Catalyst SD-WAN devices.
-

After the feature certificate installation is successful, the [Revoke Feature Certificate](#) and [View Feature Certificate](#) options are available on Cisco SD-WAN Manager.

What to do next

[Verify Trustpoint Configuration on Cisco IOS XE Catalyst SD-WAN Device, on page 74](#)

Verify Trustpoint Configuration on Cisco IOS XE Catalyst SD-WAN Device

To display the contents of syslog file with trustpoint information for Cisco IOS XE Catalyst SD-WAN device, use the **show crypto pki trustpoints status** command.

Examples

Server authentication

```
Cisco XE SD-WAN# show crypto pki trustpoints status

crypto pki trustpoint SYSLOG-SIGNING-CA
  enrollment url bootflash:vmanage-admin/
  fqdn none
  fingerprint xxxxxx
  revocation-check none
  subject-name CN=CSR-cbc47d9d-45bf-433a-9816-1f12a8b48223_vManage Root CA
```

Mutual authentication

```
Cisco XE SD-WAN# show crypto pki trustpoints status

crypto pki trustpoint SYSLOG-SIGNING-CA
  enrollment url bootflash:vmanage-admin/
  fqdn none
  fingerprint xxxxxx
```



```

revocation-check none
rsakeypair SYSLOG-SIGNING-CA 2048
subject-name CN=CSR-cbc47d9d-45bf-433a-9816-1f12a8b48223_vManage Root CA

```

Verify trustpoints on a device for a Syslog-signing-CA certificate

```
Cisco XE SD-WAN# show crypto pki trustpoints SYSLOG-SIGNING-CA status
```

```
Trustpoint SYSLOG-SIGNING-CA:
```

```
Issuing CA certificate not configured.
```

```
State:
```

```
Keys generated ..... No
```

```
Issuing CA authenticated ..... No
```

```
Certificate request(s) ..... None
```

Export Cisco SD-WAN Manager NMS Audit Log to Syslog Server

Table 41: Feature History

Feature Name	Release Information	Description
Export Cisco SD-WAN Manager Audit Log as Syslog	Cisco IOS XE Catalyst SD-WAN Release 17.3.1a Cisco vManage Release 20.3.1	The Cisco SD-WAN Manager exports audit logs in syslog message format to a configured external syslog server. This feature allows you to consolidate and store network activity logs in a central location.

On Cisco IOS XE Catalyst SD-WAN devices and Cisco vEdge devices, you can log event notification system log (syslog) messages to files on a local device, or to files on a remote host using CLI. These event notification logs are converted to system log files and exported to the syslog server. You can then retrieve system log information from the syslog server.

Configure System Logging Using CLI

Log Syslog Messages to a Local Device

By default, a priority level of “information” is enabled when you log syslog messages to a file on a local device. Use the following commands:

1. logging disk

Logs syslog messages on a hard disk

Example:

```
vm01(config-system)# logging disk
```

2. enable

Enables logging to a disk

Example:

```
vm01(config-logging-disk)# enable
```

3. file size *size*

Specifies the size of syslog files in megabytes (MB) By default, the syslog files are 10 MB. You can configure the size of syslog files to be 1–20 MB.

Example:

```
vm01(config-logging-disk)# file size 3
```

4. file rotate *number*

Rotates syslog files on an hourly basis based on the size of the file By default, 10 syslog files are created. You can configure the rotate command to be a number from 1 through 10.

Example:

```
vm01(config-logging-disk)# file rotate 3
```

For more information about logging disk commands, see the [logging disk](#) command.

Log Syslog Messages to a Remote Device

To log event notification system log (syslog) messages to a remote host, use the following commands:

1. logging server

Logs syslog messages to a remote host or syslog server You can configure the name of the server by DNS name, hostname, or IP address. You can configure up to four syslog servers.

Example:

```
vm01(config-system)# logging server 192.168.0.1
```

2. (Optional) vpn *vpn-id*

Specifies the VPN ID of the syslog server

3. (Optional) source interface *interface-name*

Specifies the source interface to reach the syslog server. The interface name can be a physical interface or a sub-interface (a VLAN-tagged interface). Ensure that the interface is located in the same VPN as the syslog server. Otherwise, the configuration is ignored. If you configure multiple syslog servers, the source interface must be the same for all of them.

Example:

```
vm01(config-server-192.168.0.1)# source interface eth0
```

4. priority *priority*

Specifies the severity of the syslog message to be saved. The default priority value is "informational" and by default, all syslog messages are recorded.

Example:

In the following example, set the syslog priority to log alert conditions.

```
vm01(config-server-192.168.0.1)# priority alert
```

If the syslog server is unreachable, the system suspends sending syslog messages for 180 seconds. When the server becomes reachable, logging resumes. For more information about logging server commands, see the [logging server](#) command.

View System Logging Information

To view system log settings after logging syslog messages to a remote host, use the **show logging** command. For example:

```
vm01(config-server-192.168.0.1)# show logging
```

```
System logging
  server 192.168.0.1
  source interface eth0
  exit
!
!
```

To view the contents of the syslog file, use the **show log** command. For example:

```
vm01(config-server-192.168.0.1)# show log nms/vmanage-syslog.log tail 10
```

To view the configured system logging settings from Cisco SD-WAN Manager, see [Audit Log](#).

To view device-specific syslog files from Cisco SD-WAN Manager, perform the following steps:

1. From the Cisco SD-WAN Manager menu, choose **Administration > Settings**, and ensure that you enable **Data Stream**.
2. From the Cisco SD-WAN Manager menu, choose **Monitor > Devices**, and choose a Cisco IOS XE Catalyst SD-WAN device

Cisco vManage Release 20.6.x and earlier: From the Cisco SD-WAN Manager menu, choose **Monitor > Network**, and choose a Cisco IOS XE Catalyst SD-WAN device.
3. Click **Troubleshooting**.
4. From **Logs**, click **Debug Log**.
5. From **Log Files**, select a name of the log file to view the log information.

Remote Logging Over TCP and TLS in Cisco Catalyst SD- WAN Control Components

Information About Remote Logging Over TCP and TLS

Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a and Cisco Catalyst SD-WAN Manager Release 20.13.1, support for remote logging of syslog messages has been enhanced to include TCP and TLS transport methods in addition to UDP. This enhancement applies to Cisco Catalyst SD-WAN Control

Components including Cisco Catalyst SD-WAN Controller, Cisco Catalyst SD-WAN Validator, and Cisco Catalyst SD-WAN Manager.

By default, UDP is enabled as the default transport type for remote logging. However, users now have the additional option to select either TCP or TLS as their transport method for remote logging.

For more information, see [System Logging, on page 54](#)

Benefits of Remote Logging Over TCP and TLS

- Syslog over TCP and TLS supports large-scale network environments. While TCP can handle large volumes of data, TLS can ensure that the log data is securely sent and protected from unauthorized access or tampering.
- You can configure up to four separate remote syslog servers with the option to assign each server a unique transport protocol such as UDP, TLS, or TCP. Alternatively, you can choose to use the same transport protocol for all four servers.
- For remote logging over TLS, a TLS profile supports TLS version 1.2. Also, various cipher suites can be accommodated within the TLS profile, depending on the TLS version.

Configure Remote Logging Over TCP and TLS

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**.
3. Click **Add Template** to select an appropriate device model.
4. In the left pane, from **Select Devices**, choose a Cisco Catalyst SD-WAN control component.
5. Under **OTHER TEMPLATES**, click **Logging** to select it as the feature template.

6. In the **Template Name** field, enter a name for the feature template.

This field is mandatory and can contain only uppercase and lowercase letters, the digits 0 to 9, hyphens (-), and underscores (_). It cannot contain spaces or any other characters.

7. In the **Description** field, enter a description for the feature template.

This field is mandatory, and it can contain all characters and spaces.

8. Configure Logging Attributes to Local Disk.

- a. Click **Disk** and configure the following parameters:

Table 42: Parameter Information

Parameter	Description
Enable Disk	To save syslog messages in a file on the local hard disk, click On or Off to disallow saving. By default, logging to a local disk file is enabled on all devices.

Parameter	Description
Maximum File Size	Enter the maximum size of syslog files. The syslog files are rotated on an hourly basis based on the file size. When the file size exceeds the configured value, the file is rotated and the <i>syslogd</i> process is notified. Range: 1 to 20 MB Default: 10 MB
Rotations	Enter the number of syslog files to create before discarding the earliest created files. Range: 1 to 10 MB Default: 10 MB

- b. To save the feature template, click **Save**.
- c. To associate the feature template with a device template, see [Create a Device Template from Feature Templates](#).

Configure TLS Profile for Server Authentication

1. From the **TLS Profile** section, click **New Profile** to configure a TLS profile.

Table 43: TLS Profile Parameters

Parameter Name	Description
Profile Name	Enter the TLS profile name
TLS Version	Choose TLS v1.2.

Parameter Name	Description
Ciphersuites	Choose one or more ciphersuites based on the TLS version. The following ciphersuites are supported: • ECDHE-ECDSA-AES256-GCM-SHA384 • ECDHE-RSA-AES256-GCM-SHA384 • DHE-RSA-AES256-GCM-SHA384 • ECDHE-ECDSA-AES128-GCM-SHA256 • ECDHE-RSA-AES128-GCM-SHA256 • DHE-RSA-AES128-GCM-SHA256 • ECDHE-ECDSA-AES256-SHA384 • ECDHE-RSA-AES256-SHA384 • DHE-RSA-AES256-SHA256 • ECDHE-ECDSA-AES128-SHA256 • ECDHE-RSA-AES128-SHA256 • DHE-RSA-AES128-SHA256 • RSA-PSK-AES256-GCM-SHA384 • DHE-PSK-AES256-GCM-SHA384 • AES256-GCM-SHA384 • PSK-AES256-GCM-SHA384 • RSA-PSK-AES128-GCM-SHA256 • DHE-PSK-AES128-GCM-SHA256 • AES128-GCM-SHA256 • PSK-AES128-GCM-SHA256 • AES256-SHA256 • AES128-SHA256

2. (Optional) Click **Add** to create another profile.

Configure Syslog Servers for TLS/TCP/UDP

1. From the **Server** section, click **New Server** to configure syslog servers.

Table 44: Syslog Servers Parameters

Parameter Name	Description
Hostname/IP Address	Enter the DNS name, hostname, IPv4 or IPv6 address of the server on which to store syslog messages. To add another syslog server and its details, click Add .
VPN ID	Enter the identifier of the VPN in which the syslog server is located or through which the syslog server can be reached. Range is 0 to 65530
Source Interface	Enter the specific interface to use for outgoing system log messages.
Priority	Choose a logging level for syslog messages that you want to log on the server. The logging level indicates the type of event that generated the syslog message. See priority levels .
Transport for messages logged to remote host	Choose either TCP, UDP, or TLS as the transport type for forwarding syslog messages to a remote server.
Port for transport (default 514)	Specify the port number for the transport type.
TLS Profile	This field displays if you have chosen the transport type as TLS. Choose a TLS profile to attach a TLS profile. See Configure TLS Profile for Server Authentication, on page 79

2. (Optional) Click **Add** to create another entry for a server.
3. Click **Save** to save the feature template

Apply the Logging Feature Template to a Device

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**.
3. From the **Create Template** drop-down list, choose **From Feature Template**
4. From the **Device Model** drop-down list, choose one of the devices.
5. In the **Template Name** field, enter a name for the feature template.

This field is mandatory and can contain only uppercase and lowercase letters, the digits 0 to 9, hyphens (-), and underscores (_). It cannot contain spaces or any other characters.
6. In the **Description** field, enter a description for the device template.

This field is mandatory, and it can contain all characters and spaces.

7. From the **Cisco Logging** drop-down list, choose the template that you created for remote logging.
8. Click **Create** to apply the template to a device.
9. From the list of device templates, click ... next to the device template that you created.
10. Click **Attach Devices**.
11. Choose the devices to which you want to attach the device template.
12. Click **Attach** to the template to the devices.

Configure Remote Logging Over TCP and TLS

For more information about using CLI templates, see [CLI Add-On Feature Templates](#) and [CLI Templates](#).



Note By default, CLI templates execute commands in global config mode.

This section provides sample CLI configurations to configure a remote server for logging with transport type as TCP and TLS.

Configure Remote Server with Transport Type as TCP

Configure Remote Server with Transport Type as TCP.

```
system
logging
server <server-ip-address>
transport tcp port 514
```

Configure Remote Server with Transport Type as TLS

Configuring a remote server for logging with Transport type as TLS is a three-step process.

1. Manage Certificate

To manage the Certificate Authority (CA) certificate from the syslog server, use the following commands below to install, list, and uninstall the certificates.

- Install a Certificate

A certificate can be installed from either a local directory, such as /home/admin, or remotely using secure copy (scp), HTTP, or FTP.

```
request logging ca-cert
install new syslog-ng ca
```

- List All Installed Certificates

```
show logging cacert
```

- Uninstall a Certificate


```
request logging ca-cert uninstall <cert-name>
```

2. Create a TLS Profile

Creating a TLS profile involves specifying the protocols and cipher suites that a device will use for secure communication. You can configure up to four TLS profiles.

Use the following configuration to create a TLS profile:

```
system
logging
tls-profile <profile-name>
  tls-version TLSv1.2
  ciphersuite <ciphersuite1> <ciphersuite2>
```

3. Attach a TLS Profile to a Remote Logging Server

Use the following configuration to attach a TLS profile to the remote logging servers:

```
server <server-ip-address>
vpn <vpn-instance-of-logging-server>
source-interface <interface-num>
transport tls
tls-profile <tls-profile-name>
```

Verify Remote Logging Over TCP and TLS

The following is a sample output from the **show logging cacert** command to view installed certificates.

```
Device# show logging cacert
INDEX  NAME          VALIDITY
-----
0      cert.pem     Fri Jun 21 20:35:10 2024
```

Configuration Example for Support for Syslog over TCP and TLS

Configuration Example for Logging Over TLS

This example shows the configuration for logging over TLS.

```
system
logging
  disk
  enable
  !
  tls-profile profile1
  version TLSv1.2
  ciphersuite ECDHE-ECDSA-AES128-SHA256 AES256-GCM-SHA384 PSK-AES256-GCM-SHA384
  PSK-AES128-GCM-SHA256 AES256-SHA256
exit
server 10.0.1.55
source interface 10.1.1.12
transport tls
tls-profile profile1
exit
!
!
```

Configuration Example for Logging Over TCP

This example shows the configuration for logging over TCP.

```
system
logging
  disk
    enable
  !
  server 10.0.1.56
    transport tcp
  exit
!
!
```



CHAPTER 5

Configure User Access and Authentication

Use the **Manage Users** screen to add, edit, or delete users and user groups from Cisco SD-WAN Manager.

Only a user logged in as the **admin** user or a user who has Manage Users write permission can add, edit, or delete users and user groups from Cisco SD-WAN Manager.

- [Configure Hardened Passwords , on page 86](#)
- [Configure User Login Options, on page 89](#)
- [Manage Users, on page 94](#)
- [Configure Users Using CLI, on page 95](#)
- [Manage a User Group, on page 96](#)
- [Creating Groups Using CLI, on page 97](#)
- [CiscoTAC User Access, on page 98](#)
- [Configure Sessions in Cisco SD-WAN Manager, on page 99](#)
- [Configuring RADIUS Authentication Using CLI, on page 101](#)
- [Configure SSH Authentication, on page 102](#)
- [Configure the Authentication Order, on page 103](#)
- [Role-Based Access with AAA, on page 105](#)
- [Configure AAA Using a Configuration Group, on page 114](#)
- [Configuring AAA using Cisco SD-WAN Manager Template, on page 118](#)
- [Configure IEEE 802.1X Authentication, on page 126](#)
- [Posture Assessment Support, on page 135](#)
- [Type 6 Passwords on Cisco IOS XE SD-WAN Routers, on page 138](#)

Configure Hardened Passwords

Table 45: Feature History

Feature Name	Release Information	Description
Hardened Passwords	Cisco IOS XE Catalyst SD-WAN Release 17.3.1a Cisco vManage Release 20.3.1	This feature enables password policy rules in Cisco SD-WAN Manager. After password policy rules are enabled, Cisco SD-WAN Manager enforces the use of strong passwords.
	Cisco IOS XE Catalyst SD-WAN Release 17.9.1a Cisco vManage Release 20.9.1	This feature lets you configure Cisco SD-WAN Manager to enforce predefined-medium security or high-security password criteria.

Enforce Strong Passwords

We recommend the use of strong passwords. You must enable password policy rules in Cisco SD-WAN Manager to enforce use of strong passwords.

After you enable a password policy rule, the passwords that are created for new users must meet the requirements that the rule defines. In addition, for releases from Cisco vManage Release 20.9.1, you are prompted to change your password the next time you log in if your existing password does not meet the requirements that the rule defines.

1. From the Cisco SD-WAN Manager menu, choose **Administration > Settings**.
2. Click **Password Policy**.
3. Perform one of these actions, based on your Cisco SD-WAN Manager release:
 - For releases before Cisco vManage Release 20.9.1, click **Enabled**.
 - For releases from Cisco vManage Release 20.9.1 click **Medium Security** or **High Security** to choose the password criteria.

By default, **Password Policy** is set to **Disabled**.

4. Click **Save**.

Password Requirements

Cisco SD-WAN Manager enforces the following password requirements after you have enabled the password policy rules:

- The following password requirements apply to releases before Cisco vManage Release 20.9.1:
 - Must contain a minimum of eight characters, and a maximum of 32 characters.

- Must contain at least one uppercase character.
 - Must contain at least one lowercase character.
 - Must contain at least one numeric character.
 - Must contain at least one of the following special characters: # ? ! @ \$ % ^ & * -.
 - Must not contain the full name or username of the user.
 - Must not reuse a previously used password.
 - Must contain different characters in at least four positions in the password.
- Minimum releases: Cisco IOS XE Catalyst SD-WAN Release 17.9.1a, Cisco vManage Release 20.9.1:

Password Criteria	Requirements
Medium Security	• Must contain a minimum of 8 characters • Must contain no more than 32 characters • Must contain at least 1 lowercase character • Must contain at least 1 uppercase character • Must contain at least 1 numeric character • Must contain at least 1 of the following special characters: # ? ! @ \$ % ^ & * - • Must not be identical to any of the last 5 passwords used • Must not contain the full name or username of the user
High Security	• Must contain a minimum of 15 characters • Must contain no more than 32 characters • Must contain at least 1 lowercase character • Must contain at least 1 uppercase character • Must contain at least 1 numeric character • Must contain at least 1 of the following special characters: # ? ! @ \$ % ^ & * - • Must not be identical to any of the last 5 passwords used • Must not contain the full name or username of the user • Must have at least eight characters that are not in the same position they were in the old password

Password Attempts Allowed

You are allowed five consecutive password attempts before your account is locked. After six failed password attempts, you are locked out for 15 minutes. If you enter an incorrect password on the seventh attempt, you are not allowed to log in, and the 15-minute lock timer starts again.

If your account is locked, wait for 15 minutes for the account to automatically be unlocked. Alternatively, reach out to an administrator to reset the password, or have an administrator unlock your account.



Note Your account gets locked even if no password is entered multiple times. When you do not enter anything in the password field, it is considered as invalid or wrong password.

Password Change Policy



Note You must have enabled password policy rules first for strong passwords to take effect. For more information, see [Enforce Strong Passwords, on page 86](#).

When resetting your password, you must set a new password. You cannot reset a password using an old password.



Note In Cisco vManage Release 20.6.4, Cisco vManage Release 20.9.1 and later releases, a user that is logged out, or a user whose password has been changed locally or on the remote TACACS server cannot log in using their old password. The user can log in only using their new password.

Reset a Locked User

If a user is locked out after multiple password attempts, an administrator with the required rights can update passwords for this user.

There are two ways to unlock a user account, by changing the password or by getting the user account unlocked.



Note Only a **netadmin** user or a user with the User Management Write role can perform this operation.

To reset the password of a user who has been locked out:

1. In **Users (Administration > Manage Users)**, choose the user in the list whose account you want to unlock.
2. Click **...** and choose **Reset Locked User**.
3. Click **OK** to confirm that you want to reset the password of the locked user. Note that this operation cannot be undone.

Alternatively, you can click **Cancel** to cancel the operation.

Reset a Locked User Using the CLI

You can reset a locked user using the CLI as follows:

1. Log in to the device as an `admin` user.
2. Run the following command:

```
Device# request aaa unlock-user username
```
3. When prompted, enter a new password for the user.

Configure User Login Options

Table 46: Feature History

Feature Name	Release Information	Description
Inactivity Lockout	Cisco Catalyst SD-WAN Manager Release 20.12.1	This feature lets you configure Cisco SD-WAN Manager to lock out users who have not logged in for a designated number of consecutive days.
Unsuccessful Login Attempts Lockout	Cisco Catalyst SD-WAN Manager Release 20.12.1	This feature lets you configure Cisco SD-WAN Manager to lock out users who have made a designated number of consecutive unsuccessful login attempts within a designated period.
Duo Multifactor Authentication Support	Cisco Catalyst SD-WAN Manager Release 20.12.1	This feature lets you configure Cisco SD-WAN Manager to require Duo multifactor authentication (MFA) to verify the identity of users before they can log in to Cisco SD-WAN Manager.

Beginning with Cisco Catalyst SD-WAN Manager Release 20.12.1, a `netadmin` user can enable the following Cisco SD-WAN Manager user login features:

Beginning with Cisco IOS XE Catalyst SD-WAN Release 17.12.1a, you can access Cisco SD-WAN Manager with basic privileges even if TACACS user is not mapped to a group. Prior to Cisco IOS XE Catalyst SD-WAN Release 17.12.1a, the access to Cisco SD-WAN Manager was denied.

- Inactivity lockout: You can configure Cisco SD-WAN Manager to lock out users who have not logged in for a designated number of consecutive days. Locked out users cannot log in to Cisco SD-WAN Manager until an administrator unlocks their accounts.

See [Configure Account Lockout](#) , on page 90.

- Unsuccessful login lockout: You can configure Cisco SD-WAN Manager to prevent users who make a designated number of consecutive unsuccessful login attempts within a designated time period from

logging in to Cisco SD-WAN Manager until a configured amount of time passes or an administrator unlocks their user accounts.

By default, Cisco SD-WAN Manager locks out users for 15 minutes after five consecutive unsuccessful login attempts within 15 minutes. After a lockout period expires, a user can log in with the correct user name and password.

See [Configure Unsuccessful Login Attempts Lockout](#), on page 91.

- Duo multifactor authentication: You can configure Cisco SD-WAN Manager to require the use of Duo multifactor authentication to verify identity before users can log in. Users must confirm a login attempt by using Duo multifactor authentication on their mobile devices.

See [Configure Duo Multifactor Authentication](#), on page 93.

Configure Account Lockout

Before You Begin

Beginning with Cisco Catalyst SD-WAN Manager Release 20.12.1, you can configure Cisco SD-WAN Manager to lock out users who have not logged in for a designated number of consecutive days.

Cisco SD-WAN Manager marks locked out users as inactive, and they cannot log in again until an administrator unlocks their accounts in Cisco SD-WAN Manager.



Note To unlock a user account, see [Reset a Locked User](#).

Configure Account Lockout

1. From the Cisco SD-WAN Manager menu, choose **Administration > Settings**.
2. Click **Account Lockout** and enable the **Inactive days before locked out** option.

(In Cisco Catalyst SD-WAN Manager Release 20.12.x, locate the **Account Lockout**, click **Edit**, and enable **Inactive days before locked out**.)

3. Configure the following options:

Field	Description
Inactive days before account locked out	Enable this option and enter the number of consecutive inactive days after which Cisco SD-WAN Manager locks out a user. An inactive day is defined as a day on which a user does not log in to Cisco SD-WAN Manager. Valid values are 2 through 90.

Field	Description
Number of failed login attempts before lockout	Enter the number of failed login attempts after which Cisco SD-WAN Manager locks out a user. Possible values: 1 through 3600 Default: 3600
Duration within which the failed attempts are counted (minutes)	Enter the period, in minutes, during which the system counts consecutive unsuccessful login attempts. For example, if you set this period to 10 minutes, and set the number of failed login attempts before lockout to 5, Cisco SD-WAN Manager locks out a user if the user makes 5 consecutive unsuccessful login attempts within 10 minutes. Possible values: 1 through 60 Default: 60
Cooldown or Lockout period	This option controls whether Cisco SD-WAN Manager automatically resets a user who is locked because of unsuccessful login attempts. This option is enabled by default. If you disable it, an administrator must manually unlocks the account of a locked-out user. a. Click Enabled adjacent to Cooldown or Lockout period. b. In the Lockout Interval (minutes) field, enter the number of minutes after which Cisco SD-WAN Manager automatically resets a locked out user. Possible values: 1 through 60 Default: 15

- Click **Save**.

Configure Unsuccessful Login Attempts Lockout

Before You Begin

Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.12.1



Note From Cisco Catalyst SD-WAN Manager Release 20.13.1 or later, use the procedure described in [Configure Account Lockout](#), on page 90.

You can configure Cisco SD-WAN Manager to lock out users who have made a designated number of consecutive unsuccessful login attempts within a period of time.

Cisco SD-WAN Manager prevents locked out users from logging in again until a configured amount of time has passed or an administrator unlocks their accounts in Cisco SD-WAN Manager.



Note To unlock a user account, see [Reset a Locked User](#).

Configure Unsuccessful Login Attempts Lockout

1. From the Cisco SD-WAN Manager menu, choose **Administration > Settings**.
2. Click **Account Lockout**
3. In the **Lockout on failed login attempts** row, click **Edit**.
4. Configure the following options:

Field	Description
Number of failed login attempts before lockout	Enter the number of failed login attempts after which Cisco SD-WAN Manager locks out a user. Possible values: 1 through 3600 Default: 3600
Duration within which the failed attempts are counted (minutes)	Enter the period, in minutes, during which the system counts consecutive unsuccessful login attempts. For example, if you set this period to 10 minutes, and set the number of failed login attempts before lockout to 5, Cisco SD-WAN Manager locks out a user if the user makes 5 consecutive unsuccessful login attempts within 10 minutes. Possible values: 1 through 60 Default: 60

Field	Description
Cooldown or Lockout period	This option controls whether Cisco SD-WAN Manager automatically resets a user who is locked because of unsuccessful login attempts. This option is enabled by default. If you disable it, an administrator must manually unlocks the account of a locked-out user. a. Click Enabled adjacent to Cooldown or Lockout period. b. In the Lockout Interval (minutes) field, enter the number of minutes after which Cisco SD-WAN Manager automatically resets a locked out user. Possible values: 1 through 60 Default: 15

5. Click **Save**.

Configure Duo Multifactor Authentication

Beginning with Cisco Catalyst SD-WAN Manager Release 20.12.1, you can configure Cisco SD-WAN Manager to require Duo multifactor authentication (MFA) to verify the identity of users before they can log in to Cisco SD-WAN Manager and other controllers. When you configure this feature, users are prompted on their mobile devices to authenticate with Duo after they enter a username and password and click **Log In** on the Cisco SD-WAN Manager **Login** screen.

This feature requires that you have a Duo account with local users created on that account.



Note

- Duo MFA does not apply to the admin user by default. To enable Duo MFA for the admin user, enable the **DUO MFA Configuration** option, and then enter the `admin-auth-order` command from the CLI.
- Users do not see a message in Cisco SD-WAN Manager that an MFA request has been sent to a mobile device.

1. Log into the Duo Admin Panel.
2. Create an Auth API application.
This provides the Duo integration key, secret key, and API hostname information required to complete Duo MFA configuration. Read more about [Duo Auth API](#).
3. From the Cisco SD-WAN Manager menu, choose **Administration** > **Settings**.
4. Click **DUO MFA Configuration**. (If you are using Cisco Catalyst SD-WAN Manager Release 20.12.x or earlier, click **Edit**.)

5. Click **Enabled**.
6. Configure the following options:

Field	Description
Integration Key	Enter the integration key (Ikey) for your Duo account.
Secret Key	Enter the secret key (Skey) for your Duo account.
API Hostname	Enter the API hostname (api-hostname) for your Duo account.
Server proxy	(Read only) Shows the server proxy that is used to access the Duo server if Cisco SD-WAN Manager is behind a firewall. Set this server proxy with the system http proxy or the system https proxy command. Note If Cisco SD-WAN Manager is deployed on a cloud that can be reached by an external network, a server proxy should not be set.

7. Click **Save**.
8. If a Cisco SD-WAN Validator or a Cisco SD-WAN Controller does not have internet access, use the following commands in the CLI or the device template of the device to provide access to the Duo MFA feature.

These commands configure the device with proxy information about the device on which Duo MFA is enabled.

```
vm# config
vm(config)# system aaa
vm(config-aaa)# multi-factor-auth
vm(config-multi-factor-auth)# duo
vm(config-duo)# api-hostname name
vm(config-duo)# secret-key key
vm(config-duo)# integration-key key
vm(config-duo)# proxy proxy_url
vm(config-duo)# commit
```

Manage Users

From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users** to add, edit, view, or delete users and user groups.

Please note the following:

- Only a user logged in as the **admin** user or a user who has Manage Users write permission can add, edit, or delete users and user groups from Cisco SD-WAN Manager.
- Each user group can have read or write permission for the features listed in this section. Write permission includes Read permission.

- All user groups, regardless of the read or write permissions selected, can view the information displayed in the Cisco SD-WAN Manager Dashboard.

Table 47: User Group Permissions for Different Device Types

Permissions	See This Section
User group permissions related to Cisco IOS XE Catalyst SD-WAN device configuration.	User Group Permissions: Cisco IOS XE Catalyst SD-WAN Devices
User group permissions related to Cisco Catalyst Wireless Gateway device configuration.	User Group Permissions: Cisco Catalyst Wireless Gateway Devices

Configure Users Using CLI

You can use the CLI to configure user credentials on each device. This way, you can create additional users and give them access to specific devices. The credentials that you create for a user by using the CLI can be different from the Cisco SD-WAN Manager credentials for the user. In addition, you can create different credentials for a user on each device. All Cisco IOS XE Catalyst SD-WAN device users with the **netadmin** privilege can create a new user.

To create a user account, configure the username and password, and place the user in a group:

This example, shows the addition of user, Bob, to an existing group:

```
Device(config)# system aaa user bob group basic
```

This example, shows the addition of user, Alice, to a new group **test-group**:

```
Device(config)# system aaa user test-group
Device(config)# system aaa user alice group test-group
```

The Username can be 1 to 128 characters long, and it must start with a letter. The name can contain only lowercase letters, the digits 0 through 9, hyphens (-), underscores (_), and periods (.). The name cannot contain any uppercase letters. Because some usernames are reserved, you cannot configure them. For a list of reserved usernames, see the **aaa** configuration command in the Cisco Catalyst SD-WAN Command Reference Guide.

The Password is the password for a user. Each username must have a password, and users are allowed to change their own password. The CLI immediately encrypts the string and does not display a readable version of the password. When a user logs in to a Cisco IOS XE Catalyst SD-WAN device, they have five chances to enter the correct password. After the fifth incorrect attempt, the user is locked out of the device, and must wait for 15 minutes before attempting to log in again.



Note Enclose any user passwords that contain the special character ! in double quotation marks (“ ”). If a double quotation is not included for the entire password, the config database (?) treats the special character as a space and ignores the rest of the password.

For example, if the password is C!sc0, use “C!sc0”.

Group name is the name of a standard Cisco Catalyst SD-WAN group (**basic**, **netadmin**, or **operator**) or of a group configured with the **usergroup** command (discussed below). If an admin user changes the permission

of a user by changing their group, and if that user is currently logged in to the device, the user is logged out and must log back in again.

The factory-default password for the admin username is admin. We strongly recommend that you modify this password the first time you configure a Cisco IOS XE Catalyst SD-WAN device:

```
Device(config)# username admin password
$9$3/IL3/UF2F2F3E$J9NKBekLWrq9ExmHk6F5VAiDMOFQfD.QPAmMxDdxz.c
```

Configure the password as an ASCII string. The CLI immediately encrypts the string and does not display a readable version of the password, for example:

```
Device# show run | sec username
username admin privilege 15 secret 9
$9$3F2M212G2/UM3U$TGe2kqoIibdIRDEj4cOVKbVFP/o4vnlFAwWnmzx1rRE
username appnav privilege 15 secret 9
$9$312L2V.F2VIM1k$P3MBAyBtGxKf/yBGnUSHQ1g/ae1QhfIbieg28buJJGI
username eft secret 9 $9$3FMJ3/UD2VEL2E$d.ke4.an41v7wEhrQc6k5wIfE9M9WkNAJxUvbbempS.
username lab privilege 15 secret 9
$9$31.J3FUD2F.E2.$/AiVn9PmLCpgr6ExVrE7dH979Wu8nbdAfbzUtfysg.
username test secret 9 $9$112J316D3/QL3k$7PZOXJAJOI1os5UI763G3XcpVhX1qcwJ.qEmgmX4X9g
username vbongair privilege 15 secret 9
$9$3/2K2UwF21QF3U$VbdQ5bq18590rRthF/NnNnOsw.dw1/EViMTFZ5.ctus
Device#
```

If you are using RADIUS to perform AAA authentication, you can configure a specific RADIUS server to verify the password:

```
Device(config)# radius server tag
```

The tag is a string that you defined with the **radius server tag** command, as described in the Cisco Catalyst SD-WAN Command Reference Guide.

Manage a User Group

Users are placed in groups, which define the specific configuration and operational commands that the users are authorized to view and modify. A single user can be in one or more groups. Cisco Catalyst SD-WAN software provides standard user groups, and you can create custom user groups, as needed:

- **basic**: Includes users who have permission to view interface and system information.
- **netadmin**: Includes the admin user, by default, who can perform all operations on the Cisco SD-WAN Manager. You can add other users to this group.
- **operator**: Includes users who have permission only to view information.
- Minimum supported release: Cisco vManage Release 20.9.1
- **network_operations**: Includes users who can perform non-security operations on Cisco SD-WAN Manager, such as viewing and modifying non-security policies, attaching and detaching device templates, and monitoring non-security data.
- Minimum supported release: Cisco vManage Release 20.9.1
- **security_operations**: Includes users who can perform security operations on Cisco SD-WAN Manager, such as viewing and modifying security policies, and monitoring security data.

Note: All user groups, regardless of the read or write permissions selected, can view the information displayed on the Cisco SD-WAN Manager Dashboard screen.

Delete a User Group

You can delete a user group when it is no longer needed. For example, you might delete a user group that you created for a specific project when that project ends.

1. From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users**.
2. Click **User Groups**.
3. Click the name of the user group you wish to delete.



Note You cannot delete any of the default user groups—basic, netadmin, operator, network_operations, and security_operations.

4. Click **Trash** icon.
5. To confirm the deletion of the user group, click **OK**.

Edit User Group Privileges

You can edit group privileges for an existing user group. This procedure lets you change configured feature read and write permissions for the user group needed.

1. From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users**.
2. Click **User Groups**.
3. Select the name of the user group whose privileges you wish to edit.



Note You cannot edit privileges for the any of the default user groups—basic, netadmin, operator, network_operations, and security_operations.

4. Click **Edit**, and edit privileges as needed.
5. Click **Save**.

If an **admin** user changes the privileges of a user by changing their group, and if that user is currently logged in to the device, the user is logged out and must log back in again.

Creating Groups Using CLI

The Cisco Catalyst SD-WAN software provides default user groups: **basic**, **netadmin**, **operator**, **network_operations**, and **security_operations**. The username **admin** is automatically placed in the **netadmin** usergroup.

If needed, you can create additional custom groups and configure privilege roles that the group members have. To create a custom group with specific authorization, configure the group name and privileges:

```
Device(config)# aaa authentication login user1 group radius enable
Device(config)# aaa authentication login user2 group radius enable
```

```
Device(config)# aaa authentication login user3 group radius enable
Device(config)#
```

group-name can be 1 to 128 characters long, and it must start with a letter. The name can contain only lowercase letters, the digits 0 through 9, hyphens (-), underscores (_), and periods (.). The name cannot contain any uppercase letters. Some group names are reserved, so you cannot configure them. For a list of them, see the `aaa` configuration command.

If a remote RADIUS or TACACS+ server validates authentication but does not specify a user group, the user is placed into the user group **basic**. If a remote server validates authentication and specifies a user group (say, X) using VSA Cisco SD-WAN-Group-Name, the user is placed into that user group only. However, if that user is also configured locally and belongs to a user group (say, Y), the user is placed into both the groups (X and Y).

In the **task** option, list the privilege roles that the group members have. The role can be one or more of the following: **interface**, **policy**, **routing**, **security**, and **system**.

Ciscotac User Access

The Cisco Edge software provides two users—**ciscotacro** and **ciscotacrw**—that are for use only by the Cisco Support team. These users are available for both cloud and on-premises installations. They operate on a consent-token challenge and token response authentication in which a new token is required for every new login session. The **ciscotacro** and **ciscotacrw** users can use this token to log in to Cisco SD-WAN Manager web server as well as the SSH Terminal on Cisco SD-WAN Manager. These users can also access Cisco SD-WAN Validator, Cisco SD-WAN Controllers, and Cisco vEdge devices using the SSH Terminal on Cisco SD-WAN Manager.

The default CLI templates include the **ciscotacro** and **ciscotacrw** user configuration. These users are enabled by default. However, a customer can disable these users, if needed.

- **ciscotacro User:** This user is part of the operator user group with only read-only privileges. This user can only monitor a configuration but cannot perform any operation that will modify the configuration of the network.
- **ciscotacrw User:** This user is part of the netadmin user group with read-write privileges. This user can modify a network configuration. In addition, only this user can access the root shell using a consent token.

Use the **tools consent-token** command to authenticate the network administrator of an organization to access system shell. Starting Cisco Catalyst SD-WAN Control Components Release 20.12.x, the **request support ciscotac** command is deprecated.

Limitations

- Only 16 concurrent sessions are supported for the **ciscotacro** and **ciscotacrw** users.
- The session duration is restricted to four hours. It is not configurable.
- The inactivity timer functionality closes user sessions that have been idle for a specified period of time. This feature is enabled by default and the timeout value is 30 minutes. However, the user configuration includes the option of extending the inactivity timer.
- A customer can remove these two users. If removed, the customer can open a case and share temporary login credentials or share the screen with the Cisco Support team for troubleshooting an issue.

Configure Sessions in Cisco SD-WAN Manager

Table 48: Feature History

Feature History	Release Information	Description
Configure Sessions in Cisco SD-WAN Manager	Cisco IOS XE Catalyst SD-WAN Release 17.3.1a Cisco vManage Release 20.3.1	This feature lets you see all the HTTP sessions that are open within Cisco SD-WAN Manager. It gives you details about the username, source IP address, domain of the user, and other information. A user with User Management Write access, or a netadmin user can trigger a log out of any suspicious user's session.

Set a Client Session Timeout in Cisco SD-WAN Manager

You can set a client session timeout in Cisco SD-WAN Manager. When a timeout is set, such as no keyboard or keystroke activity, the client is automatically logged out of the system.



Note You can edit Client Session Timeout in a multitenant environment only if you have a Provider access.

1. From the Cisco SD-WAN Manager menu, choose **Administration** > **Settings**.
2. Click **User Sessions**.
3. Under **Client Session Timeout**, click **Session Timeout**.
4. Specify the timeout value, in minutes.
5. Click **Save**.

Set a Session Lifetime in Cisco SD-WAN Manager

You can specify how long to keep your session active by setting the session lifetime, in minutes. A session lifetime indicates the amount of time for which a session can be active. If you keep a session active without letting the session expire, you will be logged out of the session in 24 hours, which is the default session timeout value.

The default session lifetime is 1440 minutes or 24 hours.



Note You can edit Session Lifetime in a multitenant environment only if you have a Provider access.

1. From the Cisco SD-WAN Manager menu, choose **Administration** > **Settings**.

2. Click **User Sessions**.
3. In the **SessionLifeTime Timeout (minutes)** field, specify the session timeout value, in minutes, from the drop-down list.
4. Click **Save**.

Set the Server Session Timeout in Cisco SD-WAN Manager

You can configure the server session timeout in Cisco SD-WAN Manager. The server session timeout indicates how long the server should keep a session running before it expires due to inactivity. The default server session timeout is 30 minutes.



Note Server Session Timeout is not available in a multitenant environment even if you have a Provider access or a Tenant access.

1. From the Cisco SD-WAN Manager menu, choose **Administration > Settings**.
2. Click **User Sessions**.
3. In **Server Session Timeout Timeout(minutes)** field, specify the timeout value, in minutes.
4. Click **Save**.

Set the maximum sessions per user role

You can configure the maximum number of concurrent login sessions for each configured user role. This maximum value applies to all users assigned to that role.

For example, if the “Max user sessions” value for the netadmin role is set to 3, then each user assigned to the netadmin role can have up to 3 concurrent login sessions across the platform. If a fourth session is initiated by one of those users, an error message appears.

Range for “Max user sessions”: 1 to 255. Default: If undefined, there is no limit on the number of concurrent sessions for that role. For government installations, recommended values are 3 sessions for administrative roles and 2 sessions for conventional roles.

The value also applies to CLI sessions.

Change maximum sessions value for a user role

1. From the Cisco SD-WAN Manager menu, choose **Administration > Users and Access > Roles**.
The “Max user sessions value” is displayed for each defined role, including both default and custom roles.
2. Click the role name, edit the **Max user sessions** value, and click **Update**.



Note As of Release 20.18.1, the previous method for changing this value in **Settings > Trust and Privacy** is no longer supported. If a “Max user sessions” value has been defined on your system by that method, that value serves as the maximum value for all user roles until changed.

Configuring RADIUS Authentication Using CLI

The Remote Authentication Dial-In User Service (RADIUS) is a distributed client/server system that secures networks against unauthorized access. RADIUS clients run on supported Cisco devices and send authentication requests to a central RADIUS server, which contains all user authentication and network service access information.

To have a Cisco IOS XE Catalyst SD-WAN device use RADIUS servers for user authentication, configure one or up to 8 servers:

```
Device(config)# transaction
Device(config)# radius server test address ipv4 10.1.1.55 acct-port 110
Device(config-radius-server)# key 33
Device(config-radius-server)# exit
Device(config)# radius server test address ipv4 10.1.1.55 auth-port 330
Device(config-radius-server)# key 55
Device(config-radius-server)#
```

For each RADIUS server, you must configure, at a minimum, its IP address and a password, or key. You can specify the key as a clear text string up to 31 characters long or as an AES 128-bit encrypted key. The local device passes the key to the RADIUS server. The password must match the one used on the server. To configure more than one RADIUS server, include the **server** and **secret-key** commands for each server.

The remaining RADIUS configuration parameters are optional.

To set the priority of a RADIUS server, as a means of choosing or load balancing among multiple RADIUS servers, set a priority value for the server. The priority can be a value from 0 through 7. A server with a lower priority number is given priority over one with a higher number.

By default, the Cisco IOS XE Catalyst SD-WAN device uses port 1812 for authentication connections to the RADIUS server and port 1813 for accounting connections. To change these port numbers, use the **auth-port** and **acct-port** commands.

If the RADIUS server is reachable via a specific interface, configure that interface with the **source-interface** command.

You can tag RADIUS servers so that a specific server or servers can be used for AAA, IEEE 802.1X, and IEEE 802.11i authentication and accounting. Define the tag here, with a string from 4 to 16 characters long. Then associate the tag with the **radius-servers** command when you configure AAA, and when you configure interfaces for 802.1X and 802.11i.

If the RADIUS server is located in a different VPN from the Cisco IOS XE Catalyst SD-WAN device, configure the server's VPN number so that the Cisco IOS XE Catalyst SD-WAN device can locate it. If you configure multiple RADIUS servers, they must all be in the same VPN.

When waiting for a reply from the RADIUS server, a Cisco IOS XE Catalyst SD-WAN device waits 3 seconds before retransmitting its request. To change this time interval, use the **timeout** command, setting a value from 1 to 1000 seconds:

```
Device# config-transaction
Device(config)# aaa group server radius server-10.99.144.201
Device(config-sg-radius)# server-private 10.99.144.201 auth-port 1812 timeout 5 retransmit
3
```

Configure SSH Authentication

Table 49: Feature History

Feature Name	Release Information	Description
Secure Shell Authentication Using RSA Keys	Cisco IOS XE Catalyst SD-WAN Release 16.12.1b	This feature helps configure RSA keys by securing communication between a client and a Cisco Catalyst SD-WAN server.

The Secure Shell (SSH) protocol provides secure remote access connection to network devices.

SSH supports user authentication using public and private keys. To enable SSH authentication, public keys of the users are stored in the home directory of authenticating user in the following location:

```
~<user>/.ssh/authorized_keys
```

A new key is generated on the client machine which owns the private-key. Any message encrypted using the public key of the SSH server is decrypted using the private key of the client.

Restrictions for SSH Authentication on Cisco Catalyst SD-WAN

- The range of SSH RSA key size supported by Cisco IOS XE Catalyst SD-WAN devices is from 2048 to 4096. SSH RSA key size of 1024 and 8192 are not supported.
- A maximum of two keys per user are allowed on Cisco IOS XE Catalyst SD-WAN devices.

SSH Authentication using Cisco SD-WAN Manager on Cisco IOS XE Catalyst SD-WAN Devices

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**, and click **Add Template**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

3. From **Select Devices**, select the type of device for which you are creating the template.
4. From **Basic Information**, choose **CISCO AAA** template.
5. From **Local**, click **New User** and enter the details.
6. Enter **SSH RSA Key**.



Note You must enter the complete public key from the `id_rsa.pub` file in **SSH RSA Key**.

Configure SSH Authentication using CLI on Cisco IOS XE Catalyst SD-WAN Devices

SSH key based login is supported on IOS. Per user a maximum of 2 keys can be supported. Also, IOS only supports RSA based keys.

Traditional IOS CLI, allow support for:

- Key-string
- Key-hash – The key-string is base64 decoded and MD5 hash is run on it.

However, the transaction yang model has provision to only copy the key-hash (instead of the entire key-string). Cisco SD-WAN Manager does this conversion and pushes the configuration to the device.

Public Keys supported on Cisco IOS XE Catalyst SD-WAN Devices

- SSH-RSA

Configure the Authentication Order

The authentication order dictates the order in which authentication methods are tried when verifying user access to a Cisco IOS XE Catalyst SD-WAN device through an SSH session or a console port. The default authentication order is **local**, then **radius**, and then **tacacs**. With the default authentication order, the authentication process occurs in the following sequence:

- The authentication process first checks whether a username and matching password are present in the running configuration on the local device.
- If the RADIUS server is unreachable (or all the servers are unreachable), the authentication process checks the TACACS+ server. For this method to work, you must configure one or more TACACS+ servers with the **system tacacs server** command. If a TACACS+ server is reachable, the user is authenticated or denied access based on that server's TACACS+ database. If a TACACS+ server is unreachable and if you have configured multiple TACACS+ servers, the authentication process checks each server sequentially, stopping when it is able to reach one of them. The user is then authenticated or denied access based on that server's TACACS+ database.
- If the TACACS+ server is unreachable (or all TACACS+ servers are unreachable), user access to the local Cisco IOS XE Catalyst SD-WAN device is denied.

To modify the default order, use the **auth-order** command:

Specify one, two, or three authentication methods in the preferred order, starting with the one to be tried first. If you configure only one authentication method, it must be **local**.

To have the "admin" user use the authentication order configured in the **auth-order** command, use the following command:

```
Device(config-system-aaa) # admin-auth-order
```

If you do not include this command, the "admin" user is always authenticated locally.

You can configure authentication to fall back to a secondary or tertiary authentication mechanism when the higher-priority authentication method fails to authenticate a user, either because the user has entered invalid credentials or because the authentication server is unreachable (or all the servers are unreachable):

- If the authentication order is configured as **radius local**:
 - With the default authentication, local authentication is used only when all RADIUS servers are unreachable. If an authentication attempt via a RADIUS server fails, the user is not allowed to log in even if they have provided the correct credentials for local authentication.
- If the authentication order is configured as **local radius**:
 - With the default authentication, RADIUS authentication is tried when a username and matching password are not present in the running configuration on the local device.
- If the authentication order is configured as **radius tacacs local**:
 - With the default authentication, TACACS+ is tried only when all RADIUS servers are unreachable, and local authentication is tried only when all TACACS+ servers are unreachable. If an authentication attempt via a RADIUS server fails, the user is not allowed to log in even if they have provided the correct credentials for the TACACS+ server. Similarly, if a TACACS+ server denies access, the user cannot log via local authentication.

If a remote server validates authentication but does not specify a user group, the user is placed into the user group **basic**.

If a remote server validates authentication and specifies a user group (say, X), the user is placed into that user group only. However, if that user is also configured locally and belongs to a user group (say, Y), the user is placed into both the groups (X and Y).

If a remote server validates authentication and that user is not configured locally, the user is logged in to the vshell as the user **basic**, with a home directory of /home/basic.

If a remote server validates authentication and that user is configured locally, the user is logged in to the vshell under their local username (say, eve) with a home direction of /home/username (so, /home/eve).



Note Tags are used for grouping, describing, or finding devices. You can tag RADIUS and TACAC servers for authentication and accounting. You can add more than one tag to a device. Starting from Cisco vManage Release 20.9.1, following new tags are used in authentication:

- Viptela-User-Group: for user group definitions instead of Viptela-Group-Name.
 - Viptela-Resource-Group: for resource group definitions.
-

The authentication fails if there is any space between keys and the values. For example, **key=value**.

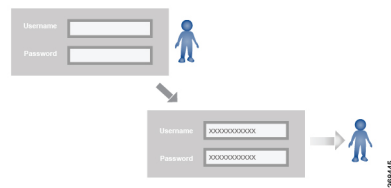
Role-Based Access with AAA

The Cisco Catalyst SD-WAN AAA software implements role-based access to control the authorization permissions for users on Cisco IOS XE Catalyst SD-WAN devices. Role-based access consists of three components:

- Users are those who are allowed to log in to a Cisco IOS XE Catalyst SD-WAN device.
- User groups are collections of users.
- Privileges are associated with each group. They define the commands that the group's users are authorized to issue.

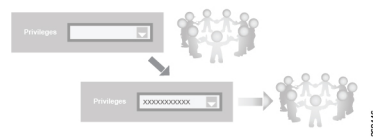
Users and User Groups

All users who are permitted to perform operations on a Cisco IOS XE Catalyst SD-WAN device must have a login account. For the login account, you configure a username and a password on the device itself. These allow the user to log in to that device. A username and password must be configured on each device that a user is allowed to access.

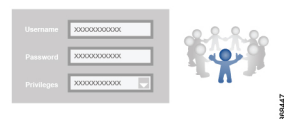


The Cisco Catalyst SD-WAN software provides one standard username, **admin**, which is a user who has full administrative privileges, similar to a UNIX superuser. By default, the **admin** username password is **admin**. You cannot delete or modify this username, but you can and should change the default password.

User groups pool together users who have common roles, or privileges, on the Cisco IOS XE Catalyst SD-WAN device. As part of configuring the login account information, you specify which user group or groups that user is a member of. You do not need to specify a group for the **admin** user, because this user is automatically in the user group **netadmin** and is permitted to perform all operations on the Cisco IOS XE Catalyst SD-WAN device.



The user group itself is where you configure the privileges associated with that group. These privileges correspond to the specific commands that the user is permitted to execute, effectively defining the role-based access to the Cisco Catalyst SD-WAN software elements.



The Cisco Catalyst SD-WAN software provides the following standard user groups:

- **basic**: The basic group is a configurable group and can be used for any users and privilege levels. This group is designed to include users who have permission to both view and modify information on the device.
- **operator**: The operator group is also a configurable group and can be used for any users and privilege levels. This group is designed to include users who have permission only to view information.
- **netadmin**: The netadmin group is a non-configurable group. By default, this group includes the **admin** user. You can add other users to this group. Users in this group are permitted to perform all operations on the device.
- Minimum supported release: Cisco vManage Release 20.9.1

network_operations: The **network_operations** group is a non-configurable group. Users in this group can perform all non-security-policy operations on the device and only view security policy information. For example, users can create or modify template configurations, manage disaster recovery, and create non-security policies such as application aware routing policy or CFlowD policy.

- Minimum supported release: Cisco vManage Release 20.9.1

security_operations: The **security_operations** group is a non-configurable group. Users in this group can perform all security operations on the device and only view non-security-policy information. For example, users can manage umbrella keys, licensing, IPS signatures auto update, TLS/SSL proxy settings, and so on.

Users of the **network_operations** group are authorized to apply policies to a device, revoke applied policies, and edit device templates. Users of the **security_operations** group require **network_operations** users to intervene on day-0 to deploy security policy on a device and on day-N to remove a deployed security policy. However, after a security policy is deployed on a device, **security_operations** users can modify the security policy without needing the **network_operations** users to intervene.



Note Only admin users can view running and local configuration. Users associated with predefined operator user group do not have access to the running and local configurations. The predefined user group operator has only read access for the template configuration. If you need only a subset of admin user privileges, then you need to create a new user group with the selected features from the features list with both read and write access and associate the group with the custom user.

Privileges for Role-Based Access

Role-based access privileges are arranged into five categories, which are called *tasks*:

- **Interface**—Privileges for controlling the interfaces on the Cisco IOS XE Catalyst SD-WAN device.
- **Policy**—Privileges for controlling control plane policy, OMP, and data plane policy.
- **Routing**—Privileges for controlling the routing protocols, including BFD, BGP, OMP, and OSPF.
- **Security**—Privileges for controlling the security of the device, including installing software and certificates. Only users belonging to the **netadmin** group can install software on the system.
- **System**—General systemwide privileges.

The tables in the following sections detail the AAA authorization rules for users and user groups. These authorization rules apply to commands issued from the CLI and to those issued from Netconf.

User Authorization Rules for Operational Commands

The user authorization rules for operational commands are based simply on the username. Any user who is allowed to log in to the Cisco IOS XE Catalyst SD-WAN device can execute most operational commands. However, only the **admin** user can issue commands that affect the fundamental operation of the device, such as installing and upgrading the software and shutting down the device.

Note that any user can issue the **config** command to enter configuration mode, and once in configuration mode, they are allowed to issue any general configuration command. Also, any user is allowed to configure their password by issuing the **system aaa user self password password** command and then committing that configuration change. For the actual commands that configure device operation, authorization is defined according to user group membership. See User Group Authorization Rules for Configuration Commands.

The following tables lists the AAA authorization rules for general CLI commands. All the commands are operational commands except as noted. Also, some commands available to the "admin" user are available only if that user is in the "netadmin" user group.

CLI Command	Any User	Admin User
clear history	X	X
commit confirm	X	X
complete-on-space	X	X
config	X	X
exit	X	X
file	X	X
help	X	X
[no] history	X	X
idle-timeout	X	X
job	X	X
logout	—	X (users in netadmin group only)
monitor	X	X
nslookup	X	X
paginate	X	X
ping	X (users in netadmin group only)	X (users in netadmin group only)
poweroff	—	X(users in netadmin group only)
prompt1	X	X
prompt2	X	X
quit	X	X

CLI Command	Any User	Admin User
reboot	—	X (users in netadmin group only)
request aaa request admin-tech request firmware request interface-reset request nms request reset request software	—	X (users in netadmin group only)
request execute request download request upload	X	X
request (everything else)	—	X
rollback (configuration mode command)	—	X (users in netadmin group only)
screen-length	X	X
screen-width	X	X
show cli	X	X
show configuration commit list	X	X
show history	X	X
show jobs	X	X
show parser dump	X	X
show running-config	X	X
show users	X	X
system aaa user <i>self</i> password <i>password</i> (configuration mode command) (Note: A user cannot delete themselves)		
tcpdump	X	X
timestamp	X	X
tools ip-route	X	X
tools netstat	X	X
tools nping	X	X
traceroute	X	X

CLI Command	Any User	Admin User
vshell	X (The availability of vshell command is unavailable to all users that are not in netadmin group in Cisco vManage Release 20.9.5.)	X (The vshell AAA authorized access is limited only to users that are in netadmin group.)

User Group Authorization Rules for Operational Commands

The following table lists the user group authorization roles for operational commands.

Operational Command	Interface	Policy	Routing	Security	System
clear app		X			
clear app-route		X			
clear arp	X				
clear bfd			X		X
clear bgp			X		X
clear bridge	X				
clear cellular	X				
clear control				X	
clear crash					X
clear dhcp					X
clear dns					X
clear igmp			X		
clear installed-certificates				X	
clear interface	X				
clear ip			X		
clear notification					X
clear omp			X		
clear orchestrator				X	
clear ospf			X		
clear pim			X		

Operational Command	Interface	Policy	Routing	Security	System
clear policy		X			
clear pppoe	X				
clear system					X
clear tunnel				X	
clear wlan	X				
clear ztp				X	X
clock					X
debug bgp			X		
debug cellular	X				
debug cflowd		X			
debug chmgr					X
debug config-mgr					X
debug dhcp-client					X
debug dhcp-helper					X
debug dhcp-server					X
debug fpm		X			
debug ftm					X
debug igmp			X		
debug netconf					X
debug omp			X		
debug ospf			X		
debug pim			X		
debug resolver			X		
debug snmp					X
debug sysmgr					X
debug transport					X
debug ttm					X
debug vdaemon				X	X

Operational Command	Interface	Policy	Routing	Security	System
debug vrrp				X	
debug wlan	X				
request certificate				X	
request control-tunnel				X	
request controller				X	
request controller-upload				X	
request csr				X	
request device				X	
request device-upload				X	
request on-vbond-controller				X	
request port-hop				X	
request root-cert-chain				X	
request security				X	
request vedge				X	
request vedge-upload				X	
request vsmart-upload				X	
show aaa					X
show app		X			
show app-route		X			
show arp	X				
show bfd			X		X
show bgp			X		
show boot-partition					X
show bridge	X				
show cellular	X				
show certificate				X	
show clock					X

Operational Command	Interface	Policy	Routing	Security	System
show control				X	X
show crash					X
show debugs—same as debug commands					
show dhcp					X
show external-nat				X	X
show hardware					X
show igmp			X		
show interface	X				
show ip			X		X
show ipsec				X	
show licenses					X
show logging					X
show multicast			X		
show nms-server					X
show notification					X
show ntp					X
show omp		X	X		X
show orchestrator				X	
show ospf			X		
show pim			X		
show policer		X			
show policy		X			
show ppp	X				
show pppoe	X				
show reboot					X
show security-info				X	
show software					X

Operational Command	Interface	Policy	Routing	Security	System
show system					X
show transport					X
show tunnel				X	
show uptime					X
show users					X
show version					X
show vrrp	X				
show wlan	X				
show ztp				X	

User Group Authorization Rules for Configuration Commands

The following table lists the user group authorization rules for configuration commands.

Configuration Command	Interface	Policy	Routing	Security	System
apply-policy		X			
banner					X
bfd			X		X
bridge	X				
omp		X	X		X
policy		X			
security				X	X
snmp					X
system					X
vpn interface	X				
vpn ip			X		
vpn router			X		
vpn service			X		
vpn (everything else, including creating, deleting, and naming)					X

Configuration Command	Interface	Policy	Routing	Security	System
wlan	X				

Configure AAA Using a Configuration Group

Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

Step 1 From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

Step 2 Create and configure a AAA feature in a System profile.

- a. Configure users.

Table 50: Local

Field	Description
Enable AAA Authentication	Enable authentication parameters.
Accounting Group	Enable accounting parameters.
Add AAA User	
Name	Enter a name for the user. It can be 1 to 128 characters long, and it must start with a letter. The name can contain only lowercase letters, the digits 0 through 9, hyphens (-), underscores (_), and periods (.). The name cannot contain any uppercase letters. The following usernames are reserved, so you cannot configure them: backup, basic, bin, daemon, games, gnats, irc, list, lp, mail, man, news, nobody, proxy, quagga, root, sshd, sync, sys, uucp, and www-data. Also, names that start with viptela-reserved are reserved.
Password	Enter a password for the user. The password is an MD5 digest string, and it can contain any characters, including tabs, carriage returns, and linefeeds. For more information, see Section 9.4 in RFC 7950, The YANG 1.1 Data Modeling Language. Each username must have a password. Users are allowed to change their own passwords. The default password for the admin user is admin. We strongly recommended that you change this password.
Confirm Password	Re-enter the password for the user.

Field	Description
Privilege	Select between privilege level 1 or 15. • Level 1: User EXEC mode. Read-only, and access to limited commands, such as the ping command. • Level 15: Privileged EXEC mode. Full access to all commands, such as the reload command, and the ability to make configuration changes. By default, the EXEC commands at privilege level 15 are a superset of those available at privilege level 1.
Add Public Key Chain	
Key String*	Enter the authentication string for a key.
Key Type	Choose ssh-rsa .

b. Configure RADIUS servers.

Table 51: RADIUS

Field	Description
Address*	Enter the IP address of the RADIUS server host.
Acct Port	Enter the UDP port to use to send 802.1X and 802.11i accounting information to the RADIUS server. Range: 1 - 65534. Default: 1813
Auth Port	Enter the UDP destination port to use for authentication requests to the RADIUS server. Default: 1812 Range: 1 - 65534
Retransmit	Enter the number of times the device transmits each RADIUS request to the server before giving up. Default: 3 Range: 0 - 100
Timeout	Enter the number of seconds a device waits for a reply to a RADIUS request before retransmitting the request. Default: 5 seconds Range: 1 through 1000
Key*	Enter the key the Cisco IOS XE Catalyst SD-WAN device passes to the RADIUS server for authentication and encryption.
Key Type	Choose Protected Access Credential (PAC) key.

c. Configure TACACS servers.

Table 52: TACACS Server

Field	Description
Address*	Enter the IP address of the TACACS+ server host.
Port	Enter the UDP destination port to use for authentication requests to the TACACS+ server. If the server is not used for authentication, configure the port number to be 0. Default: 49
Timeout	Enter the number of seconds a device waits for a reply to a TACACS+ request before retransmitting the request. Default: 5 seconds Range: 1 through 1000
Key*	Enter the key the Cisco IOS XE Catalyst SD-WAN device passes to the TACACS+ server for authentication and encryption. You can type the key as a text string from 1 to 31 characters long, and it is immediately encrypted, or you can type an AES 128-bit encrypted key. The key must match the AES encryption key used on the TACACS+ server.

- d. Configure accounting rules.

Table 53: Accounting

Field	Description
Rule Id*	Enter the accounting rule ID.
Method*	Specifies the accounting method list. Choose one of the following: • commands: Provides accounting information about specific, individual EXEC commands associated with a specific privilege level. • exec: Provides accounting records about user EXEC terminal sessions on the network access server, including username, date, and start and stop times. • network: Runs accounting for all network-related service requests. • system: Performs accounting for all system-level events not associated with users, such as reloads. Note When system accounting is used and the accounting server is unreachable at system startup time, the system will not be accessible for approximately two minutes.
Level	Choose the privilege level (1 or 15). Accounting records are generated only for commands entered by users with this privilege level.
Start Stop	Enable this option to if you want the system to send a start accounting notice at the beginning of an event and a stop record notice at the end of the event.
Use Server-group*	Choose a previously configured TACACS group. The parameters that this accounting rule defines are used by the TACACS servers that are associated with this group.

- e. Configure authorization parameters.

Table 54: Authorization

Field	Description
Server Auth Order*	Choose the authentication order. It dictates the order in which authentication methods are tried when verifying user access to a Cisco IOS XE Catalyst SD-WAN device through an SSH session or a console port.
Authorization Console	Enable this option to perform authorization for console access commands.
Authorization Config Commands	Enable this option to perform authorization for configuration commands.
Add Authorization Rule	
Rule Id*	Enter the authorization rule ID.
Method*	Choose Commands , which causes commands that a user enters to be authorized.
Level	Choose the privilege level (1 or 15) for commands to be authorized. Authorization is provided for commands entered by users with this privilege level.
If Authenticated	Enable this option to apply the authorization rule parameters only to the authenticated users. If you do not enable this option, the rule is applied to all users.
Use Server-group*	Choose a previously configured TACACS group. The parameters that this authorization rule defines are used by the TACACS servers that are associated with this group.

- f. Configure 802.1x parameters.

What to do next

Also see [Deploy a configuration group](#) .

Configuring AAA using Cisco SD-WAN Manager Template

Table 55: Feature History

Feature Name	Release Information	Description
Authorization and Accounting	Cisco IOS XE Catalyst SD-WAN Release 17.5.1a Cisco vManage Release 20.5.1	This feature provides for the configuration of authorization, which authorizes commands that a user enters on a device before the commands can be executed, and accounting, which generates a record of commands that a user executes on a device.

Configuring AAA by using the Cisco SD-WAN Manager template lets you make configuration setting in Cisco SD-WAN Manager and then push the configuration to selected devices of the same type. This procedure is a convenient way to configure several of the same type of devices at one time.

Use the AAA template for Cisco Catalyst SD-WAN Validators, Cisco SD-WAN Manager instances, Cisco Catalyst SD-WAN Controllers, and Cisco IOS XE Catalyst SD-WAN devices.

Cisco IOS XE Catalyst SD-WAN devices support configuration of authentication, authorization, and accounting (AAA) in combination with RADIUS and TACACS+.



Note You must configure a local user with a secret key via the template if you are using PPP or using MLPPP with CHAP.

Navigating to the Template Screen and Naming the Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and click **Create Template**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. From the **Create Template** drop-down list, select **From Feature Template**.
4. From the **Device Model** drop-down list, select the type of device for which you are creating the template.
5. Select **Basic Information**.
6. To create a custom template for AAA, select **Factory_Default_AAA_CISCO_Template** and click **Create Template**. The AAA template form is displayed. The top of the form contains fields for naming the template, and the bottom contains fields for defining AAA parameters.
7. In the **Template Name** field, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.

8. In the **Template Description** field, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default or to enter a value, click the **Scope** drop-down list to the left of the parameter field and select one of the following:

Table 56:

Parameter Scope	Scope Description
Device Specific (indicated by a host icon)	Use a device-specific value for the parameter. For device-specific parameters, you cannot enter a value in the feature template. You enter the value when you attach a Cisco IOS XE Catalyst SD-WAN device to a device template . When you click Device Specific, the Enter Key box opens. This box displays a key, which is a unique string that identifies the parameter in a CSV file that you create. This file is an Excel spreadsheet that contains one column for each key. The header row contains the key names (one key per column), and each row after that corresponds to a device and defines the values of the keys for that device. You upload the CSV file when you attach a Cisco IOS XE Catalyst SD-WAN device to a device template. For more information, see Create a Template Variables Spreadsheet . To change the default key, type a new string and move the cursor out of the Enter Key box. Examples of device-specific parameters are system IP address, hostname, GPS location, and site ID.
Global (indicated by a globe icon)	Enter a value for the parameter, and apply that value to all devices. Examples of parameters that you might apply globally to a group of devices are DNS server, syslog server, and interface MTUs.

Configuring Local Access for Users and User Groups

You can configure local access to a device for users and user groups. Local access provides access to a device if RADIUS or TACACS+ authentication fails.

To configure local access for individual users, select **Local**.

To add a new user, from **Local** click + **New User**, and configure the following parameters:

Table 57:

Parameter Name	Description
Name	Enter a name for the user. It can be 1 to 128 characters long, and it must start with a letter. The name can contain only lowercase letters, the digits 0 through 9, hyphens (-), underscores (_), and periods (.). The name cannot contain any uppercase letters. The following usernames are reserved, so you cannot configure them: backup, basic, bin, daemon, games, gnats, irc, list, lp, mail, man, news, nobody, proxy, quagga, root, sshd, sync, sys, uucp, and www-data. Also, names that start with viptela-reserved are reserved. Note From Cisco Catalyst SD-WAN Manager Release 20.18.1, the character limit for local user accounts remains restricted to 32 characters. However, for TACACS users, usernames extend up to 128 characters.
Password	Enter a password for the user. Each username must have a password. Users are allowed to change their own passwords. The default password for the admin user is admin. We strongly recommended that you change this password. Note When configuring local users using a Cisco SD-WAN Manager AAA template, Cisco SD-WAN Manager uses a Cisco type 9 password type. The Cisco type 9 password type uses the scrypt algorithm for hashing the passwords of local users. The Cisco SD-WAN Manager AAA template uses only the Cisco type 9 password type for hashing of local user passwords. If you configure local users using a device CLI template or a CLI add-on template, you can choose other Cisco password types for hashing of local user passwords. For more information, see Configure Type 6 Passwords Using CLI Add-On Template.
Privilege Level 1 OR 15	Select between privilege level 1 or 15. • Level 1: User EXEC mode. Read-only, and access to limited commands, such as the <code>ping</code> command. • Level 15: Privileged EXEC mode. Full Access to all commands, such as the <code>reload</code> command, and the ability to make configuration changes. By default, the EXEC commands at privilege level 15 are a superset of those available at privilege level 1
SSH RSA Key(s)	Add SSH RSA Keys by clicking the + Add button. A new field is displayed in which you can paste your SSH RSA key. To remove a key, click the - button. Devices support a maximum of 2 SSH RSA keys.

Click **Add** to add the new user. Click + **New User** again to add additional users.

To configure local access for user groups, you first place the user into either the basic or operator group. The admin is automatically placed in the netadmin group. Then you configure user groups. To make this configuration, from **Local** select **User Group**.

Click + **New User Group**, and configure the following parameters:

Table 58:

Parameter Name	Description
Name	Name of an authentication group. It can be 1 to 128 characters long, and it must start with a letter. The name can contain only lowercase letters, the digits 0 through 9, hyphens (-), underscores (_), and periods (.). The name cannot contain any uppercase letters. The Cisco Catalyst SD-WAN software provides three standard user groups, basic, netadmin, and operator. The user admin is automatically placed in the group netadmin and is the only user in this group. All users learned from a RADIUS or TACACS+ server are placed in the group basic. All users in the basic group have the same permissions to perform tasks, as do all users in the operator group. The following groups names are reserved, so you cannot configure them: adm, audio, backup, bin, cdrom, dialout, dip, disk, fax, floppy, games, gnats, input, irc, kmem, list, lp, mail, man, news, nogroup, plugdev, proxy, quagga, quaggavty, root, sasl, shadow, src, sshd, staff, sudo, sync, sys, tape, tty, uucp, users, utmp, video, voice, and www-data. Also, group names that start with the string viptela-reserved are reserved.
Feature Type	Click Preset to display a list of preset roles for the user group. Click Custom to display a list of authorization tasks that have been configured.
Feature	The feature table lists the roles for the user group. These roles are Interface, Policy, Routing, Security, and System. Each role allows the user group to read or write specific portions of the device's configuration and to execute specific types of operational commands. Click the appropriate boxes for Read, Write, and None to assign privileges to the group for each role.

Click **Add** to add the new user group.

To add another user group, click + **New User Group** again.

To delete a user group, click the trash icon at the right side of the entry. You cannot delete the three standard user groups, basic, netadmin, and operator.

Configuring RADIUS Authentication

Configure RADIUS authentication if you are using RADIUS in your deployment.

To configure a connection to a RADIUS server, from **RADIUS**, click + **New Radius Server**, and configure the following parameters:

Table 59:

Parameter Name	Description
Address	Enter the IP address of the RADIUS server host.
Authentication Port	Enter the UDP destination port to use for authentication requests to the RADIUS server. If the server is not used for authentication, configure the port number to be 0. <i>Default:</i> Port 1812
Accounting Port	Enter the UDP port to use to send 802.1X and 802.11i accounting information to the RADIUS server. <i>Range:</i> 0 through 65535. <i>Default:</i> 1813.

Parameter Name	Description
Timeout	Enter the number of seconds a device waits for a reply to a RADIUS request before retransmitting the request. <i>Default:</i> 5 seconds. <i>Range:</i> 1 through 1000
Retransmit Count	Enter the number of times the device transmits each RADIUS request to the server before giving up. <i>Default:</i> 5 seconds.
Key (Deprecated)	Enter the key the Cisco IOS XE Catalyst SD-WAN device passes to the RADIUS server for authentication and encryption. You can type the key as a text string from 1 to 31 characters long, and it is immediately encrypted, or you can type an AES 128-bit encrypted key. The key must match the AES encryption key used on the RADIUS server.

Click **Add** to add the new RADIUS server.

To add another RADIUS server, click + **New RADIUS Server** again.

To remove a server, click the trash icon.

CLI equivalent:

```
Device(config)# radius server 10.99.144.201
Device1(config-radius-server)# retransmit 5
Device(config-radius-server)# timeout 10
```

Configuring TACACS+ Authentication

Configure TACACS+ authentication if you are using TACACS+ in your deployment.

To configure a connection to a TACACS+ server, from **TACACS**, click + **New TACACS Server**, and configure the following parameters:

Table 60:

Parameter Name	Description
Address	Enter the IP address of the TACACS+ server host.
Port	Enter the UDP destination port to use for authentication requests to the TACACS+ server. If the server is not used for authentication, configure the port number to be 0. <i>Default:</i> Port 49
Timeout	Enter the number of seconds a device waits for a reply to a TACACS+ request before retransmitting the request. <i>Default:</i> 5 seconds. <i>Range:</i> 1 through 1000
Key	Enter the key the Cisco IOS XE Catalyst SD-WAN device passes to the TACACS+ server for authentication and encryption. You can type the key as a text string from 1 to 31 characters long, and it is immediately encrypted, or you can type an AES 128-bit encrypted key. The key must match the AES encryption key used on the TACACS+ server.

Click **Add** to add the new TACACS server.

To add another TACACS server, click + **New TACACS Server** again.

To remove a server, click the trash icon.

Configuring 802.1X

For information on configuring 802.1X, see [Configure IEEE 802.1X Authentication, on page 126](#).

Configuring Authentication Order

You can configure the authentication order for devices. The authentication order specifies the order in which the system attempts to authenticate user, and provides a way to proceed with authentication if the current authentication method is unavailable.

To configure AAA authentication order on a Cisco IOS XE Catalyst SD-WAN device, select the Authentication tab and configure the following parameters:

Table 61:

Parameter Name	Description
Server Group Order	Configuring a device to use AAA server groups provides a way to group existing server hosts. Grouping existing server hosts allows you to select a subset of the configured server hosts and use them for a particular service To change the default order of authentication methods that the software tries when verifying user access to a Cisco IOS XE Catalyst SD-WAN device: 1. Click the ServerGroups priority order field to display the drop-down list of server groups. The list displays groups from local, RADIUS, and TACACS authentication methods. 2. From the list, select the groups in the order that you want the software to verify a user trying to access a Cisco IOS XE Catalyst SD-WAN device. You must select at least one group from the list.

Configure Authorization and Accounting

Table 62: Feature History

Feature Name	Release Information	Description
Authorization and Accounting	Cisco IOS XE Catalyst SD-WAN Release 17.5.1a Cisco vManage Release 20.5.1	This feature provides for the configuration of authorization, which authorizes commands that a user enters on a device before the commands can be executed, and accounting, which generates a record of commands that a user executes on a device.

Restrictions to Configure Authorization and Accounting

Minimum supported release: Cisco IOS XE Catalyst SD-WAN Release 17.18.1a

- If you enter a configuration and press enter before selecting a value from an enumeration, the CLI presents a choice sub-menu. In this scenario, the final chosen value is not sent for authorization.
- The load merge and load override commands are not available when authorization is configured.
- The individual commands configured via load or rollback are not authorized or accounted.

Configuring Authorization

You can configure authorization, which causes a TACACS+ server to authorize commands that users enter on a device before the commands can be executed. Authorization is based on the policies that are configured in the TACACS+ server and on the parameters that you configure on the Authorization tab.

Prerequisites

- The TACACS+ server and the local server must be configured as first in the authentication order on the **Authentication** tab.

To configure authorization, choose the **Authorization** tab, click + **New Authorization Rule**, and configure the following parameters:

Parameter Name	Description
Console	Enable this option to perform authorization for console access commands.
Config Command	Enable this option to perform authorization for configuration commands.
Method	Choose Command , which causes commands that a user enters to be authorized.
Privilege Level 1 or 15	Choose the privilege level (1 or 15) for commands to be authorized. Authorization is provided for commands entered by users with this privilege level.
Groups	Choose a previously configured TACACS group. The parameters that this authorization rule defines are used by the TACACS servers that are associated with this group.
Authenticated	Enable this option to apply only to authenticated users the parameters that this authorization rule defines. If you do not enable this option, the rule is applied to all users.

Click Add to **add** the new authorization rule.

To add another authorization rule, click + **New Accounting Rule** again.

To remove an authorization rule, click the trash icon on the right side of the line.

CLI equivalent:

```

system
aaa
aaa authorization console
aaa authorization config-commands
aaa authorization exec default list-name method
aaa authorization commands level default list-name method

```

Configuring Accounting

You can configure accounting, which causes a TACACS+ server to generate a record of commands that a user executes on a device.

Prerequisite

- The TACACS+ server and the local server must be configured as first and second, respectively, in the authentication order on the **Authentication** tab. See [Configuring Authentication Order](#).

To configure accounting, choose the **Accounting** tab, click + **New Accounting Rule**, and configure the following parameters:

Table 63:

Parameter Name	Description
Method	Choose Command , which causes commands that a user executes to be logged.
Privilege Level 1 or 15	Choose the privilege level (1 or 15). Accounting records are generated only for commands entered by users with this privilege level.
Enable Start-Stop	Click On if you want the system to send a start accounting notice at the beginning of an event and a stop record notice at the end of the event.
Groups	Choose a previously configured TACACS group. The parameters that this accounting rule defines are used by the TACACS servers that are associated with this group.

Click **Add** to add the new accounting rule.

To add another accounting rule, click + **New Accounting Rule** again.

To remove an accounting rule, click the trash icon on the right side of the line.

CLI equivalent:

```

system
aaa
aaa accounting exec default start-stop group group-name
aaa accounting commands level default start-stop group group-name
aaa accounting network default start-stop group group-name
aaa accounting system default start-stop group group-name

```

Configure IEEE 802.1X Authentication

Table 64: Feature History

Feature Name	Release Information	Description
802.1X Support for SD-WAN	Cisco IOS XE Catalyst SD-WAN Release 17.2.1r	This feature lets you enable the IEEE 802.1X authentication on Cisco IOS XE Catalyst SD-WAN devices. To be able to configure this feature using Cisco SD-WAN Manager, ensure that Cisco SD-WAN Manager is running Cisco SD-WAN Release 20.1.1.

Starting from Cisco IOS XE Catalyst SD-WAN Release 17.2.1r, IEEE 802.1X is supported based on Identity-Based Networking Services (IBNS) 1.0 IOS-XE CLIs. This feature is supported on both LAN and WAN interfaces.

IEEE 802.1X Open Authentication and Host Modes

Any of the four host modes (single-host mode, multiple-host mode, multi-domain authentication mode, and multiauthentication mode) may be configured to allow a device to gain network access before authentication.

Open authentication is enabled by entering the **authentication open** command after host mode configuration, and acts as an extension to the configured host mode. For example, if open authentication is enabled with single-host mode, then the port will allow only one MAC address. When preauthentication open access is enabled, initial traffic on the port is restricted and independent of 802.1X is configured on the port. If no access restriction other than 802.1X is configured on the port, then a client device will have a full access on the configured VLAN. You can configure open authentication using CLI template only. You cannot configure open authentication using dot1x feature template on Cisco SD-WAN Manager.

Prerequisites

- Enable RADIUS authentication servers to authenticate IEEE 802.1x services.
- Enable IEEE 802.1X configuration on switch-port interface.
- Enable the following VLAN configurations for authenticated and unauthenticated clients:
 - Restricted VLAN (or authentication rejected VLAN)
 - Guest VLAN
 - Critical VLAN (or authentication failed VLAN)
 - Critical Voice VLAN
- Enable one of the following host-mode authentication:
 - Single-host mode
 - Multiple-host mode
 - Multiple-authentication mode

- Multi-domain mode
- Configure RADIUS Accounting attributes.
- IEEE 802.1X Authentication event using VLAN ID has to be enabled in the Add-on template, if required.

Restrictions

- IEEE 802.1X Authentication, Authorization, and Accounting (AAA) is not supported on multiple groups.
- Authentication order IEEE 802.1X MAB CLI cannot be disabled through Cisco SD-WAN Manager. The presence of this authentication order CLI results in a 60 second delay in MAB authentication when MAB client is online.
- Authentication open is not supported in feature templates but can be deployed with a CLI add on template.

Configure IEEE 802.1X Authentication using Cisco SD-WAN Manager

IEEE 802.1X is a port-based network access control (PNAC) protocol that prevents unauthorized network devices from gaining access to wired networks by providing authentication for devices that want to connect to a wired network.

A RADIUS authentication server must authenticate each client connected to a port before that client can access any services offered by network.

To configure IEEE 802.1X authentication on the interface, first create a **Cisco AAA** feature template:

1. In Cisco SD-WAN Manager, select **Configuration > Templates**
2. Click **Feature Templates**, and then click **Add Template**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

3. Select your device from the list on the left panel.
4. Select the **Cisco AAA** template.
5. Enter the **Template Name** and **Description**.
6. Select the **RADIUS** tab and under **RADIUS SERVER** click on **New RADIUS Server**.
7. Configure the following parameters:

Parameter Name	Description
Mark as Optional Row	Check the Mark as Optional Row check box to mark your configuration as device-specific.
Address	Enter IP Address of the RADIUS server.

Parameter Name	Description
Authentication Port	Click Authentication , then click Add New Authentication Entry to configure RADIUS authentication attribute–value (AV) pairs to send to the RADIUS server during an IEEE 802.1X session. To save the entry, click Add .
Accounting Port	Click Accounting , then click Add New Accounting Entry to configure RADIUS accounting attribute–value (AV) pairs to send to the RADIUS server during an IEEE 802.1X session. To save the entry, click Add .
Timeout	Configure how long to wait for replies from the RADIUS server.
Retransmit Count	Configure how many times this RADIUS server is contacted.
Key	Enter the RADIUS server shared key.

8. Click **Add**.
9. Select **RADIUS GROUP** and click on **New RADIUS Group** to configure these parameters:

Parameter Name	Description
VPN-ID	Enter the VPN through which the RADIUS or other authentication server is reachable.
Source Interface	Enter the interface that will be used to reach the RADIUS server.
Radius Server	Configure the Radius server.

10. Click **Add**.
11. Select the **802.1X** tab and enter these parameters:

Parameter Name	Description
Authentication Param	Click On to enable authentication parameters.
Accounting Param	Click On to enable accounting parameters.

12. To save this feature template, click **Save**.
13. To enable this feature on your device, ensure to add these feature templates to your device template.



Note You need to recreate the AAA feature templates as the templates created prior to Cisco vManage Release 20.5 fails when attached to the device.

Next create a **Switch Port** template that can be used for the Switch Port device:

1. To create a **Switch Port** template, repeat steps 1 to 3 from above.
2. Select the **Switch Port** template.

3. Enter the **Template Name** and **Description**.
4. Select the **Interface** tab click on **New Interface**.
5. Configure the following parameters:

Parameter Name	Description
Interface name	Enter the interface name.
Speed	Enter the interface speed.
VLAN Name	Enter the VLAN name.
VLAN ID	Enter the VLAN identifier associated with the bridging domain.
802.1X	Enable IEEE 802.1X authentication on this interface. Select "On". This will provide a further set of parameters listed below.
Interface PAE Type	Enter the IEEE 802.1x Interface PAE type.
Control Direction	Enter unidirectional or bidirectional authorization mode.
Host Mode	Select whether an IEEE 802.1X interface grants access to a single host (client) or to multiple hosts (clients): • Multi Auth—Grant access to one host on a voice VLAN and multiple hosts on data VLANs. • Multi Host—Grant access to multiple hosts • Single Host—Grant access only to the first authenticated host. This is the default. • Multi-Domain—Grant access to both a host and a voice device, such as an IP phone on the same switch port. Note These options are available only in the 'Global' Host Mode settings.
Periodic Reauthentication	Enter how often to reauthenticate IEEE 802.1X clients. By default, no reauthentication attempts are made after the initial LAN access request. Range: 0 to 1440 minutes

6. Click on **Advanced Options** and enter the following:

Parameter Name	Description
Authentication Order	Enter the order of authentication methods to use when authenticating devices for connection to the IEEE 802.1X interface. The default authentication order is RADIUS, then MAC authentication bypass (MAB).
MAC Authentication Bypass	Select to enable MAC authentication bypass (MAB) on the RADIUS server and to authenticate non-IEEE 802.1X-compliant clients using a RADIUS server.

Parameter Name	Description
Port Control Mode	Enter the port control mode to enable IEEE 802.1X port-based authentication on the interface. Auto- Configure this to enable IEEE 802.1X authentication and start the port in unauthorized state. This allows only EAPOL frames to be sent and received through the port.
Voice VLAN ID	Configure the Voice VLAN ID.
Critical VLAN	Enter the critical VLAN (or authentication failed VLAN) for IEEE 802.1x-compliant clients. Configure network access when RADIUS authentication or the RADIUS server fails.
Critical Voice VLAN	Enable the critical voice VLAN.
Guest VLAN	Configure guest VLAN to drop non-IEEE 802.1X enabled clients, if the client is not in the MAB list.
Restricted VLAN	Enter the restricted VLAN (or authentication failed VLAN) for IEEE 802.1x-compliant clients. Configure limited services to IEEE 802.1X-compliant clients that failed RADIUS authentication.

7. Click on **Add**.
8. To save this feature template, click **Save**.
9. To enable this feature on your device, ensure to add these feature templates to your device template.

Configure Switch Port Using a Configuration Group

Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

- Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.
- Step 2** Create and configure a Switch Port feature in a Service profile.
 - a. Configure Switch Port.

Table 65: Switch Port

Field	Description
Age Out Time	Enter how long an entry is in the MAC table before it ages out. Set the value to 0 to prevent entries from timing out. Range: 0, 10 through 1000000 seconds Default: 300 seconds
Configure Interface	
Interface Name	Enter the name of the interface to associate with the bridging domain, in the format geslot/port .
Mode	Choose the switch port mode. • access: Configure the interface as an access port. You can configure only one VLAN on an access port, and the port can carry traffic only for one VLAN. When you choose access, the following field appears: Switchport Access Vlan: Enter the VLAN number, which can be a value from 1 through 4094. • trunk: Configure the interface as a trunk port. You can configure one or more VLANs on a trunk port, and the port can carry traffic for multiple VLANs. When you choose trunk, the following fields appear: • Allowed Vlans: Enter the number of the VLANs for which the trunk can carry traffic and a description for the VLAN. • Switchport Trunk Native Vlan: Enter the number of the VLAN allowed to carry untagged traffic.
Shutdown	Enable the interface. By default, an interface is disabled.
Speed	Enter the speed of the interface.
Duplex	Choose full or half to specify whether the interface runs in full-duplex or half-duplex mode.

Field	Description
Port Control	Choose the port control mode to enable IEEE 802.1X port-based authentication on the interface. • auto: Enables IEEE 802.1X authentication and starts the port in the unauthorized state, allowing only EAPOL frames to be sent and received through the port. The authentication process begins when the link state of the port changes from down to up or when an EAPOL-start frame is received. The device requests the identity of the supplicant and starts relaying authentication messages between the supplicant and the authentication server. Each supplicant attempting to access the network is uniquely identified by the device by using the supplicant MAC address. • force-unauthorized: Causes the port to remain in the unauthorized state, ignoring all attempts by the supplicant to authenticate. The device cannot provide authentication services to the supplicant through the port. • force-authorized: Disables IEEE 802.1X authentication and causes the port to change to the authorized state without any authentication exchange required. The port sends and receives normal traffic without IEEE 802.1X-based authentication of the client.
Voice VLAN	Enter the Voice VLAN ID.
Pae Enable	The Cisco Catalyst SD-WAN device acts as a port access entity (PAE), allowing authorized network traffic and preventing unauthorized network traffic ingressing to and egressing from the controlled port.
MAC Authentication Bypass	Enable this option to allow MAC authentication bypass (MAB) on the RADIUS server and to authenticate non-IEEE 802.1X-compliant clients using a RADIUS server.
Host Mode	Choose whether an IEEE 802.1X interface grants access to a single host (client) or to multiple hosts (clients). • single-host: Grant access only to the first authenticated host. This is the default. • multi-auth: Grant access to one host on a voice VLAN and multiple hosts on data VLANs. • multi-host: Grant access to multiple hosts. • multi-domain: Grant access to both a host and a voice device, such as an IP phone on the same switch port.
Enable Periodic Reauth	Enable periodic re-authentication. By default, this option is enabled.
Inactivity	Enter the inactivity timeout time in seconds. Default: 60 seconds
Reauthentication	Enter the re-authentication interval in seconds.

Field	Description
Control Direction	Choose both (bidirectional) or in (unidirectional) authorization mode.
Restricted VLAN	Enter the restricted VLAN (or authentication-failed VLAN) for IEEE 802.1x-compliant clients. Configure limited services to IEEE 802.1X-compliant clients that failed RADIUS authentication.
Guest VLAN	Enter the guest VLAN to drop non-IEEE 802.1X enabled clients, if the client is not in the MAB list.
Critical VLAN	Enter the critical VLAN (or authentication-failed VLAN) for IEEE 802.1x-compliant clients. Configure network access when RADIUS authentication or the RADIUS server fails.
Enable Voice	Enable the critical voice VLAN.
Configure Static Mac Address	
MAC Address	Enter the static MAC address to map to the switch port interface.
Interface Name	Enter the name of the switch port interface.
VLAN ID	Enter the number of the VLAN for the switch port.

What to do next

Also see [Deploy a configuration group](#).

Configure IEEE 802.1X Open Authentication

You can configure IEEE 802.1X open authentication using the CLI add-on template.

```
Device# config-transaction
Device(config)# interface GigabitEthernet2
Device(config-if)# authentication open
```

Configure IEEE 802.1X Authentication using CLIs

Configuration

For this feature, two sets of configurations are required-

1. Configure the Global AAA commands:

- a. Enable or disable IEEE 802.1X globally

```
Device(config)# aaa authentication dot1x default group radius-0
Device(config)# aaa authorization network default group radius-0
Device(config)# dot1x system-auth-control
```

```
Device(config)# radius-server dead-criteria time 10 tries 3
Device(config)# radius-server deadtime 15
```

b. Enable accounting

```
Device(config)# aaa accounting dot1x default start-stop group radius-0
```

2. Configure the Interface Level commands:

a. Enable or disable IEEE 802.1X on port-basis

```
Device(config-if)# dot1x pae authenticator
Device(config-if)# authentication port-control auto
```

b. Enable or disable MAB on port-basis

```
Device(config-if)# mab
```

c. Select host-mode

```
Device(config-if)# authentication host-mode <multi-auth | multi-domain | multi-host
| single-host>
```

d. Configure voice vlan

```
Device(config-if)# switchport voice vlan <vlan-id>
```

e. Select IEEE 802.1X control direction

```
Device(config-if)# authentication control-direction <both | in>
```

f. Enable periodic re-authentication and corresponding re-authentication interval and inactivity timeout time

```
Device(config-if)# authentication periodic
Device(config-if)# authentication timer reauthenticate <interval-in-sec>
Device(config-if)# authentication timer inactivity <timeout-in-sec>
```

g. Configurable authentication orders on per-port basis

```
Device(config-if)# authentication order dot1x mab
```

h. Specify the restricted VLAN

```
Device(config-if)# authentication event fail action authorize vlan <vlan-id>
```

i. Specify the guest VLAN

```
Device(config-if)# authentication event no-response action authorize vlan <vlan-id>
```

j. Specify the critical VLAN

```
Device(config-if)# authentication event server dead action authorize vlan <vlan-id>
```

k. Enable the critical voice VLAN feature

```
Device(config-if)# authentication event server dead action authorize voice
```

Posture Assessment Support

Table 66: Feature History

Feature Name	Release Information	Description
Posture Assessment Support	Cisco IOS XE Catalyst SD-WAN Release 17.3.1a Cisco vManage Release 20.3.1	This feature enables you to utilize Posture Assessment capabilities to validate the compliance of endpoints according to security policies of your enterprise. Identity Services Engine (ISE) Posture functions are integrated into Cisco 1100 Integrated Services Routers. This feature can only be configured using the Add-On feature template in Cisco SD-WAN Manager.

In a network, endpoint validation is necessary to ensure compliance with security policies of the company and posture assessment enables you to validate this. The posture module enforces security policies on endpoints that are connected to a network. For a connection between the endpoints of Cisco 1100 Integrated Services Router and ISE (Identity Services Engine), authentication interaction between them is required. IEEE 802.1X is the recommended standard authentication process for posture assessment, MAC Authentication Bypass (MAB) can be used as well.

The posture agent software used for this is Cisco AnyConnect Posture Assessment. The Cisco AnyConnect software is installed on the endpoint and has a module called posture. Cisco AnyConnect downloads security policies from ISE server and then checks the conditions (anti-malware condition, anti-spyware condition, anti-virus condition, application condition, USB condition) of the endpoints. If all conditions are met, Cisco AnyConnect gives a 'Compliant' result to the ISE server. If not, Cisco AnyConnect gives a 'NonCompliant' result. After authorization and authentication of the endpoints by authentication and redirect Access Control Lists (ACL), Cisco AnyConnect posture module on the client end initiates posture assessment with the posture-policy server.

After posture assessment is completed and authenticated, the RADIUS CoA (Change of Authorization) process is initiated by a policy set on ISE, from RADIUS servers to re-authenticate or re-authorize new policies. Once posture assessment is successful, access to the entire network is pushed down to the Cisco ISR 1100 router and to the client, through CoA re-authentication command.

Prerequisites for Posture Assessment

- Basic IEEE 802.1x authentication process should be functional.
- Change of Authorization (CoA) should be supported.
- Redirect ACL, downloadable ACL (dACL) and critical ACL should be available.
- Device tracking policy (for identity) should be supported.
- URL redirect should be supported.

Restrictions for Posture Assessment

- Only 8 port Cisco 1100 Integrated Services Routers support ACL functions such as dACL and redirect ACL.
- ACL and Access Control Entry (ACE) rules do not support compare operations, such as >, <, >=, <=
- Up to 120 dACL ACEs are supported, and 64 Redirect ACL ACEs are supported.
- Port ACL and IPv6 ACL are not supported.
- IP option and IP fragment ACL are not supported.
- Per-VLAN device-tracking is not supported.
- Only limited per-port device tracking policy options such as glean and address tracking are allowed.

Configuring Posture Assessment on Cisco Catalyst SD-WAN

1. Use the CLI Add-on template in Cisco SD-WAN Manager to configure AAA, IEEE 802.1x, posture assessment and redirect ACL and device-tracking.

Example configurations are given below.



Note `aaa new-model` is enabled by default on Cisco Catalyst SD-WAN and is not configurable by the user. However, it must be configured on a non SD-WAN image.

a. Configure AAA

```
aaa new-model
radius server ISE1

address ipv4 198.51.100.255 auth-port 1812 acct-port 1813
key cisco

aaa group server radius ISE
server name ISE1
!
aaa authentication dot1x default group ISE
aaa authorization network default group ISE
aaa accounting dot1x default start-stop group ISE

interface vlan 15
ip address 198.51.100.1 198.51.100.254

interface GigabitEthernet0/1/0
switchport mode access
switchport access vlan 15

ip radius source-interface vlan 15
```

b. Configure IEEE 802.1x authentication and authorization

```
policy-map type control subscriber simple_dot1x
event session-started match-all
10 class always do-until-failure
10 authenticate using dot1x
!
interface GigabitEthernet0/1/7
switchport access vlan 22
switchport mode access
```

```

access-session closed
access-session port-control auto
dot1x pae authentication
service-policy type control subscriber simple_dot1x
!
interface Vlan22
 ip address 198.51.100.1 198.51.100.254

```



Note The IEEE 802.1x endpoint is connected to GigabitEthernet0/1/7.

c. Configure posture assessment and redirect ACL

```

ip http server
ip http secure-server

ip access-list extended ACL-POSTAUTH-REDIRECT
10 deny tcp any host 192.0.2.255
20 deny tcp any any eq domain
30 deny udp any any eq domain
40 deny udp any any eq bootpc
50 deny udp any any eq bootps
60 permit tcp any any eq www
70 permit tcp any any eq 443

```

d. Configure device tracking

```

!
device-tracking policy tracking_test
 security-level glean
 no protocol ndp
 no protocol dhcp6
 tracking enable
!
interface GigabitEthernet0/1/7
 device-tracking attach-policy tracking_test

```



Note The IP address mentioned belongs to ISE.

The steps you have to perform to add this configuration into the CLI Add-On template on Cisco SD-WAN Manager are documented [here](#).

2. To Configure CoA reauthentication and dACL on ISE:
 - a. Create a downloadable ACL and define the ACEs in it.
 ACL name: TEST_IP_PERMIT_ALL
 ACEs: permit ip any any
 - b. Create an authorization result and choose the downloadable ACL as dACL.
 - c. Navigate to **Administration > System > Settings > Policy Settings**, and in **Policy Sets** configuration select the authorization result as authorization policy.
3. After creating the CLI Add-On template, attach it to a device template and then Cisco SD-WAN Manager pushes all the configuration in the device template onto your device.

Type 6 Passwords on Cisco IOS XE SD-WAN Routers

Table 67: Feature History

Feature Name	Release Information	Description
Type 6 Passwords on Cisco IOS XE SD-WAN Routers	Cisco IOS XE Catalyst SD-WAN Release 17.4.1a Cisco vManage Release 20.4.1	This feature allows you to use type 6 passwords that use secure reversible encryption. This encryption provides enhanced security by using more secure algorithms to encrypt your passwords. These passwords are supported for the templates detailed in Supported Templates , on page 139.

Overview of Type 6 Passwords

The Type 6 Passwords feature enables secure reversible encryption for authentication, authorization, and accounting (AAA) and Simple Network Management Protocol (SNMP) configurations based on the advanced encryption scheme (AES) algorithm.

Reversible encryption is the process by which a password is encrypted with a reversible, symmetric encryption algorithm. To check if the password entered by the user is valid, the password is decrypted and compared to the user-input password. To perform this encryption, the symmetric encryption algorithm requires a key which you can provide. The encryption algorithm used is advanced encryption scheme (AES) algorithm in Cipher Block Chaining (CBC) mode with a PKCS#5 padding. This algorithm is used for AAA features such as RADIUS, TACACS+, SNMP, and TrustSec.

When you create a supported template in Cisco vManage Release 20.4.1 and later releases, by default type 6 passwords are used. Cisco SD-WAN Manager encrypts the passwords and sends the passwords to the router over a secure tunnel. The router then encrypts the passwords into the type 6 format and stores the password on the device. The Type 6 Passwords feature is not supported on Viptela software.



Note Cisco SD-WAN Manager encrypted passwords show up as either \$6\$ or \$8\$. Where as, Cisco IOS XE devices have encryption streams defined as type 0, type 5, type 6, type 8, and so on. On the other hand, Cisco SD-WAN Manager runs on Viptela OS which is based on Linux. Linux uses hashing and encryption schemes. Encrypted passwords on Cisco SD-WAN Manager starting with \$6\$ refer to sha512-crypt. Passwords beginning with \$8\$ represent aes-cfb 128 encryption.



Note On Cisco IOS XE Catalyst SD-WAN devices, an admin user with privilege 15 is created by default during day-0 bringup of the device. It is recommended that users don't delete this admin user.



Note We recommend using type 6 passwords to reduce the vulnerability of a malicious attack against password integrity. On upgrading your device to Cisco IOS XE Catalyst SD-WAN Release 17.4.1a, all AAA, RADIUS key, and TACACS+ keys are encrypted to type 6.

Supported Platforms

Cisco IOS XE Catalyst SD-WAN devices.

Supported Templates

The following templates support Type 6 passwords:

- RADIUS and TACACS authentication using the Cisco AAA template.
- SNMP template.
- CLI add-on template.

Restrictions

- For SNMP templates, the community name is encrypted by default. Therefore, to upgrade existing SNMP templates to type 6 passwords, delete and re-create the community and trap target.
- When using type 6 passwords with the **keychain key-string** command, the maximum password length for a clear text is 38 characters.

Configure Type 6 Passwords Using Cisco SD-WAN Manager

Upgrade Existing Templates to Type 6 Passwords

To upgrade passwords in your existing templates on Cisco SD-WAN Manager to type 6 passwords, do the following:



Note When you upgrade your routers to Cisco IOS XE Catalyst SD-WAN Release 17.4.1a, all supported passwords are automatically upgraded to type 6 passwords.

1. Navigate to **Configuration > Templates**
2. Click **Feature Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

3. For the template that you want to upgrade to type 6 passwords, click the ... button.

4. Click **Edit**.
5. Click **Save**.



Note To update the passwords, you do not need to make any other changes to the template. When you click **Save**, Cisco SD-WAN Manager automatically upgrades the passwords to type 6 passwords.

Configure Type 6 Passwords Using CLI Add-On Template

You can configure type 6 passwords when using CLI add-on feature templates by doing the following:

1. Navigate to **Configuration > Templates**.
2. Click **Feature Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

3. Click **Add Template**.
4. Under the Select Devices pane, select the devices for which you are creating the template.
5. Under the Select Template pane, scroll down to the Other Templates section.
6. Click **CLI Add-On Template**. For information on CLI add-on feature templates, see [CLI Add-On Feature Templates](#).
7. Enter a Template Name and Description.
8. Enter a CLI command to run on your device.



Note Only supported CLIs including passwords, keys, secret text, community name, and other strings can be encrypted. Refer [Commands supported for Type 6 encryption](#) for more information.

9. Select the plaintext password, key, or similar in the CLI and click the **Encrypt Type 6** button.
10. Click **Save**.

Verify Type 6 Passwords

To verify that your passwords are upgraded to type 6 passwords, you can do one of the following:

- On Cisco SD-WAN Manager, when you attach a configuration that supports type 6 passwords to your device the configuration preview displays the encrypted password. For example:

```
snmp-server community 0 $CRYPT_CLUSTER$ptqX7nQr6QvC8YZuoMGOkw==$6cVCeSpOf0vFe5iqhJqvQQ==
ro
```

Despite the command displaying the type as 0, the

\$CRYPT_CLUSTER\$ptqX7nQr6QvC8YZuoMGOkw== \$6cVCeSpOfoVFe5iqhJqvQQ== string represents your encrypted password. If your password is encrypted, it will begin with \$CRYPT_CLUSTER\$.

- On your device, you can run the following command to display your encrypted passwords:

```
Device#show run | sec aaa
aaa new-model
aaa group server tacacs+ tacacs-0
server-private 10.0.0.1 key 6 BibgKcVeWF]^aK[XfEIICXMCbdScBYAAB
aaa group server radius radius-0
server-private 10.0.0.2 timeout 5 retransmit 3 key 6 CHd_VK[]NHedcVCWGCaENGINQHLBEhDBe
```

The output displays that the password is type 6 and also displays your encrypted password.



CHAPTER 6

Role Based Access Control

Table 68: Feature History

Feature Name	Release Information	Feature Description
Co-Management: Granular Role-Based Access Control	Cisco Catalyst SD-WAN Manager Release 20.13.1	This feature introduces role-based access control (RBAC) based on sites, scope, or roles. It is a method of authorizing system access for users based on a combination of role and scope of a user. You can create scope, users and roles with required read and write permissions for Cisco SD-WAN Manager policies. RBAC prevents unauthorized access and reduces the risk of data breaches and other security incidents.
Canadian French Language Support on Cisco Catalyst SD-WAN Manager	Cisco Catalyst SD-WAN Manager Release 20.13.1	Added support for using Canadian French for the Cisco Catalyst SD-WAN Manager user interface.

- [Information About RBAC, on page 143](#)
- [Benefits of RBAC, on page 158](#)
- [Restrictions for Role Based Access Control, on page 159](#)
- [Use Cases for RBAC, on page 160](#)
- [Configure Role Based Access Control, on page 160](#)
- [Verify RBAC, on page 164](#)
- [Monitor RBAC, on page 165](#)

Information About RBAC

Information About Role Based Access Control

Role-Based Access Control (RBAC) is a method of restricting or authorizing system access for users based on user roles and scope. A role defines the privileges of a user in the system and the locale defines the

organizations (domains) that a user is allowed access. Because users are not directly assigned privileges, management of individual user privileges is simply a matter of assigning the appropriate roles and scopes. A user is granted write access to desired system resources only if the assigned role grants the access privileges and the assigned locale allows access.

User: is the entity that performs different actions in Cisco SD-WAN Manager. A user belongs to a role.

Roles: define the permissions (Read, Write or Deny) allowed for a user for different APIs or functionalities.

Scope: define the set of objects (sites, devices or templates) on which a user can perform actions.

When **Read** or **Write** is selected, the user can view and make changes for the selected features. When **Read** is selected, the user can only view information. When **Deny** is selected, the user can neither view or make changes to the Cisco IOS XE Catalyst SD-WAN.

System default roles cannot be changed or modified. The Cisco IOS XE Catalyst SD-WAN software provides the following system default roles:

- **basic:** The basic role is a system default role and is pre-built-in Cisco SD-WAN Manager. You cannot modify or delete. If you want to modify the role, you must make a copy of it and then modify it as a new customer role.
- **operator:** The operator role is also a configurable role and can be used for any users and privilege levels. This role is designed to include users who have permission only to view information.
- **netadmin:** The netadmin role is a non-configurable role. By default, this role includes the **admin** user. You can add other users to this role. Users with this role are permitted to perform all operations on the device.
- **network_operations:** The **network_operations** role is a non-configurable role. Users in this role can perform all non-security-policy operations on the device and only view security policy information. For example, users can create or modify template configurations, manage disaster recovery, and create non-security policies such as an application aware routing policy or Cflowd policy.
- **security_operations:** The **security_operations** role is a non-configurable role. Users in this role can perform all security operations on the device and only view non-security-policy information. For example, users can manage umbrella keys, licensing, IPS signatures auto update, TLS/SSL proxy settings, and so on.

Users of the **network_operations** role are authorized to apply policies to a device, revoke applied policies, and edit device templates. Users of the **security_operations** role require **network_operations** users to intervene on day-0 to deploy a security policy on a device and on day-N to remove a deployed security policy. However, after a security policy is deployed on a device, **security_operations** users can modify the security policy without needing the **network_operations** users to intervene.



Note Only netadmin users can view the running and local configuration. Users associated with a predefined operator role do not have access to the running and local configurations. The predefined role operator has only read access for the template configuration. If you need only a subset of admin user privileges, then you need to create a new role with the selected features from the features list with both read and write access and associate the role with the custom user.

Privileges for Role-Based Access

Role-based access privileges are arranged into five categories, which are called *tasks*:

- Interface—Privileges for controlling the interfaces on the Cisco IOS XE Catalyst SD-WAN device.
- Policy—Privileges for controlling the control plane policy, OMP, and data plane policy.
- Routing—Privileges for controlling the routing protocols, including BFD, BGP, OMP, and OSPF.
- Security—Privileges for controlling the security of the device, including installing software and certificates. Only users belonging to the **netadmin** group can install software on the system.
- System—General system-wide privileges.

Role-Based Access Control by VPN

Role-based access control (RBAC) is the process of restricting user access to network configurations and resources. In RBAC, users are assigned roles depending on the resources they need access to. The RBAC by VPN feature helps you to manage and control access to your network based on the VPNs. It involves setting permissions and privileges to enable access to authorized users.

RBAC by VPN

Role-based access by VPN allows a network administrator to define VPN groups with one or more network segments. The network administrator can associate a user with a VPN group that restricts user access to devices in the network and features of Cisco SD-WAN Manager.

RBAC by VPN provides the following restricted access to users configured with a VPN group:

- Access to VPN Dashboard
- Monitor devices, network, and application status via VPN dashboard
- VPN dashboard information restricted to devices with segments in the VPN group
- Monitor option restricted to devices with segments in the VPN group
- Interface monitoring on each device restricted to interfaces of segments in the VPN group

VPN Dashboard Overview

Users configured with VPN group can access only the VPN Dashboard, and it is read-only access. User with Admin access can create the VPN groups and has access to both Admin Dashboard and VPN Dashboard(s). Admin user can access these dashboards by choosing **Dashboard** from the Cisco SD-WAN Manager menu.

Role-Based Access with AAA

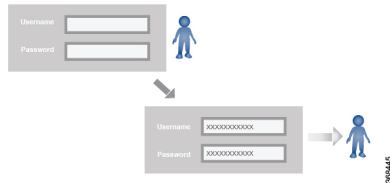
The Cisco Catalyst SD-WAN AAA software implements role-based access to control the authorization permissions for users on Cisco IOS XE Catalyst SD-WAN devices. Role-based access consists of three components:

- Users are those who are allowed to log in to a Cisco IOS XE Catalyst SD-WAN device.
- User groups are collections of users.

- Privileges are associated with each group. They define the commands that the group's users are authorized to issue.

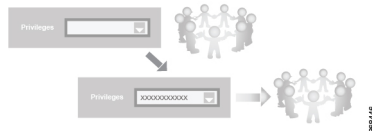
Users and User Groups

All users who are permitted to perform operations on a Cisco IOS XE Catalyst SD-WAN device must have a login account. For the login account, you configure a username and a password on the device itself. These allow the user to log in to that device. A username and password must be configured on each device that a user is allowed to access.

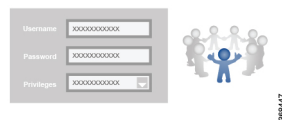


The Cisco Catalyst SD-WAN software provides one standard username, **admin**, which is a user who has full administrative privileges, similar to a UNIX superuser. By default, the **admin** username password is **admin**. You cannot delete or modify this username, but you can and should change the default password.

User groups pool together users who have common roles, or privileges, on the Cisco IOS XE Catalyst SD-WAN device. As part of configuring the login account information, you specify which user group or groups that user is a member of. You do not need to specify a group for the **admin** user, because this user is automatically in the user group **netadmin** and is permitted to perform all operations on the Cisco IOS XE Catalyst SD-WAN device.



The user group itself is where you configure the privileges associated with that group. These privileges correspond to the specific commands that the user is permitted to execute, effectively defining the role-based access to the Cisco Catalyst SD-WAN software elements.



The Cisco Catalyst SD-WAN software provides the following standard user groups:

- **basic**: The basic group is a configurable group and can be used for any users and privilege levels. This group is designed to include users who have permission to both view and modify information on the device.
- **operator**: The operator group is also a configurable group and can be used for any users and privilege levels. This group is designed to include users who have permission only to view information.
- **netadmin**: The netadmin group is a non-configurable group. By default, this group includes the **admin** user. You can add other users to this group. Users in this group are permitted to perform all operations on the device.
- Minimum supported release: Cisco vManage Release 20.9.1

network_operations: The **network_operations** group is a non-configurable group. Users in this group can perform all non-security-policy operations on the device and only view security policy information. For example, users can create or modify template configurations, manage disaster recovery, and create non-security policies such as application aware routing policy or CFlowD policy.

- Minimum supported release: Cisco vManage Release 20.9.1

security_operations: The **security_operations** group is a non-configurable group. Users in this group can perform all security operations on the device and only view non-security-policy information. For example, users can manage umbrella keys, licensing, IPS signatures auto update, TLS/SSL proxy settings, and so on.

Users of the **network_operations** group are authorized to apply policies to a device, revoke applied policies, and edit device templates. Users of the **security_operations** group require **network_operations** users to intervene on day-0 to deploy security policy on a device and on day-N to remove a deployed security policy. However, after a security policy is deployed on a device, **security_operations** users can modify the security policy without needing the **network_operations** users to intervene.



Note Only admin users can view running and local configuration. Users associated with predefined operator user group do not have access to the running and local configurations. The predefined user group operator has only read access for the template configuration. If you need only a subset of admin user privileges, then you need to create a new user group with the selected features from the features list with both read and write access and associate the group with the custom user.

Privileges for Role-Based Access

Role-based access privileges are arranged into five categories, which are called *tasks*:

- Interface—Privileges for controlling the interfaces on the Cisco IOS XE Catalyst SD-WAN device.
- Policy—Privileges for controlling control plane policy, OMP, and data plane policy.
- Routing—Privileges for controlling the routing protocols, including BFD, BGP, OMP, and OSPF.
- Security—Privileges for controlling the security of the device, including installing software and certificates. Only users belonging to the **netadmin** group can install software on the system.
- System—General systemwide privileges.

The tables in the following sections detail the AAA authorization rules for users and user groups. These authorization rules apply to commands issued from the CLI and to those issued from Netconf.

User Authorization Rules for Operational Commands

The user authorization rules for operational commands are based simply on the username. Any user who is allowed to log in to the Cisco IOS XE Catalyst SD-WAN device can execute most operational commands. However, only the **admin** user can issue commands that affect the fundamental operation of the device, such as installing and upgrading the software and shutting down the device.

Note that any user can issue the **config** command to enter configuration mode, and once in configuration mode, they are allowed to issue any general configuration command. Also, any user is allowed to configure their password by issuing the **system aaa user self password password** command and then committing that

configuration change. For the actual commands that configure device operation, authorization is defined according to user group membership. See User Group Authorization Rules for Configuration Commands.

The following tables lists the AAA authorization rules for general CLI commands. All the commands are operational commands except as noted. Also, some commands available to the "admin" user are available only if that user is in the "netadmin" user group.

CLI Command	Any User	Admin User
clear history	X	X
commit confirm	X	X
complete-on-space	X	X
config	X	X
exit	X	X
file	X	X
help	X	X
[no] history	X	X
idle-timeout	X	X
job	X	X
logout	—	X (users in netadmin group only)
monitor	X	X
nslookup	X	X
paginate	X	X
ping	X (users in netadmin group only)	X (users in netadmin group only)
poweroff	—	X(users in netadmin group only)
prompt1	X	X
prompt2	X	X
quit	X	X
reboot	—	X (users in netadmin group only)
request aaa request admin-tech request firmware request interface-reset request nms request reset request software	—	X (users in netadmin group only)
request execute request download request upload	X	X

CLI Command	Any User	Admin User
request (everything else)	—	X
rollback (configuration mode command)	—	X (users in netadmin group only)
screen-length	X	X
screen-width	X	X
show cli	X	X
show configuration commit list	X	X
show history	X	X
show jobs	X	X
show parser dump	X	X
show running-config	X	X
show users	X	X
system aaa user <i>self</i> password <i>password</i> (configuration mode command) (Note: A user cannot delete themselves)		
tcpdump	X	X
timestamp	X	X
tools ip-route	X	X
tools netstat	X	X
tools nping	X	X
traceroute	X	X
vshell	X (The availability of vshell command is unavailable to all users that are not in netadmin group in Cisco vManage Release 20.9.5.)	X (The vshell AAA authorized access is limited only to users that are in netadmin group.)

User Group Authorization Rules for Operational Commands

The following table lists the user group authorization roles for operational commands.

Operational Command	Interface	Policy	Routing	Security	System
clear app		X			
clear app-route		X			
clear arp	X				
clear bfd			X		X
clear bgp			X		X
clear bridge	X				
clear cellular	X				
clear control				X	
clear crash					X
clear dhcp					X
clear dns					X
clear igmp			X		
clear installed-certificates				X	
clear interface	X				
clear ip			X		
clear notification					X
clear omp			X		
clear orchestrator				X	
clear ospf			X		
clear pim			X		
clear policy		X			
clear pppoe	X				
clear system					X
clear tunnel				X	
clear wlan	X				
clear ztp				X	X
clock					X

Operational Command	Interface	Policy	Routing	Security	System
debug bgp			X		
debug cellular	X				
debug cflowd		X			
debug chmgr					X
debug config-mgr					X
debug dhcp-client					X
debug dhcp-helper					X
debug dhcp-server					X
debug fpm		X			
debug ftm					X
debug igmp			X		
debug netconf					X
debug omp			X		
debug ospf			X		
debug pim			X		
debug resolver			X		
debug snmp					X
debug sysmgr					X
debug transport					X
debug ttm					X
debug vdaemon				X	X
debug vrrp				X	
debug wlan	X				
request certificate				X	
request control-tunnel				X	
request controller				X	
request controller-upload				X	

Operational Command	Interface	Policy	Routing	Security	System
request csr				X	
request device				X	
request device-upload				X	
request on-vbond-controller				X	
request port-hop				X	
request root-cert-chain				X	
request security				X	
request vedge				X	
request vedge-upload				X	
request vsmart-upload				X	
show aaa					X
show app		X			
show app-route		X			
show arp	X				
show bfd			X		X
show bgp			X		
show boot-partition					X
show bridge	X				
show cellular	X				
show certificate				X	
show clock					X
show control				X	X
show crash					X
show debugs—same as debug commands					
show dhcp					X
show external-nat				X	X
show hardware					X

Operational Command	Interface	Policy	Routing	Security	System
show igmp			X		
show interface	X				
show ip			X		X
show ipsec				X	
show licenses					X
show logging					X
show multicast			X		
show nms-server					X
show notification					X
show ntp					X
show omp		X	X		X
show orchestrator				X	
show ospf			X		
show pim			X		
show policer		X			
show policy		X			
show ppp	X				
show pppoe	X				
show reboot					X
show security-info				X	
show software					X
show system					X
show transport					X
show tunnel				X	
show uptime					X
show users					X
show version					X
show vrrp	X				

Operational Command	Interface	Policy	Routing	Security	System
show wlan	X				
show ztp				X	

User Group Authorization Rules for Configuration Commands

The following table lists the user group authorization rules for configuration commands.

Configuration Command	Interface	Policy	Routing	Security	System
apply-policy		X			
banner					X
bfd			X		X
bridge	X				
omp		X	X		X
policy		X			
security				X	X
snmp					X
system					X
vpn interface	X				
vpn ip			X		
vpn router			X		
vpn service			X		
vpn (everything else, including creating, deleting, and naming)					X
wlan	X				

RBAC By Resource Group Overview

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.5.1a and Cisco vManage Release 20.5.1

RBAC by resource groups is a method of restricting or authorizing system access for users based on user groups and resource groups. A user group defines the privileges of a user in the system and the resource group defines the organizations (domains) to which a user is allowed access. Because users are not directly assigned privileges, management of individual user privileges is simply a matter of assigning the appropriate user and resource groups.

For large Cisco Catalyst SD-WAN deployments across multiple geographical locations, you can split the network administration among different regional administrators.

Based on the user groups and resources groups to which network administrators are assigned, we can broadly classify them as Global Administrators and Regional Administrators. Global administrators have access to resources in every resource group and have full read-write privileges for all the features. Regional Administrators group have full read-write privileges for all the features, but the resources they can access is controlled by the resource groups to which they are assigned.

Global Admin

User accounts in the global resource group have access to all resources. A global admin is responsible for overseeing the entire network, but not involved in the operations of the individual devices on a daily basis. The global admin can assign devices to their corresponding regions, assign the regional admin accounts, manage the controllers, maintain sharable and centralized configurations, and when necessary, operate on the individual devices.

Any user in a single tenant setup with netadmin privileges and also part of global resource group is considered as global admin. Default admin user on Cisco SD-WAN Manager is also a global-admin, and that user can assign more global-admins. Global resource group encompasses all the WAN edges, controllers in the single view.

Global admin can switch to view only a specific resource group and can create templates. Local resource group admins, also called regional admins can clone the global templates and reuse them within their resource groups.

Regional Admin

The regional admins are responsible for day-to-day operations (configuration, monitoring, onboarding, and so on) for devices in their corresponding regions. They should not have access to or visibility into devices outside of their region. The following user groups can be created:

- resource group admin – full read/write access to devices in the corresponding resource group, can troubleshoot, monitor, attach or detach templates for the WAN edges in their group
- resource group operator – read-only access to WAN edges within their resource group
- resource group basic – basic access

Resource group admins can create new templates and attach or detach to the WAN edges in their group. They can also copy global templates and re-use them.

Resource group decides which resources the user has access to. However, the level of access is controlled by the existing user group.

- If user is in **resource_group_a** and user group **resource_group_admin**, they have full read/write access to all resources in **resource_group_a**.
- If user is in **resource_group_a** and user group **resource_group_operator**, they have read only access to all resources in **resource_group_a**.
- If user is in **resource_group_a** and user group **resource_group_basic**, they have read only access to interface and system resources in **resource_group_a**.

Global Resource Group

Global group is a special system pre-defined resource group that has different access control rules.

- Users within this group are considered as global-admins, who can have full access to all resources (devices, templates and policies) in the system and they can manage the resource groups and assign resources and users to groups.
- All other users have read-only access to resources within this group.
- The system default admin account (or tenantadmin account in a multi-tenant setup) is always in this group. This privilege cannot be changed. However, the admin account may add/remove other user accounts to or from this group.

IdP (SSO)-Managed Group

An identity provider (IdP) is a service that stores and verifies user identity. IdPs typically work with single sign-on (SSO) providers to authenticate users. If a user is authenticated with a SSO service of an IdP, the group information is also provided and managed by the IDP. An IdP passes the information about the user, including the user name and all the group names, where the user belongs to. Cisco SD-WAN Manager matches the group names with the group names stored in the database to further distinguish if a particular group name passed from IdP is for user group or resource group or VPN group.

Multi-Tenancy Support

With Cisco Catalyst SD-WAN multitenancy, a service provider can manage multiple customers, called tenants, from Cisco SD-WAN Manager. The tenants share Cisco SD-WAN Manager instances, Cisco SD-WAN Validator, and Cisco SD-WAN Controller. The domain name of the service provider has subdomains for each tenant. Cisco SD-WAN Manager is deployed and configured by the service provider. The provider enables multitenancy and creates a Cisco SD-WAN Manager cluster to serve tenants. Only the provider can access a Cisco SD-WAN Manager instance through the SSH terminal.

Provider has the following features:

- resource group is not applicable as the provider manages only the controllers.
- when provider provisions a new tenant, the default user account for the tenant is tenantadmin.
- other user accounts created by the provider are included in the default global resource group.
- when a provider creates a template for a tenant, the template is included in to the global resource group.

RBAC for Policies Overview

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.6.1a and Cisco vManage Release 20.6.1

RBAC for policies allows a user or user group to have selective Read and Write (RW) access to Cisco SD-WAN Manager policies. For example,

- A user with RW access for Cflowd policy can only configure Cflowd policy, but cannot configure application-aware routing policy.
- A user with RW access for application aware routing policy can only configure application-aware routing policy, but cannot configure other policies.

This feature is only supported for centralized and localized policies, but not supported for security policies.

Information About Granular RBAC for Templates

Minimum supported release: Cisco vManage Release 20.7.1

When setting user group permissions, you can use the following template permissions to provide an RBAC user with a specific degree of access to different types of templates. This gives you control over the types of device configurations that an RBAC user can apply.

Permission	Description
CLI Add-On Template	Provides access to the CLI add-on feature template.
Device CLI Template	Provides access to the device CLI template.
SIG Template	Provides access to the SIG feature template and SIG credential template.
Other Feature Templates	Provides access to all feature templates except the SIG feature template, SIG credential template, and CLI add-on feature template.
Feature Profile	Provides access to all feature profiles.
Config Group	Provides access to all the configuration groups.

You can specify granular RBAC for each feature profile by expanding it. With the set permissions to the usergroup, ensure that you are able to access required feature profiles from **Templates > Configuration Groups**.

Single-Tenant and Multi-Tenant Scenarios

You can use granular RBAC for feature templates in single-tenant and multi-tenant Cisco SD-WAN Manager scenarios.

You can create user groups to assign specific permissions to a tenant's various teams, enabling teams to manage only specific network services without granting permission to use device CLI templates. It might be undesirable to give a tenant permission to apply device CLI templates, as the device CLI template can override any other template or device configuration.

For example, you can create a user group for a tenant's security operations group, giving them read/write access only to the SIG Template option, which would enable the security operations group to work on security configuration.

Information About Granular Configuration Task Permissions

From Cisco vManage Release 20.9.1, numerous user permission options are available, providing you fine granularity when assigning a user with permissions to manage specific configuration tasks related to configuration groups and feature profiles.

Information About Assigning Roles Locally to a User Defined by an Identity Provider

Minimum release: Cisco vManage Release 20.11.1

When you define users in an identity provider, such as Okta, for SAML SSO, one attribute that you can define for each user is the role.

When a user logs in to a Cisco SD-WAN Manager instance, Cisco SD-WAN Manager retrieves information about the user from the identity provider, including the user's role or roles. The roles defined in the identity provider map to user group permissions in Cisco SD-WAN Manager. Based on the roles of the user, Cisco SD-WAN Manager provides the user with the permissions defined by the corresponding user group.

You can assign roles locally (not depending on the identity provider) for a user profile that does not have a role defined in the identity provider.

If you have defined roles for a user through the identity provider and have also assigned user groups locally for the same user, the roles defined through the identity provider take priority.

The following table summarizes the ways to provide a user with specific permissions:

Using or Not Using an Identity Provider for SAML SSO	Roles Defined in the Identity Provider	How User Permissions Are Defined
Not using an identity provider	Not applicable	In Cisco SD-WAN Manager, assign a user to one or more user groups locally. This provides the user with the corresponding user group permissions.
Using an identity provider	Identity provider has one or more roles defined for the user.	Define roles for the user through the identity provider. Cisco SD-WAN Manager provides the user with the user group permissions corresponding to the roles.
	Identity provider does not have a role defined for the user.	Use the Remote User option when adding a user (Administration > Manage Users > Add User). See Add a User, on page 207 . In Cisco SD-WAN Manager, assign a user to one or more user groups locally. This provides the user with the corresponding user group permissions.

Benefits of RBAC

Benefits of Granular RBAC for Feature Templates

Minimum supported release: Cisco vManage Release 20.7.1

The permissions that you add for co-management are useful for providing detailed control over access to network configuration. They are useful when using Cisco Catalyst SD-WAN with tenants, enabling you to provide a tenant access to specific types of templates. This enables you to give the tenant self-management of network configuration tasks within the tenant's VPN.

For information about the permissions added for co-management, see [Information About Granular RBAC for Templates](#), on page 157.

Restrictions for Role Based Access Control

General RBAC Restrictions

- Role and scope per user:

In Cisco Catalyst SD-WAN Manager Release 20.13.1, you can only configure one role and one scope per user.

Restrictions for Application Catalog Features

- Enabling or disabling Cloud SaaS feeds:

To enable or disable Cloud SaaS feeds, a user role requires write permission for the Application Priority Write option.

In Cisco Catalyst SD-WAN Manager Release 20.13.x and Cisco Catalyst SD-WAN Manager Release 20.14.x, a user with the security_operations role can enable or disable Cloud SaaS feeds. From Cisco Catalyst SD-WAN Manager Release 20.15.1, the security_operations role does not include write permission for the Application Priority Write option, and does not support enabling or disabling Cloud SaaS feeds.

- Viewing discovered applications

Discovered applications appear on the **Configuration > Application Catalog > Discovered Applications** page.

To enable a custom role to view discovered applications, grant

- read permission for **Cloud OnRamp**, and
- read permission for
 - **Policy Configuration**
 - **Policy Group**
 - **Security Policy Configuration**
 - **Feature Profile > Embedded Security**, or
 - **Feature Profile > Embedded Security > NgFirewall**

- Creating Custom Applications

Discovered applications appear on the **Configuration > Application Catalog > Discovered Applications** page.

To enable a custom role to create custom applications from discovered applications, grant write permission for

- **Policy Configuration**
- **Policy Group**

- **Security Policy Configuration**
- **Feature Profile** > **Embedded Security**, or
- **Feature Profile** > **Embedded Security** > **NgFirewall**.

Restrictions for Granular RBAC for Feature Templates

- Template restriction options:

To use any of the template restriction options that are provided for RBAC for co-management, provide permissions for the **Template Configuration** option. If a specific user role does not have any permissions assigned in the **Template Configuration** option, the **Templates** menu does not appear for the user in Cisco SD-WAN Manager. See [Manage Users](#).

- Applying a template to a device:

To enable an RBAC user to apply templates to devices, provide write permission to the **Template Deploy** option.

Use Cases for RBAC

Use Cases for Assigning Roles Locally to a User Defined by an Identity Provider

Minimum release: Cisco vManage Release 20.11.1

An organization uses the identity provider, Okta, to authenticate users logging in to Cisco SD-WAN Manager.

A user defined through the identity provider has not been assigned any roles. A network administrator with access to Cisco SD-WAN Manager, but no access to the identity provider, can locally assign the user to a specific user group to provide the user with specific permissions.

Configure Role Based Access Control

Configure Scope

1. From the Cisco SD-WAN Manager menu, choose **Administration** > **Users and Access**.
By default **Scope** menu is selected. The table displays the list of scopes configured in the device.
2. Click **Add Scope**.
3. Enter **Scope Name** and **Description**.
4. Click **Add Nodes**.
5. Choose the required **Nodes** and click **Save**.
(Optional) Click **Edit Nodes** to update the existing nodes in the list.
6. (Optional) In the **Associations** pane, click **Add Users** to associate users.

7. In the **Add Users** pop-up window, choose the users that you want to add.
8. Click **Save**.
The selected users are associated to a scope.
9. (Optional) In the **Configurations** tab, click **Add Configurations** to add configurations.
10. In the **Add Configurations** page, choose the available configurations from the following tabs:
 - a. **Configuration Group**
 - b. **Device Template**
 - c. **Feature Template**
 - d. **Feature Profile**
 - e. **Security Policy**
 - f. **Localized Policy**
11. Click **Save**.
A new scope with nodes, users and required configurations is created.

Configure Roles

1. From the Cisco SD-WAN Manager menu, choose **Administration > Users and Access**.
By default **Roles** menu is selected. The table displays the list of scopes configured in the device.
2. Click **Add Role**.
3. Enter **Custom Role Name** in the **Add Custom Role** page.
4. Select the **Deny**, **Read**, or **Write** check box against the feature or sub feature that you want to assign a role.
5. Click **Add**.
6. You can view the new role in the table in the **Roles** page.

Copy Custom Role

1. In the list of roles, for the role you wish to copy, click ..., and click **Copy**.
The **Copy Custom Role** page is displayed.
2. Enter **Custom Role Name**.
3. Select the **Deny**, **Read**, or **Write** check box against the feature or sub feature that you want to update for a role.
4. Click **Copy**.
5. You can view the new role in the table in the **Roles** page.

Edit Custom Role

1. In the list of roles, for the role you wish to copy, click ..., and click **Edit**.
The **Edit Custom Role** page is displayed.
2. Select the **Deny**, **Read**, or **Write** check box against the feature or sub feature that you want to update for a role.



Note Starting from Cisco Catalyst SD-WAN Manager Release 20.18.1, the permissions for a role and its descendents may differ in the Deny/Read/Write table. Therefore, do not assume the parent role as the entire role for the sub-tree under it.

If you have a role with write permissions for Configuration Groups but deny or read permission for deploy, a **Change device variables** button appears instead of **Deploy** button. This button allows you to modify device-specific values during the deploy process, without initiating the deployment.

3. Click **Update**.
4. You can view the updated role in the table in the **Roles** page.

Delete a Role

You can delete a role when it is no longer needed. For example, you might delete a role that you created for a specific project when that project ends.

1. Choose the role you wish to delete, click ..., and click **delete**.
The **Warning** page is displayed.
2. To confirm the deletion of the role, click **Delete**.

Configure Users

Add User

1. From the Cisco SD-WAN Manager menu, choose **Administration > Users and Access**.
2. Click **Users**.
3. Click **Add User**.
4. Configure the following:

Field	Description
Full Name	Enter the full name of the user.
User Name	Enter the user name.
Password	Enter a password.

Field	Description
Remote User	Enable the Remote User option for remote users. If you enable this option, enter an email for the user.
Roles	Choose roles for the user.
Scope	Choose the scope for the user.
Select Locale	(Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.13.1) Choose a locale to set the language for the Cisco SD-WAN Manager user interface.



Note In Cisco Catalyst SD-WAN Manager Release 20.12.1 and earlier releases, Cisco SD-WAN Manager only supported the English language on the user interface. From Cisco Catalyst SD-WAN Manager Release 20.13.1, Cisco SD-WAN Manager user interface supports Canadian French.

- Click **Add** to add the user.

Edit User

- In the **Users** page, for the user you wish to edit, click ..., and click **Edit**.
The **Edit User** page is displayed.
- Enter **Full Name**, **User Name**.
- Choose the role from the **Roles** drop-down list.
- Choose the scope from the **Scope** drop-down list.
- (Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.13.1) Choose the locale from the **Select Locale** drop-down list.
- Click **Update**.

Copy User

- For the user you wish to copy, click ..., and click **Copy**.
The **Copy User** page is displayed.
- Enter **Full Name**, **User Name**.
- Enter the password in the **Password** and **Confirm Password** fields.
- Choose the role from the **Roles** drop-down list.
- Choose the scope from the **Scope** drop-down list.
- (Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.13.1) Choose the locale from the **Select Locale** drop-down list.

7. Click **Copy**.

Delete User

If a user no longer needs access to devices, you can delete the user. Deleting a user does not log out the user if the user is logged in.

1. For the user you wish to delete, click ..., and click **Delete**.
2. To confirm the deletion of the user, click **OK**.

Change User Password

1. For the user you wish to change the password, click ... and click **Change Password**.
2. Enter the **Current User Password**.
3. Enter the new password in the **Password** field.
4. Enter the new password again in the **Confirm Password** field.
5. Click **Update**.

Reset Locked User

1. For the user you wish to reset the lock, click ... and click **Reset Locked User**.
2. In the **Reset Locked User** pop-up menu, click **Yes**.

Administrative Lock

1. For the user you wish to reset the lock, click ... and click **Administrative Lock**.
2. In the **Lock User** pop-up menu, click **Yes**.

Configure User Sessions

User Sessions page shows a list of all the active HTTP sessions within Cisco SD-WAN Manager, including username, domain, source IP address, and so on.

To remove a user session, choose the session from the list, and click **Remove Session**.

Verify RBAC

Verify Granular RBAC Permissions

Minimum supported release: Cisco vManage Release 20.7.1

Use this procedure to verify the permissions that you have configured for a user group.

1. From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users**.

2. Click **User Groups**.
3. In the pane that displays the user groups, select a user group to display the read and write permissions assigned to the user group.
4. Scroll to the permissions that control template access to verify your configuration for the user group.

Monitor RBAC

Monitor devices for VPN Groups

To monitor devices:

1. From the Cisco SD-WAN Manager menu, choose **Monitor > Network**.
2. Click **WAN - Edge**.
3. Select the **VPN Group** and **VPN Segment** for which you want to monitor the network.

A web page displays the list of VPN groups and segments that are configured to a device.



CHAPTER 7

Role-Based Access Control (Cisco IOS XE Catalyst SD-WAN Release 17.12.x and Earlier)

Table 69: Feature History

Feature Name	Release Information	Description
Role-Based Access Control By Resource Group	Cisco IOS XE Catalyst SD-WAN Release 17.5.1a Cisco vManage Release 20.5.1	This feature introduces role-based access control (RBAC) based on sites or resource groups. It is a method of authorizing system access for users based on a combination of user groups and resource groups. For large Cisco Catalyst SD-WAN deployments across multiple geographical locations, this feature helps you to split the network administration among different regional administrators.
RBAC for Policies	Cisco IOS XE Catalyst SD-WAN Release 17.6.1a Cisco vManage Release 20.6.1	This feature allows you to create users and user groups with required read and write permissions for Cisco SD-WAN Manager policies. RBAC for policies provides users with the access to all the details of policies to help maximize the operational efficiency. It makes it easier to meet configuration requirements and guarantees that authorized users on the system are only given access to what they need.
Co-Management: Granular Role-Based Access Control for Feature Templates	Cisco vManage Release 20.7.1	This feature introduces greater granularity in assigning RBAC permissions for template use. This enables you to give a tenant self-management of network configuration tasks. Network administrators and managed service providers can use this feature to assign permissions to their end customers.

Feature Name	Release Information	Description
Co-Management: Improved Granular Configuration Task Permissions	Cisco vManage Release 20.9.1	To enable a user to self-manage specific configuration tasks, you can assign the user permissions to perform specific configuration tasks while excluding other tasks. This feature introduces numerous new permission options, enabling fine granularity in determining which configuration task permissions to provide to a user. .
RBAC for Security Operations and Network Operations Default User Groups	Cisco vManage Release 20.9.1	This feature provides the following default user groups: • network_operations user group for non-security policies • security_operations user group for security policies RBAC for policies allows you to create users and user groups with the required read and write permissions for security and non-security policies. Users can perform configuration and monitoring actions only for the authorized policy type.
Co-Management: Improved Granular Configuration for Resource group features	Cisco vManage Release 20.11.1	To enable a user to self-manage specific configuration tasks, you can assign the user permissions to perform specific configuration tasks while excluding other tasks. This feature introduces new permission options for the following configuration groups and feature profiles. • AppQoS under other feature profile • GPS under transport feature profile • Cisco VPN Interface GRE under WAN/LAN profile. • Cisco VPN Interface IPsec under WAN profile. • Cisco Multicast under LAN profile. • UCSE under other feature profile. • IPv4 Tracker and Tracker Group under transport and service feature profiles. • IPv6 DIA Tracker and Tracker Group, under transport feature profile.

Feature Name	Release Information	Description
Assigning Roles Locally for SSO-Authenticated Users	Cisco vManage Release 20.11.1	If you are using an identity provider, such as Okta, for security assertion markup language (SAML)-based single sign-on (SSO), then in most use cases, you define user roles through the identity provider. This feature enables you to assign user groups locally in Cisco SD-WAN Manager, in case no roles are defined for the user by the identity provider.

- [Information About RBAC, on page 169](#)
- [Restrictions for RBAC, on page 183](#)
- [Use Cases for RBAC, on page 184](#)
- [Configure RBAC, on page 184](#)
- [Configure RBAC Using the CLI, on page 214](#)
- [Verify RBAC, on page 216](#)
- [Monitor RBAC, on page 216](#)

Information About RBAC

Role-Based Access Control by VPN

Role-based access control (RBAC) is the process of restricting user access to network configurations and resources. In RBAC, users are assigned roles depending on the resources they need access to. The RBAC by VPN feature helps you to manage and control access to your network based on the VPNs. It involves setting permissions and privileges to enable access to authorized users.

RBAC by VPN

Role-based access by VPN allows a network administrator to define VPN groups with one or more network segments. The network administrator can associate a user with a VPN group that restricts user access to devices in the network and features of Cisco SD-WAN Manager.

RBAC by VPN provides the following restricted access to users configured with a VPN group:

- Access to VPN Dashboard
- Monitor devices, network, and application status via VPN dashboard
- VPN dashboard information restricted to devices with segments in the VPN group
- Monitor option restricted to devices with segments in the VPN group
- Interface monitoring on each device restricted to interfaces of segments in the VPN group

VPN Dashboard Overview

Users configured with VPN group can access only the VPN Dashboard, and it is read-only access. User with Admin access can create the VPN groups and has access to both Admin Dashboard and VPN Dashboard(s). Admin user can access these dashboards by choosing **Dashboard** from the Cisco SD-WAN Manager menu.

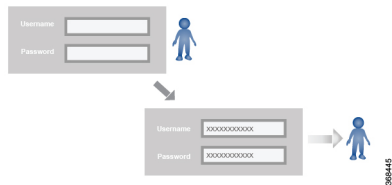
Role-Based Access with AAA

The Cisco Catalyst SD-WAN AAA software implements role-based access to control the authorization permissions for users on Cisco IOS XE Catalyst SD-WAN devices. Role-based access consists of three components:

- Users are those who are allowed to log in to a Cisco IOS XE Catalyst SD-WAN device.
- User groups are collections of users.
- Privileges are associated with each group. They define the commands that the group's users are authorized to issue.

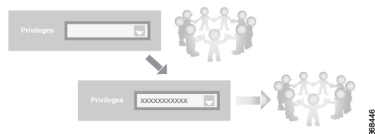
Users and User Groups

All users who are permitted to perform operations on a Cisco IOS XE Catalyst SD-WAN device must have a login account. For the login account, you configure a username and a password on the device itself. These allow the user to log in to that device. A username and password must be configured on each device that a user is allowed to access.

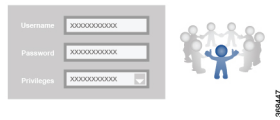


The Cisco Catalyst SD-WAN software provides one standard username, **admin**, which is a user who has full administrative privileges, similar to a UNIX superuser. By default, the **admin** username password is **admin**. You cannot delete or modify this username, but you can and should change the default password.

User groups pool together users who have common roles, or privileges, on the Cisco IOS XE Catalyst SD-WAN device. As part of configuring the login account information, you specify which user group or groups that user is a member of. You do not need to specify a group for the **admin** user, because this user is automatically in the user group **netadmin** and is permitted to perform all operations on the Cisco IOS XE Catalyst SD-WAN device.



The user group itself is where you configure the privileges associated with that group. These privileges correspond to the specific commands that the user is permitted to execute, effectively defining the role-based access to the Cisco Catalyst SD-WAN software elements.



The Cisco Catalyst SD-WAN software provides the following standard user groups:

- **basic**: The basic group is a configurable group and can be used for any users and privilege levels. This group is designed to include users who have permission to both view and modify information on the device.
- **operator**: The operator group is also a configurable group and can be used for any users and privilege levels. This group is designed to include users who have permission only to view information.
- **netadmin**: The netadmin group is a non-configurable group. By default, this group includes the **admin** user. You can add other users to this group. Users in this group are permitted to perform all operations on the device.
- Minimum supported release: Cisco vManage Release 20.9.1

network_operations: The **network_operations** group is a non-configurable group. Users in this group can perform all non-security-policy operations on the device and only view security policy information. For example, users can create or modify template configurations, manage disaster recovery, and create non-security policies such as application aware routing policy or CFlowD policy.

- Minimum supported release: Cisco vManage Release 20.9.1

security_operations: The **security_operations** group is a non-configurable group. Users in this group can perform all security operations on the device and only view non-security-policy information. For example, users can manage umbrella keys, licensing, IPS signatures auto update, TLS/SSL proxy settings, and so on.

Users of the **network_operations** group are authorized to apply policies to a device, revoke applied policies, and edit device templates. Users of the **security_operations** group require **network_operations** users to intervene on day-0 to deploy security policy on a device and on day-N to remove a deployed security policy. However, after a security policy is deployed on a device, **security_operations** users can modify the security policy without needing the **network_operations** users to intervene.



Note Only admin users can view running and local configuration. Users associated with predefined operator user group do not have access to the running and local configurations. The predefined user group operator has only read access for the template configuration. If you need only a subset of admin user privileges, then you need to create a new user group with the selected features from the features list with both read and write access and associate the group with the custom user.

Privileges for Role-Based Access

Role-based access privileges are arranged into five categories, which are called *tasks*:

- **Interface**—Privileges for controlling the interfaces on the Cisco IOS XE Catalyst SD-WAN device.
- **Policy**—Privileges for controlling control plane policy, OMP, and data plane policy.
- **Routing**—Privileges for controlling the routing protocols, including BFD, BGP, OMP, and OSPF.

- Security—Privileges for controlling the security of the device, including installing software and certificates. Only users belonging to the **netadmin** group can install software on the system.
- System—General systemwide privileges.

The tables in the following sections detail the AAA authorization rules for users and user groups. These authorization rules apply to commands issued from the CLI and to those issued from Netconf.

User Authorization Rules for Operational Commands

The user authorization rules for operational commands are based simply on the username. Any user who is allowed to log in to the Cisco IOS XE Catalyst SD-WAN device can execute most operational commands. However, only the **admin** user can issue commands that affect the fundamental operation of the device, such as installing and upgrading the software and shutting down the device.

Note that any user can issue the **config** command to enter configuration mode, and once in configuration mode, they are allowed to issue any general configuration command. Also, any user is allowed to configure their password by issuing the **system aaa user self password password** command and then committing that configuration change. For the actual commands that configure device operation, authorization is defined according to user group membership. See User Group Authorization Rules for Configuration Commands.

The following tables lists the AAA authorization rules for general CLI commands. All the commands are operational commands except as noted. Also, some commands available to the "admin" user are available only if that user is in the "netadmin" user group.

CLI Command	Any User	Admin User
clear history	X	X
commit confirm	X	X
complete-on-space	X	X
config	X	X
exit	X	X
file	X	X
help	X	X
[no] history	X	X
idle-timeout	X	X
job	X	X
logout	—	X (users in netadmin group only)
monitor	X	X
nslookup	X	X
paginate	X	X

CLI Command	Any User	Admin User
ping	X (users in netadmin group only)	X (users in netadmin group only)
poweroff	—	X(users in netadmin group only)
prompt1	X	X
prompt2	X	X
quit	X	X
reboot	—	X (users in netadmin group only)
request aaa request admin-tech request firmware request interface-reset request nms request reset request software	—	X (users in netadmin group only)
request execute request download request upload	X	X
request (everything else)	—	X
rollback (configuration mode command)	—	X (users in netadmin group only)
screen-length	X	X
screen-width	X	X
show cli	X	X
show configuration commit list	X	X
show history	X	X
show jobs	X	X
show parser dump	X	X
show running-config	X	X
show users	X	X
system aaa user <i>self</i> password <i>password</i> (configuration mode command) (Note: A user cannot delete themselves)		
tcpdump	X	X
timestamp	X	X
tools ip-route	X	X

CLI Command	Any User	Admin User
tools netstat	X	X
tools nping	X	X
tracert	X	X
vshell	X (The availability of vshell command is unavailable to all users that are not in netadmin group in Cisco vManage Release 20.9.5.)	X (The vshell AAA authorized access is limited only to users that are in netadmin group.)

User Group Authorization Rules for Operational Commands

The following table lists the user group authorization roles for operational commands.

Operational Command	Interface	Policy	Routing	Security	System
clear app		X			
clear app-route		X			
clear arp	X				
clear bfd			X		X
clear bgp			X		X
clear bridge	X				
clear cellular	X				
clear control				X	
clear crash					X
clear dhcp					X
clear dns					X
clear igmp			X		
clear installed-certificates				X	
clear interface	X				
clear ip			X		
clear notification					X
clear omp			X		

Operational Command	Interface	Policy	Routing	Security	System
clear orchestrator				X	
clear ospf			X		
clear pim			X		
clear policy		X			
clear pppoe	X				
clear system					X
clear tunnel				X	
clear wlan	X				
clear ztp				X	X
clock					X
debug bgp			X		
debug cellular	X				
debug cflowd		X			
debug chmgr					X
debug config-mgr					X
debug dhcp-client					X
debug dhcp-helper					X
debug dhcp-server					X
debug fpm		X			
debug ftm					X
debug igmp			X		
debug netconf					X
debug omp			X		
debug ospf			X		
debug pim			X		
debug resolver			X		
debug snmp					X
debug sysmgr					X

Operational Command	Interface	Policy	Routing	Security	System
debug transport					X
debug ttm					X
debug vdaemon				X	X
debug vrrp				X	
debug wlan	X				
request certificate				X	
request control-tunnel				X	
request controller				X	
request controller-upload				X	
request csr				X	
request device				X	
request device-upload				X	
request on-vbond-controller				X	
request port-hop				X	
request root-cert-chain				X	
request security				X	
request vedge				X	
request vedge-upload				X	
request vsmart-upload				X	
show aaa					X
show app		X			
show app-route		X			
show arp	X				
show bfd			X		X
show bgp			X		
show boot-partition					X
show bridge	X				

Operational Command	Interface	Policy	Routing	Security	System
show cellular	X				
show certificate				X	
show clock					X
show control				X	X
show crash					X
show debugs—same as debug commands					
show dhcp					X
show external-nat				X	X
show hardware					X
show igmp			X		
show interface	X				
show ip			X		X
show ipsec				X	
show licenses					X
show logging					X
show multicast			X		
show nms-server					X
show notification					X
show ntp					X
show omp		X	X		X
show orchestrator				X	
show ospf			X		
show pim			X		
show policer		X			
show policy		X			
show ppp	X				
show pppoe	X				

Operational Command	Interface	Policy	Routing	Security	System
show reboot					X
show security-info				X	
show software					X
show system					X
show transport					X
show tunnel				X	
show uptime					X
show users					X
show version					X
show vrrp	X				
show wlan	X				
show ztp				X	

User Group Authorization Rules for Configuration Commands

The following table lists the user group authorization rules for configuration commands.

Configuration Command	Interface	Policy	Routing	Security	System
apply-policy		X			
banner					X
bfd			X		X
bridge	X				
omp		X	X		X
policy		X			
security				X	X
snmp					X
system					X
vpn interface	X				
vpn ip			X		
vpn router			X		

Configuration Command	Interface	Policy	Routing	Security	System
vpn service			X		
vpn (everything else, including creating, deleting, and naming)					X
wlan	X				

RBAC By Resource Group Overview

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.5.1a and Cisco vManage Release 20.5.1

RBAC by resource groups is a method of restricting or authorizing system access for users based on user groups and resource groups. A user group defines the privileges of a user in the system and the resource group defines the organizations (domains) to which a user is allowed access. Because users are not directly assigned privileges, management of individual user privileges is simply a matter of assigning the appropriate user and resource groups.

For large Cisco Catalyst SD-WAN deployments across multiple geographical locations, you can split the network administration among different regional administrators.

Based on the user groups and resources groups to which network administrators are assigned, we can broadly classify them as Global Administrators and Regional Administrators. Global administrators have access to resources in every resource group and have full read-write privileges for all the features. Regional Administrators group have full read-write privileges for all the features, but the resources they can access is controlled by the resource groups to which they are assigned.

Global Admin

User accounts in the global resource group have access to all resources. A global admin is responsible for overseeing the entire network, but not involved in the operations of the individual devices on a daily basis. The global admin can assign devices to their corresponding regions, assign the regional admin accounts, manage the controllers, maintain sharable and centralized configurations, and when necessary, operate on the individual devices.

Any user in a single tenant setup with netadmin privileges and also part of global resource group is considered as global admin. Default admin user on Cisco SD-WAN Manager is also a global-admin, and that user can assign more global-admins. Global resource group encompasses all the WAN edges, controllers in the single view.

Global admin can switch to view only a specific resource group and can create templates. Local resource group admins, also called regional admins can clone the global templates and reuse them within their resource groups.

Regional Admin

The regional admins are responsible for day-to-day operations (configuration, monitoring, onboarding, and so on) for devices in their corresponding regions. They should not have access to or visibility into devices outside of their region. The following user groups can be created:

- resource group admin – full read/write access to devices in the corresponding resource group, can troubleshoot, monitor, attach or detach templates for the WAN edges in their group
- resource group operator – read-only access to WAN edges within their resource group
- resource group basic – basic access

Resource group admins can create new templates and attach or detach to the WAN edges in their group. They can also copy global templates and re-use them.

Resource group decides which resources the user has access to. However, the level of access is controlled by the existing user group.

- If user is in **resource_group_a** and user group **resource_group_admin**, they have full read/write access to all resources in **resource_group_a**.
- If user is in **resource_group_a** and user group **resource_group_operator**, they have read only access to all resources in **resource_group_a**.
- If user is in **resource_group_a** and user group **resource_group_basic**, they have read only access to interface and system resources in **resource_group_a**.

Global Resource Group

Global group is a special system pre-defined resource group that has different access control rules.

- Users within this group are considered as global-admins, who can have full access to all resources (devices, templates and policies) in the system and they can manage the resource groups and assign resources and users to groups.
- All other users have read-only access to resources within this group.
- The system default admin account (or tenantadmin account in a multi-tenant setup) is always in this group. This privilege cannot be changed. However, the admin account may add/remove other user accounts to or from this group.

IdP (SSO)-Managed Group

An identity provider (IdP) is a service that stores and verifies user identity. IdPs typically work with single sign-on (SSO) providers to authenticate users. If a user is authenticated with a SSO service of an IdP, the group information is also provided and managed by the IDP. An IdP passes the information about the user, including the user name and all the group names, where the user belongs to. Cisco SD-WAN Manager matches the group names with the group names stored in the database to further distinguish if a particular group name passed from IdP is for user group or resource group or VPN group.

Multi-Tenancy Support

With Cisco Catalyst SD-WAN multitenancy, a service provider can manage multiple customers, called tenants, from Cisco SD-WAN Manager. The tenants share Cisco SD-WAN Manager instances, Cisco SD-WAN Validator, and Cisco SD-WAN Controller. The domain name of the service provider has subdomains for each tenant. Cisco SD-WAN Manager is deployed and configured by the service provider. The provider enables multitenancy and creates a Cisco SD-WAN Manager cluster to serve tenants. Only the provider can access a Cisco SD-WAN Manager instance through the SSH terminal.

Provider has the following features:

- resource group is not applicable as the provider manages only the controllers.
- when provider provisions a new tenant, the default user account for the tenant is tenantadmin.
- other user accounts created by the provider are included in the default global resource group.
- when a provider creates a template for a tenant, the template is included in to the global resource group.

RBAC for Policies Overview

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.6.1a and Cisco vManage Release 20.6.1

RBAC for policies allows a user or user group to have selective Read and Write (RW) access to Cisco SD-WAN Manager policies. For example,

- A user with RW access for Cflowd policy can only configure Cflowd policy, but cannot configure application-aware routing policy.
- A user with RW access for application aware routing policy can only configure application-aware routing policy, but cannot configure other policies.

This feature is only supported for centralized and localized policies, but not supported for security policies.

Information About Granular RBAC for Templates

Minimum supported release: Cisco vManage Release 20.7.1

When setting user group permissions, you can use the following template permissions to provide an RBAC user with a specific degree of access to different types of templates. This gives you control over the types of device configurations that an RBAC user can apply.

Permission	Description
CLI Add-On Template	Provides access to the CLI add-on feature template.
Device CLI Template	Provides access to the device CLI template.
SIG Template	Provides access to the SIG feature template and SIG credential template.
Other Feature Templates	Provides access to all feature templates except the SIG feature template, SIG credential template, and CLI add-on feature template.
Feature Profile	Provides access to all feature profiles.
Config Group	Provides access to all the configuration groups.

You can specify granular RBAC for each feature profile by expanding it. With the set permissions to the usergroup, ensure that you are able to access required feature profiles from **Templates > Configuration Groups**.

Single-Tenant and Multi-Tenant Scenarios

You can use granular RBAC for feature templates in single-tenant and multi-tenant Cisco SD-WAN Manager scenarios.

You can create user groups to assign specific permissions to a tenant's various teams, enabling teams to manage only specific network services without granting permission to use device CLI templates. It might be undesirable to give a tenant permission to apply device CLI templates, as the device CLI template can override any other template or device configuration.

For example, you can create a user group for a tenant's security operations group, giving them read/write access only to the SIG Template option, which would enable the security operations group to work on security configuration.

Information About Granular Configuration Task Permissions

From Cisco vManage Release 20.9.1, numerous user permission options are available, providing you fine granularity when assigning a user with permissions to manage specific configuration tasks related to configuration groups and feature profiles.

Information About Assigning Roles Locally to a User Defined by an Identity Provider

Minimum release: Cisco vManage Release 20.11.1

When you define users in an identity provider, such as Okta, for SAML SSO, one attribute that you can define for each user is the role.

When a user logs in to a Cisco SD-WAN Manager instance, Cisco SD-WAN Manager retrieves information about the user from the identity provider, including the user's role or roles. The roles defined in the identity provider map to user group permissions in Cisco SD-WAN Manager. Based on the roles of the user, Cisco SD-WAN Manager provides the user with the permissions defined by the corresponding user group.

You can assign roles locally (not depending on the identity provider) for a user profile that does not have a role defined in the identity provider.

If you have defined roles for a user through the identity provider and have also assigned user groups locally for the same user, the roles defined through the identity provider take priority.

The following table summarizes the ways to provide a user with specific permissions:

Using or Not Using an Identity Provider for SAML SSO	Roles Defined in the Identity Provider	How User Permissions Are Defined
Not using an identity provider	Not applicable	In Cisco SD-WAN Manager, assign a user to one or more user groups locally. This provides the user with the corresponding user group permissions.

Using or Not Using an Identity Provider for SAML SSO	Roles Defined in the Identity Provider	How User Permissions Are Defined
Using an identity provider	Identity provider has one or more roles defined for the user.	Define roles for the user through the identity provider. Cisco SD-WAN Manager provides the user with the user group permissions corresponding to the roles.
	Identity provider does not have a role defined for the user.	Use the Remote User option when adding a user (Administration > Manage Users > Add User). See Add a User, on page 207 . In Cisco SD-WAN Manager, assign a user to one or more user groups locally. This provides the user with the corresponding user group permissions.

Benefits of RBAC

Benefits of Granular RBAC for Feature Templates

Minimum supported release: Cisco vManage Release 20.7.1

The permissions that you add for co-management are useful for providing detailed control over access to network configuration. They are useful when using Cisco Catalyst SD-WAN with tenants, enabling you to provide a tenant access to specific types of templates. This enables you to give the tenant self-management of network configuration tasks within the tenant's VPN.

For information about the permissions added for co-management, see [Information About Granular RBAC for Templates, on page 157](#).

Restrictions for RBAC

Restrictions for Granular RBAC for Feature Templates

Minimum supported release: Cisco vManage Release 20.7.1

- To use any of the template restriction options that are provided for RBAC for co-management, provide permissions for the **Template Configuration** option. If a specific user role does not have any permissions assigned in the **Template Configuration** option, the **Templates** menu does not appear for the user in Cisco SD-WAN Manager. See [Manage Users](#).
- To enable an RBAC user to apply templates to devices, provide **Write** permission to the **Template Deploy** option.

Use Cases for RBAC

Use Cases for Assigning Roles Locally to a User Defined by an Identity Provider

Minimum release: Cisco vManage Release 20.11.1

An organization uses the identity provider, Okta, to authenticate users logging in to Cisco SD-WAN Manager.

A user defined through the identity provider has not been assigned any roles. A network administrator with access to Cisco SD-WAN Manager, but no access to the identity provider, can locally assign the user to a specific user group to provide the user with specific permissions.

Configure RBAC

Manage Users

From the Cisco SD-WAN Manager menu, choose **Administration** > **Manage Users** to add, edit, view, or delete users and user groups.

Please note the following:

- Only a user logged in as the **admin** user or a user who has Manage Users write permission can add, edit, or delete users and user groups from Cisco SD-WAN Manager.
- Each user group can have read or write permission for the features listed in this section. Write permission includes Read permission.
- All user groups, regardless of the read or write permissions selected, can view the information displayed in the Cisco SD-WAN Manager Dashboard.

Table 70: User Group Permissions for Different Device Types

Permissions	See This Section
User group permissions related to Cisco IOS XE Catalyst SD-WAN device configuration.	User Group Permissions: Cisco IOS XE Catalyst SD-WAN Devices
User group permissions related to Cisco Catalyst Wireless Gateway device configuration.	User Group Permissions: Cisco Catalyst Wireless Gateway Devices

User Group Permissions: Cisco IOS XE Catalyst SD-WAN device

Table 71: User Group Permissions: Cisco IOS XE Catalyst SD-WAN devices

Feature	Read Permission	Write Permission
Alarms	Set alarm filters and view the alarms generated on the devices on the Monitor > Logs > Alarms page. Cisco vManage Release 20.6.x and earlier: Set alarm filters and view the alarms generated on the devices on the Monitor > Alarms page.	No additional permissions.
Audit Log	Set audit log filters and view a log of all the activities on the devices on the Monitor > Logs > Alarms page and the Monitor > Logs > Audit Log page. Cisco vManage Release 20.6.x and earlier: Set audit log filters and view a log of all the activities on the devices on the Monitor > Alarms page and the Monitor > Audit Log page.	No additional permissions.
Certificates	View a list of the devices in the overlay network under Configuration > Certificates > WAN Edge List. View a certificate signing request (CSR) and certificate on the Configuration > Certificates > Controllers window. Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the Controllers tab is renamed as the Control Components tab to stay consistent with Cisco Catalyst SD-WAN rebranding.	Validate and invalidate a device, stage a device, and send the serial number of valid controller devices to the Cisco Catalyst SD-WAN Validator on the Configuration > Certificates > WAN Edge List window. Generate a CSR, install a signed certificate, reset the RSA key pair, and invalidate a controller device on the Configuration > Certificates > Controllers window. Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the Controllers tab is renamed as the Control Components tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

Feature	Read Permission	Write Permission
CLI Add-On Template (Minimum supported release: Cisco vManage Release 20.7.1)	View the CLI add-on feature template on the Configuration > Templates window. Note This operation requires read permission for Template Configuration .	Create, edit, delete, and copy a CLI add-on feature template on the Configuration > Templates window. Note These operations require write permission for Template Configuration . Note For information about this option, see Information About Granular RBAC for Feature Templates
Cloud OnRamp	View the cloud applications on the Configuration > Cloud OnRamp for SaaS and Configuration > Cloud OnRamp for IaaS window.	No additional permissions.
Cluster	View information about the services running on Cisco SD-WAN Manager, a list of devices connected to a Cisco SD-WAN Manager server, and the services that are available and running on all the Cisco SD-WAN Manager servers in the cluster on the Administration > Cluster Management window.	Change the IP address of the current Cisco SD-WAN Manager, add a Cisco SD-WAN Manager server to the cluster, configure the statistics database, edit, and remove a Cisco SD-WAN Manager server from the cluster on the Administration > Cluster Management window.
Colocation	View the cloud applications on the Configuration > Cloud OnRamp for Colocation window.	No additional permissions.
Config Group > Device > Deploy (Minimum supported release: Cisco vManage Release 20.9.1)	This permission does not provide any functionality.	Deploy a configuration onto Cisco IOS XE Catalyst SD-WAN devices. Note To edit an existing feature configuration requires write permission for Template Configuration . For more details on deploying devices, see Deploy Devices .

Feature	Read Permission	Write Permission
Device CLI Template (Minimum supported release: Cisco vManage Release 20.7.1)	View the device CLI template on the Configuration > Templates window. Note This operation requires read permission for Template Configuration .	Create, edit, delete, and copy a device CLI template on the Configuration > Templates window. Note These operations require write permission for Template Configuration . Note For information about this option, see Information About Granular RBAC for Feature Templates
Device Inventory	View the running and local configuration of devices, a log of template activities, and the status of attaching configuration templates to devices on the Configuration > Devices > WAN Edge List window. View the running and local configuration of the devices and the status of attaching configuration templates to controller devices on the Configuration > Devices > Controllers window. Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the Controllers tab is renamed as the Control Components tab to stay consistent with Cisco Catalyst SD-WAN rebranding.	Upload a device's authorized serial number file to Cisco SD-WAN Manager, toggle a device from Cisco SD-WAN Manager configuration mode to CLI mode, copy a device configuration, and delete the device from the network on the Configuration > Devices > WAN Edge List window. Add and delete controller devices from the overlay network, and edit the IP address and login credentials of a controller device on the Configuration > Devices > Controllers window. Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the Controllers tab is renamed as the Control Components tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

Feature	Read Permission	Write Permission
Device Monitoring	View the geographic location of the devices on the Monitor > Geography window. View events that have occurred on the devices on the Monitor > Logs > Events page. Cisco vManage Release 20.6.x and earlier: View events that have occurred on the devices on the Monitor > Events page. View a list of devices in the network, along with device status summary, SD-WAN Application Intelligence Engine (SAIE) and Cflowd flow information, transport location (TLOC) loss, latency, and jitter information, control and tunnel connections, system status, and events on the Monitor > Devices page (only when a device is selected). Note In Cisco vManage Release 20.7.x and earlier releases, the SAIE flow is called the deep packet inspection (DPI) flow. Cisco vManage Release 20.6.x and earlier: Device information is available in the Monitor > Network page.	Ping a device, run a traceroute, and analyze the traffic path for an IP packet on the Monitor > Devices page (only when a device is selected). Note These operations require read and write permissions for Device Monitoring.
Device Reboot	View the list of devices on which the reboot operation can be performed on the Maintenance > Device Reboot window.	Reboot one or more devices on the Maintenance > Device Reboot window.
Disaster Recovery	View information about active and standby clusters running on Cisco SD-WAN Manager on the Administration > Disaster Recovery window.	No additional permissions.

Feature	Read Permission	Write Permission
Events	View the geographic location of the devices on the Monitor > Logs > Events page. View the geographic location of the devices on the Monitor > Events page.	Ping a device, run a traceroute, and analyze the traffic path for an IP packet on the Monitor > Logs > Events page (only when a device is selected).
Feature Profile > Other > Thousandeyes (Minimum supported release: Cisco vManage Release 20.9.1)	View the ThousandEyes settings on the Configuration > Templates > (View configuration group) page, in the Other Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the ThousandEyes settings on the Configuration > Templates > (Add or edit configuration group) page, in the Other Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > Service > Dhcp (Minimum supported release: Cisco vManage Release 20.9.1)	View the DHCP settings on the Configuration > Templates > (View configuration group) page, in the Service Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the DHCP settings on the Configuration > Templates > (Add or edit configuration group) page, in the Service Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > Service > Lan/Vpn (Minimum supported release: Cisco vManage Release 20.9.1)	View the LAN/VPN settings on the Configuration > Templates > (View configuration group) page, in the Service Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the LAN/VPN settings on the Configuration > Templates > (Add or edit configuration group) page, in the Service Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > Service > Lan/Vpn/Interface/Ethernet (Minimum supported release: Cisco vManage Release 20.9.1)	View the Ethernet Interface settings on the Configuration > Templates > (View configuration group) page, in the Service Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Ethernet Interface settings on the Configuration > Templates > (Add or edit configuration group) page, in the Service Profile section. Note These operations require write permission for Template Configuration .

Feature	Read Permission	Write Permission
Feature Profile > Service > Lan/Vpn/Interface/Svi (Minimum supported release: Cisco vManage Release 20.9.1)	View the SVI Interface settings on the Configuration > Templates > (View configuration group) page, in the Service Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the SVI Interface settings on the Configuration > Templates > (Add or edit configuration group) page, in the Service Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > Service > Routing/Bgp (Minimum supported release: Cisco vManage Release 20.9.1)	View the Routing/BGP settings on the Configuration > Templates > (View configuration group) page, in the Service Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Routing/BGP settings on the Configuration > Templates > (Add or edit configuration group) page, in the Service Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > Service > Routing/Ospf (Minimum supported release: Cisco vManage Release 20.9.1)	View the Routing/OSPF settings on the Configuration > Templates > (View configuration group) page, in the Service Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Routing/OSPF settings on the Configuration > Templates > (Add or edit configuration group) page, in the Service Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > Service > Switchport (Minimum supported release: Cisco vManage Release 20.9.1)	View the Switchport settings on the Configuration > Templates > (View configuration group) page, in the Service Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Switchport settings on the Configuration > Templates > (Add or edit configuration group) page, in the Service Profile section. Note These operations require write permission for Template Configuration .

Feature	Read Permission	Write Permission
Feature Profile > Service > Wirelesslan (Minimum supported release: Cisco vManage Release 20.9.1)	View the Wireless LAN settings on the Configuration > Templates > (View configuration group) page, in the Service Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Wireless LAN settings on the Configuration > Templates > (Add or edit configuration group) page, in the Service Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > System > Interface/Ethernet > Aaa (Minimum supported release: Cisco vManage Release 20.9.1)	View the AAA settings on the Configuration > Templates > (View configuration group) page, in the System Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the AAA settings on the Configuration > Templates > (Add or edit configuration group) page, in the System Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > System > Interface/Ethernet > Banner (Minimum supported release: Cisco vManage Release 20.9.1)	View the Banner settings on the Configuration > Templates > (View configuration group) page, in the System Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Banner settings on the Configuration > Templates > (Add or edit configuration group) page, in the System Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > System > Basic (Minimum supported release: Cisco vManage Release 20.9.1)	View the Basic settings on the Configuration > Templates > (View configuration group) page, in the System Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Basic settings on the Configuration > Templates > (Add or edit configuration group) page, in the System Profile section. Note These operations require write permission for Template Configuration .

Feature	Read Permission	Write Permission
Feature Profile > System > Bfd (Minimum supported release: Cisco vManage Release 20.9.1)	View the BFD settings on the Configuration > Templates > (View configuration group) page, in the System Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the BFD settings on the Configuration > Templates > (Add or edit configuration group) page, in the System Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > System > Global (Minimum supported release: Cisco vManage Release 20.9.1)	View the Global settings on the Configuration > Templates > (View configuration group) page, in the System Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Global settings on the Configuration > Templates > (Add or edit configuration group) page, in the System Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > System > Logging (Minimum supported release: Cisco vManage Release 20.9.1)	View the Logging settings on the Configuration > Templates > (View configuration group) page, in the System Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Logging settings on the Configuration > Templates > (Add or edit configuration group) page, in the System Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > System > Ntp (Minimum supported release: Cisco vManage Release 20.9.1)	View the NTP settings on the Configuration > Templates > (View configuration group) page, in the System Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the NTP settings on the Configuration > Templates > (Add or edit configuration group) page, in the System Profile section. Note These operations require write permission for Template Configuration .

Feature	Read Permission	Write Permission
Feature Profile > System > Omp (Minimum supported release: Cisco vManage Release 20.9.1)	View the OMP settings on the Configuration > Templates > (View configuration group) page, in the System Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the OMP settings on the Configuration > Templates > (Add or edit configuration group) page, in the System Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > System > Snmp (Minimum supported release: Cisco vManage Release 20.9.1)	View the SNMP settings on the Configuration > Templates > (View configuration group) page, in the System Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the SNMP settings on the Configuration > Templates > (Add or edit configuration group) page, in the System Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > Transport > Cellular Controller (Minimum supported release: Cisco vManage Release 20.9.1)	View the Cellular Controller settings on the Configuration > Templates > (View a configuration group) page, in the Transport & Management Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Cellular Controller settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Transport & Management Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > Transport > Cellular Profile (Minimum supported release: Cisco vManage Release 20.9.1)	View the Cellular Profile settings on the Configuration > Templates > (View a configuration group) page, in the Transport & Management Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Cellular Profile settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Transport & Management Profile section. Note These operations require write permission for Template Configuration .

Feature	Read Permission	Write Permission
Feature Profile > Transport > Management/Vpn (Minimum supported release: Cisco vManage Release 20.9.1)	View the Management VPN settings on the Configuration > Templates > (View configuration group) page, in the Transport & Management Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Management VPN settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Transport & Management Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > Transport > Management/Vpn/Interface/Ethernet (Minimum supported release: Cisco vManage Release 20.9.1)	View the Management Ethernet Interface settings on the Configuration > Templates > (View configuration group) page, in the Transport & Management Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Management VPN and Management Internet Interface settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Transport & Management Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > Transport > Routing/Bgp (Minimum supported release: Cisco vManage Release 20.9.1)	View the BGP Routing settings on the Configuration > Templates > (View configuration group) page, in the Transport & Management Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the BGP Routing settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Transport & Management Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > Transport > Tracker (Minimum supported release: Cisco vManage Release 20.9.1)	View the Tracker settings on the Configuration > Templates > (View configuration group) page, in the Transport & Management Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Tracker settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Transport & Management Profile section. Note These operations require write permission for Template Configuration .

Feature	Read Permission	Write Permission
Feature Profile > Transport > Wan/Vpn (Minimum supported release: Cisco vManage Release 20.9.1)	View the Wan/Vpn settings on the Configuration > Templates > (View configuration group) page, in the Transport & Management Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Wan/Vpn settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Transport & Management Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > Transport > Wan/Vpn/Interface/Cellular (Minimum supported release: Cisco vManage Release 20.9.1)	View the Wan/Vpn/Interface/Cellular settings on the Configuration > Templates > (View configuration group) page, in the Transport & Management Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Wan/Vpn/Interface/Cellular settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Transport & Management Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > Transport > Wan/Vpn/Interface/Ethernet (Minimum supported release: Cisco vManage Release 20.9.1)	View the Wan/Vpn/Interface/Ethernet settings on the Configuration > Templates > (View configuration group) page, in the Transport & Management Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Wan/Vpn/Interface/Ethernet settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Transport & Management Profile section. Note These operations require write permission for Template Configuration .
Integration Management	View information about controllers running on Cisco SD-WAN Manager, on the Administration > Integration Management window.	No additional permissions.

Feature	Read Permission	Write Permission
License Management	View license information of devices running on Cisco SD-WAN Manager, on the Administration > License Management window.	On the Administration > License Management page, configure use of a Cisco Smart Account, choose licenses to manage, and synchronize license information between Cisco SD-WAN Manager and the license server.
Interface	View information about the interfaces on a device on the Monitor > Devices > Interface page. Cisco vManage Release 20.6.x and earlier: View information about the interfaces on a device on the Monitor > Network > Interface page	Edit Chart Options to select the type of data to display, and edit the time period for which to display data on the Monitor > Devices > Interface page.
Application Monitoring (Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.12.1)	View the application health of the devices on the Monitor > Applications window.	View the application health of the devices on the Monitor > Applications window.
Manage Users	View users and user groups on the Administration > Manage Users window.	Add, edit, and delete users and user groups from Cisco SD-WAN Manager, and edit user group privileges on the Administration > Manage Users window.

Feature	Read Permission	Write Permission
Other Feature Templates (Minimum supported release: Cisco vManage Release 20.7.1)	View all feature templates except the SIG feature template, SIG credential template, and CLI add-on feature template on the Configuration > Templates window. Note This operation requires read permission for Template Configuration. Note To check the mutual authentication option, you need read permission for certificates. (Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.12.1)	Create, edit, delete, and copy all feature templates except the SIG feature template, SIG credential template, and CLI add-on feature template on the Configuration > Templates window. Note These operations require write permission for Template Configuration. Note For information about this option, see Information About Granular RBAC for Feature Templates Note To check the mutual authentication option, you need write permission for certificates. (Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.12.1)
Policy	View the common policies for all Cisco Catalyst SD-WAN Controllers or devices in the network on the Configuration > Policies window.	Create, edit, and delete the common policies for all Cisco Catalyst SD-WAN Controllers or devices in the network on the Configuration > Policies window.
Policy Configuration	View the list of policies created and details about them on the Configuration > Policies window.	Create, edit, and delete the common policies for all the Cisco Catalyst SD-WAN Controllers and devices in the network on the Configuration > Policies window.
Policy Deploy	View the current status of the Cisco Catalyst SD-WAN Controllers to which a policy is being applied on the Configuration > Policies window.	Activate and deactivate the common policies for all Cisco SD-WAN Manager servers in the network on the Configuration > Policies window.

Feature	Read Permission	Write Permission
RBAC VPN	View the VPN groups and segments based on roles on the Monitor > VPN page. Cisco vManage Release 20.6.x and earlier: View the VPN groups and segments based on roles on the Dashboard > VPN Dashboard page.	Add, edit, and delete VPNs and VPN groups from Cisco SD-WAN Manager, and edit VPN group privileges on the Administration > VPN Groups window.
Routing	View real-time routing information for a device on the Monitor > Devices > Real-Time page. Cisco vManage Release 20.6.x and earlier: View real-time routing information for a device on the Monitor > Network > Real-Time page.	Add command filters to speed up the display of information on the Monitor > Devices > Real-Time page.
Security	View the current status of the Cisco Catalyst SD-WAN Controllers to which a security policy is being applied on the Configuration > Security window.	Activate and deactivate the security policies for all Cisco SD-WAN Manager servers in the network on the Configuration > Security window.
Security Policy Configuration	Activate and deactivate the common policies for all Cisco SD-WAN Manager servers in the network on the Configuration > Security > Add Security Policy window.	Activate and deactivate the security policies for all Cisco SD-WAN Manager servers in the network on the Configuration > Security > Add Security Policy window.
Session Management	View user sessions on the Administration > Manage Users > User Sessions window.	Add, edit, and delete users and user groups from Cisco SD-WAN Manager, and edit user sessions on the Administration > Manage Users > User Sessions window.

Feature	Read Permission	Write Permission
Settings	View the organization name, Cisco Catalyst SD-WAN Validator DNS or IP address, certificate authorization settings, software version enforced on a device, custom banner on the Cisco SD-WAN Manager login page, and the current settings for collecting statistics on the Administration > Settings window.	Edit the organization name, Cisco Catalyst SD-WAN Validator DNS or IP address, certificate authorization settings, software version enforced on a device, custom banner on the Cisco SD-WAN Manager login page, current settings for collecting statistics, generate a certificate signing request (CSR) for a web server certificate, and install a certificate on the Administration > Settings window.
SIG Template (Minimum supported release: Cisco vManage Release 20.7.1)	View the SIG feature template and SIG credential template on the Configuration > Templates window. Note This operation requires read permission for Template Configuration .	Create, edit, delete, and copy a SIG feature template and SIG credential template on the Configuration > Templates window. Note These operations require write permission for Template Configuration . Note For information about this option, see Information About Granular RBAC for Feature Templates
SIG Tunnels (Minimum supported release: Cisco vManage Release 17.12)	View information about the SIG tunnels on the Monitor > Tunnels > SIG Tunnels page.	View information about the SIG tunnels on the Monitor > Tunnels > SIG Tunnels page.
Software Upgrade	View a list of devices, the custom banner on Cisco SD-WAN Manager on which a software upgrade can be performed, and the current software version running on a device on the Maintenance > Software Upgrade window.	Upload new software images on devices, upgrade, activate, and delete a software image on a device, and set a software image to be the default image on devices on the Maintenance > Software Upgrade window.

Feature	Read Permission	Write Permission
System	View system-wide parameters configured using Cisco SD-WAN Manager templates on the Configuration > Templates > Device Templates window. Note In Cisco vManage Release 20.7.x and earlier releases, Device Templates is called Device .	Configure system-wide parameters using Cisco SD-WAN Manager templates on the Configuration > Templates > Device Templates window. Note In Cisco vManage Release 20.7.x and earlier releases, Device Templates is called Device .
Template Configuration	View feature and device templates on the Configuration > Templates window.	Create, edit, delete, and copy a feature or device template on the Configuration > Templates window. Note Beginning with Cisco vManage Release 20.7.1, to create, edit, or delete a template that is already attached to a device, the user requires write permission for the Template Deploy option.
Template Deploy	View the devices attached to a device template on the Configuration > Templates window.	Attach a device to a device template on the Configuration > Templates window.
Tools	Use the admin tech command to collect the system status information for a device on the Tools > Operational Commands window.	Use the admin tech command to collect the system status information for a device, and use the interface reset command to shut down and then restart an interface on a device in a single operation on the Tools > Operational Commands window. Rediscover the network to locate new devices and synchronize them with Cisco SD-WAN Manager on the Tools > Operational Commands window. Establish an SSH session to the devices and issue CLI commands on the Tools > Operational Commands window.
vAnalytics	Launch Cisco SD-WAN Analytics on > vAnalytics window.	No additional permissions.

Feature	Read Permission	Write Permission
Workflows	Launch workflow library from > Workflows window.	No additional permissions.
Config Group > Device > Deploy (Minimum supported release: Cisco vManage Release 20.11.1)	View the devices associated to a configuration group on the Configuration > Templates > Edit Configuration Group > Associated Devices window.	Deploy a configuration onto Cisco IOS XE Catalyst SD-WAN devices. Note To edit an existing feature configuration requires write permission for Template Configuration . For more details on deploying devices, see Deploy Devices .
Feature Profile > Transport > IPv4 Tracker and Tracker Group (Minimum supported release: Cisco vManage Release 20.11.1)	View the IPv4 Tracker and Tracker Group settings on the Configuration > Templates > (View configuration group) page, in the Transport & Management Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the IPv4 Tracker and Tracker Group settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Transport & Management Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > Transport > IPv6 Tracker and Tracker Group (Minimum supported release: Cisco vManage Release 20.11.1)	View the IPv6 Tracker and Tracker Group settings on the Configuration > Templates > (View configuration group) page, in the Transport & Management Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the IPv6 Tracker and Tracker Group settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Transport & Management Profile section. Note These operations require write permission for Template Configuration .

Feature	Read Permission	Write Permission
Feature Profile > Transport > Gps (Minimum supported release: Cisco vManage Release 20.11.1)	View the GPS settings on the Configuration > Templates > (View configuration group) page, in the Transport & Management Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Gps settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Transport & Management Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > Other > APPQoE (Minimum supported release: Cisco vManage Release 20.11.1)	View the APPQoE settings on the Configuration > Templates > (View configuration group) page, in the Other section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the APPQoE settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Other section. Note These operations require write permission for Template Configuration .
Feature Profile > Other > UCSE (Minimum supported release: Cisco vManage Release 20.11.1)	View the UCSE settings on the Configuration > Templates > (View configuration group) page, in the Other section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the UCSE settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Other section. Note These operations require write permission for Template Configuration .
Feature Profile > Wan Profile > Cisco VPN Interface IPSec (Minimum supported release: Cisco vManage Release 20.11.1)	View the Cisco VPN Interface IPSec settings on the Configuration > Templates > (View configuration group) page, in the Wan Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Cisco VPN Interface IPSec settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Wan Profile section. Note These operations require write permission for Template Configuration .

Feature	Read Permission	Write Permission
Feature Profile > Wan/Lan Profile > Cisco VPN Interface GRE (Minimum supported release: Cisco vManage Release 20.11.1)	View the Cisco VPN Interface GRE settings on the Configuration > Templates > (View configuration group) page, in the Wan/Lan Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Cisco VPN Interface GRE settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Wan/Lan Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > Lan Profile > Cisco Multicast (Minimum supported release: Cisco vManage Release 20.11.1)	View the Cisco Multicast settings on the Configuration > Templates > (View configuration group) page, in the Lan Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Cisco Multicast settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Lan Profile section. Note These operations require write permission for Template Configuration .

**Note**

To create Service, System and Transport feature profiles using configuration groups, you need to provide read and write permissions on the following features to access each configuration group.

- **Feature Profile > System**
- **Feature Profile > System > AAA**
- **Feature Profile > System > BFD**
- **Feature Profile > System > Banner**
- **Feature Profile > System > Basic**
- **Feature Profile > System > Logging**
- **Feature Profile > System > NTP**
- **Feature Profile > System > OMP**
- **Feature Profile > System > SNMP**
- **Feature Profile > Service**
- **Feature Profile > Service > BFD**
- **Feature Profile > Service > LAN/VPN**
- **Feature Profile > Service > LAN/VPN/Interface/Ethernet**
- **Feature Profile > Service > Routing/BGP**
- **Feature Profile > Service > Routing/OSPF**
- **Feature Profile > Service > Routing/DHCP**
- **Feature Profile > Service > Routing/Multicast**
- **Feature Profile > Transport**
- **Feature Profile > Transport > Routing/BGP**
- **Feature Profile > Transport > WAN/VPN**
- **Feature Profile > Transport > WAN/VPN/Interface/Ethernet**

For more details on configuring features using Configuration Groups, see [Feature Management](#).

User Group Permissions: Cisco Catalyst Wireless Gateway Devices

Table 72: User Group Permissions: Cisco Catalyst Wireless Gateway Devices

Feature	Read Permission	Write Permission
Feature Profile > Teleworker > Basic (Minimum supported release: Cisco vManage Release 20.9.1)	View the basic settings on the Configuration > Templates > (View mobility configuration group) page, in the Global Profile section. Note This operation requires read permission for Template Configuration .	Configure the basic settings on the Configuration > Templates > (Add or edit mobility configuration group) page, in the Global Profile section. Note This operation requires write permission for Template Configuration .
Feature Profile > Teleworker > Cellular (Minimum supported release: Cisco vManage Release 20.9.1)	View the cellular network settings on the Configuration > Templates > (View mobility configuration group) page, in the Global Profile section. Note This operation requires read permission for Template Configuration .	Configure the cellular network settings on the Configuration > Templates > (Add or edit mobility configuration group) page, in the Global Profile section. Note This operation requires write permission for Template Configuration .
Feature Profile > Teleworker > Ethernet (Minimum supported release: Cisco vManage Release 20.9.1)	View the ethernet settings on the Configuration > Templates > (View mobility configuration group) page, in the Global Profile section. Note This operation requires read permission for Template Configuration .	Configure the ethernet settings on the Configuration > Templates > (Add or edit mobility configuration group) page, in the Global Profile section. Note This operation requires write permission for Template Configuration .
Feature Profile > Teleworker > NetworkProtocol (Minimum supported release: Cisco vManage Release 20.9.1)	View the network protocol settings on the Configuration > Templates > (View mobility configuration group) page, in the Global Profile section. Note This operation requires read permission for Template Configuration .	Configure the network protocol settings on the Configuration > Templates > (Add or edit mobility configuration group) page, in the Global Profile section. Note This operation requires write permission for Template Configuration .

Feature	Read Permission	Write Permission
Feature Profile > Teleworker > SecurityPolicy (Minimum supported release: Cisco vManage Release 20.9.1)	View the security policy settings on the Configuration > Templates > (View mobility configuration group) page, in the Global Profile section. Note This operation requires read permission for Template Configuration .	Configure the security policy settings on the Configuration > Templates > (Add or edit mobility configuration group) page, in the Global Profile section. Note This operation requires write permission for Template Configuration .
Feature Profile > Teleworker > Vpn (Minimum supported release: Cisco vManage Release 20.9.1)	View the VPN settings on the Configuration > Templates > (View mobility configuration group) page, in the Global Profile section. Note This operation requires read permission for Template Configuration .	Configure the VPN settings on the Configuration > Templates > (Add or edit mobility configuration group) page, in the Global Profile section. Note This operation requires write permission for Template Configuration .
Feature Profile > Teleworker > Wifi (Minimum supported release: Cisco vManage Release 20.9.1)	View the Wi-Fi settings on the Configuration > Templates > (View mobility configuration group) page, in the Global Profile section. Note This operation requires read permission for Template Configuration .	Configure the Wi-Fi settings on the Configuration > Templates > (Add or edit mobility configuration group) page, in the Global Profile section. Note This operation requires write permission for Template Configuration .

RBAC User Group in a Multitenant Environment

The following is the list of user group permissions for role-based access control (RBAC) in a multitenant environment:

- R stands for read permission.
- W stands for write permission.

Table 73: RBAC User Group in Multitenant Environment

Feature	Provider Admin	Provider Operator	Tenant Admin	Tenant Operator
Cloud OnRamp	RW	R	RW	R
Colocation	RW	R	RW	R

Feature	Provider Admin	Provider Operator	Tenant Admin	Tenant Operator
RBAC VPN	RW	R	RW	R
Security	RW	R	RW	R
Security Policy Configuration	RW	R	RW	R
vAnalytics	RW	R	RW	R

Add a User

1. From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users**.
2. By default **Users** is selected. The table displays the list of users configured in the device.
3. To edit, delete, or change password for an existing user, click ... and click **Edit**, **Delete**, or **Change Password** respectively.
4. To add a new user, click **Add User**.
5. Add **Full Name**, **Username**, **Password**, and **Confirm Password** details.
6. If you are using an identity provider, such as Okta, for security assertion markup language (SAML)-based single sign-on (SSO), then in most use cases, you define user roles through the identity provider. If no roles are defined for the user through the identity provider, you can enable the **Remote User** option and assign user groups locally in Cisco SD-WAN Manager. Assigning user groups locally provides an alternate method for assigning the user with permissions.

If you enable this option, enter an email address for the user.

If you have defined roles for a user through the identity provider and have also assigned user groups locally for the same user, the roles defined through the identity provider take priority.



Note This option is available from Cisco vManage Release 20.11.1.

7. In the **User Groups** drop-down list, select the user group where you want to add a user.
8. In the **Resource Group** drop-down list, select the resource group.



Note This field is available from Cisco IOS XE Catalyst SD-WAN Release 17.5.1a.

9. Click **Add**.

Delete a User

If a user no longer needs access to devices, you can delete the user. Deleting a user does not log out the user if the user is logged in.

To delete a user:

1. From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users**.
2. For the user you wish to delete, click ..., and click **Delete**.
3. To confirm the deletion of the user, click **OK**.

Edit User Details

You can update login information for a user, and add or remove a user from a user group. If you edit the details of a user who is logged in, the changes take effect after the user logs out.

To edit user details:

1. From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users**.
2. For the user you wish to edit, click ..., and click **Edit**.
3. Edit the user details.

You can also add or remove the user from user groups.

4. Click **Update**.

Change a User Password

You can update passwords for users, as needed. We recommend that you use strong passwords.

Before You Begin

If you are changing the password for an admin user, detach device templates from all Cisco SD-WAN Manager instances in the cluster before you perform this procedure. You can reattach the device templates after you complete this procedure.

To change a password for a user:

1. From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users**.
2. For the user you wish to change the password, click ... and click **Change Password**.
3. Enter the new password, and then confirm it.



Note Note that the user, if logged in, is logged out.

4. Click **Done**.

Check Users Logged In to a Device Using SSH Sessions

1. From the Cisco SD-WAN Manager menu, choose **Monitor > Devices**.

Cisco vManage Release 20.6.x and earlier: From the Cisco SD-WAN Manager menu, choose **Monitor > Network**.

2. Select the device you want to use under the **Hostname** column.
3. Click **Real Time**.
4. From **Device Options**, choose **AAA users** for Cisco IOS XE Catalyst SD-WAN devices.
A list of users logged in to this device is displayed.

Check Users Logged In to a Device Using HTTP Sessions

1. From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users**.
2. Click **User Sessions**.
A list of all the active HTTP sessions within Cisco SD-WAN Manager is displayed, including, username, domain, source IP address, and so on.

Manage a User Group

Users are placed in groups, which define the specific configuration and operational commands that the users are authorized to view and modify. A single user can be in one or more groups. Cisco Catalyst SD-WAN software provides standard user groups, and you can create custom user groups, as needed:

- **basic**: Includes users who have permission to view interface and system information.
- **netadmin**: Includes the admin user, by default, who can perform all operations on the Cisco SD-WAN Manager. You can add other users to this group.
- **operator**: Includes users who have permission only to view information.
- Minimum supported release: Cisco vManage Release 20.9.1
- **network_operations**: Includes users who can perform non-security operations on Cisco SD-WAN Manager, such as viewing and modifying non-security policies, attaching and detaching device templates, and monitoring non-security data.
- Minimum supported release: Cisco vManage Release 20.9.1
- **security_operations**: Includes users who can perform security operations on Cisco SD-WAN Manager, such as viewing and modifying security policies, and monitoring security data.

Note: All user groups, regardless of the read or write permissions selected, can view the information displayed on the Cisco SD-WAN Manager Dashboard screen.

Delete a User Group

You can delete a user group when it is no longer needed. For example, you might delete a user group that you created for a specific project when that project ends.

1. From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users**.
2. Click **User Groups**.
3. Click the name of the user group you wish to delete.



Note You cannot delete any of the default user groups—basic, netadmin, operator, network_operations, and security_operations.

4. Click **Trash** icon.
5. To confirm the deletion of the user group, click **OK**.

Edit User Group Privileges

You can edit group privileges for an existing user group. This procedure lets you change configured feature read and write permissions for the user group needed.

1. From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users**.
2. Click **User Groups**.
3. Select the name of the user group whose privileges you wish to edit.



Note You cannot edit privileges for the any of the default user groups—basic, netadmin, operator, network_operations, and security_operations.

4. Click **Edit**, and edit privileges as needed.
5. Click **Save**.

If an **admin** user changes the privileges of a user by changing their group, and if that user is currently logged in to the device, the user is logged out and must log back in again.

Create User Groups

1. From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users**.
2. Click **User Groups**.
3. Click **Add User Group**.
4. Enter **User Group Name**.
5. Select the **Read** or **Write** check box against feature that you want to assign to a user group.
6. Click **Add**.
7. You can view the new user group in the left navigation path. Click **Edit** to edit the existing read or write rules.
8. Click **Save**.

Configure and Manage VPN Segments

To configure VPN Segments:

1. From the Cisco SD-WAN Manager menu, choose **Administration > VPN Segments**. A web page displays the list of segments that are configured.
2. To edit or delete an existing segment, click ..., and click **Edit** or **Delete**.
3. To add new segment, click **Add Segment**.
4. Enter the name of the segment in the **Segment Name** field.
5. Enter the number of VPNs you want to configure in **VPN Number** field.
6. To add a new segment, click **Add**.

Configure and Manage VPN Groups

To configure VPN Groups:

1. From the Cisco SD-WAN Manager menu, choose **Administration > VPN Groups**. A web page displays the list of segments that are configured.
2. To edit or delete a VPN group, click ..., and click **Edit** or **Delete**.
3. To view the existing VPN in the dashboard, click ..., and click **View Dashboard**. The **VPN Dashboard** displays the device details of the VPN device configured.
4. To add new VPN group, click **Add Group**.
5. From **Create VPN Group**, enter VPN group name in the **VPN Group Name** field.
6. Enter a brief description of the VPN in the **Description** field.
7. Check **Enable User Group access** check box and enter the user group name.
8. From **Assign Segment**, click **Add Segment** drop-down list to add new or existing segment to the VPN group.
9. Enter the **Segment Name** and **VPN Number** in the respective fields.
10. To add the configure VPN group to a device, click **Add**.

Managing Resource Groups

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.5.1a and Cisco vManage Release 20.5.1

To configure Resource Groups:

1. From the Cisco SD-WAN Manager menu, choose **Administration > Resource Groups**. The table displays a list of resource groups that are configured in Cisco SD-WAN Manager.
2. To edit or delete a resource group, click ..., and click **Edit** or **Delete**.
3. To add new resource group, click **Add Resource Group**.
4. Enter **Resource Group Name** and the **Description**.
5. Under **Site ID**, enter **Range** or **Select ID(S)** from the drop-down list to include in the resource group.

6. To add the resource group to a device, click **Add**.

To add Users:

1. From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users**. The Manage Users screen appears.
2. By default **Users** is selected. The table displays the list of users configured in the device.
3. To edit, delete, or change password for an existing user, click **...**, and click **Edit**, **Delete**, or **Change Password** respectively.
4. To add a new user, click **Add User**.
5. Add **Full Name**, **Username**, **Password**, and **Confirm Password** details.
6. From the **User Groups** drop-down list, select the user group where you want to add a user.
7. From the **Resource Group** drop-down list, select the resource group.



Note This field is available from Cisco IOS XE Catalyst SD-WAN Release 17.5.1a.

8. Click **Add**.

Workflow to Configure RBAC for Policies

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.6.1a and Cisco vManage Release 20.6.1

To configure RBAC for policies, use the following workflow:

1. Create user groups with required Read or Write (R/W) access to selected control or data policies. For details on creating user groups, refer [Create User Groups](#).
2. Create users and assign them to required user groups. Refer [Create Users](#).
3. Create or modify or view policy configurations as required. For information about configuring policies, see [Configure Centralized Policies Using Cisco SD-WAN Manager](#).

Modify Policy Configurations

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.6.1a and Cisco vManage Release 20.6.1

1. Login to Cisco SD-WAN Manager with the new user details.
2. You can modify or update the configurations based on the requirement.

When you login to Cisco SD-WAN Manager with new user details, you can view only the user group components that are assigned to you. For more details on configuring policies, see [Cisco Catalyst SD-WAN Policies Configuration Guide](#)

Assign Users to Configure RBAC for Policies

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.6.1a and Cisco vManage Release 20.6.1

To Assign User to Create or Modify a CFlowd Data Policy

To create a CFlowd user group:

1. From Cisco SD-WAN Manager, choose **Administration > Manage Users**.
2. Click **User Groups** and **Add User Group**.
3. Enter **User Group Name**.
For example, cflowd-policy-only.
4. Check the Read or Write check box against the CFlowD Policy feature that you want to assign to a user group.
5. Click **Add**.
6. You can view the new user group in the left navigation path. Click **Edit** to edit the existing read or write rules.
7. Click **Save**.

To create a CFlowd user:

1. In Cisco SD-WAN Manager, choose **Administration > Manage Users**.
2. Click **Users**.
3. Click **Add User**.
4. In the Add New User page, enter **Full Name**, **Username**, **Password**, and **Confirm Password** details.
5. Choose **cflowd-policy-only** from the **User Groups** drop-down.
Allow the **Resource Group** to select the default resource group.
6. Click **Add**. You can view the new user in the Users window.
7. To edit the existing read or write rules for a user, click **Edit**.

To modify a Cflowd policy:

1. Login to Cisco SD-WAN Manager with the new user credentials.
You can view access only to CFlowd Policies as your login is assignend to **cflowd-policy-only** user group.
2. You can create, modify, or update the configurations based on the requirement.

Configure Granular RBAC for Feature Templates

Minimum supported release: Cisco vManage Release 20.7.1

To configure specific template access, create a user group and assign the read and write permissions using the permission types described in [Information About RBAC for Co-Management](#). The permission options for limiting template access appear with the other permission options that you choose when adding a user group.

For information about granular RBAC for feature templates, see [Information About Granular RBAC for Templates, on page 157](#).

For information about adding a user group, see [Create User Groups](#).

For a list of permission types and descriptions, see [Manage Users](#).

Configure RBAC Using the CLI

Configure Users Using CLI

You can use the CLI to configure user credentials on each device. This way, you can create additional users and give them access to specific devices. The credentials that you create for a user by using the CLI can be different from the Cisco SD-WAN Manager credentials for the user. In addition, you can create different credentials for a user on each device. All Cisco IOS XE Catalyst SD-WAN device users with the **netadmin** privilege can create a new user.

To create a user account, configure the username and password, and place the user in a group:

This example, shows the addition of user, Bob, to an existing group:

```
Device(config)# system aaa user bob group basic
```

This example, shows the addition of user, Alice, to a new group `test-group`:

```
Device(config)# system aaa user test-group
Device(config)# system aaa user alice group test-group
```

The Username can be 1 to 128 characters long, and it must start with a letter. The name can contain only lowercase letters, the digits 0 through 9, hyphens (-), underscores (_), and periods (.). The name cannot contain any uppercase letters. Because some usernames are reserved, you cannot configure them. For a list of reserved usernames, see the **aaa** configuration command in the Cisco Catalyst SD-WAN Command Reference Guide.

The Password is the password for a user. Each username must have a password, and users are allowed to change their own password. The CLI immediately encrypts the string and does not display a readable version of the password. When a user logs in to a Cisco IOS XE Catalyst SD-WAN device, they have five chances to enter the correct password. After the fifth incorrect attempt, the user is locked out of the device, and must wait for 15 minutes before attempting to log in again.



Note Enclose any user passwords that contain the special character ! in double quotation marks (" "). If a double quotation is not included for the entire password, the config database (?) treats the special character as a space and ignores the rest of the password.

For example, if the password is C!sc0, use "C!sc0".

Group name is the name of a standard Cisco Catalyst SD-WAN group (**basic**, **netadmin**, or **operator**) or of a group configured with the **usergroup** command (discussed below). If an admin user changes the permission of a user by changing their group, and if that user is currently logged in to the device, the user is logged out and must log back in again.

The factory-default password for the admin username is admin. We strongly recommend that you modify this password the first time you configure a Cisco IOS XE Catalyst SD-WAN device:

```
Device(config)# username admin password
$9$3/IL3/UF2F2F3E$J9NKBeKlWrq9ExmHk6F5VAiDMOFQFD.QPAmMxDdxz.c
```

Configure the password as an ASCII string. The CLI immediately encrypts the string and does not display a readable version of the password, for example:

```
Device# show run | sec username
username admin privilege 15 secret 9
$9$3F2M2l2G2/UM3U$TGe2kqoIibdIRDEj4cOVKbVFP/o4vnlFAwWnmzx1rRE
username appnav privilege 15 secret 9
$9$3l2L2V.F2VIM1k$P3MBayBtGxKf/yBGnUSHQ1g/ae1QhfIbieg28buJJGI
username eft secret 9 $9$3FMJ3/UD2VEL2E$d.kE4.an4lv7wEhrQc6k5wIfE9M9WkNAJxUvbbempS.
username lab privilege 15 secret 9
$9$3l.J3FUD2F.E2.$/AiVn9PmLCpgr6ExVrE7dH979Wu8nbdAfzbUtfysg.
username test secret 9 $9$1l2J3l6D3/QL3k$7PZOXJAJOIlos5UI763G3XcpVhXlqcwJ.qEmgmX4X9g
username vbonagir privilege 15 secret 9
$9$3/2K2UwF2lQF3U$VbdQ5bq18590rRthF/NnNnOsw.dwl/EViMTFZ5.ctus
Device#
```

If you are using RADIUS to perform AAA authentication, you can configure a specific RADIUS server to verify the password:

```
Device(config)# radius server tag
```

The tag is a string that you defined with the **radius server tag** command, as described in the Cisco Catalyst SD-WAN Command Reference Guide.

Creating Groups Using CLI

The Cisco Catalyst SD-WAN software provides default user groups: **basic**, **netadmin**, **operator**, **network_operations**, and **security_operations**. The username **admin** is automatically placed in the **netadmin** usergroup.

If needed, you can create additional custom groups and configure privilege roles that the group members have. To create a custom group with specific authorization, configure the group name and privileges:

```
Device(config)# aaa authentication login user1 group radius enable
Device(config)# aaa authentication login user2 group radius enable
Device(config)# aaa authentication login user3 group radius enable
Device(config)#
```

group-name can be 1 to 128 characters long, and it must start with a letter. The name can contain only lowercase letters, the digits 0 through 9, hyphens (-), underscores (_), and periods (.). The name cannot contain any uppercase letters. Some group names are reserved, so you cannot configure them. For a list of them, see the aaa configuration command.

If a remote RADIUS or TACACS+ server validates authentication but does not specify a user group, the user is placed into the user group **basic**. If a remote server validates authentication and specifies a user group (say, X) using VSA Cisco SD-WAN-Group-Name, the user is placed into that user group only. However, if that user is also configured locally and belongs to a user group (say, Y), the user is placed into both the groups (X and Y).

In the **task** option, list the privilege roles that the group members have. The role can be one or more of the following: **interface**, **policy**, **routing**, **security**, and **system**.

Verify RBAC

Verify Granular RBAC Permissions

Minimum supported release: Cisco vManage Release 20.7.1

Use this procedure to verify the permissions that you have configured for a user group.

1. From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users**.
2. Click **User Groups**.
3. In the pane that displays the user groups, select a user group to display the read and write permissions assigned to the user group.
4. Scroll to the permissions that control template access to verify your configuration for the user group.

Monitor RBAC

Monitor devices for VPN Groups

To monitor devices:

1. From the Cisco SD-WAN Manager menu, choose **Monitor > Network**.
2. Click **WAN - Edge**.
3. Select the **VPN Group** and **VPN Segment** for which you want to monitor the network.
A web page displays the list of VPN groups and segments that are configured to a device.



CHAPTER 8

Configure Devices

You can create and store configurations for all devices—the Cisco SD-WAN Manager systems themselves, Cisco Catalyst SD-WAN Controllers, Cisco Catalyst SD-WAN Validators, and routers— by using Cisco SD-WAN Manager. When the devices start up, they contact Cisco SD-WAN Manager, which then downloads the device configuration to the device. (A device that is starting up first contacts the Cisco Catalyst SD-WAN Validator, which validates the device and then sends it the IP address of Cisco SD-WAN Manager.)

The general procedure for creating configuration for all devices is the same. This section provides a high-level description of the configuration procedure. It also describes the prerequisite steps that must be performed before you can create configurations and configure devices in the overlay network.

- [Device Configuration Workflow, on page 217](#)
- [Feature Templates, on page 218](#)
- [Device Templates, on page 218](#)
- [Template Variables, on page 219](#)
- [Configuration Prerequisites, on page 219](#)
- [Create a Device Template from Feature Templates, on page 220](#)
- [Default Device Templates, on page 237](#)
- [Configuring Devices using Cisco SD-WAN Manager, on page 238](#)

Device Configuration Workflow

Devices in the overlay network that are managed by Cisco SD-WAN Manager must be configured from Cisco SD-WAN Manager. The basic configuration procedure is straightforward:

1. Create feature templates.
 - a. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
 - b. Click **Feature Templates**, and click **Add Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

2. Create device templates.
 - a. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.

- b. Click **Device Templates**, and click **Create Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. Attach device templates to individual devices.
 - a. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
 - b. Click **Device Templates**, and choose a template.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

- c. Click ..., and select **Attach Devices**.

Feature Templates

Feature templates are the building blocks of complete configuration for a device. For each feature that you can enable on a device, Cisco SD-WAN Manager provides a template form that you fill out. The form allows you to set the values for all configurable parameters for that feature.

Because device configurations vary for different device types and the different types of routers, feature templates are specific to the type of device.

Some features are mandatory for device operation, so creating templates for these features is required. Also for the same feature, you can create multiple templates for the same device type.



Note In releases prior to Cisco IOS XE Catalyst SD-WAN Release 17.7.1a, if you enter < or > special characters in a Cisco SD-WAN Manager feature template definition or description, Cisco SD-WAN Manager generates a 500 exception error while attempting to preview a Cisco SD-WAN Manager feature template.

Starting from Cisco IOS XE Catalyst SD-WAN Release 17.7.1a, if you enter < or > special characters in a Cisco SD-WAN Manager feature template definition or description, the special characters are converted to their HTML equivalents, **<** and **>**. This applies to all feature templates. You no longer receive a 500 exception error when previewing a Cisco SD-WAN Manager feature template.

Device Templates

You create and store configurations for all devices—the Cisco SD-WAN Manager systems themselves, Cisco Catalyst SD-WAN Controllers, Cisco Catalyst SD-WAN Validators, and routers—by using Cisco SD-WAN Manager. When the devices start up, they contact Cisco SD-WAN Manager, which then downloads the device configuration to the device. (A device that is starting up first contacts the Cisco Catalyst SD-WAN Validator, which validates the device and then sends it the IP address of Cisco SD-WAN Manager.)

Device templates contain complete operational configuration for a device. You create device templates by consolidating individual feature templates.

Each device template is specific for a type of device. For each device type, if multiple devices have the same configuration, you can use the same device template for them. For example, many of the routers in a network might have the same basic configuration, so you can configure them with the same templates. (You specify the differences in the templates using configuration variables, which are discussed below.) If the configurations for the same type of devices are different, you create separate device templates.

You can also create a device template by entering a CLI text-style configuration directly on Cisco SD-WAN Manager. Typically, you upload a text file containing the configuration text (or cut the configuration text from a text file and paste it into Cisco SD-WAN Manager). You can also directly type the configuration text into Cisco SD-WAN Manager.

From Cisco IOS XE Catalyst SD-WAN Release 17.5.1a and Cisco vManage Release 20.5.1, you can review your last edited configuration when your latest configuration is not being pushed to the device. For more information, see [Edit a Device Template When a Push Fails](#), on page 234.

From Cisco vManage Release 20.5.1, device variable page shows text area instead of text input field to configure CLI device template for the ease of configuration.

Template Variables

Within a feature template, some configuration commands and command options are identical across all device types. Others—such as a device system IP address, its geographic latitude and longitude, the timezone, and the overlay network site identifier—are variable, changing from device to device. When you attach the device template to a device, you are prompted to enter actual values for these command variables. You can do this either manually, by typing the values for each variable and for each device, or you can upload an Excel file in CSV format that contains the values for each device.

Configuration Prerequisites

Security Prerequisites

Before you can configure any device in the network, that device must be validated and authenticated so that Cisco SD-WAN Manager systems, Cisco Catalyst SD-WAN Controllers, and Cisco Catalyst SD-WAN Validators recognize it as being allowed in the overlay network.

To validate and authenticate the controllers in the overlay network—Cisco SD-WAN Manager, Cisco SD-WAN Controller, and Cisco Catalyst SD-WAN Validators—a signed certificate must be installed on these devices.

To validate and authenticate the routers, you receive an authorized serial number file from Cisco, which lists the serial and chassis numbers for all the routers allowed in your network. Then, you upload the serial number file to Cisco SD-WAN Manager.

Variables Spreadsheet

The feature templates that you create most likely contain variables. To have Cisco SD-WAN Manager populate the variables with actual values when you attach a device template to a device, create an Excel file that lists the variable values for each device and save the file in CSV format.

In the spreadsheet, the header row contains the variable name and each row after that corresponds to a device, defining the values of the variables. The first three columns in the spreadsheet must be the following, in this order:

- **csv-deviceId**—Serial number of the device (used to uniquely identify the device). For routers, you receive the serial numbers in the authorized serial number file sent to you from Cisco. For other devices, the serial number is included in the signed certificate you receive from Symantec or from your root CA.
- **csv-deviceIP**—System IP address of the device (used to populate the **system ip address** command).
- **csv-host-name**—Hostname of the device (used to populate the **system hostname** command).

You can create a single spreadsheet for all devices in the overlay network—Cisco Catalyst SD-WAN Controllers, Cisco Catalyst SD-WAN Validators, and routers. You do not need to specify values for all variables for all devices.

Create a Device Template from Feature Templates

Device templates define a device's complete operational configuration. A device template consists of a number of feature templates. Each feature template defines the configuration for a particular Cisco Catalyst SD-WAN software feature. Some feature templates are mandatory, indicated with an asterisk (*), and some are optional. Each mandatory feature template, and some of the optional ones, have a factory-default template. For software features that have a factory-default template, you can use either the factory-default template (named `Factory_Default_feature-name_Template`) or you can create a custom feature template.

Create a Device Template from Feature Templates

To create a device template:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. Click the **Create Template** drop-down list, and select **From Feature Template**.
4. From the **Device Model** drop-down list, select the type of device for which you wish to create the template.

Cisco SD-WAN Manager displays all the feature templates for that device type. The required feature templates are indicated with an asterisk (*), and the remaining templates are optional. The factory-default template for each feature is selected by default.

5. In the **Template Name** field, enter a name for the device template.
This field is mandatory and can contain only uppercase and lowercase letters, the digits 0 through 9, hyphens (-), and underscores (_). It cannot contain spaces or any other characters.
6. In the **Description** field, enter a description for the device template.
This field is mandatory, and it can contain any characters and spaces.

7. To view the factory-default configuration for a feature template, select the desired feature template and click **View Template**.
8. Click **Cancel** to return to the **Configuration Template** screen.
9. To create a custom template for a feature, select the desired factory-default feature template and click **Create Template**. The template form is displayed.
This form contains fields for naming the template and defining the feature parameters.
10. In the **Template Name** field, enter a name for the feature template.
This field is mandatory and can contain only uppercase and lowercase letters, the digits 0 through 9, hyphens (-), and underscores (_). It cannot contain spaces or any other characters.
11. In the **Description** field, enter a description for the feature template.
This field is mandatory, and it can contain any characters and spaces.
12. For each field, enter the desired value. You may need to click a tab or the plus sign (+) to display additional fields.
13. When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default or to enter a value, click the scope drop-down list of the parameter field and select one of the following:

Table 74:

Parameter Scope	Scope Description
Device Specific (indicated by a host icon)	Use a device-specific value for the parameter. For device-specific parameters, you cannot enter a value in the feature template. You enter the value when you attach a device to a device template. When you click Device Specific, the Enter Key box opens. This box displays a key, which is a unique string that identifies the parameter in a CSV file that you create. This file is an Excel spreadsheet that contains one column for each key. The header row contains the key names (one key per column), and each row after that corresponds to a device and defines the values of the keys for that device. You upload the CSV file when you attach a device to a device template. For more information, see Use Variable Values in Configuration Templates. To change the default key, type a new string and move the cursor out of the Enter Key box. Examples of device-specific parameters are system IP address, hostname, GPS location, and site ID.
Global (indicated by a globe icon)	Enter a value for the parameter, and apply that value to all devices. Examples of parameters that you might apply globally to a group of devices are DNS server, syslog server, and interface MTUs.

14. For some groups of parameters, you can mark the entire group as device-specific. To do this, check the **Mark as Optional Row** check box.

These parameters are then grayed out so that you cannot enter a value for them in the feature template. You enter the value or values when you attach a device to a device template.

15. Click **Save**.
16. Repeat Steps 6 through 13 to create a custom template for each additional software feature. For details on creating specific feature templates, see the templates listed in **Available Feature Templates**.
17. Click **Create**. The new configuration template is displayed in the Device Template table.

The Feature Templates column shows the number of feature templates that are included in the device template, and the Type column shows "Feature" to indicate that the device template was created from a collection of feature templates.

Another way to create device templates from feature templates is to first create one or more custom feature templates and then create device templates. You can create multiple feature templates for the same feature. For a list of feature templates, see **Available Feature Templates**.

1. Click **Feature**.
2. Click **Add Template**.
3. From **Select Devices**, select the type of device for which you wish to create a template.

You can create a single feature template for features that are available on multiple device types. You must, however, create separate feature templates for software features that are available only on the device type you are configuring.
4. Select the feature template. The template form is displayed.

This form contains fields for naming the template and fields for defining the required parameters. If the feature has optional parameters, then the template form shows a plus sign (+) after the required parameters.
5. In the **Template Name** field, enter a name for the feature template.

This field is mandatory and can contain only uppercase and lowercase letters, the digits 0 through 9, hyphens (-), and underscores (_). It cannot contain spaces or any other characters.
6. In the **Description** field, enter a description for the feature template.

This field is mandatory, and it can contain any characters and spaces.
7. For each required parameter, choose the desired value, and if applicable, select the scope of the parameter. Select the scope from the drop-down list of each parameter's value box.
8. Click the plus sign (+) from the required parameters to set the values of optional parameters.
9. Click **Save**.
10. Repeat Steps 2 to 9 for each additional feature template you wish to create.
11. Click **Device**.
12. Click the **Create Template** drop-down list and select **From Feature Template**.
13. From the **Device Model** drop-down list, select the type of device for which you wish to create the device template.

Cisco SD-WAN Manager displays the feature templates for the device type you selected. The required feature templates are indicated with an asterisk (*). The remaining templates are optional.

14. In the **Template Name** field, enter a name for the device template.
This field is mandatory and can contain only uppercase and lowercase letters, the digits 0 through 9, hyphens (-), and underscores (_). It cannot contain spaces or any other characters.
15. In the **Description** field, enter a description for the device template.
This field is mandatory, and it can contain any characters and spaces.
16. To view the factory-default configuration for a feature template, select the desired feature template and click **View Template**.
17. Click **Cancel** to return to the **Configuration Template** screen.
18. To use the factory-default configuration, click **Create** to create the device template. The new device template is displayed in the **Device Template** table. The Feature Templates column shows the number of feature templates that are included in the device template, and the Type column shows "Feature" to indicate that the device template was created from a collection of feature templates.
19. To modify the factory-default configuration, select the feature template for which you do not wish to use the factory-default template. From the drop-down list of available feature templates, select a feature template that you created.
20. Repeat Step 19 for each factory-default feature template you wish to modify.
21. Click **Create**. The new configuration template is displayed in the **Device Template** table.
The Feature Templates column shows the number of feature templates that are included in the device template, and the Type column shows "Feature" to indicate that the device template was created from a collection of feature templates.

Create a Device CLI Template

To create a device template by entering a CLI text-style configuration directly on the Cisco SD-WAN Manager:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. Click the **Create Template** drop-down list and select **CLI Template**.
4. From the **Device** Type drop-down list, select the type of device for which you wish to create the template.
5. In the **Template Name** field, enter a name for the device template.
This field is mandatory and can contain only uppercase and lowercase letters, the digits 0 through 9, hyphens (-), and underscores (_). It cannot contain spaces or any other characters.
6. In the **Description** field, enter a description for the device template.
This field is mandatory, and it can contain any characters and spaces.

7. In the CLI Configuration box, enter the configuration either by typing it, cutting and pasting it, or uploading a file.
8. To convert an actual configuration value to a variable, select the value and click **Create Variable**. Enter the variable name, and click **Create Variable**. You can also type the variable name directly, in the format `{{variable-name}}`; for example, `{{hostname}}`.
9. Click **Add**. The new device template is displayed in the Device Template table.

The **Feature Templates** column shows the number of feature templates that are included in the device template, and the Type column shows "CLI" to indicate that the device template was created from CLI text.

Manage Device Templates

Table 75: Feature History

Feature Name	Release Information	Description
Support for Draft Mode in Device Template	Cisco IOS XE Catalyst SD-WAN Release 17.5.1a Cisco vManage Release 20.5.1	This feature allows you to save the device template configuration changes in Cisco SD-WAN Manager, and then apply these configuration changes to multiple Cisco IOS XE Catalyst SD-WAN devices later. The ability to save configuration changes simplifies generating larger device template configurations and applying them to devices.

Edit a Device Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates** or **Feature Templates**, and select a template.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**, and **Feature Templates** is titled **Feature**.

3. Click ..., and click **Edit**.

You cannot change the name of a device or feature template when that is attached to a device.



Note You can edit templates simultaneously from one or more Cisco SD-WAN Manager servers. For simultaneous template edit operations, the following rules apply:

- You cannot edit the same device or feature template simultaneously.
- When you are editing a device template, all other feature templates attached to that device template are locked and you cannot perform any edit operations on them.

- When you are editing a feature template that is attached to a device template, that device template as well as all other feature templates attached to it are locked and you cannot perform any edit operations on them.

Delete a Template

Deleting a template does not remove the associated configuration from devices.

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates** or **Feature Templates**, and select a template.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**, and **Feature Templates** is titled **Feature**.

3. Click ..., and click **Delete**.
4. To confirm the deletion of the template, click **OK**.

Copy a Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates** or **Feature Templates**, and select a template.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**, and **Feature Templates** is titled **Feature**.

3. Click ..., and click **Copy**.
4. Enter a new template name and description.
5. Click **Copy**.

Edit a CLI Device Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and select a template.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. Click ..., and click **Edit**.
4. Under **Device CLI Template**, edit the template.
5. Click **Update**.

Use Variable Values in Configuration Templates

An overlay network might have multiple devices of the same type that have nearly identical configurations. This situation most commonly occurs with routers when the routers that are located in multiple stores or branch locations provide identical services, but each individual router has its own hostname, IP address, GPS location, and other site-specific properties, such as BGP neighbors. This situation also occurs in a network with redundant controller devices, such as Cisco Catalyst SD-WAN Controllers, which must all be configured with identical policies, and Cisco SD-WAN Manager systems. Again, each controller has its own individual parameters, such as hostname and IP address.

To simplify the configuration process for these devices, you can create a single configuration template that contains both static configuration values and variable values. The static values are common across all the devices, and the variable values apply only to an individual device. You provide the actual values for the variables when you attach the individual device to the device configuration template.

You can configure a variable value for a parameter in a feature configuration template in two ways:

- Select the parameter scope to be Device Specific—For an individual configuration parameter, select Device Specific to mark the parameter as a variable. Each variable must be identified by a unique text string, which is called a *key*. When you select Device Specific, an Enter Key box opens and displays the default key. You can use the default key, or you can change it by typing a new string and then moving the cursor out of the Enter Key box.
- Mark a group of related parameters as optional—For some features in some feature configuration templates, you can mark the entire feature as optional. To mark the feature in this way, click Mark as Optional Row in a section of a feature configuration template. The variable parameters are then dimmed, and you cannot configure values for them in the feature configuration template.

You enter the device-specific values for the variables when you attach the device to the configuration, in one of the following ways:

- From a file—When you are attaching a template to a device, you load a file to Cisco SD-WAN Manager. This is an Excel file in CSV format that lists all the variables and defines the variable's value for each device.
- Manually—When you attach a device template to a device, the Cisco SD-WAN Manager prompts you for the values for each of device-specific parameters, and you type in the value for each parameter.



Note Cisco Catalyst SD-WAN supports up to 500 variables in a template push operation.

Use a File for Variable Parameters

To load device-specific variable values from a file, you create a template variables file. This file is an Excel file in CSV format that lists all the variables in your the configurations of your devices and defines the values for each variable. You create this file offline and then import it into Cisco SD-WAN Manager server when you attach a device configuration to one or more devices in the overlay network.

We recommend that you create a template variables CSV file when your overlay network has more than a small number of Cisco IOS XE Catalyst SD-WAN devices.

CSV File Format

The CSV file is an Excel spreadsheet that contains one column for each variable that is required for the configuration of a device. The header row contains the variable names (one variable per column), and each row after that corresponds to a device and defines the values of the variables for that device.

You can create a single spreadsheet for all devices in the overlay network—Cisco IOS XE Catalyst SD-WAN devices, Cisco SD-WAN Manager systems, Cisco Catalyst SD-WAN Controllers, and Cisco Catalyst SD-WAN Validators—or you can create one spreadsheet for each device type. The system determines the device type from its serial number.

In the spreadsheet, for each device type and for each individual device, you specify values only for the required variables. When you do not need to specify a value for a variable, simply leave that cell blank.

The first three columns in the spreadsheet must be the following items and must be in the order shown:

Column	Column Heading	Description
1	csv-deviceId	Serial number of the device (used to uniquely identify the device). For Cisco IOS XE Catalyst SD-WAN devices, you receive the serial numbers in the authorized serial number file sent to you from Cisco. For other devices, the serial number is included in the signed certificate you receive from Symantec or from your root CA.
2	csv-deviceIP	System IP address of the device (used to populate the system ip address command).
3	csv-host-name	Hostname of the device (used to populate the system hostname command).

The headings for the remaining columns must be unique variable keys that are defined in the Enter Key box of a feature configuration template. These remaining columns can be in any order.

Generate a Skeleton CSV File

You can create a template variables CSV file manually, with the format described in the previous section, or you can have Cisco SD-WAN Manager generate a skeleton CSV file that contains all the required columns and column headings. This generated CSV file has one row for each Cisco device type, and it has the column headings for each of the variables that are required by all the feature templates included in the device configuration. The column heading text corresponds to the key string that identifies a device-specific parameter. Then you populate the rows with values for each variable.

To have Cisco SD-WAN Manager generate a skeleton CSV file:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**, and click **Add Template**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

3. Create the required feature templates for one Cisco IOS XE Catalyst SD-WAN device router, one Cisco Catalyst SD-WAN Controller, one Cisco SD-WAN Manager system, and one Cisco Catalyst SD-WAN Validator.

In each feature template:

- a. For fields that have default values, verify that you want to use that value for all devices. If you do not want to use the default, change the scope to **Global** or **Device-specific**.
 - b. For fields that apply to all devices, select the **Global** icon next to the field and set the desired global values.
 - c. For fields that are device specific, select the **Device-specific** icon next to the field and leave the field blank.
4. For each Cisco device type, create a device template.
 5. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
 6. Click **Device Templates**, and select the desired device template from the template list table.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

7. Click ..., and click **Export CSV**.
8. Repeat Steps 7 and 8 for each device template.

Edit the exported CSV file, adding at a minimum the device serial number, device system IP address, and device hostname for each device in the overlay network. Then add values for desired device-specific variables for each device. Note that variable names cannot contain forward slashes (/), backwards slashes (\), or parentheses (()).

If desired, you can combine the CSV files into a single file.

Import a CSV File

To use the device-specific variable values in the CSV file, import the file when you are attaching a device template to the Viptela device:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. For the desired template, click ..., and select **Attach Devices**.
4. In the **Attach Devices** dialog box, select the desired devices in **Available Devices** and click the arrow to move them to **Selected Devices**.
5. Click **Attach**.
6. Click the Up arrow. The Upload CSV File box displays.

7. Choose the CSV file to upload, and click **Upload**.

During the attachment process, click Import file to load the Excel file. If Cisco SD-WAN Manager detects duplicate system IP addresses for devices in the overlay network, it displays a warning message or a pop-up window. You must correct the system IP addresses to remove any duplicates before you can continue the process of attaching device templates to Viptela devices.

Manually Enter Values for Device-Specific Variables and for Optional Rows

For parameters in a feature template that you configure as device-specific, when you attach a device template to a device, Cisco SD-WAN Manager prompts you for the values to use for these parameters. Entering device-specific values in this manner is useful in test or POC networks, or if you are deploying a small network. This method generally does not scale well for larger networks.

For situations in which the configuration for many devices is identical except for a few parameters, in the feature configuration template, you can specify that the parameter be an optional row in the configuration. By selecting optional row, the feature template automatically marks the parameters as device-specific, and these parameters are dimmed so that you cannot set them in the template. You do not have to individually mark the parameters as device specific. Then, when you attach a device template to a device, Cisco SD-WAN Manager prompts you for the values to use for these parameters. Using optional rows to enter device-specific values is useful when a group of many Cisco IOS XE Catalyst SD-WAN devices provide identical services at their branch or site, but individual routers have their own hostname, IP address, GPS location, and other site or store properties, such as BGP neighbors.

Optional rows are available for some parameters in some feature configuration templates. To treat a parameter or set of parameters as an optional row, click the **Mark as Optional Row** box. For these types of parameters, the feature configuration template has a table listing all the configured parameters. The Optional column indicates which are optional rows,

To manually enter values for device-specific variables or for variables in optional rows when you attach the template to a device:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and select the desired device template.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. Click ..., and click **Attach Devices**. The **Attach Devices** dialog box opens.
4. Choose one or more devices from **Available Devices** and move them to **Selected Devices**.
5. Click **Attach**.
6. In the **Chassis Number** list, select the desired device.
7. Click ..., and click **Edit Device Template**. The **Update Device Template** dialog box opens.
8. Enter values for the optional parameters. When you are using optional rows, if you do not want to include the parameter for the specific device, do not specify a value.
9. Click **Update**.
10. Click **Next**.

If any devices have the same system IP address, a dialog box appears or an error message is displayed when you click **Next**. Modify the system IP addresses so that there are no duplicates, and click **Save**. Then click **Next** again.



Note You need to shut down the OMP on the device, before changing the system-ip on the device.

11. In the left pane, select the device. The right pane displays the device configuration and the **Config Preview** tab in the upper right corner is selected.
12. Click **Config Diff** to preview the differences between this configuration and the configuration currently running on the device, if applicable. To edit the variable values entered in the previous screen, click **Back**.
13. Click **Configure Devices** to push the configuration to the devices.
The Status column displays whether the configuration was successfully pushed. Click the **right angle bracket** to the left of the row to display details of the push operation.

View Device Templates

View a Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates** or **Feature Templates**, and select a template you wish to view.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**, and **Feature Templates** is titled **Feature**.

3. Click ..., and then click **View**.

View Device Templates Attached to a Feature Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**, and select a template you wish to view.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

3. Click ..., and click **Show Attached Device Templates**.

Device Templates dialog box opens, displaying the names of the device templates to which the feature template is attached.

View Devices Attached to a Device Template

For a device template that you created from feature templates:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and select a template you wish to view.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. Click ..., and click **Attach Devices**.
4. From **Attach Devices**, click **Attached Devices**.

For a device template that you created from a CLI template:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and select a template you wish to view.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. Click ..., and then click **Show Attached Devices**.

Attach and Detach a Device Template

To configure a device on the network, you attach a device template to the device. You can attach only one device template to a device, so the template—whether you created it by consolidating individual feature templates or by entering a CLI text-style configuration—must contain the complete configuration for the device. You cannot mix and match feature templates and CLI-style configurations.

On Cisco IOS XE Catalyst SD-WAN devices in the overlay network, you can perform the same operations, in parallel, from one or more Cisco SD-WAN Manager servers. You can perform the following template operations in parallel:

- Attach a device template to devices
- Detach a device template from a device
- Change the variable values for a device template that has devices attached to it

For template operations, the following rules apply:

- When a device template is already attached to a device, you can modify one of its feature templates. Then when you click **Update > Configure Devices**, all other template operations—including attach devices, detach devices, and edit device values—are locked on all Cisco SD-WAN Manager servers until the update operation completes. This means that a user on another Cisco SD-WAN Manager server cannot perform any template operations until the update completes.
- You can perform the attach and detach device template operations on different devices, from one or more Cisco SD-WAN Manager servers, at the same time. However, if any one of these operations is in progress

on one Cisco SD-WAN Manager server, you cannot edit any feature templates on any of the servers until the attach or detach operation completes.



Note You need to recreate the feature templates as the templates created prior to Cisco vManage Release 20.5 fails when attached to the device.

If the device being configured is present and operational on the network, the configuration is sent to the device immediately and takes effect immediately. If the device has not yet joined the network, the pushing of the configuration to the device is scheduled. When the device joins the network, Cisco SD-WAN Manager pushes the configuration immediately after it learns that the device is present in the network.

Attach a Device Template to Devices

You can attach the same templates to multiple devices, and you can do so simultaneously, in a single operation.

To attach a device template to one or more devices:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates** and select the desired template.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. Click ..., and click **Attach Devices**. The **Attach Devices** dialog box opens with the **Select Devices** tab selected
4. In the **Available Devices** column on the left, select a group and search for one or more devices, select a device from the list, or click **Select All**.
5. Click the arrow pointing right to move the device to the **Selected Devices** column on the right.
6. Click **Attach**.
7. If the template contains variables, enter the missing variable values for each device you selected in one of the following ways:
 - Enter the values manually for each device either in the table column or by clicking ... and **Edit Device Template**. When you are using optional rows, if you do not want to include the parameter for the specific device, do not specify a value.
 - Click **Import File** to upload a CSV file that lists all the variables and defines each variable's value for each device.
8. Click **Update**
9. Click **Next**.

If any devices have the same system IP address, a dialog box appears or an error message is displayed when you click **Next**. Modify the system IP addresses so that there are no duplicates, and click **Save**. Then click **Next** again.

10. In the left pane, select the device, to preview the configuration that is ready to be pushed to the device. The right pane displays the device's configuration and the **Config Preview** tab is selected. Click the **Config Diff** tab to view the differences between this configuration and the configuration currently running on the device, if applicable. Click the **Back** button to edit the variable values entered in the previous screen.
11. If you are attaching a Cisco IOS XE Catalyst SD-WAN device, click **Configure Device Rollback Timer** to configure the time interval at which the device rolls back to its previous configuration if the router loses its control connection to the overlay network. The **Configure Device Rollback Time** dialog box is displayed.
 - a. From the **Devices** drop-down list, select a device.
 - b. To enable the rollback timer, in the **Set Rollback slider**, drag the slider to the left to enable the rollback timer. When you do this, the slider changes in color from gray to green.
 - c. To disable the rollback timer, click the **Enable Rollback** slider. When you disable the timer, the Password field dialog box opens. Enter the password that you used to log in to Cisco SD-WAN Manager.
 - d. In the **Device Rollback Time slider**, drag the slider to the desired value. The default time is 5 minutes. You can configure a time from 6 to 15 minutes.
 - e. To exclude a device from the rollback timer setting, click **Add Exception** and select the devices to exclude.
 - f. The table at the bottom of the **Configure Device Rollback Time** dialog box lists all the devices to which you are attaching the template and their rollback time. To delete a configured rollback time, click the **Trash** icon from the device name.
 - g. Click **Save**.
12. Click **Configure Devices** to push the configuration to the devices. The **Status** column displays whether the configuration was successfully pushed. Click the right angle bracket to display details of the push operation.

Export a Variables Spreadsheet in CSV Format for a Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates** and select the desired template.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. Click ..., and click **Export CSV**.

Determine Why a Device Rejects a Template

When you attach a template to a device using the screen, the device might reject the template. One reason that this may occur is because the device template contains incorrect variable values. When a device rejects a template, it reverts to the previous configuration.

To determine why the device rejected the template:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. Locate the device. The **Template Status** column indicates why the device rejected the template.

Edit a Device Template When a Push Fails

Table 76: Feature History

Feature Name	Release Information	Description
Retrieve Last Edited Configuration	Cisco IOS XE Catalyst SD-WAN Release 17.5.1a Cisco vManage Release 20.5.1	This feature allows you to review the last edited configuration when a configuration push to the device fails. A copy of the last edited configuration is saved and can be retrieved to allow edits to the configuration before the next push.

If you pushed a configuration to a device, and if the push fails, you can review the configuration you last edited to identify any issues that caused a failure in pushing the configuration to the device.

Prerequisites

To review your last edited configuration, a device template must be attached to a device.

Review Last Edited Configuration in Cisco SD-WAN Manager

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and choose a device template.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. Click ..., and choose **Edit**.

The **CLI Configuration** box displays the current running configuration on the device.

4. Click **Load Last Attempted Config** to view the last edited configuration.
5. Click **Config Diff** to view the differences in the current configuration versus the last edited configuration. The **Config Diff** option is available when you modify the configuration or when you click **Load Last Attempted Config**.

6. Click **Config Preview**.



Note **Load Last Attempted Config** and the **Config Diff** option is available only when the configuration is not being pushed to the device.

7. Click **Update**.
8. Click **Configure Devices** to push the configuration to the devices. The **Status** column displays whether the configuration was successfully pushed. Click > to view the details of the push operation.

Change the Device Rollback Timer

By default, when you attach a Cisco IOS XE Catalyst SD-WAN device to a configuration template, if the router is unable to successfully start after 5 minutes, it returns to, or rolls back to, the previous configuration. For a configuration that you have created from the CLI, you can change the device's rollback timer:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and choose a device template.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. Click ..., and click **Change Device Values**.
The right pane displays the device's configuration, and the **Config Preview** tab is selected.
4. In the left pane, click the name of a device.
5. Click **Configure Device Rollback Timer**. The **Configure Device Rollback Time** pop up page is displayed.
6. From the **Devices** drop-down list, select a device.
7. To enable the rollback timer, in the **Set Rollback slider** drag the slider to the left to enable the rollback timer. When you do this, the slider changes in color from gray to green.
8. To disable the rollback timer, click **Enable Rollback slider**. When you disable the timer, the **Password** field dialog box appears. Enter the password that you used to log in to Cisco SD-WAN Manager.
9. In the **Device Rollback Time** slider, drag the slider to the desired value. The default time is 5 minutes. You can configure a time from 6 to 15 minutes.
10. To exclude a device from the rollback timer setting, click **Add Exception** and select the devices to exclude.
11. The table of the **Configure Device Rollback Time** dialog box lists all the devices to which you are attaching the template and their rollback time. To delete a configured rollback time, click the **Trash** icon of the device name.
12. Click **Save**.

13. Click **Configure Devices** to push the configuration to the devices. The **Status** column displays whether the configuration was successfully pushed. Click **(+)** to display details of the push operation.

Preview Device Configuration and View Configuration Differences

For a configuration that you have created from the CLI:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and choose the desired device template.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. Click **...**, and click **Change Device Values**.
The right pane displays the device's configuration, and **Config Preview** is selected.
4. Click the name of a device.
5. Click **Config Diff** to view the differences between this configuration and the configuration currently running on the device, if applicable. Click **Back** to edit the variable values entered in the previous screen.
6. Click **Configure Devices** to push the configuration to the devices. The Status column displays whether the configuration was successfully pushed. Click the right angle bracket to display details of the push operation.

Change Variable Values for a Device

For a configuration that you have created from device configuration templates, if the templates contain variables, Cisco SD-WAN Manager can automatically populate the variables with actual values when you attach the templates to the devices. To do this, you create an Excel file that lists the variable values for each device and save the file in CSV format. You can also enter values for these variables manually.

After you have pushed the configuration to a device, you can change the value assigned to any variable:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and choose the desired device template.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. Click **...**, and click **Change Device Values**.
The screen displays a table of all the devices that are attached to that device template.
4. For the desired device, click **...**, and click **Edit Device Template**.
5. In the **Update Device Template** dialog box, enter values for the items in the variable list.
6. Click **Update**.

7. Click **Next**.
8. Click **Configure Devices** to push the configuration to the device. The Status column displays if the configuration was successfully pushed or not. Click the right angle bracket to display the details of the push operation.

Default Device Templates

Table 77: Feature History

Feature Name	Release Information	Description
Default Device Templates	Cisco IOS XE Catalyst SD-WAN Release 17.2.1r	A default device template provides basic information that you can use to bring up devices in a deployment quickly. This feature is supported on the Cisco Cloud Services Router 1000V Series, Cisco C1111-8PLTELA Integrated Services Routers, and Cisco 4331 Integrated Services Routers.

A default device template provides basic information that you can use to bring up devices in a deployment. It provides a way for you to quickly provision devices with the minimum information that they need to operate in your network.

You cannot directly edit or update information in a device default template, but you can copy the template and then edit the copy.

To use a default device template:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. From the **Template Type** drop-down list, select **Default**.

A list of default device templates displays.

4. Perform any of these actions:

- To attach a default device template to devices, click ..., and select **Attach Devices**.

In the **Attach Devices** dialog box, select the devices that you want attach, and then click **Attach**.

- To view the configuration settings for a default device template, click ..., and choose **View**.

- To copy a default device template, click ..., and choose **View**.

In the **Template Copy** dialog box, enter a unique name and a description for the copy that you are creating, and then click **Copy**.

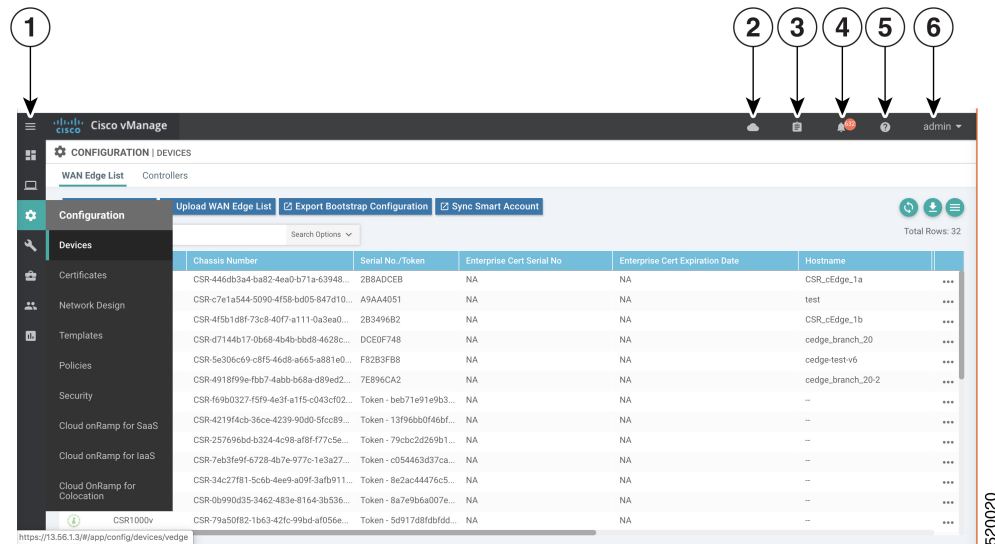
The copied version becomes a feature template that you can edit.

- To create an Excel file in CSV format that contains device-specific settings from a device template, click ..., and choose **Export CSV**. Use the dialog box that displays to open or save the CSV file.

You can use this CSV file as a reference for device-specific settings when you create other device templates.

Configuring Devices using Cisco SD-WAN Manager

Use the **Devices** screen to add and delete devices, toggle the mode of a device between CLI and Cisco SD-WAN Manager, upload the WAN Edge Serial number file, export bootstrap configuration and, and perform other device-related tasks.



1	Menu
2	CloudExpress
3	Tasks
4	Alarms
5	Help
6	User Profile

Change Configuration Modes

A device can be in either of these configuration modes:

- Cisco SD-WAN Manager mode—A template is attached to the device and you cannot change the configuration on the device by using the CLI.

- CLI mode – No template is attached to the device and the device can be configured locally by using the CLI.

When you attach a template to a device from Cisco SD-WAN Manager, it puts the device in Cisco SD-WAN Manager mode. You can change the device back to CLI mode if needed to make local changes to its configuration.

To toggle a router from Cisco SD-WAN Manager mode to CLI mode:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
2. Click **WAN Edge List**, and select a device.
3. Click the **Change Mode** drop-down list and select **CLI mode**.



Note Starting from Cisco IOS XE SD-WAN Release 17.11.1a, click the ... icon adjacent to the device that you want to change from Cisco SD-WAN Manager mode to the CLI mode and click **Config Lock (Provision Device)**.

The **Config Lock (Provision Device)** option appears only if a template is attached to the device or if a configuration group is deployed to the device.

An SSH window opens. To log in to the device, enter a username and password. You can then issue CLI commands to configure or monitor the device.

To toggle a controller device from Cisco SD-WAN Manager mode to CLI mode:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
2. Click **Controllers**, and select a device.



Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

3. Click the **Change Mode** drop-down list.
4. Select **CLI mode** and then select the device type. The **Change Mode - CLI** window opens.
5. From the **Manager mode** pane, select the device and click the right arrow to move the device to the **CLI mode** pane.
6. Click **Update to CLI Mode**.

An SSH window opens. To log in to the device, enter a username and password. You can then issue CLI commands to configure or monitor the device.



Note Starting from Cisco IOS XE SD-WAN Release 17.11.1a, click the ... icon adjacent to the device that you want to change from Cisco SD-WAN Manager mode to the CLI mode and click **Config Lock (Provision Device)**.

The **Config Lock (Provision Device)** option appears only if a template is attached to the device or if a configuration group is deployed to the device.

Upload WAN Edge Router Authorized Serial Number File

Table 78: Feature History

Feature Name	Release Information	Description
Remove Certificate SUDI requirement	Cisco IOS XE Catalyst SD-WAN Release 17.3.1a Cisco vManage Release 20.3.1	This feature allows you to use a subject SUDI serial number instead of a certificate serial number to add a device to a Cisco Catalyst SD-WAN overlay network.

The WAN eEdge router authorized serial number file contains, as applicable, the subject SUDI serial number, the chassis number, and the certificate serial numbers of all valid Cisco IOS XE Catalyst SD-WAN devices in the overlay network. You retrieve a serial number file from the Cisco Plug-and-Play (PnP) portal and upload it to Cisco SD-WAN Manager. (For more information about Cisco PnP, see [Cisco Plug and Play Support Guide for Cisco Catalyst SD-WAN Products](#).) From Cisco SD-WAN Manager, you send the file to the controllers in the network. This file is required to allow the Cisco Catalyst SD-WAN overlay network components to validate and authenticate each other and to allow the overlay network to become operational.

To upload the WAN edge router authorized serial number file to Cisco SD-WAN Manager and then download it to controllers in the network:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
2. Click **WAN Edge List**, and click **Upload WAN Edge List**.
The Quick Connect workflow opens, enabling you to upload the serial number file. See [Quick Connect Workflow](#) for details.
3. (This step applies only for releases earlier than Cisco Catalyst SD-WAN Manager Release 20.14.1.) Under **Upload WAN Edge List** screen:
 - a. Click **Choose File** and select the WAN edge router authorized serial number file you received from Cisco PnP.
 - b. To automatically validate the routers and send their chassis and serial numbers to the controllers, ensure that the **Validate the uploaded vEdge List and send to controllers** check box is selected. If you do not select this option, you must individually validate each router in **Configuration > Certificates > WAN Edge List**.
 - c. Click **Upload**.

A list of routers in the network is displayed in the router table, with details about each router.

Starting from Cisco vManage Release 20.9.2, you can monitor the newly added WAN Edge devices in the **Monitor > Devices** page.

Upload WAN Edge Router Serial Numbers from Cisco Smart Account

To allow Cisco Catalyst SD-WAN overlay network components to validate and authenticate each other and to allow the overlay network to become operational, Cisco Catalyst SD-WAN requires chassis numbers of all valid Cisco IOS XE Catalyst SD-WAN devices in the overlay network.

In addition, certificate serial numbers, subject SUDI serial numbers, or both numbers are required for all devices.

To upload the WAN edge router authorized serial numbers from a Cisco Smart account to Cisco SD-WAN Manager and then download it to all the controllers in the overlay network:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
2. Click **WAN Edge List**, and click **Sync Smart Account**.
3. In the **Sync Smart Account** window:
 - a. Enter the **Username** and **Password** for your Smart account.
 - b. To automatically validate the routers and send their chassis and serial numbers to the controllers, check the **Validate the Uploaded WAN Edge List and Send to Controllers** check box. If you do not select this option, you must individually validate each router in **Configuration > Certificates > WAN Edge List**.
 - c. Click **Sync**.

A list of routers in the network is displayed in the router table, with details about each router.

Starting from Cisco vManage Release 20.9.2, you can monitor the newly added WAN Edge devices in the **Monitor > Devices** page.

Export Device Data in CSV Format

In an overlay network, you might have multiple devices of the same type that have identical or effectively identical configurations. For example, in a network with redundant Cisco Catalyst SD-WAN Controllers, each controller must be configured with identical policies. Another example is a network with Cisco IOS XE Catalyst SD-WAN devices at multiple sites, where each Cisco IOS XE Catalyst SD-WAN device is providing identical services at each site.

Because the configurations for these devices are essentially identical, you can create one set of feature templates, which you then consolidate into one device template that you use to configure all the devices. You can create an Excel file in CSV format that lists the variables and defines each device specific variable value for each device. Then you can load the file when you attach a device template to a device.

To export data for all devices to a file in CSV format, click the Export icon. This icon, which is a downward-pointing arrow, is located to the right of the filter criteria both in the WAN Edge List and in the Controllers tab.



Note Starting from Cisco IOS XE Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

Cisco SD-WAN Manager downloads all data from the device table to an Excel file in CSV format.

View and Copy Device Configuration

View a Device's Running Configuration

Running configuration is configuration information that Cisco SD-WAN Manager obtains from the memory of a device. This information can be useful for troubleshooting.

To view a device's running configuration:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
2. Click **WAN Edge List** or **Controllers**, and select the device.



Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

3. Click ..., and click **Running Configuration**.

View a Device's Local Configuration

Local configuration is configuration that Cisco SD-WAN Manager has stored for a device. This information can be useful for troubleshooting or for determining how to access a device if, for example, a device is not reachable from Cisco SD-WAN Manager.

To view a device's local configuration created using Configuration ► Templates:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
2. Click **WAN Edge List** or **Controllers**, and select the device.



Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

3. Click ..., and click **Local Configuration**.

Copy Router Configuration

When you are replacing one router at a site with another router, you copy the old router's configuration to the new router. Then you remove the old router from the network and add the new one.

To copy the configuration from the old router to the new router:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Certificates**.
2. Mark the new Cisco IOS XE Catalyst SD-WAN device as invalid.
3. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
4. Under **WAN Edge List**, select the old router.
5. Click ..., and click **Copy Configuration**.

6. In the **Copy Configuration** window, select the new router.
7. To confirm the copy of the configuration, click **Update**.

After you have copied the configuration to the new router, you can add the new router to the network. First, delete the old router from the network, as described below. Then add the new router to the network:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Certificates**.
2. Mark the new router as valid.
3. Click **Send to Controller**.

Delete a WAN Edge Router

Delete a router if you need to remove it from your deployment. Doing so removes from the WAN edge router serial number list any of the following items that are stored for the router:

- Chassis number
- Certificate serial number
- Subject SUDI serial number



Note Deleting a router also permanently removes the router configuration from Cisco SD-WAN Manager.

To delete a router:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Certificates**.
2. Mark the WAN Edge router as invalid.
3. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
4. Click **WAN Edge List**, and select the router.
5. Click ..., and click **Delete WAN Edge**.
6. To confirm deletion of the device, click **OK**.
7. From the Cisco SD-WAN Manager menu, choose **Configuration > Certificates**.
8. Click **Send to Controller**.

Decommission a Cloud Router

Decommissioning a cloud router (such as a C8000v) removes the device's serial number from Cisco SD-WAN Manager and generates a new token for the device. To do so:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
2. Click **WAN Edge List**, and select a cloud router.
3. Click ..., and click **Decommission WAN Edge**.

4. To confirm the decommissioning of the router, click **OK**.

From Cisco Catalyst SD-WAN Manager Release 20.15.1, the process to decommission a WAN edge router has been modified. The scenarios below highlight the updates.

Table 79: Updates to Deleting a WAN Edge Router

Scenario	Action
Decommission a compromised device	Click Delete WAN Edge
Device is reachable	Perform a mandatory configuration unlock before proceeding with the decommissioning process.
Device is unreachable	The device will be unlocked after certain time when the device is unreachable. If you no longer have the device onboarded, with no Cisco SD-WAN Manager visibility, you can hold the power button for 5–10 seconds for a config reset, or 10–20 seconds for a software reset. If you have console/terminal access, you can run the request config reset command. Reuse of UUID is only possible after decommissioning a device. In this case, the Decommission option is not available since the device is offline.



Note

- The **Decommission WAN Edge** feature applies only to cloud WAN edge devices and retains the cloud WAN Edge's UUID generated on the PnP Portal.
- Physical devices do not support the **Decommission WAN Edge** functionality.

View Template Log and Device Bringup

View Log of Template Activities

A log of template activities contains information that relates to creating, editing, and deleting configuration templates, and the status of attaching configuration templates to devices. This information can be useful for troubleshooting.

To view a log of template activities:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
2. Click **WAN Edge List** or **Controllers**, and select the device.



Note Starting from Cisco IOS XE Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the Controllers tab is renamed as the Control Components tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

3. Click ..., and click **Template Log**.

View Status of Device Bringup

You can view the status of the operations involved in bringing a router or controller up in the overlay network. This information can help you monitor these operations.

To view the status of a device bringup:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
2. Click **WAN Edge List** or **Controllers**, and select the device.



Note Starting from Cisco IOS XE Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the Controllers tab is renamed as the Control Components tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

3. Click ..., and click **Device Bring Up**.

Add a Cisco SD-WAN Validator

A Cisco Catalyst SD-WAN Validator automatically orchestrates connectivity between Cisco IOS XE Catalyst SD-WAN devices and Cisco SD-WAN Manager. If any Cisco IOS XE Catalyst SD-WAN device or Cisco Catalyst SD-WAN Controller is behind a NAT, the Cisco Catalyst SD-WAN Validator also serves as an initial NAT-traversal orchestrator. To add a Cisco Catalyst SD-WAN Validator:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
2. Click **Controllers**.



Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN.

3. Click **Add Validator**.
4. In the **Add Validator** window:
 - a. Enter **Validator Management IP Address** of the Cisco SD-WAN Validator.
 - b. Enter the **Username** and **Password** to access the Cisco SD-WAN Validator.
 - c. To allow the certificate-generation process to occur automatically, check the **Generate CSR** check box.
 - d. Click **Add**.

5. Repeat Steps 2, 3 and 4 to add additional Cisco Catalyst SD-WAN Validators.

The new Cisco Catalyst SD-WAN Validator is added to the list of controllers in the Controllers screen.

Configure Cisco SD-WAN Controllers

Add a Cisco SD-WAN Controller

After the Cisco Catalyst SD-WAN Validator authenticates Cisco IOS XE Catalyst SD-WAN devices, the Cisco Catalyst SD-WAN Validator provides Cisco IOS XE Catalyst SD-WAN devices information that they need to connect to the Cisco Catalyst SD-WAN Controller. A Cisco Catalyst SD-WAN Controller controls the flow of data traffic throughout the network via data and app-route policies. To configure Cisco Catalyst SD-WAN Controllers:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
2. Click **Controllers**.



Note Cisco IOS XE Catalyst SD-WAN Release Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

3. Click **Add Controller**.
4. In the **Add Controller** window:
 - a. Enter the system IP address of the Cisco Catalyst SD-WAN Controller.
 - b. Enter the username and password to access the Cisco Catalyst SD-WAN Controller.
 - c. Select the protocol to use for control-plane connections. The default is DTLS. The DTLS (Datagram Transport Layer Security) protocol is designed to provide security for UDP communications.
 - d. If you select TLS, enter the port number to use for TLS connections. The default is 23456.
The TLS (Transport Socket Layer) protocol that provides communications security over a network.
 - e. Check the **Generate CSR** check box to allow the certificate-generation process to occur automatically.
 - f. Click **Add**.
5. Repeat Steps 2, 3 and 4 to add additional Cisco Catalyst SD-WAN Controllers. Cisco SD-WAN Manager can support up to 20 Cisco Catalyst SD-WAN Controllers in the network.

The new Cisco Catalyst SD-WAN Controller is added to the list of controllers in the Controllers screen.

Edit Controller Details

Editing controller details lets you update the IP address and login credentials of a controller device. To edit controller details:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
2. Click **Controllers**, and select the controller.



Note Cisco IOS XE Catalyst SD-WAN Release Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

3. Click ..., and click **Edit**.
4. In the **Edit** window, edit the IP address and the login credentials.
5. Click **Save**.

Delete a Controller

Deleting a controller removes it from the overlay. Delete a controller if you are replacing it or if you no longer need it in your network.

To delete a controller:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
2. Click **Controllers**, and select the controller.



Note Cisco IOS XE Catalyst SD-WAN Release Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

3. Click ..., and click **Invalidate**.
4. To confirm the removal of the device and all its control connections, click **OK**.

Configure Reverse Proxy on Controllers

To configure reverse proxy on an individual Cisco SD-WAN Manager and Cisco Catalyst SD-WAN Controller:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
2. Click **Controllers**, and select the controller.



Note Cisco IOS XE Catalyst SD-WAN Release Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

3. Click ..., and click **Add Reverse Proxy**.
The **Add Reverse Proxy** dialog box is displayed.
4. Click **Add Reverse Proxy**.
5. Configure the private IP address and port number for the device. The private IP address is the IP address of the transport interface in VPN 0. The default port number is 12346. This is the port used to establish the connections that handle control and traffic in the overlay network.

6. Configure the proxy IP address and port number for the device, to create the mapping between the private and public IP addresses and port numbers.
7. If the Cisco SD-WAN Manager NMS or Cisco Catalyst SD-WAN Controller has multiple cores, repeat Steps 5 and 6 for each core.
8. Click **Add**.

To enable reverse proxy in the overlay network, from the Cisco SD-WAN Manager menu, choose **Administration > Settings**. Then **Proxy**. Go to **Reverse Proxy**, and enable **Reverse Proxy**. Click **Save**.

Configure UCSE Using a Configuration Group

Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

Step 1 From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

Step 2 Create and configure a UCSE feature in Other profile.

- a. Configure parameter scope.

Table 80: Parameter

Parameter Scope	Scope Description
Global (Indicated by a globe icon)	Enter a value for the parameter and apply that value to all devices. Examples of parameters that you might apply globally to a group of devices are DNS server, syslog server, and interface MTUs.
Device Specific (Indicated by a host icon)	Use a device-specific value for the parameter. Choose Device Specific to provide a value for the key in the Enter Key field. The key is a unique string that helps identify the parameter. To change the default key, type a new string in the Enter Key field. Examples of device-specific parameters are system IP address, host name, GPS location, and site ID.
Default (indicated by a check mark)	The default value is shown for parameters that have a default setting.

- b. Configure options for the UCSE feature.

Table 81: Settings

Field	Description
Type	Choose a feature from the drop-down list.

Field	Description
Feature Name*	Enter a name for the feature. The name can be up to 128 characters and can contain only alphanumeric characters.
Description	Enter a description of the feature. The description can be up to 2048 characters and can contain only alphanumeric characters.

- c. Configure basic settings.

Table 82: Basic Configuration

Field	Description
Bay*	Specify the number for the SAS drive bays. The input value must be an integer.
Slot*	Specify the slot numbers for the mezzanine adapters. The input value must be an integer.

- d. Configure IMC.

Table 83: IMC

Field	Description
Access Port	Configure the interface as an access port. You can configure only one VLAN on an access port, and the port can carry traffic for only one VLAN. Not all hardware models have a dedicated access port. See the release notes for your Cisco Catalyst SD-WAN release for the supported hardware. Available options: • Dedicated • Shared Configure the appropriate port (GE or TE) based on the hardware module.
IPv4 Address*	Provide the UCS-E management port address.
Default Gateway*	Gateway tracking determine, for static routes, whether the next hop is reachable before adding that route to the device's route table. Default: Enabled.
VLAN ID	Provide the VLAN number, which can be a value from 1 through 4094.
Assign Priority	Assign the priority.

- e. Configure advanced settings.

Table 84: Advanced Configuration

Field	Description
Interface Name*	Specify the name of the interface.

Field	Description
Layer	Specify the layer details necessary for traffic exchange between different VLANs.
UCSE Interface VPN	Specify the details of the UCS-E interface VPN.
IPv4 Address	Provide the UCS-E management port address.

What to do next

Also see [Deploy a configuration group](#).

Create a UCS-E Template

Table 85: Feature History

Feature Name	Release Information	Feature Description
Create a UCS-E Template	Cisco IOS XE Catalyst SD-WAN Release 16.12.1b	This feature allows you to connect a UCS-E interface with a UCS-E server through the interface feature template.

For more information about the Cisco Unified Computing System (UCS) E-Series Servers, see the [Cisco UCS E-Series Servers and the Cisco UCS E-Series Network Compute Engine Hardware Installation Guide](#).

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

3. Click **Add Template**.
4. Select a Cisco IOS XE Catalyst SD-WAN device from the list.
5. From the **Other Templates** section, click **UCSE**.

The UCSE Feature template opens. The top of the form contains fields for naming the template, and the bottom contains fields for configuring the Integrated Management Controller (IMC).

6. In the **Template Name** field, enter a name for the template.
The name can be up to 128 characters and can contain only alphanumeric characters.
7. In the **Description** field, enter a description of the template.
The description can be up to 2048 characters and can contain only alphanumeric characters.

Configure Bay and Slot for Template

Click the Basic Configuration tab to configure the bay and the slot for the template.

Parameter Name	Description
Bay	Specify the number for the SAS drive bays.
Slot	Specify the slot numbers for the mezzanine adapters.

IMC Configuration

Click the IMC tab to configure the IMC parameters for the template.

Parameter Name	Description
Access Port	Configure the interface as an access port. You can configure only one VLAN on an access port, and the port can carry traffic for only one VLAN. Not all hardware models have a dedicated access port. See the Release Notes for your Cisco Catalyst SD-WAN release for the supported hardware. Available options: • Dedicated • Shared The type of port, GE or TE, depends on the hardware model. For example: <pre>Router(config-ucse)#imc access-port shared-lom ? GE1 GE1 TE2 TE2 TE3 TE3 console Console failover Failover</pre> Some hardware models have GE ports whereas some have TE ports. Depending on the hardware module, the appropriate port (GE or TE) needs to be configured. Otherwise you will get an error. • You can obtain the UCS-E module hardware model type by using the following commands: show inventory show platform • Failover - sub-option under Shared. For example: <pre>Router(config)#ucse subslot 1/0 Router(config-ucse)#imc access-port ? MGMT MGMT Interface shared-lom Shared LOM Router(config-ucse)#imc access-port shared-lom ? GE1 GE1 TE2 TE2 TE3 TE3 console Console failover Failover</pre>
IPv4 Address	Provide the UCS-E management port address.

Parameter Name	Description
Default Gateway	Gateway tracking determine, for static routes, whether the next hop is reachable before adding that route to the device's route table. Default: Enabled.
VLAN ID	Provide the VLAN number, which can be a value from 1 through 4094.
Assign Priority	Assign the priority.

Parameter Scope	Scope Description
Global (indicated by a globe icon)	Enter a value for the parameter and apply that value to all devices.
Device Specific (indicated by a host icon)	Use a device-specific value for the parameter. For device-specific parameters, you cannot enter a value in the feature template. You enter the value when you attach a Cisco Catalyst SD-WAN device to a device template. When you click Device Specific, the Enter Key box opens. This box displays a key, which is a unique string that identifies the parameter in a CSV file that you create. This file is an Excel spreadsheet that contains one column for each key. The header row contains the key names (one key per column), and each row after that corresponds to a device and defines the values of the keys for that device. You upload the CSV file when you attach a Cisco Catalyst SD-WAN device to a device template. To change the default key, type a new string and move the cursor out of the Enter Key box.
Default	When Default is selected, this field is not enabled.



CHAPTER 9

Device Tagging

Table 86: Feature History

Feature Name	Release Information	Description
User-Defined Device Tagging	Cisco IOS XE Catalyst SD-WAN Release 17.8.1a Cisco vManage Release 20.8.1	This feature helps you add tags to devices. You can use the tags for grouping, describing, finding, or managing devices.
Enhancements to User-Defined Device Tagging	Cisco IOS XE Catalyst SD-WAN Release 17.12.1a Cisco Catalyst SD-WAN Manager Release 20.12.1	Device tagging has the following new functionality: • When you add devices to a configuration group using rules, you can choose Match All or Match Any. • You can use Starts With and Ends With operator conditions when you add devices to a configuration group using rules. In addition, the button formerly called Add New Tag is now Create New Tag .

- [Information About Device Tagging, on page 255](#)
- [Supported Devices for Device Tagging, on page 256](#)
- [Prerequisites for Device Tagging, on page 256](#)
- [Restrictions for Device Tagging \(Cisco SD-WAN Manager 20.15.4.1 and earlier\), on page 256](#)
- [Add Tags to Devices Using Cisco SD-WAN Manager, on page 256](#)
- [Delete Tags, on page 257](#)

Information About Device Tagging

The Device Tagging feature helps you do the following:

- Add tags to devices: Tagging helps you manage devices more effectively. You can use the tags for grouping, describing, or finding devices. You can add more than one tag to a device.

- Add devices to configuration groups based on tagging: Using tags, you can create rules to define which devices need to be automatically added to a configuration group. For complete information about creating rules, see [Add Devices to a Configuration Group Using Rules](#).



Note You can use this feature in both the single-tenant and multitenant deployments.

Supported Devices for Device Tagging

This feature is supported only on Cisco IOS XE Catalyst SD-WAN devices.

Prerequisites for Device Tagging

Minimum software version for Cisco IOS XE Catalyst SD-WAN devices: Cisco IOS XE Catalyst SD-WAN Release 17.8.1a

Minimum software version for Cisco SD-WAN Manager: Cisco vManage Release 20.8.1

Restrictions for Device Tagging (Cisco SD-WAN Manager 20.15.4.1 and earlier)

Restrictions for Device Tagging

- A maximum of 25 tags can be created in a Cisco SD-WAN Manager instance.
- You can assign up to 25 tags per device.
- Only one tag rule can be added to a configuration group.

Restrictions for Tag names:

- Can be up to 25 characters long.
- Can include alphanumeric characters, hyphens (-), and underscores (_).
- Cannot include spaces or other special characters.
- The tag name is case-sensitive.

Add Tags to Devices Using Cisco SD-WAN Manager

You can add tags to devices in one of the following ways:

Use the Devices Window

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.

2. Click **WAN Edge List** and choose a device.
3. Click **Add Tags**
4. Choose a tag from the list of existing tags or click **Create New Tag** to create a new tag.
In Cisco vManage Release 20.11.1 and earlier, this was called **Add New Tag**.
5. Click **Apply**.
The specified tag is added to the device.

Use the Quick Connect Workflow

1. From the Cisco SD-WAN Manager menu, choose **Workflows > Launch Workflows**.
2. Click **Quick Connect**.
The **Quick Connect** workflow starts.
3. Click **Add Tags**
4. Follow the instructions provided in the workflow.
5. Tag the devices.
The specified tag is added to the device.

**Note**

You can edit the tags that are currently associated with a device by either adding new tags or removing unwanted tags.

Delete Tags

You can delete only those tags that are not added to a device or are not a part of a tag rule.

1. From the Cisco SD-WAN Manager menu, choose **Administration > Tag Management**.
2. Choose the tags that you want to delete.
3. Click **Delete Tags**.
4. In the confirmation dialog box, click **Yes**.



CHAPTER 10

Network Hierarchy and Resource Management

Table 87: Feature History

Feature Name	Release Information	Description
Network Hierarchy and Resource Management	Cisco IOS XE Catalyst SD-WAN Release 17.9.1a Cisco vManage Release 20.9.1	This feature enables you to create a network hierarchy in Cisco SD-WAN Manager to represent the geographical locations of your network. The network hierarchy and the associated resource IDs, including region IDs and site IDs, help you apply configuration settings to a device. In addition, the introduction of the resource manager in Cisco SD-WAN Manager automatically manages these resource IDs, thereby simplifying the overall user experience of Cisco Catalyst SD-WAN. Note that you can create a region only if you enable the Multi-Region Fabric option in Cisco SD-WAN Manager.
Network Hierarchy and Resource Management (Phase II)	Cisco IOS XE Catalyst SD-WAN Release 17.10.1a Cisco vManage Release 20.10.1	The following enhancements are introduced in the Network Hierarchy and Resource Management feature. • Creation of a system IP pool on the Configuration > Network Hierarchy page • Automatic assignment of site ID, system IP, and hostname to a device in the Quick Connect workflow • Display of detailed information on the Configuration > Network Hierarchy page, including site ID pool, region ID pool, and the list of devices associated with a site
Support for Software Defined Remote Access Pools	Cisco IOS XE Catalyst SD-WAN Release 17.11.1a Cisco vManage Release 20.11.1	Remote access refers to enabling secure access to an organization's network from devices at remote locations. The resource pool manager manages the IPv4 and IPv6 private IP address pools for Cisco Catalyst SD-WAN remote access devices. You can create a software defined remote access pool using the Configuration > Network Hierarchy page.

Feature Name	Release Information	Description
Support for Traffic Flow Collectors	Cisco IOS XE Catalyst SD-WAN Release 17.13.1a Cisco Catalyst SD-WAN Manager Release 20.13.1	This feature enables you to configure traffic flow collectors such as the Cflowd server and security logging server. Cflowd monitors service side traffic flowing through devices in the overlay network and exports flow information to the collector. Enable security logging and configure servers for high-speed logging (HSL) and collecting external syslogs. You can configure the traffic flow collectors by navigating to Configuration > Network Hierarchy > Collectors.

- [Information About Network Hierarchy and Resource Management, on page 260](#)
- [Supported Devices for Network Hierarchy and Resource Management, on page 261](#)
- [Restrictions for Network Hierarchy and Resource Management, on page 261](#)
- [Manage a Network Hierarchy, on page 262](#)
- [Assign Resource IDs to Devices, on page 271](#)
- [Configure Collectors in a Network Hierarchy, on page 274](#)

Information About Network Hierarchy and Resource Management

Overview of Network Hierarchy

You can create a network hierarchy in Cisco SD-WAN Manager to represent the geographical locations of your network. Your network hierarchy can contain three types of nodes—regions, areas, and sites. The resource IDs assigned to the nodes help you identify where to apply configuration settings later.

By default, there is one node called global in the network hierarchy.

The network hierarchy has a predetermined hierarchy with three types of nodes:

- **Region:** It represents a region in a multiregion fabric-based Cisco Catalyst SD-WAN deployment. The Multi-Region Fabric feature provides the option to divide the architecture of the Cisco Catalyst SD-WAN overlay network into multiple regional networks that operate distinctly from one another, and a central core-region network for managing inter-regional traffic.

You can create a region only if you enable the **Multi-Region Fabric** option in Cisco SD-WAN Manager. For complete information about the Multi-Region Fabric feature, see the [Cisco Catalyst SD-WAN Multi-Region Fabric \(also Hierarchical SD-WAN\) Configuration Guide](#).

- **Group/Area:** A group, also called an area, is a logical grouping of nodes in a network hierarchy. You can group sites, regions, other areas, or any combination of these into an area.
- **Site:** A site is the lowest level of node or the leaf node in a network hierarchy. You cannot create a child node under a site. You can only associate devices to a site.

For complete information about creating and managing different nodes in a network hierarchy, see [Manage a Network Hierarchy](#).

Overview of Resource Management

The resource manager in Cisco SD-WAN Manager manages the resource IDs, that is, region IDs and site IDs. It automatically generates a region ID for a region that you create on the **Configuration > Network Hierarchy** page. Similarly, it generates a site ID for a site if you do not specify it.

You can assign a site ID and a region ID to a device. For complete information about assigning resource IDs to devices, see [Assign Resource IDs to Devices](#).

If you upgrade from an earlier version of Cisco SD-WAN Manager to Cisco vManage Release 20.9.1, the resource manager in Cisco SD-WAN Manager automatically creates sites based on the site IDs of the existing devices in your setup. Sites are named as SITE_<id>. Cisco SD-WAN Manager displays these sites under the global node on the **Network Hierarchy** page. It also associates the existing devices with their sites in the network hierarchy.

Benefits of Network Hierarchy and Resource Management

- Automates the management of regions and sites.
- Saves the manual effort in an upgrade scenario when Cisco SD-WAN Manager discovers all your existing sites and displays them in the network hierarchy.
- Simplifies the onboarding and configuration of devices.
- Monitors and collects information about traffic flow.

Supported Devices for Network Hierarchy and Resource Management

This feature is supported on Cisco IOS XE Catalyst SD-WAN devices and Cisco vEdge devices.

Restrictions for Network Hierarchy and Resource Management

- You can delete a node only if it does not have any child node. For example, you can delete a site only if no devices are associated with it.
- A site is the lowest level of a node or the leaf node in a network hierarchy. You cannot create a child node under a site.
- You cannot create more than one region node between the global node and a site node.
- You cannot create a region in a multitenant deployment.
- The maximum combined number of regions and secondary regions is 63 (region ID numbers 1 through 63).

Manage a Network Hierarchy

The Network Hierarchy and Resource Management feature enables you to do the following:

- Create a region
- Create an area
- Create, edit, and delete a site

Create a Region in a Network Hierarchy

Before You Begin

(For Cisco Catalyst SD-WAN Manager Release 20.12.x or earlier) Ensure that the **Multi-Region Fabric** option in Cisco SD-WAN Manager is enabled. See [Enable Multi-Region Fabric](#) in the *Cisco Catalyst SD-WAN Multi-Region Fabric Configuration Guide*.

From Cisco Catalyst SD-WAN Manager Release 20.13.1, configuring regions is enabled by default. It does not require enabling Multi-Region Fabric.

Create a Region, Cisco Catalyst SD-WAN Manager Release 20.13.1 and Later

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy**.
2. Click ... adjacent to **Global** in the left pane and choose **Add Node**.
3. Do one of the following:
 - If Multi-Region Fabric is not enabled:

In the **Add Node** pop-up window, check the **Behave as SDWAN Region** checkbox.

If you do not check this checkbox, this procedure creates a new group within the network hierarchy instead of a region.
 - If Multi-Region Fabric is enabled:

In the **Add Node** pop-up window, choose **Region**.
4. Configure the following:

Field	Description
Name	Name for the region. The name must be unique and can contain only letters, the digits 0 through 9, hyphens (-), underscores (_), and periods (.).
Description	Description of the region.
Parent drop-down list	Choose a parent node.

5. Click **Add**.

The new region appears in the left pane.

6. (Optional) You can click a region name or a secondary region name in the left pane to display the automatically assigned region ID number. The region ID number appears above the table in the right pane. The maximum combined number of regions and secondary regions is 63 (region ID numbers 1 through 63).

Create a Region, Cisco Catalyst SD-WAN Manager Release 20.12.x or Earlier

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy**.
2. Click ... adjacent to a node (global or area) in the left pane and choose **Add MRF Region**.



Note In Cisco vManage Release 20.9.x, you can also use the **Add Node** option to add a region.

3. In the **Name** field, enter a name for the region. The name must be unique and can contain only letters, the digits 0 through 9, hyphens (-), underscores (_), and periods (.).
4. In the **Description** field, enter a description of the region.
5. From the **Parent** drop-down list, choose a parent node.
6. Click **Add**.

Create a Subregion in a Network Hierarchy



Note From Cisco IOS XE Catalyst SD-WAN Release 17.15.1a and Cisco Catalyst SD-WAN Control Components Release 20.15.1, configuration of this feature is supported only through API.

Before You Begin

Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.13.1

- From Cisco Catalyst SD-WAN Manager Release 20.13.1, configuring subregions is enabled by default. It does not require enabling Multi-Region Fabric.
- Create a region before creating a subregion. See [Create a Region in a Network Hierarchy, on page 262](#).
- For the maximum combined number of regions and secondary regions, see [Restrictions for Network Hierarchy and Resource Management, on page 261](#).

Create a Subregion

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy**.
2. Click ... adjacent to a region in the left pane and choose **Add MRF Sub Region**.
3. In the **Add Sub-Region** pop-up window, configure the following:

Field	Description
Name	Name for the region. The name must be unique and can contain only letters, the digits 0 through 9, hyphens (-), underscores (_), and periods (.).
Description	Description of the region.
Parent	This field is automatically populated with the region to which you are adding the subregion, and is not configurable.

- Click **Add**.

The new subregion appears in the left pane.

Create a Secondary Region in a Network Hierarchy



Note From Cisco IOS XE Catalyst SD-WAN Release 17.15.1a and Cisco Catalyst SD-WAN Control Components Release 20.15.1, configuration of this feature is supported only through API.

Before You Begin

Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.13.1

- Create a region before creating a subregion. See [Create a Region in a Network Hierarchy, on page 262](#).
- For the maximum combined number of regions and secondary regions, see [Restrictions for Network Hierarchy and Resource Management, on page 261](#).

Create a Secondary Region

- From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy**.
- Click ... adjacent to **Global** in the left pane and choose **Add Node**.
- In the **Add Node** pop-up window, click **Secondary Region**.
- Configure the following:

Field	Description
Name	Name for the region. The name must be unique and can contain only letters, the digits 0 through 9, hyphens (-), underscores (_), and periods (.).
Description	Description of the region.
Parent	This field shows Secondary Regions , and is not configurable.

- Click **Add**.

The new secondary region appears in the left pane, in the **Secondary Regions** section.

6. (Optional) You can click a region name or a secondary region name in the left pane to display the automatically assigned region ID number. The region ID number appears above the table in the right pane. The maximum combined number of regions and secondary regions is 63 (region ID numbers 1 through 63).

Create a Group in a Network Hierarchy

Before You Begin

In Cisco Catalyst SD-WAN Manager Release 20.12.x and earlier, a group is called an area.

Create a Group

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy**.
2. Click ... adjacent to a node (global, region, or group) in the left pane and choose **Add Node**.
3. In the **Add Node** pop-up window, in the **Type** field, choose **Group**.
(In Cisco Catalyst SD-WAN Manager Release 20.12.x and earlier, choose **Add Area**.)



Note In Cisco vManage Release 20.9.x, you can also use the **Add Node** option to add an area.

4. In the **Name** field, enter a name for the group. The name must be unique and can contain only letters, the digits 0 through 9, hyphens (-), underscores (_), and periods (.).
5. In the **Description** field, enter a description of the group.
6. From the **Parent** drop-down list, choose a parent node.
7. Click **Add**.

Create a Site in a Network Hierarchy

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy**.
2. Click ... adjacent to a node (global, region, or area) in the left pane and choose **Add Site**.



Note In Cisco vManage Release 20.9.x, you can also use the **Add Node** option to add a site.

3. In the **Name** field, enter a name for the site. The name must be unique and can contain only letters, the digits 0 through 9, hyphens (-), underscores (_), and periods (.).
4. In the **Description** field, enter a description of the site.
5. From the **Parent** drop-down list, choose a parent node.

6. In the **Site ID** field, enter a site ID.
If you do not enter the site ID, Cisco SD-WAN Manager generates a site ID for the site.
7. In the **Address** field, enter the address.
When you enter an address, latitude and longitude fields are auto populated.
8. In the **Latitude** field, enter the latitude of the site.
9. In the **Longitude** field, enter the longitude of the site.
10. Click **Add**.



Note The **Address**, **Latitude**, and **Longitude** fields are included starting from Cisco IOS XE Catalyst SD-WAN Release 17.18.1a. When you type in the address, the latitude and longitude fields are auto-populated.

Conversely, you can also enter your own latitude and longitude. If a corresponding location is found, then the address field will be auto-populated. If you do not want to provide address data for your site, then you can check the **Undisclosed Address** check box.

Field	Description
Name	Enter a name for the site. The name must be unique and can contain only letters, the digits 0 through 9, hyphens (-), underscores (_), and periods (.).
Description	Enter a description of the site.
Parent	From the drop-down list, choose a parent node.
Site ID	Enter a site ID. If you do not enter the site ID, Cisco SD-WAN Manager generates a site ID for the site.
Address	Enter the address. When you enter an address, latitude and longitude fields are auto populated.
Latitude	Specifies latitude of the site. When you enter the latitude of the site, the address field is auto populated if the location of the site is found.
Longitude	Specifies longitude of the site. When you enter the longitude of the site, the address field is auto populated if the location of the site is found.

Edit a WAN Region

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy**.
2. Click ... adjacent to the region name and choose **Edit WAN Region**.

In releases earlier than Cisco Catalyst SD-WAN Manager Release 20.15.1, choose **Edit MRF Region**.

3. Edit the options as needed. You can edit the name, description, and parent of the region.
4. Click **Save**.

Delete a WAN Region

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy**.
2. Click ... adjacent to the region name and choose **Delete WAN Region**.

In releases earlier than Cisco Catalyst SD-WAN Manager Release 20.15.1, choose **Delete MRF Region**.

3. In the confirmation dialog box, click **Yes**.

Edit a Group

Before You Begin

In Cisco Catalyst SD-WAN Manager Release 20.12.x and earlier, a group is called an area.

Edit a Group

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy**.
2. Click ... adjacent to the group name and choose **Edit Group**.
(In Cisco Catalyst SD-WAN Manager Release 20.12.x and earlier, choose **Edit Area**.)
3. Edit the options as needed. You can edit the name, description, and parent of the group.
4. Click **Save**.

Delete a Group

Before You Begin

In Cisco Catalyst SD-WAN Manager Release 20.12.x and earlier, a group is called an area.

Delete a Group

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy**.
2. Click ... adjacent to the group name and choose **Delete Group**.
(In Cisco Catalyst SD-WAN Manager Release 20.12.x and earlier, choose **Delete Area**.)
3. In the confirmation dialog box, click **Yes**.

Edit a Site

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy**.
2. Click ... adjacent to the site name and choose **Edit Site**.
3. Edit the options as needed. You can edit only the name, description, and parent of the site.
4. Click **Save**.

Delete a Site

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy**.
2. Click ... adjacent to the site name and choose **Delete Site**.
3. In the confirmation dialog box, click **Yes**.

Create a System IP Pool

Minimum releases: Cisco IOS XE Catalyst SD-WAN Release 17.10.1a, Cisco vManage Release 20.10.1

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy**.
The page displays the site pool and region pool for the Global node.
2. Click **Pools**.
3. Click **Add Pool**.
4. In the **Pool Name** field, enter a name for the pool.
5. In the **Pool Description** field, enter a description of the pool.
6. From the **Pool Type** drop-down list, choose **System IP**.
7. In the **IP Subnet*** field, enter an IP address.
8. In the **Prefix Length*** field, enter the prefix length of the system IP pool.
9. Click **Add**.



Note You can create only one system IP pool. If you want to make any changes to the pool, you must edit the existing pool.

Edit a System IP Pool

Minimum releases: Cisco IOS XE Catalyst SD-WAN Release 17.10.1a, Cisco vManage Release 20.10.1

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy**.

The page displays the site pool and region pool for the Global node. The system IP pool is also displayed if you have already created it.

2. Click ... adjacent to the system IP name and choose **Edit**.
3. Edit the options as needed.



Note You can only expand the pool range and cannot enter a lower IP address than the already specified IP address.

4. Click **Save**.

Create a Remote Access Pool

Minimum supported release: Cisco vManage Release 20.11.1

The resource pool manager supports creation of IPv4 and IPv6 private IP pools for Cisco Catalyst SD-WAN remote access devices. In the remote access configuration you can select the remote access private IP Pool by defining the number of IP addresses.

For more information on Software Defined Remote Access, see [Cisco Catalyst SD-WAN Remote Access](#).

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy**.

The page displays the site pool and region pool for the Global node.

2. Click **Add Pool**.
3. In the **Pool Name** field, enter a name for the pool.
4. In the **Pool Description** field, enter a description of the pool.
5. From the **Pool Type** drop-down list, choose **Remote Access**.
6. Choose the **IP Type** by clicking the radio button next to **IPv4** or **IPv6**.
7. In the **IP Subnet** field, enter an IP subnet.
8. In the **Prefix Length** field, enter the prefix length of the remote access pool.
9. Click **Add**.

Edit a Remote Access Pool

Minimum supported release: Cisco vManage Release 20.11.1

You can edit a remote access pool only when you want to expand the pool range.

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy**.

The page displays the site pool and region pool for the Global node. The remote access pool is also displayed if you have already created it.

2. Click ... adjacent to the remote access pool name and choose **Edit**.
3. Edit the options as needed.



Note When you edit a remote access pool, the new pool range cannot be less than the existing pool range

4. Click **Save**.

Create an IP Pool for ThousandEyes

SUMMARY STEPS

1. From the menu, choose **Configuration > Network Hierarchy**.
2. Examine the audit logs to determine if the ThousandEyes test has been posted or failed.
3. Click **Pools**.
4. Click **Add Pool**.
5. In the **Pool Name** field, enter a name for the pool.
6. In the **Pool Description** field, enter a description of the pool.
7. From the **Pool Type** drop-down list, choose **ThousadEyes**.
8. In the **IP Subnet*** field, enter an IP address.
9. In the **Prefix Length*** field, enter the prefix length of the system IP pool.
10. Click **Add**.

DETAILED STEPS

Procedure

	Command or Action	Purpose
Step 1	From the menu, choose Configuration > Network Hierarchy .	The page displays the site pool and region pool for the Global node.
Step 2	Examine the audit logs to determine if the ThousandEyes test has been posted or failed.	
Step 3	Click Pools .	
Step 4	Click Add Pool .	
Step 5	In the Pool Name field, enter a name for the pool.	
Step 6	In the Pool Description field, enter a description of the pool.	
Step 7	From the Pool Type drop-down list, choose ThousadEyes .	
Step 8	In the IP Subnet* field, enter an IP address.	
Step 9	In the Prefix Length* field, enter the prefix length of the system IP pool.	
Step 10	Click Add .	

Delete a Pool

Minimum supported release: Cisco vManage Release 20.11.1

1. From the Cisco SD-WAN Manager menu, choose **Configuration** > **Network Hierarchy**.
2. In the Global page, click ... adjacent to the pool name and choose **Delete**.
3. In the confirmation dialog box, click **Yes**.

**Note**

You can delete a pool only when the pool resources are not in use.

Assign Resource IDs to Devices

The Network Hierarchy and Resource Management feature enables you to do the following:

- Assign a site ID to a device
- Assign a region ID to a device

Assign a Site ID to a Device

You can assign a site ID to a device using one of the following ways.

Use the Quick Connect Workflow

1. From the Cisco SD-WAN Manager menu, choose **Workflows** > **Workflow Library**.
2. Start the **Quick Connect** workflow.
3. Follow the instructions provided in the workflow.
4. On the **Add and Review Device Configuration** page, enter the site ID of the device.

**Note**

- You can use any of the existing site IDs that are available in the network hierarchy or enter a new site ID. If you enter a new site ID without creating a node in the network hierarchy, the site is automatically created and listed on the **Configuration** > **Network Hierarchy** page.
- (Minimum releases: Cisco IOS XE Catalyst SD-WAN Release 17.10.1a, Cisco vManage Release 20.10.1) If you want Cisco SD-WAN Manager to automatically generate a site ID for the device, do not make any change to the default value, **AUTO**.

Use a Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration** > **Devices** > **WAN Edge List**.
2. Check if a device is attached to a device template.

3. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates > Feature Templates**.
4. Click ... adjacent to the System feature template and choose **Edit**.
5. Click the **Basic Configuration** tab and set the scope of the **Site ID** field to **Global** and enter the site ID.
6. Click **Update**.
7. Click **Configure Devices** to push the configuration to the device.

In Step 5, if you set the scope of the **Site ID** field to **Device Specific**, do the following:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates > Device Templates**.
2. Click ... adjacent to the device template and choose **Edit Device Template**.
3. In the **Site ID** field, enter the site ID.

You can use any of the existing site IDs that are available in the network hierarchy or enter a new site ID. If you enter a new site ID without creating a node in the network hierarchy, the site is automatically created and listed on the **Configuration > Network Hierarchy** page.

4. Click **Update**.
5. Click **Configure Devices** to push the configuration to the device.

Use a Configuration Group

The configuration group flow is applicable only for the Cisco IOS XE Catalyst SD-WAN devices.

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates > Configuration Groups**.
2. Click ... adjacent to the configuration group name and choose **Edit**.
3. Click **Associated Devices**.
4. Choose a device that is associated with the configuration group and click **Deploy**.

The **Deploy Configuration Group** workflow starts.

5. Follow the instructions provided in the workflow.
6. On the **Add and Review Device Configuration** page, enter the site ID of the device.

You can use any of the existing site IDs that are available in the network hierarchy or enter a new site ID. If you enter a new site ID without creating a node in the network hierarchy, the site is automatically created and listed on the **Configuration > Network Hierarchy** page.

Assign a Region ID to a Device

Before You Begin

- Have access to the **Multi-Region Fabric** feature.
- Ensure that the region is available in the network hierarchy.

Assign a Region ID

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices > WAN Edge List**.
2. Check if the corresponding device is attached to a device template.
3. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates > Feature Templates**.
4. Click ... adjacent to the System feature template and choose **Edit**.
5. Click the **Basic Configuration** tab and set the scope of the **Region ID** field to **Global** and enter the region ID.

You can use any of the existing region IDs that are available in the network hierarchy. If the specified region ID is not available in the network hierarchy, the template push operation to the devices fails.

6. Click **Update**.
7. Click **Configure Devices** to push the configuration to the device.

In Step 5, if you set the scope of the **Region ID** field to **Device Specific**, do the following:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates > Device Templates**.
2. Click ... adjacent to the device template and choose **Edit Device Template**.
3. In the **Region ID** field, enter the region ID.
4. Click **Update**.
5. Click **Configure Devices** to push the configuration to the device.

Assign a System IP to a Device

Minimum releases: Cisco IOS XE Catalyst SD-WAN Release 17.10.1a, Cisco vManage Release 20.10.1

1. From the Cisco SD-WAN Manager menu, choose **Workflows > Workflow Library**.
2. Start the **Quick Connect** workflow.
3. Follow the instructions provided in the workflow.
4. On the **Add and Review Device Configuration** page, enter the system IP of the device. If you want Cisco SD-WAN Manager to automatically generate a system IP for the device, do not make any change to the default value, **AUTO**.

Assign a Hostname to a Device

Minimum releases: Cisco IOS XE Catalyst SD-WAN Release 17.10.1a, Cisco vManage Release 20.10.1

1. From the Cisco SD-WAN Manager menu, choose **Workflows > Workflow Library**.
2. Start the **Quick Connect** workflow.
3. Follow the instructions provided in the workflow.

4. On the **Add and Review Device Configuration** page, enter the hostname of the device. If you want Cisco SD-WAN Manager to automatically generate a hostname for the device, do not make any change to the default value, **AUTO**.

Configure Collectors in a Network Hierarchy

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, Cisco Catalyst SD-WAN Manager Release 20.13.1.

Configure Cflowd and security logging servers that help monitor traffic flow and collect information about service-side traffic.

Information About Collectors

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, Cisco Catalyst SD-WAN Manager Release 20.13.1.

Collectors process traffic flowing through routers in the overlay network and export flow information to a server. The collectors maintain information about the flow and data that is extracted from the IP headers of the packets in the traffic flow.

You can configure the location of cflowd collectors, how often sets of sampled flows are sent to the collectors, and how often the samples are sent to the collectors (on Cisco SD-WAN Controllers only). You can configure a maximum of four cflowd collectors per Cisco IOS XE Catalyst SD-WAN Device. To have a cflowd configuration take effect, apply it with the appropriate data policy.

Configure Cflowd

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, Cisco Catalyst SD-WAN Manager Release 20.13.1.

Before You Begin

You can configure the location of cflowd collectors, how often sets of sampled flows are sent to the collectors, and how often the samples are sent to the collectors (on Cisco SD-WAN Controllers only). You can configure a maximum of four cflowd collectors per Cisco IOS XE Catalyst SD-WAN device. To have a cflowd configuration take effect, apply it with the appropriate data policy.

Ensure that you specify the granular role-based access control (RBAC) for Cflowd and policy groups. With specific permissions to the user group, ensure that you are able to access policy groups from **Configuration > Policy Groups**. For more information about configuring RBAC for policy groups, see [Configure RBAC for policy groups in Prerequisites for Policy Groups](#).

1. From the Cisco SD-WAN Manager menu, choose **Administration > Users and Access > Roles**.
2. Click **Edit** next to existing roles or click **Add Role** to create a new role.
3. Choose the desired permission for the **Cflowd** feature under **Network Settings** and click **Update**.

Configure Cflowd

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy > Collectors**.
2. Enable Cflowd and configure the values in the following table for the collector server:

Field	Description
Add Collector Server	
VPN ID	VPN ID of the server. Range: 0 through 65536
IPv4/IPv6 Address	IPv4 or IPv6 address of the collector server.
UDP Port	UDP port number of the collector server. Range: 1024 through 65535
Export Spreading	Toggle to enable or disable the export spreading configuration.
BFD Metrics Exporting	Toggle to enable or disable Bidirectional Forwarding Detection (BFD) metrics.
Exporting Interval	Interval in seconds for sending BFD metrics. Exporting Interval appears if you have enabled BFD Metrics Exporting . The default BFD export interval is 600 seconds.
Advanced Settings	
Active Flow Timeout (Seconds)	Active flow timeout value. Range: 30 through 3600 Default: 600 seconds.
Inactive Flow Timeout (Seconds)	Inactive flow timeout value. Range: 1 through 3600 Default: 60 seconds.
Flow Refresh Time (Seconds)	Flow refresh time in seconds. Range: 60 through 86400 seconds. Default: 600 seconds.
Sampling Rate	Sample duration in seconds. Range: 1 through 65536. Default: 1 second.
Collect TLOC Loopback	Enable to collect information about the TLOC loopback.

Field	Description
Protocol	Traffic protocol type to apply the collector to. The options are: IPv4 , IPv6 , or both . The default protocol is IPv4 .
TOS	Type of field in the IPv4 header.
Re-marked DSCP	Traffic output of the router's data policy.

You can configure up to four collector servers.

3. Click **Save**.

The Cflowd settings that you configure are applied to the application priority and SLA policy when the policy is deployed to Cisco Catalyst SD-WAN devices. You can monitor application traffic flow over IPv4, IPv6, or both network addresses. For more information about configuring additional settings, see **Monitor traffic flow** in [Application Priority and SLA](#).

Configure Security Logging

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, Cisco Catalyst SD-WAN Manager Release 20.13.1.

Configure Security Logging

You can set up security logging for Cisco IOS XE Catalyst SD-WAN devices by configuring the location of the destination IP address of the log server. You can configure up to four destination servers along with the source interface to collect the syslogs for High Speed Logging (HSL). The IP address for the destination server can be IPv4, IPv6, or both. For more information about configuring HSL, see [Configure Firewall High-Speed Logging Using the CLI Template](#). You can configure the external syslog server to export UTD logs. For more information about UTD logging, see [Create Unified Security Policy Summary](#) page.

Before You Begin

Ensure that you specify the granular role-based access control (RBAC) for security logging. Ensure that you are able to access policy groups from **Configuration > Policy Groups** by configuring specific permissions to the user group. For more information about configuring RBAC for policy groups, see "Configure RBAC for policy groups" in [Prerequisites for Policy Groups](#).

1. From the Cisco SD-WAN Manager menu, choose **Administration > Users and Access > Roles**.
2. Click **Edit** adjacent to existing roles or click **Add Role** to create a new role.
3. Choose the permission you wish to configure for the **Security Logging** feature under **Network Settings** and click **Update**.

Configure Security Logging

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy > Collectors**.
2. Enable **Security Logging** and configure the values in the following table for the high-speed logging and external syslog servers:

Field	Description
High Speed Logging	Configure the following values for the high-speed logging server: • VPN: VPN name of the high-speed logging server. The VPNs available in the drop-down list are ones that are previously configured in the configuration groups in Cisco SD-WAN Manager. • Server IP: IPv4 or IPv6 address of the log collector server. • Port: Port number on which the log collector server is listening for incoming packets.
External Syslog Server	Configure the following values for the external syslog server: • VPN: VPN name of the external syslog server. The VPNs available in the drop-down list are ones that are previously configured in the configuration groups in Cisco SD-WAN Manager. • Server IP: IPv4 or IPv6 address of the external syslog server.

You can configure up to four high-speed logging servers.



Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.16.1a and Cisco Catalyst SD-WAN Manager Release 20.16.1, server labels (**Server 1**, **Server 2**, **Server 3**, and **Server 4**) are added to the high-speed logging server and the syslog server.

For device-specific server settings, add the associated source interface in the **Additional Settings** of the NGFW policy. For more information about configuring additional settings in the policy groups, see [Configure NGFW Additional Settings](#) section.

For Global server settings, define the global NHM values. For more information about defining global values on NHM, see [Network Hierarchy and Resource Management](#).

3. Click **Save**.

The security logging settings that you configure are applied along with the embedded security policy when the policy is deployed to Cisco Catalyst SD-WAN devices. For more information about configuring the embedded security policy, see [Configure Embedded Security](#).



CHAPTER 11

Cisco Unified Communications Voice Services

Table 88: Feature History

Feature Name	Release Information	Description
Integration with Cisco Unified Communications	Cisco IOS XE Catalyst SD-WAN Release 17.3.1a Cisco vManage Release 20.3.1	This release adds support for using a feature template to enable Cisco IP-based media services.
Integration with Cisco Unified Communications	Cisco IOS XE Catalyst SD-WAN Release 17.2.1r	This feature lets you use feature templates and voice policies to enable Cisco Unified Communications (UC) voice services for supported routers. When Cisco UC voice services are enabled, routers can process calls for various endpoints, including voice ports, POTS dial peers, SIP dial peers, and phone profiles in Cisco Unified SRST mode. You can configure items for UC voice services from the Feature tab and the Voice Policy page for a supported device. Configuring UC voice services for Cisco Unified Communications requires that Cisco SD-WAN Manager runs Cisco Catalyst SD-WAN Release 20.1.1. This feature is supported on Cisco 4000 Series Integrated Services Routers.
Cisco Unified Communications Voice Services Configuration through the Workflow Library and Configuration Groups	Cisco IOS XE Catalyst SD-WAN Release 17.13.1a Cisco Catalyst SD-WAN Manager Release 20.13.1	With this release, you can configure DSP farms for UC voice services by using the Workflow Library and configuration groups

You can configure feature templates and voice policies to enable Cisco Unified Communications (UC) voice services for supported routers. These templates and policies configure parameters for FXO, FXS, and FXS/DID interfaces on these routers. Beginning with Cisco IOS XE Catalyst SD-WAN Release 17.3.1a, parameters for

PRI ISDN too can be configured. In addition, you can use the DSPFarm feature template to enable Cisco IP-based media services.

When Cisco UC voice services are enabled, routers can process calls for various endpoints, including voice ports for analog interfaces and digital interfaces, POTS dial peers, SIP dial peers, and phone profiles in Cisco Unified SRST mode.



Note Beginning with Cisco IOS XE Catalyst SD-WAN Release 17.3.1a, you also can configure and enable UC voice services by using the Workflow library or configuration groups. See [Configure UC Voice Services using the Workflow Library or Configuration Groups](#).

Configuring UC voice services for Cisco Unified Communications requires that Cisco SD-WAN Manager runs Cisco Catalyst SD-WAN Release 20.3 or later.

For more detailed information about commands to configure and maintain Cisco IOS voice applications, see [Cisco IOS Master Command List](#).

The following describe the general steps that you perform to configure, in various scenarios, voice services for Cisco Unified Communications:

- Workflow for initial configuration of Cisco Catalyst SD-WAN for Cisco Unified Communications.

Step 1	Add a voice card feature template.
Step 2	Add a call routing feature template.
Step 3	(Optional) Add an SRST feature template.
Step 4	(Optional) Add a DSPFarm Feature Template.
Step 5	(Optional) Add a voice policy.
Step 6	Provision a device template for Unified Communications.

- Workflow for adding a voice port, POTS dial peer, SIP dial peer, or SRST phone profile subpolicy to a voice policy.

Step 1	Detach the device templates that include the UC voice policy and UC-specific feature templates.
Step 2	Add the subpolicy to the voice policy.
Step 3	Map the updated voice policy to endpoints as needed.
Step 4	Attach the feature templates to a device template.

- Workflow for updating feature templates to add or delete UC endpoints.

Step 1	Detach the device templates that include the voice card UC-specific feature templates and a voice policy.
--------	---

Step 2	Update the voice card feature templates as needed.
Step 3	Map the updated voice policy to endpoints as needed.
Step 4	Attach the feature templates to a device template.

- Workflow for updating configuration parameters when the functionality of a voice port changes.

Step 1	Detach the device templates that include the voice card UC-specific feature templates and an associated voice policy mapping.
Step 2	Update the voice card feature template and voice policy as needed.
Step 3	Map the updated voice policy to endpoints as needed.
Step 4	Attach the feature templates and the voice policy to a device template.

- Workflow for changing the interface type for a T1/E1 voice module.

Step 1	Detach the device template that includes the voice card feature template that defines the T1/E1 voice module, and detach the associated mapped voice policy.
Step 2	Unmap all voice policies from the PRI ISDN voice ports that are configured for the T1/E1 voice module, and unmap the POTS dial-peers for those ports.
Step 3	In the voice card feature template, delete the PRI ISDN voice ports that are configured for the T1/E1 voice module.
Step 4	Reattach the device template to devices.
Step 5	Reload the devices.
Step 6	Detach the device template from the devices.
Step 7	In the voice card feature template create new PRI ISDN voice ports for the T1/E1 voice module as needed.
Step 8	Map the voice card feature template and voice policy to the device template.
Step 9	Map the updated voice policy to the newly created PRI ISDN voice ports as needed.

Step 10	Reattach the device template to devices.
---------	--

- Workflow for updating the clock source configuration for a T1/E1 voice module to change the primary and secondary clock sources.

Step 1	In the voice card feature template for the T1/E1 voice module that you want to update, set the clock source for each PRI ISDN voice port to Line , and push the configuration to devices.
Step 2	After the configuration is pushed successfully, in the voice card feature template set the clock source for each PRI ISDN voice port for the T1/E1 voice module to the desired values, and push the configuration to devices.

- [Configure UC Voice Services Using the Workflow Library or Configuration Groups](#), on page 282
- [Add a Voice Card Feature Template](#), on page 284
- [Add a Call Routing Feature Template](#), on page 294
- [Add an SRST Feature Template](#), on page 298
- [Add a DSPFarm Feature Template](#), on page 300
- [Add a Voice Policy](#), on page 312
- [Provision a Device Template for Unified Communications](#), on page 352
- [Dial Peer CSV File](#), on page 354
- [Translation Rules CSV File](#), on page 356
- [Monitoring UC Operations](#), on page 357
- [Cisco Unified Communications FXS and FXO Caller ID Support](#), on page 364

Configure UC Voice Services Using the Workflow Library or Configuration Groups

Beginning with Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, you can configure and enable certain UC voice service features by using the Workflow library or configuration groups.

To configure and enable UC voice services by using the Workflow Library, choose **Workflows > Configure UC Voice** from the Cisco SD-WAN Manager window and follow the prompts in the workflow.

To configure and enable UC voice services by using configuration groups, see [UC Voice Profile](#).

The following table lists the features that you can configure and enable by using these methods, and shows the first Cisco SD-WAN Manager release that supports these configuration methods.

Table 89: Supported Releases for Configuring UC Voice Services Using the Workflow Library or Configuration Groups

Feature	First Supported Release	
	Workflow Library Features	Configuration Groups Features
DSP Farm	Cisco Catalyst SD-WAN Manager Release 20.13.1	Cisco Catalyst SD-WAN Manager Release 20.13.1

Supported Devices for Cisco Unified Voice Services using the Workflow Library or Configuration Groups

- Supported platforms:

Platform	Release
Cisco ISR 4451 running Cisco IOS XE 17(12)3a	Cisco IOS XE Catalyst SD-WAN Release 17.12.3
Cisco ISR4431 running Cisco IOS XE 17(12)3a	
Cisco ISR 4351 running Cisco IOS XE 17(12)3a	
Cisco ISR4331 running Cisco IOS XE 17(12)3a	
Cisco ISR4321 running Cisco IOS XE 17(12)3a	
Cisco ISR4461	Cisco IOS XE Catalyst SD-WAN Release 17.13.1a and later
Cisco C8300-2N2S-4T2x	
Cisco C8300-2N2S-6T	
Cisco C8300-1N1S-4T2X	
Cisco C8300-1N1S-6T	
Cisco C8200-1N-4T	
Cisco C8200L-1N-4T	
Cisco Catalyst 8000V	

- Supported voice modules for analog interfaces:
 - NIM-2FXO, NIM-4FXO, NIM-2FXSP, NIM-4FXSP, NIM-2FXS/4FXOP
 - SM-X-72FXS, SM-X-24FXS/4FXO, SM-X-16FXS/2FXO, SM-X-8FXS/12FXO
- Supported voice modules for digital interfaces with Packet Voice DSP modules (PVDMS):
 - NIM-1MFT-T1/E1, NIM-2MFT-T1/E1, NIM-4MFT-T1/E1, NIM-8MFT-T1/E1
 - NIM-1CE1T1/PRI-T1/E1, NIM-2CE1T1/PRI-T1/E1, NIM-8CE1T1/PRI-T1/E1
 - PVDM4-32, PVDM4-64, PVDM4-128, PVDM4-256
- Supported PVDM modules for digital signal processor (DSP) farms:

- NIM-PVDM-32, NIM-PVDM-64, NIM-PVDM-128, NIM-PVDM-256
- PVDM4-32, PVDM4-64, PVDM4-128, PVDM4-256
- SM-X-PVDM-500, SM-X-PVDM-1000, SM-X-PVDM-2000, SM-X-PVDM-3000

Add a Voice Card Feature Template

A voice card feature template configures analog and PRI ISDN digital interfaces, which provide configuration settings for ports on voice cards in routers.

When you add a voice card feature template, for an analog interface, you configure the type of voice card you are configuring, port information for the card, and parameters for the service that you receive from your service provider. For a digital interface, you configure the type of voice card, the T1 or E1 controller, and related parameters.

When you add a module for a voice card, Cisco SD-WAN Manager assists you with the placement of the module by displaying available slots and sub-slots for the module. Cisco SD-WAN Manager determines the available slots and sub-slots based on the device model.

The following table describes options for configuring an analog interface.

Table 90: Analog Interface Configuration Options

Option	Description	Cisco IOS CLI Equivalent
Module	Select the type of voice module that is installed in the router.	—
Module Slot/Sub-slot	Enter the slot and sub-slot of the voice module.	voice-card <i>slot/subslot</i>
Use DSP	Enable this option if you want to use the built-in DSPs on the network interface module for TDM calls.	no local-bypass
Port Type	Select the type of ports on the voice module that you are configuring for this interface (FXS or FXO). You can select All to define the port type for all ports of the selected type, or Port Range to define the port type for a specified range of ports. Using Port Range, you can create analog interfaces as described later in this procedure to configure different ranges of ports.	—
Description	Enter a description of the selected port or ports. For example, fax machine or paging system.	description <i>string</i>

Option	Description	Cisco IOS CLI Equivalent
Secondary Dialtone	Available if you select FXO from the Port Type drop-down list. Set to On if you want the selected ports to generate a secondary dial tone when callers access an outside line.	secondary dialtone
Connection PLAR	Enter the Private Line Automatic Ringdown extension to which the selected ports forward inbound calls.	connection plar digits
OPX	Available if you select FXO from the Port Type drop-down list. Check this option if you want to enable Off-Premises Extension for the PLAR extension.	connection plar opx digits
Signal Type	Select the Signal Type that indicates an on-hook or off-hook condition for calls that the ports receive. Options are Loopstart , Groundstart , or DID . The DID option is available if you select FXS from the Port Type drop-down list.	signal {groundstart loopstart} signal did {delay-dial immediate wink-start}
Caller-ID Enable	Available if you select a signal type of Loopstart or Groundstart. Set to ON if you want to enable caller ID information for inbound calls.	caller-id enable
DID Signal Mode	Available if you select a signal type of DID. Choose the mode for the DID signal type (Delay Dial , Immediate , or Wink Start). Default: Wink Start.	signal did {delay-dial immediate wink-start}
Shutdown	Set to ON if you want to shut down ports that are not being used. Default: Off.	shutdown

The following table describes options for configuring a digital interface.

Table 91: Digital Interface Configuration Options

Option	Description	Cisco IOS CLI Equivalent
Digital Interface Tab Provides options for configuring parameters for a T1/E1 voice module and the clock source for the module ports. Before you configure these options, ensure that you have the appropriate DSP module installed for each T1/E1 voice module.		

Option	Description	Cisco IOS CLI Equivalent
Module	Select the type of T1/E1 voice module that is installed in the router.	—
Interface Type	Select the type of interface on the voice module: • T1 PRI—Specifies T1 connectivity of 1.544 Mbps through the telephone switching network, using AMI or B8ZS coding • E1 PRI—Specifies a wide-area digital transmission scheme used predominantly in Europe that carries data at a rate of 2.048 Mbps	card type { t1 e1 } <i>slot sub-slot</i>
Slot/Sub-slot	Enter the slot and sub-slot of the voice module.	voice-card <i>slot/sub-slot</i>
Use DSP	Enable this option if you want to use the built-in DSPs on the network interface module for TDM calls.	no local-bypass

Option	Description	Cisco IOS CLI Equivalent
Interface	Perform these actions to configure the number of T1/E1 ports to be provisioned on the module, and the clock source for each port: 1. Click Add. The Port and Clock Selector window displays. 2. Check the check box that corresponds to each port that you want to configure. The number of ports that you can configure depends on the Module type that you select. 3. For each port, select the clock source: • Line—Sets the line clock as the primary clock source. With this option, the port clocks its transmitted data from a clock that is recovered from the line receive data stream. • Primary Clock—Sets the port to be a primary clock source. • Secondary Clock—Sets the port to be a secondary clock source. • Network—Sets the backplane clock or the system oscillator clock as the module clock source. We recommend that you set one port to be the primary clock and set another port going to the same network as a secondary clock source to act as a backup. 4. Click Add.	controller {t1 e1} <i>slot/sub-slot/number</i> clock source {network line line primary line secondary}
Network Participation	This check box displays after you add an interface. Check this check box to configure the T1/E1 module to participate in the backplane clock. Uncheck this check box to remove the clock synchronization with the backplane clock for the module. By default, this check box is checked.	network-clock synchronization participate <i>slot/sub-slot</i>

Option	Description	Cisco IOS CLI Equivalent
Shutdown	Perform these actions to disable or enable the controller, serial interface, or voice port that is associated with the interface port. 1. Click Shutdown Selected. The Shutdown window displays. 2. For each port, select the item or items that you want to enable (Controller, Serial, or Voice Port). If you do not select an item, it is enabled. 3. Click Add.	controller e1/t1 slot/sub-slot/port shutdown interface serial slot/sub-slot/port: {15 23} shutdown voice-port slot/sub-slot/port: {15 23} shutdown
Time Slots	Select the number of time slots of the interface type. Valid ranges: • For T1 PRI—Time slots 1 through 24. The 24th time slot is the D channel. • For E1 PRI—Time slots 1 through 31. The 16th time slot is the D channel.	controller e1/t1 slot/sub-slot/port pri-group timeslots timeslot-range [voice-dsp]
Framing	Select the frame type for the interface type. For a T1 PRI interface type, options are: • esf—Extended super frame (default) • sf—Super frame For an E1 PRI interface type, options are: • crc4—CRC4 framing type (default) • no-crc4—No CRC4 framing type	controller t1 slot/sub-slot/port framing [esf sf] controller e1 slot/sub-slot/port framing [crc4 no-crc4] [australia]
Australia	This check box displays when you select E1 PRI for the interface type. Check this check box to use the australia framing type.	controller e1 slot/sub-slot/port framing [crc4 no-crc4] australia

Option	Description	Cisco IOS CLI Equivalent
Line Code	Select the line code type for the interface type. For a T1 PRI interface type, options are: • ami—Use Alternate Mark Inversion as the line code type • b8zs—Use binary 8-zero substitution as the line code type (default) For an E1 PRI interface type, options are: • ami—Use Alternate Mark Inversion as the line code type • hdb3—Use high-density binary 3 as the line code type (default)	controller t1 <i>slot/sub-slot/port</i> linecode [ami b8zs] controller e1 <i>slot/sub-slot/port</i> linecode [ami hdb3]
Line Termination	This check box appears only for an Interface type of E1 PRI. Select the line termination type for the E1 controller: • 75-ohm—75 ohm unbalanced termination • 120-ohm—120 ohm balanced termination (default)	controller e1 <i>slot/sub-slot/port</i> line-termination {75-ohm 120-ohm}
Cable Length Type	This check box appears only for an Interface type of T1 PRI. Select the cable length type for the T1 PRI interface type: • long—Long cable length • short—Short cable length	controller t1 <i>slot/sub-slot/port</i> cablelength {short long}
Cable Length	This check box appears only for an interface type of T1 PRI. Select the cable length for the T1 PRI interface type. Use this option to fine-tune the pulse of a signal at the receiver for a T1 cable. The default value is 0db.	controller t1 <i>slot/sub-slot/port</i> cablelength {[short [110ft 220ft 330ft 440ft 550ft 660ft]] [long [-15db -22.5db -7.5db 0db]]}

Option	Description	Cisco IOS CLI Equivalent
Network Side	Enable this option to have the device use the standard PRI network-side interface. By default, this option is disabled (set to No).	interface serial <i>slot/sub-slot/port</i> : {15 23} isdn protocol-emulate [network user]
Switch Type	Select the ISDN switch type for this interface: • primary-qsig—Supports QSIG signaling according to the Q.931 protocol. Network side functionality is assigned with the isdn protocol-emulate command. • primary-net5—NET5 ISDN PRI switch types for Asia, Australia, and New Zealand. ETSI-compliant switches for Euro-ISDN E-DSS1 signaling system. • primary-ntt—Japanese NTT ISDN PRI switches. • primary-4ess—Lucent (AT&T) 4ESS switch type for the United States. • primary-5ess—Lucent (AT&T) 5ESS switch type for the United States. • primary-dms100—Nortel DMS-100 switch type for the United States. • primary-ni—National ISDN switch type.	interface serial <i>slot/sub-slot/port</i> : {15 23} isdn switch-type [primary-4ess primary-5ess primary-dms100 primary-net5 primary-ni primary-ntt primary-qsig]

Option	Description	Cisco IOS CLI Equivalent
ISDN Timer	Perform these actions to configure the ISDN timers for the interface: - Click Add. The ISDN Timer window displays. - Configure the following timers as needed. The values are in milliseconds. T200. Valid range: integers 400 through 400000. Default: 1000. T203. Valid range: integers 400 through 400000. The default value is based on the switch type and network side configurations. T301. Valid range: integers 180000 through 86400000. The default value is based on the switch type and network side configurations. T303. Valid range: integers 400 through 86400000. The default value is based on the switch type and network side configurations. T306. Valid range: integers 400 through 86400000. Default: 30000. T309. Valid range: integers 0 through 86400000. The default value is based on the switch type and network side configurations. T310. Valid range: integers 400 through 400000. The default value is based on the switch type and network side configurations. T321. Valid range: Integers 0 through 86400000. The default value is based on the switch type and network side configurations. - Click Add.	interface serial <i>slot/sub-slot/port</i>: {15 23} isdn timer T200 <i>value</i> isdn timer T203 <i>value</i> isdn timer T301 <i>value</i> isdn timer T303 <i>value</i> isdn timer T306 <i>value</i> isdn timer T309 <i>value</i> isdn timer T310 <i>value</i> isdn timer T321 <i>value</i>
Delay Connect Timer	Select the duration, in milliseconds, to delay connect a PRI ISDN hairpin call. Valid range: integers 0 through 200. Default: 20.	voice-port <i>slot/sub-slot/port</i>: {15 23} timing delay-connect <i>value</i>

Option	Description	Cisco IOS CLI Equivalent
Clock Tab Use this tab to configure priority order for the primary and secondary clock sources that you selected for each module. This tab is available after you configure a PRI ISDN digital interface and click Add .		
Clock Priority Sorting	Configure the priority of up to six clock sources. The drop-down list displays the interface ports for which a primary or secondary clock source is defined and that is configured for network participation. Check a check box to select the port for inclusion in the priority list, and use the Up arrow next to a port to change its priority. The list displays the ports in order of priority, with the port with the highest priority at the top of the list. After you configure the priority, this field displays the selected ports in priority order. We recommend that all ports in the priority list be of the same type, either E1-PRI or T1-PRI.	network-clock input-source priority controller [t1 e1] <i>slot/sub-slot/port</i>
Automatically Sync	Select Add to enable network synchronization between all modules and the router. Default: On .	network-clock synchronization automatic
Wait to restore clock	Enter the amount of time, in milliseconds, that the router waits before including a primary clock source in the clock selection process. Valid range: 0 through 86400. Default: 300.	network-clock wait-to-restore <i>milliseconds</i>

To add a voice card feature template:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**, and click **Add Template**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is called **Feature**.

3. Select the supported device to which you want to add voice services.
4. Select **Voice Card** from the **Unified Communications** templates.

5. In **Template Name**, enter a name for the template.
This field may contain uppercase and lowercase letters, digits 0 through 9, hyphens (-), and underscores (_).
6. In **Description**, enter a description for the template.
This field can contain any characters and spaces.
7. To configure an analog interface, click **New Analog Interface** and configure interface options as described in the "Analog Configuration Options" table.

From Cisco IOS XE Catalyst SD-WAN Release 17.3.1a, click **Analog Interface** in the Interface area to access **New Analog Interface**.

You can add as many analog interfaces as needed, based on the number of interfaces that your module supports.

After you configure each analog interface, click **Add**.

If any analog interfaces are already configured, they appear in the interfaces table on this page. To edit an existing interface, click ... and click its pencil icon to edit the options in the window that pops up as described in the "Analog Configuration Options" table, and click **Save Changes**. To delete an interface, click ... and click the trash can icon.
8. To configure a PRI ISDN digital interface, in the **Interface** area, click **Digital Interface**, click **New Digital Interface**, and configure interface options as described in the "Digital Interface Configuration Options" table.

After you configure each PRI ISDN digital interface, click **Add**.

Based on the number of interfaces that your module supports, you can add as many PRI ISDN digital interfaces as needed.

If any digital interfaces are already configured, they appear in the interfaces table on this page. To edit an existing interface, click ... and click its pencil icon to edit the options in the window that pops up as described in the "Digital Interface Configuration Options" table, and click **Save Changes**. To delete an interface, click ..., and click its trash can icon.

After you save the interface configuration, you cannot change the module type, interface type, slot or sub-slot, or time slots.

If you want to change time slots, you must delete the interface and create a new one.

If you want to change the module type, interface type, and slot or sub-slot, detach the template from the device, unmap the voice policies that are associated with the interfaces, and delete all interfaces that are associated with the module and slot or sub-slot. Next, push the template to the device, reload the device, and create new required interfaces. Finally, push the new template to the device, and reattach the template to the device.
9. Click **Save**.
10. (Optional) If you want to configure more analog or PRI ISDN digital interfaces for this template, then:
 - a. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
 - b. Click **Feature Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is called **Feature**.

- c. Click ... for the template you wish to configure, and click **Edit**.
- d. Repeat Step 7 or Step 8 and Step 9.

Add a Call Routing Feature Template

A call routing feature template configures parameters for TDM-SIP trunking, including trusted IP addresses for preventing toll fraud, and a dial plan. A dial plan, made up of dial peers, defines how a router routes traffic to and from voice ports to the PSTN or to another branch.

The following table describes global options for configuring call routing.

Table 92: Global Call Routing Options

Option	Description	Cisco IOS CLI Equivalent
Trusted IPv4 Prefix List	Enter the IPv4 addresses with which the router can communicate through SIP. Enter each IPv4 address in CIDR format. For example, 10.1.2.3/32. Separate each address with a comma (,). The router does not communicate with other IPv4 addresses, which prevents fraudulent calls being placed through the router. A Trusted IPv4 Prefix is required for TDM to IP calls.	voice service voip ip address trusted list ipv4 <i>ipv4-address/ipv4-network-mask</i>
Trusted IPv6 Prefix List	Enter the IPv6 addresses with which the router can communicate through SIP. Separate each IPv6 address with a comma (,). The router does not communicate with other IPv6 addresses, which prevents fraudulent calls being placed through the router. A Trusted IPv6 Prefix is required for TDM to IP calls.	voice service voip ip address trusted list ipv6 <i>ipv6-prefix//prefix-length</i>
Source Interface	Enter the name of the source interface from which the router initiates SIP control and media traffic. This information defines how the return/response to this traffic should be sent.	voice service voip sip bind control source-interface <i>interface-id</i> bind media source-interface <i>interface-id</i>

The following table describes options for configuring dial peers.

Table 93: Dial Peer Options

Option	Description	Cisco IOS CLI Equivalent
Voice Dial Peer Tag	Enter a number to be used to reference the dial peer.	dial-peer voice <i>number</i> { pots voip }
Dial Peer Type	Select the type of dial peer that you are creating (POTS or SIP).	dial-peer voice <i>number</i> { pots voip }
Direction	Select the direction for traffic on this dial peer (Incoming or Outgoing).	Incoming: dial-peer voice <i>number</i> { pots voip } incoming called-number <i>string</i> Outgoing: dial-peer voice <i>number</i> { pots voip } destination-pattern <i>string</i>
Description	Enter a description of this dial peer.	description
Numbering Pattern	Enter a string that the router uses to match incoming calls to the dial peer. Enter the string as an E.164 format regular expression in the form [0-9,A-F#*.?+%()-]*T?.	Incoming: dial-peer voice <i>number</i> { pots voip } incoming called-number <i>string</i> Outgoing: dial-peer voice <i>number</i> { pots voip } destination-pattern <i>string</i>
Forward Digits Type	Available if you select the POTS dial peer type and the Outgoing direction. Select how the dial peer transmits digits in outgoing numbers: • All—The dial peer transmits all digits • None—The dial peer does not transmit digits that do not match the destination pattern • Some—The dial peer transmits the specified number of right-most digits Default: None.	All: dial-peer voice <i>number</i> pots forward-digits all None: dial-peer voice <i>number</i> pots forward-digits 0 Some: dial-peer voice <i>number</i> pots forward-digits <i>number</i>

Option	Description	Cisco IOS CLI Equivalent
Forward Digits	Available if you select Some for Forward Digits Type. Enter the number of right-most digits in the outgoing number to transmit. For example, if you set this value to 7 and the outgoing number is 1112223333, the dial peer transmits 2223333.	dial-peer voice <i>number</i> pots forward-digits <i>number</i>
Prefix	Available if you select the POTS dial peer type and the Outgoing direction. Enter digits to be prepended to the dial string for outgoing calls.	dial-peer voice <i>number</i> pots prefix <i>string</i>
Transport Protocol	Available if you select SIP for the Dial Peer Type. Choose the transport protocol (TCP or UDP) for SIP control signaling.	dial-peer voice <i>number</i> voip session transport { tcp udp }
Preference	Available if you select POTS or SIP for the Dial Peer Type. Select an integer from 0 to 10, where the lower the number, the higher the preference. If dial peers have the same match criteria, the system uses the one with the highest preference value. Default: 0 (highest preference).	dial-peer voice <i>number</i> voip preference <i>value</i> dial-peer voice <i>number</i> pots preference <i>value</i>
Voice Port	Available if you select the POTS dial peer type. Enter the voice port that the router uses to match calls to the dial peer. For an analog port, enter the port you want. For a digital T1 PRI ISDN port, enter a port with the suffix: 23 . For a digital E1 PRI ISDN port, enter a port with the suffix :15 . For an outgoing dial peer, the router sends calls that match the dial peer to this port. For an incoming dial peer, this port serves as an extra match criterion. The dial peers are matched only if a call comes in on this port.	dial-peer voice <i>number</i> pots For an analog port: port <i>slot/subslot/port</i> For a digital port: port <i>slot/subslot/port:15</i> port <i>slot/subslot/port:23</i>

Option	Description	Cisco IOS CLI Equivalent
Destination Address	Available if you select the SIP dial peer type and the Outgoing direction. Enter the network address of the remote voice gateway to which calls are sent after a local outgoing SIP dial peer is matched. Enter the address in one of these formats: • <i>dns:hostname.domain</i> • <i>sip-server</i> • <i>ipv4:destination-address</i> • <i>ipv6:destination-address</i>	session target {ipv4:destination-address ipv6:destination-address sip-server dns:hostname.domain}

To add a call routing feature template:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**, and click **Add Template**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is called **Feature**.

3. Select the supported device to which you want to add call routing features.
4. Click **Call Routing** from the **Unified Communications** templates.
5. In **Template Name**, enter a name for the template.
This field can contain uppercase and lowercase letters, digits 0 through 9, hyphens (-), and underscores (_).
6. In **Description**, enter a description for the template.
This field can contain any characters and spaces.
7. In **Global**, configure options as described in the "Global Call Routing Options" table.
8. In **Dial Plan**, perform one of these actions:
 - To configure a dial peer directly, configure options as described in the "Dial Peer Options" table.
 - To create or edit a dial peer CSV file, click **Download Dial Peer List** to download the system provided file named Dial-Peers.csv. The first time you download this file, it contains field names but no records. Update this file as needed by using an application such as Microsoft Excel. For detailed information about this file, see [#unique_255](#).
 - To import configuration information from a dial peer CSV file that you have created, click **Upload Dial Peer List**.

You can add as many dial peers as needed. Click **Add** after you configure each dial peer.

If any dial peers already are configured, they appear in the dial peers table on this page. To edit a configured dial peer, click ..., and click its pencil icon. Edit the options in the window that pops up as described in the table, and click **Save Changes**. To delete a dial peer, click ..., and click its trash can icon.

9. Click **Save**.

**Note**

Starting from Cisco IOS XE Catalyst SD-WAN Release 17.14.1a, you cannot configure port and trunk group together in the same configuration push. If you want to have port for dial peer, configure the port in the call routing feature template and do not configure trunk group under mapping in the device template. If you want to have trunk group for a dial peer, you should configure port to be default in the call routing feature template and configure the trunk group mapping in the device template for the dial peer.

Add an SRST Feature Template

An SRST feature template configures parameters for Cisco Unified Survivable Remote Site Telephony (SRST) for SIP. With Cisco Unified SRST, if the WAN goes down or is degraded, SIP IP phones in a branch site can register to the local gateway so that they continue to function for emergency services without requiring WAN resources that are no longer available.

The following table describes global options for configuring Cisco Unified SRST.

Table 94: Global Cisco Unified SRST Options

Option	Description	Cisco IOS CLI Equivalent
System Message	Enter a message that displays on endpoints when Cisco Unified SRST mode is in effect.	voice register global system message <i>string</i>
Max Phones	Enter the number of phones that the system can register to the local gateway when in Cisco Unified SRST mode. The available values and the maximum values that you can enter in this field depend on the device that you are configuring. Hover your mouse pointer over the Information icon next to this field to see maximum values for supported devices.	voice register global max-pool <i>max-voice-register-pools</i>
Max Directory Numbers	Enter the number of DN's that the gateway supports when in Cisco Unified SRST mode. The available values and the maximum values that you can enter in this field depend on the device that you are configuring. Hover your mouse pointer over the Information icon next to the Max phones to support field to see maximum values for supported devices.	voice register global max-dn <i>max-directory-numbers</i>

Option	Description	Cisco IOS CLI Equivalent
Music on Hold	Select Yes to play music on hold on endpoints when a caller is on hold when in Cisco Unified SRST mode. Otherwise, select No .	—
Music on Hold file	Enter the path and file name of the audio file for music on hold. The file must be in the system flash and must be in .au or .wav format. In addition, the file format must contain 8-bit 8-kHz data, for example, CCITT a-law or u-law data format.	call-manager-fallback moh <i>filename</i>

The following table describes options for configuring Cisco Unified SRST phone profiles.

Table 95: SRST Phone Profile Options

Option	Description	Cisco IOS CLI Equivalent
Voice Register Pool Tag	Enter the unique sequence number of the IP phone to be configured. The maximum value is defined by the Max phones to support option in the Global tab of the SRST feature template.	voice register pool <i>pool-tag</i>
Device Network IPv6 Prefix	Enter the IPv6 prefix of the network that contains the IP phone to support. For example, a.b.c.d/24.	voice register pool <i>pool-tag</i> id [network address mask mask]
Device Network IPv4 Prefix	Enter the IPv4 prefix of the network that contains the IP phone to support.	voice register pool <i>pool-tag</i> id [network address mask mask]

To add an SRST feature template:

1. From the Cisco SD-WAN Manager menu, Choose **Configuration > Templates**.
2. Click **Feature Templates**, and click **Add Template**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is called **Feature**.

3. Select the supported device to which you want to add Cisco Unified SRST features.
4. Click **SRST** from the Unified Communications templates.
5. In **Template Name**, enter a name for the template.
This field can contain uppercase and lowercase letters, digits 0 through 9, hyphens (-), and underscores (_).
6. In **Description**, enter a description for the template.

This field can contain any characters and spaces.

7. In **Global Settings**, configure options as described in the "Global SRST Options" table.
8. From **Phone Profile**, click **New Phone Profile** to create a phone profile, and configure options as described in the "SRST Phone Profile Options" table.

A phone profile provides pool tag and device network information for a SIP phone.

You can add as many phone profiles as needed. Click **Add** after you configure each phone profile.

If any phone profiles already are configured, they appear in the phone profiles table on this page. To edit a configured phone profile, click ..., and click its pencil icon. Edit the options in the window that pops up as described in the table, and click **Save Changes**. To delete a phone profile, click ..., and click its trash can icon.

9. Click **Save**.

Add a DSPFarm Feature Template

A DSP farm is a pool of DSP resources on a router. Cisco Catalyst SD-WAN uses DSP farm resources that are available to Cisco Unified Communications Manager for Cisco Unified Communications Manager controlled transcoding, conferencing (non-secure only), and media termination point (MTP) services. Cisco Unified Communications Manager dynamically invokes these resources as needed in a call path.

A DSPFarm feature template is used to set up and provision a DSP farm. The template supports dedicated DSP modules only. T1/E1 modules are not supported.

When you add a DSPFarm feature template, you configure options for the following items:

- Media resource modules—DSP modules and their placement on a router. You determine and build DSP farm profiles based on media resource modules.
- DSP farm profiles—Each profile defines parameters for provisioning a specific DSP farm service type. A profile includes options for provisioning a group of DSP resources that is used for transcoding, conferencing (only non-secure conferencing is supported), or MTP services. A profile is registered to a Cisco Unified Communications Manager so that the Cisco Unified Communications Manager can invoke the resources for a service as needed.
- SCCP config—Configures a local interface that is used to communicate with up to four Cisco Unified Communications Manager servers, and configures related information that is required to register the DSP farm profiles to Cisco Unified Communications Manager. Also configures one or more Cisco Unified Communications Manager groups, each of which includes up to four Cisco Unified Communications Manager servers that control the DSP farm services that, in turn, are associated with the servers.

When you add a media resource module, Cisco SD-WAN Manager assists you with the placement of the module by displaying available slots and sub-slots for the module. Cisco SD-WAN Manager determines the available slots and sub-slots based on the device model.

The following table describes options for configuring media resources.

Table 96: Media Resource Options

Option	Description	Cisco IOS CLI Equivalent
Module	Select the router resource module to carry DSP resources that are used by DSPFarm profiles.	—
Slot/sub-slot ID	Select the slot and sub-slot in which the resource module that you selected resides.	voice-card <i>slot/subslot</i> dsp service dspfarm

The following table describes options for configuring DSP farm services.

Table 97: DSP Farm Service Options

Option	Description	Cisco IOS CLI Equivalent
Profile Type	Select the type of DSP farm service that this profile is for. Options are Transcoder , Conference , and MTP	dspfarm profile <i>profile-identifier</i> { conference mtp transcode }
Profile ID	A system-generated unique identifier for the profile.	—
Universal	Available if you select Transcoder for the Profile Type When this check box is unchecked, transcoding is allowed only between the G.711 codec and other codecs. When this check box is checked, transcoding is allowed between codecs of any type.	dspfarm profile <i>profile-identifier</i> transcode [universal]

Option	Description	Cisco IOS CLI Equivalent
List Codec		codec <i>codec-name</i>

Option	Description	Cisco IOS CLI Equivalent
	Select the codecs that are available for the DSP farm service that this profile defines. The following codecs are supported. For MTP profile types, you can select one option, or you can select pass-through and one other option. If you want to change a codec, unselect the current codec before selecting a new one. • For the Transcoder profile type: • g711alaw • g711ulaw • g729abr8 • g729ar8 • g729br8 • g729r8 • g722-64 • ilbc • iSAC • pass-through • For the Conference profile type: • g711alaw • g711ulaw • g722r-64 • g729abr8 • g729ar8 • g729br8 • g729r8 • For the MTP profile type for software MTP only: • g711ulaw • g711alaw	

Option	Description	Cisco IOS CLI Equivalent
	• g722-64 • g729abr8 • g729ar8 • g729br8 • g729r8 • ilbc • iSAC • pass-through • For the MTP profile type for hardware MTP only, or for hardware and software MTP: • g711ulaw • g711alaw • pass-through	
Conference Maximum Participants	Available if you select Conference for the Profile Type. Select the maximum number of parties that can participate in a conference bridge (8, 16, or 32).	maximum conference-participants <i>number</i>
Maximum Sessions	Available if you select Transcoder or Conference for the Profile Type. Enter the maximum number of sessions that this profile can support. This value depends on the maximum number sessions that can be configured with the DSP resources that are available on the router. These resources are based on the type of modules in the router. To determine these resources, you can use a DSP calculator.	maximum sessions <i>number</i>

Option	Description	Cisco IOS CLI Equivalent
MTP Type	Available if you select MTP for the Profile Type. Select the way in which the router performs minor MTP translations such as G.711alaw to G.711ulaw, and DTMF conversions. Options are: • Hardware—MTP translations and conversions are performed by the hardware DSP resources • Software—MTP translations and conversions are performed by the router CPU	maximum session {hardware software}
MTP Maximum Hardware Sessions	Available if you select Hardware for the MTP type. Select the maximum number of hardware sessions that can be used for MTP translations and conversions. Maximum value: 4000	maximum session hardware number
MTP Maximum Software Sessions	Available if you select Software for the MTP type. Select the maximum number of CPU sessions that can be used for MTP translations and conversions. Maximum value: 6000	maximum session software number
Application	Select the type of application to which the DSP farm services that are provisioned on the device are associated.	associate application sccp
Shutdown	Enable this option to take this profile out of service.	shutdown

The following table describes options for configuring SCCP.

Table 98: SCCP Options

Option	Description	Cisco IOS CLI Equivalent
CUCM Tab Configure up to 12 Cisco Unified Communications Manager servers to which the profiles that you defined in the Profile tab register.		
Local Interface	Enter the local interface that DSP services that are associated with the SCCP application use to register with Cisco Unified Communications Manager. Enter the interface in this format: <i>interface-type/interface-number/port</i> where: • <i>interface-type</i>—Type of interface that the services use to register with Cisco Unified Communications Manager. The type can be a GigabitEthernet interface or a port channel interface. • <i>interface-number</i>—Interface number that the services use to register with Cisco Unified Communications Manager. • <i>port</i>—(Optional) Port on which the interface communicates with Cisco Unified Communications Manager. If you do not specify a port, the default value 2000 is used. For example: GigabitEthernet0/0/0.	sccp local <i>interface-type interface-number</i> [port <i>port-number</i>]

Option	Description	Cisco IOS CLI Equivalent
Server List - x	Designate a Cisco Unified Communications Manager server to which the profiles that you defined in the Profile tab register. In the first field, enter the IP address or DNS name of the Cisco Unified Communications Manager server. In the second field, enter a numerical identifier for the Cisco Unified Communications Manager server. Click the Plus Sign icon (+) to configure up to 11 additional servers. To remove a server, click its corresponding Minus Sign icon. (–).	sccp ccm { <i>ipv4-address</i> <i>ipv6-address</i> <i>dns</i> } identifier <i>identifier-number</i> version 7.0+
CUCM Groups Tab This tab is available when at least one Cisco Unified Communications Manager server is configured in the Cisco Unified Communications Manager tab. Configure a Cisco Unified Communications Manager group, which includes up to 4 Cisco Unified Communications Manager servers that control the DSP farm services that, in turn, are associated with the servers. If any Cisco Unified Communications Manager groups are already configured, they appear in the table in this tab. To edit a configured Cisco Unified Communications Manager group, click its pencil icon in the Action column, edit the options in the window that pops up as described in the following rows, and click Save Changes. To delete a Cisco Unified Communications Manager group, click its trash can icon in the Action column.		
Add New CUCM Group	Click to add a new Cisco Unified Communications Manager group.	sccp ccm group <i>group-id</i>

Option	Description	Cisco IOS CLI Equivalent
Server Groups Priority Order	Select the priority in which the Cisco Unified Communications Manager servers in this Cisco Unified Communications Manager group are used. To do so: 1. Click this field to display a list of the Cisco Unified Communications Manager servers that you configured on the Cisco Unified Communications Manager tab. 2. Select the server that you want to be the primary server. This server has the highest priority. 3. Click the field again and select the server that you want to be the redundant server with the next highest priority. Repeat this step to select other redundant servers. The servers appear in this field in priority order. To remove a server from the group, click its X icon. To change the priority order of servers, remove the servers and add them back in the desired order.	associate ccm <i>cisco-unified-communications-manager-id</i> priority priority

Option	Description	Cisco IOS CLI Equivalent
CUCM Media Resource Name Profile to be Associated	In the Cisco Unified Communications Manager Media Resource Name field, enter a unique name that is used to register a DSP farm profile to the Cisco Unified Communications Manager servers. The name must contain from 6 to 15 characters. Characters can be letter, numbers, slashes (/), hyphens (-), and underscores (_). Space characters are not allowed. In the corresponding Profile to be Associated field, select a DSP farm profile to be registered to this Cisco Unified Communications Manager group using the name that you entered. To select a profile, click this field to display a list of the profile IDs that were configured on the Profile tab, and click the ID of the profile that you want. To add another Cisco Unified Communications Manager media resource name and profile, click the plus sign (+). You can add up to 4 Cisco Unified Communications Manager media resources and profiles. To remove a Cisco Unified Communications Manager media resource name and profile, click its corresponding minus sign (-).	associate ccm <i>profile-identifier</i> register <i>device-name</i>

Option	Description	Cisco IOS CLI Equivalent
CUCM Switchback	Select the switchback method that the Cisco Unified Communications Manager servers in this Cisco Unified Communications Manager group use to switch back after a failover: • graceful—Switchback occurs after all active sessions terminate gracefully. • guard—Switchback occurs either when active sessions are terminated gracefully or when the guard timer expires, whichever happens first. • immediate—Performs the Cisco Unified Communications Manager switchback to the higher priority Cisco Unified Communications Manager immediately when the timer expires, whether there is an active connection or not. Default: graceful.	switchback method { graceful guard [<i>timeout-guard-value</i>] immediate }
CUCM Switchover	Select the switchover method that Cisco Unified Communications Manager servers in this Cisco Unified Communications Manager use group when failing over: • graceful—Switchback occurs after all active sessions terminate gracefully. • immediate—Switchover occurs immediately, whether there is an active connection or not. Default: graceful.	switchover method { graceful immediate }

To add a DSPFarm feature template:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**, and click **Add Template**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is called **Feature**.

3. Select the supported device to which you want to add a DSP farm.
4. Click **DSPFarm** from the **Unified Communications** templates.
5. In **Template Name**, enter a name for the template.

This field can contain uppercase and lowercase letters, digits 0 through 9, hyphens (-), and underscores (_).
6. In **Description**, enter a description for the template.

This field can contain any characters and spaces.
7. From **Media Resources Modules**, click **Add Media Resources**, and configure options as described in the "Media Resource Options" table.

A media resource module is a DSP module that is used by DSP Farm profiles.

You can add as many media resources interfaces as needed.

Click **Add** after you configure each media resource. After you configure a media resource, you cannot modify or delete it because other configuration items are based on the module and its placement. If you need to change a media resource configuration, you must remove the DSPFarm feature template and create a new one.

If any media resources are already configured, they appear in the table in this tab. To edit a configured media resource, click ..., and click its pencil icon. Edit the options in the window that pops up as described in the "Media Resource Options" table, and click **Save Changes**. To delete a media resource, click ..., and click its trash can icon.
8. From **Profile**, click **Add New Profile** to add a profile for a DSP farm service on a router, and configure options for the profile as described in the "DSP Farm Service Options" table.

Click **Add** after you configure a profile. You can add up to 10 DSP farm profiles for each feature template.

Before you create a profile, you must know the maximum number of sessions that can be configured with the DSP resources that are available on the router. These resources are based on the type of modules in the router. To determine these resources, you can use a DSP calculator.

After you add a profile, you can modify the List Codec, Maximum Sessions, Maximum Conference Participants, and Shutdown options. You cannot change the profile type. If you want to change the profile type, you must delete the profile and create a new one.

If any profiles are already configured, they appear in the table in this tab. To edit a configured profile, click ..., and click its pencil icon. Edit the options in the window that pops up as described in the "DSP Farm Service Options" table, and click **Save Changes**. To delete a profile, click ..., and click its trash can icon.
9. In **SCCP Config**, configure options as described in the "SCCP Options" table.
10. Click **Save**.

Add a Voice Policy

A voice policy defines how the system augments and manipulates calls for various endpoint types. Endpoints include voice ports, POTS dial peers, SIP dial peers, and Cisco Unified SRST phone profiles. A voice policy includes subpolicies for each endpoint that you want to configure.

To add a voice policy:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Unified Communications**.
2. Click **Add Voice Policy**.
3. In **Voice Policy Name**, enter a name for the policy.
4. Configure the following as required:
 - **Voice Ports**—See [Configure Voice Ports for a Voice Policy](#)
 - **POTS Dial Peers**—See [Configure POTS Dial Peers for a Voice Policy](#)
 - **SIP Dial Peers**—See [Configure SIP Dial Peers for a Voice Policy](#)
 - **SRST Phones**—See [Configure SRST Phones for a Voice Policy](#)
5. Click **Save Policy**.

Configure voice ports for a voice policy

When you configure voice ports for a voice policy, you configure options that define how the system augments and manipulates calls for the voice port endpoint type.

You can configure the following call functionality policy options, depending on the type of voice card you are using:

- **Trunk Group**— Use these options to configure voice ports as a member of a trunk group for the card. You can configure one trunk group for voice card. The following table describes these options.

Table 99: Trunk group options for voice ports

Option	Description	Cisco IOS CLI Equivalent
Add New Trunk Group	Click to add a trunk group for the selected card. You can add one trunk group for a voice port.	—
Copy from Existing	Click to copy an existing trunk group to a new trunk group. In the box that appears, change the name if desired, select a trunk group, and click Copy .	—
Name	Name of the trunk group. The name can contain up to 32 characters.	trunk group name

Option	Description	Cisco IOS CLI Equivalent
Hunt-Scheme		trunk group <i>name</i> hunt-scheme least-idle [both even odd] hunt-scheme least-used [both even odd] hunt-scheme longest-idle [both even odd] hunt-scheme round-robin [both even odd] hunt-scheme sequential [both even odd] hunt-scheme random

Option	Description	Cisco IOS CLI Equivalent
	Select the hunt scheme in the trunk group for outgoing calls: • least-idle both—Searches for an idle channel with the shortest idle time • least-idle even—Searches for an idle even-numbered channel with the shortest idle time • least-idle odd—Searches for an idle odd-numbered channel with the shortest idle time • least-used both—Searches for a trunk group member that has the highest number of available channels (applies only to PRI ISDN cards) • least-used even—Searches for a trunk group member that has the highest number of available even-numbered channels (applies only to PRI ISDN cards) • least-used odd—Searches for a trunk group member that has the highest number of available odd-numbered channels (applies only to PRI ISDN cards) • longest-idle both—Searches for an idle odd-numbered channel with the longest idle time • longest-idle even—Searches for an idle channel that has the highest number of available even-numbered channels • longest-idle odd—Searches for an idle channel that has the highest number of available odd-numbered channels • round-robin both—Searches trunk group members in turn for an idle channel, starting with the trunk group member that follows the last used member • round-robin even—Searches trunk group member in turn for an idle even-numbered channel, starting with	

Option	Description	Cisco IOS CLI Equivalent
	the trunk group member that follows the last used member • round-robin odd—Searches trunk group member in turn for an idle odd-numbered channel, starting with the trunk group member that follows the last used member • sequential-both—Searches for an idle channel, starting with the trunk group member with the highest preference within the trunk group • sequential-even—Searches for an idle even-numbered channel, starting with the trunk group member with the highest preference within the trunk group • sequential-odd—Searches for an idle odd-numbered channel, starting with the trunk group member with the highest preference within the trunk group • random—Searches for a trunk group member at random and selects a channel from the member at random Default: least-used both	
Max Calls	Enter the maximum number of calls that are allowed for the trunk group. If you do not enter a value, there is no limit on the number of calls. If the maximum number of calls is reached, the trunk group becomes unavailable for more calls. • In field—Enter the maximum number of incoming calls that are allowed for this trunk group • Out field— Enter the maximum number of outgoing calls that are allowed for this trunk group Valid range for both fields: integers 0 through 1000.	trunk group <i>name</i> max-calls <i>voice number-of-calls</i> direction [<i>in</i> <i>out</i>]

Option	Description	Cisco IOS CLI Equivalent
Max-Retry	Select the maximum number of outgoing call attempts that the trunk group makes if an outgoing call fails. If you do not enter a value and a call fails, the system does not attempt to make the call again. Valid range: integers 1 through 5.	trunk group <i>name</i> max-retry <i>attempts</i>
Save Trunk Group	Click to save the Trunk Group that you configured.	—

- **Translation Profile**—Use these options to configure translation rules for calling and called numbers. The following table describes these options.

Table 100: Translation profile options for calling and called numbers

Option	Description	Cisco IOS CLI Equivalent
Add New Translation Profile	Click to add a translation profile for the selected card. You can create up to two translation profiles for this endpoint.	voice translation-profile <i>name</i>
Copy from Existing	Click to copy an existing translation profile to a new translation profile. In the box that appears, change the name if desired, select a called translation rule and a calling translation rule, and click Copy .	—
Calling	Click to configure translation rules for the number that is calling in. The Translation Rules pane displays.	translate calling <i>translation-rule-number</i>
Called	Click to configure translation rules for the number that is being called. The Translation Rules pane displays.	translate called <i>translation-rule-number</i>

Option	Description	Cisco IOS CLI Equivalent
Translation Rules pane		voice translation-rule <i>number</i> Match and Replace Rule: rule <i>precedence</i> <i>/match-pattern/</i> <i>/ replace-pattern/</i> Reject Rule: rule <i>precedence</i> reject <i>/match-pattern/</i>

Option	Description	Cisco IOS CLI Equivalent
	- Click Add New to create a translation rule. Alternatively, you can click Copy From Existing to copy an existing translation rule to a new translation rule. In the box that appears, change the name if desired, select a called translation rule and a calling translation rule, and click Copy. - In the Translation Rule Number field, enter a unique number that designates the precedence for this rule. Valid range: integers 1 through 100. (Optional) To copy existing translation rules from a CSV file, click Import. Continue to add rules or click Finish. For detailed information about this file, see Translation Rules CSV File, on page 356. - Click Add Rule. - In the Match field, enter the string that you want the translation rule to affect. Enter the string in regular expression format beginning and ending with a slash (/). For example, <code>/^9/</code>. To include the backslash character (\) in a match string, precede the backslash with a backslash. - From the Action drop-down list, select the action that the system performs for calls that match the string in the Match field. The Reject option causes the system to reject the call. The Replace option causes the system to replace the match number with a value that you specify. - If you select the Replace action, in the Replace field that displays, enter the string to which to translate the matched string. Enter the number in regular expression format beginning and ending with a slash (/). For example, <code>//</code>, which indicates a	

Option	Description	Cisco IOS CLI Equivalent
	replacement of no string. To include the backslash character (\) in a replace string, precede the backslash with a backslash. As an example, if you specify a match string of /^9/ and a replace string of //, the system removes the leading 9 from calls with a number that begins with 9. In this case, the system translates 914085551212 to 14085551212. 8. Click Save. 9. Add more translation rules as needed. 10. (Optional) Click Export to save the translation rules that you created in a CSV file. 11. Click Finish at the bottom of the pane.	

- **Station ID**—Use these options to configure the name and number for caller ID display. The following table describes these options.

Table 101: Station ID options

Option	Description	Cisco IOS CLI Equivalent
Station Name	Enter the name of the station. The station name can contain up to 50 letters, numbers, and spaces, dashes (-), and underscores (_).	station-id name <i>name</i>
Station Number	Enter the phone number of the station in E.164 format. The station number can contain up to 15 numeric characters.	station-id number <i>number</i>

- **Line Params**—Use these options to configure line parameters on the card for voice quality. The following table describes these options.

Table 102: Line params Options

Option	Description	Cisco IOS CLI Equivalent
Gain	Enter the gain, in dB, for voice input. Valid range: -6 through 14. Default: 0	input gain <i>decibels</i>

Option	Description	Cisco IOS CLI Equivalent
Attenuation	Enter the amount of attenuation, in dB, for transmitted voice output. Valid range: –6 through 14. Default: 3.	output attenuation <i>decibels</i>
Echo Canceller	Select Enable to apply echo cancellation to voice traffic. By default, this option is enabled.	echo-cancel <i>enable</i>
Voice Activity Detection (VAD)	Select Enable to apply VAD to voice traffic. By default, this option is enabled.	vad
Compand Type	Select the companding standard to be used to convert between analog and digital signals in PCM systems (U-law or A-law). Default: U-Law.	compand-type { u-law a-law }
Impedance	This field does not apply to PRI ISDN cards. Select the terminating impedance for calls. Default: 600r.	impedance { 600c 600r 900c 900r complex1 complex2 complex3 complex4 complex5 complex6 }
Call Progress Tone	Select the locale for call progress tones.	cptone <i>locale</i>

- **Tuning Params**—Use these options to configure parameters for signaling between voice ports and another instrument. The following table describes these options.

Table 103: Tuning params options

Option	Description	Cisco IOS CLI Equivalent
Tuning Params Options for FXO Cards		
Pre Dial Delay	Enter the delay, in seconds, of the delay on the FXO interface between the beginning of the off-hook state and the initiation of DTMF signaling. Valid range: 0 through 10. Default: 1.	pre-dial-delay <i>seconds</i>

Option	Description	Cisco IOS CLI Equivalent
Supervisory Disconnect	Select the type of tone that indicates that a call has been released and that a connection should be disconnected: • Anytone—Any tone indicates a supervisory disconnect • Signal—A disconnect signal indicates a supervisory disconnect • Dualtone—A dual-tone indicates a supervisory disconnect Default: Signal.	Anytone: supervisory disconnect anytone Signal: supervisory disconnect Dualtone: supervisory disconnect dualtone {mid-call pre-connect}
Dial Type	Select the dialing method for outgoing calls: • pulse—Pulse dialer • dtmf—Dual-tone multifrequency dialer • mf—Multifrequency dialer Default: dtmf.	dial-type { dtmf pulse mf }
Timing Sup-Disconnect	Enter the minimum time, in milliseconds, that is required to ensure that an on-hook indication is intentional and not an electrical transient on the line before a supervisory disconnect occurs (based on power denial signaled by the PSTN or PBX). Valid range: 50 through 1500. Default: 350.	timing sup-disconnect <i>milliseconds</i>

Option	Description	Cisco IOS CLI Equivalent
Battery Reversal	Battery reversal reverses the battery polarity on a PBX when a call connects, then changes the battery polarity back to normal when the far-end disconnects. Select Answer to configure the port to support answer supervision by detection of battery reversal. Select Detection Delay to configure the delay time after which the card acknowledges a battery-reversal signal, then enter the delay time in milliseconds. Valid range: 0 through 800. Default: 0 (no delay). If an FXO port or its peer FXS port does not support battery reversal, do not configure battery reversal options to avoid unpredictable behavior.	battery-reversal [answer] battery-reversal-detection-delay <i>milliseconds</i>
Timing Hookflash out	Enter the duration, in milliseconds, of hookflash indications that the gateway generates on the FXO interface. Valid range: 50 through 1550. Default: 400.	timing hookflash-out <i>milliseconds</i>
Timing Guard out	Enter the number of milliseconds after a call disconnects before another outgoing call is allowed. Valid range: 300 through 3000. Default: 2000.	timing guard-out <i>milliseconds</i>
Tuning Params Options for FXS Cards		
Timing Hookflash In	Enter the minimum and maximum duration, in milliseconds, of an on-hook condition to be interpreted as a hookflash by the FXS card. Valid range for minimum duration: 0 through 400. Default minimum value: 50. Valid range for maximum duration: 50 through 1500. Default maximum value: 1000.	timing hookflash-in <i>maximum-milliseconds</i> <i>minimum-milliseconds</i>
Pulse Digit Detection	To enable pulse digit detection at the beginning of a call, select Yes. Default: Yes.	pulse-digit-detection

Option	Description	Cisco IOS CLI Equivalent
Loop Length	Select the length for signaling on FXS ports (Long or Short). Default: Short.	loop-length [long short]
Ring	• Frequency—Select the frequency, in Hz, of the alternating current that, when applied, rings a connected device. Default: 25. • DC Offset—Applies only if Loop Length is set to Long. Select the voltage threshold below which a ring does not sound on devices. Valid values: 10-volts, 20-volts, 24-volts, 30-volts, and 35-volts.	ring frequency number ring dc-offset number
Ringer Equivalence Number (REN)	Select the REN for calls that this card processes. This number specifies the loading effect of a telephone ringer on a line. Valid range: 1 through 5. Default: 1.	ren number

- **Supervisory Disconnect**—Use these options to configure parameters for supervisory disconnect events. The following table describes these options.

Table 104: Supervisory disconnect options

Option	Description	Cisco IOS CLI Equivalent
Add New Supervisory Disconnect	Click to add a supervisory disconnect event.	—
Mode	Choose the mode for the supervisory disconnect event: • Custom CPTone—Provides options for configuring cptone detection parameters for a supervisory disconnect event • Dual Tone Detection Params—Provides options for configuring dual-tone detection parameters for a supervisory disconnect event	voice class custom-cptone <i>cptone-name</i> voice class dualtone-detect-params tag

Option	Description	Cisco IOS CLI Equivalent
Supervisory Name	Applies to Custom CPTone mode. Enter a name for the supervisory disconnect event. The name can contain up to 32 characters. Valid characters are letters, numbers, dashes (-), and underscores (_).	voice class custom-cptone <i>cptone-name</i>
Dualtone	Applies to Custom CPTone mode. Select the type of dual-tone that causes a disconnect. Options are: • Busy • Disconnect • Number Unobtainable • Out of Service • Reorder • Ringback	dualtone {ringback busy reorder out-of-service number-unobtainable disconnect}
Cadence	Applies to Custom CPTone mode. Enter the cadence interval, in milliseconds, of the dual-tones that cause a disconnect. Enter the cadence as an on/off value pair, separated with a space. You can enter up to 4 on/off value pairs, separated with a space.	cadence <i>cycle-1-on-time cycle-1-off-time [cycle-2-on-time cycle-2-off-time [cycle-3-on-time cycle-3-off-time [cycle-4-on-time cycle-4-off-time]]]</i>
Dualtone Frequency	Applies to Custom CPTone mode. Enter the frequency, in Hz, of each tone in the dual-tone. Valid range for each tone is 300 through 3600.	frequency <i>frequency-1 [frequency-2]</i>
Supervisory Number	Applies to Custom Dual Tone Detection Params mode. Enter a unique number to identify dual-tone detection parameters. Valid range: 1 through 10000.	voice class dualtone-detect-params <i>tag-number</i>
Cadence-Variation	Applies to Custom Dual Tone Detection Params mode. Enter the maximum time, in milliseconds, by which the tone onset can vary from the onset time and still be detected. The system multiplies the value that you enter by 10. Valid range: 0 through 200 in units of 10. Default: 10.	cadence-variation <i>time</i>

Option	Description	Cisco IOS CLI Equivalent
Frequency	Applies to Custom Dual Tone Detection Params mode. • Max Delay—Enter the maximum delay, in milliseconds, before a supervisory disconnect is performed after the dual-tone is detected. The system multiplies the value that you enter by 10. Valid range: 0 through 100 in units of 10. Default: 10. • Max Deviation—Enter the maximum deviation, in Hz, by which each tone can deviate from configured frequencies and be detected. Valid range: 10 through 125. Default: 10. • Max Power—Enter the power of the dual-tone, in dBm0, above which a supervisory disconnect is no detected. Valid range: 0 through 20. Default: 10. • Min Power— Enter the power of the dual-tone, in dBm0, below which a supervisory disconnect is not detected. Valid range: 10 through 35. Default: 30. • Power Twist—Enter difference, in dBm0, between the minimum power and the maximum power of the dual-tone above which a supervisory disconnect is not detected. Valid range: 0 through 15. Default: 6.	freq-max-delay <i>time</i> freq-max-deviation <i>hertz</i> freq-max-power <i>dBm0</i> freq-min-power <i>dBm0</i> freq-power-twist <i>dBm0</i>
Save	Click to save the supervisory disconnect information that you configured.	—

- **DID Timers**—Use these options to configure timers for DID calls. The following table describes these options.

Table 105: DID timers options

Option	Description	Cisco IOS CLI Equivalent
Wait Before Wink	Enter the amount of time, in milliseconds, that the card waits after receiving a call before sending a wink signal to notify the remote side that it can send DNIS information. Valid range: 100 through 6500. Default: 550.	timing wait-wink <i>milliseconds</i>
Wink Duration	Enter the maximum amount of time, in milliseconds, of the wink signal for the card. Valid range: 50 through 3000. Default: 200.	timing wait-duration <i>milliseconds</i>
Clear Wait	Enter the minimum amount of time, in milliseconds, between an inactive seizure signal and the call being cleared for the card. Valid range: 200 through 2000. Default: 400.	timing clear-wait <i>milliseconds</i>
Dial Pulse Min Delay	Enter the amount of time, in milliseconds, between wink-like pulses for the card. Valid range: 0 or 140 through 5000. Default: 140.	timing dial-pulse min-delay <i>milliseconds</i>
Answer Winkwidth	Enter the minimum delay time, in milliseconds, between the start of an incoming seizure and the wink signal. Valid range: 110 through 290. Default: 210.	timing answer-winkwidth <i>milliseconds</i>

To configure voice ports for a voice policy, follow these steps:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Unified Communications**.
2. Click **Add Voice Policy**, and choose **Voice Ports** in the left pane.
3. From the **Add Voice Ports Policy Profile** drop-down list, select **Create New**.

Alternatively, you can select **Copy from Existing** to copy an existing voice policy to a new voice policy. In the box that appears, select the name of the policy profile to copy, enter a new name for the profile if desired, and click **Copy**.

4. Select **FXO**, **FXS**, **PRI ISDN**, or **FXS DID** to specify the type of voice port that the policy is for.
5. Select the types of call functionality policy options that you want to configure from the list of options that displays, and click **Next**. These option types include the following:

- **Trunk Group**—Available for FXO, FXS, FXS DID, and PRI ISDN cards.
Use these options to configure voice ports as a member of a trunk group for the card.
- **Translation Profile**—Available for FXO, FXS, PRI ISDN, and FXS DID cards.
Use these options to configure translation rules for calling and called numbers.
- **Station ID**—Available for FXO, FXS, and FXS DID cards.
Use these options to configure the name and number for caller ID display.
- **Line Params**—Available for FXO, FXS, PRI ISDN, and FXS DID cards.
Use these options to configure line parameters on the card for voice quality.
- **Tuning Params**—Available for FXO and FXS cards.
Use these options to configure parameters for signaling between voice ports and another instrument.
- **Supervisory Disconnect**—Available for FXO cards.
Use these options to configure parameters for supervisory disconnect events. These events provide an indication that a call has disconnected.
- **DID Timers**—Available for FXS DID cards.
Use these options to configure timers for DID calls.

6. In the page that displays, configure as needed the options on the tabs as needed.

The tabs that are available depend on the voice port and call functionality policy option types that you selected.

- **Trunk Group** options—For a description of these options, see the "Trunk Group Options for Voice Ports" table.

If any trunk groups are already configured for other voice cards, they appear in the trunk groups table on this page. To edit a configured trunk group, click ..., and click its pencil icon. Edit the options in the window that pops up as described in the "Trunk Group Options for Voice Ports" table, and click **Save Changes**. To delete a trunk group, click ..., and click its trash can icon.

After you click **Save Trunk Group** when saving trunk group options, configure the priority for a trunk group by double-click the Priority field for a trunk group in the Trunk Group table, entering a priority number, and pressing **Enter** or clicking outside of the Priority field. Valid priority numbers are integers 1 through 64. The number you enter is the priority of the POTS dial peer in the trunk group for incoming and outgoing calls.

- **Translation Profile** options—For a description of these options, see the "Translation Profile Options for Calling and Called Numbers" table.

After you click **Finish** when configuring translation profile options, perform these actions:

- a. Add another translation profile if needed. You can create up to two translation profiles for this endpoint.
- b. Click **Save Translation Profile**.
- c. For each translation profile that you create, double-click the dash (-) that displays in **Direction** column in the table of translation rules and select **Incoming** or **Outgoing** from the drop-down list that displays. The Incoming selection applies the corresponding translation rule to traffic

that is incoming to this endpoint. The Outgoing selection applies the corresponding translation rule to traffic that is outgoing from this endpoint.

- **Station ID** options—For a description of these options, see the "Station ID Options" table.
- **Line Params** options—For a description of these options, see the "Line Params Options" table.
- **Tuning Params** options—For a description of these options, see the "Tuning Params Options" table.
- **Supervisory Disconnect** options—For a description of these options, see the "Supervisory Disconnect Options" table.

You can configure as many supervisory disconnect events as needed.

- **DID Timers** options—For a description of these options, see the "DID Timers Options" table

7. Click **Next**
8. In **Policy Profile Name**, enter a name for this child policy.
9. In **Policy Profile Description**, enter a description for this child policy.
10. Click **Save**.

Configure POTS Dial Peers for a Voice Policy

When you configure POTS Dial Peers for a voice policy, you configure options that define how the system augments and manipulates calls for the POTS dial peer endpoint type.

You can configure the following options:

- **Trunk Groups**—The following table describes these options.

Table 106: Trunk Group Options for POTS Dial Peers

Option	Description	Cisco IOS CLI Equivalent
Add New Trunk Group	Click to add a trunk group for the selected card. You can add one trunk group for a voice port.	—

Option	Description	Cisco IOS CLI Equivalent
Copy from Existing	Click to copy an existing trunk group to a new trunk group. In the box that appears, change the name if desired, select a trunk group, and click Copy. A trunk group name whose name is preceded with "{Master}" is already associated with this voice policy. When you copy a this type of trunk group, the system reuses the existing trunk group without creating another instance of the trunk group definition. In this case, you cannot change the name.	—
Name	Name of the trunk group. The name can contain up to 32 characters.	trunk group <i>name</i>

Option	Description	Cisco IOS CLI Equivalent
Hunt-Scheme		trunk group <i>name</i> hunt-scheme least-idle [both even odd] hunt-scheme least-used [both even odd] hunt-scheme longest-idle [both even odd] hunt-scheme round-robin [both even odd] hunt-scheme sequential [both even odd] hunt-scheme random

Option	Description	Cisco IOS CLI Equivalent
	Select the hunt scheme in the trunk group for outgoing calls: • least-idle both—Searches for an idle channel with the shortest idle time • least-idle even—Searches for an idle even-numbered channel with the shortest idle time • least-idle odd—Searches for an idle odd-numbered channel with the shortest idle time • least-used both—Searches for a trunk group member that has the highest number of available channels (applies to only PRI ISDN cards) • least-used even—Searches for a trunk group member that has the highest number of available even-numbered channels (applies only to PRI ISDN cards) • least-used odd—Searches for a trunk group member that has the highest number of available odd-numbered channels (applies only to PRI ISDN cards) • longest-idle both—Searches for an idle odd-numbered channel with the longest idle time • longest-idle even—Searches for an idle channel that has the highest number of available even-numbered channels • longest-idle odd—Searches for an idle channel that has the highest number of available odd-numbered channels • round-robin both—Searches trunk group members in turn for an idle channel, starting with the trunk group member that follows the last used member • round-robin even—Searches trunk group member in turn for an idle even-numbered channel, starting with	

Option	Description	Cisco IOS CLI Equivalent
	the trunk group member that follows the last used member • round-robin odd—Searches trunk group member in turn for an idle odd-numbered channel, starting with the trunk group member that follows the last used member • sequential-both—Searches for an idle channel, starting with the trunk group member with the highest preference within the trunk group • sequential-even—Searches for an idle even-numbered channel, starting with the trunk group member with the highest preference within the trunk group • sequential-odd—Searches for an idle odd-numbered channel, starting with the trunk group member with the highest preference within the trunk group • random—Searches for a trunk group member at random and selects a channel from the member at random Default: least-used both	
Max Calls	Enter the maximum number of calls that are allowed for the trunk group. If you do not enter a value, there is no limit on the number of calls. If the maximum number of calls is reached, the trunk group becomes unavailable for more calls. • In field—Enter the maximum number of incoming calls that are allowed for this trunk group. • Out field— Enter the maximum number of outgoing calls that are allowed for this trunk group. Valid range for both fields: integers 0 through 1000.	trunk group <i>name</i> max-calls voice <i>number-of-calls</i> direction [in out]

Option	Description	Cisco IOS CLI Equivalent
Max-Retry	Select the maximum number of outgoing call attempts that the trunk group makes if an outgoing call fails. If you do not enter a value and a call fails, the system does not attempt to make the call again. Valid range: integers 1 through 5.	trunk group <i>name</i> max-retry <i>attempts</i>

- **Translation Profiles**—The following table describes these options.

Table 107: Translation Profile Options for POTS Dial Peers

Option	Description	Cisco IOS CLI Equivalent
Add New Translation Profile	Click to add a translation profile for the selected POTS dial peer. You can create up to two translation profiles for this endpoint.	—
Copy from Existing	Click to copy an existing translation profile to a new translation profile. In the box that appears, change the name if desired, select a called translation rule and a calling translation rule, and click Copy.	—
Name	Name of the translation profile. The name can contain up to 32 characters.	voice translation-profile <i>name</i>
Calling	Click to configure translation rules for the number that is calling in. The Translation Rules pane displays.	translate calling <i>translation-rule-number</i>
Called	Click to configure translation rules for the number that is being called. The Translation Rules pane displays.	translate called <i>translation-rule-number</i>

Option	Description	Cisco IOS CLI Equivalent
Translation Rules pane		voice translation-rule <i>number</i> Match and Replace Rule: rule <i>precedence</i> <i>/match-pattern/</i> <i>/ replace-pattern/</i> Reject Rule: rule <i>precedence</i> reject <i>/match-pattern/</i>

Option	Description	Cisco IOS CLI Equivalent
	- Click Add New to create a translation rule. Alternatively, you can click Copy From Existing to copy an existing translation rule to a new translation rule. In the box that appears, change the name if desired, select a called translation rule and a calling translation rule, and click Copy. - In the Translation Rule Number field, enter a unique number that designates the precedence for this rule. Valid range: integers 1 through 100. (Optional) To copy existing translation rules from a CSV file, click Import. Continue to add rules or click Finish. For detailed information about this file, see #unique_264. - Click Add Rule. - In the Match field, enter the string that you want the translation rule to affect. Enter the string in regular expression format beginning and ending with a slash (/). For example, <code>/^9/</code>. To include the backslash character (\) in a match string, precede the backslash with a backslash. - From the Action drop-down list, select the action that the system performs for calls that match the string in the Match field. The Reject option causes the system to reject the call. The Replace option causes the system to replace the match number with a value that you specify. - If you select the Replace action, in the Replace field that displays, enter the string to which to translate the matched string. Enter the number in regular expression format beginning and ending with a slash (/). For example, <code>//</code>, which indicates a replacement of no string.	

Option	Description	Cisco IOS CLI Equivalent
	To include the backslash character (\) in a replace string, precede the backslash with a backslash. As an example, if you specify a match string of /^9/ and a replace string of //, the system removes the leading 9 from calls with a number that begins with 9. In this case, the system translates 914085551212 to 14085551212. 8. Click Save. 9. Add more translation rules as needed. 10. (Optional) Click Export to save the translation rules that you created in a CSV file. 11. Click Finish at the bottom of the pane.	

To configure POTS dial peers for a voice policy:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Unified Communications**
2. Click **Add Voice Policy**, and choose **POTS Dial Peer** in the left pane.
3. From the **Add POTS Dial Peer Policy Profile** drop-down list, select **Create New**.

Alternatively, you can select **Copy from Existing** to copy an existing POTS dial peer policy to a new one. In the box that appears, select the name of the policy profile to copy, enter a new name for the profile if desired, and click **Copy**.

4. Select the types of POTS dial peers that you that you want to configure from the list of options that displays, and click **next**.

Options are **Trunk Group** (beginning with Cisco IOS XE Catalyst SD-WAN Release 17.3.1a) and **Translation Profile**.

5. To configure trunk groups, perform the following actions.

If any trunk groups are already configured, they appear in the trunk groups table on this page. To edit a configured trunk group, click ..., and click its pencil icon. Edit the options in the window that pops up as described in the "Trunk Groups for POTS Dial Peers Options" table, and click **Save Changes**. To delete a trunk group, click ..., and click its trash can icon.

- a. Configure trunk group options as described in the "Trunk Groups Options for POTS Dial Peers" table.
- b. Add another trunk group if needed.
You can create up to 64 trunk groups for this endpoint.
- c. Click **Save Trunk Group**.

- d. Configure the priority for a trunk group by double-click the Priority field for a trunk group in the Trunk Group table, entering a priority number, and pressing **Enter** or clicking outside of the Priority field. Valid priority numbers are integers 1 through 64. Repeat this process for the other trunk groups in the table. The number you enter is the priority of the POTS dial peer in the trunk group for incoming and outgoing calls.
6. To configure translation profiles, perform these actions:
 - a. Configure translation profile options as described in the "Translation Profile Options for POTS Dial Peers" table.
 - b. Add another translation profile if needed.
You can create up to two translation profiles for this endpoint.
 - c. Click **Save Translation Profile**.
 - d. For each translation profile that you create, double-click the dash (-) that displays in **Direction** column in the table of translation rules and select **Incoming** or **Outgoing** from the drop-down list that displays.

The Incoming selection applies the corresponding translation rule to traffic that is incoming to this endpoint. The Outgoing selection applies the corresponding translation rule to traffic that is outgoing from this endpoint.
7. Click **Next**.
8. In **Policy Profile Name**, enter a name for this child policy.
9. In **Policy Profile Description**, enter a description for this child policy.
10. Click **Save**.

Configure SIP Dial Peers for a Voice Policy

When you configure SIP Dial Peers for a voice policy, you configure options that define how the system augments and manipulates calls for the SIP dial peer endpoint type.

You can configure the following options, depending on the policy type for which you are configuring SIP dial peers:

- **Translation Profiles**—Use these options to configure translation rules for called and calling numbers on SIP dial peers. The following table describes these options.

Table 108: Translation Profile Options for Calling Numbers on SIP Dial Peers

Option	Description	Cisco IOS CLI Equivalent
Add New Translation Profile	Click to add a translation profile for the selected SIP dial peer. You can create up to two translation profiles for this endpoint.	voice translation-profile <i>name</i>

Option	Description	Cisco IOS CLI Equivalent
Copy from Existing	Click to copy an existing translation profile to a new translation profile. In the box that appears, change the name if desired, select a called translation rule and a calling translation rule, and click Copy .	—
Calling	Click to configure translation rules for the number that is calling in. The Translation Rules pane displays.	translate calling <i>translation-rule-number</i>
Called	Click to configure translation rules for the number that is being called. The Translation Rules pane displays.	translate called <i>translation-rule-number</i>

Option	Description	Cisco IOS CLI Equivalent
Translation Rules pane		voice translation-rule <i>number</i> Match and Replace Rule: rule <i>precedence</i> <i>/match-pattern/</i> <i>/ replace-pattern/</i> Reject Rule: rule <i>precedence</i> reject <i>/match-pattern/</i>

Option	Description	Cisco IOS CLI Equivalent
	- Click Add New to create a translation rule. Alternatively, you can click Copy From Existing to copy an existing translation rule to a new translation rule. In the box that appears, change the name if desired, select a called translation rule and a calling translation rule, and click Copy. - In the Translation Rule Number field, enter a unique number that designates the precedence for this rule. Valid range: integers 1 through 100. (Optional) To copy existing translation rules from a CSV file, click Import. Continue to add rules or click Finish. For detailed information about this file, see #unique_264. - Click Add Rule. - In the Match field, enter the string that you want the translation rule to affect. Enter the string in regular expression format beginning and ending with a slash (/). For example, <code>/^9/</code>. To include the backslash character (\) in a match string, precede the backslash with a backslash. - From the Action drop-down list, select the action that the system performs for calls that match the string in the Match field. The Reject option causes the system to reject the call. The Replace option causes the system to replace the match number with a value that you specify. - If you select the Replace action, in the Replace field that displays, enter the string to which to translate the matched string. Enter the number in regular expression format beginning and ending with a slash (/). For example, <code>//</code>, which indicates a replacement of no string.	

Option	Description	Cisco IOS CLI Equivalent
	To include the backslash character (\) in a replace string, precede the backslash with a backslash. As an example, if you specify a match string of /^9/ and a replace string of //, the system removes the leading 9 from calls with a number that begins with 9. In this case, the system translates 914085551212 to 14085551212. 8. Click Save. 9. Add more translation rules as needed. 10. (Optional) Click Export to save the translation rules that you created in a CSV file. 11. Click Finish at the bottom of the pane.	

- **Media Profiles**—Use these options to configure codecs to be available for the SIP trunk communication with remote dial peers and DTMF relay options to use for SIP calls. The following table describes these options.

Table 109: Media Profile Options

Option	Description	Cisco IOS CLI Equivalent
Add New Media Profile	Click to add a translation profile for the dial peer.	—
Copy from Existing	Click to copy an existing media profile to a new media profile. In the box that appears, enter a media profile number for the profile, and click Copy .	—
Media Profile Number	Enter a number for this SIP media profile. Valid range: Integers 1 through 10000.	voice class codec tag-number
Codec	Move from the Source list to the Target list the codecs that you want to be made available for the SIP trunk to use when communicating with the remote dial peer. Codecs in the target list are in descending order of priority, with the highest priority at the top of the list. Drag and drop items in this list to rearrange them.	voice class codec tag-number codec preference value <i>codec-type</i>

Option	Description	Cisco IOS CLI Equivalent
DTMF	Move from the Source list to the Target list the DTMF relay options that you want the system to use for SIP calls. Items in the Target list are in descending order of priority, with the highest priority at the top of the list. Drag and drop items in this list to rearrange them. If you want to include the Inband option in the Target list, it can be the only option in that list. If you want to include other options in the Target list, move the Inband option to the Source list before saving the media profile.	dtmf-relay { [[sip-notify] [sip-kpml] [rtp-nte]] }
Save	Click to save the configuration settings that you made.	—

- **Modem Pass-through**—Use these options to configure the modem pass-through feature for a SIP dial peer endpoint. The following table describes these options.

Table 110: Modem Pass-Through Options

Option	Description	Cisco IOS CLI Equivalent
Add New Modem Pass-through	Click to add a modem pass-through for this SIP dial peer endpoint.	—
Copy from Existing	Click to copy an existing modem pass-through to a new modem pass-through profile. In the box that appears, select an existing modem pass-through, enter new name if desired, and click Copy .	—
Name	Name of the modem pass-through. This name is used when you copy an existing modem pass-through profile to a new one.	—

Option	Description	Cisco IOS CLI Equivalent
Protocol	Select the protocol for the modem pass-through: • None—Modem pass-through is disabled on the device • NSE G.711ulaw—Uses named signaling events (NSEs) to communicate G.711ulaw codec switchover between gateways • NSE G.711alaw—Uses named signaling events (NSEs) to communicate G.711alaw codec switchover between gateways	None: no modem passthrough NSE G.711 ulaw: modem passthrough nse codec g711ulaw NSE G.711 alaw: modem passthrough nse codec g711alaw
Save Modem Pass-Through	Click to save the configuration settings that you made.	—

- **Fax Protocol**—Use these options to configure the fax protocol capability for a SIP dial peer endpoint. The following table describes these options.

Table 111: Fax Protocol Options

Option	Description	Cisco IOS CLI Equivalent
Add New Fax Protocol	Click to add a fax protocol for the dial peer.	—
Copy from Existing	Click to copy an existing fax protocol to a new fax protocol. In the box that appears, select an existing fax protocol, enter new name if desired, and click Copy .	—
Name	Name of the fax protocol. This name is used when you copy an existing fax profile to a new fax profile.	—

Option	Description	Cisco IOS CLI Equivalent
Primary	Select from a set of fax protocol options. Each option is a bundled set of related fax commands. For a detailed description of each bundle, see the “Primary Fax Protocol Command Bundles” table The descriptions of the bundles include the following components: • nse—Uses NSEs to switch to T.38 fax relay mode • force—Unconditionally uses Cisco Network Services Engines (NSE) to switch to T.38 fax relay • version—Specifies a version for configuring fax speed: • 0—Configures version 0, which uses T.38 version 0 (1998–G3 faxing) • 3—Configures version 3, which uses T.38 version 3 (2004–V.34 or SG3 faxing) • none—No fax pass-through or T.38 fax relay is attempted • Pass-through—The fax stream uses one of the following high-bandwidth codecs: • g711ulaw—Uses the G.711 ulaw codec • g711alaw—Uses the G.711 alaw codec	<pre>fax protocol { none pass-through {g711ulaw g711alaw} [fallback none] t38 [nse [force]] [version {0 3}] [ls-redundancy value hs-redundancy value]] [fallback {none pass-through {g711ulaw g711alaw}}]}</pre>
Fallback	Available when the primary protocol bundle name that you selected in the Primary field begins with “T.38” or with “Fax Pass-through.” Select the fallback mode for fax transmissions. This fallback mode is used if the primary fax protocol cannot be negotiated between device endpoints. For a detailed description of each option, see the “Fallback Protocol Options” table.	<pre>fax protocol {none pass-through {g711ulaw g711alaw} [fallback none] t38 [nse [force]] [version {0 3}] [ls-redundancy value hs-redundancy value]] [fallback {none pass-through {g711ulaw g711alaw}}]}</pre>

Option	Description	Cisco IOS CLI Equivalent
Low Speed	Available when the primary protocol bundle name that you selected in the Primary field begins with "T.38." Specifies the number of redundant T.38 fax packets to be sent for the low-speed V.21-based T.30 fax machine protocol. Range: varies from 0 (no redundancy) to 5. Default: 0.	ls-redundancy <i>value</i>
High Speed	Available when the primary protocol bundle name that you selected in the Primary field begins with "T.38." Specifies the number of redundant T.38 fax packets to be sent for high-speed V.17, V.27, and V.29 T.4 or T.6 fax machine image data. Range: varies from 0 (no redundancy) to 2. Default: 0	hs-redundancy <i>value</i>
Save Fax Protocol	Click to save the configuration settings that you made.	—

The following table describes the bundled sets of fax commands that are available for the Primary option when you configure the fax protocol capability for a SIP dial peer endpoint.

For low speed (ls) redundancy, the range varies from 0 (no redundancy) to 5. For high speed (HS redundancy, the range varies from 0 (no redundancy) to 2.

Table 112: Primary Fax Protocol Command Bundles

Fax Command Protocol Bundle	Description	Cisco IOS CLI Equivalent
T.38 Fax Relay Version 3	Primary fax protocol is T.38 fax relay version 3. Options for selecting the low-speed and high-speed redundancy values are available.	fax protocol t38 version 3 ls-redundancy <i>value</i> hs-redundancy <i>value</i> no fax-relay sg3-to-g3
T.38 Fax Relay Version 0	Primary fax protocol is T.38 fax relay version 0. Options for selecting the low-speed and high-speed redundancy values are available.	fax protocol t38 version 0 ls-redundancy <i>value</i> hs-redundancy <i>value</i>

Fax Command Protocol Bundle	Description	Cisco IOS CLI Equivalent
T.38 Fax Relay Version 3 NSE	Primary fax protocol is NSE based T.38 fax relay version 3. Options for selecting the low-speed and high-speed redundancy values are available.	fax protocol t38 version 3 nse ls-redundancy <i>value</i> hs-redundancy <i>value</i> no fax-relay sg3-to-g3
T.38 Fax Relay Version 3 NSE force	Primary fax protocol is NSE force option of T.38 fax relay version 3. Options for selecting the low-speed and high-speed redundancy values are available.	fax protocol t38 version 3 nse force ls-redundancy <i>value</i> hs-redundancy <i>value</i> no fax-relay sg3-to-g3
T.38 Fax Relay Version 0 NSE	Primary fax protocol is NSE option of T.38 fax relay version 0. Options for selecting the low-speed and high-speed redundancy values are available.	fax protocol t38 version 0 nse ls-redundancy <i>value</i> hs-redundancy <i>value</i>
T.38 Fax Relay Version 0 NSE force	Primary fax protocol is NSE force option of T.38 fax relay version 0. Options for selecting the low-speed and high-speed redundancy values are available.	fax protocol t38 version 0 nse force ls-redundancy <i>value</i> hs-redundancy <i>value</i>
T.38 Fax Relay Version 0 No ECM	Primary fax protocol is T.38 fax relay version 0 with ECM disabled. Options for selecting the low-speed and high-speed redundancy values are available.	fax protocol t38 version 0 ls-redundancy <i>value</i> hs-redundancy <i>value</i> fax-relay ecm disable
T.38 Fax Relay Version 0 NSE No ECM	Primary fax protocol is NSE based T.38 fax relay version 0 with ECM disabled. Options for selecting the low-speed and high-speed redundancy values are available.	fax protocol t38 version 0 nse ls-redundancy <i>value</i> hs-redundancy <i>value</i> fax-relay ecm disable
T.38 Fax Relay Version 0 NSE force No ECM	Primary fax protocol is NSE force option T.38 fax relay version 0 with ECM disabled. Options for selecting the low-speed and high-speed redundancy values are available.	fax protocol t38 version 0 nse force ls-redundancy <i>value</i> hs-redundancy <i>value</i> fax-relay ecm disable

Fax Command Protocol Bundle	Description	Cisco IOS CLI Equivalent
T.38 Fax Relay Version 0 Rate 14.4 No ECM	Primary fax protocol is T.38 fax relay version 0 with ECM disabled and fax rate of 14,400 bps. Options for selecting the low-speed and high-speed redundancy values are available.	fax protocol t38 version 0 ls-redundancy value hs-redundancy value fax-relay ecm disable fax rate 14400
T.38 Fax Relay Version 0 NSE Rate 14.4 No ECM	Primary fax protocol is NSE based T.38 fax relay version 0 with ECM disabled and fax rate of 14,400 bps. Options for selecting the low-speed and high-speed redundancy values are available.	fax protocol t38 version 0 nse ls-redundancy value hs-redundancy value fax-relay ecm disable fax rate 14400
T.38 Fax Relay Version 0 NSE force Rate 14.4 No ECM	Primary fax protocol is NSE force option T.38 fax relay version 0 with ECM disabled and fax rate of 14,400 bps. Options for selecting the low-speed and high-speed redundancy values are available.	fax protocol t38 version 0 nse force ls-redundancy value hs-redundancy value fax-relay ecm disable fax rate 14400
T.38 Fax Relay Version 0 Rate 9.6 No ECM	Primary fax protocol is T.38 fax relay version 0 with ECM disabled and fax rate of 9,600 bps Options for selecting the low-speed and high-speed redundancy values are available.	fax protocol t38 version 0 ls-redundancy value hs-redundancy value fax-relay ecm disable fax rate 9600
T.38 Fax Relay Version 0 NSE Rate 9.6 No ECM	Primary fax protocol is NSE based T.38 fax relay version 0 with ECM disabled and fax rate of 9,600 bps. Options for selecting the low-speed and high-speed redundancy values are available.	fax protocol t38 version 0 nse ls-redundancy value hs-redundancy value fax-relay ecm disable fax rate 9600
T.38 Fax Relay Version 0 NSE force Rate 9.6 No ECM	Primary fax protocol is NSE force option T.38 fax relay version 0 with ECM disabled and fax rate of 9,600 bps. Options for selecting the low-speed and high-speed redundancy values are available.	fax protocol t38 version 0 nse force ls-redundancy value hs-redundancy value fax-relay ecm disable fax rate 9600

Fax Command Protocol Bundle	Description	Cisco IOS CLI Equivalent
T.38 Fax Relay Version 0 Rate 14.4	Primary fax protocol is T.38 fax relay version 0 with ECM and fax rate of 14,400 bps. Options for selecting the low-speed and high-speed redundancy values are available.	fax protocol t38 version 0 ls-redundancy value hs-redundancy value fax rate 14400
T.38 Fax Relay Version 0 NSE Rate 14.4	Primary fax protocol is NSE based T.38 fax relay version 0 with ECM and fax rate of 14,400 bps. Options for selecting the low-speed and high-speed redundancy values are available.	fax protocol t38 version 0 nse ls-redundancy value hs-redundancy value fax rate 14400
T.38 Fax Relay Version 0 NSE force Rate 14.4	Primary fax protocol is NSE force option T.38 fax relay version 0 with ECM and fax rate of 14,400 bps. Options for selecting the low-speed and high-speed redundancy values are available.	fax protocol t38 version 0 nse force ls-redundancy value hs-redundancy value fax rate 14400
T.38 Fax Relay Version 0 Rate 9.6	Primary fax protocol is T.38 fax relay version 0 with ECM and fax rate of 9,600 bps. Options for selecting the low-speed and high-speed redundancy values are available.	fax protocol t38 version 0 ls-redundancy value hs-redundancy value fax rate 9600
T.38 Fax Relay Version 0 NSE Rate 9.6	Primary fax protocol is NSE based T.38 fax relay version 0 with ECM and fax rate of 9,600 bps. Options for selecting the low-speed and high-speed redundancy values are available.	fax protocol t38 version 0 nse ls-redundancy value hs-redundancy value fax rate 9600
T.38 Fax Relay Version 0 NSE force Rate 9.6	Primary fax protocol is NSE force option T.38 fax relay version 0 with ECM and fax rate of 9,600 bps. Options for selecting the low-speed and high-speed redundancy values are available.	fax protocol t38 version 0 nse force ls-redundancy value hs-redundancy value fax rate 9600
None	Fax protocol is disabled.	fax protocol none
Fax Pass-through G711ulaw	Primary fax protocol is fax pass-through with pass-through codec set to g711ulaw.	fax protocol pass-through g711ulaw

Fax Command Protocol Bundle	Description	Cisco IOS CLI Equivalent
Fax Pass-through G711ulaw No ECM	Primary fax protocol is fax pass-through with pass-through codec set to g711ulaw and ECM disabled.	fax protocol pass-through g711ulaw fax-relay ecm disable
Fax Pass-through G711alaw	Primary fax protocol is fax pass-through with pass-through codec set to g711alaw.	fax protocol pass-through g711alaw
Fax Pass-through G711alaw No ECM	Primary fax protocol is fax pass-through with pass-through codec set to g711alaw and ECM disabled.	fax protocol pass-through g711alaw fax-relay ecm disable

The following table describes the selections that are available for the Fallback option when you configure the fax protocol capability for a SIP dial peer endpoint.

Table 113: Fallback Protocol Options

Fallback Fax Protocol Options	Description	Cisco IOS CLI Equivalent
None	Fallback Fax Protocol is None. All special fax handling is disabled.	fax protocol t38 [nse [force]] [version {0 3}] [ls-redundancy value [hs-redundancy value]] fallback none fax protocol pass-through {g711ulaw g711alaw } fallback none
Fax Pass-through G711ulaw	Fallback Fax Protocol is Fax Pass-through with pass-through codec set to g711ulaw.	fax protocol t38 [nse [force]] [version {0 3}] [ls-redundancy value [hs-redundancy value]] fallback pass-through g711ulaw
Fax Pass-through G711alaw	Fallback Fax Protocol is Fax Pass-through with pass-through codec set to g711alaw.	fax protocol t38 [nse [force]] [version {0 3}] [ls-redundancy value [hs-redundancy value]] fallback pass-through g711alaw

To configure SIP dial peers for a voice policy:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Unified Communications**.
2. Click **SIP Dial Peer**.
3. From the **Add SIP Dial Peer Policy Profile** drop-down list, choose **Create New**.

Alternatively, you can select **Copy from Existing** to copy an existing SIP dial peer policy to a new one. In the box that appears, select the name of the policy profile to copy, enter a new name for the profile if desired, and click **Copy**.

4. Select the policy types that you want to create and click **Next**:

- **Translation Profile**—Lets you configure translation rules for calling and called numbers.
 - **Media Profile**—Lets you configure codecs to be available for the SIPtrunk communication with remote dial peers and DTMF relay options to use for SIP calls.
 - **Modem Pass-through**—Lets you configure the modem pass-through feature for a SIP dial peer endpoint.
 - **Fax Protocol**—Lets you lets you configure the fax protocol capability for a SIP dial peer endpoint. This capability is advertised and used when negotiating capabilities with the remote dial peer.
5. In the page that displays, configure options in the tabs that the following tables describe as needed.
- The tabs that are available depend on the policy types that you selected.
- **Translation Profile** options—For a description of these options, see the "Translation Profile Options for Calling Numbers on SIP Dial Peers" table.
- After you click **Finish** when configuring a translation profile, perform these actions:
- a. Add another translation profile if needed. You can create up to two translation profiles for this endpoint.
 - b. Click **Save Translation Profile**.
 - c. For each translation profile that you create, double-click the dash (-) that displays in **Direction** column in the table of translation rules and select **Incoming** or **Outgoing** from the drop-down list that displays. The Incoming selection applies the corresponding translation rule to traffic that is incoming to this endpoint. The Outgoing selection applies the corresponding translation rule to traffic that is outgoing from this endpoint.
- **Media Profile** options—For a description of these options, see the "Media Profile Options" table.
 - **Modem Pass-through** options—For a description of these options, see the "Modem Pass-Through Options" table.
 - **Fax Protocol** options—For a description of these options, see the "Fax Protocol Options" table.
6. Click **Next**.
7. In **Policy Profile Name**, enter a name for this child policy.
8. In **Policy Profile Description**, enter a description for this child policy.
9. Click **Save**.

Configure SRST Phones for a Voice Policy

When you configure SRST Phones for a voice policy, you configure options that define how the system augments and manipulates calls for the Cisco Unified SRST phone endpoint type.

The following table describes options for configuring SRST phones for a voice policy.

Table 114: SRST Phones Configuration Options

Option	Description	Cisco IOS CLI Equivalent
Media Profile Number	Enter a number for this Cisco Unified SRST media profile. Valid range: Integers 1 through 10000.	voice class codec tag-number
Codec	Move from the Source list to the Target list the codecs that you want to be available for phones when they are in Cisco Unified SRST mode and communicating with other phones that are in the same site and registered to the same gateway. Codecs in the target list are in descending order of priority, with the highest priority at the top of the list. Drag and drop items in this list to rearrange them.	voice class codec tag-number codec preference value codec-type
DTMF field	Move from the source list to the target list the DTMF relay options that you want the system to use when in Cisco Unified SRST mode. Items in the target list are in descending order of priority, with the highest priority at the top of the list. Drag and drop items in this list to rearrange them. If you want to include the Inband option in the Target list, it can be the only option in that list. If you want to include other options in the Target list, move the Inband option to the Source list before saving the media profile.	dtmf-relay { [[sip-notify] [sip-kpml] [rtp-nte]] }
Save	Click to save the configuration settings that you made.	—

To configure SRST phones for a voice policy, follow these steps:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Unified Communications**
2. Click **Add Voice Policy**, and choose **SRST Phone**.
3. From the **Add SRST Phone Policy Profile** drop-down list, select **Create New**.

Alternatively, you can select **Copy from Existing** to copy an existing policy to a new one. In the box that appears, select the name of the policy profile to copy, enter a new name for the profile if desired, and click **Copy**.
4. Click **Media Profile**, and click **Next**.
5. Click **Add New Media Profile**.

6. In the page that displays, configure options as described in the "SRST Phones Configuration Options" table.
7. Click **Next**.
8. In **Policy Profile Name**, enter a name for this child policy.
9. In **Policy Profile Description**, enter a description for this child policy.
10. Click **Save**.

Provision a Device Template for Unified Communications

When you provision a device template for Unified Communications, you select UC-specific feature templates and set up the voice policy to include with the device template.

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and click **Create Template**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is called **Device**.

3. From the **Create Template** drop-down list, select **From Feature Template**.
4. From the **Device Model** drop-down list, select the type of supported device to which you want to attach the UC-specific feature templates and map the voice policy.
5. Click **Unified Communications**.
6. To select UC-specific feature templates to include with the device template, perform these actions:
 - a. From the **Voice Card** drop-down list, select the voice card feature template that you want to attach to the device.
 - b. From the **Call Routing** drop-down list, select the call routing feature template that you want to attach to the device.
 - c. From the **SRST** drop-down list, select the SRST feature template that you want to attach to the device.
 - d. From the **DSPFarm** drop-down list, select the DSPFarm template that you want to attach to the device.
7. To set up the voice policy to include with the device template, perform these actions:
 - a. From the **Voice Policy** drop-down list, select the voice policy that you want to map to endpoints.
 - b. Click **Mapping**.
 - c. From the list of endpoint types in the left pane of the screen that displays, select the type of endpoint that contains the subpolicies that you want to map to specific endpoints.
 - d. From the list of subpolicies that displays, click ..., and click **Mapping** for the subpolicy that you want to map to specific endpoints.
 - e. In the list of endpoints that displays, select each endpoint to which you want to map the subpolicy.

f. Click **Map**.

g. Click **Save**.

8. To create the device template, click **Create**.

When you map subpolicies to endpoints, the system generates the CLI commands that the following table shows.

Table 115: Generated CLI Commands for Subpolicies to Endpoints Mapping

Endpoint	Subpolicy	Cisco IOS CLI Application Mapping	Remarks
Voice Port FXO Voice Port FXS Voice Port FXS DID Voice Port PRI ISDN POTS Dial Peer SIP Dial Peer	Translation profile	translation-profile incoming <i>profile-name</i> translation-profile outgoing <i>profile-name</i>	A translation profile policy is applied to a dial peer or a voice profile.
SRST Phone SIP Dial Peer	Media profile	voice register pool <i>number</i> voice-class codec <i>number</i> dtmf-relay {[sip-notify] [sip-kpml] [rtp-nte]}	A media profile policy includes voice class codec and DTMF relay configurations. This policy is applied to an incoming SIP dial peer, an outgoing SIP dial peer, or an SRST phone profile.
Voice Port FXO	Supervisory disconnect	voice port <i>number</i> supervisory custom-cptone <i>cptone-name</i> supervisory dualtone-detect=params <i>tag</i>	A supervisory disconnect policy such as custom-cptone or dualtone-detect-params is applied to FXO voice interfaces.

Endpoint	Subpolicy	Cisco IOS CLI Application Mapping	Remarks
Voice Port FXO Voice Port FXS Voice Port FXS DID Voice Port PRI ISDN POTS Dial Peer	Trunk group	trunk-group name <i>[preference-num]</i> voice-port number <i>trunk-group name</i> <i>[preference-num]</i> interface serial <i>slot/sub-slot/port: {15 23}</i> dial-peer voice tag pots trunkgroup name <i>preference-num</i>	If more than one interface is assigned to the same trunk group, the <i>preference-num</i> value determines the order in which the trunk group uses the interfaces. A preference-num value of 1 is the highest preference, so an interface with that value is used first. A value of 64 is the lowest preference so an interface with that value is used last.
SIP Dial Peer	Modem pass-through	None: no modem passthrough G.711 ulaw: modem passthrough nse codec g711ulaw G.711 alaw: modem passthrough nse codec g711alaw	—
SIP Dial Peer	Fax protocol	fax protocol {none pass-through {g711ulaw g711alaw} [fallback none] t38 [nse [force]] [version {0 3}] [ls-redundancy value [hs-redundancy value]] [fallback {none pass-through {g711ulaw g711alaw} }]}	—

Dial Peer CSV File

A dial peer CSV file includes information for one or more incoming and outgoing SIP and POTS dial peers. The file must be comma delimited, and each record in the file must include each field that the following table describes, in the order shown.

Table 116: Dial Peer CSV Files Fields

Field	Description
Dial Peer Tag	Number that is used to reference the dial peer.
Dial Peer Type	Type of dial peer that you are creating (pots or voip).
Direction	Direction of traffic on the dial peer (Incoming or Outgoing).
Description	Description of the dial peer.
Forward Digits	How the dial peer transmits digits in outgoing numbers: • All—The dial peer transmits all digits in the number. • None—The dial peer does not transmit digits in the number that do not match the destination pattern. • n—The dial peer transmits the number of right-most digits in the number that the integer <i>n</i> represents. For example, if <i>n</i> is 7 and the outgoing number is 1112223333, the dial peer transmits 2223333.
Preference	For POTS dial peers, a unique numeric value for the dial peer. If dial peers have the same match criteria, the system uses the one with the highest preference value.
Prefix	Digits to be prepended to outgoing POTS dial peer calls.
Numbering Pattern	String that the router uses to match incoming calls to the dial peer.
Dest. Address	Network address of the remote voice gateway to which calls are sent after a local outgoing SIP dial peer is matched.
Voice Port	Voice port that the router uses to match calls to the dial peer. For an outgoing dial peer, the router sends the calls that match the dial peer to this port. For an incoming dial peer, this port serves as an additional match criterion. The dial peer is matched only if a call comes in on this port.

Field	Description
Transport Protocol	For SIP dial peers, transport protocol (TCP or UDP) for SIP control signaling.

Example dial peer CSV file:

```
Tag,type,Direction,Description,Forward Digits,Preference,Prefix,Pattern,Dest. Address,Voice
Port,Transport
6545,voip,Outgoing,description To Voice Gateway,,1,,23456,ipv4:166.2.121.17,,udp
6756,voip,Outgoing,description ***Fax Number 6362-6362***,,0,,34567,ipv4:166.2.121.16,,tcp
768,voip,Outgoing, description Fire Alarm Dialer,,8,,5678,ipv4:166.2.121.19,,udp
10,pots,Incoming,,,5,,0115T,,1/0/1,
54,pots,Outgoing,,,6,,.T,,1/0/3,
23,pots,Incoming,,all,0,,76...,,1/0/4,
26,pots,Incoming,,5,1,55,9800.....,,1/0/5,
27,pots,Incoming,,5,1,55,9800.....,,0/1/5:15,
```

Translation Rules CSV File

When you configure translation rules for a translation profile, POTS dial peer, or SIP dial peer you can either create new translation rules or import existing translation rule information from a CSV file.

The file must be comma delimited, and each record in the file must include each field that the following table describes, in the order shown:

Table 117: Translation Rules CSV Files Fields

Field	Description
Match	String that you want the translation rule to affect. The string must be in regular expression format beginning and ending with a slash (/). For example, /^9/.
Action	Action that the system performs for calls that match the string in the Match field. Valid values are: • reject—Causes the system to reject the call • replace—Causes the system to replace the match string with the value in the Replace field
Replace	If the Action field contains replace , this field contains the string to which to translate the matched string. Enter the number in regular expression format beginning and ending with a slash (/). For example, //, which indicates a replacement of no string. As an example, if you specify a match string of /^9/ and a replace string of //, the system removes the leading 9 from calls with a number that begins with 9. In this case, the system translates 914085551212 to 14085551212.

Example translation rules CSV file:

```
Match,Action,Replace
/34/,replace,/34/
/23/,reject,
/56/,replace,/100/
/16083652563/,replace,/6083652563/
```

Monitoring UC Operations

After you enable UC voice services for supported routers, you can monitor the real-time statuses of lines, calls, interfaces, and related items that a device processes.

To monitor UC operations:

1. From the Cisco SD-WAN Manager menu, choose **Monitor** > **Devices**.

Cisco vManage Release 20.6.x and earlier: From the Cisco SD-WAN Manager menu, choose **Monitor** > **Network**.

2. In the table of devices, select the device for which you want to monitor UC operations.
3. From **Security Monitoring**, click **Real Time**.
4. In **Device Options**, select one of these options:
 - **Voice Calls**—Displays information for active voice calls. See the "Voice Call Monitoring Information" table.
 - **Voice VOIP Calls**—Displays information for active VOIP calls. See the "Voice VoIP Calls Monitoring Information" table.
 - **Voice Phone Info**—Displays information about Cisco Unified SRST registrations. See the "Voice Phone Info Monitoring Information" table.
 - **Voice Controller T1 E1 Current 15 mins Stats**—Displays configuration and status information for the T1/E1 voice module that is installed in the device, compiled over the past 15 minutes. See the "Voice Controller T1 E1 Current 15 Mins Stats Monitoring Information" table.
 - **Voice Controller T1 E1 Total Stats**—Displays configuration and status information for the T1/E1 voice module that is installed in the device, compiled since the module last started. See the "Voice Controller T1 E1 Total Stats" table.
 - **Voice ISDN Status**—Displays information about Layer 1 and Layer 2 status for the ISDN controller, and information about active calls. "See the Voice ISDN Status Information table".
 - **Voice DSPFarm SCCP CUCM Groups**—Displays detailed information about Cisco Unified Communications Manager groups that are configured for DSP farm services on a device. See the "Voice DSPFarm SCCP CUCM Groups" table.
 - **Voice DSPFarm Profile**—Displays detailed information about DSP farm service profiles and media resources that are configured on the device. See the "Voice DSPFarm Profile Monitoring Information" table.
 - **Voice DSP Farm SCCP Connections**—Displays detailed information about SCCP connections between the device and Cisco Unified Communications Manager. See the "Voice DSPFarm SCCP Connections" table.

- **Voice DSPFarm Active**—Displays operational and status information about DSP farm resources that are active on the device. See the "Voice DSPFarm Active" table.

You also can monitor operations that include UC operations by selecting the following options:

- **Interface Detail**—Displays status and statistical information for interfaces that are configured for the router.
- **Interface Statistics**—Displays statistical information for interfaces that are configured for the router
- **Interface T1/E1**—Displays information for the T1/E1 voice module that is installed in the device

The following table describes the information that you see when you monitor voice calls.

Table 118: Voice Calls Monitoring Information

Field	Description
Call ID	System assigned identifier of a telephony call leg
Voice Port	Voice port used for the call
Codec	Negotiated codec used for the call
VAD	Indicates whether VAD is enabled or disabled for the call
DSP Cannel	DSP channel used for the call
DSP Type	Type of DSP used for the call
Aborted Packets	Number of packets aborted during the call
TX Packets	Number of packets transmitted during the call
RX Packets	Number of packets received during the call
Last Updated	Date and time when the information on this page was last updated

The following table describes the information that you see when you monitor voice VoIP calls.

Table 119: Voice VoIP Calls Monitoring Information

Field	Description
Call ID	System assigned identifier of an RTP connection for a call leg
Codec	Negotiated codec used for the call
Destination Address	IP address of the destination of the call
Destination Port	RTP port of the destination of the call
TX Packets	Number of packets transmitted during the call
RX Packets	Number of packets received during the call

Field	Description
Duration (ms)	Duration of the call, in milliseconds
Last Updated	Date and time when the information on this page was last updated

The following table describes the information that you see when you monitor voice phone information.

Table 120: Voice Phone Info Monitoring Information

Field	Description
Pool Tag	Tag number that is assigned to the Cisco Unified SRST phone pool on the device
ID Network	Identifier of the network subnet that the device uses to register phones that fallback from Cisco Unified Communications Manager to this device
Registration State	Indicates whether phones that are in Cisco Unified SRST mode are registered to this device
Dialpeer Tag	System assigned tag used by the dial peer that is assigned to the directory number of phones that are in Cisco Unified SRST mode and are registered to this device
Address	IP address of the device interface that is used for SIP SRST call control when phones fail over
Directory Number	Directory number of each phone that is in Cisco Unified SRST mode
Last Updated	Date and time when the information on this page was last updated

The following table describes the information that you see when you monitor voice controller T1/E1 information for the past 15 minutes.

Table 121: Voice Controller T1 E1 Current 15 Mins Stats Monitoring Information

Field	Description
Interface-slot-num	Slot number of the controller.
Interface-subslot-num	Subslot number of the controller.
Interface-port-num	Port number of the controller.
Status	Status of the controller.
Type	Type of the controller.
Clock Source	Clock source used for the controller.
Line Code Violations	Number line code violations that have occurred.
Path Code Violations	Number path code violations that have occurred.

Field	Description
Slip Seconds	Number of slip seconds that have occurred. A slip can occur when there is a difference between the timing of a synchronous receiving terminal and the received signal.
Frame Loss Seconds	Number of seconds in which out of frame (OOF) errors have occurred.
Line Err. seconds	Number of seconds in which Line Error Seconds (LES) have occurred. A LES is a second in which one or more Line Code Violation errors are detected.
Degraded Minutes	Number of Degraded Minutes that have occurred. A Degraded Minute is one in which the estimated error rate exceeds 1E-6 but does not exceed 1E-3.
Errored Seconds	Number of Errored Seconds that have occurred.
Bursty Errored Seconds	Number of Bursty Error Seconds that have occurred. A Bursty Error Second is a second with less than 320 and more than 1 path coding violation errors, no severely errored frame defects, and no detected incoming AIS defects.
Severely Errored Seconds	Number of Severely Errored Seconds that have occurred.
Unavailable Seconds	Number of Unavailable Seconds that have occurred. This value is calculated by counting the number of seconds that the interface is unavailable.
Last Updated	Date and time when the information on this page was last updated.

The following table describes the information that you see when you monitor voice controller T1/E1 information over the period since a device last started.

Table 122: Voice Controller T1 E1 Total Stats

Field	Description
Interface-slot-num	Slot number of the controller.
Interface-subslot-num	Subslot number of the controller.
Interface-port-num	Port number of the controller.
Status	Status of the controller.
Type	Type of the controller.
Clock Source	Clock source used for the controller.
Line Code Violations	Number line code violations that have occurred.
Path Code Violations	Number path code violations that have occurred.

Field	Description
Slip Seconds	Number of slip seconds that have occurred. A slip can occur when there is a difference between the timing of a synchronous receiving terminal and the received signal.
Frame Loss Seconds	Number of seconds in which out of frame (OOF) errors have occurred
Line Err. seconds	Number of seconds in which Line Errored Seconds (LES) have occurred. A LES is a second in which one or more Line Code Violation errors are detected.
Degraded Minutes	Number of Degraded Minutes that have occurred. A Degraded Minute is one in which the estimated error rate exceeds 1E-6 but does not exceed 1E-3.
Errored Seconds	Number of Errored Seconds that have occurred.
Bursty Errored Seconds	Number of Bursty Error Seconds that have occurred. A Bursty Error Second is a second with less than 320 and more than 1 path coding violation errors, no severely errored frame defects, and no detected incoming AIS defects.
Severely Errored Seconds	Number of Severely Errored Seconds that have occurred.
Unavailable Seconds	Number of Unavailable Seconds that have occurred. This value is calculated by counting the number of seconds that the interface is unavailable.
Last Updated	Date and time when the information on this page was last updated.

The following table describes the information that you see when you monitor voice ISDN status.

Table 123: Voice ISDN Status Information

Field	Description
Key ID	Identifier of the table row
Interface	Name of the PRI ISDN digital interface
Switch Type	Switch type used for the PRI ISDN digital interface
Layer 1 Status	Layer 1 status of the PRI ISDN digital interface
Layer 2 Status	Layer 2 status of the PRI ISDN digital interface
Active Calls	Number of active calls on the PRI ISDN digital interface
Last Updated	Date and time when the information on this page was last updated

The following table describes the information that you see when you monitor Cisco Unified Communications Manager groups that are configured for DSP farm services on a device.

Table 124: Voice DSPFarm SCCP CUCM Groups Monitoring Information

Field	Description
CUCM Group ID	Identifier of the Cisco Unified Communications Manager group
Description	Description of the Cisco Unified Communications Manager group
Switchover Method	Method that the primary Cisco Unified Communications Manager server in this Cisco Unified Communications Manager group uses for failover
Switchback Method	Method that the secondary Cisco Unified Communications Manager server in this Cisco Unified Communications Manager group uses to switch back after a failover
CUCM ID	Identifier of each Cisco Unified Communications Manager server in the Cisco Unified Communications Manager group
CUCM Priority	Priority in which the Cisco Unified Communications Manager servers in this Cisco Unified Communications Manager group are used
Profile ID	Identifier of the DSP farm profile that is registered to each Cisco Unified Communications Manager server in the Cisco Unified Communications Manager group
Reg. Name	Name of the DSP farm profile that is registered to each Cisco Unified Communications Manager server in the Cisco Unified Communications Manager group
Last Updated	Date and time when the information on this page was last updated

The following table describes the information that you see when you monitor DSP farm service profiles and media resources that are configured on a device.

Table 125: Voice DSPFarm Profile Monitoring Information

Field	Description
Profile ID	Identifier of the DSP farm profile.
Service ID	Type of DSP farm service that is configured for this DSP farm profile.
Service Mode	Service mode for this DSP farm profile.
Resource ID	Resource identifier for the DSP resource group in this DSP farm profile.
Admin	Status of this DSP farm profile. If this field displays DOWN, ensure that the Shutdown option is not enabled in the Profile tab of the DSPFarm feature template that defines this DSP farm.

Field	Description
Operation	Status of the registration of the profile with Cisco Unified Communications Manager: • ACTIVE IN PROGRESS—Profile is in the process of registering with Cisco Unified Communications Manager • DOWN—Profile is unable to register with Cisco Unified Communications Manager • ACTIVE—Profile is registered with Cisco Unified Communications Manager
App. Type	Type of application with which the DSP farm services that are provisioned on the device are associated.
App. Status	Status of the association of this profile with Cisco Unified Communications Manager: • app-assoc-done—Profile is associated with Cisco Unified Communications Manager • app-assoc-not-done—Profile is not associated with Cisco Unified Communications Manager
Resource Provider	Information about the mediaresource family that relates to the profile.
Provider Status	Status of the media resources that relate to the profile.
Last Updated	Date and time when the information on this page was last updated.

The following table describes the information that you see when you monitor SCCP connections between a device and Cisco Unified Communications Manager.

Table 126: Voice DSPFarm SCCP Connections

Field	Description
Connection ID	Identifier of an SCCP connection for an active call that uses this DSP farm service
Session ID	Identifier of an SCCP session for an active call that uses this DSP farm service
Session Type	Type of DSP farm service for this SCCP connection
Mode	Mode for direction of traffic for this SCCP connection
Codec	Codec provisioned for this SCCP connection
Remote IP	IP address of the remote endpoint for this SCCP connection
Remote Port	Port number of the remote endpoint for this SCCP connection

Field	Description
Source Port	Port number of the local endpoint for this SCCP connection
Last Updated	Date and time when the information on this page was last updated

The following table describes the information that you see when you monitor DSP farm resources that are active on a device.

Table 127: Voice DSPFarm Active Monitoring Information

Field	Description
DSP	Identifier of the DSP for an active call that uses this DSP farm service
Status	Status of the DSP for an active call that uses this DSP farm service
Resource ID	Resource Identifier that is associated with the DSP that this connection uses
Bridge ID	Bridge Identifier that is associated with the DSP that this connection uses
Transmit Packets	Number of packets that this connection has transmitted
Received Packets	Number of packets that this connection has received
Last Updated	Date and time when the information on this page was last updated

Cisco Unified Communications FXS and FXO Caller ID Support

Table 128: Feature History

Feature Name	Release Information	Description
Cisco Unified Communications FXS and FXO Caller ID Support	Cisco IOS XE Catalyst SD-WAN Release 17.8.1a Cisco vManage Release 20.8.1	This feature lets you configure Foreign Exchange Station (FXS) and Foreign Exchange Office (FXO) caller ID features by using Cisco SD-WAN Manager CLI add-on feature templates.

Information About Adding Voice Features with a CLI Add-On Feature Template

You can configure FXS and FXO caller ID features in Cisco IOS XE Catalyst SD-WAN devices by using CLI add-on feature templates. For detailed information about CLI add-on feature templates, see [CLI Add-On-Feature Templates](#).

Caller ID is an analog service by which a telephone central office switch sends digital information about an incoming call. The Caller ID feature for analog FXS ports is configurable on a per-port basis to phones that

are connected to analog FXS voice ports. Caller ID also is available on analog FXO ports. Caller ID-related features are based on the identity of the calling party.

If an FXS voice port has caller-id commands configured, remove all the caller-id configurations before changing the signaling type from loop-start or ground-start to Direct Inward Dialing (DID).

If you remove a voice port from a device after a caller ID command is configured, remove the caller ID configuration from the device. Otherwise, a voice port configuration mismatch occurs between the Cisco IOS configuration and the Cisco Catalyst SD-WAN configuration.

Supported Devices for Adding Voice Features with a CLI Add-On Feature Template

- Cisco NIM-2FXO network interface module
- Cisco NIM-4FXO network interface module
- Cisco NIM-2FXSP network interface module
- Cisco NIM-4FXSP network interface module
- Cisco NIM-2FXS/4FXOP network interface module
- Cisco SM-X-72FXS double-wide service module
- Cisco SM-X-24FXS/4FXO single-wide service module
- Cisco SM-X-16FXS/2FXO single-wide service module
- Cisco SM-X-8FXS/12FXO single-wide service module

Restrictions for Adding Voice Features with a CLI Add-On Feature Template

- Caller ID must be enabled with the **caller-id enable** command before you use any of the **caller-id** commands.
- When the **caller-id alerting dsp-pre-allocate** command is used or disabled, the FXO voice port is automatically shut down and then brought up to allocate or deallocate the DSP voice channel if the FXO port is in the Idle state.

Configure Voice Features with a CLI Add-On Feature Template

The following commands provide configuration options for caller ID features:

- **caller-id alerting dsp-pre-allocate**: Statically allocates a digital signal processor (DSP) voice channel for receiving caller ID information for an on-hook (Type 1) caller ID at a receiving FXO voice port.
- **caller-id alerting line-reversal**: Sets the line-reversal alerting method for caller ID information for an on-hook (Type 1) caller ID at a sending FXS voice port and for an on-hook caller ID at a receiving FXO voice port.
- **caller-id alerting pre-ring**: Sets a 250-millisecond pre-ring alerting method for caller ID information for an on-hook (Type 1) caller ID at a sending FXS and a receiving FXO voice port.

- **caller-id alerting ring:** Sets the ring-cycle method for receiving caller ID information for an on-hook (Type 1) caller ID at a receiving FXO or a sending FXS voice port.
- **caller-id block:** Requests blocking of caller ID information display at the far end of a call that originates from an FXS port.
- **caller-id format e911:** Specifies the caller ID message type that should be the enhanced 911 format for calls that are sent on FXS voice ports.
- **caller-id mode:** Specifies a noncountry, standard caller ID mode for a receiving FXO or a sending FXS voice port.
- **clid dtmf-codes:** Specifies global caller ID DTMF start, redirect, and end codes.

Examples of Adding Voice Features with a CLI Add-On Feature Template

The following example shows caller ID configuration for FXS ports:

```
voice service pots
    clid dtmf-codes ABC
!
voice-port 1/0/0
    caller-id enable
    caller-id alerting ring 3
    station name West Wing
    station number 4085550100
!
voice-port 1/0/1
    caller-id enable
    caller-id mode DTMF start * end #
    caller-id alerting line-reversal
    station name East Wing
    station number 4085550101
!
voice-port 1/0/2
    caller-id enable
    caller-id mode BT
    caller-id alerting pre-ring
    station name Jose
    station number 4085550102
!
voice-port 1/0/3
    caller-id enable
    caller-id block
    station name a-sample
    station number 4085552000
!
voice-port 1/0/4
    caller-id enable
    caller-id format e911
    station name sample-2
    station number 4085552222
```

The following example shows caller ID configuration for FXO ports:

```
voice service pots
    clid dtmf-codes ABC
!
voice-port 2/0/0
    cptone BR
```

```
        caller-id enable
        caller-id alerting line-reversal
        caller-id alerting dsp-pre-allocate
    !
voice-port 2/0/1
    caller-id enable
    caller-id alerting ring 2
!
voice-port 2/0/2
    caller-id enable
    caller-id BT FSK
    caller-id alerting pre-ring
```




CHAPTER 12

CUBE Configuration

Table 129: Feature History

Feature Name	Release Information	Description
Cisco Unified Border Element Configuration	Cisco IOS XE Catalyst SD-WAN Release 17.7.1a Cisco vManage Release 20.7.1	This feature lets you configure Cisco Unified Border Element (CUBE) functionality by using Cisco IOS XE Catalyst SD-WAN device CLI templates or CLI add-on feature templates.
Secure SRST Support on Cisco Catalyst SD-WAN	Cisco IOS XE Catalyst SD-WAN Release 17.10.1a Cisco vManage Release 20.10.1	This feature enables you to configure Cisco Survivable Remote Site Telephony (SRST) commands on Cisco IOS XE Catalyst SD-WAN devices using Cisco SD-WAN Manager device CLI templates or CLI add-on feature templates. The feature also provides additional Cisco Unified Border Element (CUBE) commands that are qualified for use in Cisco SD-WAN Manager device CLI templates or CLI add-on feature templates.
Cisco Unified Border Element Configuration	Cisco IOS XE Catalyst SD-WAN Release 17.14.1a Cisco Catalyst SD-WAN Manager Release 20.14.1	This feature provides support for the following commands: • cipher (voice class) • nat media-keepalive • secure-ciphersuite • transport tcp tls (sip-ua) • voice-class sip nat media-keepalive

Feature Name	Release Information	Description
Survivable Remote Site Telephony (SRST) commands	Cisco IOS XE Catalyst SD-WAN Release 17.14.1a Cisco Catalyst SD-WAN Manager Release 20.14.1	This feature provides support for the following commands: • http client secure-ciphersuite • transport-tcp-tls (call-manager-fallback)

This chapter provides information about configuring devices for Cisco Unified Border Element (CUBE).

- [Information About CUBE, on page 370](#)
- [Supported Devices for CUBE Configuration, on page 370](#)
- [Restrictions for CUBE Configuration, on page 371](#)
- [Use Cases for CUBE, on page 371](#)
- [Configure CUBE, on page 371](#)
- [CUBE Commands, on page 372](#)
- [SRST Commands, on page 380](#)

Information About CUBE

CUBE bridges voice and video connectivity between two VoIP networks. It is similar to a traditional voice gateway, except for the replacement of physical voice trunks with IP-based voice trunks. Traditional gateways connect VoIP networks to telephone companies by using a circuit-switched connection, such as PRI. CUBE connects VoIP networks to other VoIP networks and enterprise networks to Internet telephony service providers (ITSPs).

CUBE provides conventional Session Border Controller (SBC) functions and a wide variety advanced features.

You can configure Cisco IOS XE Catalyst SD-WAN devices for CUBE by using device CLI templates or CLI add-on feature templates.

For more information about the CUBE setup, functionality, usage, configuration, and related topics, see the [Cisco Unified Border Element Configuration Guide](#).

Supported Devices for CUBE Configuration

- Cisco 1000 Series Integrated Services Routers
- Cisco 4000 Series Integrated Services Routers
- Cisco Catalyst 8200 Series Edge Platforms
- Cisco Catalyst 8300 Series Edge Platforms
- Cisco Catalyst 8000v Software Router
- Cisco ASR 1001-X Router
- Cisco ASR 1002-X Router

- Cisco ASR 1006-X Router with the Cisco ASR1000-RP3 Module, and the Cisco ASR1000-ESP100 or ASR1000-ESP100-X Embedded Services Processor
- Cisco ASR 1004 Router with the RP2 Route Processor and the Cisco ASR 1000-ESP40 Embedded Services Processor
- Cisco ASR 1006 Router with the RP2 Route Processor and the Cisco ASR 1000-ESP40 Embedded Services Processor
- Cisco ASR 1006-X Router with the RP2 Route Processor and the Cisco ASR 1000-ESP40 Embedded Services Processor

Restrictions for CUBE Configuration

High-availability configuration is not supported for CUBE.

Use Cases for CUBE

CUBE can be used to configure session border controller elements for a wide variety of applications, including the following:

- Enterprise premises-based collaboration capabilities using Cisco Unified Communications Manager (or another call control application) with centralized or local PSTN breakouts
- A local breakout gateway for Cisco Unified Communications Manager Cloud, which is a Cisco-hosted cloud service for large enterprises
- A local gateway to enable the Bring Your Own PSTN (BYoPSTN) option for Cisco Webex Calling
- Edge audio for Cisco Webex meetings with a direct VoIP route to the Cisco Webex cloud or through existing PSTN services

Configure CUBE

To configure a device to use the CUBE functionality, create a Cisco IOS XE Catalyst SD-WAN device CLI template or a CLI add-on feature template for the device.

For information about device CLI templates, see [CLI Templates for Cisco IOS XE Catalyst SD-WAN Device Routers](#).

For information about CLI add-on feature templates, see [CLI Add-On Feature Templates](#).

For information about CUBE configuration and usage, see [Cisco Unified Border Element Configuration Guide](#).

For information about the CUBE commands that Cisco Catalyst SD-WAN supports for use in a CLI template, see [CUBE Commands](#).

The following example shows a basic CUBE configuration using a CLI add-on template:

```
voice service voip
  ip address trusted list
```

```

ipv4 10.0.0.0.255.0.0.0
ipv6 2001:DB8:0:ABCD::1/48
!
allow-connections sip to sip
sip
no call service stop
!
dial-peer voice 100 voip
description Inbound LAN side dial-peer
session protocol sipv2
incoming called number .T
voice-class codec 1
dtmf-relay rtp-nte
!
dial-peer voice 101 voip
description Outbound LAN side dial-peer
destination pattern [2-9].....
session protocol sipv2
session target ipv4:10.10.10.1
voice-class codec 1
dtmf-relay rtp-nte
!
dial-peer voice 200 voip
description Inbound WAN side dial-peer
session protocol sipv2
incoming called-number .T
voice-class codec 1
dtmf-relay rtp-nte
!
dial-peer voice 201 voip
description Outbound WAN side dial-peer
destination pattern [2-9].....
session protocol sipv2
session target ipv4:20.20.20.1
voice-class codec 1
dtmf-relay rtp-nte

```

CUBE Commands

The following table lists the commands that are supported by Cisco Catalyst SD-WAN CLI templates for CUBE configuration. Click a command name in the **Command** column to view information about the command, its syntax, and its use.

Table 130: Cisco Catalyst SD-WAN CLI Template Commands for CUBE Configuration

Command	Description
address-hiding	Hides signaling and media peer addresses from endpoints other than the gateway.
anat	Enables Alternative Network Address Types (ANAT) on a SIP trunk.
answer-address	Specifies the full E.164 telephone number to be used to identify the dial peer of an incoming call.
application (global)	Enters application configuration mode to configure applications.

Command	Description
<code>asserted-id</code>	Enables support for the asserted ID header in incoming SIP requests or response messages, and to send the asserted ID privacy information in outgoing SIP requests or response messages.
<code>asymmetric payload</code>	Configures SIP asymmetric payload support.
<code>audio forced</code>	Allows only audio and image (for T.38 Fax) media types, and drops all other media types).
<code>authentication</code>	Enables SIP digest authentication.
<code>bind</code>	Binds the source address for signaling and media packets to the IPv4 or IPv6 address of a specific interface.
<code>block</code>	Configures global settings to drop (not pass) specific incoming SIP provisional response messages on a CUBE.
<code>call spike</code>	Configures the limit on the number of incoming calls received in a short period (a call spike).
<code>call threshold global</code>	Enables the global resources of a gateway.
<code>call treatment action</code>	Configures the action that the router takes when local resources are unavailable.
<code>call treatment cause-code</code>	Specifies the reason for the disconnection to the caller when local resources are unavailable.
<code>call treatment isdn-reject</code>	Specifies the rejection cause code for ISDN calls when all ISDN trunks are busied out, but the switch ignores the busyout trunks and still sends ISDN calls into the gateway.
<code>call treatment on</code>	Enables call treatment to process calls when local resources are unavailable.
<code>callmonitor</code>	Enables the call monitoring messaging functionality on a SIP endpoint in a VoIP network.
<code>call-route</code>	Enables header-based routing at the global configuration level.
<code>cipher (voice class)</code>	Configures the cipher setting, and associates it to a TLS profile.
<code>clid</code>	Passes the network-provided ISDN numbers in an ISDN calling party information element screening indicator field, and removes the calling party name and number from the calling-line identifier in voice service voip configuration mode. Alternatively, allows the presentation of the calling number by substituting for the missing Display Name field in the Remote-Party-ID and From headers.
<code>codec preference</code>	Specifies a list of preferred codecs to use on a dial peer.
<code>codec profile</code>	Defines audio and video capabilities that are needed for video endpoints.

Command	Description
<code>codec transparent</code>	Enables codec capabilities to be passed transparently between endpoints in a CUBE.
<code>conn-reuse</code>	Minimum supported releases: Cisco vManage Release 20.10.1 and Cisco IOS XE Catalyst SD-WAN Release 17.10.1a. Reuses the TCP connection of a SIP registration for an endpoint behind a firewall.
<code>connection-reuse</code>	Uses global listener port for sending requests over UDP.
<code>contact-passing</code>	Configures pass-through of the contact header from one leg to the other leg for 302 pass-through.
<code>cpa</code>	Enables the call progress analysis (CPA) algorithm for outbound VoIP calls and to set CPA parameters.
<code>credentials</code>	Configures a SIP TDM gateway or CUBE to send a SIP registration message when in the UP state.
<code>crypto signaling</code>	Identifies the trustpoint <i>trustpoint-name</i> keyword and argument that is used during the Transport Layer Security (TLS) handshake that corresponds to the remote device address.
<code>dial-peer cor custom</code>	Specifies that named class of restrictions (COR) apply to dial peers.
<code>dial-peer cor list</code>	Defines a class of restrictions (COR) list name.
<code>disable-early-media 180</code>	Minimum supported releases: Cisco vManage Release 20.10.1 and Cisco IOS XE Catalyst SD-WAN Release 17.10.1a. Specifies which call treatment, early media or local ringback, is provided for 180 responses with 180 responses with Session Description Protocol (SDP).
<code>dspfarm profile</code>	Enters DSP farm profile configuration mode and defines a profile for DSP farm services.
<code>dtmf-interworking</code>	Enables a delay between the dtmf-digit begin and dtmf-digit end events in the RFC 2833 packets sent from CUBE, and generates RFC 4733 compliance RTP Named Telephony Event (NTE) packets from CUBE.
<code>early-media update block</code>	Blocks the UPDATE requests with the Session Description Protocol (SDP) in an early dialog.
<code>early-offer</code>	Forces CUBE to send a SIP invite with Early Offer on the Out Leg.
<code>emergency</code>	Configures a list of emergency numbers.
<code>error-code-override</code>	Configures the SIP error code to be used at the dial peer.
<code>error-passthru</code>	Enables the passage of error messages from the incoming SIP leg to the outgoing SIP leg.

Command	Description
g729-annexb override	Configures the settings for G.729 codec interoperability and overrides the default value if the annexb attribute is not present.
gcid	Enables Global Call ID (GCID) for every call on an outbound leg of a VoIP dial peer for a SIP endpoint.
gw-accounting	Minimum supported releases: Cisco vManage Release 20.10.1 and Cisco IOS XE Catalyst SD-WAN Release 17.10.1a. Enables an accounting method for collecting call detail records (CDRs).
handle-replaces	Minimum supported releases: Cisco vManage Release 20.10.1 and Cisco IOS XE Catalyst SD-WAN Release 17.10.1a. Configures a Cisco IOS device to handle SIP INVITE with Replaces header messages at the SIP protocol level.
header-passing	Enables the passing of headers to and from SIP INVITE, SUBSCRIBE, and NOTIFY messages.
host-registrar	Populates the sip-ua registrar domain name or IP address value in the host portion of the diversion header and redirects the contact header of the 302 response.
http client connection idle timeout	Sets the number of seconds for which the HTTP client waits before terminating an idle connection.
http client connection persistent	Enables HTTP persistent connections so that multiple files can be loaded by using the same connection.
http client connection timeout	Sets the number of seconds for which the HTTP client waits for a server to establish a connection before abandoning its connection attempt.
ip qos dscp	Configures the DSCP value for QoS.
localhost	Globally configures CUBE to substitute a DNS hostname or domain as the localhost name in place of the physical IP address in the From, Call-ID, and Remote-Party-ID headers in outgoing messages.
max-conn	Specifies the maximum number of incoming or outgoing connections for a particular VoIP dial peer.
max-forwards	Minimum supported releases: Cisco vManage Release 20.10.1 and Cisco IOS XE Catalyst SD-WAN Release 17.10.1a. Globally sets the maximum number of hops, that is, proxy or redirect servers that can forward the SIP request.
media	Enables media packets to pass directly between endpoints without the intervention of CUBE, and enables signaling services.
media disable-detailed-stats	Disables the collection of detailed call statistics.

Command	Description
media profile asp	Creates a media profile to configure acoustic shock-protection parameters.
media profile nr	Creates a media profile to configure noise-reduction parameters.
media profile stream-service	Enables stream service on CUBE.
media profile video	Creates a media profile video.
media-address voice-vrf	Associates an RTP port range with VRF.
media-inactivity-criteria	Specifies the mechanism for detecting media inactivity (silence) on a voice call.
midcall-signaling	Configures the method that is used for signaling messages.
min-se	Changes the minimum session expiration (Min-SE) header value for all the calls that use the SIP session timer.
nat	Minimum supported releases: Cisco vManage Release 20.10.1 and Cisco IOS XE Catalyst SD-WAN Release 17.10.1a. Uses SIP Network Address Translation (NAT) global configuration.
nat media-keepalive	Enables media keepalive packet transmission for the specified interval of time.
notify redirect	Enables application handling of redirect requests for all VoIP dial peers.
notify ignore substate	Minimum supported releases: Cisco vManage Release 20.10.1 and Cisco IOS XE Catalyst SD-WAN Release 17.10.1a. Specifies Ignoring the Subscription-State header in a Notify message.
notify telephone-event	Minimum supported releases: Cisco vManage Release 20.10.1 and Cisco IOS XE Catalyst SD-WAN Release 17.10.1a. Configures the maximum interval between two consecutive NOTIFY messages for a particular telephone event.
num-exp	Defines how to expand a telephone extension number into a particular destination pattern.
options-ping	Enables in-dialog options.
outbound-proxy	Configures a SIP outbound proxy for outgoing SIP messages globally.
pass-thru content	Enables the pass-through of SDP from in-leg to the out-leg.
permit hostname	Minimum supported releases: Cisco vManage Release 20.10.1 and Cisco IOS XE Catalyst SD-WAN Release 17.10.1a. Stores hostnames used during validation of initial incoming INVITE messages.

Command	Description
privacy	Sets privacy support at the global level as defined in RFC 3323.
privacy-policy	Configures the privacy header policy options at the global level.
progress_ind	Configures an outbound dial peer on a CUBE to override and remove or replace the default progress indicator in specified call messages.
protocol mode	Configures the Cisco IOS SIP stack.
random-contact	Minimum supported releases: Cisco vManage Release 20.10.1 and Cisco IOS XE Catalyst SD-WAN Release 17.10.1a. Populates an outgoing INVITE message with random-contact information instead of clear-contact information.
reason-header override	Enables cause code passing from one SIP leg to another.
redirect ip2ip	Redirects SIP phone calls to SIP phone calls globally on a gateway.
redirection	Enables the handling of 3xx redirect messages
referto-passing	Disables dial peer lookup and modification of the Refer-To header when the CUBE passes across a REFER message during a call transfer.
registrar	Enables SIP gateways to register E.164 numbers on behalf of analog telephone voice ports (FXS), IP phone virtual voice ports (EFXS), and SCCP phones with an external SIP proxy or SIP registrar.
reliable	Enables SIP provisional responses (other than 100 Trying) to be sent reliably to the remote SIP endpoint.
remote-party-id	Enables translation of the Remote-Party-ID SIP header.
request-uri-passing	Enables pass-through of the host part of the Request-URI and To SIP headers.
retry bye	Configures the number of times that a BYE request is retransmitted to the other user agent.
retry invite	Minimum supported releases: Cisco vManage Release 20.10.1 and Cisco IOS XE Catalyst SD-WAN Release 17.10.1a. Configures the number of times that a SIP INVITE request is retransmitted to the other user agent.
rtcp all-pass-through	Passes through all the RTCP packets in the datapath.
rtcp keepalive	Configures RTCP keepalive report generation and generates RTCP keepalive packets.
rtp payload-type	Identifies the payload type of an RTP packet.

Command	Description
rtp-media-loop count	Configures the number of media loops before RTP voice and video media packets are dropped.
rtp-port	Configures the real-time protocol range.
rtp-ssrc multiplex	Multiplexes RTCP packets with RTP packets and sends multiple synchronization source in RTP headers (SSRCs) in an RTP session.
secure-ciphersuite	Configures the cipher suites (encryption algorithms) to be used for encryption over HTTPS for a WebSocket connection in CUBE.
session refresh	Enables SIP session refresh globally.
session transport	Configures a VoIP dial peer to use TCP or UDP as the underlying transport layer protocol for SIP messages.
set pstn-cause	Maps an incoming PSTN cause code to a SIP error status code.
set sip-status	Maps an incoming SIP error status code to a PSTN cause code.
signaling forward	Configures global settings for transparent tunneling of QSIG, Q.931, H.225, and ISUP messages.
silent discard untrusted	Discards SIP requests from untrusted sources in an incoming SIP trunk.
sip-server	Configures a network address for the SIP server interface.
srtp	Specifies that SRTP be used to enable secure calls and call fallback.
srtp negotiate	Minimum supported releases: Cisco vManage Release 20.10.1 and Cisco IOS XE Catalyst SD-WAN Release 17.10.1a. Enables the Cisco IOS Session Initiation Protocol (SIP) gateway to accept and send a Real-Time Transport Protocol (RTP) Audio/Video Profile (AVP) at the global configuration level.
stun	Enters STUN configuration mode for configuring firewall traversal parameters.
stun flowdata shared-secret	Minimum supported releases: Cisco vManage Release 20.10.1 and Cisco IOS XE Catalyst SD-WAN Release 17.10.1a. Configures a secret shared on a call control agent.
stun usage firewall-traversal flowdata	Enables firewall traversal using STUN.
supplementary-service media-renegotiate	Globally enables midcall media renegotiation for supplementary services.
timers	Configures SIP-signaling timers.

Command	Description
transport	Configures the SIP user agent (gateway) for SIP signaling messages in inbound calls through the SIP TCP, TLS over TCP, or UDP socket. This command supports TLS version 1.3 and all associated ciphers.
uc secure-wsapi	Configures a secure Cisco Unified Communication IOS services environment for a specific application.
uc wsapi	Configures a nonsecure Cisco Unified Communication IOS services environment for a specific application.
update-callerid	Enables sending updates for caller IDs.
url (SIP)	Configures URLs to either the SIP, SIP secure (SIPS), or telephone (TEL) format for your VoIP SIP calls.
vad	Enables VAD for calls using a specific dial peer.
video codec	Minimum supported releases: Cisco vManage Release 20.10.1 and Cisco IOS XE Catalyst SD-WAN Release 17.10.1a.
voice cause code	Sets the internal Q850 cause code mapping for, voice and enters voice cause configuration mode.
voice class codec	Enters voice-class configuration mode and assigns an identification tag number for a codec voice class.
voice class dpg	Creates a dial-peer group for grouping multiple outbound dial peers.
voice class e164-pattern-map	Creates an E.164 pattern map that specifies multiple destination E.164 patterns in a dial peer.
voice class media	Configures media control parameters for voice.
voice class server-group	Enters voice-class configuration mode and configures server groups (groups of IPv4 and IPv6 addresses) that can be referenced from an outbound SIP dial peer.
voice-class sip options-keepalive	Monitors connectivity between CUBE VoIP dial peers and SIP servers.
voice class sip-copylist	Configures a list of entities to be sent to the peer call leg.
voice class sip-event-list	Configures a list of SIP events to be passed through.
voice class sip-hdr-passthru-list	Configures a list of headers to be passed through the route string.
voice-class sip nat media-keepalive	Configures media keepalive to enable media keepalive packets to be transmitted for the interval specified.

Command	Description
voice class sip-profiles	Configures SIP profiles for a voice class. Configuring a sip profile, request REGISTER sip-header Authorization modify "550\"," [550@dl.ims.airtel.in mailto:550@dl.ims.airtel.in]" , on a device throws a warning message: Device# show run sec sip-profiles voice class sip-profiles 4 rule 2 request REGISTER sip-header Authorization modify "550\"," "[550@dl.ims.airtel.in mailto:550@dl.ims.airtel.in]"
voice class srtp-crypto	Enters voice class configuration mode and assigns an identification tag for an srtp-crypto voice class command.
voice class uri	Creates or modifies a voice class for matching dial peers to a SIP or TEL URI.
voice class tls-cipher	Minimum supported releases: Cisco vManage Release 20.10.1 and Cisco IOS XE Catalyst SD-WAN Release 17.10.1a. Configures an ordered set of TLS cipher suites.
voice class tls-profile	Minimum supported releases: Cisco vManage Release 20.10.1 and Cisco IOS XE Catalyst SD-WAN Release 17.10.1a. Enables voice class configuration mode, and assigns an identification tag for a TLS profile.
voice iec syslog	Enables viewing of internal error codes as they are encountered in real time.
voice statistics iec	Enables collection of internal error code statistics.
xfer target	Minimum supported releases: Cisco vManage Release 20.10.1 and Cisco IOS XE Catalyst SD-WAN Release 17.10.1a. Routes the INVITE to the refer-to destination in the REFER consume case. The routing decision is made based on the xfer target destination.

SRST Commands

The following table lists the commands that are supported by Cisco Catalyst SD-WAN CLI templates for SRST. Click a command name in the **Command** column to view information about the command, its syntax, and its use.

Table 131: Cisco Catalyst SD-WAN CLI Template Commands for SRST

Command	Description
http client secure-ciphersuite	Sets the secure encryption cipher suite for the HTTP client.

Command	Description
<code>transport-tls (call-manager-fallback)</code>	Configures a specific TLS version for Unified Secure SCCP SRST, in call-manager-fallback mode.



CHAPTER 13

Configure Network Interfaces

In the Cisco Catalyst SD-WAN overlay network design, interfaces are associated with VPNs. The interfaces that participate in a VPN are configured and enabled in that VPN. Each interface can be present only in a single VPN.

At a high level, for an interface to be operational, you must configure an IP address for the interface and mark it as operational (**no shutdown**). In practice, you always configure additional parameters for each interface.

You can configure up to 512 interfaces on a Cisco IOS XE Catalyst SD-WAN device. This number includes physical interfaces, loopback interfaces, and subinterfaces.



Note To maximize the efficiency of the load-balancing among Cisco Catalyst SD-WAN Controllers, use sequential numbers when assigning system IP addresses to the Cisco IOS XE Catalyst SD-WAN devices in the domain. Example of a sequential numbering schemes is 172.16.1.1, 172.16.1.2, 172.16.1.3, and so on.



Note Ensure that any network interface configured on a device has a unique IP address.



Note If you try to configure an interface or sub-interface beyond the supported limit, the device generates a notification to Cisco SD-WAN Manager.

- [Configure VPN, on page 384](#)
- [Configure Interfaces in the WAN Transport VPN \(VPN 0\), on page 398](#)
- [Configure the System Interface, on page 404](#)
- [Configure Control Plane High Availability, on page 405](#)
- [Configure Other Interfaces, on page 405](#)
- [Configure Interface Properties, on page 412](#)
- [Enable DHCP Server using Cisco SD-WAN Manager, on page 431](#)
- [Configuring PPPoE, on page 434](#)
- [Configure PPPoE Over ATM, on page 438](#)
- [Configuring VRRP , on page 441](#)
- [Configuring Dynamic Interfaces, on page 442](#)

- [Configure VPN Ethernet Interface Using a Configuration Group, on page 445](#)
- [Configure Management VPN Using a Configuration Group, on page 454](#)
- [Configure VPN Ethernet Interface, on page 456](#)
- [VPN Interface Bridge, on page 467](#)
- [Configure DSL IPoE Using a Configuration Group, on page 473](#)
- [VPN Interface DSL IPoE, on page 480](#)
- [Configure DSL PPPoA Using a Configuration Group, on page 491](#)
- [VPN Interface DSL PPPoA, on page 500](#)
- [Configure DSL PPPoE Using a Configuration Group, on page 509](#)
- [VPN Interface DSL PPPoE, on page 518](#)
- [Configure VPN Interface Ethernet PPPoE Using a Configuration Group, on page 528](#)
- [VPN Interface Ethernet PPPoE, on page 535](#)
- [Cisco VPN Interface GRE, on page 543](#)
- [GRE-in-UDP, on page 554](#)
- [VPN Interface IPsec , on page 556](#)
- [Configure VPN Interface Multilink Using Configuration Groups, on page 573](#)
- [VPN Interface Multilink, on page 579](#)
- [Configure VPN Interface SVI using Cisco SD-WAN Manager, on page 587](#)
- [VPN Interface T1/E1, on page 596](#)
- [Configure T1 or E1 Controller Using a Configuration Group, on page 604](#)
- [Cellular Interfaces, on page 606](#)

Configure VPN

VPN

Use the VPN template for all Cisco Catalyst SD-WAN devices running the Cisco Catalyst SD-WAN software.

To configure VPNs using Cisco SD-WAN Manager templates, follow this general workflow:

1. Create VPN feature templates to configure VPN parameters. You create a separate VPN feature template for each VPN. For example, create one feature template for VPN 0, a second for VPN 1, and a third for VPN 512.

For Cisco SD-WAN Manager Network Management Systems and Cisco Catalyst SD-WAN Controllers, you can configure only VPNs 0 and 512. Create templates for these VPNs only if you want to modify the default settings for the VPN. For Cisco IOS XE Catalyst SD-WAN devices, you can create templates for these two VPNs and for additional VPN feature templates to segment service-side user networks.

- **VPN 0—Transport VPN**, which carries control traffic via the configured WAN transport interfaces. Initially, VPN 0 contains all of a device's interfaces except for the management interface, and all interfaces are disabled.
- **VPN 512—Management VPN**, which carries out-of-band network management traffic among the Cisco IOS XE Catalyst SD-WAN devices in the overlay network. The interface used for management traffic resides in VPN 512. By default, VPN 512 is configured and enabled on all Cisco IOS XE Catalyst SD-WAN devices. For controller devices, by default, VPN 512 is not configured.

- **VPNs 1–511, 513–65527—Service VPNs**, for service-side data traffic on Cisco IOS XE Catalyst SD-WAN devices.

2. Create interface feature templates to configure the interfaces in the VPN.

Create a VPN Template



Note Cisco IOS XE Catalyst SD-WAN devices use VRFs for segmentation and network isolation. However, the following steps still apply if you are configuring segmentation for Cisco IOS XE Catalyst SD-WAN devices through Cisco SD-WAN Manager. When you complete the configuration, the system automatically converts the VPNs to VRFs for Cisco IOS XE Catalyst SD-WAN devices.



Note You can configure a static route through the VPN template.

Procedure

- Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
- Step 2** Click **Device Templates**, and click **Create Template**.

Note
In Cisco vManage Release 20.7.x and earlier releases **Device Templates** is called **Device**.
- Step 3** From the **Create Template** drop-down list, choose **From Feature Template**.
- Step 4** From the **Device Model** drop-down list, choose the type of device for which you wish to create the template.
- Step 5** To create a template for VPN 0 or VPN 512:
 - a. Click **Transport & Management VPN**, or scroll to the **Transport & Management VPN** section.
 - b. From the VPN 0 or VPN 512 drop-down list, click **Create Template**. The VPN template form appears.
The form contains fields for naming the template, and fields for defining VPN parameters.
- Step 6** To create a template for VPNs 1 through 511, and 513 through 65527:
 - a. Click **Service VPN**, or scroll to the **Service VPN** section.
 - b. Click the **Service VPN** drop-down list.
 - c. From the **VPN** drop-down list, click **Create Template**. The VPN template form displays.
The form contains fields for naming the template, and fields for defining VPN parameters.
- Step 7** In **Template Name**, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.

- Step 8** In **Template Description**, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

Configure Service VPN Using Configuration Groups

This feature helps you configure a service VPN (range 1 – 65527, except 512) or the LAN VPN.

The following tables describes the options for configuring the Service VPN feature.

Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

- Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

- Step 2** Create and configure Service VPN in a Service profile.

- a) Enter the basic configuration information.

Table 132: Basic Configuration

Field	Description
VPN*	Enter the numeric identifier of the VPN.
Name*	Enter a name for the VPN.
OMP Admin Distance IPv4	Administrative distance for OMP routes. The Cisco SD-WAN Controllers learn the topology of the overlay network and the services available in the network using OMP routes. The distance can be a value between 1–255.
OMP Admin Distance IPv6	Administrative distance for OMP routes. The Cisco SD-WAN Controllers learn the topology of the overlay network and the services available in the network using OMP routes. The distance can be a value between 1–255.

- b) Enter DNS information.

Table 133: DNS

Field	Description
Add DNS IPv4	
Primary DNS Address (IPv4)	Enter the IP address of the primary IPv4 DNS server in this VPN.
Secondary DNS Address (IPv4)	Enter the IP address of a secondary IPv4 DNS server in this VPN.
Add DNS IPv6	
Primary DNS Address (IPv6)	Enter the IP address of the primary IPv6 DNS server in this VPN.

Field	Description
Secondary DNS Address (IPv6)	Enter the IP address of a secondary IPv6 DNS server in this VPN.

- c) Enter host mapping information.

Table 134: Host Mapping

Field	Description
Add New Host Mapping	
Hostname*	Enter the hostname of the DNS server. The name can be up to 128 characters.
List of IP*	Enter up to eight IP addresses to associate with the hostname. Separate the entries with commas.

- d) Enter advertise OMP information.

Table 135: Advertise OMP

Field	Description
Add OMP Advertise IPv4	
Protocol	Choose a protocol to configure route advertisements to OMP, for this VPN: • bgp • ospf • ospfv3 • connected • static • network • aggregate Applied to Region: (Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.13.1) In a Multi-Region Fabric scenario, route aggregation is a method for reducing the number of entries that routers in a network must maintain in routing tables, for better scaling. Choose core, access, or core-and-access, to apply route aggregation only to access regions, the core region, or both. This option is applicable only to a Multi-Region Fabric border router, not an edge router or a transport gateway. • eigrp • lisp • isis

Field	Description
Select Route Policy	Enter the name of the route policy. Route policy is not supported in Cisco vManage Release 20.9.1.
Add OMP Advertise IPv6	
Protocol	Choose a protocol to configure route advertisements to OMP, for this VPN: • BGP • OSPF • Connected • Static • Network • Aggregate Applied to Region: (Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.13.1) In a Multi-Region Fabric scenario, route aggregation is a method for reducing the number of entries that routers in a network must maintain in routing tables, for better scaling. Choose core, access, or core-and-access, to apply route aggregation only to access regions, the core region, or both. This option is applicable only to a Multi-Region Fabric border router, not an edge router or a transport gateway.
Select Route Policy	Enter the name of the route policy. Route policy is not supported in Cisco vManage Release 20.9.1.
Protocol Sub Type	When you choose the OSPF protocol, specify the sub type as external.

e) Enter route information.

Table 136: Route

Field	Description
Add IPv4 Static Route	
Network Address*	Enter the IPv4 address or prefix, in decimal four-point-dotted notation, and the prefix length of the IPv4 static route to configure in the VPN.
Subnet Mask*	Enter the subnet mask.

Field	Description
Next Hop/Null 0/VPN/DHCP	Choose one of the following options to configure the next hop to reach the static route: • Next Hop: When you choose this option, the IPv4 Route Gateway Next Hop field appears. Enable this option to add the next hop. You can add a hop with and without a tracker. When you click Add Next Hop, the following fields appear: • Address*: Enter the next-hop IPv4 address. • Administrative Distance*: Enter the administrative distance for the route. When you click Add Next Hop with Tracker, the following fields appear: • Address*: Enter the next-hop IPv4 address. • Administrative Distance*: Enter the administrative distance for the route. • Tracker*: Enter the name of the gateway tracker to determine whether the next hop is reachable before adding that route to the route table of the device. • Null 0: When you choose this option, the following field appears: • IPv4 Route Null 0*: Enable this option to set the next hop to be the null interface. All packets sent to this interface are dropped without sending any ICMP messages. • VPN: When you choose this option, the following field appears: • IPv4 Route VPN*: Selects VPN as the gateway to direct packets to the transport VPN. • DHCP: When you choose this option, the following field appears: • IPv4 Route Gateway DHCP*: Assigns a static route for the default next-hop router when the DHCP server is accessed for an IP address.
Add BGP Routing	Choose a BGP route.
Add OSPF Routing	Choose an OSPF route.
Add IPv6 Static Route	
Prefix*	Enter the IPv6 address or prefix, in decimal four-point-dotted notation, and the prefix length of the IPv6 static route to configure in the VPN.

Field	Description
Next Hop/Null 0/NAT	Choose one of the following options to configure the next hop to reach the static route: • Next Hop: When you choose this option and click Add Next Hop, the following fields appear: • Address*: Enter the next-hop IPv6 address. • Administrative distance*: Enter the administrative distance for the route. • Null 0: When you choose this option, the following field appears: • IPv6 Route Null 0*: Enable this option to set the next hop to be the null interface. All packets sent to this interface are dropped without sending any ICMP messages. • NAT: When you choose this option, the following field appears: • IPv6 NAT*: Choose NAT64 or NAT66. • Interface: When you choose this option, the following fields appear: • Interface Name: Choose IPv6 interface name for the IPsec tunnel. • Next Hop: Enter the IPv6 address and the administrative distance for the next hop.

f) Enter service information.

Table 137: Service

Field	Description
Add Service	
Service Type	Choose a service available at the local site and in the VPN. Values: FW, IDS, IDP, netsvc1, netsvc2, netsvc3, netsvc4, TE, SIG
IPv4 Addresses (Maximum: 4)*	Enter up to four IP address, separated by commas. The service is advertised to the Cisco SD-WAN Controller only if one of the addresses can be resolved locally, at the local site, not via routes learned through OMP. You can configure up to four IP addresses.
Tracking*	Cisco Catalyst SD-WAN tests each service device periodically to check whether it is operational. Tracking saves the results of the periodic tests in a service log. Tracking is enabled by default.

g) Enter service route information.

Table 138: Service Route

Field	Description
Add Service Route	
Prefix*	Enter the IP address or prefix. For Umbrella SIG, use any RFC 1918 subnet for Service IP addresses.
Service*	Configure routes pointing to any service. Values: FW , IDS , IDP , netsvc1 , netsvc2 , netsvc3 , netsvc4 .
VPN*	Destination VPN to resolve the prefix.

- h) Enter GRE route information.

Table 139: GRE Route

Field	Description
Add GRE Route	
Prefix*	Enter the IP address or prefix, in decimal four-part-dotted notation, and prefix length of the GRE-specific static route.
Interface*	Enter the name of one or two GRE tunnels to use to reach the service.
VPN*	Enter the number of the VPN to reach the service. This must be VPN 0.

- i) Enter IPSEC route information.

Table 140: IPSEC Route

Field	Description
Add ipSec Route	
Prefix*	Enter the IP address or prefix, in decimal four-part-dotted notation, and prefix length of the IPsec-specific static route.
Interface*	Enter the name of one or two IPsec tunnel interfaces. If you configure two interfaces, the first is the primary IPsec tunnel, and the second is the backup. All packets are sent only to the primary tunnel. If that tunnel fails, all packets are then sent to the secondary tunnel. If the primary tunnel comes back up, all traffic is moved back to the primary IPsec tunnel.

- j) Enter NAT information

Table 141: NAT

Field	Description
Nat Pool	

Field	Description
NatPool Name*	Enter a NAT pool number configured in the centralized data policy. The NAT pool name must be unique across VPNs and VRFs. You can configure up to 31 (1–32) NAT pools per router.
Prefix Length*	Enter the NAT pool prefix length.
Range Start*	Enter a starting IP address for the NAT pool.
Range End*	Enter a closing IP address for the NAT pool.
Overload*	Enable this option to configure per-port translation. If this option is disabled, only dynamic NAT is configured on the end device. Per-port NAT is not configured. Default: Enabled
Direction*	Choose the NAT direction.
Nat64 V4 Pool	
Nat64 V4 Pool Name*	Enter a NAT pool number configured in the centralized data policy. The NAT pool name must be unique across VPNs and VRFs. You can configure up to 31 (1–32) NAT pools per router.
Nat 64 V4 Pool Range Start*	Enter a starting IP address for the NAT pool.
Nat 64 V4 Pool Range End*	Enter a closing IP address for the NAT pool.
Overload*	Enable this option to configure per-port translation. If this option is disabled, only dynamic NAT is configured on the end device. Per-port NAT is not configured. Default: Disabled

k) Enter route leak information.

Table 142: Route leak from Global VPN

Field	Description
Route Protocol*	Choose a protocol to configure leak routes from global VPN to the service VPN that you are configuring: • static • connected • bgp • ospf
Select Route Policy	Choose a route policy from the drop-down list.
Redistribution (in service VPN)	

Field	Description
Protocol*	Choose a protocol from the available options to redistribute the leaked routes: • bgp • ospf • (Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.18.1) • eigrp
Select Route Policy	Choose a route policy from the drop-down list.

- 1) Enter route target information.

Table 143: Route Target

Field	Description
IPv4 Settings	
Import Route Target List: Route Target*	Configure a route target for IPv4 interfaces. It imports routing information from the target VPN extended community.
Export Route Target List: Route Target*	Configure a route target for IPv4 interfaces. It exports routing information to the target VPN extended community.
IPv6 Settings	
Import Route Target List: Route Target*	Configure a route target for IPv6 interfaces. It imports routing information from the target VPN extended community.
Export Route Target List: Route Target*	Configure a route target for IPv6 interfaces. It exports routing information to the target VPN extended community.

What to do next

Also see [Deploy a Configuration Group](#).

Changing the Scope for a Parameter Value

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (a ) , and the default setting or value is shown. To change the default or to enter a value, click the scope drop-down list and select one of the following:

Parameter Name	Description
 Device Specific	Use a device-specific value for the parameter. For device-specific parameters, you cannot enter a value in the feature template. You enter the value when you attach a device to a device template. When you click Device Specific, the Enter Key box opens. This box displays a key, which is a unique string that identifies the parameter in a CSV file that you create. This file is an Excel spreadsheet that contains one column for each key. The header row contains the key names (one key per column), and each row after that corresponds to a device and defines the values of the keys for that device. You upload the CSV file when you attach a device to a device template. For more information, see Create a Template Variables Spreadsheet Note When you are using a CSV file for configuring device-specific variables in the device attach flow, ensure to fill all the mandatory fields before uploading. To change the default key, type a new string and move the cursor out of the Enter Key box. Examples of device-specific parameters are system IP address, hostname, GPS location, and site ID.
 Global	Enter a value for the parameter, and apply that value to all devices. Examples of parameters that you might apply globally to a group of devices are DNS server, syslog server, and interface MTUs.

Once you have created and named the template, enter the following values. Parameters marked with an asterisk are required.

Configure Basic VPN Parameters Using a Configuration Group

Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

-
- Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.
- Step 2** Create and configure a Transport VPN feature in Transport and Management profile.

a) Basic Configuration

Table 144: Basic Configuration

Field	Description
VPN	Enter the numeric identifier of the VPN.
Enhance ECMP Keying	Enable the use in the ECMP hash key of Layer 4 source and destination ports, in addition to the combination of the source IP address, destination IP address, protocol, and DSCP field, as the ECMP hash key. Default: Disabled

What to do next

Also see [Deploy a Configuration Group](#).

Configure Basic VPN Parameters

To configure basic VPN parameters, choose **Basic Configuration** and then configure the following parameters. Parameters marked with an asterisk are required to configure a VPN.

Parameter Name	Description
VPN	Enter the numeric identifier of the VPN. Range for Cisco IOS XE Catalyst SD-WAN devices: 0 through 65527 Values for Cisco Catalyst SD-WAN Controller and Cisco SD-WAN Manager devices: 0, 512
Name	Enter a name for the VPN. Note For Cisco IOS XE Catalyst SD-WAN devices, you can't enter a device-specific name for the VPN.
Enhance ECMP keying	Click On to enable the use in the ECMP hash key of Layer 4 source and destination ports, in addition to the combination of the source, and destination IP addresses, as the ECMP hash key. ECMP keying is Off by default.



Note To complete the configuration of the transport VPN on a router, you must configure at least one interface in VPN 0.

To save the feature template, click **Save**.

Configure Load-Balancing Algorithm Using the CLI



Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.8.1a, you need CLI template to configure the **src-only** load-sharing algorithm for IPv4 and IPv6 Cisco Catalyst SD-WAN and non Cisco Catalyst SD-WAN traffic. For complete details on the load-sharing algorithm CLI, see [IP Commands](#) list.

This following provides CLI configurations for selecting a Cisco Express Forwarding load-balancing algorithm for non Cisco Catalyst SD-WAN IPv4 and IPv6 traffic. You can enable ECMP keying to send the configurations for both IPv4 and IPv6.

```
Device# config-transaction
Device(config)# ip cef load-sharing algorithm {universal [id] | include-ports [ source [id]
| destination [id]] |
src-only [id]}
```

```
Device# config-transaction
Device(config)# ipv6 cef load-sharing algorithm {universal [id] | include-ports [ source
[id] | destination [id]] |
src-only [id]}
```

This following provides CLI configurations for enabling load balancing algorithm on an interface for Cisco Catalyst SD-WAN IPv4 and IPv6 traffic. You can enable ECMP keying to send the configurations for both IPv4 and IPv6.

```
Device# config-transaction
Device(config)# sdwan
Device(config-sdwan)# ip load-sharing algorithm {ip-and-ports | src-dst-ip | src-ip-only}

Device# config-transaction
Device(config)# sdwan
Device(config-sdwan)# ipv6 load-sharing algorithm {ip-and-ports | src-dst-ip | src-ip-only}
```

Configure DNS and Static Hostname Mapping Using a Configuration Group

Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

Step 1 From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

Step 2 Create and configure a Transport VPN feature in Transport and Management profile.

a) DNS

Table 145: DNS

Field	Description
Add DNS	
Primary DNS Address (IPv4)	Enter the IP address of the primary IPv4 DNS server in this VPN.

Field	Description
Secondary DNS Address (IPv4)	Enter the IP address of a secondary IPv4 DNS server in this VPN.
Add DNS IPv6	
Primary DNS Address (IPv6)	Enter the IP address of the primary IPv6 DNS server in this VPN.
Secondary DNS Address (IPv6)	Enter the IP address of a secondary IPv6 DNS server in this VPN.

b) Host Mapping

Table 146: Host Mapping

Field	Description
Add New Host Mapping	
Hostname*	Enter the hostname of the DNS server. The name can be up to 128 characters.
List of IP*	Enter up to 14 IP addresses to associate with the hostname. Separate the entries with commas.

What to do next

Also see [Deploy a Configuration Group](#).

Configure DNS and Static Hostname Mapping

To configure DNS addresses and static hostname mapping, click **DNS** and configure the following parameters:

Parameter Name	Options	Description
Primary DNS Address	Click either IPv4 or IPv6 , and enter the IP address of the primary DNS server in this VPN.	
New DNS Address	Click New DNS Address and enter the IP address of a secondary DNS server in this VPN. This field appears only if you have specified a primary DNS address.	
	Mark as Optional Row	Check the Mark as Optional Row check box to mark this configuration as device-specific. To include this configuration for a device, enter the requested variable values when you attach a device template to a device, or create a template variables spreadsheet to apply the variables.
	Hostname	Enter the hostname of the DNS server. The name can be up to 128 characters.
	List of IP Addresses	Enter up to eight IP addresses to associate with the hostname. Separate the entries with commas.
To save the DNS server configuration, click Add .		

To save the feature template, click **Save**.

Mapping Host Names to IP Addresses

```
! IP DNS-based host name-to-address translation is enabled
ip domain lookup
! Specifies hosts 192.168.1.111 and 192.168.1.2 as name servers
ip name-server 192.168.1.111 192.168.1.2
! Defines cisco.com as the default domain name the device uses to complete
! Set the name for unqualified host names
ip domain name cisco.com
```

Configure Interfaces in the WAN Transport VPN (VPN 0)

This topic describes how to configure the general properties of WAN transport and service-side network interfaces. For information about how to configure specific interface types and properties—including cellular interfaces, DHCP, PPPoE, VRRP, and WLAN interfaces.

VPN 0 is the WAN transport VPN. This VPN handles all control plane traffic, which is carried over OMP sessions, in the overlay network. For a Cisco IOS XE Catalyst SD-WAN device to participate in the overlay network, at least one interface must be configured in VPN 0, and at least one interface must connect to a WAN transport network, such as the Internet or an MPLS or a metro Ethernet network. This WAN transport interface is referred to as a tunnel interface. At a minimum, for this interface, you must configure an IP address, enable the interface, and set it to be a tunnel interface.

To configure a tunnel interface on a Cisco Catalyst SD-WAN Controller or a Cisco SD-WAN Manager, you create an interface in VPN 0, assign an IP address or configure the interface to receive an IP address from DHCP, and mark it as a tunnel interface. The IP address can be either an IPv4 or IPv6 address. To enable dual stack, configure both address types. You can optionally associate a color with the tunnel.



Note You can configure IPv6 addresses only on transport interfaces in VPN 0 and but not supported in VPN 512.

Tunnel interfaces on Cisco IOS XE Catalyst SD-WAN devices must have an IP address, a color, and an encapsulation type. The IP address can be either an IPv4 or IPv6 address. To enable dual stack in releases before Cisco IOS XE Catalyst SD-WAN Release 17.3.2, configure both address types.

To use dual stack with Cisco IOS XE Catalyst SD-WAN devices from Cisco IOS XE Catalyst SD-WAN Release 17.3.2, configure all controllers with both IPv4 and IPv6 addresses. In addition, configure DNS for the Cisco Catalyst SD-WAN Validator interface to resolve IPv4 and IPv6 address types so that controllers can reach the Cisco Catalyst SD-WAN Validator through either IP address type.



Note Starting from Cisco vManage Release 20.6.1, in case of a dual-stack configuration, if an IPv4 address or the fully qualified domain name (FQDN) is not available, but an IPv6 address is available, then the IPv6 address is used to connect to the Cisco Catalyst SD-WAN Validator.

For the tunnel interface, you can configure a static IPv4 or IPv6 address, or you can configure the interface to receive its address from a DHCP server. To enable dual stack, configure both an IPv4 and an IPv6 address on the tunnel interface.

From Cisco IOS XE Catalyst SD-WAN Release 17.3.2, Cisco IOS XE Catalyst SD-WAN devices do not support dual stack on the same TLOC or interface. Only one address type can be provisioned for a TLOC or interface. Using a second address type requires a second TLOC or interface on which it can be provisioned.

On Cisco Catalyst SD-WAN Controllers and Cisco Catalyst SD-WAN Controller NMSs, *interface-name* can be either **eth number** or **loopback number**. Because Cisco Catalyst SD-WAN Controllers and Cisco Catalyst SD-WAN Controller NMSs participate only in the overlay network's control plane, the VPNs that you can configure on these devices are VPN 0 and VPN 512. Hence, all interfaces are present only on these VPNs.

To enable the interface, include the **no shutdown** command.

Color is a Cisco Catalyst SD-WAN software construct that identifies the transport tunnel. It can be **3g**, **biz-internet**, **blue**, **bronze**, **custom1**, **custom2**, **custom3**, **default**, **gold**, **green**, **lte**, **metro-ethernet**, **mpls**, **private1** through **private6**, **public-internet**, **red**, and **silver**. The colors **metro-ethernet**, **mpls**, and **private1** through **private6** are referred to as *private colors*, because they use private addresses to connect to the remote side Cisco IOS XE Catalyst SD-WAN device in a private network. You can use these colors in a public network provided that there is no NAT device between the local and remote Cisco IOS XE Catalyst SD-WAN devices.

To limit the remote TLOCs that the local TLOC can establish BFD sessions with, mark the TLOC with the **restrict** option. When a TLOC is marked as restricted, a TLOC on the local router establishes tunnel connections with a remote TLOC only if the remote TLOC has the same color.

**Note**

When a WAN edge device is configured with two IPv6 TLOCs, one with static default route and the other one with IPv6 address autoconfig default which is the IPv6 neighbor discovery default route, the IPv6 neighbor discovery default route is not installed in the routing table. In this case, the IPv6 TLOC with IPv6 neighbor discovery default route does not work.

For IPv6 TLOC with IPv6 neighbor discovery default route to work, you can configure the static route for TLOC with IPv6 neighbor discovery to overwrite the IPv6 neighbor discovery default route and ensure that both the static routes are installed into the routing table. You can also use the IPv6 neighbor discovery default route on all interfaces.

On a Cisco Catalyst SD-WAN Controller or Cisco Catalyst SD-WAN Controller NMS, you can configure one tunnel interface. On a Cisco IOS XE Catalyst SD-WAN device, you can configure up to eight tunnel interfaces.

On Cisco IOS XE Catalyst SD-WAN devices, you must configure the tunnel encapsulation. The encapsulation can be either IPsec or GRE. For IPsec encapsulation, the default MTU is 1442 bytes, and for GRE it is 1468 bytes. These values are a function of overhead required for BFD path MTU discovery, which is enabled by default on all TLOCs. (For more information, see *Configuring Control Plane and Data Plane High Availability Parameters*.) You can configure both IPsec and GRE encapsulation by including two **encapsulation** commands under the same **tunnel-interface** command. On the remote Cisco IOS XE Catalyst SD-WAN device, you must configure the same tunnel encapsulation type or types so that the two routers can exchange data traffic. Data transmitted out of an IPsec tunnel can be received only by an IPsec tunnel, and data sent on a GRE tunnel can be received only by a GRE tunnel. The Cisco Catalyst SD-WAN software automatically selects the correct tunnel on the destination Cisco IOS XE Catalyst SD-WAN device.

A tunnel interface allows only DTLS, TLS, and, for Cisco IOS XE Catalyst SD-WAN devices, IPsec traffic to pass through the tunnel. To allow additional traffic to pass without having to create explicit policies or access lists, enable them by including one **allow-service** command for each service. You can also explicitly disallow services by including the **no allow-service** command. Note that services affect only physical interfaces. You can allow or disallow these services on a tunnel interface:

Service	Cisco Catalyst SD-WAN Controller	Cisco Catalyst SD-WAN Controller
all (Overrides any commands that allow or disallow individual services)	X	X
bgp	—	—
dhcp (for DHCPv4 and DHCPv6)	—	—
dns	—	—
https	X	—
icmp	X	X
netconf	X	—
ntp	—	—
ospf	—	—
sshd	X	X
stun	X	X

The **allow-service stun** command pertains to allowing or disallowing a Cisco IOS XE Catalyst SD-WAN device to generate requests to a generic STUN server so that the device can determine whether it is behind a NAT and, if so, what kind of NAT it is and what the device's public IP address and public port number are. On a Cisco IOS XE Catalyst SD-WAN device that is behind a NAT, you can also have tunnel interface to discover its public IP address and port number from the Cisco Catalyst SD-WAN Validator.

With this configuration, the Cisco IOS XE Catalyst SD-WAN device uses the Cisco Catalyst SD-WAN Validator as a STUN server, so the router can determine its public IP address and public port number. (With this configuration, the router cannot learn the type of NAT that it is behind.) No overlay network control traffic is sent and no keys are exchanged over tunnel interface configured to the the Cisco Catalyst SD-WAN Validator as a STUN server. However, BFD does come up on the tunnel, and data traffic can be sent on it. Because no control traffic is sent over a tunnel interface that is configured to use the Cisco Catalyst SD-WAN Validator as a STUN server, you must configure at least one other tunnel interface on the Cisco IOS XE Catalyst SD-WAN device so that it can exchange control traffic with the Cisco Catalyst SD-WAN Controller and the Cisco Catalyst SD-WAN Controller NMS.

You can log the headers of all packets that are dropped because they do not match a service configured with an **allow-service** command. You can use these logs for security purposes, for example, to monitor the flows that are being directed to a WAN interface and to determine, in the case of a DDoS attack, which IP addresses to block.

TLOC Extension

There are scenarios when Cisco IOS XE Catalyst SD-WAN devices cannot connect to a single transport directly and only one device can connect to a single transport. A switch is connected to each transport and the devices connect to each transport through the switches. To have a set-up with the switch option at a branch increases the cost of the solution and result in managing another device. TLOC extension enables a device to access the opposite WAN transport connected to the neighbouring device using a TLOC-extension interface.

TLOC Extension Over IPv6

Table 147: Feature History

Feature Name	Release Information	Description
TLOC Extension Over IPv6	Cisco IOS XE Catalyst SD-WAN Release 17.11.1a Cisco vManage Release 20.11.1	This feature enables the support of TLOC extension for IPv6. In the previous releases, TLOC extension was supported only for IPv4.

Information About TLOC Extension Over IPv6

In the earlier releases, TLOC extension was supported only over IPv4 interfaces.

This feature supports the following requirements:

- TLOC extension over IPv6 works only if the underlay supports IPv6 addressing on both the Cisco IOS XE Catalyst SD-WAN devices connecting each other.
- Implicit IPv6 ACL on TLOC tunnel interface is supported.
- IPv6 TLOC has dual stack support. When both IPv4 and IPv6 are configured, the tunnel is built on top of either IPv4 or IPv6, based on the configuration.
- TLOC interface supports NAT66. The limitations of NAT66 also applies to the TLOC extended interface.
- The following interface types supports IPv6 TLOC extension:
 - Physical interface
 - Physical sub-interface
 - Loopback interface

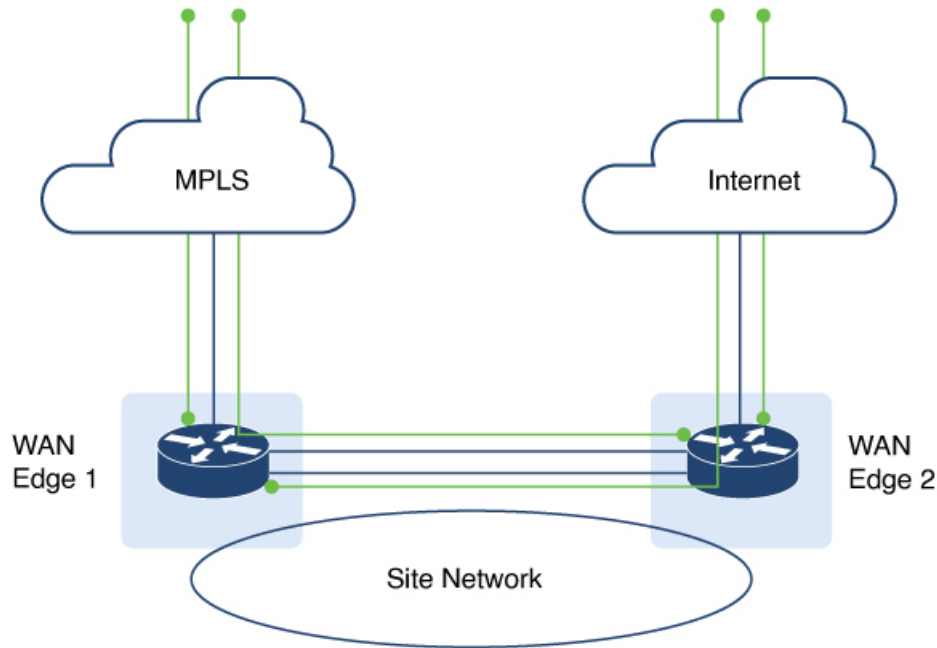


Note Only the Layer 2 setup supports IPv6 TLOC extension.

- This feature is supported for both private and public color TLOC interfaces.
- This feature supports the loopback TLOC interface that is bound to either:
 - The WAN transport circuit.
 - An extended WAN interface between two Cisco IOS XE Catalyst SD-WAN devices.

Use Case for TLOC Over IPv6 Extension

Figure 1: TLOC Extension



The TLOC extension allows each Cisco IOS XE Catalyst SD-WAN device to access the opposite transport through a TLOC-extension interface on the neighboring SD-WAN device. In the diagram, SD-WAN device 1 can access the internet through the SD-WAN device 2 TLOC extension interface in addition to the direct MPLS connection. SD-WAN device 2 can access the MPLS transport through the SD-WAN device 1 TLOC extension interface in addition to the direct internet connection. TLOC extension over IPv6 achieves redundancy in a dual-device deployment scenario with only one circuit connection on each device.

Limitations for TLOC Extension Over IPv6

- SIG is not supported on the IPv6 TLOC extension.
- NAT64 is not supported for IPv6 TLOC extension.
- TLOC extension over IPv6 is not supported for Layer 3 connections.

When a TLOC configuration is extended to a peer interface and then to ISP, the extended control connections are still up on the peer interface, even after removing TLOC Extension configuration.

In TLOC-Extension, the extender interface is part of the Cisco Catalyst SD-WAN. However, the tunnel-interface configuration under the extender interface is optional.

Configure TLOC Extension

1. Enter global configuration mode, and configure an interface.

```
Device# config-transaction
```

2. Enter SD-WAN configuration mode.


```
Device(config)# sdwan
```

3. In the SD-WAN configuration mode, configure an interface type such as, Gigabit Ethernet.

```
Device(config-sdwan)# interface GigabitEthernet3
```

4. Configure tunnel interface.

```
Device(config-interface-GigabitEthernet3)# tunnel-interface
```

5. Configure encapsulation, color, allowed services for TLOC.

```
Device(config-interface-GigabitEthernet3)# tunnel-interface
Device(config-interface-GigabitEthernet3)# encapsulation ipsec
Device(config-interface-GigabitEthernet3)# color color
Device(config-interface-GigabitEthernet3)# exit
```

6. In the global configuration mode, configure an interface.

```
Device# config-transaction
Device(config)# ip route 0.0.0.0 0.0.0.0 ip-address
```

7. On device 2, the LTE WAN connection is on GigabitEthernet1 and this transport is extended to device 1 GigabitEthernet3 TLOC interface.

```
Device(config-sdwan)# tloc-extension GigabitEthernet1
```

8. Configure NAT routes on GigabitEthernet1 for data traffic to reach back to device 1 through device 2 for GigabitEthernet3 subnet.

The following example describes how TLOC extension is configured on a network interface.

On Device1,
Configure TLOC interface on VPN 0
sdwan

```
interface GigabitEthernet3
  tunnel-interface
  encapsulation ipsec
  color custom1
  no allow-service bgp
  allow-service dhcp
  allow-service dns
  allow-service icmp
  no allow-service sshd
  no allow-service netconf
  no allow-service ntp
  no allow-service ospf
  no allow-service stun
  allow-service https
  no allow-service snmp
  no allow-service bfd
exit
```

Configure default route via this TLOC interface with nexthop to L2 connected interface of the peer (ED2 Gig3).

```
ip route 0.0.0.0 0.0.0.0 10.1.19.16
```

On Device2,
LTE WAN connection is on Gig1 and this transport is extended to ED1 Gig3 TLOC interface(custom1).
sdwan
int GigabitEthernet3
tloc-extension GigabitEthernet1

Configure NAT routes on Gig1 or appropriate routes for data traffic to reach back to ED1 via ED2 for Gig3 subnet.

Verify TLOC Extension

The following is a sample output of the commands to verify if TLOC extension is configured on a network interface.

```
Device# show sdwan control connections
PEER
CONTROLLER
PEER PEER PEER SITE DOMAIN PEER PRIV
PEER
PUB
TYPE PROT SYSTEM IP ID ID GROUP PRIVATE IP PORT
PUBLIC IP
PORT ORGANIZATION LOCAL COLOR PROXY STATE UPTIME ID
-----
vsmart dtls 172.16.255.19 100 1 2001:a0:5::13
12455 2001:a0:5::13 12455 vIPtela Inc Regression custom1
No up
0:01:23:06 0
vsmart dtls 172.16.255.20 200 1 2001:a0:c::14 12456
2001:a0:c::14 12456 vIPtela Inc Regression custom1
No up
0:01:23:06 0

Device# show sdwan bfd sessions
DST PUBLIC SOURCE TLOC REMOTE TLOC
SYSTEM IP SITE ID DST PUBLIC DETECT TX
IP STATE COLOR COLOR SOURCE IP
UPTIME PORT ENCAP MULTIPLIER INTERVAL(msec)
TRANSITIONS
-----
172.16.255.14 400 up custom1 lte 2001:a0:15::10
2001:a1:e::e 12346 ipsec 7 1000
0:00:05:50 3
```

Configure the System Interface

For each Cisco IOS XE Catalyst SD-WAN device, you configure a system interface with the **system system-ip** command. The system interface's IP address is a persistent address that identifies the Cisco IOS XE Catalyst SD-WAN device. It is similar to a router ID on a regular router, which is the address used to identify the router from which packets originated.

Specify the system IP address as an IPv4 address in decimal four-part dotted notation. Specify just the address; the prefix length (/32) is implicit.

The system IP address can be any IPv4 address except for 0.0.0.0/8, 127.0.0.0/8, and 224.0.0.0/4, and 240.0.0.0/4 and later. Each device in the overlay network must have a unique system IP address. You cannot use this same address for another interface in VPN 0.

The system interface is placed in VPN 0, as a loopback interface named **system**. Note that this is not the same as a loopback address that you configure for an interface.

To display information about the system interface, use the **show interface** command. For example:

The system IP address is used as one of the attributes of the OMP TLOC. Each TLOC is uniquely identified by a 3-tuple comprising the system IP address, a color, and an encapsulation. To display TLOC information, use the **show omp tlocs** command.

For device management purposes, it is recommended as a best practice that you also configure the same system IP address on a loopback interface that is located in a service-side VPN that is an appropriate VPN for management purposes. You use a loopback interface because it is always reachable when the router is operational and when the overlay network is up. If you were to configure the system IP address on a physical interface, both the router and the interface would have to be up for the router to be reachable. You use a service-side VPN because it is reachable from the data center. Service-side VPNs are VPNs other than VPN 0 (the WAN transport VPN) and VPN 512 (the management VPN), and they are used to route data traffic.

Configure Control Plane High Availability

A highly available Cisco Catalyst SD-WAN network contains two or more Cisco Catalyst SD-WAN Controllers in each domain. A Cisco Catalyst SD-WAN domain can have up to eight Cisco Catalyst SD-WAN Controllers, and each Cisco IOS XE Catalyst SD-WAN device, by default, connects to two of them. You change this value on a per-tunnel basis:

Configure Other Interfaces

Configure Interfaces in the Management (VRF mgmt-intf)

On all Cisco Catalyst SD-WAN devices, VPN 512 is used for out-of-band management, by default as part of the factory-default configuration. On Cisco IOS XE Catalyst SD-WAN devices the management VPN is converted to VRF Mgmt-Intf.

Cisco XE SD-WAN devices use VRFs in place of VPNs.

```
Device# show sdwan running-config | sec vrf definition Mgmt-intf
```

```
vrf definition Mgmt-intf
  address-family ipv4
    exit-address-family
  !
  address-family ipv6
    exit-address-family
  !
=====
interface GigabitEthernet0
  no shutdown
  vrf forwarding Mgmt-intf
  negotiation auto
exit
=====
config-t
ip route vrf Mgmt-intf 0.0.0.0 0.0.0.0

vrf definition Mgmt-intf
  rd 1:512
  !
  address-family ipv4
    route-target export 1:512
    route-target import 1:512
  exit-address-family
```

```

!
address-family ipv6
exit-address-family
!
!
interface GigabitEthernet1
vrf forwarding Mgmt-intf
ip address 192.168.20.11 255.255.255.0
!
ip route vrf Mgmt-intf 0.0.0.0 0.0.0.0
!

```

To display information about the configured management interfaces, use the **show interface** command. For example:

```

Device# show interface gigabitEthernet0
GigabitEthernet0 is up, line protocol is up
  Hardware is RP management port, address is d478.9bfe.9f7f (bia d478.9bfe.9f7f)
  Internet address is 10.34.9.177/16
  MTU 1500 bytes, BW 1000000 Kbit/sec, DLY 10 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full Duplex, 1000Mbps, link type is auto, media type is RJ45
  output flow-control is unsupported, input flow-control is unsupported
  ARP type: ARPA, ARP Timeout 04:00:00
  Last input 00:00:00, output 00:00:00, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/40 (size/max)
  5 minute input rate 8000 bits/sec, 12 packets/sec
  5 minute output rate 1000 bits/sec, 2 packets/sec
    4839793 packets input, 415574814 bytes, 0 no buffer
    Received 3060073 broadcasts (0 IP multicasts)
    0 runs, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
    0 watchdog, 0 multicast, 0 pause input
    82246 packets output, 41970224 bytes, 0 underruns
    Output 0 broadcasts (0 IP multicasts)
    0 output errors, 0 collisions, 0 interface resets
    0 unknown protocol drops
    0 babbles, 0 late collision, 0 deferred
    0 lost carrier, 0 no carrier, 0 pause output
    0 output buffer failures, 0 output buffers swapped out

```



Note VPN 512 is not advertised in the overlay. It is local to the device. If you need a management VPN that is reachable through the overlay, create a VPN with a number other than 512.

Configure Loopback Interfaces

Use the interface name format **loopback string**, where *string* can be any alphanumeric value and can include underscores (_) and hyphens (-). The total interface name, including the string "loopback", can be a maximum of 16 characters long. (Note that because of the flexibility of interface naming in the CLI, the interfaces **lo0** and **loopback0** are parsed as different strings and as such are not interchangeable. For the CLI to recognize as interface as a loopback interface, its name must start with the full string **loopback**.)

One special use of loopback interfaces is to configure data traffic exchange across private WANs, such as MPLS or metro Ethernet networks. To allow a router that is behind a private network to communicate directly over the private WAN with other edge routers, you direct data traffic to a loopback interface that is configured as a tunnel interface rather than to an actual physical WAN interface.

Implicit ACL on Loopback Interfaces

Table 148: Feature History

Feature Name	Release Information	Description
Implicit ACL on Loopback Interfaces	Cisco IOS XE Catalyst SD-WAN Release 17.6.1a Cisco vManage Release 20.6.1	This feature allows you to enable implicit ACL on loopback TLOC interfaces. When a loopback TLOC interface has its own implicit ACL, ACL rules are applied on the traffic destined for the interface. With implicit ACL enabled on the loopback TLOC interface, only limited services can be allowed, thereby enhancing your network security. When a loopback TLOC interface is bound to a physical interface on a Cisco IOS XE Catalyst SD-WAN device, the physical interface is treated like a physical TLOC interface.

Information About Implicit ACL on Loopback Interfaces

Access lists that you configure using localized data policy are called Explicit ACLs. Router tunnel interfaces also have implicit ACLs, which are also referred to as Services. Some of these are present by default on the tunnel interface, and they are in effect until you disable them. Through configuration, you can also enable other implicit ACLs. On Cisco IOS XE Catalyst SD-WAN devices, the following services are enabled by default: DHCP, DNS, and ICMP. You can also enable services for BGP, Netconf, NTP, OSPF, SSHD, and STUN.

You can configure and modify implicit ACLs with the **allow-service** command to allow a service. Use the **no allow-service** command to disallow a service. If both implicit ACL and explicit ACL are configured, explicit ACL takes precedence over the implicit ACL.

When Cisco IOS XE Catalyst SD-WAN device loopback interfaces are configured with a Transport Location (TLOC), implicit ACL rules are applied to the traffic destined for it. Implicit ACL on loopback interfaces are applied both in a bind mode and in an unbind mode. A bind mode is where a loopback interface is bound to a physical interface on a Cisco IOS XE Catalyst SD-WAN device to send data. In an unbind mode, a loopback interface is not bound to any physical interface.

Loopback TLOC Interface Bound to a Physical WAN Interface

When a loopback interface is a TLOC and is bound to a physical WAN interface, the corresponding implicit ACL rules are applied based on where the traffic is destined:

- If the traffic that is destined to the loopback TLOC interface is received on a physical WAN interface, the implicit ACL rules configured on the loopback TLOC interface is applied.
- If the traffic is not destined for the loopback TLOC interface, depending on whether the physical WAN interface is configured for TLOC or not, the following rules apply:
 - If the physical WAN interface is not configured with a TLOC, then routing decisions apply.



Note Use this command **implicit-acl-on-bind-intf** to enable implicit ACL protection on a physical interface in cases where a physical interface is not configured with a TLOC and bound to the loopback TLOC interface.

Forwarded or passthrough packets are dropped when a loopback TLOC interface is bound to a physical WAN interface—the same behavior as when a physical interface is configured as a TLOC. Therefore, explicit ACL must be configured on the bound physical interface to forward packets.

An explicit ACL is necessary to allow passthrough packets in the following sample scenarios:

- **Branch edge routers accessing controllers hosted in on-premises data centers:** This scenario presumes that the branch edge routers access the controllers through the data center hub, which is configured with a loopback interface bound to a physical WAN interface.
- **Branch routers accessing cloud-hosted controllers through data center internet circuits:** This scenario presumes that the branch routers are connected to the data center edge using an MPLS network. Such branch routers then access the cloud-hosted controllers through the data center edge router, which is configured with a loopback interface bound to a physical WAN interface.

-
- If a physical WAN interface is configured with TLOC, implicit ACL rules of the physical TLOC interface apply. In both these scenarios explicit ACLs on the bound physical WAN interface are necessary to allow passthrough traffic.

Loopback TLOC Interface Not Bound to a Physical WAN Interface

When a loopback interface is a TLOC, and is not bound to a physical WAN interface, implicit ACL rules are applied based on where the traffic is destined for:

- If the traffic that is destined for the loopback TLOC interface is received on a physical WAN interface, implicit ACL rules of the loopback TLOC are applied.
- If the traffic is not destined for the loopback TLOC interface, depending on whether the input physical WAN interface is configured for TLOC or not, the following rules apply:
 - If the physical WAN interface is not configured for TLOC, then routing decisions apply.

- If the physical WAN interface is configured for TLOC, the configured implicit ACL rules apply.

The difference between the bind mode and the unbind mode for loopback TLOC is that in a bind mode the passthrough traffic is dropped because the bound physical interface is treated as a TLOC by itself. In an unbind mode, the passthrough traffic is allowed.

Example Using Bind Mode and Unbind Mode

Bind Mode

A Cisco IOS XE Catalyst SD-WAN device has Loopback1 and Loopback2 configured as TLOCs and bound to the physical interface GigabitEthernet1. The device also has another interface, Loopback3, which is not configured as a TLOC.

Physical interface GigabitEthernet1 will be treated as a TLOC interface for incoming VPN 0.

To enable implicit ACL protection on physical interface GigabitEthernet1 for incoming VPN 0 traffic use the command **implicit-acl-on-bind-intf**.

In this example:

- If the traffic is destined for Loopback1, implicit ACL rules of Loopback1 are applied.
- If the traffic is destined for Loopback2, implicit ACL rules of Loopback2 are applied.
- If the traffic is destined for Loopback3 on GigabitEthernet1, traffic is allowed.
- If the traffic is destined for another device passing through GigabitEthernet1, it is dropped.

If the bound interface, GigabitEthernet1, is also configured as a TLOC, the traffic to Loopback3 will be subjected to implicit ACL rules on GigabitEthernet1.

Unbind Mode

A Cisco IOS XE Catalyst SD-WAN device has Loopback1 configured as a TLOC and is in unbind mode. Loopback2 is not configured as a TLOC. The device also has GigabitEthernet1 interface, which is configured as a TLOC, and GigabitEthernet4 interface, which is not configured as a TLOC.

In this example:

- If the traffic destined for Loopback1 arrives at GigabitEthernet1, the Loopback1 implicit ACL rules are applied. If the traffic is destined for GigabitEthernet1, the GigabitEthernet1 implicit ACL rules are applied.
- If the traffic destined for Loopback1 arrives at GigabitEthernet4, the Loopback1 implicit ACL rules are applied. If the traffic is destined for GigabitEthernet4, traffic is allowed.
- If the traffic destined for Loopback2 arrives on GigabitEthernet1, the GigabitEthernet1 implicit ACL rules are applied. If the traffic is destined for another device passing through GigabitEthernet1, it is dropped.

If the traffic is destined for another device passing through GigabitEthernet4, the traffic is forwarded.

Benefits of Implicit ACL on Loopback Interfaces

Implicit ACL on a loopback TLOC interface protects against denial of service (DoS) attacks by allowing only limited services. This enhances your network security.

Configure Implicit ACL on Loopback Interfaces

Similar to configuring physical WAN interfaces, you can configure implicit ACL on loopback interfaces using a feature template or using a CLI Add-on template in Cisco SD-WAN Manager.

For information about using a feature template to configure implicit ACL on loopback interfaces, see [Configure VPN Ethernet Interface](#).

For information on using the CLI Add-On template, see [Create a CLI Add-On Feature Template](#).

Configure Implicit ACL on Loopback Interfaces Using CLI

By default DNS, DHCP, ICMP and HTTPS services are permitted, and other services are denied.

To permit all the services, use the **allow-service all** command.



Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.16.1a, configuring **allow-service all** on Cisco Catalyst SD-WAN Controllers is only applicable for the following services **bgp**, **dhcp**, **dns**, **https**, **icmp**, **netconf**, **ntp**, **ospf**, **sshd**, and **stun**.

To permit a specific service, use the **allow-service service name** command.

To deny a service, use the **no allow-service service name** command.

Example

The following example shows implicit ACL configured on a loopback interface.

```
sdwan interface Loopback100
  tunnel-interface
    no allow-service bgp
    allow-service dhcp
    allow-service dns
    allow-service icmp
    no allow-service sshd
    no allow-service netconf
    no allow-service ntp
    no allow-service ospf
    no allow-service stun
    allow-service https
  exit
```

Configuration Examples for Implicit ACL Configured on a Loopback Interface in Bind Mode with TLOC Configured

This example shows implicit ACL configured on a loopback interface in bind mode with TLOC configured:

```
Device(config)# sdwan interface Loopback1
Device (config-interface-Loopback1)# tunnel-interface
Device (config-tunnel-interface)# encap ipsec
Device (config-tunnel-interface)# color 3g
Device (config-tunnel-interface)# bind GigabitEthernet1
Device (config-tunnel-interface)# implicit-acl-on-bind-intf
Device (config-tunnel-interface)# no allow-service bgp
Device (config-tunnel-interface)# allow-service dhcp
Device (config-tunnel-interface)# allow-service dns
Device (config-tunnel-interface)# allow-service icmp
```



```

Device (config-tunnel-interface)# no allow-service sshd
Device (config-tunnel-interface)# no allow-service netconf
Device (config-tunnel-interface)# no allow-service ntp
Device (config-tunnel-interface)# no allow-service ospf
Device (config-tunnel-interface)# no allow-service stun
Device (config-tunnel-interface)# allow-service https
Device (config-tunnel-interface)# no allow-service snmp
Device (config-tunnel-interface)# no allow-service bfd
Device (config-tunnel-interface)# exit

```

Configuration Examples for Implicit ACL Configured on a Loopback Interface in unbind Mode with TLOC Configured

This example shows implicit ACL configured on a loopback interface in unbind mode with TLOC configured:

```

Device (config)# sdwan interface Loopback1
Device (config-interface-Loopback1)# tunnel-interface
Device (config-tunnel-interface)# encaps ipsec
Device (config-tunnel-interface)# color 3g
Device (config-tunnel-interface)# no allow-service bgp
Device (config-tunnel-interface)# allow-service dhcp
Device (config-tunnel-interface)# allow-service dns
Device (config-tunnel-interface)# allow-service icmp
Device (config-tunnel-interface)# no allow-service sshd
Device (config-tunnel-interface)# no allow-service netconf
Device (config-tunnel-interface)# no allow-service ntp
Device (config-tunnel-interface)# no allow-service ospf
Device (config-tunnel-interface)# no allow-service stun
Device (config-tunnel-interface)# allow-service https
Device (config-tunnel-interface)# no allow-service snmp
Device (config-tunnel-interface)# no allow-service bfd
Device (config-tunnel-interface)# exit

```

Monitor Implicit ACL on Loopback Interfaces

Use the **show platform hardware qfp active statistics drop** command to monitor implicit ACL configuration on loopback interfaces.

Example

The following is a sample output from the **show platform hardware qfp active statistics drop** command:

```

Device# show platform hardware qfp active statistics drop
Last clearing of QFP drops statistics : never

```

```

-----
Global Drop Stats                               Packets                               Octets
-----
Disabled                                         4                                           266
Ipv4EgressIntfEnforce                           15                                          10968
Ipv6NoRoute                                     6                                           336
Nat64v6tov4                                    6                                           480

```

SVIInputInvalidMac	244	15886
SdwanImplicitAclDrop	160	27163
UnconfiguredIpv4Fia	942525	58524580
UnconfiguredIpv6Fia	77521	9587636

Configure Subinterfaces

When you create a subinterface that does not specify an IP MTU value, the subinterface inherits the IP MTU value from the parent interface. If you want the subinterface to have a different IP MTU value, use the **ip mtu** command in the subinterface configuration to set the IP MTU for the sub interface.

For example:

```
interface GigabitEthernet0/0/0
  mtu 1504
  no ip address
  !
interface GigabitEthernet0/0/0.9
  encapsulation dot1Q 9
  no shutdown
  ip address 192.168.9.32 255.255.255.0
  !
interface Tunnel9
  no shutdown
  ip unnumbered GigabitEthernet0/0/0.9
  no ip redirects
  ipv6 unnumbered GigabitEthernet0/0/0.9
  no ipv6 redirects
  tunnel source GigabitEthernet0/0/0.9
  tunnel mode sdwan
  !
sdwan
  interface GigabitEthernet0/0/0.9
    tunnel-interface
    encapsulation ipsec
    color private1
  !
  !
```

Configure Interface Properties

Set the Interface Speed

When a Cisco IOS XE Catalyst SD-WAN device comes up, the Cisco Catalyst SD-WAN software autodetects the SFPs present in the router and sets the interface speed accordingly. The software then negotiates the interface speed with the device at the remote end of the connection to establish the actual speed of the interface. To display the hardware present in the router, use the **show hardware inventory** command:

To display the actual speed of each interface, use the **show interface** command. Here, interface **ge0/0**, which connects to the WAN cloud, is running at 1000 Mbps (1Gbps; it is the 1GE PIM highlighted in the output above), and interface **ge0/1**, which connects to a device at the local site, has negotiated a speed of 100 Mbps.

For non-physical interfaces, such as those for the system IP address and loopback interfaces, the interface speed is set by default to 10 Mbps.

To override the speed negotiated by the two devices on the interface, disable autonegotiation and configure the desired speed:

For Cisco Catalyst SD-WAN Controllers and Cisco SD-WAN Manager systems, the initial interface speeds are 1000 Mbps, and the operating speed is negotiated with the device at the remote end of the interface. The controller interface speed may vary depending upon the virtualization platform, the NIC used, and the drivers that are present in the software.

Set the Interface MTU

By default, all interfaces have an MTU of 1500 bytes. You can modify this on an interface:

For releases earlier than Cisco IOS XE Catalyst SD-WAN Release 17.4.1a the MTU can range from 576 through 2000 bytes.

Starting from release Cisco IOS XE Catalyst SD-WAN Release 17.4.1a the MTU can range from 576 through 9216 bytes on 1 GE interfaces. This MTU range is also supported on 10 GE and 100 GE interfaces starting from Cisco IOS XE Catalyst SD-WAN Release 17.5.1a.

To display an interface's MTU, use the **show interface** command.

For Cisco Catalyst SD-WAN Validator, Cisco SD-WAN Manager, and Cisco Catalyst SD-WAN Controller devices, you can configure interfaces to use ICMP to perform path MTU (PMTU) discovery. When PMTU discovery is enabled, the device automatically negotiates the largest MTU size that the interface supports in an attempt to minimize or eliminate packet fragmentation:

On Cisco IOS XE Catalyst SD-WAN device, the Cisco Catalyst SD-WAN BFD software automatically performs PMTU discovery on each transport connection (that is, for each TLOC, or color). BFD PMTU discovery is enabled by default, and it is recommended that you use it and not disable it. To explicitly configure BFD to perform PMTU discovery, use the **bfd color pmtu-discovery** configuration command. However, you can choose to instead use ICMP to perform PMTU discovery: vEdge Cloud router

BFD is a data plane protocol and so does not run on Cisco Catalyst SD-WAN Validator, Cisco SD-WAN Manager, and Cisco Catalyst SD-WAN Controller devices.

VFR and Underlay Fragmentation

Table 149: Feature History

Feature Name	Release Information	Description
VFR and Underlay Fragmentation	Cisco IOS XE Catalyst SD-WAN Release 17.12.1a Cisco Catalyst SD-WAN Manager Release 20.12.1	In Cisco Catalyst SD-WAN networks, the VFR (Virtual Fragmentation Reassembly) actively fragments and reassembles packets. The packets undergo fragmentation to improve transportation efficiency while passing through a VFR-enabled Cisco IOS XE Catalyst SD-WAN device. The VFR reassembles the fragmented packets to match the original incoming packet. The reassembled packet contains critical Layer 4 or Layer 7 information necessary for proper reception by the destination device. Underlay fragmentation refers to the process of breaking down a large data packet into smaller fragments at the network layer. Underlay fragmentation allows the successful transmission of packets that exceed the MTU limitations by breaking them down into manageable fragments and ensuring their reliable delivery.

Information About VFR and Underlay Fragmentation

While transmitting data across a network, due to various network constraints, the original data packets fragment into smaller fragments to facilitate seamless transmission. While the packets travel through the Cisco IOS XE Catalyst SD-WAN device, they are fragmented. VFR allows fragmented packets to be reassembled efficiently before reaching their destination.

In Cisco Catalyst SD-WAN network, data packets undergo reassembly in two modes: the default mode and the reassembly mode.

In the default mode, packets are virtually reassembled by default. Upon the delivery of the first fragment, each feature in the network receives the entire payload of the virtually reassembled packet. When the last fragment is received, the remaining features reassemble the packet. The original packet is fragmented, and the internal fragment information structure is shared. The fragments are then queued for refragmentation based on the fragment-offset sequence. The VFR mechanism reconstructs the packets using information from the fragment headers, such as fragment identifiers, sequence numbers, and offsets.

On the other hand, in the reassembly mode, the packets undergo physical reassembly, and fragment header information isn't saved. Upon receiving the last fragment, the fragments reassemble via a metapacket, and the internal fragment information structure is released.

**Note**

- If the packets were originally fragmented using the default mode, they undergo reassembly as if they were the original incoming packets. On the other hand, when the reassembly mode is utilized to virtually fragment the packets, they experience fragmentation based on the MTU of the egress interface before reassembly.
- Some features (such as NAT, Cisco IOS XE Firewall, IPSec) automatically enable VFR to obtain Layer 4 or Layer 7 information.
- When a particular interface enables VFR, it overrides the existing firewall or NAT's VFR mode configuration by default, ensuring interoperability with the firewall or NAT.

Information About Underlay Fragmentation

Underlay fragmentation processes large data packets that exceed the MTU (Maximum Transmission Unit) size supported by the Cisco Catalyst SD-WAN network infrastructure. Each data packet has a maximum size that can transmit over the network without being fragmented. This maximum size is defined by the MTU. The process of breaking down a large data packet into smaller fragments at the network layer is known as underlay fragmentation. The underlay fragmentation enables the transmission of packets that exceed the MTU limitations by breaking them down into smaller fragments and ensuring their successful delivery.

Prerequisites For Configuring VFR and Underlay Fragmentation

The Maximum Transmission Unit (MTU) size needs to be properly configured on the network devices. The MTU defines the maximum size of a packet that can be transmitted without fragmentation. It is essential to ensure that the MTU is set appropriately on all devices involved in the network path to avoid underlay fragmentation unless it is intentionally desired.

Restrictions For Configuring VFR and Underlay Fragmentation

- The VFR process requires all fragments within an IP datagram. If fragments within an IP datagram are sent to different devices due to load balancing, VFR may fail and fragments may be dropped.
- VFR is designed to work with any feature that requires fragment reassembly (such as Cisco Catalyst SD-WAN NAT, and IPsec). By default, NAT, Crypto-based IPsec, and NAT64 enable and disable VFR internally; that is, when these features are enabled on an interface, VFR is enabled on that interface. If more than one feature attempts to enable VFR on an interface, VFR maintains a reference count to keep track of the number of features that have enabled VFR. When the reference count is zero, VFR is automatically disabled.
- The underlay fragmentation mechanism is limited to the network layer and is specific to the underlying network infrastructure. It does not handle fragmentation and reassembly across multiple network segments or end-to-end connections.

- If any of the fragments in a series of fragmented packets are lost or arrive out of order, the reassembly process may fail. This can result in incomplete or corrupted packets.
- The VFR CLIs are unavailable under port-channel sub-interfaces.

Benefits of VFR and Underlay Fragmentation

- VFR enables the Cisco IOS XE Firewall to create appropriate dynamic access control lists (ACLs) to protect the network from various fragmentation attacks.
- VFR is responsible for detecting and preventing various types of fragment attacks.
- VFR drops all fragments within a fragment chain if an overlap of a fragment is detected.

Use Cases For VFR and Underlay Fragmentation

Networks such as long-distance connections such as a connection between an airplane and airport signal towers, can experience interruptions, due to the time it takes for large packets to traverse these links. When VFR is enabled, the fragments will reassemble into a complete datagram, then are fragmented within the Cisco Catalyst SD-WAN tunnel interface. With this, the first fragment will be sent out first and there is no interruption in receiving the packets.

Underlay fragmentation helps in fragmenting large packets into smaller sizes, and reconstruct the packet back into the original one. This improves the overall application performance.

Enable Boost Mode

The boost mode helps in resolving one of the identified bottlenecks related to the memory management of fragments within the data plane of the network. Prior to Cisco IOS XE Catalyst SD-WAN Release 17.12.1a, the memory allocation to reassembly of fragments occurred from a global chunk, necessitating a lock in period for the memory until the reassembly is complete. This leads to potential competition among multiple threads for the same global chunk and results in waiting for the same memory. Starting from Cisco IOS XE Catalyst SD-WAN Release 17.12.1a, the boost mode enhances performance by utilizing CVLA, an alternative data plane memory infrastructure. Unlike the chunk mechanism, CVLA is lock-free and is an efficient memory management mechanism within Cisco IOS XE devices.



Note The boost mode is disabled by default on Cisco IOS XE Catalyst SD-WAN devices.

Enable Boost Mode Using a CLI Template

For more information about using CLI templates, see [CLI Add-On Feature Templates](#) and [CLI Templates](#).



Note By default, CLI templates execute commands in global config mode.

This section provides example CLI configurations to enable the boost mode.

1. Enable the boost mode:

platform ipreass boost-mode

Here is the complete configuration example to enable the boost mode:

```
platform ipreass boost-mode
```

Configure VFR Using a CLI Template

For more information about using CLI templates, see [CLI Add-On Feature Templates](#) and [CLI Templates](#).



Note By default, CLI templates execute commands in global config mode.

This section provides example CLI configurations to configure VFR.

Enable VFR for IPv4 packets on Inbound Interface Traffic

1. Configure an interface type and enter interface configuration mode:

```
interface interface-type interface-number
```

2. Enable VFR on the interface and specify the maximum threshold values:

```
ip virtual-reassembly [max-reassemblies number] [max-fragments number] [timeout seconds] [mode modes][drop-fragments]
```

Here is the complete configuration example to enable VFR for IPv4 packets:

```
interface GigabitEthernet5
ip virtual-reassembly max-reassemblies 64 max-fragments 16 mode default timeout 5
```

Enable VFR for IPv4 packets on Outbound Interface Traffic

1. Configure an interface type and enter interface configuration mode:

```
interface interface-type interface-number
```

2. Enable VFR for outbound interface traffic on the interface and specify the maximum threshold values:

```
ip virtual-reassembly-out [max-reassemblies number] [max-fragments number] [timeout seconds] [mode modes][drop-fragments]
```

Here is the complete configuration example to enable VFR for IPv4 packets:

```
interface GigabitEthernet 5
ip virtual-reassembly-out mode default max-fragments 64
```

Enable VFR for IPv6 packets on Inbound Interface Traffic

1. Configure an interface type and enter interface configuration mode:

```
interface interface-type interface-number
```

2. Enable VFR for IPv6 packets on inbound interface traffic

```
ipv6 virtual-reassembly [in | out][max-reassemblies number] [max-fragments number] [timeout seconds] [mode modes][drop-fragments]
```

Here is the complete configuration example to enable VFR for IPv6 packets:

```
interface GigabitEthernet 5
ipv6 virtual-reassembly in mode default max-fragments 25
max-reassemblies 1024
```

Enable VFR for IPv6 packets on Outbound Interface Traffic

1. Configure an interface type and enter interface configuration mode:

```
interface interface-type interface-number
```

2. Enable VFR for IPv6 packets on outbound interface traffic

```
ipv6 virtual-reassembly [in | out][max-reassemblies number ] [max-fragments number ] [timeout seconds ] [mode modes][drop-fragments ]
```

Here is the complete configuration example to enable VFR for IPv6 packets:

```
interface GigabitEthernet 5
ipv6 virtual-reassembly out mode default max-fragments 25
```

Configure Underlay Fragmentation Using a CLI Template

For more information about using CLI templates, see [CLI Add-On Feature Templates](#) and [CLI Templates](#).



Note By default, CLI templates execute commands in global config mode.

This section provides example CLI configurations to configure underlay fragmentation.

1. Enter the config-sdwan mode:
- ```
sdwan
```
2. Configure an interface type and enter interface configuration mode:

```
interface interface-name interface-number
```

3. Configure the tunnel interface:

```
tunnel-interface
```

4. Skip Layer 3 fragmentation and clear overlay DF bit:

```
inner-fragmentation-disable
```

5. Perform the encapsulation for the GRE interface of the TLOC:

```
encapsulation gre
```




---

**Note** Only GRE encapsulation is supported for underlay fragmentation in Cisco IOS XE Catalyst SD-WAN Release 17.12.1a.

---

Here is the complete configuration example to enable underlay fragmentation:

```
sdwan
interface GigabitEthernet1
tunnel-interface
inner-fragmentation-disable
encapsulation gre
```



## Configure Underlay Fragmentation Using Cisco SD-WAN Manager

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.15.1a

1. From the Cisco SD-WAN menu, choose **Configuration > Configuration Groups**.
2. Click **Transport & Management Profile**.
3. Select the desired transport profile and click **Edit**.
4. Click **Edit Ethernet Interface > Tunnel**.
5. Enable **Allow Fragmentation** and **MTU To Max**.
6. Click **Save**.

## Verify Boost Mode

The following is a sample output from the **show platform hardware qfp active infrastructure cvla client handles** command:

```
Device# show platform hardware qfp active infrastructure cvla client handles
Handles for cpp 0:
```

```

Entity name: IPREASS_CVLA_0
```

```
Handle: 0xeea45000
```

```
Number of allocations: 0
```

```
Memory allocated: 0
```

```
Entity name: FNF_AOR
```

```
Handle: 0xeea0d000
```

```
Number of allocations: 0
```

```
Memory allocated: 0
```

```
Entity name: NBAR_CVLA_ENTITY
```

```
Handle: 0xee946000
```

```
Number of allocations: 0
```

```
Memory allocated: 0
```

```
Entity name: FNF_Chunk 2
```

```
Handle: 0xef929000
```

```
Number of allocations: 0
```

```
Memory allocated: 0
```

```
Entity name: FNF Chunk 1
```

```
Handle: 0xef928000
```

```
Number of allocations: 0
```

```
Memory allocated: 0
```

**Note**

If there is no entity for **IPREASS\_CVLA\_\*** displayed, the boost mode is disabled. Once the boost mode is disabled, the **IPREASS\_CVLA\_\*** disappears after 64 seconds.

## Monitor VFR and Underlay Fragments Using the CLI

### Monitor VFR for IPv4 packets

The following is a sample output from the **show ip virtual-reassembly** command:

```
Device# show ip virtual-reassembly GigabitEthernet 5
GigabitEthernet5:
```

```
Virtual Fragment Reassembly (VFR) is ENABLED [out]
```

```
Concurrent reassemblies (max-reassemblies): 16
```

```
Fragments per reassembly (max-fragments): 32
```

```
Reassembly timeout (timeout): 3 seconds
```

```
Drop fragments: OFF
```

```
Current reassembly count:0
```

```
Current fragment count:0
```

```
Total reassembly count:12
```

```
Total reassembly timeout
```

The example shows if VFR for IPv4 is enabled or not. **Virtual Fragment Reassembly (VFR) is ENABLED [out]** signifies that VFR is enabled. The total packets that underwent reassembly are also displayed.

### Monitor VFR for IPv6 packets

The following is a sample output from the **show ipv6 virtual-reassembly** command:

```
Device# show ipv6 virtual-reassembly GigabitEthernet 5
GigabitEthernet5:
```

```
IPv6 Virtual Fragment Reassembly (IPv6VFR) is ENABLED [out]

IPv6 configured concurrent reassemblies (max-reassemblies): 64

IPv6 configured fragments per reassembly (max-fragments): 16

IPv6 configured reassembly timeout (timeout): 3 seconds

IPv6 configured drop fragments: OFF

IPv6 current reassembly count:0

IPv6 current fragment count:0

IPv6 total reassembly count:12

IPv6 total reassembly timeout count:0
```

The example shows if VFR for IPv6 is enabled or not. **Virtual Fragment Reassembly (VFR) is ENABLED [out]** signifies that VFR is enabled. The total packets that underwent reassembly are also displayed.

### Monitor Underlay Fragmentation

The following is a sample output from the **show ip traffic interface GigabitEthernet 1** command:

```
Device# show ip traffic interface GigabitEthernet 1
GigabitEthernet 1 statistics :
```

```
Rcvd: 11048818 total, 749458331 total_bytes

 0 format errors, 0 hop count exceeded

 0 bad header, 0 no route

 0 bad destination, 0 not a router

 0 no protocol, 0 truncated

 0 forwarded

 0 fragments, 0 total reassembled

 0 reassembly timeouts, 0 reassembly failures

 0 discards, 0 delivers

Sent: 0 total, 0 total_bytes 0 discards

 0 generated, 0 forwarded

 0 fragmented into, 0 fragments, 0 failed

Mcast: 0 received, 0 received bytes

 0 sent, 0 sent bytes
```

Bcast: 0 received, 1256 sent

The example shows the number of packets that were sent and received, including the total number of packets. A change from the previous number of packet transfer indicates that underlay fragmentation is enabled.

The following is a sample output from **show sdwan ftm tloc-list** command:

Device# **show sdwan ftm tloc-list**

--- LOCAL TLOC LIST ---

Id: 32775 (binosId=0xf808007f), Tenant Id: 0 LocalTLOC, num-nhops: 0 ,hash: 0, ref:  
1 SLA 0x0:0x0 Inner-fragmentation  
-disable: No

[TOTAL-LOCAL-TLOC:1]

--- REMOTE TLOC LIST ---

Id: 32768 (binosId=0xf808000f), Tenant Id: 0 SLAClass, num-nhops: 0 ,hash: 0, ref:  
1 SLA 0x0:0x0  
num-active-nhops: 0

Id: 32774 (binosId=0xf808006f), Tenant Id: 0 SLAClass, num-nhops: 1 ,hash: 0, ref:  
1 SLA 0x1:0x0  
[nhop1] nhop-Id: 19 , Type: IPsec , Encap: IPSEC SLA 0x1:0x0hw\_record\_index: 5  
198.100.1.5/12366->198.100.1.6/12346 pr  
oto 0x800 hash 0x13 wan-if 3 tloc 32774 R-color mpls local-tloc 32775 L-color mpls BFD UP  
tloc-capability 0 SLA 0x1:0x0 weight  
1 pref 0

num-active-nhops: 1

[TOTAL-REMOTE-TLOC:2]

--- PENDING TLOC LIST (is\_pending\_updates:FALSE)---

[TOTAL-PENDING-TLOC:0]

```
--- UNMATCHED TLOC LIST (is_pending_updates:FALSE)---
```

```
[TOTAL-UNMATCHED-TLOC:0]
```

```
--- TENANT LOCAL TLOC LIST ---
```

The example displays all the local TLOCs in the network.

The following is a sample output from **show platform software sdwan R0 next-hop overlay all** command:

```
Device# show platform software sdwan R0 next-hop overlay all
```

```
Show sdwan next-hop oce all :
```

```
OCE ID: 0xf800013f, OCE Type: SDWAN_NH_OVERLAY
```

```
Overlay: client_handle (nil), ppe addr (nil)
```

```
overlay encap: ipsec
```

```
src-ip: 198.100.1.5, src-port: 12366
```

```
dst-ip: 198.100.1.6, dst-port: 12346
```

```
flags: 0x0, linktype: MCP_LINK_IP, ifhandle: 15, encap type: MCP_ET_NULL
```

```
encap rewrite: 00
```

```
mtu: 1446, fixup: 0x0, fixup_flags_2: 0x0, color: mpls, phy_oce_handle: 31, nh_overlay_h:
0xf800013f
```

```
Overlay_CFG:
```

```
encap type: ipsec
```

```
src-ip: 198.100.1.5, src-port: 12366
```

```
dst-ip: 198.100.1.6, dst-port: 12346
```

```
local_system_ip: 1.1.1.1
```

```
remote_system_ip: 2.2.2.2
```

```
local_color: 2 [mpls], remote_color: 2 [mpls]
```

```
wan_ifindex: 8 [GigabitEthernet2], tun_ifindex: 15 [Tunnel0]
```

```
tun_adj_id: 0, l2_adj_id: 0x1f, tunnel_qos_dpidx: 0x0
```

```
bfd-ld: 20005, ipsec_flow_id: 603979786, session_id: 5
```

```
Inner-fragmentation-disable: yes
```

The example demonstrates whether the inner fragmentation is disabled or enabled in a particular next-hop overlay.

The following is a sample output from **show platform software sdwan F0 next-hop overlay all** command:

```
Device# show platform software sdwan F0 next-hop overlay all

OCE ID: 0xf800013f, OCE Type: SDWAN_NH_OVERLAY

Overlay: client_handle 0x63d321350ba0, ppe addr db910710

 overlay encap: ipsec

 src-ip: 198.100.1.5, src-port: 12366

 dst-ip: 198.100.1.6, dst-port: 12346

 flags: 0x0, linktype: MCP_LINK_SDWAN, ifhandle: 15, encap type: MCP_ET_ARPA

 encap rewrite: 00

 mtu: 1446, fixup: 0x0, fixup_flags_2: 0x800000, color: mpls, phy_oce_handle: 31,
nh_overlay_h: 0xf800013f
 Overlay_CFG:

 encap type: ipsec

 src-ip: 198.100.1.5, src-port: 12366

 dst-ip: 198.100.1.6, dst-port: 12346

 local_system_ip: 1.1.1.1

 remote_system_ip: 2.2.2.2

 local_color: 2 [mpls], remote_color: 2 [mpls]

 wan_ifindex: 8 [GigabitEthernet2], tun_ifindex: 15 [Tunnel0]

 tun_adj_id: 0, l2_adj_id: 0x1f, tunnel_qos_dpidx: 0x0

 bfd-ld: 20005, ipsec_flow_id: 603979786, session_id: 5

 Inner-fragmentation-disable: yes
```

The example demonstrates whether the inner fragmentation is disabled or enabled in all the available overlays.

## Configure TCP MSS and Clear Dont Fragment

**Table 150: Feature History**

| Feature Name      | Release Information                                                          | Description                                                                                                                                                |
|-------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Configure TCP MSS | Cisco IOS XE Catalyst SD-WAN Release 17.5.1a<br>Cisco vManage Release 20.5.1 | This feature adds support for TCP MSS adjustment on Cisco IOS XE Catalyst SD-WAN devices on both directions of the Cisco Catalyst SD-WAN tunnel interface. |

| Feature Name                          | Release Information                                                              | Description                                                                                                                                                                                                                                                                        |
|---------------------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Configure Clear Don't Fragment Option | Cisco IOS XE Catalyst SD-WAN Release 17.5.1a<br><br>Cisco vManage Release 20.5.1 | This feature provides the option to clear the Don't Fragment bit in the IPv4 packet header for packets being sent out on a Cisco Catalyst SD-WAN tunnel . When you clear the Don't Fragment configuration, packets larger than the interface MTU are fragmented before being sent. |

TCP maximum segment size (MSS) is a parameter that specifies the largest amount of data, in bytes, that a communications device can receive in a single TCP segment, without counting the TCP header or the IP header. The MSS is specified as TCP MSS, initially in the TCP SYN packet during TCP handshake. Small MSS values reduces or eliminates IP fragmentation resulting in higher overhead.

You can configure the MSS of TCP SYN packets passing through a device. By default, the MSS is dynamically adjusted based on the interface or tunnel maximum transmission unit (MTU) such that TCP SYN packets are never fragmented. For data sent over an interface, the MSS is calculated by adding the interface MTU, the IP header length, and the maximum TCP header length.

### Limitations

- TCP MSS values can be adjusted for Cisco Catalyst SD-WAN tunnel interfaces only.



**Note** Beginning with Cisco IOS XE Catalyst SD-WAN Release 17.9.1a and Cisco vManage Release 20.9.1, you can adjust the TCP MSS value for a service VPN or for Network Address Translation (NAT) Direct Internet Access (DIA) use cases. Adjusting the TCP MSS value helps prevent TCP sessions from being dropped.

For more information on NAT DIA, see the [Cisco Catalyst SD-WAN NAT Configuration Guide, Cisco IOS XE Catalyst SD-WAN Release 17.x](#).

- The option **Clear Dont Fragment** is available for Cisco Catalyst SD-WAN tunnel interfaces only.
- From Cisco IOS XE Catalyst SD-WAN Release 17.18.1a, for an IPv4 SD-WAN tunnel carrying IPv6 traffic, the system uses the IPv6 TCP MSS value. Also, for an IPv6 tunnel carrying IPv4 traffic, the system uses the IPv4 TCP MSS value. This upgrade aligns MSS adjustment with the traffic protocol and tunnel configuration, ensuring more accurate and efficient handling of traffic.

## Configure TCP MSS and Clear Dont Fragment

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**.



**Note** In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

3. Create a new CLI add-on feature template or edit one of the following templates. You can use any of the following feature templates to configure TCP MSS and clear Dont Fragment:

- [VPN Ethernet Interface](#)
- [VPN Interface DSL IPoE](#)
- [VPN Interface DSL PPoA](#)
- [VPN Interface DSL PPPoE](#)
- [VPN Interface Multilink](#)
- [VPN Interface T1/E1](#)
- [Cellular Interfaces](#)

For information on creating a new CLI add-on feature template, see [Create a CLI Add-on Feature Template](#).

4. Click **Tunnel**.
5. To configure TCP MSS, in **Tunnel TCP MSS**, specify the MSS of TPC SYN packets passing through the Cisco IOS XE Catalyst SD-WAN device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. *Range:* 552 to 1460 bytes  
*Default:* None  
  
TCP MSS affects any packet that contains an initial TCP header that flows through the router. When configured, TCP MSS is examined against the MSS exchanged in the three-way handshake. The MSS in the header is lowered if the configured setting is lower than the MSS in the header. If the MSS header value is already lower than the TCP MSS, it flows through unmodified. The host at the end of the tunnel uses the lower setting of the two hosts. If the TCP MSS is to be configured, it should be set at 40 bytes lower than the minimum path MTU.
6. Click the **Clear-Dont-Fragment** option to clear the Dont Fragment bit in the IPv4 packet header for packets being transmitted out the interface. When the Don't Fragment bit is cleared, packets larger than that interface's MTU are fragmented before being sent.




---

**Note** Clear-Dont-Fragment clears the Don't Fragment bit when there is fragmentation needed and the Don't Fragment bit is set. For packets that don't require fragmentation, the Don't Fragment bit is not affected.

---

7. Click **Save** or **Update**.

## Configure TCP MSS Using CLI

Use the following command to configure TCP MSS on the CLI:

```
Device(config)#interface Tunnel 1
Device(config-if)#ip unnumbered GigabitEthernet1
Device(config-if)#ip tcp adjust-mss 1460
```

### Verify TCP MSS Configuration

The following is sample output of the **show platform hardware qfp active feature sdwan datapath session summary** command:



```

Device#show platform hardware qfp active feature sdwan datapath session summary
Src IP Dst IP Src Port Dst Port Encap Uidb Bfd Discrim PMTU

10.1.15.25 10.1.14.14 12347 12346 IPSEC 65526 10007 1446
10.1.15.25 10.0.5.21 12347 12357 IPSEC 65526 10009 1446
10.1.15.25 10.0.5.11 12347 12347 IPSEC 65526 10008 1446
10.1.15.25 10.1.16.16 12347 12366 IPSEC 65526 10006 1446

```

## Configure Clear Dont Fragment on the CLI

Use the following command to configure **Clear Dont Fragment** option using the CLI:

```

Device(config)#interface Tunnel 1
Device(config-if)#ip unnumbered GigabitEthernet1
Device(config-if)#ip clear-dont-fragment

```

### Verify Dont Fragment Configuration on the CLI

The following is sample output of the **show platform software interface rp active name Tunnel1** command to verify if **Clear-dont-fragment** is enabled or not.

```

Device# show platform software interface rp active name Tunnel1 | include dont
IP Clear-dont-fragment: TRUE

```

The following is sample output of the **show running-config interface Tunnel1** command that displays the running configuration when **Clear-dont-fragment** is enabled.

```

Device# show running-config interface Tunnel1
Building configuration...

Current configuration : 132 bytes
!
interface Tunnel1
ip unnumbered GigabitEthernet1
ip clear-dont-fragment
tunnel source GigabitEthernet1
tunnel mode sdwan
end

```

# Configure Bandwidth Reference Values for Network Interfaces

*Table 151: Feature History*

| Feature Name                                       | Release Information                                                                                | Description                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Upstream and Downstream Bandwidth Reference Values | Cisco IOS XE Catalyst SD-WAN Release 17.16.1a<br><br>Cisco Catalyst SD-WAN Manager Release 20.16.1 | Use the upstream and downstream bandwidth reference values to govern how Cisco SD-WAN Manager displays interface utilization percentages in charts. The values also act as configurable thresholds that trigger interface-bw events when a network interface's utilization exceeds a defined point. |

## Information About Bandwidth Utilization Reference Values

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.16.1a and Cisco Catalyst SD-WAN Manager Release 20.16.1

You can configure bandwidth utilization reference values for each interface. These reference values are called upstream bandwidth and downstream bandwidth. You can configure different reference values for upstream (egress) traffic and downstream (ingress) traffic.

These values act as reference values for calculating bandwidth utilization and generating notifications. Cisco SD-WAN Manager uses these reference values to display bandwidth utilization percentages in charts.

### Information About Generating Notifications

To receive event notifications when the traffic bandwidth exceeds 85% utilization, configure reference values for both upstream (transmitted) and downstream (received) traffic. These values act as thresholds for generating the interface-bw events. The range is from 1 and 2,147,483,647 kbps.

The device samples the interface traffic each 10 seconds. If the received or transmitted bandwidth exceeds 85 percent of the configured value in 85 percent of the sampled intervals in a continuous 5-minute period, the device generates an event notification. After the first event notification is generated, sampling continues at the same frequency, but notifications are rate-limited to once per hour. A second event notification is sent, and subsequent notifications are sent, if the bandwidth exceeds 85 percent of the value in 85 percent of the 10-second sampling intervals over the next 1-hour period. If, after 1 hour, the device does not send another event notification, the interval reverts to 5 minutes.

The upstream bandwidth and downstream bandwidth settings are solely for monitoring purposes and do not impose a bandwidth limit on the traffic. For example, in some network configurations, the full bandwidth of an interface may not be available. To ensure that utilization values reflect the available network bandwidth, set bandwidth utilization reference values lower than an interface's full speed.

You can verify the configured upstream and downstream bandwidth values using the **show interface detail** command, which will display the upstream bandwidth (tx-kbps) and downstream bandwidth (rx-kbps) fields showing current usage.

You can monitor transport circuit bandwidth on Cisco IOS XE Catalyst SD-WAN devices and on Cisco SD-WAN Manager.

### Information About Monitoring Bandwidth Utilization with Interface Charts

Cisco SD-WAN Manager provides a chart showing bandwidth utilization for each interface of a device. To view the chart, from the Cisco SD-WAN Manager menu, choose **Monitor** > **Devices**, click a device and click **Interface**.

Configuring upstream or downstream reference values affects how the chart displays the percentages:

- No bandwidth utilization reference values configured: By default, devices calculate the bandwidth utilization value according to the interface speed of the connection.
- Bandwidth utilization reference values configured: If you configure bandwidth utilization reference values, devices calculate the bandwidth utilization value as a percentage of the reference point.

For example, if you configure upstream bandwidth and downstream bandwidth values of 500 megabits per second, and if the downstream utilization is 500 megabits per second, the device reports downstream utilization as 100%.

Devices limit the calculated utilization value to 100% even if the traffic utilization is more than 100% of the configured reference value.

## Restrictions for Upstream and Downstream Bandwidth Utilization

- Cisco SD-WAN Manager supports bandwidth data rate and utilization statistics only for a primary network interface, not a subinterface or loopback interface. You can still configure upstream and downstream bandwidth under a subinterface or loopback interface as a reference value or for per-tunnel QoS functionality.
- To configure upstream and downstream bandwidth references for service-side VPN interfaces, you can only use a CLI template.

## Configure Upstream and Downstream Bandwidth Reference Values Using a Feature Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration** > **Templates**.
2. Click **Feature Templates**.
3. Do one of these:
  - Click **Add Template**, choose a device, and create a Cisco VPN Interface Ethernet template.
  - Edit an existing Cisco VPN Interface Ethernet template.
4. In the **Basic Configuration** section, enter the upstream and downstream bandwidth reference values.
5. Click **Update** to save the template.

## Configure Upstream and Downstream Bandwidth Reference Values Using a CLI Add-On Template

For more information about using CLI templates, see [CLI Add-On Feature Templates](#) and [CLI Templates](#).



---

**Note** By default, CLI templates execute commands in global config mode.

---

This section provides example CLI configurations to configure upstream and downstream bandwidth reference values in Cisco SD-WAN Manager.

1. Enter the SD-WAN configuration mode.

```
sdwan
```

2. Enter interface configuration mode.

```
interface interface-name
```

3. Define the upstream or downstream bandwidth reference value in kbps.

```
bandwidth-upstream upstream-value
```

```
bandwidth-downstream downstream-value
```

Here is a complete configuration example for configuring upstream and downstream bandwidth reference values:

```
sdwan
!
interface GigabitEthernet1
bandwidth-upstream 10000
bandwidth-downstream 10000
```

## Configure Upstream and Downstream Bandwidth Reference Values Using Configuration Groups

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.16.1a and Cisco Catalyst SD-WAN Manager Release 20.16.1

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.
2. Under **Transport & Management Profile**, click ... adjacent to the transport profile, and choose **Edit**.
3. Click the edit icon adjacent to Ethernet Interface.
4. In the **Basic Configuration** section, enter the upstream and downstream bandwidth reference values.
5. Click **Save**.

## Monitor Upstream and Downstream Bandwidth Events Using the Cisco SD-WAN Manager

1. From the Cisco SD-WAN Manager menu, choose **Monitor > Logs > Events**.
2. View the details of upstream and downstream bandwidth values in the device configuration section. The default name for interface bandwidth-related events is **interface-bw**.

## Monitor Upstream and Downstream Bandwidth Reference Values Using the CLI

### Monitor Interface Alarms

Use the **show platform software sdwan interface-alarm summary** command to display the bandwidth reference values for an interface in the alarm summary. This sample output shows the summary details of a GigabitEthernet interface alarm.

See [Interface Alarm Summary](#), and [Base and High Intervals](#) for more information.

```

device# show platform software sdwan interface-alarm summary
===== Interface Alarm Summary =====
Interface Name Upstream (kbps) Downstream (kbps) Base Interval(s) High Interval(s)
GigabitEthernet1 10000 10000 300 3600

```

## Monitor Bandwidth Utilization Using the Cisco SD-WAN Manager

1. From the Cisco SD-WAN Manager menu, choose **Monitor** > **Devices**.
2. Click the **Device** tab.
3. Click the hostname of the device you want to monitor.
4. From the **Applications** section, choose **Interface**.
5. Click the **Chart Options** drop-down list and choose **Utilization**.
6. Hover over the real-time or historical data points to view the utilization percentage.

## Enable DHCP Server using Cisco SD-WAN Manager

Table 152: Feature History

| Feature Name        | Release Information                           | Feature Description                                                                                                      |
|---------------------|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| DHCP Option Support | Cisco IOS XE Catalyst SD-WAN Release 16.12.1b | This feature allows DHCP server options, 43 and 191 to configure vendor-specific information in client-server exchanges. |

Use the DHCP-Server template for all Cisco Catalyst SD-WANs.

You enable DHCP server functionality on a Cisco Catalyst SD-WAN device interface so it can assign IP addresses to hosts in the service-side network.

To configure a Cisco Catalyst SD-WAN device to act as a DHCP server using Cisco SD-WAN Manager templates:

1. Create a DHCP-Server feature template to configure DHCP server parameters, as described in this topic.
2. Create one or more interface feature templates, as described in the VPN-Interface-Ethernet and the VPN-Interface-PPP-Ethernet help topics.
3. Create a VPN feature template to configure VPN parameters. See the VPN help topic.

To configure a Cisco IOS XE Catalyst SD-WAN device interface to be a DHCP helper so that it forwards broadcast DHCP requests that it receives from DHCP servers, in the DHCP Helper field of the applicable interfaces template, enter the addresses of the DHCP servers.

### Navigate to the Template Screen and Name the Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration** > **Templates**.
2. Click **Device Templates**, and then click **Create Template**.



**Note** In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. From the **Create Template** drop-down list, choose **From Feature Template**.
4. From the **Device Model** drop-down list, select the type of device for which you are creating the template.
5. Click **Service VPN** or scroll to the **Service VPN** section.
6. Click **Service VPN** drop-down list.
7. From **Additional VPN Templates**, click **VPN Interface**.
8. From the **Sub-Templates** drop-down list, choose **DHCP Server**.
9. From the **DHCP Server** drop-down list, click **Create Template**. The DHCP-Server template form is displayed.

This form contains fields for naming the template, and fields for defining the DHCP Server parameters.

10. In **Template Name**, enter a name for the template.  
The name can be up to 128 characters and can contain only alphanumeric characters.
11. In **Template Description**, enter a description of the template.  
The description can be up to 2048 characters and can contain only alphanumeric characters.

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default or to enter a value, click the **Scope** drop-down list.

### Minimum DHCP Server Configuration

To configure DHCP server functionality, select **Basic Configuration** and configure the following parameters. Parameters marked with an asterisk as required to configure DHCP servers.

**Table 153:**

| Parameter Name    | Description                                                                                                                                                                                                   |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Address Pool*     | Enter the IPv4 prefix range, in the format <i>prefix/length</i> , for the pool of addresses in the service-side network for which the router interface acts as DHCP server.                                   |
| Exclude Addresses | Enter one or more IP addresses to exclude from the DHCP address pool. To specify multiple individual addresses, list them separated by a comma. To specify a range of addresses, separate them with a hyphen. |
| Maximum Leases    | Specify the number of IP addresses that can be assigned on this interface. <i>Range:</i> 0 through 4294967295                                                                                                 |
| Lease Time        | Specify how long a DHCP-assigned IP address is valid. <i>Range:</i> 0 through 4294967295 seconds                                                                                                              |

| Parameter Name       | Description                                                                                                                                                                                                                                                                                                                    |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Offer Time           | Specify how long the IP address offered to a DHCP client is reserved for that client. By default, an offered IP address is reserved indefinitely, until the DHCP server runs out of addresses. At that point, the address is offered to another client. <i>Range:</i> 0 through 4294967295 seconds <i>Default:</i> 600 seconds |
| Administrative State | Select Up to enable or Down to disable the DHCP functionality on the interface. By default, DHCP server functionality is disabled on an interface.                                                                                                                                                                             |

To save the feature template, click **Save**.

### Configure Static Leases

To configure a static lease to assign a static IP address to a client device on the service-side network, click **Static Lease**, and click **Add New Static Lease** and configure the following parameters:

**Table 154:**

| Parameter Name | Description                                                                           |
|----------------|---------------------------------------------------------------------------------------|
| MAC Address    | Enter the MAC address of the client to which the static IP address is being assigned. |
| IP Address     | Enter the static IP address to assign to the client.                                  |
| Hostname       | Enter the hostname of the client device.                                              |

To edit a static lease, click **pencil** icon.

To remove a static lease, click **trash** icon.

To save the feature template, click **Save**.

### Configure Advanced Options

To configure a advanced DHCP server options, click **Advanced** and then configure the following parameters:

**Table 155:**

| Parameter Name  | Description                                                                                                                                               |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Interface MTU   | Specify the maximum MTU size of packets on the interface. <i>Range:</i> 68 to 65535 bytes                                                                 |
| Domain Name     | Specify the domain name that the DHCP client uses to resolve hostnames.                                                                                   |
| Default Gateway | Enter the IP address of a default gateway in the service-side network.                                                                                    |
| DNS Servers     | Enter one or more IP address for a DNS server in the service-side network. Separate multiple entries with a comma. You can specify up to eight addresses. |
| TFTP Servers    | Enter the IP address of a TFTP server in the service-side network. You can specify one or two addresses. If two, separate them with a comma.              |

To save the feature template, click **Save**.

**Configure DHCP server using CLI**

```

Device# config-transaction
Device(dhcp-config)# ip dhcp pool DHCP-POOL
Device(dhcp-config)# network 10.1.1.1 255.255.255.0
Device(dhcp-config)# default-router 10.1.1.2
Device(dhcp-config)# dns-server 172.16.0.1
Device(dhcp-config)# domain-name DHCP-DOMAIN
Device(dhcp-config)# exit
Device(config)# ip dhcp excluded-address 10.1.1.2 10.1.1.10
Device(

```

**Release Information**

Introduced in Cisco SD-WAN Manager in Release 15.2.

## Configuring PPPoE

The Point-to-Point Protocol over Ethernet (PPPoE) connects multiple users over an Ethernet local area network to a remote site through common customer premises equipment. PPPoE is commonly used in a broadband aggregation, such as by digital subscriber line (DSL). PPPoE provides authentication with the CHAP or PAP protocol. In the Cisco Catalyst SD-WAN overlay network, Cisco Catalyst SD-WAN devices can run the PPPoE client. The PPPoE server component is not supported.

It is recommended that you configure quality of service (QoS) and shaping rate on a PPPoE Dialer interface. Queuing based QoS policies on both Dialer interface and PPPoE-enabled physical interface at the same time, is not supported.

PPPoE-enabled physical interfaces are supported on ATM PVCs and Ethernet interfaces. A dialer interface must be used for cloning virtual access. Multiple PPPoE client sessions can be configured on an Ethernet interface, but each session must use a separate dialer interface and a separate dialer pool.

The Cisco Catalyst SD-WAN implementation of PPPoE does not support the Compression Control Protocol (CCP) options, as defined in RFC 1962.

This example shows configuring PPPoE server on IPv4 interfaces:

```

!
interface Dialer100
 mtu 1492
 ip address negotiated
 encapsulation ppp
 ip tcp adjust-mss 1460
 dialer pool 100
 dialer down-with-vInterface
 ppp authentication chap callin
 ppp chap hostname cisco
 ppp chap password 7 1511021F07257A767B
 ppp ipcp route default

```





**Note** Follow these steps to replace a template configured with PPPoE as WAN interface with a regular interface in Dialer100:

1. Remove the IP address assigned to the dialer interface using the command:

```
no ip address <ip> <mask>
```

2. Add a new IP address for the dialer interface.

## Configure PPPoE from Cisco SD-WAN Manager Templates

To use Cisco SD-WAN Manager templates to configure PPPoE on Cisco IOS XE Catalyst SD-WAN device, you create three feature templates and one device template:

- Create a VPN-Interface-PPP feature template to configure PPP parameters for the PPP virtual interface.
- Create a VPN-Interface-PPP-Ethernet feature template to configure a PPPoE-enabled interface.
- Optionally, create a VPN feature template to modify the default configuration of VPN 0.
- Create a device template that incorporates the VPN-Interface-PPP, VPN-Interface-PPP-Ethernet, and VPN feature templates.

Create a VPN-Interface-PPP feature template to configure PPP parameters for the PPP virtual interface:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**, and click **Add Template**.



**Note** In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

3. Choose Cisco IOS XE Catalyst SD-WAN device Cloud or a router model.
4. Choose the **VPN-Interface-PPP** template.
5. In the template, configure the following parameters:

**Table 156:**

| Parameter Field        | Procedure                                                                           |
|------------------------|-------------------------------------------------------------------------------------|
| Template Name          | Enter a name for the template. It can be up to 128 alphanumeric characters.         |
| Description            | Enter a description for the template. It can be up to 2048 alphanumeric characters. |
| Shutdown               | Click <b>No</b> to enable the PPP virtual interface.                                |
| Interface Name         | Enter the number of the PPP interface. It can be from 1 through 31.                 |
| Description (optional) | Enter a description for the PPP virtual interface.                                  |

| Parameter Field         | Procedure                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Authentication Protocol | Select either CHAP or PAP to configure one authentication protocol, or select PAP and CHAP to configure both. For CHAP, enter the hostname and password provided by your ISP. For PAP, enter the username and password provided by your ISP. If you are configuring both PAP and CHAP, to use the same username and password for both, click Same Credentials for PAP and CHAP.                                                                                                              |
| AC Name (optional)      | Select the PPP tab, and in the AC Name field, enter the name of the the name of the access concentrator used by PPPoE to route connections to the Internet.                                                                                                                                                                                                                                                                                                                                  |
| IP MTU                  | Click <b>Advanced</b> , and in the IP MTU field, ensure that the IP MTU is at least 8 bytes less than the MTU on the physical interface. The maximum MTU for a PPP interface is 1492 bytes. If the PPPoE server does not specify a maximum receive unit (MRU), the MTU value for the PPP interface is used as the MRU.<br><br>Starting from Cisco vManage Release 20.9.1, there is 8 bytes overheads deduced based on the specified IP MTU value when configuration is pushed to the device. |
| Save                    | To save the feature template, click <b>Save</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                            |

To create a VPN-Interface-PPP-Ethernet feature template to enable the PPPoE client on the physical interfaces:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**, and click **Add Template**.



**Note** In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

3. Choose Cloud or a router model.
4. Choose the **VPN-Interface-PPP-Ethernet** template.
5. In the template, configure the following parameters:

| Parameter Field        | Procedure                                                                                                                                                                                                                                                                                           |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Template Name          | Enter a name for the template. It can be up to 128 alphanumeric characters.                                                                                                                                                                                                                         |
| Description            | Enter a description for the template. It can be up to 2048 alphanumeric characters.                                                                                                                                                                                                                 |
| Shutdown               | Click <b>No</b> to enable the PPPoE-enabled interface.                                                                                                                                                                                                                                              |
| Interface Name         | Enter the name of the physical interface in VPN 0 to associate with the PPP interface.                                                                                                                                                                                                              |
| Description (optional) | Enter a description for the PPPoE-enabled interface.                                                                                                                                                                                                                                                |
| IP Configuration       | Assign an IP address to the physical interface: <ul style="list-style-type: none"> <li>• To use DHCP, select Dynamic. The default administrative distance of routes learned from DHCP is 1.</li> <li>• To configure the IP address directly, enter of the IPv4 address of the interface.</li> </ul> |

| Parameter Field        | Procedure                                                      |
|------------------------|----------------------------------------------------------------|
| DHCP Helper (optional) | Enter up to four IP addresses for DHCP servers in the network. |
| Save                   | To save the feature template, click <b>Save</b> .              |

To create a VPN feature template to configure the PPPoE-enabled interface in VPN 0, the transport VPN:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**, and click **Add Template**.



**Note** In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

3. Choose Cloud or a router model.
4. Choose the **VPN** template.
5. In the template, configure the following parameters:

| Parameter Field            | Procedure                                                                           |
|----------------------------|-------------------------------------------------------------------------------------|
| Template Name              | Enter a name for the template. It can be up to 128 alphanumeric characters.         |
| Description                | Enter a description for the template. It can be up to 2048 alphanumeric characters. |
| VPN Identifier             | Enter VPN identifier 0.                                                             |
| Name                       | Enter a name for the VPN.                                                           |
| Other interface parameters | Configure the desired interface properties.                                         |
| Save                       | To save the feature template, click <b>Save</b> .                                   |

To create a device template that incorporates the VPN-Interface-PPP, VPN-Interface-PPP-Ethernet, and VPN feature templates:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and then click **Create Template**.



**Note** In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. From the **Create Template** drop-down list, choose **From Feature Template**.
4. From the **Device Model** drop-down list, choose the type of device for which you are creating the device template.

Cisco SD-WAN Manager displays the feature templates for the device type you selected. Required templates are indicated with an asterisk (\*).

5. Enter a name and description for the device template. These fields are mandatory. The template name cannot contain special characters.
6. In **Transport & Management VPN**, under **VPN 0**, from the drop-down list of available templates, select the desired feature template. The list of available templates are the ones that you have previously created.
7. In **Additional VPN 0 Templates**, click the plus sign (+) next to **VPN Interface PPP**.
8. From **VPN-Interface-PPP** and **VPN-Interface-PPP-Ethernet** fields, select the feature templates to use.
9. To configure multiple PPPoE-enabled interfaces in VPN 0, click the plus sign (+) next to Sub-Templates.
10. To include additional feature templates in the device template, in the remaining sections, select the feature templates in turn, and from the drop-down list of available templates, select the desired template. The list of available templates are the ones that you have previously created. Ensure that you select templates for all mandatory feature templates and for any desired optional feature templates.
11. To create the device template, click **Create**.

To attach a device template to a device:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**.



**Note** In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. Choose a template.
4. Click ..., and click **Attach Device**.
5. Search for a device or select a device from the Available Device(s) column to the left.
6. Click the arrow pointing right to move the device to the Selected Device(s) column on the right.
7. Click **Attach**.

## Configure PPPoE Over ATM

*Table 157: Feature History*

| Feature Name             | Release Information                                                          | Description                                                                                                                                                                                                 |
|--------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Configure PPPoE over ATM | Cisco IOS XE Catalyst SD-WAN Release 17.4.1a<br>Cisco vManage Release 20.4.1 | This feature provides support for configuring PPPoEoA on Cisco IOS XE Catalyst SD-WAN devices. PPPoEoA uses AAL5MUX encapsulation which delivers better efficiency compared to other encapsulation methods. |

You can configure PPPoE over ATM interfaces (PPPoEoA) on Cisco IOS XE Catalyst SD-WAN devices that support ADSL. PPPoEoA uses ATM Adaptation Layer 5 Multiplexed Encapsulation (AAL5MUX) encapsulation to carry PPPoE over ATM permanent virtual circuits (PVCs), providing efficiency gain over AAL5 LLC/SNAP encapsulation.

PPPoEoA over AAL5MUX reduces Subnetwork Access Protocol (SNAP) encapsulation bandwidth usage, using multiplexed (MUX) encapsulation to reduce the number of cells needed to carry voice packets. Deploying the PPPoEoA over ATM AAL5MUX feature in a VoIP environment results in improved throughput and bandwidth usage.

## Supported Platforms for PPPoE Over ATM

The following platforms support PPPoE over ATM:

- Cisco 1100 4G/6G Series Integrated Services routers.
- Cisco1100 Series Integrated Service routers.
- Cisco1109 Series Integrated Service routers.
- Cisco111x Series Integrated Service routers.
- Cisco1111x Series Integrated Service routers.
- Cisco1120 Series Integrated Service routers.
- Cisco1160 Series Integrated Service routers.

## Configure PPPoE Over ATM using Cisco SD-WAN Manager

You can configure PPPoE using in Cisco SD-WAN Manager using the device CLI template.

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. From **Device Templates**, click **Create Template**.



---

**Note** In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

---

3. From the **Create Template** drop-down list, select **CLI Template**.
4. From the **Device Model** drop-down list, select the type of device for which you are creating the template.
5. In **Template Name**, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.
6. In **Template Description**, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.
7. Choose **Device configuration**. Using this option, you can provide IOS-XE configuration commands that appear in the output of the `show sdwan running-config` command.
8. (Optional) To load the running config of a connected device, select it from the Load Running config from reachable device list and click **Search**.

9. In **CLI Configuration**, enter the configuration either by typing it, cutting and pasting it, or uploading a file. The configuration for PPPoEoA is available in the [Configure PPPoEoA on the CLI](#) section.
10. To convert an actual configuration value to a variable, select the value and click **Create Variable**. Enter the variable name, and click **Create Variable**. You can also type the variable name directly, in the format `{{variable-name}}`; for example, `{{hostname}}`.
11. Click **Add**. The new device template is displayed in the Device Template table. The **Type** column shows **CLI** to indicate that the device template was created from CLI text.

## Configure PPPoE Over ATM on the CLI

This section provides example CLI configurations to configure PPoE over ATM on the CLI.

```
Device(config)# interface atm number
Device(config)# no ip address
Device(config)# interface atm number point-to-point
Device(config)# no atm enable-ilmi-trap
Device(config)# encapsulation aal5mux pppoe-client
Device(config)# pppoe-client dial-pool-number number
Device(config)# interface Dialer dialer-rotary-group-number
Device(config)# mtu bytes
Device(config)# ip address negotiated
Device(config-if)# encapsulation encapsulation-type
Device(config)# load-interval seconds
Device(config)# dialer pool number
Device(config)# dialer-group group-number
Device(config)# ppp mtu adaptive
Device(config)# ppp chap hostname hostname
Device(config)# ppp chap password secret
Device(config)# ppp ipcp address required
Device(config)# ppp link reorders
```

## Configuration Example for Configuring PPPoE Over ATM Interfaces

This example shows configuring PPPoE over ATM interfaces.

```
Device(config)# interface ATM0/1/0
Device(config)# no ip address
Device(config)# no atm enable-ilmi-trap
!
Device(config)# interface ATM0/1/0.10 point-to-point
Device(config)# no atm enable-ilmi-trap
Device(config)# cdp enable
Device(config)# pvc 22/62
Device(config)# ubr 1045
Device(config-if)# encapsulation aal5mux pppoe-client
Device(config)# pppoe-client dial-pool-number 120
!
!
Device(config)# interface Dialer 120
Device(config)# mtu 1492
Device(config)# ip address negotiated
Device(config)# ip nat outside
Device(config-if)# encapsulation ppp
```

```

Device(config)# load-interval 30
Device(config)# dialer pool 120
Device(config)# dialer-group 1
Device(config)# ppp mtu adaptive
Device(config)# ppp chap hostname test@cisco.com
Device(config)# ppp chap password 0 cisco
Device(config)# ppp ipcp address required
Device(config)# ppp link reorders
!
```

## Configuring VRRP



**Note** The x710 NIC must have the **t->system-> vrrp-advt-with-phymac** command configured, for VRRP to function.

The Virtual Router Redundancy Protocol (VRRP) is a LAN-side protocol that provides redundant gateway service for switches and other IP end stations. In the Cisco Catalyst SD-WAN software, you configure VRRP on an interface, and typically on a subinterface, within a VPN.

VRRP is only supported with service-side VPNs (VPN 0 and 512 reserved) and if sub-interfaces are used, then the VRRP physical interface must be configured in VPN 0.

For each VRRP interface (or subinterface), you assign an IP address and you place that interface in a VRRP group.

The group number identifies the virtual router. You can configure a maximum of 512 groups on a router. In a typical VRRP topology, two physical routers are configured to act as a single virtual router, so you configure the same group number on interfaces on both these routers.

For each virtual router ID, you must configure an IP address.

Within each VRRP group, the router with the higher priority value is elected as primary VRRP. By default, each virtual router IP address has a default primary election priority of 100, so the router with the higher IP address is elected as primary. You can modify the priority value, setting it to a value from 1 through 254.

The primary VRRP periodically sends advertisement messages, indicating that it is still operating. If backup routers miss three consecutive VRRP advertisements, they assume that the primary VRRP is down and elect a new primary VRRP. By default, these messages are sent every second. You can change the VRRP advertisement time to be a value from 1 through 3600 seconds.

By default, VRRP uses the state of the interface on which it is running, to determine which router is the primary virtual router. This interface is on the service (LAN) side of the router. When the interface for the primary VRRP goes down, a new primary VRRP virtual router is elected based on the VRRP priority value. Because VRRP runs on a LAN interface, if a router loses all its WAN control connections, the LAN interface still indicates that it is up even though the router is functionally unable to participate in VRRP. To take WAN side connectivity into account for VRRP, you can configure one of the following:

- Track the Overlay Management Protocol (OMP) session running on the WAN connection when determining the primary VRRP virtual router.

If all OMP sessions are lost on the primary VRRP router, VRRP elects a new default gateway from among all the gateways that have one or more active OMP sessions even if the gateway chosen has a lower VRRP priority than the current primary VRRP router. With this option, VRRP failover occurs once the OMP state

changes from up to down, which occurs when the OMP hold timer expires. Until the hold timer expires and a new primary VRRP is elected, all overlay traffic is dropped. When the OMP session recovers, the local VRRP interface claims itself as primary VRRP even before it learns and installs OMP routes from the Cisco Catalyst SD-WAN Controllers. Until the routers are learned, traffic is also dropped.

- Track both the OMP session and a list of remote prefixes.

If all OMP sessions are lost, VRRP failover occurs as described for the **track-omp** option. In addition, if reachability to all the prefixes in the list is lost, VRRP failover occurs immediately, without waiting for the OMP hold timer to expire, thus minimizing the amount of overlay traffic is dropped while the router determines the primary VRRP.

As discussed above, the IEEE 802.1Q protocol adds 4 bytes to each packet's length. Hence, for packets to be transmitted, either increase the MTU size on the physical interface in VPN 0 (the default MTU is 1500 bytes) or decrease the MTU size on the VRRP interface.

For devices running on Cisco IOS XE Catalyst SD-WAN Release 17.4.1a and later, adjusting the MTU size is not required, both the physical interface and sub interface can have the same MTU size.

## Configuring Dynamic Interfaces

*Table 158: Feature History*

| Feature Name                   | Release Information                                                         | Description                                                                                                                                                                                                                 |
|--------------------------------|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Configuring Dynamic Interfaces | Cisco IOS XE Catalyst SD-WAN Release 17.3.2<br>Cisco vManage Release 20.3.2 | This feature allows you to configure dynamic interfaces for supported devices. A dynamic interface allows a device to select optimum paths in real-time.<br><br>This feature applies only to the Cisco C8500-12X4QC router. |

You can configure dynamic interfaces for supported devices. A dynamic interface allows a device to select optimum paths in real-time.

Configuring dynamic interfaces consists of these general steps:

1. Create a dynamic interface mode feature template. As part of this step, you define modes for the bays in a device.
2. Configure an Interface for Control Connections.
3. Associate the dynamic interface mode feature template with a device template.

### Create a Dynamic Interface Mode Feature Template

When you create a dynamic interface mode feature template, you create a template that defines the modes for the bays in a device.

You can configure the mode for bay 1, bay 2, or both.



The mode for bay 0 is configured automatically and cannot be changed. If you configure the mode for bay 1 as 100G, bay 0 is disabled because the 10G interfaces on bay 0 do not apply in this case.

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**.



---

**Note** In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

---

3. Click the **Create Template** drop-down list and choose **Feature Template**.
4. From the **Device Model** drop-down list, choose the device for which you wish to create the template.
5. In **Template Name**, enter a name for the template.  
This field may contain uppercase and lowercase letters, digits 0 through 9, hyphens (-), and underscores (\_).
6. In **Description**, enter a description of the template.  
This field can contain any characters and spaces.
7. From **Additional Templates**, choose the **Dynamic Interface Mode** drop-down list and click **Create Template**.
8. In **Template Name**, enter a name for the template.  
This field may contain uppercase and lowercase letters, digits 0 through 9, hyphens (-), and underscores (\_).
9. In **Description**, enter a description of the template.  
This field can contain any characters and spaces.
10. Configure the mode for bay 1, bay 2, or both bays by choosing the desired value in the **Bay 1**, **Bay 2**, or both fields.  
You cannot change the default value for bay 0.
11. Click **Save**.



---

**Note** Open a TAC case when there is a mismatch between confd and iosd configurations while changing the bay subslot mode on 8500-12X4QC.

---

### Configure an Interface for Control Connections

This section describes how to configure a new VPN 0 interface for an existing control connection to operate with the bays that you configured in “Create a Dynamic Interface Mode Feature Template.” It also describes how to configure an IPv4 route for the interface.

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**.



---

**Note** In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

---

3. Click ... of the template for which you want to configure the interface, and then choose **Edit**.
4. Click **Transport & Management VPN** and perform these actions to create interfaces for the bays:
  - a. Click **VPN Interface** in the **Additional VPN 0 Template**.
  - b. Choose the new **VPN Interface Ethernet** menu that displays, and then click **Create Template**.
  - c. In **Template Name**, enter a name for the template.

This field may contain uppercase and lowercase letters, digits 0 through 9, hyphens (-), and underscores (\_).
  - d. In **Description**, enter a description of the template.

This field can contain any characters and spaces.
  - e. Add control connections to the bays that you configured as described in “Create a Dynamic Interface Mode Feature Template.”
5. Choose **Basic Configuration** and perform these actions:
  - a. In **Interface Name**, enter a name for the interface.

Enter a name in the format that this example shows: “FortyGigabitEthernet0/1/0.”
  - b. Configure other options on this tab as needed.
6. From **Tunnel**, set **Tunnel Interface** to **On**.
7. Click **Save**.
8. Choose **IPv4 Route** and perform these actions to configure an IPv4 route for the VPN0 template:
  - a. Click **New IPv4 Route**.
  - b. In **Prefix**, enter a prefix for the IPv4 route.
  - c. In **Gateway**, choose **Next Hop**.
  - d. Configure items as needed in **Next Hop**, and then click **Add**.
  - e. Click **Save**.
9. Click **Update**.

### Associate the Dynamic Interface Mode Feature Template with a Device Template

After you create the dynamic interface mode feature template, associate it with a device template and attach the device template to a device. For instructions, see [Create a Device Template from Feature Templates](#).

# Configure VPN Ethernet Interface Using a Configuration Group

## Before you begin

On the **Configuration** > **Configuration Groups** page, choose **SD-WAN** as the solution type.

## Procedure

**Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration** > **Configuration Groups**.

**Step 2** Create and configure a Transport VPN feature in Transport and Management profile.

**Step 3** Create and configure Ethernet Interface feature in Transport VPN.

a) Configure basic VPN parameters.

| Field                        | Description                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Shutdown</b>              | Enable or disable the interface.                                                                                                                                                                                                                                                                                                                                 |
| <b>Interface Name*</b>       | Enter a name for the interface. Spell out the interface names completely (for example, GigabitEthernet0/0/0).<br><br>Configure all the interfaces of the router, even if you are not using them, so that they are configured in the shutdown state and so that all default values for them are configured.                                                       |
| <b>Description</b>           | Enter a description for the interface.                                                                                                                                                                                                                                                                                                                           |
| <b>Auto Detect Bandwidth</b> | Enable this option to automatically detect the bandwidth for WAN interfaces. The device detects the bandwidth by contacting an iPerf3 server to perform a speed test.                                                                                                                                                                                            |
| <b>IPv4 Settings</b>         | Configure an IPv4 VPN interface. <ul style="list-style-type: none"><li>• <b>Dynamic</b>: Choose <b>Dynamic</b> to set the interface as a Dynamic Host Configuration Protocol (DHCP) client so that the interface receives its IP address from a DHCP server.</li><li>• <b>Static</b>: Choose <b>Static</b> to enter an IP address that doesn't change.</li></ul> |
| <b>Dynamic DHCP Distance</b> | Enter an administrative distance value for routes learned from a DHCP server. This option is available when you choose <b>Dynamic</b> .<br><br>Default: 1                                                                                                                                                                                                        |
| <b>IP Address</b>            | Enter a static IPv4 address. This option is available when you choose <b>Static</b> .                                                                                                                                                                                                                                                                            |
| <b>Subnet Mask</b>           | Enter the subnet mask.                                                                                                                                                                                                                                                                                                                                           |

| Field                                 | Description                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Configure Secondary IP Address</b> | Enter up to four secondary IPv4 addresses for a service-side interface. <ul style="list-style-type: none"> <li>• <b>IP Address:</b> Enter the IP address.</li> <li>• <b>Subnet Mask:</b> Enter the subnet mask.</li> </ul>                                                                                                                                                                 |
| <b>DHCP Helper</b>                    | To designate the interface as a DHCP helper on a router, enter up to eight IP addresses, separated by commas, for DHCP servers in the network. A DHCP helper interface forwards BOOTP (broadcast) DHCP requests that it receives from the specified DHCP servers.                                                                                                                          |
| <b>IPv6 Settings</b>                  | Configure an IPv6 VPN interface. <ul style="list-style-type: none"> <li>• <b>Dynamic:</b> Choose <b>Dynamic</b> to set the interface as a Dynamic Host Configuration Protocol (DHCP) client so that the interface receives its IP address from a DHCP server.</li> <li>• <b>Static:</b> Choose <b>Static</b> to enter an IP address that doesn't change.</li> <li>• <b>None</b></li> </ul> |
| <b>IPv6 Address Primary</b>           | Enter a static IPv6 address. This option is available when you choose <b>Static</b> .                                                                                                                                                                                                                                                                                                      |
| <b>Add Secondary Ipv6</b>             |                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>IP Address</b>                     | Enter up to two secondary IPv6 addresses for a service-side interface.                                                                                                                                                                                                                                                                                                                     |
| <b>Bandwidth Upstream</b>             | Enter upstream bandwidth reference value.                                                                                                                                                                                                                                                                                                                                                  |
| <b>Bandwidth Downstream</b>           | Enter downstream bandwidth reference value.                                                                                                                                                                                                                                                                                                                                                |

## b) Apply Access Lists and QoS Parameters

| Field               | Description                                                                       |
|---------------------|-----------------------------------------------------------------------------------|
| <b>Adaptive QoS</b> | To enable or disable adaptive QoS on an ethernet interface on the transport side. |
| <b>Shaping Rate</b> | Enter the shaping rate to control the maximum rate of traffic sent.               |
| <b>ACL</b>          | To define IPv4 and IPv6 ACL as ingress and egress.                                |

## c) Create a tunnel interface.

| Field                   | Description                                                                          |
|-------------------------|--------------------------------------------------------------------------------------|
| <b>Tunnel Interface</b> | Enable this option to create a tunnel interface.                                     |
| <b>Per-tunnel QoS</b>   | Enable this option to apply a Quality of Service (QoS) policy on individual tunnels. |
| <b>Color</b>            | Choose a color for the TLOC.                                                         |

| Field                                | Description                                                                                                                                                                                                                                                                                                           |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Color Description                    | Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.18.1<br>Enter a description associated to the TLOC color.                                                                                                                                                                                         |
| <b>Restrict</b>                      | Enable this option to limit the remote TLOCs that the local TLOC can establish BFD sessions with. When a TLOC is marked as restricted, a TLOC on the local router establishes tunnel connections with a remote TLOC only if the remote TLOC has the same color.                                                       |
| <b>Groups</b>                        | Enter a group number.<br>Range: 1 through 4294967295                                                                                                                                                                                                                                                                  |
| <b>Border</b>                        | Enable this option to set the TLOC as a border TLOC.                                                                                                                                                                                                                                                                  |
| <b>Maximum Control Connections</b>   | Specify the maximum number of Cisco SD-WAN Controllers that the WAN tunnel interface can connect to. To have the tunnel establish no control connections, set the number to 0.<br>Range: 0 through 100<br>Default: 2                                                                                                  |
| <b>Validator As Stun Server</b>      | Enable Session Traversal Utilities for NAT (STUN) to allow the tunnel interface to discover its public IP address and port number when the Cisco IOS XE Catalyst SD-WAN device is located behind a NAT.                                                                                                               |
| <b>Exclude Controller Group List</b> | Set the identifiers of one or more Cisco SD-WAN Controller groups that this tunnel is not allowed to connect to.<br>Range: 1 through 100                                                                                                                                                                              |
| <b>Manager Connection Preference</b> | Set the preference for using a tunnel interface to exchange control traffic with Cisco SD-WAN Manager.<br>Range: 0 through 8<br>Default: 5                                                                                                                                                                            |
| <b>Full Port Hop</b>                 | Minimum release: Cisco IOS XE Catalyst SD-WAN Release 17.18.1a<br>Enable full port hopping at the TLOC level to allow devices to establish connections with controllers by switching to the next port if the current port is blocked or non-functional.<br>Default: Disabled                                          |
| <b>Port Hop</b>                      | Enable port hopping. If port hopping is enabled globally, you can disable it on an individual TLOC (tunnel interface).<br>Default: Enabled<br>Starting from Cisco IOS XE Catalyst SD-WAN Release 17.18.1a, this field is deprecated. Instead use the <b>Full Port Hop</b> option. See the <b>Full Port Hop</b> field. |
| <b>Low-Bandwidth Link</b>            | Enable this option to characterize the tunnel interface as a low-bandwidth link.                                                                                                                                                                                                                                      |

| Field                      | Description                                                                                                                                                                                                                                                                                                       |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tunnel TCP MSS</b>      | Specify the maximum segment size (MSS) of TCP SYN packets passing through the router. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented.<br><br>Range: 500 to 1460 bytes<br>Default: None                                           |
| <b>Clear-Dont-Fragment</b> | Enable this option to clear the Don't Fragment (DF) bit in the IPv4 packet header for packets being transmitted out the interface. When the DF bit is cleared, packets larger than the MTU of the interface are fragmented before being sent.                                                                     |
| <b>CTS SGT Propagation</b> | Enable CTS SGT propagation on an interface.                                                                                                                                                                                                                                                                       |
| <b>Network Broadcast</b>   | Enable this option to accept and respond to network-prefix-directed broadcasts.                                                                                                                                                                                                                                   |
| <b>Allow Service</b>       | Allow or disallow the following services on the interface: <ul style="list-style-type: none"> <li>• All</li> <li>• BGP</li> <li>• DHCP</li> <li>• NTP</li> <li>• SSH</li> <li>• DNS</li> <li>• ICMP</li> <li>• HTTPS</li> <li>• OSPF</li> <li>• STUN</li> <li>• SNMP</li> <li>• NETCONF</li> <li>• BFD</li> </ul> |
| <b>Encapsulation</b>       |                                                                                                                                                                                                                                                                                                                   |

| Field                                                                                                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Encapsulation*</b>                                                                                           | <p>Choose an encapsulation type:</p> <ul style="list-style-type: none"> <li>• <b>gre</b>: Use GRE encapsulation on the tunnel interface.</li> <li>• <b>ipsec</b>: Use IPsec encapsulation on the tunnel interface.</li> </ul> <p><b>Note</b><br/>If you select both IPsec and GRE encapsulations, two TLOCs are created for the tunnel interface that have the same IP addresses and colors, but that differ by their encapsulation.</p> <p>When you choose <b>gre</b>, the following fields appear:</p> <ul style="list-style-type: none"> <li>• <b>GRE Preference</b>: Enter a preference value for directing traffic to the tunnel. A higher value is preferred over a lower value.<br/>Range: 0 through 4294967295<br/>Default: 0</li> <li>• <b>GRE Weight</b>: Enter a weight to use to balance traffic across multiple TLOCs. A higher value sends more traffic to the tunnel.<br/>Range: 1 through 255<br/>Default: 1</li> </ul> <p>When you choose <b>ipsec</b>, the following fields appear:</p> <ul style="list-style-type: none"> <li>• <b>IPSEC Preference</b>: Enter a preference value for directing traffic to the tunnel. A higher value is preferred over a lower value.<br/>Range: 0 through 4294967295<br/>Default: 0</li> <li>• <b>IPSEC Weight</b>: Enter a weight to use to balance traffic across multiple TLOCs. A higher value sends more traffic to the tunnel.<br/>Range: 1 through 255<br/>Default: 1</li> </ul> |
| <b>Multi-Region Fabric</b><br><br><b>Note</b><br>These options appear only when Multi-Region Fabric is enabled. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Connect to Core Region</b>                                                                                   | <p>(Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.13.1)<br/>(Applicable to a border router only) In a Multi-Region Fabric scenario, enable this option to specify how to use the Ethernet interface:</p> <ul style="list-style-type: none"> <li>• <b>Share Interface with Access Region</b>: Share the interface between the access region and core region.</li> <li>• <b>Keep Exclusive to Core Region</b>: Use the interface only for the core region.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

| Field                              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Connect to Secondary Region</b> | <p>(Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.13.1)</p> <p>(Applicable to an edge router only) In a Multi-Region Fabric scenario, enable this option to specify how to use the Ethernet interface:</p> <ul style="list-style-type: none"> <li>• <b>Share Interface with Access Region:</b> Share the interface between the primary and secondary regions.</li> <li>• <b>Keep Exclusive to Secondary Region:</b> Use the interface only for the secondary region.</li> </ul> |

d) Configure an interface as a NAT device.

| Field                | Description                                                                                                                                                                                                                          |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IPv4 Settings</b> |                                                                                                                                                                                                                                      |
| <b>NAT</b>           | Enable this option to have the interface act as a NAT device.                                                                                                                                                                        |
| <b>NAT Type</b>      | <p>Choose the NAT translation type for IPv4:</p> <ul style="list-style-type: none"> <li>• <b>interface</b></li> <li>• <b>pool</b></li> <li>• <b>loopback</b></li> </ul> <p>Default: <b>interface</b>. It is supported for NAT64.</p> |
| <b>UDP Timeout</b>   | <p>Specify when NAT translations over UDP sessions time out.</p> <p>Range: 1 through 8947 minutes</p> <p>Default: 1 minute</p>                                                                                                       |
| <b>TCP Timeout</b>   | <p>Specify when NAT translations over TCP sessions time out.</p> <p>Range: 1 through 8947 minutes</p> <p>Default: 60 minutes (1 hour)</p>                                                                                            |



| Field                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Add Multiple NAT</b>         | <p>Choose the NAT type:</p> <ul style="list-style-type: none"> <li>• <b>Interface:</b> This is the default value.</li> <li>• <b>Pool:</b> Configure the following: <ul style="list-style-type: none"> <li>• <b>Pool ID:</b> Enter a NAT pool number configured in the centralized data policy. The NAT pool name must be unique across VPNs and VRFs. You can configure up to 31 (1–32) NAT pools per router.</li> <li>• <b>Range Start:</b> Enter a starting IP address for the NAT pool.</li> <li>• <b>Range End:</b> Enter a closing IP address for the NAT pool.</li> <li>• <b>Prefix length:</b> Specify the maximum number of source IP addresses that can be NATed in the NAT pool.</li> <li>• <b>Overload:</b> Enable this option to configure per-port translation. If this option is disabled, only dynamic NAT is configured on the end device. Per-port NAT is not configured.<br/>Default: Disabled</li> </ul> </li> <li>• <b>Loopback:</b> Provide a value for the NAT inside source loopback interface.</li> </ul> |
| <b>Configure New Static NAT</b> | Add a static NAT mapping                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Source IP</b>                | Enter the source IP address to be translated.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Translate IP</b>             | Enter the translated source IP address.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Direction</b>                | <p>Choose the direction in which to perform network address translation.</p> <ul style="list-style-type: none"> <li>• <b>inside:</b> Translates the IP address of packets that are coming from the service side of the device and that are destined for the transport side of the router.</li> <li>• <b>outside:</b> Translates the IP address of packets that are coming to the device from the transport side device and that are destined for a service-side device.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Source VPN</b>               | Enter the source VPN ID.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>IPv6 Settings</b>            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>IPv6 NAT</b>                 | Enable this option to have the interface act as a NAT device.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Select NAT</b>               | <p>Choose NAT64 or NAT66. When you choose NAT66, the following fields appear:</p> <ul style="list-style-type: none"> <li>• <b>Source Prefix:</b> Enter the source IPv6 prefix.</li> <li>• <b>Translated Source Prefix:</b> Enter the translated source prefix.</li> <li>• <b>Source VPN ID:</b> Enter the source VPN ID.</li> <li>• <b>Egress Interface:</b> Enable this option to have the interface act as an egress interface.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

- e) Add ARP table entries.

| Field              | Description                                                                                          |
|--------------------|------------------------------------------------------------------------------------------------------|
| <b>IP Address</b>  | Enter the IP address for the ARP entry in dotted decimal notation or as a fully qualified host name. |
| <b>MAC Address</b> | Enter the MAC address in colon-separated hexadecimal notation.                                       |

- f) Configure advanced properties.

| Field                | Description                                                                                                                                                                                                                                                         |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Duplex</b>        | Specify whether the interface runs in full-duplex or half-duplex mode.<br>Default: full                                                                                                                                                                             |
| <b>MAC Address</b>   | Specify a MAC address to associate with the interface, in colon-separated hexadecimal notation.                                                                                                                                                                     |
| <b>IP MTU</b>        | Specify the maximum MTU size of packets on the interface.<br>Range: 576 through 9216<br>Default: 1500 bytes                                                                                                                                                         |
| <b>Interface MTU</b> | Enter the maximum transmission unit size for frames received and transmitted on the interface.<br>Range: 1500 through 1518 (GigabitEthernet0), 1500 through 9216 (other GigabitEthernet)<br>Default: 1500 bytes                                                     |
| <b>TCP MSS</b>       | Specify the maximum segment size (MSS) of TCP SYN packets passing through the router. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented.<br>Range: 500 to 1460 bytes<br>Default: None |
| <b>Speed</b>         | Specify the speed of the interface, for use when the remote end of the connection does not support autonegotiation.<br>Values: 10, 100, 1000, 2500, or 10000 Mbps                                                                                                   |
| <b>ARP Timeout</b>   | ARP timeout controls how long we maintain the ARP cache on a router. Specify how long it takes for a dynamically learned ARP entry to time out.<br>Range: 0 through 2147483 seconds<br>Default: 1200 seconds                                                        |
| <b>Autonegotiate</b> | Enable this option to turn on autonegotiation.                                                                                                                                                                                                                      |

| Field                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Media Type</b>            | <p>Specify the physical media connection type on the interface. Choose one of the following:</p> <ul style="list-style-type: none"> <li>• <b>auto-select</b>: A connection is automatically selected.</li> <li>• <b>rj45</b>: Specifies an RJ-45 physical connection.</li> <li>• <b>sfp</b>: Specifies a small-form factor pluggable (SFP) physical connection for fiber media.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>TLOC Extension</b>        | <p>Enter the name of a physical interface on the same router that connects to the WAN transport. This configuration then binds this service-side interface to the WAN transport. A second router at the same site that itself has no direct connection to the WAN (generally because the site has only a single WAN connection) and that connects to this service-side interface is then provided with a connection to the WAN.</p> <p><b>Note</b><br/>TLOC extension over L3 is supported only for Cisco IOS XE Catalyst SD-WAN devices. If configuring TLOC extension over L3 for a Cisco IOS XE Catalyst SD-WAN device, enter the IP address of the L3 interface.</p>                                                                                                                                                                                                                                                                                                                |
| <b>GRE tunnel source IP</b>  | Enter the IP address of the extended WAN interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>XConnect</b>              | Enter the name of a physical interface on the same router that connects to the WAN transport.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Load Interval</b>         | Enter an interval value for interface load calculation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>IP Directed Broadcast</b> | <p>An IP directed broadcast is an IP packet whose destination address is a valid broadcast address for some IP subnet, but which originates from a node that is not itself part of that destination subnet.</p> <p>A device that is not directly connected to its destination subnet forwards an IP directed broadcast in the same way it would forward unicast IP packets destined to a host on that subnet. When a directed broadcast packet reaches a device that is directly connected to its destination subnet, that packet is broadcast on the destination subnet. The destination address in the IP header of the packet is rewritten to the configured IP broadcast address for the subnet, and the packet is sent as a link-layer broadcast.</p> <p>If directed broadcast is enabled for an interface, incoming IP packets whose addresses identify them as directed broadcasts intended for the subnet to which that interface is attached are broadcast on that subnet.</p> |
| <b>ICMP Redirect Disable</b> | <p>ICMP redirects are sent by a router to the sender of an IP packet when a packet is being routed sub-optimally. The ICMP redirect informs the sending host to forward subsequent packets to that same destination through a different gateway.</p> <p>By default, an interface allows ICMP redirect messages.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

**What to do next**

Also see [Deploy a Configuration Group](#).

# Configure Management VPN Using a Configuration Group

**Before you begin**

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

**Procedure**

**Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

**Step 2** Create and configure a management VPN feature in a Transport and Management Profile.

- a. Configure management VPN.

*Table 159: Management VPN*

| Field         | Description                                                                                |
|---------------|--------------------------------------------------------------------------------------------|
| Type          | Choose a feature from the drop-down list.                                                  |
| Feature Name* | Enter a name for the feature.                                                              |
| Description   | Enter a description of the feature. The description can contain any characters and spaces. |

- b. Configure basic settings.

*Table 160: Basic Configuration*

| Field | Description                                                                                                                                                                                                                                                                                   |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VPN   | Management VPN carries out-of-band network management traffic among the Cisco IOS XE Catalyst SD-WAN devices in the overlay network. The interface used for management traffic resides in VPN 512. By default, VPN 512 is configured and enabled on all Cisco IOS XE Catalyst SD-WAN devices. |
| Name  | Enter a name for the interface.                                                                                                                                                                                                                                                               |

- c. Configure DNS.

*Table 161: DNS*

| Field                      | Description                                                   |
|----------------------------|---------------------------------------------------------------|
| Add DNS                    |                                                               |
| Primary DNS Address (IPv4) | Enter the IPv4 address of the primary DNS server in this VPN. |

| Field                               | Description                                                   |
|-------------------------------------|---------------------------------------------------------------|
| <b>Secondary DNS Address (IPv4)</b> | Enter the IPv4 address of a secondary DNS server in this VPN. |
| <b>Add DNS IPv6</b>                 |                                                               |
| <b>Primary DNS Address (IPv6)</b>   | Enter the IPv6 address of the primary DNS server in this VPN. |
| <b>Secondary DNS Address (IPv6)</b> | Enter the IPv6 address of a secondary DNS server in this VPN. |

- d. Configure host mapping.

*Table 162: Host Mapping*

| Field                       | Description                                                                          |
|-----------------------------|--------------------------------------------------------------------------------------|
| <b>Add New Host Mapping</b> |                                                                                      |
| <b>Hostname*</b>            | Enter the hostname of the DNS server. The name can be up to 128 characters.          |
| <b>List of IP Address*</b>  | Enter IP addresses to associate with the hostname. Separate the entries with commas. |

- e. Configure IPv4/IPv6 static route.

*Table 163: IPv4/IPv6 Static Route*

| Field                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Add IPv4 Static Route</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>IP Address*</b>           | Enter the IPv4 address or prefix, in decimal four-point-dotted notation, and the prefix length of the IPv4 static route to configure in the VPN.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Subnet Mask*</b>          | Enter the subnet mask.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Gateway*</b>              | <p>Choose one of the following options to configure the next hop to reach the static route:</p> <ul style="list-style-type: none"> <li>• <b>nextHop</b>: When you choose this option and click <b>Add Next Hop</b>, the following fields appear: <ul style="list-style-type: none"> <li>• <b>Address*</b>: Enter the next-hop IPv4 address.</li> <li>• <b>Administrative distance*</b>: Enter the administrative distance for the route.</li> </ul> </li> <li>• <b>dhcp</b></li> <li>• <b>null0</b>: When you choose this option, the following field appears: <ul style="list-style-type: none"> <li>• <b>Administrative distance</b>: Enter the administrative distance for the route.</li> </ul> </li> </ul> |
| <b>Add IPv6 Static Route</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| Field                      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prefix*</b>             | Enter the IPv6 address or prefix, in decimal four-point-dotted notation, and the prefix length of the IPv6 static route to configure in the VPN.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Next Hop/Null 0/NAT</b> | <p>Choose one of the following options to configure the next hop to reach the static route:</p> <ul style="list-style-type: none"> <li>• <b>Next Hop:</b> When you choose this option and click <b>Add Next Hop</b>, the following fields appear: <ul style="list-style-type: none"> <li>• <b>Address*:</b> Enter the next-hop IPv6 address.</li> <li>• <b>Administrative distance*:</b> Enter the administrative distance for the route.</li> </ul> </li> <li>• <b>Null 0:</b> When you choose this option, the following field appears: <ul style="list-style-type: none"> <li>• <b>NULL0*:</b> Enable this option to set the next hop to be the null interface. All packets sent to this interface are dropped without sending any ICMP messages.</li> </ul> </li> <li>• <b>NAT:</b> When you choose this option, the following field appears: <ul style="list-style-type: none"> <li>• <b>IPv6 NAT:</b> Choose NAT64 or NAT66.</li> </ul> </li> </ul> |

### What to do next

Also see [Deploy a configuration group](#).

## Configure VPN Ethernet Interface

### Procedure

**Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration** > **Templates**.

**Step 2** Click **Device Templates**, and click **Create Template**.

#### Note

In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

**Step 3** From the **Create Template** drop-down list, choose **From Feature Template**.

**Step 4** From the **Device Model** drop-down list, choose the type of device for which you are creating the template.

**Step 5** To create a template for VPN 0 or VPN 512:

- Click **Transport & Management VPN** or scroll to the **Transport & Management VPN** section.
- Under **Additional VPN 0 Templates**, click **Cisco VPN Interface Ethernet**.

- c. From the **VPN Interface** drop-down list, click **Create Template**. The **Cisco VPN Interface Ethernet** template form displays.

This form contains fields for naming the template, and fields for defining the VPN Interface Ethernet parameters.

**Step 6** In **Template Name**, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.

**Step 7** In **Template Description**, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

## Configure Basic Interface Functionality

To configure basic interface functionality in a VPN, choose **Basic Configuration** and configure the following parameters:



**Note** Parameters marked with an asterisk are required to configure an interface.

| Parameter Name         | IPv4 or IPv6 | Options                  | Description                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------|--------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Shutdown*</b>       |              |                          | Click <b>No</b> to enable the interface.                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Interface name*</b> |              |                          | Enter a name for the interface.<br>For Cisco IOS XE Catalyst SD-WAN devices, you must: <ul style="list-style-type: none"> <li>• Spell out the interface names completely (for example, GigabitEthernet0/0/0).</li> <li>• Configure all the router's interfaces, even if you are not using them, so that they are configured in the shutdown state and so that all default values for them are configured.</li> </ul> |
| <b>Description</b>     |              |                          | Enter a description for the interface.                                                                                                                                                                                                                                                                                                                                                                               |
| <b>IPv4 / IPv6</b>     |              |                          | Click <b>IPv4</b> to configure an IPv4 VPN interface. Click <b>IPv6</b> to configure an IPv6 interface.                                                                                                                                                                                                                                                                                                              |
| <b>Dynamic</b>         |              |                          | Click <b>Dynamic</b> to set the interface as a Dynamic Host Configuration Protocol (DHCP) client, so that the interface receives its IP address from a DHCP server.                                                                                                                                                                                                                                                  |
|                        | <b>Both</b>  | <b>DHCP Distance</b>     | Optionally, enter an administrative distance value for routes learned from a DHCP server. Default is 1.                                                                                                                                                                                                                                                                                                              |
|                        | <b>IPv6</b>  | <b>DHCP Rapid Commit</b> | Optionally, configure the DHCP IPv6 local server to support DHCP Rapid Commit, to enable faster client configuration and confirmation in busy environments.<br>Click <b>On</b> to enable DHCP rapid commit.<br>Click <b>Off</b> to continue using the regular commit process.                                                                                                                                        |

| Parameter Name              | IPv4 or IPv6    | Options             | Description                                                                                                                                                                                                                                                       |
|-----------------------------|-----------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Static</b>               |                 |                     | Click <b>Static</b> to enter an IP address that doesn't change.                                                                                                                                                                                                   |
|                             | <b>IPv4</b>     | <b>IPv4 Address</b> | Enter a static IPv4 address.                                                                                                                                                                                                                                      |
|                             | <b>IPv6</b>     | <b>IPv6 Address</b> | Enter a static IPv6 address.                                                                                                                                                                                                                                      |
| <b>Secondary IP Address</b> | <b>IPv4</b>     |                     | Click <b>Add</b> to enter up to four secondary IPv4 addresses for a service-side interface.                                                                                                                                                                       |
| <b>IPv6 Address</b>         | <b>IPv6</b>     |                     | Click <b>Add</b> to enter up to two secondary IPv6 addresses for a service-side interface.                                                                                                                                                                        |
| <b>DHCP Helper</b>          | <b>Both</b>     |                     | To designate the interface as a DHCP helper on a router, enter up to eight IP addresses, separated by commas, for DHCP servers in the network. A DHCP helper interface forwards BootP (broadcast) DHCP requests that it receives from the specified DHCP servers. |
| <b>Block Non-Source IP</b>  | <b>Yes / No</b> |                     | Click <b>Yes</b> to have the interface forward traffic only if the source IP address of the traffic matches the interface's IP prefix range. Click <b>No</b> to allow other traffic.                                                                              |

To save the feature template, click **Save**.

## Create a Tunnel Interface

On Cisco IOS XE Catalyst SD-WAN devices, you can configure up to eight tunnel interfaces. This means that each Cisco IOS XE Catalyst SD-WAN device router can have up to eight TLOCs. On Cisco Catalyst SD-WAN Controllers and Cisco SD-WAN Manager, you can configure one tunnel interface.

For the control plane to establish itself so that the overlay network can function, you must configure WAN transport interfaces in VPN 0. The WAN interface will enable the flow of tunnel traffic to the overlay. You can add other parameters shown in the table below only after you configure the WAN interface as a tunnel interface.

To configure a tunnel interface, select **Interface Tunnel** and configure the following parameters:

| Parameter Name       | Description                                                                                                                                                                                                                                                                  |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tunnel Interface     | Click <b>On</b> to create a tunnel interface.                                                                                                                                                                                                                                |
| Color                | Select a color for the TLOC.                                                                                                                                                                                                                                                 |
| Color Description    | Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.18.1<br>Enter a description associated to the TLOC color.                                                                                                                                                |
| <b>Full Port Hop</b> | Minimum release: Cisco IOS XE Catalyst SD-WAN Release 17.18.1a<br>Enable full port hopping at the TLOC level to allow devices to establish connections with controllers by switching to the next port if the current port is blocked or non-functional.<br>Default: Disabled |



| Parameter Name      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Port Hop            | <p>Click <b>On</b> to enable port hopping, or click <b>Off</b> to disable it. If port hopping is enabled globally, you can disable it on an individual TLOC (tunnel interface). To control port hopping on a global level, use the <a href="#">System</a> configuration template.</p> <p>Default: Enabled</p> <p>Cisco SD-WAN Manager and Cisco Catalyst SD-WAN Controller default: Disabled</p> <p>Starting from Cisco IOS XE Catalyst SD-WAN Release 17.18.1a, this field is deprecated. Instead use the <b>Full Port Hop</b> option. See the <b>Full Port Hop</b> field.</p>                                                                                                                                                                                                                                                                                               |
| TCP MSS             | <p>TCP MSS affects any packet that contains an initial TCP header that flows through the router. When configured, TCP MSS is examined against the MSS exchanged in the three-way handshake. The MSS in the header is lowered if the configured TCP MSS setting is lower than the MSS in the header. If the MSS header value is already lower than the TCP MSS, the packets flow through unmodified. The host at the end of the tunnel uses the lower setting of the two hosts. If the TCP MSS is to be configured, it should be set at 40 bytes lower than the minimum path MTU.</p> <p>Specify the MSS of TPC SYN packets passing through the Cisco IOS XE Catalyst SD-WAN device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. <i>Range:</i> 552 to 1460 bytes <i>Default:</i> None</p> |
| Clear-Dont-Fragment | <p>Configure <b>Clear-Dont-Fragment</b> for packets that arrive at an interface that has Don't Fragment configured. If these packets are larger than what MTU allows, they are dropped. If you clear the Don't Fragment bit, the packets are fragmented and sent.</p> <p>Click <b>On</b> to clear the Dont Fragment bit in the IPv4 packet header for packets being transmitted out of the interface. When the Dont Fragment bit is cleared, packets larger than the MTU of the interface are fragmented before being sent.</p> <p><b>Note</b><br/><b>Clear-Dont-Fragment</b> clears the Dont Fragment bit and the Dont Fragment bit is set. For packets not requiring fragmentation, the Dont Fragment bit is not affected.</p>                                                                                                                                              |
| Allow Service       | Select <b>On</b> or <b>Off</b> for each service to allow or disallow the service on the interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

To configure additional tunnel interface parameters, click **Advanced Options**:

| Parameter Name | Description                                                                                                                                                                                                               |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Carrier        | <p>Select the carrier name or private network identifier to associate with the tunnel.</p> <p>Values: carrier1, carrier2, carrier3, carrier4, carrier5, carrier6, carrier7, carrier8, default</p> <p>Default: default</p> |

| Parameter Name       | Description                                                                                                                                                                                           |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NAT Refresh Interval | Enter the interval between NAT refresh packets sent on a DTLS or TLS WAN transport connection.<br><br>Range: 1 through 60 seconds<br><br>Default: 5 seconds                                           |
| Hello Interval       | Enter the interval between Hello packets sent on a DTLS or TLS WAN transport connection.<br><br>Range: 100 through 10000 milliseconds<br><br>Default: 1000 milliseconds (1 second)                    |
| Hello Tolerance      | Enter the time to wait for a Hello packet on a DTLS or TLS WAN transport connection before declaring that transport tunnel to be down.<br><br>Range: 12 through 60 seconds<br><br>Default: 12 seconds |

## Associate a Carrier Name with a Tunnel Interface

To associate a carrier name or private network identifier with a tunnel interface, use the **carrier** command. *carrier-name* can be **default** and **carrier1** through **carrier8**:

```
Device(config)# interface Tunnel 0
Device(config-if)# ip unnumbered GigabitEthernet1
Device(config-if)# ipv6 unnumbered GigabitEthernet2
Device(config-if)# tunnel source GigabitEthernet1
Device(config-if)# tunnel mode sdwan
Device(config-if)# exit
Device(config)# sdwan
Device(config-sdwan)# int GigabitEthernet1
Device(config-interface-GigabitEthernet1)# tunnel-interface
Device(config-tunnel-interface)# carrier default
```

## Create Tunnel Groups

By default, WAN Edge routers try to build tunnels with all other TLOCs in the network, regardless of color. When the restrict option is used with the color designation under the tunnel configuration, the TLOC is restricted to only building tunnels to TLOCs of the same color. For more information on the restrict option see, [Configure Interfaces in the WAN Transport VPN\(VPN0\)](#).

The tunnel group feature is similar to the restrict option but gives more flexibility because once a tunnel group ID is assigned under a tunnel, only TLOCs with the same tunnel group IDs can form tunnels with each other irrespective of color.

If a TLOC is associated with a tunnel group ID, it continues to form tunnels with other TLOCs in the network that are not associated with any tunnel group IDs.



**Note** The restrict option can still be used in conjunction with this feature. If used, then an interface with a tunnel group ID and restrict option defined on an interface will only form a tunnel with other interfaces with the same tunnel group ID and color.

## Configure Tunnel Groups on Cisco IOS XE Catalyst SD-WAN devices Using CLI

To configure tunnel groups on Cisco IOS XE Catalyst SD-WAN devices:

```
Device(config)# sdwan
Device(config-sdwan)# interface GigabitEthernet2

Device(config-interface-GigabitEthernet2)# tunnel-interface
Device(config-tunnel-interface)#group Group ID
```

## Limit Keepalive Traffic on a Tunnel Interface

By default, Cisco IOS XE Catalyst SD-WAN devices send a Hello packet once per second to determine whether the tunnel interface between two devices is still operational and to keep the tunnel alive. The combination of a hello interval and a hello tolerance determines how long to wait before declaring a DTLS or TLS tunnel to be down. The default hello interval is 1 second, and the default tolerance is 12 seconds. With these default values, if no Hello packet is received within 11 seconds, the tunnel is declared down at 12 seconds.

If the hello interval or the hello tolerance, or both, are different at the two ends of a DTLS or TLS tunnel, the tunnel chooses the interval and tolerance as follows:

- For a tunnel connection between two controller devices, the tunnel uses the lower hello interval and the higher tolerance interval for the connection between the two devices. (Controller devices are Cisco SD-WAN Validator, Cisco SD-WAN Manager, and Cisco SD-WAN Controller.) This choice is made in case one of the controllers has a slower WAN connection. The hello interval and tolerance times are chosen separately for each pair of controller devices.
- For a tunnel connection between a Cisco IOS XE Catalyst SD-WAN device and any controller device, the tunnel uses the hello interval and tolerance times configured on the router. This choice is made to minimize the amount traffic sent over the tunnel, to allow for situations where the cost of a link is a function of the amount of traffic traversing the link. The hello interval and tolerance times are chosen separately for each tunnel between a Cisco IOS XE Catalyst SD-WAN device and a controller device.

To minimize the amount of keepalive traffic on a tunnel interface, increase the Hello packet interval and tolerance on the tunnel interface:

```
Device(config-tunnel-interface)# hello-interval milliseconds
Device(config-tunnel-interface)# hello-tolerance seconds
```

The default hello interval is 1000 milliseconds, and it can be a time in the range 100 through 600000 milliseconds (10 minutes). The default hello tolerance is 12 seconds, and it can be a time in the range 12 through 600 seconds (10 minutes). The hello tolerance interval must be at most one-half the OMP hold time. You can configure the OMP hold time with the **omp timers holdtime** command.

## Configure an Interface as a NAT Device

For information on how to configure NAT, see the [Cisco Catalyst SD-WAN NAT Configuration Guide, Cisco IOS XE Catalyst SD-WAN Release 17.x](#).

## Apply Access Lists and QoS Parameters

Quality of service (QoS) helps determine how a service will perform. By configuring QoS, enhance the performance of an application on the WAN. To configure a shaping rate for an interface and to apply a QoS

map, a rewrite rule, access lists, and policers to a interface, click **ACL/QoS**, and configure the following parameters:

| Parameter Name     | Description                                                                                                                  |
|--------------------|------------------------------------------------------------------------------------------------------------------------------|
| Shaping rate       | Configure the aggregate traffic transmission rate on the interface to be less than line rate, in kilobits per second (kbps). |
| QoS Map            | Specify the name of the QoS map to apply to packets being transmitted out the interface.                                     |
| Rewrite Rule       | Click <b>On</b> , and specify the name of the rewrite rule to apply on the interface.                                        |
| Ingress ACL – IPv4 | Click <b>On</b> , and specify the name of the access list to apply to IPv4 packets being received on the interface.          |
| Egress ACL – IPv4  | Click <b>On</b> , and specify the name of the access list to apply to IPv4 packets being transmitted on the interface.       |
| Ingress ACL – IPv6 | Click <b>On</b> , and specify the name of the access list to apply to IPv6 packets being received on the interface.          |
| Egress ACL – IPv6  | Click <b>On</b> , and specify the name of the access list to apply to IPv6 packets being transmitted on the interface.       |
| Ingress Policer    | Click <b>On</b> , and specify the name of the policer to apply to packets received on the interface.                         |
| Egress Policer     | Click <b>On</b> , and specify the name of the policer to apply to packets being transmitted on the interface.                |

To save the feature template, click **Save**.

## Add ARP Table Entries

The Address Resolution Protocol (ARP) helps associate a link layer address (such as the MAC address of a device) to its assigned internet layer address. Configure a static ARP address when dynamic mapping is not functional. To configure static ARP table entries on the interface, select ARP. Then click **Add New ARP** and configure the following parameters:

| Parameter Name | Description                                                                                          |
|----------------|------------------------------------------------------------------------------------------------------|
| IP Address     | Enter the IP address for the ARP entry in dotted decimal notation or as a fully qualified host name. |
| MAC Address    | Enter the MAC address in colon-separated hexadecimal notation.                                       |

To save the ARP configuration, click **Add**.

To save the feature template, click **Save**.

## Configuring VRRP

To have an interface run the Virtual Router Redundancy Protocol (VRRP), which allows multiple routers to share a common virtual IP address for default gateway redundancy, select the VRRP tab. Then click **Add New VRRP** and configure the following parameters:

| Parameter Name                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Group ID                       | Enter the virtual router ID, which is a numeric identifier of the virtual router. You can configure a maximum of 24 groups.<br><br>Range: 1 through 255                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Priority                       | Enter the priority level of the router. The router with the highest priority is elected as primary VRRP router. If two routers have the same priority, the one with the higher IP address is elected as primary VRRP router.<br><br>Range: 1 through 254<br>Default: 100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Timer (milliseconds)           | Specify how often the primary VRRP router sends VRRP advertisement messages. If subordinate routers miss three consecutive VRRP advertisements, they elect a new primary VRRP router.<br><br>Range: 100 through 40950 milliseconds<br>Default: 100 msec<br><br><b>Note</b><br>When the timer is 100 ms for the VRRP feature template on Cisco IOS XE Catalyst SD-WAN devices, the VRRP fails if the traffic is high on LAN interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Track OMP<br>Track Prefix List | By default, VRRP uses the state of the service (LAN) interface on which it is running to determine which router is the primary virtual router. If a router loses all its WAN control connections, the LAN interface still indicates that it is up even though the router is functionally unable to participate in VRRP. To take WAN side connectivity into account for VRRP, configure one of the following:<br><br><b>Track OMP</b> —Click <b>On</b> for VRRP to track the Overlay Management Protocol (OMP) session running on the WAN connection. If the primary VRRP router loses all its OMP sessions, VRRP elects a new default gateway from those that have at least one active OMP session.<br><br><b>Note</b><br>From Cisco IOS XE Catalyst SD-WAN Release 17.18.1a, enabling <b>Track OMP</b> changes the device CLI command from <b>vrrp track omp shutdown</b> to <b>vrrp track omp decrement 10</b> .<br><br><b>Track Prefix List</b> —Track both the OMP session and a list of remote prefixes, which is defined in a prefix list configured on the local router. If the primary VRRP router loses all its OMP sessions, VRRP failover occurs as described for the Track OMP option. In addition, if reachability to all of the prefixes in the list is lost, VRRP failover occurs immediately, without waiting for the OMP hold timer to expire, thus minimizing the amount of overlay traffic is dropped while the routers determine the primary VRRP router. |
| IP Address                     | Enter the IP address of the virtual router. This address must be different from the configured interface IP addresses of both the local router and the peer running VRRP.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

## Configure a Prefix List for VRRP

You can configure prefix list tracking for VRRP using device and feature templates. To configure a prefix list, do the following:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Policy**.
2. Click **Localized Policy**.
3. From the **Custom Options** drop-down list, click **Lists**.
4. Click **Prefix** from the left pane, and click **New Prefix List**.
5. In **Prefix List Name**, enter a name for the prefix list.
6. Choose **IPv4** as the **Internet Protocol**.
7. In **Add Prefix**, enter the prefix entries separated by commas.
8. Click **Add**.
9. Click **Next** and configure **Forwarding Classes/QoS**.
10. Click **Next** and configure **Access Control Lists**.
11. Click **Next** and in **Route Policy** pane, select a relevant route policy and click **...**, and click **Edit** to add the newly added prefix list.
12. From the **Match** pane, click **AS Path List** and in the **Address**, choose the newly added prefix list.
13. Click **Save Match and Actions**.
14. Click **Next** and enter the **Policy Name** and **Policy Description** in the **Policy Overview** screen.
15. Click **Save Policy**.

## Configure a Prefix List for VRRP in the Device Template

To configure the Prefix List to the VRRP and the localized policy in the device template, do the following:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**.




---

**Note** In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

---

3. Select a relevant device template and click **...** and click **Edit** to edit the template details.
4. From **Policy**, select the policy with the newly added prefix list.
5. Click **Update**.
6. Click **Feature Templates**.
7. Select a relevant device template and click **...** and click **Edit** to edit the template details.
8. Click **VRRP**.

9. Select a relevant group ID and click the pen icon to associate the new prefix-list to the VRRP details.
10. Click the **Track Prefix List** drop-down list and enter the newly added prefix-list name.
11. Click **Save Changes**.
12. Click **Update** to save the changes.
13. Click **Device Templates** and select the policy with the newly added prefix list.
14. Click ... and click **Attach Devices**.
15. From **Available Devices**, double-click the relevant device to move it to **Selected Devices**, and then click **Attach**.

## Configure Prefix List For VRRP Using Configuration Groups

### Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

### Procedure

**Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

**Step 2** Create and configure Prefix List for VRRP in a Policy Object Profile.

- a. Choose the **Prefix** policy object from the **Select Policy Object** drop-down list.
- b. Enter the **Prefix List Name**.
- c. In the **Internet Protocol** field, click **IPv4** or **IPv6**.
- d. Under **Add Prefix**, enter the prefix for the list. Optionally, click the **Choose a file** link to import a prefix list.
- e. Click **Save**.

The following table describe the options for configuring the prefix.

**Table 164: Prefix List**

| Field                    | Description                                                     |
|--------------------------|-----------------------------------------------------------------|
| <b>Prefix List Name</b>  | Enter a name for the prefix list.                               |
| <b>Internet Protocol</b> | Specifies the internet protocol. The options are IPv4 and IPv6. |

### What to do next

Also see [Deploy a configuration group](#).

## Configure Advanced Properties

To configure other interface properties, select the **Advanced** tab and configure the following parameters:

| Parameter Name      | Description                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Duplex              | Choose full or half to specify whether the interface runs in full-duplex or half-duplex mode.<br>Default: full                                                                                                                                                                                                                                                                                                                    |
| MAC Address         | Specify a MAC address to associate with the interface, in colon-separated hexadecimal notation.                                                                                                                                                                                                                                                                                                                                   |
| IP MTU              | Specify the maximum MTU size of packets on the interface.<br>Range: 576 through 1804<br>Default: 1500 bytes                                                                                                                                                                                                                                                                                                                       |
| PMTU Discovery      | Click <b>On</b> to enable path MTU discovery on the interface. PMTU determines the largest MTU size that the interface supports so that packet fragmentation does not occur.                                                                                                                                                                                                                                                      |
| Flow Control        | Select a setting for bidirectional flow control, which is a mechanism for temporarily stopping the transmission of data on the interface.<br>Values: autonet, both, egress, ingress, none<br>Default: autoneg                                                                                                                                                                                                                     |
| TCP MSS             | Specify the maximum segment size (MSS) of TCP SYN packets passing through the router. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented.<br>Range: 552 to 1460 bytes<br>Default: None                                                                                                                                                               |
| Speed               | Specify the speed of the interface for use when the remote end of the connection does not support autonegotiation.<br>Values: 10, 100, 1000, or 10000 Mbps                                                                                                                                                                                                                                                                        |
| Clear-Dont-Fragment | Click <b>On</b> to clear the Don't Fragment (DF) bit in the IPv4 packet header for packets being transmitted out the interface. When the DF bit is cleared, packets larger than that interface's MTU are fragmented before being sent.<br><br><b>Note</b><br>Clear-Dont-Fragment clears the DF bit when there is fragmentation needed and the DF bit is set. For packets not requiring fragmentation, the DF bit is not affected. |



| Parameter Name               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Autonegotiation              | <p><b>Note</b></p> <p>For releases before Cisco vManage Release 20.6.1, the default value of the field is <b>On</b>. To turn autonegotiation off, click <b>Off</b>.</p> <p>From Cisco vManage Release 20.6.1, the default behavior of the field is as follows:</p> <ul style="list-style-type: none"> <li>For the Gigabit Ethernet interface type, the <b>Autonegotiation</b> field is blank by default. However, the autonegotiation is set to <b>On</b> when the field is left blank.</li> <li>For other interface types such as Ten Gigabit Ethernet and Hundred Gigabit Ethernet, the <b>Autonegotiation</b> field is blank by default. To turn autonegotiation on or off, click <b>On</b> or <b>Off</b> respectively.</li> </ul> <p>From Cisco SD-WAN Manager Release 20.12.4:</p> <p>In the Cisco Catalyst 8300 Series devices, for the TenGigabitEthernet interface type, do not leave the <b>Autonegotiation</b> field blank.</p> |
| TLOC Extension               | <p>Enter the name of a physical interface on the same router that connects to the WAN transport. This configuration then binds this service-side interface to the WAN transport. A second router at the same site that itself has no direct connection to the WAN (generally because the site has only a single WAN connection) and that connects to this service-side interface is then provided with a connection to the WAN.</p> <p>Note that TLOC extension over L3 is only supported for Cisco IOS XE routers. If configuring TLOC extension over L3 for a Cisco IOS XE router, enter the IP address of the L3 interface.</p>                                                                                                                                                                                                                                                                                                        |
| GRE Tunnel Source IP         | Enter the IP address of the extended WAN interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Xconnect (on IOS XE routers) | Enter the name of a physical interface on the same router that connects to the WAN transport.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

To save the feature template, click **Save**.

## VPN Interface Bridge

Use the VPN Interface Bridge template for all Cisco IOS XE Catalyst SD-WAN device Cloud and Cisco IOS XE Catalyst SD-WAN devices.

Integrated routing and bridging (IRB) allows Cisco IOS XE Catalyst SD-WAN devices in different bridge domains to communicate with each other. To enable IRB, create logical IRB interfaces to connect a bridge domain to a VPN. The VPN provides the Layer 3 routing services necessary so that traffic can be exchanged between different VLANs. Each bridge domain can have a single IRB interface and can connect to a single VPN, and a single VPN can connect to multiple bridge domains on a Cisco IOS XE Catalyst SD-WAN device.

To configure a bridge interface using Cisco SD-WAN Manager templates:

1. Create a VPN Interface Bridge feature template to configure parameters for logical IRB interfaces, as described in this article.

2. Create a Bridge feature template for each bridging domain, to configure the bridging domain parameters. See the Bridge help topic.

### Navigate to the Template Screen and Name the Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**.




---

**Note** In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

---

3. From the **Create Template** drop-down list, select **From Feature Template**.
4. From the **Device Model** drop-down list, select the type of device for which you are creating the template.
5. Click **Service VPN** or scroll to the **Service VPN** section.
6. Click the **Service VPN** drop-down list.
7. From **Additional VPN Templates**, click **VPN Interface Bridge**.
8. From the **VPN Interface Bridge** drop-down list, click **Create Template**.  
The VPN Interface Bridge template form is displayed. The top of the form contains fields for naming the template, and the bottom contains fields for defining VPN Interface Bridge parameters.
9. In **Template Name**, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.
10. In **Template Description**, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default or to enter a value, click the scope drop-down to the left of the parameter field and select one of the following:

Table 165:

| Parameter Scope                               | Scope Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Device Specific<br>(indicated by a host icon) | <p>Use a device-specific value for the parameter. For device-specific parameters, you cannot enter a value in the feature template. You enter the value when you attach a Viptela device to a device template .</p> <p>When you click Device Specific, the Enter Key box opens. This box displays a key, which is a unique string that identifies the parameter in a CSV file that you create. This file is an Excel spreadsheet that contains one column for each key. The header row contains the key names (one key per column), and each row after that corresponds to a device and defines the values of the keys for that device. You upload the CSV file when you attach a Viptela device to a device template. For more information, see <a href="#">Create a Template Variables Spreadsheet</a> .</p> <p>To change the default key, type a new string and move the cursor out of the Enter Key box.</p> <p>Examples of device-specific parameters are system IP address, hostname, GPS location, and site ID.</p> |
| Global (indicated by a globe icon)            | <p>Enter a value for the parameter, and apply that value to all devices.</p> <p>Examples of parameters that you might apply globally to a group of devices are DNS server, syslog server, and interface MTUs.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

### Release Information

Introduced in Cisco SD-WAN Manager NMS in Release 15.3. In Release 18.2, add support for disabling ICMP redirect messages.

## Create a Bridging Interface

To configure an interface to use for bridging servers, select **Basic Configuration** and configure the following parameters. Parameters marked with an asterisk are required to configure bridging.

Table 166:

| Parameter Name  | Description                                                                                                                                                                                                                                                      |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Shutdown*       | Click <b>No</b> to enable the interface.                                                                                                                                                                                                                         |
| Interface name* | Enter the name of the interface, in the format <b>irb number</b> . The IRB interface number can be from 1 through 63, and must be the same as the VPN identifier configured in the Bridge feature template for the bridging domain that the IRB is connected to. |
| Description     | Enter a description for the interface.                                                                                                                                                                                                                           |
| IPv4 Address*   | Enter the IPv4 address of the router.                                                                                                                                                                                                                            |

| Parameter Name                                                 | Description                                                                                                                                                                                                                                     |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DHCP Helper                                                    | Enter up to eight IP addresses for DHCP servers in the network, separated by commas, to have the interface be a DHCP helper. A DHCP helper interface forwards BOOTP (Broadcast) DHCP requests that it receives from the specified DHCP servers. |
| Block Non-Source IP                                            | Click <b>Yes</b> to have the interface forward traffic only if the source IP address of the traffic matches the interface's IP prefix range.                                                                                                    |
| Secondary IP Address (on Cisco IOS XE Catalyst SD-WAN devices) | Click <b>Add</b> to configure up to four secondary IPv4 addresses for a service-side interface.                                                                                                                                                 |

To save the template, click **Save**.

## Apply Access Lists

### Apply Access Lists

To apply access lists to IRB interfaces, select the ACL tab and configure the following parameters. The ACL filter determines what is allowed in or out of a bridging domain:

**Table 167:**

| Parameter Name     | Description                                                                                                  |
|--------------------|--------------------------------------------------------------------------------------------------------------|
| Ingress ACL – IPv4 | Click <b>On</b> , and specify the name of an IPv4 access list to packets being received on the interface.    |
| Egress ACL– IPv4   | Click <b>On</b> , and specify the name of an IPv4 access list to packets being transmitted on the interface. |

To save the feature template, click **Save**.

## Configure VRRP

To have an interface run the Virtual Router Redundancy Protocol (VRRP), which allows multiple routers to share a common virtual IP address for default gateway redundancy, choose **VRRP**. Then click **Add New VRRP** and configure the following parameters:

**Table 168:**

| Parameter Name | Description                                                                                                                                             |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Group ID       | Enter the virtual router ID, which is a numeric identifier of the virtual router. You can configure a maximum of 24 groups. <i>Range:</i> 1 through 255 |

| Parameter Name                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Priority                       | Enter the priority level of the router. The router with the highest priority is elected as primary VRRP router. If two Cisco IOS XE Catalyst SD-WAN devices have the same priority, the one with the higher IP address is elected as primary VRRP router. <i>Range:</i> 1 through 254 <i>Default:</i> 100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Timer (milliseconds)           | Specify how often the primary VRRP router sends VRRP advertisement messages. If subordinate routers miss three consecutive VRRP advertisements, they elect a new primary VRRP router.<br><br><i>Range:</i> 100 through 40950 milliseconds<br><br><i>Default:</i> 100 msec<br><br><b>Note</b><br>When the timer is 100 ms for the VRRP feature template on Cisco IOS XE Catalyst SD-WAN devices, the VRRP fails if the traffic is high on LAN interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Track OMP<br>Track Prefix List | By default, VRRP uses the state of the service (LAN) interface on which it is running to determine which Cisco IOS XE Catalyst SD-WAN device is the primary virtual router. If a Cisco IOS XE Catalyst SD-WAN device loses all its WAN control connections, the LAN interface still indicates that it is up even though the router is functionally unable to participate in VRRP. To take WAN side connectivity into account for VRRP, configure one of the following:<br><br>Track OMP—Click <b>On</b> for VRRP to track the Overlay Management Protocol (OMP) session running on the WAN connection. If the primary VRRP router loses all its OMP sessions, VRRP elects a new default gateway from those that have at least one active OMP session.<br><br>Track Prefix List—Track both the OMP session and a list of remote prefixes, which is defined in a prefix list configured on the local router. If the primary VRRP router loses all its OMP sessions, VRRP failover occurs as described for the Track OMP option. In addition, if reachability to all of the prefixes in the list is lost, VRRP failover occurs immediately, without waiting for the OMP hold timer to expire, thus minimizing the amount of overlay traffic is dropped while the Cisco IOS XE Catalyst SD-WAN devices determine the primary VRRP router. |
| IP Address                     | Enter the IP address of the virtual router. This address must be different from the configured interface IP addresses of both the local Cisco IOS XE Catalyst SD-WAN device and the peer running VRRP.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

To save the VRRP configuration, click **Add**.

To save the feature template, click **Save**.

## Add ARP Table Entries

To configure static Address Resolution Protocol (ARP) table entries on the interface, choose **ARP**. Then click **Add New ARP** and configure the following parameters:

Table 169:

| Parameter Name | Description                                                                                          |
|----------------|------------------------------------------------------------------------------------------------------|
| IP Address     | Enter the IP address for the ARP entry in dotted decimal notation or as a fully qualified host name. |
| MAC Address    | Enter the MAC address in colon-separated hexadecimal notation.                                       |

To save the ARP configuration, click **Add**.

To save the feature template, click **Save**.

## Configure Advanced Properties

To configure other interface properties, click **Advanced** and configure the following parameters:

Table 170:

| Parameter Name | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MAC Address    | <p>MAC addresses can be static or dynamic. A static MAC address is manually configured as opposed to a dynamic MAC address that is one learned via an ARP request. You can configure a static MAC on a router's interface or indicate a static MAC that identifies a router's interface.</p> <p>Specify a MAC address to associate with the interface, in colon-separated hexadecimal notation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| IP MTU         | <p>Similar to MTU, IP MTU only affects IP packets. If an IP packet exceeds the IP MTU, then the packet will be fragmented.</p> <p>Specify the maximum MTU size of packets on the interface. <i>Range:</i> 576 through 1804 <i>Default:</i> 1500 bytes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| TCP MSS        | <p>TCP MSS will affect any packet that contains an initial TCP header that flows through the router. When configured, TCP MSS will be examined against the MSS exchanged in the three-way handshake. The MSS in the header will be lowered if the configured setting is lower than what is in the header. If the header value is already lower, it will flow through unmodified. The end hosts will use the lower setting of the two hosts. If the TCP MSS is to be configured, it should be set it at 40 bytes lower than the minimum path MTU.</p> <p>Specify the maximum segment size (MSS) of TPC SYN packets passing through the Cisco IOS XE Catalyst SD-WAN device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. <i>Range:</i> 552 to 1460 bytes <i>Default:</i> None</p> |

| Parameter Name      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Clear-Dont-Fragment | <p>Configure Clear-Dont-Fragment if there are packets arriving on an interface with the DF bit set. If these packets are larger than the MTU will allow, they are dropped. If you clear the df-bit, the packets will be fragmented and sent.</p> <p>Click <b>On</b> to clear the Dont Fragment (DF) bit in the IPv4 packet header for packets being transmitted out the interface. When the DF bit is cleared, packets larger than that interface's MTU are fragmented before being sent.</p> <p><b>Note</b><br/>Clear-Dont-Fragment clears the DF bit when there is fragmentation needed and the DF bit is set. For packets not requiring fragmentation, the DF bit is not affected.</p> |
| ARP Timeout         | <p>ARP Timeout controls how long we maintain the ARP cache on a router.</p> <p>Specify how long it takes for a dynamically learned ARP entry to time out.</p> <p><i>Range: 0 through 2678400 seconds (744 hours)Default: 1200 seconds (20 minutes)</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                |
| ICMP Redirect       | <p>ICMP Redirects are sent by a router to the sender of an IP packet when a packet is being routed sub-optimally.</p> <p>The ICMP Redirect informs the sending host to forward subsequent packets to that same destination through a different gateway.</p> <p>To disable ICMP redirect messages on the interface, click <b>Disable</b>. By default, an interface allows ICMP redirect messages.</p>                                                                                                                                                                                                                                                                                      |

To save the feature template, click **Save**.

## Configure DSL IPoE Using a Configuration Group

### Before you begin

On the **Configuration** > **Configuration Groups** page, choose **SD-WAN** as the solution type.

### Procedure

- Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration** > **Configuration Groups**.
- Step 2** Create and configure the DSL IPoE parameters in a Transport and Management Profile.
- a. Configure basic configuration.

**Table 171: Basic Configuration**

| Parameter Name   | Description                                                                                                                   |
|------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Controller Slot* | <p>Enter the slot number of the controller, in the following format:</p> <p><i>slot/subslot/port</i> (for example, 0/2/0)</p> |

| Parameter Name  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Controller Mode | <p>Select the operating mode of the DSL controller from the drop-down list:</p> <ul style="list-style-type: none"> <li>• <b>ADSL1</b>: Use ITU G.992.1 Annex A full-rate mode, which provides a downstream rate of 1.3 Mbps and an upstream rate of 1.8 Mbps.</li> <li>• <b>ADSL2</b>: Use ITU G.992.3 Annex A, Annex L, and Annex M, which provides a downstream rate of 12 Mbps and an upstream rate of 1.3 Mbps.</li> <li>• <b>ADSL2+</b>: Use ITU G.992.5 Annex A and Annex M, which provides a downstream rate of 24 Mbps and an upstream rate of 3.3 Mbps.</li> <li>• <b>ANSI</b>: Operating in ADSL2/2+ mode, as defined in ITU G.991.1, G.992.3, and G.992.5, Annex A and Annex M, and in VDSL2 mode, as defined in ITU-T G.993.2.</li> <li>• <b>VDSL2</b>: Operate in VDSL2 mode, as defined in ITU-T G.993.2, which uses frequencies of up to 30 MHz to provide a downstream rate of 200 Mbps and an upstream rate of 100 Mbps.</li> </ul> |
| SRA             | Enabled by default. Click <b>No</b> to disable seamless rate adaptation on the interface. SRA adjusts the line rate based on current line conditions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

## b. Configure Ethernet.

Table 172: Ethernet

| Parameter Name               | Description                                                                                                                                                              |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ethernet Interface Name<br>* | <p>Enter the name of an ethernet interface.</p> <p>For IOS XE routers, you must spell out the interface names completely (for example, <b>GigabitEthernet0/0/0</b>).</p> |
| Description                  | Enter a description for the interface.                                                                                                                                   |
| VLAN ID                      | Enter the VLAN identifier of the Ethernet interface.                                                                                                                     |

## c. Configure Tunnel.

Table 173: Tunnel

| Parameter Name          | Description                                                                                                                                                                                            |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tunnel Interface</b> |                                                                                                                                                                                                        |
| Per Tunnel QoS          | <p>Enable per tunnel QoS and choose from the following values to configure hub-to-spoke network topologies:</p> <ul style="list-style-type: none"> <li>• <b>Spoke</b></li> <li>• <b>Hub</b></li> </ul> |
| Color                   | Select a color for the TLOC.                                                                                                                                                                           |



| Parameter Name                             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Color Description                          | Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.18.1<br>Enter a description associated to the TLOC color.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Groups                                     | Enter the list of groups in the field.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Exclude Controller Group List              | Set the Cisco SD-WAN Controllers that the tunnel interface is not allowed to connect to.<br>Range: 0 through 100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Maximum Control Connections                | Specify the maximum number of Cisco SD-WAN Controllers that the WAN tunnel interface can connect to. To have the tunnel establish no control connections, set the number to 0.<br>Range: 0 through 8<br>Default: 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Cisco SD-WAN Manager Connection Preference | Set the preference for using a tunnel interface to exchange control traffic with Cisco SD-WAN Manager.<br>Range: 0 through 8<br>Default: 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Tunnel TCP MSS                             | TCP MSS affects any packet that contains an initial TCP header that flows through the router. When configured, TCP MSS is examined against the MSS exchanged in the three-way handshake. The MSS in the header is lowered if the configured TCP MSS setting is lower than the MSS in the header. If the MSS header value is already lower than the TCP MSS, the packets flow through unmodified. The host at the end of the tunnel uses the lower setting of the two hosts. To configure TCP MSS, provide a value that is 40 bytes lower than the minimum path MTU.<br><br>Specify the MSS of TPC SYN packets passing through the Cisco IOS XE Catalyst SD-WAN. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented.<br>Range: 552 through 1460 bytes<br>Default: None |
| Border                                     | From the drop-down list, select <b>Global</b> . Click <b>On</b> to set TLOC as border TLOC.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Validator As Stun Server                   | Click <b>On</b> to enable Session Traversal Utilities for NAT (STUN) to allow the tunnel interface to discover its public IP address and port number when the router is located behind a NAT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Full Port Hop                              | Minimum release: Cisco IOS XE Catalyst SD-WAN Release 17.18.1a<br><br>Enable full port hopping at the TLOC level to allow devices to establish connections with controllers by switching to the next port if the current port is blocked or non-functional.<br>Default: Disabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Port Hop                                   | From the drop-down list, select <b>Global</b> . Click <b>Off</b> to allow port hopping on tunnel interface.<br>Default: <b>On</b> , which disallows port hopping on tunnel interface.<br><br>Starting from Cisco IOS XE Catalyst SD-WAN Release 17.18.1a, this field is deprecated. Instead use the <b>Full Port Hop</b> option. See the <b>Full Port Hop</b> field.                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| Parameter Name       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Low-Bandwidth Link   | Click <b>On</b> to set the tunnel interface as a low-bandwidth link.<br>Default: <b>Off</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Clear-Dont-Fragment  | Configure <b>Clear-Dont-Fragment</b> for packets that arrive at an interface that has Don't Fragment configured. If these packets are larger than what MTU allows, they are dropped. If you clear the Don't Fragment bit, the packets are fragmented and sent.<br><br>Click <b>On</b> to clear the Dont Fragment bit in the IPv4 packet header for packets being transmitted out of the interface. When the Dont Fragment bit is cleared, the router fragments packets larger than the MTU of the interface before sending the packets.<br><br><b>Note</b><br><b>Clear-Dont-Fragment</b> clears the Dont Fragment bit and the Dont Fragment bit is set. For packets not requiring fragmentation, the Dont Fragment bit is not affected. |
| Network Broadcast    | From the drop-down list, select <b>Global</b> . Click <b>On</b> to accept and respond to network-prefix-directed broadcasts. Enable this parameter only if the <b>Directed Broadcast</b> is enabled on the LAN interface feature template.<br>Default: <b>Off</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Carrier              | From the drop-down list, select <b>Global</b> and select the carrier name or private network identifier to associate with the tunnel.<br><br>Values: carrier1, carrier2, carrier3, carrier4, carrier5, carrier6, carrier7, carrier8, default.<br>Default: default                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Bind Loopback Tunnel | Enter the name of a physical interface to bind to a loopback interface. The interface name has the following format:<br><br><b>ge slot/port</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| NAT Refresh Interval | Set the interval between NAT refresh packets sent on a DTLS or TLS WAN transport connection.<br><br>Range: 1 through 60 seconds<br>Default: 5 seconds                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Hello Interval       | Enter the interval between Hello packets sent on a DTLS or TLS WAN transport connection.<br><br>Range: 100 through 10000 milliseconds<br>Default: 1000 milliseconds (1 second)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

| Parameter Name       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hello Tolerance      | <p>Enter the time to wait for a Hello packet on a DTLS or TLS WAN transport connection before declaring that transport tunnel to be down.</p> <p>Range: 12 through 60 seconds. Default: 12 seconds.</p> <p>The default hello interval is 1000 milliseconds, and it can be a time in the range 100 through 600000 milliseconds (10 minutes). The default hello tolerance is 12 seconds, and it can be a time in the range 12 through 600 seconds (10 minutes). To reduce outgoing control packets on a TLOC, it is recommended that on the tunnel interface you set the hello interval to 60000 milliseconds (10 minutes) and the hello tolerance to 600 seconds (10 minutes) and include the <b>no track-transport disable</b> regular checking of the DTLS connection between the edge device and the controller. For a tunnel connection between a edge device and any controller device, the tunnel uses the hello interval and tolerance times configured on the edge device. This choice is made to minimize the traffic sent over the tunnel, to allow for situations where the cost of a link is a function of the amount of traffic traversing the link. The hello interval and tolerance times are chosen separately for each tunnel between a edge device and a controller device. Another step taken to minimize the amount of control plane traffic is to not send or receive OMP control traffic over a cellular interface when other interfaces are available. This behavior is inherent in the software and is not configurable.</p> |
| Last Resort Circuit  | <p>Select to use the tunnel interface as the circuit of last resort.</p> <p><b>Note</b><br/>It is assumed that an interface configured as a circuit of last resort is unavailable and is skipped while calculating the number of control connections. As a result, the cellular modem becomes dormant, and no traffic is sent over the circuit.</p> <p>When the configurations are activated on the edge device with cellular interfaces, all the interfaces begin the process of establishing control and BFD connections. When one or more of the primary interfaces establishes a BFD connection, the circuit of last resort shuts itself down.</p> <p>If the primary interfaces lose their connections to remote edges, the circuit of last resort activates itself, triggering a BFD TLOC Down alarm and a Control TLOC Down alarm on the edge device. The last resort interfaces are a backup circuit on edge device and are activated when all other transport links BFD sessions fail. In this mode, the radio interface is turned off, and no control or data connections exist over the cellular interface.</p>                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Allow Services       | Click <b>On</b> or <b>Off</b> for each service to enable or disable the service on the cellular interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Encapsulation</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| Parameter Name | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Encapsulation  | <p>Enable atleast one of the following encapsulation methods:</p> <ul style="list-style-type: none"> <li>• <b>IPsec</b>: Enter a value to set the preference for directing traffic to the tunnel. A higher value is preferred over a lower value.<br/>Range: 0 through 4294967295<br/>Default: 0</li> <li>• <b>IPsec Preference</b>: From the drop-down list, select <b>Global</b> and enter a value to set the preference for directing traffic to the tunnel. A higher value is preferred over a lower value.<br/>Range: 0 through 4294967295<br/>Default: 0</li> <li>• <b>IPsec Weight</b>: From the drop-down list, select <b>Global</b> and enter a value to set weight for balancing traffic across multiple TLOCs. A higher value sends more traffic to the tunnel.<br/>Range: 1 through 255<br/>Default: 1</li> <li>• <b>GRE</b>: Enter a value to set GRE preference for TLOC.<br/>Range: 0 through 4294967295</li> <li>• <b>GRE Preference</b>: From the drop-down list, select <b>Global</b> and enter a value to set the preference for directing traffic to the tunnel. A higher value is preferred over a lower value.<br/>Range: 0 through 4294967295<br/>Default: 0</li> <li>• <b>GRE Weight</b>: From the drop-down list, select <b>Global</b> and enter a value to set weight for balancing traffic across multiple TLOCs. A higher value sends more traffic to the tunnel.<br/>Range: 1 through 255<br/>Default: 1</li> </ul> |

## d. Configure NAT.

Table 174: NAT

| Parameter Name        | Description                                                                                                                     |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------|
| UDP Timeout (Minutes) | <p>Specify when NAT translations over UDP sessions time out.</p> <p>Range: 1 through 65536 minutes</p> <p>Default: 1 minute</p> |

| Parameter Name        | Description                                                                                                                 |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------|
| TCP Timeout (Minutes) | Specify when NAT translations over TCP sessions time out.<br>Range: 1 through 65536 minutes<br>Default: 60 minutes (1 hour) |

e. Configure QoS.

**Table 175: QoS**

| Parameter Name      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Adaptive QoS        | Enter adaptive QoS parameters. You can leave the additional details at as default or specify your values. <ul style="list-style-type: none"> <li>• <b>Adapt Period (Minutes):</b> Choose <b>Global</b> from the drop-down list, click <b>On</b>, and enter the period in minutes.</li> <li>• <b>Shaping Rate Upstream:</b> Choose <b>Global</b> from the drop-down list, click <b>On</b>, and enter the minimum, maximum, and default upstream bandwidth in Kbps.</li> <li>• <b>Shaping Rate Downstream:</b> Choose <b>Global</b> from the drop-down list, click <b>On</b>, and enter the minimum, maximum, downstream, and upstream bandwidth in Kbps.</li> </ul> |
| Shaping Rate (kbps) | Choose <b>Global</b> from the drop-down list and configure the aggregate traffic transmission rate on the interface to be less than line rate, in kilobits per second (kbps).<br>Range: 8 through 100000000                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

f. Configure ACL.

**Table 176: ACL**

| Parameter Name           | Description                                                                          |
|--------------------------|--------------------------------------------------------------------------------------|
| IPv4 Ingress Access List | Enter the name of an IPv4 access list to packets being received on the interface.    |
| IPv4 Egress Access List  | Enter the name of an IPv4 access list to packets being transmitted on the interface. |
| IPv6 Ingress Access List | Enter the name of an IPv6 access list to packets being received on the interface.    |
| IPv6 Egress Access List  | Enter the name of an IPv6 access list to packets being transmitted on the interface. |

g. Configure advanced parameters.

**Table 177: Advanced**

| Parameter Name | Description                              |
|----------------|------------------------------------------|
| Shutdown       | Click <b>No</b> to enable the interface. |

| Parameter Name              | Description                                                                                                                                                                                                                                                                                             |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tracker / Tracker Group     | Enter the name of a tracker or tracker group to track the status of transport interfaces that connect to the internet.                                                                                                                                                                                  |
| Service Provider            | Specify the details of the service provider.                                                                                                                                                                                                                                                            |
| Bandwidth Upstream (Kbps)   | Specify the bandwidth value to generate notifications when the bandwidth of traffic transmitted on a physical interface exceeds the value.                                                                                                                                                              |
| Bandwidth Downstream (Kbps) | Specify the bandwidth value to generate notifications when the bandwidth of traffic transmitted on a physical interface exceeds the value.                                                                                                                                                              |
| IP MTU                      | Enter the maximum MTU size of packets on the interface.<br>Range: 576 through 1804<br>Default: 1500                                                                                                                                                                                                     |
| TCP MSS                     | Enter the maximum segment size (MSS) of TCP SYN packets passing through the router. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented.<br>Range: 552 through 1460 bytes<br>Default: 1500                                  |
| TLOC Extension              | Enter the name of a physical interface on the same router that connects to the WAN transport. This configuration binds the service-side interface to the WAN transport by enabling a device to access the opposite WAN transport connected to the neighbouring device using a TLOC-extension interface. |
| IP Directed Broadcast       | From the drop-down list, select <b>Global</b> to enable IP Directed Broadcast.<br><br>An IP directed broadcast is an IP packet whose destination address is a valid broadcast address for some IP subnet but which originates from a node that is not itself part of that destination subnet.           |

### What to do next

Also see [Deploy a configuration group](#).

## VPN Interface DSL IPoE

Use the IPoE template for Cisco IOS XE Catalyst SD-WAN devices.

You configure IPoE on routers with DSL interfaces, to provide support for service provider digital subscriber line (DSL) functionality.

To configure DSL interfaces on Cisco IOS XE Catalyst SD-WAN devices using Cisco SD-WAN Manager templates:

1. Create a VPN Interface DSL IPoE feature template to configure IP-over-Ethernet interface parameters, as described in this article.
2. Create a VPN feature template to configure VPN parameters. See the VPN help topic.

### Navigate to the Template Screen and Name the Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and click **Create Template**.



---

**Note** In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

---

3. From the **Create Template** drop-down list, choose **From Feature Template**.
4. From the **Device Model** drop-down list, select the type of device for which you are creating the template.
5. Click **Transport & Management VPN** or scroll to the **Transport & Management VPN** section.
6. Under **Additional VPN 0 Templates**, click **VPN Interface DSL IPoE**.
7. From the **VPN Interface DSL IPoE** drop-down list, choose **Create Template**. The **VPN Interface DSL IPoE** template form is displayed.

This form contains fields for naming the template, fields for defining the IPoE Interface parameters. 

8. In **Template Name**, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.
9. In **Template Description**, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default or to enter a value, click the **Scope** drop-down list and choose one of the following:

Table 178:

| Parameter Scope                               | Scope Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Device Specific<br>(indicated by a host icon) | <p>Use a device-specific value for the parameter. For device-specific parameters, you cannot enter a value in the feature template. You enter the value when you attach a Viptela device to a device template .</p> <p>When you click <b>Device Specific</b>, the <b>Enter Key</b> box opens. This box displays a key, which is a unique string that identifies the parameter in a CSV file that you create. This file is an Excel spreadsheet that contains one column for each key. The header row contains the key names (one key per column), and each row after that corresponds to a device and defines the values of the keys for that device. You upload the CSV file when you attach a Viptela device to a device template. For more information, see <a href="#">Create a Template Variables Spreadsheet</a> .</p> <p>To change the default key, type a new string and move the cursor out of the Enter Key box.</p> <p>Examples of device-specific parameters are system IP address, hostname, GPS location, and site ID.</p> |
| Global (indicated by a globe icon)            | <p>Enter a value for the parameter, and apply that value to all devices.</p> <p>Examples of parameters that you might apply globally to a group of devices are DNS server, syslog server, and interface MTUs.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

### Configure IPoE Functionality

To configure basic IPoE functionality, click **Basic Configuration** and configure the following parameters. Required parameters are indicated with an asterisk.

Table 179:

| Parameter Name        | Description                                                                                                          |
|-----------------------|----------------------------------------------------------------------------------------------------------------------|
| Shutdown*             | Click <b>No</b> to enable the VDSL controller interface.                                                             |
| Controller VDSL Slot* | Enter the slot number of the controller VDSL interface, in the format <i>slot/subslot/port</i> (for example, 0/2/0). |



| Parameter Name           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mode*                    | <p>Select the operating mode of the VDSL controller from the drop-down:</p> <ul style="list-style-type: none"> <li>• Auto—Default mode.</li> <li>• ADSL1—Use ITU G.992.1 Annex A full-rate mode, which provides a downstream rate of 1.3 Mbps and an upstream rate of 1.8 Mbps.</li> <li>• ADSL2—Use ITU G.992.3 Annex A, Annex L, and Annex M, which provides a downstream rate of 12 Mbps and an upstream rate of 1.3 Mbps.</li> <li>• ADSL2+— Use ITU G.992.5 Annex A and Annex M, which provides a downstream rate of 24 Mbps and an upstream rate of 3.3 Mbps.</li> <li>• ANSI—Operating in ADSL2/2+ mode, as defined in ITU G.991.1, G.992.3, and G.992.5, Annex A and Annex M, and in VDSL2 mode, as defined in ITU-T G.993.2.</li> <li>• VDSL2—Operate in VDSL2 mode, as defined in ITU-T G.993.2, which uses frequencies of up to 30 MHz to provide a downstream rate of 200 Mbps and an upstream rate of 100 Mbps..</li> </ul> |
| VDSL Modem Configuration | Enter a command to send to the DSL modem in the NIM module. If the command is valid, it is executed and the results are returned to the Cisco SD-WAN Manager NMS. If the command is not valid, it is not executed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| SRA                      | Click <b>Yes</b> to enable seamless rate adaptation on the interface. SRA adjusts the line rate based on current line conditions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

To save the feature template, click **Save**.

### Configure the Ethernet Interface

Configuring an Ethernet interface with PPPoE allows multiple users on a LAN to be connected to a remote site. To configure an Ethernet interface on the VDSL controller, click **Ethernet** and configure the following parameters. You must configure all parameters.

**Table 180:**

| Parameter Name          | Description                                                                                                                                                          |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ethernet Interface Name | Enter a name for the Ethernet interface, in the format <i>subslot/port</i> (for example 2/0). You do not need to enter the slot number, because it must always be 0. |
| VLAN ID                 | Enter the VLAN identifier of the Ethernet interface.                                                                                                                 |
| Description             | Enter a description for the interface.                                                                                                                               |
| Dynamic/Static          | Assign a dynamic or static IPv4 address to the Ethernet interface.                                                                                                   |
| IPv4 Address            | Enter the static IPv4 address of the Ethernet interface.                                                                                                             |

| Parameter Name | Description                                                                                                                                                                                                                                     |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DHCP Helper    | Enter up to eight IP addresses for DHCP servers in the network, separated by commas, to have the interface be a DHCP helper. A DHCP helper interface forwards BOOTP (Broadcast) DHCP requests that it receives from the specified DHCP servers. |

To save the feature template, click **Save**.

### Create a Tunnel Interface

On IOS XE routers, you can configure up to eight tunnel interfaces. This means that each router can have up to eight TLOCs.

For the control plane to establish itself so that the overlay network can function, you must configure WAN transport interfaces in VPN 0.

To configure a tunnel interface for the multilink interface, select the Tunnel Interface tab and configure the following parameters:

**Table 181:**

| Parameter Name    | Description                                                                                                                   |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Tunnel Interface  | Click <b>On</b> to create a tunnel interface.                                                                                 |
| Color             | Select a color for the TLOC.                                                                                                  |
| Color Description | Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.18.1<br>Enter a description associated to the TLOC color. |

| Parameter Name                                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Control Connection                             | <p>By default, Control Connection is set to <b>On</b>, which establishes a control connection for the TLOC. If the router has multiple TLOCs, click <b>No</b> to have the tunnel not establish control connection for the TLOC.</p> <p><b>Note</b><br/>We recommend a minimum of 650-700 Kbps bandwidth with default 10 msec hello-interval and 12 sec hello-tolerance parameters configured to avoid any data/packet loss in connection traffic.</p> <p>For each BFD session, an additional average sized BFD packet of 175 Bytes consumes 1.4 Kbps of bandwidth.</p> <p>A sample calculation of the required bandwidth for bidirectional BFD packet flow is given below:</p> <ul style="list-style-type: none"> <li>• 650 – 700 Kbps per device for control connections.</li> <li>• 175 Bytes (or 1.4 Kbps) per BFD session on the device (request)</li> <li>• 175 Bytes (or 1.4 Kbps) per BFD session on the device (response)</li> </ul> <p>If the path MTU discovery (PMTUD) is enabled, bandwidth for send/receive BFD packets per tunnel for every 30 secs:</p> <p>A 1500 Bytes BFD request packet is sent per tunnel every 30 secs:<br/> <math>1500 \text{ Bytes} * 8 \text{ bits/1 byte} * 1 \text{ packet} / 30 \text{ secs} = 400 \text{ bps (request)}</math></p> <p>A 147 Bytes BFD packet is sent in response:<br/> <math>147 \text{ Bytes} * 8 \text{ bits/1 byte} * 1 \text{ packet} / 30 \text{ secs} = 40 \text{ bps (response)}</math></p> <p>Therefore, a device with 775 BFD sessions (for example) requires a bandwidth of:<br/> <math>700k + (1.4k*775) + (400 * 775) + (1.4k*775) + (40 * 775) = \sim 3,5 \text{ MBps}</math></p> |
| Maximum Control Connections                    | <p>Specify the maximum number of Cisco Catalyst SD-WAN Controllers that the WAN tunnel interface can connect to. To have the tunnel establish no control connections, set the number to 0.</p> <p><i>Range: 0 through 8. Default: 2</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Cisco Catalyst SD-WAN Validator As STUN Server | <p>Click <b>On</b> to enable Session Traversal Utilities for NAT (STUN) to allow the tunnel interface to discover its public IP address and port number when the router is located behind a NAT.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Exclude Controller Group List                  | <p>Set the Cisco Catalyst SD-WAN Controllers that the tunnel interface is not allowed to connect to. <i>Range: 0 through 100</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Cisco SD-WAN Manager Connection Preference     | <p>Set the preference for using a tunnel interface to exchange control traffic with the Cisco SD-WAN Manager NMS. <i>Range: 0 through 8. Default: 5</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| Parameter Name       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Full Port Hop</b> | <p>Minimum release: Cisco IOS XE Catalyst SD-WAN Release 17.18.1a</p> <p>Enable full port hopping at the TLOC level to allow devices to establish connections with controllers by switching to the next port if the current port is blocked or non-functional.</p> <p>Default: Disabled</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Port Hop             | <p>Click <b>On</b> to enable port hopping, or click Off to disable it. When a router is behind a NAT, port hopping rotates through a pool of preselected OMP port numbers (called base ports) to establish DTLS connections with other routers when a connection attempt is unsuccessful. The default base ports are 12346, 12366, 12386, 12406, and 12426. To modify the base ports, set a port offset value. <i>Default: Enabled</i></p> <p>Starting from Cisco IOS XE Catalyst SD-WAN Release 17.18.1a, this field is deprecated. Instead use the <b>Full Port Hop</b> option. See the <b>Full Port Hop</b> field.</p>                                                                                                                                                                                                                                              |
| Low-Bandwidth Link   | Select to characterize the tunnel interface as a low-bandwidth link.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| TCP MSS              | <p>TCP MSS affects any packet that contains an initial TCP header that flows through the router. When configured, TCP MSS is examined against the MSS exchanged in the three-way handshake. The MSS in the header is lowered if the configured TCP MSS setting is lower than the MSS in the header. If the MSS header value is already lower than the TCP MSS, the packets flow through unmodified. The host at the end of the tunnel uses the lower setting of the two hosts. If the TCP MSS is to be configured, it should be set at 40 bytes lower than the minimum path MTU.</p> <p>Specify the MSS of TPC SYN packets passing through the Cisco IOS XE Catalyst SD-WAN device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. <i>Range: 552 to 1460 bytes Default: None</i></p> |
| Clear-Dont-Fragment  | <p>Configure <b>Clear-Dont-Fragment</b> for packets that arrive at an interface that has Don't Fragment configured. If these packets are larger than what MTU allows, they are dropped. If you clear the Don't Fragment bit, the packets are fragmented and sent.</p> <p>Click <b>On</b> to clear the Dont Fragment bit in the IPv4 packet header for packets being transmitted out of the interface. When the Dont Fragment bit is cleared, packets larger than the MTU of the interface are fragmented before being sent.</p> <p><b>Note</b><br/> <b>Clear-Dont-Fragment</b> clears the Dont Fragment bit and the Dont Fragment bit is set. For packets not requiring fragmentation, the Dont Fragment bit is not affected.</p>                                                                                                                                      |
| Allow Service        | Choose <b>On</b> or <b>Off</b> for each service to allow or disallow the service on the interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

To configure additional tunnel interface parameters, click **Advanced Options** and configure the following parameters:

Table 182:

| Parameter Name       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GRE                  | Use GRE encapsulation on the tunnel interface. By default, GRE is disabled.<br><br>If you select both IPsec and GRE encapsulations, two TLOCs are created for the tunnel interface that have the same IP addresses and colors, but that differ by their encapsulation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| IPsec                | Use IPsec encapsulation on the tunnel interface. By default, IPsec is enabled.<br><br>If you select both IPsec and GRE encapsulations, two TLOCs are created for the tunnel interface that have the same IP addresses and colors, but that differ by their encapsulation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| IPsec Preference     | Specify a preference value for directing traffic to the tunnel. A higher value is preferred over a lower value.<br><br><i>Range: 0 through 4294967295 Default: 0</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| IPsec Weight         | Enter a weight to use to balance traffic across multiple TLOCs. A higher value sends more traffic to the tunnel.<br><br><i>Range: 1 through 255 Default: 1</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Carrier              | Select the carrier name or private network identifier to associate with the tunnel.<br><br><i>Values: carrier1, carrier2, carrier3, carrier4, carrier5, carrier6, carrier7, carrier8, default Default: default</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Bind Loopback Tunnel | Enter the name of a physical interface to bind to a loopback interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Last-Resort Circuit  | Select to use the tunnel interface as the circuit of last resort.<br><br><b>Note</b><br>An interface configured as a circuit of last resort is expected to be down and is skipped while calculating the number of control connections, the cellular modem becomes dormant, and no traffic is sent over the circuit.<br><br>When the configurations are activated on the edge device with cellular interfaces, then all the interfaces begin the process of establishing control and BFD connections. When one or more of the primary interfaces establishes a BFD connection, the circuit of last resort shuts itself down.<br><br>Only when all the primary interfaces lose their connections to remote edges, then the circuit of last resort activates itself triggering a BFD TLOC Down alarm and a Control TLOC Down alarm on the edge device. The last resort interfaces are used as backup circuit on edge device and are activated when all other transport links BFD sessions fail. In this mode the radio interface is turned off, and no control or data connections exist over the cellular interface. |
| NAT Refresh Interval | Enter the interval between NAT refresh packets sent on a DTLS or TLS WAN transport connection. <i>Range: 1 through 60 seconds Default: 5 seconds</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| Parameter Name  | Description                                                                                                                                                                                              |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hello Interval  | Enter the interval between Hello packets sent on a DTLS or TLS WAN transport connection. <i>Range:</i> 100 through 10000 milliseconds <i>Default:</i> 1000 milliseconds (1 second)                       |
| Hello Tolerance | Enter the time to wait for a Hello packet on a DTLS or TLS WAN transport connection before declaring that transport tunnel to be down.<br><i>Range:</i> 12 through 60 seconds <i>Default:</i> 12 seconds |

### Configure the Interface as a NAT Device

To configure an interface to act as a NAT device for applications such as port forwarding, click **NAT**, click **On**, and configure the following parameters:

**Table 183:**

| Parameter Name  | Description                                                                                                                                              |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| NAT             | Click <b>On</b> to have the interface act as a NAT device.                                                                                               |
| Refresh Mode    | Select how NAT mappings are refreshed, either outbound or bidirectional (outbound and inbound). <i>Default:</i> Outbound                                 |
| UDP Timeout     | Specify when NAT translations over UDP sessions time out. <i>Range:</i> 1 through 65536 minutes <i>Default:</i> 1 minutes                                |
| TCP Timeout     | Specify when NAT translations over TCP sessions time out. <i>Range:</i> 1 through 65536 minutes <i>Default:</i> 60 minutes (1 hour)                      |
| Block ICMP      | Select <b>On</b> to block inbound ICMP error messages. By default, a router acting as a NAT device receives these error messages. <i>Default:</i> Off    |
| Respond to Ping | Select <b>On</b> to have the router respond to ping requests to the NAT interface's IP address that are received from the public side of the connection. |

To create a port forwarding rule, click **Add New Port Forwarding Rule** and configure the following parameters. You can define up to 128 port-forwarding rules to allow requests from an external network to reach devices on the internal network.

**Table 184:**

| Parameter Name   | Description                                                                                                                                                   |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Port Start Range | Enter a port number to define the port or first port in the range of interest. <i>Range:</i> 0 through 65535                                                  |
| Port End Range   | Enter the same port number to apply port forwarding to a single port, or enter a larger number to apply it to a range of ports. <i>Range:</i> 0 through 65535 |

| Parameter Name | Description                                                                                                                                                   |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Protocol       | Select the protocol to which to apply the port-forwarding rule, either TCP or UDP. To match the same ports for both TCP and UDP traffic, configure two rules. |
| VPN            | Specify the private VPN in which the internal server resides. This VPN is one of the VPN identifiers in the overlay network. <i>Range:</i> 0 through 65527    |
| Private IP     | Specify the IP address of the internal server to which to direct traffic that matches the port-forwarding rule.                                               |

To save a port forwarding rule, click **Add**.

To save the feature template, click **Save**.

### Apply Access Lists

Configure ACLs to selectively indicate what traffic will enjoy the benefits of QoS. To apply a rewrite rule, access lists, and policers to a router interface, select the ACL tab and configure the following parameters:

**Table 185:**

| Parameter Name     | Description                                                                                                                  |
|--------------------|------------------------------------------------------------------------------------------------------------------------------|
| Shaping rate       | Configure the aggregate traffic transmission rate on the interface to be less than line rate, in kilobits per second (kbps). |
| QoS map            | Specify the name of the QoS map to apply to packets being transmitted out the interface.                                     |
| Rewrite Rule       | Click <b>On</b> , and specify the name of the rewrite rule to apply on the interface.                                        |
| Ingress ACL – IPv4 | Click <b>On</b> , and specify the name of the access list to apply to IPv4 packets being received on the interface.          |
| Egress ACL – IPv4  | Click <b>On</b> , and specify the name of the access list to apply to IPv4 packets being transmitted on the interface.       |
| Ingress ACL – IPv6 | Click <b>On</b> , and specify the name of the access list to apply to IPv6 packets being received on the interface.          |
| Egress ACL – IPv6  | Click <b>On</b> , and specify the name of the access list to apply to IPv6 packets being transmitted on the interface.       |
| Ingress Policer    | Click <b>On</b> , and specify the name of the policer to apply to packets being received on the interface.                   |
| Egress Policer     | Click <b>On</b> , and specify the name of the policer to apply to packets being transmitted on the interface.                |

To save the feature template, click **Save**.

### Configure Other Interface Properties

To configure other interface properties, select the Advanced tab and configure the following properties:

Table 186:

| Parameter Name       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bandwidth Upstream   | <p>When the bandwidth of traffic transmitted on a physical interface in the WAN transport VPN (VPN 0) exceeds a specific limit by 85 percent (on Cisco IOS XE Catalyst SD-WAN devices and Cisco SD-WAN Manager NMSs only), BW Upstream issues notifications.</p> <p>For transmitted traffic, set the bandwidth above which to generate notifications. <i>Range:</i> 1 through <math>(2^{32} / 2) - 1</math> kbps</p>                                                                                                                                                                                                                                                           |
| Bandwidth Downstream | <p>When the bandwidth of traffic received on a physical interface in the WAN transport VPN (VPN 0) exceeds a specific limit by 85 percent (on Cisco IOS XE Catalyst SD-WAN devices and Cisco SD-WAN Manager NMSs only), BW Downstream issues notifications.</p> <p>For received traffic, set the bandwidth above which to generate notifications. <i>Range:</i> 1 through <math>(2^{32} / 2) - 1</math> kbps</p>                                                                                                                                                                                                                                                               |
| IP MTU               | <p>IP MTU affects IP packets. If an IP packet exceeds the IP MTU, then the packet will be fragmented.</p> <p>Specify the maximum MTU size of packets on the interface. <i>Range:</i> 576 through 1804<br/><i>Default:</i> 1500 bytes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| TCP MSS              | <p>In a single TCP/IPv4 datagram, the TCP Maximum Segment Size (MSS) defines the maximum data that a host will accept. This TCP/IPv4 datagram might be fragmented at the IPv4 layer. The MSS value is sent as a TCP header option only in TCP SYN segments.</p> <p>Specify the maximum segment size (MSS) of TPC SYN packets passing through the router. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. <i>Range:</i> 552 to 1460 bytes<br/><i>Default:</i> None</p>                                                                                                                         |
| TLOC Extension       | <p>Use a TLOC Extension to bind an interface and connect another Cisco IOS XE Catalyst SD-WAN device at the same physical site to the local router's WAN transport interface (on Cisco IOS XE Catalyst SD-WAN devices only).</p> <p>Enter the name of the physical interface on the same router that connects to the WAN transport circuit. This configuration then binds this service-side interface to the WAN transport. A second router at the same site that itself has no direct connection to the WAN (generally because the site has only a single WAN connection) and that connects to this service-side interface is then provided with a connection to the WAN.</p> |



| Parameter Name        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tracker               | <p>Tracking the interface status is useful when you enable NAT on a transport interface in VPN 0 to allow data traffic from the router to exit directly to the internet rather than having to first go to a router in a data center. In this situation, enabling NAT on the transport interface splits the TLOC between the local router and the data center into two, with one going to the remote router and the other going to the internet.</p> <p>When you enable transport tunnel tracking, the software periodically probes the path to the internet to determine whether it is up. If the software detects that this path is down, it withdraws the route to the internet destination, and traffic destined to the internet is then routed through the data center router. When the software detects that the path to the internet is again functioning, the route to the internet is reinstalled.</p> <p>Enter the name of a tracker to track the status of transport interfaces that connect to the internet.</p> |
| IP Directed-Broadcast | <p>An IP directed broadcast is an IP packet whose destination address is a valid broadcast address for some IP subnet but which originates from a node that is not itself part of that destination subnet.</p> <p>A device that is not directly connected to its destination subnet forwards an IP directed broadcast in the same way it would forward unicast IP packets destined to a host on that subnet. When a directed broadcast packet reaches a device that is directly connected to its destination subnet, that packet is broadcast on the destination subnet. The destination address in the IP header of the packet is rewritten to the configured IP broadcast address for the subnet, and the packet is sent as a link-layer broadcast.</p> <p>If directed broadcast is enabled for an interface, incoming IP packets whose addresses identify them as directed broadcasts intended for the subnet to which that interface is attached are broadcast on that subnet.</p>                                      |

To save the feature template, click **Save**.

#### Release Information

Introduced in Cisco SD-WAN Manager NMS in Release 18.4.1.

## Configure DSL PPPoA Using a Configuration Group

#### Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

#### Procedure

- Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.
- Step 2** Create and configure the DSL PPPoA parameters in a Transport and Management Profile.
  - a.** Configure basic configuration.

Table 187: Basic Configuration

| Parameter Name      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Controller Slot*    | Enter the slot number of the DSL controller, in the following format:<br><i>slot/subslot/port</i> (for example, 0/2/0)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Controller Mode     | Select the operating mode of the DSL controller from the drop-down list: <ul style="list-style-type: none"> <li>• <b>ADSL1</b>: Use ITU G.992.1 Annex A full-rate mode, which provides a downstream rate of 1.3 Mbps and an upstream rate of 1.8 Mbps.</li> <li>• <b>ADSL2</b>: Use ITU G.992.3 Annex A, Annex L, and Annex M, which provides a downstream rate of 12 Mbps and an upstream rate of 1.3 Mbps.</li> <li>• <b>ADSL2+</b>: Use ITU G.992.5 Annex A and Annex M, which provides a downstream rate of 24 Mbps and an upstream rate of 3.3 Mbps.</li> <li>• <b>ANSI</b>: Operating in ADSL2/2+ mode, as defined in ITU G.991.1, G.992.3, and G.992.5, Annex A and Annex M, and in VDSL2 mode, as defined in ITU-T G.993.2.</li> <li>• <b>VDSL2</b>: Operate in VDSL2 mode, as defined in ITU-T G.993.2, which uses frequencies of up to 30 MHz to provide a downstream rate of 200 Mbps and an upstream rate of 100 Mbps.</li> </ul> |
| SRA                 | Disabled by default. Enable SRA to disable seamless rate adaptation on the interface. SRA adjusts the line rate based on current line conditions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Dialer Pool Member* | Enter the number of the dialer pool to which the interface belongs.<br>Range: 1 through 255                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

## b. Configure ATM.

Table 188: ATM

| Parameter Name            | Description                                                                                                                                                                                                                                                                                                                  |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ATM Sub Interface Name*   | The ATM Sub interface name is auto populated based on the controller slot. Enter a value for the ATM sub interface, in the format <i>subslot/port</i> (for example ATM0/2/0.100). In this example, ".100" is the sub interface value.                                                                                        |
| Sub Interface Description | Enter a description for the interface.                                                                                                                                                                                                                                                                                       |
| VPI/VCI*                  | Create an ATM permanent virtual circuit (PVC), in the following format:<br><i>vpi/vci</i><br>Enter values for the virtual path identifier (VPI) and the virtual channel identifier (VCI).                                                                                                                                    |
| Encapsulation             | Select the encapsulation type to use on the ATM PVC from the drop-down list: <ul style="list-style-type: none"> <li>• <b>AAL5 NLPID</b>: Use NLPID multiplexing.</li> <li>• <b>AAL5 SNAP</b>: Multiplex two or more protocols on the same PVC.</li> <li>• <b>AAL5 MUX</b>: Dedicate the PVC to a single protocol.</li> </ul> |

| Parameter Name  | Description                                                                                                                                                                                                                                                                                                     |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PVC Mode</b> |                                                                                                                                                                                                                                                                                                                 |
| VBR-NRT         | Configure variable bit rate non-real-time parameters: <ul style="list-style-type: none"> <li>• Peak Cell Rate: Enter a value from 48 through 1015 Kbps.</li> <li>• Sustainable Cell Rate: Enter the sustainable cell rate, in Kbps.</li> <li>• Maximum Burst Size: This size can be 1 through 65535.</li> </ul> |
| VBR-RT          | Configure variable bit rate real-time parameters: <ul style="list-style-type: none"> <li>• Peak Cell Rate: Enter a value from 48 through 25000 Kbps.</li> <li>• Average Cell Rate: Enter the average cell rate, in Kbps.</li> <li>• Maximum Burst Size: This size can be 1 through 65535.</li> </ul>            |
| None            | Don't configure variable bit rate parameters                                                                                                                                                                                                                                                                    |

## c. Configure PPP.

Table 189: PPP

| Parameter Name              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PPP Authentication Protocol | Select the authentication protocol used by the MLP: <ul style="list-style-type: none"> <li>• <b>PAP</b>: Enter the username and password that are provided by your ISP. <i>username</i> can be up to 255 characters.</li> <li>• <b>CHAP</b>: Enter the hostname and password provided by your Internet Service Provider (ISP). <i>hostname</i> can be up to 255 characters.</li> <li>• <b>PAP</b> and <b>CHAP</b>: Configure both authentication protocols. Enter the login credentials for each protocol.</li> </ul> |
| Authentication Type         | Select the type authentication from one of the following options.: <ul style="list-style-type: none"> <li>• <b>Unidirectional</b>: Only the side receiving the call (NAS) authenticates the remote side (client). The remote client does not authenticate the server.</li> <li>• <b>Bidirectional</b>: Each side independently sends an Authenticate-Request (AUTH-REQ) and receives either an Authenticate-Acknowledge (AUTH-ACK) or Authenticate-Not Acknowledged (AUTH-NAK).</li> </ul>                            |
| CHAP Hostname*              | Enter the CHAP hostname.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| CHAP Password*              | Enter the CHAP password.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| PAP Hostname*               | Enter the PAP hostname.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| PAP Password*               | Enter the PAP password.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

## d. Configure Tunnel.

Table 190: Tunnel

| Parameter Name                             | Description                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tunnel Interface</b>                    |                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Per Tunnel QoS                             | <p>Enable per tunnel QoS and choose from the following values to configure hub-to-spoke network topologies:</p> <ul style="list-style-type: none"> <li>• <b>Spoke</b></li> <li>• <b>Hub</b></li> </ul> <p>If you select hub topology, the following option appears:</p> <ul style="list-style-type: none"> <li>• <b>Bandwidth Percentage:</b> Enter a value for the bandwidth percentage.<br/>Default: 50</li> </ul> |
| Color                                      | Choose a color for the TLOC.                                                                                                                                                                                                                                                                                                                                                                                         |
| Color Description                          | <p>Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.18.1</p> <p>Enter a description associated to the TLOC color.</p>                                                                                                                                                                                                                                                                             |
| Groups                                     | Enter the list of groups in the field.                                                                                                                                                                                                                                                                                                                                                                               |
| Exclude Controller Group List              | <p>Set the Cisco SD-WAN Controllers that the tunnel interface is not allowed to connect to.</p> <p>Range: 0 through 100</p>                                                                                                                                                                                                                                                                                          |
| Maximum Control Connections                | <p>Specify the maximum number of Cisco SD-WAN Controllers that the WAN tunnel interface can connect to. To have the tunnel establish no control connections, set the number to 0.</p> <p>Range: 0 through 8</p> <p>Default: 2</p>                                                                                                                                                                                    |
| Cisco SD-WAN Manager Connection Preference | <p>Set the preference for using a tunnel interface to exchange control traffic with Cisco SD-WAN Manager.</p> <p>Range: 0 through 8</p> <p>Default: 5</p>                                                                                                                                                                                                                                                            |

| Parameter Name           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tunnel TCP MSS           | <p>TCP MSS affects any packet containing an initial TCP header that flows through the router. When configured, TCP MSS is examined against the MSS exchanged in the three-way handshake. The MSS in the header is lowered if the configured TCP MSS setting is lower than the MSS in the header. If the MSS header value is already lower than the TCP MSS, the packets flow through unmodified. The host at the end of the tunnel uses the lower setting of the two hosts. To configure TCP MSS, provide a value that is 40 bytes lower than the minimum path MTU.</p> <p>Specify the MSS of TPC SYN packets passing through the Cisco IOS XE Catalyst SD-WAN. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented.</p> <p>Range: 552 to 1460 bytes</p>                  |
| Border                   | From the drop-down list, select <b>Global</b> . Click <b>On</b> to set TLOC as border TLOC.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Validator As Stun Server | Click <b>On</b> to enable Session Traversal Utilities for NAT (STUN) to allow the tunnel interface to discover its public IP address and port number when the router is located behind a NAT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Full Port Hop</b>     | <p>Minimum release: Cisco IOS XE Catalyst SD-WAN Release 17.18.1a</p> <p>Enable full port hopping at the TLOC level to allow devices to establish connections with controllers by switching to the next port if the current port is blocked or non-functional.</p> <p>Default: Disabled</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Port Hop                 | <p>From the drop-down list, select <b>Global</b>. Click <b>Off</b> to allow port hopping on tunnel interface.</p> <p>Default: <b>On</b>, which disallows port hopping on a tunnel interface</p> <p>Starting from Cisco IOS XE Catalyst SD-WAN Release 17.18.1a, this field is deprecated. Instead use the <b>Full Port Hop</b> option. See the <b>Full Port Hop</b> field.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Low-Bandwidth Link       | <p>Click <b>On</b> to set the tunnel interface as a low-bandwidth link.</p> <p>Default: <b>Off</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Clear-Dont-Fragment      | <p>Configure <b>Clear-Dont-Fragment</b> for packets that arrive at an interface that has Don't Fragment configured. If these packets are larger than what MTU allows, they are dropped. If you clear the Don't Fragment bit, the packets are fragmented and sent.</p> <p>Click <b>On</b> to clear the Dont Fragment bit in the IPv4 packet header for packets being transmitted out of the interface. When the Dont Fragment bit is cleared, the router fragments packets larger than the MTU of the interface before sending the packets.</p> <p>the router fragments packets larger than the MTU of the interface before sending the packets.</p> <p><b>Note</b><br/> <b>Clear-Dont-Fragment</b> clears the Dont Fragment bit and the Dont Fragment bit is set. For packets not requiring fragmentation, the Dont Fragment bit is not affected.</p> |
| Network Broadcast        | <p>From the drop-down list, select <b>Global</b>. Click <b>On</b> to accept and respond to network-prefix-directed broadcasts. Enable this parameter only if the <b>Directed Broadcast</b> is enabled on the LAN interface feature template.</p> <p>Default: <b>Off</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

| Parameter Name       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Carrier              | <p>From the drop-down list, select <b>Global</b> and select the carrier name or private network identifier to associate with the tunnel.</p> <p>Values: carrier1, carrier2, carrier3, carrier4, carrier5, carrier6, carrier7, carrier8, default.</p> <p>Default: default</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Bind Loopback Tunnel | <p>Enter the name of a physical interface to bind to a loopback interface. The interface name has the following format:</p> <p><b>ge slot/port</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| NAT Refresh Interval | <p>Set the interval between NAT refresh packets sent on a DTLS or TLS WAN transport connection.</p> <p>Range: 1 through 60 seconds</p> <p>Default: 5 seconds</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Hello Interval       | <p>Enter the interval between Hello packets sent on a DTLS or TLS WAN transport connection.</p> <p>Range: 100 through 10000 milliseconds</p> <p>Default: 1000 milliseconds (1 second)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Hello Tolerance      | <p>Enter the time to wait for a Hello packet on a DTLS or TLS WAN transport connection before declaring that transport tunnel to be down.</p> <p>Range: 12 through 60 seconds</p> <p>Default: 12 seconds</p> <p>The default hello interval is 1000 milliseconds, and it can be a time in the range 100 through 600000 milliseconds (10 minutes). The default hello tolerance is 12 seconds, and it can be a time in the range 12 through 600 seconds (10 minutes). To reduce outgoing control packets on a TLOC, it is recommended that on the tunnel interface you set the hello interval to 60000 milliseconds (10 minutes) and the hello tolerance to 600 seconds (10 minutes) and include the <b>no track-transport disable</b> regular checking of the DTLS connection between the edge device and the controller. For a tunnel connection between a edge device and any controller device, the tunnel uses the hello interval and tolerance times configured on the edge device. This choice is made to minimize the traffic sent over the tunnel, to allow for situations where the cost of a link is a function of the amount of traffic traversing the link. The hello interval and tolerance times are chosen separately for each tunnel between a edge device and a controller device. Another step taken to minimize the amount of control plane traffic is to not send or receive OMP control traffic over a cellular interface when other interfaces are available. This behavior is inherent in the software and is not configurable.</p> |

| Parameter Name       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Last Resort Circuit  | <p>Select to use the tunnel interface as the circuit of last resort.</p> <p><b>Note</b><br/>It is assumed that an interface configured as a circuit of last resort is unavailable and is skipped while calculating the number of control connections. As a result, the cellular modem becomes dormant, and no traffic is sent over the circuit.</p> <p>When the configurations are activated on the edge device with cellular interfaces, all the interfaces begin the process of establishing control and BFD connections. When one or more of the primary interfaces establishes a BFD connection, the circuit of last resort shuts itself down.</p> <p>If the primary interfaces lose their connections to remote edges, the circuit of last resort activates itself, triggering a BFD TLOC Down alarm and a Control TLOC Down alarm on the edge device. The last resort interfaces are a backup circuit on edge device and are activated when all other transport links BFD sessions fail. In this mode, the radio interface is turned off, and no control or data connections exist over the cellular interface.</p> |
| Allow Services       | Click <b>On</b> or <b>Off</b> for each service to enable or disable the service on the cellular interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Encapsulation</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

| Parameter Name | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Encapsulation  | <p>Enable at least one of the following encapsulation methods:</p> <ul style="list-style-type: none"> <li>• <b>IPsec</b>: Enter a value to set the preference for directing traffic to the tunnel. A higher value is preferred over a lower value.<br/>Range: 0 through 4294967295<br/>Default: 0</li> <li>• <b>IPsec Preference</b>: From the drop-down list, select <b>Global</b> and enter a value to set the preference for directing traffic to the tunnel. A higher value is preferred over a lower value.<br/>Range: 0 through 4294967295<br/>Default: 0</li> <li>• <b>IPsec Weight</b>: From the drop-down list, select <b>Global</b> and enter a value to set weight for balancing traffic across multiple TLOCs. A higher value sends more traffic to the tunnel.<br/>Range: 1 through 255<br/>Default: 1</li> <li>• <b>GRE</b>: Enter a value to set GRE preference for TLOC.<br/>Range: 0 through 4294967295</li> <li>• <b>GRE Preference</b>: From the drop-down list, select <b>Global</b> and enter a value to set the preference for directing traffic to the tunnel. A higher value is preferred over a lower value.<br/>Range: 0 through 4294967295<br/>Default: 0</li> <li>• <b>GRE Weight</b>: From the drop-down list, select <b>Global</b> and enter a value to set weight for balancing traffic across multiple TLOCs. A higher value sends more traffic to the tunnel.<br/>Range: 1 through 255<br/>Default: 1</li> </ul> |

## e. Configure NAT.

Table 191: NAT

| Parameter Name        | Description                                                                                                              |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------|
| UDP Timeout (Minutes) | <p>Specify when NAT translations over UDP sessions time out.<br/>Range: 1 through 8947 minutes<br/>Default: 1 minute</p> |



| Parameter Name        | Description                                                                                                                |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------|
| TCP Timeout (Minutes) | Specify when NAT translations over TCP sessions time out.<br>Range: 1 through 8947 minutes<br>Default: 60 minutes (1 hour) |

f. Configure QoS.

**Table 192: QoS**

| Parameter Name      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Adaptive QoS        | Enter adaptive QoS parameters. You can leave the additional details at as default or specify your values. <ul style="list-style-type: none"> <li>• <b>Adapt Period (Minutes):</b> Choose <b>Global</b> from the drop-down list, click <b>On</b>, and enter the period in minutes.</li> <li>• <b>Shaping Rate Upstream:</b> Choose <b>Global</b> from the drop-down list, click <b>On</b>, and enter the minimum, maximum, and default upstream bandwidth in Kbps.</li> <li>• <b>Shaping Rate Downstream:</b> Choose <b>Global</b> from the drop-down list, click <b>On</b>, and enter the minimum, maximum, downstream, and upstream bandwidth in Kbps.</li> </ul> |
| Shaping Rate (kbps) | Choose <b>Global</b> from the drop-down list and configure the aggregate traffic transmission rate on the interface to be less than line rate, in kilobits per second (kbps).<br>Range: 8 through 100000000                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

g. Configure ACL.

**Table 193: ACL**

| Parameter Name           | Description                                                                          |
|--------------------------|--------------------------------------------------------------------------------------|
| IPv4 Ingress Access List | Enter the name of an IPv4 access list to packets being received on the interface.    |
| IPv4 Egress Access List  | Enter the name of an IPv4 access list to packets being transmitted on the interface. |
| IPv6 Ingress Access List | Enter the name of an IPv6 access list to packets being received on the interface.    |
| IPv6 Egress Access List  | Enter the name of an IPv6 access list to packets being transmitted on the interface. |

h. Configure advanced parameters.

**Table 194: Advanced**

| Parameter Name | Description                              |
|----------------|------------------------------------------|
| Shutdown       | Click <b>No</b> to enable the interface. |

| Parameter Name              | Description                                                                                                                                                                                                                                                                                             |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tracker / Tracker Group     | Enter the name of a tracker or tracker group to track the status of transport interfaces that connect to the internet.                                                                                                                                                                                  |
| Service Provider            | Specify the details of the service provider.                                                                                                                                                                                                                                                            |
| Bandwidth Upstream (Kbps)   | Specify the bandwidth value to generate notifications when the bandwidth of traffic transmitted on a physical interface exceeds the value.                                                                                                                                                              |
| Bandwidth Downstream (Kbps) | Specify the bandwidth value to generate notifications when the bandwidth of traffic transmitted on a physical interface exceeds the value.                                                                                                                                                              |
| IP MTU                      | Enter the maximum MTU size of packets on the interface.<br>Range: 576 through 1804<br>Default: 1500                                                                                                                                                                                                     |
| TCP MSS                     | Enter the maximum segment size (MSS) of TCP SYN packets passing through the router. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented.<br>Range: 552 through 1460 bytes<br>Default: 1500                                  |
| TLOC Extension              | Enter the name of a physical interface on the same router that connects to the WAN transport. This configuration binds the service-side interface to the WAN transport by enabling a device to access the opposite WAN transport connected to the neighbouring device using a TLOC-extension interface. |
| IP Directed Broadcast       | From the drop-down list, select <b>Global</b> to enable IP Directed Broadcast.<br><br>An IP directed broadcast is an IP packet whose destination address is a valid broadcast address for some IP subnet but which originates from a node that is not itself part of that destination subnet.           |

### What to do next

Also see [Deploy a configuration group](#).

## VPN Interface DSL PPPoA

To provide support for service provider digital subscriber line (DSL) functionality, configure PPP-over-ATM interfaces on routers with DSL NIM modules.

Use the VPN Interface DSL PPPoA template for Cisco IOS XE Catalyst SD-WAN devices.

You configure PPP-over-ATM interfaces on routers with DSL NIM modules, to provide support for service provider digital subscriber line (DSL) functionality.

To configure DSL interfaces on Cisco routers using Cisco SD-WAN Manager templates:

1. Create a VPN Interface DSL PPPoA feature template to configure ATM interface parameters, as described in this article.
2. Create a VPN feature template to configure VPN parameters. See the VPN help topic.

### Navigate to the Template Screen and Name the Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**.



---

**Note** In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

---

3. From the **Create Template** drop-down list, select **From Feature Template**.
4. From the **Device Model** drop-down list, select the type of device for which you are creating the template.
5. Click **Transport & Management VPN** or scroll to the **Transport & Management VPN** section.
6. Under **Additional VPN 0 Templates**, click **VPN Interface DSL PPPoA**.
7. From the **VPN Interface DSL PPPoA** drop-down list, click **Create Template**. The VPN Interface DSL PPPoA template form is displayed. This form contains fields for naming the template, and fields for defining VPN Interface PPP parameters.
8. In **Template Name**, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.
9. In **Template Description**, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default or to enter a value, click the scope drop-down to the left of the parameter field and select one of the following:

Table 195:

| Parameter Scope                               | Scope Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Device Specific<br>(indicated by a host icon) | <p>Use a device-specific value for the parameter. For device-specific parameters, you cannot enter a value in the feature template. You enter the value when you attach a Viptela device to a device template .</p> <p>When you click <b>Device Specific</b>, the <b>Enter Key box</b> opens. This box displays a key, which is a unique string that identifies the parameter in a CSV file that you create. This file is an Excel spreadsheet that contains one column for each key. The header row contains the key names (one key per column), and each row after that corresponds to a device and defines the values of the keys for that device. You upload the CSV file when you attach a Viptela device to a device template. For more information, see <a href="#">Create a Template Variables Spreadsheet</a> .</p> <p>To change the default key, type a new string and move the cursor out of the Enter Key box.</p> <p>Examples of device-specific parameters are system IP address, hostname, GPS location, and site ID.</p> |
| Global (indicated by a globe icon)            | <p>Enter a value for the parameter, and apply that value to all devices.</p> <p>Examples of parameters that you might apply globally to a group of devices are DNS server, syslog server, and interface MTUs.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

### Configure VDSL Controller Functionality

To configure basic VDSL controller functionality in a VPN, select **Basic Configuration** and configure the following parameters. Required parameters are indicated with an asterisk.

Table 196:

| Parameter Name        | Description                                                                                                          |
|-----------------------|----------------------------------------------------------------------------------------------------------------------|
| Shutdown*             | Click <b>No</b> to enable the VDSL controller interface.                                                             |
| Controller VDSL Slot* | Enter the slot number of the controller VDSL interface, in the format <i>slot/subslot/port</i> (for example, 0/2/0). |

| Parameter Name           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mode*                    | <p>Select the operating mode of the VDSL controller from the drop-down:</p> <ul style="list-style-type: none"> <li>• <b>Auto</b>—Default mode.</li> <li>• <b>ADSL1</b>—Use ITU G.992.1 Annex A full-rate mode, which provides a downstream rate of 1.3 Mbps and an upstream rate of 1.8 Mbps.</li> <li>• <b>ADSL2</b>—Use ITU G.992.3 Annex A, Annex L, and Annex M, which provides a downstream rate of 12 Mbps and an upstream rate of 1.3 Mbps.</li> <li>• <b>ADSL2+</b>— Use ITU G.992.5 Annex A and Annex M, which provides a downstream rate of 24 Mbps and an upstream rate of 3.3 Mbps.</li> <li>• <b>ANSI</b>—Operate in ADSL2/2+ mode, as defined in ITU G.991.1, G.992.3, and G.992.5, Annex A and Annex M, and in VDSL2 mode, as defined in ITU-T G.993.2.</li> <li>• <b>VDSL2</b>—Operate in VDSL2 mode, as defined in ITU-T G.993.2, which uses frequencies of up to 30 MHz to provide a downstream rate of 200 Mbps and an upstream rate of 100 Mbps.</li> </ul> |
| VDSL Modem Configuration | Enter a command to send to the DSL modem in the NIM module. If the command is valid, it is executed and the results are returned to the Cisco SD-WAN Manager NMS. If the command is not valid, it is not executed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| SRA                      | Enabled by default. Click <b>No</b> to disable seamless rate adaptation on the interface. SRA adjusts the line rate based on current line conditions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

To save the feature template, click **Save**.

### Configure the ATM Interface

To configure an ATM interface on the VDSL controller, select **ATM** and configure the following parameters. You must configure all parameters.

**Table 197:**

| Parameter Name     | Description                                                                                                                                                                |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ATM Interface Name | Enter a name for the ATM interface, in the format <i>subslot/port</i> (for example 2/0). You do not need to enter the slot number, because it must always be 0.            |
| Description        | Enter a description for the interface.                                                                                                                                     |
| VPI and VCI        | Create an ATM permanent virtual circuit (PVC), in the format <i>vpi/vci</i> . Enter values for the virtual path identifier (VPI) and the virtual channel identifier (VCI). |

| Parameter Name     | Description                                                                                                                                                                                                                                                                                                                         |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Encapsulation      | Select the ATM adaptation layer (AAL) and encapsulation type to use on the ATM PVC from the drop-down list: <ul style="list-style-type: none"> <li>• AAL5 MUX—Dedicate the PVC to a single protocol.</li> <li>• AAL5 NLPID—Use NLPID multiplexing.</li> <li>• AAL5 SNAP—Multiplex two or more protocols on the same PVC.</li> </ul> |
| Dialer Pool Member | Enter the number of the dialer pool to which the interface belongs. It can be a value from 1 through 255.                                                                                                                                                                                                                           |
| VBR-NRT            | Configure variable bit rate non-real-time parameters: <ul style="list-style-type: none"> <li>• Peak Cell Rate—Enter a value from 48 through 25000 Kbps.</li> <li>• Sustainable Cell Rate—Enter the sustainable cell rate, in Kbps.</li> <li>• Maximum Burst Size—This size can be 1 cell.</li> </ul>                                |
| VBR-RT             | Configure variable bit rate real-time parameters: <ul style="list-style-type: none"> <li>• Peak Cell Rate—Enter a value from 48 through 25000 Kbps.</li> <li>• Average Cell Rate—Enter the average cell rate, in Kbps.</li> <li>• Maximum Burst Size—This size can be 1 cell.</li> </ul>                                            |

To save the feature template, click **Save**.

### Configure the PPP Authentication Protocol

To configure the PPP authentication protocol, select **PPP** and configure the following parameters:

**Table 198:**

| Parameter Name          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Authentication Protocol | Select the authentication protocol used by the MLP: <ul style="list-style-type: none"> <li>• <b>CHAP</b>—Enter the hostname and password provided by your Internet Service Provider (ISP). <i>hostname</i> can be up to 255 characters.</li> <li>• <b>PAP</b>—Enter the username and password provided by your ISP. <i>username</i> can be up to 255 characters.</li> <li>• <b>PAP</b> and <b>CHAP</b>—Configure both authentication protocols. Enter the login credentials for each protocol. To use the same username and password for both, click Same Credentials for PAP and CHAP.</li> </ul> |

To save the feature template, click **Save**.

### Create a Tunnel Interface

On Cisco IOS XE Catalyst SD-WAN devices, you can configure up to eight tunnel interfaces. This means that each Cisco IOS XE Catalyst SD-WAN device can have up to eight TLOCs.

For the control plane to establish itself so that the overlay network can function, you must configure WAN transport interfaces in VPN 0.

To configure a tunnel interface for the multilink interface, select **Tunnel Interface** and configure the following parameters:

**Table 199:**

| Parameter Name                                 | Description                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tunnel Interface                               | Click <b>On</b> to create a tunnel interface.                                                                                                                                                                                                                                                                                                                                                                    |
| Color                                          | Select a color for the TLOC.                                                                                                                                                                                                                                                                                                                                                                                     |
| Color Description                              | Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.18.1<br>Enter a description associated to the TLOC color.                                                                                                                                                                                                                                                                                    |
| Control Connection                             | If the Cisco IOS XE Catalyst SD-WAN device has multiple TLOCs, click No to have the tunnel not establish a TLOC. The default is On, which establishes a control connection for the TLOC.<br><br><b>Note</b><br>For control connection traffic without dropping any data, a minimum of 650-700 kbps bandwidth is recommended with default parameters configured for hello-interval (10) and hello-tolerance (12). |
| Maximum Control Connections                    | Specify the maximum number of Cisco Catalyst SD-WAN Controllers that the WAN tunnel interface can connect to. To have the tunnel establish no control connections, set the number to 0.<br><br><i>Range: 0 through 8 Default: 2</i>                                                                                                                                                                              |
| Cisco Catalyst SD-WAN Validator As STUN Server | Click <b>On</b> to enable Session Traversal Utilities for NAT (STUN) to allow the tunnel interface to discover its public IP address and port number when the Cisco IOS XE Catalyst SD-WAN device is located behind a NAT.                                                                                                                                                                                       |
| Exclude Controller Group List                  | Set the Cisco Catalyst SD-WAN Controllers that the tunnel interface is not allowed to connect to. <i>Range: 0 through 100</i>                                                                                                                                                                                                                                                                                    |
| Cisco SD-WAN Manager Connection Preference     | Set the preference for using a tunnel interface to exchange control traffic with the Cisco SD-WAN Manager NMS. <i>Range: 0 through 8 Default: 5</i>                                                                                                                                                                                                                                                              |
| <b>Full Port Hop</b>                           | Minimum release: Cisco IOS XE Catalyst SD-WAN Release 17.18.1a<br><br>Enable full port hopping at the TLOC level to allow devices to establish connections with controllers by switching to the next port if the current port is blocked or non-functional.<br><br>Default: Disabled                                                                                                                             |

| Parameter Name      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Port Hop            | Click <b>On</b> to enable port hopping, or click <b>Off</b> to disable it. When a router is behind a NAT, port hopping rotates through a pool of preselected OMP port numbers (called base ports) to establish DTLS connections with other routers when a connection attempt is unsuccessful. The default base ports are 12346, 12366, 12386, 12406, and 12426. To modify the base ports, set a port offset value. <i>Default:</i> Enabled<br><br>Starting from Cisco IOS XE Catalyst SD-WAN Release 17.18.1a, this field is deprecated. Instead use the <b>Full Port Hop</b> option. See the <b>Full Port Hop</b> field.                                                                                                                                                                                                                                              |
| Low-Bandwidth Link  | Select to characterize the tunnel interface as a low-bandwidth link.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Tunnel TCP MSS      | TCP MSS affects any packet that contains an initial TCP header that flows through the router. When configured, TCP MSS is examined against the MSS exchanged in the three-way handshake. The MSS in the header is lowered if the configured TCP MSS setting is lower than the MSS in the header. If the MSS header value is already lower than the TCP MSS, the packets flow through unmodified. The host at the end of the tunnel uses the lower setting of the two hosts. If the TCP MSS is to be configured, it should be set at 40 bytes lower than the minimum path MTU.<br><br>Specify the MSS of TPC SYN packets passing through the Cisco IOS XE Catalyst SD-WAN device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. <i>Range:</i> 552 to 1460 bytes <i>Default:</i> None |
| Clear-Dont-Fragment | Configure <b>Clear-Dont-Fragment</b> for packets that arrive at an interface that has Don't Fragment configured. If these packets are larger than what MTU allows, they are dropped. If you clear the Don't Fragment bit, the packets are fragmented and sent.<br><br>Click <b>On</b> to clear the Dont Fragment bit in the IPv4 packet header for packets being transmitted out of the interface. When the Dont Fragment bit is cleared, packets larger than the MTU of the interface are fragmented before being sent.<br><br><b>Note</b><br><b>Clear-Dont-Fragment</b> clears the Dont Fragment bit and the Dont Fragment bit is set. For packets not requiring fragmentation, the Dont Fragment bit is not affected.                                                                                                                                               |
| Allow Service       | Select <b>On</b> or <b>Off</b> for each service to allow or disallow the service on the interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

To configure additional tunnel interface parameters, click **Advanced Options** and configure the following parameters:

**Table 200:**

| Parameter Name | Description                                                                                                                                                                                                                                                            |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GRE            | Use GRE encapsulation on the tunnel interface. By default, GRE is disabled.<br><br>If you select both IPsec and GRE encapsulations, two TLOCs are created for the tunnel interface that have the same IP addresses and colors, but that differ by their encapsulation. |



| Parameter Name       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IPsec                | Use IPsec encapsulation on the tunnel interface. By default, IPsec is enabled.<br><br>If you select both IPsec and GRE encapsulations, two TLOCs are created for the tunnel interface that have the same IP addresses and colors, but that differ by their encapsulation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| IPsec Preference     | Specify a preference value for directing traffic to the tunnel. A higher value is preferred over a lower value.<br><br><i>Range:</i> 0 through 4294967295. <i>Default:</i> 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| IPsec Weight         | Enter a weight to use to balance traffic across multiple TLOCs. A higher value sends more traffic to the tunnel.<br><br><i>Range:</i> 1 through 255. <i>Default:</i> 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Carrier              | Select the carrier name or private network identifier to associate with the tunnel.<br><br><i>Values:</i> carrier1, carrier2, carrier3, carrier4, carrier5, carrier6, carrier7, carrier8, default. <i>Default:</i> default                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Bind Loopback Tunnel | Enter the name of a physical interface to bind to a loopback interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Last-Resort Circuit  | Select to use the tunnel interface as the circuit of last resort.<br><br><b>Note</b><br>An interface configured as a circuit of last resort is expected to be down and is skipped while calculating the number of control connections, the cellular modem becomes dormant, and no traffic is sent over the circuit.<br><br>When the configurations are activated on the edge device with cellular interfaces, then all the interfaces begin the process of establishing control and BFD connections. When one or more of the primary interfaces establishes a BFD connection, the circuit of last resort shuts itself down.<br><br>Only when all the primary interfaces lose their connections to remote edges, then the circuit of last resort activates itself triggering a BFD TLOC Down alarm and a Control TLOC Down alarm on the edge device. The last resort interfaces are used as backup circuit on edge device and are activated when all other transport links BFD sessions fail. In this mode the radio interface is turned off, and no control or data connections exist over the cellular interface. |
| NAT Refresh Interval | Enter the interval between NAT refresh packets sent on a DTLS or TLS WAN transport connection. <i>Range:</i> 1 through 60 seconds. <i>Default:</i> 5 seconds.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Hello Interval       | Enter the interval between Hello packets sent on a DTLS or TLS WAN transport connection. <i>Range:</i> 100 through 10000 milliseconds. <i>Default:</i> 1000 milliseconds (1 second).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Hello Tolerance      | Enter the time to wait for a Hello packet on a DTLS or TLS WAN transport connection before declaring that transport tunnel to be down.<br><br><i>Range:</i> 12 through 60 seconds. <i>Default:</i> 12 seconds.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

### Apply Access Lists

To apply a rewrite rule, access lists, and policers to a router interface, select **ACL** and configure the following parameters:

**Table 201:**

| Parameter Name     | Description                                                                                                                  |
|--------------------|------------------------------------------------------------------------------------------------------------------------------|
| Shaping rate       | Configure the aggregate traffic transmission rate on the interface to be less than line rate, in kilobits per second (kbps). |
| QoS map            | Specify the name of the QoS map to apply to packets being transmitted out the interface.                                     |
| Rewrite Rule       | Click <b>On</b> , and specify the name of the rewrite rule to apply on the interface.                                        |
| Ingress ACL – IPv4 | Click <b>On</b> , and specify the name of the access list to apply to IPv4 packets being received on the interface.          |
| Egress ACL – IPv4  | Click <b>On</b> , and specify the name of the access list to apply to IPv4 packets being transmitted on the interface.       |
| Ingress ACL – IPv6 | Click <b>On</b> , and specify the name of the access list to apply to IPv6 packets being received on the interface.          |
| Egress ACL – IPv6  | Click <b>On</b> , and specify the name of the access list to apply to IPv6 packets being transmitted on the interface.       |
| Ingress Policer    | Click <b>On</b> , and specify the name of the policer to apply to packets being received on the interface.                   |
| Egress Policer     | Click <b>On</b> , and specify the name of the policer to apply to packets being transmitted on the interface.                |

To save the feature template, click **Save**.

### Configure Other Interface Properties

To configure other interface properties, select **Advanced** and configure the following properties:

**Table 202:**

| Parameter Name      | Description                                                                                                                                                                                                                                                                                                |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PMTU Discovery      | Click <b>On</b> to enable path MTU discovery on the interface, to allow the router to determine the largest MTU size supported without requiring packet fragmentation.                                                                                                                                     |
| TCP MSS             | Specify the maximum segment size (MSS) of TCP SYN packets passing through the Cisco IOS XE Catalyst SD-WAN device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. <i>Range:</i> 552 to 1460 bytes. <i>Default:</i> None. |
| Clear Dont Fragment | Click <b>On</b> to clear the Don't Fragment bit in the IPv4 packet header for packets being transmitted out the interface. When the DF bit is cleared, packets larger than that interface's MTU are fragmented before being sent.                                                                          |

| Parameter Name     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Static Ingress QoS | Select a queue number to use for incoming traffic. <i>Range:0 through 7</i>                                                                                                                                                                                                                                                                                                                                                                                         |
| Autonegotiate      | Click <b>Off</b> to turn off autonegotiation. By default, an interface runs in autonegotiation mode.                                                                                                                                                                                                                                                                                                                                                                |
| TLOC Extension     | Enter the name of the physical interface on the same router that connects to the WAN transport circuit. This configuration then binds this service-side interface to the WAN transport. A second Cisco IOS XE Catalyst SD-WAN device at the same site that itself has no direct connection to the WAN (generally because the site has only a single WAN connection) and that connects to this service-side interface is then provided with a connection to the WAN. |

To save the feature template, click **Save**.

#### Release Information

Introduced in Cisco SD-WAN Manager NMS in Release 18.3.

## Configure DSL PPPoE Using a Configuration Group

#### Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

#### Procedure

**Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

**Step 2** Create and configure the DSL PPPoE parameters in a Transport and Management Profile.

- a. Configure basic configuration.

**Table 203: Basic Configuration**

| Parameter Name   | Description                                                                                                        |
|------------------|--------------------------------------------------------------------------------------------------------------------|
| Controller Slot* | Enter the slot number of the controller, in the following format:<br><i>slot/subslot/port</i> (for example, 0/2/0) |

| Parameter Name      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Controller Mode     | <p>Select the operating mode of the DSL controller from the drop-down list:</p> <ul style="list-style-type: none"> <li>• <b>ADSL1</b>: Use ITU G.992.1 Annex A full-rate mode, which provides a downstream rate of 1.3 Mbps and an upstream rate of 1.8 Mbps.</li> <li>• <b>ADSL2</b>: Use ITU G.992.3 Annex A, Annex L, and Annex M, which provides a downstream rate of 12 Mbps and an upstream rate of 1.3 Mbps.</li> <li>• <b>ADSL2+</b>: Use ITU G.992.5 Annex A and Annex M, which provides a downstream rate of 24 Mbps and an upstream rate of 3.3 Mbps.</li> <li>• <b>ANSI</b>: Operating in ADSL2/2+ mode, as defined in ITU G.991.1, G.992.3, and G.992.5, Annex A and Annex M, and in VDSL2 mode, as defined in ITU-T G.993.2.</li> <li>• <b>VDSL2</b>: Operate in VDSL2 mode, as defined in ITU-T G.993.2, which uses frequencies of up to 30 MHz to provide a downstream rate of 200 Mbps and an upstream rate of 100 Mbps.</li> </ul> |
| SRA                 | Disabled by default. Enable SRA to disable seamless rate adaptation on the interface. SRA adjusts the line rate based on current line conditions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Dialer Pool Member* | <p>Enter the number of the dialer pool to which the interface belongs.</p> <p>Range: 1 through 255</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

## b. Configure Ethernet.

**Table 204: Ethernet**

| Parameter Name           | Description                                                                                                                                                              |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ethernet Interface Name* | <p>Enter the name of an ethernet interface.</p> <p>For IOS XE routers, you must spell out the interface names completely (for example, <b>GigabitEthernet0/0/0</b>).</p> |
| Description              | Enter a description for the interface.                                                                                                                                   |
| VLAN ID                  | Enter the VLAN identifier of the Ethernet interface.                                                                                                                     |

## c. Configure PPP.

Table 205: PPP

| Parameter Name              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PPP Authentication Protocol | <p>Select the authentication protocol used by the MLP:</p> <ul style="list-style-type: none"> <li>• <b>PAP</b>: Enter the username and password that are provided by your ISP. <i>username</i> can be up to 255 characters.</li> <li>• <b>CHAP</b>: Enter the hostname and password provided by your Internet Service Provider (ISP). <i>hostname</i> can be up to 255 characters.</li> <li>• <b>PAP</b> and <b>CHAP</b>: Configure both authentication protocols. Enter the login credentials for each protocol.</li> </ul> |
| Authentication Type         | <p>Select the type authentication from one of the following options:</p> <ul style="list-style-type: none"> <li>• <b>Unidirectional</b>: Only the side receiving the call (NAS) authenticates the remote side (client). The remote client does not authenticate the server.</li> <li>• <b>Bidirectional</b>: Each side independently sends an Authenticate-Request (AUTH-REQ) and receives either an Authenticate-Acknowledge (AUTH-ACK) or Authenticate-Not Acknowledged (AUTH-NAK).</li> </ul>                             |
| CHAP Hostname*              | Enter the CHAP hostname.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| CHAP Password*              | Enter the CHAP password.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| PAP Hostname*               | Enter the PAP hostname.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| PAP Password*               | Enter the PAP password.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

## d. Configure Tunnel.

Table 206: Tunnel

| Parameter Name          | Description                                                                                                                                                                                            |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tunnel Interface</b> |                                                                                                                                                                                                        |
| Per Tunnel QoS          | <p>Enable per tunnel QoS and choose from the following values to configure hub-to-spoke network topologies:</p> <ul style="list-style-type: none"> <li>• <b>Spoke</b></li> <li>• <b>Hub</b></li> </ul> |
| Color                   | Select a color for the TLOC.                                                                                                                                                                           |
| Color Description       | <p>Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.18.1</p> <p>Enter a description associated to the TLOC color.</p>                                                               |
| Groups                  | Enter the list of groups in the field.                                                                                                                                                                 |

| Parameter Name                             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Exclude Controller Group List              | Set the Cisco SD-WAN Controllers that the tunnel interface is not allowed to connect to.<br>Range: 0 through 100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Maximum Control Connections                | Specify the maximum number of Cisco SD-WAN Controllers that the WAN tunnel interface can connect to. To have the tunnel establish no control connections, set the number to 0.<br>Range: 0 through 8<br>Default: 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Cisco SD-WAN Manager Connection Preference | Set the preference for using a tunnel interface to exchange control traffic with Cisco SD-WAN Manager.<br>Range: 0 through 8<br>Default: 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Tunnel TCP MSS                             | TCP MSS affects any packet that contains an initial TCP header that flows through the router. When configured, TCP MSS is examined against the MSS exchanged in the three-way handshake. The MSS in the header is lowered if the configured TCP MSS setting is lower than the MSS in the header. If the MSS header value is already lower than the TCP MSS, the packets flow through unmodified. The host at the end of the tunnel uses the lower setting of the two hosts. To configure TCP MSS, provide a value that is 40 bytes lower than the minimum path MTU.<br><br>Specify the MSS of TPC SYN packets passing through the Cisco IOS XE Catalyst SD-WAN. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented.<br>Range: 552 through 1460 bytes<br>Default: None |
| Border                                     | From the drop-down list, select <b>Global</b> . Click <b>On</b> to set TLOC as border TLOC.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Validator As Stun Server                   | Click <b>On</b> to enable Session Traversal Utilities for NAT (STUN) to allow the tunnel interface to discover its public IP address and port number when the router is located behind a NAT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Full Port Hop</b>                       | Minimum release: Cisco IOS XE Catalyst SD-WAN Release 17.18.1a<br><br>Enable full port hopping at the TLOC level to allow devices to establish connections with controllers by switching to the next port if the current port is blocked or non-functional.<br>Default: Disabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Port Hop                                   | From the drop-down list, select <b>Global</b> . Click <b>Off</b> to allow port hopping on tunnel interface.<br>Default: <b>On</b> , which disallows port hopping on tunnel interface.<br><br>Starting from Cisco IOS XE Catalyst SD-WAN Release 17.18.1a, this field is deprecated. Instead use the <b>Full Port Hop</b> option. See the <b>Full Port Hop</b> field.                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Low-Bandwidth Link                         | Click <b>On</b> to set the tunnel interface as a low-bandwidth link.<br>Default: <b>Off</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

| Parameter Name       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Clear-Dont-Fragment  | <p>Configure <b>Clear-Dont-Fragment</b> for packets that arrive at an interface that has Don't Fragment configured. If these packets are larger than what MTU allows, they are dropped. If you clear the Don't Fragment bit, the packets are fragmented and sent.</p> <p>Click <b>On</b> to clear the Dont Fragment bit in the IPv4 packet header for packets being transmitted out of the interface. When the Dont Fragment bit is cleared, the router fragments packets larger than the MTU of the interface before sending the packets.</p> <p><b>Note</b><br/> <b>Clear-Dont-Fragment</b> clears the Dont Fragment bit and the Dont Fragment bit is set. For packets not requiring fragmentation, the Dont Fragment bit is not affected.</p> |
| Network Broadcast    | <p>From the drop-down list, select <b>Global</b>. Click <b>On</b> to accept and respond to network-prefix-directed broadcasts. Enable this parameter only if the <b>Directed Broadcast</b> is enabled on the LAN interface feature template.</p> <p>Default: <b>Off</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Carrier              | <p>From the drop-down list, select <b>Global</b> and select the carrier name or private network identifier to associate with the tunnel.</p> <p>Values: carrier1, carrier2, carrier3, carrier4, carrier5, carrier6, carrier7, carrier8, default.</p> <p>Default: default</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Bind Loopback Tunnel | <p>Enter the name of a physical interface to bind to a loopback interface. The interface name has the following format:</p> <p><i>geslot/port</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| NAT Refresh Interval | <p>Set the interval between NAT refresh packets sent on a DTLS or TLS WAN transport connection.</p> <p>Range: 1 through 60 seconds</p> <p>Default: 5 seconds</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Hello Interval       | <p>Enter the interval between Hello packets sent on a DTLS or TLS WAN transport connection.</p> <p>Range: 100 through 10000 milliseconds</p> <p>Default: 1000 milliseconds (1 second)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

| Parameter Name       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hello Tolerance      | <p>Enter the time to wait for a Hello packet on a DTLS or TLS WAN transport connection before declaring that transport tunnel to be down.</p> <p>Range: 12 through 60 seconds</p> <p>Default: 12 seconds</p> <p>The default hello interval is 1000 milliseconds, and it can be a time in the range 100 through 600000 milliseconds (10 minutes). The default hello tolerance is 12 seconds, and it can be a time in the range 12 through 600 seconds (10 minutes). To reduce outgoing control packets on a TLOC, it is recommended that on the tunnel interface you set the hello interval to 60000 milliseconds (10 minutes) and the hello tolerance to 600 seconds (10 minutes) and include the <b>no track-transport disable</b> regular checking of the DTLS connection between the edge device and the controller. For a tunnel connection between a edge device and any controller device, the tunnel uses the hello interval and tolerance times configured on the edge device. This choice is made to minimize the traffic sent over the tunnel, to allow for situations where the cost of a link is a function of the amount of traffic traversing the link. The hello interval and tolerance times are chosen separately for each tunnel between a edge device and a controller device. Another step taken to minimize the amount of control plane traffic is to not send or receive OMP control traffic over a cellular interface when other interfaces are available. This behavior is inherent in the software and is not configurable.</p> |
| Last Resort Circuit  | <p>Select to use the tunnel interface as the circuit of last resort.</p> <p><b>Note</b></p> <p>It is assumed that an interface configured as a circuit of last resort is unavailable and is skipped while calculating the number of control connections. As a result, the cellular modem becomes dormant, and no traffic is sent over the circuit.</p> <p>When the configurations are activated on the edge device with cellular interfaces, all the interfaces begin the process of establishing control and BFD connections. When one or more of the primary interfaces establishes a BFD connection, the circuit of last resort shuts itself down.</p> <p>If the primary interfaces lose their connections to remote edges, the circuit of last resort activates itself, triggering a BFD TLOC Down alarm and a Control TLOC Down alarm on the edge device. The last resort interfaces are a backup circuit on edge device and are activated when all other transport links BFD sessions fail. In this mode, the radio interface is turned off, and no control or data connections exist over the cellular interface.</p>                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Allow Services       | Click <b>On</b> or <b>Off</b> for each service to allow or disallow the service on the cellular interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Encapsulation</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |



| Parameter Name | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Encapsulation  | <p>Enable atleast one of the following encapsulation methods:</p> <ul style="list-style-type: none"> <li>• <b>IPsec</b>: Enter a value to set the preference for directing traffic to the tunnel. A higher value is preferred over a lower value.<br/>Range: 0 through 4294967295<br/>Default: 0</li> <li>• <b>IPsec Preference</b>: From the drop-down list, select <b>Global</b> and enter a value to set the preference for directing traffic to the tunnel. A higher value is preferred over a lower value.<br/>Range: 0 through 4294967295<br/>Default: 0</li> <li>• <b>IPsec Weight</b>: From the drop-down list, select <b>Global</b> and enter a value to set weight for balancing traffic across multiple TLOCs. A higher value sends more traffic to the tunnel.<br/>Range: 1 through 255<br/>Default: 1</li> <li>• <b>GRE</b>: Enter a value to set GRE preference for TLOC.<br/>Range: 0 through 4294967295</li> <li>• <b>GRE Preference</b>: From the drop-down list, select <b>Global</b> and enter a value to set the preference for directing traffic to the tunnel. A higher value is preferred over a lower value.<br/>Range: 0 through 4294967295<br/>Default: 0</li> <li>• <b>GRE Weight</b>: From the drop-down list, select <b>Global</b> and enter a value to set weight for balancing traffic across multiple TLOCs. A higher value sends more traffic to the tunnel.<br/>Range: 1 through 255<br/>Default: 1</li> </ul> |

e. Configure NAT.

**Table 207: NAT**

| Parameter Name        | Description                                                                                                               |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------|
| UDP Timeout (Minutes) | <p>Specify when NAT translations over UDP sessions time out.<br/>Range: 1 through 65536 minutes<br/>Default: 1 minute</p> |

| Parameter Name        | Description                                                                                                                 |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------|
| TCP Timeout (Minutes) | Specify when NAT translations over TCP sessions time out.<br>Range: 1 through 65536 minutes<br>Default: 60 minutes (1 hour) |

## f. Configure QoS.

Table 208: QoS

| Parameter Name      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Adaptive QoS        | Enter adaptive QoS parameters. You can leave the additional details at as default or specify your values. <ul style="list-style-type: none"> <li>• <b>Adapt Period (Minutes):</b> Choose <b>Global</b> from the drop-down list, click <b>On</b>, and enter the period in minutes.</li> <li>• <b>Shaping Rate Upstream:</b> Choose <b>Global</b> from the drop-down list, click <b>On</b>, and enter the minimum, maximum, and default upstream bandwidth in Kbps.</li> <li>• <b>Shaping Rate Downstream:</b> Choose <b>Global</b> from the drop-down list, click <b>On</b>, and enter the minimum, maximum, downstream, and upstream bandwidth in Kbps.</li> </ul> |
| Shaping Rate (kbps) | Choose <b>Global</b> from the drop-down list and configure the aggregate traffic transmission rate on the interface to be less than line rate, in kilobits per second (kbps).<br>Range: 8 through 100000000                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

## g. Configure ACL.

Table 209: ACL

| Parameter Name           | Description                                                                          |
|--------------------------|--------------------------------------------------------------------------------------|
| IPv4 Ingress Access List | Enter the name of an IPv4 access list to packets being received on the interface.    |
| IPv4 Egress Access List  | Enter the name of an IPv4 access list to packets being transmitted on the interface. |
| IPv6 Ingress Access List | Enter the name of an IPv6 access list to packets being received on the interface.    |
| IPv6 Egress Access List  | Enter the name of an IPv6 access list to packets being transmitted on the interface. |

## h. Configure advanced parameters.

Table 210: Advanced

| Parameter Name | Description                              |
|----------------|------------------------------------------|
| Shutdown       | Click <b>No</b> to enable the interface. |

| Parameter Name              | Description                                                                                                                                                                                                                                                                                             |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tracker / Tracker Group     | Enter the name of a tracker or tracker group to track the status of transport interfaces that connect to the internet.                                                                                                                                                                                  |
| PPP Maximum Payload         | Enter the maximum receive unit (MRU) value to be negotiated during PPP-over-Ethernet negotiation.<br>Range: 64 through 1792 bytes                                                                                                                                                                       |
| Service Provider            | Specify the details of the service provider.                                                                                                                                                                                                                                                            |
| Bandwidth Upstream (Kbps)   | Specify the bandwidth value to generate notifications when the bandwidth of traffic transmitted on a physical interface exceeds the value.                                                                                                                                                              |
| Bandwidth Downstream (Kbps) | Specify the bandwidth value to generate notifications when the bandwidth of traffic transmitted on a physical interface exceeds the value.                                                                                                                                                              |
| IP MTU                      | Enter the maximum MTU size of packets on the interface.<br>Range: 576 through 1804<br>Default: 1500.                                                                                                                                                                                                    |
| TCP MSS                     | Enter the maximum segment size (MSS) of TCP SYN packets passing through the router. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented.<br>Range: 552 through 1460 bytes<br>Default: 1500                                  |
| TLOC Extension              | Enter the name of a physical interface on the same router that connects to the WAN transport. This configuration binds the service-side interface to the WAN transport by enabling a device to access the opposite WAN transport connected to the neighbouring device using a TLOC-extension interface. |
| IP Directed Broadcast       | From the drop-down list, select <b>Global</b> to enable IP Directed Broadcast.<br><br>An IP directed broadcast is an IP packet whose destination address is a valid broadcast address for some IP subnet but which originates from a node that is not itself part of that destination subnet.           |
| Tracker / Tracker Group     | Enter the name of a tracker or tracker group to track the status of transport interfaces that connect to the internet.                                                                                                                                                                                  |

### What to do next

Also see [Deploy a configuration group](#).

# VPN Interface DSL PPPoE

Use the VPN Interface DSL PPPoE template for Cisco IOS XE Catalyst SD-WAN devices.

You configure PPP-over-Ethernet interfaces on routers with DSL NIM modules, to provide support for service provider digital subscriber line (DSL) functionality.

To configure DSL interfaces on Cisco routers using Cisco SD-WAN Manager templates:

1. Create a VPN Interface DSL PPPoE feature template to configure PPP-over-Ethernet interface parameters, as described in this article.
2. Create a VPN feature template to configure VPN parameters. See the VPN help topic.

## Navigate to the Template Screen and Name the Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**.



---

**Note** In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

---

3. From the **Create Template** drop-down list, select **From Feature Template**.
4. From the **Device Model** drop-down list, select the type of device for which you are creating the template.
5. Click **Transport & Management VPN** or scroll to the **Transport & Management VPN** section.
6. Under **Additional VPN 0 Templates**, click **VPN Interface DSL PPPoE**.
7. From the **VPN Interface DSL PPPoE** drop-down list, click **Create Template**. The VPN Interface DSL PPPoE template form is displayed. This form contains fields for naming the template, and fields for defining PPPoE Interface parameters.
8. In **Template Name**, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.
9. In **Template Description**, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default or to enter a value, click the scope drop-down to the left of the parameter field and select one of the following:

Table 211:

| Parameter Scope                               | Scope Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Device Specific<br>(indicated by a host icon) | <p>Use a device-specific value for the parameter. For device-specific parameters, you cannot enter a value in the feature template. You enter the value when you attach a Viptela device to a device template .</p> <p>When you click <b>Device Specific</b>, the <b>Enter Key</b> box opens. This box displays a key, which is a unique string that identifies the parameter in a CSV file that you create. This file is an Excel spreadsheet that contains one column for each key. The header row contains the key names (one key per column), and each row after that corresponds to a device and defines the values of the keys for that device. You upload the CSV file when you attach a Viptela device to a device template. For more information, see <a href="#">Create a Template Variables Spreadsheet</a> .</p> <p>To change the default key, type a new string and move the cursor out of the Enter Key box.</p> <p>Examples of device-specific parameters are system IP address, hostname, GPS location, and site ID.</p> |
| Global (indicated by a globe icon)            | <p>Enter a value for the parameter, and apply that value to all devices.</p> <p>Examples of parameters that you might apply globally to a group of devices are DNS server, syslog server, and interface MTUs.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

### Configure VDSL Controller Functionality

To configure basic VDSL controller functionality in a VPN, select **Basic Configuration** and configure the following parameters. Required parameters are indicated with an asterisk.



**Note** If your deployment includes devices with DSL, you must include DSL interface templates in Cisco SD-WAN Manager, even if these templates are not used.

Table 212:

| Parameter Name        | Description                                                                                                          |
|-----------------------|----------------------------------------------------------------------------------------------------------------------|
| Shutdown*             | Click <b>No</b> to enable the VDSL controller interface.                                                             |
| Controller VDSL Slot* | Enter the slot number of the controller VDSL interface, in the format <i>slot/subslot/port</i> (for example, 0/2/0). |

| Parameter Name           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mode*                    | <p>Select the operating mode of the VDSL controller from the drop-down:</p> <ul style="list-style-type: none"> <li>• <b>Auto</b>—Default mode.</li> <li>• <b>ADSL1</b>—Use ITU G.992.1 Annex A full-rate mode, which provides a downstream rate of 1.3 Mbps and an upstream rate of 1.8 Mbps.</li> <li>• <b>ADSL2</b>—Use ITU G.992.3 Annex A, Annex L, and Annex M, which provides a downstream rate of 12 Mbps and an upstream rate of 1.3 Mbps.</li> <li>• <b>ADSL2+</b>— Use ITU G.992.5 Annex A and Annex M, which provides a downstream rate of 24 Mbps and an upstream rate of 3.3 Mbps.</li> <li>• <b>ANSI</b>—Operating in ADSL2/2+ mode, as defined in ITU G.991.1, G.992.3, and G992.5, Annex A and Annex M, and in VDSL2 mode, as defined in ITU-T G993.2.</li> <li>• <b>VDSL2</b>—Operate in VDSL2 mode, as defined in ITU-T G.993.2, which uses frequencies of up to 30 MHz to provide a downstream rate of 200 Mbps and an upstream rate of 100 Mbps..</li> </ul> |
| VDSL Modem Configuration | Enter a command to send to the DSL modem in the NIM module. If the command is valid, it is executed and the results are returned to the Cisco SD-WAN Manager NMS. If the command is not valid, it is not executed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| SRA                      | Click <b>Yes</b> to enable seamless rate adaptation on the interface. SRA adjusts the line rate based on current line conditions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

To save the feature template, click **Save**.

### Configure the Ethernet Interface on VDSL Controller

To configure an Ethernet interface on the VDSL controller, select **Ethernet** and configure the following parameters. You must configure all parameters.

Table 213: Feature History

| Feature Name                        | Release Information                                             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Support for Dialer Interface in DSL | Cisco IOS XE Release 17.3.2<br><br>Cisco vManage Release 20.3.1 | <p>This feature enables tracking of a Point-to-Point Protocol (PPP) session over a dialer interface on Cisco IOS XE Catalyst SD-WAN devices.</p> <p>Dialer interface is used in Digital Subscriber Line (DSL) in the deployments of Point-to-Point Protocol over Ethernet (PPPoE), Point-to-Point Protocol over Asynchronous Transfer Mode (PPPoA). Dialer interface always stay up irrespective of the PPP session status. This helps to avoid the need for additional configuration such as IP SLA and tracking for routing failover to work while using dialer interfaces.</p> <p>The following command is added to configure dialer down-with-vInterface which brings the dialer interface down when the PPP session goes down.</p> |

Table 214:

| Parameter Name          | Description                                                                                                                                                                                                                                                 |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ethernet Interface Name | Enter a name for the Ethernet interface, in the format <i>subslot/port</i> (for example 2/0). You do not need to enter the slot number, because it must always be 0.                                                                                        |
| VLAN ID                 | Enter the VLAN identifier of the Ethernet interface.                                                                                                                                                                                                        |
| Description             | Enter a description for the interface.                                                                                                                                                                                                                      |
| Dialer Pool Member      | Enter the number of the dialer pool to which the interface belongs. It can be a value from 1 through 255.                                                                                                                                                   |
| PPP Max Payload         | Enter the maximum receive unit (MRU) value to be negotiated during PPP Link Control Protocol (LCP) negotiation. <i>Range:</i> 64 through 1792 bytes                                                                                                         |
| Dialer IP               | <p>Configure the IP prefix of the dialer interface. This prefix is that of the node in the destination that the interface calls.</p> <ul style="list-style-type: none"> <li>Negotiated—Use the address that is obtained during IPCP negotiation.</li> </ul> |

To save the feature template, click **Save**.

### Configure the PPP Authentication Protocol

To configure the PPP authentication protocol, select **PPP** and configure the following parameters:

Table 215:

| Parameter Name          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Authentication Protocol | <p>Select the authentication protocol used by the MLP:</p> <ul style="list-style-type: none"> <li>• <b>CHAP</b>—Enter the hostname and password provided by your Internet Service Provider (ISP). <i>hostname</i> can be up to 255 characters.</li> <li>• <b>PAP</b>—Enter the username and password that are provided by your ISP. <i>username</i> can be up to 255 characters.</li> <li>• <b>PAP</b> and <b>CHAP</b>—Configure both authentication protocols. Enter the login credentials for each protocol. To use the same username and password for both, click Same Credentials for PAP and CHAP.</li> </ul> |

To save the feature template, click **Save**.

### Create a Tunnel Interface

On IOS XE routers, you can configure up to eight tunnel interfaces. This means that each router can have up to eight TLOCs.

For the control plane to establish itself so that the overlay network can function, you must configure WAN transport interfaces in VPN 0.

To configure a tunnel interface for the multilink interface, select the Tunnel Interface tab and configure the following parameters:

Table 216:

| Parameter Name    | Description                                                                                                                              |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Tunnel Interface  | Click <b>On</b> to create a tunnel interface.                                                                                            |
| Color             | Select a color for the TLOC.                                                                                                             |
| Color Description | <p>Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.18.1</p> <p>Enter a description associated to the TLOC color.</p> |



| Parameter Name                                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Control Connection                             | <p>By default, Control Connection is set to <b>On</b>, which establishes a control connection for the TLOC. If the router has multiple TLOCs, click <b>No</b> to have the tunnel not establish control connection for the TLOC.</p> <p><b>Note</b><br/>We recommend a minimum of 650-700 Kbps bandwidth with default 1 sec hello-interval and 12 sec hello-tolerance parameters configured to avoid any data/packet loss in connection traffic.</p> <p>For each BFD session, an additional average sized BFD packet of 175 Bytes consumes 1.4 Kbps of bandwidth.</p> <p>A sample calculation of the required bandwidth for bidirectional BFD packet flow is given below:</p> <ul style="list-style-type: none"> <li>• 650 – 700 Kbps per device for control connections.</li> <li>• 175 Bytes (or 1.4 Kbps) per BFD session on the device (request)</li> <li>• 175 Bytes (or 1.4 Kbps) per BFD session on the device (response)</li> </ul> <p>If the path MTU discovery (PMTUD) is enabled, bandwidth for send/receive BFD packets per tunnel for every 30 secs:</p> <p>A 1500 Bytes BFD request packet is sent per tunnel every 30 secs:<br/> <math>1500 \text{ Bytes} * 8 \text{ bits/1 byte} * 1 \text{ packet} / 30 \text{ secs} = 400 \text{ bps (request)}</math></p> <p>A 147 Bytes BFD packet is sent in response:<br/> <math>147 \text{ Bytes} * 8 \text{ bits/1 byte} * 1 \text{ packet} / 30 \text{ secs} = 40 \text{ bps (response)}</math></p> <p>Therefore, a device with 775 BFD sessions (for example) requires a bandwidth of:<br/> <math>700k + (1.4k*775) + (400 * 775) + (1.4k*775) + (40 * 775) = \sim 3.5 \text{ MBps}</math></p> <ul style="list-style-type: none"> <li>• STATE—specifies the vdaemon control state.</li> </ul> <p>Last Connection—If no control connection on that WAN interface, the uptime of the device is lifted.</p> <p>SPI Time Remaining—countdown to the next change in SPI for IPSec. The countdown starts at half of the rekey time.</p> |
| Maximum Control Connections                    | <p>Specify the maximum number of Cisco Catalyst SD-WAN Controllers that the WAN tunnel interface can connect to. To have the tunnel establish no control connections, set the number to 0.</p> <p><i>Range: 0 through 8 Default: 2</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Cisco Catalyst SD-WAN Validator As STUN Server | <p>Click <b>On</b> to enable Session Traversal Utilities for NAT (STUN) to allow the tunnel interface to discover its public IP address and port number when the router is located behind a NAT.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Exclude Controller Group List                  | <p>Set the Cisco Catalyst SD-WAN Controllers that the tunnel interface is not allowed to connect to. <i>Range: 0 through 100</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Parameter Name                             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cisco SD-WAN Manager Connection Preference | Set the preference for using a tunnel interface to exchange control traffic with the Cisco SD-WAN Manager NMS. <i>Range:</i> 0 through 8 <i>Default:</i> 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Full Port Hop</b>                       | <p>Minimum release: Cisco IOS XE Catalyst SD-WAN Release 17.18.1a</p> <p>Enable full port hopping at the TLOC level to allow devices to establish connections with controllers by switching to the next port if the current port is blocked or non-functional.</p> <p>Default: Disabled</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Port Hop                                   | <p>Click <b>On</b> to enable port hopping, or click Off to disable it. When a router is behind a NAT, port hopping rotates through a pool of preselected OMP port numbers (called base ports) to establish DTLS connections with other routers when a connection attempt is unsuccessful. The default base ports are 12346, 12366, 12386, 12406, and 12426. To modify the base ports, set a port offset value. <i>Default:</i> Enabled.</p> <p>Starting from Cisco IOS XE Catalyst SD-WAN Release 17.18.1a, this field is deprecated. Instead use the <b>Full Port Hop</b> option. See the <b>Full Port Hop</b> field.</p>                                                                                                                                                                                                                                                    |
| Low-Bandwidth Link                         | Select to characterize the tunnel interface as a low-bandwidth link.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Tunnel TCP MSS                             | <p>TCP MSS affects any packet that contains an initial TCP header that flows through the router. When configured, TCP MSS is examined against the MSS exchanged in the three-way handshake. The MSS in the header is lowered if the configured TCP MSS setting is lower than the MSS in the header. If the MSS header value is already lower than the TCP MSS, the packets flow through unmodified. The host at the end of the tunnel uses the lower setting of the two hosts. If the TCP MSS is to be configured, it should be set at 40 bytes lower than the minimum path MTU.</p> <p>Specify the MSS of TPC SYN packets passing through the Cisco IOS XE Catalyst SD-WAN device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. <i>Range:</i> 552 to 1460 bytes <i>Default:</i> None</p> |
| Clear-Dont-Fragment                        | <p>Configure <b>Clear-Dont-Fragment</b> for packets that arrive at an interface that has Don't Fragment configured. If these packets are larger than what MTU allows, they are dropped. If you clear the Don't Fragment bit, the packets are fragmented and sent.</p> <p>Click <b>On</b> to clear the Dont Fragment bit in the IPv4 packet header for packets being transmitted out of the interface. When the Dont Fragment bit is cleared, packets larger than the MTU of the interface are fragmented before being sent.</p> <p><b>Note</b><br/><b>Clear-Dont-Fragment</b> clears the Dont Fragment bit and the Dont Fragment bit is set. For packets not requiring fragmentation, the Dont Fragment bit is not affected.</p>                                                                                                                                              |
| Allow Service                              | Select <b>On</b> or <b>On</b> for each service to allow or disallow the service on the interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

To configure additional tunnel interface parameters, click **Advanced Options** and configure the following parameters:

Table 217:

| Parameter Name       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GRE                  | Use GRE encapsulation on the tunnel interface. By default, GRE is disabled.<br><br>If you select both IPsec and GRE encapsulations, two TLOCs are created for the tunnel interface that have the same IP addresses and colors, but that differ by their encapsulation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| IPsec                | Use IPsec encapsulation on the tunnel interface. By default, IPsec is enabled.<br><br>If you select both IPsec and GRE encapsulations, two TLOCs are created for the tunnel interface that have the same IP addresses and colors, but that differ by their encapsulation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| IPsec Preference     | Specify a preference value for directing traffic to the tunnel. A higher value is preferred over a lower value.<br><br><i>Range: 0 through 4294967295 Default: 0</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| IPsec Weight         | Enter a weight to use to balance traffic across multiple TLOCs. A higher value sends more traffic to the tunnel.<br><br><i>Range: 1 through 255 Default: 1</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Carrier              | Select the carrier name or private network identifier to associate with the tunnel.<br><br><i>Values: carrier1, carrier2, carrier3, carrier4, carrier5, carrier6, carrier7, carrier8, default Default: default</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Bind Loopback Tunnel | Enter the name of a physical interface to bind to a loopback interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Last-Resort Circuit  | Select to use the tunnel interface as the circuit of last resort.<br><br><b>Note</b><br>An interface configured as a circuit of last resort is expected to be down and is skipped while calculating the number of control connections, the cellular modem becomes dormant, and no traffic is sent over the circuit.<br><br>When the configurations are activated on the edge device with cellular interfaces, then all the interfaces begin the process of establishing control and BFD connections. When one or more of the primary interfaces establishes a BFD connection, the circuit of last resort shuts itself down.<br><br>Only when all the primary interfaces lose their connections to remote edges, then the circuit of last resort activates itself triggering a BFD TLOC Down alarm and a Control TLOC Down alarm on the edge device. The last resort interfaces are used as backup circuit on edge device and are activated when all other transport links BFD sessions fail. In this mode the radio interface is turned off, and no control or data connections exist over the cellular interface. |
| NAT Refresh Interval | Enter the interval between NAT refresh packets that are sent on a DTLS or TLS WAN transport connection. <i>Range: 1 through 60 seconds. Default: 5 seconds.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

| Parameter Name  | Description                                                                                                                                                                                                |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hello Interval  | Enter the interval between Hello packets sent on a DTLS or TLS WAN transport connection. <i>Range:</i> 100 through 10000 milliseconds. <i>Default:</i> 1000 milliseconds (1 second).                       |
| Hello Tolerance | Enter the time to wait for a Hello packet on a DTLS or TLS WAN transport connection before declaring that transport tunnel to be down.<br><i>Range:</i> 12 through 60 seconds. <i>Default:</i> 12 seconds. |

### Configure the Interface as a NAT Device

To configure an interface to act as a NAT device for applications such as port forwarding, select **NAT**, click **On** and configure the following parameters:

**Table 218:**

| Parameter Name  | Description                                                                                                                                              |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| NAT             | Click <b>On</b> to have the interface act as a NAT device.                                                                                               |
| Refresh Mode    | Select how NAT mappings are refreshed, either outbound or bidirectional (outbound and inbound). <i>Default:</i> Outbound                                 |
| UDP Timeout     | Specify when NAT translations over UDP sessions time out. <i>Range:</i> 1 through 65536 minutes <i>Default:</i> 1 minutes                                |
| TCP Timeout     | Specify when NAT translations over TCP sessions time out. <i>Range:</i> 1 through 65536 minutes <i>Default:</i> 60 minutes (1 hour)                      |
| Block ICMP      | Select <b>On</b> to block inbound ICMP error messages. By default, a router acting as a NAT device receives these error messages. <i>Default:</i> Off    |
| Respond to Ping | Select <b>On</b> to have the router respond to ping requests to the NAT interface's IP address that are received from the public side of the connection. |

To create a port forwarding rule, click **Add New Port Forwarding Rule** and configure the following parameters. You can define up to 128 port-forwarding rules to allow requests from an external network to reach devices on the internal network.

**Table 219:**

| Parameter Name   | Description                                                                                                                                                   |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Port Start Range | Enter a port number to define the port or first port in the range of interest. <i>Range:</i> 0 through 65535                                                  |
| Port End Range   | Enter the same port number to apply port forwarding to a single port, or enter a larger number to apply it to a range of ports. <i>Range:</i> 0 through 65535 |

| Parameter Name | Description                                                                                                                                                   |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Protocol       | Select the protocol to which to apply the port-forwarding rule, either TCP or UDP. To match the same ports for both TCP and UDP traffic, configure two rules. |
| VPN            | Specify the private VPN in which the internal server resides. This VPN is one of the VPN identifiers in the overlay network. <i>Range:</i> 0 through 65527    |
| Private IP     | Specify the IP address of the internal server to which to direct traffic that matches the port-forwarding rule.                                               |

To save a port forwarding rule, click **Add**.

To save the feature template, click **Save**.

### Apply Access Lists

To apply a rewrite rule, access lists, and policers to a router interface, select **ACL** and configure the following parameters:

**Table 220:**

| Parameter Name     | Description                                                                                                                  |
|--------------------|------------------------------------------------------------------------------------------------------------------------------|
| Shaping rate       | Configure the aggregate traffic transmission rate on the interface to be less than line rate, in kilobits per second (kbps). |
| QoS map            | Specify the name of the QoS map to apply to packets being transmitted out the interface.                                     |
| Rewrite Rule       | Click <b>On</b> , and specify the name of the rewrite rule to apply on the interface.                                        |
| Ingress ACL – IPv4 | Click <b>On</b> , and specify the name of the access list to apply to IPv4 packets being received on the interface.          |
| Egress ACL – IPv4  | Click <b>On</b> , and specify the name of the access list to apply to IPv4 packets being transmitted on the interface.       |
| Ingress ACL – IPv6 | Click <b>On</b> , and specify the name of the access list to apply to IPv6 packets being received on the interface.          |
| Egress ACL – IPv6  | Click <b>On</b> , and specify the name of the access list to apply to IPv6 packets being transmitted on the interface.       |
| Ingress Policer    | Click <b>On</b> , and specify the name of the policer to apply to packets being received on the interface.                   |
| Egress Policer     | Click <b>On</b> , and specify the name of the policer to apply to packets being transmitted on the interface.                |

To save the feature template, click **Save**.

### Configure Other Interface Properties

To configure other interface properties, select the Advanced tab and configure the following properties:

Table 221:

| Parameter Name       | Description                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bandwidth Upstream   | For transmitted traffic, set the bandwidth above which to generate notifications. <i>Range:</i> 1 through $(2^{32} / 2) - 1$ kbps                                                                                                                                                                                                                                                                                                      |
| Bandwidth Downstream | For received traffic, set the bandwidth above which to generate notifications. <i>Range:</i> 1 through $(2^{32} / 2) - 1$ kbps                                                                                                                                                                                                                                                                                                         |
| IP MTU               | Specify the maximum MTU size of packets on the interface. <i>Range:</i> 576 through 1804. <i>Default:</i> 1500 bytes.                                                                                                                                                                                                                                                                                                                  |
| TCP MSS              | Specify the maximum segment size (MSS) of TCP SYN packets passing through the router. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. <i>Range:</i> 552 to 1460 bytes. <i>Default:</i> None.                                                                                                                                                          |
| Clear Dont Fragment  | Click <b>On</b> to clear the Don't Fragment bit in the IPv4 packet header for packets being transmitted out the interface. When the DF bit is cleared, packets larger than that interface's MTU are fragmented before being sent.                                                                                                                                                                                                      |
| TLOC Extension       | Enter the name of the physical interface on the same router that connects to the WAN transport circuit. This configuration then binds this service-side interface to the WAN transport. A second router at the same site that itself has no direct connection to the WAN (generally because the site has only a single WAN connection) and that connects to this service-side interface is then provided with a connection to the WAN. |
| Tracker              | Enter the name of a tracker to track the status of transport interfaces that connect to the internet.                                                                                                                                                                                                                                                                                                                                  |

To save the feature template, click **Save**.

#### Release Information

Introduced in Cisco SD-WAN Manager NMS in Release 18.3.

## Configure VPN Interface Ethernet PPPoE Using a Configuration Group

#### Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

#### Procedure

**Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

**Step 2** Create and configure a Ethernet PPPoE feature under Transport VPN in a Transport and Management profile.

a) Configure basic PPPoE functionality.

| Parameter Name            | Description                                                                                                                                                        |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ethernet Interface Name * | Enter the name of an ethernet interface.<br><br>For IOS XE routers, you must spell out the interface names completely (for example, <b>GigabitEthernet0/0/0</b> ). |
| Description               | Enter a description for the ethernet interface.                                                                                                                    |
| VLAN ID                   | Enter the VLAN identifier of the Ethernet interface.                                                                                                               |
| Dialer Pool Member *      | Enter the number of the dialer pool to which the interface belongs.<br><br>Range: 1 through 255                                                                    |

b) Configure the PPP Authentication Protocol.

| Parameter Name               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PPP Authentication Protocol* | Select the authentication protocol used by the MLP: <ul style="list-style-type: none"> <li>• <b>PAP</b>: Enter the username and password that are provided by your ISP. <i>username</i> can be up to 255 characters.</li> <li>• <b>CHAP</b>: Enter the hostname and password provided by your Internet Service Provider (ISP). <i>hostname</i> can be up to 255 characters.</li> <li>• <b>PAP</b> and <b>CHAP</b>: Configure both authentication protocols. Enter the login credentials for each protocol.</li> </ul> |
| Authentication Type          | Select the type authentication from one of the following options.: <ul style="list-style-type: none"> <li>• <b>Unidirectional</b>: Only the side receiving the call (NAS) authenticates the remote side (client). The remote client does not authenticate the server.</li> <li>• <b>Bidirectional</b>: Each side independently sends an Authenticate-Request (AUTH-REQ) and receives either an Authenticate-Acknowledge (AUTH-ACK) or Authenticate-Not Acknowledged (AUTH-NAK).</li> </ul>                            |
| CHAP Hostname*               | Enter the CHAP hostname.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| CHAP Password*               | Enter the CHAP password.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| PAP Hostname*                | Enter the PAP hostname.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| PAP Password*                | Enter the PAP password.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

c) Configure a tunnel interface for the multilink interface.

| Parameter Name          | Description |
|-------------------------|-------------|
| <b>Tunnel Interface</b> |             |

| Parameter Name                             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Per Tunnel QoS                             | Enable per tunnel QoS and choose <b>Spoke</b> to configure the spoke network topology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Color                                      | Select a color for the TLOC.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Color Description                          | Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.18.1<br>Enter a description associated to the TLOC color.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Groups                                     | Enter the list of groups in the field.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Exclude Controller Group List              | Set the Cisco SD-WAN Controllers that the tunnel interface is not allowed to connect to.<br>Range: 0 through 100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Maximum Control Connections                | Specify the maximum number of Cisco SD-WAN Controllers that the WAN tunnel interface can connect to. To have the tunnel establish no control connections, set the number to 0.<br>Range: 0 through 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Cisco SD-WAN Manager Connection Preference | Set the preference for using a tunnel interface to exchange control traffic with Cisco SD-WAN Manager.<br>Range: 0 through 8<br>Default: 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Tunnel TCP MSS                             | TCP MSS affects any packet that contains an initial TCP header that flows through the router. When configured, TCP MSS is examined against the MSS exchanged in the three-way handshake. The MSS in the header is lowered if the configured TCP MSS setting is lower than the MSS in the header. If the MSS header value is already lower than the TCP MSS, the packets flow through unmodified. The host at the end of the tunnel uses the lower setting of the two hosts. To configure TCP MSS, provide a value that is 40 bytes lower than the minimum path MTU.<br><br>Specify the MSS of TPC SYN packets passing through the Cisco IOS XE Catalyst SD-WAN. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented.<br><br>Range: 552 through 1460 bytes<br>Default: None |
| Border                                     | From the drop-down list, select <b>Global</b> . Click <b>On</b> to set TLOC as border TLOC.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Validator As Stun Server                   | Click <b>On</b> to enable Session Traversal Utilities for NAT (STUN) to allow the tunnel interface to discover its public IP address and port number when the router is located behind a NAT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Full Port Hop                              | Minimum release: Cisco IOS XE Catalyst SD-WAN Release 17.18.1a<br><br>Enable full port hopping at the TLOC level to allow devices to establish connections with controllers by switching to the next port if the current port is blocked or non-functional.<br><br>Default: Disabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |



| Parameter Name       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Port Hop             | <p>From the drop-down list, select <b>Global</b>. Click <b>Off</b> to allow port hopping on tunnel interface.</p> <p>Default: <b>On</b>, which disallows port hopping on tunnel interface.</p> <p>Starting from Cisco IOS XE Catalyst SD-WAN Release 17.18.1a, this field is deprecated. Instead use the <b>Full Port Hop</b> option. See the <b>Full Port Hop</b> field.</p>                                                                                                                                                                                                                                                                                                                                                                    |
| Low-Bandwidth Link   | <p>Click <b>On</b> to set the tunnel interface as a low-bandwidth link.</p> <p>Default: <b>Off</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Clear-Dont-Fragment  | <p>Configure <b>Clear-Dont-Fragment</b> for packets that arrive at an interface that has Don't Fragment configured. If these packets are larger than what MTU allows, they are dropped. If you clear the Don't Fragment bit, the packets are fragmented and sent.</p> <p>Click <b>On</b> to clear the Dont Fragment bit in the IPv4 packet header for packets being transmitted out of the interface. When the Dont Fragment bit is cleared, the router fragments packets larger than the MTU of the interface before sending the packets.</p> <p><b>Note</b><br/> <b>Clear-Dont-Fragment</b> clears the Dont Fragment bit and the Dont Fragment bit is set. For packets not requiring fragmentation, the Dont Fragment bit is not affected.</p> |
| Network Broadcast    | <p>From the drop-down list, select <b>Global</b>. Click <b>On</b> to accept and respond to network-prefix-directed broadcasts. Enable this parameter only if the <b>Directed Broadcast</b> is enabled on the LAN interface feature template.</p> <p>Default: <b>Off</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Carrier              | <p>From the drop-down list, select <b>Global</b> and select the carrier name or private network identifier to associate with the tunnel.</p> <p>Values: carrier1, carrier2, carrier3, carrier4, carrier5, carrier6, carrier7, carrier8, default.</p> <p>Default: default</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Bind Loopback Tunnel | <p>Enter the name of a physical interface to bind to a loopback interface. The interface name has the following format:</p> <p><b>ge slot/port</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| NAT Refresh Interval | <p>Set the interval between NAT refresh packets sent on a DTLS or TLS WAN transport connection.</p> <p>Range: 1 through 60 seconds</p> <p>Default: 5 seconds</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Hello Interval       | <p>Enter the interval between Hello packets sent on a DTLS or TLS WAN transport connection.</p> <p>Range: 100 through 10000 milliseconds</p> <p>Default: 1000 milliseconds (1 second)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

| Parameter Name       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hello Tolerance      | <p>Enter the time to wait for a Hello packet on a DTLS or TLS WAN transport connection before declaring that transport tunnel to be down.</p> <p>Range: 12 through 60 seconds</p> <p>Default: 12 seconds</p> <p>The default hello interval is 1000 milliseconds, and it can be a time in the range 100 through 600000 milliseconds (10 minutes). The default hello tolerance is 12 seconds, and it can be a time in the range 12 through 600 seconds (10 minutes). To reduce outgoing control packets on a TLOC, it is recommended that on the tunnel interface you set the hello interval to 60000 milliseconds (10 minutes) and the hello tolerance to 600 seconds (10 minutes) and include the <b>no track-transport disable</b> regular checking of the DTLS connection between the edge device and the controller. For a tunnel connection between a edge device and any controller device, the tunnel uses the hello interval and tolerance times configured on the edge device. This choice is made to minimize the traffic sent over the tunnel, to allow for situations where the cost of a link is a function of the amount of traffic traversing the link. The hello interval and tolerance times are chosen separately for each tunnel between a edge device and a controller device. Another step taken to minimize the amount of control plane traffic is to not send or receive OMP control traffic over a cellular interface when other interfaces are available. This behavior is inherent in the software and is not configurable.</p> |
| Last Resort Circuit  | <p>Select to use the tunnel interface as the circuit of last resort.</p> <p><b>Note</b></p> <p>It is assumed that an interface configured as a circuit of last resort is unavailable and is skipped while calculating the number of control connections. As a result, the cellular modem becomes dormant, and no traffic is sent over the circuit.</p> <p>When the configurations are activated on the edge device with cellular interfaces, all the interfaces begin the process of establishing control and BFD connections. When one or more of the primary interfaces establishes a BFD connection, the circuit of last resort shuts itself down.</p> <p>If the primary interfaces lose their connections to remote edges, the circuit of last resort activates itself, triggering a BFD TLOC Down alarm and a Control TLOC Down alarm on the edge device. The last resort interfaces are a backup circuit on edge device and are activated when all other transport links BFD sessions fail. In this mode, the radio interface is turned off, and no control or data connections exist over the cellular interface.</p>                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Allow Services       | Click <b>On</b> or <b>Off</b> for each service to allow or disallow the service on the cellular interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Encapsulation</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

| Parameter Name | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Encapsulation  | <p>Enable at least one of the following encapsulation methods:</p> <ul style="list-style-type: none"> <li>• <b>IPsec</b>: Enter a value to set the preference for directing traffic to the tunnel. A higher value is preferred over a lower value.<br/>Range: 0 through 4294967295<br/>Default: 0</li> <li>• <b>IPsec Preference</b>: From the drop-down list, select <b>Global</b> and enter a value to set the preference for directing traffic to the tunnel. A higher value is preferred over a lower value.<br/>Range: 0 through 4294967295<br/>Default: 0</li> <li>• <b>IPsec Weight</b>: From the drop-down list, select <b>Global</b> and enter a value to set weight for balancing traffic across multiple TLOCs. A higher value sends more traffic to the tunnel.<br/>Range: 1 through 255<br/>Default: 1</li> <li>• <b>GRE</b>: Enter a value to set GRE preference for TLOC.<br/>Range: 0 through 4294967295</li> <li>• <b>GRE Preference</b>: From the drop-down list, select <b>Global</b> and enter a value to set the preference for directing traffic to the tunnel. A higher value is preferred over a lower value.<br/>Range: 0 through 4294967295<br/>Default: 0</li> <li>• <b>GRE Weight</b>: From the drop-down list, select <b>Global</b> and enter a value to set weight for balancing traffic across multiple TLOCs. A higher value sends more traffic to the tunnel.<br/>Range: 1 through 255<br/>Default: 1</li> </ul> |

d) Configure an interface to act as a NAT device for applications such as port forwarding.

| Parameter Name        | Description                                                                                                              |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------|
| UDP Timeout (Minutes) | <p>Specify when NAT translations over UDP sessions time out.<br/>Range: 1 through 8947 minutes<br/>Default: 1 minute</p> |

| Parameter Name        | Description                                                                                                                |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------|
| TCP Timeout (Minutes) | Specify when NAT translations over TCP sessions time out.<br>Range: 1 through 8947 minutes<br>Default: 60 minutes (1 hour) |

## e) Configure QoS.

| Parameter Name      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Adaptive QoS        | Enter adaptive QoS parameters. You can leave the additional details at as default or specify your values. <ul style="list-style-type: none"> <li>• <b>Adapt Period (Minutes):</b> Choose <b>Global</b> from the drop-down list, click <b>On</b>, and enter the period in minutes.</li> <li>• <b>Shaping Rate Upstream:</b> Choose <b>Global</b> from the drop-down list, click <b>On</b>, and enter the minimum, maximum, and default upstream bandwidth in Kbps.</li> <li>• <b>Shaping Rate Downstream:</b> Choose <b>Global</b> from the drop-down list, click <b>On</b>, and enter the minimum, maximum, downstream, and upstream bandwidth in Kbps.</li> </ul> |
| Shaping Rate (kbps) | Choose <b>Global</b> from the drop-down list and configure the aggregate traffic transmission rate on the interface to be less than line rate, in kilobits per second (kbps).<br>Range: 8 through 100000000                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

## f) Configure ACL.

| Parameter Name           | Description                                                                          |
|--------------------------|--------------------------------------------------------------------------------------|
| IPv4 Ingress Access List | Enter the name of an IPv4 access list to packets being received on the interface.    |
| IPv4 Egress Access List  | Enter the name of an IPv4 access list to packets being transmitted on the interface. |
| IPv6 Ingress Access List | Enter the name of an IPv6 access list to packets being received on the interface.    |
| IPv6 Egress Access List  | Enter the name of an IPv6 access list to packets being transmitted on the interface. |

## g) Configure additional tunnel interface parameters.

| Parameter Name          | Description                                                                                                                       |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Shutdown                | Choose <b>No</b> to enable the interface.                                                                                         |
| Tracker / Tracker Group | Enter the name of a tracker or tracker group to track the status of transport interfaces that connect to the internet.            |
| Maximum Payload         | Enter the maximum receive unit (MRU) value to be negotiated during PPP-over-Ethernet negotiation.<br>Range: 64 through 1792 bytes |

| Parameter Name          | Description                                                                                                                                                                                                                                                                                             |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IP MTU                  | Enter the maximum MTU size of packets on the interface.<br>Range: 576 through 1804<br>Default: 1500                                                                                                                                                                                                     |
| TCP MSS                 | Enter the maximum segment size (MSS) of TCP SYN packets passing through the router. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented.<br>Range: 552 through 1460 bytes<br>Default: 1500                                  |
| TLOC Extension          | Enter the name of a physical interface on the same router that connects to the WAN transport. This configuration binds the service-side interface to the WAN transport by enabling a device to access the opposite WAN transport connected to the neighbouring device using a TLOC-extension interface. |
| IP Directed Broadcast   | From the drop-down list, select <b>Global</b> to enable IP Directed Broadcast.<br>An IP directed broadcast is an IP packet whose destination address is a valid broadcast address for some IP subnet but which originates from a node that is not itself part of that destination subnet.               |
| Tracker / Tracker Group | Enter the name of a tracker or tracker group to track the status of transport interfaces that connect to the internet.                                                                                                                                                                                  |

### What to do next

Also see [Deploy a configuration group](#).

## VPN Interface Ethernet PPPoE

Use the PPPoE template for Cisco IOS XE Catalyst SD-WAN devices.

You configure PPPoE over GigabitEthernet interfaces on Cisco IOS XE routers, to provide PPPoE client support.

To configure interfaces on Cisco routers using Cisco SD-WAN Manager templates:

1. Create a VPN Interface Ethernet PPPoE feature template to configure Ethernet PPPoE interface parameters, as described in this section.
2. Create a VPN feature template to configure VPN parameters. See VPN help topic.

### Navigate to the Template Screen and Name the Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and click **Create Template**.



**Note** In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. From the **Create Template** drop-down list, choose **From Feature Template**.
4. From the **Device Model** drop-down list, select the type of device for which you are creating the template.
5. Click **Transport & Management VPN** or scroll to the **Transport & Management VPN** section.
6. Under **Additional VPN 0 Templates**, click **VPN Interface Ethernet PPPoE**.
7. From the **VPN Interface Ethernet PPPoE** drop-down list, click **Create Template**. The VPN Interface Ethernet PPPoE template form is displayed.

This form contains fields for naming the template, and fields for defining the Ethernet PPPoE parameters.



8. In **Template Name**, enter a name for the template.  
The name can be up to 128 characters and can contain only alphanumeric characters.
9. In **Template Description**, enter a description of the template.  
The description can be up to 2048 characters and can contain only alphanumeric characters.

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default or to enter a value, click the **Scope** drop-down list and select one of the following:

**Table 222:**

| Parameter Scope                               | Scope Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Device Specific<br>(indicated by a host icon) | <p>Use a device-specific value for the parameter. For device-specific parameters, you cannot enter a value in the feature template. You enter the value when you attach a Cisco Catalyst SD-WAN device to a device template .</p> <p>When you click <b>Device Specific</b>, the <b>Enter Key</b> box opens. This box displays a key, which is a unique string that identifies the parameter in a CSV file that you create. This file is an Excel spreadsheet that contains one column for each key. The header row contains the key names (one key per column), and each row after that corresponds to a device and defines the values of the keys for that device. You upload the CSV file when you attach a Cisco Catalyst SD-WAN device to a device template. For more information, see <a href="#">Create a Template Variables Spreadsheet</a> .</p> <p>To change the default key, type a new string and move the cursor out of the Enter Key box.</p> <p>Examples of device-specific parameters are system IP address, hostname, GPS location, and site ID.</p> |
| Global (indicated by a globe icon)            | <p>Enter a value for the parameter, and apply that value to all devices.</p> <p>Examples of parameters that you might apply globally to a group of devices are DNS server, syslog server, and interface MTUs.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

### Configure PPPoE Functionality

To configure basic PPPoE functionality, click **Basic Configuration** and configure the following parameters. Required parameters are indicated with an asterisk.

**Table 223:**

| Parameter Name          | Description                                                                                                                                                          |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Shutdown*               | Click <b>No</b> to enable the GigabitEthernet interface.                                                                                                             |
| Ethernet Interface Name | Enter the name of a GigabitEthernet interface.<br>For IOS XE routers, you must spell out the interface names completely (for example, <b>GigabitEthernet0/0/0</b> ). |
| VLAN ID                 | VLAN tag of the sub-interface.                                                                                                                                       |
| Description             | Enter a description of the Ethernet-PPPoE-enabled interface.                                                                                                         |
| Dialer Pool Member      | Enter the number of the dialer pool to which the interface belongs.<br><i>Range:</i> 100 to 255.                                                                     |
| PPP Maximum Payload     | Enter the maximum receive unit (MRU) value to be negotiated during PPP Link Control Protocol (LCP) negotiation. <i>Range:</i> 64 through 1792 bytes                  |

To save the feature template, click **Save**.

### Configure the PPP Authentication Protocol

To configure the PPP Authentication Protocol, click **PPP** and configure the following parameters. Required parameters are indicated with an asterisk.

**Table 224:**

| Parameter Name              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PPP Authentication Protocol | Select the authentication protocol used by the MLP: <ul style="list-style-type: none"> <li>• <b>CHAP</b>—Enter the hostname and password provided by your Internet Service Provider (ISP). <i>hostname</i> can be up to 255 characters.</li> <li>• <b>PAP</b>—Enter the username and password provided by your ISP. <i>username</i> can be up to 255 characters.</li> <li>• <b>PAP and CHAP</b>—Configure both authentication protocols. Enter the login credentials for each protocol. To use the same username and password for both, click Same Credentials for PAP and CHAP.</li> </ul> |

To save the feature template, click **Save**.

### Create a Tunnel Interface

On IOS XE routers, you can configure up to eight tunnel interfaces. This means that each router can have up to eight TLOCs.

For the control plane to establish itself so that the overlay network can function, you must configure WAN transport interfaces in VPN 0.

To configure a tunnel interface for the multilink interface, select **Tunnel Interface** and configure the following parameters:

**Table 225:**

| Parameter Name     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tunnel Interface   | Click <b>On</b> to create a tunnel interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Color              | Select a color for the TLOC.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Color Description  | Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.18.1<br>Enter a description associated to the TLOC color.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Control Connection | <p>By default, Control Connection is set to <b>On</b>, which establishes a control connection for the TLOC. If the router has multiple TLOCs, click <b>No</b> to have the tunnel not establish control connection for the TLOC.</p> <p><b>Note</b><br/>We recommend a minimum of 650-700 Kbps bandwidth with default 1 sec hello-interval and 12 sec hello-tolerance parameters configured to avoid any data/packet loss in connection traffic.</p> <p>For each BFD session, an additional average sized BFD packet of 175 Bytes consumes 1.4 Kbps of bandwidth.</p> <p>A sample calculation of the required bandwidth for bidirectional BFD packet flow is given below:</p> <ul style="list-style-type: none"> <li>• 650 – 700 Kbps per device for control connections.</li> <li>• 175 Bytes (or 1.4 Kbps) per BFD session on the device (request)</li> <li>• 175 Bytes (or 1.4 Kbps) per BFD session on the device (response)</li> </ul> <p>If the path MTU discovery (PMTUD) is enabled, bandwidth for send/receive BFD packets per tunnel for every 30 secs:</p> <p>A 1500 Bytes BFD request packet is sent per tunnel every 30 secs:<br/> <math>1500 \text{ Bytes} * 8 \text{ bits/1 byte} * 1 \text{ packet} / 30 \text{ secs} = 400 \text{ bps (request)}</math></p> <p>A 147 Bytes BFD packet is sent in response:<br/> <math>147 \text{ Bytes} * 8 \text{ bits/1 byte} * 1 \text{ packet} / 30 \text{ secs} = 40 \text{ bps (response)}</math></p> <p>Therefore, a device with 775 BFD sessions (for example) requires a bandwidth of:</p> $700k + (1.4k*775) + (400 *775) + (1.4k*775) + (40 *775) = \sim 3,5 \text{ MBps}$ |



| Parameter Name                                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Maximum Control Connections                    | Specify the maximum number of Cisco Catalyst SD-WAN Controllers that the WAN tunnel interface can connect to. To have the tunnel establish no control connections, set the number to 0.<br><i>Range: 0 through 8 Default: 2</i>                                                                                                                                                                                                                                                                                                                                                                                           |
| Cisco Catalyst SD-WAN Validator As STUN Server | Click <b>On</b> to enable Session Traversal Utilities for NAT (STUN) to allow the tunnel interface to discover its public IP address and port number when the router is located behind a NAT.                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Exclude Controller Group List                  | Set the Cisco Catalyst SD-WAN Controllers that the tunnel interface is not allowed to connect to. <i>Range: 0 through 100</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Cisco SD-WAN Manager Connection Preference     | Set the preference for using a tunnel interface to exchange control traffic with the Cisco SD-WAN Manager NMS. <i>Range: 0 through 8 Default: 5</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Full Port Hop</b>                           | Minimum release: Cisco IOS XE Catalyst SD-WAN Release 17.18.1a<br><br>Enable full port hopping at the TLOC level to allow devices to establish connections with controllers by switching to the next port if the current port is blocked or non-functional.<br><br>Default: Disabled                                                                                                                                                                                                                                                                                                                                      |
| Port Hop                                       | Click <b>On</b> to enable port hopping, or click <b>Off</b> to disable it. When a router is behind a NAT, port hopping rotates through a pool of preselected OMP port numbers (called base ports) to establish DTLS connections with other routers when a connection attempt is unsuccessful. The default base ports are 12346, 12366, 12386, 12406, and 12426. To modify the base ports, set a port offset value. <i>Default: Enabled</i><br><br>Starting from Cisco IOS XE Catalyst SD-WAN Release 17.18.1a, this field is deprecated. Instead use the <b>Full Port Hop</b> option. See the <b>Full Port Hop</b> field. |
| Low-Bandwidth Link                             | Select to characterize the tunnel interface as a low-bandwidth link.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Allow Service                                  | Select <b>On</b> or <b>Off</b> for each service to allow or disallow the service on the interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

To configure additional tunnel interface parameters, click **Advanced Options** and configure the following parameters:

**Table 226:**

| Parameter Name | Description                                                                                                                                                                                                                                                            |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GRE            | Use GRE encapsulation on the tunnel interface. By default, GRE is disabled.<br><br>If you select both IPsec and GRE encapsulations, two TLOCs are created for the tunnel interface that have the same IP addresses and colors, but that differ by their encapsulation. |

| Parameter Name       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IPsec                | Use IPsec encapsulation on the tunnel interface. By default, IPsec is enabled.<br><br>If you select both IPsec and GRE encapsulations, two TLOCs are created for the tunnel interface that have the same IP addresses and colors, but that differ by their encapsulation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| IPsec Preference     | Specify a preference value for directing traffic to the tunnel. A higher value is preferred over a lower value.<br><br><i>Range:</i> 0 through 4294967295. <i>Default:</i> 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| IPsec Weight         | Enter a weight to use to balance traffic across multiple TLOCs. A higher value sends more traffic to the tunnel.<br><br><i>Range:</i> 1 through 255. <i>Default:</i> 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Carrier              | Select the carrier name or private network identifier to associate with the tunnel.<br><br><i>Values:</i> carrier1, carrier2, carrier3, carrier4, carrier5, carrier6, carrier7, carrier8, default.<br><i>Default:</i> default                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Bind Loopback Tunnel | Enter the name of a physical interface to bind to a loopback interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Last-Resort Circuit  | Select to use the tunnel interface as the circuit of last resort.<br><br><b>Note</b><br>An interface configured as a circuit of last resort is expected to be down and is skipped while calculating the number of control connections, the cellular modem becomes dormant, and no traffic is sent over the circuit.<br><br>When the configurations are activated on the edge device with cellular interfaces, then all the interfaces begin the process of establishing control and BFD connections. When one or more of the primary interfaces establishes a BFD connection, the circuit of last resort shuts itself down.<br><br>Only when all the primary interfaces lose their connections to remote edges, then the circuit of last resort activates itself triggering a BFD TLOC Down alarm and a Control TLOC Down alarm on the edge device. The last resort interfaces are used as backup circuit on edge device and are activated when all other transport links BFD sessions fail. In this mode the radio interface is turned off, and no control or data connections exist over the cellular interface.<br><br><b>Note</b><br>Configuring administrative distance values on primary interface routes is not supported. |
| NAT Refresh Interval | Enter the interval between NAT refresh packets sent on a DTLS or TLS WAN transport connection. <i>Range:</i> 1 through 60 seconds. <i>Default:</i> 5 seconds                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Hello Interval       | Enter the interval between Hello packets sent on a DTLS or TLS WAN transport connection. <i>Range:</i> 100 through 10000 milliseconds. <i>Default:</i> 1000 milliseconds (1 second)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| Parameter Name  | Description                                                                                                                                                                                               |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hello Tolerance | Enter the time to wait for a Hello packet on a DTLS or TLS WAN transport connection before declaring that transport tunnel to be down.<br><i>Range:</i> 12 through 60 seconds. <i>Default:</i> 12 seconds |

### Configure the Interface as a NAT Device

To configure an interface to act as a NAT device for applications such as port forwarding, select **NAT**, click **On** and configure the following parameters:

**Table 227:**

| Parameter Name  | Description                                                                                                                                              |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| NAT             | Click <b>On</b> to have the interface act as a NAT device.                                                                                               |
| Refresh Mode    | Select how NAT mappings are refreshed, either outbound or bidirectional (outbound and inbound). <i>Default:</i> Outbound                                 |
| UDP Timeout     | Specify when NAT translations over UDP sessions time out. <i>Range:</i> 1 through 65536 minutes. <i>Default:</i> 1 minutes                               |
| TCP Timeout     | Specify when NAT translations over TCP sessions time out. <i>Range:</i> 1 through 65536 minutes. <i>Default:</i> 60 minutes (1 hour)                     |
| Block ICMP      | Select <b>On</b> to block inbound ICMP error messages. By default, a router acting as a NAT device receives these error messages. <i>Default:</i> Off    |
| Respond to Ping | Select <b>On</b> to have the router respond to ping requests to the NAT interface's IP address that are received from the public side of the connection. |

To create a port forwarding rule, click **Add New Port Forwarding Rule** and configure the following parameters. You can define up to 128 port-forwarding rules to allow requests from an external network to reach devices on the internal network.

**Table 228:**

| Parameter Name   | Description                                                                                                                                                   |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Port Start Range | Enter a port number to define the port or first port in the range of interest. <i>Range:</i> 0 through 65535                                                  |
| Port End Range   | Enter the same port number to apply port forwarding to a single port, or enter a larger number to apply it to a range of ports. <i>Range:</i> 0 through 65535 |
| Protocol         | Select the protocol to which to apply the port-forwarding rule, either TCP or UDP. To match the same ports for both TCP and UDP traffic, configure two rules. |
| VPN              | Specify the private VPN in which the internal server resides. This VPN is one of the VPN identifiers in the overlay network. <i>Range:</i> 0 through 65527    |

| Parameter Name | Description                                                                                                     |
|----------------|-----------------------------------------------------------------------------------------------------------------|
| Private IP     | Specify the IP address of the internal server to which to direct traffic that matches the port-forwarding rule. |

To save a port forwarding rule, click **Add**.

To save the feature template, click **Save**.

### Apply Access Lists

To apply a rewrite rule, access lists, and policers to a router interface, click **ACL** and configure the following parameters:

**Table 229:**

| Parameter Name     | Description                                                                                                                  |
|--------------------|------------------------------------------------------------------------------------------------------------------------------|
| Shaping rate       | Configure the aggregate traffic transmission rate on the interface to be less than line rate, in kilobits per second (kbps). |
| QoS map            | Specify the name of the QoS map to apply to packets being transmitted out the interface.                                     |
| Rewrite Rule       | Click <b>On</b> , and specify the name of the rewrite rule to apply on the interface.                                        |
| Ingress ACL – IPv4 | Click <b>On</b> , and specify the name of the access list to apply to IPv4 packets being received on the interface.          |
| Egress ACL – IPv4  | Click <b>On</b> , and specify the name of the access list to apply to IPv4 packets being transmitted on the interface.       |
| Ingress ACL – IPv6 | Click <b>On</b> , and specify the name of the access list to apply to IPv6 packets being received on the interface.          |
| Egress ACL – IPv6  | Click <b>On</b> , and specify the name of the access list to apply to IPv6 packets being transmitted on the interface.       |
| Ingress Policer    | Click <b>On</b> , and specify the name of the policer to apply to packets being received on the interface.                   |
| Egress Policer     | Click <b>On</b> , and specify the name of the policer to apply to packets being transmitted on the interface.                |

To save the feature template, click **Save**.

### Configure Other Interface Properties

To configure other interface properties, click **Advanced** and configure the following properties:

Table 230:

| Parameter Name        | Description                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bandwidth Upstream    | For transmitted traffic, set the bandwidth above which to generate notifications. <i>Range:</i> 1 through $(2^{32} / 2) - 1$ kbps                                                                                                                                                                                                                                                                                                      |
| Bandwidth Downstream  | For received traffic, set the bandwidth above which to generate notifications. <i>Range:</i> 1 through $(2^{32} / 2) - 1$ kbps                                                                                                                                                                                                                                                                                                         |
| IP MTU                | Specify the maximum MTU size of packets on the interface. <i>Range:</i> 576 through 1804. <i>Default:</i> 1500 bytes                                                                                                                                                                                                                                                                                                                   |
| TCP MSS               | Specify the maximum segment size (MSS) of TCP SYN packets passing through the router. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. <i>Range:</i> 552 to 1460 bytes. <i>Default:</i> None                                                                                                                                                           |
| TLOC Extension        | Enter the name of the physical interface on the same router that connects to the WAN transport circuit. This configuration then binds this service-side interface to the WAN transport. A second router at the same site that itself has no direct connection to the WAN (generally because the site has only a single WAN connection) and that connects to this service-side interface is then provided with a connection to the WAN. |
| Tracker               | Enter the name of a tracker to track the status of transport interfaces that connect to the internet.                                                                                                                                                                                                                                                                                                                                  |
| IP Directed-Broadcast | Enables translation of a directed broadcast to physical broadcasts. An IP directed broadcast is an IP packet whose destination address is a valid broadcast address for some IP subnet but which originates from a node that is not itself part of that destination subnet.                                                                                                                                                            |

To save the feature template, click **Save**.

### Release Information

Introduced in Cisco SD-WAN Manager NMS in Release 18.4.1.

## Cisco VPN Interface GRE

When a service, such as a firewall, is available on a device that supports only GRE tunnels, you can configure a GRE tunnel on the device to connect to the remote device by configuring a logical GRE interface. You then advertise that the service is available via a GRE tunnel, and you can create data policies to direct the appropriate traffic to the tunnel. GRE interfaces come up as soon as they are configured, and they stay up as long as the physical tunnel interface is up.

To configure GRE interfaces using Cisco SD-WAN Manager templates:

1. Create a Cisco VPN Interface GRE feature template to configure a GRE interface.
2. Create a Cisco VPN feature template to advertise a service that is reachable via a GRE tunnel, to configure GRE-specific static routes, and to configure other VPN parameters.

3. Create a data policy on the Cisco Catalyst SD-WAN Controller that applies to the service VPN, including a **set-service service-name local** command.

### Navigate to the Template Screen and Name the Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and click **Create Template**.



**Note** In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. From the **Create Template** drop-down list, select **From Feature Template**.
4. From the **Device Model** drop-down list, select the type of device for which you are creating the template.
5. To create a template for VPN 0 or VPN 512:
  - a. Click **Transport & Management VPN** or scroll to the **Transport & Management VPN** section.
  - b. Under **Additional VPN 0 Templates**, click **VPN Interface GRE**.
  - c. From the **VPN Interface GRE** drop-down list, click **Create Template**. The VPN Interface GRE template form is displayed.  
  
This form contains fields for naming the template, and fields for defining the VPN Interface GRE parameters.
6. In **Template Name**, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.
7. In **Template Description**, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default or to enter a value, click the scope drop-down to the left of the parameter field and select the parameter scope.

### Configuring a Basic GRE Interface

To configure a basic GRE interface, click **Basic Configuration** and then configure the following parameters. Parameters marked with an asterisk are required to configure a GRE interface.

**Table 231:**

| Parameter Name  | Description                                                                                                     |
|-----------------|-----------------------------------------------------------------------------------------------------------------|
| Shutdown*       | Click <b>Off</b> to enable the interface.                                                                       |
| Interface Name* | Enter the name of the GRE interface, in the format <b>gre number</b> . <i>number</i> can be from 1 through 255. |
| Description     | Enter a description of the GRE interface.                                                                       |

| Parameter Name              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source*                     | <p>Enter the source of the GRE interface:</p> <ul style="list-style-type: none"> <li>• GRE Source IP Address—Enter the source IP address of the GRE tunnel interface. This address is on the local router. This address is on the local router. GRE keepalives can not be configured when source configured as IP address.</li> <li>• Tunnel Source Interface—Enter the physical interface that is the source of the GRE tunnel. GRE keepalives can not be configured when source configured as loopback interface.</li> <li>• If you selected the Source as <b>Interface</b>, enter the name of the source interface. If you enter a loopback interface, an additional field <b>Tunnel Route-via Interface</b> displays where you enter the egress interface name.</li> </ul> |
| Destination*                | Enter the destination IP address of the GRE tunnel interface. This address is on a remote device. If this tunnel connects to a Secure Internet Gateway (SIG), specify the URL for the SIG.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| GRE Destination IP Address* | Enter the destination IP address of the GRE tunnel interface. This address is on a remote device                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| IPv4 Address                | Enter an IPv4 address for the GRE tunnel.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| IP MTU                      | Specify the maximum MTU size of packets on the interface. <i>Range: 576 through 1804 Default: 1500 bytes</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Clear-Dont-Fragment         | Click <b>On</b> to clear the Don't Fragment bit in the IPv4 packet header for packets being transmitted out the interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| TCP MSS                     | Specify the maximum segment size (MSS) of TCP SYN packets passing through the Cisco vEdge device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. <i>Range: 552 to 1460 bytes Default: None</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

To save the feature template, click **Save**.

### Configure Interface Access Lists

To configure access lists on a GRE interface, click **ACL** and configure the following parameters:

**Table 232:**

| Parameter Name     | Description                                                                                                            |
|--------------------|------------------------------------------------------------------------------------------------------------------------|
| Rewrite Rule       | Click <b>On</b> , and specify the name of the rewrite rule to apply on the interface.                                  |
| Ingress ACL – IPv4 | Click <b>On</b> , and specify the name of the access list to apply to IPv4 packets being received on the interface.    |
| Egress ACL – IPv4  | Click <b>On</b> , and specify the name of the access list to apply to IPv4 packets being transmitted on the interface. |

### Configure Tracker Interface

To configure a tracker interface to track the status of a GRE interface, select **Advanced** and configure the following parameter:

*Table 233:*

| Parameter Name | Description                                                                                     |
|----------------|-------------------------------------------------------------------------------------------------|
| Tracker        | Enter the name of a tracker to track the status of GRE interfaces that connect to the Internet. |

## Configure VPN Interface GRE on Transport Side Using a Configuration Group

### Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

### Procedure

**Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

**Step 2** Create and configure GRE.

- a. Configure GRE parameters.

*Table 234: Basic Configuration*

| Field                           | Description                                                                                                                                                                                                                                                     |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Interface Name (1..255)*</b> | Enter the name of the GRE interface.<br>Range: 1 through 255.                                                                                                                                                                                                   |
| <b>Interface Description</b>    | Enter a description of the GRE interface.                                                                                                                                                                                                                       |
| <b>Tunnel Mode</b>              | Choose from one of the following GRE tunnel modes: <ul style="list-style-type: none"> <li>• <b>ipv4 underlay</b>: GRE tunnel with IPv4 underlay. IPv4 underlay is the default value.</li> <li>• <b>ipv6 underlay</b>: GRE tunnel with IPv6 underlay.</li> </ul> |
| <b>Multiplexing</b>             | Choose <b>Yes</b> to enable multiplexing, in case of a tunnel in the transport VPN.<br>Default: No                                                                                                                                                              |
| <b>Preshared Key for IKE</b>    | Enter the preshared key (PSK) for authentication.                                                                                                                                                                                                               |

- b. Configure Tunnel fields.



Table 235: Tunnel

| Field       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source      | <p>Enter the source of the GRE interface:</p> <ul style="list-style-type: none"> <li>• <b>IP Address:</b> Enter the source IP address of the GRE tunnel interface. Based on the option you selected in the <b>Tunnel Mode</b> drop-down list, enter an IPv4 or an IPv6 address. This address is on the local router.</li> <li>• <b>Interface:</b> Enter the egress interface name for the GRE tunnel.</li> <li>• <b>Tunnel Route Via*:</b> Specify the tunnel route details to steer the GRE tunnel traffic through.</li> </ul> <p><b>Note</b><br/>If the Tunnel Source Interface type is a loopback interface, enter the interface for traffic to be routed to. You cannot use the tunnel route via option to configure IPsec tunnels on a cellular interface because cellular interfaces do not include a next hop IP address for the default route.</p> |
| Destination | <p>Enter the source of the GRE interface:</p> <ul style="list-style-type: none"> <li>• <b>GRE Destination IP Address*:</b> Enter the destination IP address of the GRE tunnel interface. This address is on a remote device.</li> <li>• <b>IP Address:</b> Based on the option you selected in the <b>Tunnel Mode</b> drop-down list, enter an IPv4 or an IPv6 address for the GRE tunnel. <ul style="list-style-type: none"> <li>• <b>Mask*:</b> Enter the subnet mask.</li> </ul> </li> <li>• <b>IPv6 Address:</b> Enter the destination IPv6 or address for the GRE tunnel.</li> </ul>                                                                                                                                                                                                                                                                  |

## c. Configure IKE fields.

Table 236: IKE

| Field                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IKE Version            | <p>Enter 1 to choose IKEv1.</p> <p>Enter 2 to choose IKEv2.</p> <p>Default: IKEv1</p>                                                                                                                                                                                                                                                                                                                                                                                              |
| IKE Integrity Protocol | <p>Choose one of the following modes for the exchange of keying information and setting up IKE security associations:</p> <ul style="list-style-type: none"> <li>• <b>Main:</b> Establishes an IKE SA session before starting IPsec negotiations.</li> <li>• <b>Aggressive:</b> Negotiation is quicker, and the initiator and responder ID pass in the clear. Aggressive mode does not provide identity protection for communicating parties.</li> </ul> <p>Default: Main mode</p> |

| Field                              | Description                                                                                                                                                                                                           |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IKE Rekey Interval</b>          | Specify the interval for refreshing IKE keys.<br>Range: 3600 through 1209600 seconds (1 hour through 14 days)<br>Default: 14400 seconds (4 hours)                                                                     |
| <b>IKE Cipher Suite</b>            | Specify the type of authentication and encryption to use during IKE key exchange.<br>Values: aes128-cbc-sha1, aes128-cbc-sha2, aes256-cbc-sha1, aes256-cbc-sha2<br>Default: aes256-cbc-sha1                           |
| <b>IKE Diffie-Hellman Group</b>    | Specify the Diffie-Hellman group to use in IKE key exchanges.<br>Values: 2, 14, 15, 16, 19, 20, 21, 24<br>Default: 16                                                                                                 |
| <b>IKE ID for Local End Point</b>  | If the remote IKE peer requires a local endpoint identifier, specify it.<br>Range: 1 through 64 characters<br>Default: Source IP address of the tunnel                                                                |
| <b>IKE ID for Remote End Point</b> | If the remote IKE peer requires a remote end point identifier, specify it.<br>Range: 1 through 64 characters<br>Default: Destination IP address of the tunnel<br>There is no default option if you have chosen IKEv2. |

## d. Configure IPSEC fields.

Table 237: IPSEC

| Field                       | Description                                                                                                                                    |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IPsec Rekey Interval</b> | Specify the interval for refreshing IKE keys.<br>Range: 3600 through 1209600 seconds (1 hour through 14 days)<br>Default: 3600 seconds         |
| <b>IPsec Replay Window</b>  | Specify the replay window size for the IPsec tunnel.<br>Values: 64, 128, 256, 512, 1024, 2048, 4096, 8192 bytes<br>Default: 512 bytes          |
| <b>IPsec Cipher Suite</b>   | Specify the authentication and encryption to use on the IPsec tunnel.<br>Values: aes256-cbc-sha1, aes256-gcm, null-sha1<br>Default: aes256-gcm |

| Field                          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Perfect Forward Secrecy</b> | Specify the PFS settings to use on the IPsec tunnel by choosing one of the following values: <ul style="list-style-type: none"> <li>• <b>group-2</b>: Use the 1024-bit Diffie-Hellman prime modulus group</li> <li>• <b>group-14</b>: Use the 2048-bit Diffie-Hellman prime modulus group</li> <li>• <b>group-15</b>: Use the 3072-bit Diffie-Hellman prime modulus group</li> <li>• <b>group-16</b>: Use the 4096-bit Diffie-Hellman prime modulus group</li> <li>• <b>none</b>: Disable PFS</li> </ul> Default: <b>group-16</b> |
| <b>DPD Interval</b>            | Specify the interval for IKE to send Hello packets on the connection.<br>Range: 10 through 3600 seconds (1 hour)<br>Default: 10 seconds                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>DPD Retries</b>             | Specify how many unacknowledged packets to accept before declaring an IKE peer to be dead and then removing the tunnel to the peer.<br>Range: 2 through 60<br>Default: 3                                                                                                                                                                                                                                                                                                                                                          |
| <b>Application</b>             | Choose an application from the drop-down list: <ul style="list-style-type: none"> <li>• None</li> <li>• Sig</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                            |

e. Configure advanced fields.

**Table 238: Advanced**

| Field           | Description                                                                                                                                                                                                                                                                                                                                                  |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Shutdown</b> | Click <b>Off</b> to enable the interface.                                                                                                                                                                                                                                                                                                                    |
| <b>IP MTU</b>   | Based on your choice in the <b>Tunnel Mode</b> option, specify the maximum MTU size of the IPv6 packets on the interface.<br>Range: 576 through 9216<br>Default: 1500 bytes                                                                                                                                                                                  |
| <b>TCP MSS</b>  | Based on your choice in the <b>Tunnel Mode</b> option, specify the maximum segment size (MSS) of TCP SYN packets passing through the Cisco IOS XE Catalyst SD-WAN device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented.<br>Range: 552 through 1460 bytes<br>Default: None |

| Field                      | Description                                                                                                                |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Clear-Dont-Fragment</b> | Click <b>On</b> to clear the Don't Fragment bit in the IPv4 packet header for packets being transmitted out the interface. |
| <b>Tunnel Protection</b>   | Choose <b>Yes</b> to enable tunnel protection.<br>Default: No                                                              |

### What to do next

Also see [Deploy a configuration group](#).

## Configure GRE on Service Side Using a Configuration Group

### Before you begin

On the **Configuration** > **Configuration Groups** page, choose **SD-WAN** as the solution type.

### Procedure

**Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration** > **Configuration Groups**.

**Step 2** Create and configure GRE in Service Profile.

- a. Configure Basic Configuration fields.

*Table 239: Basic Configuration*

| Field                           | Description                                                                                                                                                                                                                                                     |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Interface Name (1..255)*</b> | Enter the name of the GRE interface, in the format <b>grenumber</b> . The value for <b>number</b> can be from 1 through 255.                                                                                                                                    |
| <b>Interface Description</b>    | Enter a description of the GRE interface.                                                                                                                                                                                                                       |
| <b>Tunnel Mode</b>              | Choose from one of the following GRE tunnel modes: <ul style="list-style-type: none"> <li>• <b>ipv4 underlay</b>: GRE tunnel with IPv4 underlay. IPv4 underlay is the default value.</li> <li>• <b>ipv6 underlay</b>: GRE tunnel with IPv6 underlay.</li> </ul> |
| <b>Preshared Key for IKE</b>    | Enter the preshared key (PSK) for authentication.                                                                                                                                                                                                               |

- b. Configure Tunnel Fields.

Table 240: Tunnel

| Field       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source      | <p>Enter the source of the GRE interface:</p> <ul style="list-style-type: none"> <li>• <b>IP Address:</b> Enter the source IP address of the GRE tunnel interface. Based on the option you selected in the <b>Tunnel Mode</b> drop-down list, enter an IPv4 or an IPv6 address. This address is on the local router.</li> <li>• <b>Interface:</b> Enter the egress interface name for the GRE tunnel.</li> <li>• <b>Tunnel Route Via*:</b> Specify the tunnel route details to steer the GRE tunnel traffic through.</li> </ul> <p><b>Note</b><br/>If the Tunnel Source Interface type is a loopback interface, enter the interface for traffic to be routed to. You cannot use the tunnel route via option to configure IPsec tunnels on a cellular interface because cellular interfaces do not include a next hop IP address for the default route.</p> |
| Destination | <p>Enter the source of the GRE interface:</p> <ul style="list-style-type: none"> <li>• <b>GRE Destination IP Address*:</b> Enter the destination IP address of the GRE tunnel interface. This address is on a remote device.</li> <li>• <b>IP Address:</b> Based on the option you selected in the <b>Tunnel Mode</b> drop-down list, enter an IPv4 or an IPv6 address for the GRE tunnel. <ul style="list-style-type: none"> <li>• <b>Mask*:</b> Enter the subnet mask.</li> </ul> </li> <li>• <b>IPv6 Address:</b> Enter the destination IPv6 or address for the GRE tunnel.</li> </ul>                                                                                                                                                                                                                                                                  |

## c. Configure IKE fields.

Table 241: IKE

| Field                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IKE Version            | <p>Enter 1 to choose IKEv1.</p> <p>Enter 2 to choose IKEv2.</p> <p>Default: IKEv1</p>                                                                                                                                                                                                                                                                                                                                                                                              |
| IKE Integrity Protocol | <p>Choose one of the following modes for the exchange of keying information and setting up IKE security associations:</p> <ul style="list-style-type: none"> <li>• <b>Main:</b> Establishes an IKE SA session before starting IPsec negotiations.</li> <li>• <b>Aggressive:</b> Negotiation is quicker, and the initiator and responder ID pass in the clear. Aggressive mode does not provide identity protection for communicating parties.</li> </ul> <p>Default: Main mode</p> |

| Field                              | Description                                                                                                                                                                                                           |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IKE Rekey Interval</b>          | Specify the interval for refreshing IKE keys.<br>Range: 3600 through 1209600 seconds (1 hour through 14 days)<br>Default: 14400 seconds (4 hours)                                                                     |
| <b>IKE Cipher Suite</b>            | Specify the type of authentication and encryption to use during IKE key exchange.<br>Values: aes128-cbc-sha1, aes128-cbc-sha2, aes256-cbc-sha1, aes256-cbc-sha2<br>Default: aes256-cbc-sha1                           |
| <b>IKE Diffie-Hellman Group</b>    | Specify the Diffie-Hellman group to use in IKE key exchanges.<br>Values: 2, 14, 15, 16, 19, 20, 21, 24<br>Default: 16                                                                                                 |
| <b>IKE ID for Local End Point</b>  | If the remote IKE peer requires a local endpoint identifier, specify it.<br>Range: 1 through 64 characters<br>Default: Source IP address of the tunnel                                                                |
| <b>IKE ID for Remote End Point</b> | If the remote IKE peer requires a remote end point identifier, specify it.<br>Range: 1 through 64 characters<br>Default: Destination IP address of the tunnel<br>There is no default option if you have chosen IKEv2. |

## d. Configure IPSEC fields.

Table 242: IPSEC

| Field                                 | Description                                                                                                                                    |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IPsec Rekey Interval (Seconds)</b> | Specify the interval for refreshing IKE keys.<br>Range: 3600 through 1209600 seconds (1 hour through 14 days)<br>Default: 3600 seconds         |
| <b>IPsec Replay Window</b>            | Specify the replay window size for the IPsec tunnel.<br>Values: 64, 128, 256, 512, 1024, 2048, 4096, 8192 bytes<br>Default: 512 bytes          |
| <b>IPsec Cipher Suite</b>             | Specify the authentication and encryption to use on the IPsec tunnel.<br>Values: aes256-cbc-sha1, aes256-gcm, null-sha1<br>Default: aes256-gcm |

| Field                          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Perfect Forward Secrecy</b> | <p>Specify the PFS settings to use on the IPsec tunnel by choosing one of the following values:</p> <ul style="list-style-type: none"> <li>• <b>group-2</b>: Use the 1024-bit Diffie-Hellman prime modulus group</li> <li>• <b>group-14</b>: Use the 2048-bit Diffie-Hellman prime modulus group</li> <li>• <b>group-15</b>: Use the 3072-bit Diffie-Hellman prime modulus group</li> <li>• <b>group-16</b>: Use the 4096-bit Diffie-Hellman prime modulus group</li> <li>• <b>none</b>: Disable PFS</li> </ul> <p>Default: <b>group-16</b></p> |
| <b>DPD Interval</b>            | <p>Specify the interval for IKE to send Hello packets on the connection.</p> <p>Range: 10 through 3600 seconds (1 hour)</p> <p>Default: 10 seconds</p>                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>DPD Retries</b>             | <p>Specify how many unacknowledged packets to accept before declaring an IKE peer to be dead and then removing the tunnel to the peer.</p> <p>Range: 2 through 60</p> <p>Default: 3</p>                                                                                                                                                                                                                                                                                                                                                         |
| <b>Application</b>             | <p>Choose an application from the drop-down list:</p> <ul style="list-style-type: none"> <li>• None</li> <li>• Sig</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                   |

e. Configure advanced fields.

**Table 243: Advanced**

| Field           | Description                                                                                                                                                                                                                                                                                                  |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Shutdown</b> | Click <b>Off</b> to enable the interface.                                                                                                                                                                                                                                                                    |
| <b>IP MTU</b>   | <p>Based on your choice in the <b>Tunnel Mode</b> option, specify the maximum MTU size of the IPv4 or IPv6 packets on the interface.</p> <p>Range: 576 through 9216</p> <p>Default: 1500 bytes</p>                                                                                                           |
| <b>TCP MSS</b>  | <p>Specify the maximum segment size (MSS) of the IPv4 TCP SYN packets passing through the Cisco vEdge device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented.</p> <p>Range: 552 through 1460 bytes</p> <p>Default: None</p> |

| Field                      | Description                                                                                                                                                                                                                                                                                           |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IPv6 TCP MSS</b>        | Specify the maximum segment size (MSS) of the IPv6 TCP SYN packets passing through the Cisco vEdge device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented.<br><br>Range: 552 through 1460 bytes<br><br>Default: None |
| <b>Clear-Dont-Fragment</b> | Click <b>On</b> to clear the Don't Fragment bit in the IPv4 packet header for packets being transmitted out the interface.                                                                                                                                                                            |
| <b>Tunnel Protection</b>   | Choose <b>Yes</b> to enable tunnel protection.<br><br>Default: No                                                                                                                                                                                                                                     |

#### What to do next

Also see [Deploy a configuration group](#).

## GRE-in-UDP

*Table 244: Feature History*

| Feature Name | Release Information                                                                                           | Description                                            |
|--------------|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| GRE-in-UDP   | Cisco IOS XE Catalyst SD-WAN Release 17.11.1a<br><br>Cisco Catalyst SD-WAN Control Components Release 20.11.1 | You can configure GRE encapsulation for UDP transport. |

## Information About GRE-in-UDP

Cisco Catalyst SD-WAN supports generic routing encapsulation (GRE) with UDP for IPv4 and IPv6 traffic.

With a GRE-in-UDP tunnel, a router encapsulates GRE packets, containing information such as the source and destination ports, within a UDP header. The router sends the UDP packet through the tunnel. The destination device de-encapsulates the UDP packet.

## Supported Devices for GRE-in-UDP

Cisco IOS XE Catalyst SD-WAN devices.



## Prerequisites for GRE-in-UDP

Configure GRE encapsulation.

## Restrictions for GRE-in-UDP

Any restrictions that apply to GRE encapsulation apply to GRE-in-UDP.

## Configure GRE-in-UDP Using a CLI Template

For more information about using CLI templates, see [CLI Add-On Feature Templates](#) and [CLI Templates](#).



---

**Note** By default, CLI templates execute commands in global config mode.

---

You can configure GRE-in-UDP tunnel only through a CLI template.

1. For the desired interface, enter interface configuration mode.

```
sdwan
interface interface
```

2. Enter tunnel interface mode.

```
tunnel-interface
```

3. Configure GRE encapsulation.

```
encapsulation gre
```

4. Configure GRE-in-UDP as the encapsulation mode.

```
gre-in-udp
```

### Example

Here is a complete example of configuring GRE-in-UDP.

```
interface GigabitEthernet1
 tunnel-interface
 encapsulation gre
 color lte
 gre-in-udp
 no allow-service bgp
 allow-service dhcp
 allow-service dns
 allow-service icmp
 no allow-service sshd
 no allow-service netconf
 no allow-service ntp
 no allow-service ospf
 no allow-service stun
 allow-service https
 no allow-service snmp
 no allow-service bfd
exit
```

# VPN Interface IPsec

Use the VPN Interface IPsec feature template to configure IPsec tunnels on Cisco IOS XE service VPNs that are being used for Internet Key Exchange (IKE) sessions. You can configure IPsec on tunnels for VPN 1 through 65527, except for 512.

Cisco Cisco IOS XE Catalyst SD-WAN devices use VRFs in place of VPNs. However, the following steps still apply to configure Cisco IOS XE Catalyst SD-WAN devices through Cisco SD-WAN Manager. In Cisco SD-WAN Manager, the system automatically maps the VPN configurations to VRF configurations.

In controller mode, only Route based IPsec tunnels are supported.

## Create VPN IPsec Interface Template

### Procedure

**Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.

**Step 2** Click **Feature Templates**.

#### Note

In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

**Step 3** Click **Add Template**.

**Step 4** Choose a Cisco IOS XE Catalyst SD-WAN device from the list.

**Step 5** From the VPN section, click **VPN Interface IPsec**. The Cisco VPN Interface IPsec template displays.

**Step 6** In **Template Name**, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.

**Step 7** In **Template Description**, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

## Basic Configuration

To configure a basic IPsec tunnel interface select **Basic Configuration** and configure the following parameters:

| Parameter Name  | Options/Format                              | Description                                                                        |
|-----------------|---------------------------------------------|------------------------------------------------------------------------------------|
| Shutdown*       | <b>Yes / No</b>                             | Click <b>No</b> to enable the interface; click <b>Yes</b> to disable.              |
| Interface Name* | <b>ipsec number</b> (1...255)               | Enter the name of the IPsec interface. <i>Number</i> can be from 1 through 255.    |
| Description     | Enter a description of the IPsec interface. |                                                                                    |
| IPv4 Address*   | <i>ipv4-prefix/length</i>                   | Enter the IPv4 address of the IPsec interface. The address must have a /30 subnet. |

| Parameter Name | Options/Format                                                                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source *       | Set the source of the IPsec tunnel that is being used for IKE key exchange:      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                | IP Address                                                                       | Click and enter the IPv4 address that is the source tunnel interface. This address must be configured in <b>VPN 0</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                | Interface                                                                        | <p>Click and enter the name of the physical interface that is the source of the IPsec tunnel. This interface must be configured in <b>VPN 0</b>.</p> <ul style="list-style-type: none"> <li>If you selected the Source as <b>Interface</b>, enter the name of the source interface. If you enter a loopback interface, an additional field <b>Tunnel Route-via Interface</b> displays where you enter the egress interface name.</li> </ul> <p><b>Note</b><br/>You cannot use the tunnel route via option to configure IPSec tunnels on a cellular interface because cellular interfaces do not include a next hop IP address for the default route.</p> |
| Destination*   | Set the destination of the IPsec tunnel that is being used for IKE key exchange. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                | IPsec Destination IP Address                                                     | Enter an IPv4 address that points to the destination.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                | TCP MSS                                                                          | <p>Specify the maximum segment size (MSS) of TCP SYN packets passing through the router. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented.</p> <p><i>Range:</i> 552 to 1960 bytes</p> <p><i>Default:</i> None</p>                                                                                                                                                                                                                                                                                                                                                         |
|                | IP MTU                                                                           | <p>Specify the maximum transmission unit (MTU) size of packets on the interface.</p> <p><i>Range:</i> 576 through 2000</p> <p><i>Default:</i> 1500 bytes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

### CLI Equivalent

```
crypto
 interface tunnel ifnum
 no shutdown
 vrf forwarding vrf_id
 ip address ip_address[mask]
 tunnel source wanif_ip
 tunnel mode {ipsec ipv4 | gre ip}
 tunnel destination gateway_ip
 tunnel protection ipsec profile ipsec_profile_name
```

## Configure Dead-Peer Detection

To configure Internet key exchange (IKE) dead-peer detection (DPD) to determine whether the connection to an IKE peer is functional and reachable, click DPD and configure the following parameters:

| Parameter Name | Description                                                                                                                                                                          |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DPD Interval   | Specify the interval for IKE to send Hello packets on the connection.<br><br>Range: 10 through 3600 seconds<br><br>Default: Disabled                                                 |
| DPD Retries    | Specify how many unacknowledged packets to accept before declaring an IKE peer to be dead and then tearing down the tunnel to the peer.<br><br>Range: 2 through 60<br><br>Default: 3 |

To save the feature template, click **Save**.

### CLI Equivalent

```
crypto
 ikev2
 profile ikev2_profile_name
 dpd 10-3600 2-60 {on-demand | periodic}
```

## Configure IKE

Table 245: Feature History

| Feature Name                     | Release Information                          | Description                                                                 |
|----------------------------------|----------------------------------------------|-----------------------------------------------------------------------------|
| SHA256 Support for IPSec Tunnels | Cisco IOS XE Catalyst SD-WAN Release 17.2.1r | This feature adds support for HMAC_SHA256 algorithms for enhanced security. |

To configure IKE, click **IKE** and configure the following parameters:



**Note** When you create an IPsec tunnel on a Cisco IOS XE Catalyst SD-WAN device, IKE Version 1 is enabled by default on the tunnel interface.

### IKE Version 1 and IKE Version 2

To configure the IPsec tunnel that carries IKEv1 and IKEv2 traffic, click **IPSEC** and configure the following parameters:

| Parameter Name              | Options                                    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IKE Version</b>          | <b>1</b> IKEv1<br><b>2</b> IKEv2           | Enter <b>1</b> to choose IKEv1.<br>Enter <b>2</b> to choose IKEv2.<br><i>Default:</i> IKEv1<br><b>Note</b><br>In IKEv2 Preshared Keys (PSK), the '\ ' character is not supported and should not be used.                                                                                                                                                                                                                                                                                                       |
| <b>IKE Mode</b>             | <b>Aggressive mode</b><br><b>Main mode</b> | For IKEv1 only, specify one of the following modes: <ul style="list-style-type: none"> <li>• Aggressive mode - Negotiation is quicker, and the initiator and responder ID pass in the clear.</li> <li>• Establishes an IKE SA session before starting IPsec negotiations.</li> </ul> <b>Note</b><br>For IKEv2, there is no mode.<br><b>Note</b><br>IKE aggressive mode with pre-shared keys should be avoided where possible. Otherwise a strong pre-shared key should be chosen.<br><i>Default:</i> Main mode |
| <b>IPsec Rekey Interval</b> | 3600 - 1209600 seconds                     | Specify the interval for refreshing IKE keys.<br><i>Range:</i> 1 hour through 14 days<br><i>Default:</i> 14400 seconds (4 hours)                                                                                                                                                                                                                                                                                                                                                                               |

| Parameter Name                  | Options                                                                                                                                                                                                                                                                             | Description                                                                                                                                                                                                                                                                    |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IKE Cipher Suite</b>         | <ul style="list-style-type: none"> <li>• AES 256 CBC SHA 256</li> <li>• AES 256 CBC SHA 384</li> <li>• AES 256 CBC SHA 512</li> <li>• AES 256 CBC SHA 1</li> <li>• AES 256 GCM</li> <li>• Nul SHA 256</li> <li>• Nul SHA 384</li> <li>• Nul SHA 512</li> <li>• Nul SHA 1</li> </ul> | Specify the type of authentication and encryption to use during IKE key exchange.<br><br><i>Default:</i> AES 256 CBC SHA 1                                                                                                                                                     |
| <b>IKE Diffie-Hellman Group</b> | 2<br>14<br>15<br>16                                                                                                                                                                                                                                                                 | Specify the Diffie-Hellman group to use in IKE key exchange, whether IKEv1 or IKEv2. <ul style="list-style-type: none"> <li>• 1024-bit modulus</li> <li>• 2048-bit modulus</li> <li>• 3072-bit modulus</li> <li>• 4096-bit modulus</li> </ul> <i>Default:</i> 4096-bit modulus |
| <b>IKE Authentication</b>       | Configure IKE authentication.                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                |
|                                 | <b>Preshared Key</b>                                                                                                                                                                                                                                                                | Enter the password to use with the preshared key.                                                                                                                                                                                                                              |
|                                 | <b>IKE ID for Local End Point</b>                                                                                                                                                                                                                                                   | If the remote IKE peer requires a local end point identifier, specify it.<br><br><i>Range:</i> 1 through 64 characters<br><i>Default:</i> Tunnel's source IP address                                                                                                           |
|                                 | <b>IKE ID for Remote End Point</b>                                                                                                                                                                                                                                                  | If the remote IKE peer requires a remote end point identifier, specify it.<br><br><i>Range:</i> 1 through 64 characters<br><i>Default:</i> Tunnel's destination IP address                                                                                                     |



**Note** When you are pushing authentication from Cisco SD-WAN Manager, use the authentication string configured for the source and destination stations in double quotes as special characters are not supported. The string can be up to eight characters long.

To save the feature template, click **Save**.

### Change the IKE Version from IKEv1 to IKEv2

To change the IKE version, do the following:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**, and then click **Add Template**.



**Note** In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is called **Feature**.

3. Choose the device for which you are creating the template.
4. Click **Basic Configuration**.
5. Use the **shutdown** parameter with the **yes** option (**yes shutdown**) to shut down the tunnel.
6. Remove the ISAKMP profile from the IPsec profile.
7. Attach the IKEv2 profile with the IPsec profile.



**Note** Perform this step if you already have an IKEv2 profile. Otherwise, create an IKEv2 profile first.

8. Use the **shutdown** parameter with the **no** option (**no shutdown**) to start up the tunnel.



**Note** You must issue the **shutdown** operations in two separate operations.



**Note** There is no single CLI for changing the IKE version. You need to follow the sequence of steps listed in the Change the IKE Version from IKEv1 to IKEv2 section.

### CLI Equivalents for IKEv1

#### ISAKMP CLI Configuration for IKEv1

```
crypto
 isakmp
 keepalive 60-86400 2-60 {on-demand | periodic}
 policy policy_num
 encryption {AES128-CBC-SHA1 | AES256-CBC-SHA1}
```

```

 hash {sha384 | sha256 | sha}
 authentication pre-share
 group {2 | 14 | 16 | 19 | 20 | 21}
 lifetime 60-86400
 profile ikev1_profile_name
 match identity address ip_address [mask]
 keyring keyring_name

```

### IPsec CLI Configuration for IKEv1

```

profile ipsec_profile_name
 set transform-set transform_set_name
 set isakmp-profile ikev1_profile_name
 set security-association
 lifetime {kilobytes disable | seconds 120-2592000}
 replay {disable | window-size [64 | 128 | 256 | 512 | 1024]}
 set pfs group {14 | 16 | 19 | 20 | 21}
 keyring keyring_name
 pre-shared-key address ip_address [mask] key key_string
 ipsec transform-set transform_set_name {esp-gcm 256 | esp-aes 256 [esp-sha384-hmac |
esp-sha256-hmac] mode tunnel

```

### Summary Steps

1. enable
2. configure terminal
3. crypto isakmp policy *priority*
4. encryption {des | 3des | aes | aes 192 | aes 256 }
5. hash {sha | sha256 | sha384 | md5 }
6. authentication {rsa-sig | rsa-encr | pre-share }
7. group {1 | 2 | 5 | 14 | 15 | 16 | 19 | 20 | 24 }
8. lifetime *seconds*
9. exit
10. exit

### CLI Equivalent for IKE2

```

crypto
 ikev2
 proposal proposal_name
 encryption {3des | aes-cbc-128 | aes-cbc-192 | aes-cbc-256 | des}
 integrity {sha256 | sha384 | sha512}
 group {2 | 14 | 15 | 16}
 keyring idev2_keyring_name
 peer peer_name
 address tunnel_dest_ip [mask]
 pre-shared-key key_string
 profile ikev2_profile_name
 match identity remote address ip_address
 authentication {remote | local} pre-share
 keyring local ikev2_keyring_name
 lifetime 120-86400

```



## Configure IPsec Tunnel Parameters

To configure the IPsec tunnel that carries Internet Key Exchange (IKE) traffic, click IPsec and configure the following parameters:

| Parameter Name                 | Options                                                                                         | Description                                                                                                                                                                                                                                                    |
|--------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IPsec Rekey Interval</b>    | 3600 - 1209600 seconds                                                                          | Specify the interval for refreshing IKE keys.<br>Range: 1 hour through 14 days<br>Default: 3600 seconds                                                                                                                                                        |
| <b>IKE Replay Window</b>       | 64, 128, 256, 512, 1024, 2048, 4096, 8192                                                       | Specify the replay window size for the IPsec tunnel.<br>Default: 512                                                                                                                                                                                           |
| <b>IPsec Cipher Suite</b>      | aes256-cbc-sha1<br>aes256-gcm<br>null-sha1                                                      | Specify the authentication and encryption to use on the IPsec tunnel<br>Default: aes256-gcm                                                                                                                                                                    |
| <b>Perfect Forward Secrecy</b> | 2 1024-bit modulus<br>14 2048-bit modulus<br>15 3072-bit modulus<br>16 4096-bit modulus<br>none | Specify the PFS settings to use on the IPsec tunnel.<br>Choose one of the following Diffie-Hellman prime modulus groups:<br>1024-bit – group-2<br>2048-bit – group-14<br>3072-bit – group-15<br>4096-bit – group-16<br>none –disable PFS.<br>Default: group-16 |



**Note** Starting from Cisco IOS XE Catalyst SD-WAN Release 17.11.1a, as part of the security hardening, the weaker ciphers are deprecated. As part of this change, the option to configure Diffie-Hellman (DH) groups 1, 2, and 5 is no longer supported. DH groups are used in IKE to establish session keys and are also available in IPsec as support for perfect forward secrecy.

To save the feature template, click **Save**.

### CLI Equivalent

```
crypto
 ipsec
 profile ipsec_profile_name
 set ikev2-profile ikev2_profile_name
 set security-association
 lifetime {seconds 120-2592000 | kilobytes disable}
 replay {disable | window-size {64 | 128 | 256 | 512 | 1024 | 4096 | 8192}}
 set pfs group {2 | 14 | 15 | 16 | none}
 set transform-set transform_set_name
```

## Configure IPSEC on Transport Side Using a Configuration Group

**Note**

If...

- you are running SD-WAN Manager releases from SD-WAN Manager 20.15.1 to SD-WAN Manager 20.15.3, and
- you are using the IPSEC feature to configure an edge device using Cisco IOS XE Catalyst SD-WAN Release 17.12.x or earlier,

then you must also configure a command using a CLI add-on profile. This command provides backward compatibility for edge devices using Cisco IOS XE Catalyst SD-WAN Release 17.12.x or earlier. Without this, the tunnel does not operate correctly.

To do this, create the CLI add-on profile and add it to the configuration group that you are using the configure the device. In the profile, include the **tunnel mode ipsec ipv4-old** command.

Using the CLI add-on profile with the **tunnel mode ipsec ipv4-old** command is not necessary in these releases:

- SD-WAN Manager 20.15.4 and later releases of 20.15.x
- SD-WAN Manager 20.18.1 and later releases

**Before you begin**

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

**Procedure**

**Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

**Step 2** Create and configure the IPSEC..

- a. Configure the VPN Interface IPsec feature.

| Field         | Description                                                                                |
|---------------|--------------------------------------------------------------------------------------------|
| Type          | Choose a feature from the drop-down list.                                                  |
| Feature Name* | Enter a name for the feature.                                                              |
| Description   | Enter a description of the feature. The description can contain any characters and spaces. |

- b. Configure Basic Configuration parameters.

*Table 246: Basic Configuration*

| Field          | Description                            |
|----------------|----------------------------------------|
| Interface Name | Enter the name of the IPsec interface. |

| Field                                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                        | Enter a description of the IPsec interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Tunnel Mode</b>                        | Choose from one of the following IPsec tunnel modes: <ul style="list-style-type: none"> <li>• <b>ipv4</b>: IPsec tunnel with IPv4 overlay and IPv4 underlay. IPv4 underlay is the default value.</li> <li>• <b>ipv6</b>: IPsec tunnel with IPv6 overlay and IPv6 underlay.</li> <li>• <b>ipv4-v6overlay</b>: IPsec tunnel with IPv6 overlay and IPv4 underlay.</li> </ul>                                                                                                                            |
| <b>Multiplexing</b>                       | Choose <b>Yes</b> to enable multiplexing, if there is a tunnel in the transport VPN.<br>Default: No                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Interface Address</b>                  | Enter the IPv4 or IPv6 address of the IPsec interface, based on your choice from the <b>Tunnel Mode</b> drop-down list.                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Mask</b>                               | Enter the subnet mask.                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Preshared Key for IKE</b>              | Enter the preshared key (PSK) for authentication.                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Associated Tracker / Tracker Group</b> | Choose a tracker or a tracker group from the drop-down list to associate with the IPsec tunnel.                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Tunnel Source</b>                      | Enter the source of the IPsec interface: <ul style="list-style-type: none"> <li>• <b>IP Address</b>: Enter the source IP address of the IPsec tunnel interface. Enter an IPv4 or IPv6 address that is based on your selection in the <b>Tunnel Mode</b> option. This address is on the local router.</li> <li>• <b>Interface</b>: Enter the physical interface in the <b>IPsec Source Interface</b> field, which is the source of the IPsec tunnel.</li> </ul>                                       |
| <b>Tunnel Destination</b>                 | Enter the destination IP address of the IPsec tunnel interface. This address is on a remote device. <ul style="list-style-type: none"> <li>• <b>Address</b>: Enter the destination IP address of the IPsec tunnel interface. Enter an IPv4 or IPv6 address based on your selection in the <b>Tunnel Mode</b> option.</li> <li>• <b>Application</b>: Choose an application from the drop-down list. <ul style="list-style-type: none"> <li>• <b>None</b></li> <li>• <b>Sig</b></li> </ul> </li> </ul> |

c. Configure Internet Key Exchange fields.

Table 247: Internet Key Exchange

| Field                              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IKE Version</b>                 | Enter 1 to choose IKEv1.<br>Enter 2 to choose IKEv2.<br>Default: IKEv1                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>IKE Integrity Protocol</b>      | Choose one of the following modes for the exchange of keying information and setting up IKE security associations: <ul style="list-style-type: none"> <li>• <b>Main:</b> Establishes an IKE SA session before starting IPsec negotiations.</li> <li>• <b>Aggressive:</b> Negotiation is quicker, and the initiator and responder ID pass in the clear. Aggressive mode does not provide identity protection for communicating parties.</li> </ul> Default: Main mode |
| <b>IPsec Rekey Interval</b>        | Specify the interval for refreshing IKE keys.<br>Range: 3600 through 1209600 seconds (1 hour through 14 days)<br>Default: 14400 seconds (4 hours)                                                                                                                                                                                                                                                                                                                    |
| <b>IKE Cipher Suite</b>            | Specify the type of authentication and encryption to use during IKE key exchange.<br>Values: aes128-cbc-sha1, aes128-cbc-sha2, aes256-cbc-sha1, aes256-cbc-sha2<br>Default: aes256-cbc-sha1                                                                                                                                                                                                                                                                          |
| <b>IKE Diffie-Hellman Group</b>    | Specify the Diffie-Hellman group to use in IKE key exchanges.<br>Values: 2, 14, 15, 16, 19, 20, 21, 24<br>Default: 16                                                                                                                                                                                                                                                                                                                                                |
| <b>IKE ID for Local End Point</b>  | If the remote IKE peer requires a local endpoint identifier, specify it.<br>Range: 1 through 64 characters<br>Default: Source IP address of the tunnel                                                                                                                                                                                                                                                                                                               |
| <b>IKE ID for Remote End Point</b> | If the remote IKE peer requires a remote endpoint identifier, specify it.<br>Range: 1 through 64 characters<br>Default: Destination IP address of the tunnel<br>There is no default option if you choose IKEv2.                                                                                                                                                                                                                                                      |

d. Configure IPSEC fields.

Table 248: IPSEC

| Field                          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IPsec Rekey Interval</b>    | Specify the interval for refreshing IKE keys.<br>Range: 3600 through 1209600 seconds (1 hour through 14 days)<br>Default: 3600 seconds (1 hour)                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>IPsec Replay Window</b>     | Specify the replay window size for the IPsec tunnel.<br>Values: 64, 128, 256, 512, 1024, 2048, 4096, 8192 bytes<br>Default: 512 bytes                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>IPsec Cipher Suite</b>      | Specify the authentication and encryption to use on the IPsec tunnel.<br>Values: <b>aes256-cbc-sha1</b> , <b>aes256-gcm</b> , <b>null-sha1</b><br>Default: <b>aes256-gcm</b>                                                                                                                                                                                                                                                                                                                                                      |
| <b>Perfect Forward Secrecy</b> | Specify the PFS settings to use on the IPsec tunnel by choosing one of the following values: <ul style="list-style-type: none"> <li>• <b>group-2</b>: Use the 1024 bit Diffie-Hellman prime modulus group</li> <li>• <b>group-14</b>: Use the 2048 bit Diffie-Hellman prime modulus group</li> <li>• <b>group-15</b>: Use the 3072 bit Diffie-Hellman prime modulus group</li> <li>• <b>group-16</b>: Use the 4096 bit Diffie-Hellman prime modulus group</li> <li>• <b>none</b>: Disable PFS</li> </ul> Default: <b>group-16</b> |

- e. Configure Advanced IPSEC fields.

Table 249: Advanced

| Field                   | Description                                                                                                                                                                                                                                                                            |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Associated VPN</b>   | Select a VPN from the drop-down list to associate with the IPsec tunnel.                                                                                                                                                                                                               |
| <b>Tunnel Route Via</b> | Specify the tunnel route details to steer the application traffic through.<br><br><b>Note</b><br>You cannot use the tunnel route via option to configure IPSec tunnels on a cellular interface because cellular interfaces do not include a next hop IP address for the default route. |
| <b>DPD Interval</b>     | Specify the interval for IKE to send Hello packets on the connection.<br>Range: 10 through 3600 seconds (1 hour)<br>Default: 10 seconds                                                                                                                                                |

| Field                      | Description                                                                                                                                                                                                                                                                                               |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DPD Retries</b>         | Specify how many unacknowledged packets to accept before declaring an IKE peer to be dead and then removing the tunnel to the peer.<br><br>Range: 2 through 60<br>Default: 3                                                                                                                              |
| <b>TCP MSS</b>             | Specify the maximum segment size (MSS) of TCP SYN packets passing through the Cisco IOS XE Catalyst SD-WAN device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented.<br><br>Range: 552 through 1460 bytes<br>Default: None |
| <b>Clear-Dont-Fragment</b> | Click <b>On</b> to clear the Don't Fragment bit in the IPv4 packet header for packets being transmitted out the interface.                                                                                                                                                                                |
| <b>IP MTU</b>              | Based on your choice in the <b>Tunnel Mode</b> option, specify the maximum MTU size of the IPv4 or IPv6 packets on the interface.<br><br>Range: 576 through 9216<br>Default: 1500 bytes                                                                                                                   |
| <b>Shutdown</b>            | Click <b>Off</b> to enable the interface.                                                                                                                                                                                                                                                                 |

**What to do next**

Also see [Deploy a configuration group](#).

## Configure IPSEC on Service Side Using a Configuration Group

**Before you begin**

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

**Procedure**

**Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

**Step 2** Create and Configure the VPN interface IPSEC feature.

- a. Configure the VPN interface IPSEC fields.

| Field       | Description                               |
|-------------|-------------------------------------------|
| <b>Type</b> | Choose a feature from the drop-down list. |

| Field                | Description                                                                                |
|----------------------|--------------------------------------------------------------------------------------------|
| <b>Feature Name*</b> | Enter a name for the feature.                                                              |
| <b>Description</b>   | Enter a description of the feature. The description can contain any characters and spaces. |

b. Configure Basic Configuration fields.

*Table 250: Basic Configuration*

| Field                     | Description                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Interface Name</b>     | Enter the name of the IPsec interface.                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Description</b>        | Enter a description of the IPsec interface.                                                                                                                                                                                                                                                                                                                                                            |
| <b>Tunnel Mode</b>        | Choose from one of the following IPsec tunnel modes: <ul style="list-style-type: none"> <li>• <b>ipv4</b>: IPsec tunnel with IPv4 overlay and IPv4 underlay. IPv4 underlay is the default value.</li> <li>• <b>ipv6</b>: IPsec tunnel with IPv6 overlay and IPv6 underlay.</li> <li>• <b>ipv4-v6overlay</b>: IPsec tunnel with IPv6 overlay and IPv4 underlay.</li> </ul>                              |
| <b>Interface Address</b>  | Enter the IPv4 or IPv6 address of the IPsec interface, based on your choice from the <b>Tunnel Mode</b> drop-down list.                                                                                                                                                                                                                                                                                |
| <b>Mask</b>               | Enter the subnet mask.                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Tunnel Source</b>      | Enter the source of the IPsec interface: <ul style="list-style-type: none"> <li>• <b>IP Address</b>: Enter the IPv4 or IPv6 address of the IPsec interface, based on your choice from the <b>Tunnel Mode</b> drop-down list.. This address is on the local router.</li> <li>• <b>Interface</b>: Enter the physical interface that is the source of the IPsec tunnel.</li> </ul>                        |
| <b>Tunnel Destination</b> | Enter the destination of the IPsec interface: <ul style="list-style-type: none"> <li>• <b>Address</b>: Enter the destination IPv4 or IPv6 address of the IPsec interface, based on your choice from the <b>Tunnel Mode</b> drop-down list. This address is on a remote device.</li> <li>• <b>Application</b>: Choose an application from the drop-down list.</li> <li>• None</li> <li>• Sig</li> </ul> |

| Field                      | Description                                                                                                                                                                                                                                                                        |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>TCP MSS</b>             | Specify the maximum segment size (MSS) of TCP SYN packets passing through the vEdge router. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented.<br><br>Range: 552 through 1460 bytes<br>Default: None |
| <b>Clear-Dont-Fragment</b> | Click On to clear the Don't Fragment bit in the IPv4 packet header for packets being transmitted out the interface.                                                                                                                                                                |
| <b>IP MTU</b>              | Specify the maximum MTU size of packets on the interface.<br><br>Range: 576 through 1804<br>Default: 1500 bytes                                                                                                                                                                    |

## c. Configure IKE fields.

*Table 251: Internet Key Exchange*

| Field                                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IKE Version</b>                    | Enter 1 to choose IKEv1.<br>Enter 2 to choose IKEv2.<br>Default: IKEv1                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>IKE Integrity Protocol</b>         | Choose one of the following modes for the exchange of keying information and setting up IKE security associations: <ul style="list-style-type: none"> <li>• <b>Main:</b> Establishes an IKE SA session before starting IPsec negotiations.</li> <li>• <b>Aggressive:</b> Negotiation is quicker, and the initiator and responder ID pass in the clear. Aggressive mode does not provide identity protection for communicating parties.</li> </ul> Default: Main mode |
| <b>IPsec Rekey Interval (Seconds)</b> | Specify the interval for refreshing IKE keys.<br>Range: 3600 through 1209600 seconds (1 hour through 14 days)<br>Default: 14400 seconds (4 hours)                                                                                                                                                                                                                                                                                                                    |
| <b>IKE Cipher Suite</b>               | Specify the type of authentication and encryption to use during IKE key exchange.<br>Values: aes128-cbc-sha1, aes128-cbc-sha2, aes256-cbc-sha1, aes256-cbc-sha2<br>Default: aes256-cbc-sha1                                                                                                                                                                                                                                                                          |
| <b>IKE Diffie-Hellman Group</b>       | Specify the Diffie-Hellman group to use in IKE key exchanges.<br>Values: 2, 14, 15, 16, 19, 20, 21, 24<br>Default: 16                                                                                                                                                                                                                                                                                                                                                |



| Field                              | Description                                                                                                                                                                                                           |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IKE ID for Local End Point</b>  | If the remote IKE peer requires a local endpoint identifier, specify it.<br>Range: 1 through 64 characters<br>Default: Source IP address of the tunnel                                                                |
| <b>IKE ID for Remote End Point</b> | If the remote IKE peer requires a remote end point identifier, specify it.<br>Range: 1 through 64 characters<br>Default: Destination IP address of the tunnel<br>There is no default option if you have chosen IKEv2. |

d. Configure IPSEC fields.

**Table 252: IPSEC**

| Field                          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IPsec Rekey Interval</b>    | Specify the interval for refreshing IKE keys.<br>Range: 3600 through 1209600 seconds (1 hour through 14 days)<br>Default: 3600 seconds                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>IPsec Replay Window</b>     | Specify the replay window size for the IPsec tunnel.<br>Values: 64, 128, 256, 512, 1024, 2048, 4096, 8192 bytes<br>Default: 512 bytes                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>IPsec Cipher Suite</b>      | Specify the authentication and encryption to use on the IPsec tunnel.<br>Values: <b>aes256-cbc-sha1</b> , <b>aes256-gcm</b> , <b>null-sha1</b><br>Default: <b>aes256-gcm</b>                                                                                                                                                                                                                                                                                                                                                      |
| <b>Perfect Forward Secrecy</b> | Specify the PFS settings to use on the IPsec tunnel by choosing one of the following values: <ul style="list-style-type: none"> <li>• <b>group-2</b>: Use the 1024-bit Diffie-Hellman prime modulus group</li> <li>• <b>group-14</b>: Use the 2048-bit Diffie-Hellman prime modulus group</li> <li>• <b>group-15</b>: Use the 3072-bit Diffie-Hellman prime modulus group</li> <li>• <b>group-16</b>: Use the 4096-bit Diffie-Hellman prime modulus group</li> <li>• <b>none</b>: Disable PFS</li> </ul> Default: <b>group-16</b> |

e. Configure advanced fields.

Table 253: Advanced

| Field                      | Description                                                                                                                                                                                                                                                                            |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Associated VPN</b>      | Select a VPN from the drop-down list to associate with the IPsec tunnel.                                                                                                                                                                                                               |
| <b>Tunnel Route Via</b>    | Specify the tunnel route details to steer the application traffic through.<br><br><b>Note</b><br>You cannot use the tunnel route via option to configure IPsec tunnels on a cellular interface because cellular interfaces do not include a next hop IP address for the default route. |
| <b>DPD Interval</b>        | Specify the interval for IKE to send Hello packets on the connection.<br>Range: 10 through 3600 seconds (1 hour)<br>Default: 10 seconds                                                                                                                                                |
| <b>DPD Retries</b>         | Specify how many unacknowledged packets to accept before declaring an IKE peer to be dead and then removing the tunnel to the peer.<br>Range: 2 through 60<br>Default: 3                                                                                                               |
| <b>TCP MSS</b>             | Specify the maximum segment size (MSS) of TCP SYN packets passing through the Cisco vEdge device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented.<br>Range: 552 through 1460 bytes<br>Default: None   |
| <b>Clear-Dont-Fragment</b> | Click <b>On</b> to clear the Don't Fragment bit in the IPv4 packet header for packets being transmitted out the interface.                                                                                                                                                             |
| <b>IP MTU</b>              | Based on your choice in the <b>Tunnel Mode</b> option, specify the maximum MTU size of the IPv4 or IPv4 packets on the interface.<br>Range: 576 through 9216<br>Default: 1500 bytes                                                                                                    |
| <b>Shutdown</b>            | Click <b>Off</b> to enable the interface.                                                                                                                                                                                                                                              |

**Step 3****What to do next**

Also see [Deploy a configuration group](#).

# Configure VPN Interface Multilink Using Configuration Groups

Use the VPN Interface Multilink feature to configure multilink interface properties for Cisco SD-WAN Manager devices.

## Before you begin

On the **Configuration** > **Configuration Groups** page, choose **SD-WAN** as the solution type.

## Procedure

**Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration** > **Configuration Groups**.

**Step 2** Create and configure VPN Interface Multilink in a service profile.

a) Enter the basic configuration information.

**Table 254: Basic Configuration**

| Parameter Name              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Interface Name              | Enter the name of the multilink interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Multilink Group Number *    | Enter the number of the multilink group. It must be the same as the number you enter in the multilink interface name parameter.<br><br>Range: 1 through 65535                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| PPP Authentication Protocol | Select the authentication protocol used by the multilink interface: <ul style="list-style-type: none"><li>• <b>CHAP</b>: Enter the hostname and password provided by your Internet Service Provider (ISP). <i>hostname</i> can be up to 255 characters.</li><li>• <b>PAP</b>: Enter the username and password provided by your ISP. <i>username</i> can be up to 255 characters.</li><li>• <b>PAP</b> and <b>CHAP</b>: Configure both authentication protocols. Enter the login credentials for each protocol. To use the same username and password for both, click Same Credentials for PAP and CHAP.</li></ul> |
| Hostname *                  | Enter hostname for PPP CHAP Authentication.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| CHAP Password *             | Enter password for PPP CHAP Authentication.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| IPv4 Address *              | To configure a static address, click <b>Static</b> and enter an IPv4 address.<br><br>To set the interface as a DHCP client so that the interface to receive its IP address from a DHCP server, click Dynamic. You can optionally set the DHCP distance to specify the administrative distance of routes learned from a DHCP server.<br><br>Default: 1                                                                                                                                                                                                                                                             |
| Mask                        | Choose a value for the subnet mask.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| Parameter Name | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IPv6 Address * | <p>To configure a static address for an interface in VPN 0, click Static and enter an IPv6 address.</p> <p>To set the interface as a DHCP client so that the interface to receive its IP address from a DHCP server, click Dynamic. You can optionally set the DHCP distance to specify the administrative distance of routes learned from a DHCP server. The default DHCP distance is 1. You can optionally enable DHCP rapid commit, to speed up the assignment of IP addresses.</p> |

b) Enter multilink information

**Table 255: Multilink**

| Parameter Name             | Description                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Add T1/E1 Interface</b> |                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>T1</b>                  |                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Description                | Enter a description for the T1 controller.                                                                                                                                                                                                                                                                                                                                                                                   |
| Slot*                      | Enter the number of the slot in slot/subslot/port format, where the T1 NIM is installed. For example, 0/1/0.                                                                                                                                                                                                                                                                                                                 |
| Framing                    | <p>Enter the T1 frame type:</p> <ul style="list-style-type: none"> <li>• <b>esf</b>: Send T1 frames as extended superframes. This is the default.</li> <li>• <b>sf</b>: Send T1 frames as superframes. Superframing is sometimes called D4 framing.</li> </ul>                                                                                                                                                               |
| Clock Source               | <p>Select the clock source:</p> <ul style="list-style-type: none"> <li>• <b>line</b>: Use phase-locked loop (PLL) on the interface. This is the default. When both T1 ports use line clocking and neither port is configured as the primary, by default, port 0 is the primary clock source and port 1 is the secondary clock source.</li> <li>• <b>internal</b>: Use the controller framer as the primary clock.</li> </ul> |
| Line Code                  | <p>Select the line encoding to use to send T1 frames:</p> <ul style="list-style-type: none"> <li>• <b>ami</b>: Use alternate mark inversion (AMI) as the linecode. AMI signaling uses frames grouped into superframes.</li> <li>• <b>b8zs</b>: Use bipolar 8-zero substitution as the linecode. This is the default. B8ZS uses frames that are grouped into extended superframes.</li> </ul>                                 |

| Parameter Name                      | Description                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cable Length                        | <p>Select the cable length to configure the attenuation</p> <ul style="list-style-type: none"> <li>• <b>short</b>: Set the transmission attenuation for cables that are 660 feet or shorter.</li> <li>• <b>long</b>: Attenuate the pulse from the transmitter using pulse equalization and line buildout. You can configure a long cable length for cables longer than 660 feet.</li> </ul> <p>There is no default length.</p> |
| <b>E1</b>                           |                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Description                         | Enter a description for the E1 controller.                                                                                                                                                                                                                                                                                                                                                                                     |
| Slot*                               | Enter the number of the slot in slot/subslot/port format, where the E1 NIM is installed. For example, 0/1/0.                                                                                                                                                                                                                                                                                                                   |
| Framing                             | <p>Enter the E1 frame type:</p> <ul style="list-style-type: none"> <li>• <b>crc4</b>: Use cyclic redundancy check 4 (CRC4). This is the default.</li> <li>• <b>no-crc4</b>: Do not use CRC4.</li> </ul>                                                                                                                                                                                                                        |
| Clock Source                        | <p>Select the clock source:</p> <ul style="list-style-type: none"> <li>• <b>line</b>: Use phase-locked loop (PLL) on the interface. This is the default. When both E1 ports use line clocking and neither port is configured as the primary, by default, port 0 is the primary clock source and port 1 is the secondary clock source.</li> <li>• <b>internal</b>: Use the controller framer as the primary clock.</li> </ul>   |
| Line Code                           | <p>Select the line encoding to use to send E1 frames:</p> <ul style="list-style-type: none"> <li>• <b>ami</b>: Use alternate mark inversion (AMI) as the linecode.</li> <li>• <b>hdb3</b>: Use high-density bipolar 3 as the linecode. This is the default.</li> </ul>                                                                                                                                                         |
| <b>Add Channel Group</b>            |                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Channel Group                       | <p>To configure the serial WAN on the interface, enter a channel group number.</p> <p>Range: 0 through 30</p>                                                                                                                                                                                                                                                                                                                  |
| Time Slot                           | <p>To configure the serial WAN on the interface, enter a value for the timeslot.</p> <p>Range: 0 through 31</p>                                                                                                                                                                                                                                                                                                                |
| <b>Add New A/S Serial Interface</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Interface Name                      | Enter the name of the serial interface.                                                                                                                                                                                                                                                                                                                                                                                        |
| Description                         | Enter a description for the serial interface.                                                                                                                                                                                                                                                                                                                                                                                  |
| Bandwidth                           | For transmitted traffic, set the bandwidth above which to generate notifications.                                                                                                                                                                                                                                                                                                                                              |

| Parameter Name | Description                                                       |
|----------------|-------------------------------------------------------------------|
| Clock Rate     | Specify a value for the clock rate.<br>Range: 1200 through 800000 |

- c) Enter tunnel information

**Table 256: Tunnel**

| Parameter Name                             | Description                                                                                                                                                                                                                                                                  |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Color                                      | Choose a color for the TLOC.                                                                                                                                                                                                                                                 |
| Color Description                          | Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.18.1<br>Enter a description associated to the TLOC color.                                                                                                                                                |
| Restrict                                   | Enable this option to drop packets when a tunnel to the service is unreachable.                                                                                                                                                                                              |
| Groups                                     | Enter the list of groups in the field.                                                                                                                                                                                                                                       |
| Border                                     | From the drop-down list, select <b>Global</b> . Click <b>On</b> to set TLOC as border TLOC.                                                                                                                                                                                  |
| Maximum Control Connections                | Specify the maximum number of Cisco SD-WAN Controllers that the WAN tunnel interface can connect to. To have the tunnel establish no control connections, set the number to 0.<br>Range: 0 through 8<br>Default: 2                                                           |
| Validator As Stun Server                   | Click <b>On</b> to enable Session Traversal Utilities for NAT (STUN) to allow the tunnel interface to discover its public IP address and port number when the router is located behind a NAT.                                                                                |
| Exclude Controller Group List              | Set the Cisco SD-WAN Controllers that the tunnel interface is not allowed to connect to.<br>Range: 0 through 100                                                                                                                                                             |
| Cisco SD-WAN Manager Connection Preference | Set the preference for using a tunnel interface to exchange control traffic with Cisco SD-WAN Manager.<br>Range: 0 through 8<br>Default: 5                                                                                                                                   |
| <b>Full Port Hop</b>                       | Minimum release: Cisco IOS XE Catalyst SD-WAN Release 17.18.1a<br>Enable full port hopping at the TLOC level to allow devices to establish connections with controllers by switching to the next port if the current port is blocked or non-functional.<br>Default: Disabled |

| Parameter Name     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Port Hop           | <p>From the drop-down list, select <b>Global</b>. Click <b>Off</b> to allow port hopping on tunnel interface.</p> <p>Default: <b>On</b>, which disallows port hopping on tunnel interface</p> <p>Starting from Cisco IOS XE Catalyst SD-WAN Release 17.18.1a, this field is deprecated. Instead use the <b>Full Port Hop</b> option. See the <b>Full Port Hop</b> field.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Low-Bandwidth Link | <p>Click <b>On</b> to set the tunnel interface as a low-bandwidth link.</p> <p>Default: <b>Off</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Network Broadcast  | <p>From the drop-down list, select <b>Global</b>. Click <b>On</b> to accept and respond to network-prefix-directed broadcasts. Enable this parameter only if the <b>Directed Broadcast</b> is enabled on the LAN interface feature template.</p> <p>Default: <b>Off</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Tunnel TCP MSS     | <p>TCP MSS affects any packet that contains an initial TCP header that flows through the router. When configured, TCP MSS is examined against the MSS exchanged in the three-way handshake. The MSS in the header is lowered if the configured TCP MSS setting is lower than the MSS in the header. If the MSS header value is already lower than the TCP MSS, the packets flow through unmodified. The host at the end of the tunnel uses the lower setting of the two hosts. To configure TCP MSS, provide a value that is 40 bytes lower than the minimum path MTU.</p> <p>Specify the MSS of TPC SYN packets passing through the Cisco IOS XE Catalyst SD-WAN. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented.</p> <p>Range: 552 through 1460 bytes</p> |

d) Enter ACL information.

**Table 257: ACL**

| Parameter Name     | Description                                                                          |
|--------------------|--------------------------------------------------------------------------------------|
| Ingress ACL - IPv4 | Enter the name of an IPv4 access list to packets being received on the interface.    |
| Egress ACL - IPv4  | Enter the name of an IPv4 access list to packets being transmitted on the interface. |
| Igress ACL - IPv6  | Enter the name of an IPv6 access list to packets being received on the interface.    |
| Egress ACL - IPv6  | Enter the name of an IPv6 access list to packets being transmitted on the interface. |

e) Enter advanced information.

Table 258: Advanced

| Parameter Name          | Description                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Shutdown                | Click <b>No</b> to enable the multilink interface.                                                                                                                                                                                                                                                                                                                                                          |
| Description             | Enter a description for the multilink interface.                                                                                                                                                                                                                                                                                                                                                            |
| PPP Authentication Type | Select the type authentication from one of the following options.: <ul style="list-style-type: none"> <li>• <b>Unidirectional</b>: The server initiates the authentication.</li> <li>• <b>Bidirectional</b>: Both the client and the server can initiate the authentication.</li> </ul>                                                                                                                     |
| TCP MSS                 | Specify the maximum segment size (MSS) of TPC SYN packets passing through the Cisco Catalyst SD-WAN device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented.<br><br>Range: 500 through 1460 bytes<br><br>Default: 536                                                                                                       |
| Disable Fragmentation   | Click <b>On</b> to disable fragmentation for PPP Multilink Protocol data units (PDUs).                                                                                                                                                                                                                                                                                                                      |
| Fragment Max Delay      | Configure the delay between the transmission of fragments in a PPP Multilink Protocol link.<br><br>Range: 0 through 1000<br><br>Default: No CLI Command                                                                                                                                                                                                                                                     |
| Interleaving Fragments  | Enable interleave fragmentation for PPP Multilink Protocol data units (PDUs).                                                                                                                                                                                                                                                                                                                               |
| TLOC Extension          | Enter the name of a physical interface on the same router that connects to the WAN transport. This configuration binds the service-side interface to the WAN transport by enabling a device to access the opposite WAN transport connected to the neighbouring device using a TLOC-extension interface.                                                                                                     |
| IP MTU                  | Specify the maximum MTU size of packets on the interface. MLP encapsulation adds 6 extra bytes (4 header, 2 checksum) to each outbound packet. These overhead bytes reduce the effective bandwidth on the connection; therefore, the throughput for an MLP bundle is slightly less than an equivalent bandwidth connection that is not using MLP.<br><br>Range: 576 through 1804<br><br>Default: 1500 bytes |
| IP Directed-Broadcast   | Enable the translation of a directed broadcast to physical broadcasts.                                                                                                                                                                                                                                                                                                                                      |
| Shaping Rate (Kbps)     | Configure the aggregate traffic transmission rate on the interface to be less than line rate, in kilobits per second (kbps).                                                                                                                                                                                                                                                                                |

**What to do next**

Also see [Deploy a Configuration Group](#).



# VPN Interface Multilink

Use the VPN Interface Multilink template for Cisco IOS XE Catalyst SD-WAN devices running the Cisco Catalyst SD-WAN software.



**Note** Cisco IOS XE Catalyst SD-WAN devices use VRFs in place of VPNs. However, the following steps still apply to configure Cisco IOS XE Catalyst SD-WAN devices through Cisco SD-WAN Manager. When you complete the configuration, the system automatically maps the VPN configurations to VRF configurations.

Multilink Point-to-Point Protocol (MLP) is used to combine multiple physical links into a single logical connection, called an MLP bundle.

To configure multilink on Cisco IOS XE Catalyst SD-WAN Device using Cisco SD-WAN Manager templates:

1. Create a VPN Interface Multilink feature template to configure multilink interface properties.
2. Optionally, create a VPN feature template to modify the default configuration of VPN 0.

## Navigate to the Template Screen and Name the Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**.



**Note** In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. From the **Create Template** drop-down list, select **From Feature Template**.
4. From the **Device Model** drop-down list, select the type of device for which you are creating the template.
5. If you are configuring the multilink interface in the transport VPN (VPN 0):
  - a. Click **Transport & Management VPN** or scroll to the **Transport & Management VPN** section.
  - b. Under Additional VPN 0 Templates, located to the right of the screen, click **VPN Interface Multilink Controller**.
6. If you are configuring the multilink interface in a service VPN (VPNs other than VPN 0):
  - a. Click **Service VPN** or scroll to the **Service VPN** section.
  - b. In the Service **VPN** drop-down list, enter the number of the service VPN.
  - c. Under Additional VPN Templates, located to the right of the screen, click **VPN Interface Multilink Controller**.
7. From the **VPN Interface Multilink Controller** drop-down list, click **Create Template**. The VPN Multilink template form is displayed. This form contains fields for naming the template, and fields for defining multilink Interface parameters.

8. In **Template Name**, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.
9. In **Template Description**, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default or to enter a value, click the scope drop-down to the left of the parameter field and select one of the following:

**Table 259:**

| Parameter Scope                               | Scope Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Device Specific<br>(indicated by a host icon) | <p>Use a device-specific value for the parameter. For device-specific parameters, you cannot enter a value in the feature template. You enter the value when you attach a Viptela device to a device template .</p> <p>When you click <b>Device Specific</b>, the <b>Enter Key</b> box opens. This box displays a key, which is a unique string that identifies the parameter in a CSV file that you create. This file is an Excel spreadsheet that contains one column for each key. The header row contains the key names (one key per column), and each row after that corresponds to a device and defines the values of the keys for that device. You upload the CSV file when you attach a Viptela device to a device template. For more information, see <a href="#">Create a Template Variables Spreadsheet</a> .</p> <p>To change the default key, type a new string and move the cursor out of the Enter Key box.</p> <p>Examples of device-specific parameters are system IP address, hostname, GPS location, and site ID.</p> |
| Global (indicated by a globe icon)            | <p>Enter a value for the parameter, and apply that value to all devices.</p> <p>Examples of parameters that you might apply globally to a group of devices are DNS server, syslog server, and interface MTUs.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

### Configure a Multilink Interface

To configure a multilink interface, select **Basic Configuration** and configure the following parameters. Parameters marked with an asterisk are required to configure the interface.



**Note** If you are creating a VPN Interface Multilink template, you do not need to create a T1/E1 Controller template or a VPN Interface T1/E1 template.

**Table 260:**

| Parameter Name  | Description                                                                      |
|-----------------|----------------------------------------------------------------------------------|
| Shutdown*       | Click <b>No</b> to enable the multilink interface.                               |
| Interface Name* | Enter the number of the MLP interface. It can be a number from 1 through 65,535. |

| Parameter Name          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description             | Enter a description for the multilink interface.                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Multilink Group Number* | Enter the number of the multilink group. It can be a number from 1 through 65,535 but it must be the same as the number you enter in the Multilink Interface Name parameter.                                                                                                                                                                                                                                                                                                                   |
| IPv4 Address*           | To configure a static address, click <b>Static</b> and enter an IPv4 address.<br><br>To set the interface as a DHCP client so that the interface to receive its IP address from a DHCP server, click <b>Dynamic</b> . You can optionally set the DHCP distance to specify the administrative distance of routes learned from a DHCP server. The default DHCP distance is 1.                                                                                                                    |
| IPv6 Address*           | To configure a static address for an interface in VPN 0, click <b>Static</b> and enter an IPv6 address.<br><br>To set the interface as a DHCP client so that the interface to receive its IP address from a DHCP server, click <b>Dynamic</b> . You can optionally set the DHCP distance to specify the administrative distance of routes learned from a DHCP server. The default DHCP distance is 1. You can optionally enable DHCP rapid commit, to speed up the assignment of IP addresses. |
| Bandwidth Upstream      | For transmitted traffic, set the bandwidth above which to generate notifications.<br><i>Range:</i> 1 through $(2^{32} / 2) - 1$ kbps                                                                                                                                                                                                                                                                                                                                                           |
| Bandwidth Downstream    | For received traffic, set the bandwidth above which to generate notifications.<br><i>Range:</i> 1 through $(2^{32} / 2) - 1$ kbps                                                                                                                                                                                                                                                                                                                                                              |
| IP MTU                  | Specify the maximum MTU size of packets on the interface. MLP encapsulation adds 6 extra bytes (4 header, 2 checksum) to each outbound packet. These overhead bytes reduce the effective bandwidth on the connection; therefore, the throughput for an MLP bundle is slightly less than an equivalent bandwidth connection that is not using MLP. <i>Range:</i> 576 through 1804 <i>Default:</i> 1500 bytes                                                                                    |

To save the feature template, click **Save**.

### Configure the PPP Authentication Protocol

To configure the PPP authentication protocol, select **PPP** and configure the following parameters:

Table 261:

| Parameter Name          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Authentication Protocol | <p>Select the authentication protocol used by the MLP:</p> <ul style="list-style-type: none"> <li>• <b>CHAP</b>—Enter the hostname and password provided by your Internet Service Provider (ISP). <i>hostname</i> can be up to 255 characters.</li> <li>• <b>PAP</b>—Enter the username and password provided by your ISP. <i>username</i> can be up to 255 characters.</li> <li>• <b>PAP and CHAP</b>—Configure both authentication protocols. Enter the login credentials for each protocol. To use the same username and password for both, click Same Credentials for PAP and CHAP.</li> </ul> |

To save the feature template, click **Save**.

### Create a Tunnel Interface

You can configure up to eight tunnel interfaces. This means that each device can have up to eight TLOCs.

For the control plane to establish itself so that the overlay network can function, you must configure WAN transport interfaces in VPN 0.

To configure a tunnel interface for the multilink interface, select the **Tunnel Interface** tab and configure the following parameters:

Table 262:

| Parameter Name    | Description                                                                                                                              |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Tunnel Interface  | Click <b>On</b> to create a tunnel interface.                                                                                            |
| Color             | Select a color for the TLOC.                                                                                                             |
| Color Description | <p>Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.18.1</p> <p>Enter a description associated to the TLOC color.</p> |

| Parameter Name                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Control Connection            | <p>By default, Control Connection is set to <b>On</b>, which establishes a control connection for the TLOC. If the router has multiple TLOCs, click <b>No</b> to have the tunnel not establish control connection for the TLOC.</p> <p><b>Note</b><br/>We recommend a minimum of 650-700 Kbps bandwidth with default 1 sec hello-interval and 12 sec hello-tolerance parameters configured to avoid any data/packet loss in connection traffic.</p> <p>For each BFD session, an additional average sized BFD packet of 175 Bytes consumes 1.4 Kbps of bandwidth.</p> <p>A sample calculation of the required bandwidth for bidirectional BFD packet flow is given below:</p> <ul style="list-style-type: none"> <li>• 650 – 700 Kbps per device for control connections.</li> <li>• 175 Bytes (or 1.4 Kbps) per BFD session on the device (request)</li> <li>• 175 Bytes (or 1.4 Kbps) per BFD session on the device (response)</li> </ul> <p>If the path MTU discovery (PMTUD) is enabled, bandwidth for send/receive BFD packets per tunnel for every 30 secs:</p> <p>A 1500 Bytes BFD request packet is sent per tunnel every 30 secs:<br/> <math>1500 \text{ Bytes} * 8 \text{ bits/1 byte} * 1 \text{ packet} / 30 \text{ secs} = 400 \text{ bps (request)}</math></p> <p>A 147 Bytes BFD packet is sent in response:<br/> <math>147 \text{ Bytes} * 8 \text{ bits/1 byte} * 1 \text{ packet} / 30 \text{ secs} = 40 \text{ bps (response)}</math></p> <p>Therefore, a device with 775 BFD sessions (for example) requires a bandwidth of:<br/> <math>700k + (1.4k*775) + (400 * 775) + (1.4k*775) + (40 * 775) = \sim 3,5 \text{ MBps}</math></p> |
| Maximum Control Connections   | <p>Specify the maximum number of Cisco Catalyst SD-WAN Controller that the WAN tunnel interface can connect to. To have the tunnel establish no control connections, set the number to 0.</p> <p><i>Range: 0 through 8 Default: 2</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| vBond As STUN Server          | <p>Click <b>On</b> to enable Session Traversal Utilities for NAT (STUN) to allow the tunnel interface to discover its public IP address and port number when the device is located behind a NAT.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Exclude Controller Group List | <p>Set the Cisco Catalyst SD-WAN Controller that the tunnel interface is not allowed to connect to. <i>Range: 0 through 100</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| vManage Connection Preference | <p>Set the preference for using a tunnel interface to exchange control traffic with Cisco SD-WAN Manager. <i>Range: 0 through 8 Default: 5</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| Parameter Name       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Full Port Hop</b> | <p>Minimum release: Cisco IOS XE Catalyst SD-WAN Release 17.18.1a</p> <p>Enable full port hopping at the TLOC level to allow devices to establish connections with controllers by switching to the next port if the current port is blocked or non-functional.</p> <p>Default: Disabled</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Port Hop             | <p>Click <b>On</b> to enable port hopping, or click <b>Off</b> to disable it. When a router is behind a NAT, port hopping rotates through a pool of preselected OMP port numbers (called base ports) to establish DTLS connections with other routers when a connection attempt is unsuccessful. The default base ports are 12346, 12366, 12386, 12406, and 12426. To modify the base ports, set a port offset value. <i>Default:</i> Enabled</p> <p>Starting from Cisco IOS XE Catalyst SD-WAN Release 17.18.1a, this field is deprecated. Instead use the <b>Full Port Hop</b> option. See the <b>Full Port Hop</b> field.</p>                                                                                                                                                                                                                                              |
| Low-Bandwidth Link   | Select to characterize the tunnel interface as a low-bandwidth link.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Tunnel TCP MSS       | <p>TCP MSS affects any packet that contains an initial TCP header that flows through the router. When configured, TCP MSS is examined against the MSS exchanged in the three-way handshake. The MSS in the header is lowered if the configured TCP MSS setting is lower than the MSS in the header. If the MSS header value is already lower than the TCP MSS, the packets flow through unmodified. The host at the end of the tunnel uses the lower setting of the two hosts. If the TCP MSS is to be configured, it should be set at 40 bytes lower than the minimum path MTU.</p> <p>Specify the MSS of TPC SYN packets passing through the Cisco IOS XE Catalyst SD-WAN device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. <i>Range:</i> 552 to 1460 bytes <i>Default:</i> None</p> |
| Clear-Dont-Fragment  | <p>Configure <b>Clear-Dont-Fragment</b> for packets that arrive at an interface that has Don't Fragment configured. If these packets are larger than what MTU allows, they are dropped. If you clear the Don't Fragment bit, the packets are fragmented and sent.</p> <p>Click <b>On</b> to clear the Dont Fragment bit in the IPv4 packet header for packets being transmitted out of the interface. When the Dont Fragment bit is cleared, packets larger than the MTU of the interface are fragmented before being sent.</p> <p><b>Note</b><br/> <b>Clear-Dont-Fragment</b> clears the Dont Fragment bit and the Dont Fragment bit is set. For packets not requiring fragmentation, the Dont Fragment bit is not affected.</p>                                                                                                                                             |
| Allow Service        | Select <b>On</b> or <b>Off</b> for each service to allow or disallow the service on the interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

To configure additional tunnel interface parameters, click **Advanced Options** and configure the following parameters:

Table 263:

| Parameter Name       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GRE                  | Use GRE encapsulation on the tunnel interface. By default, GRE is disabled.<br><br>If you select both IPsec and GRE encapsulations, two TLOCs are created for the tunnel interface that have the same IP addresses and colors, but that differ by their encapsulation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| IPsec                | Use IPsec encapsulation on the tunnel interface. By default, IPsec is enabled.<br><br>If you select both IPsec and GRE encapsulations, two TLOCs are created for the tunnel interface that have the same IP addresses and colors, but that differ by their encapsulation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| IPsec Preference     | Specify a preference value for directing traffic to the tunnel. A higher value is preferred over a lower value.<br><br><i>Range:</i> 0 through 4294967295. <i>Default:</i> 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| IPsec Weight         | Enter a weight to use to balance traffic across multiple TLOCs. A higher value sends more traffic to the tunnel.<br><br><i>Range:</i> 1 through 255. <i>Default:</i> 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Carrier              | Select the carrier name or private network identifier to associate with the tunnel.<br><br><i>Values:</i> carrier1, carrier2, carrier3, carrier4, carrier5, carrier6, carrier7, carrier8, default.<br><i>Default:</i> default                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Bind Loopback Tunnel | Enter the name of a physical interface to bind to a loopback interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Last-Resort Circuit  | Select to use the tunnel interface as the circuit of last resort.<br><br><b>Note</b><br>An interface configured as a circuit of last resort is expected to be down and is skipped while calculating the number of control connections, the cellular modem becomes dormant, and no traffic is sent over the circuit.<br><br>When the configurations are activated on the edge device with cellular interfaces, then all the interfaces begin the process of establishing control and BFD connections. When one or more of the primary interfaces establishes a BFD connection, the circuit of last resort shuts itself down.<br><br>Only when all the primary interfaces lose their connections to remote edges, then the circuit of last resort activates itself triggering a BFD TLOC Down alarm and a Control TLOC Down alarm on the edge device. The last resort interfaces are used as backup circuit on edge device and are activated when all other transport links BFD sessions fail. In this mode the radio interface is turned off, and no control or data connections exist over the cellular interface. |
| NAT Refresh Interval | Enter the interval between NAT refresh packets sent on a DTLS or TLS WAN transport connection. <i>Range:</i> 1 through 60 seconds. <i>Default:</i> 5 seconds                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Parameter Name  | Description                                                                                                                                                                                               |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hello Interval  | Enter the interval between Hello packets sent on a DTLS or TLS WAN transport connection. <i>Range:</i> 100 through 10000 milliseconds. <i>Default:</i> 1000 milliseconds (1 second)                       |
| Hello Tolerance | Enter the time to wait for a Hello packet on a DTLS or TLS WAN transport connection before declaring that transport tunnel to be down.<br><i>Range:</i> 12 through 60 seconds. <i>Default:</i> 12 seconds |

### Apply Access Lists

To apply a rewrite rule, access lists, and policers to a router interface, select **ACL** and configure the following parameters:

**Table 264:**

| Parameter Name     | Description                                                                                                                  |
|--------------------|------------------------------------------------------------------------------------------------------------------------------|
| Shaping rate       | Configure the aggregate traffic transmission rate on the interface to be less than line rate, in kilobits per second (kbps). |
| QoS map            | Specify the name of the QoS map to apply to packets being transmitted out the interface.                                     |
| Rewrite Rule       | Click <b>On</b> , and specify the name of the rewrite rule to apply on the interface.                                        |
| Ingress ACL – IPv4 | Click <b>On</b> , and specify the name of the access list to apply to IPv4 packets being received on the interface.          |
| Egress ACL – IPv4  | Click <b>On</b> , and specify the name of the access list to apply to IPv4 packets being transmitted on the interface.       |
| Ingress ACL – IPv6 | Click <b>On</b> , and specify the name of the access list to apply to IPv6 packets being received on the interface.          |
| Egress ACL – IPv6  | Click <b>On</b> , and specify the name of the access list to apply to IPv6 packets being transmitted on the interface.       |
| Ingress Policer    | Click <b>On</b> , and specify the name of the policer to apply to packets being received on the interface.                   |
| Egress Policer     | Click <b>On</b> , and specify the name of the policer to apply to packets being transmitted on the interface.                |

To save the feature template, click **Save**.

### Configure Other Interface Properties

To configure other interface properties, select the Advanced tab and configure the following properties:



Table 265:

| Parameter Name      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PMTU Discovery      | Click <b>On</b> to enable path MTU discovery on the interface, to allow the router to determine the largest MTU size supported without requiring packet fragmentation.                                                                                                                                                                                                                                                                                       |
| TCP MSS             | Specify the maximum segment size (MSS) of TCP SYN packets passing through the Cisco Catalyst SD-WAN device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented.<br><i>Range: 552 to 1460 bytes. Default: None</i>                                                                                                                                                               |
| Clear Dont Fragment | Click <b>On</b> to clear the Don't Fragment bit in the IPv4 packet header for packets being transmitted out the interface. When the DF bit is cleared, packets larger than that interface's MTU are fragmented before being sent.                                                                                                                                                                                                                            |
| Static Ingress QoS  | Select a queue number to use for incoming traffic. <i>Range: 0 through 7</i>                                                                                                                                                                                                                                                                                                                                                                                 |
| Auto negotiate      | Click <b>Off</b> to turn off autonegotiation. By default, an interface runs in autonegotiation mode.                                                                                                                                                                                                                                                                                                                                                         |
| TLOC Extension      | Enter the name of the physical interface on the same router that connects to the WAN transport circuit. This configuration then binds this service-side interface to the WAN transport. A second Cisco Catalyst SD-WAN device at the same site that itself has no direct connection to the WAN (generally because the site has only a single WAN connection) and that connects to this service-side interface is then provided with a connection to the WAN. |

To save the feature template, click **Save**.

#### Release Information

Introduced in Cisco SD-WAN Manager in Release 18.3.

## Configure VPN Interface SVI using Cisco SD-WAN Manager

Use the VPN Interface SVI template to configure SVI for Cisco IOS XE Catalyst SD-WAN devices. You configure a switch virtual interface (SVI) to configure a VLAN interface.

To configure DSL interfaces on Cisco routers using Cisco SD-WAN Manager templates, create a VPN Interface SVI feature template to configure VLAN interface parameters.

#### Create VPN Interface SVI Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. In **Device Templates**, click **Create Template**.



**Note** In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. From the **Create Template** drop-down, choose **From Feature Template**.

4. From the **Device Model** drop-down, choose the type of device for which you are creating the template.
5. If you are configuring the SVI in the transport VPN (VPN 0):
  - a. Click **Transport & Management VPN**, or scroll to the Transport & Management VPN section.
  - b. Under Additional VPN 0 Templates, click **VPN Interface SVI**.
6. If you are configuring the SVI in a service VPN (VPNs other than VPN 0):
  - a. Click **Service VPN**, or scroll to the Service VPN section.
  - b. In the **Service VPN** drop-down list, enter the number of the service VPN.
  - c. Under **Additional VPN Templates**, click **VPN Interface SVI**.
7. From the **VPN Interface SVI** drop-down, click **Create Template**. The VPN Interface SVI template form is displayed.  
The form contains fields for naming the template, and fields for defining VLAN Interface parameters.
8. In **Template Name**, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.
9. In **Template Description**, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

When you open a feature template initially, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default or to enter a value, click the **scope** drop-down next to the parameter field.



**Note** To get the SVI interface up and functional, ensure that the appropriate VLAN is explicitly configured on the Switch Port Access or Trunk interface.

### Configure Basic Interface Functionality

*Table 266: Feature History*

| Feature Name                                 | Release Information                          | Description                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Support for Configuring Secondary IP Address | Cisco IOS XE Catalyst SD-WAN Release 17.2.1r | You can configure up to four secondary IPv4 or IPv6 addresses, and up to four DHCP helpers. Secondary IP addresses can be useful for forcing unequal load sharing between different interfaces, for increasing the number of IP addresses in a LAN when no more IPs are available from the subnet, and for resolving issues with discontinuous subnets and classful routing protocol. |

To configure basic VLAN interface functionality in a VPN, choose **Basic Configuration** and configure the following parameters. Parameters marked with an asterisk are required to configure an interface.

**Table 267:**

| Parameter Name                | Description                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Shutdown*                     | Click <b>No</b> to enable the VLAN interface.                                                                                                                                                                                                                                                                                                                                            |
| VLAN Interface Name*          | Enter the VLAN identifier of the interface. <i>Range:</i> 1 through 1094.                                                                                                                                                                                                                                                                                                                |
| Description                   | Enter a description for the interface.                                                                                                                                                                                                                                                                                                                                                   |
| IP MTU                        | Specify the maximum MTU size of packets on the interface. <i>Range:</i> 576 through 1500. <i>Default:</i> 2000 bytes                                                                                                                                                                                                                                                                     |
| IPv4* or IPv6                 | Click to configure one or more IPv4 or IPv6 addresses for the interface. (Beginning with Cisco IOS XE SD-WAN Release 17.2.)                                                                                                                                                                                                                                                              |
| IPv4 Address*<br>IPv6 Address | Enter the IPv4 address for the interface.                                                                                                                                                                                                                                                                                                                                                |
| Secondary IP Address          | Click <b>Add</b> to enter up to four secondary IP addresses. (Beginning with Cisco IOS XE SD-WAN Release 17.2.)                                                                                                                                                                                                                                                                          |
| DHCP Helper*                  | Enter up to eight IP addresses for DHCP servers in the network to have the interface be a DHCP helper. Separate each address with a comma. A DHCP helper interface forwards BOOTP (Broadcast) DHCP requests that it receives from the specified DHCP servers.<br><br>Click <b>Add</b> to configure up to four DHCP helpers. (Beginning with Cisco IOS XE SD-WAN Release 17.2, for IPv6.) |

To save the feature template, click **Save**.

### Apply Access Lists

To apply a rewrite rule, access lists, and policers to a router interface, choose **ACL** and configure the following parameters:

**Table 268:**

| Parameter Name     | Description                                                                                                          |
|--------------------|----------------------------------------------------------------------------------------------------------------------|
| Ingress ACL – IPv4 | Click <b>On</b> and specify the name of the access list to apply to IPv4 packets being received on the interface.    |
| Egress ACL – IPv4  | Click <b>On</b> and specify the name of the access list to apply to IPv4 packets being transmitted on the interface. |
| Ingress Policer    | Click <b>On</b> and specify the name of the policer to apply to packets being received on the interface.             |

| Parameter Name | Description                                                                                                 |
|----------------|-------------------------------------------------------------------------------------------------------------|
| Egress Policer | Click <b>On</b> and specify the name of the policer to apply to packets being transmitted on the interface. |

To save the feature template, click **Save**.

### Configure VRRP

To have an interface run the Virtual Router Redundancy Protocol (VRRP), which allows multiple routers to share a common virtual IP address for default gateway redundancy, choose **VRRP**. Then click **Add New VRRP** and configure the following parameters:

**Table 269:**

| Parameter Name                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Group ID                       | Enter the virtual router ID, which is a numeric identifier of the virtual router. You can configure a maximum of 24 groups. <i>Range:</i> 1 through 255                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Priority                       | Enter the priority level of the router. The router with the highest priority is elected as the primary router. If two Cisco IOS XE Catalyst SD-WAN devices have the same priority, the one with the higher IP address is elected as the primary one. <i>Range:</i> 1 through 254<br><i>Default:</i> 100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Timer                          | Specify how often the primary VRRP router sends VRRP advertisement messages. If the subordinate routers miss three consecutive VRRP advertisements, they elect a new primary router. <i>Range:</i> 1 through 3600 seconds <i>Default:</i> 1 second                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Track OMP<br>Track Prefix List | By default, VRRP uses the state of the service (LAN) interface on which it is running to determine which Cisco IOS XE Catalyst SD-WAN device is the primary virtual router. If a Cisco IOS XE Catalyst SD-WAN device loses all its WAN control connections, the LAN interface still indicates that it is up even though the router is functionally unable to participate in VRRP. To take WAN side connectivity into account for VRRP, configure one of the following:<br><br>Track OMP—Click <b>On</b> for VRRP to track the Overlay Management Protocol (OMP) session running on the WAN connection. If the primary VRRP router loses all its OMP sessions, VRRP elects a new default gateway from those that have at least one active OMP session.<br><br>Track Prefix List—Track both the OMP session and a list of remote prefixes, which is defined in a prefix list configured on the local router. If the primary VRRP router loses all its OMP sessions, VRRP failover occurs as described for the Track OMP option. In addition, if reachability to all of the prefixes in the list is lost, VRRP failover occurs immediately, without waiting for the OMP hold timer to expire, thus minimizing the amount of overlay traffic is dropped while the Cisco IOS XE Catalyst SD-WAN device determines the primary VRRP router. |
| IP Address                     | Enter the IP address of the virtual router. This address must be different from the configured interface IP addresses of both the local Cisco IOS XE Catalyst SD-WAN device and the peer running VRRP.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

### Add ARP Table Entries

To configure static Address Resolution Protocol (ARP) table entries on the interface, choose **ARP**. Then click **Add New ARP** and configure the following parameters:

Table 270:

| Parameter Name | Description                                                                                          |
|----------------|------------------------------------------------------------------------------------------------------|
| IP Address     | Enter the IP address for the ARP entry in dotted decimal notation or as a fully qualified host name. |
| MAC Address    | Enter the MAC address in colon-separated hexadecimal notation.                                       |

To save the ARP configuration, click **Add**.

To save the feature template, click **Save**.

### Configure Other Interface Properties

To configure other interface properties, choose **Advanced** and configure the following properties:

Table 271:

| Parameter Name | Description                                                                                                                                                                                                                                                                                              |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TCP MSS        | Specify the maximum segment size (MSS) of TCP SYN packets passing through the Cisco IOS XE Catalyst SD-WAN device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. <i>Range:</i> 552 to 1460 bytes <i>Default:</i> None |
| ARP Timeout    | Specify how long it takes for a dynamically learned ARP entry to time out. <i>Range:</i> 0 through 2678400 seconds (744 hours) <i>Default:</i> 1200 (20 minutes)                                                                                                                                         |

To save the feature template, click **Save**.

## Configure SVI Interface Using a Configuration Group

### Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

### Procedure

- Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.
- Step 2** Create and configure the SVI Interface feature.
  - a. Configure the SVI Interface feature.

| Field                | Description                                                                                |
|----------------------|--------------------------------------------------------------------------------------------|
| Type                 | Choose a feature from the drop-down list.                                                  |
| Feature Name*        | Enter a name for the feature.                                                              |
| Description          | Enter a description of the feature. The description can contain any characters and spaces. |
| Associated VPN: VPN* | Choose a VPN.                                                                              |

- b. Configure basic configuration fields.

Table 272: Basic Configuration

| Field                                   | Description                                                                                                                                                                                                                                                   |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Shutdown                                | Enable or disable the VLAN interface.                                                                                                                                                                                                                         |
| VLAN Interface Name*                    | Enter a name for the VLAN interface.<br>The name must contain a minimum of five characters. The name must be in the following format:<br><code>^vlan ([1-9]\d \d) / {0,2} (0 [1-9]\d*) ([:\.\.][1-9]\d*) ?</code>                                             |
| Interface Description                   | Enter a description for the interface.                                                                                                                                                                                                                        |
| Interface MTU                           | Enter the maximum transmission unit size for frames received and transmitted on the interface.<br>Range: 1500 through 9216<br>Default: 1500 bytes                                                                                                             |
| IP MTU                                  | Enter the maximum transmission unit (MTU) size of IP packets sent on an interface.<br>Range: 576 through 9216<br>Default: 1500 bytes                                                                                                                          |
| <b>Configure IPV4 Address</b>           |                                                                                                                                                                                                                                                               |
| IPv4 Address Prefix*                    | Enter the IPv4 address for the interface.                                                                                                                                                                                                                     |
| List of DHCP helper addresses*          | Enter up to eight IP addresses for DHCP servers in the network to have the interface be a DHCP helper. Separate each address with a comma. A DHCP helper interface forwards BOOTP (Broadcast) DHCP requests that it receives from the specified DHCP servers. |
| <b>Configure IPV4 Secondary Address</b> |                                                                                                                                                                                                                                                               |
| Secondary IP Address*                   | Enter up to four secondary IP addresses.                                                                                                                                                                                                                      |
| <b>Configure IPV6 Address</b>           |                                                                                                                                                                                                                                                               |
| IPv6 address*                           | Enter the IPv6 address for the interface.                                                                                                                                                                                                                     |

| Field                                   | Description                                                                                                                                                                                                    |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Configure IPV6 Secondary Address</b> |                                                                                                                                                                                                                |
| <b>Address*</b>                         | Enter up to four secondary IP addresses.                                                                                                                                                                       |
| <b>Configure IPV6 DHCP Helper</b>       |                                                                                                                                                                                                                |
| <b>Address*</b>                         | Enter an IP address for DHCP servers in the network to have the interface be a DHCP helper. A DHCP helper interface forwards BOOTP (Broadcast) DHCP requests that it receives from the specified DHCP servers. |
| <b>VPN</b>                              | VPN ID for the DHCP helper address.                                                                                                                                                                            |

- c. Configure ACL fields.

*Table 273: ACL*

| Field                           | Description                                              |
|---------------------------------|----------------------------------------------------------|
| <b>Configure Access List V4</b> |                                                          |
| <b>Direction*</b>               | Choose a direction of the ACL: <b>in</b> or <b>out</b> . |
| <b>Name of ACL*</b>             | Enter the name of the access list.                       |
| <b>Configure Access List V6</b> |                                                          |
| <b>Direction*</b>               | Choose a direction of the ACL: <b>in</b> or <b>out</b> . |
| <b>Name of ACL*</b>             | Enter the name of the access list.                       |

- d. Configure VRRP fields.

*Table 274: VRRP*

| Field                 | Description                                                                                                                                                                                                                                                        |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Configure VRRP</b> |                                                                                                                                                                                                                                                                    |
| <b>Group ID*</b>      | Enter the virtual router ID, which is a numeric identifier of the virtual router. You can configure a maximum of 24 groups.<br>Range: 1 through 255                                                                                                                |
| <b>Priority*</b>      | Enter the priority level of the router. The router with the highest priority is elected as the primary router. If two routers have the same priority, the one with the higher IP address is elected as the primary router.<br>Range: 1 through 254<br>Default: 100 |

| Field                                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Timer*</b>                        | Specify how often the primary VRRP router sends VRRP advertisement messages. If secondary routers miss three consecutive VRRP advertisements, they elect a new primary router .<br><br>Range: 100 through 40950 seconds<br>Default: 100 seconds                                                                                                                                                                                                                                                                                                        |
| <b>Track OMP</b>                     | When you enable this option, VRRP tracks the Overlay Management Protocol (OMP) session running on the WAN connection. If the primary VRRP router loses all its OMP sessions, VRRP elects a new default gateway from those that have at least one active OMP session.                                                                                                                                                                                                                                                                                   |
| <b>Prefix List*</b>                  | Track both the OMP session and a list of remote prefixes, which is defined in a prefix list configured on the local router. If the primary VRRP router loses all its OMP sessions, VRRP failover occurs as described for the <b>Track OMP</b> option. In addition, if the reachability to one of the prefixes in the list is lost, VRRP failover occurs immediately, without waiting for the OMP hold timer to expire, thus minimizing the amount of overlay traffic while the Cisco IOS XE Catalyst SD-WAN device determines the primary VRRP router. |
| <b>IP Address</b>                    | Enter the IP address of the virtual router. This address must be different from the configured interface IP addresses of both the local router and the peer running VRRP.                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Add VRRP IP Address Secondary</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Address*</b>                      | Enter an IP address for the secondary VRRP router.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>TLOC Preference Change</b>        | Enable or disable this option to set whether the TLOC preference can be changed or not.                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Add VRRP Tracking Object</b>      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Tracker Id*</b>                   | Enter the interface object ID or object group tracker ID.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Track Action*</b>                 | Choose one of the options: <ul style="list-style-type: none"> <li>• <b>decrement</b></li> <li>• <b>shutdown</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Decrement Value</b>               | Enter a decrement value.<br><br>Range: 1-255<br><br>From Cisco vManage Release 20.10.1, this option is enabled only when you choose <b>decrement</b> in <b>Track Action</b> .                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Configure VRRP IPv6</b>           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Group ID*</b>                     | Enter the virtual router ID, which is a numeric identifier of the virtual router. You can configure a maximum of 24 groups.<br><br>Range: 1 through 255                                                                                                                                                                                                                                                                                                                                                                                                |



| Field                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Priority*</b>             | Enter the priority level of the router. The router with the highest priority is elected as the primary router. If two routers have the same priority, the one with the higher IP address is elected as the primary router.<br><br>Range: 1 through 254<br><br>Default: 100                                                                                                                                                                                                                                                                             |
| <b>Timer*</b>                | Specify how often the primary VRRP router sends VRRP advertisement messages. If secondary routers miss three consecutive VRRP advertisements, they elect a new primary router .<br><br>Range: 100 through 40950 seconds<br><br>Default: 100 seconds                                                                                                                                                                                                                                                                                                    |
| <b>Track OMP*</b>            | When you enable this option, VRRP tracks the Overlay Management Protocol (OMP) session running on the WAN connection. If the primary VRRP router loses all its OMP sessions, VRRP elects a new default gateway from those that have at least one active OMP session.                                                                                                                                                                                                                                                                                   |
| <b>Track Prefix List</b>     | Track both the OMP session and a list of remote prefixes, which is defined in a prefix list configured on the local router. If the primary VRRP router loses all its OMP sessions, VRRP failover occurs as described for the <b>Track OMP</b> option. In addition, if the reachability to one of the prefixes in the list is lost, VRRP failover occurs immediately, without waiting for the OMP hold timer to expire, thus minimizing the amount of overlay traffic while the Cisco IOS XE Catalyst SD-WAN device determines the primary VRRP router. |
| <b>Add VRRP IPv6 Primary</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>IPv6 Link Local*</b>      | Enter a virtual link local IPv6 address, which represents the link local address of the group. The address should be in standard link local address format. For example, FE80::AB8.                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Prefix</b>                | Enter the IPv6 address of the primary VRRP router.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

- e. Configure ARP fields.

**Table 275: ARP**

| Field                | Description                                                                                          |
|----------------------|------------------------------------------------------------------------------------------------------|
| <b>Configure ARP</b> |                                                                                                      |
| <b>IP Address*</b>   | Enter the IP address for the ARP entry in dotted decimal notation or as a fully qualified host name. |
| <b>MAC Address*</b>  | Enter the MAC address in colon-separated hexadecimal notation.                                       |

- f. Configure advanced fields.

Table 276: Advanced

| Field                               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>TCP MSS</b>                      | Specify the maximum segment size (MSS) of TCP SYN packets passing through the router. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented.<br><br>Range: 552 to 1960 bytes<br><br>Default: None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>ARP Timeout</b>                  | Specify how long it takes for a dynamically learned ARP entry to time out.<br><br>Range: 0 through 2678400 seconds (744 hours)<br><br>Default: 1200 (20 minutes)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>IP Directed-Broadcast</b>        | An IP directed broadcast is an IP packet whose destination address is a valid broadcast address for some IP subnet but which originates from a node that is not itself part of that destination subnet.<br><br>A device that is not directly connected to its destination subnet forwards an IP directed broadcast in the same way it would forward unicast IP packets destined to a host on that subnet. When a directed broadcast packet reaches a device that is directly connected to its destination subnet, that packet is broadcast on the destination subnet. The destination address in the IP header of the packet is rewritten to the configured IP broadcast address for the subnet, and the packet is sent as a link-layer broadcast.<br><br>If directed broadcast is enabled for an interface, incoming IP packets whose addresses identify them as directed broadcasts intended for the subnet to which that interface is attached are broadcast on that subnet. |
| <b>ICMP/ICMPv6 Redirect Disable</b> | ICMP redirects are sent by a router to the sender of an IP packet when a packet is being routed sub-optimally. The ICMP redirect informs the sending host to forward subsequent packets to that same destination through a different gateway.<br><br>By default, an interface allows ICMP redirect messages.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

**What to do next**

Also see [Deploy a configuration group](#).

## VPN Interface T1/E1

Use the VPN Interface T1/E1 template for Cisco Catalyst SD-WANs running the Cisco Catalyst SD-WAN software.

To configure the T1/E1 interfaces in a VPN using Cisco SD-WAN Manager templates:

1. Create a VPN Interface T1/E1 feature template to configure T1/E1 interface parameters, as described in this article.
2. Create a T1/E1 Controller template to configure the T1 or E1 network interface module (NIM) parameters.
3. Create a VPN feature template to configure VPN parameters.

### Navigate to the Template Screen and Name the Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**.



**Note** In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. From the **Create Template** drop-down list, select **From Feature Template**.
4. From the **Device Model** drop-down list, select the type of device for which you are creating the template.
5. To create a template for VPN 0 or VPN 512:



**Note** **Note:** Cisco IOS XE Catalyst SD-WAN devices use VRFs in place of VPNs. However, the following steps still apply to configure Cisco IOS XE Catalyst SD-WAN devices through Cisco SD-WAN Manager. When you complete the configuration, the system automatically maps the VPN configurations to VRF configurations.

- a. Click **Transport & Management VPN** or scroll to the **Transport & Management VPN** section.
  - b. Under **Additional VPN 0 Templates**, click **VPN Interface T1/E1 Serial**.
  - c. From the **VPN Interface T1/E1 Serial** drop-down list, click **Create Template**. The **VPN Interface T1/E1** template form is displayed. This form contains fields for naming the template, and fields for defining VPN Interface Ethernet parameters.
6. To create a template for VPNs 1 through 511, and 513 through 65527:
    - a. Click **Service VPN** or scroll to the **Service VPN** section.
    - b. Click the **Service VPN** drop-down list.
    - c. Under **Additional VPN** templates, click **VPN Interface**.
    - d. From the **VPN Interface** drop-down list, click **Create Template**. The **VPN Interface Ethernet** template form is displayed. This form contains fields for naming the template, and fields for defining VPN Interface Ethernet parameters.
  7. In **Template Name**, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.
  8. In **Template Description**, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default or to enter a value, click the scope drop-down to the left of the parameter field.

### Configure Basic Interface Functionality

To configure basic interface functionality in a VPN, select **Basic Configuration** and configure the following parameters. Parameters marked with an asterisk are required to configure an interface.

**Table 277:**

| Parameter Name       | Description                                                                                                                                                                                                                                  |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Shutdown*            | Click <b>No</b> to enable the interface.                                                                                                                                                                                                     |
| Interface name*      | Enter a name for the interface. The name should be in the format <b>serial slot / subslot / port : channel-group</b> .<br><br>You must also configure a number for the channel group in the T1/E1 Controller feature configuration template. |
| Description          | Enter a description for the interface.                                                                                                                                                                                                       |
| IPv4 Address*        | Enter an IPv4 address.                                                                                                                                                                                                                       |
| IPv6 Address*        | Enter an IPv6 address.                                                                                                                                                                                                                       |
| Bandwidth Upstream   | For transmitted traffic, set the bandwidth above which to generate notifications. <i>Range:</i> 1 through $(2^{32} / 2) - 1$ kbps                                                                                                            |
| Bandwidth Downstream | For received traffic, set the bandwidth above which to generate notifications. <i>Range:</i> 1 through $(2^{32} / 2) - 1$ kbps                                                                                                               |
| IP MTU               | Specify the maximum MTU size of packets on the interface. <i>Range:</i> 576 through 1804 <i>Default:</i> 1500 bytes                                                                                                                          |

### Create a Tunnel Interface

On Cisco IOS XE routers, you can configure up to eight tunnel interfaces. This means that each router can have up to eight TLOCs.

For the control plane to establish itself so that the overlay network can function, you must configure WAN transport interfaces in VPN 0.

To configure a tunnel interface for the multilink interface, select **Tunnel Interface** and configure the following parameters:

**Table 278:**

| Parameter Name   | Description                                   |
|------------------|-----------------------------------------------|
| Tunnel Interface | Click <b>On</b> to create a tunnel interface. |
| Color            | Select a color for the TLOC.                  |

| Parameter Name                                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Color Description                              | Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.18.1<br>Enter a description associated to the TLOC color.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Control Connection                             | <p>By default, Control Connection is set to <b>On</b>, which establishes a control connection for the TLOC. If the router has multiple TLOCs, click <b>No</b> to have the tunnel not establish control connection for the TLOC.</p> <p><b>Note</b><br/>We recommend a minimum of 650-700 Kbps bandwidth with default 1 sec hello-interval and 12 sec hello-tolerance parameters configured to avoid any data/packet loss in connection traffic.</p> <p>For each BFD session, an additional average sized BFD packet of 175 Bytes consumes 1.4 Kbps of bandwidth.</p> <p>A sample calculation of the required bandwidth for bidirectional BFD packet flow is given below:</p> <ul style="list-style-type: none"> <li>• 650 – 700 Kbps per device for control connections.</li> <li>• 175 Bytes (or 1.4 Kbps) per BFD session on the device (request)</li> <li>• 175 Bytes (or 1.4 Kbps) per BFD session on the device (response)</li> </ul> <p>If the path MTU discovery (PMTUD) is enabled, bandwidth for send/receive BFD packets per tunnel for every 30 secs:</p> <p>A 1500 Bytes BFD request packet is sent per tunnel every 30 secs:<br/> <math>1500 \text{ Bytes} * 8 \text{ bits/1 byte} * 1 \text{ packet} / 30 \text{ secs} = 400 \text{ bps (request)}</math></p> <p>A 147 Bytes BFD packet is sent in response:<br/> <math>147 \text{ Bytes} * 8 \text{ bits/1 byte} * 1 \text{ packet} / 30 \text{ secs} = 40 \text{ bps (response)}</math></p> <p>Therefore, a device with 775 BFD sessions (for example) requires a bandwidth of:<br/> <math>700k + (1.4k*775) + (400 * 775) + (1.4k*775) + (40 * 775) = \sim 3,5 \text{ MBps}</math></p> |
| Maximum Control Connections                    | <p>Specify the maximum number of Cisco Catalyst SD-WAN Controllers that the WAN tunnel interface can connect to. To have the tunnel establish no control connections, set the number to 0.</p> <p><i>Range: 0 through 8 Default: 2</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Cisco Catalyst SD-WAN Validator As STUN Server | Click <b>On</b> to enable Session Traversal Utilities for NAT (STUN) to allow the tunnel interface to discover its public IP address and port number when the router is located behind a NAT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Exclude Controller Group List                  | Set the Cisco Catalyst SD-WAN Controllers that the tunnel interface is not allowed to connect to. <i>Range: 0 through 100</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Cisco SD-WAN Manager Connection Preference     | Set the preference for using a tunnel interface to exchange control traffic with the Cisco SD-WAN Manager NMS. <i>Range: 0 through 8 Default: 5</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| Parameter Name      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Full Port Hop       | <p>Minimum release: Cisco IOS XE Catalyst SD-WAN Release 17.18.1a</p> <p>Enable full port hopping at the TLOC level to allow devices to establish connections with controllers by switching to the next port if the current port is blocked or non-functional.</p> <p>Default: Disabled</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Port Hop            | <p>Click <b>On</b> to enable port hopping, or click <b>Off</b> to disable it. When a router is behind a NAT, port hopping rotates through a pool of preselected OMP port numbers (called base ports) to establish DTLS connections with other routers when a connection attempt is unsuccessful. The default base ports are 12346, 12366, 12386, 12406, and 12426. To modify the base ports, set a port offset value. <i>Default: Enabled</i></p> <p>Starting from Cisco IOS XE Catalyst SD-WAN Release 17.18.1a, this field is deprecated. Instead use the <b>Full Port Hop</b> option. See the <b>Full Port Hop</b> field.</p>                                                                                                                                                                                                                                       |
| Low-Bandwidth Link  | Select to characterize the tunnel interface as a low-bandwidth link.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Tunnel TCP MSS      | <p>TCP MSS affects any packet that contains an initial TCP header that flows through the router. When configured, TCP MSS is examined against the MSS exchanged in the three-way handshake. The MSS in the header is lowered if the configured TCP MSS setting is lower than the MSS in the header. If the MSS header value is already lower than the TCP MSS, the packets flow through unmodified. The host at the end of the tunnel uses the lower setting of the two hosts. If the TCP MSS is to be configured, it should be set at 40 bytes lower than the minimum path MTU.</p> <p>Specify the MSS of TPC SYN packets passing through the Cisco IOS XE Catalyst SD-WAN device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. <i>Range: 552 to 1460 bytes Default: None</i></p> |
| Clear-Dont-Fragment | <p>Configure <b>Clear-Dont-Fragment</b> for packets that arrive at an interface that has <b>Dont Fragment</b> configured. If these packets are larger than what MTU allows, they are dropped. If you clear the Don't Fragment bit, the packets are fragmented and sent.</p> <p>Click <b>On</b> to clear the <b>Dont Fragment</b> bit in the IPv4 packet header for packets being transmitted out of the interface. When the <b>Dont Fragment</b> bit is cleared, packets larger than the MTU of the interface are fragmented before being sent.</p> <p><b>Note</b><br/> <b>Clear-Dont-Fragment</b> clears the <b>Dont Fragment</b> bit and the <b>Dont Fragment</b> bit is set. For packets not requiring fragmentation, the <b>Dont Fragment</b> bit is not affected.</p>                                                                                             |
| Allow Service       | Select <b>On</b> or <b>Off</b> for each service to allow or disallow the service on the interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

To save the feature template, click **Save**.

### Release Information

Introduced in Cisco vManage Release 18.2.

## T1/E1 Controller

Use the T1/E1 Controller template for Cisco IOS XE Catalyst SD-WAN devices running the Cisco Catalyst SD-WAN software.

To configure the T1/E1 interfaces in a VPN using Cisco SD-WAN Manager templates:

1. Create a T1/E1 Controller template to configure the T1 or E1 network interface module (NIM) parameters, as described in this article.
2. Create a VPN Interface T1/E1 feature template to configure T1/E1 interface parameters.
3. Create a VPN feature template to configure VPN parameters.

### Navigate to the Template Screen and Name the Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**.



---

**Note** In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

---

3. From the **Create Template** drop-down list, select **From Feature Template**.
4. From the **Device Model** drop-down list, select the type of device for which you are creating the template.
5. To create a template for VPN 0 or VPN 512:
  - a. Click **Transport & Management VPN** or scroll to the **Transport & Management VPN** section.
  - b. Under **Additional VPN 0 Templates**, click **VPN Interface**.
  - c. From the **VPN Interface** drop-down list, click **Create Template**. The **VPN Interface T1/E1** template form is displayed. This form contains fields for naming the template, and fields for defining VPN Interface Ethernet parameters.
6. To create a template for VPNs 1 through 511, and 513 through 65527:
  - a. Click **Service VPN** or scroll to the **Service VPN** section.
  - b. Click the **Service VPN** drop-down list.
  - c. Under **Additional VPN** templates, click **VPN Interface**.
  - d. From the **VPN Interface** drop-down list, click **Create Template**. The VPN Interface Ethernet template form is displayed. This form contains fields for naming the template, and fields for defining VPN Interface Ethernet parameters.
7. In **Template Name**, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.
8. In **Template Description**, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default or to enter a value, click the **Scope** drop-down list and select one of the following:

- Device Specific (indicated by a host icon)
- Global (indicated by a globe icon)

### Configure a T1 Controller

To configure a T1 controller, click **T1** and configure the following parameters. Parameters marked with an asterisk are required to configure an interface.

**Table 279:**

| Parameter Name | Description                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Slot*          | Enter the number of the slot in slot/subslot/port format, where the T1 NIM is installed. For example, 0/1/0.                                                                                                                                                                                                                                                                                                        |
| Framing*       | Enter the T1 frame type: <ul style="list-style-type: none"> <li>• <b>esf</b>—Send T1 frames as extended superframes. This is the default.</li> <li>• <b>sf</b>—Send T1 frames as superframes. Superframing is sometimes called D4 framing.</li> </ul>                                                                                                                                                               |
| Line Code      | Select the line encoding to use to send T1 frames: <ul style="list-style-type: none"> <li>• <b>ami</b>—Use alternate mark inversion (AMI) as the linecode. AMI signaling uses frames grouped into superframes.</li> <li>• <b>b8zs</b>—Use bipolar 8-zero substitution as the linecode. This is the default. B8ZS uses frames that are grouping into extended superframes</li> </ul>                                 |
| Clock Source   | Select the clock source: <ul style="list-style-type: none"> <li>• <b>internal</b>—Use the controller framer as the primary clock.</li> <li>• <b>line</b>—Use phase-locked loop (PLL) on the interface. This is the default. When both T1 ports use line clocking and neither port is configured as the primary, by default, port 0 is the primary clock source and port 1 is the secondary clock source.</li> </ul> |
| Line Mode      | If you choose the Line clock source, select whether the line is a primary or a secondary line.                                                                                                                                                                                                                                                                                                                      |
| Description    | Enter a description for the controller.                                                                                                                                                                                                                                                                                                                                                                             |
| Channel Group  | Enter the number of the channel group. If you do so, you must enter a time slot in the Time Slot field. <i>Range:</i> 0 through 30                                                                                                                                                                                                                                                                                  |
| Time Slot      | Enter the time slot or time slots that are part of the channel group. <i>Range:</i> 1 through 24                                                                                                                                                                                                                                                                                                                    |



| Parameter Name | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cable Length   | <p>Select the cable length to configure the attenuation</p> <ul style="list-style-type: none"> <li>• long—Attenuate the pulse from the transmitter using pulse equalization and line buildout. You can configure a long cable length for cables longer than 660 feet.</li> <li>• short—Set the transmission attenuation for cables that are 660 feet or shorter.</li> </ul> <p>There is no default length.</p>                                                                                                                                                                                                                                               |
| Length         | <p>If you specify a value in the <b>Cable Length Field</b>, enter the length of the cable.</p> <p>For short cables, the length values can be:</p> <ul style="list-style-type: none"> <li>• 110—Length from 0 through 110 feet</li> <li>• 220—Length from 111 through 220 feet</li> <li>• 330—Length from 221 through 330 feet</li> <li>• 440—Length from 331 through 440 feet</li> <li>• 550—Length from 441 through 550 feet</li> <li>• 660—Length from 551 through 660 feet</li> </ul> <p>For long cables, the length values can be:</p> <ul style="list-style-type: none"> <li>• 0 dB</li> <li>• -7.5 dB</li> <li>• -15 dB</li> <li>• -22.5 dB</li> </ul> |

To save the feature template, click **Save**.

### Configure an E1 Controller

To configure an E1 controller, click **E1** and configure the following parameters. Parameters marked with an asterisk are required to configure an interface.

**Table 280:**

| Parameter Name | Description                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Slot*          | Enter the number of the slot in slot/subslot/port format, where the E1 NIM is installed. For example, 0/1/0.                                                                                          |
| Framing*       | <p>Enter the E1 frame type:</p> <ul style="list-style-type: none"> <li>• <b>crc4</b>—Use cyclic redundancy check 4 (CRC4). This is the default.</li> <li>• <b>no-crc4</b>—Do not use CRC4.</li> </ul> |

| Parameter Name | Description                                                                                                                                                                                                                                                                                                                                         |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Line Code*     | Select the line encoding to use to send E1 frames: <ul style="list-style-type: none"> <li>• ami—Use alternate mark inversion (AMI) as the linecode.</li> <li>• hdb3—Use high-density bipolar 3 as the linecode. This is the default.</li> </ul>                                                                                                     |
| Clock Source   | Select the clock source: <ul style="list-style-type: none"> <li>• internal—Use the controller framer as the primary clock.</li> <li>• line—Use phase-locked loop (PLL) on the interface. This is the default.</li> </ul>                                                                                                                            |
| Line Mode      | If you choose the Line clock source, select whether the line is a primary or secondary line. If you configure both a primary and a secondary line, if the primary line fails, the PLL automatically switches to the secondary line. When the PLL on the primary line becomes active again, the PLL automatically switches back to the primary line. |
| Description    | Enter a description for the controller.                                                                                                                                                                                                                                                                                                             |
| Channel Group  | To configure the serial WAN on the E1 interface, enter a channel group number. <i>Range:</i> 0 through 30                                                                                                                                                                                                                                           |
| Time Slot      | For a channel group, configure the timeslot. <i>Range:</i> 1 through 31                                                                                                                                                                                                                                                                             |

To save the feature template, click **Save**.

#### Release Information

Introduced in Cisco vManage Release 18.1.1.

## Configure T1 or E1 Controller Using a Configuration Group

#### Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

#### Procedure

- 
- Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.
- Step 2** Create and configure the T1 or E1 network interface module (NIM) parameters in a Transport and Management Profile.
- Configure a T1 Controller.
- Parameters marked with an asterisk are mandatory.

Table 281: Configure a T1 Controller

| Parameter Name | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Slot*          | Enter the number of the slot in slot/subslot/port format, where the T1 NIM is installed. For example, 0/1/0.                                                                                                                                                                                                                                                                                                                                                                   |
| Description    | Enter a description for the controller.                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Framing        | It is an optional field. Enter the T1 frame type: <ul style="list-style-type: none"> <li>• <b>esf</b>: Send T1 frames as extended superframes. This is the default.</li> <li>• <b>sf</b>: Send T1 frames as superframes. Superframing is sometimes called D4 framing.</li> </ul>                                                                                                                                                                                               |
| Line Code      | It is an optional field. Select the line encoding to use to send T1 frames: <ul style="list-style-type: none"> <li>• <b>ami</b>: Use alternate mark inversion (AMI) as the linecode. AMI signaling uses frames grouped into superframes.</li> <li>• <b>b8zs</b>: Use bipolar 8-zero substitution as the linecode. This is the default. B8ZS uses frames that are grouping into extended superframes</li> </ul>                                                                 |
| Cable Length   | Select the cable length to configure the attenuation <ul style="list-style-type: none"> <li>• <b>short</b>: Set the transmission attenuation for cables that are 660 feet or shorter.</li> <li>• <b>long</b>: Attenuate the pulse from the transmitter using pulse equalization and line buildout. You can configure a long cable length for cables longer than 660 feet.</li> </ul> <p>There is no default length.</p>                                                        |
| Clock Source   | Select the clock source: <ul style="list-style-type: none"> <li>• <b>line</b>: Use phase-locked loop (PLL) on the interface. This is the default. When both T1 ports use line clocking and neither port is configured as the primary, by default, port 0 is the primary clock source and port 1 is the secondary clock source.</li> <li>• <b>internal</b>: Use the controller framer as the primary clock.</li> <li>• <b>loop-timed</b>:</li> <li>• <b>network</b>:</li> </ul> |

## b. Configure an E1 Controller.

Parameters marked with an asterisk are mandatory.

Table 282: Configure an E1 Controller

| Parameter Name | Description                                                                                                  |
|----------------|--------------------------------------------------------------------------------------------------------------|
| Slot*          | Enter the number of the slot in slot/subslot/port format, where the E1 NIM is installed. For example, 0/1/0. |

| Parameter Name | Description                                                                                                                                                                                                                                                     |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description    | Enter a description for the controller.                                                                                                                                                                                                                         |
| Framing        | Enter the E1 frame type: <ul style="list-style-type: none"> <li>• <b>crc4</b>: Use cyclic redundancy check 4 (CRC4). This is the default.</li> <li>• <b>no-crc4</b>: Do not use CRC4.</li> </ul>                                                                |
| Line Code      | Choose the line encoding to use to send E1 frames: <ul style="list-style-type: none"> <li>• <b>ami</b>: Use alternate mark inversion (AMI) as the linecode.</li> <li>• <b>hdb3</b>: Use high-density bipolar 3 as the linecode. This is the default.</li> </ul> |
| Clock Source   | Choose the clock source: <ul style="list-style-type: none"> <li>• <b>internal</b>: Use the controller framer as the primary clock.</li> <li>• <b>line</b>: Use phase-locked loop (PLL) on the interface. This is the default.</li> </ul>                        |

c. Configure Channel Group.

*Table 283: Channel Group*

| Parameter Name    | Description                                                                                                                                                                                                                                                                                                                            |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Add Channel Group | To configure the serial WAN on the E1 interface, enter a channel group number and a value for the timeslot. <ul style="list-style-type: none"> <li>• <b>Channel Group</b>: Enter a value for the channel group.<br/>Range: 0 through 30</li> <li>• <b>Time Slot</b>: Type a value for the timeslot.<br/>Range: 0 through 31</li> </ul> |

### What to do next

Also see [Deploy a configuration group](#).

## Cellular Interfaces

To enable LTE connectivity, configure cellular interfaces on a router that has a cellular module. The cellular module provides wireless connectivity over a service provider's cellular network. One use case is to provide wireless connectivity for branch offices.

A cellular network is commonly used as a backup WAN link, to provide network connectivity if all the wired WAN tunnel interfaces on the router become unavailable. You can also use a cellular network as the primary WAN link for a branch office, depending on usage patterns within the branch office and the data rates supported by the core of the service provider's cellular network.

When you configure a cellular interface on a device, you can connect the device to the Internet or another WAN by plugging in the power cable of the device. The device then automatically begins the process of joining the overlay network, by contacting and authenticating with Cisco Catalyst SD-WAN Validators, Cisco Catalyst SD-WAN Controllers, and Cisco SD-WAN Manager systems.

## Configure Cellular Interfaces Using a Configuration Groups

### Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

### Procedure

**Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

**Step 2** Create and configure a Transport VPN feature in Transport and Management profile.

**Step 3** Create and configure Cellular Interface feature in Transport VPN.

a) Configure basic cellular interface functionality.

| Field                  | Description                                                                                                                                                                                                                                    |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Shutdown*</b>       | Enable or disable the interface.                                                                                                                                                                                                               |
| <b>Interface Name*</b> | Enter the name of the interface.                                                                                                                                                                                                               |
| <b>Description*</b>    | Enter a description of the cellular interface.                                                                                                                                                                                                 |
| <b>DHCP Helper</b>     | Enter up to four IP addresses for DHCP servers in the network, separated by commas, to have the interface be a DHCP helper. A DHCP helper interface forwards BOOTP (Broadcast) DHCP requests that it receives from the specified DHCP servers. |

b) Configure a tunnel interface.

| Field                   | Description                                                                                                                                                                                                        |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tunnel Interface</b> | Enable this option to create a tunnel interface.                                                                                                                                                                   |
| <b>Carrier</b>          | Choose the carrier name or private network identifier to associate with the tunnel.<br><br>Values: carrier1, carrier2, carrier3, carrier4, carrier5, carrier6, carrier7, carrier8, default<br><br>Default: default |
| <b>Color</b>            | Choose a color for the TLOC.                                                                                                                                                                                       |

| Field                                | Description                                                                                                                                                                                                                                                     |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Color Description                    | Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.18.1<br>Enter a description associated to the TLOC color.                                                                                                                                   |
| <b>Hello Interval</b>                | Enter the interval between Hello packets sent on a DTLS or TLS WAN transport connection.<br><br>Range: 100 through 600000 milliseconds<br>Default: 1000 milliseconds (1 second)                                                                                 |
| <b>Hello Tolerance</b>               | Enter the time to wait for a Hello packet on a DTLS or TLS WAN transport connection before declaring that transport tunnel to be down.<br><br>Range: 12 through 6000 seconds<br>Default: 12 seconds                                                             |
| <b>Last-Resort Circuit</b>           | Enable this option to use the tunnel interface as the circuit of last resort.                                                                                                                                                                                   |
| <b>Restrict</b>                      | Enable this option to limit the remote TLOCs that the local TLOC can establish BFD sessions with. When a TLOC is marked as restricted, a TLOC on the local router establishes tunnel connections with a remote TLOC only if the remote TLOC has the same color. |
| <b>Group</b>                         | Enter a group number.<br><br>Range: 1 through 4294967295                                                                                                                                                                                                        |
| <b>Border</b>                        | Enable this option to set the TLOC as a border TLOC.                                                                                                                                                                                                            |
| <b>Maximum Control Connections</b>   | Specify the maximum number of Cisco SD-WAN Controllers that the WAN tunnel interface can connect to. To have the tunnel establish no control connections, set the number to 0.<br><br>Range: 0 through 100<br>Default: 2                                        |
| <b>NAT Refresh Interval</b>          | Enter the interval between NAT refresh packets sent on a DTLS or TLS WAN transport connection.<br><br>Range: 1 through 60 seconds<br>Default: 5 seconds                                                                                                         |
| <b>Validator As Stun Server</b>      | Enable Session Traversal Utilities for NAT (STUN) to allow the tunnel interface to discover its public IP address and port number when the Cisco IOS XE Catalyst SD-WAN device is located behind a NAT.                                                         |
| <b>Exclude Controller Group List</b> | Set the identifiers of one or more Cisco SD-WAN Controller groups that this tunnel is not allowed to connect to.<br><br>Range: 1 through 100                                                                                                                    |

| Field                                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Manager Connection Preference</b> | Set the preference for using a tunnel interface to exchange control traffic with Cisco SD-WAN Manager.<br><br>Range: 0 through 8<br><br>Default: 5                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Full Port Hop</b>                 | Minimum release: Cisco IOS XE Catalyst SD-WAN Release 17.18.1a<br><br>Enable full port hopping at the TLOC level to allow devices to establish connections with controllers by switching to the next port if the current port is blocked or non-functional.<br><br>Default: Disabled                                                                                                                                                                                                                                                                                |
| <b>Port Hop</b>                      | Enable port hopping. When a router is behind a NAT, port hopping rotates through a pool of preselected OMP port numbers (called base ports) to establish DTLS connections with other routers when a connection attempt is unsuccessful. The default base ports are 12346, 12366, 12386, 12406, and 12426. To modify the base ports, set a port offset value.<br><br>Default: Enabled<br><br>Starting from Cisco IOS XE Catalyst SD-WAN Release 17.18.1a, this field is deprecated. Instead use the <b>Full Port Hop</b> option. See the <b>Full Port Hop</b> field. |
| <b>Low-Bandwidth Link</b>            | Enable the low-bandwidth feature.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Tunnel TCP MSS</b>                | Specify the maximum segment size (MSS) of TCP SYN packets passing through the router. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented.<br><br>Range: 500 to 1460 bytes<br><br>Default: None                                                                                                                                                                                                                                                                                         |
| <b>Clear-Dont-Fragment</b>           | Enable this option to clear the Don't Fragment (DF) bit in the IPv4 packet header for packets being transmitted out the interface. When the DF bit is cleared, packets larger than the MTU of the interface are fragmented before being sent.                                                                                                                                                                                                                                                                                                                       |
| <b>Network Broadcast</b>             | Enable this option to accept and respond to network-prefix-directed broadcasts.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| Field                 | Description                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Allow Service</b>  | <p>Allow or disallow the following services on the interface:</p> <ul style="list-style-type: none"> <li>• <b>All</b></li> <li>• <b>BGP</b></li> <li>• <b>DHCP</b></li> <li>• <b>NTP</b></li> <li>• <b>SSH</b></li> <li>• <b>DNS</b></li> <li>• <b>ICMP</b></li> <li>• <b>HTTPS</b></li> <li>• <b>OSPF</b></li> <li>• <b>STUN</b></li> <li>• <b>SNMP</b></li> <li>• <b>NETCONF</b></li> <li>• <b>BFD</b></li> </ul> |
| <b>Encapsulation</b>  |                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>GRE</b>            | <p>Use GRE encapsulation on the tunnel interface. By default, GRE is disabled.</p> <p>If you select both IPsec and GRE encapsulations, two TLOCs are created for the tunnel interface that have the same IP addresses and colors, but that differ by their encapsulation.</p>                                                                                                                                       |
| <b>GRE Preference</b> | <p>Specify a preference value for directing traffic to the tunnel. A higher value is preferred over a lower value.</p> <p>Range: 0 through 4294967295</p> <p>Default: 0</p>                                                                                                                                                                                                                                         |
| <b>GRE Weight</b>     | <p>Enter a weight to use to balance traffic across multiple TLOCs. A higher value sends more traffic to the tunnel.</p> <p>Range: 1 through 255</p> <p>Default: 1</p>                                                                                                                                                                                                                                               |
| <b>IPsec</b>          | <p>Use IPsec encapsulation on the tunnel interface. By default, IPsec is enabled.</p> <p>If you select both IPsec and GRE encapsulations, two TLOCs are created for the tunnel interface that have the same IP addresses and colors, but that differ by their encapsulation.</p>                                                                                                                                    |



| Field                   | Description                                                                                                                                                  |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IPsec Preference</b> | Specify a preference value for directing traffic to the tunnel. A higher value is preferred over a lower value.<br>Range: 0 through 4294967295<br>Default: 0 |
| <b>IPsec Weight</b>     | Enter a weight to use to balance traffic across multiple TLOCs. A higher value sends more traffic to the tunnel.<br>Range: 1 through 255<br>Default: 1       |

- c) Configure a cellular interface to act as a NAT device for applications such as port forwarding.

| Field               | Description                                                                                                                |
|---------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>NAT</b>          | Enable this option to have the interface act as a NAT device.                                                              |
| <b>UDP Timeout*</b> | Specify when NAT translations over UDP sessions time out.<br>Range: 1 through 8947 minutes<br>Default: 1 minutes           |
| <b>TCP Timeout*</b> | Specify when NAT translations over TCP sessions time out.<br>Range: 1 through 8947 minutes<br>Default: 60 minutes (1 hour) |

- d) Configure static Address Resolution Protocol (ARP) table entries on the interface.

| Field               | Description                                                                                          |
|---------------------|------------------------------------------------------------------------------------------------------|
| <b>IP Address*</b>  | Enter the IP address for the ARP entry in dotted decimal notation or as a fully qualified host name. |
| <b>MAC Address*</b> | Enter the MAC address in colon-separated hexadecimal notation.                                       |

- e) Configure other interface properties.

| Field              | Description                                                                                                 |
|--------------------|-------------------------------------------------------------------------------------------------------------|
| <b>MAC Address</b> | Specify a MAC address to associate with the interface, in colon-separated hexadecimal notation.             |
| <b>IP MTU</b>      | Specify the maximum MTU size of packets on the interface.<br>Range: 576 through 9216<br>Default: 1500 bytes |

| Field                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Interface MTU</b>  | <p>Enter the maximum transmission unit size for frames received and transmitted on the interface.</p> <p>Range: 1500 through 9216</p> <p>Default: 1500 bytes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>TCP MSS</b>        | <p>Specify the maximum segment size (MSS) of TPC SYN packets passing through the router. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented.</p> <p>Range: 500 to 1460 bytes</p> <p>Default: None</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>TLOC Extension</b> | <p>Enter the name of a physical interface on the same router that connects to the WAN transport. This configuration then binds this service-side interface to the WAN transport. A second router at the same site that itself has no direct connection to the WAN (generally because the site has only a single WAN connection) and that connects to this service-side interface is then provided with a connection to the WAN.</p> <p><b>Note</b><br/>TLOC extension over L3 is supported only for Cisco IOS XE Catalyst SD-WAN devices. If configuring TLOC extension over L3 for a Cisco IOS XE Catalyst SD-WAN device, enter the IP address of the L3 interface.</p>                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Tracker</b>        | <p>Tracking the interface status is useful when you enable NAT on a transport interface in VPN 0 to allow data traffic from the router to exit directly to the internet rather than having to first go to a router in a data center. In this situation, enabling NAT on the transport interface splits the TLOC between the local router and the data center into two, with one going to the remote router and the other going to the internet.</p> <p>When you enable transport tunnel tracking, Cisco Catalyst SD-WAN periodically probes the path to the internet to determine whether it is up. If Cisco Catalyst SD-WAN detects that this path is down, it withdraws the route to the internet destination, and traffic destined to the internet is then routed through the data center router. When Cisco Catalyst SD-WAN detects that the path to the internet is again functioning, the route to the internet is reinstalled.</p> <p>Enter the name of a tracker to track the status of transport interfaces that connect to the internet.</p> |

| Field                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IP Directed-Broadcast</b> | <p>An IP directed broadcast is an IP packet whose destination address is a valid broadcast address for some IP subnet but which originates from a node that is not itself part of that destination subnet.</p> <p>A device that is not directly connected to its destination subnet forwards an IP directed broadcast in the same way it would forward unicast IP packets destined to a host on that subnet. When a directed broadcast packet reaches a device that is directly connected to its destination subnet, that packet is broadcast on the destination subnet. The destination address in the IP header of the packet is rewritten to the configured IP broadcast address for the subnet, and the packet is sent as a link-layer broadcast.</p> <p>If directed broadcast is enabled for an interface, incoming IP packets whose addresses identify them as directed broadcasts intended for the subnet to which that interface is attached are broadcast on that subnet.</p> |

### What to do next

Also see [Deploy a Configuration Group](#).

## Configure Cellular Interfaces Using Templates

To configure cellular interfaces using Cisco SD-WAN Manager templates:

1. Create a VPN Interface Cellular feature template to configure cellular module parameters, as described in this section.
2. Create a Cellular Profile template to configure the profiles used by the cellular modem.
3. Create a VPN feature template to configure VPN parameters.



**Note** If your deployment includes devices with cellular interface, you must include cellular controller templates in Cisco SD-WAN Manager, even if these templates are not used.

If the device has the LTE or cellular controller module configured and the cellular controller feature template does not exist, then the device tries to remove the cellular controller template. For releases earlier than Cisco IOS XE Release 17.4.2, the following error message is displayed.

```
bad-cli - No controller Cellular 0/2/0, parser-context - No controller Cellular 0/2/0,
parser-response % Cannot remove controllers this way
```

For devices running on Cisco IOS XE Release 17.4.2 and later, the device will return an access-denied error message.

### Create VPN Interface Cellular

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.

2. Click **Device Templates**.



**Note** In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. From the **Create Template** drop-down list, choose **From Feature Template**.
4. From the **Device Model** drop-down list, select the type of device for which you are creating the template.
5. Click **Transport & Management VPN** or scroll to the **Transport & Management VPN** section.
6. Under **Additional Cisco VPN 0 Templates**, click **VPN Interface Cellular**.
7. From the **VPN Interface Cellular** drop-down list, click **Create Template**. The VPN Interface Cellular template form is displayed.  
  
This form contains fields for naming the template, and fields for defining the VPN Interface Cellular parameters.
8. In **Template Name**, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.
9. In **Template Description**, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default or to enter a value, click the scope drop-down list.

### Configure Basic Cellular Interface Functionality

To configure basic cellular interface functionality, click **Basic Configuration** and configure the following parameters. Parameters marked with an asterisk are required to configure an interface. You must also configure a tunnel interface for the cellular interface.

**Table 284:**

| Parameter Name       | Description                                                                                                                                                                                                                                    |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Shutdown*            | Click <b>No</b> to enable the interface.                                                                                                                                                                                                       |
| Interface Name*      | Enter the name of the interface. It must be <b>cellular0</b> .                                                                                                                                                                                 |
| Description          | Enter a description of the cellular interface.                                                                                                                                                                                                 |
| DHCP Helper          | Enter up to four IP addresses for DHCP servers in the network, separated by commas, to have the interface be a DHCP helper. A DHCP helper interface forwards BOOTP (Broadcast) DHCP requests that it receives from the specified DHCP servers. |
| Bandwidth Upstream   | For transmitted traffic, set the bandwidth above which to generate notifications. Range: 1 through $(2^{32} / 2) - 1$ kbps                                                                                                                     |
| Bandwidth Downstream | For received traffic, set the bandwidth above which to generate notifications. Range: 1 through $(2^{32} / 2) - 1$ kbps                                                                                                                        |

| Parameter Name | Description                                                                                          |
|----------------|------------------------------------------------------------------------------------------------------|
| IP MTU*        | Enter 1428 to set the MTU size, in bytes. This value must be 1428. You cannot use a different value. |

To save the feature template, click **Save**.

### Create a Tunnel Interface

To configure an interface in VPN 0 to be a WAN transport connection, you must configure a tunnel interface on the cellular interface. The tunnel, which provides security from attacks, is used to send the phone number. At a minimum, select **On** and select a color for the interface, as described in the previous section. You can generally accept the system defaults for the remainder of the tunnel interface settings.

To configure a tunnel interface, click **Tunnel**, and configure the following parameters. Parameters marked with an asterisk (\*) are required to configure a cellular interface.

| Parameter Name              | Description                                                                                                                                                                                                            |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tunnel Interface*           | From the drop-down, select <b>Global</b> . Click <b>On</b> to create a tunnel interface.                                                                                                                               |
| Per-tunnel QoS              | From the drop-down, select <b>Global</b> . Click <b>On</b> to create per-tunnel QoS. You can apply a Quality of Service (QoS) policy on individual tunnels, and is only supported for hub-to-spoke network topologies. |
| Per-tunnel QoS Aggregator   | From the drop-down, select <b>Global</b> . Click <b>On</b> to create per-tunnel QoS.<br><b>Note</b><br>'bandwidth downstream' is required for per-Tunnel QoS feature to take effect as spoke role.                     |
| Color*                      | From the drop-down, select <b>Global</b> . Select a color for the TLOC. The color typically used for cellular interface tunnels is <b>lte</b> .                                                                        |
| Color Description           | Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.18.1<br>Enter a description associated to the TLOC color.                                                                                          |
| Groups                      | From the drop-down, select <b>Global</b> . Enter the list of groups in the field.                                                                                                                                      |
| Border                      | From the drop-down, select <b>Global</b> . Click <b>On</b> to set TLOC as border TLOC.                                                                                                                                 |
| Maximum Control Connections | Set the maximum number of Cisco SD-WAN Controller that the WAN tunnel interface can connect to. To have the tunnel establish no control connections, set the number to 0. Range: 0 through 8<br>Default: 2             |
| vBond As STUN Server        | Click <b>On</b> to enable Session Traversal Utilities for NAT (STUN) to allow the tunnel interface to discover its public IP address and port number when the router is located behind a NAT.                          |

| Parameter Name                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Exclude Control Group List    | <p>Set the identifiers of one or more Cisco SD-WAN Controller groups that this tunnel is not allows to establish control connections with.</p> <p>Range: 0 through 100</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| vManage Connection Preference | <p>Set the preference for using the tunnel to exchange control traffic with the Cisco SD-WAN Manager.</p> <p>Range: 0 through 9</p> <p>Default: 5</p> <p>If the edge device has two or more cellular interfaces, you can minimize the amount of traffic between the Cisco SD-WAN Manager and the cellular interfaces by setting one of the interfaces to be the preferred one to use when sending updates to the Cisco SD-WAN Manager and receiving configurations from the Cisco SD-WAN Manager.</p> <p>To have a tunnel interface never connect to the Cisco SD-WAN Manager, set the number to 0. At least one tunnel interface on the edge device must have a nonzero Cisco SD-WAN Manager connection preference.</p>                                                                                                                                         |
| Full Port Hop                 | <p>Minimum release: Cisco IOS XE Catalyst SD-WAN Release 17.18.1a</p> <p>Enable full port hopping at the TLOC level to allow devices to establish connections with controllers by switching to the next port if the current port is blocked or non-functional.</p> <p>Default: Disabled</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Port Hop                      | <p>From the drop-down, select <b>Global</b>. Click <b>Off</b> to allow port hopping on tunnel interface.</p> <p>Default: <b>On</b>, which disallows port hopping on tunnel interface.</p> <p>Starting from Cisco IOS XE Catalyst SD-WAN Release 17.18.1a, this field is deprecated. Instead use the <b>Full Port Hop</b> option. See the <b>Full Port Hop</b> field.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Low-Bandwidth Link            | <p>Click <b>On</b> to set the tunnel interface as a low-bandwidth link.</p> <p>Default: <b>Off</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Tunnel TCP MSS                | <p>TCP MSS affects any packet that contains an initial TCP header that flows through the router. When configured, TCP MSS is examined against the MSS exchanged in the three-way handshake. The MSS in the header is lowered if the configured TCP MSS setting is lower than the MSS in the header. If the MSS header value is already lower than the TCP MSS, the packets flow through unmodified. The host at the end of the tunnel uses the lower setting of the two hosts. If the TCP MSS is to be configured, it should be set at 40 bytes lower than the minimum path MTU.</p> <p>Specify the MSS of TPC SYN packets passing through the Cisco IOS XE Catalyst SD-WAN device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. Range: 552 to 1460 bytes. Default: None</p> |

| Parameter Name      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Clear-Dont-Fragment | <p>Configure <b>Clear-Dont-Fragment</b> for packets that arrive at an interface that has Don't Fragment configured. If these packets are larger than what MTU allows, they are dropped. If you clear the Don't Fragment bit, the packets are fragmented and sent.</p> <p>Click <b>On</b> to clear the Dont Fragment bit in the IPv4 packet header for packets being transmitted out of the interface. When the Dont Fragment bit is cleared, packets larger than the MTU of the interface are fragmented before being sent.</p> <p><b>Note</b><br/> <b>Clear-Dont-Fragment</b> clears the Dont Fragment bit and the Dont Fragment bit is set. For packets not requiring fragmentation, the Dont Fragment bit is not affected.</p> |
| Network Broadcast   | <p>From the drop-down, select <b>Global</b>. Click <b>On</b> to accept and respond to network-prefix-directed broadcasts. Turn this <b>On</b> only if the <b>Directed Broadcast</b> is enabled on the LAN interface feature template.</p> <p>Default: <b>Off</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Allow Service       | <p>Click <b>On</b> or <b>Off</b> for each service to allow or disallow the service on the cellular interface.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

To configure additional tunnel interface parameters, click **Advanced Options** and configure the following parameters:

**Table 285:**

| Parameter Name | Description                                                                                                                                                                                                                                                                                                                                   |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GRE            | <p>From the drop-down, select <b>Global</b>. Click <b>On</b> to use GRE encapsulation on the tunnel interface. By default, GRE is disabled.</p> <p>If you select both IPsec and GRE encapsulations, two TLOCs are created for the tunnel interface that have the same IP addresses and colors, but that differ by their encapsulation.</p>    |
| GRE Preference | <p>From the drop-down, select <b>Global</b>. Enter a value to set GRE preference for TLOC.</p> <p>Range: 0 to 4294967295</p>                                                                                                                                                                                                                  |
| GRE Weight     | <p>From the drop-down, select <b>Global</b>. Enter a value to set GRE weight for TLOC.</p> <p>Default: 1</p>                                                                                                                                                                                                                                  |
| IPsec          | <p>From the drop-down, select <b>Global</b>. Click <b>On</b> to use IPsec encapsulation on the tunnel interface. By default, IPsec is enabled.</p> <p>If you select both IPsec and GRE encapsulations, two TLOCs are created for the tunnel interface that have the same IP addresses and colors, but that differ by their encapsulation.</p> |

| Parameter Name       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IPsec Preference     | From the drop-down, select <b>Global</b> . Enter a value to set the preference for directing traffic to the tunnel. A higher value is preferred over a lower value.<br>Range: 0 through 4294967295. Default: 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| IPsec Weight         | From the drop-down, select <b>Global</b> . Enter a value to set weight for balancing traffic across multiple TLOCs. A higher value sends more traffic to the tunnel.<br>Range: 1 through 255. Default: 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Carrier              | From the drop-down, select <b>Global</b> . From the <b>Carrier</b> drop-down, select the carrier name or private network identifier to associate with the tunnel. Values: carrier1, carrier2, carrier3, carrier4, carrier5, carrier6, carrier7, carrier8, default. Default: default                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Bind Loopback Tunnel | Enter the name of a physical interface to bind to a loopback interface. The interface name has the format <b>ge slot/port</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Last-Resort Circuit  | From the drop-down, select <b>Global</b> . Click <b>On</b> to use the tunnel interface as the circuit of last resort. By default, it is disabled.<br><br><b>Note</b><br>An interface configured as a circuit of last resort is expected to be down and is skipped while calculating the number of control connections, the cellular modem becomes dormant, and no traffic is sent over the circuit.<br><br>When the configurations are activated on the edge device with cellular interfaces, then all the interfaces begin the process of establishing control and BFD connections. When one or more of the primary interfaces establishes a BFD connection, the circuit of last resort shuts itself down.<br><br>Only when all the primary interfaces lose their connections to remote edges, then the circuit of last resort activates itself triggering a BFD TLOC Down alarm and a Control TLOC Down alarm on the edge device. The last resort interfaces are used as backup circuit on edge device and are activated when all other transport links BFD sessions fail. In this mode the radio interface is turned off, and no control or data connections exist over the cellular interface. |
| NAT Refresh Interval | Set the interval between NAT refresh packets sent on a DTLS or TLS WAN transport connection. Range: 1 through 60 seconds. Default: 5 seconds.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Hello Interval       | Enter the interval between Hello packets sent on a DTLS or TLS WAN transport connection. Range: 100 through 10000 milliseconds. Default: 1000 milliseconds (1 second).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |



| Parameter Name  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hello Tolerance | <p>Enter the time to wait for a Hello packet on a DTLS or TLS WAN transport connection before declaring that transport tunnel to be down.</p> <p>Range: 12 through 60 seconds. Default: 12 seconds.</p> <p>The default hello interval is 1000 milliseconds, and it can be a time in the range 100 through 600000 milliseconds (10 minutes). The default hello tolerance is 12 seconds, and it can be a time in the range 12 through 600 seconds (10 minutes). To reduce outgoing control packets on a TLOC, it is recommended that on the tunnel interface you set the hello interval to 60000 milliseconds (10 minutes) and the hello tolerance to 600 seconds (10 minutes) and include the <b>no track-transport disable</b> regular checking of the DTLS connection between the edge device and the controller. For a tunnel connection between a edge device and any controller device, the tunnel uses the hello interval and tolerance times configured on the edge device. This choice is made to minimize the traffic sent over the tunnel, to allow for situations where the cost of a link is a function of the amount of traffic traversing the link. The hello interval and tolerance times are chosen separately for each tunnel between a edge device and a controller device. Another step taken to minimize the amount of control plane traffic is to not send or receive OMP control traffic over a cellular interface when other interfaces are available. This behavior is inherent in the software and is not configurable.</p> |

To save the feature template, click **Save**.

### Configure the Cellular Interface as a NAT Device

To configure a cellular interface to act as a NAT device for applications such as port forwarding, click **NAT**, and configure the following parameters:

**Table 286:**

| Parameter Name  | Description                                                                                                                                              |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| NAT             | Click <b>On</b> to have the interface act as a NAT device.                                                                                               |
| Refresh Mode    | Select how NAT mappings are refreshed, either outbound or bidirectional (outbound and inbound). Default: Outbound                                        |
| UDP Timeout     | Specify when NAT translations over UDP sessions time out. Range: 1 through 65536 minutes. Default: 1 minute                                              |
| TCP Timeout     | Specify when NAT translations over TCP sessions time out. Range: 1 through 65536 minutes. Default: 60 minutes (1 hour)                                   |
| Block ICMP      | Select <b>On</b> to block inbound ICMP error messages. By default, a router acting as a NAT device receives these error messages. Default: Off           |
| Respond to Ping | Select <b>On</b> to have the router respond to ping requests to the NAT interface's IP address that are received from the public side of the connection. |

To create a port forwarding rule, click **Add New Port Forwarding Rule** and configure the following parameters. You can define up to 128 port-forwarding rules to allow requests from an external network to reach devices on the internal network.

**Table 287:**

| Parameter Name   | Description                                                                                                                                                   |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Port Start Range | Enter a port number to define the port or first port in the range of interest. Range: 0 through 65535                                                         |
| Port End Range   | Enter the same port number to apply port forwarding to a single port, or enter a larger number to apply it to a range of ports. Range: 0 through 65535        |
| Protocol         | Select the protocol to which to apply the port-forwarding rule, either TCP or UDP. To match the same ports for both TCP and UDP traffic, configure two rules. |
| VPN              | Specify the private VPN in which the internal server resides. This VPN is one of the VPN identifiers in the overlay network. Range: 0 through 65527           |
| Private IP       | Specify the IP address of the internal server to which to direct traffic that matches the port-forwarding rule.                                               |

To save a port forwarding rule, click **Add**.

To save the feature template, click **Save**.

### Apply Access Lists

To configure a shaping rate to a cellular interface and to apply a QoS map, a rewrite rule, access lists, and policers to a router interface, click **ACL/QoS** and configure the following parameters:

**Table 288: Access Lists Parameters**

| Parameter Name     | Description                                                                                                                  |
|--------------------|------------------------------------------------------------------------------------------------------------------------------|
| Shaping rate       | Configure the aggregate traffic transmission rate on the interface to be less than line rate, in kilobits per second (kbps). |
| QoS map            | Specify the name of the QoS map to apply to packets being transmitted out the interface.                                     |
| Rewrite rule       | Click <b>On</b> , and specify the name of the rewrite rule to apply on the interface.                                        |
| Ingress ACL – IPv4 | Click <b>On</b> , and specify the name of an IPv4 access list to packets being received on the interface.                    |
| Egress ACL– IPv4   | Click <b>On</b> , and specify the name of an IPv4 access list to packets being transmitted on the interface.                 |
| Ingress ACL – IPv6 | Click <b>On</b> , and specify the name of an IPv6 access list to packets being received on the interface.                    |
| Egress ACL– IPv6   | Click <b>On</b> , and specify the name of an IPv6 access list to packets being transmitted on the interface.                 |

| Parameter Name  | Description                                                                                                   |
|-----------------|---------------------------------------------------------------------------------------------------------------|
| Ingress policer | Click <b>On</b> , and specify the name of the policer to apply to packets being received on the interface.    |
| Egress policer  | Click <b>On</b> , and specify the name of the policer to apply to packets being transmitted on the interface. |

To save the feature template, click **Save**.

### Add ARP Table Entries

To configure static Address Resolution Protocol (ARP) table entries on the interface, click **ARP**. Then click **Add New ARP** and configure the following parameters:

*Table 289:*

| Parameter Name | Description                                                                                          |
|----------------|------------------------------------------------------------------------------------------------------|
| IP Address     | Enter the IP address for the ARP entry in dotted decimal notation or as a fully qualified host name. |
| MAC Address    | Enter the MAC address in colon-separated hexadecimal notation.                                       |

To save the ARP configuration, click **Add**.

To save the feature template, click **Save**.

### Configure Other Interface Properties

To configure other interface properties, click **Advanced** and configure the following parameters.

*Table 290: Cellular Interfaces Advanced Parameters*

| Parameter Name      | Description                                                                                                                                                                                                                                                     |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PMTU Discovery      | Click <b>On</b> to enable path MTU discovery on the interface, to allow the router to determine the largest MTU size supported without requiring packet fragmentation.                                                                                          |
| TCP MSS             | Specify the maximum segment size (MSS) of TCP SYN packets passing through the router. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. Range: 552 to 1460 bytes. Default: None. |
| Clear-Dont-Fragment | Click <b>On</b> to clear the Don't Fragment (DF) bit in the IPv4 packet header for packets being transmitted out the interface. When the DF bit is cleared, packets larger than that interface's MTU are fragmented before being sent.                          |
| Static Ingress QoS  | Select a queue number to use for incoming traffic. Range: 0 through 7                                                                                                                                                                                           |
| Autonegotiate       | Click <b>Off</b> to turn off autonegotiation. By default, an interface runs in autonegotiation mode.                                                                                                                                                            |

| Parameter Name        | Description                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TLOC Extension        | Enter the name of a physical interface on the same router that connects to the WAN transport. This configuration then binds this service-side interface to the WAN transport. A second router at the same site that itself has no direct connection to the WAN (generally because the site has only a single WAN connection) and that connects to this service-side interface is then provided with a connection to the WAN. |
| Tracker               | Enter the name of a tracker to track the status of transport interfaces that connect to the internet.                                                                                                                                                                                                                                                                                                                        |
| IP Directed-Broadcast | From the drop-down, select <b>Global</b> . Click <b>On</b> for IP directed-broadcast.<br>Default: <b>Off</b>                                                                                                                                                                                                                                                                                                                 |

To save the feature template, click **Save**.

## Configure Cellular Interfaces Using CLI

The following example enables a cellular interface:

```
interface Cellular0/2/0
 description Cellular interface
 no shutdown
 ip address negotiated
 ip mtu 1428
 mtu 1500
 exit

controller Cellular 0/2/0
 lte sim max-retry 1
 lte failovertimer 7
 profile id 1 apn Broadband authentication none pdn-type ipv4
```

## Data Profile

**Table 291: Feature History**

| Feature Name                                                                    | Release Information                                                          | Description                                                             |
|---------------------------------------------------------------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Ability to Configure APNs under Running Configurations for Single and Dual SIMs | Cisco IOS XE Catalyst SD-WAN Release 17.8.1a<br>Cisco vManage Release 20.8.1 | This feature allows you to create a data profile for a cellular device. |

A data profile for a cellular device defines the following parameters, which the device uses for communication with the service provider. You can configure the following parameters by using the **profile id** command in cellular configuration mode. For more information about the following parameters, see [profile id](#).

- Identification number of the data profile
- Name of the access point network of the service provider
- Authentication type used for APN access: No authentication, CHAP authentication only, PAP authentication only, or either CHAP or PAP authentication

- Username and password that are provided by the service provider for APN access authentication, if authentication is used
- Type of packet data matching that is used for APN access: IPv4 type bearer, IPv6 type bearer, or IPv4v6 type bearer
- SIM slot that contains the SIM to configure

## Best Practices for Configuring Cellular Interfaces

Cellular technology on edge devices can be used in a number of ways:

- Circuit of last resort: An interface configured as a circuit of last resort is expected to be down and is skipped while calculating the number of control connections, the cellular modem becomes dormant, and no traffic is sent over the circuit.

When the configurations are activated on the edge device with cellular interfaces, then all the interfaces begin the process of establishing control and BFD connections. When one or more of the primary interfaces establishes a BFD connection, the circuit of last resort shuts itself down.

Only when all the primary interfaces lose their connections to remote edges, then the circuit of last resort activates itself triggering a BFD TLOC Down alarm and a Control TLOC Down alarm on the edge device. The last resort interfaces are used as backup circuit on edge device and are activated when all other transport links BFD sessions fail. In this mode the radio interface is turned off, and no control or data connections exist over the cellular interface.

Use the **last-resort-circuit** command to configure a cellular interface to be a circuit of last resort.



---

**Note** **last-resort-circuit** is not limited to cellular interfaces.

The operating principle for cellular interfaces also applies to GigabitEthernet interfaces.

---

- Active circuit: You can choose to use a cellular interface as an active circuit, perhaps because it is the only last-mile circuit or to always keep the cellular interface active so that you can measure the performance of the circuit. In this scenario the amount of bandwidth utilized to maintain control and data connections over the cellular interface can become a concern. Here are some best practices to minimize bandwidth usage over a cellular interface:
  - Increase control packet timers—To minimize control traffic on a cellular interface, you can decrease how often protocol update messages are sent on the interface. OMP sends Update packets every second, by default. You can increase this interval to a maximum of 65535 seconds (about 18 hours) by including the **omp timers advertisement-interval** configuration command. BFD sends Hello packets every second, by default. You can increase this interval to a maximum of 5 minutes (300000 milliseconds) by including the **bfd color hello-interval** configuration command. (Note that you specify the OMP Update packet interval in seconds and the BFD Hello packet interval in milliseconds.)
  - Prioritize Cisco SD-WAN Manager control traffic over a non-cellular interface: When a edge device has both cellular and non-cellular transport interfaces, by default, the edge device chooses one of the interfaces to use to exchange control traffic with the Cisco SD-WAN Manager. You can configure the edge device to never use the cellular interface to exchange traffic with the Cisco SD-WAN

Manager, or you can configure a lower preference for using the cellular interface for this traffic. You configure the preference by including the **vmanage-connection-preference** command when configuring the tunnel interface. By default, all tunnel interface have a Cisco SD-WAN Manager connection preference value of 5. The value can range from 0 through 8, where a higher value is more preferred. A tunnel with a preference value of 0 can never exchange control traffic with the Cisco SD-WAN Manager.



---

**Note** At least one tunnel interface on the edge device must have a non-0 Cisco SD-WAN Manager connection preference value. Otherwise, the device has no control connections.

---



# CHAPTER 14

## Cellular Interfaces

**Table 292: Feature History**

| Feature Name                                                     | Release Information                                                                            | Description                                                                                                                                                  |
|------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Configure Cellular Interfaces                                    | Cisco Catalyst SD-WAN Manager Release 18.1.1                                                   | You can configure cellular interfaces on devies with cellular modules to enable LTE connectivity.                                                            |
| Cellular Module Support for Cisco Catalyst Rugged Series Routers | Cisco IOS XE Catalyst SD-WAN Release 17.16.1a<br>Cisco Catalyst SD-WAN Manager Release 20.16.1 | Support for cellular modules on Cisco Cisco Catalyst IR1101, IR1800 and IR18340 Rugged Series Routers.                                                       |
| Reset the profile configuration of a cellular modem              | Cisco IOS XE Catalyst SD-WAN Release 17.18.1a<br>Cisco Catalyst SD-WAN Manager Release 20.18.1 | This sub-feature of the Cellular Controller feature enables you to reset the configuration of a cellular modem operating on a device using CLI.              |
| Band select support in Controller Mode                           | Cisco Catalyst SD-WAN Manager Release 20.18.1                                                  | This sub-feature of the Cellular Controller feature introduces the Cellular Band Select feature-parcel, enabling you to specify which cellular bands to use. |

- [Cellular Interfaces, on page 626](#)
- [Supported Devices, on page 626](#)
- [Configure Cellular Interfaces using a Configuration Group, on page 627](#)
- [Configure Cellular Interfaces Using Templates, on page 637](#)
- [Configure Cellular Interfaces Using CLI, on page 646](#)
- [Reset the profile configuration of a cellular modem using the CLI, on page 647](#)
- [LTE Modem Crash Diagnostics, on page 647](#)
- [Data Profile, on page 648](#)
- [Best Practices for Configuring Cellular Interfaces, on page 648](#)

# Cellular Interfaces

To enable LTE connectivity, configure cellular interfaces on a router that has a cellular module. The cellular module provides wireless connectivity over a service provider's cellular network. One use case is to provide wireless connectivity for branch offices.

A cellular network is commonly used as a backup WAN link, to provide network connectivity if all the wired WAN tunnel interfaces on the router become unavailable. You can also use a cellular network as the primary WAN link for a branch office, depending on usage patterns within the branch office and the data rates supported by the core of the service provider's cellular network.

When you configure a cellular interface on a device, you can connect the device to the Internet or another WAN by plugging in the power cable of the device. The device then automatically begins the process of joining the overlay network, by contacting and authenticating with Cisco Catalyst SD-WAN Validators, Cisco Catalyst SD-WAN Controllers, and Cisco SD-WAN Manager systems.

## Supported Devices

**Table 293: Supported Platforms and Modules**

| Platform                    | Minimum Supported Release                                                                      | Supported Modules                                                                                          |
|-----------------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| IR1101 Rugged Series Router | Cisco IOS XE Catalyst SD-WAN Release 17.15.1a<br>Cisco Catalyst SD-WAN Manager Release 20.15.1 | P-LTE-MNA, P-LTE-VZ, P-LTE-US, P-LTE-GB, P-LTE-IN, P-LTE-JN, P-LTEA-EA, P-LTEA-LA, P-LTEAP18-GL, P-5GS6-GL |
|                             | Cisco IOS XE Catalyst SD-WAN Release 17.16.1a<br>Cisco Catalyst SD-WAN Manager Release 20.16.1 | P-LTEA7-JP, P-LTEA7-NA, P-LTEA7-EAL, P-5GS6-R16SA-GL                                                       |
|                             | Cisco IOS XE Catalyst SD-WAN Release 17.18.1a<br>Cisco Catalyst SD-WAN Manager Release 20.18.1 | P-LTE-450                                                                                                  |
| IR8100 Rugged Series Router | Cisco IOS XE Catalyst SD-WAN Release 17.15.1a<br>Cisco Catalyst SD-WAN Manager Release 20.15.1 | IRMH-LTE-MNA, IRMH-LTEA-EA, IRMH-LTEA-LA, IRMH-LTEAP18-GL, IRMH-5GS6-GL                                    |
|                             | Cisco IOS XE Catalyst SD-WAN Release 17.16.1a<br>Cisco Catalyst SD-WAN Manager Release 20.16.1 | IRMH-LTE7-NA-900, IRMH-5GR16SA, IRMH-LTE7-EAL                                                              |



| Platform                    | Minimum Supported Release                                                                      | Supported Modules                                                                                          |
|-----------------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| IR1800 Rugged Series Router | Cisco IOS XE Catalyst SD-WAN Release 17.15.1a<br>Cisco Catalyst SD-WAN Manager Release 20.15.1 | P-LTE-MNA, P-LTE-VZ, P-LTE-US, P-LTE-GB, P-LTE-IN, P-LTE-JN, P-LTEA-EA, P-LTEA-LA, P-LTEAP18-GL, P-5GS6-GL |
|                             | Cisco IOS XE Catalyst SD-WAN Release 17.16.1a<br>Cisco Catalyst SD-WAN Manager Release 20.16.1 | P-LTEA7-NA, P-LTEA7-JP, P-LTEA7-EAL, P-5GS6-R16SA-GL                                                       |
|                             | Cisco IOS XE Catalyst SD-WAN Release 17.18.1a<br>Cisco Catalyst SD-WAN Manager Release 20.18.1 | P-LTE-450                                                                                                  |
| IR8340 Rugged Series Router | Cisco IOS XE Catalyst SD-WAN Release 17.15.1a<br>Cisco Catalyst SD-WAN Manager Release 20.15.1 | P-LTE-MNA, P-LTEA-EA, P-LTEA-LA, P-LTEAP18-GL, P-5GS6-GL                                                   |
|                             | Cisco IOS XE Catalyst SD-WAN Release 17.16.1a<br>Cisco Catalyst SD-WAN Manager Release 20.16.1 | P-LTEA7-NA, P-LTEA7-JP, P-LTEA7-EAL, P-5GS6-R16SA-GL                                                       |
| Cisco ESR 6300 Series       | Cisco IOS XE Catalyst SD-WAN Release 17.15.1a<br>Cisco Catalyst SD-WAN Manager Release 20.15.1 | P-LTE-MNA, P-LTEA-EA, P-LTEA-LA, P-LTEAP18-GL                                                              |

## Configure Cellular Interfaces using a Configuration Group

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.16.1a, Cisco Catalyst SD-WAN Manager Release 20.16.1

### Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

### SUMMARY STEPS

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.
2. Create and configure a [Transport VPN](#) feature in a Transport & Management profile.
3. Create a configure a [Cellular Interface](#) feature to associate with the Transport VPN.
4. Click **Add New Feature** to create and configure a [Cellular Controller](#) feature.

5. Configure a [Cellular Profile](#) feature to associate with the Cellular Controller.

## DETAILED STEPS

### Procedure

**Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

**Step 2** Create and configure a [Transport VPN](#) feature in a Transport & Management profile.

- a. Configure the basic configuration parameters, such as the VPN.

*Table 294: Basic Configuration*

| Field                      | Description                                                                                                                                                                                                                            |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>VPN</b>                 | Enter the numeric identifier of the VPN.                                                                                                                                                                                               |
| <b>Enhance ECMP Keying</b> | Enable the use in the ECMP hash key of Layer 4 source and destination ports, in addition to the combination of the source IP address, destination IP address, protocol, and DSCP field, as the ECMP hash key.<br><br>Default: Disabled |

- b. Configure DNS.

*Table 295: DNS*

| Field                               | Description                                                      |
|-------------------------------------|------------------------------------------------------------------|
| <b>Add DNS</b>                      |                                                                  |
| <b>Primary DNS Address (IPv4)</b>   | Enter the IP address of the primary IPv4 DNS server in this VPN. |
| <b>Secondary DNS Address (IPv4)</b> | Enter the IP address of a secondary IPv4 DNS server in this VPN. |
| <b>Add DNS IPv6</b>                 |                                                                  |
| <b>Primary DNS Address (IPv6)</b>   | Enter the IP address of the primary IPv6 DNS server in this VPN. |
| <b>Secondary DNS Address (IPv6)</b> | Enter the IP address of a secondary IPv6 DNS server in this VPN. |

- c. Configure host mapping.

*Table 296: Host Mapping*

| Field                       | Description                                                                                   |
|-----------------------------|-----------------------------------------------------------------------------------------------|
| <b>Add New Host Mapping</b> |                                                                                               |
| <b>Hostname*</b>            | Enter the hostname of the DNS server. The name can be up to 128 characters.                   |
| <b>List of IP*</b>          | Enter up to 14 IP addresses to associate with the hostname. Separate the entries with commas. |

## d. Configure routes.

Table 297: Route

| Field                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Add IPv4 Static Route</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Network address*</b>      | Enter the IPv4 address or prefix, in decimal four-point-dotted notation, and the prefix length of the IPv4 static route to configure in the VPN.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Subnet Mask*</b>          | Enter the subnet mask.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Gateway*</b>              | <p>Choose one of the following options to configure the next hop to reach the static route:</p> <ul style="list-style-type: none"> <li>• <b>nextHop</b>: When you choose this option and click <b>Add Next Hop</b>, the following fields appear: <ul style="list-style-type: none"> <li>• <b>Address*</b>: Enter the next-hop IPv4 address.</li> <li>• <b>Administrative distance*</b>: Enter the administrative distance for the route.</li> </ul> </li> <li>• <b>dhcp</b></li> <li>• <b>null0</b>: When you choose this option, the following field appears: <ul style="list-style-type: none"> <li>• <b>Administrative distance</b>: Enter the administrative distance for the route.</li> </ul> </li> </ul>                                                                                                                                                                                                                                                        |
| <b>Add IPv6 Static Route</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Prefix*</b>               | Enter the IPv6 address or prefix, in decimal four-point-dotted notation, and the prefix length of the IPv6 static route to configure in the VPN.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Next Hop/Null 0/NAT</b>   | <p>Choose one of the following options to configure the next hop to reach the static route:</p> <ul style="list-style-type: none"> <li>• <b>Next Hop</b>: When you choose this option and click <b>Add Next Hop</b>, the following fields appear: <ul style="list-style-type: none"> <li>• <b>Address*</b>: Enter the next-hop IPv6 address.</li> <li>• <b>Administrative distance*</b>: Enter the administrative distance for the route.</li> </ul> </li> <li>• <b>Null 0</b>: When you choose this option, the following field appears: <ul style="list-style-type: none"> <li>• <b>IPv6 Route Null 0*</b>: Enable this option to set the next hop to be the null interface. All packets sent to this interface are dropped without sending any ICMP messages.</li> </ul> </li> <li>• <b>NAT</b>: When you choose this option, the following field appears: <ul style="list-style-type: none"> <li>• <b>IPv6 NAT*</b>: Choose NAT64 or NAT66.</li> </ul> </li> </ul> |

| Field           | Description         |
|-----------------|---------------------|
| Add BGP Routing | Choose a BGP route. |

- e. Configure services.

*Table 298: Service*

| Field        | Description                                           |
|--------------|-------------------------------------------------------|
| Add Service  |                                                       |
| Service Type | Choose the service available in the VPN.<br>Value: TE |

**Step 3** Create a configure a [Cellular Interface](#) feature to associate with the Transport VPN.

- a. Adjacent to the Transport VPN feature, click + and add a Cellular Interface feature as a subfeature.  
b. Configure the basic parameters.

| Field           | Description                                                                                                                                                                                                                                    |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Shutdown*       | Enable or disable the interface.                                                                                                                                                                                                               |
| Interface Name* | Enter the name of the interface.                                                                                                                                                                                                               |
| Description*    | Enter a description of the cellular interface.                                                                                                                                                                                                 |
| DHCP Helper     | Enter up to four IP addresses for DHCP servers in the network, separated by commas, to have the interface be a DHCP helper. A DHCP helper interface forwards BOOTP (Broadcast) DHCP requests that it receives from the specified DHCP servers. |

- c. Configure the tunnel parameters.

| Field             | Description                                                                                                                                                                                                |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tunnel Interface  | Enable this option to create a tunnel interface.                                                                                                                                                           |
| Carrier           | Choose the carrier name or private network identifier to associate with the tunnel.<br>Values: carrier1, carrier2, carrier3, carrier4, carrier5, carrier6, carrier7, carrier8, default<br>Default: default |
| Color             | Choose a color for the TLOC.                                                                                                                                                                               |
| Color Description | Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.18.1<br>Enter a description associated to the TLOC color.                                                                              |

| Field                                | Description                                                                                                                                                                                                                                                     |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hello Interval</b>                | Enter the interval between Hello packets sent on a DTLS or TLS WAN transport connection.<br><br>Range: 100 through 600000 milliseconds<br>Default: 1000 milliseconds (1 second)                                                                                 |
| <b>Hello Tolerance</b>               | Enter the time to wait for a Hello packet on a DTLS or TLS WAN transport connection before declaring that transport tunnel to be down.<br><br>Range: 12 through 6000 seconds<br>Default: 12 seconds                                                             |
| <b>Last-Resort Circuit</b>           | Enable this option to use the tunnel interface as the circuit of last resort.                                                                                                                                                                                   |
| <b>Restrict</b>                      | Enable this option to limit the remote TLOCs that the local TLOC can establish BFD sessions with. When a TLOC is marked as restricted, a TLOC on the local router establishes tunnel connections with a remote TLOC only if the remote TLOC has the same color. |
| <b>Group</b>                         | Enter a group number.<br><br>Range: 1 through 4294967295                                                                                                                                                                                                        |
| <b>Border</b>                        | Enable this option to set the TLOC as a border TLOC.                                                                                                                                                                                                            |
| <b>Maximum Control Connections</b>   | Specify the maximum number of Cisco SD-WAN Controllers that the WAN tunnel interface can connect to. To have the tunnel establish no control connections, set the number to 0.<br><br>Range: 0 through 100<br>Default: 2                                        |
| <b>NAT Refresh Interval</b>          | Enter the interval between NAT refresh packets sent on a DTLS or TLS WAN transport connection.<br><br>Range: 1 through 60 seconds<br>Default: 5 seconds                                                                                                         |
| <b>Validator As Stun Server</b>      | Enable Session Traversal Utilities for NAT (STUN) to allow the tunnel interface to discover its public IP address and port number when the Cisco IOS XE Catalyst SD-WAN device is located behind a NAT.                                                         |
| <b>Exclude Controller Group List</b> | Set the identifiers of one or more Cisco SD-WAN Controller groups that this tunnel is not allowed to connect to.<br><br>Range: 1 through 100                                                                                                                    |
| <b>Manager Connection Preference</b> | Set the preference for using a tunnel interface to exchange control traffic with Cisco SD-WAN Manager.<br><br>Range: 0 through 8<br>Default: 5                                                                                                                  |

| Field                      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Full Port Hop</b>       | <p>Minimum release: Cisco IOS XE Catalyst SD-WAN Release 17.18.1a</p> <p>Enable full port hopping at the TLOC level to allow devices to establish connections with controllers by switching to the next port if the current port is blocked or non-functional.</p> <p>Default: Disabled</p>                                                                                                                                                                                                                                                                                |
| <b>Port Hop</b>            | <p>Enable port hopping. When a router is behind a NAT, port hopping rotates through a pool of preselected OMP port numbers (called base ports) to establish DTLS connections with other routers when a connection attempt is unsuccessful. The default base ports are 12346, 12366, 12386, 12406, and 12426. To modify the base ports, set a port offset value.</p> <p>Default: Enabled</p> <p>Starting from Cisco IOS XE Catalyst SD-WAN Release 17.18.1a, this field is deprecated. Instead use the <b>Full Port Hop</b> option. See the <b>Full Port Hop</b> field.</p> |
| <b>Low-Bandwidth Link</b>  | Enable the low-bandwidth feature.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Tunnel TCP MSS</b>      | <p>Specify the maximum segment size (MSS) of TCP SYN packets passing through the router. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented.</p> <p>Range: 500 to 1460 bytes</p> <p>Default: None</p>                                                                                                                                                                                                                                                                                         |
| <b>Clear-Dont-Fragment</b> | Enable this option to clear the Don't Fragment (DF) bit in the IPv4 packet header for packets being transmitted out the interface. When the DF bit is cleared, packets larger than the MTU of the interface are fragmented before being sent.                                                                                                                                                                                                                                                                                                                              |
| <b>Network Broadcast</b>   | Enable this option to accept and respond to network-prefix-directed broadcasts.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

| Field                 | Description                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Allow Service</b>  | <p>Allow or disallow the following services on the interface:</p> <ul style="list-style-type: none"> <li>• <b>All</b></li> <li>• <b>BGP</b></li> <li>• <b>DHCP</b></li> <li>• <b>NTP</b></li> <li>• <b>SSH</b></li> <li>• <b>DNS</b></li> <li>• <b>ICMP</b></li> <li>• <b>HTTPS</b></li> <li>• <b>OSPF</b></li> <li>• <b>STUN</b></li> <li>• <b>SNMP</b></li> <li>• <b>NETCONF</b></li> <li>• <b>BFD</b></li> </ul> |
| <b>Encapsulation</b>  |                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>GRE</b>            | <p>Use GRE encapsulation on the tunnel interface. By default, GRE is disabled.</p> <p>If you select both IPsec and GRE encapsulations, two TLOCs are created for the tunnel interface that have the same IP addresses and colors, but that differ by their encapsulation.</p>                                                                                                                                       |
| <b>GRE Preference</b> | <p>Specify a preference value for directing traffic to the tunnel. A higher value is preferred over a lower value.</p> <p>Range: 0 through 4294967295</p> <p>Default: 0</p>                                                                                                                                                                                                                                         |
| <b>GRE Weight</b>     | <p>Enter a weight to use to balance traffic across multiple TLOCs. A higher value sends more traffic to the tunnel.</p> <p>Range: 1 through 255</p> <p>Default: 1</p>                                                                                                                                                                                                                                               |
| <b>IPsec</b>          | <p>Use IPsec encapsulation on the tunnel interface. By default, IPsec is enabled.</p> <p>If you select both IPsec and GRE encapsulations, two TLOCs are created for the tunnel interface that have the same IP addresses and colors, but that differ by their encapsulation.</p>                                                                                                                                    |

| Field                   | Description                                                                                                                                                  |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IPsec Preference</b> | Specify a preference value for directing traffic to the tunnel. A higher value is preferred over a lower value.<br>Range: 0 through 4294967295<br>Default: 0 |
| <b>IPsec Weight</b>     | Enter a weight to use to balance traffic across multiple TLOCs. A higher value sends more traffic to the tunnel.<br>Range: 1 through 255<br>Default: 1       |

## d. Configure NAT.

| Field               | Description                                                                                                                |
|---------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>NAT</b>          | Enable this option to have the interface act as a NAT device.                                                              |
| <b>UDP Timeout*</b> | Specify when NAT translations over UDP sessions time out.<br>Range: 1 through 8947 minutes<br>Default: 1 minutes           |
| <b>TCP Timeout*</b> | Specify when NAT translations over TCP sessions time out.<br>Range: 1 through 8947 minutes<br>Default: 60 minutes (1 hour) |

## e. Configure ACL/QoS.

## f. Configure the advanced parameters.

| Field                | Description                                                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MAC Address</b>   | Specify a MAC address to associate with the interface, in colon-separated hexadecimal notation.                                                   |
| <b>IP MTU</b>        | Specify the maximum MTU size of packets on the interface.<br>Range: 576 through 9216<br>Default: 1500 bytes                                       |
| <b>Interface MTU</b> | Enter the maximum transmission unit size for frames received and transmitted on the interface.<br>Range: 1500 through 9216<br>Default: 1500 bytes |



| Field                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>TCP MSS</b>               | <p>Specify the maximum segment size (MSS) of TCP SYN packets passing through the router. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented.</p> <p>Range: 500 to 1460 bytes</p> <p>Default: None</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>TLOC Extension</b>        | <p>Enter the name of a physical interface on the same router that connects to the WAN transport. This configuration then binds this service-side interface to the WAN transport. A second router at the same site that itself has no direct connection to the WAN (generally because the site has only a single WAN connection) and that connects to this service-side interface is then provided with a connection to the WAN.</p> <p><b>Note</b><br/>TLOC extension over L3 is supported only for Cisco IOS XE Catalyst SD-WAN devices. If configuring TLOC extension over L3 for a Cisco IOS XE Catalyst SD-WAN device, enter the IP address of the L3 interface.</p>                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Tracker</b>               | <p>Tracking the interface status is useful when you enable NAT on a transport interface in VPN 0 to allow data traffic from the router to exit directly to the internet rather than having to first go to a router in a data center. In this situation, enabling NAT on the transport interface splits the TLOC between the local router and the data center into two, with one going to the remote router and the other going to the internet.</p> <p>When you enable transport tunnel tracking, Cisco Catalyst SD-WAN periodically probes the path to the internet to determine whether it is up. If Cisco Catalyst SD-WAN detects that this path is down, it withdraws the route to the internet destination, and traffic destined to the internet is then routed through the data center router. When Cisco Catalyst SD-WAN detects that the path to the internet is again functioning, the route to the internet is reinstalled.</p> <p>Enter the name of a tracker to track the status of transport interfaces that connect to the internet.</p> |
| <b>IP Directed-Broadcast</b> | <p>An IP directed broadcast is an IP packet whose destination address is a valid broadcast address for some IP subnet but which originates from a node that is not itself part of that destination subnet.</p> <p>A device that is not directly connected to its destination subnet forwards an IP directed broadcast in the same way it would forward unicast IP packets destined to a host on that subnet. When a directed broadcast packet reaches a device that is directly connected to its destination subnet, that packet is broadcast on the destination subnet. The destination address in the IP header of the packet is rewritten to the configured IP broadcast address for the subnet, and the packet is sent as a link-layer broadcast.</p> <p>If directed broadcast is enabled for an interface, incoming IP packets whose addresses identify them as directed broadcasts intended for the subnet to which that interface is attached are broadcast on that subnet.</p>                                                                 |

**Step 4** Click **Add New Feature** to create and configure a **Cellular Controller** feature.

Configure the basic parameters, mostly related to SIMs.

*Table 299: Basic Settings*

| Field                | Description                                                                                                                                                                                                                                                |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Type                 | Choose a feature from the drop-down list.                                                                                                                                                                                                                  |
| Feature Name         | Enter a name for the feature. The name can be up to 128 characters and can contain only alphanumeric characters.                                                                                                                                           |
| Description          | Enter a description of the feature. The description can be up to 2048 characters and can contain only alphanumeric characters.                                                                                                                             |
| Cellular ID          | Enter the interface slot and port number in which the cellular NIM card is installed. Currently, it can be 0/1/0 or 0/2/0.                                                                                                                                 |
| Primary SIM slot     | Enter the number of the primary SIM slot. It can be 0 or 1. The other slot is automatically set to be the secondary. If there is a single SIM slot, this parameter is not applicable.                                                                      |
| SIM Failover Retries | Specify the maximum number of times to retry connecting to the secondary SIM when service on the primary SIM becomes unavailable. If there is a single SIM slot, this parameter is not applicable.<br><br>Range: 0 through 65535<br>Default: 10            |
| SIM Failover Timeout | Specify how long to wait before switching from the primary SIM to the secondary SIM if service on the primary SIM becomes unavailable. If there is a single SIM slot, this parameter is not applicable.<br><br>Range: 3 to 7 minutes<br>Default: 3 minutes |
| Firmware Auto Sim    | By default, this option is enabled. AutoSIM analyzes any active SIM card and determines which service provider network is associated with that SIM. Based on that analysis, AutoSIM automatically loads the appropriate firmware.                          |

**Step 5** Configure a **Cellular Profile** feature to associate with the Cellular Controller.

- a. Adjacent to the Cellular Controller feature, click + and add a Cellular Profile feature as a subfeature.
- b. Configure the basic parameters.

*Table 300: Basic Settings*

| Field        | Description                                                                                                      |
|--------------|------------------------------------------------------------------------------------------------------------------|
| Type         | Choose a feature from the drop-down list.                                                                        |
| Feature Name | Enter a name for the feature. The name can be up to 128 characters and can contain only alphanumeric characters. |

| Field                           | Description                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>              | Enter a description of the feature. The description can be up to 2048 characters and can contain only alphanumeric characters.                                                                                                                                                                                                                                                                          |
| <b>Profile ID</b>               | Enter the identification number of the profile to use on the router.<br>Range: 1 through 15                                                                                                                                                                                                                                                                                                             |
| <b>Access Point Name</b>        | Enter the name of the gateway between the service provider network and the public internet. It can be up to 32 characters long.                                                                                                                                                                                                                                                                         |
| <b>Authentication</b>           | Choose the authentication method used for the connection to the cellular network. It can be <b>none</b> , <b>pap</b> , <b>chap</b> , or <b>pap_chap</b> .                                                                                                                                                                                                                                               |
| <b>Profile Username</b>         | Enter the username to use when making cellular connections for web services. It can be 1 to 32 characters. It can contain any alphanumeric characters, including spaces.                                                                                                                                                                                                                                |
| <b>Profile Password</b>         | Enter the user password to use when making cellular connections for web services. The password is case-sensitive and can be clear text, or an AES-encrypted key.<br><br>From Cisco Catalyst SD-WAN Manager Release 20.15.1, when you enter the password as clear text, Cisco SD-WAN Manager encrypts the password. When you view the configuration preview, the password appears in its encrypted form. |
| <b>Packet Data Network Type</b> | Choose the packet data network (PDN) type of the cellular network. It can be IPv4, IPv6, or IPv4v6.                                                                                                                                                                                                                                                                                                     |
| <b>No Overwrite</b>             | Enable this option to overwrite the profile on the cellular modem. By default, this option is disabled.                                                                                                                                                                                                                                                                                                 |

#### What to do next

Also see [Deploy a configuration group](#).

## Configure Cellular Interfaces Using Templates

To configure cellular interfaces using Cisco SD-WAN Manager templates:

1. Create a VPN Interface Cellular feature template to configure cellular module parameters, as described in this section.
2. Create a Cellular Profile template to configure the profiles used by the cellular modem.
3. Create a VPN feature template to configure VPN parameters.



**Note** If your deployment includes devices with cellular interface, you must include cellular controller templates in Cisco SD-WAN Manager, even if these templates are not used.

If the device has the LTE or cellular controller module configured and the cellular controller feature template does not exist, then the device tries to remove the cellular controller template. For releases earlier than Cisco IOS XE Release 17.4.2, the following error message is displayed.

```
bad-cli - No controller Cellular 0/2/0, parser-context - No controller Cellular 0/2/0,
parser-response % Cannot remove controllers this way
```

For devices running on Cisco IOS XE Release 17.4.2 and later, the device will return an access-denied error message.

### Create VPN Interface Cellular

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**.



**Note** In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. From the **Create Template** drop-down list, choose **From Feature Template**.
4. From the **Device Model** drop-down list, select the type of device for which you are creating the template.
5. Click **Transport & Management VPN** or scroll to the **Transport & Management VPN** section.
6. Under **Additional Cisco VPN 0 Templates**, click **VPN Interface Cellular**.
7. From the **VPN Interface Cellular** drop-down list, click **Create Template**. The VPN Interface Cellular template form is displayed.  
  
This form contains fields for naming the template, and fields for defining the VPN Interface Cellular parameters.
8. In **Template Name**, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.
9. In **Template Description**, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default or to enter a value, click the scope drop-down list.

### Configure Basic Cellular Interface Functionality

To configure basic cellular interface functionality, click **Basic Configuration** and configure the following parameters. Parameters marked with an asterisk are required to configure an interface. You must also configure a tunnel interface for the cellular interface.

Table 301:

| Parameter Name       | Description                                                                                                                                                                                                                                    |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Shutdown*            | Click <b>No</b> to enable the interface.                                                                                                                                                                                                       |
| Interface Name*      | Enter the name of the interface. It must be <b>cellular0</b> .                                                                                                                                                                                 |
| Description          | Enter a description of the cellular interface.                                                                                                                                                                                                 |
| DHCP Helper          | Enter up to four IP addresses for DHCP servers in the network, separated by commas, to have the interface be a DHCP helper. A DHCP helper interface forwards BOOTP (Broadcast) DHCP requests that it receives from the specified DHCP servers. |
| Bandwidth Upstream   | For transmitted traffic, set the bandwidth above which to generate notifications. Range: 1 through $(2^{32} / 2) - 1$ kbps                                                                                                                     |
| Bandwidth Downstream | For received traffic, set the bandwidth above which to generate notifications. Range: 1 through $(2^{32} / 2) - 1$ kbps                                                                                                                        |
| IP MTU*              | Enter 1428 to set the MTU size, in bytes. This value must be 1428. You cannot use a different value.                                                                                                                                           |

To save the feature template, click **Save**.

### Create a Tunnel Interface

To configure an interface in VPN 0 to be a WAN transport connection, you must configure a tunnel interface on the cellular interface. The tunnel, which provides security from attacks, is used to send the phone number. At a minimum, select **On** and select a color for the interface, as described in the previous section. You can generally accept the system defaults for the remainder of the tunnel interface settings.

To configure a tunnel interface, click **Tunnel**, and configure the following parameters. Parameters marked with an asterisk (\*) are required to configure a cellular interface.

| Parameter Name            | Description                                                                                                                                                                                                            |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tunnel Interface*         | From the drop-down, select <b>Global</b> . Click <b>On</b> to create a tunnel interface.                                                                                                                               |
| Per-tunnel QoS            | From the drop-down, select <b>Global</b> . Click <b>On</b> to create per-tunnel QoS. You can apply a Quality of Service (QoS) policy on individual tunnels, and is only supported for hub-to-spoke network topologies. |
| Per-tunnel QoS Aggregator | From the drop-down, select <b>Global</b> . Click <b>On</b> to create per-tunnel QoS.<br><b>Note</b><br>'bandwidth downstream' is required for per-Tunnel QoS feature to take effect as spoke role.                     |
| Color*                    | From the drop-down, select <b>Global</b> . Select a color for the TLOC. The color typically used for cellular interface tunnels is <b>lte</b> .                                                                        |

| Parameter Name                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Color Description             | <p>Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.18.1</p> <p>Enter a description associated to the TLOC color.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Groups                        | From the drop-down, select <b>Global</b> . Enter the list of groups in the field.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Border                        | From the drop-down, select <b>Global</b> . Click <b>On</b> to set TLOC as border TLOC.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Maximum Control Connections   | <p>Set the maximum number of Cisco SD-WAN Controller that the WAN tunnel interface can connect to. To have the tunnel establish no control connections, set the number to 0. Range: 0 through 8</p> <p>Default: 2</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| vBond As STUN Server          | Click <b>On</b> to enable Session Traversal Utilities for NAT (STUN) to allow the tunnel interface to discover its public IP address and port number when the router is located behind a NAT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Exclude Control Group List    | <p>Set the identifiers of one or more Cisco SD-WAN Controller groups that this tunnel is not allows to establish control connections with.</p> <p>Range: 0 through 100</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| vManage Connection Preference | <p>Set the preference for using the tunnel to exchange control traffic with the Cisco SD-WAN Manager.</p> <p>Range: 0 through 9</p> <p>Default: 5</p> <p>If the edge device has two or more cellular interfaces, you can minimize the amount of traffic between the Cisco SD-WAN Manager and the cellular interfaces by setting one of the interfaces to be the preferred one to use when sending updates to the Cisco SD-WAN Manager and receiving configurations from the Cisco SD-WAN Manager.</p> <p>To have a tunnel interface never connect to the Cisco SD-WAN Manager, set the number to 0. At least one tunnel interface on the edge device must have a nonzero Cisco SD-WAN Manager connection preference.</p> |
| Full Port Hop                 | <p>Minimum release: Cisco IOS XE Catalyst SD-WAN Release 17.18.1a</p> <p>Enable full port hopping at the TLOC level to allow devices to establish connections with controllers by switching to the next port if the current port is blocked or non-functional.</p> <p>Default: Disabled</p>                                                                                                                                                                                                                                                                                                                                                                                                                              |

| Parameter Name      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Port Hop            | <p>From the drop-down, select <b>Global</b>. Click <b>Off</b> to allow port hopping on tunnel interface.</p> <p>Default: <b>On</b>, which disallows port hopping on tunnel interface.</p> <p>Starting from Cisco IOS XE Catalyst SD-WAN Release 17.18.1a, this field is deprecated. Instead use the <b>Full Port Hop</b> option. See the <b>Full Port Hop</b> field.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Low-Bandwidth Link  | <p>Click <b>On</b> to set the tunnel interface as a low-bandwidth link.</p> <p>Default: <b>Off</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Tunnel TCP MSS      | <p>TCP MSS affects any packet that contains an initial TCP header that flows through the router. When configured, TCP MSS is examined against the MSS exchanged in the three-way handshake. The MSS in the header is lowered if the configured TCP MSS setting is lower than the MSS in the header. If the MSS header value is already lower than the TCP MSS, the packets flow through unmodified. The host at the end of the tunnel uses the lower setting of the two hosts. If the TCP MSS is to be configured, it should be set at 40 bytes lower than the minimum path MTU.</p> <p>Specify the MSS of TPC SYN packets passing through the Cisco IOS XE Catalyst SD-WAN device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. Range: 552 to 1460 bytes. Default: None</p> |
| Clear-Dont-Fragment | <p>Configure <b>Clear-Dont-Fragment</b> for packets that arrive at an interface that has Don't Fragment configured. If these packets are larger than what MTU allows, they are dropped. If you clear the Don't Fragment bit, the packets are fragmented and sent.</p> <p>Click <b>On</b> to clear the Dont Fragment bit in the IPv4 packet header for packets being transmitted out of the interface. When the Dont Fragment bit is cleared, packets larger than the MTU of the interface are fragmented before being sent.</p> <p><b>Note</b><br/> <b>Clear-Dont-Fragment</b> clears the Dont Fragment bit and the Dont Fragment bit is set. For packets not requiring fragmentation, the Dont Fragment bit is not affected.</p>                                                                                                                                |
| Network Broadcast   | <p>From the drop-down, select <b>Global</b>. Click <b>On</b> to accept and respond to network-prefix-directed broadcasts. Turn this <b>On</b> only if the <b>Directed Broadcast</b> is enabled on the LAN interface feature template.</p> <p>Default: <b>Off</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Allow Service       | <p>Click <b>On</b> or <b>Off</b> for each service to allow or disallow the service on the cellular interface.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

To configure additional tunnel interface parameters, click **Advanced Options** and configure the following parameters:

Table 302:

| Parameter Name       | Description                                                                                                                                                                                                                                                                                                                                   |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GRE                  | <p>From the drop-down, select <b>Global</b>. Click <b>On</b> to use GRE encapsulation on the tunnel interface. By default, GRE is disabled.</p> <p>If you select both IPsec and GRE encapsulations, two TLOCs are created for the tunnel interface that have the same IP addresses and colors, but that differ by their encapsulation.</p>    |
| GRE Preference       | <p>From the drop-down, select <b>Global</b>. Enter a value to set GRE preference for TLOC.</p> <p>Range: 0 to 4294967295</p>                                                                                                                                                                                                                  |
| GRE Weight           | <p>From the drop-down, select <b>Global</b>. Enter a value to set GRE weight for TLOC.</p> <p>Default: 1</p>                                                                                                                                                                                                                                  |
| IPsec                | <p>From the drop-down, select <b>Global</b>. Click <b>On</b> to use IPsec encapsulation on the tunnel interface. By default, IPsec is enabled.</p> <p>If you select both IPsec and GRE encapsulations, two TLOCs are created for the tunnel interface that have the same IP addresses and colors, but that differ by their encapsulation.</p> |
| IPsec Preference     | <p>From the drop-down, select <b>Global</b>. Enter a value to set the preference for directing traffic to the tunnel. A higher value is preferred over a lower value.</p> <p>Range: 0 through 4294967295. Default: 0</p>                                                                                                                      |
| IPsec Weight         | <p>From the drop-down, select <b>Global</b>. Enter a value to set weight for balancing traffic across multiple TLOCs. A higher value sends more traffic to the tunnel.</p> <p>Range: 1 through 255. Default: 1</p>                                                                                                                            |
| Carrier              | <p>From the drop-down, select <b>Global</b>. From the <b>Carrier</b> drop-down, select the carrier name or private network identifier to associate with the tunnel. Values: carrier1, carrier2, carrier3, carrier4, carrier5, carrier6, carrier7, carrier8, default. Default: default</p>                                                     |
| Bind Loopback Tunnel | <p>Enter the name of a physical interface to bind to a loopback interface. The interface name has the format <b>ge slot/port</b>.</p>                                                                                                                                                                                                         |



| Parameter Name       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Last-Resort Circuit  | <p>From the drop-down, select <b>Global</b>. Click <b>On</b> to use the tunnel interface as the circuit of last resort. By default, it is disabled.</p> <p><b>Note</b><br/>An interface configured as a circuit of last resort is expected to be down and is skipped while calculating the number of control connections, the cellular modem becomes dormant, and no traffic is sent over the circuit.</p> <p>When the configurations are activated on the edge device with cellular interfaces, then all the interfaces begin the process of establishing control and BFD connections. When one or more of the primary interfaces establishes a BFD connection, the circuit of last resort shuts itself down.</p> <p>Only when all the primary interfaces lose their connections to remote edges, then the circuit of last resort activates itself triggering a BFD TLOC Down alarm and a Control TLOC Down alarm on the edge device. The last resort interfaces are used as backup circuit on edge device and are activated when all other transport links BFD sessions fail. In this mode the radio interface is turned off, and no control or data connections exist over the cellular interface.</p>                                                                                                                                                                                                                                                                                                                                           |
| NAT Refresh Interval | Set the interval between NAT refresh packets sent on a DTLS or TLS WAN transport connection. Range: 1 through 60 seconds. Default: 5 seconds.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Hello Interval       | Enter the interval between Hello packets sent on a DTLS or TLS WAN transport connection. Range: 100 through 10000 milliseconds. Default: 1000 milliseconds (1 second).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Hello Tolerance      | <p>Enter the time to wait for a Hello packet on a DTLS or TLS WAN transport connection before declaring that transport tunnel to be down.</p> <p>Range: 12 through 60 seconds. Default: 12 seconds.</p> <p>The default hello interval is 1000 milliseconds, and it can be a time in the range 100 through 600000 milliseconds (10 minutes). The default hello tolerance is 12 seconds, and it can be a time in the range 12 through 600 seconds (10 minutes). To reduce outgoing control packets on a TLOC, it is recommended that on the tunnel interface you set the hello interval to 60000 milliseconds (10 minutes) and the hello tolerance to 600 seconds (10 minutes) and include the <b>no track-transport disable</b> regular checking of the DTLS connection between the edge device and the controller. For a tunnel connection between a edge device and any controller device, the tunnel uses the hello interval and tolerance times configured on the edge device. This choice is made to minimize the traffic sent over the tunnel, to allow for situations where the cost of a link is a function of the amount of traffic traversing the link. The hello interval and tolerance times are chosen separately for each tunnel between a edge device and a controller device. Another step taken to minimize the amount of control plane traffic is to not send or receive OMP control traffic over a cellular interface when other interfaces are available. This behavior is inherent in the software and is not configurable.</p> |

To save the feature template, click **Save**.

### Configure the Cellular Interface as a NAT Device

To configure a cellular interface to act as a NAT device for applications such as port forwarding, click **NAT**, and configure the following parameters:

**Table 303:**

| Parameter Name  | Description                                                                                                                                              |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| NAT             | Click <b>On</b> to have the interface act as a NAT device.                                                                                               |
| Refresh Mode    | Select how NAT mappings are refreshed, either outbound or bidirectional (outbound and inbound). Default: Outbound                                        |
| UDP Timeout     | Specify when NAT translations over UDP sessions time out. Range: 1 through 65536 minutes. Default: 1 minute                                              |
| TCP Timeout     | Specify when NAT translations over TCP sessions time out. Range: 1 through 65536 minutes. Default: 60 minutes (1 hour)                                   |
| Block ICMP      | Select <b>On</b> to block inbound ICMP error messages. By default, a router acting as a NAT device receives these error messages. Default: Off           |
| Respond to Ping | Select <b>On</b> to have the router respond to ping requests to the NAT interface's IP address that are received from the public side of the connection. |

To create a port forwarding rule, click **Add New Port Forwarding Rule** and configure the following parameters. You can define up to 128 port-forwarding rules to allow requests from an external network to reach devices on the internal network.

**Table 304:**

| Parameter Name   | Description                                                                                                                                                   |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Port Start Range | Enter a port number to define the port or first port in the range of interest. Range: 0 through 65535                                                         |
| Port End Range   | Enter the same port number to apply port forwarding to a single port, or enter a larger number to apply it to a range of ports. Range: 0 through 65535        |
| Protocol         | Select the protocol to which to apply the port-forwarding rule, either TCP or UDP. To match the same ports for both TCP and UDP traffic, configure two rules. |
| VPN              | Specify the private VPN in which the internal server resides. This VPN is one of the VPN identifiers in the overlay network. Range: 0 through 65527           |
| Private IP       | Specify the IP address of the internal server to which to direct traffic that matches the port-forwarding rule.                                               |

To save a port forwarding rule, click **Add**.

To save the feature template, click **Save**.

### Apply Access Lists

To configure a shaping rate to a cellular interface and to apply a QoS map, a rewrite rule, access lists, and policers to a router interface, click **ACL/QoS** and configure the following parameters:

**Table 305: Access Lists Parameters**

| Parameter Name     | Description                                                                                                                  |
|--------------------|------------------------------------------------------------------------------------------------------------------------------|
| Shaping rate       | Configure the aggregate traffic transmission rate on the interface to be less than line rate, in kilobits per second (kbps). |
| QoS map            | Specify the name of the QoS map to apply to packets being transmitted out the interface.                                     |
| Rewrite rule       | Click <b>On</b> , and specify the name of the rewrite rule to apply on the interface.                                        |
| Ingress ACL – IPv4 | Click <b>On</b> , and specify the name of an IPv4 access list to packets being received on the interface.                    |
| Egress ACL– IPv4   | Click <b>On</b> , and specify the name of an IPv4 access list to packets being transmitted on the interface.                 |
| Ingress ACL – IPv6 | Click <b>On</b> , and specify the name of an IPv6 access list to packets being received on the interface.                    |
| Egress ACL– IPv6   | Click <b>On</b> , and specify the name of an IPv6 access list to packets being transmitted on the interface.                 |
| Ingress policer    | Click <b>On</b> , and specify the name of the policer to apply to packets being received on the interface.                   |
| Egress policer     | Click <b>On</b> , and specify the name of the policer to apply to packets being transmitted on the interface.                |

To save the feature template, click **Save**.

### Add ARP Table Entries

To configure static Address Resolution Protocol (ARP) table entries on the interface, click **ARP**. Then click **Add New ARP** and configure the following parameters:

**Table 306:**

| Parameter Name | Description                                                                                          |
|----------------|------------------------------------------------------------------------------------------------------|
| IP Address     | Enter the IP address for the ARP entry in dotted decimal notation or as a fully qualified host name. |
| MAC Address    | Enter the MAC address in colon-separated hexadecimal notation.                                       |

To save the ARP configuration, click **Add**.

To save the feature template, click **Save**.

### Configure Other Interface Properties

To configure other interface properties, click **Advanced** and configure the following parameters.

**Table 307: Cellular Interfaces Advanced Parameters**

| Parameter Name        | Description                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PMTU Discovery        | Click <b>On</b> to enable path MTU discovery on the interface, to allow the router to determine the largest MTU size supported without requiring packet fragmentation.                                                                                                                                                                                                                                                       |
| TCP MSS               | Specify the maximum segment size (MSS) of TCP SYN packets passing through the router. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. Range: 552 to 1460 bytes. Default: None.                                                                                                                                                              |
| Clear-Don't-Fragment  | Click <b>On</b> to clear the Don't Fragment (DF) bit in the IPv4 packet header for packets being transmitted out the interface. When the DF bit is cleared, packets larger than that interface's MTU are fragmented before being sent.                                                                                                                                                                                       |
| Static Ingress QoS    | Select a queue number to use for incoming traffic. Range: 0 through 7                                                                                                                                                                                                                                                                                                                                                        |
| Autonegotiate         | Click <b>Off</b> to turn off autonegotiation. By default, an interface runs in autonegotiation mode.                                                                                                                                                                                                                                                                                                                         |
| TLOC Extension        | Enter the name of a physical interface on the same router that connects to the WAN transport. This configuration then binds this service-side interface to the WAN transport. A second router at the same site that itself has no direct connection to the WAN (generally because the site has only a single WAN connection) and that connects to this service-side interface is then provided with a connection to the WAN. |
| Tracker               | Enter the name of a tracker to track the status of transport interfaces that connect to the internet.                                                                                                                                                                                                                                                                                                                        |
| IP Directed-Broadcast | From the drop-down, select <b>Global</b> . Click <b>On</b> for IP directed-broadcast.<br>Default: <b>Off</b>                                                                                                                                                                                                                                                                                                                 |

To save the feature template, click **Save**.

## Configure Cellular Interfaces Using CLI

The following example enables a cellular interface:

```
interface Cellular0/2/0
 description Cellular interface
 no shutdown
 ip address negotiated
 ip mtu 1428
 mtu 1500
 exit

controller Cellular 0/2/0
 lte sim max-retry 1
 lte failovertimer 7
 profile id 1 apn Broadband authentication none pdn-type ipv4
```

# Reset the profile configuration of a cellular modem using the CLI

Starting from Cisco IOS XE Catalyst SD-WAN Release 17.18.1a, when you need to switch from one cellular provider to another, you can reset the cellular modem to clear existing profile configurations and apply new settings using the **cellularslot lte profile reset** command.

Resetting the profile configuration of a modem prevents remnants of previous carrier settings (such as Access Point Name (APN) or authentication details) from interfering with new configurations, which is vital for successful network attachment and operation.

The following examples reset the profile configurations of a cellular modem:

- For routers, use the command with a 3-tuple slot identifier.

```
CellularGateway# cellular 0/2/0 profile-reset
```

- For cellular gateways, the *slot* variable is 1.

```
CellularGateway# cellular 1 profile-reset
```



---

**Note** When the modem's activated firmware is set to **Generic**, executing the command **cellularslot lte profile reset** will not perform any action

---

## LTE Modem Crash Diagnostics

In the event that an LTE modem crashes, *crash dump* is the information stored in the device bootflash that:

- makes troubleshooting easier by appearing in the admin-tech file
- helps diagnose the cause of the crash

Use the **lte modem crash-action auto-collect** command to automatically gather and store detailed operational data in the event of an LTE modem crash.

To disable, use the **no lte modem crash-action auto-collect** command.

When enabled, this command instructs the device to automatically collect a crash dump from the LTE modem upon a crash event. This dump contains various diagnostic logs and memory states that are essential for analyzing why the modem crashed.



---

**Note** Ensure that the device has sufficient bootflash storage available to accommodate the large amount of data collected.

---

### Verify the Crash-Dump Progress

Use the following command to verify the crash-dump progress.

```
Device#show cellular 0/2/0 logs modem-crashdump
Modem crashdump logging = off
Device#
```

When a crash-dump is in progress, it is displayed as follows:

```
Router#show cellular 0/3/0 logs modem-crashdump
Modem crashdump logging = on
Progress = 44%
Router#
Router#show cellular 0/3/0 logs modem-crashdump
Modem crashdump logging = on
Progress = 98%
Router#show cellular 0/3/0 logs modem-crashdump
Modem crashdump logging = off
```

Once the collection is complete, the status will return to off.

## Data Profile

**Table 308: Feature History**

| Feature Name                                                                    | Release Information                                                          | Description                                                             |
|---------------------------------------------------------------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Ability to Configure APNs under Running Configurations for Single and Dual SIMs | Cisco IOS XE Catalyst SD-WAN Release 17.8.1a<br>Cisco vManage Release 20.8.1 | This feature allows you to create a data profile for a cellular device. |

A data profile for a cellular device defines the following parameters, which the device uses for communication with the service provider. You can configure the following parameters by using the **profile id** command in cellular configuration mode. For more information about the following parameters, see [profile id](#).

- Identification number of the data profile
- Name of the access point network of the service provider
- Authentication type used for APN access: No authentication, CHAP authentication only, PAP authentication only, or either CHAP or PAP authentication
- Username and password that are provided by the service provider for APN access authentication, if authentication is used
- Type of packet data matching that is used for APN access: IPv4 type bearer, IPv6 type bearer, or IPv4v6 type bearer
- SIM slot that contains the SIM to configure

## Best Practices for Configuring Cellular Interfaces

Cellular technology on edge devices can be used in a number of ways:

- Circuit of last resort: An interface configured as a circuit of last resort is expected to be down and is skipped while calculating the number of control connections, the cellular modem becomes dormant, and no traffic is sent over the circuit.

When the configurations are activated on the edge device with cellular interfaces, then all the interfaces begin the process of establishing control and BFD connections. When one or more of the primary interfaces establishes a BFD connection, the circuit of last resort shuts itself down.

Only when all the primary interfaces lose their connections to remote edges, then the circuit of last resort activates itself triggering a BFD TLOC Down alarm and a Control TLOC Down alarm on the edge device. The last resort interfaces are used as backup circuit on edge device and are activated when all other transport links BFD sessions fail. In this mode the radio interface is turned off, and no control or data connections exist over the cellular interface.

Use the **last-resort-circuit** command to configure a cellular interface to be a circuit of last resort.



---

**Note** **last-resort-circuit** is not limited to cellular interfaces.

The operating principle for cellular interfaces also applies to GigabitEthernet interfaces.

---

- **Active circuit:** You can choose to use a cellular interface as an active circuit, perhaps because it is the only last-mile circuit or to always keep the cellular interface active so that you can measure the performance of the circuit. In this scenario the amount of bandwidth utilized to maintain control and data connections over the cellular interface can become a concern. Here are some best practices to minimize bandwidth usage over a cellular interface:
  - **Increase control packet timers**—To minimize control traffic on a cellular interface, you can decrease how often protocol update messages are sent on the interface. OMP sends Update packets every second, by default. You can increase this interval to a maximum of 65535 seconds (about 18 hours) by including the **omp timers advertisement-interval** configuration command. BFD sends Hello packets every second, by default. You can increase this interval to a maximum of 5 minutes (300000 milliseconds) by including the **bfd color hello-interval** configuration command. (Note that you specify the OMP Update packet interval in seconds and the BFD Hello packet interval in milliseconds.)
  - **Prioritize Cisco SD-WAN Manager control traffic over a non-cellular interface:** When a edge device has both cellular and non-cellular transport interfaces, by default, the edge device chooses one of the interfaces to use to exchange control traffic with the Cisco SD-WAN Manager. You can configure the edge device to never use the cellular interface to exchange traffic with the Cisco SD-WAN Manager, or you can configure a lower preference for using the cellular interface for this traffic. You configure the preference by including the **vmanage-connection-preference** command when configuring the tunnel interface. By default, all tunnel interface have a Cisco SD-WAN Manager connection preference value of 5. The value can range from 0 through 8, where a higher value is more preferred. A tunnel with a preference value of 0 can never exchange control traffic with the Cisco SD-WAN Manager.



---

**Note** At least one tunnel interface on the edge device must have a non-0 Cisco SD-WAN Manager connection preference value. Otherwise, the device has no control connections.

---







## CHAPTER 15

# Hot Standby Router Protocol (HSRP)

Table 309: Feature History

| Feature Name                                                                     | Release Information                                                                                         | Description                                                                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Support for HSRP and HSRP Authentication on Cisco IOS XE Catalyst SD-WAN Devices | Cisco IOS XE Catalyst SD-WAN Release 17.7.1a<br>Cisco vManage Release 20.7.1<br>Cisco SD-WAN Release 20.7.1 | This feature allows you to configure HSRPv2 and HSRP authentication on Cisco IOS XE Catalyst SD-WAN platforms via CLI template. HSRP is a long-standing Cisco proprietary First Hop Redundancy Protocol (FHRP) to support version 2 of the protocol and authentication. |

- [Information About HSRP, on page 651](#)
- [Supported Devices for HSRP, on page 653](#)
- [Configure HSRP Using CLI, on page 654](#)
- [Verify HSRP Configurations Using CLI, on page 656](#)

## Information About HSRP

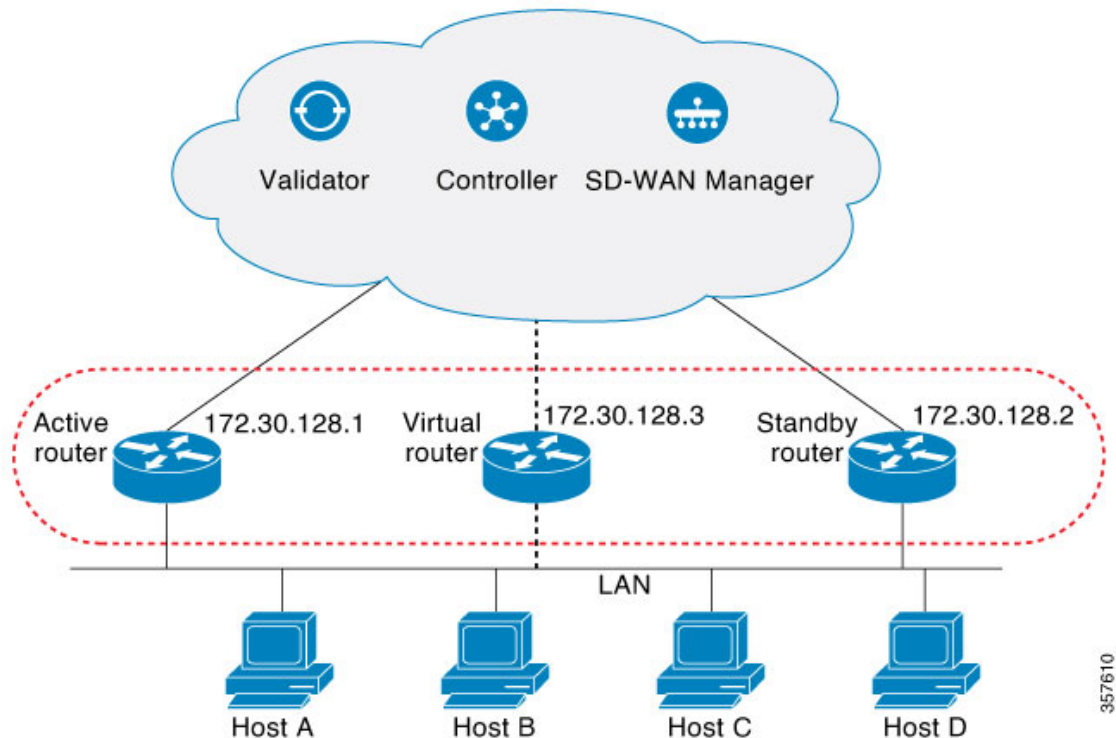
The Hot Standby Router Protocol (HSRP) is a First Hop Redundancy Protocol (FHRP) designed to allow transparent failover of the first-hop IP device. HSRP provides high network availability by providing first-hop routing redundancy for IP hosts on networks configured with a default gateway IP address. For identifying an active and standby device in a group of routers, HSRP is used. In a group of device interfaces, the active device is the device of choice for routing packets; the standby device is the device that takes over if the active device fails or if preset conditions are met.

You can configure multiple hot standby groups on an interface, thereby making full use of redundant devices and load sharing.

The following figure shows a network configured for HSRP. By sharing a virtual MAC address and IP address, two or more devices can act as a single virtual router. The virtual device represents the common default gateway for devices that are configured to provide backup to each other. You don't need to configure the hosts on the LAN with the IP address of the active device. Instead, you can configure them with the IP address (virtual IP address) of the virtual device as their default gateway. If the active device fails to send a hello

message within a configurable time period, the standby device takes over and responds to the virtual addresses and becomes the active device, taking over the duties of the active device.

**Figure 2: HSRP Topology**



### HSRP Version 2 Support

Following are the HSRP version 2 (HSRPv2) features:

- HSRPv2 advertises and learns millisecond timer values. This change ensures stability of the HSRP groups in all cases.
- HSRPv2 expands the group number range from 0 to 4095.
- HSRPv2 provides improved management and troubleshooting. The HSRPv2 packet format includes a 6-byte identifier field that is used to uniquely identify the sender of the message. Typically, this field is populated with the interface MAC address.
- HSRPv2 uses the IP multicast address 224.0.0.102 to send hello packets. This multicast address allows Cisco Group Management Protocol (CGMP) leave processing to be enabled at the same time as HSRP.
- HSRPv2 has a different packet format that uses a type-length-value (TLV) format.

### HSRP MD5 Authentication

HSRP supports simple plain text string and message digest 5 (MD5) schemes of protocol packets authentication. HSRP MD5 authentication is an advanced type of authentication that generates an MD5 digest for the HSRP portion of the multicast HSRP protocol packet. This functionality provides added security and protects against the threat from HSRP-spoofing software.

MD5 authentication provides greater security than the alternative plain text authentication scheme. MD5 authentication allows each HSRP group member to use a secret key to generate a keyed MD5 hash that is part of the outgoing packet. A keyed hash of an incoming packet is generated, and if the hash within the incoming packet doesn't match the generated hash, the packet is ignored.

The key for the MD5 hash can be either given directly in the configuration using a key string or supplied indirectly through a key chain.

HSRP packets will be rejected in any of the following cases:

- The authentication schemes differ on the device and in the incoming packets.
- MD5 digests differ on the device and in the incoming packets.
- Text authentication strings differ on the device and in the incoming packets.

### HSRP Object Tracking

Object tracking separates the tracking mechanism from HSRP and creates a separate standalone tracking process that can be used by any other process and HSRP. The priority of a device can change dynamically when it has been configured for object tracking, and the object that is being tracked goes down. Examples of objects that can be tracked are the line protocol state of an interface or the reachability of an IP route. If the specified object goes down, the HSRP priority is reduced.

### HSRP Benefits

- Redundancy: HSRP employs a redundancy scheme that is time proven and deployed extensively in large networks.
- Fast Failover: HSRP provides transparent fast failover of the first-hop device.
- Preemption: Preemption allows a standby device to delay becoming active for a configurable amount of time.
- Authentication: HSRP MD5 algorithm authentication protects against HSRP-spoofing software and uses the industry-standard MD5 algorithm for improved reliability and security.

## Supported Devices for HSRP

Cisco Catalyst 8500 Series Edge Platforms

Cisco Catalyst 8300 Series Edge Platforms

Cisco Catalyst 8200 Series Edge Platforms

Cisco Catalyst 8200 uCPE Series Edge Platforms

Cisco ASR 1000 Series Aggregation Services Routers

Cisco ISR 1000 and ISR 4000 Series Integrated Services Routers (ISRs)

Cisco ISR 1100 and ISR 1100X Series Integrated Services Routers (ISRs)

Cisco IR1101 Integrated Services Router Rugged

Cisco Catalyst 8000v Series Cloud Services Router

For details on supported models for each of these device families, refer to [Cisco Catalyst SD-WAN Device Compatibility](#) page.

## Configure HSRP Using CLI

You can configure HSRP using the Cisco SD-WAN Manager CLI Add-on feature templates and CLI device templates. For more information on configuring using CLI templates, see [CLI Templates](#).



**Note** The following commands can be used in any order.

The following list provides information about HSRP configuration on Cisco IOS XE Catalyst SD-WAN devices.

- Enable HSRP.

Create (or enable) the HSRP group in IPv4 using its number and virtual IP address:

```
Device(config)# interface interface-type
Device(config-if)# standby group-number ip [ip-address [secondary]]
```

Activate HSRP in IPv6:

```
Device(config)# interface interface-type
Device(config-if)# standby group-number ipv6 {link-local-address | autoconfig }
```

- Change to Version 2.

Change the HSRP version. Note that the **nostandby** or **nostandby version 2** commands are rejected when the interface has IPv6 groups.



**Note** **nostandby** or **nostandby version 2** command is rejected when the interface has IPv6 groups.

```
Device(config)# interface interface-type
Device(config-if)# standby version {1|2}
```

- Configure HSRP priority and preemption.

Set the priority value used in choosing the active router, and configure HSRP preemption and preemption delay:

```
Device(config)# interface interface-type
Device(config-if)# standby group-number ip [ip-address [secondary]]
Device(config-if)# standby group-number priority [priority]
Device(config-if)# standby group-number preempt [delay{ [minimum seconds] [reload seconds] [sync seconds] }]
```

- Configure HSRP Authentication.

Configure HSRP MD5 authentication using a key chain.

Key chains allow a different key string to be used at different times according to the key chain configuration. HSRP queries the appropriate key chain to obtain the current live key and key ID for the specified key chain.

```

Device(config)# interface interface-type
Device(config-if)# ip address ip-address mask [secondary]
Device(config-if)# standby group-number priority [priority]
Device(config-if)# standby group-number preempt [delay{ [minimum seconds] [reload
seconds] [sync seconds]]]
Device(config-if)# standby group-number authentication md5 key-chain key-chain-name
Device(config-if)# standby group-number ip [ip-address [secondary]]

```

Configure HSRP text authentication.

The authentication string can be up to eight characters in length; the default string is Cisco.

```

Device(config)# interface interface-type
Device(config-if)# ip address ip-address mask [secondary]
Device(config-if)# standby group-number priority [priority]
Device(config-if)# standby group-number preempt [delay{ [minimum seconds] [reload
seconds] [sync seconds]]]
Device(config-if)# standby group-number authentication text string
Device(config-if)# standby group-number ip [ip-address [secondary]]

```

- Configure HSRP timers.

Configure the time between the hello packets and the time before other routers declare the active router to be inactive:

```

Device(config)# interface interface-type
Device(config-if)# standby group-number ip [ip-address [secondary]]
Device(config-if)# standby group-number timers hellotime holdtime

```

- Configure HSRP object tracking.

Configure HSRP to track an object and change the HSRP priority based on the state of the object:

```

Device(config)# interface interface-type
Device(config-if)# standby group-number track object-number [decrement priority-decrement]
[shutdown]

```

- Improve CPU and network performance with HSRP multiple group optimization.

Configure an HSRP group as a client group:

```

Device(config)# interface interface-type
Device(config-if)# standby group-number follow group-name

```

Configure the HSRP client group refresh interval:

```

Device(config)# interface interface-type
Device(config-if)# standby group-number mac-refresh seconds

```

- Configure an HSRP virtual MAC address.

Specify a virtual MAC address for HSRP:

```

Device(config)# interface interface-type
Device(config-if)# standby group-number mac-address mac-address

```

- Link IP redundancy clients to HSRP groups.

Configure the name of a standby group:

```

Device(config)# interface interface-type
Device(config-if)# standby group-number name [redundancy-name]

```

The following is a complete HSRP configuration example on Cisco IOS XE Catalyst SD-WAN devices through CLI:

```

config-transaction
!
 interface GigabitEthernet0/0/1.94
 encapsulation dot1Q 94
 vrf forwarding 509
 ip address 10.96.194.2 255.255.255.0
 ip directed-broadcast
 ip mtu 1500
 ip nbar protocol-discovery
 standby version 2
 standby 1 preempt
 standby 94 ip 10.96.194.1
 standby 94 timers 1 4
 standby 94 priority 110
 standby 94 preempt delay minimum 180
 standby 94 authentication md5 key-string 7 094F471A1A0A
 standby 94 track 8 shutdown
 standby 194 ipv6 2001:10:96:194::1/64
 standby 194 timers 1 4
 standby 194 priority 110
 standby 194 preempt delay minimum 180
 standby 194 authentication md5 key-string 7 094F471A1A0A
 standby 194 track 80 shutdown
 ip policy route-map clear-df
 ipv6 address 2001:10:96:194::2/64
 ipv6 mtu 1500
 arp timeout 1200
 end

```

## Verify HSRP Configurations Using CLI

The following is a sample output from the **show standby** command displaying the standby router information:

```

Device# show standby
GigabitEthernet0/0/1.94 - Group 94 (version 2)
 State is Standby
 1 state change, last state change 01:06:09
 Track object 8 state Up
 Virtual IP address is 10.96.194.1
 Active virtual MAC address is 0000.0c9f.f05e (MAC Not In Use)
 Local virtual MAC address is 0000.0c9f.f05e (v2 default)
 Hello time 1 sec, hold time 4 sec
 Next hello sent in 0.688 secs
 Authentication MD5, key-string
 Preemption enabled, delay min 180 secs
 Active router is 10.96.194.2, priority 110 (expires in 4.272 sec)
 MAC address is ccl6.7e8c.6ddl
 Standby router is local
 Priority 105 (configured 105)
 Group name is "hsrp-Gi0/0/1.94-94" (default)
 FLAGS: 0/1
GigabitEthernet0/0/1.94 - Group 194 (version 2)
 State is Standby
 1 state change, last state change 01:06:07
 Track object 80 state Up
 Link-Local Virtual IPv6 address is FE80::5:73FF:FEA0:C2 (impl auto EUI64)
 Virtual IPv6 address 2001:10:96:194::1/64
 Active virtual MAC address is 0005.73a0.00c2 (MAC Not In Use)
 Local virtual MAC address is 0005.73a0.00c2 (v2 IPv6 default)
 Hello time 1 sec, hold time 4 sec
 Next hello sent in 0.480 secs

```

```

Authentication MD5, key-string
Preemption enabled, delay min 180 secs
Active router is FE80::CE16:7EFF:FE8C:6DD1, priority 110 (expires in 4.032 sec)
 MAC address is ccl6.7e8c.6ddl
Standby router is local
Priority 105 (configured 105)
Group name is "hsrp-Gi0/0/1.94-194" (default)
FLAGS: 0/1

```

The following is a sample output from the **show standby** command displaying HSRP Version 2 information if HSRP Version 2 is configured:

```

Device# show standby
Ethernet0/1 - Group 1 (version 2)
 State is Speak
 Virtual IP address is 10.21.0.10
 Active virtual MAC address is unknown
 Local virtual MAC address is 0000.0c9f.f001 (v2 default)
 Hello time 3 sec, hold time 10 sec
 Next hello sent in 1.804 secs
 Preemption enabled
 Active router is unknown
 Standby router is unknown
 Priority 20 (configured 20)
 Group name is "hsrp-Et0/1-1" (default)
Ethernet0/2 - Group 1
 State is Speak
 Virtual IP address is 10.22.0.10
 Active virtual MAC address is unknown
 Local virtual MAC address is 0000.0c07.ac01 (v1 default)
 Hello time 3 sec, hold time 10 sec
 Next hello sent in 1.804 secs
 Preemption disabled
 Active router is unknown
 Standby router is unknown
 Priority 90 (default 100)
 Track interface Serial2/0 state Down decrement 10
 Group name is "hsrp-Et0/2-1" (default)

```

The following is a sample output from the **show standby** command displaying HSRP authentication information if HSRP MD5 authentication is configured:

```

Device# show standby
Ethernet0/1 - Group 1
 State is Active
 5 state changes, last state change 00:17:27
 Virtual IP address is 10.21.0.10
 Active virtual MAC address is 0000.0c07.ac01
 Local virtual MAC address is 0000.0c07.ac01 (default)
 Hello time 3 sec, hold time 10 sec
 Next hello sent in 2.276 secs
 Authentication MD5, key-string, timeout 30 secs
 Preemption enabled
 Active router is local
 Standby router is unknown
 Priority 110 (configured 110)
 Group name is "hsrp-Et0/1-1" (default)

```

The following is a sample output from the **show standby brief** command displaying HSRP information for a specific interface:

```

Device# show standby brief
Interface Grp Pri P State Active Standby Virtual IP
Gi0/0/1.94 94 105 P Standby 10.96.194.2 local 10.96.194.1
Gi0/0/1.94 194 105 P Standby FE80::CE16:7EFF:FE8C:6DD1 local FE80::5:73FF:FEA0:C2

```

The following is a sample output from the **show standby neighbors** command displaying the HSRP neighbors on Ethernet interface 0/0. Neighbor 10.0.0.250 is active for group 2 and standby for groups 1 and 8, and is registered with BFD:

```
Device# show standby neighbors Ethernet0/0
HSRP neighbors on Ethernet0/0
 10.0.0.250
 Active groups: 2
 Standby groups: 1, 8
 BFD enabled
 10.0.0.251
 Active groups: 5, 8
 Standby groups: 2
 BFD enabled
 10.0.0.253
 No Active groups
 No Standby groups
 BFD enabled
```

The following is a sample output from the **show standby neighbors** command displaying information for all HSRP neighbors:

```
Device# show standby neighbors
HSRP neighbors on FastEthernet2/0
 10.0.0.2
 No active groups
 Standby groups: 1
 BFD enabled
HSRP neighbors on FastEthernet2/0
 10.0.0.1
 Active groups: 1
 No standby groups
 BFD enabled
```





## CHAPTER 16

# Configure a Cellular Gateway

- [Configure a Cellular Gateway](#), on page 659
- [Information About Configuring a Cellular Gateway](#), on page 659
- [Supported Cellular Gateway Devices](#), on page 660
- [Configure a Cellular Gateway Using a Feature Template in Cisco SD-WAN Manager](#), on page 660
- [Configure a Cellular Gateway Using a Configuration Group in Cisco SD-WAN Manager](#), on page 663

## Configure a Cellular Gateway

*Table 310: Feature History*

| Feature Name                                               | Release Information                                                                       | Feature Description                                                                                                                                                                        |
|------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cellular Gateway Configuration                             | Cisco vManage Release 20.4.1<br>Cisco IOS XE Catalyst SD-WAN Release 17.4.1a (on devices) | This feature provides templates for configuring a supported cellular gateway as an IP pass-through device. This release supports the Cisco Cellular Gateway CG418-E and CG522-E.           |
| Cellular Gateway Configuration Using a Configuration Group | Cisco Catalyst SD-WAN Manager Release 20.13.1<br>Cisco IOS CG Release 17.13.1             | Added support for configuring cellular gateways using configuration groups. A new Create Cellular Gateway Group workflow creates a configuration group specifically for cellular gateways. |

## Information About Configuring a Cellular Gateway

You can configure a supported cellular gateway as an IP pass-through device. By positioning the configured device in an area in your facility that has a strong LTE signal, the signal can be extended over an Ethernet connection to a routing infrastructure in a location with a weaker LTE signal.

### Secure Communication with Devices through a vmanage-admin Account

Cisco SD-WAN Manager communicates with devices, such as Cisco Catalyst Cellular Gateways, using a secure channel—either a datagram transport layer security (DTLS) tunnel or transport layer security (TLS)

tunnel. Within this secure channel, it communicates with the devices or controllers using the NETCONF protocol, within an SSH session. It uses an internal-use-only passwordless "vmanage-admin" user account on the device or controller. The vmanage-admin account is created during the initial device setup. Cisco SD-WAN Manager uses this secure channel for monitoring, configuring, and managing devices.

As noted, the vmanage-admin user accounts do not have any password associated with them, so Cisco SD-WAN Manager uses a passwordless procedure to log in to the account. To accomplish this, Cisco SD-WAN Manager generates an asymmetric encryption public-private key pair. During deployment of a device, Cisco SD-WAN Manager copies the public key that it has generated to the device. It sends the public key using a proprietary protocol, within a secure channel—a DTLS or TLS tunnel.

The activity that Cisco SD-WAN Manager performs using the vmanage-admin account appears in syslog messages and in the output of certain show commands. The syslog messages are logged with the same level of detail as activities performed through any other user account. The level of syslog detail depends on the syslog configuration of the device.

Cisco SD-WAN Manager requires the vmanage-admin account on devices in order to monitor, configure, and manage the devices. Removing, disabling, or altering this account on a device would prevent Cisco SD-WAN Manager from performing these activities, and is not supported.

## Supported Cellular Gateway Devices

Cisco Catalyst Cellular Gateway models:

- CG418-E
- CG522-E

## Configure a Cellular Gateway Using a Feature Template in Cisco SD-WAN Manager

### Before You Begin

This procedure configures a cellular gateway using a feature template. For information about using a configuration group, see [Configure a Cellular Gateway Using a Configuration Group in Cisco SD-WAN Manager, on page 663](#).

### Configure a Cellular Gateway Using a Feature Template

1. Create a device template for Cisco Cellular Gateway CG418-E devices.

See [Configure Devices](#) in the *Cisco Catalyst SD-WAN Systems and Interfaces Configuration Guide, Cisco IOS XE Catalyst SD-WAN Release 17.x*.

After you enter a description for the feature template:

- a. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
- b. Click **Device Templates**.



**Note** In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

- c. From the **Create Template** drop-down list choose **From Feature Template**.
- d. From the **Device Model** drop-down list select the type of device for which you are creating the template.
- e. Choose **Cellular Gateway > Cellular Gateway Platform > Create Template**. Then configure the Cellular Gateway Platform feature template as shown in the following table.

**Table 311: Cellular Gateway Platform Template Parameters**

| Parameter Name             | Description                                                                                                                                                                                                                                 |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Basic Configuration Tab    |                                                                                                                                                                                                                                             |
| Time Zone                  | Choose the time zone to use for the device. The device uses this time zone for clock synchronization when NTP is configured.                                                                                                                |
| Management Interface       | Enter the IPv4 address of the management interface for accessing the device.                                                                                                                                                                |
| Admin-Password             | Enter the admin user password for logging in to the device by using an SSH client or a console port.                                                                                                                                        |
| NTP-Servers                | Configure one or more NTP servers to which the device synchronizes its clock.                                                                                                                                                               |
| Cellular Configuration Tab |                                                                                                                                                                                                                                             |
| IP-Src-Violation           | Choose <b>v4 only</b> , <b>v6 only</b> , or <b>v4 and v6</b> to enable the IP source violation feature for the corresponding IP address types. Choose <b>None</b> if you do not want to enable this feature.                                |
| Auto-SIM                   | Choose <b>On</b> to enable the auto-SIM feature. When this feature is enabled, the device automatically detects the service provider to which SIMs in the device belong and automatically loads the appropriate firmware for that provider. |
| Primary SIM Slot           | Choose the slot that contains the primary SIM card for the device. If the device loses service to this slot, it fails over to the secondary slot.                                                                                           |
| Failover-Timer (minutes)   | Enter the number of minutes that the device waits before trying to communicate with the primary SIM slot after the device detects loss of service to this slot.                                                                             |

| Parameter Name | Description                                                                                                                                       |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Max-Retry      | Enter the number of consecutive unsuccessful attempts by the device to communicate with the primary SIM before failing over to the secondary slot |

- f. Choose **Cellular Gateway** > **Cellular Gateway Profile** and choose **Create Template** from the Cellular Gateway Profile drop-down list. Then configure the Cellular Gateway Profile feature template as shown in the following table.

**Table 312: Cellular Gateway Profile Template Parameters**

| Parameter Name           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Basic Configuration Tab  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| SIM                      | <p>Choose a SIM slot and configure the following options to create a profile for the SIM in that slot. This profile indicates to the service provider which of its cellular networks the SIM should attach to.</p> <ul style="list-style-type: none"> <li>• Profile ID: Enter a unique ID for the profile</li> <li>• Access Point Name: Enter the name of the access point for this profile</li> <li>• Packet Data Network Type: Choose the type of network for data services for this profile (<b>IPv4</b>, <b>IPv6</b>, or <b>IPv4v6</b>)</li> <li>• Authentication: Choose the authentication method that this profile uses for data, and enter the user name and password for this method in the Profile Username and Profile Password fields that display</li> </ul> <p>You can configure one profile for each SIM slot in the device.</p> |
| Add Profile              | <p>Click to add an access point name (APN) profile that the cellular device uses to attach to a cellular network.</p> <p>You can add up to 16 profiles.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Profile ID               | <p>Enter a unique identifier for the profile.</p> <p>Valid values: Integers 1 through 16.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Access Point Name        | Enter a name to identify the cellular access point.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Packet Data Network Type | Choose the packet data network (PDN) type of the cellular network ( <b>IPv4</b> , <b>IPv6</b> , or <b>IPv4v6</b> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| Parameter Name             | Description                                                                                                                                                |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Authentication             | Choose the authentication method that is used to attach to the cellular access point ( <b>none</b> , <b>pap</b> , <b>chap</b> , <b>pap_chap</b> ).         |
| Profile Username           | If you choose an authentication method other than <b>none</b> , enter the user name to use for authentication when attaching to the cellular access point. |
| Password                   | If you choose an authentication method other than <b>none</b> , enter the password to use for authentication when attaching to the cellular access point.  |
| Add                        | Click to add the profile your are configuring.                                                                                                             |
| Advanced Configuration Tab |                                                                                                                                                            |
| Attach Profile             | Choose the profile that the device uses to connect to the cellular network.                                                                                |
| Cellular 1/1 Profile       | Choose the profile that the device uses for data connectivity over the cellular network.                                                                   |

2. Attach the device template to the device.

For information, see [Attach and Detach a Device Template](#) in the *Cisco Catalyst SD-WAN Systems and Interfaces Configuration Guide, Cisco IOS XE Catalyst SD-WAN Release 17.x*.

# Configure a Cellular Gateway Using a Configuration Group in Cisco SD-WAN Manager

## Before You Begin

Create a configuration group for Cisco Catalyst Cellular Gateways using **Workflows > Create Cellular Gateway Group**. On the **Configuration Groups** page, the resulting configuration group is labelled **cellulargateway** in the **Device Solution** column.

For information about creating configuration groups and applying them to devices, see the [Using Configuration Groups](#) section of *Cisco Catalyst SD-WAN Configuration Groups, Cisco IOS XE Catalyst SD-WAN Release 17x*.

## Configure a Cellular Gateway Using a Configuration Group

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.
2. Click ... adjacent to a configuration group for a Cisco Catalyst Cellular Gateway and choose **Edit**.
3. Open the **Global Profile** section and add (click **Add Global Profile Feature**) or edit (click ... and **Edit**) any of the following features.

- **AAA** feature:

Table 313: Local

| Parameter Name  | Description                                                       |
|-----------------|-------------------------------------------------------------------|
| <b>Name</b>     | The account name is preset to <b>admin</b> and cannot be changed. |
| <b>Password</b> | Enter a password for login.                                       |

Table 314: TACACS

| Parameter Name              | Description                                                                                                                                                                                                                                                                                    |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>TACACS Configuration</b> | Enable TACACS configuration.<br>Click <b>Add TACACS</b> to add one or more TACACS servers.                                                                                                                                                                                                     |
| <b>Authentication</b>       | TACACS authentication option: <ul style="list-style-type: none"> <li>• <b>tacacs_ascii</b>: Send authentication information in ASCII format.</li> <li>• <b>tacacs_pap</b>: Send authentication information using the password authentication protocol (PAP).</li> </ul>                        |
| <b>Timeout</b>              | Timeout for TACACS authentication.<br>Range: 1 through 1000 seconds                                                                                                                                                                                                                            |
| <b>TACACS</b>               |                                                                                                                                                                                                                                                                                                |
| <b>IP Address</b>           | IP address of the TACACS server.                                                                                                                                                                                                                                                               |
| <b>Auth Port</b>            | TCP port number to connect to the TACACS server.<br>Default: 49                                                                                                                                                                                                                                |
| <b>Secret Key</b>           | Encryption key for encrypting and decrypting traffic between the cellular gateway and the TACACS server. Configure the same key on the TACACS server.                                                                                                                                          |
| <b>Source Interface</b>     | Preconfigured as Cellular1/0, and cannot be changed.<br>This is the only interface that the cellular gateway can use for communication with the TACACS server.                                                                                                                                 |
| <b>Priority</b>             | Priority level of the TACACS server. Zero is a default priority value and indicates the highest priority. If a cellular gateway is unable to establish a connection with the highest priority server, it attempts to connect to the server of the next highest priority.<br>Range: 0 through 7 |

- **Cellular** feature:

Table 315: Cellular Settings

| Parameter Name                           | Description                                                                                                               |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Primary Slot</b>                      | Choose a SIM slot to designate it as primary.<br>Range: 0, 1<br>Default: 0                                                |
| <b>SIM SLOT 0 Cellular Profile</b>       |                                                                                                                           |
| <b>Profile Id</b>                        | Profile ID.<br>You can click <b>Add</b> to add multiple profiles.                                                         |
| <b>Access Point Name</b>                 | Access point name, from your service provider.                                                                            |
| <b>Authentication Method</b>             | Authentication method ( <b>none</b> , <b>pap</b> , <b>chap</b> , <b>pap_or_chap</b> ) indicated by your service provider. |
| <b>Username</b>                          | Username for authentication, as indicated by your service provider.                                                       |
| <b>Password</b>                          | Password for authentication, as indicated by your service provider.                                                       |
| <b>Packet Data Network Type</b>          | Packet data network type ( <b>IPv4</b> , <b>IPv6</b> , <b>IPv4v6</b> ), as indicated by your service provider.            |
| <b>Attach Profile</b>                    | Choose the attach profile from the defined profiles.                                                                      |
| <b>Data Profile</b>                      | Choose the data profile from the defined profiles. You can use the same profile for the attach profile and data profile.  |
| <b>SIM SLOT 1 Cellular Profile</b>       |                                                                                                                           |
| See the fields described for SIM slot 0. |                                                                                                                           |

- **Logging** feature:

Table 316: Disk

| Parameter Name          | Description                                                                                                                                                                                                                                                                                         |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Disk File Rotate</b> | Maximum number of log files to store locally.<br><br>The device collects diagnostic monitor log files, which have a maximum size of 20 MB each, until the number of files reaches the rotate value. Then the device deletes the oldest file to make room for a new file.<br><br>Range: 1 through 10 |
| <b>Disk File Size</b>   | Maximum file size for each log file that the device stores locally. After reaching the maximum size, the device creates a new log file, with a numerically sequenced filename.<br><br>Range: 1 through 20 megabytes                                                                                 |

Table 317: Servers

| Parameter Name    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Server Name Type  | Choose <b>ipv4</b> or <b>ipv6</b> , according to the server address type, or choose <b>dns</b> if you enter a server domain name in the <b>Server Name Value</b> field.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Server Name Value | IP address or domain name of the server.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Source IP         | By default, this is the system IP address. You can choose the <b>Device Specific</b> option to specify per device.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Priority          | <p>Filter the type of log messages saved using one of the following priority options, listed from lowest to highest priority.</p> <p>Each priority option configures the device to save log messages of that priority and all higher priorities.</p> <p>For example, information is the lowest priority of message, so choosing <b>information</b> includes information log messages and all other log messages too. Choosing <b>error</b> excludes information, notice, and warn log messages, but includes error messages and all other log messages of higher priority (critical, alert, and emergency).</p> <p>From lowest to highest priority, the options are the following:</p> <ul style="list-style-type: none"> <li>• information</li> <li>• notice</li> <li>• warn</li> <li>• error</li> <li>• critical</li> <li>• alert</li> <li>• emergency</li> </ul> |

- **Network Protocol** feature:



Table 318: Basic Configuration

| Parameter Name           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Passthrough</b>       | <p>The cellular gateway operates in one of two modes: IP passthrough and NAT.</p> <p>In IP passthrough mode, the cellular gateway passes the public IP address assigned by the internet service provider (ISP) to a downstream device attached to the cellular gateway.</p> <p>Disabling the <b>Passthrough</b> option enables NAT, which gives the devices that are connected to the cellular gateway access to a DHCP server and to the local gateway.</p> <p><b>Note</b><br/>Enabling passthrough mode disables and hides the other fields in the <b>Basic Configuration</b> section.</p> |
| <b>DHCP Pool</b>         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>DHCP Pool</b>         | Enable a DHCP pool for NAT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>DHCP Network Pool</b> | IP address pool, in classless interdomain routing (CIDR) format.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Lease Days</b>        | <p>Days for DHCP lease time</p> <p>Range: 0 to 365</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Lease Hours</b>       | <p>Hours for DHCP lease time.</p> <p>Range: 0 to 23</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Lease Minutes</b>     | <p>Minutes for DHCP lease time.</p> <p>Range: 0 to 59</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>PAT Configuration</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>PAT Configuration</b> | Enable port address translation (PAT).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Add PAT Config</b>    | Click this to add one or more PAT configurations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Description</b>       | Description of the PAT configuration.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Protocol</b>          | Choose <b>TCP</b> or <b>UDP</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>LocalAddress</b>      | IPv4 format address.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>LocalPort</b>         | <p>Port number.</p> <p>Range: 0 to 65535</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>InterfaceName</b>     | Preconfigured as Cellular1/0, which is the WAN interface for the cellular gateway.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

| Parameter Name | Description                              |
|----------------|------------------------------------------|
| GlobalPort     | Global port number.<br>Range: 1 to 65535 |

*Table 319: NTP Servers*

| Parameter Name | Description                                                                                                                 |
|----------------|-----------------------------------------------------------------------------------------------------------------------------|
| NTP            | To configure a network time protocol (NTP) server, enter an IPv4 address or a DNS name.<br>Maximum number of NTP servers: 4 |

4. (Optional) To add CLI configuration commands, do the following:
  - a. Open the **CLI Add-on Profile**.
  - b. Click **Add Feature**.
  - c. In the **Type** drop-down list, choose **Config**.
  - d. Enter a name for the feature.
  - e. Enter a CLI configuration.
  - f. Click **Save**.



---

**Note** CLI configuration commands in the CLI Add-on Profile override any configuration done using the Global Profile.

---



## CHAPTER 17

# Network Slicing

Table 320: Feature History

| Feature Name             | Release Information                                                                            | Description                                                                                                                      |
|--------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Cellular Network Slicing | Cisco IOS XE Catalyst SD-WAN Release 17.16.1a<br>Cisco Catalyst SD-WAN Manager Release 20.16.1 | This feature allows multiple networks to exist on the same physical network to optimize the network for different traffic types. |

- [Information about Network Slicing, on page 669](#)
- [Restrictions for Network Slicing, on page 670](#)
- [Configure Network Slicing, on page 670](#)
- [Configure Cellular Network Slicing Using CLI, on page 672](#)
- [Verify Cellular Network Slicing Configuration, on page 673](#)

## Information about Network Slicing

Network slicing refers to partitioning a physical network into multiple virtual parts, called slices. The network slicing solution allows service providers to partition a 5G physical network into multiple virtual network slices. Using this feature, a host platform can set up multiple simultaneous network connections, each with its own IP address, using slice types that are suited to each of the host requirements. This approach allows service providers to allocate dedicated resources for different types of services for optimal performance and efficiency.

5G network slicing is supported only on 5G standalone networks. It is not available on 5G non-standalone (NSA) networks. The slice or service types have a specific set of network functions and resources defined by the network provider. Service type defines the expected behavior of a network slice in terms of its features and services. These service types are supported in network slicing:

- Enhanced Mobile Broadband (eMBB): for high-bandwidth and low-latency applications.
- Ultra-reliable low latency communications (URLLC): for high reliability and low latency transmission requirements.
- Massive Internet of Things (MIoT): for sending small quantities of data.

## Restrictions for Network Slicing

- You can enable network slicing for cellular profile only on P-5GS6-R16SA-GL module in 5G standalone mode.
- Configuring network slicing parameters on profile one or the default profile for the carrier is not allowed.

## Configure Network Slicing

### Before You Begin

Create a configuration group for Cisco Catalyst Cellular Gateways using **Workflows > Create Cellular Gateway Group**. On the **Configuration Groups** page, the resulting configuration group is labelled cellular gateway in the **Device Solution** column.

For information about creating configuration groups and applying them to devices, see the [Using Configuration Groups](#) section of *Cisco Catalyst SD-WAN Configuration Groups, Cisco IOS XE Catalyst SD-WAN Release 17.x*.

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.
2. Create a [Configuration Group](#). A new configuration group is created.
3. In the configuration group, click **Add Profile**.
4. Choose **Transport & Management Profile** from the drop-down list.
5. Click **Edit** to add features to the Transport and Management profile.
6. Click **Add New Feature**.
7. Choose **Cellular Controller** from the **Add Feature** drop-down list.
8. Choose **Cellular Profile** from the **Add Feature** drop-down list.

**Table 321: Cellular Settings**

| Field                              | Description                                                                |
|------------------------------------|----------------------------------------------------------------------------|
| <b>Primary Slot</b>                | Choose a SIM slot to designate it as primary.<br>Range: 0, 1<br>Default: 0 |
| <b>SIM SLOT 0 Cellular Profile</b> |                                                                            |
| <b>Profile Id</b>                  | Profile ID.<br>You can click <b>Add</b> to add multiple profiles.          |
| <b>Access Point Name</b>           | Access point name, from your service provider.                             |

| Field                                    | Description                                                                                                               |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Authentication Method</b>             | Authentication method ( <b>none</b> , <b>pap</b> , <b>chap</b> , <b>pap_or_chap</b> ) indicated by your service provider. |
| <b>Username</b>                          | Username for authentication, as indicated by your service provider.                                                       |
| <b>Password</b>                          | Password for authentication, as indicated by your service provider.                                                       |
| <b>Packet Data Network Type</b>          | Packet data network type ( <b>IPv4</b> , <b>IPv6</b> , <b>IPv4v6</b> ), as indicated by your service provider.            |
| <b>Attach Profile</b>                    | Choose a attach profile from the defined profiles.                                                                        |
| <b>Data Profile</b>                      | Choose a data profile from the defined profiles. You can use the same profile for the attach profile and data profile.    |
| <b>SIM SLOT 1 Cellular Profile</b>       |                                                                                                                           |
| See the fields described for SIM slot 0. |                                                                                                                           |

9. For the configured **Cellular Controller**, choose the **Cellular Profile** or click **Add New** to add a new cellular profile.
10. Configure **Cellular Profile** using the following table.

*Table 322: Cellular Profile Settings*

| Field                    | Description                                                                                                                                                              |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Type</b>              | Choose a feature from the drop-down list.                                                                                                                                |
| <b>Feature Name</b>      | Enter a name for the feature. The name can be up to 128 characters and can contain only alphanumeric characters.                                                         |
| <b>Description</b>       | Enter a description of the feature. The description can be up to 2048 characters and can contain only alphanumeric characters.                                           |
| <b>Profile ID</b>        | Enter the identification number of the profile to use on the router.<br>Range: 1 through 15                                                                              |
| <b>Access Point Name</b> | Enter the name of the gateway between the service provider network and the public internet. It can be up to 32 characters long.                                          |
| <b>Authentication</b>    | Choose the authentication method used for the connection to the cellular network. It can be <b>none</b> , <b>pap</b> , <b>chap</b> , or <b>pap_chap</b> .                |
| <b>Profile Username</b>  | Enter the username to use when making cellular connections for web services. It can be 1 to 32 characters. It can contain any alphanumeric characters, including spaces. |

| Field                           | Description                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Profile Password</b>         | Enter the user password to use when making cellular connections for web services. The password is case-sensitive and can be clear text, or an AES-encrypted key.<br><br>From Cisco Catalyst SD-WAN Manager Release 20.15.1, when you enter the password as clear text, Cisco SD-WAN Manager encrypts the password. When you view the configuration preview, the password appears in its encrypted form.  |
| <b>Packet Data Network Type</b> | Choose the packet data network (PDN) type of the cellular network. It can be IPv4, IPv6, or IPv4v6.                                                                                                                                                                                                                                                                                                      |
| <b>No Overwrite</b>             | Enable this option to overwrite the profile on the cellular modem. By default, this option is disabled.                                                                                                                                                                                                                                                                                                  |
| <b>Slice Type</b>               | Choose the network slice type (SST) for the profile. The options are: <ul style="list-style-type: none"> <li>• eMBB: Enhanced Mobile Broadband, used for high data throughput.</li> <li>• URLLC: Ultra-Reliable Low Latency Communication, used for high reliability and low latency transmission.</li> <li>• MIoT: Massive IoT, used for many devices transmitting small quantities of data.</li> </ul> |
| <b>Slice Differentiator</b>     | Enter the slice differentiator (SD) for the profile. This is an optional value that enables the user equipment (UE) to use multiple slice instances of the same SST.<br><br>Range: 0 through 16777214                                                                                                                                                                                                    |
| <b>Slot</b>                     | Enter the associated SIM slot for the profile.                                                                                                                                                                                                                                                                                                                                                           |

## Configure Cellular Network Slicing Using CLI

The following example shows how to configure a cellular network slicing using a CLI.

```
Device(config-controller)# profile id 3 apn apn-ns authentication none pdn-type ipv4
no-overwrite
slot Associated sim slot
slice-type Associated network slice type(SST)
```

```
Device(config-controller)# profile id 3 apn apn-ns authentication none pdn-type ipv4 slice-
type
<1-3> Slice type number: 1(eMBB), 2(URLLC), 3(MIoT)
Device(config-controller)# profile id 3 apn apn-ns authentication none pdn-type ipv4 slice-
type embb
slice-differentiator Associated Slice Differentiator(SD) <--- if SD is bypassed here,"FF
FF FF" will be used internally as the SD to indicate that there is no SD associated with
this SST.
slot Associated sim slot
```

```
Device(config-controller)# profile id 3 apn apn-ns authentication none pdn-type ipv4 slice
type
embb slice-differentiator
<0-16777214> Slice Differentiator
Device(config-controller)#
Device(config-controller)#$ pdn ipv6 slice-type mlt slice-differentiator 6
slot Associated sim slot
```

## Verify Cellular Network Slicing Configuration

The following is a sample output from the **show cellular** command that displays network slicing configuration and slice type details:

```
Device# show cellular profile 6

Profile 6 = INACTIVE

PDP Type = IPv6
Access Point Name (APN) = TestNSSAI6_URLLC1
Authentication = None
S-NSSAI Slice Type is URLLC
S-NSSAI Slice Differentiator = 1
```







# CHAPTER 18

## Configure Geofencing

*Table 323: Feature History*

| Feature Name                                                                          | Release Information                                                          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Geofencing                                                                            | Cisco IOS XE Catalyst SD-WAN Release 17.6.1a<br>Cisco vManage Release 20.6.1 | This feature provides a way to restrict a device's location to an operational geographical boundary, and to identify a device's location and report any violations of the configured boundary. If the device is identified to be in violation, you can restrict network access to the device using Cisco SD-WAN Manager operational commands.<br><br>In the CLI or a CLI template, configure geofencing coordinates for establishing the location of the device. You can also register for SMS alerts. |
| Added Support for Configuring Geofencing Using a <b>Cisco System</b> Feature Template | Cisco IOS XE Catalyst SD-WAN Release 17.7.1a<br>Cisco vManage Release 20.7.1 | This feature adds support for configuring the geographical boundary of a device using a <b>Cisco System</b> feature template.<br><br>With this feature, you can also configure automatic geolocation detection, where the device determines its own location, while configuring geofencing. A new parameter <b>auto-detect-geofencing-location</b> is added to the <b>geolocation (system)</b> command.                                                                                                |
| Added Support for LTE Advanced NIM Modules                                            | Cisco IOS XE Catalyst SD-WAN Release 17.8.1a                                 | Added support for Long-Term Evolution (LTE) Advanced Network Interface Modules (NIMs) for Cisco ISR 4000 routers.                                                                                                                                                                                                                                                                                                                                                                                      |

- [Information About Geofencing](#), on page 676
- [Supported Devices for Geofencing](#), on page 677
- [Prerequisites for Geofencing](#), on page 678
- [Restrictions for Geofencing](#), on page 678
- [Configure Geofencing Using a Cisco System Template](#), on page 678
- [Configure Geofencing Using the CLI](#), on page 680
- [Verify Geofencing Configuration](#), on page 681
- [Monitor Geofencing Alarms](#), on page 683

- [Configuration Example for Geofencing, on page 684](#)

## Information About Geofencing

Geofencing allows you to define a geographical boundary within which a device can be deployed. When devices are detected outside of the boundary, SMS alerts as well as critical-event alarms can be generated to Cisco SD-WAN Manager.

Global Positioning System (GPS) within a Long-Term Evolution Pluggable Interface Module (PIM) is used for device detection and monitoring in Cisco IOS XE Catalyst SD-WAN devices.

On the device CLI or through a Cisco SD-WAN Manager CLI template, you can configure the following settings:

- Base location (latitude and longitude) and a geofence range for device detection
- Short-message service (SMS) alert registration for sending SMS messages to a mobile number
- GPS enablement on a Long-Term Evolution PIM in the controller cellular 0/x/0 section



---

**Note** You can also enable GPS on a Long-Term Evolution PIM using a feature template.

---

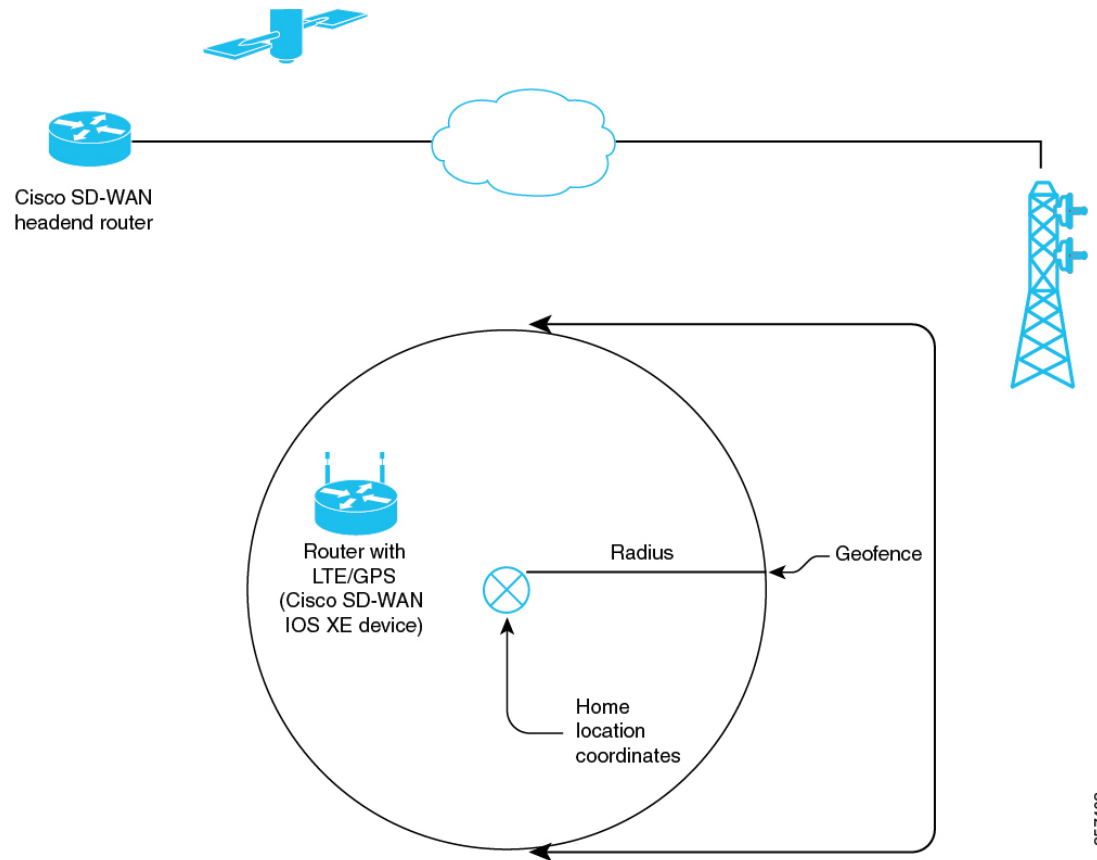
Starting from Cisco vManage Release 20.7.1, you can configure geofencing using a **Cisco System** feature template. You can also enable automatic geolocation detection of a device where the device determines its own base location.

In Cisco SD-WAN Manager, you can use operational commands for restricting network access if a device exceeds its geographical boundary.

For more information on the operational commands for restricting network access, see the [Cisco Catalyst SD-WAN Monitor and Maintain Configuration Guide](#).

Geofencing status alerts are sent to Cisco SD-WAN Manager upon detection of device boundary violations.

Figure 3: Overview of Geofencing



357403

## Benefits of Geofencing

- Protects against inappropriate access to an organization's network if a device is beyond its geographical boundary
- Notifies end users of any displaced devices
- Supports a geofence radius for specifying the target location of the device
- Supports SMS alerts for mobile phone alerts

## Supported Devices for Geofencing

Supported Devices:

- Cisco ISR 1000 with Long-Term Evolution (fixed and pluggable)
- Cisco Catalyst 8K with Long-Term Evolution Pluggable Interface Module (PIM)
- Cisco ISR 4000 with Long-Term Evolution Advanced Network Interface Modules (NIMs)

Supported Long-Term Evolution PIMs:

- P-LTE-VZ(WP7601)
- P-LTE-US(WP7603)
- P-LTE-JN(WP7605)
- P-LTE-MNA(WP7610)
- P-LTE-GB(WP7607)
- P-LTE-IN(WP7608)
- P-LTE-AU(WP7609)
- P-LTEA-EA(EM7455)
- P-LTEA-LA(EM7430)

Supported Long-Term Evolution Advanced NIMs:

- NIM-LTEA-EA(EM7455)
- NIM-LTEA-LA(EM7430)

## Prerequisites for Geofencing

- Ensure that your Cisco IOS XE Catalyst SD-WAN C1100 series router has a built-in Long-Term Evolution interface.
- Enable geofencing using the CLI or a CLI template. From Cisco vManage Release 20.7.1, you can also enable geofencing using a feature template.

For more information, see [Cisco IOS XE SD-WAN Qualified Command Reference](#).

- A SIM card is mandatory in the Long-Term Evolution PIM for receiving SMS alerts.

## Restrictions for Geofencing

- Geofencing can be used only in Cisco Catalyst SD-WAN controller mode.

## Configure Geofencing Using a Cisco System Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**.



---

**Note** In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

---

3. Click **Add Template**.
4. Choose a device.
5. In the **Select Template > Basic Information** section, click **Cisco System**.
6. In the **Template Name** field, enter a name for the template.  
The name can be up to 128 characters and can contain only alphanumeric characters.
7. In the **Template Description** field, enter a description of the template.  
The description can be up to 2048 characters and can contain only alphanumeric characters.
8. In the **Basic Configuration** section of the **Cisco System** template, choose a value from the drop-down list for **Console Baud Rate (bps)**.  
**Console Baud Rate (bps)** is a mandatory field for configuring geofencing.
9. Click **GPS** or navigate to the **GPS** section of the **Cisco System** template.
10. In the **Latitude** field, leave the field set to **Default** for automatic detection of a device.  
The following are the allowed values: -90.0 - 90.0.
11. In the **Longitude** field, leave the field set to **Default** for automatic detection of a device.  
The following are the allowed values: -180.0 - 180.0.

**Caution**

If you manually specify **Latitude** and **Longitude** coordinates, you disable automatic detection of a device. Automatic detection of a device can fail if a device does not have a last-known valid location.

12. In the **Geo Fencing Enable** field, change the scope from **Default** to **Global**, and click **Yes** to enable geofencing.  
The **Geo Fencing Enable** field is not enabled by default.
13. (Optional) In the **Geo Fencing Range in meters** field, specify a geofencing range unit in meters.  
The geofencing range specifies the radius from the base target location in meters.  
The default geofencing range is 100 meters. You can configure a geofencing range of 100 to 10,000 meters.
14. (Optional) In the **Enable SMS** drop-down list, change the scope to **Global**, and click **Yes** to enable SMS alerts.  
An SMS alert is delivered when a device is determined to be outside the configured geofencing radius of its target location.

**Note**

The presence of a SIM card is mandatory in the Long-Term Evolution PIM for receiving SMS alerts.

15. (Optional) In the **Mobile Number 1** field, add a mobile number for receiving SMS alerts.



**Note** Mobile numbers must start with a + sign, include a country code, an area code, with no spaces between the country code and the area code, and the remaining digits.

The following is a sample mobile number: +12344567236.

You can configure additional mobile phone numbers by clicking the + icon.

You can configure up to a maximum of four mobile numbers.

16. Click **Save**.

## Configure Geofencing Using the CLI

### Configure Latitude, Longitude, a Geofence Range, and Enable SMS Alerts

This section provides example CLI configurations for the following:

- Configure a base location, latitude and longitude.
- Enable automatic detection of a device where the device determines its own location.
- Enable, configure, and specify a geofence range.



**Note**

- Geofencing range unit is in meters.
- Geofencing range is an optional configuration parameter, and if not configured, it takes the default value of 100 meters.

- Add mobile numbers for receiving SMS alerts.

1. Configure a base location:

```
Device(config)# system
Device(config-system)# gps-location latitude 37.317342 longitude -122.218170
```

2. Enable automatic detection of a device:

```
Router(config)# system
Router(config-system)# no gps-location latitude
Router(config-system)# no gps-location longitude
Router(config-system)# gps-location auto-detect-geofencing-location
```



**Note** Do not configure latitude and longitude coordinates when using the auto-detect-geofencing-location parameter. You can choose to either configure a base location using latitude and longitude coordinates, or you can enable automatic detection of a device.

3. Enable, configure, and specify a geofence range:

```
Device(config-system)# gps-location geo-fencing-enable
Device(config-system)# gps-location geo-fencing-config
Device(config-geo-fencing-config)# geo-fencing-range 1000
```

4. Set up an SMS alert by adding the cell phone numbers for the users of the device:

```
Device(config-geo-fencing-config)# sms

Device(config-sms)# sms-enable
Device(config-sms)# mobile-number +12344567234
Device(config-mobile-number-+12344567234)# exit
Device(config-mobile-number-+12344567234)# mobile-number +12344567235
Device(config-mobile-number-+12344567235)# exit
Device(config-mobile-number-+12344567235)# mobile-number +12344567236
Device(config-mobile-number-+12344567236)# exit
Device(config-mobile-number-+12344567236)# mobile-number +12344567237
Device(config-mobile-number-+12344567237)# exit
Device(config-sms)# commit
```

5. Commit your changes.

### Enable GPS on a Long-Term Evolution PIM in the Controller Cellular Section

This section provides sample CLI configurations for enabling GPS on the Long-Term Evolution PIM in the 0/x/0 section of the configuration.

1. Enable GPS on a Long-Term Evolution PIM in the controller cellular section:

```
Device(config)# controller Cellular 0/2/0
Device(config-Cellular-0/2/0)# lte gps enable
```

2. Enable ms-based mode with a SIM card present in a Long-Term Evolution PIM. We recommend that you use ms-based with a SIM card present.

Mobile station-based assistance refers to the case where the Global Navigation Satellite System (GNSS-enabled) mobile device computes its own position locally.

```
Device(config-Cellular-0/2/0)# lte gps mode ms-based
```

3. Enable National Marine Electronics Association (NMEA) streaming:

```
Device(config-Cellular-0/2/0)# lte gps nmea
```

4. Commit your changes.

## Verify Geofencing Configuration

The following is a sample output from the **show sdwan geofence-status** command:

```
Device# show sdwan geofence-status
geofence-status
 Geofence Config Status = Geofencing-Enabled
 Target Latitude = 37.317342
 Target Longitude = -122.218170
 Geofence Range(in m) = 100
 Current Device Location Status = Location-Valid
 Current Latitude = 37.317567
 Current Longitude = -122.218170
 Current Device Status = Within-defined-fence
 Distance from target location(in m) = 30
```

```
Last updated device location timestamp = 2021-05-06T22:58:34+00:00
Auto-Detect Geofencing Enabled = true
```

In this output, Geofence Config Status = Geofencing-Enabled, so geofencing is enabled.

In this output, Auto-Detect Geofencing Enabled = true. Therefore, automatic detection of the device is enabled. If automatic detection of the device is not enabled, Auto-Detect Geofencing Enabled = false is displayed in the output.

The following is a sample output from the **show cellular 0/x/0 gps** command:

```
Device# show cellular 0/2/0 gps
GPS Feature = enabled
GPS Mode Configured = ms-based
GPS Port Selected = Dedicated GPS port
GPS Status = GPS coordinates acquired
Last Location Fix Error = Offline [0x0]
=====
GPS Error Count = 0
NMEA packet count = 17899
NMEA unknown packet count = 0

Per talker traffic count =
 US-GPS = 5982
 GLONASS = 2560
 GALILEO = 3505
 BEIDOU = 0
 GNSS = 3409
 Unknown talker = 2443
=====
Speed over ground in km/hr = 0
=====

Latitude = 31 Deg 19 Min 14.6203 Sec North
Longitude = 122 Deg 58 Min 32.8164 Sec West
*Apr 15 23:58:45.298: GPS Mode Configured =Timestamp (GMT) = Thu Apr 15 23:57:21 2021

Fix type index = 0, Height = 18 m
Satellite Info

Satellite #2, elevation 51, azimuth 42, SNR 24 *
Satellite #5, elevation 36, azimuth 144, SNR 34 *
Satellite #6, elevation 14, azimuth 45, SNR 24 *
Satellite #12, elevation 72, azimuth 146, SNR 33 *
Satellite #25, elevation 60, azimuth 305, SNR 25 *
=====
Total Satellites in view = 5
Total Active Satellites = 5
GPS Quality Indicator = 1
Total satellites from each constellation:
 US-GPS = 3
 GLONASS = 1
 GALILEO = 1
 BEIDOU = 0
=====
```

In this output, GPS Feature = enabled and GPS Mode Configured = ms-based. Therefore, GPS for controller cellular is enabled, and ms-based is configured.

The following is a sample output from the **show sdwan notification stream vptela** command:

```
Device# show sdwan notification stream vptela
notification
 eventTime 2021-04-13T23:05:02.881093+00:00
```



```

system-logout-change
severity-level minor
host-name pm5
system-ip 172.16.255.15
user-name admin
user-id 0
!
!
notification
eventTime 2021-04-14T00:36:31.344117+00:00
geo-fence-alert-status
severity-level major
host-name pm5
system-ip 172.16.255.15
alert-type device-location-inside
alert-msg Device Locking started for Geofencing Mode and device is within range

```

## Monitor Geofencing Alarms

You can monitor geofencing alarms based on severity or based on time.

The following are the types of geofencing alarms.

**Table 324: Geofencing Alarm Types**

| Type                           | Severity        | Description                                                                                                                                                                                                                                                   |
|--------------------------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Device Location Outside</b> | <b>Critical</b> | This notification is sent when the device location is outside the defined geofencing range.                                                                                                                                                                   |
| <b>Device Location Inside</b>  | <b>Major</b>    | This notification is sent when the device location is determined to be inside the defined geofence range when it was previously determined to be outside the defined geofence range, or the device location could not be obtained due to a GPS signal outage. |
| <b>Device Location Lost</b>    | <b>Major</b>    | This notification is sent when the device location cannot be determined due to a GPS outage.                                                                                                                                                                  |
| <b>Device Location Update</b>  | <b>Major</b>    | This notification is sent when the device location changes by more than 20 meters either when geofencing is enabled or not. If geofencing is not enabled, this notification is sent only if the device location is available.                                 |

You can monitor geofencing alarms using Cisco SD-WAN Manager.

1. From the Cisco SD-WAN Manager menu, choose **Monitor** > **Logs**.

Cisco vManage Release 20.6.1 and earlier: From the Cisco SD-WAN Manager menu, choose **Monitor** > **Alarms**.

2. If there are geofencing alarms, the alarms display in the form of a chart, followed by a table.

You can filter the data for a specified time range: (1h, 3h, 6h, and so on), or click **Custom** to define a time range.

3. To view the alarm details, click . . . and choose **Alarm Details** to view information about the device.

## Configuration Example for Geofencing

### End-to-End Configuration for Geofencing and Controller Cellular

The following is an end-to-end sample output that displays the configuration process for geofencing and controller cellular when configuring automatic detection of a device:

```
system
 gps-location auto-detect-geofencing-location
 gps-location geo-fencing-enable
 gps-location geo-fencing
 geo-fencing-range 1000
 sms
 sms-enable
 mobile-number +112312345676
 !
 mobile-number +112312345677
 !
 mobile-number +112312345678
 !
 mobile-number +112312345679
 !
 !
system-ip 10.1.1.35
site-id 273
admin-tech-on-failure
organization-name LTE-Test
vbond vbond-dummy.test.info port 12346
!
controller Cellular 0/2/0
lte gps enable
lte gps mode ms-based
lte gps nmea
!
```

The following is an end-to-end sample output that displays the configuration process for geofencing and controller cellular when manually configuring latitude and longitude coordinates:

```
system
 gps-location latitude 37.317342
 gps-location longitude -122.218170
 gps-location geo-fencing-enable
 gps-location geo-fencing-config
 geo-fencing-range 1000
 sms
 sms-enable
 mobile-number +112312345676
 !
 mobile-number +112312345677
```

```
!
mobile-number +112312345678
!
mobile-number +112312345679
!
!
!
```





## CHAPTER 19

# Configuring and Monitoring P-LTE-450 MHz Modules

*Table 325: Feature History*

| Feature Name                                                            | Release Information                                                                             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Configuring and Monitoring P-LTE-450 modules using Cisco SD-WAN Manager | Cisco IOS XE Catalyst SD-WAN Release 17.18.1a and Cisco Catalyst SD-WAN Manager Release 20.18.1 | <p>You can configure category P Long-Term Evolution (LTE) 450MHz Pluggable Interface Module (PIM), referred to as P-LTE-450, on rugged series routers using the Ethernet interface in Cisco SD-WAN Manager. Additionally, Cisco SD-WAN Manager enables you to monitor the module's performance through the monitoring dashboard.</p> <p>Cisco IR1101 Rugged Series Router and Cisco IR1800 Rugged Series Router support P-LTE-450 modules.</p> |

- [Configuring and monitoring P-LTE-450 modules, on page 687](#)
- [Prerequisite for configuring P-LTE-450 MHz modules, on page 688](#)
- [Restrictions for configuring P-LTE-450 modules, on page 688](#)
- [Configure a P-LTE-450 module using Configuration Group and CLI Add-on Profile, on page 689](#)
- [Monitor the P-LTE-450 module using Cisco Catalyst SD-WAN Manager, on page 690](#)
- [Monitor a P-LTE-450 module using CLI , on page 690](#)

## Configuring and monitoring P-LTE-450 modules

A P-LTE-450 module configuration and monitoring is a network management solution that:

- Enables configuration of P-LTE-450 modules through the Ethernet Interface option in Cisco SD-WAN Manager.
- Support real-time monitoring of the P-LTE-450 module's performance on Cisco SD-WAN Manager monitoring dashboard, and
- Simplifies usage of LTE 450 MHz for industrial applications.

**P-LTE-450 Pluggable Interface Module**

P-LTE-450 PIM enables reliable Long-Term Evolution (LTE) connectivity, specifically tailored for the 450 MHz frequency band, which is widely used in critical communication scenarios such as utility networks, public safety, and rural broadband.

P-LTE-450 module is similar to a Catalyst Cellular Gateway with the following features:

- The modem is an IP passthrough for both IPv4 and IPv6 traffic.
- The management interface, GigabitEthernet 0/1/0 (Base chassis), or GigabitEthernet 0/4/0 (Expansion module), is configured under Cisco IOS XE.

**Key difference with LTE modules**

P-LTE-450 MHz modules is different from regular LTE modules on the IOS XE platform with a few notable differences. Some of the key differences are:

- IP passthrough is configured on the Gigabit Ethernet interface instead of the cellular interface.
- Allows minimum P-LTE-450 cellular configuration through Cisco IOS XE config mode under Gigabit Ethernet 0/x/0 interface through `lte 450 xxx` commands.
- Supports GPS connectivity through SMA(f) connector.

There is no Cisco IOS XE support equivalent to Cellular PIM for GPS features.

**Note**

For more information about , see [P-LTE 450 Support](#) and [450MHz Category-4 LTE PIM](#).

## Prerequisite for configuring P-LTE-450 MHz modules

Minimum supported release for configuring P-LTE-450 MHz modules using Cisco SD-WAN Manager is Cisco Catalyst SD-WAN Manager Release 20.18.1 and Cisco IOS XE Catalyst SD-WAN Release 17.18.1a.

- The P-LTE-450 requires Network Advantage license to be recognized under Cisco IOS XE. For more information about licensing policy, see [Smart Licensing Using Policy \(SLP\)](#).
- Thermal Considerations: The host router and the P-LTE-450 module is able to boot up and operate at a lower temperature than the radio module on the P-LTE-450. The radio module will not operate at a temperature below -30° C (-22° F).

## Restrictions for configuring P-LTE-450 modules

**Avoid Using P-LTE-450 as a Primary Link for PNP**

Do not configure the P-LTE-450 module as a primary link for Plug and Play (PNP) deployments.

**Platform supports only one P-LTE-450 module**

Configure only a single P-LTE-450 module on each platform.

**Place P-LTE-450 Only in Supported Slots**

Configure the P-LTE-450 module only in slot 0/1 or slot 0/4 due to hardware limitations.

## Configure a P-LTE-450 module using Configuration Group and CLI Add-on Profile

To configure a P-LTE-450 module with the **Ethernet Interface** option in Configuration Group and CLI Add-on Profile, follow these steps:

**Before you begin**

- You must note the username and password that can be found on the sticker label that comes with the P-LTE-450 module. Before performing any P-LTE-450 parameter configuration, set the username and password using CLI.

**Procedure**

**Step 1** Add the following commands in the [add-on CLI](#) profile:

```
lte450 band 31
lte450 profile id 2 apn <apn_name> authentication none pdn-type ipv4 vlan
<vlan_number>
```

In the preceding command, do the following:

- Configure the username and password of the P-LTE-450 module using the Add-on Profile.
- Provide an appropriate apn\_name such as 'internet'. It should be communicated by your service provider.
- Provide an appropriate vlan\_number such as '2'.
- Use band '31' or '72'.

**Example:**

```
Device:interface GigabitEthernet0/4/0
no shutdown
ip address dhcp
negotiation auto
lte450 credential username admin password VzbbPxZJ42WUhPHB
lte450 band 31
lte450 profile id 2 apn internet authentication none pdn-type ipv4 vlan 2
exit
```

**Note**

Commands in the CLI add-on Profile override configurations specified by corresponding configuration group features.

**Step 2** Configure **Ethernet Interface** in Configuration Group using the **Transport VPN** profile.

For more information about configuring Ethernet interface in Configuration Group, see [Ethernet Interface](#).

**Note**

For the the P-LTE-450 module, in the **Interface Name\*** field, enter either GigabitEthernet0/4/0.2 or GigabitEthernet0/1/0.2.

---

**What to do next**

After configuring the **Ethernet Interface** option, associate the Configuration Group containing the Transport profile to the **Ethernet Interface** option and attach it to the device. For more information, see [Deploy a Configuration Group](#).

## Monitor the P-LTE-450 module using Cisco Catalyst SD-WAN Manager

You can monitor the detailed hardware, radio-related, and network-specific information about the P-LTE-450 module using Cisco SD-WAN Manager.

### Procedure

---

**Step 1** From the Cisco SD-WAN Manager menu, choose **Monitor > Devices**.

**Step 2** Choose a device from the list of devices that appear on the dashboard.

**Step 3** Click **Real Time** in the left pane.

**Step 4** Click **Device Options**, and choose a command.

*Table 326: Device Options*

| Device Option            | Command                                 | Description                                       |
|--------------------------|-----------------------------------------|---------------------------------------------------|
| Cellular LTE450 Hardware | <code>show lte450 0/1/0 hardware</code> | Displays the details of modem and PIM information |
| Cellular LTE450 Radio    | <code>show lte450 0/1/0 radio</code>    | Displays the signal quality                       |
| Cellular LTE450 Network  | <code>show lte450 0/1/0 network</code>  | Displays the mobile network status                |

---

## Monitor a P-LTE-450 module using CLI

You can monitor the hardware, radio, and network-specific details of the P-LTE-450 module using the following CLI commands.



### show lte450 0/1/0 radio

The following is a sample output from the show lte450 0/1/0 radio command using the keyword lte450. The example shows

#### Example

```
Device# show lte450 0/1/0 radio
Signal quality
=====
RSSI = -77 dBm
RSRP = -99 dBm
RSRQ = -8 dB
SINR = 5 dB

Antenna signal quality
=====
Main RSSI = -78 dBm
Main RSRP = -101 dBm
Aux RSSI = -77 dB
Aux RSRP = -99 dB
```

### show lte450 0/1/0 hardware

The following is a sample output from the show lte450 0/1/0 hardware hardware command using the keyword lte450. The example shows

#### Example

```
Device# show lte450 0/1/0 <hardware>

System Status
=====
System time = Wed May 21 05:31:08 2025
Uptime = 2h 33m 48s
Load Average = 2.67 2.56 2.55
Free Disk space = 89204 KB
Free RAM = 54944 KB

Modem information
=====
Modem MAC = 88:5D:90:EF:FF:FF
Modem hardware version = ML620EU
Modem firmware version = 0.3.4.1/ML620EUV13_RELEASE_20240305
Modem temperature = 48.0 deg C

PIM information
=====
PIM MAC = FC:C2:3D:30:E3:F2
PIM hardware version = IPS-701-V3
PIM firmware version = IPS701-v1.3.0-Secured
PIM serial number = 50030301564
Management IPv4 Address = 192.168.200.1
Management IPv6 Address = fd00:ffff:c0:a8c8::1/64
PIM temperature = 49.5 deg C

GNSS Location Data
=====
Latitude =
Longitude =
Altitude =
Time =

Watchdog Statistics
```

```
=====
Hardware watchdog counter = 0
Modem watchdog counter = 0
```

### show lte450 0/1/0 network

The following is a sample output from the show lte450 0/1/0 network command using the keyword lte450. The example shows

#### Example

```
Device# show lte450 0/1/0 network

Mobile network status
=====
Connection Status = Online
Connection time = 2h 31m 21s
IMSI = 001010123456058
ICCID = 89860001010123456789
MSISDN =
APN-1 IPv4 address = 192.168.4.38
APN-1 IPv6 address = 2001:0468:3000:0008:2001:0468:3000:0008
APN-2 IPv4 address =
APN-2 IPv6 address =
APN-3 IPv4 address =
APN-3 IPv6 address =

Mobile connection information
=====
IMEI = 862128050313077
System mode = LTE
EPS State = ATTACHED
RRC State = RRC IDLE
CEREG = 1
APN-1 APN name = ims.mnc001.mcc001.gprs
APN-2 APN name =
APN-3 APN name =
Band = 31
PLMN = 00101
Cell ID = 27447298
TAC = 0(1)
PCI = 2
TX power = 0
Bandwidth = 5MHz
LTE RX channel = 9895
LTE TX channel = 27785
Current upload = 0.0 Kbps
Current download = 0.0 Kbps
```

### show lte450 0/1/0 all

The following is a sample output from the show lte450 0/1/0 all command using the keyword lte450. The example shows

#### Example

```
Device# show lte450 0/1/0 all
System Status
=====
System time = Wed May 21 05:31:44 2025
Uptime = 2h 34m 24s
Load Average = 2.47 2.53 2.54
Free Disk space = 89204 KB
```

```
Free RAM = 54944 KB

Modem information
=====
Modem MAC = 88:5D:90:EF:FF:FF
Modem hardware version = ML620EU
Modem firmware version = 0.3.4.1/ML620EUV13_RELEASE_20240305
Modem temperature = 48.0 deg C

PIM information
=====
PIM MAC = FC:C2:3D:30:E3:F2
PIM hardware version = IPS-701-V3
PIM firmware version = IPS701-v1.3.0-Secured
PIM serial number = 50030301564
Management IPv4 Address = 192.168.200.1
Management IPv6 Address = fd00:ffff:c0:a8c8::1/64
PIM temperature = 50.0 deg C

GNSS Location Data
=====
Latitude =
Longitude =
Altitude =
Time =

Watchdog Statistics
=====
Hardware watchdog counter = 0
Modem watchdog counter = 0

Mobile network status
=====
Connection Status = Online
Connection time = 2h 31m 54s
IMSI = 001010123456058
ICCID = 89860001010123456789
MSISDN =
APN-1 IPv4 address = 192.168.4.38
APN-1 IPv6 address = 2001:0468:3000:0008:2001:0468:3000:0008
APN-2 IPv4 address =
APN-2 IPv6 address =
APN-3 IPv4 address =
APN-3 IPv6 address =

Mobile connection information
=====
IMEI = 862128050313077
System mode = LTE
EPS State = ATTACHED
RRC State = RRC IDLE
CEREG = 1
APN-1 APN name = ims.mnc001.mcc001.gprs
APN-2 APN name =
APN-3 APN name =
Band = 31
PLMN = 00101
Cell ID = 27447298
TAC = 0(1)
PCI = 2
TX power = 0
Bandwidth = 5MHz
LTE RX channel = 9895
LTE TX channel = 27785
```

```
Current upload = 0.0 Kbps
Current download = 0.0 Kbps
```

```
Signal quality
```

```
=====
```

```
RSSI = -77 dBm
```

```
RSRP = -99 dBm
```

```
RSRQ = -8 dB
```

```
SINR = 5 dB
```

```
Antenna signal quality
```

```
=====
```

```
Main RSSI = -78 dBm
```

```
Main RSRP = -101 dBm
```

```
Aux RSSI = -77 dB
```

```
Aux RSRP = -99 dB
```



## CHAPTER 20

# VRRP Interface Tracking

Table 327: Feature History

| Feature Name                                                     | Release Information                                                              | Description                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------------------------------------|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VRRP Interface Tracking for Cisco IOS XE Catalyst SD-WAN Devices | Cisco IOS XE Catalyst SD-WAN Release 17.7.1a<br><br>Cisco vManage Release 20.7.1 | This feature enables VRRP to set the edge as active or standby based on the WAN Interface or SIG tracker events and increase the TLOC preference value on a new VRRP active to ensure traffic symmetry, for Cisco IOS XE Catalyst SD-WAN Devices.<br><br>Starting this release, you can configure VRRP interface tracking through Cisco SD-WAN Manager feature template and CLI template on Cisco IOS XE Catalyst SD-WAN Devices. |

- [Information About VRRP Interface Tracking](#), on page 695
- [Restrictions and Limitations](#), on page 696
- [VRRP Tracking Use Cases](#), on page 696
- [Configure Object Tracker Using a Configuration Group](#), on page 697
- [Workflow to Configure VRRP Tracking](#), on page 698
- [Configure an Object Tracker](#), on page 698
- [Configure VRRP for a VPN Interface Template and Associate Interface Object Tracker](#), on page 699
- [Configure VRRP Tracking Using CLI Templates](#), on page 700
- [Configuration Example for VRRP Object Tracking Using CLI](#), on page 701
- [Configuration Examples for SIG Object Tracking](#), on page 702
- [Monitor VRRP Configuration](#), on page 702
- [Verify VRRP Tracking](#), on page 703

## Information About VRRP Interface Tracking

The Virtual Router Redundancy Protocol (VRRP) is a LAN-side protocol that provides redundant gateway service for switches and other IP end stations. In Cisco IOS XE Catalyst SD-WAN devices, you can configure VRRP on interfaces and subinterfaces using Cisco SD-WAN Manager templates and CLI add-on templates.

For more information, see [Configuring VRRP](#).

## Restrictions and Limitations

- VRRP is only supported with service-side VPNs. If you are using subinterfaces, configure VRRP physical interfaces in VPN 0.
- VRRP tracking is enabled on either a physical uplink interface or a logical tunnel interface (IPSEC or GRE or both).
- The VRRP Tracking feature does not support IP prefix as an object.
- You can use the same tracker under multiple VRRP groups or VPNs.
- You cannot use the same track object to track multiple interfaces.
- You can group a maximum of 16 track objects under a list track object.
- You cannot configure **tloc-change** and **increase-preference** on more than one VRRP group.

## VRRP Tracking Use Cases

The VRRP state is determined based on the tunnel link status. If the tunnel or interface is down on the primary VRRP, then the traffic is directed to the secondary VRRP. The secondary VRRP router in the LAN segment becomes primary VRRP to provide gateway for the service-side traffic.

### Zscaler Tunnel Use Case 1—Primary VRRP, Single Internet Provider

The primary and secondary Zscaler tunnels are connected through a single internet provider to the primary VRRP. The primary and secondary VRRP routers are connected through using TLOC extension. In this scenario, the VRRP state transition occurs if the primary and secondary tunnels go down on primary VRRP. The predetermined priority value decrements when the tracking object is down, which triggers the VRRP state transition. To avoid asymmetric routing, VRRP notifies this change to the Overlay through OMP.

### Zscaler Tunnel Use Case 2—VRRP Routers in TLOC Extension, Dual Internet Providers

The primary and secondary VRRP routers are configured in TLOC extension high availability mode. The primary and secondary Zscaler tunnels are directly connected with primary and secondary VRRP routers, respectively, using dual internet providers. In this scenario too, the VRRP state transition occurs if the primary and secondary tunnels go down on primary VRRP. The predetermined priority value decrements when the tracking object is down, which triggers the VRRP state transition. VRRP notifies this change to the Overlay through OMP.

### TLOC Preference

Transport Locators (TLOCs) connect an OMP route to a physical location. A TLOC is directly reachable using an entry in the routing table of the physical network, or represented by a prefix beyond a NAT device.

In Cisco IOS XE Catalyst SD-WAN devices, the TLOC change increase preference value increases based on the configured value. You can configure the TLOC change increase preference value on both the active and the backup nodes.

# Configure Object Tracker Using a Configuration Group

## Before you begin

On the **Configuration** > **Configuration Groups** page, choose **SD-WAN** as the solution type.

## Procedure

**Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration** > **Configuration Groups**.

**Step 2** Create and configure a object tracker in a Service Profile.

Configure basic settings.

*Table 328: Basic Settings*

| Parameter Name       | Description                                                                                                                                                                                                                                                                                                                                            |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tracker Type*</b> |                                                                                                                                                                                                                                                                                                                                                        |
| Interface            | Configure the following interface values: <ul style="list-style-type: none"><li>• <b>Object tracker ID*</b>: Enter the object tracker ID number.<br/>Range: 1-1000</li><li>• <b>Interface name*</b>: Enter the global or device-specific tracker interface name. For example, Gigabitethernet1 or Gigabitethernet2.</li></ul>                          |
| SIG                  | <b>Object tracker ID*</b> : Enter the object tracker ID number.                                                                                                                                                                                                                                                                                        |
| Route                | Configure the route details: <ul style="list-style-type: none"><li>• <b>Object tracker ID*</b>: Enter the object tracker ID number.<br/>Range: 1-1000</li><li>• <b>Route IP*</b>: Enter the IPv4 address of the route.</li><li>• <b>Route IP Mask*</b>: Select a value for the subnet mask.</li><li>• <b>VPN</b>: Enter a value for the VPN.</li></ul> |

## What to do next

Also see [Deploy a configuration group](#).

## Workflow to Configure VRRP Tracking

1. Configure an object tracker. For more information, see [Configure an Object Tracker, on page 698](#).
2. Configure VRRP for a VPN Interface template and associate the object tracker with the template. For more information, see [Configure VRRP for a VPN Interface Template and Associate Interface Object Tracker, on page 699](#).

## Configure an Object Tracker

Use the **Cisco System** template to configure an object tracker.

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**.



---

**Note** In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

---

3. Navigate to the **Cisco System** template for the device.



---

**Note** To create a **System** template, see [Create System Template](#)

---

4. Click **Tracker** and choose **New Object Tracker** to configure the tracker parameters.

*Table 329: Tracker Parameters*

| Field                | Description                                                                            |
|----------------------|----------------------------------------------------------------------------------------|
| <b>Tracker Type</b>  | Choose <b>Interface</b> or <b>SIG</b> or <b>Route</b> to configure the object tracker. |
| <b>Object ID</b>     | Enter the object ID number.                                                            |
| <b>Interface</b>     | Choose global or device-specific tracker interface name.                               |
| <b>Route IP</b>      | Enter the IP route prefix to track the state of an IP route.                           |
| <b>Route IP Mask</b> | Enter the prefix mask.                                                                 |
| <b>VPN</b>           | Enter the VPN number.                                                                  |

5. Click **Add**.
6. Optionally, to create a tracker group, click **Tracker**, and click **Tracker Groups > New Object Tracker Groups** to configure the tracker parameters.





**Note** Ensure that you have created two trackers to create a track group.

*Table 330: Object Tracker Group Parameters*

| Field                   | Description                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Group Tracker ID</b> | Enter the name of the tracker group.                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Tracker ID</b>       | Enter the name of the object tracker that you want to group.                                                                                                                                                                                                                                                                                                                                                        |
| <b>Criteria</b>         | Choose <b>AND</b> or <b>OR</b> explicitly.<br><br><b>OR</b> ensures that the transport interface status is reported as active if either one of the associated trackers of the tracker group reports that the route is active.<br><br>If you choose <b>AND</b> operation, the transport interface status is reported as active if both the associated trackers of the tracker group report that the route is active. |



**Note** Provide information in all the mandatory fields before you save the template.

7. Click **Add**.
8. Click **Save**.

## Configure VRRP for a VPN Interface Template and Associate Interface Object Tracker

To configure VRRP for a **Cisco VPN** template, do the following:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**.



**Note** In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

3. Navigate to the **Cisco VPN Interface Ethernet** template for the device.



**Note** For information about creating a new **Cisco VPN Interface Ethernet** template, see [Configure VPN Ethernet Interface](#).

4. Click **VRRP** and choose **IPv4**.

- Click **New VRRP** to create a new VRRP or edit the existing VRRP and configure the following parameters:

| Parameter Name               | Description                                                                                         |
|------------------------------|-----------------------------------------------------------------------------------------------------|
| TLOC Preference Change       | (Optional) Choose <b>On</b> or <b>Off</b> to set whether the TLOC preference can be changed or not. |
| TLOC Preference Change Value | (Optional) Enter the TLOC preference change. Range: 1 to 4294967295.                                |

- Click the **Add Tracking Object** link, and in the **Tracking Object** dialog box that is displayed, click **Add Tracking Object**.
- In the **Tracker ID** field, enter the Interface Object ID or Object Group Tracker ID.
- From the **Action** drop-down list, choose **Decrement** and enter the **Decrement Value** as 1. Cisco vEdge Devices supports decrement value of 1.  
Or  
Choose **Shutdown**.
- Click **Add**.
- Click **Add** to save the VRRP details.
- Click **Save**.

## Configure VRRP Tracking Using CLI Templates

You can configure VRRP tracking using the CLI add-on feature templates and CLI device templates. For more information, see [CLI Templates](#).

## VRRP Object Tracking Using CLI

### Interface Object Tracking using CLI

Use the following configuration to add an interface to a track list using Cisco SD-WAN Manager device CLI template:

```
Device(config)# track <object-id1> interface <interface-type-number> [line-protocol]
Device(config-tracker)# exit
Device(config)# track < object-id2> interface <interface-type-number> [line-protocol]
Device(config-tracker)# exit
Device(config)# track <group-object-id> list boolean [and | Or]
Device(config-tracker)# object <object-id1>
Device(config-tracker)# object <object-id2>
Device(config-tracker)# exit
Device(config)# interface GigabitEthernet2
```

```
Device(config-if)# vrf forwarding <vrf-number>
```

```
Device(config-if)# ipv4 address <ip-address> <subnet-mask>
```

```

Device(config-if)# negotiation auto
Device(config-if)# vrrp <vrrp-number> address-family ipv4
Device(config-if-vrrp)# address <ipv4-address> [primary | secondary]
Device(config-if-vrrp)# track <object-id> [decrement <dec-value> | shutdown]
Device(config-if-vrrp)# tloc-change increase-preference <value>
Device(config-if-vrrp)# exit

```

## SIG Container Tracking

The following example shows how to configure a track list and tracking for SIG containers using the Cisco SD-WAN Manager device CLI template.



**Note** In Cisco IOS XE Catalyst SD-WAN Release 17.7.1a SIG Object Tracking, you can only set *global* as the variable for Service Name.

### SIG Object Tracking Using CLI

```

Device(config)# track <object-id1> service global

Device(config-tracker)# exit
Device(config)# track <object-id2> service global
Device(config-tracker)# exit
Device(config)# track <group-object-id> list boolean [and | Or]
Device(config-tracker)# object <object-id1>
Device(config-tracker)# object <object-id2>
Device(config-tracker)# exit

Device(config)# interface GigabitEthernet2

Device(config-if)# vrf forwarding <vrf-number>

Device(config-if)# ip address <ip-address> <subnet-mask>
Device(config-if)# negotiation auto
Device(config-if)# vrrp <vrrp-number> address-family ipv4
Device(config-if-vrrp)# address <ipv4-address> [primary | secondary]
Device(config-if-vrrp)# track <object-id> [decrement <dec-value> | shutdown]
Device(config-if-vrrp)# tloc-change increase-preference <value>
Device(config-if-vrrp)#exit

```

## Configuration Example for VRRP Object Tracking Using CLI

### Interface Object Tracking Using CLI

```

config-transaction
 track 100 interface Tunnel123 line-protocol
 exit
 track 200 interface GigabitEthernet5 line-protocol
 exit
track 400 list boolean and
 object 100
 object 200
 exit

```

```

interface GigabitEthernet2
 vrf forwarding 1
 ip address 10.10.1.1 255.255.255.0
 negotiation auto
 vrrp 1 address-family ipv4
 address 10.10.1.10 primary
 track 400 decrement 10
 tloc-change increase-preference 333
 exit

```

## Configuration Examples for SIG Object Tracking

### SIG Object Tracking Using CLI

```

config-transaction
 track 1 service global
 exit
 exit
 track 2 service global
track 3 list boolean and
 object 1
 object 2
 exit

interface GigabitEthernet2
 vrf forwarding 1
 ip address 10.10.1.1 255.255.255.0
 negotiation auto
 vrrp 1 address-family ipv4
 address 10.10.1.10 primary
 track 3 decrement 10
 tloc-change increase-preference 333
 exit

```

## Monitor VRRP Configuration

To view information about VRRP configuration:

1. From the Cisco SD-WAN Manager menu, choose **Monitor > Devices**.  
Cisco vManage Release 20.6.x and earlier: From the Cisco SD-WAN Manager menu, choose **Monitor > Network**.
2. Choose a device from the list of devices.
3. Click **Real Time**.
4. From the **Device Options** drop-down list, choose **VRRP Information**.




---

**Note** You can view the status of the VRRP configuration in **Track State**.

---

# Verify VRRP Tracking

Device# **show vrrp**

The following is a sample output for the **show vrrp** command:

```
GigabitEthernet2 - Group 1 - Address-Family IPv4
 State is MASTER
 State duration 37 mins 52.978 secs
 Virtual IP address is 10.10.1.10
 Virtual MAC address is 0000.5E00.0101
 Advertisement interval is 1000 msec
 Preemption enabled
 Priority is 100
 State change reason is VRRP_TRACK_UP
Tloc preference configured, value 333
Track object 400 state UP decrement 10
 Master Router is 10.10.1.1 (local), priority is 100
 Master Advertisement interval is 1000 msec (expires in 607 msec)
 Master Down interval is unknown
 FLAGS: 1/1
```

Device# **show track brief**

The following is a sample output for the **show track brief** command:

| Track | Type      | Instance         | Parameter     | State | Last Change |
|-------|-----------|------------------|---------------|-------|-------------|
| 100   | interface | Tunnel123        | line-protocol | Up    | 00:12:48    |
| 200   | interface | GigabitEthernet5 | line-protocol | Up    | 00:49:57    |
| 400   | list      |                  | boolean       | Up    | 00:12:47    |

Device# **show track list**

The following is a sample output for the **show track list** command:

```
Track 400
 List boolean and
 Boolean AND is Up
 6 changes, last change 00:12:58
 object 100 Up
 object 200 Up
 Tracked by:
 VRRPv3 GigabitEthernet2 IPv4 group 1
```

Device# **show track list brief**

The following is a sample output for the **show track brief** command:

| Track | Type | Instance | Parameter | State | Last Change |
|-------|------|----------|-----------|-------|-------------|
| 400   | list |          | boolean   | Up    | 00:13:02    |





## CHAPTER 21

# Configure VDSL and G.SHDSL

This chapter provides usage information and guidelines for configuring very-high-data-rate DSL (VDSL) and G.symmetric high bit rate DSL (G.SHDSL) in SD-WAN mode.

- [Configure VDSL, on page 705](#)
- [Configure G.SHDSL, on page 709](#)

## Configure VDSL

The following table provides usage information and guidelines for configuring asymmetric DSL (ADSL2/2+) and VDSL for supported Integrated Services Router Network Interface Modules (ISR NIMs) in SD-WAN mode. VDSL2 and ADSL2/2+ provide highly reliable WAN connections for remote sites.

For related information, see [VDSL Commands](#).

| Function                 | Command                                                                                                                                                         | Guidelines                                                                                                                                                                                                                                                     |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Configure operating mode | Device# <b>configure terminal</b><br>Device(config)# <b>controller</b><br><b>VDSL slot/subslot/port</b><br>Device(config)# <b>operating</b><br><b>mode auto</b> | To switch from operating mode auto adsl1 (adsl2+/ or vdsl2) to operating mode auto ads2+ (adsl1 or vdsl2), switch to operating mode auto first.<br><br>Before you change the operating mode, ensure that line-mode is changed to line-mode single-wire line 0. |
| Enable DSL on a line     | Device(config)# <b>line-mode</b><br><b>single-wire</b><br><b>line</b> <i>line-number</i>                                                                        | This command is supported only on DSL NIM-VAB-A.                                                                                                                                                                                                               |
| Enable bonding           | Device(config)# <b>line-mode</b><br><b>bonding</b>                                                                                                              | This command is supported only on DSL NIM-VAB-A.                                                                                                                                                                                                               |

| Function                                            | Command                                                                                                                                                           | Guidelines                                                                                                                                                                                                 |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Load firmware on a device                           | Device# <b>configure terminal</b><br>Device(config)# <b>controller VDSL slot/subslot/port</b><br>Device(config-controller)# <b>firmware phy filename filename</b> | The Cisco Catalyst SD-WAN CLI template does not support specifying the file location. Prepend the file name with flash: or with bootflash:, depending on its location.                                     |
| Enable or disable SRA                               | Device(config-controller)# <b>sra</b>                                                                                                                             | The Cisco Catalyst SD-WAN CLI template does not support the sra <i>line number</i> command. In line-mode bonding, sra enables sra on both lines and no sra disables sra on both lines.                     |
| Enable or disable bitswap                           | Device(config-controller)# <b>bitswap</b>                                                                                                                         | The Cisco Catalyst SD-WAN CLI template does not support the bitswap <i>line number</i> command. In line-mode bonding, bitswap enables bitswap on both lines and no bitswap disables bitswap on both lines. |
| Enable modem features                               | Device(config-controller)# <b>modemkeyword</b>                                                                                                                    | —                                                                                                                                                                                                          |
| Display a description of a controller               | Device(config-controller)# <b>description string</b>                                                                                                              | —                                                                                                                                                                                                          |
| Enable dual ended line testing                      | Device(config-controller)# <b>diagnostics DELT</b>                                                                                                                | —                                                                                                                                                                                                          |
| Modify the file in which the training log is stored | Device(config-controller)# <b>training log filename flash: filename</b>                                                                                           | The Cisco Catalyst SD-WAN CLI template does not support specifying the file location. Prepend the file name with flash: or with bootflash:, depending where the file should be stored.                     |
| Enable sync mode                                    | Device(config-controller)# <b>sync mode mode</b>                                                                                                                  | To switch from one sync mode to another, delete the existing sync mode, then configure the new one.                                                                                                        |
| Enable sync interval                                | Device(config-controller)# <b>sync interval seconds</b>                                                                                                           | —                                                                                                                                                                                                          |

### Command Examples

```
Device# config-transaction
Device(config)# controller VDSL 0/0/0
Device(config)# operating mode auto
```



```

Device# config-transaction
Device(config)# line-mode single-wire line 1

Device# config-transaction
Device(config)# line-mode bonding

Device# config-transaction
Device(config)# controller VDSL 0/0/0
Device(config-controller)# firmware phy filename flash:IDC_1.7.2.6_DFE_FW_BETA_120111A.pkg

Device# config-transaction
Device(config-controller)# sra

Device# config-transaction
Device(config-controller)# bitswap

Device# config-transaction
Device(config)# controller VDSL 0/0/0
Device(config-controller)# modem customUKAnnexM

Device# config-transaction
Device(config)# controller VDSL 0/0/0
Device(config-controller)# description to ISP 1

Device# config-transaction
Device(config)# controller VDSL 0/0/0
Device(config-controller)# diagnostics DELT

Device# config-transaction
Device(config)# controller VDSL 0/0/0
Device(config-controller)# training log filename bootflash:VDSLLOG.log

Device# config-transaction
Device(config)# controller VDSL 0/0/0
Device(config-controller)# sync mode ansi previous

Device# configure terminal
Device(config)# ptp clock ordinary domain 0
Device(config-ptp-clk)# clock-port slave slaveport
Device(config-ptp-port)# sync interval -4
Device(config-ptp-port)# end

```

### Configuration Example

```

Device(config)# show controllers vdsL 0/2/0
Controller VDSL 0/2/0 is UP

Daemon Status: UP

 XTU-R (DS) XTU-C (US)
Chip Vendor ID: 'BDCM' 'BDCM'
Chip Vendor Specific: 0x0000 0xA39A
Chip Vendor Country: 0xB500 0xB500
Modem Vendor ID: 'CSCO' 'BDCM'
Modem Vendor Specific: 0x4602 0x0000
Modem Vendor Country: 0xB500 0xB500
Serial Number Near: FGL2149956Y C1117-4P 16.7.20180

```

```

Serial Number Far:
Modem Version Near: 16.7.20180709:09395
Modem Version Far: 0xa39a

Modem Status: TC Sync (Showtime!)
DSL Config Mode: AUTO
Trained Mode: G.993.2 (VDSL2) Profile 17a

TC Mode: PTM
Selftest Result: 0x00
DELT configuration: disabled
DELT state: not running

Failed full inits: 0
Short inits: 0
Failed short inits: 0

Modem FW Version: 4.14L.04
Modem PHY Version: A2pv6F039t.d26d

```

Line 0:

|                         | XTU-R (DS)    |      |      | XTU-C (US)    |      |      |      |
|-------------------------|---------------|------|------|---------------|------|------|------|
| Trellis:                | ON            |      |      | ON            |      |      |      |
| SRA:                    | enabled       |      |      | enabled       |      |      |      |
| SRA count:              | 0             |      |      | 0             |      |      |      |
| Bit swap:               | enabled       |      |      | enabled       |      |      |      |
| Bit swap count:         | 1             |      |      | 3             |      |      |      |
| Line Attenuation:       | 18.4 dB       |      |      | 0.0 dB        |      |      |      |
| Signal Attenuation:     | 0.0 dB        |      |      | 0.0 dB        |      |      |      |
| Noise Margin:           | 5.2 dB        |      |      | 6.0 dB        |      |      |      |
| Attainable Rate:        | 46022 kbits/s |      |      | 18866 kbits/s |      |      |      |
| Actual Power:           | 14.5 dBm      |      |      | 10.4 dBm      |      |      |      |
| Per Band Status:        | D1            | D2   | D3   | U0            | U1   | U2   | U3   |
| Line Attenuation(dB):   | 13.9          | 32.7 | 50.1 | N/A           | 25.6 | 37.7 | 42.3 |
| Signal Attenuation(dB): | 13.5          | 32.4 | N/A  | N/A           | 25.0 | 36.9 | 41.9 |
| Noise Margin(dB):       | 5.3           | 5.1  | N/A  | N/A           | 6.0  | 6.0  | 5.9  |
| Total FECC:             | 446           |      |      | 0             |      |      |      |
| Total ES:               | 3             |      |      | 0             |      |      |      |
| Total SES:              | 0             |      |      | 0             |      |      |      |
| Total LOSS:             | 0             |      |      | 0             |      |      |      |
| Total UAS:              | 50            |      |      | 50            |      |      |      |
| Total LPRS:             | 0             |      |      | 0             |      |      |      |
| Total LOFS:             | 0             |      |      | 0             |      |      |      |
| Total LOLS:             | 0             |      |      | 0             |      |      |      |

|                     | DS Channel1 | DS Channel0 | US Channel1 | US Channel0 |
|---------------------|-------------|-------------|-------------|-------------|
| Speed (kbps):       | NA          | 47610       | NA          | 18859       |
| SRA Previous Speed: | NA          | 0           | NA          | 0           |
| Previous Speed:     | NA          | 0           | NA          | 0           |
| Reed-Solomon EC:    | NA          | 446         | NA          | 0           |
| CRC Errors:         | NA          | 51          | NA          | 0           |
| Header Errors:      | NA          | 3935        | NA          | 0           |
| Interleave (ms):    | NA          | 1.00        | NA          | 1.00        |
| Actual INP:         | NA          | 0.00        | NA          | 0.00        |

```

Training Log : Stopped
Training Log Filename : flash:vdslllog.bin

```

# Configure G.SHDSL

## Overview

G.SHDSL is an international standard that allows devices to send and receive high-speed symmetrical data streams over a single pair of copper wires. This section provides information about the Cisco G.SHDSL EFM/ATM NIM and provides guidelines for configuring G.SHDSL in SD-WAN mode.

For related information, see [Configuring Cisco G.SHDSL HWICs in Cisco Access Routers](#) and [VDSL Commands](#).

## Cisco G.SHDSL EFM/ATM NIM

The Cisco G.SHDSL EFM/ATM NIM connects Cisco 4000 Series Integrated Services Routers with central office Digital Subscriber Line Access Multiplexers (DSLAMs) and supports up to four DSL pairs. The DSL pairs are bundled in groups and configured in the Cisco IOS CLI by using the `dsl-group` command. Use the mode command to choose the mode (ATM or EFM).

The NIM supports the following configuration:

- You can configure up to four DSL groups.
- You can configure auto mode on only one DSL group. For example, DSL group 0.
- In ATM Mode, you can configure the lines to use 2-wire, 4-wire (standard or enhanced), or m-pair.
- In EFM mode, you can configure a DSL group with any one of the lines in 2-wire non-bonding mode or with multiple lines in bonding mode.
- Depending on the mode (ATM or EFM), the corresponding interface (ATM or EFM) is automatically created.

## Cisco G.SHDSL Configuration Guidelines

The following table provides usage information and guidelines that apply when you configure the Cisco G.SHDSL EFM/ATM in CPE or CO mode.

| Function                                                        | Command                                             | Guidelines                                                                                                                                                                                                      |
|-----------------------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Configure a device with the <code>dsl-group auto</code> command | Device(config-controller)#<br><b>dsl-group auto</b> | Use customer premises equipment (CPE) mode when configuring a device with the <code>dsl-group auto</code> command. If you use this command in Central Office (CO) mode, the configuration does not take effect. |
| Add or delete a link                                            | —                                                   | The <code>efm-grp</code> command is not supported. To add or delete a link to a dsl-group, delete the dsl-group, then create a new dsl-group.                                                                   |

| Function                          | Command                                                                                                                                | Guidelines                                                                                                                                                                                                                                                      |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Load firmware on a device         | Device(config-controller)#<br><b>firmware phy filename</b><br><i>location</i>                                                          | File name location options are not supported when using the firmware phy command. Prepend the file name with flash: or with bootflash:, depending on the location.                                                                                              |
| Create or delete an annex         | Device(config-controller-dsl-group)#<br><b>no shdsl annex</b><br><br>Device(config-controller-dsl-group)#<br><b>no shdsl rate rate</b> | To avoid Cisco IOS and Cisco Catalyst SD-WAN configuration from going out of sync when you create or delete an annex, create or delete the rate in the same transaction.                                                                                        |
| Enable SHDSL to use enhanced mode | (config-controller-dsl-group)#<br><b>shdsl 4-wire mode enhanced</b>                                                                    | To enable SHDSL to use the enhanced mode in a 2-pair digital subscriber line (DSL) group, use the shdsl 4-wire mode enhanced command in configuration controller DSL group mode.                                                                                |
| Ignore CRC errors                 | (config-controller-dsl-group)#<br><b>ignoreseconds</b>                                                                                 | To configure a device to ignore CRC errors, use the ignore command. Replace <i>timeout</i> with a value from 0 through 60, which indicates the number of seconds that the device ignores CRC errors that do not resolve before the device terminates an action. |
| Shutdown a DSL group              | (config-controller-dsl-group)#<br><b>shutdown</b>                                                                                      | To shut down a DSL group, use the shutdown command.                                                                                                                                                                                                             |

## Examples

```
Device# config-transaction
Device(config)# controller SHDSL 0/0/0
Device(config-controller)# dsl-group auto
```

```
Device# config-transaction
Device(config)# controller VDSL 0/0/0
Device(config-controller)# firmware phy filename bootflash:IDC_1.1.1.0_DFE_1.1-1.8.1__001.pkg
```

```
Device# config-transaction
Device(config)# controller SHDSL 0/0/0
Device(config-controller)# dsl-group 0 pairs 0
Device(config-controller-dsl-group)# no shdsl annex
Device(config-controller-dsl-group)# no shdsl rate 5696
```

```
Device# config-transaction
Device(config)# controller SHDSL 0/0/0
Device(config-controller)# termination cpe
```

```
Device(config-controller)# dsl-group 0 pairs 0
(config-controller-dsl-group)# shdsl 4-wire mode enhanced
```

```
Device# config-transaction
Device(config)# controller SHDSL 0/0/0
Device(config-controller)# termination cpe
Device(config-controller)# dsl-group 0 pairs 0
config-controller-dsl-group)# ignore 30
```

```
Device# config-transaction
Device(config)# controller SHDSL 0/0/0
Device(config-controller)# termination cpe
Device(config-controller)# dsl-group 0 pairs 0
config-controller-dsl-group)# shutdown
```

### Configuration Example

```
Device# sh controllers shDSL 0/1/0
Controller SHDSL 0/1/0 is UP
 Hardware is NIM-SHDSL-EA, on slot 0,bay 0
 Capabilities: EFM: 2-wire, EFM-Bond, Annex A, B, F & G
 ATM: 2-wire, Mpair, Annex A, B, F & G
 CPE termination
 cdb=0x7F7EB723D8A8
 Vendor: Intel, Chipset: SOCRATES-4e
 PHY Source: System
 IDC Firmware version: 0.0.0.0
 DFE Firmware version:
 Group 0 info:
 Type: EFM Auto status: Down
 Ethernet Interface: Ethernet0/1/0, hwidb: 0x7F7EB723B648
 ATM Interface: ATM0/1/0, hwidb: 0x7F7EB724CE08
 Configured/active num links: 4/0, bit map: 0xF/0x0
 Line termination: CPE, Annex: auto
 PMMS disabled,Line coding: AUTO-TCPAM
 Configured/actual rate: AUTO/0 kbps
 Dying Gasp: Present
 SHDSL wire-pair (0) is in DSL DOWN state
 LOSWS Defect alarm: none
 SNR Margin alarm: none
 Loop Attenuation alarm: none
 Termination: CPE, Line mode: EFM Auto, Annex: auto
 Line coding: AUTO-TCPAM
 Configured/actual rate: AUTO/0 kbps
 Modem status: DOWN_NOT_READY,Condition: NO_COND_
 DSL Stats:
 Power Back Off: 0dB
 LoopAttn: 0dB, SnrMargin: 0dB
 Current 15 minute statistics (Time elapsed 1 seconds)
 ES:0, SES:0, CRC:0, LOSWS:0, UAS:0
 Previous 15 minute statistics
 ES:0, SES:0, CRC:0, LOSWS:0, UAS:0
 Current 24 hr statistics
 ES:0, SES:0, CRC:0, LOSWS:0, UAS:0
 Previous 24 hr statistics
 ES:0, SES:0, CRC:0, LOSWS:0, UAS:0
 EFM Stats:
 EFM-TC Tx: data frames: 0
 EFM-TC Rx: data frames: 0
 SHDSL wire-pair (1) is in DSL DOWN state
 LOSWS Defect alarm: none
 SNR Margin alarm: none
```

```

Loop Attenuation alarm: none
Termination: CPE, Line mode: EFM Auto, Annex: auto
Line coding: AUTO-TCPAM
Configured/actual rate: AUTO/0 kbps
Modem status: DOWN_NOT_READY,Condition: NO_COND_
DSL Stats:
 Power Back Off: 0dB
 LoopAttn: 0dB, SnrMargin: 0dB
 Current 15 minute statistics (Time elapsed 1 seconds)
 ES:0, SES:0, CRC:0, LOSWS:0, UAS:0
 Previous 15 minute statistics
 ES:0, SES:0, CRC:0, LOSWS:0, UAS:0
 Current 24 hr statistics
 ES:0, SES:0, CRC:0, LOSWS:0, UAS:0
 Previous 24 hr statistics
 ES:0, SES:0, CRC:0, LOSWS:0, UAS:0
EFM Stats:
 EFM-TC Tx: data frames: 0
 EFM-TC Rx: data frames: 0
SHDSL wire-pair (2) is in DSL DOWN state
 LOSWS Defect alarm: none
 SNR Margin alarm: none
 Loop Attenuation alarm: none
 Termination: CPE, Line mode: EFM Auto, Annex: auto
 Line coding: AUTO-TCPAM
 Configured/actual rate: AUTO/0 kbps
 Modem status: DOWN_NOT_READY,Condition: NO_COND_
DSL Stats:
 Power Back Off: 0dB
 LoopAttn: 0dB, SnrMargin: 0dB
 Current 15 minute statistics (Time elapsed 1 seconds)
 ES:0, SES:0, CRC:0, LOSWS:0, UAS:0
 Previous 15 minute statistics
 ES:0, SES:0, CRC:0, LOSWS:0, UAS:0
 Current 24 hr statistics
 ES:0, SES:0, CRC:0, LOSWS:0, UAS:0
 Previous 24 hr statistics
 ES:0, SES:0, CRC:0, LOSWS:0, UAS:0
EFM Stats:
 EFM-TC Tx: data frames: 0
 EFM-TC Rx: data frames: 0
SHDSL wire-pair (3) is in DSL DOWN state
 LOSWS Defect alarm: none
 SNR Margin alarm: none
 Loop Attenuation alarm: none
 Termination: CPE, Line mode: EFM Auto, Annex: auto
 Line coding: AUTO-TCPAM
 Configured/actual rate: AUTO/0 kbps
 Modem status: DOWN_NOT_READY,Condition: NO_COND_
DSL Stats:
 Power Back Off: 0dB
 LoopAttn: 0dB, SnrMargin: 0dB
 Current 15 minute statistics (Time elapsed 1 seconds)
 ES:0, SES:0, CRC:0, LOSWS:0, UAS:0
 Previous 15 minute statistics
 ES:0, SES:0, CRC:0, LOSWS:0, UAS:0
 Current 24 hr statistics
 ES:0, SES:0, CRC:0, LOSWS:0, UAS:0
 Previous 24 hr statistics
 ES:0, SES:0, CRC:0, LOSWS:0, UAS:0
EFM Stats:
 EFM-TC Tx: data frames: 0
 EFM-TC Rx: data frames: 0
Group 1 is not configured

```

```
Group 2 is not configured
Group 3 is not configured
```







## CHAPTER 22

# Dynamic On-Demand Tunnels

Table 331: Feature History

| Feature Name                                      | Release Information                                                                                       | Description                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dynamic On-Demand Tunnels                         | Cisco IOS XE Catalyst SD-WAN Release 17.3.1a<br>Cisco vManage Release 20.3.1                              | This feature enables you to configure an Inactive state for tunnels between edge devices, reducing performance demands on devices and reducing network traffic.                                                                                                                                                                                                   |
| Dynamic On-Demand Tunnels with Transport Gateways | Cisco Catalyst SD-WAN Control Components Release 20.12.1<br>Cisco IOS XE Catalyst SD-WAN Release 17.12.1a | <p>A transport gateway can serve as the hub between two spoke devices, providing the backup route that is necessary for spoke-to-spoke on-demand tunnels to operate.</p> <p>Using a transport gateway as the hub simplifies the process of enabling on-demand tunnels. This method does not require any change to control policy on Cisco SD-WAN Controllers.</p> |

- [Information About On-Demand Tunnels, on page 715](#)
- [Prerequisites for On-Demand Tunnels, on page 718](#)
- [Restrictions for On-Demand Tunnels, on page 722](#)
- [Configure On-Demand Tunnels, on page 723](#)
- [Monitor the Status of On-Demand Tunnels, on page 727](#)

## Information About On-Demand Tunnels

Cisco Catalyst SD-WAN supports dynamic on-demand tunnels between any two Cisco Catalyst SD-WAN spoke devices. These tunnels are triggered to be set up only when there is traffic between the two devices. After the flow of traffic between the devices stops, a user-configurable inactivity timer starts, and after the configured time, the tunnel between the devices is removed. The on-demand link between the two devices is

then considered to be inactive. In this inactive state, it does not use network bandwidth and does not affect device performance.

### Backup Route and Reactivating the Tunnel

To enable two spoke device peers to use on-demand tunnels, they must have an alternate route, a backup route, through a hub. Using the backup route, either spoke device can resume traffic flow between the two spokes, which reactivates the tunnel to handle the traffic directly from peer to peer.

### Advantages

On-demand tunnels offer the following advantages:

- Improved performance, especially for less-powerful platforms operating in a full-mesh network.
- Improved latency in hub-and-spoke deployments when on-demand tunnels are used between spokes.
- Reduced bandwidth use in the network because tunnels in Inactive state do not require Bidirectional Forwarding Detection (BFD) probes, so there is less BFD traffic produced in the network.
- Direct tunnels between spokes, while also optimizing CPU and memory usage.

### Mechanism

When you configure a site to use dynamic tunnels, the on-demand functionality is enabled. In this mode of operation, Cisco Catalyst SD-WAN edge routers do not bring up direct tunnels to other sites that are also enabled with on-demand functionality.

Cisco Catalyst SD-WAN selects one or more edge routers (typically centrally located routers) to act as backup forwarding node(s), providing a secondary path for traffic between two nodes. The backup node(s) are not enabled for on-demand. All on-demand sites form static tunnels with the backup node(s). The backup node(s) provide a static backup route for traffic between two nodes that have on-demand enabled.

The first packet of traffic between two nodes is routed through the static backup path, and triggers the on-demand tunnel to become active between the sites. The backup path continues to forward traffic until the direct path becomes active.

All on-demand sites learn the TLOCs and prefixes of all other on-demand remote sites. The prefixes also have a backup path set up through Cisco Catalyst SD-WAN Controller control policy. So in the control plane, the on-demand tunnel network has the same state as a full-mesh tunnel network, including a backup path.

The control plane downloads to the data plane, routes, with the backup path and remote TLOCs that represent a potential direct path between any two sites, but it does not set up a direct path tunnel to remote TLOCs.

All prefixes learned from remote sites must have a backup path. A less-specific aggregate route from a hub site is not a valid backup path. The specific prefix advertised from the remote branch must also be advertised as a backup from the hub.

If this specific prefix is missing as a backup, the on-demand tunnel setup will fail.

Therefore, the backup path must be a static tunnel that is always UP. This static tunnel should include both the specific prefixes from the remote branch and any necessary aggregate routes to ensure proper tunnel establishment and maintenance.

Traffic from either end of the on-demand tunnel triggers setting up the tunnel. This enables on-demand tunnels to accommodate network address translation (NAT) traversal.

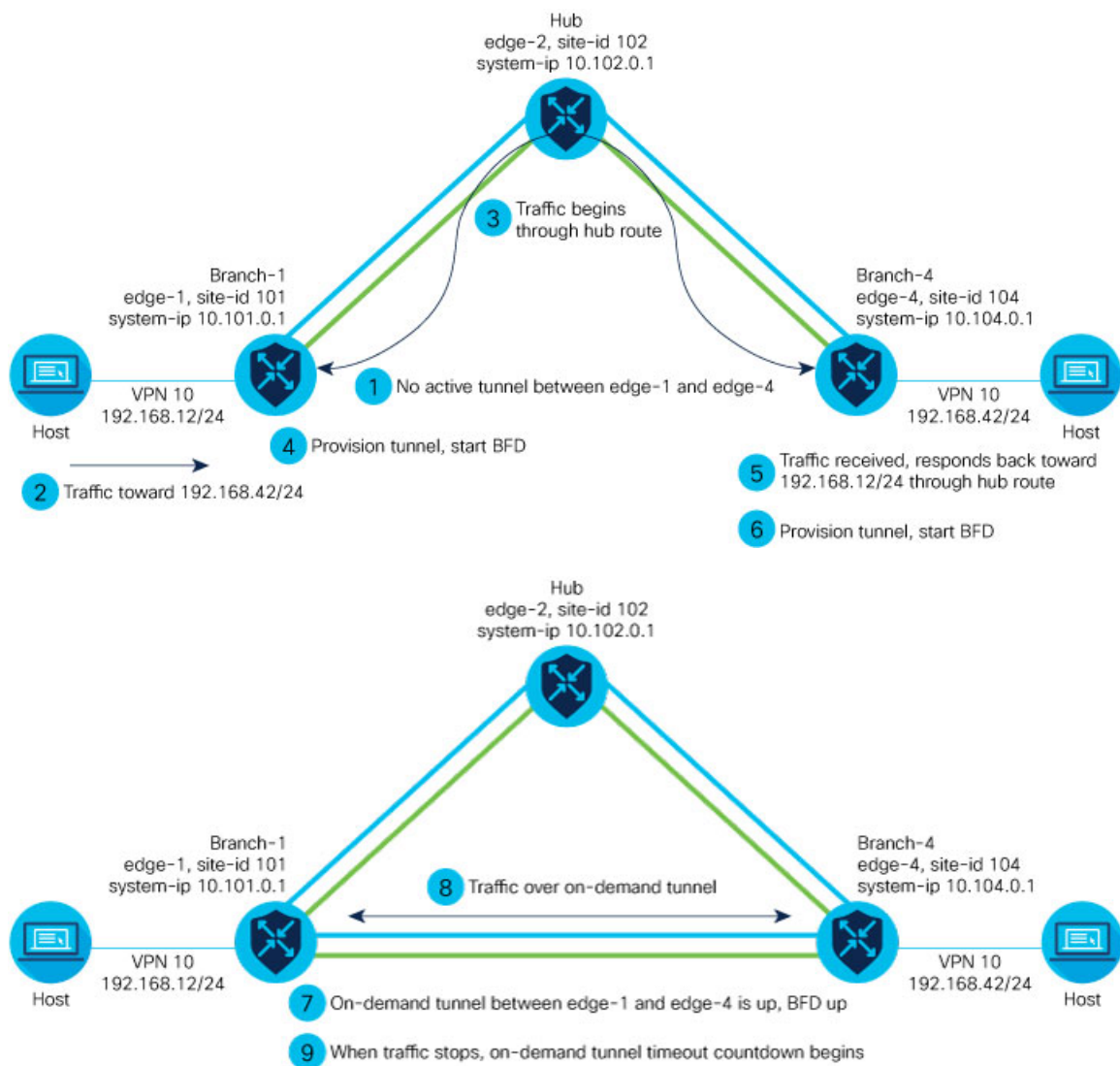
The on-demand tunnel feature introduces two states for the on-demand branch site:

- **Inactive:** The on-demand tunnel is not set up with the remote site. There is no active traffic to or from the remote site. Remote site TLOCs are inactive - no bidirectional forwarding detection (BFD) is set up, the prefix is installed with the inactive paths, and the backup path is set as the path to forward any traffic. The inactive path detects flows and triggers a direct site-to-site tunnel to be set up.
- **Active:** The on-demand direct site-to-site tunnel is set up to the remote site. There is active traffic to or from the remote site. This state is identical to the case of a typical tunnel, where the remote TLOCs have BFD set up, and the prefix is installed with the direct path tunnel. In this state, tunnel activity is tracked. If there is no traffic for the “idle time” duration (default 10 minutes), the direct site-to-site tunnel is removed and the state changes to Inactive.

### Steps in Illustrations

The figures below show the following steps that occur between two edge routers with an on-demand tunnel configured.

1. There is no active tunnel between the two edge routers. edge-1 and edge-4 are in Inactive state.
  2. The host behind edge-1 initiates traffic toward the host behind edge-4.
  3. edge-1 forwards the traffic through the backup path using the hub or backup node to edge-4.
  4. edge-1 provisions the on-demand tunnel and begins bidirectional forwarding detection (BFD). edge-4 is now in Active state on edge-1.
  5. When edge-4 receives the return traffic for the host behind edge-1, it forwards the traffic through the backup path using the hub or backup node to edge-1.
  6. edge-4 provisions the on-demand tunnel and begins BFD. edge-1 is now in Active state on edge-4.
  7. At this point, the on-demand tunnel between edge-1 and edge-4 is up, and BFD is up.
  8. Traffic between the two edge devices takes the direct route through the on-demand tunnel.
  9. Both edge-1 and edge-4 track the traffic activity on the on-demand tunnel in both directions.
- If there is no traffic for the idle timeout duration, the on-demand tunnel is deleted, and the edge-1 and edge-4 devices go back to the Inactive state.



### On-Demand Tunnels with a Transport Gateway

A transport gateway can serve as the hub between two spoke devices, providing the backup route that is necessary for spoke-to-spoke on-demand tunnels to operate. Using a transport gateway as the hub simplifies the process of enabling on-demand tunnels. This method does not require configuring control policy on Cisco SD-WAN Controllers.

For information about configuration, see [Configure On-Demand Tunnels Using a Transport Gateway, on page 725](#).

## Prerequisites for On-Demand Tunnels

There are several prerequisites for using on-demand tunnels:

- [Configure a Centralized Control Policy for On-Demand Tunnels, on page 723](#)

- [Prerequisites: OMP Settings, on page 719](#)
- [Prerequisites: Hub Device Traffic Engineering Service, on page 720](#)
- [Prerequisites: Spoke Device ECMP Limit, on page 721](#)

## Prerequisites: OMP Settings

The Cisco Catalyst SD-WAN Controller send-path-limit must be more than the default 4.

When on-demand tunnels are enabled, spokes use backup paths through the hub, so a higher path limit is necessary. The direct paths as well as the backup paths need to be advertised. To accommodate this, increase the Cisco Catalyst SD-WAN Controller send-path-limit to advertise all available paths. We recommend to use the maximum possible value.



**Note** If there are too many Hub TLOCs configured in the on-demand tunnel control policy, the recommended value for send-path-limit is not enough always. In such cases, the on-demand tunnel feature will not work at all.

Starting from Cisco vManage Release 20.8.1 and Cisco IOS XE Catalyst SD-WAN Release 17.8.1a, the maximum send-path-limit is 32. In Cisco vManage Release 20.7.x and earlier releases, the maximum send-path-limit is 16.

For information about configuring Cisco SD-WAN Controller send-path-limit, see the routing configuration guides on the [Cisco Catalyst SD-WAN Configuration Guides](#) page.

### Configure the OMP Send Path Limit Using a Feature Template

For a Cisco Catalyst SD-WAN Controller in Managed mode, use this procedure to configure the OMP send path limit.

To confirm that it is in Managed mode, from the Cisco SD-WAN Manager menu, choose **Configuration > Control Components**. For a **Controller** row, the **Managed By** column shows a template name.

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**.



**Note** In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

3. Edit or create an OMP template for device type Controller. In some earlier releases, the device type is called vSmart.



**Note** Any functioning Cisco Catalyst SD-WAN deployment has at least one vSmart OMP template configured.

4. In **Basic Configuration**, set the **Number of Paths Advertised per Prefix** to 16 (recommended).  
From Cisco SD-WAN Release 20.8.1, the maximum is 32.

### Configure the OMP Send Path Limit for an Edge Device, Using CLI Commands

Use a CLI add-on profile, CLI add-on feature template, or CLI template to execute these CLI commands. By default, CLI templates and the CLI add-on profile execute commands in global configuration mode. For more information about using CLI templates, see [CLI Add-On Feature Templates](#) and [CLI Templates](#).

For an edge device, use these commands to configure an OMP send path limit.

1. Enter OMP configuration mode.

```
omp
```

2. Set the send path limit. We recommend 16.

From Cisco SD-WAN Release 20.8.1, the maximum is 32.

```
send-path-limit 16
```

Example configuration:

```
sdwan
omp
send-path-limit 16
```

## Prerequisites: Hub Device Traffic Engineering Service

On the hub device, the Traffic Engineering service (service TE) must be enabled.

This ensures that the Cisco Catalyst SD-WAN Overlay Management Protocol (OMP) on the spoke devices accepts the backup path through the hub, which is being added as an intermediate path between the two spoke devices. Without this, the backup path through the hub would be considered invalid and unresolved by the spoke devices.

### Enable the Traffic Engineering Service Using Cisco SD-WAN Manager

1. In Cisco SD-WAN Manager, open **Configuration > Templates**.
2. Click **Feature Templates**.




---

**Note** In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

---

3. Click **Add Template**.
4. Select a platform.
5. From **VPN**, select **VPN**.
6. Ensure that in **Basic Configuration**, the **VPN** field is set to 0.
7. From **Service**, click **New Service** and select **TE**.
8. Click **Add**, and then click **Update**. The TE service appears in the table of services.
9. Apply the VPN-0 template to the hub.

### Enable the Traffic Engineering Service, Using CLI Commands (Cisco IOS XE Catalyst SD-WAN Devices)

Use a CLI add-on profile, CLI add-on feature template, or CLI template to execute these CLI commands. By default, CLI templates and the CLI add-on profile execute commands in global configuration mode. For more information about using CLI templates, see [CLI Add-On Feature Templates](#) and [CLI Templates](#).

```
sdwan
 service TE vrf global
exit
```

### Enable the Traffic Engineering Service, Using CLI Commands (Cisco vEdge Devices)

```
vpn 0
 service TE
exit
```

## Prerequisites: Spoke Device ECMP Limit

On spoke devices, the ECMP limit must be more than the default 4. Recommended: 16

When on-demand tunnels are enabled, spoke devices create both direct and backup paths. To accommodate the need for more paths, increase the ECMP limit .

### Configure the ECMP Limit Using a Feature Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**.



---

**Note** In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

---

3. Click **Add Template**.
4. Select a device and click **Cisco OMP**.
5. In **Basic Configuration**, set the **ECMP Limit** field to 16 (recommended).

### Configure the ECMP Limit Using a Configuration Group

On the **Configuration > Configuration Groups** page, choose the SD-WAN solution type.

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.
2. Do one of these:
  - Edit a profile directly:  
In the System Profile tab, create (**Add New**) or edit a System profile.
  - Edit a profile in a configuration group:  
Open a configuration group and edit the System profile.
3. In the system profile, create or edit an [OMP](#) feature.

4. In the **Basic Configuration** section, use the **ECMP Limit** field to configure the equal-cost multi-path (ECMP) limit. We recommend 16.

### Configure the ECMP Limit, Using CLI Commands

Use a CLI add-on profile, CLI add-on feature template, or CLI template to execute these CLI commands. By default, CLI templates and the CLI add-on profile execute commands in global configuration mode. For more information about using CLI templates, see [CLI Add-On Feature Templates](#) and [CLI Templates](#).

```
sdwan
omp
no shutdown
ecmp-limit 16
```



**Note** You can view the current `ecmp-limit` using the **show running-config omp** command.

## Restrictions for On-Demand Tunnels

- On-demand tunnel Performance Routing (PfR) statistics collection starts fresh every time an on-demand tunnel is setup. PfR statistics are not cached for deleted on-demand tunnels after idle timeout.
- Out of order (OOO) packets may occur when traffic moves from the backup path to the direct on-demand tunnel. Packets are forwarded by the Cisco Catalyst SD-WAN router as they are received.
- Unidirectional flows do not trigger on-demand tunnel setup. They continue to use the backup path.
- Multicast traffic does not trigger on-demand tunnel setup. It continues to use the backup path.
- Do not configure a data policy that applies a **set tloc-list** action to an on-demand site TLOC. If configured, traffic will be dropped.
- On-demand tunnels are not supported when the Pair Wise Key (PWK) IPsec feature is enabled.
- All TLOCs in the system are reset (disabled and then enabled) when you execute **on-demand enable** or **no on-demand enable**.
- When an edge device provisions on-demand tunnels, it provisions to all the TLOCs on the remote edge device.
- For a multi-home site to be in on-demand mode, you must configure on-demand enable on all of the systems at the site.
- All edge devices using on-demand tunnels are kept active if there is a service or user traffic on any on-demand tunnel in either direction.
- On-demand tunnels can be enabled between two sites only if both sites are enabled with on-demand mode.
- The first packet to any host behind a remote site triggers on-demand tunnel setup to that remote site. Return traffic from that host triggers tunnel setup in the opposite direction.
- All prefixes from on-demand remote sites must also have a backup path configured. If not, sites will not be able set up on-demand tunnels. The backup path is a static tunnel and must be always UP.



- The setup or removal of on-demand tunnels does not affect overlay route (OMP) updates by Cisco Catalyst SD-WAN Controller, or service/LAN-side route updates (examples: OSPF or BGP).
- If either the local site or the remote site is not in on-demand mode, static tunnels are set up between the sites.

## Configure On-Demand Tunnels

The following procedures describe how to configure on-demand tunnels using different methods, including using control policy, or a simpler method using a transport gateway as a hub.

### Configure On-Demand Tunnels Using Control Policy

To configure on-demand tunnels using the control policy method, do the following:

1. Configure a control policy, as described in [Configure a Centralized Control Policy for On-Demand Tunnels, on page 723](#).
2. On spoke devices, enable on-demand tunnels, as described in [Enable On-Demand Tunnels on a Spoke Device Using a Template, on page 726](#) and [Enable On-Demand Tunnels Using a CLI Template, on page 727](#).

### Configure a Centralized Control Policy for On-Demand Tunnels

#### Before You Begin

This procedure configures a centralized control policy on a Cisco Catalyst SD-WAN Controller to enable on-demand tunnels.

- The Cisco Catalyst SD-WAN Controller centralized control policy must include the **tloc-action backup** action.

**Explanation:** This ensures that the backup path through the hub for communication between all of the spoke devices.

- The Cisco Catalyst SD-WAN Controller centralized control policy must accept all spoke prefix routes.
- The Cisco Catalyst SD-WAN Controller centralized control policy must accept TLOCs of all spokes.

For information about configuring a Cisco SD-WAN Controller **centralized control policy**, see the policies configuration guides on the [Cisco Catalyst SD-WAN Configuration Guides page](#).

- When configuring on-demand tunnels using a transport gateway, do not use the control policy procedure described here. For information, see [Configure On-Demand Tunnels Using a Transport Gateway, on page 725](#).

#### Configure a Centralized Control Policy for On-Demand Tunnels Using Cisco SD-WAN Manager

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Policies**.
2. Select **Centralized Policy**.

3. Click **Add Policy**.
4. In the left pane, click **Site**.
5. Click **Next**.
6. Click **Add Topology** and select **Custom Control (Route & TLOC)**.
7. Enter a name and description for the topology.
8. Click **Sequence Type**.
9. In the **Add Control Policy** pop-up window, choose **Route**.
10. Click **Sequence Rule** to create a sequence.
11. Click **Match**.
12. Among the match conditions, click **Site**.
13. In the **Match Conditions** area, click the **Site List** menu and choose a site list.
14. Click **Actions**.
15. Click **Accept**.
16. Among the actions, click **TLOC Action**.
17. In the **Actions** area, click the **TLOC Action** menu and choose **Backup**.
18. Among the actions, click **TLOC**.
19. In the **Actions** area, click the **TLOC List** menu and choose or create a TLOC list.
20. Click **Save Match and Actions**.
21. Click **Default Action**.
22. In the **Default Action** area, click the pencil icon to edit.
23. Near the **Actions** label, click **Accept**.
24. Click **Save Control Policy**.
25. Click **Next** twice.
26. In the Topology tab, click **New Site/WAN Region List**.
27. Click **Outbound Site List** and choose a site list that defines the sites at which you are enabling on-demand tunnels.
28. Adjacent to the site list, click **Add**.
29. Enter a name and description for the policy.
30. Click **Save Policy**.

### Configure Centralized Control Policy for On-Demand Tunnels Using a CLI Policy

Prerequisite: The Cisco SD-WAN Controller must be managed by Cisco SD-WAN Manager.

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Classic > Policies**.

2. Open **Centralized Policy**.
3. From **Custom Options**, choose **Centralized Policy > CLI Policy**.
4. Click **Add Policy**.
5. Enter the CLI commands for the policy.

Example:

```
control-policy Dynamic-Tunnel-Control-Policy
sequence 100
 match route
 site-list Branches
 !
 action accept
 set
 tloc-action backup
 tloc-list Hub-TLOCs
 !
 !
sequence 200
 match tloc
 !
 action accept
 !
default-action accept
!
lists
site-list Branches
 site-id 200
 site-id 300
!
tloc-list Hub-TLOCs
 tloc 10.0.0.1 color mpls encap ipsec
 tloc 10.0.0.1 color public-internet encap ipsec
!
!
apply-policy
 site-list Branches
 control-policy Dynamic-Tunnel-Control-Policy out
!
!
```

## Configure On-Demand Tunnels Using a Transport Gateway

### Before You Begin

- On Cisco SD-WAN Controllers, configure the send path limit, as described in [Prerequisites: OMP Settings, on page 719](#).
- On spoke devices, configure the ECMP limit, as described in [Prerequisites: Spoke Device ECMP Limit, on page 721](#).
- When using a transport gateway as a hub to support on-demand tunnels, there is no need to create or modify a control policy. Do not use the procedure described in [Configure a Centralized Control Policy for On-Demand Tunnels, on page 723](#).

### Configure On-Demand Tunnels Using Transport Gateways

1. On a router serving as the hub, providing a backup route between spokes, enable transport gateway functionality, as described in , in the [Transport Gateway](#) section of the *Cisco Catalyst SD-WAN Routing Configuration Guide*.
2. On spoke devices, enable on-demand tunnels and configure the idle timeout, as described in [Enable On-Demand Tunnels on a Spoke Device Using a Template](#), on page 726.

## Enable On-Demand Tunnels on a Spoke Device Using a Configuration Group

### Before you begin

On the **Configuration > Configuration Groups** page, choose the SD-WAN solution type.

### Procedure

- 
- |               |                                                                                                                                                                                                                                                                                             |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Step 1</b> | From the Cisco SD-WAN Manager menu, choose <b>Configuration &gt; Configuration Groups</b> .                                                                                                                                                                                                 |
| <b>Step 2</b> | Do one of these: <ul style="list-style-type: none"><li>• Edit a profile directly:<br/>In the System Profile tab, create (<b>Add New</b>) or edit a System profile.</li><li>• Edit a profile in a configuration group:<br/>Open a configuration group and edit the System profile.</li></ul> |
| <b>Step 3</b> | In the System profile, create ( <b>Add New</b> ) or edit a <a href="#">Basic</a> feature.                                                                                                                                                                                                   |
| <b>Step 4</b> | In the <b>Advanced</b> section, use the <b>On Demand Tunnel</b> control to enable on-demand tunnels.                                                                                                                                                                                        |
- 

## Enable On-Demand Tunnels on a Spoke Device Using a Template

### Before You Begin

- See the [Prerequisites for On-Demand Tunnels](#).
- Do not enable on-demand on the hub device.
- On the spoke devices, enable on-demand at the system level. In the case of multi-homed sites, enable on-demand on all systems at the site.

### Enable On-Demand Tunnels on a Spoke Device

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**.



**Note** In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

3. Click **Add Template**.
4. Select a device.
5. From **Basic Information**, select **Cisco System**.
6. Click **Advanced**.
7. Enable **On-demand Tunnel**.
8. (optional) Configure the **On-demand Tunnel Idle Timeout** time. The default idle timeout value is 10 minutes. Range: 1 to 65535 minutes
9. Attach the System feature template to the device template for the spoke device.

## Enable On-Demand Tunnels Using a CLI Template

For more information about using CLI templates, see [CLI Add-On Feature Templates](#) and [CLI Templates](#). By default, CLI templates execute commands in global configuration mode.

### Before You Begin

- See [Prerequisites for On-Demand Tunnels, on page 718](#).
- Do not enable on-demand on the hub device

### Enable On-Demand Tunnels

On the spoke devices, enable on-demand tunnels at the system level. In the case of multi-homed sites, enable on-demand on all systems in the site.

The default idle timeout value is 10 minutes. Range: 1 to 65535 minutes

### Example

```
system
 on-demand enable
 on-demand idle-timeout 10
```

## Monitor the Status of On-Demand Tunnels

The following sections describe procedures for monitoring the status of on-demand tunnels.

## View the Current Status of On-Demand Tunnels Using Cisco SD-WAN Manager

1. From the Cisco SD-WAN Manager menu, choose **Monitor** > **Devices**.

Cisco vManage Release 20.6.x and earlier: From the Cisco SD-WAN Manager menu, choose **Monitor** > **Network**.

2. Select a device.
3. Select **Real Time**.
4. For **Device Options**, select one of the following:
  - **On Demand Local**: Displays the status of on-demand tunnels on the specified device.
  - **On Demand Remote**: Displays the status of on-demand tunnels on the specified device, and on all connected devices.

The output is equivalent to executing the `show [sdwan] system on-demand [remote-system] [system-ip ip-address]` CLI command. It displays the status of on-demand tunnels.

## View a Chart of the On-Demand Tunnel Status Over Time in Cisco SD-WAN Manager

1. From the Cisco SD-WAN Manager menu, choose **Monitor** > **Devices**.  
Cisco vManage Release 20.6.x and earlier: From the Cisco SD-WAN Manager menu, choose **Monitor** > **Network**.
2. Select a device.
3. From **WAN**, choose **Tunnel**.
4. From the **Chart Options** drop-down list, select **On-Demand Tunnel Status**. The chart shows the status of tunnels as ACTIVE or INACTIVE. INACTIVE indicates that an on-demand tunnel is in Inactive mode.

## View the Route to a Destination Device

Viewing the route between routers A and B can show whether the route is using an on-demand tunnel. On router A, use the **traceroute** command and enter router B as the destination. The command output shows whether the current route includes a hop at a hub device or whether the route is directly to the destination.

In the following examples, the router IP addresses are as follows:

- Router A: 10.1.1.1
- Router B: 10.1.1.2
- Hub device: 10.100.1.100

### No Active On-Demand Tunnel

In the following example, there is no active on-demand tunnel between routers A and B, so the route includes the hub device. Note that it takes two hops to reach router B.

```
RouterA#traceroute vrf 1 10.1.1.2 numeric
Type escape sequence to abort.
Tracing the route to 10.1.1.2
VRF info: (vrf in name/id, vrf out name/id)
```

```

1 10.100.1.100 10 msec 8 msec 0 msec
2 10.1.1.2 2 msec * 1 msec

```

### Active On-Demand Tunnel

In the following example, there is an active on-demand tunnel between routers A and B, so the route from router A and to router B is direct.

```

RouterA#traceroute vrf 1 10.1.1.2 numeric
Type escape sequence to abort.
Tracing the route to 10.1.1.2
VRF info: (vrf in name/id, vrf out name/id)
 1 10.1.1.2 1 msec

```

## View OMP Routes

Viewing OMP routes can show the status of on-demand tunnels between two routers. Use the **show sdwan omp routes** command and view the **STATUS** column. The following table shows the possible values for this column, depending on whether an on-demand tunnel is active or not between two routers:

**Table 332: Status of Routes, with or without an Active On-Demand Tunnel Between Two Routers**

| On-Demand Tunnel Between Routers A and B | STATUS for OMP Routes Between Routers A and B     | STATUS for Backup Routes (through the Hub)   |
|------------------------------------------|---------------------------------------------------|----------------------------------------------|
| Not active                               | I, U, IA<br>(installed, unresolved, and inactive) | C, I, R<br>(chosen, installed, and resolved) |
| Active                                   | C, I, R<br>(chosen, installed, and resolved)      | R<br>(resolved)                              |







## CHAPTER 23

# Track Static Routes for Service VPNs

Table 333: Feature History

| Feature Name                                                                                             | Release Information                                                          | Description                                                                                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Static Route Tracker for Service VPNs                                                                    | Cisco IOS XE Catalyst SD-WAN Release 17.3.1a<br>Cisco vManage Release 20.3.1 | This feature enables you to configure IPv4 static route endpoint tracking for service VPNs.<br><br>For static routes, endpoint tracking determines whether the configured endpoint is reachable before adding that route to the route table of the device. |
| TCP/UDP Endpoint Tracker and Dual Endpoint Static Route Tracker for Cisco IOS XE Catalyst SD-WAN devices | Cisco IOS XE Catalyst SD-WAN Release 17.7.1a<br>Cisco vManage Release 20.7.1 | This feature enables you to configure the TCP/UDP static route endpoint trackers. Using this feature you can also configure IPv4, TCP/UDP dual endpoint static-route tracker groups for service VPNs to enhance the reliability of probes.                 |

- [Information About Static Route Tracking, on page 731](#)
- [Supported Platforms, on page 732](#)
- [Restrictions for IPv4 Static Route Tracking, on page 732](#)
- [Configure Tracker Group Using a Configuration Group, on page 733](#)
- [Workflow to Configure IPv4 Static Route Tracking, on page 733](#)
- [Configure Static Routes Using CLI, on page 737](#)
- [Configuration Examples for Static Route Tracking Using the CLI, on page 739](#)
- [Verify Static Route Tracking Configuration Using CLI, on page 741](#)

## Information About Static Route Tracking

Static-route tracking for service VPNs enables you to track the availability of the configured endpoint address to determine if the static route can be included in the routing table of a device. This is applicable when a site

uses a static route in a service VPN to advertise its route over Overlay Management Protocol (OMP). The static route tracker periodically sends ICMP ping probes to the configured endpoint. If the tracker does not receive a response, the static route is not included in the routing table and is not advertised to OMP. You can configure an alternative next-hop address or a static route with a higher administrative distance to provide a backup path. This path is advertised over OMP.

**Note**

- From Cisco IOS XE Catalyst SD-WAN Release 17.7.1a, you can configure TCP/UDP individual endpoint trackers and configure a tracker group with dual endpoints (using two trackers), and associate the trackers and tracker group to a static route. Dual endpoints help in avoiding false negatives that might be introduced because of the unavailability of the routes.
- From Cisco IOS XE Catalyst SD-WAN Release 17.15.1a only UP/DOWN status changes are reported to Cisco SD-WAN Manager, while RTT value changes are not. This optimization improves efficiency in extensive networks.

## Supported Platforms

- Cisco ASR 1000 Series Aggregated Services Routers
- Cisco ISR 1000 Series-Integrated Services Routers
- Cisco ISR 4000 Series Integrated Services Routers
- Cisco CSR 1000 Series Cloud Service Routers

## Restrictions for IPv4 Static Route Tracking

- Only one endpoint tracker is supported per static route per next-hop address.
- IPv6 static routes are not supported.
- To configure a static route with tracker:
  1. Delete any existing static route, if it is already configured without a tracker. Plan for any connectivity downtime that might occur during this step for static route advertisement.
  2. Configure a new static route with tracker using the same prefix and next-hop as the deleted static route.
- To add a new tracker after you reach maximum tracker limit per router:
  1. Delete an old tracker and attach the template to the device.
  2. Add a new tracker and attach the device to the template again.
- UDP tracker endpoint enabled with IP SLA UDP packet responder is supported only on Cisco IOS XE Catalyst SD-WAN devices.

- You cannot link the same endpoint-tracker to static routes in different VPNs. Endpoint-tracker is identified by a name and can be used for multiple static routes in a single VPN.

## Configure Tracker Group Using a Configuration Group

### Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

### Procedure

**Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

**Step 2** Create and configure a tracker group in a Service Profile.

*Table 334: Tracker Group*

| Field                    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tracker Elements*</b> | This field is displayed only if you chose <b>Tracker Type</b> as the <b>Tracker Group</b> . Add the existing interface tracker names, separated with a space. When you add this tracker to the template, the tracker group is associated with these individual trackers, and you can then associate the tracker group to an interface.                                                                                                                                                                                                                                |
| <b>Tracker Boolean</b>   | <p>This field is displayed only if you chose <b>Tracker Type</b> as the <b>Tracker Group</b>. Select <b>AND</b> or <b>OR</b>.</p> <p><b>OR</b> is the default boolean operation. An <b>OR</b> ensures that the transport interface status is reported as active if either one of the associated trackers of the tracker group reports that the interface is active.</p> <p>If you select the <b>AND</b> operation, the transport-interface status is reported as active if both the associated trackers of the tracker group report that the interface is active.</p> |

### What to do next

Also see [Deploy a configuration group](#).

## Workflow to Configure IPv4 Static Route Tracking

1. Configure an endpoint tracker using the System template.
2. Configure a static route using the VPN template.
3. Apply the tracker to the next-hop address.

## Create a Static Route Tracker

Use the **System Template** to create a tracker for static routes.



**Note** Delete existing static routes, if any, before you create a static route tracker. Configure a new static route tracker using the same prefix and next hop as the deleted static route.

1. From Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**.



**Note** In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

3. Navigate to the **Cisco System** template for the device.



**Note** For information about creating a System template, see [Create System Template](#).

4. Click **Tracker**. Click **New Endpoint Tracker** to configure the tracker parameters.

**Table 335: Tracker Parameters**

| Field        | Description                                                                                                                                                                                                                                                                                                |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name         | Name of the tracker. The name can be up to 128 alphanumeric characters.                                                                                                                                                                                                                                    |
| Threshold    | Wait time for the probe to return a response before declaring that the configured endpoint is down. Range is from 100 to 1000 milliseconds. Default is 300 milliseconds.                                                                                                                                   |
| Interval     | Time interval between probes to determine the status of the configured endpoint. Default is 60 seconds (1 minute).<br>Range is from 20 to 600 seconds.                                                                                                                                                     |
| Multiplier   | Number of times probes are sent before declaring that the endpoint is down. Range is from 1 to 10. Default is 3.                                                                                                                                                                                           |
| Tracker Type | From the drop-down, choose Global. From the Tracker Type field drop-down, choose Static Route.<br><br>From Cisco IOS XE Catalyst SD-WAN Release 17.7.1a, you can configure a tracker group with dual endpoints on Cisco IOS XE Catalyst SD-WAN devices and associate this tracker group to a static route. |

| Field                      | Description                                                                                                                                                |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Endpoint Type              | Choose endpoint type IP Address.<br><br><b>Note</b><br>Configuring the tracker type Static Route using endpoint URL or endpoint DNS name is not supported. |
| End-Point Type: IP Address | IP address of the static route end point. This is the destination on the internet to which the router sends probes to determine the status of the route.   |

5. Click **Add**.
6. Click **Save**.
7. To create a tracker group, click **Tracker Groups > New Endpoint Tracker Groups** and configure the tracker parameters.



**Note** Ensure that you have created two trackers to form a tracker group.

**Table 336: Tracker Group Parameters**

| Fields           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name             | Name of the tracker group.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Tracker Type     | From the drop-down, choose <b>Global</b> . From the Tracker Type field drop-down, choose <b>Static Route</b> .<br><br>From Cisco IOS XE Catalyst SD-WAN Release 17.7.1a, you can configure a tracker group with dual endpoints on Cisco IOS XE Catalyst SD-WAN devices and associate this tracker group to a static route.                                                                                                                                                                                                                                                  |
| Tracker Elements | This field is displayed only if you chose <b>Tracker-group</b> as the tracker type. Add the existing interface tracker names (separated by a space). When you add this tracker to the template, the tracker group is associated with these individual trackers, and you can then associate the tracker group to a static route.                                                                                                                                                                                                                                             |
| Tracker Boolean  | From the drop-down list, choose <b>Global</b> . This field is displayed only if you chose <b>tracker-group</b> as the <b>Tracker Type</b> . By default, the <b>OR</b> option is selected. Choose <b>AND</b> or <b>OR</b> .<br><br><b>OR</b> ensures that the static route status is reported as active if either one of the associated trackers of the tracker group report that the route is active.<br><br>If you select <b>AND</b> , the static route status is reported as active if both the associated trackers of the tracker group report that the route is active. |

8. Click **Add**.

9. Click **Save**.



**Note** Complete all the mandatory actions before you save the template.

## Configure a Next Hop Static Route with Tracker

Use the **VPN** template to associate a tracker to a static route next hop.



**Note** You can apply only one tracker per static route next hop.

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**.



**Note** In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

3. Navigate to the **Cisco VPN Template** for the device.



**Note** For information about creating a VPN template, see [Create VPN Template](#).

4. Enter **Template Name** and **Description** as required.
5. In Basic Configuration, by default, VPN is set to 0. Set a VPN value within (1–511, 513–65530) range for service VPNs, for service-side data traffic on Cisco IOS XE Catalyst SD-WAN devices.



**Note** You can configure static route tracker only on service VPNs.

6. Click **IPv4 Route**.
7. Click **New IPv4 Route**.
8. In the **IPv4 Prefix** field, enter a value.
9. Click **Next Hop**.
10. Click **Add Next Hop with Tracker** and enter values for the fields listed in the table.

| Parameter Name | Description                                        |
|----------------|----------------------------------------------------|
| Address        | Specify the next-hop IPv4 address.                 |
| Distance       | Specify the administrative distance for the route. |

| Parameter Name            | Description                                                                                                                                                               |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tracker                   | Enter the name of the gateway tracker to determine whether the next hop is reachable before adding that route to the route table of the device.                           |
| Add Next Hop with Tracker | Enter the name of the gateway tracker with the next hop address to determine whether the next hop is reachable before adding that route to the route table of the device. |

11. Click **Add** to create the static route with the next-hop tracker.
12. Click **Save**.



**Note** You need to fill all the mandatory fields in the form to save the VPN template.

## Monitor Static Route Tracker Configuration

### View Static Route Tracker

To view information about a static tracker on a transport interface:

1. From the Cisco SD-WAN Manager menu, choose **Monitor** > **Devices**.  
Cisco vManage Release 20.6.x and earlier: From the Cisco SD-WAN Manager menu, choose **Monitor** > **Network**.
2. Choose a device from the list of devices.
3. Click **Real Time**.
4. From the **Device Options** drop-down list, choose **Endpoint Tracker Info**.

## Configure Static Routes Using CLI

The following sections provide information about how to configure static routes using the CLI.

### Configure a Static Route Tracker



**Note** You can configure static route tracking using the Cisco SD-WAN Manager CLI Add-on feature templates and CLI device templates. For more information on configuring using CLI templates, see [CLI Templates](#).

```
Device# config-transaction
Device(config)# endpoint-tracker <tracker-name>
Device(config-endpoint-tracker)# tracker-type <tracker-type>
```

```

Device(config-endpoint-tracker)# endpoint-ip <ip-address>
Device(config-endpoint-tracker)# threshold <value>
Device(config-endpoint-tracker)# multiplier <value>
Device(config-endpoint-tracker)# interval <value>
Device(config-endpoint-tracker)# exit
Device(config)# track <tracker-name> endpoint-tracker

```

### Configure a Static Route Tracker with TCP Port as the Endpoint

```

Device# config-transaction
Device(config)# endpoint-tracker <tracker-name>
Device(config-endpoint-tracker)# tracker-type <tracker-type>
Device(config-endpoint-tracker)# endpoint-ip <ip-address> tcp <port-number>
Device(config-endpoint-tracker)# threshold <value>
Device(config-endpoint-tracker)# multiplier <value>
Device(config-endpoint-tracker)# interval <value>
Device(config-endpoint-tracker)# exit
Device(config)# track <tracker-name> endpoint-tracker

```

### Configure a Static Route Tracker with UDP Port as the Endpoint

```

Device# config-transaction
Device(config)# endpoint-tracker <tracker-name>
Device(config-endpoint-tracker)# tracker-type <tracker-type>
Device(config-endpoint-tracker)# endpoint-ip <ip-address> udp <port-number>
Device(config-endpoint-tracker)# threshold <value>
Device(config-endpoint-tracker)# multiplier <value>
Device(config-endpoint-tracker)# interval <value>
Device(config-endpoint-tracker)# exit
Device(config)# track <tracker-name> endpoint-tracker

```

### Configure Tracker Groups



**Note** You can create tracker groups to probe static routes from Cisco IOS XE Catalyst SD-WAN Release 17.7.1a and Cisco vManage Release 20.7.1.

```

Device# config-transaction
Device(config)# endpoint-tracker <tracker-name1>
Device(config-endpoint-tracker)# tracker-type <tracker-type>
Device(config-endpoint-tracker)# endpoint-ip <ip-address> tcp <port-number>
Device(config-endpoint-tracker)# threshold <value>
Device(config-endpoint-tracker)# multiplier <value>
Device(config-endpoint-tracker)# interval <value>
Device(config-endpoint-tracker)# exit
Device(config)# track <tracker-name1> endpoint-tracker

```

```

Device# config-transaction
Device(config)# endpoint-tracker <tracker-name2>
Device(config-endpoint-tracker)# tracker-type <tracker-type>
Device(config-endpoint-tracker)# endpoint-ip <ip-address> udp <port-number>
Device(config-endpoint-tracker)# threshold <value>
Device(config-endpoint-tracker)# multiplier <value>
Device(config-endpoint-tracker)# interval <value>
Device(config-endpoint-tracker)# exit
Device(config)# track <tracker-name2> endpoint-tracker

```



```

Device(config)# endpoint-tracker <static-tracker-group>
Device(config-endpoint-tracker)# tracker-type tracker-group
Device(config-endpoint-tracker)# tracker-elements <tracker-name1> <tracker-name2>
Device(config-endpoint-tracker)# boolean {and | or}
Device(config-endpoint-tracker)# exit
Device(config)# track <static-tracker-group> endpoint-tracker

Device(config)# ip route vrf <vrf-name> <prefix> <mask> <next-hop-ipaddress>
<administrative-distance> track name <static-tracker-group>

```

**Note**

- Use the **ip route** command to bind a tracker or tracker group with a static route and to configure a backup route for administrative distance that is higher than the default value of 1.
- You can apply only one tracker to an endpoint.
- A tracker group can have a mix of endpoint trackers. For example, you can create a tracker group with an IP address tracker and UDP tracker.

## Configuration Examples for Static Route Tracking Using the CLI

### Configure Tracker

This example shows how to configure a single static route tracker:

```

config-transaction
!
 endpoint-tracker tracker1
!
 tracker-type static-route
 endpoint-ip 10.1.1.1
 threshold 100
 multiplier 5
 interval 20
 exit
!
track tracker1 endpoint-tracker
!
ip route vrf 1 192.168.0.0 255.255.0.0 10.1.19.16 100 track name tracker1

```

This example shows how to configure a tracker with TCP port as endpoint:

```

config-transaction
!
 endpoint-tracker tcp-10001
!
 tracker-type static-route
 endpoint-ip 10.0.0.1 tcp 10001
 threshold 100
 interval 10
 multiplier 1
 exit
!
track tcp-10001 endpoint-tracker

```

```
!
ip route vrf 1 192.168.0.0 255.255.0.0 10.1.19.16 100 track name tcp-10001
```

This example shows how to configure a tracker with UDP port as endpoint:

```
config-transaction
!
 endpoint-tracker udp-10001
!
 tracker-type static-route
 endpoint-ip 10.0.0.1 udp 10001
 threshold 100
 interval 10
 multiplier 1
 exit
!
track udp-10001 endpoint-tracker
!
ip route vrf 1 192.168.0.0 255.255.0.0 10.1.19.16 100 track name udp-10001
```

### Configure Tracker Groups

This example shows how to configure a tracker group with two trackers (two endpoints). You can create tracker groups to probes static routes from Cisco IOS XE Catalyst SD-WAN Release 17.7.1a.

```
config-transaction
!
 endpoint-tracker tcp-10001
!
 tracker-type static-route
 endpoint-ip 10.1.1.1 tcp 10001
 threshold 100
 multiplier 5
 interval 20
 track tcp-10001 endpoint-tracker
!
 endpoint-tracker udp-10002
!
 tracker-type static-route
 endpoint-ip 10.2.2.2 udp 10002
 threshold 100
 multiplier 5
 interval 20
 track udp-10002 endpoint-tracker
!
 endpoint-tracker static-tracker-group
!
 tracker-type tracker-group
 tracker-elements tcp-10001 udp-10002
 boolean and
 track static-tracker-group endpoint-tracker
!
ip route vrf 1 192.168.0.0 255.255.0.0 10.1.19.16 100 track name static-tracker-group
```

**Note**

- You must configure an administrative distance when you are configuring through CLI templates.
- Use the **ip route** command to bind the tracker or tracker group with a static route and to configure a backup route for administrative distance when it is higher than the default value of 1.
- You can apply only one tracker to an endpoint.

## Verify Static Route Tracking Configuration Using CLI

### Command Verification

Use the following command to verify if the configuration is committed. The following sample configuration shows tracker definition for a static route tracker and its application to an IPv4 static route:

```
Device# show running-config | sec endpoint-tracker
endpoint-tracker tracker1
endpoint-ip 10.1.1.1
interval 60
multiplier 5
tracker-type static-route
endpoint-tracker tracker2
endpoint-ip 10.1.1.12
interval 40
multiplier 2
tracker-type static-route
track tracker2 endpoint-tracker
track tracker1 endpoint-tracker
```

Use the following command to verify the IPv4 route:

```
Device# show running-config | inc ip route
ip route vrf 1 10.1.1.11 255.255.0.0 10.20.2.17 track name tracker2
ip route vrf 1 10.1.1.12 255.255.0.0 10.20.24.17 track name tracker1
```

The following is a sample output from the **show endpoint-tracker static-route** command displaying individual static route tracker status:

```
Device# show endpoint-tracker static-route
Tracker Name Status RTT (in msec) Probe ID
tcp-10001 UP 3 1
udp-10002 UP 1 6
```

The following is a sample output from the **show endpoint-tracker tracker-group** command displaying tracker group status:

```
Device# show endpoint-tracker group
Tracker Name Element trackers name Status RTT in msec Probe ID
group-tcp-10001-udp-10002 tcp-10001, udp-10002 UP (UP AND UP) 5, 1 9, 10
```

The following is a sample output from the **show endpoint-tracker records** command displaying tracker/tracker group configuration:

```
Device# show endpoint-tracker records
Record Name Endpoint EndPoint Type Threshold(ms) Multiplier
Interval(s) Tracker-Type
```

```

group-tcp-10001-udp-10002 tcp-10001 AND udp-10002 N/A N/A N/A
N/A static-tracker-group
tcp-10001 10.1.1.1 TCP 100 1
20 static-route
udp-10002 10.2.2.2 UDP 100 1
20 static-route

```

The following is a sample output from the **show ip static route vrf** command:

```

Device# show ip static route vrf 1
Codes: M - Manual static, A - AAA download, N - IP NAT, D - DHCP,
 G - GPRS, V - Crypto VPN, C - CASA, P - Channel interface processor,
 B - BootP, S - Service selection gateway
 DN - Default Network, T - Tracking object
 L - TL1, E - OER, I - iEdge
 D1 - Dot1x Vlan Network, K - MWAM Route
 PP - PPP default route, MR - MRIPv6, SS - SSLVPN
 H - IPe Host, ID - IPe Domain Broadcast
 U - User GPRS, TE - MPLS Traffic-eng, LI - LIIN
 IR - ICMP Redirect, Vx - VXLAN static route
 LT - Cellular LTE, Ev - L2EVPN static route
Codes in []: A - active, N - non-active, B - BFD-tracked, D - Not Tracked, P - permanent,
-T Default Track
Codes in (): UP - up, DN - Down, AD-DN - Admin-Down, DL - Deleted
Static local RIB for 1
T 192.168.0.0 [1/0] via 10.1.19.16 [A]

```



## CHAPTER 24

# NAT DIA Tracker for Cisco IOS XE Catalyst SD-WAN Devices

---

For information on the NAT DIA tracker for Cisco IOS XE Catalyst SD-WAN devices, see the [NAT DIA Tracker](#) section of the *Cisco Catalyst SD-WAN NAT Configuration Guide, Cisco IOS XE Release 17.x*.





## CHAPTER 25

# Service-Side NAT on Cisco IOS XE Catalyst SD-WAN Devices

---

For information on service-side NAT on Cisco IOS XE Catalyst SD-WAN devices, see the [Service-Side NAT](#) section in the *Cisco Catalyst SD-WAN NAT Configuration Guide, Cisco IOS XE Catalyst SD-WAN Release 17.x*.







## CHAPTER 26

# DHCP Vendor Option Support

*Table 337: Feature History*

| Feature Name               | Release Information                                                            | Description                                                                                                                                                                                                            |
|----------------------------|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DHCP Vendor Option Support | Cisco IOS XE Catalyst SD-WAN Release 17.10.1a<br>Cisco vManage Release 20.10.1 | This feature allows DHCP client options, 124 and 125 to configure vendor-specific information in client-server exchanges.<br><br>Configure this feature using the CLI Add-on feature template in Cisco SD-WAN Manager. |

- [Information about DHCP Vendor Option Support, on page 747](#)
- [DHCPv6 Client Options, on page 748](#)
- [Configure DHCP Vendor Option Using a CLI Template, on page 749](#)
- [Configure DHCPv6 Client Option Using a CLI Template, on page 750](#)

## Information about DHCP Vendor Option Support

The configurable dynamic host configuration protocol client functionality allows a DHCP client to use a user-specified client identifier, class identifier, or suggested lease time when requesting an address from a DHCP server.

Configuration parameters and other control information are carried in tagged data items that are stored in the options field of the DHCP message. The DHCP client provides flexibility by allowing the following options to be configured for a DHCP client:

- Option 124—This option is used by DHCP clients and servers to exchange vendor-class information.
- Option 125—This option is used by DHCP clients and servers to exchange vendor-specific information.

In the DHCP address assignment workflow, Option 124 (Vendor-Identifying Vendor Class) and Option 125 (Vendor-Identifying Information) are used to provide differential services. These options are used by Zero-Touch Provisioning (ZTP), Cisco Plug-and-Play (PnP), and Identity Services Engine (ISE) across solutions to enable several use cases. For example, the content of Option 124 is used for device classification, enable solution specific feature and so on.

By default, Cisco IOS XE DHCP client sends the following data:

| Attribute                              | IPv4 DHCP Option | Default Value |
|----------------------------------------|------------------|---------------|
| Vendor-Identifying Vendor Class Option | 124              | PID           |



**Note** The **ip dhcp client vendor-class** <mac-address | ascii | disable | hex> command overrides PID with MAC Address or user defined string or disable Option 124.

The DHCP Vendor Option Support feature introduces new CLI parameters to make Option 124 and Option 125 flexible. You can modify and customize enabling vendor specific options to carry different values for different customer features. The combination of Option 124 and Option 125 enables various features.

The **ip dhcp client vendor-class** command provides flexibility to pack either Device PID or MAC Address of the DHCP client or any user configurable string in option-124. The default behavior for this command is to continue to send Device PID when you choose option 124. The default behavior can be overridden to carry MAC Address in Day 1 configuration mode by explicitly requesting option-125 from the server using the **ip dhcp client vendor-class** command.

## DHCPv6 Client Options

Configuration parameters and other control information are carried in tagged data items that are stored in the options field of the DHCP message. The DHCP client provides flexibility by allowing the following options to be configured for a DHCP client:

- Option 16—This option is used by DHCP clients and servers to exchange vendor-class information.
- Option 17—This option is used by DHCP clients and servers to exchange vendor-specific information.

In DHCPv6, option-16 and option-17 is used by DHCP clients and servers to exchange vendor-specific information.

By default, Cisco IOS XE DHCP client sends the following data:

| Attribute           | IPv6 DHCP Option | default value |
|---------------------|------------------|---------------|
| Vendor Class Option | 16               | PID           |



**Note** The **ipv6 dhcp client vendor-class** <mac-address | hex | ascii | disable> command can be used to override default value of PID with MAC Address or User defined string or disable the option.

The **ipv6 dhcp client vendor-class** command provides flexibility to pack either Device PID or MAC Address of the DHCP Client or any user configurable string in option-16. The default behavior for this command is to continue to send Device PID when you choose option 16 but it can be overridden to carry MAC Address in Day 1 configuration mode using the **ipv6 dhcp client vendor-class** command.

# Configure DHCP Vendor Option Using a CLI Template

For more information about using CLI templates, see [CLI Add-On Feature Templates](#).



---

**Note** By default, CLI templates execute commands in global config mode.

---

The section provides a sample CLI configurations to configure DHCP vendor option.

1. Configure an interface type and enter the interface configuration mode.

```
interface type number
```

2. Acquire an IP address on an interface from DHCP.

```
ip address dhcp
```

3. Configure the DHCP vendor-class option.

```
ip dhcp client vendor-class [mac-address | ascii | hex | disable]
```



---

**Note** You must first configure the **no ip dhcp-client** command before configuring the IP address.

---

The following example shows the configuration to override the device PID with MAC address:

```
interface GigabitEthernet 0/0/0
ip address dhcp
ip dhcp client vendor-class mac-address
!
```

The DHCP vendor-class option, overrides the Device PID with MAC Address.

The following example shows the configuration to override the device PID with user defined string in hex or in ascii format:

```
interface GigabitEthernet 0/0/0
ip address dhcp
ip dhcp client vendor-class hex aabbcc
!
```

```
interface GigabitEthernet 0/0/0
ip address dhcp
ip dhcp client vendor-class ascii cisco
!
```

The following example shows the configuration to disable option-124 in DHCP messages:

```
interface GigabitEthernet 0/0/0
ip address dhcp
ip dhcp client vendor-class disable
!
```

# Configure DHCPv6 Client Option Using a CLI Template

For more information about using CLI templates, see [CLI Add-On Feature Templates](#).



**Note** By default, CLI templates execute commands in global config mode.

The section provides a sample CLI configurations to configure DHCP vendor option.

1. Configure an interface type and enter the interface configuration mode.

```
interface type number
```

2. Acquire an IPv6 address on an interface from DHCP.

```
ipv6 address dhcp
```

3. Configure the DHCP vendor-class option.

```
ipv6 dhcp client vendor-class {mac-address | ascii | hex | disable}
```

By default DHCPv6 client carries device PID of the device in option-16. This default behaviour can be overridden by configuring the **ipv6 dhcp client vendor-class** command.

The following example shows the configuration to override the device PID with MAC address:

```
interface GigabitEthernet 0/0/0
 ipv6 address dhcp
 ipv6 dhcp client vendor-class mac-address
 !
```

The DHCPv6 vendor-class option, overrides the Device PID with MAC Address.

The following example shows the configuration to override the device PID with user defined string in hex or in ascii format:

```
interface GigabitEthernet 0/0/0
 ipv6 address dhcp
 ipv6 dhcp client vendor-class hex aabbcc
 !
```

```
interface GigabitEthernet 0/0/0
 ipv6 address dhcp
 ipv6 dhcp client vendor-class ascii cisco
 !
```

The following example shows the configuration to disable option-16 in DHCP messages:

```
interface GigabitEthernet 0/0/0
 ipv6 address dhcp
 ipv6 dhcp client vendor-class disable
 !
```



## CHAPTER 27

# IP DHCP Smart-Relay

Table 338: Feature History

| Feature Name        | Release Information                                                                            | Feature Description                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IP DHCP Smart-Relay | Cisco IOS XE Catalyst SD-WAN Release 17.13.1a<br>Cisco Catalyst SD-WAN Manager Release 20.13.1 | With this feature, you can set the gateway address to the secondary IP address using the DHCP relay agent, when there is no IP address and DHCP service information from the DHCP server. A DHCP relay agent is any host or IP router that forwards DHCP packets between clients and servers.<br><br>This functionality is useful when the DHCP server cannot be configured to use secondary pools. |

- [Information About the IP DHCP Smart-Relay, on page 751](#)
- [Prerequisites for IP DHCP Smart-Relay, on page 752](#)
- [Configure IP DHCP Smart-Relay Agent Using a CLI Template, on page 752](#)

## Information About the IP DHCP Smart-Relay

A Dynamic Host Configuration Protocol (DHCP) relay agent is any host that forwards DHCP packets between clients and servers. Relay agents are used to forward requests and replies between clients and servers when they are not on the same physical subnet. Relay-agent forwarding is distinct from the normal forwarding of an IP router, where IP datagrams are switched between networks transparently. Relay agents receive DHCP messages and then generate a new DHCP message to send out on another interface. The relay agent sets the gateway IP address and, if configured, adds the relay agent information option (option 82) in the packet and forwards it to the DHCP server. The reply from the server is forwarded back to the client after removing option 82.

The Cisco IOS XE DHCP relay agent supports the use of unnumbered interfaces. An unnumbered interface can borrow the IP address of another interface already configured on the router, which conserves network and address space. For DHCP clients connected through the unnumbered interfaces, the DHCP relay agent automatically adds a static host route once the DHCP client obtains an address, specifying the unnumbered interface as the outbound interface. The route is automatically removed once the lease time expires or when the client releases the address.

## Benefits of IP DHCP Smart-Relay

- Using automatic IP address assignment at each remote site substantially reduces the internet access cost. Static IP addresses are considerably more expensive to purchase than are automatically allocated IP addresses.
- Enables easier configuration and minimizes operational overhead and costs associated with device configuration tasks and eases deployment by nontechnical users.
- DHCP server maintains configurations for several subnets. An administrator only needs to update a single, central server when configuration parameters change.

## Prerequisites for IP DHCP Smart-Relay

- To configure the IP DHCP smart-relay feature, configure the IP helper address on desired interfaces using **ip helper-address** command. You can use the **service dhcp** command to enable the DHCP service and **no service dhcp** command, if the service is disabled depending on the requirement.
- The Cisco DHCP relay agent is enabled on an interface only when the **ip helper-address** command is configured. This command enables the DHCP broadcast to be forwarded to the configured DHCP server.

## Configure IP DHCP Smart-Relay Agent Using a CLI Template

To forward UDP broadcasts to the DHCP server, configure helper addresses on the interface. If you have configured the secondary addresses on that interface and you want the router to step through each IP network when forwarding DHCP requests, use the **ip dhcp smart-relay** command. If smart relay agent forwarding is not configured, all requests are forwarded using the primary IP address on the interface. If smart relay agent forwarding is not configured, all requests are forwarded using only the primary IP address on the interface.

If the **ip dhcp smart-relay** command is configured, the relay agent counts the number of times that the client retries sending a request to the DHCP server when there is no DHCPOFFER message from the DHCP server. After three retries, the relay agent sets the gateway address to the secondary address. If the DHCP server still does not respond after three more retries, then the next secondary address is used as the gateway address.

For more information about using CLI templates, see [CLI Add-On Feature Templates](#) and [CLI Templates](#).

1. Enter SD-WAN configuration mode.

```
sdwan
```

2. Enable DHCP server.

```
service dhcp
```

3. In the SD-WAN configuration mode, configure an interface type such as, Gigabit Ethernet.

```
interface GigabitEthernet0/0
```

4. Enable the DHCP broadcast to be forwarded to the configured DHCP server.

```
ip helper-address
```

5. Configure the DHCP relay agent to switch the gateway address to a secondary address when there is no DHCPOFFER message from a DHCP server.

```
ip dhcp smart-relay
```

The following is a DHCP smart-relay CLI configuration. In the following example, the device forwards the DHCP broadcast received on GigabitEthernet interface 0/0 to the DHCP server (10.0.0.1), by inserting 192.168.255.254 in the gateway address field of the DHCP packet.

```
service dhcp
ip address 172.16.0.1 255.255.0.0
secondary ip address 192.168.255.254 255.255.0.0

interface GigabitEthernet0/0
ip helper-address 10.0.0.1
ip dhcp smart-relay
end
```







## CHAPTER 28

# IPv6 Functionality

This chapter describes the options for enabling IPv6 functionality for Cisco Catalyst SD-WAN templates and policies. Use the information in this chapter if your deployment uses IPv6.

### Configure IPv6 Functionality for an Interface or Subinterface Template

To configure IPv6 functionality for an interface or subinterface template, perform the following steps.

Cisco Catalyst SD-WAN supports dual stack: you can configure IPv4 and IPv6 in the same deployment. You can configure up to three global IPv6 addresses per interface.

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**, and click **Add Template** to select an appropriate device model.



**Note** In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled as **Feature**.

3. Select **Cisco VPN Interface Ethernet** from the list of templates.
4. From **Basic Configuration**, click **IPv6** and configure the parameters that the following table describes:

| Parameter Name | Description                                                                 |
|----------------|-----------------------------------------------------------------------------|
| Static         | This radio button is selected by default because IPv6 addresses are static. |
| IPv6 Address   | Enter the IPv6 address of the interface or subinterface.                    |

*CLI equivalent:*

```
interface GigabitEthernet1
 no shutdown
 ipv6 address 2001:DB8:1::1/64
 ipv6 enable
```

### Configure IPv6 Functionality for an OMP Template

To configure IPv6 functionality for an Overlay Management Protocol (OMP) template, follow these steps:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.

- Click **Feature Templates**, and click **Add Template** to select an appropriate device model.



**Note** In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled as **Feature**.

- Select **Cisco OMP** from the list of templates.
- Click **Advertise** and choose **IPv6** to configure the parameters that the following table describes:

| Parameter Name | Description                                                                                                             |
|----------------|-------------------------------------------------------------------------------------------------------------------------|
| Connected      | Click <b>Off</b> to disable advertising connected routes to OMP.<br>By default, connected routes are advertised to OMP. |
| Static         | Click <b>Off</b> to disable advertising static routes to OMP.<br>By default static routes are advertised to OMP.        |
| BGP            | Click <b>On</b> to advertise BGP routes to OMP. By default, BGP routes are not advertised to OMP.                       |

*CLI equivalent:*

First enable Service VRF for IPv6:

```
config-transaction
vrf definition 1
 rd 1:1
 address-family ipv6
```

Next enable OMP.

OMP supports global IPv6 configuration. In addition, per VRF level configuration is allowed. Per VRF level configuration overrides global configuration.

```
config-transaction
sdwan
 omp
 !
 address-family ipv6
 advertise bgp
 advertise connected

 address-family ipv6 vrf 1
 advertise static
```

Global configuration is the default configuration, so IPv6 is enabled by default for OMP. To disable IPv6 OMP route redistribution for a particular VRF, configure the redistribution protocol to no as follows:

```
config-transaction
sdwan
 omp
 !
 address-family ipv6
 advertise bgp
 advertise connected

 address-family ipv6 vrf 1
```

```
no advertise connected
no advertise static
no advertise bgp
```

### Configure IPv6 Functionality for a BGP Template

To configure IPv6 functionality for a Border Gateway Protocol (BGP) template, follow these steps:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**, and click **Add Template** to select an appropriate device model.



**Note** In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled as **Feature**.

3. Select **Cisco BGP** from the list of templates.
4. Click **Unicast Address Family** and choose **IPv6** to configure the parameters that the following table describes:

| Tab               | Parameter Name | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   | Maximum Paths  | Specify the maximum number of parallel IBGP paths that can be installed into a route table to enable IBGP multipath load sharing. <i>Range:</i> 0 to 32                                                                                                                                                                                                                                                                                                                                                                     |
|                   | Address Family | Enter the BGP IPv6 unicast address family.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| RE-DISTRIBUTE     |                | Click the <b>Redistribute</b> tab, and then click <b>Add New Redistribute</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                   | Protocol       | Select the protocols from which to redistribute routes into BGP, for all BGP sessions. Options are Connected, NAT, OMP, OSPF, and Static. At a minimum, select the following: <ul style="list-style-type: none"> <li>• For service-side BGP routing, select OMP. By default, OMP routes are not redistributed into BGP.</li> <li>• For transport-side BGP routing, select Connected, and then under Route Policy, specify a route policy that has BGP advertise the loopback interface address to its neighbors.</li> </ul> |
|                   | Route Policy   | Enter the name of the route policy to apply to redistributed routes.                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                   |                | Click <b>Add</b> to save the redistribution information.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| NETWORK           |                | Click the <b>Network</b> tab, and then click <b>Add New Network</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                   | Network Prefix | Enter a network prefix, in the format of <i>prefix/length</i> , to be advertised by BGP.                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                   |                | Click <b>Add</b> to save the network prefix.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| AGGREGATE ADDRESS |                | Click the <b>Aggregate Address</b> tab, and then click <b>Add New Aggregate Address</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                   |

| Tab | Parameter Name   | Description                                                                                       |
|-----|------------------|---------------------------------------------------------------------------------------------------|
|     | Aggregate Prefix | Enter the prefix of the addresses to aggregate for all BGP sessions, in the format prefix/length. |
|     | AS Set Path      | Click <b>On</b> to generate set path information for the aggregated prefixes.                     |
|     | Summary Only     | Click <b>On</b> to filter out more specific routes from BGP updates.                              |
|     |                  | Click <b>Add</b> to save the aggregate address.                                                   |

1. In the Neighbor area, click **IPv6**, create a new neighbor or edit an existing one, and then configure the parameters that the following table describes.

Parameters marked with an asterisk are required.

| Parameter Name | Description                                                                                                                                          |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| IPv6 Address*  | Specify the IPv6 address of the BGP neighbor.                                                                                                        |
| Description    | Enter a description of the BGP neighbor.                                                                                                             |
| Remote AS*     | Enter the AS number of the remote BGP peer.                                                                                                          |
| Address Family | Select <b>Global</b> from the drop-down list, click <b>On</b> and select the address family. Enter the address family information.                   |
| Shutdown       | To shut down a BGP neighbor when you push the template, select <b>Global</b> from the drop-down list and then click <b>Yes</b> . <i>Default: Off</i> |

*CLI equivalent:*

```
config-transaction
router bgp 1
 bgp log-neighbor-changes
 address-family ipv6 unicast vrf 1
 neighbor 2001:DB8:19::1 remote-as 2
 neighbor 2001:DB8:19::1 activate
 neighbor 2001:DB8:19::1 advertisement-interval 1
 neighbor 2001:DB8:19::1 password cisco
 redistribute omp
 redistribute static
 exit-address-family
```

### Configure IPv6 Functionality for a VRRP Template

To configure IPv6 functionality for a Virtual Router Redundancy Protocol (VRRP) template, follow these steps:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**, and click **Add Template** to select an appropriate device model.



**Note** In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled as **Feature**.

3. Select **Cisco VPN Interface Ethernet** from the list of templates.
4. Click **VRRP** and choose **IPv6**.
5. Click **New VRRP**.
6. Configure the parameters that the following table describes:

| Parameter Name          | Description                                                                                                                                                                                                                                                                                                                            |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Group ID                | Enter a virtual router ID, which represents a group of routers.<br>Range:<br>1 through 255                                                                                                                                                                                                                                             |
| Priority                | Enter the priority level of the router within a VRRP group.<br><ul style="list-style-type: none"> <li>• <i>Range:</i> 1 through 254</li> <li>• <i>Default:</i> 100</li> </ul>                                                                                                                                                          |
| Timer                   | Not used.                                                                                                                                                                                                                                                                                                                              |
| Track OMP               | Select <b>On</b> to track the Overlay Management Protocol (OMP) session running on the WAN connection when determining the primary VRRP virtual router. <i>Default:</i> Off                                                                                                                                                            |
| Track Prefix List       | Enter a value to track a list of IPv6 remote prefixes. This value is an alphanumeric string that is configured under Policy.                                                                                                                                                                                                           |
| Link Local IPv6 Address | Enter a virtual link local IPv6 address, which represents the link local address of the group. The address should be in standard link local address format. For example, FE80::AB8.                                                                                                                                                    |
| Global IPv6 Address     | Enter a virtual global unicast IPv6 address, which represents the global address of the group. The address should be an IPv6 global prefix address that has the same mask as the interface forwarding address on which the VRRP group is configured. For example, 2001::2/124.<br><br>You can configure up to 3 global IPv6 addresses. |

*CLI equivalent:*

```

config-transaction
interface GigabitEthernet1

 vrrp 10 address-family ipv6
 priority 20
 track omp shutdown
 address FE80::10:100:1 primary
 address 2001:10:100::1/64

```

```

Prefix-list tracking
track 1 ipv6 route 1:1::1/128
 reachability
 ipv6 vrf 1

track 2 ipv6 route 2:2::2/128
 reachability
 ipv6 vrf 2

track 20 list boolean or
 object 1
 object 2

vrrp 10 address-family ipv6
 track 20 shutdown

```

### Configure IPv6 Functionality for an SNMP Template

To configure IPv6 functionality for an SNMP template, follow these steps:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**, and click **Add Template** to select an appropriate device model.




---

**Note** In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled as **Feature**.

---

3. Click **Cisco SNMP** from the list of templates.
4. Choose **SNMP Version > TRAP TARGET SERVER** and create or edit an SNMP trap target.
5. Configure the parameters that the following table describes:

| Parameter Name   | Description                                                                                         |
|------------------|-----------------------------------------------------------------------------------------------------|
| VPN ID           | Enter the number of the VPN to use to reach the trap server. Range: 0 through 65530.                |
| IP Address       | Enter the IP address of the SNMP server.                                                            |
| UDP Port         | Enter the UDP port number for connecting to the SNMP server. Range: 1 though 65535.                 |
| Trap Group Name  | Select the name of a trap group that was configured under the Group tab.                            |
| User Name        | Select the name of a community that was configured under the Community tab.                         |
| Source Interface | Enter the interface to use to send traps to the SNMP server that is receiving the trap information. |




---

**Note** Make sure that you have already configured the SNMP community and trap target group.

---

*CLI equivalent:*

The following example permits any SNMP to access all objects with read-only permission using the community string named public. The device also will send Border Gateway Protocol(BGP) traps IPv6 host 3ffe:b00:c18:1::3/127 using SNMP v1. The community string named public will be sent with the traps.

```
Device# config-transaction
Device(config)# snmp-server community public
Device(config)# snmp-server enable traps bgp
Device(config)# snmp-server host 3ffe:b00:c18:1::3/127 public
```

In the following example, the SNMP context A is associated with the views in SNMPv2c group GROUP1 and the IPv6 named access list public2:

```
Device# config-transaction
Device(config)# snmp-server context A
Device(config)# snmp mib community-map commA context A target-list comm AVpn
Device(config)# snmp mib target list commAVpn vrf CustomerA
Device(config)# snmp-server view viewA ciscoPingMIB included
Device(config)# snmp-server view viewA ipForward included
Device(config)# snmp-server group GROUP1 v2c contextA read viewA write viewA notify access
ipv6 public2
```

The following example configures the IPv6 host as the notification server:

```
Device> enable
Device# config-transaction
Device(config)# snmp-server community mgr view restricted rw ipv6 mgr2
Device(config)# snmp-server engineID remote 3ffe:b00:c18:1::3/127 remotev6
Device(config)# snmp-server group publicv2c access ipv6 public2
Device(config)# snmp-server hosthost1.com2c vrf trap-vrf mgr
Device(config)# snmp-server user user1 bldg1 remote3ffe:b00:c18:1::3/127 v2c access ipv6
public2
Device(config)# snmp-server enable traps bgp
Device(config)# exit
```

### Configure IPv6 Functionality for a DHCP Relay Agent Template

To configure IPv6 functionality for a DHCP Relay Agent template, follow these steps:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**, and click **Add Template** to select an appropriate device model.




---

**Note** In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled as **Feature**.

---

3. Select **Cisco VPN Interface Ethernet** from the list of templates.
4. From **Basic Configuration**, click **IPv6**.
5. Click **Add** next to **DHCP Helper**.
6. Configure the parameters that the following table describes.

Table 339:

| Parameter Name    | Description                                             |
|-------------------|---------------------------------------------------------|
| DHCPv6 Helper #   | IP address of the DHCP helper                           |
| DHCPv6 Helper VPN | VPN ID of the VPN source interface for the DHCP helper. |

*CLI equivalent:*

```
device-configuration
interface GigabitEthernet8
 vrf forwarding 2
 no ip address
 ipv6 address 2001:A14:99::F/64
 ipv6 dhcp relay destination vrf 1 2001:A14:19::12 GigabitEthernet2
```

### Configure IPv6 Functionality for an ACL Template or a QoS Template

To configure IPv6 functionality for an ACL and QoS template, follow these steps:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**, and click **Add Template** to select an appropriate device model.



**Note** In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled as **Feature**.

3. Select **Cisco VPN Interface Ethernet** from the list of templates.
4. From **ACL/QoS**, configure the parameters that the following table describes:

| Parameter Name           | Description                                             |
|--------------------------|---------------------------------------------------------|
| Ingress ACL – IPv6       | Click <b>On</b> to enable the IPv6 ingress access list. |
| IPv6 Ingress Access List | Enter the name of the IPv6 ingress access list.         |
| Egress ACL – IPv6        | Click <b>On</b> to enable the IPv6 egress access list.  |
| IPv6 Egress Access List  | Enter the name of the IPv6 egress access list.          |

*CLI Equivalent for Configuring IPv6 Functionality for an ACL Template:*

```
Device(config)# policy
Device(config-policy)# ipv6
Device(config-ipv6)# access-list ipv6_acl
Device(config-access-list-ipv6_acl)# sequence 11
Device(config-sequence-11)# match
Device(config-match)# source-ip 2001:380:1::64/128
Device(config-match)# destination-ip 2001:3c0:1::64/128
Device(config-match)# source-port 4000
Device(config-match)# destination-port 3000
Device(config-match)# traffic-class 6
Device(config-match)# next-header 6
```



```

Device(config-match)# packet-length 1000
Device(config-match)# action accept
Device(config-action)#

Device(config)# sdwan interface GigabitEthernet6 ipv6 access-list ipv6_acl in
Device(config-interface-GigabitEthernet6)#
Device(config-interface-GigabitEthernet6)#

Device(config)# policy lists data-ipv6-prefix-list source_ipv6_list
Device(config-data-ipv6-prefix-list-source_ipv6_list)# ipv6-prefix 2001:380:1::/64

Device(config)# policy
Device(config-policy)# ipv6
Device(config-ipv6)# access-list ipv_ipv6_prefix
Device(config-access-list-ipv_ipv6_prefix)# sequence 11
Device(config-sequence-11)# match
Device(config-match)# source-data-prefix-list data-ipv6-prefix-list
Device(config-match)# destination-data-prefix-list source_ipv6_list
Device(config-match)# destination-ip 2001:3c0:1::64/128
Device(config-match)# source-port 4000
Device(config-match)# destination-port 3000
Device(config-match)# traffic-class 6
Device(config-match)# next-header 6
Device(config-match)# packet-length 1000
Device(config-match)# !
Device(config-match)# action accept

```

### *CLI Equivalent for Configuring IPv6 Functionality for a QoS Template:*

```

Device(config)# class-map match-any class0
Device(config-cmap)# match qos-group 0
Device(config-cmap)# class-map match-any class1
Device(config-cmap)# match qos-group 1
Device(config-cmap)# !
Device(config-cmap)# policy-map qos_map_for_data_policy
Device(config-pmap)# class class0
Device(config-pmap-c)# bandwidth percent 10
Device(config-pmap-c)# random-detect
Device(config-pmap-c)# class class1
Device(config-pmap-c)# bandwidth percent 10
Device(config-pmap-c)# random-detect
Device(config-pmap-c)#
Device(config-pmap-c)# policy
Device(config-policy)# no app-visibility
Device(config-policy)# class-map
Device(config-class-map)# class class0 queue 0
Device(config-class-map)# class class1 queue 1
Device(config-class-map)# !
Device(config-class-map)# ipv6
Device(config-ipv6)# access-list fwd_class_data_policy
Device(config-access-list-fwd_class_data_policy)# sequence 5
Device(config-sequence-5)# match
Device(config-match)# traffic-class 0
Device(config-match)# !
Device(config-match)# action accept
Device(config-action)# count fwd_class_data_policycnt_5
Device(config-action)# class class0
Device(config-action)# !
Device(config-action)# !
Device(config-action)# sequence 6
Device(config-sequence-6)# match
Device(config-match)# traffic-class 1
Device(config-match)# !
Device(config-match)# action accept
Device(config-action)# count fwd_class_data_policycnt_6

```

```

Device(config-action)# class class1
Device(config-action)# !
Device(config-action)# !
Device(config-action)# !
Device(config-action)# default-action drop

class-map match-any class0
match qos-group 0
class-map match-any class1
match qos-group 1
!
policy-map qos_map_for_data_policy
class class0
 bandwidth percent 10
 random-detect
class class1
 bandwidth percent 10
 random-detect

policy
no app-visibility
class-map
 class class0 queue 0
 class class1 queue 1
!
ipv6
 access-list fwd_class_data_policy
 sequence 5
 match
 traffic-class 0
 !
 action accept
 count fwd_class_data_policycnt_5
 class class0
 !
 sequence 6
 match
 traffic-class 1
 !
 action accept
 count fwd_class_data_policycnt_6
 class class1
!
default-action drop

```

### Configure IPv6 Functionality for a Logging Template

To configure IPv6 functionality for a Logging template, follow these steps:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**, and click **Add Template** and then select an appropriate device model.




---

**Note** In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled as **Feature**.

---

3. Select **Cisco Logging** from the list of templates.
4. From **Server**, click **IPv6**.
5. Configure the parameters that the following table describes.

| Parameter Name             | Description                                                              |
|----------------------------|--------------------------------------------------------------------------|
| IPv6 Hostname/IPv6 Address | Host name or IP address of the server to direct the logging information. |
| VPN ID                     | VPN ID of the VPN source interface.                                      |
| Source Interface           | Name of the source interface.                                            |
| Priority                   | Choose the maximum severity of messages that are logged.                 |

*CLI equivalent:*

```
config-transaction
Device(config)# logging host ipv6
AAAA:BBBB:CCCC:DDDD::FFFF
```



**Note** Creating and deleting the logging host configurations in same transaction causes unexpected behaviour. For example, deleting **logging host ipv6-address** and creating **logging host ipv6-address vrf vrf-name** configuration in same transaction causes both configurations to disappear from the device. We recommend you to send the two requests in separate transactions.

### Configure IPv6 Functionality for a New Prefix List

To configure an IPv6 address for a new prefix list, follow these steps:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Policies**.
2. From the **Custom Options** drop-down list, select **Lists**. You can make this selection for a Centralized Policy or a Localized Policy
3. Select **Prefix** from the list on the left and then select **New Prefix List**.
4. Click **IPv6** and enter the IPv6 address in **Add Prefix**.

*CLI equivalent:*

```
config-transaction
Device(config)# policy
Device(config-policy)# ipv6
Device(config-ipv6)# access-list ipv6_acl
Device(config-access-list-ipv6_acl)# sequence 11
Device(config-sequence-11)# match
Device(config-match)# source-ip 2001:380:1::64/128
Device(config-match)# destination-ip 2001:3c0:1::64/128
```

### Configure IPv6 Functionality for a Data Prefix

To configure an IPv6 address for a new prefix list, follow these steps:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Policies**.
2. From the Custom Options drop-down list, select **Lists**. You can make this selection for a Centralized Policy or a Localized Policy
3. Select **Data Prefix** from the list on the left and then select **New Data Prefix List**.

4. From **Internet Protocol**, click **IPv6** and enter the IPv6 address in **Add Prefix**.

*CLI equivalent:*

```
Device(config)# policy lists data-ipv6-prefix-list source_ipv6_list
Device(config-data-ipv6-prefix-list-source_ipv6_list)# ipv6-prefix 2001:380:1::/64
```

### Configure IPv6 Functionality for a Centralized Policy

To configure a centralized policy to apply to IPv6 address families, follow these steps:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Policies**.
2. From the Custom Options drop-down menu, select **Traffic Policy** under Centralized Policy.
3. Select **Traffic Data**.
4. Select **Add Policy** and click **Create New**.
5. Click **Sequence Type** and then select **Traffic Engineering**.
6. Click **Sequence Rule**.
7. From the **Protocol** drop-down list, select **IPv6** to apply the policy only to IPv6 address families, or select **Both** to apply the policy IPv4 and IPv6 address families.
8. Click **Sequence Type** and then select **QoS**.
9. Click **Sequence Rule**.
10. From the Protocol drop-down list, click **IPv6** to apply the policy only to IPv6 address families, or select **Both** to apply the policy IPv4 and IPv6 address families.

*CLI equivalent:*

```
config-transaction
(config)# policy
(config-policy)# lists ipv6-prefix-list foo ipv6-prefix 1::1/64
 ipv6-prefix-list ipv6-1
 ipv6-prefix 1::1/128
```

### Configure IPv6 Functionality for a Localized Policy

To configure a localized policy to apply to IPv6 address families, follow these steps:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Policies**.
2. From the **Custom Options** drop-down list, select **Access Control Lists** under Localized Policy.
3. Click **Add Access Control List Policy** and choose **Add IPv6 ACL Policy**. The policy you create will apply only to IPv6 address families.

*CLI equivalent:*

In the following example, IPv6 routes that have addresses specified by the prefix list named marketing are matched:

```
config-transaction
Device(config)# route-map name
Device(config-route-map)# match ipv6 address prefix-list marketing
```

- [DHCP for IPv6, on page 767](#)
- [IPv6 as Preferred Address Family in a Dual Stack Environment, on page 777](#)
- [Information About IPv6 as Preferred Address Family in a Dual Stack Environment, on page 778](#)
- [Benefits of IPv6 as Preferred Address Family in a Dual Stack Environment, on page 778](#)
- [Use Cases for IPv6 as Preferred Address Family in a Dual Stack Environment, on page 779](#)
- [Configure IPv6 as Preferred Address Family in a Dual Stack Environment, on page 779](#)
- [Configure IPv6 as Preferred Address Family in a Dual Stack Environment Using a CLI Template, on page 782](#)
- [Monitor IPv6 as Preferred Address Family in a Dual Stack Environment, on page 783](#)
- [Monitor IPv6 as Preferred Address Family in a Dual Stack Environment Using the CLI , on page 783](#)
- [Troubleshooting , on page 784](#)
- [Configuration Example for IPv6 as Preferred Address Family in a Dual Stack Environment, on page 784](#)

## DHCP for IPv6

*Table 340: Feature History*

| Feature Name  | Release Information                                                          | Description                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DHCP for IPv6 | Cisco IOS XE Catalyst SD-WAN Release 17.7.1a<br>Cisco vManage Release 20.7.1 | <p>This feature allows you to configure DHCP for IPv6 (DHCPv6) on Cisco IOS XE Catalyst SD-WAN devices to assign IPv6 addresses to hosts on an IPv6-enabled network.</p> <p>Assigning of IPv6 addresses is accomplished using SLAAC, DHCPv6, DHCPv6 Prefix Delegation, or DHCPv6 Relay.</p> <p>A Cisco IOS XE Catalyst SD-WAN device can be configured for DHCPv6 as a DHCP server, DHCP client, or as a DHCP relay agent.</p> |

## Prerequisites for DHCPv6

- Basic IPv6 connectivity for assigning IPv6 addresses to hosts connected to the Cisco IOS XE Catalyst SD-WAN devices.

## Restrictions For DHCPv6

- This feature is supported only through CLI configuration.
- A unique DHCPv6 pool name must be provided for each VRF.

## Information About DHCPv6

You can configure Dynamic Host Configuration Protocol (DHCP) for IPv6 to assign addresses on an IPv6-enabled network. Alternatively, you can also configure Stateless Address Autoconfiguration (SLAAC) to assign addresses on an IPv6-enabled network.

### SLAAC

The most common method for IPv6 client address assignment is SLAAC. SLAAC provides simple plug-and-play connectivity where hosts self-assign an address based on the IPv6 prefix.

SLAAC is configured as follows:

- Host sends a router solicitation message.
- Hosts wait for a Router Advertisement (RA) message.
- Hosts take the first 64 bits of the IPv6 prefix from the RA message and combine it with the 64-bit EUI-64 address (in the case of Ethernet, this is created from the MAC Address) to create a global unicast address. The host also uses the source IP address, in the IP header, of the RA message, as its default gateway.
- Duplicate Address Detection (DAD) is performed by IPv6 clients in order to ensure that random addresses that are picked do not collide with other clients.
- The choice of algorithm is up to the client and is often configurable.

The last 64 bits of the IPv6 address can be learned based on the following 2 algorithms:

- EUI-64 which is based on the MAC address of the interface, or
- Private addresses that are randomly generated.

### SLAAC and DHCPv6

#### DHCPv6

IPv6 devices use multicast to acquire IP addresses and to find DHCPv6 servers. The basic DHCPv6 client-server concept is similar to DHCP for IPv4. If a client wants to receive configuration parameters, it sends out a request on the attached local network to detect available DHCPv6 servers. The server responds with the requested information in a Reply message.

The DHCPv6 client knows whether to use DHCPv6 based upon the instruction from a router on its link-local network. The default gateway has two configurable bits in its RA available for this purpose:

- O bit—When this bit is set, the client can use DHCPv6 to retrieve other configuration parameters (for example, TFTP server address or DNS server address) but not the client's IP address.
- M bit—When this bit is set, the client can use DHCPv6 to retrieve a managed IPv6 address and other configuration parameters from a DHCPv6 server.

#### Stateless DHCP

Stateless DHCPv6 is a combination of SLAAC and DHCPv6. With this option SLAAC is still used to retrieve an IP address while DHCP is used to obtain additional information such as TFTP server address, DNS server address. In this case, the device sends an RA with the O bit set but does not set the M bit. This is known as Stateless DHCPv6 because the DHCPv6 server does not have to track the client address bindings.

### Stateful DHCP

Stateful DHCPv6 functions exactly the same as DHCP IPv4 in which hosts receive both their IPv6 address and additional parameters from the DHCP server. When a device sends an RA with the M bit set, this indicates that clients must use DHCP to obtain their IP addresses. When the M bit is set, the setting of the O bit is irrelevant because the DHCP server also returns other configuration information together with the addresses. This is known as Stateful DHCPv6 because the DHCPv6 server tracks the client address bindings.

### DHCPv6 Prefix Delegation

The DHCPv6 prefix delegation feature is a stateful mode of operation for simple delegation of prefixes from a delegating edge device (DHCP server) to requesting edge device (DHCP clients).

DHCPv6 prefix delegation feature is ideal for the following situations where:

- A delegating edge device that does not have the information about the topology of the networks to which the requesting edge device is attached to.
- A delegating edge device does not require other information apart from the identity of the requesting edge device to choose a prefix for delegation. This mechanism is appropriate for use by an ISP to delegate a prefix to a subscriber. After the ISP has delegated prefixes to a subscriber, the subscriber may further subnet and assign prefixes to the links within the subscriber's network.

### DHCPv6 Relay

A DHCPv6 relay agent is an edge device, residing on the client's network, is used to relay messages between the client and the server when a DHCPv6 server is not in the same network as the DHCPv6 clients.

## Benefits of DHCPv6

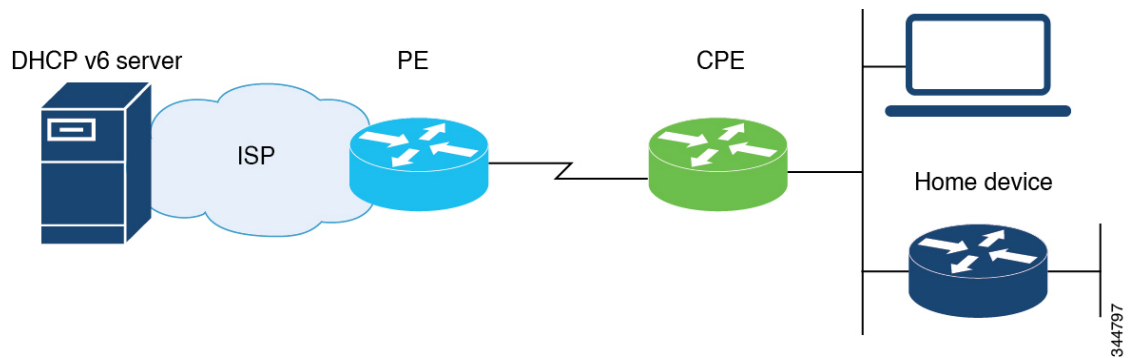
Configuring DHCP for IPv6 allows you to have more IP address compared to IPv4. With IPv6, there can be no depletion of IP addresses.

## Use Cases For DHCPv6

Cisco IOS XE Catalyst SD-WAN devices can be configured for DHCPv6 as a server, client, or a relay agent. As a server, a Cisco IOS XE Catalyst SD-WAN device can be configured for SLAAC, Stateless DHCP or for prefix delegation.

### SLAAC with DHCP

The figure below shows a typical broadband deployment.



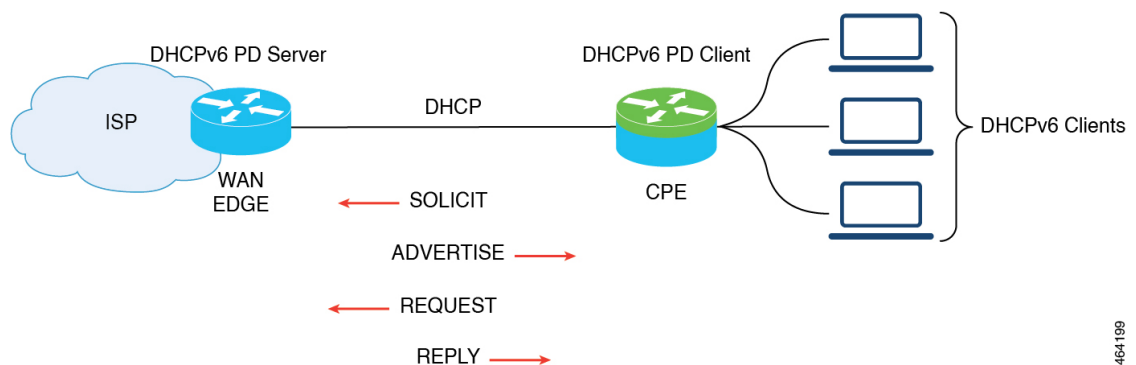
A Cisco IOS XE Catalyst SD-WAN device deployed on a customer premises (CPE) and connected to a ISP edge (PE) device can be a stateless or stateful DHCPv6 client. In either case, the ISP-side DHCPv6 server might provide configuration parameters such as Domain Name System (DNS) server addresses, domain names, and Simple Network Time Protocol (SNTP) servers to the DHCP client on the CPE. Such information can be specific to ISPs.

In addition to being a DHCPv6 client (toward the ISP), the CPE can act as a DHCPv6 server to the home network. For example, neighbor discovery followed by a stateless or stateful DHCPv6 client can occur on the link between the CPE and the home devices. In some cases, the information to be provided to the home network is the same as that obtained from the ISP-side DHCPv6 server. Therefore, the DHCPv6 component on the CPE allows automatic importing of configuration parameters from the DHCPv6 client to the DHCPv6 server pool.

### DHCPv6 Prefix Delegation

The model of operation for prefix delegation is as follows. In this sample topology, an edge device is configured as a DHCP server which is provisioned with prefixes to be delegated to a DHCP client. A Cisco IOS XE Catalyst SD-WAN device is configured as a DHCP client and requests prefix(es) from the server. The server chooses prefix(es) for delegation and responds with prefix(es) to the DHCP client. The DHCP client is then responsible for the delegated prefix(es).

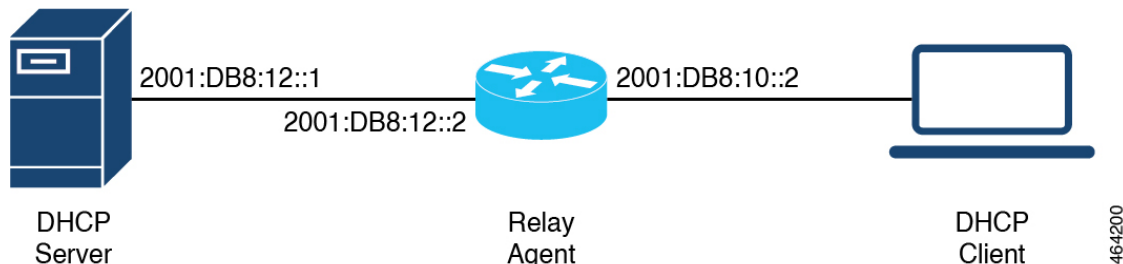
For example, the client might assign a subnet from a delegated prefix to one of its interfaces and begin sending Router Advertisements for the prefix on that link. Each prefix has an associated preferred lifetime and valid lifetime, which constitute an agreement about the length of time over which the client is allowed to use the prefix. A client can request an extension of the lifetimes on a delegated prefix and is required to terminate the use of a delegated prefix if the valid lifetime of the prefix expires.





### DHCPv6 Relay

In this sample topology, the DHCP server is not in the same network as DHCP client. A Cisco IOS XE Catalyst SD-WAN device residing on the client's network acts as a relay agent to relay messages between the client and the server.



## Configure DHCPv6

1. From Cisco SD-WAN Manager menu, choose **Configuration** > **Templates**.
2. Click **Device Templates**.



**Note** In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled as **Device**

3. From **Create Template** drop-down, choose **CLI Template**.



**Note** You can also use the CLI Add-on template to configure DHCP for IPv6 for client and server. For more information, see [Create a CLI Add-On Feature Template](#).

4. From **Device Model**, choose a device model for which you are creating the template.
5. In the **Template Name** field, enter a name for the device template. This field is mandatory and can contain only uppercase and lowercase letters, the digits 0 to 9, hyphens (-), and underscores (\_). It cannot contain spaces or any other characters.
6. In the **Description** field, enter a description for the device template. This field is mandatory, and it can contain any character and spaces.
7. In the **CLI Configuration** field, enter the DHCP configuration for IPv6 for client and server by typing it, cutting and pasting it, or uploading a file.
8. Click **Save**.

## Configure SLAAC

This example shows how to configure SLAAC on the client side.

```

device(config)# interface GigabitEthernet0/0/2
device(config-if)# ipv6 address autoconfig

```

```
device(config-if)# ipv6 enable
device(config-if)# end
```

This example shows how to configure SLAAC on the server side.

```
device(config)# interface GigabitEthernet1
device(config-if)# ipv6 address 2010:AB8:0:1::1/64
device(config-if)# ipv6 enable
device(config-if)# end
```

## Configure SLAAC and DHCPv6 Pool for Options

This example shows how to configure SLAAC and DHCPv6 pool on the client side.

```
device(config)# interface GigabitEthernet0/0/2
device(config-if)# ipv6 address autoconfig
device(config-if)# ipv6 enable
device(config-if)# ipv6 nd autoconfig default-route
device(config-if)# ipv6 dhcp client request vendor
device(config-if)# end
```

This example shows how to configure SLAAC and DHCPv6 pool on the server side.

```
device(config)# interface GigabitEthernet1
device(config-if)# ipv6 address 2010:AB8:0:1::1/64
device(config-if)# ipv6 enable
device(config-if)# ipv6 nd autoconfig default-route
device(config-if)# ipv6 nd other-config-flag
device(config-if)# ipv6 dhcp server dhcpv6
device(config-if)# end

device(config)# ipv6 dhcp pool dhcpv6
device(config-dhcpv6)# dns-server 2001:DB8:3000:3000::42
device(config-dhcpv6)# domain-name example.com
device(config-dhcpv6)# vendor-specific 100
device(config-dhcpv6)# suboption 1 address 2001:CC:1234:44::10
device(config-dhcpv6)# suboption 2 ascii "ip phone"
```

## Configure DHCPv6 (stateful) Address Assignment

This example shows how to configure DHCPv6 address assignment on the client side.

```
device(config)# interface GigabitEthernet0/0/2
device(config-if)# ipv6 address dhcp
device(config-if)# ipv6 enable
device(config-if)# ipv6 nd autoconfig default-route
device(config-if)# ipv6 dhcp client request vendor
device(config-if)# end
```

This example shows how to configure DHCPv6 address assignment on the server side.

```
device(config)# interface GigabitEthernet1
device(config-if)# ipv6 address 2010:AB8:0:1::1/64
```

```

device(config-if)# ipv6 enable
device(config-if)# ipv6 nd autoconfig default-route
device(config-if)# ipv6 nd managed-config-flag
device(config-if)# ipv6 dhcp server dhcpv6
device(config-if)# end

device(config)# ipv6 dhcp pool dhcpv6
device(config-dhcpv6)# address prefix 2010:AB8:0:1::1/64 lifetime 200 200
device(config-dhcpv6)# dns-server 2001:DB8:3000:3000::42
device(config-dhcpv6)# domain-name example.com
device(config-dhcpv6)# vendor-specific 100
device(config-dhcpv6)# suboption 1 address 2001:CC:1234:44::10
device(config-dhcpv6)# suboption 2 ascii "ip phone"

```

## Configure DHCPv6 with Prefix Delegation (stateful)

This example shows how to configure DHCPv6 with prefix delegation on the client side.

```

device(config)# interface GigabitEthernet0/0/2
device(config-if)# ipv6 enable
device(config-if)# ipv6 nd autoconfig default-route
device(config-if)# ipv6 dhcp client pd prefix_from_provider
device(config-if)# ipv6 dhcp client request vendor
device(config-if)# end

```

This example shows how to configure DHCPv6 with prefix delegation on the server side.

```

device(config)# interface GigabitEthernet1
device(config-if)# ipv6 address 2010:AB8:0:1::1/64
device(config-if)# ipv6 enable
device(config-if)# ipv6 nd autoconfig default-route
device(config-if)# ipv6 nd managed-config-flag
device(config-if)# ipv6 nd ra interval 20
device(config-if)# ipv6 dhcp server dhcpv6
device(config-if)# end

device(config)# ipv6 dhcp pool dhcpv6
device(config-dhcpv6)# prefix-delegation pool dhcpv6-pool1 lifetime 200 200
device(config-dhcpv6)# dns-server 2001:DB8:3000:3000::42
device(config-dhcpv6)# domain-name example.com
device(config-dhcpv6)# vendor-specific 100
device(config-dhcpv6)# suboption 1 address 2001:CC:1234:44::10
device(config-dhcpv6)# suboption 2 ascii "ip phone"
device(config)# ipv6 local pool dhcpv6-pool1 2001:DB8:1200::/40 48

```

## Configure DHCPv6 with Relay

This example shows how to configure DHCPv6 with relay on the client side.

```

device(config)# interface GigabitEthernet3
device(config-if)# ipv6 address dhcp
device(config-if)# ipv6 enable
device(config-if)# ipv6 dhcp client pd pr-from-pd
device(config-if)# ipv6 dhcp client request vendor
device(config-if)# no mop enabled

```

```
device(config-if)# no mop sysid
device(config-if)# end
```

This example shows the configurations on the client facing WAN edge device that acts as the relay agent.

```
device(config)# interface TenGigabitEthernet0/0/5
device(config-if)# vrf forwarding 10
device(config-if)# load-interval 30
device(config-if)# ipv6 address 2001:BB:1000::10/64
device(config-if)# ipv6 enable
device(config-if)# ipv6 dhcp relay destination 2001:BB8:1200::2
device(config-if)# ipv6 dhcp relay option vpn
device(config-if)# end
```

This example shows the configurations on the server facing WAN edge device.

```
device(config)# interface GigabitEthernet0/0/3
device(config-if)# vrf forwarding 10
device(config-if)# no ip address
device(config-if)# negotiation auto
device(config-if)# ipv6 address 2001:BB8:1200::1/64
device(config-if)# ipv6 enable
device(config-if)# end
```

This example shows how to configure DHCPv6 with relay on the server side.

```
device(config)# interface GigabitEthernet2
device(config-if)# ipv6 address 2001:BB8:1200::2/64
device(config-if)# ipv6 enable
device(config-if)# ipv6 dhcp server dhcpv6
device(config-if)# end

device(config)# ipv6 dhcp pool dhcpv6
device(config-dhcpv6)# prefix-delegation pool dhcpv6-pool10 lifetime infinite infinite
device(config-dhcpv6)# address prefix 2001:BB:1000::/64 lifetime 200 200
device(config-dhcpv6)# dns-server 2001:BB:1200::42
device(config-dhcpv6)# domain-name relay.com
device(config)# ipv6 local pool dhcpv6-pool10 8001:ABCD::/40 48
```

## Verify DHCPv6 Client and Server Configuration

### Verify DHCPv6 Interface Information

The following is a sample output from the **show ipv6 dhcp interface** command that provides details about DHCPv6 address allocation.

```
Device# show ipv6 dhcp interface GigabitEthernet0/0/2
GigabitEthernet0/0/2 is in client mode
Prefix State is IDLE
Address State is OPEN
Renew for address will be sent in 00:01:09
List of known servers:
 Reachable via address: FE80::250:56FF:FEBD:DBD1
 DUID: 00030001001EBD43F800
 Preference: 0
Configuration parameters:
 IA NA: IA ID 0x00080001, T1 100, T2 160
 Address: 2010:AB8:0:1:95D1:CFC:F227:23FB/128
 preferred lifetime 200, valid lifetime 200
 expires at Oct 26 2021 07:28 AM (170 seconds)
```

```

DNS server: 2001:DB8:3000:3000::42
Domain name: example.com
Information refresh time: 0
Vendor-specific Information options:
 Enterprise-ID: 100
Prefix Rapid-Commit: disabled
Address Rapid-Commit: disabled

```

The following is a sample output from the **show ipv6 dhcp interface** command that provides details about DHCPv6 prefix delegation.

```

Device# show ipv6 dhcp interface GigabitEthernet0/0/2
GigabitEthernet0/0/2 is in client mode
Prefix State is OPEN
Renew will be sent in 00:01:34
Address State is IDLE
List of known servers:
 Reachable via address: FE80::250:56FF:FEBD:BD1
 DUID: 00030001001EBD43F800
 Preference: 0
Configuration parameters:
 IA PD: IA ID 0x00080001, T1 100, T2 160
 Prefix: 2001:DB8:1202::/48
 preferred lifetime 200, valid lifetime 200
 expires at Oct 26 2021 07:30 AM (194 seconds)
 DNS server: 2001:DB8:3000:3000::42
 Domain name: example.com
 Information refresh time: 0
Prefix name: prefix_from_server
Prefix Rapid-Commit: disabled
Address Rapid-Commit: disabled

```

The following is a sample output from the **show ipv6 dhcp interface** command that provides details about SLAAC with DHCP.

```

Device# show ipv6 dhcp interface GigabitEthernet0/0/2
GigabitEthernet0/0/2 is in client mode
Prefix State is IDLE (0)
Information refresh timer expires in 23:59:49
Address State is IDLE
List of known servers:
 Reachable via address: FE80::250:56FF:FEBD:BD1
 DUID: 00030001001EBD43F800
 Preference: 0
Configuration parameters:
 DNS server: 2001:DB8:3000:3000::42
 Domain name: example.com
 Information refresh time: 0
 Vendor-specific Information options:
 Enterprise-ID: 100
Prefix Rapid-Commit: disabled
Address Rapid-Commit: disabled

```

### View DHCPv6 Pool Information

The following is a sample output from the **show ipv6 dhcp pool** command that provides details about DHCPv6 address allocation.

```

Device# show ipv6 dhcp pool
DHCPv6 pool: relay_server
VRF 10
Prefix pool: dhcpv6-pool2
Address allocation prefix: 5001:DB8:1234:42::/64 valid 20000 preferred 20000 (1 in use,
0 conflicts)

```

```

 preferred lifetime 200, valid lifetime 200
DNS server: 2001:BB8:3000:3000::42
Domain name: relay.com
Information refresh: 60
Vendor-specific Information options:
Enterprise-ID: 10
 suboption 1 address 2001:DB8:1234:42::10
 suboption 2 ascii 'ip phone'
Active clients: 1
Pool is configured to include all configuration options in REPLY

```

The following is a sample output from the **show ipv6 dhcp pool** command that provides details about DHCPv6 prefix delegation.

```

Device# show ipv6 dhcp pool
DHCPv6 pool: relay_server
VRF 10
Prefix pool: dhcpv6-pool2
Address allocation prefix: 5001:DB8:1234:42::/64 valid 20000 preferred 20000 (0 in use,
0 conflicts)
 preferred lifetime 200, valid lifetime 200
DNS server: 2001:BB8:3000:3000::42
Domain name: relay.com
Information refresh: 60
Vendor-specific Information options:
Enterprise-ID: 10
 suboption 1 address 2001:DB8:1234:42::10
 suboption 2 ascii 'ip phone'
Active clients: 1
Pool is configured to include all configuration options in REPLY

```

### View DHCPv6 Bindings

The following is a sample output from the **show ipv6 dhcp binding** command that provides details about DHCPv6 address allocation.

```

Device# show ipv6 dhcp binding
Client: FE80::250:56FF:FEBD:8261
DUID: 00030001001EE6DBF500
Username : unassigned
VRF : 10
IA NA: IA ID 0x00080001, T1 10000, T2 16000
 Address: 5001:DB8:1234:42:500C:B3FA:54A7:F63D
 preferred lifetime 20000, valid lifetime 20000
 expires at Oct 26 2021 01:17 PM (19925 seconds)

```

The following is a sample output from the **show ipv6 dhcp binding** command that provides details about DHCPv6 prefix delegation.

```

Device# show ipv6 dhcp binding
Client: FE80::250:56FF:FEBD:8261
DUID: 00030001001EE6DBF500
Username : unassigned
VRF : 10
Interface : GigabitEthernet0/0/3
IA PD: IA ID 0x00080001, T1 100, T2 160
 Prefix: 2001:BB8:1602::/48
 preferred lifetime 200, valid lifetime 200
 expires at Oct 26 2021 08:01 AM (173 seconds)

```

### View DHCPv6 Database

The following is a sample output from the **show ipv6 dhcp database** command.

```

Device# show ipv6 dhcp database
Database agent bootflash:
 write delay: 300 seconds, transfer timeout: 300 seconds
 last written at Oct 26 2021 08:01 AM, write timer expires in 250 seconds
 last read at never
 successful read times 0
 failed read times 0
 successful write times 2
 failed write times 0

```

### View DHCPv6 Relay Bindings

The following is a sample output from the **show ipv6 dhcp relay bindings** command that provides details about DHCPv6 relay.

```

Device# show ipv6 dhcp relay binding

Relay Bindings associated with default vrf:

Relay Bindings associated with vrf 10:
Prefix: 2001:AA8:1100::/48 (GigabitEthernet3)
 DUID: 00030001001E49674C00
 IAID: 851969
 lifetime: INFINITE
 expiration: INFINITE
Summary:
 Total number of Relay bindings = 1
 Total number of IAPD bindings = 1
 Total number of IANA bindings = 0
 Total number of Relay bindings added by Bulk lease = 0

```

## IPv6 as Preferred Address Family in a Dual Stack Environment

**Table 341: Feature History**

| Feature Name                                                 | Release Information                                                                                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IPv6 as Preferred Address Family in a Dual Stack Environment | <p>Cisco IOS XE Catalyst SD-WAN Release 17.10.1a</p> <p>Cisco Catalyst SD-WAN Control Components Release 20.10.1</p> | <p>This feature allows you to select IPv6 as the preferred address family for control and data connections in a dual stack network environment.</p> <p>For Cisco SD-WAN Manager and Cisco Catalyst SD-WAN Controller, configure IPv6 as the preferred address family by using the feature template or the CLI template. For Cisco IOS XE Catalyst SD-WAN devices, configure IPv6 as the preferred address family using the Configuration Groups, Quick Connect or a CLI template.</p> |

## Information About IPv6 as Preferred Address Family in a Dual Stack Environment

Cisco Catalyst SD-WAN provides you the option to select a preferred address family—IPv4 or IPv6—to establish control and data connections in a dual stack network environment. Use the **Dual Stack IPv6 Default** drop-down list in Cisco SD-WAN Manager to set IPv6 or IPv4.

On a Cisco IOS XE Catalyst SD-WAN device, when you choose the **True** option from the **Dual Stack IPv6 Default** drop-down list, the device establishes an IPv6 control connection with Cisco SD-WAN Manager and Cisco Catalyst SD-WAN Controller it is connected to. When you choose the **False** option from the **Dual Stack IPv6 Default** drop-down list, an IPv4 connection is established to connect to Cisco SD-WAN Manager and Cisco Catalyst SD-WAN Controller.

Data connections or Bidirectional Forwarding Detection (BFD) sessions are established based on the IPv6 option set in local, remote Cisco IOS XE Catalyst SD-WAN devices. In a dual stack environment, when the **True** option is chosen in a local or remote Cisco IOS XE Catalyst SD-WAN device, the BFD session is an IPv6 connection. Otherwise, it is IPv4.

When you choose the **True** option from the **Dual Stack IPv6 Default** drop-down list in Cisco SD-WAN Manager or Cisco Catalyst SD-WAN Controller, IPv6 connections to other Cisco SD-WAN Manager and Cisco Catalyst SD-WAN Controller instances are established. When you choose the **False** option from the **Dual Stack IPv6 Default** drop-down list, an IPv4 connection is established.



### Note

- The connections from Cisco SD-WAN Manager, Cisco Catalyst SD-WAN Controller and Cisco IOS XE Catalyst SD-WAN devices to the Cisco Catalyst SD-WAN Validator is always dual (IPv4 and IPv6) in a dual stack network environment whether the **Dual Stack IPv6 Default** drop-down list options set to **True** or **False**.
- The **Dual Stack IPv6 Default** drop-down list options applies to Cisco IOS XE Catalyst SD-WAN devices, Cisco SD-WAN Manager, and Cisco Catalyst SD-WAN Controller, and not to Cisco Catalyst SD-WAN Validator.
- An IPv6 connection can be configured on Cisco IOS XE Catalyst SD-WAN devices in sites that are behind NAT44 and NAT66.

## Benefits of IPv6 as Preferred Address Family in a Dual Stack Environment

You have the option to migrate from IPv4 to IPv6, which allows you to have more IP addresses compared to IPv4. With IPv6, there can be no depletion of IP addresses.



# Use Cases for IPv6 as Preferred Address Family in a Dual Stack Environment

From Cisco IOS XE Catalyst SD-WAN Release 17.10.1a, Cisco Catalyst SD-WAN Control Components Release 20.10.1—to migrate from IPv4 to IPv6, you have the option to select a default connectivity option—IPv4 or IPv6—for control connections and data connections.

## Configure IPv6 as Preferred Address Family in a Dual Stack Environment

Using Cisco SD-WAN Manager, you can configure Cisco IOS XE Catalyst SD-WAN devices, Cisco SD-WAN Manager, and Cisco Catalyst SD-WAN Controller to set IPv6 as the default connectivity option for control and data connections.

### Configure Cisco IOS-XE SD-WAN Devices for IPv6 Connectivity

You can use one of these options to configure an IPv6 connection on Cisco IOS XE Catalyst SD-WAN devices:

- CLI template and CLI add-on template
- Configuration groups
- Quick connect

#### CLI Template and CLI Add-On Template

Use the CLI template or the CLI add-on template to configure IPv6 for a Cisco IOS XE Catalyst SD-WAN device. The CLI configuration for Cisco IOS XE Catalyst SD-WAN devices is provided in [Configure IPv6 as Preferred Address Family in a Dual Stack Environment Using a CLI Template](#) section. For more information about using CLI templates, see [CLI Templates](#) and [CLI Add-On Feature Templates](#).

#### Configuration Groups

To configure an IPv6 connection on Cisco IOS XE Catalyst SD-WAN devices using configuration groups, perform this procedure:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates > Configuration Groups**.
2. Click ... adjacent to the configuration group name and choose **Edit**.
3. Click **Associated Devices**.
4. Choose one or more Cisco IOS XE Catalyst SD-WAN devices, and then click **Deploy**.
5. In the **Process Overview** window, click **Next**.
6. The **Selected Devices to Deploy** page displays the Cisco IOS XE Catalyst SD-WAN devices you selected previously. Check or uncheck one or more Cisco IOS XE Catalyst SD-WAN devices and then click **Next**.

7. From the **Dual Stack IPv6 Default** drop-down list, choose **True** to set IPv6 as a default connection, and click **Next**.

The **True** option enables Cisco IOS XE Catalyst SD-WAN devices to establish an IPv6 connection with Cisco SD-WAN Manager and Cisco Catalyst SD-WAN Controller it is connected to. When you choose **False**, an IPv4 connection is established.

BFD sessions are established based on the IPv6 option set in local, remote Cisco IOS XE Catalyst SD-WAN devices. In a dual stack environment, when the **True** option is chosen in a local or remote Cisco IOS XE Catalyst SD-WAN device, the BFD session is an IPv6 connection. Otherwise, it is IPv4.




---

**Note** The connections from the Cisco IOS XE Catalyst SD-WAN devices to Cisco Catalyst SD-WAN Validator is always dual (IPv4 and IPv6) in a dual IP stack environment whether the **Dual Stack IPv6 Default** drop-down list options set to **True** or **False**.

---

8. In the Summary window, click **Deploy**.

For more information on using configuration groups, see [Configuration Groups and Feature Profiles](#).

### Quick Connect

To configure an IPv6 connection on Cisco IOS XE Catalyst SD-WAN devices using the quick connect workflow, perform this procedure:

1. From the Cisco SD-WAN Manager menu, choose **Workflows > Quick Connect**.
2. In the **Process Overview** window, click **Next**.

3. Choose an option to sync your devices, and then click **Next**

For more information, see [Quick Connect Workflow](#)

4. In the **Selected devices to bring up** window, check one or more Cisco IOS XE Catalyst SD-WAN devices, and then click **Next**.
5. From the **Dual Stack IPv6 Default** drop-down list, choose **True** to set IPv6 as a default connection and click **Apply**, and then click **Next**.

The **True** option enables Cisco IOS XE Catalyst SD-WAN devices to establish an IPv6 connection with Cisco SD-WAN Manager and Cisco Catalyst SD-WAN Controller it is connected to. When you choose **False**, an IPv4 connection is established.

BFD sessions are established based on the IPv6 option set in local, remote Cisco IOS XE Catalyst SD-WAN devices. In a dual stack environment, If you choose the **True** option in a local or remote Cisco IOS XE Catalyst SD-WAN device, the BFD session is an IPv6 connection. Otherwise, it is IPv4.




---

**Note** The connections from the Cisco IOS XE Catalyst SD-WAN devices to Cisco Catalyst SD-WAN Validator is always dual (IPv4 and IPv6) in a dual IP stack environment whether you choose the **True** or the **False** option.

---

6. In the Summary window, click **Deploy**.

## Configure Cisco SD-WAN Manager and Cisco SD-WAN Controller for IPv6 Connectivity

You can use one of these options to configure an IPv6 connection on Cisco SD-WAN Manager and Cisco Catalyst SD-WAN Controller:

- CLI template and CLI add-on template
- Feature template

### CLI Template

Use the CLI template to configure IPv6 in Cisco SD-WAN Manager and Cisco Catalyst SD-WAN Controller. The CLI configuration for Cisco SD-WAN Manager and Cisco Catalyst SD-WAN Controller is provided in [Configure IPv6 as Preferred Address Family in a Dual Stack Environment Using a CLI Template](#). For more information about using CLI templates, see [CLI Templates](#).

### Feature Template

To configure an IPv6 connection in Cisco SD-WAN Manager using the feature template, perform this procedure:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates** and choose **Add Template**.
3. Choose a Cisco SD-WAN controller.
4. Under **BASIC INFORMATION**, click **System**.
5. In the **Template Name** field, enter a name for the feature template. This field is mandatory and can contain only uppercase and lowercase letters, the digits 0 to 9, hyphens (-), and underscores (\_). It cannot contain spaces or any other characters.
6. In the **Description** field, enter a description for the feature template. This field is mandatory, and it can contain all characters and spaces.
7. Under the **Basic Information** tab, click the **On** radio button adjacent to **Dual Stack IPv6 Default** field to set IPv6 as a default connection.

The **On** option sets Cisco SD-WAN Manager and Cisco Catalyst SD-WAN Controller to establish an IPv6 connection with all other Cisco SD-WAN Manager and Cisco Catalyst SD-WAN Controller instances. When you click the **Off** radio button, an IPv4 connection is established.



---

**Note**

The connections from Cisco SD-WAN Manager and Cisco Catalyst SD-WAN Controller to Cisco Catalyst SD-WAN Validator is always dual (IPv4 and IPv6) in a dual IP stack environment irrespective of whether you click the **On** or **Off** radio button.

---

8. Click **Save**.

# Configure IPv6 as Preferred Address Family in a Dual Stack Environment Using a CLI Template

## Configure Cisco IOS-XE SD-WAN Devices for IPv6 in Dual IP Stack Using a CLI Template

For more information about using CLI templates, see [CLI Add-On Feature Templates](#) and [CLI Templates](#).



**Note** By default, CLI templates execute commands in global config mode.

This section provides sample CLI configurations of IPv6 as the preferred address family in Cisco IOS XE Catalyst SD-WAN devices:

1. Enable IPv6 on the tunnel interface:

```
interface tunnell
no shutdown
ipv6 enable
```

2. Enable IPv6:

```
system
ipv6-strict-control true
```

The following example shows how to configure IPv6 as the preferred address family in Cisco IOS XE Catalyst SD-WAN devices.

```
interface Tunnell
no shutdown
ip unnumbered GigabitEthernet1
tunnel source GigabitEthernet1
tunnel mode sdwan
ipv6 enable
exit
```

```
system
gps-location latitude 32.0
gps-location longitude -100.0
system-ip 10.16.255.14
domain-id 1
site-id 400
ipv6-strict-control true
admin-tech-on-failure
organization-name "Cisco"
vbond vbond
```

## Configure Cisco SD-WAN Manager and Cisco SD-WAN Controller for IPv6 in a Dual IP Stack Using a CLI Template

For more information about using CLI templates, see [CLI Add-On Feature Templates](#) and [CLI Templates](#).



**Note** By default, CLI templates execute commands in global config mode.

The following example shows how to configure IPv6 as the preferred address family in a Cisco Catalyst SD-WAN Controller and Cisco SD-WAN Manager:

Enable IPv6:

```
system
ipv6-strict-control true
```

Here is the complete configuration example for configuring IPv6 as the preferred address family on a Cisco Catalyst SD-WAN Controller and Cisco SD-WAN Manager.

```
system
host-name vm9
system-ip 10.16.255.19
site-id 400
ipv6-strict-control true
port-offset 9
no daemon-restart
admin-tech-on-failure
no vrrp-advt-with-phymac
organization-name "Cisco"
vbond vbond
```

## Monitor IPv6 as Preferred Address Family in a Dual Stack Environment

After you successfully configure an IPv6 connection, the BFD connections will be up and running in Cisco SD-WAN Manager. To view the BFD connections in Cisco SD-WAN Manager:

1. From the Cisco SD-WAN Manager menu, choose **Monitor** > **Devices**
2. Verify the status of the connection under the **BFD** column.

## Monitor IPv6 as Preferred Address Family in a Dual Stack Environment Using the CLI

Use the following **show** commands to view control and data connection information for IPv4 and IPv6.

### Cisco IOS XE Catalyst SD-WAN Devices

- **show sdwan control connections**
- **show sdwan control local-properties**
- **show sdwan bfd sessions**
- **show sdwan omp tlocs**

- **show sdwan bfd tloc-summary-list**

For more information on these **show** commands, see the chapter [Troubleshooting Commands](#) in the Cisco IOS XE SD-WAN Qualified Command Reference guide.

#### Cisco SD-WAN Manager and Cisco Catalyst SD-WAN Controller

- **show control connections**
- **show control local-properties**

For more information on these **show** commands, see the chapter [Operational Commands](#).

## Troubleshooting

### Problem

BFD sessions are down.

### Possible Causes

- Verify the IP address connections.
- IPv6 might not be enabled on a tunnel interface.

### Solution

- Check and confirm the IP configuration settings for both IPv4 and IPv6 on Cisco IOS XE Catalyst SD-WAN devices, as well as within the Cisco SD-WAN Manager and the Cisco Catalyst SD-WAN Controller. For more information, see [Troubleshoot Common BFD Errors](#).
- When BFD sessions are in a down state and you manually enable IPv6 on the tunnel interface, some sessions may persist in the down state. To resolve this, execute the command **clear sdwan omp all** on each of the Cisco IOS XE Catalyst SD-WAN devices to re-establish the sessions.

## Configuration Example for IPv6 as Preferred Address Family in a Dual Stack Environment

### Configuration Example for IPv6 configured on a Cisco IOS-XE SD-WAN Device

This example shows how to configure IPv6 as the preferred address family on a Cisco IOS XE Catalyst SD-WAN device.

```
show sdwan running-config system
system
gps-location latitude 32.0
gps-location longitude -100.0
system-ip 10.16.255.14
domain-id 1
```

```
site-id 400
ipv6-strict-control true
admin-tech-on-failure
organization-name "Cisco"
vbond vbond
```

### Configuration Example for IPv6 configured on Cisco SD-WAN Manager and Cisco Catalyst SD-WAN Controller

This example shows how to configure IPv6 as the preferred address family on Cisco SD-WAN Manager and Cisco Catalyst SD-WAN Controller.

```
show running-config system
system
host-name vm9
system-ip 10.16.255.19
site-id 400
ipv6-strict-control true
port-offset 9
no daemon-restart
admin-tech-on-failure
no vrrp-advt-with-phymac
organization-name "Cisco"
vbond vbond
```







## CHAPTER 29

# IP Directed Broadcast

An IP directed broadcast is an IP packet whose destination address is a valid broadcast address for some IP subnet but which originates from a node that is not itself part of that destination subnet.

A device that is not directly connected to its destination subnet forwards an IP directed broadcast in the same way it would forward unicast IP packets destined to a host on that subnet. When a directed broadcast packet reaches a device that is directly connected to its destination subnet, that packet is broadcast on the destination subnet. The destination address in the IP header of the packet is rewritten to the configured IP broadcast address for the subnet, and the packet is sent as a link-layer broadcast.

If directed broadcast is enabled for an interface, incoming IP packets whose addresses identify them as directed broadcasts intended for the subnet to which that interface is attached are broadcast on that subnet.



**Note** The access control list (ACL) option for directed broadcast is not supported in Cisco SD-WAN Manager.

To enable the translation of a directed broadcast to physical broadcasts, use the `ip directed-broadcast` command. To disable this function, use the `no` form of this command. By default, `ip directed-broadcast` is disabled and all IP directed broadcasts are dropped.

**ip directed-broadcast** and **no ip directed-broadcast**

### Example

This example shows how to enable forwarding of IP directed broadcasts on Ethernet interface 2/1:

```
device# configure-transaction
device(config)# interface ethernet 2/1
device(config-if)# ip address 10.114.114.1 255.255.255.0
device(config-if)# ip directed-broadcast
device(config-if)# end
```





## CHAPTER 30

# Migrate Shared Templates to Cisco IOS XE Catalyst SD-WAN Templates

### Overview

In Cisco vManage 20.1.1, support is added for additional feature templates exclusively for Cisco IOS XE Catalyst SD-WAN devices.

In releases before Cisco vManage 20.1.1, when you created a template for both Cisco vEdge and Cisco IOS XE Catalyst SD-WAN devices, the same template is shared for both device types. For these templates, the configuration is specified using Cisco vEdge commands. If the template is then used with a Cisco IOS XE device, the configuration was converted for Cisco IOS XE devices. Due to this conversion of Cisco vEdge commands, some functionality was not unavailable for Cisco IOS XE Catalyst SD-WAN devices. For example, NAT DIA.

In these releases, there are two types of shared templates:

- Shared feature templates: If you specify a Cisco IOS XE Catalyst SD-WAN device when creating a feature template, a shared feature template is created.
- Shared device templates: A device template that contains a shared feature template.

In Cisco vManage 20.1.1 and onwards, feature templates have been separated for Cisco vEdge devices and Cisco IOS XE Catalyst SD-WAN devices. These feature templates that are exclusively created for Cisco IOS XE Catalyst SD-WAN devices enable support for additional features. To use these feature templates, you can migrate your shared feature templates to the exclusive templates.

### List of Migrated Templates

The following table lists the shared templates and their corresponding exclusive templates for Cisco IOS XE Catalyst SD-WAN devices available in Cisco vManage 20.1.1 and onwards.



**Note** The AAA feature template is not supported with the exclusive Cisco IOS XE Catalyst SD-WAN device feature templates.

If your existing template contains an AAA feature template, you can replace it either before migration or after migration:

- Before migration—Replace it with the AAA-Cisco template that was introduced in 19.1.
- or
- After migration—After the migration is complete, manually create a Cisco AAA template and attach it to your device template.

| Shared Feature Template | Shared Template Type      | Exclusive Cisco IOS XE Catalyst SD-WAN Device Feature Template | Exclusive Cisco IOS XE Catalyst SD-WAN Device Feature Template Type |
|-------------------------|---------------------------|----------------------------------------------------------------|---------------------------------------------------------------------|
| Banner                  | banner                    | Cisco Banner                                                   | cisco_banner                                                        |
| BFD                     | bfd-vedge                 | Cisco BFD                                                      | cisco_bfd                                                           |
| BGP                     | bgp                       | Cisco BGP                                                      | cisco_bgp                                                           |
| DHCP Server             | dhcp-server               | Cisco DHCP Server                                              | cisco_dhcp_server                                                   |
| Logging                 | logging                   | Cisco Logging                                                  | cisco_logging                                                       |
| NTP                     | ntp                       | Cisco NTP                                                      | cisco_ntp                                                           |
| OMP                     | omp-vedge                 | Cisco OMP                                                      | cisco_omp                                                           |
| OSPF                    | ospf                      | Cisco OSPF                                                     | cisco_ospf                                                          |
| Security                | security-vedge            | Cisco Security                                                 | cisco_security                                                      |
| SNMP                    | snmp                      | Cisco SNMP                                                     | cisco_snmp                                                          |
| System                  | system-vedge              | Cisco System                                                   | cisco_system                                                        |
| VPN Interface GRE       | vpn-vedge-interface-gre   | Cisco VPN Interface GRE                                        | cisco_vpn_interface_gre                                             |
| VPN Interface IPsec     | vpn-vedge-interface-ipsec | Cisco VPN Interface IPsec                                      | cisco_vpn_interface_ipsec                                           |
| VPN Interface Ethernet  | vpn-vedge-interface       | Cisco VPN Interface Ethernet                                   | cisco_vpn_interface                                                 |
| VPN                     | vpn-vedge                 | Cisco VPN                                                      | cisco_vpn                                                           |

### Migrate Shared Templates

You can continue using the older shared templates, however the shared templates may not have access to the latest features. We recommend migrating existing templates to enable access to the latest features. For example, if you are using the `VPN Interface Ethernet` shared template, the template still continues to work. However to use new features, such as NAT DIA, you must migrate to the exclusive feature template called `Cisco VPN Interface Ethernet`.

## Migrate Shared Templates Using the Cisco SD-WAN Manager Migration Tool

Prerequisites:

- At least one Cisco IOS XE Catalyst SD-WAN device template should exist with shared Cisco IOS XE Catalyst SD-WAN device feature templates attached prior to upgrading to Cisco vManage 20.1.1 or higher.

To migrate existing shared templates using Cisco SD-WAN Manager, perform the following steps:

1. From the Cisco SD-WAN Manager menu, choose **Tools > Template Migration**.
2. Click **Migrate All Templates**.
3. Enter a prefix for the new migrated templates. For example `Migrated_`. All migrated templates are prefixed with this identifier.
4. To migrate the templates, click **OK**.
5. Once the migration begins, click **Tasks** to track the status of the migration.
6. Once the migration is complete, you must manually attach the migrated templates to your devices. For each of the migrated templates, click ... and choose **Attach Devices to Migrated Template**.





## CHAPTER 31

# CLI Templates for Cisco IOS XE Catalyst SD-WAN Devices

You can configure CLI templates for Cisco IOS XE Catalyst SD-WAN devices in the following ways.



### Note

If you generate a CLI template in a higher version of Cisco SD-WAN Manager and then try to apply it in a lower version, it may not be supported depending on the configuration. In this case, Cisco SD-WAN Manager might also deny access and generate an error message. We recommend that you use a CLI template generated in an earlier version of Cisco SD-WAN Manager. For example, if you are using Cisco vManage Release 20.7.x, you can use a CLI template generated in Cisco vManage Release 20.6.x and earlier releases.

- [Device Configuration-Based CLI Templates for Cisco IOS XE Catalyst SD-WAN Devices, on page 793](#)
- [Intent-Based CLI Templates for Cisco IOS XE Catalyst SD-WAN Devices, on page 795](#)

## Device Configuration-Based CLI Templates for Cisco IOS XE Catalyst SD-WAN Devices

Cisco SD-WAN Manager configures Cisco IOS XE Catalyst SD-WAN devices using a combination of feature templates and policies (localized policies, security policies). In Cisco vManage 20.1.1 and onwards, Cisco SD-WAN Manager allows you to specify CLI templates that use the device configuration with Cisco IOS XE Catalyst SD-WAN devices. You can use these templates to push the device configuration (yang-cli) to devices directly.

In a single operation, Cisco SD-WAN Manager pushes the difference between the device configuration and configuration provided by the user in the template directly to the Cisco IOS XE Catalyst SD-WAN devices. Cisco SD-WAN Manager also displays a preview of the configuration before it is pushed to the device, as it does with other templates. The described workflow also applies if you want to make any additions, changes, or removals to the template.



- Note** To configure features not accessible using Cisco SD-WAN Manager, we recommend doing the following:
1. Use the relevant feature template in addition to a CLI add-on feature template. For more information, see [Qualified CLIs for CLI Add-On Feature Templates, on page 822](#).
  2. For situations where the previous option is not sufficient, use the device configuration-based CLI templates as described in this section.

## Feature Information for CLI Template for Cisco XE SD-WAN Routers

*Table 342: Feature History*

| Feature Name                       | Release Information                                                  | Description                                                                                                                                                                         |
|------------------------------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Device Configuration CLI Templates | Cisco IOS XE Catalyst SD-WAN Release 17.2.1r<br>Cisco vManage 20.1.1 | The CLI Templates feature has been updated to support device configuration-based CLIs. You can use these templates to push the device configuration (yang-cli) to devices directly. |

## Limitations

**Auxiliary ports:** When using a CLI template for Cisco Integrated Services Routers that have an auxiliary port, do not include commands for auxiliary ports, such as **line aux 0**. Doing so results in an error. These commands may be executed directly on the device.

When you import the CLI template configuration using the command, `show sdwan running-config`, you need to add quotes manually for the CLI template on the Cisco SD-WAN Manager.

From Cisco Catalyst SD-WAN Manager Release 20.12.x, policies configured using a Cisco SD-WAN Controller template are ignored. To configure policies, navigate to **Configuration > Policies > Custom Options > CLI policy**, add the policy and activate it for Cisco SD-WAN Controllers.

## Configure CLI Templates in Cisco SD-WAN Manager

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**.



**Note** In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled as **Device**.

3. From the **Create Template** drop-down list, select **CLI Template**.
4. From the **Device Model** drop-down list, select the type of device for which you are creating the template.
5. In **Template Name**, enter a name for the template.

The name can be up to 128 characters and can contain only alphanumeric characters.



6. In **Template Description**, enter a description of the template.  
The description can be up to 2048 characters and can contain only alphanumeric characters.
7. Choose **Device configuration**. Using this option, you can provide IOS-XE configuration commands that appear in the output of the `show sdwan running-config` command.
8. (Optional) To load the running config of a connected device, select it from the Load Running config from reachable device list and click **Search**.
9. In **CLI Configuration**, enter the configuration either by typing it, cutting and pasting it, or uploading a file.
10. To convert an actual configuration value to a variable, select the value and click **Create Variable**. Enter the variable name, and click **Create Variable**. You can also type the variable name directly, in the format `{{variable-name}}`; for example, `{{hostname}}`.  
These variables can be filled in device variables page per device after attaching the template. Values can be entered manually or can be uploaded via a csv file.
11. To save the feature template, click **Add**. The new device template is displayed in the Device Template table.

## Intent-Based CLI Templates for Cisco IOS XE Catalyst SD-WAN Devices

The CLI Templates for Cisco IOS XE Catalyst SD-WAN device features allows you to configure intent-based CLI templates for Cisco IOS XE Catalyst SD-WAN devices using Cisco SD-WAN Manager. Intent-based CLI template refer to the command line interface configuration that are based on the Cisco vEdge device syntax. Using CLI templates, Cisco SD-WAN Manager enables pushing Cisco vEdge syntax-based commands to Cisco IOS XE Catalyst SD-WAN device in Cisco IOS XE Syntax.



**Note** With the support of device configuration-based CLI templates, the intent-based CLI templates will be deprecated. We recommend using the device configuration-based CLI templates as described in [Device Configuration-Based CLI Templates for Cisco IOS XE Catalyst SD-WAN Devices, on page 793](#).

Using Cisco SD-WAN Manager CLI templates significantly reduces the effort to configure feature templates.

## Feature Information for CLI Template for Cisco IOS XE Catalyst SD-WAN devices

Table 343: Feature History

| Feature Name                             | Release Information                                        | Description                                                                                                                                                                   |
|------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CLI Template for Cisco XE SD-WAN Routers | Cisco IOS XE Release 16.11.1a<br>Cisco SD-WAN release 19.1 | The CLI Templates for Cisco IOS XE Catalyst SD-WAN device features allows to you configure intent-based CLI templates for Cisco XE SD-WAN routers using Cisco SD-WAN Manager. |
| VRF Configuration                        | Cisco IOS XE Catalyst SD-WAN Release 17.2.1r               | Support for VRF configuration increased from a total of 100 to a total of 300 VRFs. Supported on: Cisco ASR 1001-HX and Cisco ASR 1002-HX                                     |

### Benefits of CLI Templates

- You can reuse any Cisco vEdge-specific Cisco SD-WAN Manager feature templates for Cisco IOS XE Routers. When you create a device template using Cisco XE SDWAN Feature Templates, Cisco SD-WAN Manager displays the intent-based configuration (vEdge CLI syntax) and the corresponding device-based (Cisco XE SDWAN Routers) configuration. You can examine the intent-based configuration and repurpose that to create a separate CLI template for XE SDWAN routers.
- You can make multiple changes to a CLI template in a single edit.
- You can use a single configuration across multiple devices of the same device models. Variables can be used for rapid bulk configuration rollout with unique per-device settings. Common configurations like system-IP, site-id, hostname, IP addresses, and so on, can be defined as editable variables in the template and the same template can be attached to multiple devices.
- You can define custom length for variables in CLI Templates.
- You can use any existing IOS-XE device intent configuration as input for CLI template.
- Content of a CLI template can be used across multiple IOS-XE device types (common CLIs like VPN, VPN interface, BGP, OSPF and so on).

### Limitations

**Auxiliary ports:** When using a CLI template for Cisco Integrated Services Routers that have an auxiliary port, do not include commands for auxiliary ports, such as **line aux 0**. Doing so results in an error. These commands may be executed directly on the device.

### Configuring CLI Templates in Cisco SD-WAN Manager

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and click **Create Template**.



**Note** In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled as **Device**.

3. From the **Create Template** drop-down list, select **CLI Template**.
4. From the **Device Model** drop-down list, select the type of device for which you are creating the template.
5. In **Template Name**, enter a name for the template.  
The name can be up to 128 characters and can contain only alphanumeric characters.
6. In **Template Description**, enter a description of the template.  
The description can be up to 2048 characters and can contain only alphanumeric characters.
7. The configuration of the CLI template can either be intent-based or based on the device configuration.
  - **Intent:** If you specify **Intent**, you specify commands in the Cisco vEdge format. If the device you've selected is a Cisco IOS XE Catalyst SD-WAN device, Cisco SD-WAN Manager converts the configuration for the device.
  - **Device configuration:** This option is available from Cisco IOS XE Catalyst SD-WAN Release 17.2.1r and onwards and only for Cisco IOS XE Catalyst SD-WAN devices. For this option, you must specify the entire device configuration as it appears in `show sd-wan running config`.



**Note** You can only use this feature with the qualified CLIs detailed in [Qualified CLIs for CLI Add-On Feature Templates, on page 822](#).

You can upload a configuration file using **Select a File** or copy and paste the CLI configuration. Following is an example of an intent-based CLI with variables.

```
system

 host-name {{hostname}}
 system-ip {{system_ip}}
 domain-id 1

 site-id {{site_id}}
 port-offset 1
 admin-tech-on-failure
 organization-name "XYZ"
 logging
 disk
 enable
!!
```

These variables can be filled in device variables page per device after attaching the template. Values can be entered manually or can be uploaded via a csv file.

8. To save the feature template, click **Add**.



**Note** See the Attach Devices to a Device Template section in this topic to know more about attaching a device to a template and reusing a template for multiple devices of the same device model.

## Sample Configurations for CLI Template

### System Level Configuration

**Table 344: System Level Parameters**

| CLI Template Configuration                                                                                                                                                                                                                                                | Configuration on the Device                                                                                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> system  host-name pm4  system-ip 172.16.255.14  overlay-id 1  site-id 400  control-session-pps 300  admin-tech-on-failure  sp-organization-name "XYZ Inc Regression" organization-name "XYZ Regression" console-baud-rate 115200 vbond 10.0.12.26 port 12346 </pre> | <pre> system  host-name  pm4 system-ip  172.16.255.14 overlay-id  1 site-id 400 control-session-pps 300 admin-tech-on-failure sp-organization-name "XYZ Inc Regression" organization-name "XYZ Inc Regression" console-baud-rate 11520 vbond 10.0.12.26 port 12346 </pre> |

**AAA Configuration - Authentication, authorization, and accounting (AAA) with RADIUS and TACACS+****Table 345: AAA Configuration**

| CLI Template Configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Configuration on the Device                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> aaa auth- order local radius tacacs usergroup basic  task system read write task interface read write !  usergroup netadmin !  usergroup operator task system read task interface read task policy read task routing read task security read ! user admin password  \$6\$nbblkA==Sae/DO78l/wluPUohhBU2L6h/ Q.PLkurGvxjRlS9OWB9iTtFwSGNQcABV6F MW57vuEHvo3zp3qdYVinLmMIu/p/ secret \$9\$3/IL3/UF2F2F3\$J9NKBKlWrc9ExtHk6F5VAiDMOFQfD.QPmMddcz.c  ! ! radius  server 10.99.144.200 source-interface GigabitEthernet0/0/1 exit server 10.99.144.201  source-interface GigabitEthernet0/1/0 exit ! tacacs  server 10.0.1.1 auth-port 50  vpn 0  source-interface GigabitEthernet0/0/1  key 1  secret-key \$8\$Kcuva0CM871E8czESwV5g/YX4Q8pY1LSNk/+PIDRpCg=  exit ! ! </pre> | <pre> aaa group server tacacs+ server-10.0.1.1 server-private 10.0.1.1 timeout 5 key \$8\$vs5hzVg/Z6EeuUdNHTzOwWPsUv9V/50xmcRfShWp3YI=  ip tacacs source-interface GigabitEthernet0/0/1 ! aaa group server radius server-10.99.144.200  server-private 10.99.144.200 auth-port 1812 timeout 5 retransmit 3 ip radius source-interface GigabitEthernet0/0/1 !  aaa group server radius server-10.99.144.201  server-private 10.99.144.201 auth-port 1812 timeout 5 retransmit 3 ip radius source-interface GigabitEthernet0/1/0 ! aaa authentication login default local group radius group tacacs+ aaa authorization exec default local group radius group tacacs+ a aa session-id common --- added by default  username admin privilege 15 secret 9 \$9\$3/IL3/UF2F2F3\$J9NKBKlWrc9ExtHk6F5VAiDMOFQfD.QPmMddcz.c </pre> |

**Logging configuration - Configures logging to either the local hard drive or a remote host****Table 346: Logging Configuration**

| CLI Template Configuration                                                                                                                                                 | Configuration on the Device                                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> logging   disk     enable     file size 12     file rotate 6   !   server 192.168.13.1     vpn 0     source-interface Loopback1     priority alert   exit   ! </pre> | <pre> logging   disk     enable   !   ! logging persistent size 75497472 filesize 12582912   logging buffered 512000 --- added by default  logging host 192.168.13.1 no logging rate-limit logging source-interface Loopback1 logging persistent </pre> |

**Switch Port and VLAN configuration****Table 347: Switch Port Configuration**

| CLI Template Configuration                                                                                                                                                                                                                                                                                      | Configuration on the Device                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> interface GigabitEthernet0/1/4   switchport   mode trunk   access vlan vlan 10   access vlan name "DHCP Vlan"   trunk allowed vlan 10   !   no shutdown  vpn 10   name "DHCP VPN"   interface Vlan10     description "Vlan 10 Mgmt interface"     ip address 10.29.35.1/24     no shutdown   !   ! </pre> | <pre> interface GigabitEthernet0/1/4   switchport ios-sw:mode trunk   switchport ios-sw:trunk allowed vlan 10   no shutdown   no ip address   exit  interface Vlan10   description Vlan 10 Mgmt interface   no shutdown   arp timeout 1200   vrf forwarding 10   ip address 10.29.35.1 255.255.255.0   ip mtu 1500   exit </pre> |

## Cellular Configuration

**Table 348: Cellular Configuration - Configures cellular controllers and cellular interfaces**

| CLI Template Configuration                                                                                                                                                                                | Configuration on the Device                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>vpn 0   interface Cellular0/2/0     description "Cellular interface"     no shutdown   !  controller cellular 0/2/0   lte sim max-retry 1   lte failovertimer 7   profile id 1 apn Broadband !</pre> | <pre>interface Cellular0/2/0   description Cellular interface   no shutdown   ip address negotiated   ip mtu 1428   mtu 1500   exit  controller Cellular 0/2/0   lte sim max-retry 1   lte failovertimer 7   profile id 1 apn Broadband authentication   none pdn-type ipv4</pre> |

**BGP, OSPF, and EIGRP - Configures BGP, OSPF, and EIGRP Routing Protocols under Transport or Service VPN***Table 349: BGP, OSPF, and EIGRP Configuration*

| CLI Template Configuration | Configuration on the Device |
|----------------------------|-----------------------------|
|----------------------------|-----------------------------|



| CLI Template Configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Configuration on the Device |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <pre> vpn1   bgp 2     shutdown     distance external 30     distance internal 250     distance local 10     address-family ipv4-unicast       network 10.0.100.0/24     redistribute static route-policy   route_map     redistribute connected route-policy   route_map     !     neighbor 10.0.100.1     no shutdown     remote-as 3     timers       keepalive 12       holdtime 20       connect-retry 300       advertisement-interval 123     !     update-source GigabitEthernet0/0/1     ebgp-multihop 1     password \$8\$9pou4PH9b60B072hcw3MmSSdLCfJk8bVys12lLVb+08=      address-family ipv4-unicast  vpn 1   router     ospf       router-id 172.16.255.15       compatible rfc1583       timers spf 200 1000 10000       redistribute connected route-policy     route_map       max-metric router-lsa administrative       area 23       stub       interface GigabitEthernet0/0/1       cost 23       authentication type message-digest       authentication authentication-key key1      exit   exit   !  vpn 1   router     eigrp 1       af-interface GigabitEthernet0/0/2       no split-horizon       exit-af-interface     !     address-family ipv4 network 10.1.10.1/32     address-family ipv4 topology base     redistribute omp     exit-af-topology </pre> |                             |

| CLI Template Configuration | Configuration on the Device                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | <pre> router bgp 2   bgp log-neighbor-changes   distance bgp 30 250 10   address-family ipv4 unicast vrf 1     neighbor 10.0.100.1 remote-as 3     neighbor 10.0.100.1 activate     neighbor 10.0.100.1 ebgp-multihop 1     neighbor 10.0.100.1 maximum-prefix 2147483647 100     neighbor 10.0.100.1 password 0 password     neighbor 10.0.100.1 send-community both     neighbor 10.0.100.1 timers 12 20     neighbor 10.0.100.1 update-source GigabitEthernet0/0/1   network 10.0.100.0 mask 255.255.255.0   redistribute connected   redistribute static route-map route_map   exit-address-family ! timers bgp 60 180  router ospf 1 vrf 1   auto-cost reference-bandwidth 100   max-metric router-lsa   timers throttle spf 200 1000 10000   router-id 172.16.255.15   default-information originate   distance ospf external 110   distance ospf inter-area 110   distance ospf intra-area 110   redistribute connected subnets route-map route_map ! interface GigabitEthernet0/0/1   no shutdown   arp timeout 1200   vrf forwarding 1   ip address 10.1.100.14 255.255.255.0   ip redirects   ip mtu 1500   ip ospf 1 area 23   ip ospf network broadcast   mtu 1500   negotiation auto   exit ! router eigrp eigrp-name   address-family ipv4 vrf 1 autonomous-system 1   af-interface GigabitEthernet0/0/2   hello-interval 5   hold-time 15   no split-horizon exit-af-interface ! network 10.1.10.1 0.0.0.0   topology base   redistribute omp   exit-af-topology ! exit-address-family </pre> |

| CLI Template Configuration | Configuration on the Device |
|----------------------------|-----------------------------|
|                            | !<br>!                      |

## VPN, Interface, and Tunnel Configuration for WAN and LAN interfaces

**Table 350: VPN, Interface, and Tunnel Configuration**

| CLI Template Configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Configuration on the Device                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> vpn 0   interface GigabitEthernet0/2/0     ip address 10.1.14.14/24     tunnel-interface       encapsulation ipsec       color lte       no allow-service bgp       allow-service dhcp       allow-service dns       allow-service icmp       no allow-service sshd       no allow-service netconf       no allow-service ntp       no allow-service ospf       no allow-service stun       allow-service https     !     autonegotiate     no shutdown   !   ip route 0.0.0.0/0 10.1.14.13  vpn 512 interface GigabitEthernet0   ip dhcp-client ipv6 dhcp-client autonegotiate no shutdown ! ! </pre> | <pre> ip route 0.0.0.0 0.0.0.0 10.1.14.13 1  interface GigabitEthernet0/2/0   no shutdown   arp timeout 1200 - added by default   ip address 10.1.14.14 255.255.255.0   ip redirects --&gt; added by default   ip mtu 1500   mtu 1500   negotiation auto --&gt; added by default   exit  interface Tunnel20 ---&gt; based on the interface 0/2/0   no shutdown   ip unnumbered GigabitEthernet0/2/0   no ip redirects   ipv6 unnumbered GigabitEthernet0/2/0   no ipv6 redirects   tunnel source GigabitEthernet0/2/0   tunnel mode sdwan  sdwan   interface GigabitEthernet0/2/0     tunnel-interface       encapsulation ipsec weight 1       color lte       no last-resort-circuit       vmanage-connection-preference 5       no allow-service all       no allow-service bgp       allow-service dhcp       allow-service dns       allow-service icmp       no allow-service sshd       no allow-service netconf       no allow-service ntp       no allow-service ospf       no allow-service stun  interface GigabitEthernet0   no shutdown   arp timeout 1200   vrf forwarding Mgmt-intf   ip address dhcp client-id GigabitEthernet0 ip   redirects   ip dhcp client default-router distance 1 ip   mtu 1500   mtu 1500   negotiation auto </pre> |

**Network Address Translation (NAT) over Direct Internet Access (DIA)****Table 351: NAT over DIA**

| CLI Template Configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Configuration on the Device                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> vpn 201   interface GigabitEthernet0/0/2.2901   description gigi21   ip address 10.201.201.1/24   mtu 1496   no shutdown   vrrp 100     track-omp     ipv4 10.201.201.3   !   !   !   dhcp-server     address-pool 10.201.201.0/24     exclude 10.201.201.1-10.201.201.10     10.201.201.20-10.201.201.22     offer-time 600     lease-time 86400     admin-state up     options       default-gateway 10.201.201.1       dns-servers 10.99.139.201       tftp-servers 10.99.139.201     !     !     !   ip route 0.0.0.0/0 vpn 0   !   vpn 0     interface GigabitEthernet0/0/0     ip address 172.16.10.1/24     nat       udp-timeout 3       tcp-timeout 40       respond-to-ping     !     ! </pre> | <pre> interface GigabitEthernet0/0/2.2901   no shutdown   encapsulation dot1Q 2901   vrf forwarding 201   ip address 10.201.201.1 255.255.255.0   ip mtu 1496   vrrp 100 address-family ipv4     vrrpv2     address 10.201.201.3     priority 100     track omp shutdown   exit   exit  ip dhcp excluded-address vrf 201 10.201.201.1 10.201.201.10 ip dhcp excluded-address vrf 201 10.201.201.20 10.201.201.22 ip dhcp pool vrf-201-GigabitEthernet0/0/2.2901   option 150 ip 10.99.139.201   vrf 201   lease 1 0 0   default-router 10.201.201.1   dns-server 10.99.139.201   network 10.201.201.0 255.255.255.0   exit ip dhcp use hardware-address client-id no ip dhcp use class ip dhcp use vrf remote  ip nat inside source list nat-dia-vpn-hop-access-list interface GigabitEthernet0/0/0 overload   ip nat translation tcp-timeout 40   ip nat translation udp-timeout 3   ip nat route vrf 201 0.0.0.0 0.0.0.0 global  interface GigabitEthernet1/0/2   no shutdown   arp timeout 1200   ip address 10.1.15.15 255.255.255.0   ip nat outside   ip redirects   ip mtu 1500   mtu 1500   negotiation auto </pre> |

## NAT64 Configuration

**Table 352: NAT64 Configuration**

|                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> vpn 1   nat64     v4 pool pool1 start-address 10.1.1.10     v4 pool pool1 end-address 10.1.1.100   !   interface GigabitEthernet3     ip address 10.1.19.15/24     nat64     !     autonegotiate     no shutdown   ! </pre> | <pre> interface GigabitEthernet3   no shutdown   arp timeout 1200   vrf forwarding 1   ip address 10.1.19.15 255.255.255.0   negotiation auto   nat64 enable   nat64 prefix stateful 2001::F/64 vrf 1    nat64 v4 pool pool1 10.1.1.10 10.1.1.100   nat64 v6v4 list global-list pool pool1 vrf   1   nat64 translation timeout tcp 60   nat64 translation timeout udp 1 </pre> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Multilink and T1/E1 - Configures T1/E1 Controller and Serial, Multilink Interfaces

**Table 353: Configuring Multilink**

| CLI Template Configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Configuration on the Device                                                                                                                                                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> card type t1 0 2 controller T1 0/2/0   framing esf   clock source internal   linecode b8zs   cablelength long 0db   channel-group 1 timeslots 15   channel-group 2 timeslots 12   channel-group 3 timeslots 10   channel-group 4 timeslots 10 ! interface Multilink1   no shutdown   encapsulation ppp   ip address 10.1.10.30 255.255.255.0   ppp pap sent-username admin password admin   ppp authentication pap   ppp multilink   ppp multilink links minimum 1   ppp multilink fragment disable   ppp multilink group 1 exit interface Serial0/2/0:1   no shutdown   encapsulation ppp   bandwidth 1536   no ip address   load-interval 30   ppp pap sent-username admin password admin   ppp authentication pap   ppp multilink   ppp multilink group 1 exit </pre> | <pre> interface Multilink1   ip address 10.1.10.30/24 shutdown   controller T1 0/2/0     linecode b8zs     channel-group 1     channel-group 3 ! ppp pap sent-username admin password admin ppp authentication pap ppp multilink ppp multilink group 1 </pre> |

## Local QoS Policy

*Table 354: Local QoS Policy*

| CLI Template Configuration | Configuration on the Device |
|----------------------------|-----------------------------|
|----------------------------|-----------------------------|

| CLI Template Configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Configuration on the Device                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> vpn 1 interface GigabitEthernet0/0/1 ip address 10.2.54.15/24 no shutdown access-list MyACL in ! policy class-map class best-effort queue 3 class bulk-data queue 2 class critical-data queue 1 class voice queue 0 ! access-list MyACL sequence 10 match dscp 46 ! action accept class voice ! ! sequence 20 match source-ip      10.1.1.0/24 destination-ip 192.168.10.0/24 ! action accept class bulk-data set dscp 32 ! ! ! sequence 30 match destination-ip 192.168.20.0/24 ! action accept class critical-data set dscp 22 ! ! ! sequence 40 action accept class best-effort set dscp 0 ! ! ! default-action accept ! qos-scheduler be-scheduler class      best-effort bandwidth-percent 20 buffer-percent 20 drops      red-drop ! qos-scheduler bulk-scheduler </pre> | <pre> interface GigabitEthernet0/0/1 access-list MyACL in exit class-map match-any best-effort match qos-group 3 ! class-map match-any bulk-data match qos-group 2 ! class-map match-any critical-data match qos-group 1 ! class-map match-any voice match qos-group 0 ! policy-map MyQoSMap class best-effort random-detect bandwidth percent 20 ! class bulk-data random-detect bandwidth percent 20 ! class critical-data random-detect bandwidth percent 40 ! class voice priority percent 20 ! ! policy no app-visibility no flow-visibility no implicit-acl-logging log-frequency      1000 class-map class best-effort queue 3 class bulk-data queue 2 class critical-data queue 1 class voice queue 0 ! access-list MyACL sequence 10 match dscp 46 ! action accept class voice ! ! sequence 20 match source-ip      10.1.1.0/24 destination-ip 192.168.10.0/24 ! action accept class bulk-data set dscp 32 ! </pre> |

| CLI Template Configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Configuration on the Device                                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> class                bulk-data bandwidth-percent 20 buffer-percent      20 drops               red-drop ! qos-scheduler critical-scheduler class                critical-data bandwidth-percent 40 buffer-percent      40 drops               red-drop ! qos-scheduler voice-scheduler class                voice bandwidth-percent 20 buffer-percent      20 scheduling          llq ! qos-map MyQoSMap qos-scheduler be-scheduler qos-scheduler bulk-scheduler qos-scheduler critical-scheduler qos-scheduler voice-scheduler ! ! ! ! </pre> | <pre> ! ! sequence 30 match   destination-ip 192.168.20.0/24 ! action accept class critical-data set   dscp 22 ! ! ! sequence 40 action accept class best-effort set   dscp 0 ! ! ! default-action accept ! ! ! ! </pre> |



**Security Policy (ZBFW, IPS/IDS, URL-Filtering) Configuration****Table 355: Security Policy (ZBFW, IPS/IDS, URL-Filtering)**

| CLI Template Configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Configuration on the Device |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <pre> policy   zone internet     vpn 0   !   zone zone1     vpn 1   !   zone zone2     vpn 2   !   zone-pair ZP_zone1_internet_fw_policy     source-zone      zone1     destination-zone internet     zone-policy      fw_policy   !   zone-pair ZP_zone1_zone2_fw_policy     source-zone      zone1     destination-zone zone2     zone-policy      fw_policy   !   zone-based-policy fw_policy     sequence 1     match       source-data-prefix-list subnet1     !     action inspect     !     !     default-action pass   !   zone-to-nozone-internet deny lists   data-prefix-list subnet1     ip-prefix 10.0.10.0/24   !   !   url-filtering url_filter     web-category-action block     web-categories      games     block-threshold      moderate-risk     block text     "&lt;![CDATA[&lt;h3&gt;Access" to the requested     page has been denied]]&gt;"     target-vpns          1   !   intrusion-prevention intrusion_policy     security-level connectivity     inspection-mode protection     log-level          err     target-vpns        1   !   failure-mode          open   !   !   ! </pre> |                             |

| CLI Template Configuration | Configuration on the Device                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | <pre> ip access-list extended fw_policy-seq-1-acl_    11 permit object-group fw_policy-seq-1-service-og_ object-group subnet1 any ! ip access-list extended utd-nat-acl   10 permit ip any any ! class-map type inspect match-all fw_policy-seq-1-cm_   match access-group name fw_policy-seq-1-acl_ ! policy-map type inspect fw_policy   class fw_policy-seq-1-cm_     inspect   !   class class-default     pass   ! ! object-group service fw_policy-seq-1-service-og_   ip ! parameter-map type inspect-global   alert on   log dropped-packets   multi-tenancy   vpn zone security ! parameter-map type umbrella global   token A5EA676087BF66A42DC4F722C2AFD10D00256274 dnscrypt vrf 1   dns-resolver                      umbrella   match-local-domain-to-bypass ! ! zone security internet   vpn 0 ! zone security zone1   vpn 1 ! zone security zone2   vpn 2 ! zone-pair security ZP_zone1_internet_fw_policy source zone1 destination internet   service-policy type inspect fw_policy ! zone-pair security ZP_zone1_zone2_fw_policy source zone1 destination zone2   service-policy type inspect fw_policy ! app-hosting appid utd app-resource package-profile cloud-low app-vnic gateway0 virtualportgroup 0 </pre> |

| CLI Template Configuration | Configuration on the Device                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | <pre> guest-interface 0     guest-ipaddress 192.168.1.2 netmask 255.255.255.252 !     app-vnic gateway1 virtualportgroup 1 guest-interface 1     guest-ipaddress 192.0.2.2 netmask 255.255.255.252 !     start !     utd multi-tenancy     utd engine standard multi-tenancy     web-filter block page profile block-url_filter     text &lt;![CDATA[&lt;h3&gt;Access to the requested page has been denied&lt;/h3&gt;&lt;p&gt;Please contact your Network Administrator&lt;/p&gt;]]&gt; !     web-filter url profile url_filter     categories block     games !     block page-profile block-url_filter     log level error     reputation     block-threshold moderate-risk ! ! threat-inspection profile intrusion_policy      threat protection     policy connectivity     logging level err ! utd global ! policy utd-policy-vrf-1     all-interfaces     vrf 1     threat-inspection profile intrusion_policy      web-filter url profile url_filter exit ! </pre> |

## Configuring NTP

**Table 356: Configuring NTP**

| CLI Template Configuration                                                                      | Configuration on the Device                                            |
|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| <pre>ntp   server 10.29.43.1     source-interface GigabitEthernet1     version 4   exit !</pre> | <pre>ntp server 198.51.241.229 source GigabitEthernet1 version 4</pre> |

## IPv6 Configuration

**Table 357: IPv6 Configuration**

| CLI Template Configuration                                                                         | Configuration on the Device                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>vpn 1   interface GigabitEthernet3     ipv6 address 2671:123A::1/128     shutdown   ! !</pre> | <pre>interface GigabitEthernet3   shutdown   arp timeout 1200   vrf forwarding 1   no ip address   ip redirects   ip mtu 1500   ipv6 address 2671:123A::1/128   ipv6 redirects   mtu 1500   negotiation auto   exit   vrf definition 1     rd 1:1     address-family ipv4       exit-address-family     !     address-family ipv6       exit-address-family     !   !</pre> |

## Service Configuration

In Cisco IOS XE Catalyst SD-WAN Release 17.7.1a and earlier, only the following configurations under **service** can be configured via CLI templates:

```
service pad
service config
service tcp-keepalives-in
service tcp-keepalives-out
service tcp-small-servers
service udp-small-servers
```

## VRF Configuration

Configure up to 300 VRFs, with a corresponding subinterface for each VRF. The example configures two VRFs.



---

**Note** Do not configure VLAN 1. It is reserved for the native VLAN.

---

| CLI Template Configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Configuration on the Device |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <pre> ! vpn 2 router bgp 1000   address-family ipv4-unicast     redistribute omp   address-family ipv6-unicast     redistribute omp   !   neighbor 192.0.2.2     no shutdown     remote-as 2   !   ipv6-neighbor 2001:DB8:2::2     remote-as 2   ! ! interface GigabitEthernet0/0/0.2   ip address 192.0.2.1/24   ipv6 address 2001: DB8:2::1/64   mtu 1496   no shutdown ! ! vpn 3 router bgp 1000   address-family ipv4-unicast     redistribute omp   address-family ipv6-unicast     redistribute omp   !   neighbor 192.0.3.2     no shutdown     remote-as 3   !   ipv6-neighbor 2001: DB8:3::2     remote-as 3   ! ! interface GigabitEthernet0/0/0.3   ip address 192.0.3.1/24   ipv6 address 2001: DB8:3::1/64   mtu 1496   no shutdown ! </pre> |                             |

| CLI Template Configuration | Configuration on the Device                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | <pre> vrf definition 2 rd 1:2 address-family ipv4   route-target export 1000:2   route-target import 1000:2   exit-address-family ! address-family ipv6   exit-address-family ! ! router bgp 1000 bgp log-neighbor-changes distance bgp 20 200 20 ! address-family ipv4 vrf 2   redistribute omp   neighbor 192.0.2.2 remote-as 2   neighbor 192.0.2.2 activate   neighbor 192.0.2.2 send-community both exit-address-family ! address-family ipv6 vrf 2   redistribute omp   neighbor 2001:DB8:2::2 remote-as 2   neighbor 2001: DB8:2::2 activate   neighbor 2001: DB8:2::2 send-community both exit-address-family ! interface GigabitEthernet0/0/0.2 encapsulation dot1Q 2 vrf forwarding 2 ip address 192.0.2.1 255.255.255.0 ip mtu 1496 ipv6 address 2001:DB8:2::1/64 end  vrf definition 3 rd 1:3 address-family ipv4   route-target export 1000:3   route-target import 1000:3   exit-address-family ! address-family ipv6   exit-address-family ! ! router bgp 1000 bgp log-neighbor-changes distance bgp 20 200 20 ! address-family ipv4 vrf 3   redistribute omp   neighbor 192.0.3.2 remote-as 3   neighbor 192.0.3.2 activate   neighbor 192.0.3.2 send-community both exit-address-family ! address-family ipv6 vrf 3   redistribute omp   neighbor 2001:DB8:3::2 remote-as 3 </pre> |

| CLI Template Configuration | Configuration on the Device                                                                                                                                                                                                                                            |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | <pre>neighbor 2001: DB8:3::2 activate neighbor 2001: DB8:3::2 send-community both exit-address-family ! interface GigabitEthernet0/0/0.3 encapsulation dot1Q 3 vrf forwarding 3 ip address 192.0.3.1 255.255.255.0 ip mtu 1496 ipv6 address 2001:DB8:3::1/64 end</pre> |





## CHAPTER 32

# CLI Add-On Feature Templates

Table 358: Feature History Table

| Feature Name                                                   | Release Information                                                    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CLI Add-On Feature Templates                                   | Cisco IOS XE Catalyst SD-WAN Release 17.2.1r<br>Cisco vManage 20.1.1   | <p>This feature adds a new feature template called the CLI add-on feature template. You can use this feature template to attach specific CLI configurations to a device. If a configuration cannot be specified using Cisco SD-WAN Manager but can be configured using the CLI on the device, then you can use this feature template to specify such configurations. You can also use CLI add-on feature templates to add small pieces of CLI configuration, instead of an entire running configuration.</p> <p>This feature is not intended to replace existing feature templates but instead to enhance their functionality. Note that not all CLIs are qualified. For more information, see <a href="#">Qualified CLIs for Cisco IOS XE Release 17.2.1r</a>.</p> |
| Additional Commands Qualified for CLI Add-On Feature Templates | Cisco IOS XE Release Amsterdam 17.2.1v<br>Cisco SD-WAN Release 20.1.12 | <p>With each release, we qualify commands for use with the CLI add-on feature templates feature. In this release, we qualified additional commands. See the <a href="#">Appendix</a> in Cisco IOS XE SD-WAN Qualified Command Reference.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

- [Overview of the CLI Add-On Feature Templates, on page 820](#)

- [Restrictions for CLI Add-On Feature Templates, on page 820](#)
- [Create a CLI Add-On Feature Template, on page 821](#)
- [Qualified CLIs for CLI Add-On Feature Templates, on page 822](#)

## Overview of the CLI Add-On Feature Templates

If you attach a device template containing both a feature template and the new CLI add-on feature template, the configurations are merged. The merge gives priority to the new CLI add-on feature templates. Cisco SD-WAN Manager first generates the configurations based on the feature template. After the configuration is generated, it uses the configuration from the CLI add-on feature templates to merge it into the feature template config output that was previously generated. Hence, using this feature, you can add specific device configurations that are not provided by the existing feature templates or you can override the configurations of existing feature templates.

When you specify commands using the template, use the commands as per the syntax displayed in the `show sdwan running-config` output. When you attach the template to the device, Cisco SD-WAN Manager takes the information from all feature templates and also takes the data you specified using the CLI add-on feature template to create the device configuration. The commands that you specify in the CLI add-on feature template overwrites any equivalent commands in the corresponding feature template.

In addition to changing existing commands, the CLI add-on feature template can also be used to specify commands that are not available in Cisco SD-WAN Manager but are qualified for the device. For example, for Cisco AAA, the `attempts login` command is not available in Cisco SD-WAN Manager. By using a CLI add-on feature template, you can specify the `aaa authentication attempts login number` command for a device. After you create the feature template, ensure that you add it to the device template.



---

**Note** You must define the CLI add-on feature template before you use it in a device template.

For a list of CLIs that are qualified, see [Qualified CLIs for CLI Add-on Feature Templates](#).

---

## Restrictions for CLI Add-On Feature Templates

The following restrictions apply when using the CLI add-on feature templates:

- This feature is only supported on Cisco IOS XE Catalyst SD-WAN devices running Cisco IOS XE Catalyst SD-WAN Release 17.2.1r or onwards.
- Only one CLI add-on template can be attached per device template.
- Ensure that you only use configuration commands as they appear in the output of the `show sdwan running-config` command. Before using a command in the CLI add-on feature template, verify the command by logging in and running it on the intended device.
- Unsupported commands in your configuration cause errors and results in a failure when pushing the configuration to the device. For example, "login local" is an unsupported command.

For a release-wise list of commands qualified for use in the CLI add-on feature template, see [Qualified CLI Commands for CLI Add-on Feature templates](#).

# Create a CLI Add-On Feature Template

To create a CLI add-on feature template, do the following:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**, and click **Add Template** to select an appropriate device model..




---

**Note** In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled as **Feature**.

---

3. From **Select Devices**, select the devices for which you are creating the template.
4. From **Select Template**, scroll down to the **OTHER TEMPLATES** section.
5. Click **CLI Add-On Template**.
6. In **Template Name**, enter a name for the feature template.  
This field is mandatory and can contain only uppercase and lowercase letters, the digits 0 through 9, hyphens (-), and underscores (\_). It cannot contain spaces or any other characters.
7. In **Description**, enter a description for the device template.  
This field is mandatory, and it can contain any characters and spaces.
8. In **CLI Configuration**, enter the configuration either by typing it, cutting and pasting it, or uploading a file.
9. To convert an actual configuration value to a variable, select the value and click **Create Variable**. Enter the variable name, and click **Create Variable**. You can also type the variable name directly, in the format `{{variable-name}}`. For example: `{{hostname}}`.
10. Click **Save**.  
The new feature template is displayed in the Feature Template table.
11. To use the CLI add-on feature template, edit the device template as follows:
  - a. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
  - b. Click **Device Templates**.




---

**Note** In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled as **Device**.

---

- c. Select the device template for which you want to add the CLI add-on feature template.
- d. Click ..., and choose **Edit**.
- e. Scroll to **Additional Templates**.
- f. From **CLI Add-On Template**, select the CLI add-on feature template that you previously created.
- g. Click **Update**.



---

**Note** In Cisco IOS XE Catalyst SD-WAN Release 17.7.x, while creating a CLI template, if the following CLIs are visible in the template, then ensure that you manually delete the CLIs from the template before attaching the template to the device:

licensing config enable false

licensing config privacy hostname false

licensing config privacy version false

licensing config utility utility-enable false

---

## Qualified CLIs for CLI Add-On Feature Templates

For a release-wise list of CLI commands that are qualified for use in Cisco SD-WAN Manager CLI templates, see the [Appendix](#) in Cisco IOS XE SD-WAN Qualified Command Reference.



## CHAPTER 33

# Cisco Catalyst SD-WAN EtherChannel

Table 359: Feature History

| Feature Name                       | Release Information                                                          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cisco Catalyst SD-WAN EtherChannel | Cisco IOS XE Catalyst SD-WAN Release 17.6.1a<br>Cisco vManage Release 20.6.1 | <p>This feature allows you to configure EtherChannels on Cisco IOS XE Catalyst SD-WAN devices on the service side.</p> <p>An EtherChannel provides fault-tolerant high speed link, redundancy, and increased bandwidth between Cisco IOS XE Catalyst SD-WAN devices and other devices such as routers, switches, or servers connected in a network.</p> <p>You can configure EtherChannels only using the CLI device templates and CLI add-on feature templates.</p> |

| Feature Name                                                 | Release Information                                                                                       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EtherChannels on the Transport Side                          | <p>Cisco IOS XE Catalyst SD-WAN Release 17.13.1a</p> <p>Cisco Catalyst SD-WAN Manager Release 20.13.1</p> | <p>Adds support for configuring EtherChannels on the transport side of a Cisco IOS XE Catalyst SD-WAN device.</p> <p>This feature also introduces support for aggregate EtherChannel Quality of Service (QoS) on the transport side.</p> <p>By combining EtherChannel and QoS, you can optimize network utilization, enhance performance, and maintain quality for specific traffic types.</p> <p><b>Note</b><br/>This feature has limited availability.</p> |
| Load Balancing for EtherChannels on the Transport Side       | <p>Cisco IOS XE Catalyst SD-WAN Release 17.14.1a</p> <p>Cisco Catalyst SD-WAN Manager Release 20.14.1</p> | This feature adds the ability to configure load balancing for EtherChannels on the transport side for Cisco IOS XE Catalyst SD-WAN devices.                                                                                                                                                                                                                                                                                                                  |
| Configure EtherChannels using Configuration Groups           | <p>Cisco IOS XE Catalyst SD-WAN Release 17.15.1a</p> <p>Cisco Catalyst SD-WAN Manager Release 20.15.1</p> | With this feature you can configure EtherChannels on service and transport side using configuration groups.                                                                                                                                                                                                                                                                                                                                                  |
| Load Balancing for EtherChannels on Individual Port Channels | <p>Cisco IOS XE Catalyst SD-WAN Release 17.15.1a</p> <p>Cisco Catalyst SD-WAN Manager Release 20.15.1</p> | With this feature you can load balance EtherChannels for individual port channels on service and transport side using CLI templates.                                                                                                                                                                                                                                                                                                                         |

- [Information About Cisco Catalyst SD-WAN EtherChannel, on page 824](#)
- [Benefits of Cisco Catalyst SD-WAN EtherChannel, on page 829](#)
- [EtherChannels on the Service Side, on page 829](#)
- [EtherChannels on the Transport Side, on page 836](#)
- [Monitor Configured EtherChannel Using CLI, on page 841](#)
- [Aggregate EtherChannel Quality of Service, on page 842](#)

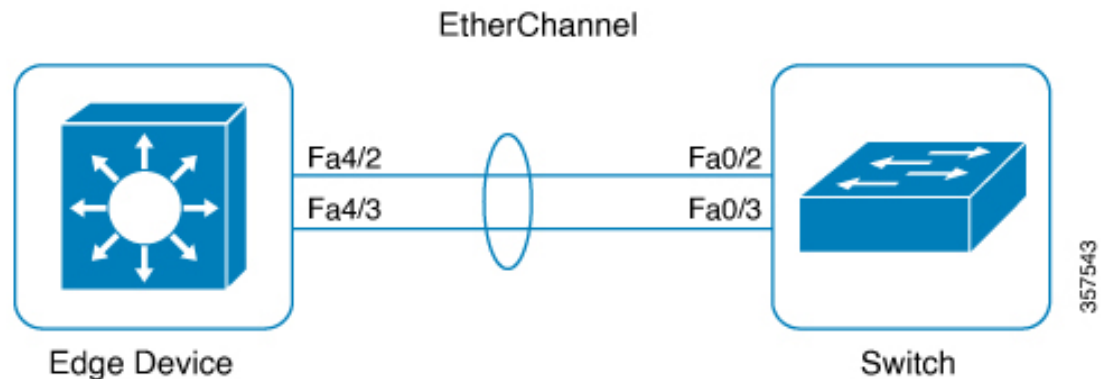
## Information About Cisco Catalyst SD-WAN EtherChannel

An EtherChannel provides fault-tolerant high-speed links between switches, routers, and servers. You can use the EtherChannel to increase bandwidth between the wiring closets and the data center, and also deploy it at any place in a network where bottlenecks are likely to occur. An EtherChannel provides automatic recovery

for the loss of a link by redistributing the load across the remaining links. If a link fails, an EtherChannel redirects traffic from the failed link to the remaining links in the channel.

An EtherChannel comprises a channel group and a port-channel interface. The channel group binds physical ports to the port-channel interface. Configuration changes applied to the port-channel interface apply to all the physical ports bound together in the channel group.

**Figure 4: EtherChannel**



- Using EtherChannels in a network provides increased bandwidth and resilience.
  - **Bandwidth:** An EtherChannel allows multiple links to be combined into one logical link. Because an EtherChannel offers redundancy of links, you can configure EtherChannels to increase the speed in a network.
  - **Resilience:** An EtherChannel also provides network resilience. Even if a link within an EtherChannel fails, traffic that is previously carried over the failed link switches to the remaining links within the EtherChannel. Thus, EtherChannel provides automatic recovery for the loss of a link by redistributing the load across the remaining links.
- The number of supported port channels differs based on the specific device model.
- The number of supported member interfaces for a port channel differs based on the specific device model.
- EtherChannel supports the following combinations:
  - Two active links
  - Active and passive links
  - Single member link
  - Loopback interface in bind or unbind mode to the port channel

Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a and Cisco Catalyst SD-WAN Manager Release 20.13.1, EtherChannels configured on the transport side support the following:

- Control and management connections (DTLS, OMP) to Cisco Catalyst SD-WAN Manager, Cisco Catalyst SD-WAN Validator, and Cisco Catalyst SD-WAN Controller
- IPSEC tunnels for data traffic
- IPv4 forwarding

- L2 TLOC extension
- Explicit ACL (Access Control Lists)
- Implicit ACL on a port channel TLOC
- IPv4 static routing
- Loopback TLOC (ability to bind loopback to port channel)
- Port channel sub-interfaces
- Control policies on Cisco Catalyst SD-WAN Controller

Starting from Cisco IOS XE Catalyst SD-WAN Release 17.14.1a and Cisco Catalyst SD-WAN Manager Release 20.14.1, EtherChannels configured on the transport side support the following:

- IPv6 for EtherChannels, allowing for the transmission of IPv6 traffic across aggregated links.
- Handling traffic through Generic Routing Encapsulation (GRE) tunnels, facilitating the encapsulation of various network protocols.
- Advanced routing protocols such as OSPF and BGP over EtherChannels enables dynamic routing in Cisco Catalyst SD-WAN.
- NAT-DIA across EtherChannels, providing direct internet access by converting private IP addresses to public ones for efficient internet-bound traffic routing. For more information, see [Configure NAT](#).

### EtherChannel in Cisco Catalyst SD-WAN

To create an EtherChannel, begin by configuring a port channel. A port channel is a logical interface on a Cisco IOS XE Catalyst SD-WAN device. After you create an EtherChannel, the configuration changes that are applied to the port-channel interface are also applied to all the physical ports assigned to the port-channel interface.

The maximum number of interfaces that can be combined into a single EtherChannel using LACP is eight, although the actual limit may depend on the specific model of the device.

You can configure an EtherChannel using one these methods:

- Link Aggregation Control Protocol (LACP) mode
- Static mode

Use the LACP mode to configure an EtherChannel if it is supported on both ends of a device. If either of the device does not support LACP mode, use a static mode to configure an EtherChannel.

### LACP Mode

LACP facilitates the automatic creation of EtherChannels by exchanging LACP packets between the Ethernet ports.

This table shows the user-configurable EtherChannel LACP modes.



Table 360: EtherChannel LACP Modes

| Mode           | Description                                                                                                                                                                                                   |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>active</b>  | Places a port in an active negotiating state in which the port starts negotiations with other ports by sending LACP packets.                                                                                  |
| <b>passive</b> | Places a port in a passive negotiating state in which the port responds to the packets that it receives, but does not start LACP packet negotiation. This setting minimizes the transmission of LACP packets. |

Both the **active** and **passive** modes enable ports to negotiate with partner ports based on port speed.

Ports can form an EtherChannel when they are in different LACP modes as long as the modes are compatible. For example:

- A port in the **active** mode can form an EtherChannel with another port that is in the **active** or **passive** mode.
- A port in the **passive** mode cannot form an EtherChannel with another port that is also in the **passive** mode because neither port starts LACP negotiation.

In addition to the standard LACP configuration, the following LACP-related commands are supported:

- **lacp min-bundle**
- **lacp max-bundle**
- **lacp system-priority**
- **lacp port-priority**
- **lacp fast-switchover**
- **lacp rate fast**

### Static Mode

You can manually create an EtherChannel by using the **interface port-channel** command in the global configuration mode. You then use the **channel-group interface** command in the global configuration mode to assign an interface to the EtherChannel. After you configure an EtherChannel, the configuration changes applied to the port-channel interface are applied to all the physical ports assigned to the port-channel interface. Unlike an LACP mode, in a static mode, no packets are sent for negotiations with the other ports. Instead, you must manually configure the ports as part of an EtherChannel.

Information related to LACP on port-channel interfaces can be obtained using the **show lacp** command. See [show lacp](#).

### EtherChannel Load Balancing

An EtherChannel balances traffic load across the links in a channel. You can specify one of several different load-balancing modes. EtherChannels can use either dynamic flow-based load balancing or virtual LAN (VLAN) manual load balancing.

You can configure the load-balancing method globally for all the port channels or directly on specific port channels. The global configuration applies only to those port channels for which you have not explicitly configured load balancing. The port-channel configuration overrides the global configuration.

The following load-balancing methods are supported on Cisco IOS XE Catalyst SD-WAN devices:

- Flow-Based
- VLAN-Based

### Flow-Based Load Balancing

Flow-based load balancing is the default load-balancing method, and is enabled by default at the global level. Flow-based load balancing identifies different flows of traffic based on the key fields in the data packet. For example, IPv4 source and destination IP addresses can be used to identify a flow. The various data traffic flows are then mapped to the different member links of a port channel. After the mapping is done, the data traffic for a flow is transmitted through the assigned member link. The flow mapping is dynamic and changes when there is any change in the state of a member link to which a flow is assigned. The flow mapping is dynamic when member links are added or deleted.

### VLAN-Based Load Balancing

VLAN-based load balancing allows you to configure static assignment of user traffic, as identified by a VLAN ID, to a given member link of an EtherChannel. You can manually assign VLAN subinterfaces to a primary and secondary link. This feature allows load balancing to downstream equipment regardless of vendor equipment capabilities, and provides failover protection by redirecting traffic to the secondary member link if the primary link fails. Member links are supported with up to 16 bundles per chassis.

### EtherChannels Load Balancing on the Transport Side of Cisco IOS XE Catalyst SD-WAN Devices

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.14.1a and Cisco Catalyst SD-WAN Manager Release 20.14.1.

Load balancing for EtherChannels on the transport side is achieved by using the inner IP headers, which include the original source and destination IP addresses found in encapsulated packets. Cisco IOS XE Catalyst SD-WAN devices use a hash algorithm to analyze the inner IP addresses for distribution of network traffic across available paths.

Configure load balancing for EtherChannels on the transport side using the **port-channel load-balance-hash-algo sdwan** command. With load balancing configured, a router distributes network traffic among all available paths within the EtherChannel. By default, **sdwan** uses the inner packet source and destination IP address.

## Information About Configuring EtherChannels using Configuration Groups

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.15.1a and Cisco Catalyst SD-WAN Manager Release 20.15.1

You can configure port channel interfaces and member links using configuration groups in Cisco SD-WAN Manager.

## Load Balancing on the Transport Side for Individual Port Channels

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.15.1a and Cisco Catalyst SD-WAN Manager Release 20.15.1

You can enable load balancing on per port channel in the interface using the **load-balance-hash-algo sdwan** command.

## Benefits of Cisco Catalyst SD-WAN EtherChannel

- Provides fault-tolerance. If any one of the links in an EtherChannel fail, the EtherChannel automatically redistributes traffic across the remaining links.
- Helps increase bandwidth between Cisco IOS XE Catalyst SD-WAN devices and other devices such as switches and servers that are connected in a network.

## EtherChannels on the Service Side

### Supported Devices for Cisco Catalyst SD-WAN EtherChannel on the Service Side

#### Service Side

The following platforms support EtherChannel and also offer load balancing for EtherChannel on the service side:

- Cisco 4000 Series Integrated Services Routers
  - Cisco 4451-X Integrated Services Router
  - Cisco 4461 Integrated Services Router
  - Cisco 4431 Integrated Services Router
  - Cisco 4331 Integrated Services Router
  - Cisco 4351 Integrated Services Router
- Cisco ASR 1000 Series Aggregation Services Routers
  - Cisco ASR 1001-X Router
  - Cisco ASR 1006-X Router
  - Cisco ASR 1001-HX Router
  - Cisco ASR 1002-HX Router
  - Cisco ASR 1002-X Router
- Cisco Catalyst 8000V Edge Software
- Cisco Catalyst 8200 Router
- Cisco Catalyst 8300 Router
- Cisco Catalyst 8500 Series Edge Router

**Supported NIMs**

Any L3 Ethernet interface on Network Interface Modules (NIMs) or Service Modules (SMs) can support EtherChannels on the service side.




---

**Note** Network Interface Modules (NIMs) with L2 ports do not support EtherChannels on the service side.

---

## Prerequisites for Cisco Catalyst SD-WAN EtherChannel on the Service Side

- All the LAN ports in each EtherChannel must be of the same speed.
- All the LAN ports must be configured on Layer 3 service-side ports.
- All member interfaces in a portchannel must have the same speed and duplex, when using platforms that support multiple rate SFPs on the same port.

## Restrictions for Cisco Catalyst SD-WAN EtherChannel on the Service Side

- The maximum number of port channel interfaces that a device can support varies, depending on the particular model of the device.
- You can configure EtherChannels on a device by using the CLI, or using only the CLI templates or CLI add-on feature templates in Cisco SD-WAN Manager.
- Network Interface Modules (NIMs) with L2 ports do not support EtherChannels on the service side.
- The EtherChannel Quality of Service (QoS) feature on port channels is not supported on the service side.
- The Aggregate EtherChannel QoS feature on port channels is not supported on the service side.
- An EtherChannel does not support Digital Signal Processor (DSP) farm services and voice services.
- Sub interfaces cannot be added as member of EtherChannel.

## Configure Load Balancing for EtherChannels on the Service Side Using CLI Commands

For more information about using CLI templates, see [CLI Templates](#).



- 
- Note**
- From Cisco Catalyst SD-WAN Manager Release 20.15.1, you can configure any other hash algorithms for flow-based load balancing on per port-channel interface on the service side.
  - Load balancing uses a flow-based method by default, with the default hash algorithm being **src-dst-ip**.
  - The Hash Algorithms For Flow-Based Load Balancing feature is supported on Cisco Aggregation Services Routers platforms, and Cisco Catalyst Router platforms, where the hardware load-balancing for Etherchannel is supported. This command is not supported on Cisco Integrated Services Routers.
-

## Enable Load Balancing on an Individual Port Channel

1. Enter the port channel interface configuration mode.

```
interface Port-channel channel-number
```

2. Enable load balancing on an individual port channel.

```
load-balancing flow
```

This example shows how to set the load-balancing method to flow, when VLAN-manual method is configured globally:

```
Device# config-transaction
Device(config)# interface port-channel 1
Device(config-if)# load-balancing flow
```

This example shows how to set the load-balancing method to VLAN:

```
Device# config-transaction
Device(config)# interface port-channel 1
Device(config-if)# load-balancing vlan
```

This example shows a configuration where flow-based load balancing is configured on port channel 2 while the VLAN-manual method is configured globally:

```
port-channel load-balancing vlan-manual
interface Port-channel2
 ip address 10.0.0.1 255.255.255.0
 load-balancing flow

interface GigabitEthernet2/1/0
 no ip address
 channel-group 2

interface GigabitEthernet2/1/1
 no ip address
 channel-group 2
```

This example shows configuration for VLAN when the load balancing is set to default on the global level:

```
port-channel load-balancing vlan-manual

interface Port-channel1
interface Port-channel1.100
 encapsulation dot1Q 100 primary GigabitEthernet 1/1/1
 secondary GigabitEthernet 1/2/1
 ip address 10.16.2.100 255.255.255.0

interface Port-channel1.200
 encapsulation dot1Q 200 primary GigabitEthernet 1/2/1
 ip address 10.16.3.200 255.255.255.0
interface Port-channel1.300
 encapsulation dot1Q 300
 ip address 10.16.4.300 255.255.255.0

interface GigabitEthernet 1/1/1
 no ip address
 channel-group 1!
interface GigabitEthernet 1/2/1
 no ip address
 channel-group 1
```



**Note** Interface 1 and interface 2 must be member ports of a port channel when **encapsulation dot1q** is configured.

## Enable Hash Algorithms for Flow-Based Load Balancing on a Global level

To configure specific flow-based hash algorithms on a global level use:

**port-channel load-balance-hash-algo** *hash-algo*

This example shows configuration for enabling a hash algorithm on a global level flow-based load balancing:

```
device(config)# port-channel load-balance-hash-algo src-mac
```

## Enable Flow-Based Load Balancing on an Individual Port Channel Interface

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.15.1a and Cisco Catalyst SD-WAN Manager Release 20.15.1

1. Enter the port channel interface configuration mode.

```
interface Port-channel channel-number
```

2. Enable flow-based load balancing hash algorithm.

```
load-balance-hash-algo dst-ip
```

This example shows configuration of hash algorithms for flow-based load balancing on an individual port channel interface. When **sdwan** hash algorithm is configured on the transport side, you can enable different hash algorithm options on the service side.

```
device(config)# interface Port-channel 1
device(config-if)# load-balance-hash-algo sdwan
device(config-if)# exit
device(config)# interface Port-channel 2
device(config-if)# load-balance-hash-algo src-dst-mixed-ip-port
device(config-if)# exit
device(config)# interface Port-channel 3
device(config-if)# no shut
device(config-if)# commit
device(config-if)# end
```

## Enable VLAN Load Balancing Per Port Channel on the Service Side

1. Enter the port channel interface configuration mode.

```
interface Port-channel channel-number
```

2. Enable vlan on per port channel.

```
load-balancing vlan
```

This example shows configuration for VLAN load balancing on the service side, when the flow-based load balancing is set to default on the global level:

```
interface Port-channel 1
load-balancing vlan
```

## Configure a Service-Side Port Channel and Member Link Interface using Configuration Groups in Cisco SD-WAN Manager

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.
2. Under **Service Profile**, click **Add New** to add a new service profile or select any existing service profiles, and click **Edit**.
3. Select **+**, and click on the **Ethernet Interface** to add a new interface. See [Ethernet Interface](#) for more details.
4. Select **Add New** from the Ethernet Interface drop-down menu.
5. Under **Basic Configurations**, enter the interface name and description.
6. Click **Ether Channel**, and assign the EtherChannel to a **member interface** or a **port channel** interface from the drop down menu.

If you configure the EtherChannel interface as a port channel, then the default port is LACP. You can assign a different port channel mode using the **Port Channel Mode** option from Cisco SD-WAN Manager.




---

**Note** You can create multiple interfaces which can be either member or port channel interfaces.

---

7. Click **Save**.
8. Click **Configuration** tab, and deploy the newly created port channels.

## Configure Load Balancing for EtherChannels on the Service Side Using CLI Commands

For more information about using CLI templates, see [CLI Add-On Feature Templates](#) and [CLI Templates](#).

### Enable Flow Based Load Balancing on a Global Level




---

**Note** The default hash algorithm for flow-based load balancing is **src-dst-ip**.

---

#### Port-channel load-balancing flow

#### Enable Flow Based Load Balancing Per Port Channel

```
interface Port-channel channel-number
load-balancing flow
```

#### Enable Hash Algorithms for Flow based Load Balancing for Each Portchannel Interface

```
interface Port-channel 1
load-balance-hash-algo [src-dst-ip dst-ip | dst-mac | src-dst-ip |
src-dst-mac | src-dst-mixed-ip-port | src-ip | src-mac]
```

**Enable Hash Algorithms for Flow based Load Balancing on a Global level**

```
port-channel load-balance-hash-algo {dst-ip | dst-mac | src-dst-ip |
src-dst-mac | src-dst-mixed-ip-port | src-ip | src-mac}
```



**Note** From Cisco Catalyst SD-WAN Manager Release 20.15.1, you can use any other hash algorithms for load balancing on the service side.

The **Hash Algorithms For Flow-based Load Balancing** feature is supported only on Cisco Aggregation Services Routers platforms, where the hardware load-balancing for Etherchannel is supported. This command is not supported on Cisco Integrated Services Routers and Cisco Catalyst Router platforms.

This example shows a configuration where flow-based load balancing is configured on port channel 2 while the VLAN manual method is configured globally:

```
!
port-channel load-balancing vlan-manual

.
.
.
interface Port-channel2
 ip address 10.0.0.1 255.255.255.0
 no negotiation auto
 load-balancing flow
!

interface GigabitEthernet2/1/0
 no ip address
 negotiation auto
 cdp enable
 channel-group 2
!
interface GigabitEthernet2/1/1
 no ip address
 negotiation auto
 cdp enable
 channel-group 2
!
```

This example shows a configuration for each port channel interface where port-channel1 uses **sdwan** hash algorithm for the transport side, Port-channel2 uses the **src-dst-mixed-ip-port** for the service side, and Port-channel3 uses the globally default **src-dst-ip** hash algorithm for the service side. :

```
device(config)# interface Port-channel 1
device(config-if)# load-balance-hash-algo sdwan
device(config-if)# exit

device(config)# interface Port-channel 2
device(config-if)# load-balance-hash-algo src-dst-mixed-ip-port
device(config-if)# exit

device(config)# interface Port-channel 3
device(config-if)# no shut
device(config-if)# commit
device(config-if)# end
```



The following is a sample output to view the configuration for each interface port channel using **show etherchannel load-balancing** command.

```
device# show etherchannel load-balancing
flow-based
LB Algo type: Source Destination IP

Port-Channel: LB Method
Port-channel1 : flow-based (SDWAN Inner packet LB)
Port-channel2 : flow-based (Source Destination Port, IP addr)
Port-channel3 : flow-based (Source Destination IP)
```

### Manual Traffic Distribution Based on VLAN ID

**port-channel load-balancing vlan-manual**



**Note** This command is available for configuration in the global configuration mode, and applies to all the port-channel configured on the device.

This example shows how the load-balancing configuration can be globally applied to define policies for handling traffic by using the **port-channel load-balancing** command.

```
port-channel load-balancing vlan-manual

!
interface Port-channel1
!
interface Port-channel1.100
 encapsulation dot1Q 100 primary GigabitEthernet 1/1/1
 secondary GigabitEthernet 1/2/1
 ip address 10.16.2.100 255.255.255.0
!
interface Port-channel1.200
 encapsulation dot1Q 200 primary GigabitEthernet 1/2/1
 ip address 10.16.3.200 255.255.255.0
!
interface Port-channel1.300
 encapsulation dot1Q 300
 ip address 10.16.4.300 255.255.255.0
!
interface GigabitEthernet 1/1/1
 no ip address
 channel-group 1!
interface GigabitEthernet 1/2/1
 no ip address
 channel-group 1
```

### Enable VLAN Load Balancing Per Port Channel on the Service Side

```
interface Port-channel channel-number
load-balancing vlan
```

This example shows configuration for VLAN load balancing on the service side, when the flow-based load balancing is set to default on the global level:

```
interface Port-channel channel-number
interface GigabitEthernet slot/subslot/port
channel-group channel-group-number
```

```
interface GigabitEthernet slot/subslot/port
channel-group channel-group-number
interface Port-channel channel-number
load-balancing vlan
interface Port-channel channel-number
encapsulation dot1q vlan_id primary interface1 secondary interface2
```



---

**Note** Interface 1 and interface 2 must be member ports of a port channel when **encapsulation dot1q** is configured.

---

## EtherChannels on the Transport Side

### Supported Devices for Cisco Catalyst SD-WAN EtherChannel on the Transport Side

From Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the following platforms support EtherChannels on the transport side. From Cisco IOS XE Catalyst SD-WAN Release 17.14.1a, the following platforms support load balancing:

- Cisco 4000 Series Integrated Services Routers
  - Cisco 4461 Integrated Services Router
- Cisco ASR 1000 Series Aggregation Services Routers
  - Cisco ASR 1001-HX Router
  - Cisco ASR 1002-HX Router
- Cisco Catalyst 8200 Series Edge Routers
- Cisco Catalyst 8300 Series Routers
- Cisco Catalyst 8500 Series Edge Routers



---

**Note** Starting with Cisco IOS XE Catalyst SD-WAN Release 17.14.1a, the load balancing configuration command **portchannel load-balance-hash-algo sdwan** is supported only on the Cisco 4461 Integrated Services Router and Cisco Catalyst 8300 Series routers.

---

### Prerequisites for Cisco Catalyst SD-WAN EtherChannel on the Transport Side

- All the member links in each EtherChannel must be of the same speed.
- All the member links must be configured on Layer 3 transport side ports.

- All member interfaces in a portchannel must have the same speed and duplex, when using platforms that support multiple rate SFPs on the same port.

## Restrictions for Cisco Catalyst SD-WAN EtherChannel on the Transport Side

- The maximum number of port channel interfaces that a device can support varies, depending on the particular model of the device.
- You can configure EtherChannels on a device by using the CLI, or using only the CLI templates or CLI add-on feature templates in Cisco SD-WAN Manager.
- Network Interface Modules (NIMs) with L2 ports do not support EtherChannels on the transport side.
- In a deployment involving an EtherChannel Link Aggregation Group (LAG) from a Cisco IOS XE Catalyst SD-WAN device to a multichassis LAG (MC-LAG) between two upstream paths, SLA-based Application-Aware Routing (AAR) forwarding can be inaccurate if the traffic load on the two upstream paths is not symmetric.
- The use of port channel on virtual devices such as Cisco Catalyst 8000V is not supported.
- Cisco IOS XE Catalyst SD-WAN Release 17.13.1a does not include support for an endpoint tracker on port-channel TLOCs.
- Platforms such as the Cisco Catalyst 8500 Series Edge Routers support multi-rate interfaces, allowing 1G SFP modules to be used in default 10G interfaces. Despite this, in the output of **show** commands, the interfaces appear as TenGigabitEthernet x/x/x. You can bundle the 1G SFP interfaces together to form a port channel.

## Configure a Transport Side EtherChannel Using a CLI Template

In Cisco Catalyst SD-WAN Manager, you can configure EtherChannels on the transport side using CLI templates. For more information about using CLI templates, see [CLI Add-On Feature Templates](#) and [CLI Templates](#).



---

**Note** By default, CLI templates execute commands in global config mode.

---

1. Configure a Layer 3 port channel.

```
interface Port-channel channel-number
ip address ip-address mask
ipv6 address ipv6-address/prefix-length
```

2. Assign Interfaces to a Layer 3 port channel with LACP active or passive options.

a.

```
interface GigabitEthernet slot/subslot/port
no ip address
channel-group channel-group-number mode {active passive}
exit
```

- b. Configure EtherChannel with LACP Parameters.

```
lacp system-priority priority
interface GigabitEthernet slot/subslot/port
lacp port-priority priority
```

- c. Configure a static EtherChannel.

```
interface GigabitEthernet slot/subslot/port
no ip address
channel-group channel-group-number
```

3. Configure tunnels.

```
interface Tunnel tunnel-number
ip unnumbered Port-channel channel-group-number
no ip redirects
tunnel source Port-channel channel-group-number
tunnel mode sdwan
```

```
sdwan
interface Port-channel channel-group-number
tunnel-interface
encapsulation {ipsec gre}
color color-type
```

This example shows how to configure a Layer 3 EtherChannel, and how to assign two ports to channel 1 with the LACP mode as active and passive:

```
interface Port-channel1
ip address 10.48.48.15 255.255.255.0
ip ospf priority 0
ip ospf 65535 area 51
load-interval 30
no negotiation auto
```

```
interface GigabitEthernet0/0/0
no ip address
negotiation auto
lacp rate fast
channel-group 1 mode active
end
```

```
interface GigabitEthernet0/0/4
no ip address
negotiation auto
lacp rate fast
channel-group 1 mode passive
end
```

The following is a configuration example for creating an EtherChannel on the transport side.

```
interface Tunnel2
ip unnumbered Port-channel1
tunnel source Port-channel1
tunnel mode sdwan
```

```
interface Port-channell
 tunnel-interface
 encapsulation ipsec
 color lte
```

## Configure a Transport-Side Port Channel and Member Link Interface using Configuration Groups in Cisco SD-WAN Manager

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.
2. Under **Transport and Management**, click **Add New** to add a new service profile or select any existing service profiles, and click **Edit**.
3. Select **+**, and click on the **Ethernet Interface** to add a new interface. See [Ethernet Interface](#) for more details.
4. Select **Add New** from the Ethernet Interface drop-down menu.
5. Under **Basic Configurations**, enter the interface name and description.
6. Click **Ether Channel**, and assign the EtherChannel to a port channel or member interface from the drop down menu.




---

**Note** You can create multiple interfaces which can be either member or port channel interfaces.

---

7. Select **Global** from the **Tunnel** field, to enable a tunnel for the port channel interface.




---

**Note** To reassign the ethernet interface type, you must delete the existing ethernet interface feature and create a new one.

---

8. Click **Save**.
9. Click **Configuration** tab, and deploy the newly created port channels.

## Configure Load Balancing for EtherChannels on the Transport Side Using CLI Commands

### Enable Load Balancing on Individual Portchannel Interface

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.15.1a Cisco Catalyst SD-WAN Manager Release 20.15.1




---

**Note** We recommend using this method to configure load balancing for EtherChannels on the transport side.

---

1. Enter the port channel interface configuration mode.

**interface Portchannel** *channel number*

2. Enable load balancing on an individual port channel.

**load-balance-hash-algo** **sdwan**

### Enable Load Balancing Globally for EtherChannels on the Transport Side

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.14.1a and Cisco Catalyst SD-WAN Manager Release 20.14.1

For more information about using CLI templates, see [CLI Add-On Feature Templates](#) and [CLI Templates](#).

Enable load balancing globally for EtherChannels on the transport side.

**port-channel load-balance-hash-algo** **sdwan**



**Note** In this command, **port-channel load-balance-hash-algo sdwan**, the **sdwan** option was added in Cisco IOS XE Catalyst SD-WAN Release 17.14.1a.

### Enable Hash Algorithms Globally for EtherChannels on the Transport Side

1. Configure the algorithm used for load balancing.

To configure load balancing for IPv4 addresses, which is the default setting, use the following configuration:

```
sdwan
ip load-sharing algorithm {src-dst-ip|ip-and-ports|src-ip-only}
```

To configure load balancing for IPv6 addresses, use the following configuration:

```
sdwan
ipv6 load-sharing algorithm {src-dst-ip|ip-and-ports|src-ip-only}
```

- **src-dst-ip**: Balances traffic based on both source and destination IP addresses.
- **ip-and-ports**: Balances traffic using a combination of IP addresses and port numbers.
- **src-ip-only**: Balances traffic based solely on the source IP address.

The **ip load-sharing algorithm** command is a global configuration that applies to all Cisco Catalyst SD-WAN tunnels. Changing the algorithm with options such as **src-dst-ip** or **src-dst-mixed-ip-port** affects the load-sharing mechanism for other Cisco Catalyst SD-WAN tunnel traffic as well.

When you configure a port channel on both the service side and the transport side, using the **port-channel load-balance-hash-algo sdwan** command applies load balancing to the transport side. For the Service side, the port channel defaults to the **src-dst-ip** load balancing mode. For more information, see [Configure Network Interfaces](#).

To change the load-balancing algorithm for the Service side when a Transport-VPN port-channel is also configured, use the **port-channel load-balance-hash-algo** command. This command allow you to switch from the default **sdwan** mode to alternative modes such as **dst-ip**, **dst-mac**, **src-dst-ip**, **src-dst-mac**, **src-dst-mixed-ip-port**, **src-ip**, or **src-mac**. However, this change disables the SD-WAN-based load balancing for the transport side.

Here's the complete configuration for enabling load balancing and apply the desired hash algorithm for traffic distribution on the transport side of Cisco IOS XE Catalyst SD-WAN devices.

```
port-channel load-balance-hash-algo sdwan
sdwan
ip load-sharing algorithm src-dst-ip
```

```
port-channel load-balance-hash-algo sdwan
sdwan
ipv6 load-sharing algorithm src-dst-ip
```

This example shows configuration enabling load balancing for each port channel interface. When **sdwan** hash algorithm is configured on the transport side, you can enable different hash algorithm options on the service side.

```
device(config)# interface Port-channel 1
device(config-if)# load-balance-hash-algo sdwan
device(config-if)# exit

device(config)# interface Port-channel 2
device(config-if)# load-balance-hash-algo src-dst-mixed-ip-port
device(config-if)# exit

device(config)# interface Port-channel 3
device(config-if)# no shut
device(config-if)# commit
device(config-if)# end
```

The following is a sample output to view the configuration for per-interface port channel using **show etherchannel load-balancing** command.

```
device# show etherchannel load-balancing
flow-based
LB Algo type: Source Destination IP

Port-Channel: LB Method
Port-channel1 : flow-based (SDWAN Inner packet LB)
Port-channel2 : flow-based (Source Destination Port, IP addr)
Port-channel3 : flow-based (Source Destination IP)
```

## Monitor Configured EtherChannel Using CLI

### Example 1

The following is a sample output from the **show etherchannel summary** command. This example shows summary for each channel group.

```
Device# show etherchannel summary

Flags: D - down P/bndl - bundled in port-channel
 I - stand-alone s/susp - suspended
 H - Hot-standby (LACP only)
 R - Layer3 S - Layer2
 U - in use f - failed to allocate aggregator

 M - not in use, minimum links not met
 u - unsuitable for bundling
 w - waiting to be aggregated
 d - default port
```

```

Number of channel-groups in use: 1
Number of aggregators: 1

Group Port-channel Protocol Ports
-----+-----+-----+-----
1 Po1 (RU) LACP Te0/3/0 (bndl) Te0/3/1 (hot-sby)

RU - L3 port-channel UP State
SU - L2 port-channel UP state
P/bndl - Bundled
S/susp - Suspended

```

### Example 2

The following is a sample output from the **show etherchannel load-balancing** command. This example displays the load-balancing method that is applied to each port channel.

```

Device# show etherchannel load-balancing

EtherChannel Load-Balancing Method:
Global LB Method: flow-based
LB Algo type: SDWAN Inner packet LB

Port-Channel: LB Method
Port-channel1 : flow-based (SDWAN Inner packet LB)

```

## Aggregate EtherChannel Quality of Service

The Aggregate EtherChannel Quality of Service (QoS) feature improves the quality of service by effectively managing various network parameters, such as delay, jitter (or delay variation), bandwidth, and packet loss. Its primary function is to offer improved services for specific types of network traffic. The feature allows the application of an aggregate egress-queuing policy-map on the main or sub-interface of a port channel. Furthermore, it facilitates QoS support on the aggregate port channel's main interface on Cisco IOS XE Catalyst SD-WAN devices.

### Prerequisites for Aggregate EtherChannel Quality of Service

- Identify aggregate port channel interfaces before creating them using the **platform qos port-channel-aggregate** command.
- In a port channel, all member links must be of the same speed.

### Restrictions for Aggregate EtherChannel Quality of Service

- The aggregate port channel can support four member links and eight aggregate port channel interfaces.
- You can apply a policy map to the aggregate a port channel's main interface or sub-interface only. Member link QoS is not supported.
- You cannot spontaneously convert port channels to and from the aggregate status. You must delete the interface port-channel from the configurations before adding or removing the matching **platform qos port-channel-aggregate** command.



- QoS applications which are used to manage, prioritize and control the behavior of data transmission over a network are not supported on port channel member links.

QoS policies applied to aggregate port channel main interfaces and port channel sub-interfaces are not supported.

- When you enable aggregate QoS, it is not possible to directly modify a channel group on a member link. To make changes, the old channel group needs to be removed and the new one must be added. First push one template to remove the old member link and port channel configuration, then another template to add the new configuration.

## Configure Aggregate EtherChannel Quality of Service Using a CLI Template

In Cisco Catalyst SD-WAN Manager, you can configure aggregate EtherChannel QoS using the CLI templates. For more information about using CLI templates, see [CLI Add-On Feature Templates](#) and [CLI Templates](#).




---

**Note** By default, CLI templates execute commands in global config mode.

---

1. Create the aggregated port channel.

```
platform qos port-channel-aggregate port-channel-number
interface Port-channel channel-number
no shutdown
ip address ip-address mask
```

2. Assign member links to port channel.

```
interface GigabitEthernet slot/subslot/port
no negotiation auto
channel-group channel-group-number mode {active passive}
exit
```

3. Configure tunnels.

```
interface Tunnel tunnel-number
no shutdown
ip unnumbered port-channel-interface
tunnel source port-channel-interface
tunnel mode sdwan
```

```
sdwan
interface channel-group-number
tunnel-interface
encapsulation ipsec
color public-internet
```

4. Configure QoS.

```
interface channel-group-number
```

**service-policy output** *pre-defined qos policy-map*

Here's the complete configuration example for configuring aggregate EtherChannel QoS.

```

!
class-map match-any Best-Effort
 match qos-group 2
!
class-map match-any Bulk
 match qos-group 3
!
class-map match-any Business
 match qos-group 1
!
class-map match-any Critical
 match qos-group 0
!
policy-map qos_template
 class Critical
 police rate percent 15
 !
 priority level 1
 !
 class Business
 bandwidth remaining percent 55
 !
 class Best-Effort
 bandwidth remaining percent 10
 !
 class Bulk
 bandwidth remaining percent 20
 !
!
policy-map shape_Port-channel1
 class class-default
 service-policy qos_template
 shape average 100000000
 !
!
interface TenGigabitEthernet0/1/6
 no shutdown
 no negotiation auto
 channel-group 1 mode active
 lacp rate fast
exit
interface TenGigabitEthernet0/1/7
 no shutdown
 no negotiation auto
 channel-group 1 mode active
 lacp rate fast
exit
interface Port-channel1
 no shutdown
 ip address 10.1.15.15 255.255.255.0
 ipv6 nd ra suppress all
 service-policy output shape_Port-channel1
exit
interface Tunnel1
 no shutdown
 ip unnumbered Port-channel1
 tunnel source Port-channel1
 tunnel mode sdwan
exit

```

```

!
sdwan
interface Port-channel1
 tunnel-interface
 encapsulation ipsec
 color lte
 no allow-service bgp
 allow-service dhcp
 allow-service dns
 allow-service icmp
 no allow-service sshd
 no allow-service netconf
 no allow-service ntp
 no allow-service ospf
 no allow-service stun
 allow-service https
 no allow-service snmp
 no allow-service bfd
 exit
exit

```

## Verify Aggregate EtherChannel Quality of Service

To view QoS issues on a port channel interface, use the **show policy-map interface Port-channel** command.

```

Device# show policy-map interface Port-channel 1
Port-channel1

```

```

Service-policy output: shape_Port-channel1

```

```

Class-map: class-default (match-any)
 121 packets, 20797 bytes
 5 minute offered rate 2000 bps, drop rate 0000 bps
Match: any
Queueing
queue limit 416 packets
(queue depth/total drops/no-buffer drops) 0/0/0
(pkts output/bytes output) 121/20797
shape (average) cir 100000000, bc 400000, be 400000
target shape rate 100000000

```

```

Service-policy : qos_template

```

```

queue stats for all priority classes:
Queueing
priority level 1
queue limit 512 packets
(queue depth/total drops/no-buffer drops) 0/0/0
(pkts output/bytes output) 121/20797

```

```

Class-map: Critical (match-any)
 121 packets, 20797 bytes
 5 minute offered rate 2000 bps, drop rate 0000 bps
Match: qos-group 0
police:
 rate 15 %
 rate 15000000 bps, burst 468750 bytes
 conformed 121 packets, 20797 bytes; actions:
 transmit
 exceeded 0 packets, 0 bytes; actions:
 drop
 conformed 2000 bps, exceeded 0000 bps
Priority: Strict, b/w exceed drops: 0

```

```
Priority Level: 1

Class-map: Business (match-any)
 0 packets, 0 bytes
 5 minute offered rate 0000 bps, drop rate 0000 bps
 Match: qos-group 1
 Queueing
 queue limit 416 packets
 (queue depth/total drops/no-buffer drops) 0/0/0
 (pkts output/bytes output) 0/0
 bandwidth remaining 55%

Class-map: Best-Effort (match-any)
 0 packets, 0 bytes
 5 minute offered rate 0000 bps, drop rate 0000 bps
 Match: qos-group 2
 Queueing
 queue limit 416 packets
 (queue depth/total drops/no-buffer drops) 0/0/0
 (pkts output/bytes output) 0/0
 bandwidth remaining 10%

Class-map: Bulk (match-any)
 0 packets, 0 bytes
 5 minute offered rate 0000 bps, drop rate 0000 bps
 Match: qos-group 3
 Queueing
 queue limit 416 packets
 (queue depth/total drops/no-buffer drops) 0/0/0
 (pkts output/bytes output) 0/0
 bandwidth remaining 20%

Class-map: class-default (match-any)
 0 packets, 0 bytes
 5 minute offered rate 0000 bps, drop rate 0000 bps
 Match: any

 queue limit 416 packets
 (queue depth/total drops/no-buffer drops) 0/0/0
 (pkts output/bytes output) 0/0
```



## CHAPTER 34

# Cisco Catalyst SD-WAN Multitenancy

---

- [Cisco Catalyst SD-WAN Multitenancy, on page 848](#)
- [Information about Cisco Catalyst SD-WAN multitenancy, on page 850](#)
- [Supported Devices and Controller Specifications, on page 856](#)
- [Restrictions, on page 857](#)
- [Initial Setup for Multitenancy, on page 858](#)
- [Expand a Multitenant Deployment to Support More Tenants and Tenant Devices, on page 866](#)
- [Manage Tenants, on page 869](#)
- [Cisco SD-WAN Manager Dashboard for Multitenancy, on page 877](#)
- [Manage Tenant WAN Edge Devices, on page 882](#)
- [Tenant-Specific Policies on Cisco SD-WAN Controller, on page 883](#)
- [Manage Tenant Data, on page 884](#)
- [View OMP Statistics per Tenant on a Cisco SD-WAN Controller, on page 887](#)
- [View Tenants Associated with a Cisco SD-WAN Controller, on page 888](#)
- [Migrate Single-Tenant Cisco Catalyst SD-WAN Overlay to Multitenant Cisco Catalyst SD-WAN Deployment, on page 888](#)
- [Migrate a Tenant from a Multitenant Cisco Catalyst SD-WAN Overlay to Single-Tenant Cisco Catalyst SD-WAN Deployment , on page 892](#)
- [Migrate Multitenant Cisco Catalyst SD-WAN Overlay, on page 896](#)
- [Upgrade Cisco Catalyst SD-WAN Controller and Edge Device Software, on page 899](#)
- [Multitenant Cisco SD-WAN Manager: Disaster Recovery, on page 900](#)
- [Multitenant Cisco SD-WAN Manager: Disaster Recovery in an Overlay Network with Virtual Routers, on page 905](#)
- [Multitenant Cisco SD-WAN Manager: Disaster Recovery After a Failed Data Center Becomes Operational, on page 911](#)
- [Replace Faulty Cisco SD-WAN Controller, on page 916](#)
- [RADIUS and TACACS Support for Multitenancy, on page 917](#)

# Cisco Catalyst SD-WAN Multitenancy

**Table 361: Feature History**

| Feature Name                                                      | Release Information                                                                                       | Description                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tenant Device Forecasting                                         | Cisco IOS XE Catalyst SD-WAN Release 17.6.1a<br>Cisco vManage Release 20.6.1                              | With this feature, a service provider can control the number of WAN edge devices a tenant can add to their overlay network. By doing so, the provider can utilize Cisco Catalyst SD-WAN controller resources efficiently. For more information, see <a href="#">Tenant Device Forecasting</a> .                                                                                                                                          |
| Migrate Multitenant Cisco Catalyst SD-WAN Overlay                 | Cisco IOS XE Catalyst SD-WAN Release 17.6.1a<br>Cisco vManage Release 20.6.1                              | This feature enables you to migrate a multitenant Cisco Catalyst SD-WAN overlay comprising shared Cisco SD-WAN Manager instances and Cisco SD-WAN Validator, and tenant-specific Cisco SD-WAN Controllers to a multitenant overlay comprising shared Cisco SD-WAN Manager instances, Cisco SD-WAN Validator, and Cisco SD-WAN Controllers. For more information, see <a href="#">Migrate Multitenant Cisco Catalyst SD-WAN Overlay</a> . |
| Enhanced Cisco Catalyst SD-WAN Manager Dashboard for Multitenancy | Cisco IOS XE Catalyst SD-WAN Release 17.12.1a<br>Cisco Catalyst SD-WAN Manager Release 20.12.1            | This feature ensures a consistent user experience across tenant and service provider dashboards.<br><br>The Cisco Catalyst SD-WAN Manager dashboard provides visibility into the available resources on shared devices.                                                                                                                                                                                                                  |
| RADIUS and TACACS Support for Multitenancy                        | Cisco IOS XE Catalyst SD-WAN Release 17.12.1a<br>Cisco Catalyst SD-WAN Control Components Release 20.12.1 | This feature enables support for Remote Authentication Dial-In User Service (RADIUS) and Terminal Access Controller Access Control System (TACACS) authentication in a multitenant deployment on WAN edge devices. For more information, see <a href="#">Information about RADIUS and TACACS Support for Multitenancy</a> .                                                                                                              |

| Feature Name                                                                                   | Release Information                                                                                       | Description                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Migration of a Tenant from a Multitenant Overlay to a Single-Tenant Deployment                 | Cisco IOS XE Catalyst SD-WAN Release 17.13.1a<br>Cisco Catalyst SD-WAN Manager Release 20.13.1            | This feature supports the migration of a tenant from a multitenant overlay to a single-tenant deployment. To migrate a tenant between two Cisco Catalyst SD-WAN deployments, move the tenant configurations, statistical data, and WAN edge devices from one deployment to another.                                   |
| Multitenancy Support for Cisco Catalyst Cellular Gateways                                      | Cisco IOS CG Release 17.14.1<br>Cisco Catalyst SD-WAN Control Components Release 20.14.1                  | Added multitenancy support for Cisco Catalyst Cellular Gateways.                                                                                                                                                                                                                                                      |
| Support for Remote AAA and Webhook Notifications Over Management VPN 512                       | Cisco IOS XE Catalyst SD-WAN Release 17.16.1a<br>Cisco Catalyst SD-WAN Manager Release 20.16.1            | This feature provides increased security by allowing you to configure tenants to receive AAA traffic and webhook notifications over management VPN 512.                                                                                                                                                               |
| Configure RADIUS and TACACS Servers to Receive Authentication Requests Over Management VPN 512 | Cisco IOS XE Catalyst SD-WAN Release 17.16.1a<br>Cisco Catalyst SD-WAN Manager Release 20.16.1            | This feature provides increased security by allowing you to configure tenants to send and receive AAA traffic over WAN transport VPN 0 or management VPN 512. For more information, see <a href="#">Configure Remote AAA</a> .                                                                                        |
| Support for Multitenancy Providers to View Tenant's Alarms and Events                          | Cisco IOS XE Catalyst SD-WAN Release 17.16.1a<br>Cisco Catalyst SD-WAN Manager Release 20.16.1            | This feature allows providers to view tenant alarms and events. Multitenancy providers can enable or disable this feature for each tenant.                                                                                                                                                                            |
| Support for Tenant to Access to Controller-Specific Information                                | Cisco IOS XE Catalyst SD-WAN Release 17.16.1a<br>Cisco Catalyst SD-WAN Control Components Release 20.16.1 | This feature enables tenants to access their controller-specific restricted information, which enhances visibility and simplifies management and troubleshooting within the Cisco Catalyst SD-WAN network. This improved visibility aids in addressing issues related to OMP, routing, control connections, and more. |

# Information about Cisco Catalyst SD-WAN multitenancy

With Cisco Catalyst SD-WAN multitenancy, a service provider can manage multiple customers, called tenants, from Cisco SD-WAN Manager. The tenants share the same set of underlying Cisco SD-WAN controllers: Cisco SD-WAN Manager, Cisco Catalyst SD-WAN Validator, and Cisco Catalyst SD-WAN Controller. The tenant data is logically isolated on these shared controllers.

The service provider accesses Cisco SD-WAN Manager using a domain name mapped to the IP address of a Cisco SD-WAN Manager cluster and manages the multitenant deployment. Each tenant is provided a subdomain to access a tenant-specific Cisco SD-WAN Manager view and manage the tenant deployment. For example, a service provider using the domain name `managed-sp.com`, can assign tenants Customer1 and Customer2 the subdomains `customer1.managed-sp.com` and `customer2.managed-sp.com` and manage them on the same set of Cisco SD-WAN controllers, instead of providing each customer a single tenant setup with a dedicated set of Cisco SD-WAN controllers.

Key features of Cisco Catalyst SD-WAN multitenancy include:

- Full enterprise multitenancy:
  - Cisco Catalyst SD-WAN supports multitenancy and offers enterprises the flexibility of segregated roles such as service provider and tenants. Service providers can use multitenancy to provide Cisco Catalyst SD-WAN service offerings to their customers.
- Multitenancy:
  - Multitenant Cisco SD-WAN Manager
  - Multitenant Cisco Catalyst SD-WAN Validators
  - Multitenant Cisco Catalyst SD-WAN Controllers
- Tenant-specific WAN edge devices.
- Security:
  - Send and receive AAA traffic over management VPN 512 from Cisco IOS XE Catalyst SD-WAN Release 17.16.1a.
- Overlapping VPN numbers:
  - A particular VPN or a set of common VPNs is assigned to a specific tenant, with their own configurations and monitoring dashboard environment. These VPN numbers can overlap where they are used by other tenants.
- On-prem and cloud deployment models:
  - Cisco Catalyst SD-WAN controllers can be deployed in an organization data center on servers running the VMware ESXi 6.7 or later, or the Kernel-based Virtual Machine (KVM) hypervisor. Cisco Catalyst SD-WAN controllers can also be hosted on Amazon Web Services (AWS) servers by Cisco CloudOps.
- Tenant-specific Cisco SD-WAN Analytics:
  - Cisco SD-WAN Analytics is a cloud-based service that offers insights into the performance of applications and the underlying SD-WAN network infrastructure. Each tenant can obtain Cisco



SD-WAN Analytics insights for their overlay network by requesting a tenant-specific Cisco SD-WAN Analytics instance and enabling data collection on Cisco SD-WAN Manager. The service provider must enable cloud services on Cisco SD-WAN Manager in the provider view to facilitate the onboarding of the Cisco SD-WAN Analytics instance for the tenant overlay network.



**Note** Starting from Cisco SD-WAN Manager Release 20.13.1, a single node multitenancy is supported. Expand a single node cluster into a multitenant deployment.

### Multitenant Cisco SD-WAN Manager

Cisco SD-WAN Manager is deployed and configured by the service provider. The provider enables multitenancy and creates a Cisco SD-WAN Manager cluster to serve tenants. Only the provider can access a Cisco SD-WAN Manager instance through the SSH terminal.

In the Provider view, Cisco SD-WAN Manager offers service providers an overall view of the SD-WAN multitenant deployment and allows a provider to manage all the Cisco Catalyst SD-WAN Validator and Cisco Catalyst SD-WAN Controller devices. Cisco SD-WAN Manager also allows service providers to monitor and manage the deployments of each tenant through Provider-as-Tenant view.

In the Tenant view, Cisco SD-WAN Manager allows individual tenants to:

- Monitor and manage their own deployment through a dashboard.
- Deploy and configure WAN edge devices.
- Configure custom policies on Cisco Catalyst SD-WAN Controllers.

Cisco Catalyst SD-WAN Control Component infrastructure settings are not displayed in Tenant view.

### Multitenant Cisco Catalyst SD-WAN Validators

Cisco Catalyst SD-WAN Validators are deployed and configured by the service provider. Only the provider can access a Cisco Catalyst SD-WAN Validator through the SSH terminal.

Cisco Catalyst SD-WAN Validators serve WAN edge devices of multiple tenants as the devices are added to the overlay network.

### Multitenant Cisco Catalyst SD-WAN Controllers

Cisco Catalyst SD-WAN Controllers are deployed by the service provider. Only the provider can create and attach device and feature templates to Cisco Catalyst SD-WAN Controllers, and can access a Cisco Catalyst SD-WAN Controller through the SSH terminal.

- When a tenant is created, Cisco SD-WAN Manager assigns two Cisco Catalyst SD-WAN Controllers for the tenant. The Cisco Catalyst SD-WAN Controllers form an active-active cluster.

Each tenant is assigned only two Cisco Catalyst SD-WAN Controllers. Before a tenant is created, two Cisco Catalyst SD-WAN Controllers must be available to serve the tenant.

- When more than one pair of Cisco Catalyst SD-WAN Controllers are available to serve a tenant, Cisco SD-WAN Manager assigns to the tenant the pair of Cisco Catalyst SD-WAN Controllers connected to the lowest number of forecast devices. If two pairs of Cisco SD-WAN Controllers are connected to the

same number of devices, Cisco SD-WAN Manager assigns to the tenant the pair of Cisco SD-WAN Controllers serving the lowest number of tenants.

- From Cisco vManage Release 20.9.1, while onboarding a tenant to a multitenant deployment, you can choose the pair of multitenant Cisco SD-WAN Controllers that serve the tenant. After onboarding a tenant, you can migrate the tenant to a different pair of multitenant Cisco SD-WAN Controllers, if necessary. For more information, see [Flexible Tenant Placement on Multitenant Cisco SD-WAN Controllers](#).
- Each pair of Cisco Catalyst SD-WAN Controllers can serve up to 24 tenants.
- Tenants can configure custom policies on the Cisco Catalyst SD-WAN Controllers assigned to them. Cisco SD-WAN Manager notifies the Cisco Catalyst SD-WAN Controllers to pull the policy templates. Cisco Catalyst SD-WAN Controllers pull the templates and deploy the policy configuration for the specific tenant.
- Only the provider can view events, audit logs, and OMP alarms for a Cisco Catalyst SD-WAN Controller on Cisco SD-WAN Manager.



---

**Note** Starting from Cisco SD-WAN Manager Release 20.16.1, a provider can view alarms and events for the sites and devices in its tenancy.

---

### Tenant-specific WAN edge devices

A tenant or the provider acting on behalf of a tenant can:

- Add WAN edge devices to the tenant network.
- Configure the devices.
- Remove the devices from the tenant network.
- Access the device through the SSH terminal.

A provider can manage the WAN edge devices only from [provider-as-tenant](#) view. In the [provider](#) view, Cisco SD-WAN Manager does not show any WAN edge device information.

Cisco SD-WAN Manager reports WAN edge device events, logs, and alarms only in the [tenant](#) and the provider-as-tenant views.

### Single tenant environments

A single tenant environment exclusively manages, and is responsible for, its own Cisco Catalyst SD-WAN Control Components and devices. All configured resources are visible to the single tenant administrator in the Cisco SD-WAN Manager interface.

### Cloud-delivered Catalyst SD-WAN

Cloud-delivered Catalyst SD-WAN operates as a tenant within a multitenant environment rather than as a single tenant. Cloud-delivered Catalyst SD-WAN users do not see controller infrastructure settings in Cisco SD-WAN Manager. Their available information is limited to their own components and WAN edge devices.

For more information on Cloud-delivered Catalyst SD-WAN, consult the [Cloud-delivered Cisco SD-WAN Getting Started Guide](#).

### Feature availability

Some Cisco Catalyst SD-WAN features are available only in specific tenancy configurations.

**Table 362: SD-WAN feature availability**

| Feature                                                        | Single Tenant | Multitenant                     | Cloud-delivered Catalyst SD-WAN |
|----------------------------------------------------------------|---------------|---------------------------------|---------------------------------|
| Identity Services Engine (ISE) integration                     | Yes           | No                              | No                              |
| Intent-based hub and spoke topology                            | Yes           | No                              | No                              |
| Controller Group Affinity                                      | Yes           | No                              | No                              |
| Cisco Duo multifactor authentication (MFA) support             | Yes           | No                              | No                              |
| SD-Routing support without an SD-WAN Controller                | Yes           | No                              | No                              |
| Data stream and packet capture setting changes at tenant level | —             | No, supported at provider level | No, supported at provider level |

## Information About Tenant Migration in a Multitenant Deployment

(Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.6.1a and Cisco vManage Release 20.6.1) Migration of a single-tenant overlay to a multitenant deployment is only supported with the Cisco Catalyst SD-WAN controllers deployed on-premises.

(Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.13.1a and Cisco Catalyst SD-WAN Manager Release 20.13.1) Migration of a tenant from a multitenant overlay to a single-tenant deployment is supported.

The tenant migration involves export of tenant data from the source Cisco SD-WAN Manager instance and import of data to the destination Cisco SD-WAN Manager instance. After the data migration is complete, the tenant WAN edge devices with active control connections with the source Cisco SD-WAN Manager migrate and form connections with the destination Cisco SD-WAN Manager.

## User Roles in Multitenant Environment

A multi-tenant environment includes the service provider and tenant roles. Each role has distinct privileges, views, and functions.

## Provider Role

The provider role entitles system-wide administrative privileges. A user with the provider role has the default username **admin**. The provider user can access Cisco SD-WAN Manager using the domain name of the service provider or by using the Cisco SD-WAN Manager IP address. When using a domain name, the domain name has the format `https://managed-sp.com`.

The **admin** user is part of the user group **netadmin**. Users in this group are permitted to perform all operations on the controllers and the WAN edge devices of the tenants. You can add additional users to the **netadmin** group.

You cannot modify the privileges of the **netadmin** group. On Cisco SD-WAN Manager, you can view the privileges of the user group from the **Administration > Manage Users > User Groups** page.



**Note** When you create a new provider user in Cisco SD-WAN Manager, including a **netadmin** user, by default, the user is not allowed SSH access to the Cisco SD-WAN Manager VM. To enable SSH access, configure SSH authentication using a AAA template and push the template to Cisco SD-WAN Manager. For more information on enabling SSH authentication, see [SSH Authentication using Cisco SD-WAN Manager on Cisco IOS XE Catalyst SD-WAN Devices](#).

For more information about configuring users and user groups, see [Configure User Access and Authentication](#).

Cisco SD-WAN Manager offers two views to a provider:

### • Provider View

When a provider user logs in to multi-tenant Cisco SD-WAN Manager as **admin** or another **netadmin** user, Cisco SD-WAN Manager presents the provider view and displays the provider dashboard.

You can perform the following functions from the provider view:

- Provision and manage Cisco SD-WAN Manager, Cisco SD-WAN Validators and Cisco SD-WAN Controllers.
- Add, modify, or delete tenants.
- Monitor the overlay network.
- Starting from Cisco Catalyst SD-WAN Manager Release 20.16.1, view alarms and events for the sites and devices of its tenants.

### • Provider-as-Tenant View

When a provider user selects a specific tenant from the **Select Tenant** drop-down list at the top of the provider dashboard, Cisco SD-WAN Manager presents the provider-as-tenant view and displays the tenant dashboard for the selected tenant. The provider user has the same view of Cisco SD-WAN Manager as a tenant user would when logged in as **tenantadmin**. From this view, the provider can manage the tenant deployment on behalf of the tenant.

In the provider dashboard, a table of tenants presents a status summary for each tenant. A provider user can also launch the provider-as-tenant view by clicking on a tenant name in this table.

### Tenant Role

The tenant role entitles tenant administrative privileges. A user with the tenant role has the default username **tenantadmin**. The default password is **Cisco#123@Viptela**. We recommend that you change the default password on first login. For information on changing the default password, see [Hardware and Software Installation](#).

The **tenantadmin** user is part of the user group **tenantadmin**. Users in this group are permitted to perform all operations on the WAN edge devices of the tenants. You can add additional users to the **tenantadmin** group.

You cannot modify the privileges of the **tenantadmin** group. On Cisco SD-WAN Manager, you can view the privileges of the user group from the **Administration > Manage Users > User Groups** page.

For more information about configuring users and user groups, see [Configure User Access and Authentication](#).

A tenant user can log in to Cisco SD-WAN Manager using a dedicated URL and the default username **tenantadmin**. For example, the dedicated URL of a tenant could be `https://customer1.managed-sp.com` for a provider using the domain name `https://managed-sp.com`. When the user logs in, Cisco SD-WAN Manager presents the tenant view and displays the tenant dashboard.



---

**Tip** If you cannot access the dedicated tenant URL, update the subdomain details in the `/etc/hosts` file on the local machine. Alternatively, if you use an external DNS server, add a DNS entry for the tenant subdomain.

---

A tenant user with administrative privileges can perform the following functions:

- Provision and manage tenant routers
- Monitor overlay network of the tenant
- Create custom policies on the assigned Cisco SD-WAN Controller
- Upgrade the software on the tenant routers.
- Starting from Cisco Catalyst SD-WAN Manager Release 20.16.1, view tenant-specific information of controller connections and OMP statistics in a Cisco Catalyst SD-WAN network.

## Provider and Tenant Remote Servers and Images

### Cisco Catalyst SD-WAN Manager Release 20.14.1 and Earlier Releases

In these releases, remote servers and images operate as follows:

- Only the provider can add remote servers and images.
- The remote servers and images are visible to all tenants. Tenants can use the remote servers and images but can't edit them.

### Cisco Catalyst SD-WAN Manager Release 20.15.1

In these releases, remote servers and images operate as follows:

- A tenant can add a remote server and remote image for both software images and virtual images. The remote server and image are visible only to the corresponding tenant and not to the provider or other tenants.
- The provider can add a remote server, a remote image, and a local image for both software images and virtual images in Cisco SD-WAN Manager.

## Supported Devices and Controller Specifications

The following Cisco Catalyst SD-WAN edge devices support multitenancy.

**Table 363: Supported Devices**

| Platform                            | Device Models                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cisco IOS XE Catalyst SD-WAN device | <ul style="list-style-type: none"> <li>• Cisco ASR 1000 Series Aggregation Services Routers</li> <li>• Cisco ISR 1000 Series Integrated Services Routers</li> <li>• Cisco ISR 4000 Series Integrated Services Routers</li> <li>• Cisco Catalyst 8200 Series Edge Platforms</li> <li>• Cisco Catalyst 8300 Series Edge Platforms</li> <li>• Cisco Catalyst 8500 Series Edge Platforms</li> <li>• Cisco Catalyst 8000V Edge Software</li> <li>• Cisco ENCS Platforms</li> </ul> |
| Cisco Catalyst Cellular Gateways    | (From Cisco IOS CG Release 17.14.1 and Cisco Catalyst SD-WAN Control Components Release 20.14.1) <ul style="list-style-type: none"> <li>• CG418-E</li> <li>• CG522-E</li> </ul>                                                                                                                                                                                                                                                                                               |

The following hypervisors are supported for multitenancy:

- VMware ESXi 6.7 or later
- KVM
- AWS (cloud-hosted and managed by Cisco CloudOps)
- Microsoft Azure (cloud-hosted and managed by Cisco CloudOps)

From Cisco vManage Release 20.6.1, a multitenant Cisco SD-WAN Manager instance can have one of the following three personas. The personas enable a predefined set of services on the Cisco SD-WAN Manager instance.

**Table 364: Cisco SD-WAN Manager Personas**

| Persona      | Services                                                                                                                                                           |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Compute+Data | Cluster Oracle, Service Proxy, Messaging Service, Coordination Service, Configuration Database, Data Collection Agent, Statistics Database, and Application Server |
| Data         | Cluster Oracle, Service Proxy, Application Server, Data Collection Agent, and Statistics Database                                                                  |
| Compute      | Cluster Oracle, Service Proxy, Messaging Service, Coordination Service, Configuration Database, and Application Server                                             |

The supported hardware specifications for the Cisco SD-WAN Validator, Cisco SD-WAN Manager, and the Cisco SD-WAN Controllers are as follows:

#### Hardware Specifications to Support 50 Tenants and 1000 Devices

For more information on supported hardware specifications for the Cisco SD-WAN Validator, Cisco SD-WAN Manager, and the Cisco SD-WAN Controllers see, [Cisco Catalyst SD-WAN Controller Compatibility Matrix and Recommended Computing Resources](#).

#### Hardware Specifications to Support 75 Tenants and 2500 Devices

For more information on supported hardware specifications for the Cisco SD-WAN Validator, Cisco SD-WAN Manager, and the Cisco SD-WAN Controllers see, [Cisco Catalyst SD-WAN Controller Compatibility Matrix and Recommended Computing Resources](#).

#### Hardware Specifications to Support 100 Tenants and 5000 Devices

For more information on supported hardware specifications for the Cisco SD-WAN Validator, Cisco SD-WAN Manager, and the Cisco SD-WAN Controllers see, [Cisco Catalyst SD-WAN Controller Compatibility Matrix and Recommended Computing Resources](#).

#### Hardware Specifications to Support 150 Tenants and 7500 Devices

For more information on supported hardware specifications for the Cisco SD-WAN Validator, Cisco SD-WAN Manager, and the Cisco SD-WAN Controllers see, [Cisco Catalyst SD-WAN Controller Compatibility Matrix and Recommended Computing Resources](#).

## Restrictions

- Do not use a user-configured system IP address to connect to a device through SSH. Instead, use the IP address of the `vmanage_system` interface; this IP address is assigned by Cisco SD-WAN Manager.

To find the IP address of the `vmanage_system` interface, use one of the following methods:

- Launch the device SSH terminal from Cisco SD-WAN Manager and find the `vmanage_system` IP address from the first line of the log-in prompt.
- Run the **show interface description** command and find the `vmanage_system` IP address from the command output.
- If you add a second tenant immediately after adding a tenant, Cisco SD-WAN Manager adds them sequentially, and not in parallel.
- If you are adding a WAN edge device that you had previously invalidated and deleted from an overlay network, you must reset the device software after adding the device. To reset the software on a Cisco IOS XE Catalyst SD-WAN device, use the command **request platform software sdwan software reset**.
- For Cisco IOS XE Catalyst SD-WAN Release 17.12.1a and earlier releases, single node Cisco SD-WAN Manager is not supported on a multitenant deployment. A minimum of 3-Node Cisco SD-WAN Manager cluster is required for a multitenant deployment.
- When a Cisco SD-WAN Controller or Cisco SD-WAN Validator upgrade is in progress, upgrade of tenant edge devices is not supported.
- Controller group feature is not supported in multitenant mode.
- The WAN edge device's site ID must be different from the Cisco SD-WAN Control Components site ID, when the Cisco SD-WAN Manager has different public and private IP addresses.

### Restrictions for Migration of a Tenant from a Multitenant Overlay to a Single-Tenant Deployment

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.13.1a and Cisco Catalyst SD-WAN Manager Release 20.13.1

- Change in the tenant organization name is not supported when the tenant moves from the Cisco Catalyst SD-WAN source to destination deployment.
- Tenant migration with multitenant WAN edge devices is not supported.
- Data traffic loss is expected during migration as devices are migrating from one set of Cisco SD-WAN Controllers to another.
- All user passwords are set to the default Cisco password on the destination overlay. The default password is **Cisco#123@Viptela**.
- Statistical data of the tenant that can be relearned by destination Cisco SD-WAN Manager is not migrated.
- The migration procedure does not support multiple imports on the same destination Cisco SD-WAN Manager. Reinitialize the destination Cisco SD-WAN Manager to allow import again.

## Initial Setup for Multitenancy

### Prerequisites

- Download and install software versions as recommended in the following table:



**Table 365: Minimum Software Prerequisites for Cisco Catalyst SD-WAN Multitenancy**

| Device                              | Software Version                             |
|-------------------------------------|----------------------------------------------|
| Cisco SD-WAN Manager                | Cisco vManage Release 20.6.1                 |
| Cisco SD-WAN Validator              | Cisco SD-WAN Release 20.6.1                  |
| Cisco SD-WAN Controller             | Cisco SD-WAN Release 20.6.1                  |
| Cisco IOS XE Catalyst SD-WAN Device | Cisco IOS XE Catalyst SD-WAN Release 17.6.1a |

A configuration in which one or more controllers, or WAN edge devices, are running software versions earlier than those mentioned in the table above is not supported.

- Do not migrate an existing single-tenant Cisco SD-WAN Manager instance into multitenant mode, even if you invalidate or delete all devices from the existing Cisco SD-WAN Manager instance. Instead, download and install a new Cisco SD-WAN Manager software image.



**Note** After you enable Cisco SD-WAN Manager for multitenancy, you cannot migrate it back to single tenant mode.

- Follow the recommended hardware specifications in the *Supported Devices and Controller Specifications* section of this document.
- Log in to Cisco SD-WAN Manager as the provider **admin** user.

1. Create Cisco SD-WAN Manager cluster.

- a. To support 50 tenants and 1000 devices across all tenants, [create a 3-node Cisco SD-WAN Manager Multitenant cluster](#).
- b. To support 100 tenants and 5000 devices across all tenants, [create a 6-node Cisco SD-WAN Manager Multitenant cluster](#).
- c. From Cisco IOS XE Release 17.6.3a, Cisco vManage Release 20.6.3, to support 150 tenants and 7500 devices across all tenants, [create a 6-node Cisco SD-WAN Manager Multitenant cluster](#).

2. Create and configure Cisco SD-WAN Validator instances. See [Deploy Cisco SD-WAN Validator](#).

While configuring Cisco SD-WAN Validator instances, configure the service provider organization name (`sp-organization-name`) and the organization name (`organization-name`). See [Configure Organization Name in Cisco SD-WAN Validator](#).

```
sp-organization-name multitenancy
organization-name multitenancy
```

3. Create Cisco SD-WAN Controller instances. See [Deploy the Cisco SD-WAN Controller](#).

- To support 50 tenants and 1000 devices across all tenants, deploy 6 Cisco SD-WAN Controller instances.
- To support 100 tenants and 5000 devices across all tenants, deploy 10 Cisco SD-WAN Controller.

- From Cisco IOS XE Release 17.6.3a, Cisco vManage Release 20.6.3, to support 150 tenants and 7500 devices across all tenants, deploy 16 Cisco SD-WAN Controllers.
- a. [Add Cisco SD-WAN Controllers](#) to the overlay network.
- 4. Onboard new tenants. See [Add a New Tenant, on page 871](#).

## Create a 3-Node Cisco SD-WAN Manager Multitenant Cluster

1. Download the Cisco vManage Release 20.6.1 or later software image from [Cisco Software Download](#).
2. Create three Cisco SD-WAN Manager instances (say vManage1, vManage2, and vManage3) by installing the downloaded software image file. See [Deploy Cisco SD-WAN Manager](#).



### Important

- Deploy Cisco SD-WAN Manager servers having the hardware specifications in the table *Hardware Specifications to Support 50 Tenants and 1000 Devices* from the *Supported Devices and Controller Specifications* section of this document.
- Choose the **Compute+Data** persona for each Cisco SD-WAN Manager instance.

3. Complete the following operations on vManage1:
  - a. Configure the following using the CLI:
    - System IP address
    - Site ID
    - Service Provider organization name (`sp-organization-name`)
    - Organization-name
    - Cisco SD-WAN Validator IP address
    - VPN 0 Transport/Tunnel interface
    - VPN 0 Out-of-band (OOB) interface: Ensure that you assign a static IP address to this interface. Do not enable DHCP.
    - VPN 512 Management interface



### Note

Configure only one default route in VPN 0.

- b. [Enable Multitenancy on Cisco SD-WAN Manager, on page 864](#).
- c. (Optional) Using the CLI, install the Root CA certificate for vManage1.



### Note

Skip this step if you are using a Symantec or Cisco PKI certificate.

- d. Complete the following through the Cisco SD-WAN Manager:
    1. [Generate a Certificate Signing Request](#)
    2. After Symantec or your enterprise root CA has signed the certificate, [install the signed certificate](#).
  - e. [Configure the Cluster IP Address of the Cisco SD-WAN Manager Server](#).

Before proceeding to the next step, ensure that the **Manager IP Address** field on the **Administration > Cluster Management** page shows the OOB interface address.
4. Complete the following operations on vManage2 and vManage3:



---

**Important** Do not enable multitenancy on vManage2 and vManage3.

---

- a. Configure the following using the CLI:
  - System IP address
  - Site ID
  - Service Provider organization name (sp-organization-name)
  - Organization-name
  - Cisco SD-WAN Validator IP address
  - VPN 0 Transport/Tunnel interface
  - VPN 0 Out-of-band (OOB) interface: Ensure that you assign a static IP address to this interface. Do not enable DHCP.
  - VPN 512 Management interface
- b. (Optional) Using the CLI, install the Root CA certificate for vManage1.



---

**Note** Skip this step if you are using a Symantec or Cisco PKI certificate.

---

- c. Complete the following through the Cisco SD-WAN Manager:
  1. [Generate a Certificate Signing Request](#)
  2. After Symantec or your enterprise root CA has signed the certificates, [install signed certificate](#).
- d. [Log in to the Cisco SD-WAN Manager Web Application Server](#).
- e. Ping the OOB interfaces on the other two Cisco SD-WAN Manager instances and ensure they are reachable.
- f. [Configure the Cluster IP Address of the Cisco SD-WAN Manager Server](#).

Before proceeding to the next step, ensure that the **Manager IP Address** field on the **Administration > Cluster Management** page shows the OOB interface address.

5. Log in to the vManage1 GUI and [add vManage2 to the cluster](#).

vManage2 reboots before being added to the cluster.

While vManage2 is being added to the cluster, on the **Administration > Cluster Management** page, the **Configure Status** for vManage2 shows **Pending**. You can monitor the System Generated Cluster Sync transaction to check the progress of the adding vManage2 to the cluster.

When the operation is completed, on the **Administration > Cluster Management** page, you can view both vManage1 and vManage2, and their node personas.

6. Repeat **Step 5** and add vManage3 to the cluster.



**Note** After rebooting, you have to select persona (non-cloud setup) from CLI and services starts running on the node according to the selected persona.

## Create a 6-Node Cisco SD-WAN Manager Multitenant Cluster

1. Download the Cisco vManage Release 20.6.1 or later software image from [Cisco Software Download](#).
2. Create six Cisco SD-WAN Manager instances by installing the downloaded software image file. See [Deploy Cisco SD-WAN Manager](#).



### Important

- To support 100 tenants and 5000 devices across all tenants, deploy Cisco SD-WAN Manager servers having the hardware specifications in the table *Hardware Specifications to Support 100 Tenants and 5000 Devices* from the *Supported Devices and Controller Specifications* section of this document.

From Cisco IOS XE Release 17.6.3a, Cisco vManage Release 20.6.3, to support 150 tenants and 7500 devices across all tenants, deploy Cisco SD-WAN Manager servers having the hardware specifications in the table *Hardware Specifications to Support 150 Tenants and 7500 Devices* from the *Supported Devices and Controller Specifications* section of this document.

- Choose the **Compute+Data** persona for three Cisco SD-WAN Manager instances (say vManage1, vManage2, and vManage 3). Choose the **Data** persona for the other three Cisco SD-WAN Manager instances (say vManage4, vManage5, and vManage6).

3. Complete the following operations on vManage1:
  - a. Configure the following using the CLI:
    - System IP address
    - Site ID
    - Service Provider organization name (`sp-organization-name`)
    - Organization-name
    - Cisco SD-WAN Validator IP address
    - VPN 0 Transport/Tunnel interface

- VPN 0 Out-of-band (OOB) interface: Ensure that you assign a static IP address to this interface. Do not enable DHCP.
- VPN 512 Management interface



---

**Note** Configure only one default route in VPN 0.

---

- [Enable Multitenancy on Cisco SD-WAN Manager, on page 864.](#)
- (Optional) Using the CLI, install the Root CA certificate for vManage1.



---

**Note** Skip this step if you are using a Symantec or Cisco PKI certificate.

---

- Complete the following through the Cisco SD-WAN Manager:
    - [Generate a Certificate Signing Request](#)
    - After Symantec or your enterprise root CA has signed the certificate, [install the signed certificate](#).
  - [Configure the Cluster IP Address of the Cisco SD-WAN Manager Server.](#)

Before proceeding to the next step, ensure that the **Manager IP Address** field on the **Administration > Cluster Management** page shows the OOB interface address.
4. Complete the following operations on vManage2 through vManage6:



---

**Important** Do not enable multitenancy on vManage2 through vManage6.

---

- Configure the following using the CLI:
  - System IP address
  - Site ID
  - Service Provider organization name (`sp-organization-name`)
  - Organization-name
  - Cisco SD-WAN Validator IP address
  - VPN 0 Transport/Tunnel interface
  - VPN 0 Out-of-band (OOB) interface: Ensure that you assign a static IP address to this interface. Do not enable DHCP.
  - VPN 512 Management interface
- (Optional) Using the CLI, install the Root CA certificate for vManage1.




---

**Note** Skip this step if you are using a Symantec or Cisco PKI certificate.

---

- c. Complete the following through the Cisco SD-WAN Manager:
  1. [Generate a Certificate Signing Request](#)
  2. After Symantec or your enterprise root CA has signed the certificate, [install the signed certificate](#).
- d. [Log in to the Cisco SD-WAN Manager Web Application Server](#).
- e. Ping the OOB interfaces on the other Cisco SD-WAN Manager instances and ensure they are reachable.
- f. [Configure the Cluster IP Address of the Cisco SD-WAN Manager Server](#).

Before proceeding to the next step, ensure that the **Manager IP Address** field on the **Administration > Cluster Management** page shows the OOB interface address.

5. Log in to the vManage1 GUI and [add vManage2 to the cluster](#).

vManage2 reboots before being added to the cluster.

While vManage2 is being added to the cluster, on the **Administration > Cluster Management** page, the **Configure Status** for vManage2 shows **Pending**. You can monitor the System Generated Cluster Sync transaction to check the progress of the adding vManage2 to the cluster.

When the operation is completed, on the **Administration > Cluster Management** page, you can view both vManage1 and vManage2, and their node personas.

6. Repeat **Step 5** and add vManage3 through vManage6 to the cluster.

## Enable Multitenancy on Cisco SD-WAN Manager




---

**Note** Administrator triggered disaster recovery is supported for multitenant clusters from Cisco vManage Release 20.6.1 or later releases.

---

### Prerequisites

Do not migrate an existing single-tenant Cisco SD-WAN Manager into multitenant mode, even if you invalidate or delete all devices from the existing Cisco SD-WAN Manager. Instead, download and install a new software image of Cisco vManage Release 20.6.1 or a later release.




---

**Note** After you enable multitenancy on Cisco SD-WAN Manager, you cannot migrate it back to single tenant mode.

---

1. Launch Cisco SD-WAN Manager using the URL `https://vmanage-ip-address:port`. Log in as the provider **admin** user.

2. From the Cisco SD-WAN Manager menu, choose **Administration > Settings > Tenancy Mode**. If you are using Cisco Catalyst SD-WAN Manager Release 20.12.x or earlier, click **Edit**.
3. In the **Tenancy** field, click **Multitenant**.
4. In the **Domain** field, enter the domain name of the service provider (for example, managed-sp.com).
5. Enter a **Cluster Id** (for example, cluster-1 or 123456).
6. Click **Save**. If you are using Cisco Catalyst SD-WAN Manager Release 20.12.x or earlier, click **Proceed** to confirm that you want to change the tenancy mode.

Cisco SD-WAN Manager reboots in multitenant mode and when a provider user logs in to Cisco SD-WAN Manager, the provider dashboard appears.



**Note** The **Domain** and **Cluster Id** values created in steps 5 and 6 serve as the Provider FQDN. Ensure these values conform to current DNS naming conventions. You can not modify these values after the configuration is saved. To change these values, a new Cisco SD-WAN Manager cluster need to be deployed. For more details on Provider and Tenant DNS requirements, see step 3.d in [Add a New Tenant](#).

## Add Cisco SD-WAN Controller

1. Log in to Cisco SD-WAN Manager as the provider **admin** user.
2. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
3. Click **Controllers**.



**Note** Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

4. Click **Add Controller**.
5. In the **Add Controller** dialog box, do the following:
  - a. In the **Controller Management IP Address** field, enter the system IP address of the Cisco SD-WAN Controller.
  - b. Enter the **Username** and **Password** required to access the Cisco SD-WAN Controller.
  - c. Select the protocol to use for control-plane connections. The default is **DTLS**.  
If you select **TLS**, enter the port number to use for TLS connections. The default is 23456.
  - d. Check the **Generate CSR** check box for Cisco SD-WAN Manager to create a Certificate Signing Request.
  - e. Click **Add**.
6. From the Cisco SD-WAN Manager menu, choose **Configuration > Certificates**.  
For the newly added Cisco SD-WAN Controller, the **Operation Status** reads **CSR Generated**.

- a. For the newly added Cisco SD-WAN Controller, click **More Options** icon and click **View CSR**.
  - b. Submit the CSR to the Certificate Authority (CA) and obtain a signed certificate.
7. From the Cisco SD-WAN Manager menu, choose **Configuration > Certificates**.
  8. Click **Install Certificate**.
  9. In the **Install Certificate** dialog box, paste the **Certificate Text** or click **Select a file** upload the certificate file. Click **Install**.

Cisco SD-WAN Manager installs the certificate on the Cisco SD-WAN Controller. Cisco SD-WAN Manager also sends the serial number of the certificate to other controllers.

On the **Configuration > Certificates** page, the **Operation Status** for the newly added Cisco SD-WAN Controller reads as **Validator Updated**.

On the **Configuration > Devices** page, the new controller is listed in the Controller table with the controller type, hostname of the controller, IP address, site ID, and other details. The **Mode** is set to **CLI**.

10. Change the mode of the newly added Cisco SD-WAN Controller to **Manager Mode** by attaching a template to the device.
  - a. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.  
For more information on configuration using CLI template, see [Device Configuration-Based CLI Templates](#).
  - b. Click **Device Templates**.




---

**Note** In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled as **Device**

---

- c. Find the template to be attached to the Cisco SD-WAN Controller.
- d. Click ..., and click **Attach Devices**.
- e. In the **Attach Devices** dialog box, move the new controller to the **Selected Device** list and click **Attach**.
- f. Verify the **Config Preview** and click **Configure Devices**.

Cisco SD-WAN Manager pushes the configuration from the template to the new controller.

In the **Configuration > Devices** page, the **Mode** for the Cisco SD-WAN Controller shows **Manager Mode**. The new Cisco SD-WAN Controller is ready to be used in your multitenant deployment.

## Expand a Multitenant Deployment to Support More Tenants and Tenant Devices

As a service provider, suppose you have deployed a C to the overlay to support up to 100 tenants and 5000 devices. From Cisco IOS XE Release 17.6.3a, Cisco vManage Release 20.6.3, you can expand the Cisco



SD-WAN Manager cluster and add additional Cisco SD-WAN Controllers to the overlay to support up to 150 tenants and 7500 devices.

### Prerequisites

A multitenant Cisco Catalyst SD-WAN overlay that supports up to 50 tenants and 1000 devices, deployed according to the steps outlined in the *Initial Setup for Multitenancy* section of this document.

1. [Expand the existing 3-node Cisco SD-WAN Manager cluster to a 6-node cluster.](#)
2. To support up to 100 tenants and 5000 devices, you must have 10 Cisco SD-WAN Controllers in the overlay. So, deploy 4 Cisco SD-WAN Controllers in addition to the 6 existing Cisco SD-WAN Controllers in the overlay.

To support up to 150 tenants and 7500 devices, you must have 16 Cisco SD-WAN Controllers in the overlay. So, deploy 10 Cisco SD-WAN Controllers in addition to the 6 existing Cisco SD-WAN Controllers in the overlay.

- a. Create Cisco SD-WAN Controller instances. See [Deploy the Cisco SD-WAN Controller](#).
- b. [Add Cisco SD-WAN Controllers](#) to the overlay network.

You can now add more tenants or allow your existing tenants to add more devices subject to the relevant limits.



---

**Note** Starting from Cisco SD-WAN Manager Release 20.13.1, you can expand a single node cluster into 3 or 6 node clusters.

---

## Expand a 3-Node Cluster to a 6-node Cluster



---

**Note** You can only expand a 3-node Cisco SD-WAN Manager cluster to a 6-node Cisco SD-WAN Manager cluster. Expansion of the 3-node cluster to other cluster sizes is not supported.

---

1. To support 100 tenants and 5000 devices: Upgrade the three Cisco SD-WAN Manager servers in the existing 3-node cluster to the hardware specifications in the table *Hardware Specifications to Support 100 Tenants and 5000 Devices* from the *Supported Devices and Controller Specifications* section of this document.

From Cisco IOS XE Release 17.6.3a, Cisco vManage Release 20.6.3, to support 150 tenants and 7500 devices: Upgrade the three Cisco SD-WAN Manager servers in the existing 3-node cluster to the hardware specifications in the table *Hardware Specifications to Support 150 Tenants and 7500 Devices* from the *Supported Devices and Controller Specifications* section of this document.

2. Download the Cisco vManage Release 20.6.1 or a later release software image from [Cisco Software Download](#).
3. Create three Cisco SD-WAN Manager instances (say vManage1, vManage2, and vManage3) by installing the downloaded software image file. See [Deploy Cisco SD-WAN Manager](#).

**Important**

- Deploy Cisco SD-WAN Manager servers having the hardware specifications in the table *Hardware Specifications to Support 100 Tenants and 5000 Devices* from the *Supported Devices and Controller Specifications* section of this document.

From Cisco IOS XE Release 17.6.3a, Cisco vManage Release 20.6.3, to support 150 tenants and 7500 devices, deploy Cisco SD-WAN Manager servers having the hardware specifications in the table *Hardware Specifications to Support 150 Tenants and 7500 Devices* from the *Supported Devices and Controller Specifications* section of this document.

- Choose the **Data** persona for each Cisco SD-WAN Manager instance.

#### 4. Complete the following operations on vManage1 through vManage3:

**Important**

Do not enable multitenancy on vManage1 through vManage3.

##### a. Configure the following using the CLI:

- System IP address
- Site ID
- Service Provider organization name (`sp-organization-name`)
- Organization-name
- Cisco SD-WAN Validator IP address
- VPN 0 Transport/Tunnel interface
- VPN 0 Out-of-band (OOB) interface: Ensure that you assign a static IP address to this interface. Do not enable DHCP.
- VPN 512 Management interface

**Note**

Configure only one default route in VPN 0.

##### b. (Optional) Using the CLI, install the Root CA certificate for vManage1.

**Note**

Skip this step if you are using a Symantec or Cisco PKI certificate.

##### c. Complete the following through the Cisco SD-WAN Manager:

1. [Generate a Certificate Signing Request](#)
2. After Symantec or your enterprise root CA has signed the certificate, [install the signed certificate](#).

##### d. [Log in to the Cisco SD-WAN Manager Web Application Server](#).

- e. Ping the OOB interfaces on the other Cisco SD-WAN Manager instances and ensure they are reachable.
- f. [Configure the Cluster IP Address of the Cisco SD-WAN Manager Server.](#)

Before proceeding to the next step, ensure that the **Manager IP Address** field on the **Administration > Cluster Management** page shows the OOB interface address.

5. Log in to the GUI of the existing 3-node Cisco SD-WAN Manager cluster and [add vManage1 to the cluster](#).

vManage1 reboots before being added to the cluster.

While vManage1 is being added to the cluster, on the **Administration > Cluster Management** page, the **Configure Status** for vManage1 shows **Pending**. You can monitor the System Generated Cluster Sync transaction to check the progress of the adding vManage1 to the cluster.

When the operation is completed, on the **Administration > Cluster Management** page, you can view vManage1 and its node persona listed along with the three Cisco SD-WAN Manager instances that were part of the original 3-node cluster.

6. Repeat **Step 4** and add vManage2 and vManage3 to the cluster.

## Manage Tenants

### Tenant Device Forecasting

While adding a new tenant to the multitenant Cisco Catalyst SD-WAN deployment, a service provider can forecast the number of WAN edge devices that the tenant may deploy in their overlay network. Cisco SD-WAN Manager enforces this forecast limit. If the tenant tries to add devices beyond this limit, Cisco SD-WAN Manager responds with an appropriate error message and the device addition fails.

In a multitenant deployment, a tenant can add a maximum of 1000 devices to their overlay network.



**Note** From Cisco IOS XE Release 17.6.2, Cisco vManage Release 20.6.2, you can modify the device forecast for a tenant after the tenant is added. This modification is not supported in Cisco IOS XE Release 17.6.1a, Cisco vManage Release 20.6.1.

### Benefits:

- The service provider can ensure that the Cisco Catalyst SD-WAN controller resources are used more efficiently.
- Depending on the configuration, a multitenant deployment can support a fixed number of WAN edge devices across all tenants. By forecasting the number of devices a tenant may add, the service provider can assign a quota for each tenant from the overall pool of edge devices that the deployment can support.

## View Specific Cisco Catalyst SD-WAN Controller Information in Tenant Dashboard

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.16.1a and Cisco Catalyst SD-WAN Manager Release 20.16.1

When adding a new tenant or modifying an existing tenant's information, providers can allow tenants to view their specific Cisco SD-WAN Controller information. The enhanced visibility into controller information allows tenants more efficient monitoring, management, and troubleshooting using Cisco Catalyst SD-WAN Manager. For more information about configuring tenant's access to Cisco Catalyst SD-WAN Controller information, see the following sections:

- [Add a New Tenant, on page 871](#)
- [Modify Tenant Information, on page 875](#)

## View Tenant Alarms in Provider Dashboard

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.16.1a and Cisco Catalyst SD-WAN Manager Release 20.16.1

1. Log in to Cisco SD-WAN Manager as the provider **admin** user.
2. From the Cisco SD-WAN Manager menu, choose **Operated as Provider > Provider > Logs**.
3. In the **Alarms** tab, toggle **View tenant alarms**.
4. Optional: Filter alarms using **Advanced Filter** based on **Tenant**, **Object Type**, **Object List**, **Severity**, and **Type**.
5. Optional: Click the bell icon on top to view all the notifications for tenant's alarms.

### What to do next:

Subscribe to tenant's alarm notifications.

## View Tenant Events in Provider Dashboard

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.16.1a and Cisco Catalyst SD-WAN Manager Release 20.16.1

1. Log in to Cisco SD-WAN Manager as the provider **admin** user.
2. From the Cisco SD-WAN Manager menu, choose **Operated as Provider > Provider > Logs**.
3. In the **Events** tab, toggle **View tenant events**.
4. Optional: Filter events using **Advanced Filter** based on **Tenant**, **Object Type**, **Object List**, **Severity**, and **Type**.

# Add a New Tenant

## Prerequisites

- At least two Cisco SD-WAN Controllers must be operational and in the `Manager` mode before you can add new tenants.

A Cisco SD-WAN Controller enters the `Manager` mode when you push a template onto the controller from Cisco SD-WAN Manager. A Cisco SD-WAN Controller in the `CLI` mode cannot serve multiple tenants.

- Each pair of Cisco SD-WAN Controllers can serve a maximum of 24 tenants and a maximum of 1000 tenant devices. Ensure that there at least two Cisco SD-WAN Controllers that can serve a new tenant. If no pair of Cisco SD-WAN Controllers in the deployment can serve a new tenant, add two Cisco SD-WAN Controllers and change their mode to `Manager`.
- If you add a second tenant immediately after adding a tenant, Cisco SD-WAN Manager adds them sequentially, and not in parallel.
- Each tenant must have a unique Virtual Account (VA) on **Plug and Play Connect** on [Cisco Software Central](#). The tenant VA should belong to the same Smart Account (SA) as the provider VA.
- For an on-premises deployment, create a **Validator** controller profile for the tenant on **Plug and Play Connect**. The fields in the following table are mandatory.

**Table 366: Controller Profile Fields**

| Field                       | Description/Value                                                                                                                                                                                                                                                                                                                   |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Profile Name</b>         | Enter a name for the controller profile.                                                                                                                                                                                                                                                                                            |
| <b>Multi-Tenancy</b>        | From the drop-down list, select <b>Yes</b> .                                                                                                                                                                                                                                                                                        |
| <b>SP Organization Name</b> | Enter the provider organization name.                                                                                                                                                                                                                                                                                               |
| <b>Organization Name</b>    | <p>Enter the tenant organization name in the format &lt;SP Org Name&gt;-&lt;Tenant Org Name&gt;.</p> <p><b>Note</b><br/>The organization name can be up to 50 characters.</p> <p><b>Note</b><br/>A mismatch of organization name format of the controller profile and the tenant creation leads to a failure in device sync up.</p> |
| <b>Primary Controller</b>   | Enter the host details for the primary Cisco SD-WAN Validator.                                                                                                                                                                                                                                                                      |

If you are using a Cisco-hosted multitenant environment, see [Add a new tenant in Cisco-hosted multitenant environment](#)

- Log in to Cisco SD-WAN Manager as the provider **admin** user.
- From the Cisco SD-WAN Manager menu, choose **Administration > Tenant Management**.

3. Click **Add Tenant**. In the **Add Tenant** dialog box:

- a. Enter a name for the tenant.

- b. Enter a description of the tenant.

The description can be up to 256 characters and can contain only alphanumeric characters.

- c. Enter the name of the organization.

The organization name is case-sensitive. Each tenant or customer must have a unique organization name.

Enter the organization name in the following format:

<SP Org Name>-<Tenant Org Name>

For example, if the provider organization name is 'multitenancy' and the tenant organization name is 'Customer1', while adding the tenant, enter the organization name as **multitenancy-Customer1**.



**Note** The organization name can be up to 50 characters.

A mismatch of organization name format of the controller profile and the tenant creation leads to a failure in device sync up.

- d. In the **URL Subdomain Name** field, enter the fully qualified sub-domain name of the tenant.

- The sub-domain name must include the domain name of the service provider. For example, for the `managed-sp.com` service provider, a valid domain name can be `customer1.managed-sp.com`.



**Note** The service provider name is shared amongst all tenants. Hence, ensure that the URL naming convention follows the same domain name convention that was provided while enabling multitenancy from **Administration > Settings > Tenancy Mode**.

- For an on-premises deployment, add the fully qualified sub-domain name of the tenant to the DNS. Map the fully qualified sub-domain name to the IP addresses of the three Cisco SD-WAN Manager instances in the Cisco SD-WAN Manager cluster.

- **Provider Level:** Create DNS A record and map it to the IP addresses of the Cisco SD-WAN Manager instances running in the Cisco SD-WAN Manager cluster. The A record is derived from the domain and Cluster ID that was created in steps 5 and 6 in [Enable Multitenancy on Cisco SD-WAN Manager](#). For example, if domain is **sdwan.cisco.com** and Cluster ID is **vmanage123**, then A record will need to be configured as **vmanage123.sdwan.cisco.com**.



**Note** If you fail to update DNS entries, it will result in authentication errors when logging in to Cisco SD-WAN Manager. Validate DNS is configured correctly by executing **nslookup vmanage123.sdwan.cisco.com**.

- **Tenant Level:** Create DNS CNAME records for each tenant created and map them to the FQDN created at the Provider Level. For example, if domain is **sdwan.cisco.com** and tenant name is **customer1** the CNAME record will need to be configured as **customer1.sdwan.cisco.com**.



**Note** Cluster ID is not required for CNAME record. Validate DNS is configured correctly by executing **nslookup customer1.sdwan.cisco.com**.

For a cloud deployment, the fully qualified sub-domain name of the tenant is automatically added to the DNS as part of the tenant creation process. After you add a tenant, it could take up to an hour before the fully qualified sub-domain name of the tenant can be resolved by the DNS.

- e. In the **Number of Devices** field, enter the number of WAN edge devices that the tenant can deploy. If the tenant tries to add WAN edge devices beyond this number, Cisco SD-WAN Manager reports an error and the device addition fails.
- f. Starting from Cisco Catalyst SD-WAN Manager Release 20.16.1, toggle **Controller Visibility** to enable the following in your tenant's dashboard:
  - In the **Monitor** dashboard, the **Controller Components** card with the **Controller** option that has an active hyperlink to view tenant-specific controller information
  - In the **Devices** page, controller details relevant to tenant
  - In the **Certificates** page, certificate details relevant to tenant
  - In **Devices > Real-time, Control Connections** and OMP statistics relevant to tenant.
- g. Click **Save**.

The **Create Tenant** screen appears, and the **Status** of the tenant creation reads **In progress**. To view status messages related to the creation of a tenant, click the > button to the left of the status.

Cisco SD-WAN Manager does the following:

- creates the tenant
- assigns two Cisco SD-WAN Controllers to serve the tenant and pushes a CLI template to these controllers to configure tenant information
- sends the tenant and Cisco SD-WAN Controller information to Cisco SD-WAN Validator.

#### What to do next:

After the **Status** column changes to **Success**, you can view the tenant information on the **Administration > Tenant Management** page.

## Add a new tenant in Cisco-hosted multitenant environment

1. Log in to Cisco SD-WAN Manager as the provider **admin** user.
2. From the Cisco SD-WAN Manager menu, choose **Administration > Tenant Management**.

3. Click **Add Tenant**. In the **Add Tenant** dialog box:

- a. Enter a name for the tenant. It must be the same as the Virtual Account used for the tenant.
- b. Enter a description.
- c. Enter the name of the organization in the following format:

<Provider Org Name>-<Tenant Org Name>

For example, if the provider organization name is 'EFT20.17-VA-Main – 841534' and the tenant organization name is 'T1', enter the organization name as **EFT20.17-VA-Main – 841534-T1**. The tenant organization name can be T1 for Tenant 1, T2 for Tenant 2, and so on.



**Note** The organization name can be up to 50 characters.

- d. Enter the sub-domain URL in FQDN format.
  - The sub-domain name must include `sdwan.cisco.com`. For example, a sub-domain name can be `Eftt1.sdwan.cisco.com`.
  - Ensure the sub-domain name is unique by performing a `nslookup` or `ping` on the expected domain. If the domain can be found, use a new URL.

The FQ sub-domain name of the tenant is automatically added to the DNS as part of the tenant creation process. After you add a tenant, it could take up to an hour before the fully qualified sub-domain name of the tenant is resolved by the DNS.

- e. In the **Number of Devices** field, enter the number of WAN edge devices that the tenant can deploy. If the tenant tries to add WAN edge devices beyond this number, Cisco SD-WAN Manager reports an error and the device addition fails.
- f. Choose Auto placement or manual option for controller assignment.
- g. Click **Save**.

**What to do next:**

After the process is complete, Cisco SD-WAN Manager automatically generates the controller profile in the tenant Virtual Account and the FQDN. You will receive an email after completion of the process.

## View Tenant Information

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.12.1a and Cisco Catalyst SD-WAN Manager Release 20.12.1

You can view the following tenant information from the **Tenant Management > Tenants** page:

- **Tenant Name**
- **Description**
- **Controllers**
- **Forecasted Edge Count**



- **Total Edge Count:** Total number of both multitenant and single-tenant edge devices.
- **Multi Tenant WAN Edge Devices:** To view the number of multitenant edge device, click the non-zero number.
- **Tenant-Provider VPN Mapping:** To view the tenant and device VPN mappings for the tenant, click the non-zero number.
- **Service Connector:** View the multitenant edge device that provides a VXLAN connectivity to tenants.
- **Notifications:** View whether webhook notifications are managed by the tenant or a provider.
- **AAA:** View whether remote AAA is managed by the tenant or a provider.
- **Controller Visibility:** View whether controller visibility is enabled or disabled.

## Modify Tenant Information

1. Log in to Cisco SD-WAN Manager as the provider **admin** user.
2. From the Cisco SD-WAN Manager menu, choose **Administration > Tenant Management**.
3. In the left pane, click the name of the tenant.  
The tenant information is displayed in a pane on the right.
4. To modify tenant data, do as follows:
  - a. In the right pane, click the pencil icon.
  - b. In the **Edit Tenant** dialog box, you can modify the following:
    - **Description:** The description can be up to 256 characters and can contain only alphanumeric characters.
    - **Forecasted Edge Count:** The number of WAN edge devices that the tenant can deploy.  
A tenant can add a maximum of 1000 devices.



### Note

This option is available from Cisco IOS XE Release 17.6.2, Cisco vManage Release 20.6.2.

If you increase the number of devices that a tenant can deploy, you must add the required number of device licenses to the tenant virtual account on **Plug and Play Connect** on [Cisco Software Central](#).

Before you increase the number of devices that a tenant can deploy, ensure that the Cisco SD-WAN Controller pair assigned to the tenant can support this increased number. A pair of Cisco SD-WAN Controllers can support a maximum of 24 tenants and 1000 devices across all these tenants.

- **Subdomain URL:** Modify the fully qualified sub-domain name of the tenant.

- c. (Optional) **Configure Services**: Set up service connectors to connect Cisco SD-WAN Manager to on-premises services such as AAA or custom webhook servers via the Cisco Catalyst SD-WAN overlay.
  - **Service Connector IP**: Choose the service connector IP address from the drop-down list.
  - **Service Connector Interface Name**: Choose the VXLAN tunnel endpoint (interface name) from the drop-down list.
  - **Provider Managed - External Services**: Enable services to enforce provider-managed or tenant-managed configuration.
    - **Notifications**: Configure alarm notifications through custom webhook channels. Choose **Yes** to enable notifications and configure a provider-managed multitenant setup. The default value is **No**, making it a tenant-managed setup.
    - **AAA**: Configure remote AAA for managing tenants. Choose **Yes** to enable remote AAA and configure a provider-managed multitenant setup. The default value is **No**, making it a tenant-managed setup.
- d. Starting from Cisco Catalyst SD-WAN Manager Release 20.16.1, toggle **Controller Visibility** to enable the following in your tenant's dashboard:
  - In the **Monitor** dashboard, the **Controller Components** card with the **Controller** option that has an active hyperlink to view tenant-specific controller information
  - In the **Devices** page, controller details relevant to tenant
  - In the **Certificates** page, certificate details relevant to tenant
  - In **Devices > Real-time, Control Connections** and OMP statistics relevant to tenant.
- e. Click **Save**

## Delete a Tenant

Before you delete a tenant, delete all tenant WAN edge devices. See [Delete a WAN Edge Device from a Tenant Network, on page 883](#).

1. Log in to Cisco SD-WAN Manager as the provider **admin** user.
2. From the Cisco SD-WAN Manager menu, choose **Administration > Tenant Management**.
3. In the left pane, click the name of the tenant.  
The tenant information is displayed in a pane on the right.
4. To delete the tenant, do as follows:
  - a. In the right pane, click the trash icon.
  - b. In the **Delete Tenant** dialog box, enter the provider **admin** password and click **Save**.

# Cisco SD-WAN Manager Dashboard for Multitenancy

After enabling Cisco SD-WAN Manager for multitenancy, you can view the multitenant dashboard when you log in to Cisco SD-WAN Manager. Cisco SD-WAN Manager multitenant dashboard is a portal where the provider or tenant can view and provision the underlying system.

The bar at the top of every Cisco SD-WAN Manager multitenant screen includes icons that allow smooth navigation.

## View Cisco Catalyst SD-WAN Validator Health Dashlet

Minimum supported releases: Cisco Catalyst SD-WAN Manager Release 20.12.1

You can view the state of each Cisco Catalyst SD-WAN Validator, which is the cumulative state of the Cisco Catalyst SD-WAN Validator in a selected time window, and the number of Cisco Catalyst SD-WAN Validators in each state in the **Validator Health** dashlet on **Monitor Overview** dashboard.

You can filter the Cisco Catalyst SD-WAN Validator dashlet view based on the health status using the drop-down list for **Good Devices**, **Fair Devices**, and **Poor Devices** and also for **CPU Load**.

Click **View Details** to open the **Monitor > Devices** page to view the device health in table view.

## View Cisco SD-WAN Manager Health Dashlet

Minimum supported releases: Cisco Catalyst SD-WAN Manager Release 20.12.1

You can view the state of Cisco SD-WAN Manager in the **Manager Health** dashlet on **Monitor Overview** dashboard.

You can filter the Cisco SD-WAN Manager dashlet view based on the health status using the drop-down list for **Good Devices**, **Fair Devices**, and **Poor Devices** and also for **CPU Load**.

Click **View Details** to open the **Monitor > Devices** page to view the device health in table view.

## View Cisco Catalyst SD-WAN Controller Health Dashlet

Minimum supported releases: Cisco Catalyst SD-WAN Manager Release 20.12.1

You can view the list of tenants hosted on a particular device by clicking the **Controller** bar. You can view the state of each Cisco Catalyst SD-WAN Controller, which is the cumulative state of the Cisco Catalyst SD-WAN Controller in a selected time window, and the number of Cisco Catalyst SD-WAN Controllers in each state in the **Controller Health** dashlet on **Monitor Overview** dashboard.

You can filter the Cisco Catalyst SD-WAN Controller dashlet view based on the health status using the drop-down list for **Good Devices**, **Fair Devices**, and **Poor Devices** and also for **CPU Load**.

Click **View Details** to open the **Monitor > Devices** page to view the device health in table view.

## View Multi Tenant WAN Edge Health Dashlet

Minimum supported releases: Cisco Catalyst SD-WAN Manager Release 20.12.1

You can view the state of each WAN edge device, which is the cumulative state of the devices in a selected time window, and the number of WAN edge devices in each state in the **Multi Tenant WAN Edge Health** dashlet on **Monitor Overview** dashboard.

You can view the list of tenants hosted on a particular device by clicking the multi-tenant WAN edge device bar. You can filter the Multi Tenant WAN Edge Health dashlet view based on the health status using the drop-down list for **Good Devices**, **Fair Devices**, and **Poor Devices** and also for **CPU Load**.

Click **View Details** to open the **Monitor > Devices** page to view the device health in table view.

## View Tenant Activity, Device, and Network Information

When you log in to a multitenant Cisco SD-WAN Manager as an administrator, the provider dashboard displays the following components. To return to the provider dashboard from other Cisco SD-WAN Manager screens, click **Dashboard**.

- **Device pane** — runs across the top of the multitenant dashboard screen. The device pane displays the number of active Cisco Catalyst SD-WAN Controllers, Cisco SD-WAN Validator, and Cisco SD-WAN Manager instances, the connectivity status of devices, and information on certificates that have expired or about to expire.
- **Tenants pane** — displays the total number of tenants and a summary of the control status, site health, router health, and Cisco Catalyst SD-WAN Controller status of all tenants.
- **Table of tenants in the overlay network** — List of individual tenants, with separate information about the control status, site health, WAN edge device health, and Cisco Catalyst SD-WAN Controller status for each tenant.

To display tenant-specific status summary information,

1. Click a tenant name from the tenant list.

A dialog box opens on the right side of the screen that provides additional information about the status of the tenant.

2. To access the tenant dashboard for the selected tenant, click **<Tenant name> Dashboard**.

Cisco SD-WAN Manager presents the provider-as-tenant view and displays the tenant dashboard. To return to the provider view, click **Provider** at the top of page.

3. To close the dialog box, click the tenant name from the tenant list.

## View Detailed Information of a Tenant Setup

Cisco SD-WAN Manager displays the tenant dashboard, which provides information about a tenant deployment when

- a provider **admin** user selects a specific tenant from the **Select Tenant** drop-down list in the provider dashboard. This view is called the provider-as-tenant view.
- a **tenantadmin** user logs in to Cisco SD-WAN Manager. This view is called the tenant view.

### View All Network Connections in the Tenant Overlay Network

The **Device** pane runs across the top of the tenant dashboard and displays the number of control connections from Cisco SD-WAN Manager to the Cisco SD-WAN Controllers and routers in the overlay network of a tenant. For each WAN edge device, the Device pane shows

- Total number of control connections between Cisco SD-WAN Controllers and WAN edge devices
- Number of valid control connections between Cisco SD-WAN Controllers and WAN edge devices
- Number of invalid control connections between Cisco SD-WAN Controllers and WAN edge devices

Click a connection number, or the **Up** or **Down** arrow, to display a table with detailed information about each connection. Click the **More Actions** icon at the right of each table row to access the **Device Dashboard** or **Real Time** view from the **Monitor > Network** screen, or access the **Tools > SSH Terminal** Screen.

Click a connection number, or the **Up** or **Down** arrow, to display a table with detailed information about each connection. Click the **More Actions** icon at the right of each table row to access the **Device Dashboard** or **Real Time** view from the **Monitor > Devices** screen, or access the **Tools > SSH Terminal** Screen.



---

**Note** In Cisco vManage Release 20.6.1 and earlier releases, you can view information related to the **Monitor > Devices** page under the **Monitor > Network** page.

---

### View Information About Device Reboots

The **Reboot** pane displays the total number of reboots in the last 24 hours for all devices in the network. It includes soft and cold reboots and reboots that occurred as a result of power-cycling a device. For each reboot, the following information is listed:

- System IP and hostname of the device that rebooted.
- Time when the device was rebooted.
- Reason for the device reboot

If the same device reboots more than once, each reboot option is reported separately.

Click the **Reboot** pane to open the **Reboot** dialog box. In the **Reboot** dialog box, click the **Crashes** tab. For all device crashes, the following information is listed:

- System IP and hostname of the device on which the crash occurred.
- Crash index of the device
- Core time when the device crashed.
- File name of the device crash log

### View Network Connections

The **Control Status** pane displays whether Cisco SD-WAN Controller and WAN edge devices are connected. Each Cisco SD-WAN Controller must connect to all other Cisco SD-WAN Controllers in the network. Each WAN edge device must connect to the maximum number of configured Cisco SD-WAN Controllers. The **Control Status** pane displays three network connection counts:

- **Control Up** — total number of devices with the required number of operational control plane connections to a Cisco SD-WAN Controller
- **Partial** — total number of devices with some, but not all, operational control plane connection to Cisco SD-WAN Controllers.
- **Control Down** — total number of devices with no control plane connection to a Cisco SD-WAN Controller

To display a table with device details, click a row from the **Control Status** dialog box. Click the **More Actions** icon at the right of each table row to access the **Device Dashboard** or **Real Time** view from the **Monitor > Devices** screen.




---

**Note** In Cisco vManage Release 20.6.1 and earlier releases, you can view information related to the **Monitor > Devices** page under the **Monitor > Network** page.

---

### View State of Data Connections for a Site

The **Site Health** pane displays the state of data connections for a site. When a site has multiple WAN edge devices, this pane displays the state for the entire site and not for individual devices. The Site Health pane displays three connectivity states:

- **Full WAN Connectivity** — total number of sites where all BFD sessions on all routers are in the up state.
- **Partial WAN Connectivity** — total number of sites where tunnel and all BFD sessions on all routers are in the down state. These sites still have limited data plane connectivity.
- **No WAN Connectivity** — total number of sites where all BFD sessions on all routers are in the down state. These sites have no data plane connectivity.

To display a table with detailed information about each site, node, or tunnel, click a row from the **Site Health** dialog box. Click the **More Actions** icon at the right of each row in the table to access the **Device Dashboard** or **Real Time** view from the **Monitor > Devices** screen, or access the **Tools > SSH Terminal** screen.




---

**Note** In Cisco vManage Release 20.6.1 and earlier releases, you can view information related to the **Monitor > Devices** page under the **Monitor > Network** page.

---

### View Interface Usage for WAN Edge Interfaces

The **Transport Interface Distribution** pane displays interface usage in the last 24 hours for all WAN edge interfaces in VPN 0. It includes all TLOC interfaces. Click the pane to view details of interface usage in the **Transport Interface Distribution** dialog box.

### View WAN Edge Device Counts

The **WAN Edge Inventory** pane provides four WAN edge device counts:

- **Total** — total number of authorized serial numbers for WAN edge devices that have been uploaded on Cisco SD-WAN Manager. The serial number is uploaded on the **Configuration > Devices** screen.

- **Authorized** — total number of authorized WAN edge devices in the overlay network. These WAN edge devices are marked as **Valid** in the **Configuration > Certificates > WAN Edge List** screen.
- **Deployed** — total number of deployed WAN edge devices. These are WAN edge devices that are marked as **Valid** and are now operational in the network.
- **Staging** — total number of WAN edge devices you configure at a staging site before they are made a part of the overlay network. These routers do not take part in any routing decisions and do not affect network monitoring through Cisco SD-WAN Manager.

Click the pane to view hostname, system IP, site ID, and other details of each router from the **WAN Edge Inventory** dialog box.

### View Aggregated State of WAN Edge Devices

The **WAN Edge Health** pane offers an aggregated view of the state of WAN edge devices by providing a count of the number of devices in each state, therefore describing the health of the hardware nodes. The three WAN edge device states are:

- **Normal** — number of WAN edge devices with memory, hardware, and CPU in normal state. Using less than 70% of total memory or total CPU is classified as, normal.
- **Warning** — number of WAN edge devices with memory, hardware, or CPU in warning state. Using between 70% and 90% of total memory or total CPU is classified as, warning
- **Error** — number of WAN edge devices with memory, hardware, or CPU in error state. Using more than 90% of total memory or total CPU is classified as, error.

Click a number or the WAN edge device state to display a table with the last 12 or 24 hours of memory usage, CPU utilization, and hardware-related alarms, including temperature, power supply, and PIM modules. Click the **More Actions** icon at the right of each row in the table to access the following:

- **Hardware Environment**
- **Real Time** view from the **Monitor > Network** screen



**Note** InCisco vManage Release 20.6.1 and earlier releases, you can view information related to the **Monitor > Devices** page under the **Monitor > Network** page.

- **Tools > SSH Terminal** screen.

### View WAN Edge Device Loss, Latency, Jitter

The **Transport Health** pane displays the aggregated average loss, latency, and jitters for all links and all combinations of colors (for example, all LTE-to-LTE links, all LTE-to-3G links).

From the **Type** drop-down arrow, choose loss, latency, or jitter.

Click the  icon to select a time period for which to display the transport health.

Click the  icon to open the **Transport Health** dialog box. This dialog box displays a more detailed view. To display information in a tabular format, click the **Details** tab. You can choose to change the displayed health type and time period.

### View SAIE Flow Information of WAN Edge Devices

The **Top Applications** pane displays SD-WAN Application Intelligence Engine (SAIE) flow information for traffic transiting routers in the overlay network.



#### Note

- In Cisco vManage Release 20.7.x and earlier releases, the SAIE flow is known as deep packet inspection (DPI).
- The SAIE flow information is shown only for the last 24 hours. To view SAIE flow information for a time before the last 24 hours, you must check the information for the specific device.

Click the  icon to select a time period for which to display data. From the **VPN** drop-down list, select a VPN to display SAIE information for all flows in that VPN.

Click the  icon to open the **Top Applications** dialog box. This dialog box displays a more detailed view of the same information. You can change the VPN and time period.

### View Tunnels Data

The **Application-Aware Routing** pane allows you to choose the following tunnel criteria from the **Type** drop-down arrow:

- Loss
- Latency
- Jitter

Based on the tunnel criteria, the pane displays the 10 worst tunnels. For example, if you choose loss, the pane shows 10 tunnels with the greatest average loss over the last 24 hours.

Click the  icon against a row to display a graphical representation of the data. Select a time period for which to display data or click **Custom** to display a drop-down arrow for specifying a custom time period.

Click the  icon to open the **Application-Aware Routing** dialog box. This dialog box displays the 25 worst tunnels based on criteria you choose from the **Type** drop-down arrow, the criteria being loss, latency, and jitter.

## Manage Tenant WAN Edge Devices

### Add a WAN Edge Device to a Tenant Network



#### Note

If you are adding a WAN edge device that you had previously invalidated and deleted from an overlay network, you must reset the device software after adding the device. To reset the software on a Cisco IOS XE Catalyst SD-WAN device, use the command **request platform software sdwan software reset**.

1. Log in to Cisco SD-WAN Manager.



If you're a provider user, log in as the **admin**. In the provider dashboard, choose a tenant from the drop-down list to enter the provider-as-tenant view.

If you're a tenant user, log in as the **tenantadmin**.

2. Upload the device serial number file to Cisco SD-WAN Manager.
3. Validate the device and send details to controllers.
4. Create a configuration template for the device and attach the device to the template.

While configuring the device, configure the service provider organization name and the tenant organization name as in the following example:

```
sp-organization-name multitenancy
organization-name multitenancy-Customer1
```




---

**Note** Enter the `organization-name` in the format `<SP Org Name>-<Tenant Org Name>`.

---

5. Bootstrap the device using bootstrap configuration generated through Cisco SD-WAN Manager or manually create the initial configuration on the device.
6. If you are using Enterprise Certificates to authenticate the device, download the CSR from Cisco SD-WAN Manager and get the CSR signed by the Enterprise CA. Install the certificate on Cisco SD-WAN Manager.

## Delete a WAN Edge Device from a Tenant Network

1. Log in to Cisco SD-WAN Manager.

If you're a provider user, log in as the **admin**. In the provider dashboard, choose a tenant from the drop-down list to enter the provider-as-tenant view.

If you're a tenant user, log in as the **tenantadmin**.

2. Detach the device from any configuration templates.
3. [Delete a WAN Edge Router](#).

## Tenant-Specific Policies on Cisco SD-WAN Controller

A provider **admin** user (from the Cisco SD-WAN Manager provider-as-tenant view) or a **tenantadmin** user (from the Cisco SD-WAN Manager tenant view) can create and deploy tenant-specific policies on the Cisco SD-WAN Controller serving the tenant. The user can configure a CLI policy or create the policy using the UI policy configuration wizard.

When you activate or deactivate a policy,

1. Cisco SD-WAN Manager identifies the Cisco SD-WAN Controllers serving the tenant.
2. Cisco SD-WAN Manager notifies the Cisco SD-WAN Controllers to pull the policy configuration.
3. Cisco SD-WAN Controllers pull and deploy the policy configuration.
4. Cisco SD-WAN Manager reports the status of the policy pull by the Cisco SD-WAN Controllers.

# Manage Tenant Data

## Back Up Tenant Data

The tenant data backup solution of Cisco SD-WAN Manager multitenancy provides the following functionalities:

- [Create, list, and extract configuration backup files.](#)
- Back up configuration database of a specific tenant with an option to restore it later. See [Back up and restore configuration database.](#)
- Delete back up files of a tenant stored in Cisco SD-WAN Manager. For deleting tenant data backup files, see [Delete tenant data backup file.](#)

The following factors are applicable when using data backup solution:

- The tenant data backup solution operations can be performed by a tenant administrator in the tenant view and or by a provider administrator in the provider-as-tenant view. To know how to access tenant dashboard through different views, see [User Roles in Multitenant Environment, on page 853.](#)
- A tenant is allowed to perform the following backup operations at a particular time and must complete an operation before starting a new operation:
  - Back up a single configuration database
  - Download the backup file.
  - Restore or import backup files
  - Delete backup files.
  - List backup files

- A tenant backup file format is as follows:

```
Bkup_tenantId_MMDDYY-HHMMSS_taskIdWithoutDash.tar.gz
```

- The tenant data backup operation is a readonly operation on the configuration database. However, to ensure data consistency and prevent data loss, do not perform any major changes on the network while the operation is in progress.
- Multiple tenants can perform back-up and restore operations in parallel.
- A tenant is not allowed to perform other backup operations when the restore operation of the tenant database is in-progress. So, a tenant can perform a single backup operation and when this operation is in-progress, all new backup operation requests are rejected.

The remaining tenants can continue with their backup operations.

- A tenant must perform backup and restore operations on Cisco SD-WAN Manager instances running identical Cisco SD-WAN Manager software versions.
- A tenant can store a maximum of three backup files in Cisco SD-WAN Manager and can download to store them outside Cisco SD-WAN Manager repository. If the tenant already has three backup files, a

subsequent backup operation results in the earliest backup file being deleted and a new backup file being generated.

- Ensure that the following parameter values match in both the backup file and the setup where tenant has requested for a restore operation:
  - Tenant Id
  - Organization Name
  - SP Organization Name
- The tenant data backup solution creates a task in the tenant view of Cisco SD-WAN Manager. Therefore, the tenant can monitor the progress of the operation from the task view of the tenant dashboard.
- A provider cannot back up provider data using this solution. Therefore, the provider can back up all tenants information at once by backing up all tenants configuration database using CLI.

## Create, Extract, and List Configuration Data Backup File

1. Log in to Cisco SD-WAN Manager.

If you're a provider user, log in as the **admin**. In the provider dashboard, choose a tenant from the drop-down list to enter the provider-as-tenant view.

If you're a tenant user, log in as the **tenantadmin**.

2. In the address bar, modify the URL path with `dataservice` for the REST API connection.

**Example:** `https://<tenant_URL>/dataservice`

3. Create a configuration backup file by using the following API:

`https://<tenant_URL>/dataservice/tenantbackup/export.`

4. If the configuration backup file has been created successfully, Cisco SD-WAN Manager task view indicates that the backup file has been generated. You can view the process identifier of the created process or task.

**Example:**

```
{
 "processId": "72d69805-b987-436f-9b7a-afef2f3f9061",
 "status": "in-progress"
}
```

5. Verify the task status using the obtained process identifier.

**Example:**

`https://<tenant_URL>/dataservice/device/action/status/72d69805-b987-436f-9b7a-afef2f3f9061`

The verification generates the details of the task in the JSON file format.

6. After the task is completed, extract or download the backed-up file available under the **data** section of the JSON task file.

**Example:** To extract or download the backup file, use the following API:

`https://<tenant_URL>/dataservice/tenantbackup/download/1570057020772/backup_1570057020772_100919-181838.tar.gz`

7. List backup files stored in Cisco SD-WAN Manager using the following API.

**Example:** `https://<tenant_URL>/dataservice/tenantbackup/list`

## Restore and Delete Tenant Data Backup File

### Before you begin:

To run the restore and delete tenant data backup files API, you can download and install Postman tool or any other alternative tool for testing http applications and services. In this document, the procedure to restore and delete tenant data backup files has been explained using the Postman tool. Postman is a software tool used as an API development environment. You can download the tool from the Postman website.

1. Open Google Chrome, or another browser, and enable developer mode on it.
2. Log in to Cisco SD-WAN Manager.

If you're a provider user, log in as the **admin**. In the provider dashboard, choose a tenant from the drop-down list to enter the provider-as-tenant view.

If you're a tenant user, log in as the **tenantadmin**.

3. To get header information of the restore API, do as follows:
  - a. On the right side of the screen, click the **Network** tab to get the network capture view.
  - b. In the network capture view, click the **Name** column to sort the listed items.
  - c. Search and click **index.html**.
  - d. Click the **Headers** tab and expand **Request Headers**.
  - e. Choose all text under **Request Headers** and copy it to the clipboard.
4. Import backup files through the Postman UI:
  - a. Open the Postman UI.
  - b. To disable SSL certificate verification, click **Postman > Preferences > General > Request**. Turn off **SSL Certificate Verification**.
  - c. In the Postman UI, create a new tab.
  - d. Click **Headers** and then click **Bulk Edit**.
  - e. Paste the text that was copied in step 3 from the **Request Headers** block into an editable form.
  - f. From the **GET** method drop-down list, choose **POST**.
  - g. In the **Paste request URL** field, paste the dedicated URL of the tenant and include `dataservice/tenantbackup/import`.  
**Example:** `https://customer1.managed-sp.com/dataservice/tenantbackup/import`
  - h. Click the **Body** tab and select **form-data**.
  - i. Under **KEY** column, enter `bakup.tar.gz`
  - j. Under **VALUE** column, click **Select Files** and select a backup file to be imported.
  - k. To run the API, click **Send**.

In the **Response** section of the Postman UI, you can view the JSON information that indicates the file that was restored.

5. Monitor the restoration of backup files in either of the following ways:
  - a. Use Cisco SD-WAN Manager task view that indicates if backup file has been imported successfully. You can view the process identifier of the created process or task.

**Example:**

```
{
 "processId": "40adb6c0-eacc-4ad4-ba6c-2c2da2e96d1d",
 "status": "Import Successfully Submitted for tenant 1579026919487"
}
```

- b. Use the following URL to get the status,

`https://<tenant_URL>/dataservice/device/action/status/<processId>`

**Example:**

`https://customer1.managed-sp.com/dataservice/device/action/status/40adb6c0-eacc-4ad4-ba6c-2c2da2e96d1d`

6. Delete tenant data backup file through Postman UI.
  - a. In the Postman UI, create a new tab.
  - b. Click **Headers** and then click **Bulk Edit**.
  - c. Paste the text that was copied in step 3 from the **Request Headers** block into an editable form.
  - d. From the **GET** method drop-down list, choose **DELETE**.
  - e. In the **Paste request URL** field, paste the dedicated URL of the tenant and include `dataservice/tenantbackup/delete?fileName=' filename '`. The filename can either be name of the backup file or all.

**Example:**

`https://customer1.managed-sp.com/dataservice/tenantbackup/delete?fileName=bkup_1579026919487_012820-180712_c09230904dfc40edb0d1e50b68b03002.tar.gz`

**Example:** `https://customer1.managed-sp.com/dataservice/tenantbackup/delete?fileName=all`

- f. To run the API, click **Send**.

In the **Response** section of the Postman UI, you can view the JSON information that indicates the files that were deleted.

**Example:**

```
{
 "Deleted": [
 "bkup_1579026919487_012820-180712_c09230904dfc40edb0d1e50b68b03002.tar.gz"
]
}
```

## View OMP Statistics per Tenant on a Cisco SD-WAN Controller

1. Log in to Cisco SD-WAN Manager as the provider **admin** user.
2. From the Cisco SD-WAN Manager menu, choose **Monitor > Devices**.  
Cisco vManage Release 20.6.x and earlier: From the Cisco SD-WAN Manager menu, choose **Monitor > Network**.
3. In the table of devices, click on the hostname of a Cisco SD-WAN Controller.

4. In the left pane, click **Real Time**.
5. In the **Device Options** field, enter **OMP** and select the OMP statistics you wish to view.
6. In the **Select Filters** dialog box, click **Show Filters**.
7. Enter the **Tenant Name** and click **Search**.

Cisco SD-WAN Manager displays the selected OMP statistics for the particular tenant.

## View Tenants Associated with a Cisco SD-WAN Controller

1. Log in to Cisco SD-WAN Manager as the provider **admin** user.
2. Click a **Controller** connection number to display a table with detailed information about each connection.  
Cisco SD-WAN Manager displays a table that provides a summary of the Cisco SD-WAN Controllers and their connections.
3. For a Cisco SD-WAN Controller, click ... and click **Tenant List**.

Cisco SD-WAN Manager displays a summary of tenants associated with the Cisco SD-WAN Controller.

## Migrate Single-Tenant Cisco Catalyst SD-WAN Overlay to Multitenant Cisco Catalyst SD-WAN Deployment

### Before You Begin

- Before you begin the migration,
  - Migration of a single-tenant overlay to a multitenant deployment is only supported with the Cisco Catalyst SD-WAN controllers deployed on-premises. Migration is yet to be supported with cloud-hosted Cisco Catalyst SD-WAN controllers.
  - Ensure that the edge devices in the single-tenant deployment can reach the Cisco SD-WAN Validator in the multitenant deployment
  - Ensure that the template, routing, and policy configuration on the edge devices is synchronized with the current configuration on Cisco SD-WAN Manager
  - Configure a maintenance window for the single-tenant overlay before performing this procedure. See [Configure or Cancel SD-WAN Manager Server Maintenance Window](#).
- Minimum software requirements for the single-tenant overlay to be migrated:

| Device                           | Software Version             |
|----------------------------------|------------------------------|
| Cisco SD-WAN Manager             | Cisco vManage Release 20.6.1 |
| Cisco SD-WAN Validator           | Cisco SD-WAN Release 20.6.1  |
| Cisco Catalyst SD-WAN Controller | Cisco SD-WAN Release 20.6.1  |

| Device                              | Software Version                             |
|-------------------------------------|----------------------------------------------|
| Cisco IOS XE Catalyst SD-WAN device | Cisco IOS XE Catalyst SD-WAN Release 17.6.1a |

- Minimum software requirements for the multitenant deployment to which the single-tenant overlay must be migrated:

| Device                              | Software Version                             |
|-------------------------------------|----------------------------------------------|
| Cisco SD-WAN Manager                | Cisco vManage Release 20.6.1                 |
| Cisco SD-WAN Validator              | Cisco SD-WAN Release 20.6.1                  |
| Cisco Catalyst SD-WAN Controller    | Cisco SD-WAN Release 20.6.1                  |
| Cisco IOS XE Catalyst SD-WAN device | Cisco IOS XE Catalyst SD-WAN Release 17.6.1a |

- The software versions of the Cisco Catalyst SD-WAN controllers and WAN edge devices must be identical in both the single-tenant and multitenant deployments.
- We recommend that you use a custom script or a third-party application like Postman to execute the API calls.

### Migration Procedure

1. Export the single-tenant deployment and configuration data from a Cisco SD-WAN Manager instance controlling the overlay.

|               |                                                  |
|---------------|--------------------------------------------------|
| Method        | POST                                             |
| URL           | <code>https://ST-vManage-IP-address</code>       |
| Endpoint      | <code>/dataservice/tenantmigration/export</code> |
| Authorization | Admin user credentials.                          |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Body     | <p>Required</p> <p>Format: Raw JSON</p> <pre>{   "desc": &lt;tenant_description&gt;,   "name": &lt;tenant_name&gt;,   "subdomain": &lt;tenant_name&gt;.&lt;domain&gt;,   "orgName": &lt;tenant_orgname &gt; }</pre> <p>Field Description:</p> <ul style="list-style-type: none"> <li>• <b>desc:</b> A description of the tenant. The description can be up to 256 characters and can contain only alphanumeric characters.</li> <li>• <b>name:</b> Unique name for the tenant in the multitenant deployment.</li> <li>• <b>subdomain:</b> Fully qualified sub-domain name of the tenant. The sub-domain name must include the domain name of the service provider. For example, if <code>managed-sp.com</code> is the domain name of service provider, and the tenant name is <code>Customer1</code>, the tenant sub-domain name would be <code>customer1.managed-sp.com</code>.</li> <li>• <b>orgName:</b> Name of the tenant organization. The organization name is case-sensitive.</li> </ul> |
| Response | <p>Format: JSON</p> <pre>{   "processId": &lt;vManage_process_ID&gt;, }</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

While exporting the data, Cisco SD-WAN Manager attempts to detach any CLI templates from the edge devices in preparation for the migration to the multitenant deployment. If prompted by Cisco SD-WAN Manager, detach CLI templates from the edge devices and execute the export API call again.

2. Check the status of the data export task in Cisco SD-WAN Manager. When the task succeeds, download the data using the URL  
<https://ST-vManage-IP-address/dataservice/tenantmigration/download/default.tar.gz>
3. On a multitenant Cisco SD-WAN Manager instance, import the data exported from the single-tenant overlay.

|               |                                                                                             |
|---------------|---------------------------------------------------------------------------------------------|
| Method        | POST                                                                                        |
| URL           | <a href="https://MT-vManage-IP-address">https://MT-vManage-IP-address</a>                   |
| Endpoint      | /dataservice/tenantmigration/import                                                         |
| Authorization | Provider Admin user credentials.                                                            |
| Body          | <p>Required</p> <p>Format: form-data</p> <p>Key Type: File</p> <p>Value: default.tar.gz</p> |



|          |                                                                                                                     |
|----------|---------------------------------------------------------------------------------------------------------------------|
| Response | Format: JSON<br><pre>{   "processId": &lt;vManage_process_ID&gt;,   "migrationTokenURL": &lt;token_URL&gt;, }</pre> |
|----------|---------------------------------------------------------------------------------------------------------------------|

When the task succeeds, on the multitenant Cisco SD-WAN Manager, you can view the devices, templates, and policies imported from the single-tenant overlay.

- Obtain the migration token using the token URL obtained in response to the API call in **Step 3**.

|               |                                                      |
|---------------|------------------------------------------------------|
| Method        | GET                                                  |
| URL           | <code>https://MT-vManage-IP-address</code>           |
| Endpoint      | migrationTokenURL obtained in <b>Step 3</b> .        |
| Authorization | Provider Admin user credentials.                     |
| Response      | The migration token as a large blob of encoded text. |

- On the single-tenant Cisco SD-WAN Manager instance, initiate the migration of the overlay to the multitenant deployment.

|               |                                                                                      |
|---------------|--------------------------------------------------------------------------------------|
| Method        | POST                                                                                 |
| URL           | <code>https://ST-vManage-IP-address</code>                                           |
| Endpoint      | <code>dataservice/tenantmigration/networkMigration</code>                            |
| Authorization | Admin user credentials.                                                              |
| Body          | Required<br>Format: Raw text<br>Content: Migration token obtained in <b>Step 4</b> . |
| Response      | Format: JSON<br><pre>{   "processId": &lt;vManage_process_ID&gt;, }</pre>            |

In Cisco SD-WAN Manager, check the status of the migration task. As part of the migration task, the address of the multitenant Cisco SD-WAN Validator, and the service provider and tenant organization names are pushed to the WAN edge devices of the single-tenant overlay. If the task succeeds, WAN edge devices form control connections to controllers in the multitenant deployment; the WAN edge devices are no longer connected to the controllers of the single-tenant overlay.

Attach any CLI templates detached from the edge devices (in Step 1) after migration to the multitenant deployment. Before you attach the templates, update the Cisco SD-WAN Validator IP address and the Organization name to match the configuration of the multitenant deployment.



**Note** In the single-tenant deployment, if Cisco SD-WAN Manager-signed certificates are installed on cloud-based WAN edge devices, the certificates are cleared when the devices are migrated to the multitenant deployment. You must re-certify the devices on the multitenant Cisco SD-WAN Manager. If enterprise certificates are installed on the cloud-based WAN edge devices, the certificates are not affected by the migration. For more information, see [Enterprise Certificates](#).

# Migrate a Tenant from a Multitenant Cisco Catalyst SD-WAN Overlay to Single-Tenant Cisco Catalyst SD-WAN Deployment

## Before You Begin

- Manually migrate the serial number of the WAN edge device associated to a virtual account on the source Cisco SD-WAN Manager overlay in Cisco PNP to the destination virtual account.
- Ensure that you manually create the controller profile on the destination virtual account for on-prem to on-prem or cloud to on-prem deployments.
- Ensure that the source and destination Cisco SD-WAN Manager instances have the same Certificate Authority (CA). If not, recertify the devices after the migration is complete.
- Ensure that you check the CPU, memory, and disk size requirements of the destination overlay Cisco SD-WAN Controllers before the migration to meet the WAN edge forecast requirements.
- Ensure that there is no overlap between the configured system IP addresses of edge devices and the destination overlay controllers.
- Ensure that the destination single-tenant Cisco SD-WAN Manager does not have any configurations before migration. You can configure only mandatory admin settings and all other configurations can be done after data import.
- Ensure that the Cisco SD-WAN Control Components in the source and destination overlays are using the same software release. The migration process does not check for a software release mismatch and a mismatch blocks the import of tenant data, causing the migration to fail.
- Ensure that all devices in a tenant have connectivity to the Cisco SD-WAN Validator in the destination single-tenant overlay. The migration procedure supports a Cisco SD-WAN Validator on the single-tenant deployment configured either with IP or DNS.

Push any required static route configuration to the devices before initiating any of the migration steps.

- Ensure that the WAN edge devices that are configured using CLI, device template, or configuration groups, have an IP host mapping to the Cisco SD-WAN Validator in the destination single-tenant overlay.
- Ensure that there are valid control connections from Cisco SD-WAN Manager to the WAN edge devices in the source overlay.
- Configure a maintenance window for the multitenant overlay before performing this procedure. See [Configure or Cancel SD-WAN Manager Server Maintenance Window](#).

- We recommend that you use a custom script or a third-party application like Postman to execute the API calls.

### Migration Procedure

1. Export the multitenant deployment configuration and statistical data from a Cisco SD-WAN Manager instance controlling the source overlay.

|               |                                                  |
|---------------|--------------------------------------------------|
| Method        | POST                                             |
| URL           | <code>https://MT-vManage-IP-address</code>       |
| Endpoint      | <code>/dataservice/tenantmigration/export</code> |
| Authorization | Administrator user credentials.                  |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Body     | <p>Required</p> <p>Format: Raw JSON</p> <p>Example:</p> <pre>{   "name": "tenant1",   "desc": "This is tenant1",   "orgName": "vIPtela Inc MT to ST Migration Regression-Tenant1 Inc",   "subDomain": "tenant1.mtreg.com",   "wanEdgeForecast": 100,   "migrationKey": "tenant1TenantMigrationKey123",   "isDestinationOverlayMT": false }</pre> <p>Field descriptions:</p> <p><b>Note</b><br/>Ensure that the <code>name</code>, <code>desc</code>, <code>orgName</code>, <code>subdomain</code>, and <code>wanEdgeForecast</code> match the tenant you wish to migrate.</p> <ul style="list-style-type: none"> <li>• <code>name</code>: Unique name for the tenant in the multitenant deployment. The name should be between 8-32 characters and can contain only alphanumeric characters.</li> <li>• <code>desc</code>: Description of the tenant. The description can be up to 256 characters and can contain only alphanumeric characters.</li> <li>• <code>orgName</code>: Name of the tenant organization. The organization name is case-sensitive.</li> <li>• <code>subdomain</code>: Fully qualified sub-domain name of the tenant. The sub-domain name must include the domain name of the service provider. For example, if <code>managed-sp.com</code> is the domain name of service provider, and the tenant name is <code>Customer1</code>, the tenant sub-domain name would be <code>customer1.managed-sp.com</code>.</li> <li>• <code>wanEdgeForecast</code>: Number of WAN edge devices that the tenant can deploy.</li> <li>• <code>migrationKey</code>: Migration key which is used to encrypt sensitive data during migration. The migration key should be between 8-32 characters and can contain only alphanumeric characters.</li> <li>• <code>isDestinationOverlayMT</code>: Boolean variable which specifies if the migration is happening to a multitenant overlay or not.</li> </ul> |
| Response | <p>Format: JSON</p> <pre>{   "processId": &lt;vManage_process_ID&gt;, }</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

2. Check the status of the data export task in Cisco SD-WAN Manager. When the task is successfully complete, download the data from the following URL:

<https://MT-vManage-IP-address/dataservice/tenantmigration/download/default.tar.gz>

3. Import the data to the single-tenant instance, as follows:

- a. Execute the following API:

|               |                                                                                                                     |
|---------------|---------------------------------------------------------------------------------------------------------------------|
| Method        | POST                                                                                                                |
| URL           | <code>https://ST-vManage-IP-address</code>                                                                          |
| Endpoint      | <code>/dataservice/tenantmigration/import/{migrationKey}</code><br>Use the same migration key specified earlier.    |
| Authorization | Provider administrator user credentials.                                                                            |
| Body          | Required<br>Format: form-data<br>Key Type: File<br>Value: <code>default.tar.gz</code>                               |
| Response      | Format: JSON<br><pre>{   "processId": &lt;vManage_process_ID&gt;,   "migrationTokenURL": &lt;token_URL&gt;, }</pre> |

- b. When the task is complete, on the single-tenant Cisco SD-WAN Manager, you can view the devices, templates, and policies imported from the multitenant overlay.

- After the import, update information related to the device templates, policies, and other deployment-specific parameters. Check and update the administrator settings as some of the administrator settings specific to the source overlay are not exported. The import does not override the administrator settings that are already configured in destination Cisco SD-WAN Manager.
- If a centralized policy is present on the source tenant, the migration copies the policy to the destination overlay. We recommend creating Cisco SD-WAN Controller templates and attaching them to the devices. Apply the centralized policy to devices in the destination overlay before proceeding.
- Obtain the migration token using the token URL from the previous step.

|               |                                                               |
|---------------|---------------------------------------------------------------|
| Method        | GET                                                           |
| URL           | <code>https://ST-vManage-IP-address</code>                    |
| Endpoint      | <code>migrationTokenURL</code> obtained in the previous step. |
| Authorization | Provider administrator user credentials.                      |
| Response      | The migration token as a large encoded text.                  |

- On the multitenant Cisco SD-WAN Manager instance, initiate the migration of the overlay to the single-tenant deployment.

|               |                                                           |
|---------------|-----------------------------------------------------------|
| Method        | POST                                                      |
| URL           | <code>https://MT-vManage-IP-address</code>                |
| Endpoint      | <code>dataservice/tenantmigration/networkMigration</code> |
| Authorization | Administrator user credentials.                           |

|          |                                                                                         |
|----------|-----------------------------------------------------------------------------------------|
| Body     | Required<br>Format: Raw text<br>Content: Migration token obtained in the previous step. |
| Response | Format: JSON<br><pre>{   "processId": &lt;vManage_process_ID&gt;, }</pre>               |

In Cisco SD-WAN Manager, check the status of the migration task. When the task succeeds, WAN edge devices form control connections to controllers in the single-tenant deployment; the WAN edge devices are no longer connected to the controllers of the multitenant overlay.

**8.** After the migration is successfully complete, perform the following tasks:

- If WAN edge devices have Cisco SD-WAN Manager signed certificates in the source setup, the certificates are cleared from the device during migration and control connections are lost. Recertify the devices in the destination.
- The passwords are updated to the default password in the destination overlay for users created on a tenant in the source overlay. Make any configuration changes specific to the destination overlay.
- Delete the tenant on the source overlay after migration and verification is complete.

## Migrate Multitenant Cisco Catalyst SD-WAN Overlay

### Prerequisites

Minimum software requirements for Cisco Catalyst SD-WAN controllers and WAN edge devices in the multitenant overlay to be migrated:

| Device                              | Software Version             |
|-------------------------------------|------------------------------|
| Cisco SD-WAN Manager                | Cisco vManage Release 20.3.3 |
| Cisco SD-WAN Validator              | Cisco SD-WAN Release 20.3.3  |
| Cisco SD-WAN Controller             | Cisco SD-WAN Release 20.3.3  |
| Cisco IOS XE Catalyst SD-WAN device | Cisco IOS XE Release 17.3.3  |

### Restrictions

- This migration procedure applies only to Cisco Catalyst SD-WAN controllers deployed on premises.
- The multitenant overlay can only be migrated to a setup in which Cisco SD-WAN Manager instances run Cisco vManage Release 20.6.1 software and Cisco Catalyst SD-WAN controllers run Cisco SD-WAN Release 20.6.1 software.
- This migration procedure cannot be used to merge two or more multitenant overlays. Only one multitenant overlay can be migrated to the new setup at a time.

## Migration Procedure

1. Upgrade the software on the three Cisco SD-WAN Manager instances in the cluster to Cisco vManage Release 20.6.1. For more information, see [Upgrade Cisco SD-WAN Manager Cluster](#).



**Note** Run the command **request nms configuration-db upgrade** on only one of the Cisco SD-WAN Manager instances.

2. After the Cisco SD-WAN Manager software is upgraded to Cisco vManage Release 20.6.1, log in to the Cisco SD-WAN Manager.  
You're prompted to set a new password.
  - a. Enter a new password that adheres to the password guidelines.
3. Upload the Cisco SD-WAN Release 20.6.1 software to Cisco SD-WAN Manager. For more information, see [Add an Image to the Software Repository](#).
4. Upgrade the Cisco SD-WAN Validator software to Cisco SD-WAN Release 20.6.1. For more information, see [Upgrade the Software Image on a Device](#).
5. Create two Cisco SD-WAN Controller instances running Cisco SD-WAN Release 20.6.1 software. See [Deploy the Cisco SD-WAN Controller](#).



**Note** With two Cisco SD-WAN Controller instances, you can support up to 24 tenants. To support up to 50 tenants, create six Cisco SD-WAN Controller instances.

- a. [Add Cisco SD-WAN Controllers](#) to the overlay network.

The **Provider Dashboard** shows the new Cisco SD-WAN Controllers running Cisco SD-WAN Release 20.6.1 software. The **Tenant Dashboard** shows the older Cisco SD-WAN Controllers running Cisco SD-WAN Release 20.3.3 software.

6. Enable the maintenance window on Cisco SD-WAN Manager. For more information, see [Configure or Cancel SD-WAN Manager Server Maintenance Window](#).  
A maintenance window of 3 to 4 hours is recommended.
7. Migrate the tenant configuration from the older tenant-specific Cisco SD-WAN Controllers running Cisco SD-WAN Release 20.3.3 software to the new shared Cisco SD-WAN Controllers running Cisco SD-WAN Release 20.6.1 software.

|               |                                                     |
|---------------|-----------------------------------------------------|
| Method        | POST                                                |
| URL           | <code>https://&lt;vmanageip&gt;:&lt;port&gt;</code> |
| Endpoint      | <code>dataservice/tenant/vsmart-mt/migrate</code>   |
| Authorization | Provider <b>admin</b> user credentials.             |

|          |                                                              |
|----------|--------------------------------------------------------------|
| Body     | Required<br>Format: Raw JSON<br>{ }                          |
| Response | Format: JSON<br>{<br>"processId": <vManage_process_ID>,<br>} |

In Cisco SD-WAN Manager, check the status of the migration task using the `processId` from the API response. During the migration task, the following changes are affected:

- a. The older Cisco SD-WAN Controllers are invalidated and deleted from the overlay network.
  - b. In the tenant view, the older Cisco SD-WAN Controllers are removed from the **Tenant Dashboard**, and the **Devices** and the **Certificates** page.
  - c. The tenant WAN edge devices are connected to the new Cisco SD-WAN Controllers.
8. (Optional) Upgrade the Cisco IOS XE Catalyst SD-WAN device software to Cisco IOS XE Catalyst SD-WAN Release 17.6.1a. For more information, see [Upgrade the Software Image on a Device](#) and [Activate a New Software Image](#).



**Tip** It is not necessary to upgrade the tenant WAN edge device software in the same maintenance window in which you migrate the multitenant overlay. However, we recommend that you upgrade the tenant WAN edge device software within a few weeks of the migration.

### Verify the Migration

1. In the provider view, perform the following checks:
  - a. From the **Main Dashboard** page, verify whether the tenant WAN edge devices are connected to the new multitenant Cisco SD-WAN Controllers.
  - b. [View Tenants Associated with a Cisco SD-WAN Controller, on page 888](#).
  - c. On the Cisco SD-WAN Controller CLI, run the command **show control connections**. In the command output, verify that control connections are established between the Cisco SD-WAN Controller and the tenant WAN edge devices.
2. In the provider-as-tenant view, verify whether the multitenant Cisco SD-WAN Controllers appear on the **Tenant Dashboard**.



# Upgrade Cisco Catalyst SD-WAN Controller and Edge Device Software

## Prerequisites

Minimum software requirements for Cisco Catalyst SD-WAN controllers and WAN edge devices:

| Device                              | Software Version                      |
|-------------------------------------|---------------------------------------|
| Cisco SD-WAN Manager                | Cisco vManage Release 20.4.1 or later |
| Cisco SD-WAN Validator              | Cisco SD-WAN Release 20.4.1 or later  |
| Cisco Catalyst SD-WAN Controller    | Cisco SD-WAN Release 20.4.1 or later  |
| Cisco IOS XE Catalyst SD-WAN device | Cisco IOS XE Release 17.4.1 or later  |

## Upgrade Procedure

1. Upgrade the software on the three Cisco SD-WAN Manager instances in the cluster to Cisco vManage Release 20.6.1 or a later release. For more information, see [Upgrade Cisco SD-WAN Manager Cluster](#).



---

**Note** Skip the step to upgrade the configuration-db service using the command **request nms configuration-db upgrade**.

---

2. After the Cisco SD-WAN Manager software is upgraded to Cisco vManage Release 20.6.1 or a later release, log in to the Cisco SD-WAN Manager.
3. Upload the Cisco SD-WAN Release 20.6.1 or a later release and the Cisco IOS XE Catalyst SD-WAN Release 17.6.1a or a later release software to Cisco SD-WAN Manager. For more information, see [Add an Image to the Software Repository](#).
4. Upgrade the Cisco SD-WAN Validator software to Cisco SD-WAN Release 20.6.1 or a later release. For more information, see [Upgrade the Software Image on a Device](#) and [Activate a New Software Image](#).
5. Enable maintenance window on Cisco SD-WAN Manager. For more information, see [Configure or Cancel SD-WAN Manager Server Maintenance Window](#).
6. Upgrade the Cisco Catalyst SD-WAN Controller software to Cisco SD-WAN Release 20.6.1 or a later release. For more information, see [Upgrade the Software Image on a Device](#) and [Activate a New Software Image](#).
7. Upgrade the Cisco IOS XE Catalyst SD-WAN device software to Cisco IOS XE Catalyst SD-WAN Release 17.6.1a or a later release. For more information, see [Upgrade the Software Image on a Device](#) and [Activate a New Software Image](#).



---

**Tip** We recommend that you upgrade the WAN edge device software in the same maintenance window. If the WAN edge device software is not upgraded within the OMP graceful restart window, traffic may be lost.

---

# Multitenant Cisco SD-WAN Manager: Disaster Recovery



**Note** Administrator triggered disaster recovery is supported for multitenant clusters from Cisco vManage Release 20.6.1 or later releases.

If a Multitenant Cisco SD-WAN Manager cluster or the data center hosting the Cisco SD-WAN Manager nodes in the cluster fail, you can recover from the failure by activating a standby Cisco SD-WAN Manager cluster. You can perform disaster recovery as follows:

1. Deploy and configure a standby Cisco SD-WAN Manager cluster.

The standby Cisco SD-WAN Manager cluster is not part of the overlay network and is not active.

2. Back up the configuration database of the active Cisco SD-WAN Manager cluster periodically.

Choose a Cisco SD-WAN Manager node in the cluster that hosts the configuration database service and back up the configuration database.

3. If the active Cisco SD-WAN Manager cluster fails, restore the most recent configuration database on the standby Cisco SD-WAN Manager cluster, activate the standby Cisco SD-WAN Manager cluster, and remove the previously active Cisco SD-WAN Manager cluster from the overlay network.

Choose a Cisco SD-WAN Manager node in the cluster that hosts the configuration database service and restore the configuration database backed up from the previously active Cisco SD-WAN Manager cluster.

To test disaster recovery, you can simulate a scenario in which the active Cisco SD-WAN Manager cluster fails. One way to simulate such a failure would be by disabling the tunnel interface as described in this document.

## Prerequisites

- The number of Cisco SD-WAN Manager nodes in the active and standby clusters must be identical.
- Each Cisco SD-WAN Manager node in the active and standby clusters must run the same Cisco SD-WAN Manager software release.
- Each Cisco SD-WAN Manager node in the active and standby clusters must be able to connect to the WAN transport IP address of the Cisco SD-WAN Validator in the overlay network.
- Initially, the tunnel interfaces of the Cisco SD-WAN Manager nodes in the standby cluster must be disabled.
- The Cisco SD-WAN Manager nodes in the standby cluster must be certified.
- The clock of every Cisco SD-WAN Manager node in the standby cluster must be synchronized with the clocks of the Cisco Catalyst SD-WAN controllers and WAN edge devices in the overlay network. If NTP is configured on the overlay, configure the same on the standby Cisco SD-WAN Manager nodes.
- The Cisco SD-WAN Manager nodes in the active and standby clusters should use identical neo4j credentials.

## Restrictions

- Do not interrupt any active processes while backing up the configuration database.
- If you wish to enable SD-AVC, you must do so before the restoring the configuration database on standby Cisco SD-WAN Manager node.

## Configure a Standby Cisco SD-WAN Manager Cluster

1. Configure the standby Cisco SD-WAN Manager nodes with a similar running configuration as the active Cisco SD-WAN Manager nodes. Install local certificates on the standby Cisco SD-WAN Manager nodes.



**Note** The running configuration on a standby Cisco SD-WAN Manager is usually identical to that of an active Cisco SD-WAN Manager node. However, you must ensure that settings such as the system IP address and the tunnel interface IP address are unique.

2. On the standby Cisco SD-WAN Manager nodes, shut down the transport interface in VPN 0: On the CLI, include the **shutdown** command in the transport interface configuration.
3. Create a standby cluster using the standby Cisco SD-WAN Manager nodes.

With the standby Cisco SD-WAN Manager nodes configured in this manner, the overlay network is not aware of the standby Cisco SD-WAN Manager cluster.

## Back Up the Active Cisco SD-WAN Manager Cluster Configuration

Back up the full configuration database of the active Cisco vManage cluster periodically. Additionally, take snapshots of the active Cisco SD-WAN Manager virtual machines.

1. Choose an active Cisco SD-WAN Manager node that hosts the configuration database service and export a backup the configuration database. On the CLI of the Cisco SD-WAN Manager node, run the following command: **request nms configuration-db backup path file-path**

The command backs up the configuration database in a `.tar.gz` file and saves the file in the specified *file-path*.

In the following example, the database is backed up to a file named `db_backup.tar.gz` in the `/home/admin/` directory.

```
Active-vManage# request nms configuration-db backup path /home/admin/db_backup
Successfully saved database to /home/admin/db_backup.tar.gz
```

2. Choose a standby Cisco SD-WAN Manager node that hosts the configuration database service and copy the configuration database backup to this node.

In the following example, `db_backup.tar.gz` is copied from the active Cisco SD-WAN Manager node to the `/home/admin/` directory of a standby Cisco SD-WAN Manager node.

```
Active-vManage# request execute vpn 512 scp /home/admin/db_backup.tar.gz
admin@10.126.93.92:/home/admin
The authenticity of host '10.126.93.92 (10.126.93.92)' can't be established.
ECDSA key fingerprint is SHA256:jTjJWQ0UNHv1rBUxWzNjd8mUz819gPf51MeopsgDlAc.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '10.126.93.92' (ECDSA) to the list of known hosts.
viptela 18.4.5
```

```
admin@10.126.93.92's password:
db_backup.tar.gz 100% 399KB 4.4MB/s 00:00
```

### Restore Cisco SD-WAN Manager Cluster Using the Configuration Database Backup

Restore the most recent backup of the configuration database from the active Cisco SD-WAN Manager cluster on the standby Cisco SD-WAN Manager node to which you copied this backup.



#### Note

- The restore operation does not restore all the information included in the configuration database. Cisco SD-WAN Manager configurations such as users and repositories must be configured on the standby Cisco SD-WAN Manager node after the configuration database is restored using the backup.
- When you complete the steps that follow, the previously active Cisco SD-WAN Manager nodes cannot be reused. To reuse the nodes, you must perform additional steps that are beyond the scope of this document.

1. On the CLI of the standby Cisco SD-WAN Manager node, run the following command: **request nms configuration-db restore path file-path**

In the following example, the configuration database is restored using the backup file `db_backup.tar.gz`.

```
Standby-vManage# request nms configuration-db restore path /home/admin/db_backup.tar.gz
Configuration database is running in a standalone mode
Importing database...Successfully restored database
```

2. Verify that the appropriate services are running on the standby Cisco SD-WAN Manager nodes: On the CLI of each standby Cisco SD-WAN Manager node, run the **request nms all status** command. From the command output, verify the services running on the node.
3. Verify that every standby Cisco SD-WAN Manager node has a list of all the active and standby Cisco SD-WAN Manager nodes.
  - a. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices > Controllers**.



#### Note

Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

- b. Verify that the page displays all active and standby Cisco SD-WAN Manager nodes.
4. On the standby Cisco SD-WAN Manager nodes, enable the transport interface on VPN 0.

Use one of the following two methods:

- a. Enable the transport interface in VPN 0: On the CLI of each standby Cisco SD-WAN Manager node, run the **no shutdown** command.

```
Active-vManage# config
Active-vManage(config)# vpn 0 interface interface-name
Active-vManage(config-interface)# no shutdown
Active-vManage(config-interface)# commit and-quit
```

- b. Activate the tunnel interface in VPN 0: On the CLI of each standby Cisco SD-WAN Manager node, run the **tunnel-interface** command.

```
Active-vManage# config
Active-vManage(config)# vpn 0 interface interface-name
Active-vManage(config-interface)# tunnel-interface
Active-vManage(config-interface)# commit and-quit
```

5. Add each standby Cisco SD-WAN Manager node to the overlay network.
  - a. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
  - b. Click **Controllers**.



**Note** Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

- c. For a Cisco SD-WAN Validator, click ... and click **Edit**.
  - d. In the **Edit** dialog box, enter the following details of the Cisco SD-WAN Validator: WAN transport IP address, username, and password.
  - e. Repeat **Step 5c** and **Step 5d** for every Cisco SD-WAN Validator.

6. Disconnect the active Cisco SD-WAN Manager nodes from the overlay network.



**Note** In a lab environment, where you are simulating a disaster scenario, you can perform this step. However, if you cannot reach Cisco SD-WAN Manager instances in an actual disaster scenario, you may not be able to perform this step and can omit the step.

Use one of the following two methods:

- a. Shut down the transport interface in VPN 0: On the CLI of each active Cisco SD-WAN Manager node, run the **shutdown** command.

```
Active-vManage# config
Active-vManage(config)# vpn 0 interface interface-name
Active-vManage(config-interface)# shutdown
Active-vManage(config-interface)# commit and-quit
```

- b. Deactivate the tunnel interface in VPN 0: On the CLI of each active Cisco SD-WAN Manager node, run the **no tunnel-interface** command.

```
Active-vManage# config
Active-vManage(config)# vpn 0 interface interface-name
Active-vManage(config-interface)# no tunnel-interface
Active-vManage(config-interface)# commit and-quit
```

7. From the standby Cisco SD-WAN Manager, send the updated controller and device list to the Cisco SD-WAN Validator.

Send the list of controllers:

- a. From the Cisco SD-WAN Manager menu, choose **Configuration > Certificates**.

- b. Click **Controllers**.



---

**Note** Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

---

- c. Click **Send to Validator**.

Wait for the configuration task to complete. When the task is complete,

- The standby Cisco SD-WAN Manager nodes become the active Cisco SD-WAN Manager nodes.
- The previously active Cisco SD-WAN Manager nodes are no longer part of the overlay network.
- The active Cisco SD-WAN Manager nodes have the configuration from the most recent configuration database backup.
- Every controller establishes connection with the other controllers in the network.

- d. Click **WAN Edge List**.

- e. Click **Send to Controllers**.

8. Verify that the following are intact:

- Policies
- Templates
- Controller and WAN edge device lists

9. Verify the valid Cisco SD-WAN Manager nodes.

- a. Log in to the CLI of each Cisco SD-WAN Validator and run the **show orchestrator valid-vmanage-id** command.

In the command output, verify that the chassis numbers of the active and previously active Cisco SD-WAN Manager nodes are listed.

- b. Log in to the CLI of a WAN edge device and run the **show control valid-vmanage-id** command.

In the command output, verify that the chassis numbers of the active and previously active Cisco SD-WAN Manager nodes are listed. Also, check whether the device is connected to the active Cisco SD-WAN Manager nodes and the Cisco Catalyst SD-WAN Controller.

10. Invalidate the previously active Cisco SD-WAN Manager nodes.



---

**Note** After you invalidate the Cisco SD-WAN Manager nodes, the nodes cannot be reused without performing additional steps that are beyond the scope of this document.

---

- a. From the Cisco SD-WAN Manager menu, choose **Configuration > Certificates**.
- b. Click **Controllers**.



**Note** Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

c. For each previously active Cisco SD-WAN Manager node, click ... and click **Invalidate**.

11. Verify the valid Cisco SD-WAN Manager nodes.

a. Log in to the CLI of each Cisco SD-WAN Validator and run the **show orchestrator valid-vmanage-id** command.

In the command output, verify that the chassis numbers of only the active Cisco SD-WAN Manager nodes are listed.

b. Log in to the CLI of WAN edge device and run the **show control valid-vmanage-id** command.

In the command output, verify that the chassis numbers of only the active Cisco SD-WAN Manager nodes are listed. Also, check whether the device is connected to the active Cisco SD-WAN Manager nodes and the Cisco Catalyst SD-WAN Controller.

The Cisco SD-WAN Manager cluster that was initially the standby cluster is now the active Cisco SD-WAN Manager cluster.

## Multitenant Cisco SD-WAN Manager: Disaster Recovery in an Overlay Network with Virtual Routers

If a Multitenant Cisco SD-WAN Manager cluster or the data center hosting the Cisco SD-WAN Manager nodes in the cluster fail, you can recover from the failure by activating a standby Cisco SD-WAN Manager cluster. You can perform disaster recovery as follows:

1. Deploy and configure a standby Cisco SD-WAN Manager cluster.

The standby Cisco SD-WAN Manager cluster is not part of the overlay network and is not active.

2. Back up the configuration database of the active Cisco SD-WAN Manager cluster periodically.

Choose a Cisco SD-WAN Manager node in the cluster that hosts the configuration database service and back up the configuration database.

3. If the active Cisco SD-WAN Manager cluster fails, restore the most recent configuration database on the standby Cisco SD-WAN Manager cluster, activate the standby Cisco SD-WAN Manager cluster, and remove the previously active Cisco SD-WAN Manager cluster from the overlay network.

Choose a Cisco SD-WAN Manager node in the cluster that hosts the configuration database service and restore the configuration database backed up from the previously active Cisco SD-WAN Manager cluster.

To test disaster recovery, you can simulate a scenario in which the active Cisco SD-WAN Manager cluster fails. One way to simulate such a failure would be by disabling the tunnel interface as described in this document.

The following disaster recovery procedure applies to an overlay network in which Cisco vEdge Cloud routers are deployed at branch locations.

### Prerequisites

- The number of Cisco SD-WAN Manager nodes in the active and standby clusters must be identical.
- Each Cisco SD-WAN Manager node in the active and standby clusters must run the same Cisco SD-WAN Manager software release.
- Each Cisco SD-WAN Manager node in the active and standby clusters must be able to connect to the WAN transport IP address of the Cisco SD-WAN Validator in the overlay network.
- Initially, the tunnel interfaces of the Cisco SD-WAN Manager nodes in the standby cluster must be disabled.
- The Cisco SD-WAN Manager nodes in the standby cluster must be certified.
- The clock of every Cisco SD-WAN Manager node in the standby cluster must be synchronized with the clocks of the Cisco Catalyst SD-WAN controllers and WAN edge devices in the overlay network. If NTP is configured on the overlay, configure the same on the standby Cisco SD-WAN Manager nodes.
- The Cisco SD-WAN Manager nodes in the active and standby clusters should use identical neo4j credentials.

### Restrictions

- Do not interrupt any active processes while backing up the configuration database.
- If you wish to enable SD-AVC, you must do so before the restoring the configuration database on standby Cisco SD-WAN Manager node.

### Configure a Standby Cisco SD-WAN Manager Cluster

1. Configure the standby Cisco SD-WAN Manager nodes with a similar running configuration as the active Cisco SD-WAN Manager nodes. Install local certificates on the standby Cisco SD-WAN Manager nodes.



**Note** The running configuration on a standby Cisco SD-WAN Manager is usually identical to that of an active Cisco SD-WAN Manager node. However, you must ensure that settings such as the system IP address and the tunnel interface IP address are unique.

2. On the standby Cisco SD-WAN Manager nodes, shut down the transport interface in VPN 0: On the CLI, include the **shutdown** command in the transport interface configuration.
3. Create a standby cluster using the standby Cisco SD-WAN Manager nodes.

With the standby Cisco SD-WAN Manager nodes configured in this manner, the overlay network is not aware of the standby Cisco SD-WAN Manager cluster.

### Back Up the Active Cisco SD-WAN Manager Cluster Configuration

Back up the full configuration database of the active Cisco vManage cluster periodically. Additionally, take snapshots of the active Cisco SD-WAN Manager virtual machines.



1. Choose an active Cisco SD-WAN Manager node that hosts the configuration database service and export a backup the configuration database. On the CLI of the Cisco SD-WAN Manager node, run the following command: **request nms configuration-db backup path file-path**

The command backs up the configuration database in a `.tar.gz` file and saves the file in the specified *file-path*.

In the following example, the database is backed up to a file named `db_backup.tar.gz` in the `/home/admin/` directory.

```
Active-vManage# request nms configuration-db backup path /home/admin/db_backup
Successfully saved database to /home/admin/db_backup.tar.gz
```

2. Choose a standby Cisco SD-WAN Manager node that hosts the configuration database service and copy the configuration database backup to this node.

In the following example, `db_backup.tar.gz` is copied from the active Cisco SD-WAN Manager node to the `/home/admin/` directory of a standby Cisco SD-WAN Manager node.

```
Active-vManage# request execute vpn 512 scp /home/admin/db_backup.tar.gz
admin@10.126.93.92:/home/admin
The authenticity of host '10.126.93.92 (10.126.93.92)' can't be established.
ECDSA key fingerprint is SHA256:jTjJWQ0UNHv1rBUxWzNjd8mUz819gPf51MeopsgDlAc.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '10.126.93.92' (ECDSA) to the list of known hosts.
viptela 18.4.5

admin@10.126.93.92's password:
db_backup.tar.gz 100% 399KB 4.4MB/s 00:00
```

### Restore Cisco SD-WAN Manager Cluster Using the Configuration Database Backup

Restore the most recent backup of the configuration database from the active Cisco SD-WAN Manager cluster on the standby Cisco SD-WAN Manager node to which you copied this backup.



#### Note

- The restore operation does not restore all the information included in the configuration database. Cisco SD-WAN Manager configurations such as users and repositories must be configured on the standby Cisco SD-WAN Manager node after the configuration database is restored using the backup.
- When you complete the steps that follow, the previously active Cisco SD-WAN Manager nodes cannot be reused. To reuse the nodes, you must perform additional steps that are beyond the scope of this document.

1. On the CLI of the standby Cisco SD-WAN Manager node, run the following command: **request nms configuration-db restore path file-path**

In the following example, the configuration database is restored using the backup file `db_backup.tar.gz`.

```
Standby-vManage# request nms configuration-db restore path /home/admin/db_backup.tar.gz
Configuration database is running in a standalone mode
Importing database...Successfully restored database
```

2. Verify that the appropriate services are running on the standby Cisco SD-WAN Manager nodes: On the CLI of each standby Cisco SD-WAN Manager node, run the **request nms all status** command. From the command output, verify the services running on the node.

3. Verify that every standby Cisco SD-WAN Manager node has a list of all the active and standby Cisco SD-WAN Manager nodes.
  - a. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices > Controllers**.




---

**Note** Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

---

- b. Verify that the page displays all active and standby Cisco SD-WAN Manager nodes.
4. Log in to the CLI of each Cisco SD-WAN Validator and run the **show orchestrator valid-vmanage-id** command.

In the command output, verify that the chassis numbers of the active and previously active Cisco SD-WAN Manager nodes are listed.

5. Log in to the CLI of Cisco vEdge Cloud router and run the **show control valid-vmanage-id** command.
- In the command output, verify that the chassis numbers of the active and previously active Cisco SD-WAN Manager nodes are listed.

6. On the standby Cisco SD-WAN Manager nodes, enable the transport interface on VPN 0.

Use one of the following two methods:

- a. Enable the transport interface in VPN 0: On the CLI of each standby Cisco SD-WAN Manager node, run the **no shutdown** command.

```
Active-vManage# config
Active-vManage(config)# vpn 0 interface interface-name
Active-vManage(config-interface)# no shutdown
Active-vManage(config-interface)# commit and-quit
```

- b. Activate the tunnel interface in VPN 0: On the CLI of each standby Cisco SD-WAN Manager node, run the **tunnel-interface** command.

```
Active-vManage# config
Active-vManage(config)# vpn 0 interface interface-name
Active-vManage(config-interface)# tunnel-interface
Active-vManage(config-interface)# commit and-quit
```

7. Add each standby Cisco SD-WAN Manager node to the overlay network.
  - a. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
  - b. Click **Controllers**.




---

**Note** Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

---

- c. For a Cisco SD-WAN Validator, click ... and click **Edit**.
  - d. In the **Edit** dialog box, enter the following details of the Cisco SD-WAN Validator: WAN transport IP address, username, and password.

- e. Repeat **Step 7c** and **Step 7d** for every Cisco SD-WAN Validator.

8. Disconnect the active Cisco SD-WAN Manager nodes from the overlay network.



**Note** In a lab environment, where you are simulating a disaster scenario, you can perform this step. However, if you cannot reach Cisco SD-WAN Manager instances in an actual disaster scenario, you may not be able to perform this step and can omit this step.

Use one of the following two methods:

- a. Shut down the transport interface in VPN 0: On the CLI of each active Cisco SD-WAN Manager node, run the **shutdown** command.

```
Active-vManage# config
Active-vManage(config)# vpn 0 interface interface-name
Active-vManage(config-interface)# shutdown
Active-vManage(config-interface)# commit and-quit
```

- b. Deactivate the tunnel interface in VPN 0: On the CLI of each active Cisco SD-WAN Manager node, run the **no tunnel-interface** command.

```
Active-vManage# config
Active-vManage(config)# vpn 0 interface interface-name
Active-vManage(config-interface)# no tunnel-interface
Active-vManage(config-interface)# commit and-quit
```

9. From the standby Cisco SD-WAN Manager, send the updated controller and device list to the Cisco SD-WAN Validator.

Send the list of controllers:

- a. From the Cisco SD-WAN Manager menu, choose **Configuration > Certificates**.
- b. Click **Controllers**.



**Note** Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

- c. Click **Send to Validator**.

Wait for the configuration task to complete. When the task is complete,

- The standby Cisco SD-WAN Manager nodes become the active Cisco SD-WAN Manager nodes.
- The previously active Cisco SD-WAN Manager nodes are no longer part of the overlay network.
- The active Cisco SD-WAN Manager nodes have the configuration from the most recent configuration database backup.
- Every controller establishes connection with the other controllers in the network.

- d. Click **WAN Edge List**.

- e. Click **Send to Controllers**.
10. Verify that the following are intact:
    - Policies
    - Templates
    - Controller and WAN edge device lists
  11. Verify the valid Cisco SD-WAN Manager nodes.
    - a. Log in to the CLI of each Cisco SD-WAN Validator and run the **show orchestrator valid-vmanage-id** command.  
 In the command output, verify that the chassis numbers of the active and previously active Cisco SD-WAN Manager nodes are listed.
    - b. Log in to the CLI of a Cisco vEdge Cloud router and run the **show control valid-vmanage-id** command.  
 In the command output, verify that the chassis numbers of the active and previously active Cisco SD-WAN Manager nodes are listed. Also, check whether the device is connected to the active Cisco SD-WAN Manager nodes and the Cisco Catalyst SD-WAN Controller.
  12. Invalidate the previously active Cisco SD-WAN Manager nodes.  
 The previously active Cisco SD-WAN Manager is the certificate issuer for the cloud WAN edge devices. The active Cisco SD-WAN Manager issues certificates to the cloud WAN edge devices only after the previously active Cisco SD-WAN Manager nodes are invalidated.

**Note**

- After you invalidate the Cisco SD-WAN Manager nodes, the nodes cannot be reused without performing additional steps that are beyond the scope of this document.
- When you invalidate the previously active Cisco SD-WAN Manager nodes, Cisco SD-WAN Manager marks the nodes as invalid and sends an update to all controllers. However, Cisco SD-WAN Manager does not send an updated list of valid Cisco SD-WAN Manager UUIDs to Cisco SD-WAN Validator immediately because the previously active Cisco SD-WAN Manager is the CA for the cloud WAN edge devices. So, the output of the **show orchestrator valid-vmanage-id** command on a Cisco SD-WAN Validator includes the UUIDs of the invalidated Cisco SD-WAN Manager nodes.

Cisco SD-WAN Manager has a scheduled task that runs every 24 hours and checks to see if all the cloud WAN edges have been moved to the active Cisco SD-WAN Manager. Cisco SD-WAN Manager sends the updated list of valid Cisco SD-WAN Manager UUIDs to Cisco SD-WAN Validator only after the cloud WAN edge devices have been moved to the active Cisco SD-WAN Manager. After this list is received, the output of the **show orchestrator valid-vmanage-id** command on a Cisco SD-WAN Validator does not include the UUIDs of the invalidated Cisco SD-WAN Manager nodes.

- a. From the Cisco SD-WAN Manager menu, choose **Configuration > Certificates**.
- b. Click **Controllers**.



**Note** Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

c. For each previously active Cisco SD-WAN Manager node, click ... and click **Invalidate**.

13. Verify the valid Cisco SD-WAN Manager nodes after 24 hours.

a. Log in to the CLI of each Cisco SD-WAN Validator and run the **show orchestrator valid-vmanage-id** command.

In the command output, verify that the chassis numbers of only the active Cisco SD-WAN Manager nodes are listed.

b. Log in to the CLI of WAN edge device and run the **show control valid-vmanage-id** command.

In the command output, verify that the chassis numbers of only the active Cisco SD-WAN Manager nodes are listed. Also, check whether the device is connected to the active Cisco SD-WAN Manager nodes and the Cisco Catalyst SD-WAN Controllers.

The Cisco SD-WAN Manager cluster that was initially the standby cluster is now the active Cisco SD-WAN Manager cluster.

## Multitenant Cisco SD-WAN Manager: Disaster Recovery After a Failed Data Center Becomes Operational

If a Multitenant Cisco SD-WAN Manager cluster or the data center hosting the Cisco SD-WAN Manager nodes in the cluster fail, you can recover from the failure by activating a standby Cisco SD-WAN Manager cluster. You can perform disaster recovery as follows:

1. Deploy and configure a standby Cisco SD-WAN Manager cluster.

The standby Cisco SD-WAN Manager cluster is not part of the overlay network and is not active.

2. Back up the configuration database of the active Cisco SD-WAN Manager cluster periodically.

Choose a Cisco SD-WAN Manager node in the cluster that hosts the configuration database service and back up the configuration database.

3. If the active Cisco SD-WAN Manager cluster fails, restore the most recent configuration database on the standby Cisco SD-WAN Manager cluster, activate the standby Cisco SD-WAN Manager cluster, and remove the previously active Cisco SD-WAN Manager cluster from the overlay network.

Choose a Cisco SD-WAN Manager node in the cluster that hosts the configuration database service and restore the configuration database backed up from the previously active Cisco SD-WAN Manager cluster.

To test disaster recovery, you can simulate a scenario in which the active Cisco SD-WAN Manager cluster fails. One way to simulate such a failure would be by disabling the tunnel interface as described in this document.

The following procedure applies to a scenario in which an initially active Cisco SD-WAN Manager cluster or the data center hosting the cluster failed and the standby Cisco SD-WAN Manager cluster was configured

to be the active Cisco SD-WAN Manager cluster. If the cluster that was initially active becomes operational again, it serves as a standby cluster. By completing the following procedure, you can turn this standby cluster into the active cluster.

### Check the Configuration of the Standby Cisco SD-WAN Manager

1. Check whether the running configuration of the standby Cisco SD-WAN Manager nodes is similar to the running configuration of the active Cisco SD-WAN Manager nodes. Local certificates must be installed on the standby Cisco SD-WAN Manager nodes.



**Note** The running configuration on a standby Cisco SD-WAN Manager is usually identical to that of an active Cisco SD-WAN Manager node. However, you must ensure that settings such as the system IP address and the tunnel interface IP address are unique.

2. On the standby Cisco SD-WAN Manager nodes, shut down the transport interface in VPN 0: On the CLI, include the **shutdown** command in the transport interface configuration.

With the standby Cisco SD-WAN Manager nodes configured in this manner, the overlay network is not aware of the standby Cisco SD-WAN Manager cluster.

### Back Up the Active Cisco SD-WAN Manager Cluster Configuration

Back up the full configuration database of the active Cisco vManage cluster periodically. Additionally, take snapshots of the active Cisco SD-WAN Manager virtual machines.

1. Choose an active Cisco SD-WAN Manager node that hosts the configuration database service and export a backup the configuration database. On the CLI of the Cisco SD-WAN Manager node, run the following command: **request nms configuration-db backup path file-path**

The command backs up the configuration database in a `.tar.gz` file and saves the file in the specified *file-path*.

In the following example, the database is backed up to a file named `db_backup.tar.gz` in the `/home/admin/` directory.

```
Active-vManage# request nms configuration-db backup path /home/admin/db_backup
Successfully saved database to /home/admin/db_backup.tar.gz
```

2. Choose a standby Cisco SD-WAN Manager node that hosts the configuration database service and copy the configuration database backup to this node.

In the following example, `db_backup.tar.gz` is copied from the active Cisco SD-WAN Manager node to the `/home/admin/` directory of a standby Cisco SD-WAN Manager node.

```
Active-vManage# request execute vpn 512 scp /home/admin/db_backup.tar.gz
admin@10.126.93.92:/home/admin
The authenticity of host '10.126.93.92 (10.126.93.92)' can't be established.
ECDSA key fingerprint is SHA256:jTjJWQUNHvLrBUxWzNjd8mUz819gPf5lMeopsgDlAc.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '10.126.93.92' (ECDSA) to the list of known hosts.
viptela 18.4.5

admin@10.126.93.92's password:
db_backup.tar.gz 100% 399KB 4.4MB/s 00:00
```

## Restore Cisco SD-WAN Manager Cluster Using the Configuration Database Backup

Restore the most recent backup of the configuration database from the active Cisco SD-WAN Manager cluster on the standby Cisco SD-WAN Manager node to which you copied this backup.



### Note

- The restore operation does not restore all the information included in the configuration database. Cisco SD-WAN Manager configurations such as users and repositories must be configured on the standby Cisco SD-WAN Manager node after the configuration database is restored using the backup.
- When you complete the steps that follow, the previously active Cisco SD-WAN Manager nodes cannot be reused. To reuse the nodes, you must perform additional steps that are beyond the scope of this document.

1. On the CLI of the standby Cisco SD-WAN Manager node, run the following command: **request nms configuration-db restore path *file-path***

In the following example, the configuration database is restored using the backup file `db_backup.tar.gz`.

```
Standby-vManage# request nms configuration-db restore path /home/admin/db_backup.tar.gz
Configuration database is running in a standalone mode
Importing database...Successfully restored database
```

2. Verify that the appropriate services are running on the standby Cisco SD-WAN Manager nodes: On the CLI of each standby Cisco SD-WAN Manager node, run the **request nms all status** command. From the command output, verify the services running on the node.
3. Verify that every standby Cisco SD-WAN Manager node has a list of all the active and standby Cisco SD-WAN Manager nodes.
  - a. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices > Controllers**.



### Note

Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

- b. Verify that the page displays all active and standby Cisco SD-WAN Manager nodes.
4. On the standby Cisco SD-WAN Manager nodes, enable the transport interface on VPN 0. Use one of the following two methods:
    - a. Enable the transport interface in VPN 0: On the CLI of each standby Cisco SD-WAN Manager node, run the **no shutdown** command.

```
Active-vManage# config
Active-vManage(config)# vpn 0 interface interface-name
Active-vManage(config-interface)# no shutdown
Active-vManage(config-interface)# commit and-quit
```

- b. Activate the tunnel interface in VPN 0: On the CLI of each standby Cisco SD-WAN Manager node, run the **tunnel-interface** command.

```
Active-vManage# config
Active-vManage(config)# vpn 0 interface interface-name
```

```
Active-vManage(config-interface) # tunnel-interface
Active-vManage(config-interface) # commit and-quit
```

5. Add each standby Cisco SD-WAN Manager node to the overlay network.
  - a. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
  - b. Click **Controllers**.



**Note** Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

- c. For a Cisco SD-WAN Validator, click ... and click **Edit**.
  - d. In the **Edit** dialog box, enter the following details of the Cisco SD-WAN Validator: WAN transport IP address, username, and password.
  - e. Repeat **Step 5c** and **Step 5d** for every Cisco SD-WAN Validator.
6. Disconnect the active Cisco SD-WAN Manager nodes from the overlay network.



**Note** In a lab environment, where you are simulating a disaster scenario, you can perform this step. However, if you cannot reach Cisco SD-WAN Manager instances in an actual disaster scenario, you may not be able to perform this step and can omit the step.

Use one of the following two methods:

- a. Shut down the transport interface in VPN 0: On the CLI of each active Cisco SD-WAN Manager node, run the **shutdown** command.

```
Active-vManage# config
Active-vManage(config) # vpn 0 interface interface-name
Active-vManage(config-interface) # shutdown
Active-vManage(config-interface) # commit and-quit
```

- b. Deactivate the tunnel interface in VPN 0: On the CLI of each active Cisco SD-WAN Manager node, run the **no tunnel-interface** command.

```
Active-vManage# config
Active-vManage(config) # vpn 0 interface interface-name
Active-vManage(config-interface) # no tunnel-interface
Active-vManage(config-interface) # commit and-quit
```

7. From the standby Cisco SD-WAN Manager, send the updated controller and device list to the Cisco SD-WAN Validator.

Send the list of controllers:

- a. From the Cisco SD-WAN Manager menu, choose **Configuration > Certificates**.
- b. Click **Controllers**.





**Note** Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

c. Click **Send to Validator**.

Wait for the configuration task to complete. When the task is complete,

- The standby Cisco SD-WAN Manager nodes become the active Cisco SD-WAN Manager nodes.
- The previously active Cisco SD-WAN Manager nodes are no longer part of the overlay network.
- The active Cisco SD-WAN Manager nodes have the configuration from the most recent configuration database backup.
- Every controller establishes connection with the other controllers in the network.

d. Click **WAN Edge List**.

e. Click **Send to Controllers**.

8. Verify that the following are intact:

- Policies
- Templates
- Controller and WAN edge device lists

9. Verify the valid Cisco SD-WAN Manager nodes.

a. Log in to the CLI of each Cisco SD-WAN Validator and run the **show orchestrator valid-vmanage-id** command.

In the command output, verify that the chassis numbers of the active and previously active Cisco SD-WAN Manager nodes are listed.

b. Log in to the CLI of a WAN edge device and run the **show control valid-vmanage-id** command.

In the command output, verify that the chassis numbers of the active and previously active Cisco SD-WAN Manager nodes are listed. Also, check whether the device is connected to the active Cisco SD-WAN Manager nodes and the Cisco Catalyst SD-WAN Controller.

10. Invalidate the previously active Cisco SD-WAN Manager nodes.



**Note** After you invalidate the Cisco SD-WAN Manager nodes, the nodes cannot be reused without performing additional steps that are beyond the scope of this document.

a. From the Cisco SD-WAN Manager menu, choose **Configuration > Certificates**.

b. Click **Controllers**.



**Note** Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

- c. For each previously active Cisco SD-WAN Manager node, click ... and click **Invalidate**.
11. Verify the valid Cisco SD-WAN Manager nodes.
  - a. Log in to the CLI of each Cisco SD-WAN Validator and run the **show orchestrator valid-vmanage-id** command.  
In the command output, verify that the chassis numbers of only the active Cisco SD-WAN Manager nodes are listed.
  - b. Log in to the CLI of WAN edge device and run the **show control valid-vmanage-id** command.  
In the command output, verify that the chassis numbers of only the active Cisco SD-WAN Manager nodes are listed. Also, check whether the device is connected to the active Cisco SD-WAN Manager nodes and the Cisco Catalyst SD-WAN Controllers.

The Cisco SD-WAN Manager cluster that was initially the standby cluster is now the active Cisco SD-WAN Manager cluster.

## Replace Faulty Cisco SD-WAN Controller

To replace a faulty Cisco SD-WAN Controller with a new instance, follow these steps:

1. Create a Cisco SD-WAN Controller instance. See [Deploy the Cisco SD-WAN Controller](#).
2. [Add the Cisco SD-WAN Controller](#) to the overlay network.
3. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
4. Click **Controllers**.



**Note** Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

5. For the faulty Cisco SD-WAN Controllers, click ... and click **Invalidate**.  
The **Invalidate** dialog box appears.



**Note** If you have not added a new Cisco SD-WAN Controller that can replace the faulty Cisco SD-WAN Controller, Cisco SD-WAN Manager indicates this through an error message. Click **Cancel** in the **Invalidate** dialog box and add a new Cisco SD-WAN Controller before invalidating the faulty instance.

6. In the **Invalidate** dialog box, do the following:

- a. Check the **Replace Controller** check box.
- b. From the **Select Controller** drop-down list, choose the new Cisco SD-WAN Controller that should replace the faulty instance.
- c. Click **Invalidate**.

Cisco SD-WAN Manager launches the **Invalidate Device** and **Push CLI Template Configuration** task. When these tasks are completed, the faulty Cisco SD-WAN Controller is invalidated and removed from the overlay network. The tenants that were served by the faulty Cisco SD-WAN Controller are now served by the new Cisco SD-WAN Controller that you chose as the replacement.

# RADIUS and TACACS Support for Multitenancy

## Information about RADIUS and TACACS Support for Multitenancy

From Cisco IOS XE Catalyst SD-WAN Release 17.12.1a, Cisco SD-WAN Manager supports for RADIUS and TACACS servers in a multitenant deployment.

### RADIUS

RADIUS is a distributed client and server system that secures networks that have authorized access. RADIUS clients run on supported Cisco devices and send authentication requests to a central RADIUS server, which contains all user authentication and network service access information.

### TACACS

TACACS is a security application that provides centralized validation of users attempting to gain access to an access point. Unlike RADIUS, TACACS does not authenticate wireless client devices accessing the network through an access point.

TACACS provides for separate and modular authentication, authorization, and accounting. Each service can be tied into its own database to take advantage of other services available on that server or on the network.

TACACS administered through the AAA security services can provide these services:

- **Authentication:** Provides complete control of authentication through login and password dialog, challenge and response, and messaging support.
- **Authorization:** Provides fine-grained control of user privileges for the duration of the session, including access control, session duration, or protocol support. You can also enforce restrictions on the commands to execute a TACACS authorization feature.
- **Accounting:** Collects and sends information used for billing, auditing, and reporting to the TACACS daemon. Network managers can use the accounting feature to track administrator activity for security audit or to provide information for user billing. Accounting records include administrator identities, start and stop times, executed commands, number of packets, and number of bytes.



**Note** NAT is not supported between Cisco SD-WAN Manager and the multitenant connector.

# Prerequisites for Cloud Multitenant with On-Prem Per-tenant AAA and Provider AAA

## Prerequisites for Cloud Multitenant with On-Prem Per-tenant AAA and Provider AAA

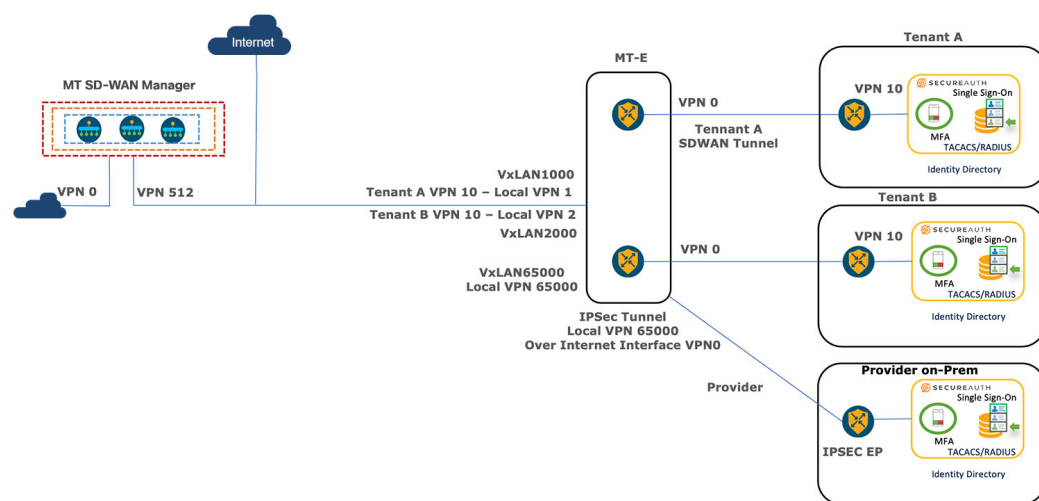
- A Multitenant edge connector is onboarded in Cisco SD-WAN Manager.
- Provider can have tenant configurations only through a device or feature template.
- The edge connector is on the same premises as the controllers.
- Cisco SD-WAN Manager is configured with VPN 512 interface.
- VxLAN tunnels must use the VPN 512 interface as the underlay.
- In a Cisco SD-WAN Manager cluster, there is a VxLAN tunnel created between each Cisco SD-WAN Manager node and the edge connector.
- A provider's RADIUS and TACACS server cannot be shared with the tenant.
- RADIUS and TACACS server authentication is within the tenant network.
- Multiple RADIUS and TACACS servers are used for the same tenant.
- A tenant's RADIUS and TACACS server is on-prem or cloud-hosted.



**Note** You must configure an external AAA server and provide mapping between the user and the Viptela groups to authentication. For example, Viptela-Group-Name as basic, tenantadmin, or operator.

The following illustration shows the architecture of the cloud multitenancy with on-prem per tenant and provider AAA.

**Figure 5: Cloud Multitenant with On-Prem Per-tenant and Provider AAA**



## Workflows to Configure Remote AAA

### Enable Multitenancy

1. From the Cisco SD-WAN Manager menu, choose **Administration** > **Settings**.
2. Click **Edit** adjacent to the **Tenancy Mode Tenancy Mode**. If you are using Cisco Catalyst SD-WAN Manager Release 20.12.x or earlier, click **Edit**.
3. Click **Multitenant**.
4. In the **Domain** field, enter the domain name of the service provider (for example, managed-sp.com).
5. Enter a **Cluster Id** (for example, cluster-1 or 123456).
6. Click **Save**. If you are using Cisco Catalyst SD-WAN Manager Release 20.12.x or earlier, click **Proceed** to confirm that you want to change the tenancy mode.

Cisco SD-WAN Manager reboots in multitenant mode and when a provider user logs in to Cisco SD-WAN Manager, the provider dashboard appears.

For information about configuring AAA using feature templates for a single tenant, see [Configuring AAA using Cisco SD-WAN Manager Template](#).

### Configure the Tenant

To on board the Edge Connector in a Cisco SD-WAN Manager provider, perform the following steps:

1. From the Cisco SD-WAN Manager menu, choose **Administration** > **Tenant Management**.
2. Click **Edit** adjacent to the **Tenant**.  
The **Edit Tenant** window is displayed.
3. Enter the description in the **Description** field.
4. Enter the edge number in the **Forecasted Edge** field.
5. Enter the sub-domain URL in the **URL Subdomain** field.
6. Enable the **Edge Connector** option.
7. Choose the **Edge Connector IP** from the drop-down list.
8. Choose the VXLAN tunnel endpoint from the **Edge Connector VTEP Interface Name** drop-down list.  
For a cloud deployment, it is essential to provision the Cloud-hosted SD-WAN Control Components and gateway with the third interface **GigabitEthernet3** for AAA. When selecting the VXLAN tunnel endpoint from the **Edge Connector VTEP Interface Name** drop-down list, choose **GigabitEthernet3**. Selecting any other interface, such as **VPN 0** or **VPN 512**, might result in misconfiguration.
9. Click **Save**.

## Configure Remote AAA

Cisco SD-WAN Manager reboots in multitenant mode and when a provider user logs in to Cisco SD-WAN Manager, the provider dashboard appears.

1. From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users**.
2. Click **Remote AAA**.
3. Expand the **Remote AAA** tab to configure remote AAA.
4. Enter the order in which to attempt different authentication methods in the **Authentication Order** field.
5. Choose the option in **Authentication Fallback** to fallback if higher-priority authentication fails.
6. Choose the **Admin Authentication Order** to authenticate a tenantadmin user according to the authentication order.
7. Enable or disable audit logs in the **Disable Audit Logs** field.
8. Enable or disable user accounting in the **Enable/disable user accounting** field.
9. Click **Save** to save the changes.

## Configure RADIUS

1. From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users**.
2. Click **Remote AAA**.
3. Expand the **RADIUS** tab to configure a RADIUS server.
4. Enter the duration to wait for replies from the RADIUS server in the **Timeout** field.
5. Enter the number of times you want to contact a RADIUS server in the **Retransmit Count** field.
6. Click **New RADIUS Server** to add a new RADIUS server.

| Field                        | Description                                                                                                                                            |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Radius Server Details</b> |                                                                                                                                                        |
| <b>Address</b>               | Enter the IP address of the RADIUS server.                                                                                                             |
| <b>Authentication Port</b>   | Enter the port to connect to the RADIUS server.                                                                                                        |
| <b>Accounting Port</b>       | Enter the port used to connect to the server.                                                                                                          |
| <b>Key</b>                   | Enter the password to access the RADIUS server.                                                                                                        |
| <b>Secret Key</b>            | Enter the AES encrypted key to access the RADIUS server.                                                                                               |
| <b>Priority</b>              | Enter the server priority.                                                                                                                             |
| <b>VPN ID</b>                | Enter the VPN in which the RADIUS server.<br><br><b>Note</b><br>This option is available in Cisco Catalyst SD-WAN Manager Release 20.15.x and earlier. |

| Field                                                                                                                                               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>VPN IP Subnet</b>                                                                                                                                | Enter the VPN IP subnet (VXLAN tunnel VPN IP subnet) in which the RADIUS server is located.<br><br><b>Note</b><br>This option is available in Cisco Catalyst SD-WAN Manager Release 20.15.x and earlier.                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Network Connectivity</b><br><br><b>Note</b><br>This Network Connectivity option is available from Cisco Catalyst SD-WAN Manager Release 20.16.1. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Source VPN</b>                                                                                                                                   | Choose from these VPN options, the location of the RADIUS server: <ul style="list-style-type: none"> <li>• <b>0</b>: Transport VPN.</li> <li>• <b>512</b>: Management VPN.</li> </ul> Configure these fields only if you have specified the tenant to manage the AAA server. These fields do not appear if the multitenant setup is configured to be managed by the provider. <ul style="list-style-type: none"> <li>• (Optional) <b>Source Subnet</b>: Enter the VPN IP subnet that is unique within the tenant network.</li> <li>• (Optional) <b>Destination VPN</b>: Enter the VPN in which the RADIUS server is located.</li> </ul> |

- Click **Save** and **Add**.

## Configure TACACS

- From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users**.
- Click **Remote AAA**.
- Expand the **TACACS** tab to configure TACACS.
- Enter the duration to wait for replies from the TACACS server in the **Timeout** field.
- Choose the TACACS authentication type from the **Authentication** drop-down list.
  - ASCII
  - PAP
- Click **New TACACS Server** to add a new TACACS server.

| Field                        | Description                                |
|------------------------------|--------------------------------------------|
| <b>TACACS Server Details</b> |                                            |
| <b>Address</b>               | Enter the IP address of the TACACS server. |

| Field                                                                                                                                              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Authentication Port</b>                                                                                                                         | Enter the port to connect to a TACACS server.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Key</b>                                                                                                                                         | Enter the password to access the TACACS server.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Secret Key</b>                                                                                                                                  | Enter the AES encrypted key to access the TACACS server.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Priority</b>                                                                                                                                    | Enter the TACACS server priority.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>VPN ID</b>                                                                                                                                      | Enter the VPN in which the TACACS server.<br><br><b>Note</b><br>This option is available in Cisco Catalyst SD-WAN Manager Release 20.15.x and earlier.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>VPN IP Subnet</b>                                                                                                                               | Enter the VPN IP subnet in which the TACACS server is located.<br><br><b>Note</b><br>This option is available in Cisco Catalyst SD-WAN Manager Release 20.15.x and earlier.                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Network Connectivity</b><br><br><b>Note</b><br>The Network Connectivity option is available from Cisco Catalyst SD-WAN Manager Release 20.16.1. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Source VPN</b>                                                                                                                                  | Choose from these VPN options, the location of the TACACS server: <ul style="list-style-type: none"> <li>• <b>0</b>: Transport VPN.</li> <li>• <b>512</b>: Management VPN.</li> </ul> Configure these fields only if you have specified the tenant to manage the AAA server. These fields do not appear if the multitenant setup is configured to be managed by the provider. <ul style="list-style-type: none"> <li>• (Optional) <b>Source Subnet</b>: Enter the VPN IP subnet that is unique within the tenant network.</li> <li>• (Optional) <b>Destination VPN</b>: Enter the VPN in which the TACACS server is located.</li> </ul> |

7. Click **Add**.

## Verify RADIUS and TACACS Configuration for Multitenancy

The following is a RADIUS and TACACS configuration example on Cisco IOS XE Catalyst SD-WAN devices through CLI:

```
Device# interface GigabitEthernet4
description VTEP Interface
```



```
no shutdown

arp timeout 1200

ip address 172.1.1.101 255.255.255.0

no ip redirects

ip mtu 1500

load-interval 30

mtu 1500

negotiation auto

exit
```

Use the **show ip interface brief** command to show the AAA configuration:

```
Device# show ip interface brief | i <VPN IP SUBNET of VxTunnel>
```

Where <VPN IP SUBNET of VXTunnel> is one that is configured under Tenant - > Administration  
-> Remote AAA

The output shows one tunnel per one node. If there are three nodes in a cluster, the output displays three tunnels in the subnet.





## CHAPTER 35

# Flexible Tenant Placement on Multitenant Cisco Catalyst SD-WAN Controllers

*Table 367: Feature History*

| Feature Name                                                               | Release Information          | Description                                                                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Flexible Tenant Placement on Multitenant Cisco Catalyst SD-WAN Controllers | Cisco vManage Release 20.9.1 | With this feature, while onboarding a tenant to a multitenant deployment, you can choose the pair of multitenant Cisco SD-WAN Controllers that serve the tenant. After onboarding a tenant, you can migrate the tenant to a different pair of multitenant Cisco SD-WAN Controller, if necessary. |

- [Information About Flexible Tenant Placement on Multitenant Cisco SD-WAN Controllers, on page 925](#)
- [Restrictions for Flexible Tenant Placement on Multitenant Cisco SD-WAN Controllers, on page 927](#)
- [Assign Cisco SD-WAN Controllers to Tenants During Onboarding, on page 927](#)
- [Update Cisco SD-WAN Controllers Placement For a Tenant, on page 932](#)

## Information About Flexible Tenant Placement on Multitenant Cisco SD-WAN Controllers

### Automatic Tenant Placement by Cisco SD-WAN Manager

In Cisco vManage Release 20.8.x and earlier releases, when you onboard a tenant, Cisco SD-WAN Manager assigns a pair of multitenant Cisco SD-WAN Controllers to the tenant based on an internal algorithm that considers factors such as the following:

- number of tenant WAN edge devices that you forecast for the tenant
- number of tenants served by a pair of multitenant Cisco SD-WAN Controllers
- number of WAN edge devices connected to a pair of multitenant Cisco SD-WAN Controllers

After the tenant is onboarded, if the tenant needs to add more devices than you originally forecast, you can modify the forecast if the pair of multitenant Cisco SD-WAN Controllers serving the tenant can accommodate these additional WAN edge devices. If the Cisco SD-WAN Controllers cannot accommodate the additional WAN edge devices, you must delete the tenant and onboard the tenant again with the revised device forecast so that Cisco SD-WAN Manager assigns a suitable pair of Cisco SD-WAN Controllers. If none of the pairs of multitenant Cisco SD-WAN Controllers can accommodate the revised device forecast, add a new pair of Cisco SD-WAN Controllers and then onboard the tenant.

### Flexible Tenant Placement by Provide Admin User

From Cisco vManage Release 20.9.1, while onboarding a tenant, you have the flexibility to choose the pair of multitenant Cisco SD-WAN Controllers that are assigned to the tenant. Automatic tenant placement by Cisco SD-WAN Manager continues to be the default behavior with flexible tenant placement as an optional configuration.

To help you with flexible tenant placement, Cisco SD-WAN Manager lists available multitenant Cisco SD-WAN Controllers and provides the following details, as a percentage, for each controller:

- number of tenants assigned
- number of tenant WAN edge devices connected
- memory utilized
- CPU utilized

A multitenant Cisco SD-WAN Controller can serve a maximum of 24 tenants and 1000 tenant WAN edge devices across all the tenants. While onboarding a tenant, choose a pair of controllers that can be assigned one more tenant and can also connect to the number of WAN edge devices forecast for the tenant.

After the tenant is onboarded, if the tenant needs to add more devices than you originally forecast and the assigned pair of multitenant Cisco SD-WAN Controllers cannot connect to these additional WAN edge devices, you can migrate the tenant to another pair of Cisco SD-WAN Controllers that can serve one more tenant and accommodate the revised WAN edge device forecast for the tenant. If none of the multitenant Cisco SD-WAN Controllers pairs can accommodate the revised device forecast, you can migrate other tenants to alternative Cisco SD-WAN Controllers so that the controller utilization is efficient and the tenant assignment is optimal. If the optimization doesn't create the capacity required to accommodate the revised device forecast for the tenant, add a new pair of Cisco SD-WAN Controllers and then migrate the tenant.

## Benefits of Flexible Tenant Placement on Multitenant Cisco SD-WAN Controllers

- Choose Cisco SD-WAN Controllers deployed in different failure zones to reduce the probability of both the controllers failing simultaneously. In a cloud environment, choose controllers deployed in different regions.
- Choose Cisco SD-WAN Controllers deployed in the same geographical region as the tenant WAN edge devices to reduce latency.
- Choose Cisco SD-WAN Controllers based on the CPU, DRAM, and hard disk resources allocated, and the utilization of these resources.
- Migrate a tenant to a different Cisco SD-WAN Controller to accommodate changes in the tenant device forecast.

# Restrictions for Flexible Tenant Placement on Multitenant Cisco SD-WAN Controllers

If you wish to migrate a tenant to different pair of Cisco SD-WAN Controllers, you must change the Cisco SD-WAN Controllers assigned to the tenant one at a time. Doing so ensures that one of the Cisco SD-WAN Controllers is available to the tenant WAN edge devices during the migration and prevents disruptions in traffic.

## Assign Cisco SD-WAN Controllers to Tenants During Onboarding

### Prerequisites

- At least two Cisco SD-WAN Controllers must be operational and in Cisco SD-WAN Manager before you can add new tenants.

A Cisco SD-WAN Controller enters the **Manager** mode when you push a template to the controller from Cisco SD-WAN Manager. A Cisco SD-WAN Controller in the **CLI** mode cannot serve multiple tenants.

- Each pair of Cisco SD-WAN Controllers can serve a maximum of 24 tenants and a maximum of 1000 tenant devices. Ensure that there are at least two Cisco SD-WAN Controllers that can serve a new tenant. If no pair of Cisco SD-WAN Controllers in the deployment can serve a new tenant, add two Cisco SD-WAN Controllers and change their mode to **Manager**.
- Add up to 16 tenants in a single operation. If you add more than one tenant, during the **Add Tenant** task, Cisco SD-WAN Manager adds the tenants one after another and not in parallel.

While an **Add Tenant** task is in progress, do not perform a second tenant addition operation. If you do so, the second Add Tenant task fails.

- Each tenant must have a unique Virtual Account (VA) on Plug and Play Connect on [Cisco Software Central](#). The tenant VA should belong to the same Smart Account (SA) as the provider VA.
- For an on-premises deployment, create a Cisco SD-WAN Validator controller profile for the tenant on Plug and Play Connect. The fields in the following table are mandatory.

| Field                | Description                                                                                                                         |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Profile Name         | Enter a name for the controller profile.                                                                                            |
| Multi-Tenancy        | From the drop-down list, select <b>Yes</b> .                                                                                        |
| SP Organization Name | Enter the provider organization name.                                                                                               |
| Organization Name    | Enter the tenant organization name in the format <SP Org Name>-<Tenant Org Name>. The organization name can be up to 50 characters. |
| Primary Controller   | Enter the host details for the primary Cisco SD-WAN Validator.                                                                      |

For a cloud deployment, the Cisco SD-WAN Validator controller profile is created automatically as part of the tenant creation process.

1. Log in to Cisco SD-WAN Manager as the provider **admin** user.
2. From the Cisco SD-WAN Manager menu, choose **Administration > Tenant Management**.
3. Click **Add Tenant**.
4. In the **Add Tenant** slide-in pane, click **New Tenant**.
5. Configure the following tenant details:

| Field                    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>              | Enter a name for the tenant.<br><br>For a cloud deployment, the tenant name should be same as the tenant VA name on Plug and Play Connect.                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Description</b>       | Enter a description for the tenant.<br><br>The description can have up to 256 characters and can contain only alphanumeric characters.                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Organization Name</b> | Enter the name of the tenant organization. The organization name can have up to 50 characters.<br><br>The organization name is case-sensitive. Each tenant or customer must have a unique organization name.<br><br>Enter the organization name in the following format:<br><br><SP Org Name>-<Tenant Org Name><br><br>For example, if the provider organization name is 'managed-sp' and the tenant organization name is 'customer1', while adding the tenant, enter the organization name as 'managed-sp-customer1'. |

| Field         | Description |
|---------------|-------------|
| URL Subdomain |             |

| Field | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | <p>Enter the fully qualified subdomain name of the tenant.</p> <ul style="list-style-type: none"> <li>The subdomain name must include the domain name of the service provider. For example, for the managed-sp.com service provider, a valid domain name for customer1 is customer1.managed-sp.com.</li> </ul> <p><b>Note</b><br/>The service provider name is shared amongst all tenants. Ensure that the URL naming convention follows the same domain name convention that was followed while enabling multitenancy using <b>Administration &gt; Settings &gt; Tenancy Mode</b>.</p> <ul style="list-style-type: none"> <li>For an on-premises deployment, add the fully qualified subdomain name of the tenant to the DNS. Map the fully qualified subdomain name to the IP addresses of the three Cisco SD-WAN Manager instances in the Cisco SD-WAN Manager cluster. <ul style="list-style-type: none"> <li>Provider DNS: Create a DNS A record and map it to the IP addresses of the Cisco SD-WAN Manager instances running in the Cisco SD-WAN Manager cluster. The A record is derived from the provider's domain name and the cluster ID that was created while enabling multitenancy on Cisco SD-WAN Manager. For example, if the provider's domain name is <code>sdwan.cisco.com</code> and the cluster ID is <code>vmanage123</code>, configure the A record as <code>vmanage123.sdwan.cisco.com</code>.</li> </ul> <p><b>Note</b><br/>If you fail to add the DNS A record, you will experience authentication errors when logging in to Cisco SD-WAN Manager.</p> <p>Validate that the DNS is configured correctly by using the <b>nslookup</b> command. Example: <code>nslookup vmanage123.sdwan.cisco.com</code>.</p> <li>Tenant DNS: Create DNS CNAME records for each tenant that you created and map them to the provider FQDN. For example, if the provider's domain name is <code>sdwan.cisco.com</code> and tenant name is <code>customer1</code>, configure the CNAME record as <code>customer1.sdwan.cisco.com</code>.</li> </li></ul> <p>Cluster ID is not required in the CNAME record.</p> <p>Validate that the DNS is configured correctly by using the <b>nslookup</b> command. Example: <code>nslookup customer1.sdwan.cisco.com</code>.</p> <ul style="list-style-type: none"> <li>For a cloud deployment, the fully qualified subdomain name</li> </ul> |



| Field                         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                         |                                                                                                                                                                                                                                                                                       |                      |                                                                                                                                                                                                                                                                                                                                                                                                                             |                 |                                                           |              |                                                  |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------------------------------------------------------|--------------|--------------------------------------------------|
|                               | of the tenant is automatically added to the DNS as part of the tenant creation process. After you add a tenant, it could take up to an hour before the fully qualified subdomain name of the tenant can be resolved by the DNS.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                         |                                                                                                                                                                                                                                                                                       |                      |                                                                                                                                                                                                                                                                                                                                                                                                                             |                 |                                                           |              |                                                  |
| <b>Forecasted Devices</b>     | <p>Enter the number of WAN edge devices that the tenant can add to the overlay.</p> <p>If the tenant tries to add WAN edge devices beyond this number, Cisco SD-WAN Manager reports an error and the device addition fails.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                         |                                                                                                                                                                                                                                                                                       |                      |                                                                                                                                                                                                                                                                                                                                                                                                                             |                 |                                                           |              |                                                  |
| <b>Select two Controllers</b> | <ul style="list-style-type: none"> <li>Automatic tenant placement: Ensure that the <b>Select two Controllers</b> field has the value <b>Autoplacement</b>. This is the default configuration.</li> <li>Flexible tenant placement: <ul style="list-style-type: none"> <li>a. Click the <b>Select two Controllers</b> drop-down list.</li> </ul> <p>Cisco SD-WAN Manager lists the hostnames of the available Cisco SD-WAN Controllers. For each Cisco SD-WAN Controller, Cisco SD-WAN Manager shows whether the controller is reachable and reports the following utilization details:</p> <table border="1"> <tr> <td>Tenant hosting capacity</td><td>Each Cisco SD-WAN Controller can serve a maximum of 24 tenants. Tenant hosting capacity represents the number of tenants to which the Cisco SD-WAN Controller is assigned in the form of a percentage. This value indicates whether you can assign another tenant to this controller.</td></tr> <tr> <td>Used device capacity</td><td>Each Cisco SD-WAN Controller can support a maximum of 1000 tenant WAN edge devices. Used device capacity represents the number of tenant WAN edge devices connected to the Cisco SD-WAN Controller in the form of a percentage of the maximum capacity (1000 WAN edge devices). This value indicates whether the Cisco SD-WAN Controller can support the number of devices forecast for the tenant that you are onboarding.</td></tr> <tr> <td>Memory utilized</td><td>This value represents memory consumption as a percentage.</td></tr> <tr> <td>CPU utilized</td><td>This value represents CPU usage as a percentage.</td></tr> </table> <li>b. Select two Cisco SD-WAN Controllers to assign to the tenant based on the utilization details.</li> </li></ul> <p>To select a Cisco SD-WAN Controller, check the check box adjacent to its hostname.</p> | Tenant hosting capacity | Each Cisco SD-WAN Controller can serve a maximum of 24 tenants. Tenant hosting capacity represents the number of tenants to which the Cisco SD-WAN Controller is assigned in the form of a percentage. This value indicates whether you can assign another tenant to this controller. | Used device capacity | Each Cisco SD-WAN Controller can support a maximum of 1000 tenant WAN edge devices. Used device capacity represents the number of tenant WAN edge devices connected to the Cisco SD-WAN Controller in the form of a percentage of the maximum capacity (1000 WAN edge devices). This value indicates whether the Cisco SD-WAN Controller can support the number of devices forecast for the tenant that you are onboarding. | Memory utilized | This value represents memory consumption as a percentage. | CPU utilized | This value represents CPU usage as a percentage. |
| Tenant hosting capacity       | Each Cisco SD-WAN Controller can serve a maximum of 24 tenants. Tenant hosting capacity represents the number of tenants to which the Cisco SD-WAN Controller is assigned in the form of a percentage. This value indicates whether you can assign another tenant to this controller.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                         |                                                                                                                                                                                                                                                                                       |                      |                                                                                                                                                                                                                                                                                                                                                                                                                             |                 |                                                           |              |                                                  |
| Used device capacity          | Each Cisco SD-WAN Controller can support a maximum of 1000 tenant WAN edge devices. Used device capacity represents the number of tenant WAN edge devices connected to the Cisco SD-WAN Controller in the form of a percentage of the maximum capacity (1000 WAN edge devices). This value indicates whether the Cisco SD-WAN Controller can support the number of devices forecast for the tenant that you are onboarding.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                         |                                                                                                                                                                                                                                                                                       |                      |                                                                                                                                                                                                                                                                                                                                                                                                                             |                 |                                                           |              |                                                  |
| Memory utilized               | This value represents memory consumption as a percentage.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                         |                                                                                                                                                                                                                                                                                       |                      |                                                                                                                                                                                                                                                                                                                                                                                                                             |                 |                                                           |              |                                                  |
| CPU utilized                  | This value represents CPU usage as a percentage.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                         |                                                                                                                                                                                                                                                                                       |                      |                                                                                                                                                                                                                                                                                                                                                                                                                             |                 |                                                           |              |                                                  |

6. To save the tenant configuration, click **Save**.
7. To add another tenant, repeat Step 4 to Step 6.
8. To onboard tenants to the deployment, click **Add**.

Cisco SD-WAN Manager initiates the Create Tenant Bulk task to onboard the tenants.

As part of this task, Cisco SD-WAN Manager performs the following activities:

- creates the tenant
- assigns two Cisco SD-WAN Controllers to serve the tenant and pushes a CLI template to these controllers to configure tenant information
- sends the tenant and Cisco SD-WAN Controller information to Cisco SD-WAN Validator

When the task is successfully completed, you can view the tenant information, including the Cisco SD-WAN Controller and Cisco SD-WAN Validators assigned to the tenant, on the **Administration > Tenant Management** page.

## Update Cisco SD-WAN Controllers Placement For a Tenant

You can migrate a tenant to a different pair of Cisco SD-WAN Controllers from the controllers that are currently assigned to the tenant. For instance, if you need to increase the tenant WAN edge device forecast and the controllers assigned to the tenant cannot connect to these revised number of tenant WAN edge devices, you can migrate the tenant to a pair of controllers that can accommodate the revised forecast.

If you wish to migrate a tenant to different pair of Cisco SD-WAN Controllers, you must change the Cisco SD-WAN Controllers that are assigned to the tenant one at a time. Doing so ensures that one of the Cisco SD-WAN Controllers is available to the tenant WAN edge devices during the migration and prevents disruptions in traffic.

1. Log in to Cisco SD-WAN Manager as the provider **admin** user.
2. From the Cisco SD-WAN Manager menu, choose **Administration > Tenant Management**.
3. For the tenant you wish to migrate to a different controller, click ... adjacent to the tenant organization name.
4. Click **Update Controller Placement**.
5. In the **Update Controller Placement** slide-in pane, configure the following:

| Field                                 | Description                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source Controller (currently applied) | a. Click the <b>Source Controller (currently applied)</b> drop-down list.<br><br>Cisco SD-WAN Manager lists the hostnames of the Cisco SD-WAN Controllers assigned to the tenant. For each Cisco SD-WAN Controller, Cisco SD-WAN Manager shows whether the controller is reachable and reports the following utilization details: |                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                       | Tenant hosting capacity                                                                                                                                                                                                                                                                                                           | Each Cisco SD-WAN Controller can serve a maximum of 24 tenants. Tenant hosting capacity represents the number of tenants to which the Cisco SD-WAN Controller is assigned in the form of a percentage. This value indicates whether you can assign another tenant to this controller.                                                                                                                              |
|                                       | Used device capacity                                                                                                                                                                                                                                                                                                              | Each Cisco SD-WAN Controller can support a maximum of 1000 tenant WAN edge devices. Used device capacity represents the number of tenant WAN edge devices connected to the Cisco SD-WAN Controller in the form of a percentage of the maximum capacity (1000 devices). This value indicates whether the Cisco SD-WAN Controller can support the number of devices forecast for the tenant that you are onboarding. |
|                                       | Memory utilized                                                                                                                                                                                                                                                                                                                   | This value represents memory consumption as a percentage.                                                                                                                                                                                                                                                                                                                                                          |
|                                       | CPU utilized                                                                                                                                                                                                                                                                                                                      | This value represents CPU usage as a percentage.                                                                                                                                                                                                                                                                                                                                                                   |
|                                       | b. Check the check box adjacent to the hostname of one of the Cisco SD-WAN Controllers assigned to the tenant.                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                    |

| Field                         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                         |                                                                                                                                                                                                                                                                                       |                      |                                                                                                                                                                                                                                                                                                                                                                                                                    |                 |                                                           |              |                                                  |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------------------------------------------------------|--------------|--------------------------------------------------|
| <b>Destination Controller</b> | <p><b>a.</b> Click the <b>Destination Controller</b> drop-down list.</p> <p>Cisco SD-WAN Manager lists the hostnames of the available Cisco SD-WAN Controllers that are not assigned to the tenant. For each Cisco SD-WAN Controller, Cisco SD-WAN Manager shows whether the controller is reachable and reports the following utilization details:</p> <table> <tr> <td>Tenant hosting capacity</td><td>Each Cisco SD-WAN Controller can serve a maximum of 24 tenants. Tenant hosting capacity represents the number of tenants to which the Cisco SD-WAN Controller is assigned in the form of a percentage. This value indicates whether you can assign another tenant to this controller.</td></tr> <tr> <td>Used device capacity</td><td>Each Cisco SD-WAN Controller can support a maximum of 1000 tenant WAN edge devices. Used device capacity represents the number of tenant WAN edge devices connected to the Cisco SD-WAN Controller in the form of a percentage of the maximum capacity (1000 devices). This value indicates whether the Cisco SD-WAN Controller can support the number of devices forecast for the tenant that you are onboarding.</td></tr> <tr> <td>Memory utilized</td><td>This value represents memory consumption as a percentage.</td></tr> <tr> <td>CPU utilized</td><td>This value represents CPU usage as a percentage.</td></tr> </table> <p><b>b.</b> Check the check box adjacent to the hostname of the Cisco SD-WAN Controller you want to assign to the tenant.</p> <p>If you select a Cisco SD-WAN Controller that does not have the required capacity to serve the tenant devices, the update operation fails.</p> | Tenant hosting capacity | Each Cisco SD-WAN Controller can serve a maximum of 24 tenants. Tenant hosting capacity represents the number of tenants to which the Cisco SD-WAN Controller is assigned in the form of a percentage. This value indicates whether you can assign another tenant to this controller. | Used device capacity | Each Cisco SD-WAN Controller can support a maximum of 1000 tenant WAN edge devices. Used device capacity represents the number of tenant WAN edge devices connected to the Cisco SD-WAN Controller in the form of a percentage of the maximum capacity (1000 devices). This value indicates whether the Cisco SD-WAN Controller can support the number of devices forecast for the tenant that you are onboarding. | Memory utilized | This value represents memory consumption as a percentage. | CPU utilized | This value represents CPU usage as a percentage. |
| Tenant hosting capacity       | Each Cisco SD-WAN Controller can serve a maximum of 24 tenants. Tenant hosting capacity represents the number of tenants to which the Cisco SD-WAN Controller is assigned in the form of a percentage. This value indicates whether you can assign another tenant to this controller.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                         |                                                                                                                                                                                                                                                                                       |                      |                                                                                                                                                                                                                                                                                                                                                                                                                    |                 |                                                           |              |                                                  |
| Used device capacity          | Each Cisco SD-WAN Controller can support a maximum of 1000 tenant WAN edge devices. Used device capacity represents the number of tenant WAN edge devices connected to the Cisco SD-WAN Controller in the form of a percentage of the maximum capacity (1000 devices). This value indicates whether the Cisco SD-WAN Controller can support the number of devices forecast for the tenant that you are onboarding.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                         |                                                                                                                                                                                                                                                                                       |                      |                                                                                                                                                                                                                                                                                                                                                                                                                    |                 |                                                           |              |                                                  |
| Memory utilized               | This value represents memory consumption as a percentage.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                         |                                                                                                                                                                                                                                                                                       |                      |                                                                                                                                                                                                                                                                                                                                                                                                                    |                 |                                                           |              |                                                  |
| CPU utilized                  | This value represents CPU usage as a percentage.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                         |                                                                                                                                                                                                                                                                                       |                      |                                                                                                                                                                                                                                                                                                                                                                                                                    |                 |                                                           |              |                                                  |

6. Click **Update**.

7. To change the other Cisco SD-WAN Controller that is assigned to the tenant, repeat Step 3 to Step 6.

Cisco SD-WAN Manager initiates the **Tenant Controller Update** task to assign the selected Cisco SD-WAN Controller to the tenant, migrating the tenant details from the Cisco SD-WAN Controller that was previously assigned. When the task is successfully completed, you can view the tenant information, including the Cisco SD-WAN Controllers assigned to the tenant, on the **Administration > Tenant Management** page.



## CHAPTER 36

# Multitenant WAN Edge Devices

**Table 368: Feature History**

| Feature Name                                                                         | Release Information                                                            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Multitenant WAN Edge Devices                                                         | Cisco IOS XE Catalyst SD-WAN Release 17.7.1a<br>Cisco vManage Release 20.7.1   | With this feature, a service provider can deploy, configure, and manage multitenant WAN edge devices in a multitenant Cisco Catalyst SD-WAN deployment.                                                                                                                                                                                                                                                                                                                                                                             |
| Distribute Device Resources Among Tenants Using Tiers                                | Cisco IOS XE Catalyst SD-WAN Release 17.8.1a<br>Cisco vManage Release 20.8.1   | <p>With this feature, you can define tiers and assign tenants to tiers. While defining a tier, you limit the amount of a multitenant WAN edge resource that is allocated to a tenant in the tier, when the tenant is onboarded to a multitenant WAN edge device.</p> <p>From this release, you can specify how many tenant VPNs can be created for a tenant belonging to a tier. In subsequent releases, tiers will be enhanced to support limits on the usage of additional device resources such as firewall, NAT, and TLOCs.</p> |
| Enhanced Multitenant Tier Definition to include Route and TLOC Resource-Usage Limits | Cisco IOS XE Catalyst SD-WAN Release 17.10.1a<br>Cisco vManage Release 20.10.1 | This feature is enhanced to support route and TLOC resource-usage limits. A service provider can assign a tier to limit the routes and TLOC resource-usage to the tenant based on the service agreement.                                                                                                                                                                                                                                                                                                                            |

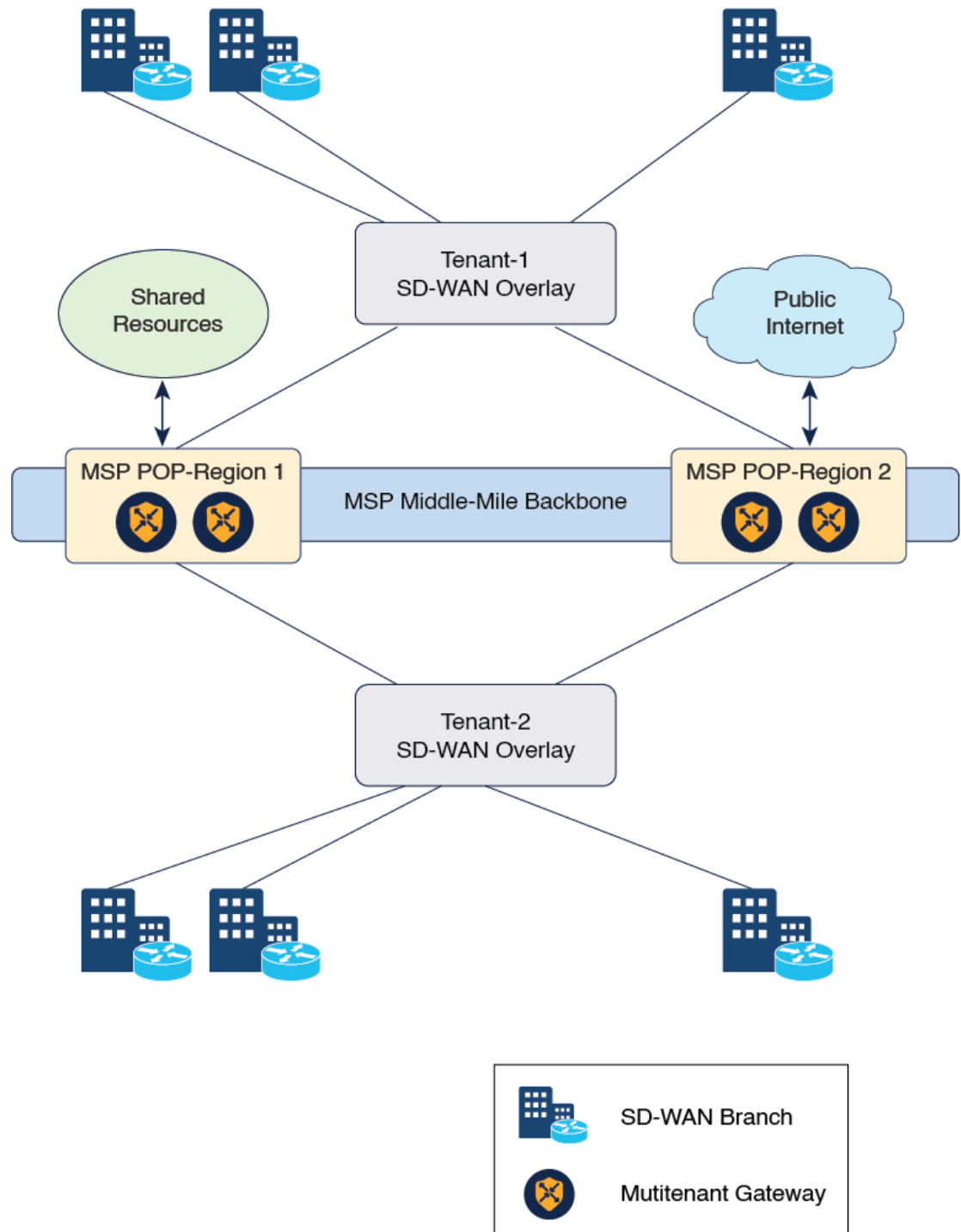
| Feature Name                                               | Release Information                                                                                | Description                                                                                                                                                                        |
|------------------------------------------------------------|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Enhanced Multitenant Tier Definition to include NAT Limits | Cisco IOS XE Catalyst SD-WAN Release 17.12.1a<br><br>Cisco Catalyst SD-WAN Manager Release 20.12.1 | With this feature, you can configure maximum limit on NAT translations per tenant.<br><br>From this release <b>Tier</b> is called <b>Resource Profile</b> in Cisco SD-WAN Manager. |

- [Information About Multitenant WAN Edge Devices, on page 936](#)
- [Supported Devices for Multitenant WAN Edge Devices, on page 939](#)
- [Prerequisites for Multitenant WAN Edge Devices, on page 940](#)
- [Restrictions for multitenant WAN edge devices, on page 940](#)
- [Configure Multitenant WAN Edge Devices, on page 941](#)
- [Verify Configuration and Operation of Multitenant WAN Edge Devices, on page 950](#)
- [Troubleshoot Multitenant WAN Edge Device Errors, on page 963](#)

## Information About Multitenant WAN Edge Devices

As a service provider managing a multitenant Cisco Catalyst SD-WAN deployment, you may wish to deploy a multitenant WAN edge device in the overlay network to serve as a shared gateway for traffic belonging to multiple tenants. For example, you can deploy such a shared gateway in each regional point of presence (PoP). You can carry inter-region traffic belonging to multiple tenants through these shared gateways and the transport backbone linking the PoPs.

Figure 6: Multitenant WAN Edge Devices as Shared Gateways



Multitenant WAN edge devices isolate traffic belonging to different tenants by mapping a tenant service VPN (referred to as tenant VPN) to a device VPN (also referred to as the device VRF). Cisco SD-WAN Manager performs the mapping between the tenant and device VPNs when you onboard a tenant on a multitenant WAN edge device.

Multitenant WAN edge devices establish control connections with the Cisco SD-WAN Validator nodes specified in the bootstrap configuration, and then connect to nodes in the Cisco SD-WAN Manager cluster. When you onboard a tenant to a multitenant WAN edge device, the device establishes control connections to the Cisco SD-WAN Controller assigned to the tenant.

The service provider must deploy, configure, and manage multitenant WAN edge devices. The devices and their states are displayed only in the Cisco SD-WAN Manager provider view. The provider, acting on behalf of the tenant, must deploy, configure, and manage single-tenant WAN edge devices owned by a tenant. The devices and their states are displayed in the tenant view or the provider-as-tenant view. When a tenant is onboarded to a multitenant WAN edge device, the multitenant WAN edge device can interoperate with single-tenant WAN edge devices owned by the tenant and other multitenant WAN edge devices to which the tenant is onboarded.

## Resource Profiles (Tiers)

When you onboard many tenants on a multitenant WAN edge device, you may need to distribute the limited device resources among the tenants to ensure fair usage of resources or to implement different service-level agreements (SLAs). A tier lets you define and limit how much of each device resource a tenant assigned to the tier can consume. After creating a tier, when you onboard a tenant, you assign a tenant to a particular tier to apply the resource-usage limits to the tenant.

### Usage Notes

- After you create a tier, you cannot modify the device-resource-usage limits specified in the tier. To apply a different set of limits to tenants, you must create a new tier and assign the relevant tenants to the new tier.
- You can delete a tier only when no tenants are assigned to it.

### Resource Usage Limits in Resource Profiles (Tiers)

*Table 369: Resource Usage Limits in Tiers*

| Resource Usage Limit | Description                                                                                                                                                                                                                                                                                                                                                                                                     | Available From                                               |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| Number of VPNs       | <p>Maximum number of tenant VPNs that can be created for a tenant belonging to the tier.</p> <p>Cisco SD-WAN Manager enforces the limit when you create a new tenant VPN for a tenant.</p> <ul style="list-style-type: none"> <li>• If you have already created the maximum number of tenant VPNs specified in the tier, Cisco SD-WAN Manager reports the error and doesn't apply the configuration.</li> </ul> | Cisco IOS XE Release 17.8.1 and Cisco vManage Release 20.8.1 |



| Resource Usage Limit | Description                                                                                                                                                                               | Available From                                                                                     |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Route-limit          | The number of IPv4 unicast and IPv6 unicast routes that can be created for a tenant belonging to the tier.<br><br>Route limit on a tenant is the sum of routes from all VRFs.             | Cisco IOS XE Catalyst SD-WAN Release 17.10.1a<br><br>Cisco vManage Release 20.10.1                 |
| TLOC                 | TLOC allows you to map transport interfaces to tenants. At least one TLOC needs to be selected per tier and you can include up to 16 TLOCs in a tier.                                     | Cisco IOS XE Catalyst SD-WAN Release 17.10.1a<br><br>Cisco vManage Release 20.10.1                 |
| NAT limit            | The maximum limit on the number of NAT translations per tenant.<br><br>Once the maximum limit has reached for a tenant, the packets are dropped and further translations are not allowed. | Cisco IOS XE Catalyst SD-WAN Release 17.12.1a<br><br>Cisco Catalyst SD-WAN Manager Release 20.12.1 |

## Benefits of Multitenant WAN Edge Devices

As a managed service provider, by deploying multitenant WAN edge devices, you can

- reuse the edge devices and the interconnecting transport backbone to serve multiple tenants
- lower capital and operational expenditure
- provide faster access to tenants to shared resources, SaaS, and IaaS through the shared transport backbone
- manage tenant association with the devices, tenant-specific policies, and QoS requirements with Cisco SD-WAN Manager as the unified management interface

## Supported Devices for Multitenant WAN Edge Devices

| Device Family                                      | Device Model                                                                                                                                                                           |
|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cisco ASR 1000 Series Aggregation Services Routers | ASR 1001-HX<br>ASR 1001-X<br>ASR 1002-HX<br>ASR 1002-X<br><br><b>Note</b><br>Cisco IOS XE Catalyst SD-WAN Release 17.9.1a is the last supported release for ASR 1001-X and ASR 1002-X. |

|                                                   |                                                                      |
|---------------------------------------------------|----------------------------------------------------------------------|
| Cisco Catalyst 8000V Edge Software                | Catalyst 8000V                                                       |
| Cisco Catalyst 8300 Series Edge Platforms         | C8300-1N1S-4T2X<br>C8300-1N1S-6T<br>C8300-2N2S-4T2X<br>C8300-2N2S-6T |
| Cisco Catalyst 8500 Series Edge Platforms         | C8500-12X<br>C8500-12X4QC<br>C8500L-8S4X                             |
| Cisco ISR 4000 Series Integrated Services Routers | ISR 4461                                                             |

## Prerequisites for Multitenant WAN Edge Devices

- You must have completed the initial setup for a multitenant Cisco Catalyst SD-WAN deployment.
  - Multitenant Cisco SD-WAN Validator and Multitenant Cisco SD-WAN Controller must run Cisco SD-WAN Release 20.7.1 or a later release software.
  - Multitenant Cisco SD-WAN Manager must run Cisco vManage Release 20.7.1 or a later release software.
  - Cisco IOS XE Catalyst SD-WAN devices must run Cisco IOS XE Release 17.7.1 or a later release software.

## Restrictions for multitenant WAN edge devices

### Unsupported services and technologies

Multitenant WAN edge devices do not support these services and technologies:

- Cloud Express and Multicloud workflows
- Zone-Based Firewall (ZBFW) and advanced security features
- Per-tenant DPI statistics
- Dynamic on-demand tunnels
- SNMP
- Per-tenant management of NAT resources
- OMP IPv6 route filtering
- OMP notifications

**Provider ownership of devices**

The provider must own, deploy, and manage all multitenant WAN edge devices in the deployment. The provider must also deploy and manage any single-tenant device owned by a specific tenant.

**Unique system IP addresses**

Configure a unique system IP address for each WAN edge device in the multitenant Cisco Catalyst SD-WAN deployment, irrespective of whether the device is a multitenant device owned and managed by the provider or a single-tenant device owned by a tenant and managed by the provider on behalf of the tenant.

**Maximum SLA classes**

You can configure a maximum of 16 SLA classes. You can either assign specific SLA classes to tenants or share SLA classes among tenants.

**Migration of device from the tenant level to be multitenant device**

You cannot migrate a single-tenant WAN edge device from the tenant level to serve as a multitenant WAN edge device at the provider level. You must decommission the single-tenant device and delete it from Cisco SD-WAN Manager, perform a factory reset on the device to erase the existing configuration, and onboard the device at the provider level.

**Precedence in VRF limits**

Tenant limits takes precedence when VRF limits are also configured.

## Configure Multitenant WAN Edge Devices

**Configuration Workflow**

Perform the following configuration procedures as the Provider admin user.

1. Complete [Initial Setup for Multitenancy](#).
  - a. [Add Tenants](#).
  - b. (Optional) [Onboard Tenant-Owned WAN Edge Devices](#).



---

**Note** As the provider admin, you must onboard the devices from the provider-as-tenant view. Configure unique system IP address for each WAN edge device in the deployment across all tenant overlay networks.

---

2. Enable Multitenant WAN Edge Deployment.
3. Onboard WAN Edge Devices at the Provider Level.

**Note**

- Importing WAN edge device details from the Plug and Play (PnP) portal to Cisco SD-WAN Manager using **Sync Smart Account** is not supported. Export the device serial file from the PnP portal and import the file to Cisco SD-WAN Manager.
- Configure unique system IP address for each WAN edge device in the deployment across all tenant overlay networks.

4. Enable Multitenancy on Provider-Managed WAN Edge Devices.
5. Create Tiers.
6. Onboard Tenants to a Multitenant WAN Edge Device.
7. Create Tenant VPN for Onboarded Tenants.
8. (Optional) Configure Required Policies.

## Enable Multitenant WAN Edge Deployment

### Before You Begin

Ensure that every WAN edge device in the deployment, across tenants, is configured with a unique system IP address.

1. Log in to Cisco SD-WAN Manager as the Provider admin user.
2. From the Cisco SD-WAN Manager menu, choose **Administration > Settings**.
3. Find **MT Edge Deployment Settings** and click **Edit**.
4. For **Enable MT Edge Deployment**, click **Enabled**.  
By default, **Enable MT Edge Deployment** is **Disabled**.
5. Click **Save**.

If two or more WAN edge devices in the deployment are configured with the same system IP address, Cisco SD-WAN Manager reports an error. Modify the configuration of the WAN edge devices and try to enable multitenant WAN edge deployment.

## Onboard WAN Edge Devices at the Provider Level

1. Log in to Cisco SD-WAN Manager as the Provider admin user.
2. Upload the device serial number file to Cisco SD-WAN Manager. While uploading the file, choose the option to validate and send the device list to controllers.
3. Bootstrap the device using bootstrap configuration generated through Cisco SD-WAN Manager or manually create the initial configuration on the device.

4. If you are using Enterprise Certificates to authenticate the device, download the Certificate Signing Request (CSR) from Cisco SD-WAN Manager and get the CSR signed by the Enterprise CA. Install the certificate on Cisco SD-WAN Manager.
5. Create a configuration template for the device and attach the device to the template.

While configuring the device, configure the service provider organization name as `sp-organization-name` and the tenant `organization-name`.

## Enable Multitenancy on Provider-Level WAN Edge Devices

You can enable multitenancy on a provider-level WAN edge using the Multi Tenant parameter in the System template.

1. Log in to Cisco SD-WAN Manager as the Provider admin user.
2. From the Cisco SD-WAN Manager menu, choose **Configuration** > **Templates**.
3. Click **Feature**.
4. Find the System template of the provider-level WAN edge device for which you wish to enable multitenancy.
5. For the System template, click ... and click **Edit**.
6. In the Basic Configuration area, find the **Multi Tenant** parameter. Initially, the parameter has a default scope and the default value **Off**. For the **Multi Tenant** parameter,
  - a. Click the scope drop-down list and choose **Global** scope.
  - b. Click **On** to enable multitenancy.
7. Click **Update** to save and apply the modified configuration.

The provider-level WAN edge device can serve more than one tenant.

## Create a Resource Profile (Tier)

1. Log in to Cisco SD-WAN Manager as the Provider admin user.
2. From the Cisco SD-WAN Manager menu, choose **Administration** > **Tenant Management**.
3. Click **Resource Profiles**.

In Cisco vManage Release 20.11.1 and earlier releases, **Resource Profiles** is called **Tiers**.

Any existing tiers are displayed in a table.

4. Click **Add a Resource Profile**.

In Cisco vManage Release 20.11.1 and earlier releases, **Add a Resource Profile** is called **Add Tier**.

5. In the **Add Tier** slide-in pane, do the following:
  - a. Enter the following details:

| Field                                                                                                                                            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Resource Profile Name</b><br>In Cisco vManage Release 20.11.1 and earlier releases, <b>Resource Profile Name</b> is called <b>Tier Name</b> . | Enter a unique name for the tier.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Maximum VPN</b>                                                                                                                               | Enter the maximum number of VPNs that can be created on a multitenant WAN edge device for a tenant assigned to this tier.<br><br>Minimum value: 1<br><br>Maximum value: The maximum number of VPNs that you can specify for a tier depends on the device model. See <a href="#">Table 370: Maximum Number of VPNs Supported by Each Device Model, on page 945</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>NAT Limit (Optional)</b>                                                                                                                      | Minimum supported release: Cisco IOS XE Catalyst SD-WAN Release 17.12.1a and Cisco Catalyst SD-WAN Manager Release 20.12.1<br><br>Enter the maximum number of NAT translations that are allowed on each tenant.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Route Limit (Optional)</b>                                                                                                                    | Minimum supported release: Cisco IOS XE Catalyst SD-WAN Release 17.10.1a and Cisco vManage Release 20.10.1<br><br>(Optional) Specify IPv4 unicast or IPv6 unicast route limits. Route limit on a tenant is the sum of routes from all VRFs.<br><br>Default value is 0.<br><br><b>Note</b><br>The value 0 means there is no route limit configured in the tier definition.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Route Limit Type</b>                                                                                                                          | Minimum supported release: Cisco IOS XE Catalyst SD-WAN Release 17.10.1a and Cisco vManage Release 20.10.1<br><br>The following two route limit types can be configured for IPv4 or IPv6 routes on the device: <ul style="list-style-type: none"> <li>• <b>Warning-only:</b> This option allows to install new tenant IPv4 or IPv6 routes even after total exceeds their respective limit. A warning message is shown on device console when IPv4 route limit or IPv6 route limit is exceeded.</li> <li>• <b>Warning with threshold:</b> This option allows to configure a warning threshold, which is the percentage of the route limit. A warning message is shown on device console when the threshold percentage of IPv4 route limit or IPv6 route limit is reached. When a tenant's total IPv4 or IPv6 routes exceed the configured IPv4 or IPv6 route limit, routes are rejected.</li> </ul> |

| Field                                                                                                                               | Description                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Threshold</b>                                                                                                                    | <p>Minimum supported release: Cisco IOS XE Catalyst SD-WAN Release 17.10.1a and Cisco vManage Release 20.10.1</p> <p>Specify the route limit threshold value when you chose <b>Warning with threshold</b> option. When threshold percentage of IPv4 or IPv6 route limit is reached, a warning message is displayed on the device console.</p> <p>Range: 1 to 100.</p> |
| <b>Allowed Transport</b><br>In Cisco vManage Release 20.11.1 and earlier releases, <b>Allowed Transport</b> is called <b>TLOC</b> . | <p>Minimum supported release: Cisco IOS XE Catalyst SD-WAN Release 17.10.1a and Cisco vManage Release 20.10.1</p> <p>Add TLOC details for the tier. For a tier, add at least one TLOC and up to 16 TLOCs. A TLOC definition includes the TLOC color and the encapsulation type.</p> <p>Range: 1 to 16</p>                                                             |
| <b>Color</b>                                                                                                                        | <p>Minimum supported release: Cisco IOS XE Catalyst SD-WAN Release 17.10.1a and Cisco vManage Release 20.10.1</p> <p>Select the color from the drop-down list. The color attribute helps to identify an individual WAN transport tunnel.</p>                                                                                                                          |
| <b>Encapsulation</b>                                                                                                                | <p>Minimum supported release: Cisco IOS XE Catalyst SD-WAN Release 17.10.1a and Cisco vManage Release 20.10.1</p> <p>Select encapsulation type as either GRE or IPSec per TLOC configuration.</p>                                                                                                                                                                     |

Table 370: Maximum Number of VPNs Supported by Each Device Model

| Device Model    | Maximum Number of VPNs |
|-----------------|------------------------|
| ASR1001-X       | 80                     |
| ASR1001-HX      | 336                    |
| ASR1002-X       | 336                    |
| ASR1002-HX      | 336                    |
| C8500-12X4QC    | 336                    |
| C8500-12X       | 336                    |
| C8500L-8S4X     | 336                    |
| C8300-1N1S-6T   | 200                    |
| C8300-1N1S-4T2X | 200                    |
| C8300-2N2S-6T   | 200                    |
| C8300-2N2S-4T2X | 200                    |
| Catalyst 8000V  | 300                    |

| Device Model | Maximum Number of VPNs |
|--------------|------------------------|
| ISR4461      | 80                     |

- b. To add the tier, click **Save**. To discard your entries and close the slide-in pane, click **Cancel**.

After you click **Save**, the slide-in pane is closed and the new tier is listed in the table along with any existing tiers.

## Onboard Tenants to a Multitenant WAN Edge Device

You can onboard tenants to a multitenant WAN edge device using the Tenant template. If you haven't onboarded a tenant to the device, create a tenant template, add tenants, and attach the tenant template to the device template. If you have onboarded tenants to the device, to onboard a new tenant, update the tenant template attached to the device.

When a new tenant is onboarded to the multitenant WAN edge device, the device establishes control connections to the Cisco SD-WAN Controllers assigned to the tenant.

### Before You Begin

Before onboarding the tenant to a multitenant WAN edge device, [add the tenant](#) to the multitenant deployment and create the tier with which you wish to associate the tenant.

### Create a Tenant Template

1. Log in to Cisco SD-WAN Manager as the Provider admin user.
2. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
3. Find the device template for the multitenant WAN edge device to which you wish to onboard tenants.
4. For the device template, click ... and click **Edit**.  
The device template is displayed.
5. Click **Additional Templates**.
6. In the **Additional Templates** area, click the **Tenant** template drop-down list and then click **Create Template**.
7. In the Tenant template form, do as follows:
  - a. Enter a unique **Template Name**. The template name can contain up to 128 alphanumeric characters.
  - b. Enter a **Description** for the template. The description can contain up to 2048 alphanumeric characters.
  - c. In the **Tenant** area, click **New Tenant**.
  - d. From the **Tenant Name** drop-down list, choose the tenant organization name.  
In Cisco vManage Release 20.11.1 and earlier releases, **Tenant Name** is called **Org Name**.
  - e. From the **Resource Profile Name** drop-down list, choose a tier for the tenant.  
In Cisco vManage Release 20.11.1 and earlier releases, **Resource Profile Name** is called **Tier Name**.



- f. Click **Add**.
  - g. Repeat **Step c** to **Step f** to add additional tenants.
8. Click **Save**.
9. For the device template, click **Update** to save and apply the modified configuration.
10. Select the target device in the left pane and click **Configure Devices**.

## Update a Tenant Template

1. Log in to Cisco SD-WAN Manager as the Provider admin user.
2. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
3. Click **Feature**.
4. For the tenant template attached to the device, click ... and click **Edit**.  
The tenant template is displayed.
5. In the Tenant template form, do the following:
  - a. In the **Tenant** area, click **New Tenant**.
  - b. From the **Org Name** drop-down list, choose the tenant organization name.
  - c. From the **Tier Name** drop-down list, choose a tier for the tenant.
  - d. Click **Add**.
  - e. Repeat **Step a** to **Step d** to add additional tenants.
6. Click **Update**.

## Create Tenant VPN for Onboarded Tenants

After onboarding a tenant to a multitenant WAN edge device, use the Cisco VPN template to create tenant VPNs. To isolate VPN traffic of one tenant from the VPN traffic of other tenants onboarded on the multitenant WAN edge device, Cisco SD-WAN Manager maps a tenant VPN ID to a device VPN ID while you create the tenant VPN.

## Create a Cisco VPN Template

1. Log in to Cisco SD-WAN Manager as the Provider admin user.
2. From the Cisco SD-WAN Manager menu, choose **Configuration** > **Templates**.
3. Find the Device template for the multitenant WAN edge device to which you wish to onboard tenants.
4. For the device template, click ... and click **Edit**.  
The device template is displayed.
5. Click **Service VPN**.

6. In the **Service VPN** area, click **Add VPN**.
7. In the **Add VPN** slide-in pane, click **Create VPN Template**.
8. In the **Create VPN Template** slide-in pane, do the following:
  - a. Enter a unique **Template Name**. The template name can contain up to 128 alphanumeric characters.
  - b. Enter a **Description** for the template. The description can contain up to 2048 alphanumeric characters.
  - c. In the **Basic Configuration** area, map the tenant VPN ID to a device VPN ID:
    1. From drop-down list corresponding to **Tenant VPN**, choose the tenant organization name.
    2. In the text field corresponding to **Tenant VPN**, enter the tenant VPN ID.
    3. Click **Generate VPN ID**.

A read-only **VPN** field displays the device VPN ID for the tenant VPN ID. This mapping is performed by Cisco SD-WAN Manager. For a tenant, Cisco SD-WAN Manager maps a particular tenant VPN ID to the same device VPN ID on all the multitenant WAN edge devices.
  - d. Configure other properties of the tenant VPN in the template.
  - e. Click **Save**.
9. In the **Add VPN** slide-in pane, move the template created in **Step 8** from **Available VPN Templates** to **Selected VPN Templates**.
10. Click **Next**.
11. Add any additional Cisco VPN templates as needed.
12. Click **Add**.
13. For the device template, click **Update** to save and apply the modified configuration.
14. Select the target device in the left pane and click **Configure Devices**.

### Update a Cisco VPN Template

If you made a copy of an existing Cisco VPN template, you must modify the template for the new tenant VPN that you wish to create.

1. Log in to Cisco SD-WAN Manager as the Provider admin user.
2. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
3. Click **Feature**.
4. For the copied Cisco VPN template, click ... and click **Edit**.

The Cisco VPN template is displayed.
5. In the Cisco VPN template form, do the following:
  - a. In the **Basic Configuration** area, map the tenant VPN ID to a device VPN ID:
    1. From drop-down list corresponding to **Tenant VPN**, choose the tenant organization name.
    2. In the text field corresponding to **Tenant VPN**, enter the tenant VPN ID.

3. Click **Update VPN ID**.

A read-only **VPN** field displays the device VPN ID for the tenant VPN ID. This mapping is performed by Cisco SD-WAN Manager. For a tenant, Cisco SD-WAN Manager maps a particular tenant VPN ID to the same device VPN ID on all the multitenant WAN edge devices.

- b. Configure other properties of the tenant VPN in the template.

- c. Click **Update**.

6. Attach the Cisco VPN template to the device template of the target multitenant WAN edge device.

When you try to apply the tenant VPN configuration to the device, Cisco SD-WAN Manager checks the following:

1. The number of tenant VPNs that can be created for a tenant is restricted by the maximum number of the VPNs that is specified in the tier to which the tenant belongs. If the maximum number of tenant VPNs is already created for the tenant, Cisco SD-WAN Manager reports an error and does not apply the VPN configuration to the device.
2. Each device model supports a certain maximum number of device VPNs. On a multitenant WAN edge device, each tenant VPN is mapped to device VPN. If the maximum number of device VPNs supported by the device are already created and mapped to tenant VPNs, Cisco SD-WAN Manager reports an error and does not apply the configuration to the device.

## Remove Tenant from a Multitenant WAN Edge Device

To remove a tenant from a multitenant WAN Edge Device, you must detach the tenant service VPN template from the device template and delete the tenant from the Tenant template.

1. Log in to Cisco SD-WAN Manager as the Provider admin user.
2. Remove the tenant service VPN template from the Device template:
  - a. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
  - b. Find the device template for the multitenant WAN edge device to which you wish to onboard tenants.
  - c. For the device template, click ... and click **Edit**.

The device template is displayed.
  - d. Click **Service VPN**.
  - e. In the **Service VPN** area, check the check box for the VPN template to be removed.
  - f. Click **Remove VPN**.
  - g. Click **Update** to save and apply the modified configuration.
3. Delete tenant from the Tenant template:
  - a. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
  - b. Click **Feature**.
  - c. Find the Tenant template from which you should delete the tenant.

- d. For the Tenant template, click ... and click **Edit**.
- e. In the Tenant section, find the organization name of the tenant you wish to delete.
- f. Click the Trash icon corresponding to the tenant organization name.
- g. Click **Update** to save and apply the modified configuration.

## Delete a Tier

1. Log in to Cisco SD-WAN Manager as the Provider admin user.
2. From the Cisco SD-WAN Manager menu, choose **Administration > Tenant Management**.
3. Click **Tiers**.  
Existing tiers are displayed in a table.
4. For the desired tier, click ... in the **Actions** column, and then, click **Delete**.
5. In the **Delete Tier** dialog box, confirm that you wish to delete the tier.  
The tier is deleted and is no longer listed in the table.

# Verify Configuration and Operation of Multitenant WAN Edge Devices

## View Tenants Onboarded to Multitenant WAN Edge Device

- The following is a sample output of the **show sdwan running-config tenant** command.

```
Device# show sdwan running-config tenant

tenant "multitenancy-Customer1"
 tier
 tier-name tier_tenant1
 max-vpn 10
 !
 tenant-vpn 1
 device-vpn 1
 !
 tenant-tloc mpls ipsec
 !
 tenant-tloc public-internet ipsec
 !
!
tenant "multitenancy-Customer2"
 tier
 tier-name tier_tenant2
 max-vpn 12
 !
 tenant-vpn 1
 device-vpn 2
 !
!
```

```

tenant-tloc mpls ipsec
!
tenant-tloc public-internet ipsec
!
!
tenant "multitenancy-Customer3"
tier
 tier-name tier_tenant3
 max-vpn 10
!
tenant-vpn 1
 device-vpn 3
!
tenant-tloc mpls ipsec
!
tenant-tloc public-internet ipsec
!
!
tenant "multitenancy-Customer4"
tier
 tier-name tier_tenant4
 max-vpn 10
!
tenant-vpn 1
 device-vpn 4
!
tenant-tloc mpls ipsec
!
tenant-tloc public-internet ipsec
!
!

```

- The following is a sample output of the **show sdwan tenant-summary** command.

```

Device# show sdwan tenant-summary
tenants-summary max-tenants 30
tenants-summary num-active-tenants 4
GLOBAL
ORG NAME ID UUID

multitenancy-Customer1 16880 774cf81a-1d35-47f3-8c3f-ccb12506e09c
multitenancy-Customer2 23216 62c614be-fc18-4ed0-8f77-ddcd5196a412
multitenancy-Customer3 22400 48ba0449-f177-49c3-926c-a6d5077e34ae
multitenancy-Customer4 14624 61684731-4bda-40b9-9067-c2c9b846f8e8

```

- The following is a sample output of the **show tenant all** command.

```

Device# show tenant all

```

| Tenant  | ID    | Tier         | Tenant VPNs |
|---------|-------|--------------|-------------|
| Tenant1 | 16880 | tier_tenant1 | 1           |
| Tenant2 | 23216 | tier_tenant2 | 1           |
| Tenant3 | 22400 | tier_tenant3 | 1           |
| Tenant4 | 14624 | tier_tenant4 | 1           |

- The following is a sample output of the **show tenant mapping table** command.

```

Device# show tenant mapping table

```

| Tenant | Tenant VPN | Device VPN | Active |
|--------|------------|------------|--------|
|--------|------------|------------|--------|

---

|         |   |   |     |
|---------|---|---|-----|
| Tenant1 | 1 | 1 | YES |
| Tenant2 | 1 | 2 | YES |
| Tenant3 | 1 | 3 | YES |
| Tenant4 | 1 | 4 | YES |

---

- The following is a sample output of the **show tenant Tenant1** command.

```
Device# show tenant Tenant1

Tenant Tenant1
 Tenant ID: 30176
 UUID: 5a8b858d-d090-4cc3-8321-a663b08043d3
 Flags: 0x0000
 Resource Limits (Tier "tier_tenant1"):
 Maximum IPv4 Routes 100 (warning-threshold:50)
 Maximum IPv6 Routes 100 (warning-only)
 Maximum NAT Sessions 3

 Mapping Entries:
 Tenant VPN -> Device VPN
 1 -> 1 (Active)
```

- The following is a sample output of the **show ip route tenant Tenant1** command.

```
Device# show ip route tenant Tenant1

Tenant name is Tenant1 (id:30176)
 route_limit: 100, warning_limit_percent: 50%
 route_count: 7, rejected_routes: 0

 vrf_name: 1, vrf_id: 4, tenant_vpn_id: 1 route_count: 7, rejected_routes: 0

Routing Table: 1
 Tenant Name: Tenant1, Tenant ID: 30176
 Rejected Routes in tenant: 0, Rejected Routes in this routing table: 0
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
 n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 H - NHRP, G - NHRP registered, g - NHRP registration summary
 o - ODR, P - periodic downloaded static route, l - LISP
 a - application route
 + - replicated route, % - next hop override, p - overrides from PfR
 & - replicated local route overrides by connected

Gateway of last resort is not set

 172.16.0.0/16 is variably subnetted, 4 subnets, 2 masks
C 172.16.11.0/24 is directly connected, GigabitEthernet4.101
L 172.16.11.2/32 is directly connected, GigabitEthernet4.101
m 172.16.21.0/24 [251/0] via 172.16.255.16, 2w4d, Sdwan-system-intf
m 172.16.31.0/24 [251/0] via 172.16.255.14, 2w4d, Sdwan-system-intf
S 192.168.11.0/24 [1/0] via 172.16.11.1
m 192.168.21.0/24 [251/0] via 172.16.255.16, 2w4d, Sdwan-system-intf
m 192.168.31.0/24 [251/0] via 172.16.255.14, 2w4d, Sdwan-system-intf
```

- The following is a sample output of the **show ipv6 route tenant Tenant1** command.

```

Device# show ipv6 route tenant Tenant1

Tenant name is Tenant1 (id:30176)
 route_limit: 100, warning_only: True
 route_count: 1, rejected_routes: 0

 vrf_name: 1, vrf_id: 4, tenant_vpn_id: 1 route_count: 1, rejected_routes: 0

IPv6 Routing Table - 1 - 1 entries
 Tenant Name: Tenant1, Tenant ID: 30176
 Rejected Routes in tenant: 0, Rejected Routes in this routing table: 0
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
 B - BGP, R - RIP, H - NHRP, I1 - ISIS L1
 I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP
 EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE - Destination
 NDr - Redirect, RL - RPL, O - OSPF Intra, OI - OSPF Inter
 OE1 - OSPF ext 1, OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1
 ON2 - OSPF NSSA ext 2, la - LISP alt, lr - LISP site-registrations
 ld - LISP dyn-eid, lA - LISP away, le - LISP extranet-policy
 lp - LISP publications, ls - LISP destinations-summary, a - Application
 m - OMP
L FF00::/8 [0/0]
 via Null0, receive

```

- The following is a sample output of the **Show run | sec tenant-definition** command.

```

Device# show run | sec tenant-definition

tenant-definition "Tenant1"
 global-tenant-id 45516
 universal-unique-id 696e6fa0-078c-47fb-81b1-40df3b04c8e1
 !
 tier tier_tenant6
 max nat-session 10
 max routes
 !
 address-family ipv4
 unicast-route-limit 15000 warning-threshold 80
 !
 address-family ipv6
 unicast-route-limit 15000 warning-threshold 80
 !
 tenant-vpn-id 1
 device-vpn 1

```

- The following is a sample output of the **show ip nat translations tenant Tenant1 total** command.

```

Device# show ip nat translations tenant Tenant1 total

Total number of translations: 2

```

- The following is a sample output of the **show logging | i TENANT** command.

```

Device# show logging | i TENANT

*Feb 4 21:10:33.625: %IOSXE-4-PLATFORM: R0/0: cpp_cp: QFP:0.0 Thread:000
TS:00000004456610265827 %NAT-4-PER_TENANT_MAX_ENTRIES: per-tenant maximum limit of 10
reached for 26144.

```

- The following is a sample output IPv4 WARNING-ONLY Syslog Messages.

```
Device# show logging process ios start last boot | i route limit
```

```
2022/11/18 09:16:23.973714834 {iosrp_R0-0}{255}: [iosrp] [16608]: (ERR): Nov 18
09:16:23.973: %RIBTENANT-3-ROUTELIMITWARNING_ON: tenant(name:Tenant1, id:16880) ipv4
unicast route limit warning threshold: alarm_on
```




---

**Note** ROUTELIMITWARNING\_ON: “alarm\_on” means the route count has crossed the route limit.

---

```
Device# show ip route tenant "Tenant2"
```

```
2022/11/18 09:33:40.651174649 {iosrp_R0-0}{255}: [iosrp] [16608]: (ERR): Nov 18
09:33:40.651: %RIBTENANT-3-ROUTELIMITWARNING_OFF: tenant(name:Tenant1, id:16880) ipv4
unicast route limit warning threshold:
alarm_off
```




---

**Note** ROUTELIMITWARNING\_OFF: “alarm\_off” means for the time being, route count is reduced and it has not crossed the warning/route limit.

---

- The following is a sample output of IPv6 WARNING-ONLY Syslog Messages.

```
Device# show logging process ios start last boot | i route limit
```

```
2022/11/18 09:11:23.589778787 {iosrp_R0-0}{255}: [iosrp] [16608]: (ERR): Nov 18
09:11:23.589: %RIBTENANT-3-ROUTELIMITWARNING_ON: tenant(name:Tenant1, id:16880) ipv6
unicast route limit warning threshold: alarm_on
```




---

**Note** ROUTELIMITWARNING\_ON: "alarm\_on" means the route count has crossed the route limit.

---

```
Device# show ip route tenant "Tenant2 Inc"
```

```
2022/11/18 09:33:40.661037261 {iosrp_R0-0}{255}: [iosrp] [16608]: (ERR): Nov 18
09:33:40.661: %RIBTENANT-3-ROUTELIMITWARNING_OFF: tenant(name:Tenant1, id:16880) ipv6
unicast route limit warning threshold:
alarm_off
```




---

**Note** ROUTELIMITWARNING\_OFF: “alarm\_off” means for the time being, route count is reduced and it has not crossed the warning/route limit.

---

- The following is a sample output IPv4 Warning Threshold Syslog Messages.

```
Device# show logging process ios start last boot | i route limit
```

```
2022/11/17 19:07:04.330712142 {iosrp_R0-0}{255}: [iosrp] [16608]: (ERR): Nov 17
```



```
19:07:04.330: %RIBTENANT-3-ROUTE LIMIT WARNING_ON: tenant(name:Tenant2, id:23216) ipv4
unicast route limit warning threshold: alarm_on
```



**Note** ROUTELIMITWARNING\_ON: “alarm\_on” means the route count has crossed the warning threshold

```
Device# show ip route tenant "Tenant2"
```

```
2022/11/18 01:36:02.083288966 {iosrp_R0-0}{255}: [iosrp] [16608]: (ERR): Nov 18
01:36:02.083: %RIBTENANT-3-ROUTE LIMIT WARNING_OFF: tenant(name:Tenant2, id:23216) ipv4
unicast route limit warning threshold: alarm_off
```



**Note** ROUTELIMITWARNING\_OFF: “alarm\_off” means means for the time being, route count is reduced and it has not crossed the warning threshold.

- The following is a sample output IPv4 Warning Threshold Route-Limit Exceeded Syslog Messages.

```
Device# show logging process ios start last boot | i route limit
```

```
2022/11/18 10:06:35.698972324 {iosrp_R0-0}{255}: [iosrp] [16608]: (ERR): Nov 18
10:06:35.698: %RIBTENANT-3-ROUTE LIMIT EXCEEDED_ON: tenant(name:Tenant2, id:23216) ipv4
unicast route limit exceeded: alarm_on
```

```
2022/11/18 10:06:35.699002244 {iosrp_R0-0}{255}: [ribcmn] [16608]: (info): Failed to
add static route 192.168.202.0/24 to table(name:2, id:0x5) due to tenant(name:Tenant2,
id:23216) route limit exceeded
```



**Note** ROUTELIMITEXCEEDED\_ON: “alarm\_on” means the route count has exceeded tenant route limit

```
Device# show logging process ios start last boot | i route limit
```

```
2022/11/17 20:12:02.090953653 {iosrp_R0-0}{255}: [iosrp] [16608]: (ERR): Nov 17
20:12:02.090: %RIBTENANT-3-ROUTE LIMIT EXCEEDED_OFF: tenant(name:Tenant2, id:23216) ipv4
unicast route limit exceeded: alarm_off
```



**Note** ROUTELIMITEXCEEDED\_OFF: “alarm\_off” means for the time being, route count is reduced and it has not exceeded the tenant route limit.

- The following is a sample output IPv6 Warning Threshold Syslog Messages.

```
Device# show logging process ios start last boot | i route limit
```

```
2022/11/17 19:41:31.886639286 {iosrp_R0-0}{255}: [iosrp] [16608]: (ERR): Nov 17
19:41:31.886: %RIBTENANT-3-ROUTE LIMIT WARNING_ON: tenant(name:Tenant2, id:23216) ipv6
unicast route limit warning threshold: alarm_on
```



**Note** ROUTELIMITWARNING\_ON: “alarm\_on” means the route count has crossed the warning threshold

```
Device# show logging process ios start last boot | i route limit
```

```
2022/11/17 19:49:06.553836193 {iosrp_R0-0}{255}: [iosrp] [16608]: (ERR): Nov 17
19:49:06.553: %RIBTENANT-3-ROUTELIMITWARNING_OFF: tenant(name:Tenant2, id:23216) ipv6
unicast route limit warning threshold: alarm_off
```



**Note** ROUTELIMITWARNING\_OFF: “alarm\_off” means means for the time being, route count is reduced and it has not crossed the warning threshold.

- The following is a sample output IPv6 Warning Threshold Route-Limit Exceeded Syslog Messages.

```
Device# show logging process ios start last boot | i route limit
```

```
2022/11/17 05:07:19.326942097 {iosrp_R0-0}{255}: [iosrp] [17094]: (ERR): *Nov 17
05:07:19.326: %RIBTENANT-3-ROUTELIMITEXCEEDED_ON: tenant(name:Tenant2, id:7888) ipv6
unicast route limit exceeded: alarm_on
```

```
2022/11/17 05:07:19.327070153 {iosrp_R0-0}{255}: [ribcmn] [17094]: (info): Failed to
add static route 2001:C0A8:29C::/64 to table(name:2, id:0x1E000002) due to
tenant(name:Tenant2, id:7888) route limit exceeded
```



**Note** ROUTELIMITEXCEEDED\_ON: “alarm\_on” means the route count has exceeded tenant route limit.

```
Device# show logging process ios start last boot | i route limit
```

```
2022/11/17 05:15:32.132557582 {iosrp_R0-0}{255}: [iosrp] [17094]: (ERR): *Nov 17
05:15:32.132: %RIBTENANT-3-ROUTELIMITEXCEEDED_OFF: tenant(name:Tenant2, id:7888) ipv6
unicast route limit exceeded: alarm_off
```



**Note** ROUTELIMITEXCEEDED\_OFF: “alarm\_off” means for the time being, route count is reduced and it has not exceeded the tenant route limit.

### View Tenant-Device Mapping on Cisco SD-WAN Validator

The following is a sample output of the **show support orchestrator tenant-uuid-map** command.

```
vBond# show support orchestrator tenant-uuid-map
```

| Type   | Chassis-num/uuid                     | Tenant id list   |
|--------|--------------------------------------|------------------|
| vSmart | 0c90593a-0f40-4890-a980-5e14907482f7 | 18624,6672,27120 |

```

| vSmart | 8a083d5e-1350-4946-932e-237758bb2280 | 12448,6672,10384 |

| vSmart | c99c50da-3dff-4951-a598-b3cf71530e99 | 18624,12448,27120,10384 |

| vEdge | c8k-24d9f68c-8c01-4e6c-813a-959752f30e73 | 18624,12448,6672,27120,10384 |

| vEdge | c8k-fdd9c202-8756-4079-9a56-278e6635412b | 18624,12448,6672 |

```

You can view the tenant global ID on a multitenant WAN edge device using the **show sdwan tenant-summary** command. On a Cisco SD-WAN Controller, you can use the **show tenant-summary** command.

### View Tenant-Cisco SD-WAN Controller Mapping on Cisco SD-WAN Validator

The following is a sample output of the **show tenant-mapping** command.

```
vBond# show tenant-mapping
```

```
VSMART
```

```
SERIAL
```

```
NUM TENANT NAMES TENANT COUNT

12345990 ["multitenancy-Customer6" "multitenancy-Customer4" "multitenancy-Customer3"
"multitenancy-Customer1"] 4
12345992 -
 0
12345994 ["multitenancy-Customer6" "multitenancy-Customer5" "multitenancy-Customer3"
"multitenancy-Customer2"] 4
12345997 -
 0
12345998 -
 0
12346001 ["multitenancy-Customer5" "multitenancy-Customer4" "multitenancy-Customer2"
"multitenancy-Customer1"] 4
```

### View Tenant-Mapping on Cisco SD-WAN Controller

The following is a sample output of the **show tenant-summary** command.

```
vSmart# show tenant-summary
```

```
tenant-summary max-tenants 24
```

```
tenant-summary num-active-tenants 4
```

```
TENANT ORG NAME TENANT TENANT
 ID VPN ID

multitenancy-Customer1 1 1003
multitenancy-Customer2 2 1004
multitenancy-Customer3 3 1005
multitenancy-Customer4 4 1006
```

### View Multitenant WAN Edge Device to Cisco SD-WAN Controller Connections

The following is a sample output of the **show sdwan control tenant-connections** command.

```
Device# show sdwan control tenant-connections
```

```
PEER LOCAL TENANT
```

```
SYSTEM IP COLOR NAME
```

```

172.16.255.19 mpls multitenancy-Customer3
172.16.255.19 mpls multitenancy-Customer2
172.16.255.19 public-internet multitenancy-Customer3
172.16.255.19 public-internet multitenancy-Customer2
```

```

172.16.255.20 mpls multitenancy-Customer4
172.16.255.20 mpls multitenancy-Customer1 Inc
172.16.255.20 mpls multitenancy-Customer2 Inc
172.16.255.20 public-internet multitenancy-Customer4 Inc
172.16.255.20 public-internet multitenancy-Customer1 Inc
172.16.255.20 public-internet multitenancy-Customer2 Inc
172.16.255.24 mpls multitenancy-Customer4 Inc
172.16.255.24 mpls multitenancy-Customer1 Inc
172.16.255.24 mpls multitenancy-Customer3 Inc
172.16.255.24 public-internet multitenancy-Customer4 Inc
172.16.255.24 public-internet multitenancy-Customer1 Inc
172.16.255.24 public-internet multitenancy-Customer3 Inc

```

The PEER SYSTEM IP column shows the IP address of the Cisco SD-WAN Controller the multitenant WAN edge device is connected to. The TENANT NAME entry for a PEER SYSTEM IP shows the name of the tenant organization for which the connection to the Cisco SD-WAN Controller is established.

### View OMP Information on a Multitenant WAN Edge Device

- The following is a sample output of the **show sdwan tenant *tenant-name* omp peers** command.

```

Device# show sdwan tenant multitenancy-Customer1 omp peers
R -> routes received
I -> routes installed
S -> routes sent

TENANT DOMAIN OVERLAY SITE REGION
ID PEER TYPE ID ID ID ID STATE UPTIME R/I/S

23216 172.16.255.19 vsmart 1 1 101 None up 1:13:42:12 24/24/22
23216 172.16.255.20 vsmart 1 1 102 None up 1:13:42:12 24/0/22

```

The output shows the Cisco SD-WAN Controllers to which the multitenant WAN edge device is connected for the particular tenant and summarizes the OMP exchanges between the Cisco SD-WAN Controllers and the device.

- The following is a sample output of the **show sdwan tenant *tenant-name* omp routes** command.

```

Device# show sdwan tenant multitenancy-Customer1 omp routes

Code:

C -> chosen
I -> installed
Red -> redistributed
Rej -> rejected
L -> looped
R -> resolved
S -> stale
Ext -> extranet
Inv -> invalid
Stg -> staged
IA -> On-demand inactive

```

U -> TLOC unresolved

Reo -> reoriginated

|               |     | PATH            |               |            |           | ATTRIBUTE |           |
|---------------|-----|-----------------|---------------|------------|-----------|-----------|-----------|
| TENANT        | VPN | PREFIX          | FROM PEER     | ID         | LABEL     | STATUS    | TYPE      |
| TLOC          | IP  | COLOR           | ENCAP         | PREFERENCE | REGION ID | REGION    | PATH      |
| 23184         | 1   | 172.16.11.0/24  | 0.0.0.0       | 66         | 1003      | C,Red,R   | installed |
| 172.16.255.15 |     | mpls            | ipsec -       | None       |           | 65534     |           |
| 172.16.255.15 |     | public-internet | 0.0.0.0       | 69         | 1003      | C,Red,R   | installed |
|               |     |                 | ipsec -       | None       |           | 65534     |           |
| 23184         | 1   | 172.16.31.0/24  | 172.16.255.19 | 5          | 1003      | C,I,R     | installed |
| 172.16.255.14 |     | mpls            | ipsec -       | None       |           | 65534     |           |
| 172.16.255.14 |     | public-internet | 172.16.255.19 | 6          | 1003      | C,I,R     | installed |
|               |     |                 | ipsec -       | None       |           | 65534     |           |
| 23184         | 1   | 192.168.11.0/24 | 0.0.0.0       | 66         | 1003      | C,Red,R   | installed |
| 172.16.255.15 |     | mpls            | ipsec -       | None       |           | 65534     |           |
| 172.16.255.15 |     | public-internet | 0.0.0.0       | 69         | 1003      | C,Red,R   | installed |
|               |     |                 | ipsec -       | None       |           | 65534     |           |
| 23184         | 1   | 192.168.31.0/24 | 172.16.255.19 | 7          | 1003      | C,I,R     | installed |
| 172.16.255.14 |     | mpls            | ipsec -       | None       |           | 65534     |           |
| 172.16.255.14 |     | public-internet | 172.16.255.19 | 8          | 1003      | C,I,R     | installed |
|               |     |                 | ipsec -       | None       |           | 65534     |           |

- The following is a sample output of the **show sdwan tenant *tenant-name* omp services** command.

Device# **show sdwan tenant multitenancy-Customer1 omp services**

C -> chosen

I -> installed

Red -> redistributed

Rej -> rejected

L -> looped

R -> resolved

S -> stale

```

Ext -> extranet

Stg -> staged

IA -> On-demand inactive

Inv -> invalid

Reo -> reoriginated

```

| ADDRESS              |               |         |         |               |           | PATH | REGION |
|----------------------|---------------|---------|---------|---------------|-----------|------|--------|
| FAMILY LABEL         | TENANT STATUS | VPN VRF | SERVICE | ORIGINATOR    | FROM PEER | ID   | ID     |
| ipv4<br>1003         | 23184         | 1       | VPN     | 172.16.255.15 | 0.0.0.0   | 66   | NA     |
|                      | C,Red,R       | 1       |         |               | 0.0.0.0   | 69   | NA     |
| 1003<br>ipv6<br>1003 | C,Red,R       | 1       | VPN     | 172.16.255.15 | 0.0.0.0   | 66   | NA     |
|                      | 23184         | 1       |         |               | 0.0.0.0   | 69   | NA     |
| 1003<br>1003         | C,Red,R       | 1       |         |               | 0.0.0.0   | 66   | NA     |
|                      | C,Red,R       | 1       |         |               | 0.0.0.0   | 69   | NA     |
| 1003                 | C,Red,R       | 1       |         |               |           |      |        |

Related per-Tenant OMP commands:

- **show sdwan tenant *tenant-name* omp tlocs**
- **show sdwan tenant *tenant-name* omp multicast-routes**
- **show sdwan tenant *tenant-name* omp ipv6-routes**

Related global OMP commands:

- **show sdwan omp tloc-paths**
- **show sdwan omp summary**

### View OMP Information on a Cisco SD-WAN Controller

- The following is a sample output of the **show tenant *tenant-name* omp peers** command.

```

vSmart# show tenant multitenancy-Customer1 omp peers
R -> routes received

I -> routes installed

S -> routes sent

```

|               |       | DOMAIN | OVERLAY | SITE |       |             |       |
|---------------|-------|--------|---------|------|-------|-------------|-------|
| PEER          | TYPE  | ID     | ID      | ID   | STATE | UPTIME      | R/I/S |
| -----         |       |        |         |      |       |             |       |
| 172.16.255.14 | vedge | 1      | 1       | 400  | up    | 23:09:40:04 | 4/0/0 |
| 172.16.255.15 | vedge | 1      | 1       | 500  | up    | 0:14:33:55  | 0/0/0 |

```
172.16.255.24 vsmart 1 1 103 up 44:06:36:31 4/0/4
```

The output shows the other Cisco SD-WAN Controller serving the tenant and the multitenant or tenant-managed WAN edge devices connected to the Cisco SD-WAN Controller.

- The following is a sample output of the **show tenant *tenant-name* omp routes** command.

```
vSmart# show tenant multitenancy-Customer1 omp routes
```

```

omp route entries for vpn 1 route 172.16.33.0/24

```

RECEIVED FROM:

```
peer 172.16.255.14
path-id 66
label 1005
status C,R
loss-reason not set
lost-to-peer not set
lost-to-path-id not set
```

Attributes:

```
originator 172.16.255.14
type installed
tloc 172.16.255.14, mpls, ipsec
ultimate-tloc not set
domain-id not set
overlay-id 1
site-id 400
region-id None
region-path 65534
preference not set
tag not set
origin-proto connected
origin-metric 0
as-path not set
```

```

community not set
unknown-attr-len not set
.
.
.

omp route entries for vpn 1 route 192.168.33.0/24

RECEIVED FROM:

peer 172.16.255.14
path-id 66
label 1005
status C,R
loss-reason not set
lost-to-peer not set
lost-to-path-id not set

Attributes:

originator 172.16.255.14
type installed
tloc 172.16.255.14, mpls, ipsec
ultimate-tloc not set
domain-id not set
overlay-id 1
site-id 400
region-id None
region-path 65534
preference not set
tag not set
origin-proto static
origin-metric 0
as-path not set
community not set
unknown-attr-len not set

```



The command output shows the routes advertised by multitenant and tenant-managed WAN edge devices for the tenant VPNs.

### View Per Tenant Policy Configuration on a Multitenant WAN Edge Device

To view per tenant policy configuration, use the following commands:

- **show sdwan tenant *tenant-name* policy from-vsmart**
- **show sdwan tenant *tenant-name* policy data-policy-filter**
- **show sdwan tenant *tenant-name* policy app-route-policy-filter**
- **show sdwan tenant *tenant-name* policy from-vsmart policy data-policy**
- **show sdwan tenant *tenant-name* policy from-vsmart policy app-route-policy**

## Troubleshoot Multitenant WAN Edge Device Errors

| Error Scenario            | Log File                                                                                                                          |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Device Onboarding         | Cisco SD-WAN Manager: /var/log/nms/vmanage-server.log                                                                             |
| Device Configuration Pull | Cisco SD-WAN Manager:<br>/var/log/nms/vmanage-server-deviceconfig-template.log<br>WAN edge device: /bootflash/sdwan/cfgloader.log |





## CHAPTER 37

# Cisco Catalyst SD-WAN Multitenancy (Cisco IOS XE Releases 17.4.x and 17.5.x)

Table 371: Feature History

| Feature Name                       | Release Information                                                          | Feature Description                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cisco Catalyst SD-WAN Multitenancy | Cisco IOS XE Catalyst SD-WAN Release 17.4.1a<br>Cisco vManage Release 20.4.1 | With Cisco Catalyst SD-WAN multitenancy, a service provider can manage multiple customers, called tenants, from Cisco SD-WAN Manager. In a multitenant Cisco Catalyst SD-WAN deployment, tenants share Cisco SD-WAN Manager instances, Cisco Catalyst SD-WAN Validators and Cisco Catalyst SD-WAN Controllers. Tenant data is logically isolated on these shared resources. |

- [Overview of Cisco Catalyst SD-WAN Multitenancy, on page 965](#)
- [User Roles in Multitenant Environment, on page 968](#)
- [Hardware Supported and Specifications, on page 969](#)
- [Initial Setup for Multitenancy, on page 971](#)
- [Manage Tenants, on page 974](#)
- [Cisco SD-WAN Manager Dashboard for Multitenancy, on page 977](#)
- [Manage Tenant WAN Edge Devices, on page 982](#)
- [Tenant-Specific Policies on Cisco Catalyst SD-WAN Controllers, on page 983](#)
- [Manage Tenant Data, on page 983](#)
- [View OMP Statistics per Tenant on a Cisco SD-WAN Controller, on page 987](#)
- [View Tenants Associated with a Cisco SD-WAN Controller, on page 987](#)
- [Migrate Single-Tenant Cisco Catalyst SD-WAN Overlay to Multitenant Cisco Catalyst SD-WAN Deployment, on page 988](#)

## Overview of Cisco Catalyst SD-WAN Multitenancy

With Cisco Catalyst SD-WAN multitenancy, a service provider can manage multiple customers, called tenants, from Cisco SD-WAN Manager. The tenants share Cisco SD-WAN Manager instances, Cisco Catalyst SD-WAN Validators, and Cisco Catalyst SD-WAN Controllers. The domain name of the service provider has subdomains

for each tenant. For example, the `multitenancy.com` service provider can manage the tenants Customer1 (`Customer1.multitenancy.com`) and Customer2 (`Customer2.multitenancy.com`).

Following are the key features of Cisco Catalyst SD-WAN multitenancy:

- Full enterprise multitenancy: Cisco Catalyst SD-WAN supports multitenancy and offers enterprises the flexibility of segregated roles such as service provider and tenants. Service providers can use multitenancy to provide Cisco Catalyst SD-WAN service offerings to their customers.
- Multi-tenant Cisco SD-WAN Manager:
  - Cisco SD-WAN Manager is deployed and configured by the service provider. The provider enables multitenancy and creates a Cisco SD-WAN Manager cluster to serve tenants. Only the provider can access a Cisco SD-WAN Manager instance through the SSH terminal.



#### Note

To connect to a device through SSH, use the IP address of the `vmanage_system` interface; this IP address is assigned by Cisco SD-WAN Manager. Do not use a user-configured system IP address to connect to a device through SSH.

You can find the IP address of the `vmanage_system` interface from the output of the **show interface description** command. Alternatively, you can launch the device SSH terminal from Cisco SD-WAN Manager and find the `vmanage_system` IP address from the first line of the log-in prompt.

- Cisco SD-WAN Manager offers service providers an overall view of the SD-WAN multi-tenant deployment and allows a provider to manage the shared Cisco Catalyst SD-WAN Validator and Cisco Catalyst SD-WAN Controller devices. Cisco SD-WAN Manager also allows service providers to monitor and manage the deployments of each tenant.
- Cisco SD-WAN Manager allows tenants to monitor and manage their deployment. Through Cisco SD-WAN Manager, tenants can deploy and configure WAN edge devices. Tenants can also configure custom policies on assigned Cisco Catalyst SD-WAN Controllers.
- Multi-tenant Cisco Catalyst SD-WAN Validators:
  - Cisco Catalyst SD-WAN Validators are deployed and configured by the service provider. Only the provider can access a Cisco Catalyst SD-WAN Validator through the SSH terminal.



#### Note

To connect to a device through SSH, use the IP address of the `vmanage_system` interface; this IP address is assigned by Cisco SD-WAN Manager. Do not use a user-configured system IP address to connect to a device through SSH.

You can find the IP address of the `vmanage_system` interface from the output of the **show interface description** command. Alternatively, you can launch the device SSH terminal from Cisco SD-WAN Manager and find the `vmanage_system` IP address from the first line of the log-in prompt.

- Cisco Catalyst SD-WAN Validators serve WAN edge devices of multiple tenants as the devices are added to the overlay network.
- Multi-tenant Cisco Catalyst SD-WAN Controllers:

- Cisco Catalyst SD-WAN Controllers are deployed by the service provider. Only the provider can create and attach device and feature templates to Cisco Catalyst SD-WAN Controllers, and can access a Cisco Catalyst SD-WAN Controller through the SSH terminal.

**Note**

To connect to a device through SSH, use the IP address of the `vmanage_system` interface; this IP address is assigned by Cisco SD-WAN Manager. Do not use a user-configured system IP address to connect to a device through SSH.

You can find the IP address of the `vmanage_system` interface from the output of the **show interface description** command. Alternatively, you can launch the device SSH terminal from Cisco SD-WAN Manager and find the `vmanage_system` IP address from the first line of the log-in prompt.

- When a tenant is created, Cisco SD-WAN Manager assigns two Cisco Catalyst SD-WAN Controllers for the tenant. The Cisco Catalyst SD-WAN Controllers form an active-active cluster.

Each tenant is assigned only two Cisco Catalyst SD-WAN Controllers. Before a tenant is created, two Cisco Catalyst SD-WAN Controllers must be available to serve the tenant.

- Each pair of Cisco Catalyst SD-WAN Controllers can serve a maximum of 24 tenants.
- Tenants can configure custom policies on the Cisco Catalyst SD-WAN Controllers assigned to them. Cisco SD-WAN Manager notifies the Cisco Catalyst SD-WAN Controllers to pull the policy templates. Cisco Catalyst SD-WAN Controllers pull the templates and deploy the policy configuration for the specific tenant.
- Only the provider can view events, audit logs, and OMP alarms for a Cisco Catalyst SD-WAN Controller on Cisco SD-WAN Manager.

- WAN Edge Devices:

- A tenant or the provider acting on behalf of a tenant can add WAN edge devices to the tenant network, configure the devices, and remove the devices from the tenant network, or access the device through the SSH terminal.

**Note**

To connect to a device through SSH, use the IP address of the `vmanage_system` interface; this IP address is assigned by Cisco SD-WAN Manager. Do not use a user-configured system IP address to connect to a device through SSH.

You can find the IP address of the `vmanage_system` interface from the output of the **show interface description** command. Alternatively, you can launch the device SSH terminal from Cisco SD-WAN Manager and find the `vmanage_system` IP address from the first line of the log-in prompt.

- A provider can manage the WAN edge devices only from [provider-as-tenant](#) view. In the [provider](#) view, Cisco SD-WAN Manager does not present any WAN edge device information.
- Cisco SD-WAN Manager reports WAN edge device events, logs, and alarms only in the [tenant](#) and the provider-as-tenant views.

- Overlapping VPN numbers: A particular VPN or a set of common VPNs is assigned to a specific tenant, with their own configurations and monitoring dashboard environment. These VPN numbers can overlap where they are used by other tenants.
- On-prem and cloud deployment models: Cisco Catalyst SD-WAN controllers can be deployed in an organization data center on servers running the VMware vSphere ESXi or the Kernel-based Virtual Machine (KVM) hypervisor. Cisco Catalyst SD-WAN controllers can also be deployed in the cloud on Amazon Web Services (AWS) servers.

## User Roles in Multitenant Environment

A multi-tenant environment includes the service provider and tenant roles. Each role has distinct privileges, views, and functions.

### Provider Role

The provider role entitles system-wide administrative privileges. A user with the provider role has the default username **admin**. The provider user can access Cisco SD-WAN Manager using the domain name of the service provider or by using the Cisco SD-WAN Manager IP address. When using a domain name, the domain name has the format `https://multitenancy.com`.

The **admin** user is part of the user group **netadmin**. Users in this group are permitted to perform all operations on the controllers and the Cisco Catalyst SD-WAN devices of the tenants. You can add additional users to the **netadmin** group.

You cannot modify the privileges of the **netadmin** group. On Cisco SD-WAN Manager, you can view the privileges of the user group from the **Administration > Manage Users > User Groups** page.



**Note** When you create a new provider user in Cisco SD-WAN Manager, including a **netadmin** user, by default, the user is not allowed SSH access to the Cisco SD-WAN Manager VM. To enable SSH access, configure SSH authentication using a AAA template and push the template to Cisco SD-WAN Manager. For more information on enabling SSH authentication, see [SSH Authentication using Cisco SD-WAN Manager on Cisco IOS XE Catalyst SD-WAN Devices](#).

For more information about configuring users and user groups, see [Configure User Access and Authentication](#).

Cisco SD-WAN Manager offers two views to a provider:

#### • Provider View

When a provider user logs in to multi-tenant Cisco SD-WAN Manager as **admin** or another **netadmin** user, Cisco SD-WAN Manager presents the provider view and displays the provider dashboard.

You can perform the following functions from the provider view:

- Provision and manage Cisco SD-WAN Manager, Cisco Catalyst SD-WAN Validators and Cisco Catalyst SD-WAN Controllers.
- Add, modify, or delete tenants.
- Monitor the overlay network.

- **Provider-as-Tenant View**

When a provider user selects a specific tenant from the **Select Tenant** drop-down list at the top of the provider dashboard, Cisco SD-WAN Manager presents the provider-as-tenant view and displays the tenant dashboard for the selected tenant. The provider user has the same view of Cisco SD-WAN Manager as a tenant user would when logged in as **tenantadmin**. From this view, the provider can manage the tenant deployment on behalf of the tenant.

In the provider dashboard, a table of tenants presents a status summary for each tenant. A provider user can also launch the provider-as-tenant view by clicking on a tenant name in this table.

### Tenant Role

The tenant role entitles tenant administrative privileges. A user with the tenant role has the default username **tenantadmin**. The default password is **Cisco#123@Viptela**. We recommend that you change the default password on first login. For information on changing the default password, see [Hardware and Software Installation](#).

The **tenantadmin** user is part of the user group **tenantadmin**. Users in this group are permitted to perform all operations on the WAN edge devices of the tenants. You can add additional users to the **tenantadmin** group.

You cannot modify the privileges of the **tenantadmin** group. On Cisco SD-WAN Manager, you can view the privileges of the user group from the **Administration > Manage Users > User Groups** page.

For more information about configuring users and user groups, see [Configure User Access and Authentication](#).

A tenant user can log in to Cisco SD-WAN Manager using a dedicated URL and the default username **tenantadmin**. For example, the dedicated URL of a tenant could be `https://Customer1.multitenancy.com` for a provider using the domain name `https://multitenancy.com`. When the user logs in, Cisco SD-WAN Manager presents the tenant view and displays the tenant dashboard.

A tenant user with administrative privileges can perform the following functions:

- Provision and manage tenant routers
- Monitor overlay network of the tenant
- Create custom policies on the assigned Cisco Catalyst SD-WAN Controllers
- Upgrade the software on the tenant routers.

## Hardware Supported and Specifications

The following platforms support multitenancy.

**Table 372: Router Models**

| Platform                            | Router Models                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cisco IOS XE Catalyst SD-WAN device | <ul style="list-style-type: none"> <li>• Cisco ASR 1000 Series Aggregation Services Routers</li> <li>• Cisco ISR 1000 Series Integrated Services Routers</li> <li>• Cisco ISR 4000 Series Integrated Services Routers</li> <li>• Cisco Catalyst 8300 Series Edge Platforms</li> <li>• Cisco Catalyst 8500 Series Edge Platforms</li> <li>• Cisco Catalyst 8000V Edge Software</li> </ul> |

The following hypervisors and deployment model are supported for multitenancy.

**Table 373: Deployment Model**

| Specification                         | Description                                                                                             |
|---------------------------------------|---------------------------------------------------------------------------------------------------------|
| Supported hypervisors                 | VMware, KVM, AWS (cloud-hosted by Cisco)                                                                |
| Cisco SD-WAN Manager Deployment Model | Cluster, 3 Cisco SD-WAN Manager instances with each instance running all Cisco SD-WAN Manager services. |

The supported hardware specifications for the Cisco Catalyst SD-WAN Validator, Cisco SD-WAN Manager, and the Cisco Catalyst SD-WAN Controller are as follows:

**Table 374: On-prem Deployment**

| Server                     | Cisco SD-WAN Manager | Cisco Catalyst SD-WAN Validator | Cisco Catalyst SD-WAN Controller |
|----------------------------|----------------------|---------------------------------|----------------------------------|
| <b>Deployment Model</b>    | Cluster              | N/A                             | Non-containerized                |
| <b>Number of Instances</b> | 3                    | 2                               | 2 per 24 tenants                 |
| <b>CPU</b>                 | 32 vCPU              | 4 vCPU                          | 8 vCPU                           |
| <b>DRAM</b>                | 72 GB                | 4 GB                            | 16 GB                            |
| <b>Hard Disk</b>           | 1 TB                 | 10 GB                           | 16 GB                            |



|                                 |                                                                                                                                                                                                                |     |     |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|
| <b>NMS Service Distribution</b> | Some services run on all three Cisco SD-WAN Manager instances in the cluster, while some services run on only one of the three instances in the cluster. Therefore, the CPU load may vary among the instances. | N/A | N/A |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|



**Note** If DPI is enabled, we recommend that the aggregated DPI data across all Cisco SD-WAN Manager instances not exceed 350 GB per day.

## Initial Setup for Multitenancy

### Prerequisites

- Download and install software versions as recommended in the following table:

*Table 375: Software Prerequisites for Cisco Catalyst SD-WAN Multitenancy*

| Device                              | Software Version                             |
|-------------------------------------|----------------------------------------------|
| Cisco SD-WAN Manager                | Cisco vManage Release 20.4.1                 |
| Cisco Catalyst SD-WAN Validator     | Cisco SD-WAN Release 20.4.1                  |
| Cisco Catalyst SD-WAN Controller    | Cisco SD-WAN Release 20.4.1                  |
| Cisco IOS XE Catalyst SD-WAN device | Cisco IOS XE Catalyst SD-WAN Release 17.4.1a |

A configuration in which one or more controllers, or WAN edge devices, are running software versions earlier than those mentioned in the table above is not supported.

- Do not migrate an existing single-tenant Cisco SD-WAN Manager instance into multitenant mode, even if you invalidate or delete all devices from the existing Cisco SD-WAN Manager instance. Instead, download and install a new Cisco SD-WAN Manager software image.



**Note** After you enable Cisco SD-WAN Manager for multitenancy, you cannot migrate it back to single tenant mode.

- Log in to Cisco SD-WAN Manager as the provider **admin** user.
- Create three Cisco SD-WAN Manager instances and associated configuration templates. See [Deploy Cisco SD-WAN Manager](#).

- a. While configuring Cisco SD-WAN Manager instances, configure the service provider organization name (`sp-organization-name`) and the organization name (`organization-name`).

**Example:**

```
sp-organization-name multitenancy
organization-name multitenancy
```

2. Configure one of the Cisco SD-WAN Manager instances to support multitenancy. See [Enable Multitenancy on Cisco SD-WAN Manager, on page 972](#)
3. Create a Cisco SD-WAN Manager cluster consisting of three Cisco SD-WAN Manager instances. See [Cluster Management](#).
  - The Cisco SD-WAN Manager cluster must have three Cisco SD-WAN Manager instances. A cluster with more than three instances or fewer than three instances is not a supported configuration for Cisco Catalyst SD-WAN multitenancy.
  - While creating the Cisco SD-WAN Manager cluster, add the Cisco SD-WAN Manager instance configured to support multitenancy before adding the other two Cisco SD-WAN Manager instances.
4. Certify all instances of Cisco SD-WAN Manager. See [Generate Cisco SD-WAN Manager Certificate](#).
5. Create and configure Cisco SD-WAN Validator instances. See [Deploy Cisco SD-WAN Validator](#).  
While configuring Cisco SD-WAN Validator instances, configure the service provider organization name (`sp-organization-name`) and the organization name (`organization-name`). See [Configure Organization Name in Cisco SD-WAN Validator](#).  

```
sp-organization-name multitenancy
organization-name multitenancy
```
6. Create Cisco SD-WAN Controller instances. See [Deploy the Cisco SD-WAN Controllers](#).  
To support 50 tenants and 1000 devices across all tenants, deploy 6 Cisco SD-WAN Controller instances. To support 100 tenants and 5000 devices across all tenants, deploy 12 Cisco SD-WAN Controllers.
  - a. [Add Cisco SD-WAN Controllers](#) to the overlay network.
7. Onboard new tenants. See [Add a New Tenant](#).

## Enable Multitenancy on Cisco SD-WAN Manager

1. Launch Cisco SD-WAN Manager using the URL `https://vmanage-ip-address:port`. Log in as the provider **admin** user.
2. From the Cisco SD-WAN Manager menu, choose **Administration > Settings > Tenancy Mode**. If you are using Cisco Catalyst SD-WAN Manager Release 20.12.x or earlier, click **Edit**.
3. In the **Tenancy** field, click **Multitenant**.
4. In the **Domain** field, enter the domain name of the service provider (for example, `multitenancy.com`).
5. Enter a **Cluster Id** (for example, `cluster-1` or `123456`).
6. Click **Save**. If you are using Cisco Catalyst SD-WAN Manager Release 20.12.x or earlier, click **Proceed** to confirm that you want to change the tenancy mode.

Cisco SD-WAN Manager reboots in multitenant mode and when a provider user logs in to Cisco SD-WAN Manager, the provider dashboard appears.

## Add Cisco SD-WAN Controller

1. Log in to Cisco SD-WAN Manager as the provider **admin** user.
2. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
3. Click **Controllers**.



---

**Note** Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

---

4. Click **Add Controller**.
5. In the **Add Controller** dialog box, do the following:
  - a. In the **Controller Management IP Address** field, enter the system IP address of the Cisco SD-WAN Controller.
  - b. Enter the **Username** and **Password** required to access the Cisco SD-WAN Controller.
  - c. Select the protocol to use for control-plane connections. The default is **DTLS**.  
If you select **TLS**, enter the port number to use for TLS connections. The default is 23456.
  - d. Check the **Generate CSR** check box for Cisco SD-WAN Manager to create a Certificate Signing Request.
  - e. Click **Add**.
6. From the Cisco SD-WAN Manager menu, choose **Configuration > Certificates**.  
For the newly added Cisco SD-WAN Controller, the **Operation Status** reads **CSR Generated**.
  - a. For the newly added Cisco SD-WAN Controller, click **More Options** icon and click **View CSR**.
  - b. Submit the CSR to the Certificate Authority (CA) and obtain a signed certificate.
7. From the Cisco SD-WAN Manager menu, choose **Configuration > Certificates**.
8. Click **Install Certificate**.
9. In the **Install Certificate** dialog box, paste the **Certificate Text** or click **Select a file** upload the certificate file. Click **Install**.

Cisco SD-WAN Manager installs the certificate on the Cisco SD-WAN Controller. Cisco SD-WAN Manager also sends the serial number of the certificate to other controllers.

On the **Configuration > Certificates** page, the **Operation Status** for the newly added Cisco SD-WAN Controller reads as **Validator Updated**.

On the **Configuration > Devices** page, the new controller is listed in the Controller table with the controller type, hostname of the controller, IP address, site ID, and other details. The **Mode** is set to **CLI**.

10. Change the mode of the newly added Cisco SD-WAN Controller to **Manager** by attaching a template to the device.
  - a. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
  - b. Click **Device Templates**.




---

**Note** In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled as **Device**

---

- c. Find the template to be attached to the Cisco SD-WAN Controller.
- d. Click ..., and click **Attach Devices**.
- e. In the **Attach Devices** dialog box, move the new controller to the **Selected Device** list and click **Attach**.
- f. Verify the **Config Preview** and click **Configure Devices**.

Cisco SD-WAN Manager pushes the configuration from the template to the new controller.

In the **Configuration > Devices** page, the **Mode** for the Cisco SD-WAN Controller shows **Manager**. The new Cisco SD-WAN Controller is ready to be used in your multitenant deployment.

## Manage Tenants

### Add a New Tenant

#### Prerequisites

- At least two Cisco SD-WAN Controllers must be operational and in the **Manager** mode before you can add new tenants.

A Cisco SD-WAN Controller enters the **Manager** mode when you push a template onto the controller from Cisco SD-WAN Manager. A Cisco SD-WAN Controller in the **CLI** mode cannot serve multiple tenants.

- Each pair of Cisco SD-WAN Controllers can serve a maximum of 24 tenants. Ensure that there at least two Cisco SD-WAN Controllers that can serve a new tenant. If no pair of Cisco SD-WAN Controllers in the deployment can serve a new tenant, add two Cisco SD-WAN Controllers and change their mode to **Manager**.
- If you add a second tenant immediately after adding a tenant, Cisco SD-WAN Manager adds them sequentially, and not in parallel.
- Each tenant must have a unique Virtual Account (VA) on **Plug and Play Connect** on [Cisco Software Central](#). The tenant VA should belong to the same Smart Account (SA) as the provider VA.

- For an on-premises deployment, create a Cisco SD-WAN Validator controller profile for the tenant on **Plug and Play Connect**. The fields in the following table are mandatory.

**Table 376: Controller Profile Fields**

| Field                       | Description/Value                                                                                                                                          |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Profile Name</b>         | Enter a name for the controller profile.                                                                                                                   |
| <b>Multi-Tenancy</b>        | From the drop-down list, select <b>Yes</b> .                                                                                                               |
| <b>SP Organization Name</b> | Enter the provider organization name.                                                                                                                      |
| <b>Organization Name</b>    | Enter the tenant organization name in the format <SP Org Name>-<Tenant Org Name>. <p><b>Note</b><br/>The organization name can be up to 50 characters.</p> |
| <b>Primary Controller</b>   | Enter the host details for the primary Cisco SD-WAN Validator.                                                                                             |

For a cloud deployment, the Cisco SD-WAN Validator controller profile is created automatically as part of the tenant creation process.

- Log in to Cisco SD-WAN Manager as the provider **admin** user.
- From the Cisco SD-WAN Manager menu, choose **Administration > Tenant Management**.
- Click **Add Tenant**. In the **Add Tenant** dialog box:
  - Enter a name for the tenant.  
For a cloud deployment, the tenant name should be same as the tenant VA name on **Plug and Play Connect**.
  - Enter a description of the tenant.  
The description can be up to 256 characters and can contain only alphanumeric characters.
  - Enter the name of the organization.  
The organization name is case-sensitive. Each tenant or customer must have a unique organization name.  
Enter the organization name in the following format:  
<SP Org Name>-<Tenant Org Name>  
For example, if the provider organization name is 'multitenancy' and the tenant organization name is 'Customer1', while adding the tenant, enter the organization name as **multitenancy-Customer1**.



**Note** The organization name can be up to 50 characters.

- In the **URL Subdomain Name** field, enter the fully qualified sub-domain name of the tenant.

- The sub-domain name must include the domain name of the service provider. For example, for the `multitenancy.com` service provider, a valid domain name can be `Customer1.multitenancy.com`.



**Note** The service provider name is shared amongst all tenants. Hence, ensure that the URL naming convention follows the same domain name convention that was provided while enabling multitenancy from the Cisco SD-WAN Manager **Administration > Settings > Tenancy Mode** GUI navigation path.

- For an on-premises deployment, add the fully qualified sub-domain name of the tenant to the DNS. Map the fully qualified sub-domain name to the IP addresses of the three Cisco SD-WAN Manager instances in the Cisco SD-WAN Manager cluster.

When creating fully qualified domain names (FQDN) the following DNS entries are required:

- **Provider Level:** Create DNS A record and map it to the IP addresses of the Cisco SD-WAN Manager instances running in the Cisco SD-WAN Manager cluster. The A record is derived from the domain and Cluster ID that was created in steps 5 and 6 in [Enable Multitenancy on Cisco SD-WAN Manager](#). For example, if domain is `sdwan.cisco.com` and Cluster ID is `vmanage123`, then A record will need to be configured as `vmanage123.sdwan.cisco.com`.



**Note** If you fail to update DNS entries, it will result in authentication errors when logging in to Cisco SD-WAN Manager. Validate DNS is configured correctly by executing `nslookup vmanage123.sdwan.cisco.com`.

- **Tenant Level:** Create DNS CNAME records for each tenant created and map them to the FQDN created at the Provider Level. For example, if domain is `sdwan.cisco.com` and tenant name is `customer1` the CNAME record will need to be configured as `customer1.sdwan.cisco.com`.



**Note** Cluster ID is not required for CNAME record. Validate DNS is configured correctly by executing `nslookup customer1.sdwan.cisco.com`.

For a cloud deployment, the fully qualified sub-domain name of the tenant is automatically added to the DNS as part of the tenant creation process. After you add a tenant, it could take up to an hour before the fully qualified sub-domain name of the tenant can be resolved by the DNS.

e. Click **Save**.

The **Create Tenant** screen appears, and the **Status** of the tenant creation reads **In progress**. To view status messages related to the creation of a tenant, click the > button to the left of the status.

Cisco SD-WAN Manager does the following:

- creates the tenant
- assigns two Cisco SD-WAN Controllers to serve the tenant and pushes a CLI template to these controllers to configure tenant information

- sends the tenant and Cisco SD-WAN Controller information to Cisco SD-WAN Validators.

**What to do next:**

After the **Status** column changes to **Success**, you can view the tenant information on the **Administration > Tenant Management** page.

## Modify Tenant Information

1. Log in to Cisco SD-WAN Manager as the provider **admin** user.
2. From the Cisco SD-WAN Manager menu, choose **Administration > Tenant Management**.
3. In the left pane, click the name of the tenant.  
The tenant information is displayed in a pane on the right.
4. To modify tenant data, do as follows:
  - a. In the right pane, click the pencil icon.
  - b. In the **Edit Tenant** dialog box, modify the tenant name, description, or domain name.
  - c. Click **Save**

## Delete a Tenant

Before you delete a tenant, delete all tenant WAN edge devices. See [Delete a WAN Edge Device from a Tenant Network, on page 983](#).

1. Log in to Cisco SD-WAN Manager as the provider **admin** user.
2. From the Cisco SD-WAN Manager menu, choose **Administration > Tenant Management**.
3. In the left pane, click the name of the tenant.  
The tenant information is displayed in a pane on the right.
4. To delete the tenant, do as follows:
  - a. In the right pane, click the trash icon.
  - b. In the **Delete Tenant** dialog box, enter the provider **admin** password and click **Save**.

## Cisco SD-WAN Manager Dashboard for Multitenancy

After enabling Cisco SD-WAN Manager for multitenancy, you can view the multitenant dashboard when you log in to Cisco SD-WAN Manager. Cisco SD-WAN Manager multitenant dashboard is a portal where the provider or tenant can view and provision the underlying system.

The bar at the top of every Cisco SD-WAN Manager multitenant screen includes icons that allow smooth navigation.

## View Tenant Activity, Device, and Network Information

When you log in to a multitenant Cisco SD-WAN Manager as an administrator, the provider dashboard displays the following components. To return to the provider dashboard from other Cisco SD-WAN Manager screens, click **Dashboard** at the left bar.

- **Device pane** — runs across the top of the multitenant dashboard screen. The device pane displays the number of active Cisco Catalyst SD-WAN Controllers, Cisco Catalyst SD-WAN Validators, and Cisco SD-WAN Manager instances, the connectivity status of devices, and information on certificates that have expired or about to expire.
- **Tenants pane** — displays the total number of tenants and a summary of the control status, site health, router health, and Cisco Catalyst SD-WAN Controller status of all tenants.
- **Table of tenants in the overlay network** — List of individual tenants, with separate information about the control status, site health, WAN edge device health, and Cisco Catalyst SD-WAN Controller status for each tenant.

To display tenant-specific status summary information,

1. Click a tenant name from the tenant list.

A dialog box opens on the right side of the screen that provides additional information about the status of the tenant.

2. To access the tenant dashboard for the selected tenant, click <Tenant name> **Dashboard**.

Cisco SD-WAN Manager presents the provider-as-tenant view and displays the tenant dashboard. To return to the provider view, click **Provider** at the top of page.

3. To close the dialog box, click the tenant name from the tenant list.

## View Detailed Information of a Tenant Setup

Cisco SD-WAN Manager displays the tenant dashboard, which provides information about a tenant deployment when

- a provider **admin** user selects a specific tenant from the **Select Tenant** drop-down list in the provider dashboard. This view is called the provider-as-tenant view.
- a **tenantadmin** user logs in to Cisco SD-WAN Manager. This view is called the tenant view.

### View All Network Connections in the Tenant Overlay Network

The **Device** pane runs across the top of the tenant dashboard and displays the number of control connections from Cisco SD-WAN Manager to the Cisco SD-WAN Controllers and routers in the overlay network of a tenant. For each WAN edge device, the Device pane shows

- Total number of control connections between Cisco SD-WAN Controllers and WAN edge devices
- Number of valid control connections between Cisco SD-WAN Controllers and WAN edge devices
- Number of invalid control connections between Cisco SD-WAN Controllers and WAN edge devices



Click a connection number, or the **Up** or **Down** arrow, to display a table with detailed information about each connection. Click the **More Actions** icon at the right of each table row to access the **Device Dashboard** or **Real Time** view from the **Monitor > Devices** screen, or access the **Tools > SSH Terminal** Screen.



**Note** In Cisco vManage Release 20.6.x and earlier releases, **Real Time** view is part of the **Monitor > Network** screen.

### View Information About Device Reboots

The **Reboot** pane displays the total number of reboots in the last 24 hours for all devices in the network. It includes soft and cold reboots and reboots that occurred as a result of power-cycling a device. For each reboot, the following information is listed:

- System IP and hostname of the device that rebooted.
- Time when the device was rebooted.
- Reason for the device reboot

If the same device reboots more than once, each reboot option is reported separately.

Click the **Reboot** pane to open the **Reboot** dialog box. In the **Reboot** dialog box, click **Crashes**. For all device crashes, the following information is listed:

- System IP and hostname of the device on which the crash occurred.
- Crash index of the device
- Core time when the device crashed.
- File name of the device crash log

### View Network Connections

The **Control Status** pane displays whether Cisco SD-WAN Controller and WAN edge devices are connected. Each Cisco SD-WAN Controller must connect to all other Cisco SD-WAN Controllers in the network. Each WAN edge device must connect to the maximum number of configured Cisco SD-WAN Controllers. The **Control Status** pane displays three network connection counts:

- Control Up — total number of devices with the required number of operational control plane connections to a Cisco SD-WAN Controller
- Partial — total number of devices with some, but not all, operational control plane connection to Cisco SD-WAN Controllers.
- Control Down — total number of devices with no control plane connection to a Cisco SD-WAN Controller

To display a table with device details, click a row from the **Control Status** dialog box. Click the **More Actions** icon at the right of each table row to access the **Device Dashboard** or **Real Time** view from the **Monitor > Devices** screen.



**Note** In Cisco vManage Release 20.6.x and earlier releases, **Real Time** view is part of the **Monitor > Network** screen.

### View State of Data Connections for a Site

The **Site Health** pane displays the state of data connections for a site. When a site has multiple WAN edge devices, this pane displays the state for the entire site and not for individual devices. The Site Health pane displays three connectivity states:

- Full WAN Connectivity — total number of sites where all BFD sessions on all routers are in the up state.
- Partial WAN Connectivity — total number of sites where tunnel and all BFD sessions on all routers are in the down state. These sites still have limited data plane connectivity.
- No WAN Connectivity — total number of sites where all BFD sessions on all routers are in the down state. These sites have no data plane connectivity.

To display a table with detailed information about each site, node, or tunnel, click a row from the **Site Health** dialog box. Click the **More Actions** icon at the right of each row in the table to access the **Device Dashboard** or **Real Time** view from the **Monitor > Devices** screen, or access the **Tools > SSH Terminal** screen.



**Note** In Cisco vManage Release 20.6.x and earlier releases, **Real Time** view is part of the **Monitor > Network** screen.

### View Interface Usage for WAN Edge Interfaces

The **Transport Interface Distribution** pane displays interface usage in the last 24 hours for all WAN edge interfaces in VPN 0. It includes all TLOC interfaces. Click the pane to view details of interface usage in the **Transport Interface Distribution** dialog box.

### View WAN Edge Device Counts

The **WAN Edge Inventory** pane provides four WAN edge device counts:

- Total — total number of authorized serial numbers for WAN edge devices that have been uploaded on Cisco SD-WAN Manager. The serial number is uploaded on the **Configuration > Devices** screen.
- Authorized — total number of authorized WAN edge devices in the overlay network. These WAN edge devices are marked as **Valid** in the **Configuration > Certificates > WAN Edge List** screen.
- Deployed — total number of deployed WAN edge devices. These are WAN edge devices that are marked as **Valid** and are now operational in the network.
- Staging — total number of WAN edge devices you configure at a staging site before they are made a part of the overlay network. These routers do not take part in any routing decisions and do not affect network monitoring through Cisco SD-WAN Manager.

Click the pane to view hostname, system IP, site ID, and other details of each router from the **WAN Edge Inventory** dialog box.

## View Aggregated State of WAN Edge Devices

The **WAN Edge Health** pane offers an aggregated view of the state of WAN edge devices by providing a count of the number of devices in each state, therefore describing the health of the hardware nodes. The three WAN edge device states are:

- **Normal** — number of WAN edge devices with memory, hardware, and CPU in normal state. Using less than 70% of total memory or total CPU is classified as, normal.
- **Warning** — number of WAN edge devices with memory, hardware, or CPU in warning state. Using between 70% and 90% of total memory or total CPU is classified as, warning
- **Error** — number of WAN edge devices with memory, hardware, or CPU in error state. Using more than 90% of total memory or total CPU is classified as, error.

Click a number or the WAN edge device state to display a table with the last 12 or 24 hours of memory usage, CPU utilization, and hardware-related alarms, including temperature, power supply, and PIM modules. Click the **More Actions** icon at the right of each row in the table to access the following:

- **Hardware Environment**
- **Real Time** view from the **Monitor > Devices** screen  
Cisco vManage Release 20.6.x and earlier: **Real Time** view from the **Monitor > Network** screen
- **Tools > SSH Terminal** screen.

## View WAN Edge Device Loss, Latency, Jitter

The **Transport Health** pane displays the aggregated average loss, latency, and jitters for all links and all combinations of colors (for example, all LTE-to-LTE links, all LTE-to-3G links).

From the **Type** drop-down arrow, choose loss, latency, or jitter.

Click the  icon to select a time period for which to display the transport health.

Click the  icon to open the **Transport Health** dialog box. This dialog box displays a more detailed view. To display information in a tabular format, click **Details**. You can choose to change the displayed health type and time period.

## View DPI Flow Information of WAN Edge Devices

The **Top Applications** pane displays DPI flow information for traffic transiting routers in the overlay network.



**Note** DPI flow information is shown only for the last 24 hours. To view DPI flow information for a time before the last 24 hours, you must check the information for the specific device.

Click the  icon to select a time period for which to display data. From the **VPN** drop-down list, select a VPN to display DPI information for all flows in that VPN.

Click the  icon to open the **Top Applications** dialog box. This dialog box displays a more detailed view of the same information. You can change the VPN and time period.

### View Tunnels Data

The **Application-Aware Routing** pane allows you to choose the following tunnel criteria from the **Type** drop-down arrow:

- Loss
- Latency
- Jitter

Based on the tunnel criteria, the pane displays the 10 worst tunnels. For example, if you choose loss, the pane shows 10 tunnels with the greatest average loss over the last 24 hours.

Click the  icon against a row to display a graphical representation of the data. Select a time period for which to display data or click **Custom** to display a drop-down arrow for specifying a custom time period.

Click the  icon to open the **Application-Aware Routing** dialog box. This dialog box displays the 25 worst tunnels based on criteria you choose from the **Type** drop-down arrow, the criteria being loss, latency, and jitter.

## Manage Tenant WAN Edge Devices

### Add a WAN Edge Device to a Tenant Network

1. Log in to Cisco SD-WAN Manager.

If you're a provider user, log in as the **admin**. In the provider dashboard, choose a tenant from the drop-down list to enter the provider-as-tenant view.

If you're a tenant user, log in as the **tenantadmin**.

2. Upload the device serial number file to Cisco SD-WAN Manager.
3. Validate the device and send details to controllers.
4. Create a configuration template for the device and attach the device to the template.

While configuring the device, configure the service provider organization name and the tenant organization name as in the following example:

```
sp-organization-name multitenancy
organization-name multitenancy-Customer1
```




---

**Note** Enter the `organization-name` in the format `<SP Org Name>-<Tenant Org Name>`.

---

5. Bootstrap the device using bootstrap configuration generated through Cisco SD-WAN Manager or manually create the initial configuration on the device.
6. If you are using Enterprise Certificates to authenticate the device, download the CSR from Cisco SD-WAN Manager and get the CSR signed by the Enterprise CA. Install the certificate on Cisco SD-WAN Manager.

## Delete a WAN Edge Device from a Tenant Network

1. Log in to Cisco SD-WAN Manager.  
If you're a provider user, log in as the **admin**. In the provider dashboard, choose a tenant from the drop-down list to enter the provider-as-tenant view.  
If you're a tenant user, log in as the **tenantadmin**.
2. Detach the device from any configuration templates.
3. [Delete a WAN Edge Router](#).

## Tenant-Specific Policies on Cisco Catalyst SD-WAN Controllers

A provider **admin** user (from the Cisco SD-WAN Manager provider-as-tenant view) or a **tenantadmin** user (from the Cisco SD-WAN Manager tenant view) can create and deploy tenant-specific policies on the Cisco Catalyst SD-WAN Controllers serving the tenant. The user can configure a CLI policy or create the policy using the UI policy configuration wizard.

When you activate or deactivate a policy,

1. Cisco SD-WAN Manager identifies the Cisco Catalyst SD-WAN Controllers serving the tenant.
2. Cisco SD-WAN Manager notifies the Cisco Catalyst SD-WAN Controllers to pull the policy configuration.
3. Cisco Catalyst SD-WAN Controllers pull and deploy the policy configuration.
4. Cisco SD-WAN Manager reports the status of the policy pull by the Cisco Catalyst SD-WAN Controllers.

## Manage Tenant Data

### Back Up Tenant Data

The tenant data backup solution of Cisco SD-WAN Manager multitenancy provides the following functionalities:

- [Create, list, and extract configuration backup files](#).
- Back up configuration database of a specific tenant with an option to restore it later. See [Back up and restore configuration database](#).
- Delete back up files of a tenant stored in Cisco SD-WAN Manager. For deleting tenant data backup files, see [Delete tenant data backup file](#).

The following factors are applicable when using data backup solution:

- The tenant data backup solution operations can be performed by a tenant administrator over tenant view and as a provider. To know how to access tenant dashboard through different views, see [User Roles in Multitenant Environment](#), on page 968.

- A tenant is allowed to perform the following backup operations at a particular time and must complete an operation before starting a new operation:
  - Back up a single configuration database
  - Download the backup file.
  - Restore or import backup files
  - Delete backup files.
  - List backup files

- A tenant backup file format is as follows:

```
Bkup_tenantId_MMDDYY-HHMMSS_taskIdWithoutDash.tar.gz
```

- The tenant data backup operation is a readonly operation on the configuration database. However, to ensure data consistency and prevent data loss, do not perform any major changes on the network.
- When a backup or restore operation for a specific tenant is in-progress, other tenants are allowed to perform the backup and restore operations smoothly.
- A tenant is not allowed to perform other backup operations when the restore operation of the tenant database is in-progress. So, a tenant can perform a single backup operation and when this operation is in-progress, all new backup operation requests are rejected.

The remaining tenants can continue with their backup operations.

- A tenant must use the same Cisco SD-WAN Manager version for backup generation and restore operation.
- A tenant can store a maximum of three backup files in Cisco SD-WAN Manager and can download to store them outside Cisco SD-WAN Manager repository. If the tenant already has three backup files, a subsequent backup operation results in the earliest backup file being deleted and a new backup file being generated.
- Ensure that the following parameter values match in both the backup file and the setup where tenant has requested for a restore operation:
  - Tenant Id
  - Organization Name
  - SP Organization Name
- The tenant data backup solution creates a task in the tenant view of Cisco SD-WAN Manager. Therefore, the tenant can monitor the progress of the operation from the task view of the tenant dashboard.
- A provider cannot back up provider data using this solution. Therefore, the provider can back up all tenants information at once by backing up all tenants configuration database using CLI.

## Create, Extract, and List Configuration Data Backup File

1. Log in to Cisco SD-WAN Manager.

If you're a provider user, log in as the **admin**. In the provider dashboard, choose a tenant from the drop-down list to enter the provider-as-tenant view.

If you're a tenant user, log in as the **tenantadmin**.

2. In the address bar, modify the URL path with `dataservice` for the REST API connection.

**Example:** `https://<tenant_URL>/dataservice`

3. Create a configuration backup file by using the following API:

`https://<tenant_URL>/dataservice/tenantbackup/export.`

4. If the configuration backup file has been created successfully, Cisco SD-WAN Manager task view indicates that the backup file has been generated. You can view the process identifier of the created process or task.

**Example:**

```
{
 "processId": "72d69805-b987-436f-9b7a-afef2f3f9061",
 "status": "in-progress"
}
```

5. Verify the task status using the obtained process identifier.

**Example:**

`https://<tenant_URL>/dataservice/device/action/status/72d69805-b987-436f-9b7a-afef2f3f9061`

The verification generates the details of the task in the JSON file format.

6. After the task is completed, extract or download the backed-up file available under the **data** section of the JSON task file.

**Example:** To extract or download the backup file, use the following API:

`https://<tenant_URL>/dataservice/tenantbackup/download/1570057020772/backup_1570057020772_100919-181838.tar.gz`

7. List backup files stored in Cisco SD-WAN Manager using the following API.

**Example:** `https://<tenant_URL>/dataservice/tenantbackup/list`

## Restore and Delete Tenant Data Backup File

### Before you begin:

To run the restore and delete tenant data backup files API, you can download and install Postman tool or any other alternative tool for testing http applications and services. In this document, the procedure to restore and delete tenant data backup files has been explained using the Postman tool. Postman is a software tool used as an API development environment. You can download the tool from the Postman website.

1. Open Google Chrome, or another browser, and enable developer mode on it.
2. Log in to Cisco SD-WAN Manager.

If you're a provider user, log in as the **admin**. In the provider dashboard, choose a tenant from the drop-down list to enter the provider-as-tenant view.

If you're a tenant user, log in as the **tenantadmin**.

3. To get header information of the restore API, do as follows:
  - a. On the right side of the screen, click the **Network** tab to get the network capture view.
  - b. In the network capture view, click the **Name** column to sort the listed items.
  - c. Search and click **index.html**.

- d. Click the **Headers** tab and expand **Request Headers**.
  - e. Choose all text under **Request Headers** and copy it to the clipboard.
4. Import backup files through the Postman UI:
  - a. Open the Postman UI.
  - b. To disable SSL certificate verification, click **Postman > Preferences > General > Request**. Turn off **SSL Certificate Verification**.
  - c. In the Postman UI, create a new tab.
  - d. Click **Headers** and then click **Bulk Edit**.
  - e. Paste the text that was copied in step 3 from the **Request Headers** block into an editable form.
  - f. From the **GET** method drop-down list, choose **POST**.
  - g. In the **Paste request URL** field, paste the dedicated URL of the tenant and include `dataservice/tenantbackup/import`.  
**Example:** `https://Customer1.multitenancy.com/dataservice/tenantbackup/import`
  - h. Click the **Body** tab and select **form-data**.
  - i. Under **KEY** column, enter `bakup.tar.gz`
  - j. Under **VALUE** column, click **Select Files** and select a backup file to be imported.
  - k. To run the API, click **Send**.  
 In the **Response** section of the Postman UI, you can view the JSON information that indicates the file that was restored.
5. Monitor the restoration of backup files in either of the following ways:
  - a. Use Cisco SD-WAN Manager task view that indicates if backup file has been imported successfully. You can view the process identifier of the created process or task.  
**Example:**

```
{
 "processId": "40adb6c0-eacc-4ad4-ba6c-2c2da2e96d1d",
 "status": "Import Successfully Submitted for tenant 1579026919487"
}
```
  - b. Use the following URL to get the status,  
`https://<tenant_URL>/dataservice/device/action/status/<processId>`  
**Example:**  
`https://Customer1.multitenancy.com/dataservice/device/action/status/40adb6c0-eacc-4ad4-ba6c-2c2da2e96d1d`
6. Delete tenant data backup file through Postman UI.
  - a. In the Postman UI, create a new tab.
  - b. Click **Headers** and then click **Bulk Edit**.
  - c. Paste the text that was copied in step 3 from the **Request Headers** block into an editable form.
  - d. From the **GET** method drop-down list, choose **DELETE**.



- e. In the **Paste request URL** field, paste the dedicated URL of the tenant and include `dataservice/tenantbackup/delete?fileName='filename'`. The filename can either be name of the backup file or all.

**Example:**

```
https://Customer1.multitenancy.com/dataservice/tenantbackup/delete?fileName=bkup_1579026919487_012820-180712_c09230904dfc40edb0d1e50b68b03002.tar.gz
```

**Example:**

```
https://Customer1.multitenancy.com/dataservice/tenantbackup/delete?fileName=all
```

- f. To run the API, click **Send**.

In the **Response** section of the Postman UI, you can view the JSON information that indicates the files that were deleted.

**Example:**

```
{
 "Deleted": [
 "bkup_1579026919487_012820-180712_c09230904dfc40edb0d1e50b68b03002.tar.gz"
]
}
```

## View OMP Statistics per Tenant on a Cisco SD-WAN Controller

1. Log in to Cisco SD-WAN Manager as the provider **admin** user.
2. From the Cisco SD-WAN Manager menu, choose **Monitor > Devices**.  
Cisco vManage Release 20.6.x and earlier: From the Cisco SD-WAN Manager menu, choose **Monitor > Network**.
3. In the table of devices, click on the hostname of a Cisco SD-WAN Controller.
4. In the left pane, click **Real Time**.
5. In the **Device Options** field, enter **OMP** and select the OMP statistics you wish to view.
6. In the **Select Filters** dialog box, click **Show Filters**.
7. Enter the **Tenant Name** and click **Search**.

Cisco SD-WAN Manager displays the selected OMP statistics for the particular tenant.

## View Tenants Associated with a Cisco SD-WAN Controller

1. Log in to Cisco SD-WAN Manager as the provider **admin** user.
2. Click a **Controller** connection number to display a table with detailed information about each connection.  
Cisco SD-WAN Manager displays a table that provides a summary of the Cisco SD-WAN Controllers and their connections.
3. For a Cisco SD-WAN Controller, click ... and click **Tenant List**.  
Cisco SD-WAN Manager displays a summary of tenants associated with the Cisco SD-WAN Controller.

# Migrate Single-Tenant Cisco Catalyst SD-WAN Overlay to Multitenant Cisco Catalyst SD-WAN Deployment

**Table 377: Feature History**

| Feature Name                                                                                        | Release Information                                                          | Description                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Migrate Single-Tenant Cisco Catalyst SD-WAN Overlay to Multitenant Cisco Catalyst SD-WAN Deployment | Cisco IOS XE Catalyst SD-WAN Release 17.5.1a<br>Cisco vManage Release 20.5.1 | This feature enables you to migrate a single-tenant Cisco Catalyst SD-WAN overlay to a multitenant deployment using a sequence of Cisco SD-WAN Manager API calls. |

## Before You Begin

- Before you begin the migration,
  - Ensure that the edge devices in the single-tenant deployment can reach the Cisco SD-WAN Validator in the multitenant deployment
  - Ensure that the template, routing, and policy configuration on the edge devices is synchronized with the current configuration on Cisco SD-WAN Manager
  - Ensure that the Certificate Authority (CA) on both single-tenant and multitenant Cisco SD-WAN Managers are same.
  - Configure a maintenance window for the single-tenant overlay before performing this procedure. See [Configure or Cancel Cisco SD-WAN Manager Server Maintenance Window](#).
- Minimum software requirements for the single-tenant overlay to be migrated:

| Device                              | Software Version                             |
|-------------------------------------|----------------------------------------------|
| Cisco SD-WAN Manager                | Cisco vManage Release 20.5.1                 |
| Cisco SD-WAN Validator              | Cisco SD-WAN Release 20.5.1                  |
| Cisco SD-WAN Controller             | Cisco SD-WAN Release 20.5.1                  |
| Cisco IOS XE Catalyst SD-WAN device | Cisco IOS XE Catalyst SD-WAN Release 17.4.1a |

- Minimum software requirements for the multitenant deployment to which the single-tenant overlay must be migrated:

| Device                              | Software Version                             |
|-------------------------------------|----------------------------------------------|
| Cisco SD-WAN Manager                | Cisco vManage Release 20.5.1                 |
| Cisco SD-WAN Validator              | Cisco SD-WAN Release 20.5.1                  |
| Cisco SD-WAN Controller             | Cisco SD-WAN Release 20.5.1                  |
| Cisco IOS XE Catalyst SD-WAN device | Cisco IOS XE Catalyst SD-WAN Release 17.5.1a |

- We recommend that you use a custom script or a third-party application like Postman to execute the API calls.

### Migration Procedure

1. Export the single-tenant deployment and configuration data from a Cisco SD-WAN Manager instance controlling the overlay.

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Method        | POST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| URL           | <code>https://ST-vManage-IP-address</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Endpoint      | <code>/dataservice/tenantmigration/export</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Authorization | Admin user credentials.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Body          | <p>Required</p> <p>Format: Raw JSON</p> <pre>{   "desc": &lt;tenant_description&gt;,   "name": &lt;tenant_name&gt;,   "subdomain": &lt;tenant_name&gt;.&lt;domain&gt;,   "orgName": &lt;tenant_orgname &gt; }</pre> <p>Field Description:</p> <ul style="list-style-type: none"> <li>• <b>desc:</b> A description of the tenant. The description can be up to 256 characters and can contain only alphanumeric characters.</li> <li>• <b>name:</b> Unique name for the tenant in the multitenant deployment.</li> <li>• <b>subdomain:</b> Fully qualified sub-domain name of the tenant. The sub-domain name must include the domain name of the service provider. For example, if <code>multitenancy.com</code> is the domain name of service provider, and the tenant name is <code>Customer1</code>, the tenant sub-domain name would be <code>Customer1.multitenancy.com</code>.</li> <li>• <b>orgName:</b> Name of the tenant organization. The organization name is case-sensitive.</li> </ul> |
| Response      | <p>Format: JSON</p> <pre>{   "processId": &lt;vManage_process_ID&gt;, }</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

While exporting the data, Cisco SD-WAN Manager attempts to detach any CLI templates from the edge devices in preparation for the migration to the multitenant deployment. If prompted by Cisco SD-WAN Manager, detach CLI templates from the edge devices and execute the export API call again.

2. Check the status of the data export task in Cisco SD-WAN Manager. When the task succeeds, download the data using the URL  
`https://ST-vManage-IP-address/dataservice/tenantmigration/download/default.tar.gz`
3. On a multitenant Cisco SD-WAN Manager instance, import the data exported from the single-tenant overlay.

|               |                                                                                                                     |
|---------------|---------------------------------------------------------------------------------------------------------------------|
| Method        | POST                                                                                                                |
| URL           | <code>https://MT-vManage-IP-address</code>                                                                          |
| Endpoint      | <code>/dataservice/tenantmigration/import</code>                                                                    |
| Authorization | Provider Admin user credentials.                                                                                    |
| Body          | Required<br>Format: form-data<br>Key Type: File<br>Value: <code>default.tar.gz</code>                               |
| Response      | Format: JSON<br><pre>{   "processId": &lt;vManage_process_ID&gt;,   "migrationTokenURL": &lt;token_URL&gt;, }</pre> |

When the task succeeds, on the multitenant Cisco SD-WAN Manager, you can view the devices, templates, and policies imported from the single-tenant overlay.

- Obtain the migration token using the token URL obtained in response to the API call in **Step 3**.

|               |                                                            |
|---------------|------------------------------------------------------------|
| Method        | GET                                                        |
| URL           | <code>https://MT-vManage-IP-address</code>                 |
| Endpoint      | <code>migrationTokenURL</code> obtained in <b>Step 3</b> . |
| Authorization | Provider Admin user credentials.                           |
| Response      | The migration token as a large blob of encoded text.       |

- On the single-tenant Cisco SD-WAN Manager instance, initiate the migration of the overlay to the multitenant deployment.

|               |                                                                                      |
|---------------|--------------------------------------------------------------------------------------|
| Method        | POST                                                                                 |
| URL           | <code>https://ST-vManage-IP-address</code>                                           |
| Endpoint      | <code>dataservice/tenantmigration/networkMigration</code>                            |
| Authorization | Admin user credentials.                                                              |
| Body          | Required<br>Format: Raw text<br>Content: Migration token obtained in <b>Step 4</b> . |
| Response      | Format: JSON<br><pre>{   "processId": &lt;vManage_process_ID&gt;, }</pre>            |

In Cisco SD-WAN Manager, check the status of the migration task. As part of the migration task, the address of the multitenant Cisco SD-WAN Validator, and the service provider and tenant organization

names are pushed to the WAN edge devices of the single-tenant overlay. If the task succeeds, WAN edge devices form control connections to controllers in the multitenant deployment; the WAN edge devices are no longer connected to the controllers of the single-tenant overlay.

Attach any CLI templates detached from the edge devices (in Step 1) after migration to the multitenant deployment. Before you attach the templates, update the Cisco SD-WAN Validator IP address and the Organization name to match the configuration of the multitenant deployment.



---

**Note** In the single-tenant deployment, if Cisco SD-WAN Manager-signed certificates are installed on cloud-based WAN edge devices, the certificates are cleared when the devices are migrated to the multitenant deployment. You must re-certify the devices on the multitenant Cisco SD-WAN Manager. If enterprise certificates are installed on the cloud-based WAN edge devices, the certificates are not affected by the migration. For more information, see [Enterprise Certificates](#).

---





## CHAPTER 38

# Cisco Catalyst SD-WAN Carrier Supporting Carrier

Table 378: Feature History

| Feature Name                                                              | Release Information                                                          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cisco Catalyst SD-WAN Support for Carrier Supporting Carrier Connectivity | Cisco IOS XE Catalyst SD-WAN Release 17.6.1a<br>Cisco vManage Release 20.6.1 | The feature adds support for carrier supporting carrier (CSC) connectivity on Cisco IOS XE Catalyst SD-WAN devices.<br><br>CSC enables you to interconnect IP or multiprotocol label switching (MPLS) networks operating at different sites over an MPLS backbone network. Using CSC requires an edge router that supports CSC functionality, called a carrier edge (CE) device, at each site. This feature enables a Cisco IOS XE Catalyst SD-WAN device to serve as a CE device, making it unnecessary to have a separate dedicated CE device at each site managed by Cisco Catalyst SD-WAN. |

- [Prerequisites for Cisco Catalyst SD-WAN Carrier Supporting Carrier, on page 993](#)
- [Restrictions for Cisco Catalyst SD-WAN Carrier Supporting Carrier, on page 994](#)
- [Information About Cisco Catalyst SD-WAN Carrier Supporting Carrier, on page 994](#)
- [Benefits of Cisco Catalyst SD-WAN Carrier Supporting Carrier, on page 996](#)
- [Use Cases for Cisco Catalyst SD-WAN Carrier Supporting Carrier, on page 996](#)
- [Configure Carrier Supporting Carrier, on page 996](#)
- [Verify That a Device is Configured for Carrier Supporting Carrier, on page 1000](#)

## Prerequisites for Cisco Catalyst SD-WAN Carrier Supporting Carrier

A Cisco IOS XE Catalyst SD-WAN device that functions as a CSC customer edge (CSC-CE) device must have an external border gateway protocol (eBGP) peer connection with the CSC provider edge (CSC-PE) router.

## Restrictions for Cisco Catalyst SD-WAN Carrier Supporting Carrier

- IPv6 addressing is not supported.
- Network address translation (NAT) for the MPLS link is not supported.
- Firewall services on the MPLS link are not supported.
- Cloud OnRamp for SaaS is not supported.
- VPN route leak is not supported.

## Information About Cisco Catalyst SD-WAN Carrier Supporting Carrier

### Carrier Supporting Carrier

Carrier supporting carrier (CSC) is a hierarchical VPN model that allows organizations to interconnect their IP or MPLS networks located at different sites over an MPLS backbone network. This eliminates the need for the organizations to build and maintain their own MPLS backbone.

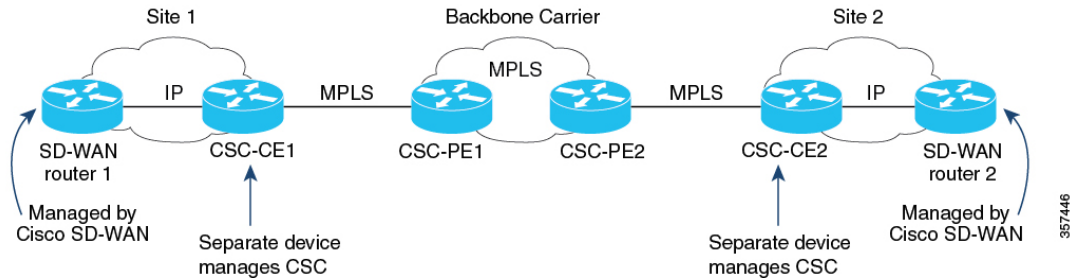
The following are components of CSC:

- **Backbone carrier:** The service provider that provides the backbone network. Typically, the backbone carrier network employs multiple segments to segregate the traffic of different customer carriers that share the backbone carrier network. The backbone carrier may be managed by the same organization or by a different organization as the customer carriers.
- **Customer carrier:** An organization that uses the backbone network to route traffic from one site to another. The customer carrier may be part of the organization that operates the backbone network, or may be independent.
- **CSC-CE:** Customer edge (CE) device. This device operates within a local site network and connects the site to the backbone carrier, using an MPLS connection. It utilizes the backbone carrier to connect to other sites.
- **CSC-PE:** Provider edge (PE) device. This device operates within the backbone carrier network and connects to CSC-CE devices at customer sites, using an MPLS connection.

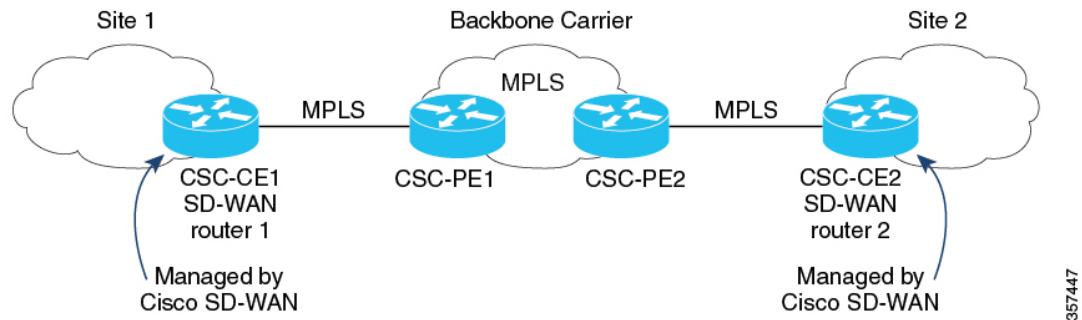
### Cisco Catalyst SD-WAN Carrier Supporting Carrier

The following illustration shows a CSC network topology with a Cisco IOS XE Catalyst SD-WAN device at each site, using a release earlier than Cisco IOS XE Catalyst SD-WAN Release 17.6.1a. Because the Cisco IOS XE Catalyst SD-WAN devices cannot function as a CSC-CE when using these releases, the topology requires two separate devices at each site: an edge device managed by Cisco Catalyst SD-WAN and a separate CSC-CE device.



**Figure 7: Carrier Supporting Carrier with Cisco Catalyst SD-WAN, Before Cisco IOS XE Catalyst SD-WAN Release 17.6.1a**

From Cisco IOS XE Catalyst SD-WAN Release 17.6.1a, a Cisco IOS XE Catalyst SD-WAN device can serve as a CSC-CE device, making it unnecessary to have a separate dedicated CSC-CE device. As compared with the previous illustration, the following illustration shows a simpler CSC network topology, with Cisco IOS XE Catalyst SD-WAN devices providing CSC-CE functionality.

**Figure 8: Carrier Supporting Carrier with Cisco Catalyst SD-WAN, Cisco IOS XE Catalyst SD-WAN Release 17.6.1a and Later**

### Traffic Flow

If a CSC-CE device only has an MPLS connection to the neighbor CSC-PE device, then all traffic from the CSC-CE device uses the MPLS connection, including the following traffic types:

- Service VPN traffic
- Control traffic
- Cisco Catalyst SD-WAN bidirectional forwarding detection (BFD) probe traffic

If a CSC-CE device has an MPLS connection to the neighbor CSC-PE device and also has a separate connection to the internet, then the traffic from the CSC-CE device may use different connections, as follows:

- Based on the configured traffic policy, control traffic and BFD probe traffic can use the internet and MPLS connections.
- Service VPN traffic uses only the MPLS connection.

### Label Switching

For traffic that uses an MPLS connection between a CSC device and the backbone carrier, the backbone carrier manages the traffic using label-switched paths, and has no information about the customer carrier routes.

# Benefits of Cisco Catalyst SD-WAN Carrier Supporting Carrier

Cisco Catalyst SD-WAN support for CSC enables a Cisco IOS XE Catalyst SD-WAN device to serve as an edge device at a site where CSC is required. With the Cisco IOS XE Catalyst SD-WAN device providing CSC-CE functionality, it is not necessary to have a separate router serving the CE role.

## Use Cases for Cisco Catalyst SD-WAN Carrier Supporting Carrier

Cisco Catalyst SD-WAN support for CSC is useful for global organizations that use CSC with a backbone carrier to support multiple, separate divisions of the organization. Each division's traffic is private but shares a common backbone carrier.

Service providers that use a CSC topology may benefit from Cisco Catalyst SD-WAN support for CSC. Carrier edge devices managed by Cisco Catalyst SD-WAN can support CSC, making it unnecessary to have a separate device to manage CSC functionality.

## Configure Carrier Supporting Carrier

You can configure the CE devices for CSC in the following ways:

- (Recommended) In Cisco SD-WAN Manager, use a BGP feature template.
- In Cisco SD-WAN Manager, use a CLI template to configure CSC by CLI.

## Configure Carrier Supporting Carrier

Perform the following steps to configure a CE device for CSC using a new feature template.

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and click **Create Template**. From the drop-down, choose **From Feature Template**.



---

**Note** In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled as **Device**.

---

3. In the **Device Model** field, choose the correct device model.
4. In the **Device Role** field, choose **SDWAN Edge**.
5. In the **Template Name** field, enter a name for the template.
6. In the **Transport & Management VPN** section, in the **Cisco VPN 0** field, choose a template to configure VPN 0 according to the network architecture.

For information about configuring VPN 0, see [Configure Interfaces in the WAN Transport VPN \(VPN 0\)](#) in the Cisco Catalyst SD-WAN Systems and Interfaces Configuration Guide, Cisco IOS XE Release 17.x.

7. In the **Cisco VPN Interface Ethernet** field, choose a template to configure the interface.  
For information about configuring this field, see [Configure VPN Ethernet Interface](#) in the Cisco Catalyst SD-WAN Systems and Interfaces Configuration Guide, Cisco IOS XE Release 17.x.
8. In the **Transport & Management VPN** section, click **Cisco BGP** to add the Cisco BGP field.  
For information about configuring a BGP template, see [Configure BGP Using SD-WAN Manager Templates](#) in the Cisco Catalyst SD-WAN Routing Configuration Guide, Cisco IOS XE Release 17.x.
9. In the **MPLS Interface** section, in the **Interface Name 1** field, enter the interface used to connect the device to the backbone carrier.
10. In the **Neighbor** section, click **Advanced Options** to display CSC options.
11. Configure the following fields, which are specific to CSC support:

| Field         | Description                                                                                                                                  |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Send Label    | Choose <b>On</b> to enable CSC support.                                                                                                      |
| Explicit Null | If the device uses a loopback WAN interface, choose <b>On</b> .                                                                              |
| As Override   | If the two CE devices (CE1 and CE2) that connect through the backbone carrier use the same autonomous system (AS) number, choose <b>On</b> . |
| Allowas In    | Similarly to <b>As Override</b> , if the two CE sites use the same AS number, choose <b>On</b> .                                             |

12. Click **Save** to save the BGP configuration.
13. Click **Create** to create the feature template.  
The **Configuration > Templates** page appears, showing available templates.
14. Attach the template to the device.
  - a. On the **Configuration > Templates** page.
  - b. Click **Device Templates**.



**Note** In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled as **Device**.

- c. For the new template, click ... and choose **Attach Devices**.
- d. Move a device to the **Selected Devices** column and click **Attach**.

## Configure Carrier Supporting Carrier Using the CLI

We recommend that you use the BGP feature template in Cisco SD-WAN Manager to configure Cisco IOS XE Catalyst SD-WAN devices for use with CSC. If it is necessary to configure a device by CLI, use a CLI template in Cisco SD-WAN Manager.

## Before You Begin

Before you configure a Cisco IOS XE Catalyst SD-WAN device to provide CSC-CE functionality, apply a BGP configuration to the device. The following steps add CSC functionality.

## Configure Carrier Supporting Carrier the CLI

### 1. Configure the following on CSC-CE1:

- a. Configure the device to map MPLS labels to VRFs. For incoming traffic, the router checks the MPLS label of the traffic and uses the IP lookup table of the VRF mapped to that label. For example, if MPLS label 10 is mapped to VRF 1, then for incoming traffic with the MPLS label 10, the router uses the IP lookup table of VRF 1. For information about mapping MPLS labels to VRFs, see the Cisco documentation for MPLS forwarding commands.

```
Device# config-transaction
Device(config)# mpls label mode all-vrfs protocol bgp-vpnv4 per-vrf
Device(config)# mpls label mode all-vrfs protocol bgp-vpnv6 per-vrf
Device(config)# mpls label range min-label max-label static min-static-label
max-static-label
```

- b. Enable multiprotocol label switching (MPLS) on the interface.

```
Device(config)# interface interface
Device(config-if)# mpls bgp forwarding
```

- c. Enter router configuration mode and configure the router to run a BGP process.

```
Device(config-if)# router bgp bgp-number
```

- d. Configure a CSC-PE device as the neighbor, where *neighbor-ip* is the address of the neighbor CSC-PE device.

```
Device(config-router)# neighbor neighbor-ip allowas-in
```

- e. If the device uses a loopback WAN interface, advertise the ability of the router to send MPLS labels with BGP routes. The **explicit-null** keyword enables a CSC-CE router to send labels with a value of 0 to its neighbor.




---

**Note** If you include the **neighbor neighbor-ip send-label explicit-null** command on a device that does not use a loopback WAN interface, it does not adversely impact performance.

---

```
Device(config-router)# neighbor neighbor-ip send-label explicit-null
```

---

### 2. Configure the following on CSC-CE2:

- a. Configure the device to map MPLS labels to VRFs. For incoming traffic, the router checks the MPLS label of the traffic and uses the IP lookup table of the VRF mapped to that label. For example, if MPLS label 10 is mapped to VRF 1, then for incoming traffic with the MPLS label 10, the router uses the IP lookup table of VRF 1. For information about mapping MPLS labels to VRFs, see the Cisco documentation for MPLS forwarding commands.

```
Device# config-transaction
Device(config)# mpls label mode all-vrfs protocol bgp-vpnv4 per-vrf
Device(config)# mpls label mode all-vrfs protocol bgp-vpnv6 per-vrf
Device(config)# mpls label range min-label max-label static min-static-label
max-static-label
```

- b. Enable multiprotocol label switching (MPLS) on the interface.

```
Device(config)# interface interface
Device(config-if)# mpls bgp forwarding
```

- c. Enter router configuration mode and configure the router to run a BGP process.

```
Device(config-if)# router bgp bgp-number
```

- d. Configure a CSC-PE device as the neighbor, where *neighbor-ip* is the address of the neighbor CSC-PE device.

```
Device(config-router)# neighbor neighbor-ip as-override
```

- e. If the device uses a loopback WAN interface, advertise the ability of the router to send MPLS labels with BGP routes.

```
Device(config-router)# neighbor neighbor-ip send-label explicit-null
```

### Example

The following examples show a complete BGP configuration, including CSC functionality, for two devices: CSC-CE1 and CSC-CE2.

- CSC-CE1 has the address 10.1.1.10.
- CSC-CE2 has the address 10.1.1.20.
- CSC-PE1 (the neighbor of CSC-CE1) has the address 10.2.2.10.
- CSC-PE2 (the neighbor of CSC-CE2) has the address 10.2.2.20.

The following is the configuration for CSC-CE1:

```
mpls label mode all-vrfs protocol bgp-vpnv4 per-vrf
mpls label mode all-vrfs protocol bgp-vpnv6 per-vrf
mpls label range 100000 1048575 static 16 99
interface GigabitEthernet2
 no shutdown
 mpls bgp forwarding
 ip address 10.1.1.15 255.255.255.0

router bgp 10
 bgp log-neighbor-changes
 bgp router-id 172.16.255.15
 neighbor 10.1.1.20 remote-as 100
 neighbor 10.1.1.20 fall-over bfd
 address-family ipv4 unicast
 maximum-paths 4
 neighbor 10.1.1.20 activate
 neighbor 10.1.1.20 advertisement-interval 30
 neighbor 10.2.2.10 allowas-in
 neighbor 10.2.2.10 send-label explicit-null
 neighbor 10.1.1.20 send-community both
 exit-address-family
!
timers bgp 60 180
```

The following is the configuration for CSC-CE2:

```
mpls label mode all-vrfs protocol bgp-vpnv4 per-vrf
mpls label mode all-vrfs protocol bgp-vpnv6 per-vrf
mpls label range 100000 1048575 static 16 99
```

```

interface GigabitEthernet5
 ip address 10.0.6.11 255.255.255.0
 negotiation auto
 mpls bgp forwarding

router bgp 10
 bgp log-neighbor-changes
 bgp router-id 172.16.255.11
 neighbor 10.1.1.10 remote-as 200
 address-family ipv4 unicast
 neighbor 10.1.1.10 activate
 neighbor 10.1.1.10 advertisement-interval 30
 neighbor 10.2.2.20 as-override
 neighbor 10.2.2.20 send-label explicit-null
 network 10.0.7.0 mask 255.255.255.0
 redistribute connected
 redistribute static
 exit-address-family

```

## Verify That a Device is Configured for Carrier Supporting Carrier

To verify that a device is configured correctly to reach a remote CSC-CE device, execute the **show ip route remote-csc-ce-device-address** command on the device. Verify that the command output shows the following:

- A routing entry for the remote site IP address.
- One or more routing descriptor blocks describing the next-hop addresses for the path to the remote CSC-CE device. Verify that each descriptor block includes an MPLS label.

### Example

```

Device# show ip route 10.0.1.100
Routing entry for 10.0.1.0/24
...
Routing Descriptor Blocks:
* 10.1.1.100, from 10.1.1.100, 00:00:50 ago
...
MPLS label: 26
...

```

If the device is not configured correctly, the output displays the following:

```
% Subnet not in table
```



## CHAPTER 39

# Wireless Management on Cisco 1000 Series Integrated Services Routers

**Table 379: Feature History**

| Feature Name                                                                         | Release Information                                                          | Description                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Wireless Management on Cisco ISR 1000 Series Routers (supporting WiFi 5 WLAN module) | Cisco IOS XE Catalyst SD-WAN Release 17.6.1a<br>Cisco vManage Release 20.6.1 | This feature enables you to configure wireless LAN settings on WiFi 5-capable Cisco 1000 Series Integrated Services Routers using Cisco SD-WAN Manager.<br><br>With Cisco SD-WAN Manager, you can automate the wireless LAN controller configuration and provide wireless connectivity without the need for another external controller to configure and manage the wireless settings on the routers.   |
| Wireless Management on Cisco ISR 1000 Series Routers (supporting WiFi 6 WLAN module) | Cisco IOS XE Catalyst SD-WAN Release 17.9.1a<br>Cisco vManage Release 20.9.1 | This feature enables you to configure wireless LAN settings on WiFi 6-capable Cisco 1000 Series Integrated Services Routers using Cisco SD-WAN Manager.<br><br>The Embedded Wireless Controller on Cisco 1000 Series Integrated Services Routers helps you provide wireless connectivity without the need for another external controller to configure and manage the wireless settings on the routers. |

- [Supported Devices for Wireless Management on Cisco ISR 1000 Series Routers, on page 1002](#)
- [Prerequisites for Wireless Management on Cisco ISR 1000 Series Routers, on page 1003](#)
- [Restrictions for Wireless Management on Cisco ISR 1000 Series Routers, on page 1004](#)

- [Information About Wireless Management on Cisco ISR 1000 Series Routers](#), on page 1004
- [Configure wireless management on Cisco ISR 1000 series routers using a configuration group](#), on page 1004
- [Configure Wireless Management on Cisco ISR 1000 Series Routers](#), on page 1006
- [Configure Wireless Management on Cisco ISR 1000 Series Routers Using a CLI Template](#), on page 1009
- [Monitor Wireless Configuration on Cisco ISR 1000 Series Routers](#), on page 1010
- [Configuration Example for Wireless Configuration on Cisco ISR 1000 Series Routers](#), on page 1011
- [Troubleshooting Wireless Configuration on Cisco ISR 1000 Series Routers](#), on page 1011

## Supported Devices for Wireless Management on Cisco ISR 1000 Series Routers

The following table displays a list of Cisco ISR 1000 Series routers that include the WLAN module and supporting WiFi 5.



Table 380: Cisco ISR 1000 Series Routers

| Device Family                                                    | Device Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Release Version                                                                  |
|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Cisco ISR 1000 Series Routers with WLAN module supporting WiFi 5 | <ul style="list-style-type: none"> <li>• C1101-4PLTEPW</li> <li>• C1109-4PLTE2PW</li> <li>• C1111-4PW</li> <li>• C1111-8PLTEEA</li> <li>• C1111-8PW</li> <li>• C1112-8PLTEEA</li> <li>• C1112-8PW</li> <li>• C1113-8PLTEEA</li> <li>• C1113-8PMW</li> <li>• C1113-8PW</li> <li>• C1116-4PLTEEA</li> <li>• C1116-4PW</li> <li>• C1117-4PLTEEA</li> <li>• C1117-4PLTELA</li> <li>• C1117-4PMLTEEA</li> <li>• C1117-4PMW</li> <li>• C1117-4PW</li> <li>• C1121-8PLTEPW</li> <li>• C1121X-8PLTEPW</li> </ul> | Cisco IOS XE Catalyst SD-WAN Release 17.6.1a<br><br>Cisco vManage Release 20.6.1 |
| Cisco ISR 1000 Series Routers with WLAN module supporting WiFi 6 | <ul style="list-style-type: none"> <li>• C1131X-8PLTEPW</li> <li>• C1131-8PLTEPW</li> <li>• C1131X-8PW</li> <li>• C1131-8PW</li> </ul>                                                                                                                                                                                                                                                                                                                                                                   | Cisco IOS XE Catalyst SD-WAN Release 17.9.1a<br><br>Cisco vManage Release 20.9.1 |

## Prerequisites for Wireless Management on Cisco ISR 1000 Series Routers

- Add the management interface of the Wireless LAN (WLAN) module to a specific VLAN in order to access servers such as DHCP and RADIUS.

- Configure a DHCP server to assign the IP address for the access point.
- Configure a switch virtual interface (SVI) on the Cisco ISR 1000 Services Router for virtual WLAN controller management.

## Restrictions for Wireless Management on Cisco ISR 1000 Series Routers

- You can configure only one access point on the LAN side of the router that is configured with Cisco Mobility Express. However, you can connect other external access points to the router that are not configured with Cisco Mobility Express.
- Ensure that there are no other accessible wireless controllers on the LAN side.

## Information About Wireless Management on Cisco ISR 1000 Series Routers

A WLAN module supporting WiFi 5 is provisioned on a Cisco ISR 1000 Series Routers for wireless connectivity. Cisco Mobility Express, a virtual wireless LAN controller, is installed in the WLAN module to provide wireless LAN access. The wireless settings for wireless LAN access are available on Cisco Mobility Express, and these settings can be configured and managed using Cisco SD-WAN Manager

C1131 Cisco IOS XE Catalyst SD-WAN devices includes an Embedded Wireless Controller (EWC) that supports WiFi 6. The EWC also serves as a virtual wireless controller that is installed on the WLAN module. The wireless settings for wireless LAN access are available on the EWC; these settings can be configured and managed using Cisco SD-WAN Manager.

## Configure wireless management on Cisco ISR 1000 series routers using a configuration group

### Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

### Procedure

- 
- Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.
- Step 2** Create and configure a Wireless LAN feature in Service profile.
- a) Configure the basic details for the wireless LAN.

| Field               | Description                                                                     |
|---------------------|---------------------------------------------------------------------------------|
| <b>Enable 2.4G*</b> | Disable this option to shut down the radio type of 2.4 GHz.<br>Default: Enabled |
| <b>Enable 5G*</b>   | Disable this option to shut down the radio type of 5 GHz.<br>Default: Enabled   |
| <b>Country*</b>     | Choose the country where the router is installed.                               |
| <b>Username*</b>    | Specify the username of Cisco Mobility Express.                                 |
| <b>Password*</b>    | Specify the password of Cisco Mobility Express.                                 |

- b) Configure ME IP address.

| Field                  | Description                                                                                      |
|------------------------|--------------------------------------------------------------------------------------------------|
| <b>ME Dynamic IP*</b>  | Enable this option so that the interface receives its IP address dynamically from a DHCP server. |
| <b>ME IP Address</b>   | Specify the IP address of Cisco Mobility Express.                                                |
| <b>Subnet Mask</b>     | Specify the subnet mask of Cisco Mobility Express.                                               |
| <b>Default Gateway</b> | Specify the default gateway address of Cisco Mobility Express.                                   |

- c) Configure the Wi-Fi SSID details for setting up a wireless LAN.

| Field                       | Description                                                                                                                                      |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Add SSID</b>             |                                                                                                                                                  |
| <b>SSID Name*</b>           | Enter a name for the wireless SSID.<br>It can be a string from 4 to 32 characters. The SSID must be unique.                                      |
| <b>Admin State*</b>         | Enable this option to indicate that the interface has been configured.                                                                           |
| <b>Broadcast SSID*</b>      | Enable this option if you want to broadcast the SSID. Disable this option if you do not want the SSID to be visible to all the wireless clients. |
| <b>VLAN (Range 1-4094)*</b> | Enter a VLAN ID for the wireless LAN traffic.                                                                                                    |
| <b>Radio Type</b>           | Choose one of the following radio types: <ul style="list-style-type: none"> <li>• 2.4GHz</li> <li>• 5GHz</li> <li>• All</li> </ul>               |

| Field          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Security Type* | Choose a security type: <ul style="list-style-type: none"> <li>• <b>WPA2 Enterprise</b>: Choose this option for an enterprise where you authenticate and authorize network users with a remote RADIUS server.</li> <li>• <b>WPA2 Personal</b>: Choose this option to authenticate users who want to access the wireless network using a passphrase.</li> <li>• <b>Open</b>: Choose this option to allow access to the wireless network without authentication.</li> </ul> |
| Passphrase*    | This field is available if you choose <b>WPA2 Personal</b> as the security type. Set a pass phrase. This pass phrase provides users access to the wireless network.                                                                                                                                                                                                                                                                                                       |
| QoS Profile    | Choose a QoS profile.                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

**What to do next**

Also see [Deploy a Configuration Group](#).

## Configure Wireless Management on Cisco ISR 1000 Series Routers

To configure and manage wireless settings on Cisco ISR 1000 Series Routers:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**.
3. Click **Add Template** to select an appropriate device model.



**Note** In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled as **Feature**.

4. In the left pane, from **Select Devices**, choose a Cisco ISR 1000 Series Router for which you are creating a template.
5. Under **OTHER TEMPLATES**, click **ISR1K Wireless** to select it as the feature template.
6. In the **Template Name** field, enter a name for the feature template.  
This field is mandatory and can contain only uppercase and lowercase letters, the digits 0 to 9, hyphens (-), and underscores (\_). It cannot contain spaces or any other characters.
7. In the **Description** field, enter a description for the feature template.  
This field is mandatory, and it can contain all characters and spaces.

## 8. Enter the Wi-Fi SSID details for setting up a wireless LAN:

| Parameter Name                      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Wireless Network Name (SSID)</b> | Enter a name for the wireless SSID.<br><br>It can be a string from 4 to 32 characters. The SSID must be unique.                                                                                                                                                                                                                                                                                                                                                           |
| <b>VLAN (Range 1-4094)</b>          | Enter a VLAN ID for the wireless LAN traffic.                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Security Type</b>                | Choose a security type: <ul style="list-style-type: none"> <li>• <b>WPA2 Enterprise</b>: Choose this option for an enterprise where you authenticate and authorize network users with a remote RADIUS server.</li> <li>• <b>WPA2 Personal</b>: Choose this option to authenticate users who want to access the wireless network using a passphrase.</li> <li>• <b>Open</b>: Choose this option to allow access to the wireless network without authentication.</li> </ul> |
| <b>RADIUS Server IP</b>             | (Optional) This field is available if you choose the <b>WPA2 Enterprise</b> option as the security type. Enter the IP address of the RADIUS server.                                                                                                                                                                                                                                                                                                                       |
| <b>Authentication Port</b>          | (Optional) This field is available if you choose the <b>WPA2 Enterprise</b> option as the security type. Enter the authentication port number of the RADIUS server.                                                                                                                                                                                                                                                                                                       |
| <b>Shared Secret</b>                | (Optional) This field is available if you choose the <b>WPA2 Enterprise</b> option as the security type. Enter the shared secret key of the RADIUS server.                                                                                                                                                                                                                                                                                                                |
| <b>Passphrase</b>                   | (Optional) This field is available if you choose the <b>WPA2 Personal</b> option as the security type. Set a pass phrase. This pass phrase provides users with access to the wireless network.                                                                                                                                                                                                                                                                            |
| <b>Admin State</b>                  | Choose an admin state.                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Radio Type</b>                   | Choose one of the following radio types: <ul style="list-style-type: none"> <li>• <b>2.4GHz</b></li> <li>• <b>5GHz</b></li> <li>• <b>Both</b></li> </ul>                                                                                                                                                                                                                                                                                                                  |
| <b>Broadcast SSID</b>               | Choose <b>On</b> to broadcast the SSID. Choose <b>Off</b> if you do not want the SSID to be visible to all the wireless clients.                                                                                                                                                                                                                                                                                                                                          |

| Parameter Name | Description           |
|----------------|-----------------------|
| QoS Profile    | Choose a QoS profile. |

9. Enter the **General** details for the wireless LAN:

| Parameter Name | Description                                                                                                                                       |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Country        | Choose the country where the ISR is installed.                                                                                                    |
| Username       | Specify the username of Cisco Mobility Express.<br>If you are using a C1131 Cisco IOS XE Catalyst SD-WAN device specify the username for the EWC. |
| Password       | Specify the password for Cisco Mobility Express or the EWC.                                                                                       |

10. Enter the **Advanced** details for the wireless LAN:

| Parameter Name        | Description                                                                                                                                                                                                                                          |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Controller IP Address | <b>Note</b><br>For Cisco IOS XE Catalyst SD-WAN Release 17.6.1a, and Cisco vManage Release 20.6.1 and earlier releases, this field is displayed as <b>ME IP Address</b> .<br><br>Specify the Management IP address of Cisco Mobility Express or EWC. |
| Subnet Mask           | Specify the subnet mask for the Management IP address.                                                                                                                                                                                               |
| Default Gateway       | Specify the default gateway address of Cisco Mobility Express or EWC.                                                                                                                                                                                |
| 2.4GHz Shutdown       | Click <b>Yes</b> to shut down the 2.4 GHz radio type.<br>Click <b>No</b> to not shut down this radio type.                                                                                                                                           |
| 5GHz Shutdown         | Click <b>Yes</b> to shut down the 5 GHz radio type. Click <b>No</b> to not shut down this radio type.                                                                                                                                                |

11. Click **Save** to save your wireless configuration.

# Configure Wireless Management on Cisco ISR 1000 Series Routers Using a CLI Template

This section provides sample CLI configurations to configure and manage wireless settings on Cisco ISR 1000 Series Routers using the CLI templates.

## Configure Radio Profile Using a CLI Template

For more information about using CLI templates, see [CLI Add-On Feature Templates](#) and [CLI Templates](#).



**Note** By default, CLI templates execute commands in global config mode.

```
radio-profile 24ghz
shutdown
exit
radio-profile 5ghz
no shutdown
```

## Configure WLAN Profile Using a CLI Template

For more information about using CLI templates, see [CLI Add-On Feature Templates](#) and [CLI Templates](#).



**Note** By default, CLI templates execute commands in global config mode.

```
wlan-profile wlan-profile-sample-1
vlan-id 100
ssid sample-ssid-1
data-security personal
passphrase 0 Pass-Phrase-Sample123#
qos-type silver
wlan-profile wlan-profile-sample-2
vlan-id 200
ssid sample-ssid-2
data-security enterprise
aaa radius-server 10.2.3.4 auth-port 1812 shared-secret 0 EsrdT_23sss

qos-type gold
nobroadcast-ssid
```

## Configure General WLAN Settings Using a CLI Template

For more information about using CLI templates, see [CLI Add-On Feature Templates](#) and [CLI Templates](#).



**Note** By default, CLI templates execute commands in global config mode.

```
wireless-lan country US
wireless-lan mgmt ip address 10.16.1.100 255.255.255.0 default-gateway 192.168.1.1
wireless-lan mgmt credential username admin password 0 sRe32dfst#asd
```

Here is the complete configuration example that shows that show how to configure and manage wireless settings on Cisco ISR 1000 Series Routers.

```
wlan-profile TEST-Enterprise
radio-band all
vlan-id 300
ssid TEST-Enterprise
data-security enterprise
aaa radius-server 192.168.100.20 auth-port 1812 shared-secret 6 EsrdT_23sss
qos-type silver
```

```
wlan-profile TEST-Personal
radio-band all
ssid TEST-Personal
data-security personal
passphrase 0 IdSvs23452#
qos-type silver
```

```
radio-profile 24ghz
channel auto
channel-bandwidth auto
```

```
radio-profile 5ghz
channel auto
channel-bandwidth auto
```

```
wireless-lan mgmt ip address 192.168.1.11 255.255.255.0 default-gateway 192.168.1.1
wireless-lan mgmt credential username admin password 6 sRe32dfst#asd
wireless-lan country US
```

## Monitor Wireless Configuration on Cisco ISR 1000 Series Routers

To monitor the wireless settings that are configured on Cisco ISR 1000 Series Routers using Cisco SD-WAN Manager, perform this procedure :

1. From the Cisco SD-WAN Manager menu, navigate to **Monitor > Network**.
2. Choose a router from the list of the routers.
3. Click **Real Time** in the left pane.
4. From the **Device Options** drop-down list, choose one of the following options:

| Device Option  | Description                                        |
|----------------|----------------------------------------------------|
| Wireless Radio | Displays the radio parameters of the wireless LAN. |
| Wireless SSID  | Displays information about the wireless SSID.      |



| Device Option    | Description                                                          |
|------------------|----------------------------------------------------------------------|
| Wireless Clients | Displays information about the wireless clients in the wireless LAN. |

## Configuration Example for Wireless Configuration on Cisco ISR 1000 Series Routers

The following is an example of wireless configuration of a Cisco ISR 1000 Series Routers:

```
wlan-profile TEST-Enterprise
radio-band all
vlan-id 300
ssid TEST-Enterprise
data-security enterprise
aaa radius-server 192.168.100.20 auth-port 1812 shared-secret 6 EsrdT_23sss
qos-type silver
```

```
wlan-profile TEST-Personal
radio-band all
ssid TEST-Personal
data-security personal
passphrase 0 IdSvs23452#
qos-type silver
```

```
radio-profile 24ghz
channel auto
channel-bandwidth auto
```

```
radio-profile 5ghz
channel auto
channel-bandwidth auto
```

```
wireless-lan mgmt ip address 192.168.1.11 255.255.255.0 default-gateway 192.168.1.1
wireless-lan mgmt credential username admin password 6 sRe32dfst#asd
wireless-lan country US
```

## Troubleshooting Wireless Configuration on Cisco ISR 1000 Series Routers

### Access Point Cannot Connect to Cisco Mobility Express or EWC

#### Problem

An access point is not able to connect to the Cisco Mobility Express or EWC.

**Possible Causes**

This problem is most likely to occur when there is no DHCP server in the management VLAN (the native VLAN of interface Wlan-GigabitEthernet).

**Solution**

Add the management interface of the WLAN module to a specific VLAN in order to access servers like DHCP and RADIUS. See [Prerequisites for Wireless Management on Cisco ISR 1000 Series Routers, on page 1003](#)

A DHCP server is required in the native VLAN of the WiFi module to assign IP address for the access point. Without IP address, the access point will not be able to connect to Cisco Mobility Express or EWC.



## CHAPTER 40

# Simplified Workflow for Onboarding Cisco ThousandEyes Agent and Configuring Tests

- [Simplified Workflow for Onboarding Cisco ThousandEyes Agent and Configuring Tests](#), on page 1013
- [Information About Simplified Workflow for Onboarding Cisco ThousandEyes Agent and Configuring Tests](#), on page 1014
- [Prerequisites for Simplified Workflow to Onboard Cisco ThousandEyes Agent and Configure Tests using Cisco Catalyst SD-WAN Manager](#), on page 1014
- [Create an IP Pool for the Cisco ThousandEyes Agents](#), on page 1015
- [Restrictions for Simplified Workflow to Onboard Cisco ThousandEyes Agent and Configure Tests Using Cisco Catalyst SD-WAN Manager](#), on page 1016
- [Configure Cisco ThousandEyes Agent and Tests Using Workflows](#), on page 1016
- [Monitor Onboarded Cisco ThousandEyes Agents and Configured Tests](#), on page 1017
- [Troubleshoot Cisco ThousandEyes Enterprise Agent on Cisco IOS XE Catalyst SD-WAN Device](#), on page 1017

## Simplified Workflow for Onboarding Cisco ThousandEyes Agent and Configuring Tests

*Table 381: Feature History*

| Feature Name                                                                      | Release Information                                                                            | Description                                                                                                      |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Simplified Workflow for Onboarding Cisco ThousandEyes Agent and Configuring Tests | Cisco IOS XE Catalyst SD-WAN Release 17.16.1a<br>Cisco Catalyst SD-WAN Manager Release 20.16.1 | A simplified workflow to easily onboard Cisco ThousandEyes agent and configure tests using Cisco SD-WAN Manager. |

# Information About Simplified Workflow for Onboarding Cisco ThousandEyes Agent and Configuring Tests

Minimum supported release: Cisco IOS XE Catalyst SD-WAN Release 17.16.1a and Cisco Catalyst SD-WAN Manager Release 20.16.1

The workflow simplifies the process of onboarding Cisco ThousandEyes agents and configuration of tests in Cisco SD-WAN Manager. Using the simplified workflow, you can easily onboard Cisco ThousandEyes agents and configure application monitoring tests directly from Cisco SD-WAN Manager in a few steps. The streamlined workflow involves the following steps:

- Establish a connection to the Cisco ThousandEyes platform through Cisco SD-WAN Manager.
- Configure tests in Cisco ThousandEyes directly from Cisco SD-WAN Manager.
- Choose applications, sites, and devices for testing and configure the necessary parameters in Cisco SD-WAN Manager.
- Monitor the agents and tests from the **ThousandEyes Monitoring** dashboard in Cisco SD-WAN Manager.

## Prerequisites for Simplified Workflow to Onboard Cisco ThousandEyes Agent and Configure Tests using Cisco Catalyst SD-WAN Manager

Minimum supported release: Cisco IOS XE Catalyst SD-WAN Release 17.16.1a and Cisco Catalyst SD-WAN Manager Release 20.16.1

- Ensure the following before initiating the simplified workflow:
  - You have a Cisco ThousandEyes account before using the simplified workflow to configure a Cisco ThousandEyes agent in Cisco SD-WAN Manager.
    - You must be a user with the Organization Admin role to access the Cisco ThousandEyes account.
  - An appropriate NAT configuration exists to enable the Cisco ThousandEyes Enterprise agent to discover and connect to the Cisco ThousandEyes application. For more information, see [NAT Direct Internet Access](#)
  - Create an IP pool in the global node of Network Hierarchy Manager.
- Complete the following tasks before initiating the simplified workflow.

**Table 382: Tasks Before Running the Simplified Workflow**

| Configuration Task                                                         | Information                                                                      |
|----------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Deploy Cisco IOS XE Catalyst SD-WAN devices using the configuration group. | See <a href="#">Deploy Devices Using the Deploy Configuration Group Workflow</a> |

| Configuration Task                                                                                                                                                                                  | Information                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Upload Cisco ThousandEyes Enterprise agent software.                                                                                                                                                | <ul style="list-style-type: none"> <li>To upload software on the Cisco SD-WAN Manager software repository, see <a href="#">Upload Cisco ThousandEyes Enterprise Agent Software to Cisco SD-WAN Manager</a></li> <li>To add Virtual Images from a remote server, see <a href="#">Add Virtual Images to the Repository</a></li> </ul> |
| Create an IP pool in the global node of Network Hierarchy Manager. When deploying Cisco ThousandEyes agent through the workflow, the Virtual Port Group Interface IP address contains this IP pool. | See <a href="#">Create an IP Pool for the ThousandEyes Agents</a>                                                                                                                                                                                                                                                                   |
| Add IPv4 DNS server details in the Transport VPN feature of the Configuration Group associated with the device.                                                                                     | See <a href="#">DNS</a>                                                                                                                                                                                                                                                                                                             |

## Create an IP Pool for the Cisco ThousandEyes Agents

Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.16.1 and Cisco IOS XE Catalyst SD-WAN Release 17.16.1a

1. From the menu, choose **Configuration > Network Hierarchy**.



**Note** The page displays the site pool and region pool for the Global node.

2. Click **Pools**.
3. Click **Add Pool**.
4. In the **Pool Name** field, enter a name for the pool.
5. In the **Pool Description** field, enter a description of the pool.
6. From the **Pool Type** drop-down list, choose **ThousandEyes**.
7. In the **IP Subnet\*** field, enter an IP address.
8. In the **Prefix Length\*** field, enter the prefix length of the system IP pool.
9. Click **Add**.

# Restrictions for Simplified Workflow to Onboard Cisco ThousandEyes Agent and Configure Tests Using Cisco Catalyst SD-WAN Manager

Minimum supported releases: Cisco Catalyst SD-WAN Manager Release 20.16.1 and Cisco IOS XE Catalyst SD-WAN Release 17.16.1a

- The workflow does not support editing tests or updating agent parameters for sites with deployed Cisco ThousandEyes agents.
- The simplified workflow supports only devices that are associated with configuration groups.
- In the workflow, you can select only one device in a site to deploy the Cisco ThousandEyes Agent and configure tests associated with it. To deploy Cisco ThousandEyes Agent on multiple devices in a site, use the Configuration Group method, see [Using Configuration Groups](#).

**Note**

To add new devices to the same Configuration Group to deploy ThousandEyes agent or to update the existing Configuration Group, use Configuration Group flows.

## Configure Cisco ThousandEyes Agent and Tests Using Workflows

Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.16.1 and Cisco IOS XE Catalyst SD-WAN Release 17.16.1a

1. From the Cisco SD-WAN Manager menu, choose **Workflows > Workflow Library**.
2. Start the **Configure ThousandEyes** workflow.
3. Read the terms and conditions. After accepting the terms and conditions, click **Connect ThousandEyes** to start the workflow.
4. When the Cisco ThousandEyes portal opens, log in to your account using your login credentials. The page redirects to Cisco SD-WAN Manager.
5. Follow the instructions provided in the workflow.

**Note**

In the workflow, select all sites associated with a specific Configuration Group simultaneously. This ensures that the Configuration Group with Cisco ThousandEyes agent applies to all sites associated with the Configuration Group.



**Note** The time required to onboard Cisco ThousandEyes agent and configure tests varies based on the size of your network.

**What to do next:**

- Monitor the status of agents and enable or disable agents in the **ThousandEyes Monitoring** dashboard in the Cisco SD-WAN Manager.
- Launch **ThousandEyes Monitoring** dashboard to view test results.

## Monitor Onboarded Cisco ThousandEyes Agents and Configured Tests

Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.16.1 and Cisco IOS XE Catalyst SD-WAN Release 17.16.1a

To monitor the status of Cisco ThousandEyes agents and configured tests in the Cisco SD-WAN Manager monitoring dashboard, navigate to **Tools > ThousandEyes Monitoring**.

You can view the online, offline, and disabled agents, and launch Cisco ThousandEyes from the dashboard.

The monitoring dashboard includes two sections:

- **Tests:** Provides information about the tests configured using the simplified workflow.
- **Agents:** Provides information about the status of Cisco ThousandEyes agents configured using the simplified workflow. It also provides a list of configured devices and sites for the test, allowing you to enable, disable, or delete a Cisco ThousandEyes agent.

## Troubleshoot Cisco ThousandEyes Enterprise Agent on Cisco IOS XE Catalyst SD-WAN Device

Minimum supported release: Cisco IOS XE Catalyst SD-WAN Release 17.16.1a and Cisco Catalyst SD-WAN Manager Release 20.16.1

1. Review the following log files in Cisco SD-WAN Manager for error messages or relevant information:  
/var/log/nms/vmanage-server.log and  
/var/log/nms/vmanage-server-deviceconfig-template.log.
2. Examine the audit logs to determine if the ThousandEyes test was posted or it failed.







# CHAPTER 41

## Extended Visibility with Cisco SD-WAN and Cisco ThousandEyes

- [Extended Visibility with Cisco Catalyst SD-WAN and Cisco ThousandEyes, on page 1019](#)

## Extended Visibility with Cisco Catalyst SD-WAN and Cisco ThousandEyes

**Table 383: Feature History**

| Feature Name                                                                 | Release Information                                                                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Extended Visibility with Cisco Catalyst SD-WAN and Cisco ThousandEyes        | <p>Cisco IOS XE Release 17.6.1a</p> <p>Cisco vManage Release 20.6.1</p>                 | <p>You can deploy Cisco ThousandEyes Enterprise agent natively as a container application on supported Cisco IOS XE Catalyst SD-WAN devices to integrate Cisco Catalyst SD-WAN with Cisco ThousandEyes. You can install and activate the Cisco ThousandEyes Enterprise agent through Cisco SD-WAN Manager.</p> <p>By integrating Cisco Catalyst SD-WAN with Cisco ThousandEyes, you can gain granular insights into network and application performance with full hop-by-hop path analysis across the Internet, and isolate fault domains for expedited troubleshooting and resolution.</p> |
| Cisco ThousandEyes Support for Cisco 1000 Series Integrated Services Routers | <p>Cisco IOS XE Catalyst SD-WAN Release 17.7.1a</p> <p>Cisco vManage Release 20.7.1</p> | <p>You can deploy Cisco ThousandEyes Enterprise agent natively as a container application on Cisco ISR 1100X-6G devices. You can install and activate the Cisco ThousandEyes Enterprise agent through Cisco SD-WAN Manager.</p>                                                                                                                                                                                                                                                                                                                                                             |

| Feature Name                                                                                                                    | Release Information                                                                            | Description                                                                                                                                                                                                                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cisco ThousandEyes Support for Cisco Catalyst 8500 Series Edge Platforms and Cisco ASR 1000 Series Aggregation Services Routers | Cisco IOS XE Catalyst SD-WAN Release 17.8.1a<br>Cisco vManage Release 20.8.1                   | You can deploy Cisco ThousandEyes Enterprise agent natively as a container application on Cisco Catalyst 8500 Series Edge Platforms and Cisco ASR 1000 Series Aggregation Services Routers. You can install and activate the Cisco ThousandEyes Enterprise agent through Cisco SD-WAN Manager. |
| Extended Device Support for the Cisco ThousandEyes Agent in Cisco ISR 1100 Series                                               | Cisco Catalyst SD-WAN Manager Release 20.16.1<br>Cisco IOS XE Catalyst SD-WAN Release 17.16.1a | Support for the Cisco ThousandEyes Enterprise agent is extended to additional routers in Cisco ISR 1100 Series.                                                                                                                                                                                |

## Information About for Extending Visibility with Cisco Catalyst SD-WAN and Cisco ThousandEyes

Cisco ThousandEyes is a SaaS application that provides you an end-to-end view across networks and services that impact your business. It monitors the network traffic paths across internal, external, and carrier networks and the Internet in realtime to provide network performance data. Cisco ThousandEyes provides intelligent insights into your WAN and the cloud and helps you optimize application delivery and end-user experience.

### Deploy and Configure the ThousandEyes Agent

From Cisco IOS XE Release 17.6.1, you can deploy and configure the Cisco ThousandEyes Enterprise agent on Cisco IOS XE Catalyst SD-WAN devices to enable extensive monitoring of the WAN traffic for enhanced visibility within and beyond the Cisco Catalyst SD-WAN fabric. The Cisco ThousandEyes Enterprise agent is an embedded Docker-based application that runs on Cisco IOS XE Catalyst SD-WAN devices as a docker-type container application using the IOx Docker application-hosting capability.

### Services in the Cisco Networking Cloud

From Cisco Catalyst SD-WAN Manager Release 20.14.1, you can navigate to Cisco services hosted in the Cisco Networking Cloud, such as Cisco ThousandEyes. Enable the cross platform navigator in **Administration > Settings > Cross Platform Navigator**.

### More Information

For more information on Cisco ThousandEyes and on configuring tests and viewing results in the Cisco ThousandEyes portal, see the Cisco ThousandEyes *Getting Started* documentation.

## Supported Devices for Extended Visibility with Cisco Catalyst SD-WAN and Cisco ThousandEyes

| Platform                                           | Device Model     | Architecture | Release                                                |
|----------------------------------------------------|------------------|--------------|--------------------------------------------------------|
| Cisco Catalyst 8300 Series Edge Platforms          | C8300-1N1S-6T    | x86_64       | Cisco vManage Release 20.6.1                           |
|                                                    | C8300-1N1S-4T2X  |              | Cisco IOS XE Release 17.6.1a and later                 |
|                                                    | C8300-2N2S-6T    |              |                                                        |
|                                                    | C8300-2N2S-4T2X  |              |                                                        |
| Cisco Catalyst 8200 Series Edge Platforms          | C8200-1N-4T      | x86_64       |                                                        |
|                                                    | C8200L-1N-4T     |              |                                                        |
| Cisco 4000 Series Integrated Services Routers      | ISR4461          | x86_64       |                                                        |
|                                                    | ISR4451          |              |                                                        |
|                                                    | ISR4431          |              |                                                        |
|                                                    | ISR4351          |              |                                                        |
|                                                    | ISR4331          |              |                                                        |
|                                                    | ISR4321          |              |                                                        |
|                                                    | ISR4221X         |              |                                                        |
| Cisco 1000 Series Integrated Services Routers      | ISR1100X-6G      | x86_64       | Cisco vManage Release 20.7.1                           |
|                                                    |                  |              | Cisco IOS XE Catalyst SD-WAN Release 17.7.1a and later |
| Cisco Catalyst 8500 Series Edge Platforms          | C8500-12X        | x86_64       | Cisco vManage Release 20.8.1                           |
|                                                    | C8500-12X4QC     |              | Cisco IOS XE Catalyst SD-WAN Release 17.8.1a and later |
|                                                    | C8500L-8S4X      |              |                                                        |
| Cisco ASR 1000 Series Aggregation Services Routers | ASR 1001-HX      | x86_64       |                                                        |
|                                                    | ASR 1001-X       |              |                                                        |
|                                                    | ASR 1002-HX      |              |                                                        |
|                                                    | ASR 1002-X       |              |                                                        |
|                                                    | ASR 1006-X (RP3) |              |                                                        |

| Platform                                      | Device Model   | Architecture | Release                                                                                            |
|-----------------------------------------------|----------------|--------------|----------------------------------------------------------------------------------------------------|
| Cisco 1000 Series Integrated Services Routers | C1111X-8P      | aarch64      | Cisco IOS XE Catalyst SD-WAN Release 17.16.1a<br><br>Cisco Catalyst SD-WAN Manager Release 20.16.1 |
|                                               | C1121X-8P      |              |                                                                                                    |
|                                               | C1121X-8PLTEP  |              |                                                                                                    |
|                                               | C1121X-8PLTEPW |              |                                                                                                    |
|                                               | C1126X-8PLTEP  |              |                                                                                                    |
|                                               | C1127X-8PLTEP  |              |                                                                                                    |
|                                               | C1131X-8PW     |              |                                                                                                    |
|                                               | C1131X-8PLTEPW |              |                                                                                                    |
|                                               | C1161X-8P      |              |                                                                                                    |
|                                               | C1161X-8PLTEP  |              |                                                                                                    |

### Supported ThousandEyes Enterprise Agent Version

We recommend you to update to the latest version of the ThousandEyes Enterprise agent. Download the latest version of the ThousandEyes Enterprise agent from [Cisco ThousandEyes Agent](#) page.

### Storage and DRAM Requirements

- **External Storage:** On devices that are equipped with external storage (SSD M.2 NVMe), the Cisco ThousandEyes Enterprise agent is installed in the external storage. The minimum external storage capacity that is required to install the Cisco ThousandEyes Enterprise agent is 8 GB. If the device does not have sufficient external storage capacity, upgrade the storage capacity to meet the minimum requirement.

Although the minimum external storage capacity that is required is 8 GB, we recommend that you provision the device with an external storage capacity of 16 GB or more. With the minimum external storage capacity, you may need to manually clean up files while upgrading the software image on the device.

- **Bootflash:** On devices that are not equipped with external storage, the Cisco ThousandEyes Enterprise agent is installed on the bootflash. The minimum bootflash capacity that is required to install the Cisco ThousandEyes Enterprise agent is 8 GB. If the device does not have sufficient bootflash capacity, upgrade the storage capacity to meet the minimum requirement.



#### Important

On the ISR1100X-6G, the Cisco ThousandEyes Enterprise agent is installed on the bootflash. For this particular device model, the minimum bootflash capacity that is required to install the agent is 16 GB.

Although the minimum bootflash capacity that is required is 8 GB, we recommend that you provision the device with a bootflash capacity of 16 GB or more. With the minimum bootflash capacity, you may need to manually clean up files while upgrading the software image on the device.

- **DRAM:** The minimum DRAM capacity that is required to install the Cisco ThousandEyes Enterprise agent is 8 GB. If a device does not have the minimum DRAM capacity that is required to install the Cisco ThousandEyes Enterprise agent, upgrade the DRAM to meet the minimum requirement.
- Cisco ThousandEyes Enterprise agent can be deployed with other applications if the device has the resources (CPU, memory, and storage) to run the other applications. If the available resources are not sufficient to run the other applications, IOX generates an error message and does not run the other applications.

To host the Cisco ThousandEyes Enterprise agent, a Cisco IOS XE Catalyst SD-WAN device must have a minimum of 8 GB DRAM. If you wish host additional applications such as UTD and DRE on the same device, we recommend that you provision the device with at least a 16 GB DRAM.

## Prerequisites for Extending Visibility with Cisco Catalyst SD-WAN and Cisco ThousandEyes

- Before deploying the Cisco ThousandEyes Enterprise agent, you must create an account on the Cisco ThousandEyes portal and obtain an account group token. The agent uses the token to authenticate itself with Cisco ThousandEyes and check-in to the right Cisco ThousandEyes account.

For information on obtaining the account group token, see *Where Can I Get the Account Group Token?* on Cisco ThousandEyes Documentation portal.

- The Cisco ThousandEyes Enterprise agent requires DNS name resolution and HTTP/HTTPS connectivity to discover and register with the Cisco ThousandEyes portal. Ensure that this connectivity exists before deploying the agent by configuring the appropriate firewall rules, NAT settings, upstream routing, and other related settings.

For more information on the required firewall configuration, see *Firewall Configuration for Enterprise Agents* on Cisco ThousandEyes Documentation portal.

## Restrictions for Extending Visibility with Cisco Catalyst SD-WAN and Cisco ThousandEyes

- Cisco ThousandEyes Enterprise agent probes are sourced from Virtual Port-Group interfaces and are not affected by AppRoute data policies.
- The Cisco ThousandEyes Enterprise agent, hosted natively as a container application on Cisco IOS XE Catalyst SD-WAN devices, does not support browser-based application tests, such as page load test and transaction test.
- For every changes to the ThousandEyes instance to reflect the changes done, you need deactivate and reactivate it.

For Cisco IOS XE Catalyst SD-WAN devices prior to Cisco IOS XE Release 17.6.1, the Docker image can be installed either directly from the ThousandEyes download servers, or by downloading the container image to a local machine and uploading it to the router via SCP, FTP, TFTP, or USB storage, depending on whether the router has direct internet access or not.

For Cisco IOS XE Catalyst SD-WAN devices after Cisco IOS XE Release 17.6.1, in addition to the previous methods, you can install the Cisco ThousandEyes Enterprise agent using bootflash.

# Configure Cisco ThousandEyes Enterprise Agent on Cisco IOS XE Catalyst SD-WAN Devices

## Configure Cisco ThousandEyes Enterprise Agent Using a Configuration Group

### Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

### Procedure

**Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

**Step 2** Create and configure a ThousandEyes feature in Other profile.

- a. Configure ThousandEyes parameters.

**Table 384: Parameters**

| Parameter Scope                               | Scope Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Device Specific<br>(indicated by a host icon) | <p>Use a device-specific value for the parameter. For device-specific parameters, you cannot enter a value in the feature template. You enter the value when you attach a Cisco Catalyst SD-WAN device to a device template.</p> <p>When you click Device Specific, the Enter Key box opens. This box displays a key, which is a unique string that identifies the parameter in a CSV file that you create. This file is an Excel spreadsheet that contains one column for each key. The header row contains the key names (one key per column), and each row after that corresponds to a device and defines the values of the keys for that device. You upload the CSV file when you attach a Cisco Catalyst SD-WAN device to a device template.</p> <p>To change the default key, type a new string and move the cursor out of the Enter Key box.</p> <p>Examples of device-specific parameters are system IP address, host name, GPS location, and site ID.</p> |
| Global (indicated by a globe icon)            | <p>Enter a value for the parameter and apply that value to all devices.</p> <p>Examples of parameters that you might apply globally to a group of devices are DNS server, syslog server, and interface MTUs.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

- b. Configure the ThousandEyes settings.

**Table 385: Settings**

| Field        | Description                               |
|--------------|-------------------------------------------|
| Type         | Choose a feature from the drop-down list. |
| Feature Name | Enter a name for the feature.             |

| Field                        | Description                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>           | Enter a description of the feature. The description can contain any characters and spaces.                                                                                                                                                                                                                                                                      |
| <b>Account Group Token</b>   | Enter the Cisco ThousandEyes Account Group Token.                                                                                                                                                                                                                                                                                                               |
| <b>VPN</b>                   | Service VPN. The <b>Global</b> or the <b>Device Specific</b> setting indicates service VPN.<br><br>When you set the VPN configuration as a <b>Global</b> or a <b>Device Specific</b> setting, enter the ID of the service VPN in which you want to provision the Cisco ThousandEyes Enterprise agent.                                                           |
| <b>Management IP</b>         | Enter an IP address for the Cisco ThousandEyes Enterprise agent. This field is available only when you specify the service VPN.                                                                                                                                                                                                                                 |
| <b>Management Subnet</b>     | Choose a subnet mask from the drop-down list for the Cisco ThousandEyes Enterprise agent. This field is available only when you specify the service VPN.<br><br><b>Note</b><br>This IP-prefix address ( <b>Management IP</b> and <b>Management Subnet</b> ) must be unique within the fabric and must not overlap with the IP addresses of other branch agents. |
| <b>Agent Default Gateway</b> | Enter a default gateway address. This IP address is assigned to the virtual port group of the router. This field is available only when you specify the service VPN.                                                                                                                                                                                            |
| <b>Name Server IP</b>        | Enter the IP address of your preferred DNS server.<br><br>This server can exist within or outside the Cisco Catalyst SD-WAN fabric but must be reachable from the service VPN.                                                                                                                                                                                  |
| <b>Host Name</b>             | Enter the hostname that the agent must use when registering with the Cisco ThousandEyes portal. By default, the agent uses the hostname of the Cisco IOS XE Catalyst SD-WAN device.                                                                                                                                                                             |

| Field             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Proxy Type</b> | <p>If the Cisco ThousandEyes Enterprise agent must use proxy server for external access, choose one of the following as proxy type:</p> <ul style="list-style-type: none"> <li>• <b>static</b></li> <li>• <b>pac</b></li> <li>• <b>none</b></li> </ul> <p>Static proxy settings:</p> <ul style="list-style-type: none"> <li>• <b>Proxy Host:</b> Set the configuration as a <b>Global</b> setting and enter the hostname of the proxy server.</li> <li>• <b>Proxy Port:</b> Set the configuration as a <b>Global</b> setting and enter the port number of the proxy server.</li> </ul> <p>PAC settings:</p> <ul style="list-style-type: none"> <li>• <b>PAC URL:</b> Set the configuration as a <b>Global</b> setting and enter the URL of the proxy auto-configuration (PAC) file.</li> </ul> |

### What to do next

Also see [Deploy a configuration group](#).

## Upload Cisco ThousandEyes Enterprise Agent Software to Cisco SD-WAN Manager

1. Download the latest version of Cisco ThousandEyes Enterprise agent software from the [Cisco ThousandEyes Agent Settings](#) page.
2. From the Cisco SD-WAN Manager menu, choose **Maintenance > Software Repository**.
3. Click **Virtual Images**.
4. Click **Upload Virtual Image** and click **Manager**.
5. In the **Upload VNF's Package to Manager** dialog box, browse to the location of the downloaded Cisco ThousandEyes Enterprise agent software file and select the file.  
Alternatively, drag and drop the Cisco ThousandEyes Enterprise agent software file.
6. Enter a description for the file.
7. (Optional) Add desired tags.
8. Click **Upload**.

## Provision Cisco ThousandEyes Enterprise Agent in a Service VPN

You can provision the Cisco ThousandEyes Enterprise agent in a service VPN for more visibility into the performance of the Cisco Catalyst SD-WAN overlay and underlay networks.



### Prerequisites

- Ensure that the appropriate DNS and NAT configuration exists to enable the Cisco ThousandEyes Enterprise agent to discover and connect to the Cisco ThousandEyes application.
- Upload Cisco ThousandEyes Enterprise agent software to Cisco SD-WAN Manager.



**Note** If you have uploaded more than one version of the Cisco ThousandEyes Enterprise agent software to the Cisco SD-WAN Manager software repository, while provisioning the agent, Cisco SD-WAN Manager installs and activates the latest version of the agent software.

- For application monitoring use cases, deploy the ThousandEyes agent in the Service VPN and use an SD-WAN data policy to send ThousandEyes probes across transports

### Procedure

1. Create feature template for the Cisco ThousandEyes Enterprise agent:
  - a. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
  - b. Click **Feature Templates**, and click **Add Template** to select an appropriate device model.



**Note** In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled as **Feature**.

- c. Choose the supported devices to which you want to apply this template.
- d. In the **Other Templates** section, click **ThousandEyes Agent**.
- e. **Template Name:** Enter a name for the template. Ensure that the template name is unique.
- f. **Description:** Enter a description for the template.
- g. In the **BASIC CONFIGURATION** section, configure the following:

|                       |                                                                                                                                                                                                                                                            |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Account Group Token   | Enter the Cisco ThousandEyes Account Group Token.                                                                                                                                                                                                          |
| VPN                   | <ol style="list-style-type: none"> <li>1. Set the VPN configuration as a <b>Global</b> or a <b>Device Specific</b> setting.</li> <li>2. Enter the ID of the service VPN in which you want to provision the Cisco ThousandEyes Enterprise agent.</li> </ol> |
| Agent IP Address      | Enter an IP address for the Cisco ThousandEyes Enterprise agent.<br>This IP Address should be unique within the fabric and should not overlap with the IP addresses of other branch agents.                                                                |
| Agent Default Gateway | Enter a default gateway address. This IP address is assigned to the virtual port group of the router.                                                                                                                                                      |



**Tip** You can create and allocate a service subnet for the agent network. Two usable IP addresses are required to provision the Cisco ThousandEyes Enterprise agent on each Cisco IOS XE Catalyst SD-WAN device. One of the IP addresses must be assigned to the agent and second IP address to the router virtual port group.

h. In the **ADVANCED** section, configure the following:

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name Server    | (Optional parameter from Cisco vManage Release 20.7.1 and Cisco IOS XE Release 17.7.1a)<br><br>Enter the IP address of your preferred DNS server.<br><br>This server can exist within or outside the Cisco Catalyst SD-WAN fabric but must be reachable from the service VPN.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Hostname       | (Optional) Enter the hostname that the agent must use when registering with the Cisco ThousandEyes portal. By default, the agent uses the Cisco IOS XE Catalyst SD-WAN device's hostname.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Web Proxy Type | (Optional) If the Cisco ThousandEyes Enterprise agent must use proxy server for external access, choose one of the following as proxy type: <ul style="list-style-type: none"> <li>• <b>Static</b></li> <li>• <b>PAC</b></li> </ul> Static proxy settings: <ul style="list-style-type: none"> <li>• <b>Proxy Host</b>: Set the configuration as a <b>Global</b> setting and enter the hostname of the proxy server.</li> <li>• <b>Proxy Port</b>: Set the configuration as a <b>Global</b> setting and enter the port number of the proxy server.</li> </ul> PAC settings: <ul style="list-style-type: none"> <li>• <b>PAC URL</b>: Set the configuration as a <b>Global</b> setting and enter the URL of the proxy auto-configuration (PAC) file.</li> </ul> |

i. Click **Save**.

2. Attach the ThousandEyes Agent feature template to device template:

- a. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
- b. Click **Device Templates**.



**Note** In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled as **Device**

- c. Find the device template for the target device.
- d. For this template, click ..., and click **Edit**.

- e. Click **Additional Templates**.
- f. In the **Additional Templates** section, choose the **ThousandEyes Agent** feature template created earlier.
- g. Click **Update**.
- h. Update necessary variables, if any, and click **Next**.
- i. Review the configuration and click **Configure Devices**.

3. Repeat **Step 2** for each device on which you want to deploy the Cisco ThousandEyes Enterprise agent.

The Cisco ThousandEyes Enterprise agent is deployed on the chosen devices. The agent registers with and establishes secure communication with the cloud-based Cisco ThousandEyes application to receive necessary updates and configuration. You can configure various tests and see resultant network and application telemetry data on the [Cisco ThousandEyes](#) portal.

When you host the ThousandEyes agent on a router, you can deploy it in either a Service VPN or the Transport VPN (VPN0), based on your functional requirements and network topology. Deploying the ThousandEyes agent in VPN0 offers a simple routing approach when you need to send probe traffic over a single available underlay WAN link. In environments with multiple WAN links, Cisco recommends deploying the ThousandEyes agent in a Service VPN. This deployment method gives you granular control and enables you to create policies that direct probe traffic over specific underlay WAN paths, effectively simulating real-world traffic flows. If you deploy the ThousandEyes agent in VPN0 in a multi-WAN environment, the router load-balances probe traffic across all available WAN links, which may result in non-actionable monitoring insights.

## Provision Cisco ThousandEyes Enterprise Agent in a Service VPN Using CLI

This section provides example command sequences to provision the Cisco ThousandEyes Enterprise agent on Cisco IOS XE Catalyst SD-WAN devices using a device CLI template or an add-on CLI template.

### Prerequisites

- Ensure that the appropriate DNS and NAT configuration exists to enable the Cisco ThousandEyes Enterprise agent to discover and connect to the Cisco ThousandEyes application.
- Upload Cisco ThousandEyes Enterprise agent software to Cisco SD-WAN Manager.



**Note** If you have uploaded more than one version of the Cisco ThousandEyes Enterprise agent software to the Cisco SD-WAN Managersoftware repository, while provisioning the agent, Cisco SD-WAN Manager installs and activates the latest version of the agent software.

This section provides example CLI configurations to provision the Cisco ThousandEyes Enterprise agent in a service VPN.

1. Enable IOX on the device.

```
iox
```

2. Configure virtual port group. The virtual port group acts as the gateway for the Cisco ThousandEyes Enterprise agent.

```
interface VirtualPortGroup4
 vrf forwarding 100
 ip address 192.168.61.1 255.255.255.252
```

3. Configure app-hosting parameters for the Cisco ThousandEyes Enterprise agent.

```
app-hosting appid te
app-vnic gateway0 virtualportgroup 4 guest-interface 0
 guest-ipaddress 192.168.61.2 netmask 255.255.255.252
app-default-gateway 192.168.61.1 guest-interface 0
app-resource docker
 prepend-pkg-opts
 run-opts 1 "-e TEAGENT_ACCOUNT_TOKEN=z0kemf"
 run-opts 2 "--hostname ISR4461TE"
 run-opts 3 "-e TEAGENT_PROXY_TYPE=STATIC -e TEAGENT_PROXY_LOCATION=proxy-exmaple.com:80"
name-server0 192.168.168.183
start
```



#### Note

- You can use the proxy configuration only if the Cisco ThousandEyes agent does not have an Internet access without a proxy. Also, the hostname is optional and if you do not provide the hostname during the installation, the device hostname will be used as the Cisco ThousandEyes agent hostname. The device hostname will be displayed on the Cisco ThousandEyes portal. From Cisco IOS XE Release 17.7.1a, the DNS name server information is optional.
- If the Cisco ThousandEyes agent uses a private IP address, establish a connection to the device through NAT.

## Upgrade Cisco ThousandEyes Enterprise Agent Software

1. Download a new version of Cisco ThousandEyes Enterprise agent software and upload the software to Cisco SD-WAN Manager. See *Upload Cisco ThousandEyes Enterprise Agent Software to Cisco SD-WAN Manager*.
2. From the Cisco SD-WAN Manager menu, choose **Maintenance > Software Upgrade**.
3. Select the Cisco IOS XE Catalyst SD-WAN devices on which you want to upgrade the Cisco ThousandEyes Enterprise agent software.
4. Click **Upgrade Virtual Image**.
5. In the **Virtual Image Upgrade** dialog box, choose the new version of the Cisco ThousandEyes Enterprise agent software from the drop-down list. Click **Upgrade**.
6. On the **Maintenance > Software Upgrade** page, select the Cisco IOS XE Catalyst SD-WAN devices on which you upgraded the Cisco ThousandEyes Enterprise agent software.
7. Click **Activate Virtual Image**.

## Uninstall Cisco ThousandEyes Enterprise Agent Software

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.

2. Click **Device Templates**.



---

**Note** In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled as **Device**

---

3. Find the device template for the device from which the Cisco ThousandEyes agent software must be removed.
4. For this template, click ... and then click **Edit**.
5. Click **Additional Templates**.
6. In the **Additional Templates** section, for **ThousandEyes Agent** choose **None** from the drop-down list.
7. Click **Update**.
8. Update necessary variables, if any, and click **Next**.
9. Review the configuration and click **Configure Devices**.

## Troubleshoot Cisco ThousandEyes Enterprise Agent on Cisco IOS XE Catalyst SD-WAN Devices

1. Connect to Cisco ThousandEyes Enterprise agent.  
`Device#app-hosting connect appid Appid session /bin/bash`
2. To verify the agent configuration, check the following CFG file: `/etc/te-agent.cfg`
3. To view the agent logs, check the following file: `/var/log/agent/te-agent.log`





## CHAPTER 42

# Layer 2 VPN

---

- [Layer 2 VPN, on page 1034](#)
- [Information About Layer 2 VPN Support within the Cisco Catalyst SD-WAN Overlay Network, on page 1034](#)
- [Network Topology for Layer 2 Connections, on page 1035](#)
- [Layer 2 Control Protocol packet forwarding, on page 1036](#)
- [Multihoming, on page 1037](#)
- [L2VPN Hub-and-Spoke Support, on page 1038](#)
- [Supported Platforms for Layer 2 VPN, on page 1039](#)
- [Restrictions for Layer 2 VPN, on page 1039](#)
- [Configure Layer 2 VPN Using CLI Template, on page 1040](#)
- [Configure an L2VPN on a Cisco IOS XE Catalyst SD-WAN Device Using CLI Template, on page 1040](#)
- [Configure Point-to-Point Layer 2 VPN Using CLI Template, on page 1041](#)
- [Configure an Edge Router at Site A for Point-to-Point Layer 2 VPN Using CLI Template, on page 1042](#)
- [Configure an edge Router at Site B for Point-to-Point Layer 2 VPN Using CLI Template, on page 1043](#)
- [Configure Point-to-Multipoint Layer 2 VPN Using CLI Template, on page 1044](#)
- [Configure an Edge Router at Sites A, B, and C, on page 1044](#)
- [Configure an edge Router at Site B for Point-to-Point Layer 2 VPN Using CLI Template, on page 1045](#)
- [Configure an edge router at Site C for point-to-point Layer 2 VPN using CLI Template, on page 1047](#)
- [Configure Layer 2 VPN Switchport Using CLI Template, on page 1048](#)
- [Verify Layer 2 VPN Using CLI, on page 1050](#)
- [Monitor Configured Layer 2 VPN Using CLI, on page 1055](#)

# Layer 2 VPN

**Table 386: Feature History**

| Feature Name                                           | Release Information                                                                                       | Description                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Layer 2 (L2) VPN                                       | Cisco IOS XE Catalyst SD-WAN Release 17.14.1a<br>Cisco Catalyst SD-WAN Control Components Release 20.14.x | The feature adds Layer 2 VPN support on the Cisco Catalyst SD-WAN overlay network.<br><br>It allows you to configure Layer 2 point-to-point and point-to-multipoint connections within the Cisco Catalyst SD-WAN fabric.                                                                                                                  |
| Layer 2 (L2) VPN Multihoming and Hub-and-Spoke Support | Cisco IOS XE Catalyst SD-WAN Release 17.15.1a<br>Cisco Catalyst SD-WAN Manager Release 20.15.x            | With this feature, you can configure Layer 2 VPN on multiple devices on the same site in an active-standby configuration.<br><br>This feature also enables Layer 2 connections using an indirect path, such as a hub, for point-to-multipoint connections within the Cisco Catalyst SD-WAN fabric.                                        |
| Layer 2 VPN Packet Forwarding                          | Cisco IOS XE Catalyst SD-WAN Release 17.18.1a<br>SD-WAN Control Components 20.18.1                        | Layer 2 Control Protocol (L2CP) packet forwarding in Cisco Catalyst SD-WAN Manager is supported for both point-to-point (P2P) and point-to-multipoint (P2MP) modes. In P2P mode, you can control which L2CP packets to block or use locally, while in P2MP mode, you can forward specific L2CP packets transparently to remote endpoints. |

## Information About Layer 2 VPN Support within the Cisco Catalyst SD-WAN Overlay Network

The Cisco Catalyst SD-WAN solution provides Layer 3 services with security, segmentation, and scalability across the overlay network. Considering the importance of Layer 2 (L2) connectivity, particularly for legacy systems and non-IP applications, Layer 2 services are supported within the Cisco Catalyst SD-WAN overlay network. L2VPN support enables using legacy applications that require Layer 2 connectivity across the Cisco Catalyst SD-WAN fabric.

Starting from Cisco IOS XE Catalyst SD-WAN Release 17.14.1a, the following L2VPN features are supported:

- Point-to-point L2VPN Service (P2P)
- Point-to-Multipoint L2VPN Service (P2MP)
- Single homing
- Flood and Learn in WAN and LAN
- Ingress replication for Broadcast, Unknown-unicast and Multicast (BUM)



- Full mesh topology only

Starting from Cisco IOS XE Catalyst SD-WAN Release 17.15.1a, the following L2VPN features are supported:

- Multihoming for P2P and P2MP
- Hub-and-spoke topology support for L2VPN services
- The MAC learning mode (previously the Flood and Learn in WAN and LAN) is changed to learning through OMP protocol (that is, Control Plane).

**Note**

From Cisco IOS XE Catalyst SD-WAN Release 17.18.1a, you can adjust the TCP Maximum Segment Size (MSS) even for a TCP packet encapsulated in an MPLS label. You can set the TCP MSS per the Path Maximum Transmission Unit (PMTU) with 30 bytes to account for Layer 2 headers, such as Ethernet, VLAN tags and MPLS headers.

For IPv4, the TCP MSS is set per PMTU with 80 bytes for IPv4 and TCP headers and an additional 30 bytes for Layer 2 headers. For example, if the PMTU is 1438, the TCP MSS is set as 1328 (1438 - 80 - 30).

For IPv6, the TCP MSS is set per PMTU with 100 bytes for IPv6 and TCP headers and an additional 30 bytes for Layer 2 headers. For example, if the PMTU is 1438, the TCP MSS is set as 1308 (1438-100-30).

For more information about configuring TCP MSS, see [Configure TCP MSS Using CLI](#)

This change helps prevent drop in TCP sessions, improving overall network performance and reliability.

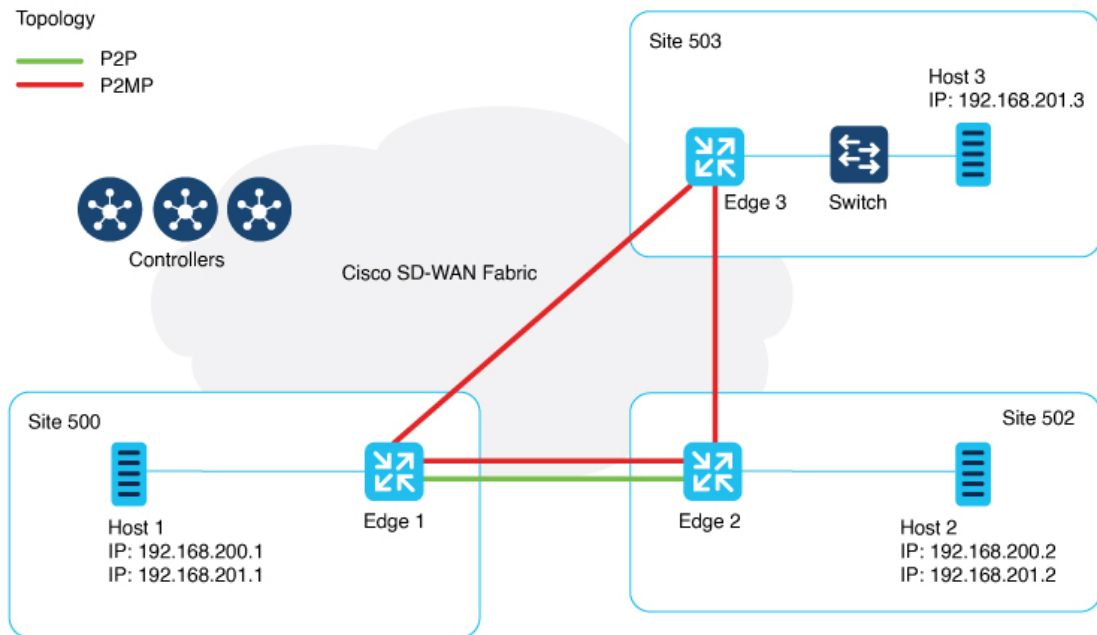
## Network Topology for Layer 2 Connections

This illustration shows three sites and shows P2P (green line) and P2MP (red lines) connections between edge routers at the sites.

- Point-to-Point (P2P): Connects sites 500 and 502 with a dedicated Layer 2 VPN. The L2VPN connection between the two sites allows Host 1 and Host 2 to interact.
- Point-to-Multipoint (P2MP): Connects sites 500, 502, and 503 with Layer 2 VPN. Host 1 communicates with both Host 2 and Host 3 across a Layer 2 multipoint network.

The L2VPN connections use existing Cisco Catalyst SD-WAN tunnels.

Figure 9: Topology



## Layer 2 Control Protocol packet forwarding

Layer 2 Control Protocol (L2CP) forwarding in SD-WAN Manager.

- supports controlling of L2CP packets forwarded in P2P mode, and
- enables forwarding of L2CP packets in P2MP mode.

The support for L2CP packet forwarding in SD-WAN Manager includes, R4, R5, R6, R8, R9, RA, RB, RC, RD, RF, cdp, dot1x, dtp, elmi, esmc, lacp, lldp, pagp, ptpdp, stp, udld, and vtp packets.

### Controlling L2CP packet forwarding in P2P mode

From Cisco IOS XE Catalyst SD-WAN Release 17.18.1a, in P2P mode, you can control the Control Protocol (L2CP) packets that you want to block or use locally.

Here is an example of how you can control forwarding of L2CP packets in P2P mode in a P2P setup.

```
Device#2vpn sdwan instance 222 point-to-point
Device#l2protocol peer cdp stp
```

After applying the **l2protocol peer cdp stp** command within the l2vpn sdwan instance 222 point-to-point configuration, the SD-WAN instance 222 will no longer forward Cisco Discovery Protocol (CDP) and Spanning Tree Protocol (STP) packets.

### Forwarding L2CP packets in P2MP mode

From Cisco IOS XE Catalyst SD-WAN Release 17.18.1a, in P2MP mode, you can forward specific L2CP packets in a multipoint setup.

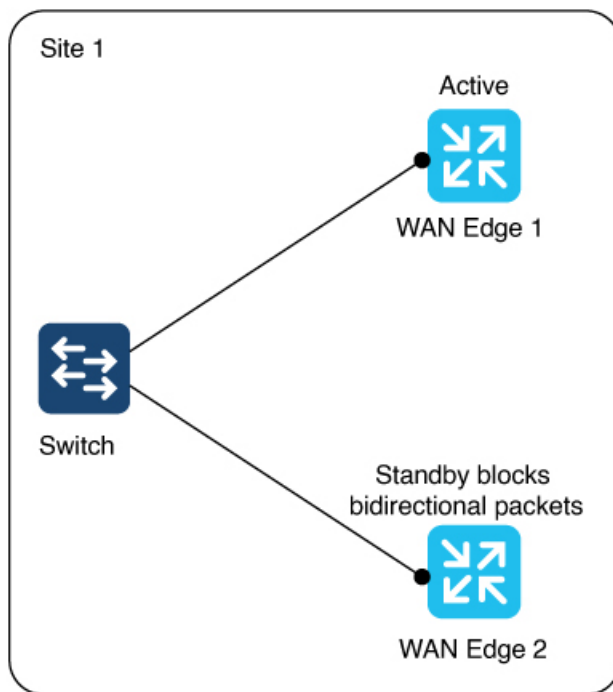
Here is an example of how the command can be used to forward L2CP packets in P2MP mode:

```
Device#l2vpn sdwan instance 111 multipoint
Device#l2protocol forward cdp stp
```

After applying the **l2protocol forward cdp stp** command within the l2vpn sdwan instance 111 multipoint configurations, the SD-WAN instance 111 will actively forward CDP and STP packets. This means these packets will be transparently passed through the L2VPN to the remote endpoints.

## Multihoming

Figure 10: Multihoming



The illustration shows two edge routers on the same site connected to a switch. For an (instance-id + vc), one router is active and the other is on standby. (instance-id +vc) maps to a bridge domain and a bridge-domain maps to a VLAN (or a VLAN range).

The router on standby blocks bidirectional traffic for that VLAN.

Multihoming supports L2VPN configuration on up to two edge devices on the same site, thereby providing redundancy for L2VPN service over SD-WAN.

Multihoming allows an active-standby scenario where one device is chosen as active and the other as standby. This provides automated failover. It determines which of the two edge devices should be active and which

one should be on standby. When the OMP timer expires on the controller, it marks the L2VPN status route as stale, and notifies other edges.

### Active and Standby Device Role Determination

The active and standby roles are decided automatically based on the following algorithm:

$(\text{SDWAN-Instance-ID} + \text{VC-ID}) \bmod 2$

If the modular result is 0, the edge with lower system-ip is selected as the active device. The edge with the higher system-ip is selected as the standby device.

If the modular result is 1, the edge with higher system-ip is selected as the active device. The edge with the lower system-ip is selected as the standby device.

Example:

There are two WAN edge devices. WAN edge 1 has a system-ip of 172.16.255.10. WAN edge 2 has a system-ip of 172.16.255.11.

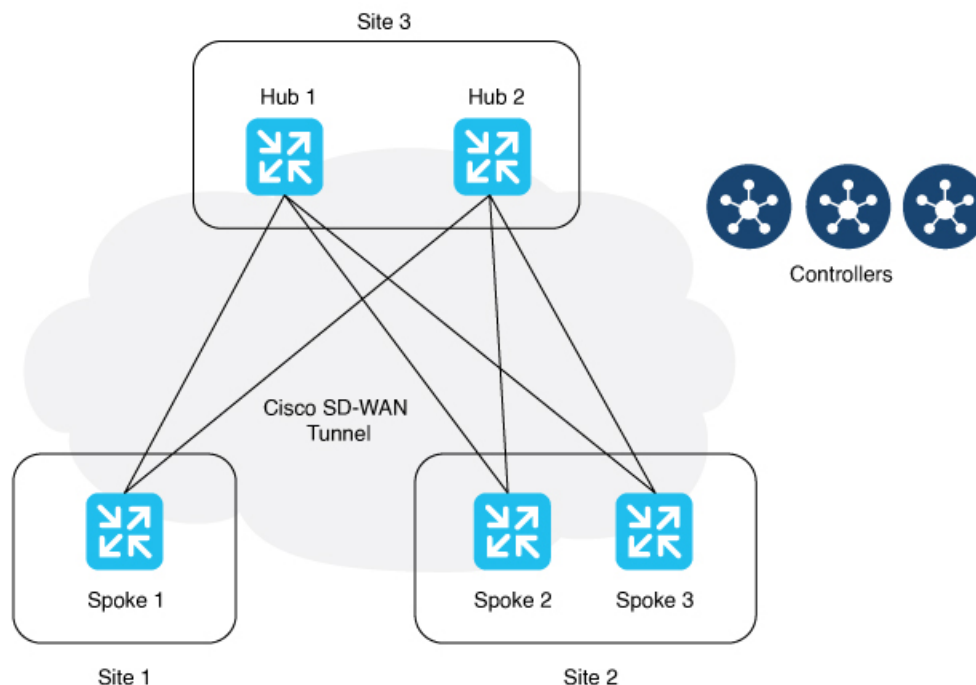
For sdwan-instance-id 100, vc-id 2, WAN edge 1 with the lower system-ip is selected as the active device. WAN edge 2 is the standby device.

For sdwan-instance-id 100, vc-id 1, WAN edge 2 with the higher system-ip is selected as the active device. WAN edge 1 is the standby device.

If a failure occurs on the service side of one of the edge devices, the controller is notified about a change to the L2VPN status route, and other edge routers can switchover traffic to the new active device.

## L2VPN Hub-and-Spoke Support

Minimum software releases: Cisco Catalyst SD-WAN Manager Release 20.15.1

**Figure 11: Hub-and-Spoke**

The preceding illustration shows P2MP Layer 2 VPN hub-and-spoke topology. In this configuration, spokes communicate with each other through the hubs. Layer 2 VPN hub-and-spoke supports Layer 2 connections using an indirect path, such as a hub.

You can enable Layer 2 VPN with only intent-based hub-and-spoke topology introduced in Cisco Catalyst SD-WAN Manager Release 20.12.1. It is used to build the hub-and-spoke topology in the network.

Layer 2 VPN hub-and-spoke supports P2MP. For more information about the intent-based hub-and-spoke feature, see [Hub-and-Spoke](#).

## Supported Platforms for Layer 2 VPN

Minimum software releases: Cisco IOS XE Catalyst SD-WAN Release 17.14.1a

All Cisco IOS XE Catalyst SD-WAN devices.

## Restrictions for Layer 2 VPN

Minimum software releases: Cisco IOS XE Catalyst SD-WAN Release 17.14.1a and Cisco Catalyst SD-WAN Control Components Release 20.14.1

- Only CLI template or CLI add-on template configuration is supported for Layer 2 VPN.
- For both single homing and multihoming, only one LAN side interface is supported in a bridge-domain.
- P2P configuration between two spokes is not supported. In such cases, use P2MP instead of P2P.



**Note** P2P configuration between hub and spoke is supported.

- Starting from Cisco IOS XE Catalyst SD-WAN Release 17.15.1a, multihoming only supports dual homing.
- Starting from Cisco IOS XE Catalyst SD-WAN Release 17.15.1a, hub-and-spoke topology is supported for Layer 2 VPN. It is limited by:
  - No support for Point-to-Point Layer 2 VPN service between spokes.
  - Support for up to 6000 spokes and 6000 sites within the same Layer 2 VPN in hub-and-spoke topology, and
  - Support for only 256 sites within the same Layer 2 VPN in a non-hub-and-spoke design.
- When upgrading to Cisco IOS XE Catalyst SD-WAN Release 17.15.1a or Cisco Catalyst SD-WAN Manager Release 20.15.1, you might experience minor outages on the Layer 2 VPN functionality until all participating edge routers and controllers are upgraded.
- Due to the change of the MAC learning mode from Flood and Learn in WAN and LAN to OMP protocol (Control Plane), there is no L2VPN interconnectivity between devices running both Cisco IOS XE Catalyst SD-WAN Release 17.14.1a and Cisco IOS XE Catalyst SD-WAN Release 17.15.1a.

## Configure Layer 2 VPN Using CLI Template

Follow these procedures to configure a Layer 2 VPN on a Cisco Catalyst SD-WAN overlay network.

- [Configure an L2VPN on a Cisco IOS XE Catalyst SD-WAN Device Using CLI Template, on page 1040](#)
- [Configure Point-to-Point Layer 2 VPN Using CLI Template, on page 1041](#)
- [Configure Point-to-Multipoint Layer 2 VPN Using CLI Template, on page 1044](#)
- [Configure Layer 2 VPN Switchport Using CLI Template, on page 1048](#)

## Configure an L2VPN on a Cisco IOS XE Catalyst SD-WAN Device Using CLI Template

### Before you begin

For more information about using CLI templates, see [CLI Add-On Feature Templates](#) and [CLI Templates](#).

### Procedure

**Step 1** Configure an L2VPN instance for P2P and P2MP connections.

```
l2vpn sdwan instance instance-id point-to-point
l2vpn sdwan instance instance-id multipoint
```

The instance ID is a unique identifier for each L2VPN connection, and must not overlap or be shared with any Layer 3 VRFs in the Cisco Catalyst SD-WAN fabric. For example, you cannot use L2VPN instance 10 and vrf definition 10.

**Step 2** Configure a bridge-domain.

```
bridge-domain bridge-id
```

**Step 3** Configure a Layer 2 interface on a Cisco IOS XE Catalyst SD-WAN device.

```
interface vlan-id
 service instance instance-id ethernet
 encapsulation dot1q vlan-id
 no shutdown
```

**Note**

A rewrite is used to modify the default VLAN tag. If you have not configured rewrite under service instance, dot1q must be the same at all sites participating in the Layer 2 network. The rewrite option in a Layer 2 configuration modifies the VLAN tags of packets as they ingress or egress an interface. To use the rewrite option, you need to configure Ethernet Virtual Connections (EVCs) on edge routers (Cisco ASR 1000 Series). For more information about configuring an EVC, see [Configuring Ethernet Virtual Connections on a Cisco Router](#).

---

## Configure Point-to-Point Layer 2 VPN Using CLI Template

### Before You Begin

- You can use one L2VPN instance ID for one or more bridge domains. It must be the same at both ends of the circuit.  
To identify a particular bridge-domain, use Virtual Circuit (VC) ID. This ID is the identifier of the virtual circuit between the Cisco IOS XE Catalyst SD-WAN devices.
- To create a P2P pseudowire, L2VPN instance ID, and VC ID must be the same on different Cisco IOS XE Catalyst SD-WAN devices.
- Remote-site-id is only supported for P2P configuration.

This following section provides the CLI configuration to configure P2P L2VPN services between two sites (Site A and Site B) on the Cisco Catalyst SD-WAN overlay network.

1. [Configure an Edge Router at Site A for Point-to-Point Layer 2 VPN Using CLI Template, on page 1042](#)
2. [Configure an edge Router at Site B for Point-to-Point Layer 2 VPN Using CLI Template, on page 1043](#)

For more information about using CLI templates, see [CLI Add-On Feature Templates](#) and [CLI Templates](#).



---

**Note** By default, CLI templates execute commands in global config mode.

---

# Configure an Edge Router at Site A for Point-to-Point Layer 2 VPN Using CLI Template

Site A uses an edge router and connects the Ethernet interface to the L2 network that bridges to Site B.

## Procedure

**Step 1** Define the L2VPN instance for point-to-point service:

```
l2vpn sdwan instance instance-id point-to-point
```

**Step 2** Configure the Ethernet interface:

```
interface interface-name
 service instance instance-id ethernet
 encapsulation dot1q vlan-id
```

**Step 3** Define the bridge domain and associate it with the interface and L2VPN instance:

```
bridge-domain bridge-id
 member vlan-name service-instance instance-id
 member sdwan instance instance-id remote-site remote-site-id vc-id virtual-circuit-id single-homing
```

Starting from Cisco IOS XE Catalyst SD-WAN Release 17.15.1a, you can specify the homing type as dual homing to enable multihoming.

```
bridge-domain bridge-id
 member vlan-name service-instance instance-id
 member sdwan instance instance-id remote-site remote-site-id vc-id virtual-circuit-id
 dual-homing
```

## Example

The following configures Site A using Cisco Catalyst 8000V Edge Software to manage traffic through GigabitEthernet5, which is linked to the Layer 2 network that provides connectivity to Site B.

```
l2vpn sdwan instance 100 point-to-point

interface GigabitEthernet5
 service instance 100 ethernet
 encapsulation dot1q 2002
 !
bridge-domain 100
 member GigabitEthernet5 service-instance 100
 member sdwan-instance 100 remote-site 502 vc-id 100 single-homing
```



# Configure an edge Router at Site B for Point-to-Point Layer 2 VPN Using CLI Template

Site B uses an edge router and Switchport Ethernet interface.

Follow these steps to configure an edge Router at Site B for point-to-point Layer 2 VPN.

## Procedure

**Step 1** Define the L2VPN instance for point-to-point service.

```
l2vpn sdwan instance instance-id point-to-point
```

**Step 2** Define the VLAN for the L2VPN.

```
vlan vlan-id
 name l2vpn
```

**Step 3** Configure the VLAN interface.

```
interface interface-name
 service instance instance-id ethernet
 encapsulation dot1q vlan-id
 no shutdown
```

**Step 4** Configure the Ethernet interface as an access port for VLAN.

```
interface interface-name
 switchport access vlan vlan-id
```

**Step 5** Define the bridge-domain for site B and associate it with the VLAN and L2VPN instance.

```
bridge-domain bridge-id
 member vlan-name service-instance instance-id
 member sdwan instance instance-id remote-site remote-site-id vc-id virtual-circuit-id single-homing
```

Starting from Cisco IOS XE Catalyst SD-WAN Release 17.15.1a, specify the homing type as dual homing to enable multihoming.

```
bridge-domain bridge-id
 member vlan-name service-instance instance-id
 member sdwan instance instance-id remote-site remote-site-id vc-id virtual-circuit-id
 dual-homing
```

**Example**

The following configures Switchport GigabitEthernet 0/1/7 at Site B to connect to the interface with a Cisco ISR1100-8P device.

```
l2vpn sdwan instance 100 point-to-point
vlan 2002
 name L2vpn
interface Vlan2002
 service instance 100 ethernet
 encapsulation dot1q 2002
 no shutdown
 !
interface GigabitEthernet 0/1/7
 switchport access vlan 2002
bridge-domain 100
 member Vlan2002 service-instance 100
 member sdwan-instance 100 remote-site 500 vc-id 100 single-homing
```

After configuring the point-to-point L2VPN service on both sites, you can integrate these configuration blocks into your CLI Template or CLI Add-On Feature Template. This template can then be used to deploy the configuration across the relevant devices in the Cisco Catalyst SD-WAN fabric. Verify the connectivity and functionality of the L2VPN service following the deployment to confirm that the bridge between site A and site B is operational.

## Configure Point-to-Multipoint Layer 2 VPN Using CLI Template

- For more information about using CLI templates, see [CLI Add-On Feature Templates](#) and [CLI Templates](#).  
By default, CLI templates execute commands in global config mode.
- One L2VPN instance ID can be used by one or more bridge domains. VC ID is used to identify a particular bridge-domain.
- L2VPN instance ID and VC ID must be the same on different edge devices.

This following section provides steps for configuring P2MP L2VPN over Cisco Catalyst SD-WAN overlay, connecting a local Layer 2 network at site A to multiple remote sites (B and C). Site A uses Gigabit Ethernet interface to connect to the Layer 2 network for bridging.

1. [Configure an Edge Router at Sites A, B, and C, on page 1044](#)
2. [Configure an Edge router at site B](#)
3. [Configure an Edge router at branch router site C](#)

## Configure an Edge Router at Sites A, B, and C

Site A is using an edge router, where an Ethernet interface is connected to the Layer 2 network that bridges to Site B and Site C.

Follow these steps to configure an Edge Router at Sites A, B, and C.

### Procedure

**Step 1** Define the L2VPN instance for the multipoint service on the data center router:

```
l2vpn sdwan instance instance-id multipoint
```

**Step 2** Configure the Ethernet interface on the data center router:

```
interface interface-name
service instance instance-id ethernet
encapsulation dot1q vlan-id
```

**Step 3** Define the bridge-domain on the data center route and associate it with the interface and L2VPN instance:

```
bridge-domain bridge-id
member vlan-name service-instance instance-id
member sdwan instance instance-id remote-site remote-site-id vc-id virtual-circuit-id
single-homing
```

Starting from Cisco IOS XE Catalyst SD-WAN Release 17.15.1a, specify the homing type as dual homing to enable multihoming.

```
bridge-domain bridge-id
member vlan-name service-instance instance-id
member sdwan instance instance-id remote-site remote-site-id vc-id virtual-circuit-id
dual-homing
```

## Configure an edge Router at Site B for Point-to-Point Layer 2 VPN Using CLI Template

Site B uses an edge router and Switchport Ethernet interface.

Follow these steps to configure an edge Router at Site B for point-to-point Layer 2 VPN.

### Procedure

**Step 1** Define the L2VPN instance for point-to-point service.

```
l2vpn sdwan instance instance-id point-to-point
```

**Step 2** Define the VLAN for the L2VPN.

```
vlan vlan-id
name l2vpn
```

**Step 3** Configure the VLAN interface.

```
interface interface-name
 service instance instance-id ethernet
 encapsulation dot1q vlan-id
 no shutdown
```

**Step 4** Configure the Ethernet interface as an access port for VLAN.

```
interface interface-name
 switchport access vlan vlan-id
```

**Step 5** Define the bridge-domain for site B and associate it with the VLAN and L2VPN instance.

```
bridge-domain bridge-id
 member vlan-name service-instance instance-id
 member sdwan instance instance-id remote-site remote-site-id vc-id virtual-circuit-id single-homing
```

Starting from Cisco IOS XE Catalyst SD-WAN Release 17.15.1a, specify the homing type as dual homing to enable multihoming.

```
bridge-domain bridge-id
 member vlan-name service-instance instance-id
 member sdwan instance instance-id remote-site remote-site-id vc-id virtual-circuit-id
 dual-homing
```

### Example

The following configures Switchport GigabitEthernet 0/1/7 at Site B to connect to the interface with a Cisco ISR1100-8P device.

```
l2vpn sdwan instance 100 point-to-point
vlan 2002
 name L2vpn
interface Vlan2002
 service instance 100 ethernet
 encapsulation dot1q 2002
 no shutdown
 !
interface GigabitEthernet 0/1/7
 switchport access vlan 2002
bridge-domain 100
 member Vlan2002 service-instance 100
 member sdwan-instance 100 remote-site 500 vc-id 100 single-homing
```

After configuring the point-to-point L2VPN service on both sites, you can integrate these configuration blocks into your CLI Template or CLI Add-On Feature Template. This template can then be used to deploy the configuration across the relevant devices in the Cisco Catalyst SD-WAN fabric. Verify the connectivity and functionality of the L2VPN service following the deployment to confirm that the bridge between site A and site B is operational.

# Configure an edge router at Site C for point-to-point Layer 2 VPN using CLI Template

## Before you begin

Repeat the same steps as for branch router C, substituting the specific interface used on site B.

## Procedure

- 
- Step 1** Define the L2VPN instance for multipoint service on the branch router:
- ```
l2vpn sdwan instance instance-id multipoint
```
- Step 2** Define the VLAN for the L2VPN on the branch router:
- ```
vlan vlan-id
name L2vpn
```
- Step 3** Configure the VLAN interface on the branch router:
- ```
interface interface-name  
service instance instance-id ethernet  
encapsulation dot1q vlan-id  
no shutdown
```
- Step 4** Configure the Ethernet interface on the branch router as an access port for VLAN:
- ```
interface interface-name
switchport access vlan vlan-id
```
- Step 5** Define the bridge-domain on the branch router and associate it with the VLAN and L2VPN instance:
- ```
bridge-domain bridge-id  
member vlan-name service-instance instance-id  
member sdwan instance instance-id vc-id virtual-circuit-id single-homing
```
- Starting from Cisco IOS XE Catalyst SD-WAN Release 17.15.1a, specify the homing type as dual homing to enable multihoming.
- ```
bridge-domain bridge-id
member vlan-name service-instance instance-id
member sdwan instance instance-id remote-site remote-site-id vc-id virtual-circuit-id
dual-homing
```
- 

## Example

This section provides an example configuration for P2MP L2VPN service within the Cisco Catalyst SD-WAN overlay network, connecting a local Layer 2 network at site A to multiple remote sites (B and C). Site A uses GigabitEthernet6 interface to connect to the L2 network for bridging.

Verify the connectivity and functionality of the P2MP L2VPN service and ensure that all sites are correctly bridged.

Site A is using a Cisco Catalyst 8000V edge router, where GigabitEthernet6 is connected to the Layer 2 network that bridges to site B and site C.

```
l2vpn sdwan instance 200 multipoint

vlan 2001
 name L2MPvpn

interface Vlan2001
 service instance 200 ethernet
 encapsulation dot1q 2001
 no shutdown
 !
interface GigabitEthernet 0/1/6
 switchport access vlan 2001

bridge-domain 200
 member Vlan2001 service-instance 200
 member sdwan-instance 200 vc-id 200 single-homing
```

Configure branch router C:

Repeat the same steps as for branch router B, substituting the specific interface used on router 503. In this example, we have used the GigabitEthernet 0/1/6 interface.

```
l2vpn sdwan instance 200 multipoint

vlan 2001
 name L2MPvpn

interface Vlan2001
 service instance 200 ethernet
 encapsulation dot1q 2001
 no shutdown
 !
bridge-domain 200
 member Vlan2001 service-instance 200
 member sdwan-instance 200 vc-id 200 single-homing
```

## Configure Layer 2 VPN Switchport Using CLI Template

If your device such as Cisco ISR1121-8P or similar has embedded switchports and you want to use one of them for the L2VPN services, configure a VLAN interface first and then assign that VLAN to your switchport as described in this section.

To support a Layer 2 switchport, configure a service instance in the VLAN interface. In the VLAN interface, a packet always has the dot1q tag even when the Layer 2 switchport is configured with switchport mode access. Therefore, the dot1q tag is mandatory in the service instance of the VLAN interface.

This following section provides steps to configure a Layer 2 switchport for P2MP (applicable for devices with embedded switchports). You can also configure a Layer 2 switchport for P2P by updating the Layer 2 VPN instance command.

Site A is using an edge router, where the Ethernet interface is connected to the Layer 2 network that bridges to Site B and Site C.

## Procedure

**Step 1** Define the Layer 2 VPN instance for multipoint service on the branch routers:

```
l2vpn sdwan instance instance-id multipoint
```

**Step 2** Define the VLAN for the Layer 2 VPN on the branch routers:

```
vlan vlan-id
name l2vpn
```

**Step 3** Configure the Ethernet interface on the routers:

```
interface interface-name
```

**Step 4** Set the switch port access VLAN and switchport mode to access to accept traffic only from the specified VLAN:

```
switchport access Vlan vlan-id
```

**Step 5** Configure the VLAN interface on a router and disable the IP address assignment

```
interface interface-name
no ip address
service instance instance-id ethernet
encapsulation dot1q vlan-id
```

**Step 6** Define the bridge-domain on the data center router and associate it with the interface and L2VPN instance:

```
bridge domain bridge-id
member vlan-name service-instance instance-id
member sdwan instance instance-id vc-id virtual-circuit-id single homing
```

## Example

The following configures a Layer 2 VPN Switchport to integrate a multipoint SD-WAN instance and bridge-domain. This configuration sets up GigabitEthernet0/1/2 as an access port for VLAN 201.

```
l2vpn sdwan instance 200 multipoint

interface GigabitEthernet0/1/2
 switchport access Vlan 201
 switchport mode access

interface Vlan201
 no ip address
 service instance 200 ethernet
 encapsulation dot1q 201
 !

bridge-domain 201
 member Vlan201 service-instance 200
 member sdwan-instance 200 vc-id 201 single-homing
```

## Verify Layer 2 VPN Using CLI

Follow these procedures to verify a Layer 2 VPN configuration on a Cisco Catalyst SD-WAN overlay network.

1. [View a Layer 2 VPN Status, on page 1050](#)
2. [View L2VPN Information Learned Through OMP Route on a Cisco Catalyst SD-WAN Controller, on page 1050](#)
3. [View Bridge-Domain Information, on page 1052](#)
4. [View Cisco Catalyst SD-WAN Flood List Information and Packet Counters in Data Plane, on page 1052](#)
5. [View Packet Counters in Data Plane, on page 1053](#)

### View a Layer 2 VPN Status

Minimum Supported Releases: Cisco IOS XE Catalyst SD-WAN Release 17.14.1a, Cisco Catalyst SD-WAN Manager Release 20.14.1

#### Procedure

Use the **show l2vpn sdwan [instance *instance-id*][vc-id *vc-id*]** command to view the remote peer information, system IP, status, and so on.

#### Example

The following example is for a Cisco IOS XE Catalyst SD-WAN device.

```
Device# show l2vpn sdwan instance 13 vc-id 13
VC_ID: 13 Bridge-domain: 13
Local l2vpn status: UP
Local Pseudoports: GigabitEthernet7 service instance 13
```

### View L2VPN Information Learned Through OMP Route on a Cisco Catalyst SD-WAN Controller

#### Procedure

Use the **show sdwan omp l2-routes[vpn *vpn-id*] [vc-id *vc-id*]** command shows the specific L2-route or path learned in the specific VPN and virtual circuit. If the **vpn** and **vc-id** are not included, the command shows Layer 2 routes learned through OMP from all VPNs across the Cisco Catalyst SD-WAN fabric.



## Example

The following is a sample output from the **show omp l2-routes** command displaying Layer 2 routes learned through OMP for Cisco Catalyst SD-WAN Controllers.

Device# **show omp l2-routes | tab**

C -> chosen  
 I -> installed  
 Red -> redistributed  
 Rej -> rejected  
 L -> looped  
 R -> resolved  
 S -> stale  
 Ext -> extranet  
 Inv -> invalid  
 Stg -> staged  
 IA -> On-demand inactive  
 U -> TLLOC unresolved

| VPN           | VC ID | PATH ID | ORIGINATOR LABEL | STATUS | REMOTE ROUTE SITE |                | IP ADDRESS | VPN TYPE   | SITE ID |
|---------------|-------|---------|------------------|--------|-------------------|----------------|------------|------------|---------|
|               |       |         |                  |        | TYPE              | MAC ADDRESS    |            |            |         |
| 12            | 12    |         | 172.16.255.15    | vpn    |                   | 0000.0000.0000 | ::         | p2p        | 500     |
| 172.16.255.15 |       | 66      | 1004             | C,R    |                   | 501            |            |            |         |
| 172.16.255.15 |       | 69      | 1004             | C,R    |                   | 501            |            |            |         |
| 172.16.255.20 |       | 1       | 1004             | C,R    |                   | 501            |            |            |         |
| 172.16.255.20 |       | 2       | 1004             | C,R    |                   | 501            |            |            |         |
| 12            | 12    |         | 172.16.255.27    | vpn    |                   | 0000.0000.0000 | ::         | p2p        | 501     |
| 172.16.255.20 |       | 1       | 1014             | C,R    |                   | 500            |            |            |         |
| 172.16.255.27 |       | 70      | 1014             | C,R    |                   | 500            |            |            |         |
| 13            | 13    |         | 172.16.255.15    | vpn    |                   | 0000.0000.0000 | ::         | multipoint | 500     |
| 172.16.255.15 |       | 66      | 1006             | C,R    |                   | -              |            |            |         |
| 172.16.255.15 |       | 69      | 1006             | C,R    |                   | -              |            |            |         |
| 172.16.255.20 |       | 1       | 1006             | C,R    |                   | -              |            |            |         |
| 172.16.255.20 |       | 2       | 1006             | C,R    |                   | -              |            |            |         |
| 13            | 13    |         | 172.16.255.27    | vpn    |                   | 0000.0000.0000 | ::         | multipoint | 501     |
| 172.16.255.20 |       | 1       | 1016             | C,R    |                   | -              |            |            |         |
| 172.16.255.27 |       | 70      | 1016             | C,R    |                   | -              |            |            |         |
| 13            | 13    |         | 172.16.255.32    | vpn    |                   | 0000.0000.0000 | ::         | multipoint | 503     |
| 172.16.255.20 |       | 1       | 1007             | C,R    |                   | -              |            |            |         |
| 172.16.255.32 |       | 71      | 1007             | C,R    |                   | -              |            |            |         |
| 14            | 1     |         | 172.16.255.27    | vpn    |                   | 0000.0000.0000 | ::         | multipoint | 501     |
| 172.16.255.20 |       | 1       | 1018             | C,R    |                   | -              |            |            |         |
| 172.16.255.27 |       | 70      | 1018             | C,R    |                   | -              |            |            |         |
| 15            | 1     |         | 172.16.255.15    | vpn    |                   | 0000.0000.0000 | ::         | p2p        | 500     |
| 172.16.255.15 |       | 66      | 1020             | C,R    |                   | 501            |            |            |         |
| 172.16.255.15 |       | 69      | 1020             | C,R    |                   | 501            |            |            |         |

```

172.16.255.20 1 1020 C,R 501

172.16.255.20 2 1020 C,R 501
15 1 172.16.255.27 vpn 0000.0000.0000 :: p2p 501
172.16.255.20 1 1020 C,R 500

172.16.255.27 70 1020 C,R 500

```

## View Bridge-Domain Information

### Procedure

Use the **show platform software sdwan ftmd bridge-domain** command on a device to verify information related to bridge domains within the context of Forwarding Table Management Daemon (FTMD).

### Example

The following is a sample output from the **show platform software sdwan ftmd bridge-domain** command that displays information related to bridge domains within the context of Forwarding Table Management Daemon (FTMD).

```

Device# show platform software sdwan ftmd bridge-domain
L2vpn Bridge-domain 12 Table:
 sdwan efp dpidx: 4210708(0x404014)
 Label: 1004 lbl-nhop-id: 196611 (binosId=0xf830003f)
 Bum Label: 1005 bum-lbl-nhop-id: 196612 (binosId=0xf830004f)
 Remote Site Table(1 entries in total):
 remote-site-id: 501 sla-nhop-id: 29 (binosId=0xf80001df)

L2vpn Bridge-domain 13 Table:
 sdwan efp dpidx: 4210709(0x404015)
 Label: 1006 lbl-nhop-id: 196613 (binosId=0xf830005f)
 Bum Label: 1007 bum-lbl-nhop-id: 196614 (binosId=0xf830006f)
 Remote Site Table(2 entries in total):
 remote-site-id: 501 sla-nhop-id: 30 (binosId=0xf80001ef)
 remote-site-id: 503 sla-nhop-id: 33 (binosId=0xf800021f)

```

## View Cisco Catalyst SD-WAN Flood List Information and Packet Counters in Data Plane

### Procedure

Use the **show platform hardware qfp active feature bridge-domain datapath bridge-domain-id sdwan-flood-list** command to verify information related to Cisco Catalyst SD-WAN flood list information.

### Example

The following is a sample output from the **show platform hardware qfp active feature bridge-domain datapath bridge-domain-id sdwan-flood-list** command that displays the Cisco Catalyst SD-WAN flood list information.

```
Device#show platform software sdwan ftmd bridge-domain
L2vpn Bridge-domain 12 Table:
 sdwan efp dpidx: 4210708(0x404014)
 Label: 1004 lbl-nhop-id: 196611 (binosId=0xf830003f)
 Bum Label: 1005 bum-lbl-nhop-id: 196612 (binosId=0xf830004f)
 Remote Site Table(1 entries in total):
 remote-site-id: 501 sla-nhop-id: 29 (binosId=0xf80001df)

L2vpn Bridge-domain 13 Table:
 sdwan efp dpidx: 4210709(0x404015)
 Label: 1006 lbl-nhop-id: 196613 (binosId=0xf830005f)
 Bum Label: 1007 bum-lbl-nhop-id: 196614 (binosId=0xf830006f)
 Remote Site Table(2 entries in total):
 remote-site-id: 501 sla-nhop-id: 30 (binosId=0xf80001ef)
 remote-site-id: 503 sla-nhop-id: 33 (binosId=0xf800021f)
```

## View Packet Counters in Data Plane

### Procedure

Use the **show platform hardware qfp active feature bridge-domain datapath bridge-id** command to verify information related to a QuantumFlow Processor (QFP) hardware module packet counters for a specific bridge domain within the data path.

### Example

The following is a sample output from the **show platform hardware qfp active feature bridge-domain datapath bridge-id** command to display a QFP hardware module packet counters for a specific bridge domain within the data path.

```
Device# show platform hardware qfp active feature bridge-domain datapath 200
QFP L2BD Bridge Domain information
```

```
BD id : 200
State enabled : Yes
Aging timeout (sec) : 300
Aging active entry : Yes
Max mac limit : 65536
Unkwn mac limit flood : Yes
```

```

mac_learn_enabled : Yes
mac_learn_controlled : No
Unknown unicast olist : Yes

otv_aed_enabled : No
otv_enabled : No
mcast_snooping_enabled : No

Feature : sdwan

SISF snoop protocols : None

Sdwan instance id : 200

Mac learned : 0

BDI outer vtag : 00000000

BDI inner vtag : 00000000

```

#### Replication tree info:

```

Global replication : depth encode 0X1000001, (head 0XE4E90000)

Split-horizon-group 0 : depth encode 00000000, (head 00000000)

Split-horizon-group 1 : depth encode 00000000, (head 00000000)
Bridge Domain statistics

```

|                              |          |          |
|------------------------------|----------|----------|
| Total bridged                | pkts : 0 | bytes: 0 |
| Total unknown unicast        | pkts : 0 | bytes: 0 |
| Total broadcasted            | pkts : 0 | bytes: 0 |
| Total to BDI                 | pkts : 0 | bytes: 0 |
| Total injected               | pkts : 0 | bytes: 0 |
| Total mac-sec violation drop | pkts : 0 | bytes: 0 |
| Total mac-sec move drop      | pkts : 0 | bytes: 0 |
| Total mac-sec unknown drop   | pkts : 0 | bytes: 0 |
| Total source filter drop     | pkts : 0 | bytes: 0 |
| Total bfib policy drop       | pkts : 0 | bytes: 0 |
| Total replication start drop | pkts : 0 | bytes: 0 |
| Total recycle tail drop      | pkts : 0 | bytes: 0 |
| Total static MAC move drop   | pkts : 0 | bytes: 0 |
| Total BD disabled drop       | pkts : 0 | bytes: 0 |

```

Total STP state drop pkts : 0 bytes: 0
Total UUF suppression drop pkts : 0 bytes: 0
Total sisf ctrl punt pkts : 0 bytes: 0
Total sisf ctrl drop pkts : 0 bytes: 0
Total p2p lan to wan pkts : 0 bytes: 0
Total p2p wan to lan pkts : 0 bytes: 0

```

## Monitor Configured Layer 2 VPN Using CLI

The following is a sample output from the **show l2vpn sdwan all** command. The following examples show the configuration and status information for L2VPN instances within a Cisco Catalyst SD-WAN overlay network. The output includes details for both point-to-point (P2P) and point-to-multipoint (P2MP) topologies.

### Example 1

```

Device#show l2vpn sdwan all
L2VPN sdwan Instance : 100
VPN Type : point-to-point
 VC_ID: 100 Bridge-domain: 100 UP
 Local l2vpn status: UP
 Local Pseudoports: GigabitEthernet5 service instance 100
 Remote Site: 53
 System IP status up/down color encap label DF
 10.100.31.53 DOWN 00:15:04 public-internet ipsec 1023 N/A

```

### Example 2

```

Device#show l2vpn sdwan all
L2VPN sdwan Instance : 200
VPN Type : multipoint
IP Local-learning : Disabled
Flooding Suppression : Disabled
 VC_ID: 200 Bridge-domain: 200 UP
 Local l2vpn status: UP
 Local Pseudoports: GigabitEthernet5 service instance 200
 Remote Site: 50
 System IP status up/down color encap label DF
 10.100.31.50 UP 00:04:14 public-internet ipsec 1008 N/A

 Remote Site: 53
 System IP status up/down color encap label DF
 10.100.31.53 UP 00:15:00 public-internet ipsec 1025 N/A

```

The following is a sample output from the **show l2vpn sdwan instance instance-id vc-id vc-idpeers** command. The following examples show information about a specific Cisco Catalyst SD-WAN L2VPN instance (instance 200) and its associated virtual circuit (vc-id 200), including details about its peer connections.

```
show l2vpn sdwan instance instance-id vc-id vc-id peers
```

### Example 1

```
Device1#show l2vpn sdwan instance 200 vc-id 200 peers
```

```
Remote Site: 50 MACs Learn: 0
```

| System IP    | status | up/down  | color           | encap | label | DF  |
|--------------|--------|----------|-----------------|-------|-------|-----|
| 10.100.31.50 | UP     | 00:19:54 | public-internet | ipsec | 1008  | N/A |

```
Remote Site: 53 MACs Learn: 0
```

| System IP    | status | up/down  | color           | encap | label | DF  |
|--------------|--------|----------|-----------------|-------|-------|-----|
| 10.100.31.53 | UP     | 00:30:40 | public-internet | ipsec | 1025  | N/A |

## Example 2

```
Device#show l2vpn sdwan instance 200 vc-id 200 peers
```

```
Remote Site: 1 MACs Learn: 0
```

| System IP   | status | up/down  | color           | encap | label | DF  |
|-------------|--------|----------|-----------------|-------|-------|-----|
| 10.100.31.1 | UP     | 00:30:13 | public-internet | ipsec | 1014  | N/A |



## CHAPTER 43

# Troubleshoot Cisco Catalyst SD-WAN Systems and Interfaces

- [Overview, on page 1057](#)
- [Support Articles, on page 1057](#)
- [Feedback Request, on page 1059](#)
- [Disclaimer and Caution, on page 1059](#)

## Overview

This chapter provides links to documents authored by Cisco subject matter experts (SMEs). They aim to help you resolve technical issues without requiring a support ticket. If these documents are unable to resolve your issue, we recommend visiting the applicable [Cisco Community](#). There is a wealth of information and advice available from fellow Cisco customers who may have experienced this issue already and provided a solution. If you are not able to find a resolution on the Community, it may be best that you raise a support ticket at [Cisco Support](#). In cases where a support ticket has to be raised, these documents provide guidance about the data that should be collected and added to the support ticket. Specify the support document you referred, and TAC can create an improvement request with the document owner.

## Support Articles

The documents in this section were created using specific software and hardware listed in the Components Used section of each article. However, this does not mean that they are limited to what is listed in Components Used, and generally remain relevant for later versions of software and hardware. Note that there could be some changes in the software or hardware that can cause commands to stop working, the syntax to change, or GUIs and CLIs to look different from one release to another.

The following are the support articles associated with this technology:

| Document                                                                          | Description                                                                                  |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| <a href="#">Collect an Admin-Tech in SDWAN Environment and Upload to TAC Case</a> | This document describes how to initiate an <code>admin-tech</code> in an SD-WAN environment. |

| Document                                                                                   | Description                                                                                                                                                                |
|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Configure a Custom Cisco SD-WAN Manager Application Server Logo</a>            | This document describes the process to modify Cisco Catalyst SD-WAN Manager application server logo images. The change is made from the Cisco Catalyst SD-WAN Manager CLI. |
| <a href="#">Configure and Troubleshoot a DHCP Server on Cisco IOS XE SDWAN Router</a>      | This document describes how to configure and troubleshoot a DHCP Server on a Cisco SD-WAN IOS XE Router.                                                                   |
| <a href="#">Configure and Verify SD-WAN On-demand Tunnels</a>                              | This document describes configuration and verification steps to create SD-WAN On-demand Tunnels.                                                                           |
| <a href="#">Configure Banner Feature Templates with Special Characters</a>                 | This document describes the use of banner feature templates for the generation of banner and message of the day (MOTD) text blocks in Cisco IOS XE.                        |
| <a href="#">Configure Basic Parameters to Form Control Connections on cEdge</a>            | This document describes the basic configuration and correct commit order to onboard a Cisco IOS XE Catalyst SD-WAN device to a Cisco Catalyst SD-WAN overlay.              |
| <a href="#">Configure IPsec and GRE in the Same Tunnel Interface on XE SD-WAN</a>          | This document describes the configuration to enable IPsec and GRE encapsulation for the same tunnel interface on a Cisco IOS XE SD-WAN Router.                             |
| <a href="#">Configure L3 TLOC Extension</a>                                                | This video describes how to configure L3 TLOC Extension on SD-WAN.                                                                                                         |
| <a href="#">Configure Layer 3 TLOC Extension</a>                                           | This document describes how to configure TLOC-Extension Layer 3(L3) on Cisco SD-WAN.                                                                                       |
| <a href="#">Configure SD-WAN Edge Router for Inline Deployment</a>                         | This document describes how to configure Cisco SD-WAN Edge with MPLS transport to access Cisco SD-WAN Controller on Internet via inline DC WAN Edge.                       |
| <a href="#">Configure ThousandEyes on SD-WAN Devices</a>                                   | This video provides the procedure to configure ThousandEyes on SD-WAN Devices.                                                                                             |
| <a href="#">Configure Thousand Eyes on SD-WAN Devices</a>                                  | This document describes how to integrate ThousandEyes Endpoint Agent on Cisco Catalyst SD-WAN.                                                                             |
| <a href="#">Configure Host Entry for Cisco SD-WAN Validator</a>                            | This document describes the procedure to configure host entry for Cisco Catalyst SD-WAN Validator.                                                                         |
| <a href="#">Configure TLOC-Extension Using Cisco SD-WAN Manager Feature Template</a>       | This document describes how to configure TLOC-Extension using Cisco SD-WAN Manager feature template.                                                                       |
| <a href="#">Create a Cisco SD-WAN Controller CLI Template to Push a Centralized Policy</a> | This document describes an easy way to create a CLI Template for Cisco Catalyst SD-WAN Controllers as they are needed to push a Centralized Policy for the overlay.        |
| <a href="#">Perform Password Recovery with a Template on SD-WAN</a>                        | This document describes the steps for password recovery of a device with a template in Cisco Catalyst SD-WAN environment.                                                  |



| Document                                                                                     | Description                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Quick Start Guide - Data Collection for Various SD-WAN Issues</a>                | This document describes several Cisco Catalyst SD-WAN issues along relevant data that must be collected in advance before you open a TAC case to improve the speed of troubleshooting and/or problem resolution. |
| <a href="#">Troubleshoot SD-WAN Control Connections</a>                                      | This document describes some of the probable causes that lead to a problem with Control Connections and how to troubleshoot them.                                                                                |
| <a href="#">Troubleshoot SD-WAN Dynamic On-Demand Tunnels</a>                                | This document describes troubleshoot commands that can be used when configuring or checking an issue related to SD-WAN dynamic on-demand tunnels.                                                                |
| <a href="#">Troubleshoot "Unable to Validate Proxy Server" Error in Cisco SD-WAN Manager</a> | This document describes the "Failed to update setting Invalid request. unable to validate proxy server" error in Cisco SD-WAN Manager and how to resolve it.                                                     |
| <a href="#">Understand NTP Association Codes in SD-WAN Controllers</a>                       | This document describes how to understand NTP association status codes on SD-WAN controllers.                                                                                                                    |
| <a href="#">Verify and Identify Packet Loss in the WAN for SD-WAN</a>                        | This document describes how to identify and collect data when traffic has loss across the WAN but no drops are seen on the SD-WAN Edge.                                                                          |
| <a href="#">Cellular TLOC failing</a>                                                        | This document describes how to troubleshoot a failing cellular TLOC on a device that has both cellular and non-cellular interfaces.                                                                              |

## Feedback Request

Your input helps. A key aspect to improving these support documents is customer feedback. Note that these documents are owned and maintained by multiple teams within Cisco. If you find an issue specific to the document (unclear, confusing, information missing, etc):

- Provide feedback using the **Feedback** button located at the right panel of the corresponding article. The document owner will be notified, and will either update the article, or flag it for removal.
- Include information regarding the section, area, or issue you had with the document and what could be improved. Provide as much detail as possible.

## Disclaimer and Caution

The information in this document was created from the devices in a specific lab environment. All of the devices used in this document started with a cleared (default) configuration. If your network is live, ensure that you understand the potential impact of any command.





## CHAPTER 44

# Appendix: Cisco SD-WAN Manager How-Tos

---

- [How to Load a Custom Cisco SD-WAN Manager Application Server Logo, on page 1061](#)

## How to Load a Custom Cisco SD-WAN Manager Application Server Logo

To change the Cisco SD-WAN Manager web application server logo and load a new custom logo, use the **request nms application-server update-logo** command.

The logo image is located in the upper left corner of all Cisco SD-WAN Manager web application server screens. You can load two files, a larger version, which is displayed on wider browser screens, and a smaller version, which is displayed when the screen size narrows. Both files must be PNG files located on the local device, and both must be 1 MB or smaller in size. For best resolution, it is recommended that the image for the large logo be 180 x 33 pixels, and for the small logo 30 x 33 pixels.

